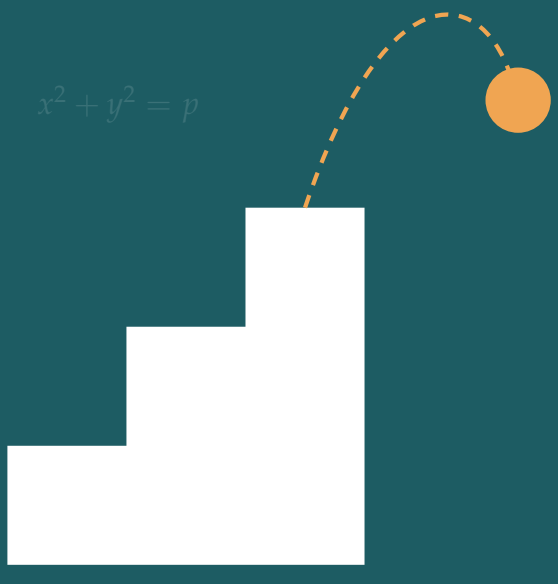
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

Past STEP questions as a doorway to university mathematics



KEITH McNULTY

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Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

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Introduction

> *I can't teach you mathematics with this book, but I believe that much hard work on your part, chipping away at the problems, will eventually reveal the mathematician that is within you.*
>
> — Stephen Siklos, *Advanced Problems in Mathematics*

This is a book of hard problems, and of what lies behind them. Every one of its 136 sections starts with a question from a past STEP paper. But the question is only the way in. Behind almost every good STEP question is a piece of mathematics that you will meet properly at university, and the aim of this book is to take you through the door and show you what is on the other side.

What STEP is, and why it exists

STEP, the Sixth Term Examination Paper, is a set of mathematics examinations taken in the summer of the final year of school. It was introduced by the University of Cambridge in 1987, and its job is to help decide who should study mathematics there: almost all Cambridge offers for Mathematics are conditional on STEP grades, and over the years other universities, notably Warwick, have used it too. At the start there were STEP papers in many subjects, but by 2001 only mathematics remained. For most of its history there were three papers, I, II and III; Paper I was last set in 2019, and today candidates take Papers II and III. The examination was run for many years by Cambridge Assessment, and since 2024 it has been administered by OCR.

A STEP paper looks nothing like a school examination. Instead of ten short, compulsory questions, there are a dozen or so long ones, of which candidates attempt six, and a strong candidate who makes real progress on four will do very well. The questions do not tell you how many marks each part is worth, and they rarely ask you to do something you have practised. They ask you to take the mathematics you already know and use it somewhere new.

Stephen Siklos, who knew STEP better than anyone, described its purpose like this. It is a hurdle: success in STEP turns out to predict success in a university mathematics degree remarkably well (he reported a correlation of 0.63 between STEP rank and first-year Cambridge examination rank, very high by the standards of such things). It is a preparation: the style of thinking that STEP rewards is the style of undergraduate mathematics. And it is a test of motivation: preparing for STEP takes real effort, and a student who is not willing to make that effort is unlikely to enjoy a demanding mathematics degree.

Stephen Siklos

It is impossible to write about STEP without writing about Stephen Siklos. He was a Cambridge mathematician, a former research student of Stephen Hawking who worked on the mathematics of the early universe and of gravitational waves, a lecturer in the Department of Applied Mathematics and Theoretical Physics, and later Senior Tutor and then President of Jesus College. He became the Senior Examiner for STEP Mathematics when it began in 1987, and held the post for thirty-two years, until his death in August 2019. In that time he drafted questions, "knocked other people's questions into shape" (his own phrase, for something he said had long been a source of interest and entertainment), scrutinised every paper and oversaw the marking. He was also the driving force behind the STEP Easter School and the STEP Support Programme, which give free help to students whose schools cannot. One obituary said that he did more than anyone to make the Cambridge entrance examination in mathematics accessible to students whatever their school background.

His book *Advanced Problems in Mathematics: Preparing for University*, published free online by Open Book Publishers, is still the best introduction to STEP, and I strongly recommend it. Two things he wrote in it have shaped this book. The first is his view of what a good STEP question is: one in which the first part shows you an unfamiliar idea, and the later parts ask you to show that you have understood it by using it again in a harder setting; or one whose answer depends on a parameter that you have to discover for yourself; or one that starts with special cases and ends with a general result. Those are exactly the questions that lead somewhere, and most of the questions in this book are of those kinds. The second is his insistence that solutions should be "springboards rather than feather beds". You learn nothing by reading a solution you have not first fought with yourself.

How I met STEP

I first came across STEP in the early 1990s, when my mathematics teacher gave me some of the papers as an extra challenge. I worked through them for the fun of it rather than for an offer, and never sat the real examination, but I remember those papers as being extraordinarily hard: questions that seemed to have nothing to do with anything I had learned, that went on for a page, and that I could start but rarely finish. I also remember the feeling, on the occasions when something finally worked, that I had seen something that school mathematics had not shown me.

Many years later I came back to STEP from the other side, helping students to prepare for it. Working through hundreds of questions with them, I came to see the papers differently. Not every question is a gem. Some are, frankly, long tests of stamina: a page of careful algebra, a demanding but routine calculation, with not much insight or pleasure at the end. Stamina matters, and university mathematics needs it, but those questions are not what make STEP special.

What makes it special is that, scattered across the more than 1,500 questions set since 1987, there are hundreds of genuinely beautiful ones. The best way to explain what I mean is to show you some.

Some gems

Here are seven of the questions you will meet later in the book, chosen to span its range: two from combinatorics, two from dynamical systems, two from mechanics and one from probability. Each starts with something you can picture, ends with something you might not have believed, and opens onto a piece of mathematics that is studied seriously at university.

Order hiding in chaos. Draw six dots on a page and join every pair with a line, fifteen lines in all, colouring each line gold or silver however you like. Now try to avoid creating a triangle whose three sides are all the same colour. You will fail, every time. With five dots it can be done (colour the sides of a pentagon gold and the five diagonals silver), but with six it is impossible, however cunning your colouring. A 1989 STEP I question asks you to prove this, with a single hint: look at the five lines that leave one dot. Three of them must share a colour, and from there the proof takes a few lines. There are $2^{15} = 32\,768$ ways to colour the lines, and the argument deals with all of them at once, without checking a single case.

The result is the smallest case of Ramsey's theorem, proved by Frank Ramsey in 1928, which says that complete disorder is impossible: any large enough structure, however haphazardly it is put together, contains a pocket of perfect order. How large is "large enough"? To guarantee four dots joined entirely in one colour you need 18 dots, a fact proved in 1955. For five dots, nobody knows: the answer lies somewhere between 43 and 46, and the upper bound was proved only in 2024. Paul Erdős joked that if aliens threatened to destroy the Earth unless we told them this number, we should set every computer and mathematician to work on it; but if they asked for the answer for six, we should try to destroy the aliens. You will meet Ramsey theory in a university combinatorics or graph theory course, and the pigeonhole principle that powers the STEP proof in almost every course you take (Section 9.2).

Strange dice. With two ordinary dice, a total of 7 is the most likely and totals of 2 and 12 the least, in the familiar pattern of $1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1$ ways. Is there any other way to number a pair of dice with positive whole numbers that gives exactly the same chances for every total? It seems unlikely, but there is exactly one: $1, 2, 2, 3, 3, 4$ on one die and $1, 3, 4, 5, 6, 8$ on the other. These are *Sicherman dice*, named after George Sicherman, who found them, and made famous by Martin Gardner in 1978. In any game that depends only on the total thrown, you could swap them for ordinary dice and nobody would ever know.

A 1993 question approaches the idea from the other end. It gives you a table showing how many ways each total can be thrown with two oddly numbered dice, and asks you to work out the numbering, like a detective reconstructing a crime from its traces. The section then shows the idea that makes the whole subject simple: turn each die into a polynomial, with a term x^k for each face showing k. Throwing two dice corresponds to multiplying their polynomials, because $x^i \times x^j = x^{i+j}$ adds the numbers on the faces. Finding new dice with the same totals becomes a matter of factorising a polynomial in a different way, rather as 36 can be split as 4×9 instead of 6×6. And because polynomials, like whole numbers, factorise in only one way into pieces that cannot be

broken down further, the factorisation also proves that Sicherman's dice are the only alternative. This trick of encoding a counting problem in a polynomial, a *generating function*, is one of the most powerful tools in combinatorics, and the unique factorisation behind it is a central theme of a first university course in algebra (Section 9.6).

Chaos with a formula. Start with a number between 0 and 1, multiply it by 4 and by one minus itself, and repeat: $u_{n+1} = 4u_n(1 - u_n)$. This is the *logistic map*, a crude model of a population that grows but is limited by its food supply. The rule could hardly be simpler, yet the numbers it produces jump around with no visible pattern, and two starting values that agree to ten decimal places typically give completely different sequences within a few dozen steps. The biologist Robert May drew attention to it in 1976, in a famous paper with the title "Simple mathematical models with very complicated dynamics", and it became the standard example of chaos, usually explored by computer.

A 2014 question reveals something unexpected: the sequence has an exact formula. If the starting value is $\sin^2\theta$, then $u_n = \sin^2(2^n\theta)$, which you can prove with nothing more than the double angle formulae and induction. (The formula was known to Stanisław Ulam and John von Neumann in the 1940s.) You might expect an exact formula to tame the chaos. Instead it explains it completely. Each step doubles the angle, so any tiny uncertainty in the angle doubles at every step, and errors in the starting value grow just as fast on average: after fifty steps an error has grown about a million billion times. Writing the angle, as a fraction of π, in binary makes this vivid: each step of the map shifts the binary digits one place to the left and throws away the first, so the sequence reads off the digits of the starting value one by one, and to predict a hundred steps ahead you need to know the starting value to a hundred binary places. The same formula also finds every cycle of the map, and shows that there are cycles of every length, packed densely into $[0, 1]$, every one of them unstable. This is the subject of dynamical systems, which you will meet in university courses on differential equations and, later, in courses on chaos itself (Section 8.7).

Period three implies chaos. Take the function $f(x) = 7 - 2|x|$, start with a number, and keep feeding the answer back in. Starting at 1 gives $1, 5, -3, 1, 5, -3, \ldots$: a cycle of length three. A 2025 question asks you to hunt for cycles like this one, by sketching the graphs of f, $f(f(x))$ and $f(f(f(x)))$ and finding where they cross the line $y = x$. Each extra application of f doubles the number of zigzags in the graph, and the section turns that observation into a proof that this function has cycles of *every* length: two, three, four, a hundred, a million.

That is not a special feature of this function. In 1975 Tien-Yien Li and James Yorke proved that *any* continuous function with a cycle of length three must have cycles of every other length, in a paper with the irresistible title "Period three implies chaos". The proof needs nothing more than the intermediate value theorem, which you already use whenever you argue that a continuous function that changes sign must cross zero, together with a clever diagram showing which intervals the function stretches over which. It later turned out that a Ukrainian mathematician, Oleksandr Sharkovsky, had proved a stronger result in 1964: he found a strange ordering of the whole numbers, beginning $3, 5, 7, 9, \ldots$, such that a cycle of any length forces cycles of every length that

comes after it. The intermediate value theorem is one of the first results proved carefully in a university analysis course; this question shows how much it can do (Section 8.8).

Why a bicycle pump gets warm. Pump up a bicycle tyre and then feel the pump: it is warm. Squeezing a gas heats it, which is how a diesel engine ignites its fuel without a spark. But why? A 2002 question answers this by following a single air molecule as it bounces back and forth between the end of the pump and the piston being pushed in. Each time the molecule bounces off the advancing piston, it picks up a little speed, like a tennis ball hit by a moving racket. The question asks you to describe one bounce exactly, turn that into a recurrence for the length of the cylinder at each collision, and then solve the recurrence.

The solution reveals a hidden law: the molecule's speed multiplied by the length of the cylinder hardly changes. Halve the length and you double the speed, and since temperature measures the energy of the molecules, the gas heats up. Quantities like this, which stay almost constant when a system is changed slowly, are called *adiabatic invariants*, and with a little more work the same argument leads to the law of adiabatic compression of a gas that you will meet in university thermodynamics. The idea reaches a long way. Enrico Fermi suggested in 1949 that cosmic rays reach their enormous energies in exactly this way, by bouncing repeatedly between moving magnetic fields in space; and Paul Ehrenfest argued in 1916 that adiabatic invariants are the quantities that quantum mechanics must quantise. A question about a bicycle pump turns out to touch statistical physics, astrophysics and the beginnings of quantum theory (Section 10.4).

A ball that comes back. Throw a ball gently forwards and it moves steadily away from you until it lands. Common sense says this is always so: the ball is always moving forwards, so how could it get closer? But throw it steeply enough, at more than about 70.5° to the horizontal, and something odd happens. For part of its flight, as it comes down, the ball gets *closer* to you, even though it is still moving away horizontally. Its height is falling faster than its horizontal distance is growing.

A 2019 question asks you to prove this and to find the critical angle exactly. The method is one you will use again and again: instead of the distance itself, differentiate its square, which avoids square roots; the derivative turns out to be a quadratic in time, and the question of whether the distance ever decreases becomes the question of whether that quadratic has real roots. Its discriminant gives the answer, and the critical angle turns out to be the angle whose sine is exactly $\frac{2\sqrt{2}}{3}$. The lesson goes well beyond projectiles. Many quantities that "obviously" increase do not, and at university, showing that something is monotonic, by finding the sign of a derivative, is a key step in proving inequalities, in proving that sequences converge, and in showing that the solutions of a differential equation settle down (Section 11.10).

The last passenger. A train is fully booked. The first passenger to board ignores their ticket and sits in a seat chosen at random. Everyone after that sits in their own seat if it is free, and in a randomly chosen free seat if it is not. What is the chance that the last passenger to board finds their own seat free? With a hundred seats the possibilities branch bewilderingly: one displaced passenger can displace another, who displaces another, in a chain of any length. A direct calculation looks hopeless. Yet the answer is

exactly $\frac{1}{2}$, whatever the number of seats.

A 2021 question reaches this answer step by step: it asks you to find the answer for two and three seats, to set up a recurrence by conditioning on where the first passenger sits, and then to prove by induction that the answer is always $\frac{1}{2}$. That is a perfectly good proof, but it leaves you wondering *why* the answer is so simple, and the section then gives a one-line argument that explains it. Whenever a passenger has to choose at random, the first passenger's seat and the last passenger's seat are treated in exactly the same way; the process cannot tell them apart until one of them is taken. So the last free seat is equally likely to be either, and it must be one of them, because every other seat is taken by its owner, or by someone displaced, before the last passenger boards. Arguments from symmetry like this, which replace pages of calculation with a single observation, run all through university probability (Section 12.5).

Taken together, questions like these touch every one of the main areas of a first-year mathematics degree: number theory, analysis, calculus, complex numbers, algebra, linear algebra, differential equations, combinatorics, mechanics, probability and statistics.

What this book is for

That observation is the reason for this book. Most STEP preparation, quite reasonably, aims at passing the examination: learn the techniques, practise the papers, build up speed and stamina. This book aims at that too, and working through it will make you better at STEP. But its real purpose is different. It is to help you prepare for STEP *with university mathematics in mind*: to see each question not as an obstacle to be cleared but as an invitation, and to follow the invitation as far as it goes.

The students who do best at STEP, and who flourish at university afterwards, are almost always the ones who love mathematics for its own sake. They enjoy the moment when a hard problem gives way. They want to know why a trick works, not just that it does. When an answer comes out to be suspiciously simple, like $\frac{1}{2}$ or $\frac{1}{e}$ or $\pi^2/6$, they suspect that something deeper is going on, and they go looking for it. Those are the habits this book tries to encourage. Each question is chosen because it leads to a discovery: an idea, a theorem, a way of thinking, that is bigger than the question itself.

The ways of thinking matter as much as the theorems. University mathematics asks you to think in ways that school mathematics rarely does: to prove things rather than calculate them, to argue by contradiction or by infinite descent, to count the same thing in two ways, to find a quantity that never changes, to construct a counterexample, to show that something exists without finding it. STEP questions train exactly these habits, often without saying so. In this book every section names the thinking skill at its heart, so that you can recognise it when you meet it again.

How the book is organised

The book has thirteen chapters, organised by the area of university mathematics that the questions lead to, rather than by the STEP paper they came from.

- **Pure mathematics** (Chapters 1 to 9): number theory and proof; sequences, series and convergence; functions, inequalities and rigour; calculus done properly; complex numbers; algebra and structure; linear algebra and geometry; differential equations and dynamical systems; and counting and chance, which also draws on the combinatorial questions from the probability papers.
- **Mechanics** (Chapters 10 and 11): conservation laws, collisions and gravity; and oscillations, rotation and moving frames, ending with a set of surprising trajectories.
- **Probability and statistics** (Chapters 12 and 13): probability, from clear thinking about conditioning to the central limit theorem; and statistics and randomness in the world.

Each chapter opens with an essay: why the area matters, what you will meet in it at university, and three big questions that run through the chapter. Each closes with a wrap-up that returns to those questions, a toolkit of the methods used, pointers to other chapters, and suggestions for reading before university. At the end of each chapter are brief answers to the exercises. Before Chapter 1 there is a glossary of the terms in the book that are unlikely to be familiar from school, and a table of the mathematical symbols it uses; if a word or a symbol stops you, look it up there.

Every section follows the same pattern. It begins with *the question*, exactly as it was set. *The hook* explains why the question is more interesting than it looks. *Before you start* gives you a first foothold. *Working it through* is a guided solution in stages, interrupted by boxes headed *Ask yourself* and *Pause and try*, which are the questions a good teacher would ask you at that point; they are there to be answered, not skipped. *Looking back* picks out the key moves. *What's really going on* is the heart of the section: the university mathematics behind the question, its history and its surprises. *The thinking skill* names the habit of mind the question exercises. *Try this next* gives follow-up problems, often including related STEP questions, and *Further reading* suggests books and articles at a level a strong school student or first-year undergraduate can manage.

A few sections use topics that are not on the current STEP specification, such as groups, moments of inertia and moment generating functions. They are included because the questions are too good to leave out, and each gives the facts it needs.

How to use this book as a student

- **Do the question first.** Give each question at least half an hour of honest effort before you read further. Being stuck is not a sign of failure; it is where the learning happens. If you are completely stuck, read *Before you start*, then try again.
- **Answer the boxes.** When the text asks you a question, stop and answer it, on paper. The guided solutions are written to be read with a pencil in your hand.
- **Do not skip the "What's really going on" sections.** They are not examinable, but they are the reason the book exists, and they are what you will remember.
- **Read the sections in any order.** Each section stands on its own, and cross-references point you to related ideas elsewhere. Follow your curiosity.

- **Try the exercises, then check the answers.** The answers at the end of each chapter are brief: they give the result and the key step, not a full write-up.
- **Use the worksheets.** The website has a printable worksheet for every section, with the question, the hints in order and the exercises, for working away from a screen.
- **Read beyond the book.** Siklos's book, the STEP Support Programme and the past papers themselves are all freely available. The further reading in each section is chosen to be accessible before university.

How to use this book as a teacher

- **As a source of extension work.** Each section is a self-contained session: the question, a guided path through it, and somewhere interesting to go afterwards. The thinking skills make good themes for a sequence of sessions.
- **As a discussion guide.** The *Ask yourself* and *Pause and try* boxes are natural prompts for a class or small group. Pose them before revealing the next stage.
- **With the worksheets.** Each worksheet gives the question, the hints in order and the follow-up exercises; students can choose how many hints to take.
- **For STEP preparation.** Questions from all three papers, from the specimen papers to 2025, are included, many with related questions in the exercises. The chapter wrap-ups and toolkits help students to see connections across topics.
- **For students without STEP support.** The book is written so that a determined student can work through it alone. If your school does not offer STEP preparation, it can do much of that job.

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I have tried hard to make the mathematics correct, but a book of this size will certainly contain errors. If you find one, however small, please report it as an issue on the book's GitHub repository, `github.com/keithmcnulty/step_book_mcnulty`. Corrections will be made on the website, `beyondstep.keithmcnulty.org`, as they come in.

A last word before you start

Siklos ended the introduction to his book with the story of Michelangelo's *David*, carved from a block of marble that had lain untouched for decades, and said that he could not teach you mathematics, but that hard work chipping away at problems would reveal the mathematician within you. I cannot improve on that. This book offers you the marble, and some tools, and a few pictures of what other people have found inside. The chipping is up to you. I hope you enjoy it as much as I have enjoyed putting the book together.

Keith McNulty
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Glossary

University mathematics has its own vocabulary, and much of it will be new to you. This glossary collects the words in this book that you are unlikely to have met at school, with a short explanation of each and, where it helps, a pointer to the section where the idea is used or explained properly. The explanations are meant to get you started, not to be precise definitions; where a word has a careful definition, the section it points to gives it.

A few words, such as *set*, *function* and *proof*, will be familiar, but are used at university with more care than at school, and they are included for that reason. Words in *italics* within an entry have entries of their own.

A

Adiabatic invariant A quantity that stays almost constant when a system is changed slowly. A molecule bouncing in a slowly compressed cylinder keeps its speed multiplied by the length of the cylinder nearly fixed (Section 10.4).

Algebraic number A number that is a root of a polynomial equation with integer coefficients (positive, negative or zero), not all zero. $\sqrt{2}$, i and $\frac{1}{2}(1+\sqrt{5})$ are algebraic; numbers that are not algebraic are *transcendental* (Section 6.3).

Analysis The branch of mathematics that puts calculus on a rigorous footing: limits, continuity, differentiation, integration and infinite series, all defined precisely and proved from first principles. Chapters 2 to 4 are an introduction to it.

Angular momentum The rotational counterpart of momentum. For a body spinning about an axis it is the *moment of inertia* multiplied by the angular velocity, and it is conserved when there is no turning effect from outside (Section 11.6).

Arithmetic–geometric mean The common limit of two sequences formed by repeatedly replacing a pair of numbers by their arithmetic mean and their geometric mean. It converges extraordinarily fast (Section 2.6).

Associative An operation $*$ is associative if $(a*b)*c = a*(b*c)$ for all a, b, c, so that brackets can be left out. Addition and multiplication of numbers are associative; subtraction is not.

Asymptotic Two expressions are asymptotic, written $f(n) \sim g(n)$, if their ratio tends to 1 as $n \to \infty$. Stirling's formula says that $n! \sim \sqrt{2\pi n}\,(n/\mathrm{e})^n$, even though the difference between the two sides grows without limit (Section 12.23).

Axiom A statement taken as a starting point, without proof, from which everything else

is deduced. Much of university algebra starts from a short list of axioms and asks what follows from them (Section 7.3, Section 6.5).

B

Base case The first step of a proof by *induction*: checking that the statement is true for the smallest value, usually $n = 0$ or $n = 1$. The *inductive step* then shows that truth for one value implies truth for the next.

Base-rate fallacy The mistake of judging the probability of something from the evidence alone while ignoring how common it was to begin with. If a disease is rare enough, even a positive result from an accurate test can be more likely to be a false alarm than not; whether it is depends on how rare the disease is and how accurate the test is (Section 12.2).

Basis A set of vectors such that every vector in the space can be written in exactly one way as a combination of them. In the plane, $\mathbf{i}$ and $\mathbf{j}$ form a basis; so does any pair of vectors that are not parallel. The number of vectors in a basis is the *dimension* of the space (Section 6.3).

Bayes' theorem The rule for turning one conditional probability round: $\mathrm{P}(A \mid B) = \mathrm{P}(B \mid A)\,\mathrm{P}(A)/\mathrm{P}(B)$. It tells you how to update a belief in the light of evidence (Section 12.2).

Bias An *estimator* is biased if, on average over many samples, it comes out too high or too low; it is *unbiased* if its expectation is exactly the quantity it estimates (Section 13.3).

Bijection A one-to-one correspondence: a rule that pairs off every member of one set with exactly one member of another, with nothing left over on either side. Two finite sets have the same size exactly when there is a bijection between them, and this is how the size of an infinite set is defined (Section 1.8, Section 9.10).

Binary operation A rule for combining two objects to give a third, such as addition of numbers, multiplication of matrices, or composition of functions.

Bound An upper bound for a set or sequence of numbers is a number that none of them exceeds; a lower bound is one that none of them falls below. A sequence is *bounded* if it has both (Section 2.2).

Branching process A model of a population in which each individual independently has a random number of offspring. The central question is whether the family line dies out, and the answer comes from a *probability generating function* (Section 12.15).

C

Calculus of variations The study of problems in which the unknown is a whole function or curve, chosen to make some quantity as small or as large as possible: the shortest path, the quickest slide, the curve enclosing the most area (Section 4.7).

Catenary The shape of a hanging chain, the graph of $y = a\cosh(x/a)$. It looks like a parabola but is not one (Section 11.14).

Cauchy distribution A continuous distribution, with a bell-shaped graph, whose tails are so heavy that it has no mean. The average of a sample from it is no more reliable than a single observation (Section 12.22).

Central limit theorem The theorem that the sum of a large number of independent random quantities with the same distribution and finite variance, suitably shifted and scaled, has approximately a normal distribution, whatever that common distribution is. Versions exist for quantities with different distributions, provided no few of them dominate the sum. It explains why the bell curve appears everywhere (Section 12.24).

Chaos Behaviour of a deterministic system that is so sensitive to its starting point that long-term prediction is impossible in practice: two starting points that are very close together lead to completely different futures (Section 8.7, Section 8.8).

Closed A set is closed under an operation if combining two members of the set always gives another member. The even numbers are closed under addition; the odd numbers are not.

Commutative An operation $*$ is commutative if $a * b = b * a$ always. Multiplication of numbers is commutative; multiplication of matrices and of *quaternions* is not (Section 7.1, Section 6.7).

Completeness The property of the real numbers that they have no gaps: every increasing sequence that is bounded above converges to a real number. The rational numbers lack it, since a sequence of fractions can close in on $\sqrt{2}$. Much of *analysis* rests on it (Section 2.2).

Congruence $a \equiv b \pmod{n}$, read "a is congruent to b modulo n", means that a and b leave the same remainder on division by n; equivalently, n divides $a - b$. Arithmetic with congruences is called *modular arithmetic* (Section 1.5).

Conjecture A statement that is believed to be true but has not been proved.

Conservation law A statement that some quantity, such as energy, momentum or angular momentum, does not change as a system evolves. Conservation laws often let you solve a problem without following its details (Section 10.1).

Continuous A function is continuous if small changes in the input produce small changes in the output; informally, its graph can be drawn without lifting the pen. Precisely, f is continuous at a if $f(x) \to f(a)$ as $x \to a$ (Section 3.1, Section 8.8).

Contradiction, proof by A proof that assumes the statement is false and deduces something impossible. The classic example is the proof that $\sqrt{2}$ is irrational (Section 1.3).

Contrapositive The contrapositive of "if P then Q" is "if not Q then not P". The two statements are logically equivalent, and proving the contrapositive is often easier. It is not the same as the *converse*, "if Q then P", which may well be false.

Converge A sequence converges if its terms get, and stay, as close as you like to a fixed number, its *limit*. A series converges if its *partial sums* converge. A sequence or series that does not converge *diverges* (Section 2.1, Section 2.3).

Convex A function is convex if the chord joining any two points on its graph lies on or above the graph, as for x^2 or e^x. Many inequalities, including AM–GM, are consequences of convexity (Section 3.3).

Corollary A result that follows quickly from a *theorem* just proved.

Correlation A number between -1 and 1 measuring how close two random quantities are to having a straight-line relationship. It is the *covariance* scaled by the two standard deviations (Section 13.7).

Countable A set is countable if its members can be listed as a first, second, third, and so on, possibly for ever. The whole numbers and, surprisingly, the fractions are countable; the real numbers are not (Section 1.8).

Counterexample A single example showing that a general statement is false. One counterexample is enough to disprove a claim, however many examples support it.

Covariance $\mathrm{Cov}(X, Y) = \mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y)$, a measure of how two random quantities vary together. It is zero for independent quantities, though zero covariance does not imply independence (Section 12.4, Section 13.7).

Cycle A set of points that a repeated function visits in turn: $x_1 \mapsto x_2 \mapsto \cdots \mapsto x_n \mapsto x_1$. The number n is its *period*, and the points are *periodic points* (Section 8.8).

Cycloid The curve traced by a point on the rim of a rolling wheel. Turned upside down, it is both the curve of quickest descent and the curve along which every slide takes the same time (Section 11.4).

Cyclotomic polynomial The polynomial $\Phi_n(x)$ whose roots are exactly the *primitive* nth *roots of unity*. It has integer coefficients and is *irreducible*, and $x^n - 1$ is the product of the $\Phi_d(x)$ for all divisors d of n: for example, $x^4 - 1 = (x-1)(x+1)(x^2+1)$ (Section 6.1).

D

Derangement A rearrangement of a list in which nothing stays in its original position (Section 9.8).

Derivation An operation D on functions (or on other objects) with $D(f+g) = Df + Dg$ and the product rule $D(fg) = f\,Dg + g\,Df$. Differentiation is the standard example, and much of its behaviour follows from these two rules alone (Section 6.5).

Determinant A number calculated from a square matrix, which measures the factor by which the corresponding transformation scales areas or volumes. It is zero exactly when the matrix has no inverse (Section 7.3).

Differentiable A function is differentiable at a point if its graph has a well-defined tangent there, so that the limit defining the derivative exists. Every differentiable function is *continuous*, but not the other way round: $|x|$ is continuous but not differentiable at 0.

Dimension The number of vectors in a *basis*: the number of independent directions in a *vector space*.

Diverge See *converge*.

Dynamical system A rule describing how the state of a system changes with time, either step by step (a sequence defined by $x_{n+1} = f(x_n)$) or continuously (a differential equation). Chapter 8 is about them.

E

Eigenvalue, eigenvector A non-zero vector $\mathbf{v}$ is an eigenvector of a matrix $\mathbf{M}$ if $\mathbf{Mv} = \lambda\mathbf{v}$ for some number λ, its eigenvalue: the transformation stretches $\mathbf{v}$ without turning it (Section 7.2).

Envelope A curve that touches every member of a family of curves. The envelope of all the trajectories of a ball thrown at a fixed speed is the *safety parabola*, the boundary of the region the ball can reach (Section 11.9).

Equilibrium A state of a system that does not change. It is *stable* if small disturbances stay small, and *asymptotically stable* if they also die away, as they do for a pendulum with friction; a frictionless pendulum hanging straight down is stable, but a small push makes it swing for ever. It is *unstable* if some small disturbances grow (Section 8.3, Section 11.8).

Estimator A rule for calculating, from data, a guess at an unknown quantity, such as the size of a population or the mean of a distribution (Section 13.3).

Euler characteristic For a polyhedron with V vertices, E edges and F faces, the number $V - E + F$, which is always 2 for a polyhedron without holes (Section 7.9).

Existence proof A proof that something exists that does not necessarily say how to find it. The *pigeonhole principle* and the *intermediate value theorem* both give existence proofs.

Expectation The mean value of a random quantity X, written $\mathrm{E}(X)$: the long-run average of many independent repetitions. Expectation is additive, $\mathrm{E}(X + Y) = \mathrm{E}(X) + \mathrm{E}(Y)$, whether or not X and Y are independent (Section 9.11).

Extremal principle The strategy of looking at the smallest, largest, first or last object of some kind, and showing that it has a property that leads either to the result or to a contradiction (Section 1.3).

F

Field A number system in which you can add, subtract, multiply and divide (except by zero), with all the usual rules holding. The rational, real and complex numbers are fields; so are the integers modulo a prime. The integers are not, because you cannot always divide (Section 6.3).

Field extension A larger field containing a smaller one, such as the numbers $a + b\sqrt{2}$ with a, b rational, which contain the rationals. Its *degree* is its *dimension* as a *vector space* over the smaller field (Section 6.3).

Fixed point A point x with $f(x) = x$, which the function leaves where it is. On a graph, fixed points are where $y = f(x)$ meets $y = x$ (Section 2.2, Section 8.8).

Frame of reference A point of view from which positions and motions are measured. An *inertial* frame is one in which Newton's first law holds; in an accelerating frame, such as a train speeding up, there appear to be extra forces (Section 11.5).

Function A rule that assigns to each member of one set, the *domain*, exactly one member of another set, the *codomain*. At university a function is not the same as a formula: many important functions have no formula, and some formulae do not define functions.

Functional equation An equation whose unknown is a function, such as $f(x + y) = f(x)f(y)$ (Section 3.8).

Fundamental theorem of algebra Every polynomial of degree at least 1 with complex coefficients has a complex root, and so factorises completely into linear factors over the complex numbers.

G

Gamma function A function Γ that extends the factorial to non-integers: $\Gamma(n+1) = n!$ for whole numbers n, and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, which in a sense says that $(-\frac{1}{2})! = \sqrt{\pi}$ (Section 4.4).

Gaussian integer A complex number $a + b\mathrm{i}$ with a and b integers. They behave much like ordinary integers, with their own primes and unique factorisation (Section 1.6).

Generating function A power series $a_0 + a_1x + a_2x^2 + \cdots$ whose coefficients are the terms of a sequence. Operations on the series, such as multiplication, correspond to useful operations on the sequence (Section 9.4). A *probability generating function* does this for the probabilities of a random whole number (Section 9.6, Section 12.15); a *moment generating function* is a close relative (Section 12.24).

Group A set with a *binary operation* that is *associative*, has an *identity* element and gives every element an *inverse*. The symmetries of a shape form a group, as do the non-zero complex numbers under multiplication. Groups are the language of symmetry (Section 6.6, Section 6.7).

H

Hyperbolic geometry A geometry in which, through a point not on a given line, there are infinitely many lines that never meet it, and the angles of a triangle add up to less than π. A disc with a suitable way of measuring distance is a model of it (Section 5.8).

I

Identity An identity element for an operation $*$ is an element e with $e * a = a * e = a$ for every a: 0 for addition, 1 for multiplication, the identity matrix $\mathbf{I}$ for matrix multiplication. Separately, an identity is an equation that holds for every value of its variables, such as $\cos^2\theta + \sin^2\theta \equiv 1$.

If and only if "P if and only if Q", often written "P iff Q" or $P \iff Q$, means that P and Q are either both true or both false. Proving it needs two arguments, one in each direction.

Independent Two events are independent if knowing that one has happened does not change the probability of the other: $\mathrm{P}(A \cap B) = \mathrm{P}(A)\mathrm{P}(B)$. Three or more events can be independent in pairs without being independent all together (Section 12.4).

Induction A method of proving a statement for every whole number n: prove it for the first value (the *base case*), then prove that whenever it is true for one value it is true for the next (the *inductive step*). In *strong induction*, the inductive step may use the statement for all smaller values, not just the previous one (Section 1.2).

Infinite descent A form of proof by contradiction: show that any solution of a problem in whole numbers would lead to a smaller one, and so to an endless decreasing sequence of positive whole numbers, which is impossible (Section 1.5).

Intermediate value theorem If a *continuous* function takes the values a and b, it takes every value in between. So a continuous function that changes sign must have a root (Section 8.8).

Invariant A quantity that does not change while a process runs, or under some transformation. Finding an invariant is one of the most powerful ways to prove that something is impossible, or to solve a problem without following every step (Section 7.7, Section 10.4).

Inverse An inverse of an element a, for an operation with *identity* e, is an element b with $a * b = b * a = e$. An inverse function undoes a function: $\mathrm{f}^{-1}(\mathrm{f}(x)) = x$.

Irrational A real number that cannot be written as a fraction $\frac{p}{q}$ with p and q integers ($q \neq 0$), such as $\sqrt{2}$, π or e (Section 1.4).

Irreducible A polynomial is irreducible if it cannot be written as the product of two polynomials of lower degree with coefficients of the same kind. $x^2 - 2$ is irreducible over the rationals, but factorises as $(x - \sqrt{2})(x + \sqrt{2})$ over the reals (Section 6.1).

Isomorphic Two structures, such as two *groups*, are isomorphic if there is a *bijection* between them that matches their operations exactly: they are the same structure with different labels (Section 6.7, Section 6.8).

Iteration Applying the same function again and again, so that $x_1 = \mathrm{f}(x_0)$, $x_2 = \mathrm{f}(x_1)$, and so on. The sequence x_0, x_1, x_2, $\ldots$ is the *orbit* of x_0 (Section 8.7).

L

Law of large numbers The theorem that the average of many independent repetitions of a random quantity tends to its *expectation*, provided the expectation exists (Section 12.22).

Lemma A result proved mainly as a step towards a larger *theorem*.

Limit The value that a sequence or function gets, and stays, arbitrarily close to. Making this idea precise was one of the great achievements of nineteenth-century mathematics, and it is the starting point of *analysis* (Section 2.1).

Limit cycle A closed loop in the motion of a system that nearby motions spiral towards (or away from), such as the steady rhythm of a heartbeat or a clock (Section 8.4).

Linear map A function T between vector spaces with $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(\lambda\mathbf{u}) = \lambda T(\mathbf{u})$. Every linear map can be represented by a matrix, and linear algebra is the study of them (Section 7.2).

Logistic map The rule $u_{n+1} = ru_n(1 - u_n)$, a simple model of a population limited by its food supply, and the standard example of *chaos* (Section 8.7).

M

Markov chain A random process that moves between states in steps, in which the probability of the next step depends only on the present state, not on how the process got there (Section 12.16).

Maximum likelihood A way of estimating an unknown quantity: choose the value that makes the observed data most probable (Section 13.2).

Memoryless A random waiting time is memoryless if, however long you have already waited, the remaining wait has the same distribution as it did at the start. The exponential distribution is the only continuous memoryless distribution (Section 12.18).

Minimal polynomial The polynomial of lowest degree, with rational coefficients and leading coefficient 1, that has a given algebraic number as a root. The minimal polynomial of $\sqrt{2}$ is $x^2 - 2$ (Section 6.3).

Modular arithmetic See *congruence*.

Moment of inertia The rotational counterpart of mass: a measure of how hard it is to change a body's rate of rotation about an axis. It depends on how far the mass is

from the axis, not just on how much mass there is (Section 11.7).

Monotonic A sequence or function is monotonic if it never goes down (increasing) or never goes up (decreasing). A monotonic sequence that is bounded must converge (Section 2.2, Section 11.10).

N

Necessary and sufficient P is a necessary condition for Q if Q cannot happen without P ($Q \Rightarrow P$); it is sufficient if P guarantees Q ($P \Rightarrow Q$). A condition that is both is equivalent to Q.

Normal modes The special patterns of vibration of a system in which every part moves in step at the same frequency. Any vibration of the system is a combination of its normal modes (Section 11.2).

O

Order The order of an element g of a *group* is the smallest positive whole number n for which g^n is the *identity*: the number of times you must apply it to get back to the start (Section 7.4). If there is no such n, as for the number 1 in the integers under addition, g has *infinite order*. The order of a finite group is the number of elements it has.

Orthogonal Perpendicular. At university the word is extended beyond vectors: two functions f and g can be called orthogonal if $\int \mathrm{f}(x)\mathrm{g}(x)\,\mathrm{d}x = 0$ over a suitable range (Section 8.6).

P

Parameter A quantity that is fixed within a particular problem but can vary from one problem to another, such as r in the *logistic map*. Many STEP questions ask how the answer depends on a parameter.

Partial sum The sum $a_1 + a_2 + \cdots + a_n$ of the first n terms of a series. The series *converges* if its partial sums do (Section 2.3).

Pigeonhole principle If more than n objects are put into n boxes, some box contains at least two of them. Simple as it is, it proves some remarkable *existence* results (Section 9.2).

Poisson process A model of events that occur at random, independently, at a constant average rate, in time or in space: radioactive decays, phone calls, stars in the sky (Section 12.19, Section 12.18).

Primitive root of unity An nth root of unity whose powers give all n of the nth roots of unity; equivalently, one that is not an mth root of unity for any smaller m (Section 6.1).

Probability density function For a continuous random quantity, the function f whose integral over an interval gives the probability of landing in that interval.

Proof A complete logical argument that a statement is true, starting from definitions, axioms and results already proved. At university, a calculation that gives the right answer is not a proof unless each step is justified.

Q

Quaternions A number system discovered by William Rowan Hamilton in 1843, extending the complex numbers with three square roots of -1, written i, j, k, which satisfy $\mathrm{i}^2 = \mathrm{j}^2 = \mathrm{k}^2 = \mathrm{ijk} = -1$. Their multiplication is not *commutative*. They are used to describe rotations in three dimensions (Section 6.7, Section 7.5).

R

Random variable A quantity whose value is determined by the outcome of a random experiment, such as the number of heads in ten tosses of a coin.

Random walk A path made of successive random steps, such as a counter that moves one place left or right on the toss of a coin (Section 9.10, Section 12.17).

Rapidity A measure of speed in special relativity that, unlike velocity, simply adds when one motion is combined with another in the same direction. If c is the speed of light, velocity v has rapidity $\operatorname{artanh}(v/c)$ (Section 7.6).

Recurrence relation An equation giving each term of a sequence in terms of earlier ones, such as $F_{n+1} = F_n + F_{n-1}$ for the Fibonacci numbers.

Resonance The large response of a system that is pushed periodically at, or near, its own natural frequency, as when a child on a swing is pushed in time with the swing (Section 11.1).

Riemann integral The precise definition of the integral as the limit of sums of areas of thin rectangles, when that limit exists (Section 4.1).

Rigour The standard of argument in which every step is justified and nothing is taken for granted because it looks obvious or is true in the examples you have tried (Chapter 3).

Root of unity A complex number z with $z^n = 1$ for some positive whole number n. The nth roots of unity are $\mathrm{e}^{2\pi \mathrm{i} k/n}$ for $k = 0, 1, \ldots, n-1$, which form a regular n-gon on the unit circle (Section 5.2).

S

Sample space The set of all possible outcomes of a random experiment. Events are subsets of it (Section 12.1).

Set A collection of objects, its *elements*, considered as a single thing. The order of the elements does not matter, and nor does repetition: $\{1,2,3\}$ and $\{3,1,2,2\}$ are the same set.

Simple harmonic motion Motion in which the acceleration is proportional to the displacement from a fixed point and directed towards it, $\ddot{x} = -\omega^2 x$. The solution is $x = A\cos(\omega t + \varepsilon)$, and the period, $2\pi/\omega$, does not depend on the size of the swing (Section 11.4).

Stable, unstable See *equilibrium*.

Stirling's formula The approximation $n! \approx \sqrt{2\pi n}\,(n/\mathrm{e})^n$, whose percentage error tends to zero as n grows (Section 12.23).

Subgroup A subset of a *group* that is itself a group under the same operation (Section 6.6).

Symmetry argument An argument that two quantities must be equal because nothing in the problem distinguishes between them. A good symmetry argument can replace pages of calculation (Section 12.5).

T

Theorem An important statement that has been proved. Lesser results are often called *propositions*, those proved on the way to a theorem *lemmas*, and quick consequences *corollaries*.

Trace The sum of the entries on the leading diagonal of a square matrix. Like the *determinant*, it does not change when the matrix is written in a different *basis*, and it equals the sum of the *eigenvalues* (Section 7.4, Section 7.7).

Transcendental A number that is not *algebraic*: it is not the root of any polynomial equation with integer coefficients, not all zero. e and π are transcendental, though proving it is hard (Section 1.4).

Triangle inequality $|a+b| \leq |a| + |b|$, for real numbers, complex numbers or vectors: one side of a triangle is never longer than the other two together.

U

Unique factorisation The property that every whole number greater than 1 can be written as a product of primes in only one way, apart from the order of the factors. It is not as obvious as it looks: there are number systems in which it fails (Section 1.7).

Uniqueness theorem A theorem saying that a problem has only one solution. For differential equations, uniqueness can fail when the equation is not well behaved, so that a single starting point leads to several different futures (Section 8.1).

V

Vector space A collection of objects, called vectors, that can be added together and multiplied by numbers, following the familiar rules for arrows in the plane. Polynomials, functions and solutions of many differential equations all form vector spaces, so the methods of linear algebra apply to them (Section 6.3).

W

Well-defined An object or rule is well-defined if its definition really does specify a single unambiguous thing: if it does not depend, for example, on an arbitrary choice made along the way.

Well-ordering principle Every non-empty set of positive whole numbers has a smallest member. It is equivalent to the principle of *induction*, and it lies behind every argument that begins "take the smallest counterexample" (Section 1.3).

Without loss of generality A phrase, often shortened to WLOG, meaning that the argument considers only one case because the other cases are the same apart from relabelling. "Without loss of generality, $a \leq b$" means that if $a > b$ you can swap their names.

Mathematical Symbols

Mathematical writing packs a lot of meaning into a few marks, and university lecturers use many symbols that school textbooks rarely do. The tables below list the symbols used in this book, grouped by topic, with what each one means and how it is read aloud. Most of them will become second nature within a few weeks of starting a degree; until then, this is somewhere to look them up.

Two conventions are worth knowing from the start. First, this book follows STEP in printing some letters upright rather than in italics when they have a fixed meaning: e for the base of natural logarithms, i for $\sqrt{-1}$, d in $\mathrm{d}y/\mathrm{d}x$, and the names of functions, as in $\mathrm{f}(x)$ and $\mathrm{P}(A)$. Italic letters are variables. Secondly, the same symbol sometimes means different things in different contexts; $\equiv$ and $\sim$ are the main culprits, and the tables point out where.

Logic and proof

Symbol	Meaning	Example or reading
$P \Rightarrow Q$	P implies Q: if P is true, so is Q	$x = 2 \Rightarrow x^2 = 4$; "$P$ implies Q"
$P \Leftarrow Q$	P is implied by Q	"P if Q"
$P \iff Q$	P if and only if Q: each implies the other	$x^2 = 4 \iff x = \pm 2$; also written "iff"
$\forall$	for all, for every	$\forall x \in \mathbb{R},\ x^2 \geq 0$
$\exists$	there exists	$\exists\, n \in \mathbb{Z}$ with $n^2 = 9$
$\neg P$	not P, the negation of P	the negation of "$\forall x,\ P(x)$" is "$\exists\, x$ with $\neg P(x)$"
$:=$	is defined to equal	$\mathrm{f}(x) := x^2 + 1$
$\blacksquare$	end of proof; also written QED	from the Latin *quod erat demonstrandum*, "which was to be shown"

Sets and number systems

Symbol	Meaning	Example or reading
$\{\dots\}$	the set whose elements are listed	$\{1, 2, 3\}$
$\{x : \dots\}$	the set of all x such that $\dots$; also written $\{x \mid \dots\}$	$\{x : x^2 < 2\}$; "the set of x such that $x^2 < 2$"
$x \in A$	x is an element of A	$\sqrt{2} \in \mathbb{R}$; "$\sqrt{2}$ is in $\mathbb{R}$"
$x \notin A$	x is not an element of A	$\sqrt{2} \notin \mathbb{Q}$
$A \subseteq B$	A is a subset of B: every element of A is in B	$\mathbb{Z} \subseteq \mathbb{Q}$
$A \cup B$	union: elements in A or B (or both)	$\{1, 2\} \cup \{2, 3\} = \{1, 2, 3\}$
$A \cap B$	intersection: elements in both A and B	$\{1, 2\} \cap \{2, 3\} = \{2\}$
$A \setminus B$	elements of A that are not in B	$\{1, 2, 3\} \setminus \{2\} = \{1, 3\}$
A'	complement: everything not in A	also written A^{c} or $\bar{A}$
$\varnothing$	the empty set, with no elements	$\{x \in \mathbb{R} : x^2 < 0\} = \varnothing$
$\lvert A \rvert$	the number of elements of a finite set A	$\lvert\{a, b, c\}\rvert = 3$
$A \times B$	the set of all pairs (a, b) with $a \in A$, $b \in B$	$\mathbb{R} \times \mathbb{R}$ is the plane, also written $\mathbb{R}^2$
$\mathbb{N}$	the natural numbers $1, 2, 3, \dots$	some authors include 0
$\mathbb{Z}$	the integers $\dots, -2, -1, 0, 1, 2, \dots$	from the German *Zahlen*, "numbers"
$\mathbb{Q}$	the rational numbers (fractions)	from *quotient*
$\mathbb{R}$	the real numbers	
$\mathbb{C}$	the complex numbers	
$\mathbb{H}$	the quaternions	after Hamilton (Section 6.7)
$\mathbb{Z}[\mathrm{i}]$	numbers $a + b\mathrm{i}$ with $a, b \in \mathbb{Z}$	the Gaussian integers (Section 1.6)
$[a, b]$	closed interval: $a \le x \le b$	includes its endpoints
(a, b)	open interval: $a < x < b$	excludes its endpoints; also a pair or a point

Numbers, divisibility and approximation

Symbol	Meaning	Example or reading
$a \mid b$	a divides b: b is an integer multiple of a	$3 \mid 12$; "3 divides 12"
$a \nmid b$	a does not divide b	$5 \nmid 12$
$a \equiv b \pmod{n}$	a and b leave the same remainder on division by n	$17 \equiv 2 \pmod{5}$; "17 is congruent to 2 modulo 5"
$\equiv$	(without a modulus) is identically equal to: true for all values	$(x+1)^2 \equiv x^2 + 2x + 1$
$\lfloor x \rfloor$	floor: the greatest integer $\leq x$	$\lfloor 3.7 \rfloor = 3$, $\lfloor -3.7 \rfloor = -4$
$\lceil x \rceil$	ceiling: the least integer $\geq x$	$\lceil 3.2 \rceil = 4$
$n!$	n factorial, $1 \times 2 \times \cdots \times n$	$0! = 1$ by convention
$\binom{n}{r}$	the binomial coefficient $\frac{n!}{r!\,(n-r)!}$	"n choose r"; also written ${}^{n}\mathrm{C}_{r}$
$\approx$	is approximately equal to	$\pi \approx 3.1416$
$\sim$	is asymptotic to: the ratio tends to 1	$n! \sim \sqrt{2\pi n}\,(n/\mathrm{e})^n$ (but see also Probability)
$\propto$	is proportional to	$v \propto 1/d$
$\pm$, $\mp$	plus or minus; minus or plus	$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
$0.\overline{01}$	recurring digits	$0.\overline{01} = 0.010101\ldots$

Functions

Symbol	Meaning	Example or reading
$\mathrm{f} : A \to B$	f is a function from the set A to the set B	$\mathrm{f} : \mathbb{R} \to \mathbb{R}$
$x \mapsto \ldots$	x is sent to $\ldots$	$x \mapsto x^2 - 2$; "x maps to $x^2 - 2$"
$\mathrm{f} \circ \mathrm{g}$	composition: do g, then f	$(\mathrm{f} \circ \mathrm{g})(x) = \mathrm{f}(\mathrm{g}(x))$
f^{-1}	the inverse function of f	not $1/\mathrm{f}$
f^n	in this book, usually f applied n times	$\mathrm{f}^3(x) = \mathrm{f}(\mathrm{f}(\mathrm{f}(x)))$; but $\sin^2 x$ means $(\sin x)^2$
$\lvert x \rvert$	modulus, or absolute value	$\lvert -3 \rvert = 3$
$\exp x$	the exponential function e^x	
$\ln x$	natural logarithm, to base e	at university often written $\log x$
sec, cosec, cot	$1/\cos$, $1/\sin$, $1/\tan$	
arcsin, arccos, arctan	inverse trigonometric functions	also written $\sin^{-1}$, etc.
sinh, cosh, tanh	hyperbolic functions: $\sinh x = \frac{1}{2}(\mathrm{e}^x - \mathrm{e}^{-x})$, $\cosh x = \frac{1}{2}(\mathrm{e}^x + \mathrm{e}^{-x})$, $\tanh = \sinh / \cosh$	read "shine" (or "sinch"), "cosh" and "than" (or "tanch")
artanh	inverse of tanh	$\operatorname{artanh} x = \frac{1}{2} \ln \frac{1+x}{1-x}$
max, min	the largest, smallest of	$\max(3, 7) = 7$
$\Gamma(x)$	the gamma function	$\Gamma(n+1) = n!$ (Section 4.4)

Sums, limits and calculus

Symbol	Meaning	Example or reading
$\sum$	sum	$\sum_{k=1}^{n} k = \frac{1}{2}n(n+1)$
$\prod$	product	$\prod_{k=1}^{n} k = n!$
∞	infinity: not a number, but shorthand for "without limit"	$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$
$x \to a$	x tends to a	$\frac{1}{n} \to 0$ as $n \to \infty$
lim	limit	$\lim_{x\to 0} \frac{\sin x}{x} = 1$
$\mathrm{f}'(x)$, $\mathrm{f}''(x)$	first and second derivatives	"f dash", or "f prime"
$\dot{x}$, $\ddot{x}$	derivatives with respect to time	$\dot{x} = \frac{\mathrm{d}x}{\mathrm{d}t}$; "$x$ dot", "x double dot"
$\frac{\partial f}{\partial x}$	partial derivative: differentiate with respect to x, holding other variables fixed	"partial f by partial x"
$\int_a^b$	definite integral from a to b	
$\oint$	integral round a closed loop	$\oint p\,\mathrm{d}q$ (Section 10.4)
δx, Δx	a small change in x; a change in x	
ε, δ	conventional names for small positive numbers in analysis	"for every $\varepsilon > 0$ there is a $\delta > 0$ such that ..."

Complex numbers

Symbol	Meaning	Example or reading
i	a square root of -1	engineers write j
$\mathrm{Re}\,z$, $\mathrm{Im}\,z$	real and imaginary parts	if $z = 3 + 4\mathrm{i}$, $\mathrm{Re}\,z = 3$, $\mathrm{Im}\,z = 4$
z^*, $\bar{z}$	complex conjugate	$(3+4\mathrm{i})^* = 3 - 4\mathrm{i}$; both notations are used
$\lvert z\rvert$	modulus: distance from 0	$\lvert 3+4\mathrm{i}\rvert = 5$
$\arg z$	argument: angle from the positive real axis	$\arg \mathrm{i} = \frac{\pi}{2}$
$\mathrm{e}^{\mathrm{i}\theta}$	$\cos\theta + \mathrm{i}\sin\theta$	$\mathrm{e}^{\mathrm{i}\pi} = -1$
ω	conventionally, a complex cube root of unity	$\omega = \mathrm{e}^{2\pi\mathrm{i}/3}$, $1 + \omega + \omega^2 = 0$

Vectors and matrices

Symbol	Meaning	Example or reading
$\mathbf{a}$, $\overrightarrow{AB}$	a vector; the vector from A to B	handwritten as $\underline{a}$
$\hat{\mathbf{n}}$	a unit vector, of length 1	"n hat"
$\lvert\mathbf{a}\rvert$	the length (magnitude) of $\mathbf{a}$	
$\mathbf{a}\cdot\mathbf{b}$	scalar (dot) product	$\mathbf{a}\cdot\mathbf{b} = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$
$\mathbf{a}\times\mathbf{b}$	vector (cross) product	perpendicular to $\mathbf{a}$ and $\mathbf{b}$
$\mathbf{M}$	a matrix	
$\mathbf{I}$	the identity matrix	$\mathbf{Iv} = \mathbf{v}$ for every $\mathbf{v}$
$\mathbf{M}^{-1}$	the inverse matrix	$\mathbf{MM}^{-1} = \mathbf{I}$
$\det\mathbf{M}$	the determinant of $\mathbf{M}$	$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$
$\operatorname{tr}\mathbf{M}$	the trace: sum of the diagonal entries	$\operatorname{tr}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a + d$
λ	conventionally, an eigenvalue	$\mathbf{Mv} = \lambda\mathbf{v}$

Probability and statistics

Symbol	Meaning	Example or reading
$\mathrm{P}(A)$	the probability of the event A	
$\mathrm{P}(A \mid B)$	the probability of A given B	$\mathrm{P}(A \mid B) = \mathrm{P}(A \cap B)/\mathrm{P}(B)$; "$A$ given B"
$\mathrm{E}(X)$	the expectation (mean) of X	also written μ
$\mathrm{Var}(X)$	the variance of X	$\mathrm{E}(X^2) - \mathrm{E}(X)^2$; also written σ^2
$\mathrm{Cov}(X, Y)$	covariance	$\mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y)$
$\mathrm{Corr}(X, Y)$	correlation	$\mathrm{Cov}(X, Y)/\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}$; also written ρ
$X \sim \ldots$	X has the distribution ...	$X \sim \mathrm{B}(16, \frac{1}{2})$; "$X$ is distributed as ..."
$\mathrm{B}(n, p)$	binomial distribution	number of successes in n trials
$\mathrm{N}(\mu, \sigma^2)$	normal distribution, mean μ, variance σ^2	
$\mathrm{f}(x)$, $\mathrm{F}(x)$	probability density function; cumulative distribution function	$\mathrm{F}(x) = \mathrm{P}(X \le x)$
$\mathrm{G}(t)$	probability generating function	$\mathrm{G}(t) = \mathrm{E}(t^X)$
$\mathrm{M}_X(\theta)$	moment generating function	$\mathrm{M}_X(\theta) = \mathrm{E}(\mathrm{e}^{\theta X})$
$\bar{x}$	the mean of a sample	$\bar{x} = \frac{1}{n}\sum x_i$
$\hat{\theta}$	an estimate of the unknown quantity θ	"theta hat"

The Greek alphabet

Greek letters are used throughout mathematics, often with conventional meanings. The table gives the lower-case and capital forms (capitals that look like Roman letters, such as A for alpha, are omitted, since they are not used as separate symbols), the name, and some typical uses.

Letter	Name	Typical uses
α	alpha	angles; roots of an equation
β	beta	angles; roots of an equation; the beta function
γ, Γ	gamma	Euler's constant $\gamma \approx 0.5772$; the gamma function Γ; in relativity, $\gamma = 1/\sqrt{1 - v^2/c^2}$
δ, Δ	delta	a small quantity; Δ for a change, or for a triangle or discriminant
ε, ϵ	epsilon	a small positive quantity; in mechanics, a phase angle
ζ	zeta	the Riemann zeta function, $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ for $s > 1$ (extended to other values, except $s = 1$, by more advanced methods)
η	eta	efficiency; a small quantity
θ, Θ	theta	angles; an unknown parameter in statistics
ι	iota	rarely used
κ	kappa	curvature; constants
λ, Λ	lambda	eigenvalues; a parameter, such as a Poisson rate; wavelength
μ	mu	a mean; the coefficient of friction
ν	nu	frequency
ξ, Ξ	xi	a variable or random quantity
π, Π	pi	$\pi \approx 3.14159$; Π for a product
ρ	rho	density; correlation; distance from an origin
σ, Σ	sigma	standard deviation; Σ for a sum
τ	tau	time; torque
υ, Υ	upsilon	rarely used
ϕ, φ, Φ	phi	angles; the golden ratio $\phi = \frac{1}{2}(1 + \sqrt{5})$; rapidity; Φ for the standard normal distribution function
χ	chi	the χ^2 (chi-squared) distribution
ψ, Ψ	psi	a wave function in quantum mechanics
ω, Ω	omega	angular velocity or frequency; a cube root of unity; Ω for a sample space

Mechanics

Symbol	Meaning	Example or reading
g	acceleration due to gravity	about $9.8\,\mathrm{m\,s^{-2}}$ at the Earth's surface
G	the gravitational constant	force Gm_1m_2/r^2 between two masses
e	coefficient of restitution (in italics, not to be confused with e)	along the line of impact, speed of separation $= e\times$ speed of approach
ω	angular velocity; angular frequency	$\ddot{x} = -\omega^2 x$ for simple harmonic motion
I	moment of inertia	rotational energy $\frac{1}{2}I\omega^2$ (Section 11.7)
c	the speed of light	about $3 \times 10^8\,\mathrm{m\,s^{-1}}$
$\hbar$	Planck's constant divided by 2π	"h bar" (Section 8.6)

CHAPTER

1

Number Theory and Proof

Whole numbers are the oldest objects in mathematics, and they still hold some of its deepest secrets.

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Why Whole Numbers?

Mathematics is the queen of the sciences, and number theory is the queen of mathematics.

— Carl Friedrich Gauss

Whole numbers are the first mathematical objects anyone meets. You learned to count before you could read. And yet the simplest questions about them can be fiendishly hard. Is every even number greater than 2 the sum of two primes? Nobody knows: Christian Goldbach asked in 1742, computers have checked it beyond 4×10^{18}, and it has still never been proved. Are there infinitely many pairs of primes that differ by 2, like 11 and 13, or 101 and 103? Nobody knows that either, although in 2013 an almost unknown mathematician called Yitang Zhang stunned the world by proving that there are infinitely many pairs of primes differing by at most 70 million, a number since brought down to 246. Pierre de Fermat scribbled in a margin in 1637 that $x^n + y^n = z^n$ has no solutions in positive whole numbers when $n \geq 3$. It took more than 350 years, and some of the most sophisticated mathematics ever created, before Andrew Wiles proved him right in 1994.

This mixture of questions a child can understand and answers that can take centuries is what makes number theory addictive. It is also why number theory has always been the training ground for *proof*. When you cannot check every case, because there are infinitely many whole numbers, you have to find an argument that covers them all at once. Most of the proof techniques you will use for the rest of your mathematical life were first sharpened on questions about whole numbers.

Pure, useless, and running the internet

In 1940 the Cambridge mathematician G. H. Hardy wrote, with some pride, that number theory was the purest of all mathematics, too remote from ordinary life to be put to any practical use, and so too remote to be put to any *harmful* use either. He could hardly have been more wrong. Every time you buy something online, send a secure message or unlock your phone, you are relying on number theory. The RSA system (1977) depends on the fact that multiplying two large primes is easy but recovering them from their product seems to be impossibly hard. Modern systems use the arithmetic of the very curves you will meet at the end of this chapter. The "useless" mathematics of the eighteenth and nineteenth centuries became the security of the twenty-first.

What you will meet at university

If you study mathematics at university, number theory will be one of your first destinations and one of your last.

- **In your first term** you will almost certainly take a course with a name like *Numbers and Sets* or *Foundations*. It will teach you to write proofs properly, and much of its raw material is number theory: induction, the Euclidean algorithm, modular arithmetic, Fermat's Little Theorem, the Chinese Remainder Theorem, and the surprising fact that some infinite sets are bigger than others.
- **In group theory**, usually in the first year, you will discover that many number-theory facts are shadows of a single theorem about symmetry, Lagrange's Theorem.
- **In the second year** a course in *Number Theory* typically covers quadratic reciprocity (Gauss's "golden theorem"), continued fractions and Pell's equation.
- **Later** you can go further: *algebraic number theory*, where new number systems such as the Gaussian integers rescue unique factorisation; *analytic number theory*, which uses calculus and complex analysis to count primes; and the theory of *elliptic curves*, which lies behind the proof of Fermat's Last Theorem and one of the seven million-dollar Millennium Prize Problems.

The questions that run through this chapter

The nine STEP questions in this chapter circle around three big questions. Keep them in mind as you work; we will come back to them at the end.

1. **How many primes are there, and how are they spread out?** Everyone knows there are infinitely many. But how thinly do they thin out, and what kind of argument can tell us?
2. **How can you prove that a number is *not* a fraction?** Showing that a number is rational just needs an example. Showing that it is irrational means ruling out infinitely many possibilities. How is that ever done?
3. **Which equations have whole-number solutions, and what happens when we change the number system?** Sometimes we can show an equation has no solutions at all. Sometimes the best way to understand whole numbers is to step outside them, into the complex numbers or onto a curve.

A chapter about proof

Running underneath those three questions is a fourth theme: *how to prove things*. Each section has been chosen partly because it shows a different method of proof in action, and between them they make a fair toolkit for a first-year undergraduate.

§	Topic	Way of thinking
1.1	Fermat numbers	An infinite family that *needs* infinitely many primes
1.2	Products of primes	Strong induction; information hidden in binomial coefficients
1.3	$\sqrt{2}$ again	"Take the smallest one": the well-ordering principle
1.4	e and e^2	Trapping a whole number strictly between 0 and 1
1.5	Impossible equations	Infinite descent; working modulo n
1.6	Sums of two squares	Changing the setting: complex numbers
1.7	A strange number system	Questioning what you take for granted
1.8	Listing the fractions	Building a bijection; descent in reverse
1.9	Cubes and curves	Geometry solving arithmetic

How to read this chapter

Each section starts with a STEP question exactly as it was set. Try it yourself first, for at least half an hour, before you read on. You will learn far more from a question you have struggled with than from one you have only read about. When you do read on, the *Ask yourself* boxes are invitations to stop and think before the answer arrives, and the *Pause and try* boxes mark the moments where the key idea is within your reach. Then the *What's really going on* part of each section looks past the exam to the mathematics you will meet at university.

The sections increase roughly in sophistication, but they are largely independent. If one doesn't grip you, move on and come back to it later.

1.1 A Prime Factory

The question

The nth *Fermat number*, F_n, is defined by

$$F_n = 2^{2^n} + 1, \qquad n = 0, 1, 2, \ldots,$$

where 2^{2^n} means 2 raised to the power 2^n.

Calculate F_0, F_1, F_2 and F_3. Show that, for $k = 1$, $k = 2$ and $k = 3$,

$$F_0 F_1 \cdots F_{k-1} = F_k - 2. \qquad (*)$$

Prove, by induction or otherwise, that $(*)$ holds for all $k \geq 1$. Deduce that no two Fermat numbers have a common factor greater than 1.

Hence show that there are infinitely many prime numbers.

The hook

You almost certainly already know that there are infinitely many primes. Euclid proved it around 300 BC, and it is one of the first proofs anyone meets. So why would a STEP question ask you to prove it *again*?

Because this proof works in a completely different way. Euclid's argument says: *give me any finite list of primes and I will show you it is incomplete.* This one says: *here is a single, explicit sequence of numbers, written down in advance, and every one of them contains a prime that none of the others contain.* It is a machine for manufacturing new primes, and the reason it works is a beautifully simple identity that the question walks you towards.

Along the way you will meet numbers that fooled Fermat himself.

Before you start

Read the question through once and notice its shape. It has four stages:

1. compute some examples;
2. spot (well, be told) a pattern, and check it on those examples;
3. prove the pattern in general;
4. use the pattern twice: first to deduce something about common factors, then something about primes.

This "examples → conjecture → proof → consequences" structure is how a great deal of mathematics is actually done, and STEP questions often follow it. When you see it, the early parts are usually not just warm-up: they are where you build the intuition you will need later.

> **Ask yourself** The last line asks about *infinitely many primes*. The middle asks about *common factors*. What could possibly connect "no two Fermat numbers share a factor" with "there are infinitely many primes"?

Hold on to that question. We will come back to it.

Working it through

STAGE 1 Getting a feel for the numbers

Be careful with the notation: 2^{2^n} means $2^{(2^n)}$, not $(2^2)^n$. The exponent is itself a power of 2, so these numbers grow *very* fast.

$$F_0 = 2^1 + 1 = 3, \qquad F_2 = 2^4 + 1 = 17,$$
$$F_1 = 2^2 + 1 = 5, \qquad F_3 = 2^8 + 1 = 257.$$

> **Ask yourself** What do you notice about these four numbers?

They are all prime. (Check 257: it is not divisible by $2, 3, 5, 7, 11$ or 13, and $17^2 = 289 > 257$, so we can stop.) The next one, $F_4 = 2^{16} + 1 = 65537$, is also prime. Pierre de Fermat noticed exactly this in 1640 and conjectured that *every* number of this form is prime. Keep that in mind; we will see later whether he was right.

STAGE 2 Checking the identity

Now check $(*)$ for small k:

- $k = 1$: the left-hand side is just $F_0 = 3$, and $F_1 - 2 = 3$. ✓
- $k = 2$: $F_0 F_1 = 3 \times 5 = 15$, and $F_2 - 2 = 15$. ✓
- $k = 3$: $F_0 F_1 F_2 = 15 \times 17 = 255$, and $F_3 - 2 = 255$. ✓

> **Ask yourself** Look at the step from $k = 2$ to $k = 3$. We took 15, multiplied by 17, and got 255. In terms of Fermat numbers, we computed $(F_2 - 2) \times F_2$ and got $F_3 - 2$. Is that a coincidence?

That observation is the whole proof in miniature. Let's see why it should be true.

STAGE 3 Proving the identity

Pause and try Before reading on, try to show directly that

$$(F_k - 2)\,F_k = F_{k+1} - 2.$$

Hint: write everything in terms of powers of 2. What does $F_k - 2$ look like?

We have $F_k - 2 = 2^{2^k} - 1$ and $F_k = 2^{2^k} + 1$, so

$$(F_k - 2)\,F_k = \left(2^{2^k} - 1\right)\left(2^{2^k} + 1\right).$$

Ask yourself This is a product of the form $(a-1)(a+1)$. What is that?

It is a difference of two squares, $a^2 - 1$. With $a = 2^{2^k}$,

$$a^2 = \left(2^{2^k}\right)^2 = 2^{2\cdot 2^k} = 2^{2^{k+1}},$$

so

$$(F_k - 2)\,F_k = 2^{2^{k+1}} - 1 = F_{k+1} - 2. \tag{$\dagger$}$$

That is the engine. Notice what squaring does: it doubles the exponent, and doubling 2^k gives 2^{k+1}. The definition of the Fermat numbers is perfectly tuned to the difference-of-squares factorisation.

Now the induction is almost automatic.

Ask yourself What exactly do we need to show in the inductive step?

We assume $(*)$ holds for some $k \geq 1$,

$$F_0 F_1 \cdots F_{k-1} = F_k - 2,$$

and we want to show it holds for $k+1$:

$$F_0 F_1 \cdots F_{k-1} F_k = F_{k+1} - 2.$$

Multiply both sides of the assumption by F_k and use $(\dagger)$:

$$F_0 F_1 \cdots F_{k-1} F_k = (F_k - 2)\,F_k = F_{k+1} - 2.$$

We have already checked $k = 1$, so by induction $(*)$ holds for every $k \geq 1$. ∎

Ask yourself Could we have avoided induction altogether?

Yes, and it is worth seeing how. Apply $(\dagger)$ repeatedly, starting from $F_0 - 2 = 1$:

$$F_k - 2 = (F_{k-1} - 2)F_{k-1} = (F_{k-2} - 2)F_{k-2}F_{k-1} = \cdots = (F_0 - 2)\,F_0 F_1 \cdots F_{k-1}.$$

Since $F_0 - 2 = 1$, we are done. This "unrolling" is really the same induction written out in full, but some people find it easier to see.

STAGE 4 No common factors

We want to show that if $m \neq n$ then F_m and F_n have no common factor greater than 1.

Ask yourself Suppose some number d divides both F_m and F_n, with $m < n$. Where does F_m appear in $(*)$?

Take $k = n$ in $(*)$:

$$F_0 F_1 \cdots F_m \cdots F_{n-1} = F_n - 2.$$

Since $m < n$, the number F_m is one of the factors on the left. So if d divides F_m, then d divides the whole left-hand side; that is, d divides $F_n - 2$.

Ask yourself Now d divides F_n *and* $F_n - 2$. What else must it divide?

Their difference: d divides $F_n - (F_n - 2) = 2$. So $d = 1$ or $d = 2$.

Ask yourself Can d be 2?

No: every Fermat number is a power of 2 plus 1, so it is odd. Hence $d = 1$, and no two distinct Fermat numbers share a factor greater than 1. ■

Mathematicians say the Fermat numbers are *pairwise coprime*.

STAGE 5 Infinitely many primes

Now we return to the question we asked at the start.

Ask yourself Every whole number greater than 1 has at least one prime factor. Pick one prime factor of each Fermat number. What can you say about the primes you have picked?

Let p_n be any prime that divides F_n. If two of these were equal, say $p_m = p_n$ with $m \neq n$, then that prime would be a common factor of F_m and F_n greater than 1, which we have just shown is impossible. So the primes

$$p_0,\ p_1,\ p_2,\ p_3,\ \ldots$$

are all *different*. There are infinitely many Fermat numbers, so there are infinitely many different primes. ■

Notice that we never needed to know *which* prime divides each Fermat number, or even whether the Fermat numbers are themselves prime. All we needed was that they never share a factor. Each one is forced to "use up" at least one prime that no other can use.

n	$F_n = 2^{2^n} + 1$	factorisation
0	3	prime
1	5	prime
2	17	prime
3	257	prime
4	65 537	prime
5	4 294 967 297	$641 \times 6\,700\,417$
6	18 446 744 073 709 551 617	$274\,177 \times 67\,280\,421\,310\,721$

Figure 1.1.1 The first Fermat numbers. Fermat guessed that they are all prime, but Euler found the factor 641 of F_5 in 1732, and no Fermat number beyond F_4 is known to be prime. Even so, the identity $(*)$ shows that the factors of different Fermat numbers never overlap: each row brings new primes.

Looking back

It is worth pausing to see what made this work, because the same ideas will come back many times in this book.

- **The examples weren't decoration.** Spotting that $15 \times 17 = 255$ was "$(F_2 - 2) \times F_2 = F_3 - 2$" *was* the key idea. When a question gives you small cases, ask what each step is doing, not just whether the numbers match.
- **A single identity did all the work.** Everything flowed from $(a-1)(a+1) = a^2 - 1$. Hard-looking STEP questions are often driven by one simple piece of algebra.
- **The final argument is about structure, not size.** We proved there are infinitely many primes without finding a single new one. Instead we showed that an infinite family of numbers *needs* infinitely many primes to build them.

What's really going on

Was Fermat right?

No. In 1732 Leonhard Euler showed that

$$F_5 = 2^{32} + 1 = 4\,294\,967\,297 = 641 \times 6\,700\,417.$$

Euler didn't find 641 by luck. He proved that any prime factor of F_n must have a very special form (for $n = 5$, it must be one more than a multiple of 64), which drastically cut down the candidates. You can prove this yourself in Exercise 4.

Since then, every Fermat number anyone has managed to test, from F_5 up to F_{32}, has turned out to be composite. As far as anyone knows, F_0 to F_4 may be the *only* Fermat primes, but nobody has been able to prove it. It is a genuinely open problem.

The delightful twist is that our proof doesn't care. Prime or composite, the Fermat numbers still produce infinitely many primes.

Fermat primes and constructing polygons

Fermat primes turn up somewhere quite unexpected. The ancient Greeks knew how to construct an equilateral triangle and a regular pentagon with ruler and compass, and notice that $3 = F_0$ and $5 = F_1$. For two thousand years nobody could construct any other regular polygon with a prime number of sides. Then in 1796, aged 19, Carl Friedrich Gauss showed how to construct a regular *17-gon*, and $17 = F_2$.

Gauss went further: a regular N-gon can be constructed with ruler and compass exactly when N is a power of 2 multiplied by *distinct Fermat primes*. (Pierre Wantzel completed the proof in 1837.) So which polygons can be constructed depends on how many Fermat primes there are. You will meet the key calculation behind Gauss's 17-gon in Section 6.2 (STEP II 1991, Question 4), and the full story belongs to *Galois theory*, one of the highlights of an undergraduate degree.

One theorem, many proofs

Why would mathematicians bother proving the same theorem more than once? Because different proofs reveal different things. This proof, usually credited to Christian Goldbach in a 1730 letter to Euler, shows that *any* infinite sequence of pairwise coprime whole numbers greater than 1 forces infinitely many primes. Fermat numbers are just one convenient example. (Exercise 3 gives another.)

It also gives more than Euclid's proof does. The primes $p_0, p_1, \ldots, p_{n-1}$ are n *different* primes, each at most F_{n-1}. So there are at least n primes no bigger than $2^{2^{n-1}} + 1$. That is a *quantitative* statement about how the primes are spread out. It is very weak (the truth is much better), but it is a first step towards one of the great results of nineteenth-century mathematics, the *Prime Number Theorem*, which describes exactly how the primes thin out. Section 1.2 takes the next step.

The thinking skill

Building an argument from a structural property

To show that infinitely many objects of type A exist, find infinitely many objects of type B, each of which needs its own object of type A.

Here A = primes and B = Fermat numbers, and "needs its own" means "has a prime factor not shared with any other". The same pattern appears in counting arguments, in linear algebra (infinitely many independent vectors force infinite dimension) and in analysis.

You also practised *induction*, and in particular the habit of asking *"what does the step from k to $k+1$ actually do?"* That question is often more useful than the formal template.

Try this next

1 **Euclid's version.** Write out Euclid's proof that there are infinitely many primes. In what sense is it "by contradiction", and in what sense is the Fermat-number proof "constructive"?

2 **The last digit.** Show that for $n \geq 2$ the Fermat number F_n ends in the digit 7.

3 **Sylvester's sequence.** Define $a_1 = 2$ and $a_{n+1} = a_n^2 - a_n + 1$, so the sequence begins $2, 3, 7, 43, 1807, \ldots$

(a) Show that $a_{n+1} - 1 = a_1 a_2 \cdots a_n$.

(b) Deduce that the terms are pairwise coprime, and hence give another proof that there are infinitely many primes.

(c) HARDER Show that $\dfrac{1}{a_1} + \dfrac{1}{a_2} + \cdots + \dfrac{1}{a_n} = 1 - \dfrac{1}{a_{n+1} - 1}$.

4 **Euler's trick.** *This needs Fermat's Little Theorem: if p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.* Suppose a prime p divides F_n.

(a) Show that $2^{2^n} \equiv -1 \pmod{p}$, and hence that $2^{2^{n+1}} \equiv 1 \pmod{p}$.

(b) Let d be the smallest positive integer with $2^d \equiv 1 \pmod{p}$. Show that d divides 2^{n+1} but not 2^n, and conclude that $d = 2^{n+1}$.

(c) Using Fermat's Little Theorem, show that d divides $p - 1$. Deduce that $p \equiv 1 \pmod{2^{n+1}}$.

(d) For $n = 5$, this says any prime factor of F_5 has the form $64k + 1$. Check that 641 does. How many primes of this form are smaller than 641?

5 **A bound on the primes.** Use the argument in *One theorem, many proofs* to show that the nth prime is at most $2^{2^{n-1}} + 1$. How good is this bound when $n = 5$?

Further reading

- **Martin Aigner & Günter M. Ziegler, *Proofs from THE BOOK*** (Springer), Chapter 1, "Six proofs of the infinity of primes". It includes this proof alongside Euclid's and four others, one of which uses topology. One of the best books you can read before university.
- **Harold Davenport, *The Higher Arithmetic*** (Cambridge University Press), Chapters 1–2. A beautifully written introduction to primes and factorisation, ideal for the summer before university.
- **Michal Křížek, Florian Luca & Lawrence Somer, *17 Lectures on Fermat Numbers*** (Springer, 2001). Everything about Fermat numbers, from Euler's factorisation to Gauss's polygons. Dip into the early chapters.
- **Ian Stewart, *Why Beauty Is Truth*** (Basic Books), the chapter on Gauss. The story of the 17-gon told for a general reader.

1.2 How Thinly Spread Are the Primes?

The question

(i) By considering the binomial expansion of $(1+x)^{2m+1}$, prove that

$$\binom{2m+1}{m} < 2^{2m}$$

for any positive integer m.

(ii) For any positive integers r and s with $r < s$, $P_{r,s}$ is defined as follows: $P_{r,s}$ is the product of all the prime numbers greater than r and less than or equal to s, if there are any such prime numbers; if there are no such prime numbers, then $P_{r,s} = 1$.

For example, $P_{3,7} = 35$, $P_{7,10} = 1$ and $P_{14,18} = 17$.

Show that, for any positive integer m, $P_{m+1,2m+1}$ divides $\binom{2m+1}{m}$, and deduce that

$$P_{m+1,2m+1} < 2^{2m}.$$

(iii) Show that, if $P_{1,k} < 4^k$ for $k = 2, 3, \ldots, 2m$, then $P_{1,2m+1} < 4^{2m+1}$.

(iv) Prove that $P_{1,n} < 4^n$ for $n \geq 2$.

The hook

In Section 1.1 we saw that there are infinitely many primes. But "infinitely many" is a blunt statement. The primes certainly get rarer: there are 25 primes below 100 but only 14 between 900 and 1000. *How* rare do they get?

This question proves something that sounds quite modest: if you multiply together all the primes up to n, the answer is less than 4^n. But that innocent inequality is the first real step towards answering "how rare?", and it is the key lemma in one of the most celebrated short proofs in mathematics, by the young Paul Erdős in 1932.

Before you start

The notation $P_{r,s}$ looks fussy, but the idea is simple: multiply together the primes in a range. Make sure you understand the examples. $P_{3,7}$ uses the primes greater than 3 and at most 7, namely 5 and 7, so it is 35.

Ask yourself Part (iv) asks you to prove a statement for *all* $n \geq 2$. Part (iii) assumes the statement for every k up to $2m$. What kind of induction is that?

It is *strong* induction: to prove the case n, you may assume *all* the earlier cases, not just the one immediately before. Keep that in mind; it is why the question is laid out the way it is.

Working it through

STAGE 1 A binomial coefficient is not too big

Ask yourself What do you get if you put $x = 1$ in the binomial expansion of $(1+x)^{2m+1}$?

You get

$$2^{2m+1} = \binom{2m+1}{0} + \binom{2m+1}{1} + \cdots + \binom{2m+1}{2m+1}.$$

All the terms are positive, so each one is less than 2^{2m+1}. That is not quite good enough; we want 2^{2m}, which is half as much.

Ask yourself Pascal's triangle is symmetric. Which other term in the sum equals $\binom{2m+1}{m}$?

By symmetry, $\binom{2m+1}{m} = \binom{2m+1}{m+1}$, and these are two *different* terms of the sum (since $m \neq m+1$). So

$$2\binom{2m+1}{m} = \binom{2m+1}{m} + \binom{2m+1}{m+1} < 2^{2m+1},$$

with strict inequality because the sum also contains other positive terms such as $\binom{2m+1}{0} = 1$. Dividing by 2 gives $\binom{2m+1}{m} < 2^{2m}$. ■

STAGE 2 Which primes divide that binomial coefficient?

Write the binomial coefficient out:

$$\binom{2m+1}{m} = \frac{(2m+1)!}{m!\,(m+1)!} = \frac{(m+2)(m+3)\cdots(2m+1)}{m!}.$$

Ask yourself Let p be a prime with $m+1 < p \leq 2m+1$. Does p divide the numerator? Can it divide the denominator?

The numerator $(m+2)(m+3)\cdots(2m+1)$ contains p itself as one of its factors, so p divides it. The denominator $m!$ is a product of numbers all smaller than p; since p is prime, it cannot divide a product of numbers none of which it divides. So p divides

the numerator but not the denominator, and therefore p divides the whole binomial coefficient (which, remember, is a whole number).

Pause and try We now know that *each* prime in the range divides $\binom{2m+1}{m}$. Why does it follow that their *product* $P_{m+1,2m+1}$ divides it too? (Careful: if 4 and 6 both divide a number, 24 need not.)

The difference is that our primes are *distinct primes*. If distinct primes $p_1, p_2, \ldots$ each divide N, then their product divides N: write $N = p_1 N_1$; then p_2 divides $p_1 N_1$ but not p_1, so it divides N_1; and so on. (This uses the fact that if a prime divides a product then it divides one of the factors. That fact is worth a moment's respect; Section 1.7 shows a world in which its analogue breaks down.)

So $P_{m+1,2m+1}$ divides $\binom{2m+1}{m}$, and since a divisor of a positive number is no bigger than it,

$$P_{m+1,\,2m+1} \le \binom{2m+1}{m} < 2^{2m}. \qquad \blacksquare$$

STAGE 3 Splitting the product

Ask yourself How can $P_{1,2m+1}$, the product of all primes up to $2m+1$, be built from products we already know about?

Split the primes up to $2m+1$ into those up to $m+1$ and those above $m+1$:

$$P_{1,2m+1} = P_{1,m+1} \times P_{m+1,2m+1}.$$

The second factor is less than $2^{2m} = 4^m$ by Stage 2. For the first, we are allowed to assume $P_{1,k} < 4^k$ for every k from 2 to $2m$, and $k = m+1$ is in that range (because $m \ge 1$ gives $2 \le m+1 \le 2m$). So

$$P_{1,2m+1} < 4^{m+1} \times 4^m = 4^{2m+1}. \qquad \blacksquare$$

This is where strong induction earns its keep: to deal with $2m+1$ we needed the case $m+1$, which is roughly *half* as big, not the case just before.

STAGE 4 All n

Ask yourself Part (iii) handles odd numbers. What happens for an even number $n = 2m+2$? Is it prime?

Check the start: $P_{1,2} = 2 < 16$ and $P_{1,3} = 6 < 64$. Now suppose $P_{1,k} < 4^k$ for all k from 2 up to $n-1$, where $n \ge 4$.

- If n is even, then n is not prime (it is bigger than 2), so the primes up to n are the same as the primes up to $n-1$. Hence $P_{1,n} = P_{1,n-1} < 4^{n-1} < 4^n$.
- If n is odd, write $n = 2m+1$ with $m \ge 2$. Then part (iii) gives $P_{1,n} < 4^n$ directly.

By strong induction, $P_{1,n} < 4^n$ for every $n \geq 2$. ■

Looking back

- **Binomial coefficients carry arithmetic information.** $\binom{2m+1}{m}$ is a single number that is small (less than 4^m) but is forced to be divisible by many primes. Squeezing many primes into a small number tells you there can't be too many of them.
- **Strong induction is the natural tool when a problem halves itself.** The case $2m+1$ depended on the case $m+1$.
- **Small cases need checking separately.** The induction only gets going once $n \geq 4$, so $n = 2$ and 3 had to be done by hand.

What's really going on

From 4^n to the Prime Number Theorem

Take logarithms of the result: $\ln P_{1,n} < n \ln 4$, that is,

$$\sum_{p \leq n} \ln p < (\ln 4)\, n \approx 1.39\, n.$$

In the 1850s Pafnuty Chebyshev proved bounds of exactly this kind, above *and below*, showing that $\sum_{p \leq n} \ln p$ grows in proportion to n. Gauss, aged about 15, had already guessed from tables that the number of primes up to n is roughly $n / \ln n$. This became the *Prime Number Theorem*, finally proved in 1896 by Jacques Hadamard and Charles de la Vallée Poussin using complex analysis. In its logarithmic form it says

$$\sum_{p \leq n} \ln p \sim n,$$

so the product of the primes up to n is roughly e^n. Our bound of 4^n is not far off at all: Figure 1.2.1 shows all three.

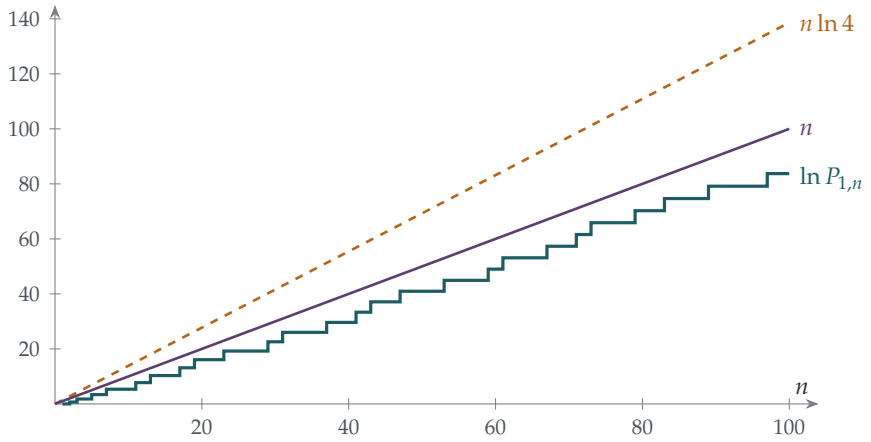


Figure 1.2.1 The logarithm of the product of all primes up to n (teal), jumping by $\ln p$ at each prime p. This section proves it stays below $n \ln 4$ (amber, dashed); the Prime Number Theorem says it grows like n (plum).

Bertrand's postulate

In 1845 Joseph Bertrand conjectured that for every $n \geq 1$ there is always a prime between n and $2n$. Chebyshev proved it in 1852. Eighty years later, Paul Erdős, aged 19, found a beautiful elementary proof. Its key ingredients are this question's lemma and a careful look at which primes divide the *middle* binomial coefficient $\binom{2n}{n}$. Martin Aigner and Günter Ziegler chose it as Chapter 2 of *Proofs from THE BOOK*, a collection of the most elegant proofs in mathematics.

Where this goes next

The precise distribution of the primes is controlled by the zeros of the Riemann zeta function,

$$\zeta(s) = 1 + \frac{1}{2^s} + \frac{1}{3^s} + \cdots,$$

a function you will meet in a much humbler guise in Section 2.4, where $\zeta(2) = \pi^2/6$. The *Riemann Hypothesis*, which predicts exactly where those zeros lie, would pin down the error in the Prime Number Theorem as tightly as possible. It is perhaps the most famous unsolved problem in mathematics, and one of the Millennium Prize Problems.

The thinking skill

Strong induction, and extracting information from a clever object

If the case n depends on cases much smaller than $n-1$, assume all earlier cases at once.

The other idea here is subtler and very characteristic of number theory: to show something about *all* primes in a range, find one object (here a binomial coefficient) that every one of those primes must divide, and that you can also show is not too large.

Try this next

1 **Feel the numbers.** Compute $P_{1,n}$ for $n = 10, 20, 30$ and compare it with 4^n and with e^n. (A calculator is fine here.)

2 **The middle binomial coefficient.** Show that $\binom{2m}{m}$ is the largest of the $2m+1$ terms in the expansion of $(1+1)^{2m}$, and deduce that $\binom{2m}{m} \geq \dfrac{4^m}{2m+1}$.

3 **Not too many primes.** Let $\pi(n)$ be the number of primes up to n. Every prime between $\sqrt{n}$ and n is bigger than $\sqrt{n}$, and their product is at most $P_{1,n} < 4^n$. Deduce that

$$\pi(n) - \pi(\sqrt{n}) < \frac{2\ln 4}{\ln n}\, n,$$

and hence that $\pi(n) < 2.8\, n/\ln n + \sqrt{n}$. Compare this with the Prime Number Theorem's $\pi(n) \approx n/\ln n$.

4 **Check Bertrand.** Verify Bertrand's postulate for $n = 1, 2, \ldots, 50$ by finding a prime between n and $2n$ in each case. Can you do it efficiently using a short list of primes, each less than twice the one before?

Further reading

- **Aigner & Ziegler, *Proofs from THE BOOK*,** Chapter 2, "Bertrand's postulate".
- **Marcus du Sautoy, *The Music of the Primes*** (Fourth Estate). A popular history of the Prime Number Theorem and the Riemann Hypothesis.
- **G. J. O. Jameson, *The Prime Number Theorem*** (Cambridge University Press). Rigorous but gentle, and written for undergraduates; the first chapters need only A-level calculus.

1.3 The Smallest Counterexample

The question

(i) In the following argument to show that $\sqrt{2}$ is irrational, give proofs appropriate for steps 3, 5 and 6.

1. Assume that $\sqrt{2}$ is rational.
2. Define the set S to be the set of positive integers with the following property: n is in S if and only if $n\sqrt{2}$ is an integer.
3. Show that the set S contains at least one positive integer.
4. Define the integer k to be the smallest positive integer in S.
5. Show that $(\sqrt{2}-1)k$ is in S.
6. Show that steps 4 and 5 are contradictory and hence that $\sqrt{2}$ is irrational.

(ii) Prove that $2^{1/3}$ is rational if and only if $2^{2/3}$ is rational.

Use an argument similar to that of part (i) to prove that $2^{1/3}$ and $2^{2/3}$ are irrational.

The hook

You have probably seen a proof that $\sqrt{2}$ is irrational: suppose $\sqrt{2} = a/b$ in lowest terms, square, deduce that a is even, then that b is even, contradiction. It is a fine proof. But it relies on a trick about *even numbers* that doesn't obviously generalise. What about $\sqrt[3]{2}$? Or $\sqrt{7}$?

This question offers a different proof, and its central idea is almost absurdly simple: *look at the smallest one*. The same idea will reappear, in disguise, several times in this chapter.

Before you start

This is an unusual STEP question: it gives you the skeleton of a proof and asks you to fill in the missing justifications. That is exactly how undergraduates learn to read mathematics. A textbook proof often says "clearly" or "it follows that", and part of your job is to check that it really does.

Ask yourself Step 4 says "define k to be the smallest positive integer in S". Why are we entitled to do that? Would it be allowed if S were a set of positive *real* numbers?

Any non-empty set of positive integers has a smallest element. This is called the *well-ordering principle*, and it is equivalent to the principle of mathematical induction. It is *false* for sets of positive reals: the set $\{x : x > 0\}$ has no smallest element. So step 4 relies on step 3 (the set must be non-empty) and on S consisting of integers.

Working it through

STAGE 1 Step 3: S is not empty

Ask yourself If $\sqrt{2} = a/b$ for positive integers a and b, can you write down an element of S?

If $\sqrt{2} = a/b$, then $b\sqrt{2} = a$, which is an integer. So b is in S, and S is not empty.

STAGE 2 Step 5: a smaller element

Now k is the smallest element of S, so $k\sqrt{2}$ is an integer; call it j.

Pause and try To show that $(\sqrt{2}-1)k$ belongs to S you must check three things. What are they?

We need $(\sqrt{2}-1)k$ to be (a) an integer, (b) positive, and (c) such that multiplying it by $\sqrt{2}$ gives an integer.

- (a) $(\sqrt{2}-1)k = k\sqrt{2} - k = j - k$, a difference of integers.
- (b) $\sqrt{2} > 1$, so $\sqrt{2} - 1 > 0$ and hence $(\sqrt{2}-1)k > 0$.
- (c) $(\sqrt{2}-1)k \cdot \sqrt{2} = 2k - k\sqrt{2} = 2k - j$, an integer.

So $(\sqrt{2}-1)k \in S$.

STAGE 3 Step 6: the contradiction

Ask yourself How big is $\sqrt{2} - 1$?

Since $1 < \sqrt{2} < 2$, we have $0 < \sqrt{2} - 1 < 1$, and so $(\sqrt{2}-1)k < k$. We have found an element of S smaller than k, the smallest element of S. That is impossible, so our assumption in step 1 was false: $\sqrt{2}$ is irrational. ■

Notice that we never needed "lowest terms" or any facts about even and odd numbers.

The whole proof rests on three facts: $\sqrt{2} \cdot \sqrt{2}$ is an integer, $\sqrt{2}$ lies strictly between two consecutive integers, and integers have a well-ordering.

STAGE 4 Cube roots: linking the two numbers

Write $c = 2^{1/3}$, so that $c^2 = 2^{2/3}$ and $c^3 = 2$.

? **Ask yourself** If c is rational, is c^2? And if c^2 is rational, can you get back to c using only multiplication and division?

If c is rational then so is $c^2 = c \times c$. Conversely, if c^2 is rational then so is $2/c^2 = c^3/c^2 = c$. So $2^{1/3}$ is rational if and only if $2^{2/3}$ is. ■

STAGE 5 Adapting the argument

? **Ask yourself** In part (i), the key was that $(\sqrt{2})^2 = 2$ is an integer, so multiplying an element of S by $\sqrt{2}$ twice takes you back to integers. For $c = 2^{1/3}$ you would need to multiply *three* times. How should the set S change?

Suppose c is rational; then c^2 is too. Define

$$S = \{n \text{ a positive integer} : nc \text{ and } nc^2 \text{ are both integers}\}.$$

- S **is not empty.** If $c = a/b$ and $c^2 = a'/b'$, then bb' is in S.
- **Let** k **be its smallest element**, and consider $(c-1)k$. We have $1 < c < 2$ (since $1^3 < 2 < 2^3$), so $0 < (c-1)k < k$.
- **It is an integer:** $(c-1)k = kc - k$.
- **Times** c**:** $(c-1)k \cdot c = kc^2 - kc$, an integer.
- **Times** c^2**:** $(c-1)k \cdot c^2 = kc^3 - kc^2 = 2k - kc^2$, an integer.

So $(c-1)k \in S$ and is smaller than k: contradiction. Hence $2^{1/3}$ is irrational, and so, by Stage 4, is $2^{2/3}$. ■

Looking back

- **"Take the smallest one" is a proof technique.** Assume a counterexample exists, take the smallest, and manufacture a smaller one. This is called the *extremal principle*, or a *minimal counterexample* argument.
- **The set** S **did the bookkeeping.** Instead of juggling numerators and denominators, we collected all the "good" multipliers into one set.
- **Generalising meant asking what really made the proof work.** For $\sqrt{2}$ one multiplication by $\sqrt{2}$ was needed; for $\sqrt[3]{2}$ we needed two. Spotting that told us exactly how to redefine S.

What's really going on

A proof you can draw

There is a lovely picture behind part (i), due to Stanley Tennenbaum. If $\sqrt{2} = a/b$ then $a^2 = 2b^2$: a square of side a has the same area as two squares of side b. Place the two b-squares in opposite corners of the a-square, as in Figure 1.3.1. They overlap in a small square, and leave two small squares uncovered. Equal areas force the overlap to equal the two uncovered squares together, giving a *smaller* pair of integers with the same property. Smaller and smaller pairs can't go on for ever, so no such squares exist. The smaller pair is exactly $(2b - a,\ a - b)$, which is our $(\sqrt{2} - 1)k$ step in disguise.

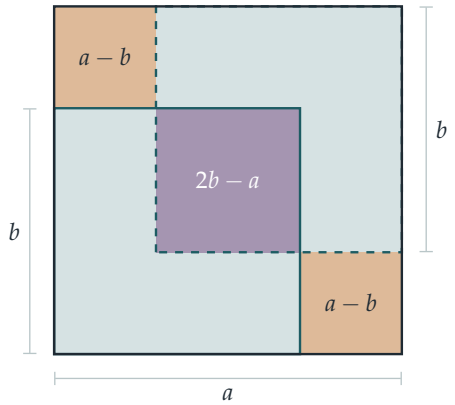


Figure 1.3.1 Tennenbaum's proof. Two squares of side b (teal) sit in opposite corners of a square of side a. If $a^2 = 2b^2$, the area counted twice (the plum overlap) must equal the area missed (the two amber corners), so $(2b - a)^2 = 2(a - b)^2$: a smaller solution.

Well-ordering, induction and descent

The well-ordering principle and the principle of induction are logically equivalent: each can be proved from the other. At university you will see both used as the foundations of the natural numbers. In Section 1.5 we will use the same principle in a different form, Fermat's *method of infinite descent*: there is no infinite strictly decreasing sequence of positive integers.

How far does it go?

The argument proves that $\sqrt{N}$ is irrational whenever N is not a perfect square (see the exercises), and more generally that the nth root of a whole number is either a whole number or irrational. At university this becomes a statement about *algebraic integers*: a rational number that satisfies a monic polynomial equation with integer coefficients, such as $x^3 - 2 = 0$, must be an integer. You will meet the idea again in Section 6.3.

The legend that the Pythagoreans drowned Hippasus for revealing the irrationality of $\sqrt{2}$ is probably untrue. What is true is that the discovery forced Greek mathematicians to separate the study of number from the study of length, a split that took until the nineteenth century, and Dedekind's construction of the real numbers, to heal.

The thinking skill

The extremal principle

If you want to show that something never happens, suppose it does, choose the smallest (or largest, or simplest) instance, and use it to build an even smaller one.

The choice of what "smallest" means is the creative step. Here it was the smallest n making $n\sqrt{2}$ an integer.

Try this next

1. **Other square roots.** Adapt the argument to show that $\sqrt{3}$ is irrational. Which number plays the role of $\sqrt{2}-1$?

2. **All non-squares.** Let N be a positive integer that is not a perfect square, and let m be the integer with $m < \sqrt{N} < m+1$. Prove that $\sqrt{N}$ is irrational by considering $(\sqrt{N}-m)k$.

3. **Spot the flaw.** Try to run the argument for $\sqrt{4}$. Exactly which step fails?

4. **Tennenbaum's picture.** Use Figure 1.3.1 to show that if $a^2 = 2b^2$ then $(2b-a)^2 = 2(a-b)^2$. Check this identity algebraically too.

Further reading

- **Daniel Velleman, *How to Prove It*** (Cambridge University Press). A thorough introduction to proof, including induction and well-ordering.
- **R. P. Burn, *A Pathway into Number Theory*** (Cambridge University Press). A book made entirely of carefully sequenced questions; ideal preparation for a first-year course.
- **John Conway & Joseph Shipman, "Extreme proofs I: the irrationality of $\sqrt{2}$"** (*Mathematical Intelligencer*, 2013). Many proofs of one fact, compared with wit.

1.4 e Is Irrational, and So Is e^2

This section uses two questions. The first, from 1997, proves that e is irrational. The second, from 2024, uses a cleverer version of the same idea to prove that e^2 is irrational, which turns out to be genuinely harder.

Warm-up: STEP III 1997, Question 7

For each positive integer n, let

$$a_n = \frac{1}{n+1} + \frac{1}{(n+1)(n+2)} + \frac{1}{(n+1)(n+2)(n+3)} + \cdots,$$
$$b_n = \frac{1}{n+1} + \frac{1}{(n+1)^2} + \frac{1}{(n+1)^3} + \cdots.$$

(i) Evaluate b_n.

(ii) Show that $0 < a_n < 1/n$.

(iii) Deduce that $a_n = n!\,e - [n!\,e]$ (where $[x]$ is the integer part of x).

(iv) Hence show that e is irrational.

Main question: STEP III 2024, Question 7

In this question, you need not consider issues of convergence.

For positive integer n let

$$f(n) = \frac{1}{n+1} + \frac{1}{(n+1)(n+2)} + \frac{1}{(n+1)(n+2)(n+3)} + \cdots$$

and

$$g(n) = \frac{1}{n+1} - \frac{1}{(n+1)(n+2)} + \frac{1}{(n+1)(n+2)(n+3)} - \cdots.$$

(i) Show, by considering a geometric series, that $0 < f(n) < \dfrac{1}{n}$.

(ii) Show, by comparing consecutive terms, that $0 < g(n) < \dfrac{1}{n+1}$.

(iii) Show, for positive integer n, that $(2n)!\,e - f(2n)$ and $\dfrac{(2n)!}{e} + g(2n)$ are both integers.

(iv) Show that if $qe = \dfrac{p}{e}$ for some positive integers p and q, then $qf(2n) + pg(2n)$ is an integer for all positive integers n.

(v) Hence show that the number e^2 is irrational.

The hook

In Section 1.3 we proved that $\sqrt{2}$ is irrational using the fact that $\sqrt{2}$ satisfies a simple equation, $x^2 = 2$. But e doesn't satisfy any such equation that we know of. All we have is its series,

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots.$$

How can a series tell you that a number is not a fraction?

The answer is one of the most useful ideas in number theory: *a whole number cannot lie strictly between* 0 *and* 1. If you can show that some quantity is a whole number, and also that it is strictly between 0 and 1, you have a contradiction. Everything in this section is a variation on that theme.

Before you start

Ask yourself Suppose $e = p/q$. Multiplying e by q gives an integer. What does multiplying by $q!$ do to each term of the series for e?

Multiplying $\dfrac{1}{k!}$ by $q!$ gives an integer whenever $k \leq q$. So $q!\,e$ splits into a whole-number part (the first $q+1$ terms) and a "tail" (everything after). If $e = p/q$ then $q!\,e$ is itself an integer, so the tail must be an integer too. The plan is to show the tail is strictly between 0 and 1.

Working it through

STAGE 1 The warm-up: e is irrational

The series b_n is geometric with first term and ratio both $\frac{1}{n+1}$, so

$$b_n = \frac{1/(n+1)}{1 - 1/(n+1)} = \frac{1}{n}.$$

Ask yourself Compare a_n and b_n term by term. Which is bigger?

The first terms are equal. After that, $(n+1)(n+2) > (n+1)^2$, and in general $(n+$

$1)(n+2)\cdots(n+k) > (n+1)^k$ for $k \geq 2$, so each later term of a_n is smaller than the corresponding term of b_n. Hence $0 < a_n < b_n = 1/n$.

Now look at what a_n really is. Multiply the series for e by $n!$:

$$n!\,e = \underbrace{\frac{n!}{0!} + \frac{n!}{1!} + \cdots + \frac{n!}{n!}}_{\text{an integer}} + \underbrace{\frac{n!}{(n+1)!} + \frac{n!}{(n+2)!} + \cdots}_{=a_n}.$$

Since $0 < a_n < 1$, the integer part of $n!\,e$ is the first bracket and a_n is the fractional part: $a_n = n!\,e - [n!\,e]$.

Pause and try Now finish the proof. Suppose $e = p/q$. Which value of n should you choose?

Take $n = q$. Then $q!\,e = (q-1)!\,p$ is an integer, so its fractional part a_q is 0. But $a_q > 0$. Contradiction: e is irrational. ■

This proof is due to Joseph Fourier, in about 1815.

STAGE 2 Why e^2 is harder

Ask yourself Try the same idea on e^2. Its series is $\sum 2^k/k!$. What goes wrong when you multiply by $n!$ and look at the tail?

The tail is now $\dfrac{2^{n+1}}{n+1} + \dfrac{2^{n+2}}{(n+1)(n+2)} + \cdots$, and the powers of 2 ruin everything: the tail is no longer small. Multiplying by $n!$ doesn't help with the 2^k in the numerators.

The 2024 question's trick is to avoid e^2 altogether. If $e^2 = p/q$, then

$$qe = \frac{p}{e}.$$

Now we have two numbers, e and $1/e$, each with a factorial series and no troublesome powers of 2:

$$e = \sum_{k=0}^{\infty} \frac{1}{k!}, \qquad \frac{1}{e} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}.$$

STAGE 3 The two tails

Part (i) is exactly the warm-up: $0 < f(n) < 1/n$.

For part (ii), $g(n)$ is an *alternating* series whose terms decrease in size.

Ask yourself For an alternating series $t_1 - t_2 + t_3 - \cdots$ with $t_1 > t_2 > t_3 > \cdots > 0$, why is the sum between 0 and t_1?

Group the terms in pairs: $(t_1 - t_2) + (t_3 - t_4) + \cdots$ is a sum of positive brackets, so it is positive. Group them the other way: $t_1 - (t_2 - t_3) - (t_4 - t_5) - \cdots$ is t_1 minus positive brackets, so it is less than t_1. Here $t_1 = \frac{1}{n+1}$, and the terms decrease because each is the previous one divided by something bigger than 1. So $0 < g(n) < \frac{1}{n+1}$.

STAGE 4 Integers again

Multiply e by $(2n)!$ and split off the tail, exactly as in the warm-up:

$$(2n)!\, e = (\text{integer}) + f(2n).$$

So $(2n)!\, e - f(2n)$ is an integer. Now do the same for $1/e$:

$$\frac{(2n)!}{e} = \sum_{k=0}^{2n} (-1)^k \frac{(2n)!}{k!} + \sum_{k=2n+1}^{\infty} (-1)^k \frac{(2n)!}{k!}.$$

> **Ask yourself** What is the sign of the first term of the tail, $k = 2n+1$? How does the tail compare with $g(2n)$?

The first term of the tail has $k = 2n+1$, which is odd, so it is negative. The tail is

$$-\frac{1}{2n+1} + \frac{1}{(2n+1)(2n+2)} - \cdots = -g(2n).$$

So $\dfrac{(2n)!}{e} + g(2n)$ is an integer. ■

This is why the question uses $2n$ rather than n: with an even multiplier, the tail of $1/e$ is *exactly* $-g$, with a predictable sign.

STAGE 5 The contradiction

Suppose $qe = p/e$. Multiply both sides by $(2n)!$ and use Stage 4:

$$q\big(\text{integer} + f(2n)\big) = p\big(\text{integer} - g(2n)\big).$$

Rearranging, $qf(2n) + pg(2n)$ equals a combination of integers, so it is an integer. That is part (iv).

> **Pause and try** Now use the bounds from Stage 3. What happens to $qf(2n) + pg(2n)$ when n is very large?

Both $f(2n)$ and $g(2n)$ are positive, so $qf(2n) + pg(2n) > 0$. And

$$qf(2n) + pg(2n) < \frac{q}{2n} + \frac{p}{2n+1} < \frac{p+q}{2n}.$$

Choose $n > (p+q)/2$. Then $qf(2n) + pg(2n)$ is an integer strictly between 0 and 1. That is impossible, so e^2 is irrational. ■

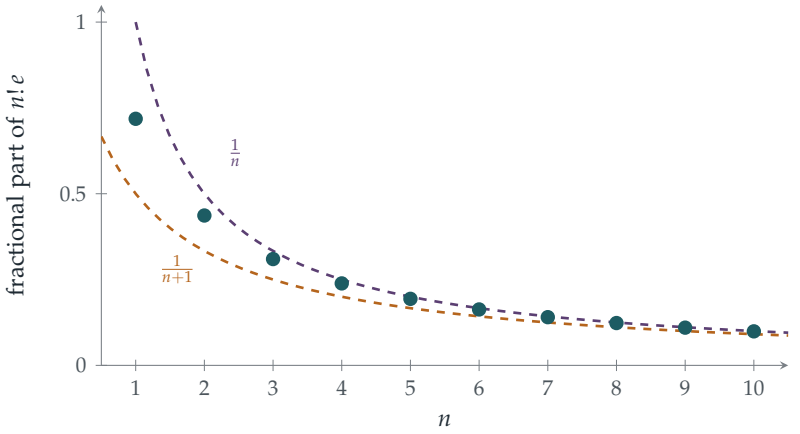


Figure 1.4.1 The tail $a_n = n!\,e - [n!\,e]$ (teal), which is the fractional part of $n!\,e$. It is always strictly between $\frac{1}{n+1}$ and $\frac{1}{n}$: never zero, but less than 1. If e were a fraction $\frac{p}{q}$, then $n!\,e$ would be a whole number for every $n \geq q$, and the teal points would be 0.

Looking back

- **The engine is "no integer lies strictly between 0 and 1".** Both proofs build a quantity that must be an integer and then trap it.
- **Factorials make tails small.** Multiplying by $n!$ turned the start of the series into integers and left a tail that shrinks like $1/n$.
- **When a direct attack fails, rearrange.** Rather than attacking $e^2 = p/q$ head-on, we rewrote it as $qe = p/e$, which spread the difficulty across two easier numbers.
- **The details matter.** The choice of $2n$ rather than n fixed the sign of a tail. Many proofs hinge on a small detail like this.

What's really going on

How irrational is e?

Joseph Liouville proved in 1840 that e^2 is irrational, and in fact that it doesn't satisfy any quadratic equation with integer coefficients. In 1873 Charles Hermite went much further: he proved that e is *transcendental*, meaning it is not a root of *any* polynomial with integer coefficients. His proof is a sophisticated version of exactly the argument above: build an integer, show it is strictly between 0 and 1. Nine years later Ferdinand von Lindemann adapted Hermite's method to show that π is transcendental, finally proving that the ancient problem of "squaring the circle" with ruler and compass is impossible.

What we still don't know

For all the power of these methods, embarrassingly simple questions remain open. Is $e + \pi$ irrational? Is $e\pi$? Nobody knows, although at least one of them must be transcendental (exercise 4). Is $\zeta(3) = 1 + \frac{1}{8} + \frac{1}{27} + \cdots$ irrational? Yes: Roger Apéry astonished the mathematical world by proving it in 1978. Is $\zeta(5)$? Nobody knows.

Approximation

Behind these proofs lies a deeper idea: a number is irrational if it can be approximated by fractions *too well*. The partial sums of the series for e are fractions that get very close to e very quickly; a fraction can't be approximated by other fractions that well. This is the starting point of *Diophantine approximation*, a branch of number theory you can explore further in Section 2.9, where a different family of integrals produces even better fractions closing in on e.

The thinking skill

Trapping an integer

To prove a number is irrational, assume it is a fraction, build from it a quantity that must be a whole number, and show that the quantity is strictly between 0 *and* 1.

The art lies in choosing the quantity. Here, multiplying by a factorial made most of a series into integers and left a tail that was small but positive.

Try this next

1 **Using $1/e$ alone.** Prove that e is irrational by supposing $1/e = p/q$ and using only the alternating series for $1/e$.

2 $\sin 1$ **is irrational.** Use $\sin 1 = 1 - \dfrac{1}{3!} + \dfrac{1}{5!} - \cdots$ to prove that $\sin 1$ is irrational. (Multiply by a suitable odd factorial.)

3 **Why not π?** Explain why this method doesn't work directly for π. Then look up Ivan Niven's one-page proof (1947) that π is irrational, which uses an integral instead of a series. It is within reach of a strong A-level student.

4 **One of them is transcendental.** HARDER Show that e and π are the roots of $x^2 - (e + \pi)x + e\pi = 0$. You may assume that a root of a polynomial whose coefficients are algebraic numbers is itself algebraic, and that π is not algebraic (Lindemann). Deduce that at least one of $e + \pi$ and $e\pi$ is transcendental.

Further reading

- **Aigner & Ziegler, *Proofs from THE BOOK*,** Chapter 8, "Some irrational numbers". Proves that e, e^2, e^4 and π^2 are irrational, using exactly these ideas.
- **Ivan Niven, *Irrational Numbers*** (Mathematical Association of America). A short classic covering everything from $\sqrt{2}$ to transcendence.
- **Julian Havil, *The Irrationals*** (Princeton University Press). A readable history of irrational numbers.

1.5 Infinite Descent

The question

(i) Suppose that there are three non-zero integers a, b and c for which $a^3 + 2b^3 + 4c^3 = 0$.

Explain why there must exist an integer p, with $|p| < |a|$, such that $4p^3 + b^3 + 2c^3 = 0$, and show further that there must exist integers p, q and r, with $|p| < |a|$, $|q| < |b|$ and $|r| < |c|$, such that $p^3 + 2q^3 + 4r^3 = 0$.

Deduce that no such integers a, b and c can exist.

(ii) Prove that there are no non-zero integers a, b and c for which $9a^3 + 10b^3 + 6c^3 = 0$.

(iii) By considering the expression $(3n \pm 1)^2$, prove that, unless an integer is a multiple of three, its square is one more than a multiple of 3. Deduce that the sum of the squares of two integers can only be a multiple of three if each of the integers is a multiple of three.

Hence prove that there are no non-zero integers a, b and c for which $a^2 + b^2 = 3c^2$.

(iv) Prove that there are no non-zero integers a, b and c for which $a^2 + b^2 + c^2 = 4abc$.

The hook

How can you prove that an equation has *no* whole-number solutions? You can't try every possibility; there are infinitely many. Pierre de Fermat, in the 1630s, found a way: show that any solution would produce another solution with *smaller* numbers, and then another, smaller still, and so on for ever. But positive whole numbers can't keep getting smaller for ever. So there was never a solution in the first place.

Fermat called this the *method of infinite descent*, and he was very proud of it. It is the reverse side of the well-ordering principle from Section 1.3, and it is still one of the main tools for proving that equations have no solutions.

Before you start

All four parts have the same shape: assume a solution, shrink it, contradict. The question is *how* to shrink.

Ask yourself In part (i), look at $a^3 + 2b^3 + 4c^3 = 0$. Two of the three terms are obviously even. What does that tell you about the third?

That is the kind of observation that starts every descent: some divisibility is forced on you.

Working it through

STAGE 1 The first descent

Since $a^3 = -2b^3 - 4c^3$, the number a^3 is even, and so a is even (the cube of an odd number is odd). Write $a = 2p$. Then $|p| = |a|/2 < |a|$, and substituting,

$$8p^3 + 2b^3 + 4c^3 = 0 \quad \Longrightarrow \quad 4p^3 + b^3 + 2c^3 = 0.$$

Ask yourself The new equation is $4p^3 + b^3 + 2c^3 = 0$. It has the same coefficients, just rotated. Can you do the same trick again?

Now $b^3 = -4p^3 - 2c^3$ is even, so $b = 2q$ with $|q| < |b|$, and dividing by 2 gives $2p^3 + 4q^3 + c^3 = 0$. Once more, c must be even: $c = 2r$ with $|r| < |c|$, and dividing by 2,

$$p^3 + 2q^3 + 4r^3 = 0.$$

This is the *original* equation, with p, q, r in place of a, b, c. Each is half the size, and none of them is zero, because halving a non-zero number can't give zero.

Pause and try Turn this into a proof that no non-zero solution exists. What, exactly, should you assume is the "smallest"?

Among all solutions in non-zero integers, choose one with $|a|$ as small as possible. (The well-ordering principle lets us do this, if any solutions exist at all.) The argument above produces another non-zero solution with $|p| < |a|$, contradicting minimality. So there are no non-zero solutions. ■

An equivalent way to say it: from one solution we could build an infinite sequence of solutions with $|a| > |p| > \cdots$, an infinite strictly decreasing sequence of positive integers, which is impossible.

STAGE 2 A different prime

Ask yourself In $9a^3 + 10b^3 + 6c^3 = 0$, parity doesn't obviously help. Look at the coefficients 9, 10, 6. Which prime divides two of them?

The prime 3 divides 9 and 6, so 3 divides $10b^3$, and hence 3 divides b. Write $b = 3q$ and

divide by 3:

$$3a^3 + 90q^3 + 2c^3 = 0.$$

Now 3 divides $2c^3$, so $c = 3r$; dividing by 3 again gives $a^3 + 30q^3 + 18r^3 = 0$. So 3 divides a: $a = 3p$, and dividing by 3 once more,

$$9p^3 + 10q^3 + 6r^3 = 0.$$

Exactly the original equation, with smaller non-zero numbers. By the same descent argument, there are no non-zero solutions. ■

STAGE 3 **Squares modulo 3**

Ask yourself Every integer is of the form $3n$, $3n+1$ or $3n-1$. What are their squares, modulo 3?

$(3n \pm 1)^2 = 9n^2 \pm 6n + 1 = 3(3n^2 \pm 2n) + 1$, one more than a multiple of 3. And $(3n)^2$ is a multiple of 3. So a square leaves remainder 0 or 1 on division by 3, never 2.

Now if $a^2 + b^2$ is a multiple of 3, the possible remainders are $0+0$, $0+1$, $1+0$ and $1+1$, which give 0, 1, 1, 2. Only the first is a multiple of 3, so both a and b must be multiples of 3.

Ask yourself If $a^2 + b^2 = 3c^2$ and $a = 3p$, $b = 3q$, what can you say about c?

We get $9p^2 + 9q^2 = 3c^2$, so $c^2 = 3(p^2 + q^2)$ is a multiple of 3, and hence $c = 3r$. Then $9p^2 + 9q^2 = 27r^2$, so $p^2 + q^2 = 3r^2$: the same equation, smaller numbers. Descent again. ■

STAGE 4 **A twist: the equation changes**

Ask yourself In $a^2 + b^2 + c^2 = 4abc$ the right-hand side is a multiple of 4. What are the squares modulo 4?

Even squares are multiples of 4 and odd squares are one more than a multiple of 4 ($(2n+1)^2 = 4n^2 + 4n + 1$). So $a^2 + b^2 + c^2$ leaves remainder 0, 1, 2 or 3 on division by 4, according to how many of a, b, c are odd. Since the right-hand side is a multiple of 4, *none* of them can be odd. Write $a = 2p$, $b = 2q$, $c = 2r$:

$$4(p^2 + q^2 + r^2) = 32\,pqr \quad \Longrightarrow \quad p^2 + q^2 + r^2 = 8\,pqr.$$

Ask yourself This is *not* the same equation: the 4 has become an 8. Does the descent still work?

Here is the trick: *prove something stronger*. Consider all equations of the form

$$x^2 + y^2 + z^2 = 2^m\,xyz \qquad \text{with } m \geq 2.$$

For any such m, the right-hand side is a multiple of 4, so exactly the same argument shows x, y, z are all even. Halving them gives a solution of the equation with 2^{m+1}, which is again of this form. So from a non-zero solution of the m equation we get a smaller non-zero solution of the $m+1$ equation. Taking the solution, of *any* equation in the family, with $|x|$ smallest gives a contradiction. In particular, $a^2+b^2+c^2=4abc$ has no non-zero solutions. ■

Sometimes the statement you want to prove is too weak to support its own induction, and the cure is to prove a more general statement that *is* self-supporting. This is a surprisingly common trick at university.

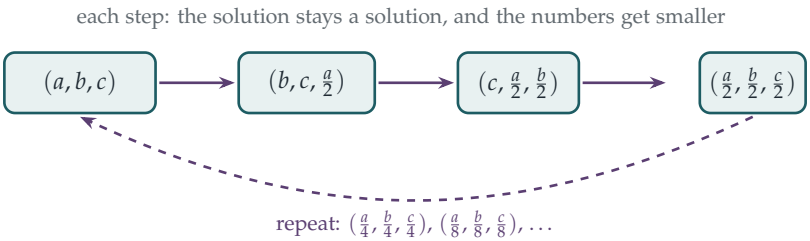


Figure 1.5.1 The descent in part (i). From a solution of $a^3+2b^3+4c^3=0$, three steps produce another solution with every entry halved. Repeating for ever would give an endless decreasing sequence of positive integers $|a|>|\frac{a}{2}|>|\frac{a}{4}|>\cdots$, which is impossible. So there was no solution to begin with.

Looking back

- **Descent needs a source of divisibility.** Each time, we found a prime p (here 2 or 3) that was forced to divide all the variables, and divided it out.
- **Working modulo n is the way to find it.** Looking at remainders mod 2, 3 or 4 turned "could anything work?" into a short finite check.
- **If the equation changes, generalise.** Part (iv) needed a whole family of equations, closed under the descent step.

What's really going on

Fermat, descent and his Last Theorem

Fermat used infinite descent to prove that $x^4+y^4=z^2$ has no solutions in positive integers. This is the only case of his famous "Last Theorem" for which he left a proof, and it settles the case $n=4$ of $x^n+y^n=z^n$. Euler later used descent, with a gap that was later filled, for $n=3$. Descent is still the starting point of the modern theory: the proof of the Mordell–Weil theorem about curves like the one in Section 1.9 is a

sophisticated descent.

Local obstructions

Every proof in this section worked by looking at remainders modulo some number. If an equation has no solutions modulo n for some n, or no solutions in real numbers, it certainly has no solutions in integers. Such reasons are called *local obstructions*, and at university you will meet the *p-adic numbers*, a way of studying all the remainders modulo $p, p^2, p^3, \ldots$ at once.

A wonderful theorem of Hasse and Minkowski says that for *quadratic* equations like $a^2 + b^2 = 3c^2$, local obstructions are the only obstructions: if there are no "local" reasons to fail, there really is a rational solution. For cubic equations this breaks down spectacularly. Ernst Selmer showed in 1951 that

$$3x^3 + 4y^3 + 5z^3 = 0$$

has solutions modulo every whole number, and in real numbers, but no solution in non-zero integers. Explaining when and why the "local-to-global" principle fails is a major theme of modern number theory.

An equation that *does* have solutions

Part (iv) has a famous cousin: $x^2 + y^2 + z^2 = 3xyz$, the *Markov equation*. It has infinitely many solutions in positive integers, starting with $(1,1,1)$, $(1,1,2)$, $(1,2,5)$, $(1,5,13)$ and $(2,5,29)$, and they are connected to how well irrational numbers can be approximated by fractions. See exercise 4.

The thinking skill

Infinite descent

To show an equation has no solutions, show that any solution would produce a strictly smaller one. Then the smallest solution can't exist.

Pair this with working modulo n to find the divisibility that drives the descent, and be ready to generalise the statement if it isn't preserved by the descent step.

Try this next

1 **A STEP I warm-up.** *(STEP I 2010 Question 8.)* Show that the only integer solution of $a^3 + 3b^3 = 9c^3$ is $a = b = c = 0$. Then, by considering final digits, show the same for $p^4 + 2q^4 = 5r^4$.

2 **A modulus with no descent needed.** Show that $x^2 + y^2 = 4z + 3$ has no integer solutions at all, just by working modulo 4.

3 **Rational points.** Use part (iii) to show that there is no point with *rational* coordinates on the circle $x^2 + y^2 = 3$. (Contrast this with $x^2 + y^2 = 2$, which contains $(1, 1)$.)

4 **Markov triples.** Show that if (x, y, z) satisfies $x^2 + y^2 + z^2 = 3xyz$, then so does $(x, y, 3xy - z)$. Starting from $(1, 1, 1)$, generate the triples listed above. (This "jumping" trick is the opposite of descent.)

Further reading

- **Harold Davenport, *The Higher Arithmetic*,** Chapters 2 and 5.
- **Joseph Silverman, *A Friendly Introduction to Number Theory*** (Pearson). Very readable; includes Fermat's descent proof for $x^4 + y^4 = z^2$.
- **Simon Singh, *Fermat's Last Theorem*** (Fourth Estate). The story, from Fermat's margin to Wiles's proof, for a general reader.

1.6 Sums of Two Squares

The question

(i) The complex numbers z and w have real and imaginary parts given by $z = a + \mathrm{i}b$ and $w = c + \mathrm{i}d$. Prove that $|zw| = |z||w|$.

(ii) By considering the complex numbers $2 + \mathrm{i}$ and $10 + 11\mathrm{i}$, find positive integers h and k such that $h^2 + k^2 = 5 \times 221$.

(iii) Find positive integers m and n such that $m^2 + n^2 = 8045$.

(iv) You are given that $102^2 + 201^2 = 50805$. Find positive integers p and q such that $p^2 + q^2 = 36 \times 50805$.

(v) Find three distinct pairs of positive integers r and s such that $r^2 + s^2 = 25 \times 1002082$ and $r < s$.

(vi) You are given that $109 \times 9193 = 1002037$. Find positive integers t and u such that $t^2 + u^2 = 9193$.

The hook

Which whole numbers can be written as the sum of two squares? $5 = 1^2 + 2^2$ and $13 = 2^2 + 3^2$ can, but 3, 7 and 21 can't. Here is a strange fact: $65 = 5 \times 13$ can be written as a sum of two squares in *two* different ways, $1^2 + 8^2$ and $4^2 + 7^2$. Why should multiplying two such numbers give another, and why in two ways?

The answer comes from an unexpected direction. The question asks you to leave the whole numbers behind and work with *complex* numbers. In the complex numbers, sums of two squares become simple, and some baffling-looking arithmetic becomes almost mechanical.

Before you start

Ask yourself If $z = a + \mathrm{i}b$ with a and b integers, what is $|z|^2$?

It is $a^2 + b^2$, a sum of two squares. So "N is a sum of two squares" means exactly "$N = |z|^2$ for some complex number z with integer real and imaginary parts". Complex numbers like this are called *Gaussian integers*. Hold on to that translation; the whole

question depends on it.

Working it through

STAGE 1 Moduli multiply

Ask yourself Write out zw in terms of a, b, c, d. What do you need to show about $|zw|^2$?

We have $zw = (ac - bd) + \mathrm{i}(ad + bc)$, so

$$|zw|^2 = (ac - bd)^2 + (ad + bc)^2 = a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2,$$

because the cross terms $-2abcd$ and $+2abcd$ cancel. And this factorises:

$$a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2 = (a^2 + b^2)(c^2 + d^2) = |z|^2|w|^2.$$

Taking square roots, $|zw| = |z||w|$. ■

Now read this as a statement about whole numbers:

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2. \qquad (\star)$$

The product of two sums of two squares is a sum of two squares, and the complex numbers tell you exactly what the two squares are.

STAGE 2 Using it

Part (ii): $5 = |2 + \mathrm{i}|^2$ and $221 = 10^2 + 11^2 = |10 + 11\mathrm{i}|^2$. So multiply:

$$(2 + \mathrm{i})(10 + 11\mathrm{i}) = 20 + 22\mathrm{i} + 10\mathrm{i} - 11 = 9 + 32\mathrm{i},$$

giving $5 \times 221 = 9^2 + 32^2$. (Check: $81 + 1024 = 1105$.)

Ask yourself What happens if you multiply $10 + 11\mathrm{i}$ by $2 - \mathrm{i}$ instead? It has the same modulus as $2 + \mathrm{i}$.

$(2 - \mathrm{i})(10 + 11\mathrm{i}) = 20 + 22\mathrm{i} - 10\mathrm{i} + 11 = 31 + 12\mathrm{i}$, so also $1105 = 31^2 + 12^2$. Choosing $2 + \mathrm{i}$ or its conjugate gives *different* representations. That explains the two ways of writing 65 in the hook (Figure 1.6.1): $65 = |(2 + \mathrm{i})(3 + 2\mathrm{i})|^2 = |4 + 7\mathrm{i}|^2$ and $65 = |(2 - \mathrm{i})(3 + 2\mathrm{i})|^2 = |8 + \mathrm{i}|^2$.

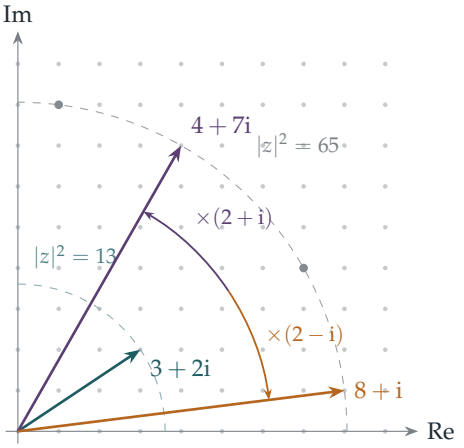


Figure 1.6.1 Multiplying by $2 \pm \mathrm{i}$ scales lengths by $\sqrt{5}$ and rotates by about $26.6°$, anticlockwise for $2+\mathrm{i}$ and clockwise for $2-\mathrm{i}$. Starting from $3+2\mathrm{i}$ on the circle $|z|^2 = 13$, the two choices land on different lattice points of the circle $|z|^2 = 65$. The grey points are their reflections in the line $y = x$.

STAGE 3 Factorise first

Ask yourself 8045 ends in 5. What does that suggest?

$8045 = 5 \times 1609$. We need 1609 as a sum of two squares; trying squares just below 1609 quickly finds $1609 = 40^2 + 3^2$. Then

$$(2+\mathrm{i})(40+3\mathrm{i}) = 77 + 46\mathrm{i}, \qquad \text{so} \qquad 8045 = 77^2 + 46^2,$$

and the conjugate gives $8045 = 83^2 + 34^2$ as well.

Part (iv) is quicker: $36 = 6^2 + 0^2$, and multiplying by 6 just scales: $36 \times 50805 = 612^2 + 1206^2$.

STAGE 4 Three ways

Ask yourself 25 can be written as a sum of two squares in more than one way. How many Gaussian integers have modulus 5, up to signs and swapping?

$25 = 5^2 + 0^2 = 3^2 + 4^2$, corresponding to 5, $3+4\mathrm{i}$ and $3-4\mathrm{i}$. Next, 1002082: since $1001^2 = 1002001$, we have $1002082 = 1001^2 + 9^2$. Multiplying $1001 + 9\mathrm{i}$ by each of the three gives:

$$\begin{aligned} 5\,(1001+9\mathrm{i}) &= 5005 + 45\mathrm{i}, \\ (3+4\mathrm{i})(1001+9\mathrm{i}) &= 2967 + 4031\mathrm{i}, \\ (3-4\mathrm{i})(1001+9\mathrm{i}) &= 3039 - 3977\mathrm{i}. \end{aligned}$$

So the three pairs (r,s) are $(45,5005)$, $(2967,4031)$ and $(3039,3977)$.

STAGE 5 Dividing

Pause and try Part (vi) runs the machine backwards. We know $109 = 10^2 + 3^2$ and $1002037 = 1001^2 + 6^2$. How might you find 9193 as a sum of two squares, without searching?

If multiplying Gaussian integers multiplies moduli, then *dividing* should divide them. Try

$$\frac{1001 + 6\mathrm{i}}{10 + 3\mathrm{i}} = \frac{(1001 + 6\mathrm{i})(10 - 3\mathrm{i})}{109} = \frac{10028 - 2943\mathrm{i}}{109} = 92 - 27\mathrm{i}.$$

It divides exactly, so $9193 = 92^2 + 27^2$. (Check: $8464 + 729 = 9193$.)

Ask yourself What happens if you divide by $10 - 3\mathrm{i}$ instead?

You get $(1001 + 6\mathrm{i})(10 + 3\mathrm{i})/109 = (9992 + 3063\mathrm{i})/109$, which is *not* a Gaussian integer, since $9992/109$ isn't a whole number. Only one of the two "factors of 109" divides $1001 + 6\mathrm{i}$, just as only some primes divide a given whole number. The Gaussian integers have their own primes and their own factorisations.

Looking back

- **Translate, solve, translate back.** "Sum of two squares" became "modulus squared of a Gaussian integer". In that language the problem was easy.
- **Factorise first.** Every part began by breaking a big number into smaller ones whose representations were easy to find.
- **Conjugates give new answers.** The number of representations depends on how many ways there are to choose a factor or its conjugate.

What's really going on

Fermat's two-squares theorem

Which primes are sums of two squares? $2 = 1^2 + 1^2$, 5, 13, 17, 29, 37, 41 are; 3, 7, 11, 19, 23, 31 are not. The pattern: apart from 2, a prime is a sum of two squares exactly when it leaves remainder 1 on division by 4. Fermat stated this in 1640; Euler proved it in 1749. In the language of Gaussian integers it says: a prime $p \equiv 1 \pmod 4$ *splits* as $p = (a + b\mathrm{i})(a - b\mathrm{i})$, while a prime $p \equiv 3 \pmod 4$ stays prime. For example, $5 = (2 + \mathrm{i})(2 - \mathrm{i})$, but 3 cannot be factorised at all.

In 1990 Don Zagier published a proof of the two-squares theorem that is famously one sentence long. It appears, with explanation, in *Proofs from THE BOOK*.

A new number system with its own primes

Gauss realised that the Gaussian integers behave remarkably like the ordinary integers. There is a division algorithm and a Euclidean algorithm, and every Gaussian integer factorises *uniquely* into Gaussian primes (up to multiplying by ± 1 or $\pm \mathrm{i}$). Combined with $(\star)$, this lets you count exactly how many ways a number can be written as a sum of two squares, just from its prime factorisation.

At university this is the gateway to *algebraic number theory*: studying number systems such as $\mathbb{Z}[\mathrm{i}]$ or $\mathbb{Z}[\sqrt{-5}]$ in order to understand ordinary whole numbers. Section 1.7 shows why the subject needed to be invented: not every such system is as well behaved as the Gaussian integers.

Four squares and beyond

Every positive integer is a sum of *four* squares (Lagrange, 1770). The proof relies on a four-square version of $(\star)$, which comes from Hamilton's *quaternions* in the same way that $(\star)$ comes from complex numbers. You will see it in Section 7.8.

The thinking skill

Changing the setting

A problem that is awkward in one number system may be natural in a larger one.

Sums of two squares are awkward in the integers but simply moduli in the complex numbers. Mathematicians constantly enlarge number systems in this way: integers to rationals, rationals to reals, reals to complex numbers. Each enlargement makes some old problems easy.

Try this next

1. **An obstruction.** Show that no number of the form $4k + 3$ is a sum of two squares. (What are the squares modulo 4?)

2. **Two ways.** Write 65 and 85 each as a sum of two squares in two different ways, and explain your answers using Gaussian integers.

3. **Four ways.** $1105 = 5 \times 13 \times 17$. Find four different ways to write 1105 as a sum of two positive squares. Why four?

4. **Gaussian primes.** HARDER Show that if $3 = (a + b\mathrm{i})(c + d\mathrm{i})$ with a, b, c, d integers, then one of the factors has modulus 1. (Take moduli squared.) This shows that 3 is a Gaussian prime.

Further reading

- **John Stillwell, *Elements of Number Theory*** (Springer). Builds up to Gaussian integers and beyond, with history.
- **Aigner & Ziegler, *Proofs from THE BOOK*,** Chapter 4, "Representing numbers as sums of two squares".
- **Joseph Silverman, *A Friendly Introduction to Number Theory*,** chapters on sums of squares and Gaussian integers.

1.7 When Factorisation Fails

The question

The set S consists of all the positive integers that leave a remainder of 1 upon division by 4. The set T consists of all the positive integers that leave a remainder of 3 upon division by 4.

(i) Describe in words the sets $S \cup T$ and $S \cap T$.

(ii) Prove that the product of any two integers in S is also in S. Determine whether the product of any two integers in T is also in T.

(iii) Given an integer in T that is not a prime number, prove that at least one of its prime factors is in T.

(iv) For any set X of positive integers, an integer in X (other than 1) is said to be *X-prime* if it cannot be expressed as the product of two or more integers *all in X* (and all different from 1).

(a) Show that every integer in T is either T-prime or is the product of an odd number of T-prime integers.

(b) Find an example of an integer in S that can be expressed as the product of S-prime integers in two distinct ways. [Note: $s_1 s_2$ and $s_2 s_1$ are not counted as distinct ways of expressing the product of s_1 and s_2.]

The hook

Every whole number greater than 1 can be written as a product of primes in exactly one way: $60 = 2 \times 2 \times 3 \times 5$, and there is no other way to do it. You have probably used this fact so often that it feels too obvious to need proof.

This question builds a small world of numbers that looks perfectly sensible but in which that "obvious" fact is false. It is a STEP I question, so the mathematics is gentle, but the lesson is a deep one. It is exactly the lesson that defeated a famous attempted proof of Fermat's Last Theorem in 1847 and led to the invention of a whole new branch of algebra.

Before you start

Ask yourself List the first few members of S and T. Which members of S are *S-prime*, according to the definition in part (iv)?

$S = \{1, 5, 9, 13, 17, 21, 25, 29, 33, \ldots\}$ and $T = \{3, 7, 11, 15, 19, 23, \ldots\}$. A number in S is S-prime if it can't be written as a product of smaller members of S (other than 1). Notice that 9 is S-prime, even though $9 = 3 \times 3$ in the ordinary integers, because 3 is not in S. "Prime" depends on which numbers you are allowed to use.

Working it through

STAGE 1 Union, intersection and products

$S \cup T$ is the set of all positive odd numbers, since every odd number leaves remainder 1 or 3 on division by 4. $S \cap T$ is empty, because no number leaves two different remainders.

Ask yourself How do you write a general member of S? What is the product of two of them?

Members of S are $4a + 1$ for $a \geq 0$, and

$$(4a + 1)(4b + 1) = 16ab + 4a + 4b + 1 = 4(4ab + a + b) + 1 \in S.$$

So S is closed under multiplication. For T:

$$(4a + 3)(4b + 3) = 16ab + 12a + 12b + 9 = 4(4ab + 3a + 3b + 2) + 1,$$

which is in S, *not* T. For example, $3 \times 7 = 21 \in S$. So T is not closed under multiplication.

STAGE 2 A prime factor in T

Ask yourself Take a number in T that isn't prime. Its prime factors are all odd. What would happen if they were all in S?

The number is odd, so all its prime factors are odd and lie in S or T. If they were *all* in S, then since S is closed under multiplication, their product would be in S. But the product is our number, which is in T. So at least one prime factor is in T. ■

STAGE 3 An odd number of factors

Ask yourself If you multiply together an even number of members of T, where does the product land? What about an odd number?

Each pair of T-numbers multiplies to an S-number, and S-numbers multiply to S-numbers, so an even number of T-factors gives a product in S. An odd number gives (something in S) $\times$ (one element of T), which is in T by the same kind of calculation.

Now take $t \in T$. If t is T-prime, we are done. Otherwise, write t as a product of members of T, each bigger than 1 and so smaller than t. Any of those that are not T-prime can be broken down further, and so on. The factors get smaller each time, so the process stops (well-ordering again), leaving t as a product of T-primes. Since the product is in T, the number of factors must be odd. ■

STAGE 4 Two factorisations

Pause and try Find a number in S with two genuinely different factorisations into S-primes. *Hint:* use numbers like 9, which are S-prime only because their ordinary prime factors are missing from S.

The S-primes begin $5, 9, 13, 17, 21, 29, 33, 37, 41, 49, \ldots$. Now $21 = 3 \times 7$ and $9 = 3 \times 3$ and $49 = 7 \times 7$ are all S-prime, because 3 and 7 are not in S. And

$$441 = 21 \times 21 = 9 \times 49.$$

Two different factorisations of 441 into S-primes (Figure 1.7.1). Unique factorisation fails in S. ■

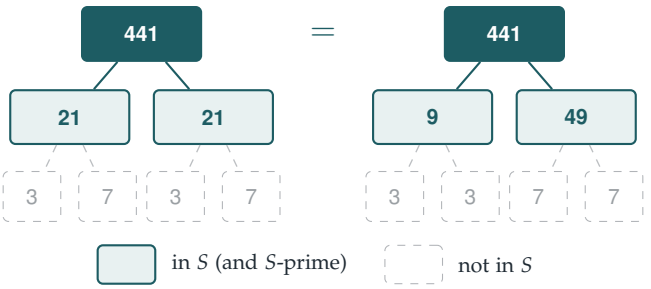


Figure 1.7.1 Two factorisations of 441 into S-primes. In the ordinary integers both trees continue down to $3 \times 3 \times 7 \times 7$, but 3 and 7 are not in S, so inside S the factorisations stop at the second row, and they are different.

Looking back

- **Being "prime" depends on the system.** 9 and 21 are composite in the integers but prime in S, because the numbers you would need to factorise them are missing.
- **Closure matters.** S is closed under multiplication; T is not. That single difference drove parts (iii) and (iv).
- **An obvious fact can be a theorem.** Unique factorisation in the integers needs a proof. This question shows why: in a system that looks very similar, it is false.

What's really going on

What makes the integers special

Why *does* factorisation work in the ordinary integers? The key fact is *Euclid's lemma*: if a prime p divides a product ab, then p divides a or p divides b. It is proved using the Euclidean algorithm, and it is exactly what fails in S. There, the S-prime 9 divides $21 \times 21 = 441$ (because $441 = 9 \times 49$), but 9 doesn't divide 21 in S. In your first year at university you will prove Euclid's lemma carefully and see that unique factorisation follows from it.

The failed proof of Fermat's Last Theorem

On 1 March 1847 Gabriel Lamé announced to the Paris Academy that he had proved Fermat's Last Theorem. His idea was to factorise $x^n + y^n$ using complex roots of unity,

$$x^n + y^n = (x+y)(x+\zeta y)(x+\zeta^2 y)\cdots(x+\zeta^{n-1}y), \qquad \zeta = e^{2\pi \mathrm{i}/n}$$

(for odd n), and then argue about the prime factors of each bracket. Joseph Liouville immediately objected: why should numbers built from ζ factorise uniquely? Weeks later it emerged that Ernst Kummer had already shown that they often don't. The first failure is at $n = 23$.

Kummer rescued what he could by inventing "ideal numbers", extra objects that restore unique factorisation, and proved Fermat's Last Theorem for a large class of exponents. Richard Dedekind turned Kummer's ideal numbers into *ideals*, one of the central concepts of modern algebra.

A smaller example to remember

In the number system $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$,

$$6 = 2 \times 3 = (1+\sqrt{-5})(1-\sqrt{-5}),$$

and all four factors are "prime" in that system (exercise 3). Compare this with the Gaussian integers of Section 1.6, where factorisation *is* unique. Which systems of this kind have unique factorisation is a deep question. Gauss conjectured that, among systems of the form $\mathbb{Z}[\sqrt{-d}]$ and their close relatives, only nine have it. This was finally proved by Heegner, Baker and Stark between 1952 and 1967.

The thinking skill

Questioning what you take for granted

When a fact feels too obvious to prove, ask what it depends on, and look for a setting where it fails.

Building a counterexample, a world where the "obvious" fact is false, is one of the best ways to find out what a proof really needs. Much of undergraduate algebra consists of asking which familiar properties of the integers survive in more general systems.

Try this next

1 **Another example.** Show that 693 has two different factorisations into S-primes. Can you find a general recipe for producing such examples?

2 **Even numbers.** Let $E = \{2, 4, 6, 8, \ldots\}$. Which numbers are E-prime? Show that 60 has more than one factorisation into E-primes.

3 $\mathbb{Z}[\sqrt{-5}]$**.** Define the *norm* $N(a + b\sqrt{-5}) = a^2 + 5b^2$. Show that $N(zw) = N(z)N(w)$. By showing that no element has norm 2 or 3, deduce that 2, 3 and $1 \pm \sqrt{-5}$ cannot be factorised further in $\mathbb{Z}[\sqrt{-5}]$.

4 **Euclid's lemma in action.** HARDER Assuming Euclid's lemma (if a prime divides ab, it divides a or b), prove that every integer greater than 1 has only one factorisation into primes, apart from the order of the factors.

Further reading

- **Ian Stewart & David Tall,** ***Algebraic Number Theory and Fermat's Last Theorem*** (CRC Press). The opening chapters tell the story of Lamé, Kummer and ideals at a level a keen first-year student can follow.
- **John Stillwell,** ***Elements of Number Theory***. Unique factorisation in the integers and in the Gaussian integers, compared side by side.
- **Harold Edwards,** ***Fermat's Last Theorem: A Genetic Introduction*** (Springer). For the ambitious: Kummer's work explained through its history.

1.8 Counting the Rationals

The question

A sequence u_k, for integer $k \geq 1$, is defined as follows.

$$u_1 = 1, \qquad u_{2k} = u_k \text{ for } k \geq 1, \qquad u_{2k+1} = u_k + u_{k+1} \text{ for } k \geq 1.$$

(i) Show that, for every pair of consecutive terms of this sequence, except the first pair, the term with odd subscript is larger than the term with even subscript.

(ii) Suppose that two consecutive terms in this sequence have a common factor greater than one. Show that there are then two consecutive terms earlier in the sequence which have the same common factor. Deduce that any two consecutive terms in this sequence are co-prime (do not have a common factor greater than one).

(iii) Prove that it is not possible for two positive integers to appear consecutively in the same order in two different places in the sequence.

(iv) Suppose that a and b are two co-prime positive integers which do not occur consecutively in the sequence with b following a. If $a > b$, show that $a - b$ and b are two co-prime positive integers which do not occur consecutively in the sequence with b following $a - b$, and whose sum is smaller than $a + b$. Find a similar result for $a < b$.

(v) For each integer $n \geq 1$, define the function f from the positive integers to the positive rational numbers by $f(n) = \dfrac{u_n}{u_{n+1}}$. Show that the range of f is all the positive rational numbers, and that f has an inverse.

The hook

Can you list all the positive fractions, one after another, so that every fraction appears exactly once? It seems impossible: between any two fractions there are infinitely many more, so there is no "next" fraction to write down. Georg Cantor showed in the 1870s that it can nevertheless be done, which means that, in a precise sense, there are no more fractions than there are whole numbers.

Cantor's list was clumsy: it included $\frac{2}{4}$ as well as $\frac{1}{2}$, and the repeats had to be crossed out. This question builds a list with no repeats at all, from a sequence with an absurdly simple rule. It was published by Neil Calkin and Herbert Wilf in 2000, though the

sequence itself was studied by Moritz Stern in 1858.

Before you start

Ask yourself Before proving anything, compute the first fifteen or so terms. What do you notice about the ratios u_n/u_{n+1}?

The sequence begins

$$1,\ 1,\ 2,\ 1,\ 3,\ 2,\ 3,\ 1,\ 4,\ 3,\ 5,\ 2,\ 5,\ 3,\ 4,\ 1,\ 5,\ \ldots$$

and the ratios u_n/u_{n+1} are

$$\tfrac{1}{1},\ \tfrac{1}{2},\ \tfrac{2}{1},\ \tfrac{1}{3},\ \tfrac{3}{2},\ \tfrac{2}{3},\ \tfrac{3}{1},\ \tfrac{1}{4},\ \tfrac{4}{3},\ \tfrac{3}{5},\ \tfrac{5}{2},\ \tfrac{2}{5},\ \tfrac{5}{3},\ \tfrac{3}{4},\ \tfrac{4}{1},\ \ldots$$

Every fraction appears in lowest terms, and none is repeated. The question asks you to prove that this goes on for ever, and that every positive fraction eventually turns up.

Working it through

STAGE 1 Odd beats even

Consecutive terms are either (u_{2k}, u_{2k+1}) or (u_{2k+1}, u_{2k+2}).

Ask yourself Write each of these pairs in terms of u_k and u_{k+1}. Which term is bigger?

$(u_{2k}, u_{2k+1}) = (u_k,\ u_k + u_{k+1})$ and $(u_{2k+1}, u_{2k+2}) = (u_k + u_{k+1},\ u_{k+1})$. Since every term is a positive integer (an easy induction), $u_k + u_{k+1}$ is bigger than both u_k and u_{k+1}. The only exception is the first pair, $(u_1, u_2) = (1, 1)$, which doesn't fit either pattern because u_1 is defined directly. ■

STAGE 2 Consecutive terms are coprime

Ask yourself Suppose $d > 1$ divides both u_{2k} and u_{2k+1}, that is, both u_k and $u_k + u_{k+1}$. What else does d divide?

d divides $(u_k + u_{k+1}) - u_k = u_{k+1}$, so d divides the consecutive pair u_k, u_{k+1}, which is earlier in the sequence. The other case is identical: if d divides $u_k + u_{k+1}$ and u_{k+1}, it divides u_k.

So a common factor of a consecutive pair is passed back to an earlier pair, then an earlier one, and so on. Positions can't decrease for ever (well-ordering yet again), so eventually d would have to divide the first pair, $u_1 = u_2 = 1$. That is impossible for $d > 1$. So every consecutive pair is coprime. ■

STAGE 3 **No pair appears twice**

Pause and try Stage 1 tells you something useful: if you see a consecutive pair (a, b), you can tell whether a sits at an odd or an even position. How? And can you then recover the pair that came before it, in the sense of Stage 2?

If $a < b$, then (by Stage 1) b is at the odd position, so the pair is $(u_{2k}, u_{2k+1}) = (u_k, u_k + u_{k+1})$ for some k, and the "parent" pair (u_k, u_{k+1}) is $(a,\ b-a)$. If $a > b$, the pair is (u_{2k+1}, u_{2k+2}) and its parent is $(a-b,\ b)$. If $a = b$, Stage 1 shows the pair must be the very first pair, $(1, 1)$.

So each consecutive pair other than the first determines its parent pair, *and the position of the parent* (you can see this structure as a tree in Figure 1.8.1): the parent is at position k when the child is at $2k$ or $2k+1$. Now suppose (a, b) appeared at two different positions. Their parents would be the same pair at two positions, and so would *their* parents, and so on. Following this back, we would eventually find $(1, 1)$ at a position other than 1, which Stage 1 rules out. ■

STAGE 4 **Every coprime pair appears**

Ask yourself Suppose some coprime pair (a, b) never appears consecutively. Which such pair should you look at?

Take the "missing" coprime pair with the smallest possible value of $a + b$: a minimal counterexample. It can't be $(1, 1)$, which does appear, so $a \neq b$ (coprime and equal forces $a = b = 1$).

- If $a > b$: the pair $(a - b, b)$ is coprime (any common factor of $a - b$ and b divides a), and its sum a is smaller than $a + b$. By minimality it does appear, say at positions $k, k+1$. But then $u_{2k+1} = u_k + u_{k+1} = a$ and $u_{2k+2} = u_{k+1} = b$, so (a, b) appears after all.
- If $a < b$: similarly $(a, b - a)$ appears at some $k, k+1$, and then $(u_{2k}, u_{2k+1}) = (a, b)$.

Either way, contradiction. So every coprime pair of positive integers appears consecutively. ■

(The question phrases part (iv) as a descent: a missing pair produces a smaller missing pair. That is the same argument, read the other way.)

STAGE 5 **A list of all fractions**

Every positive rational number can be written uniquely in lowest terms as a/b with a, b coprime. By Stage 4, (a, b) appears as some consecutive pair (u_n, u_{n+1}), so $f(n) = a/b$: the range of f is all positive rationals. By Stage 3 it appears only once, and by Stage 2 every value $f(n)$ is already in lowest terms, so two different n can't give the same fraction. Hence f is a bijection from the positive integers to the positive rationals, and has an inverse. ■

Looking back

- **Descent and its reverse.** Common factors and repeated pairs were chased *back* to the start of the sequence. Missing pairs were handled by a minimal counterexample, building *forward*.
- **A bijection needs two halves.** "Every fraction appears" (onto) and "no fraction appears twice" (one-to-one) are separate claims, proved separately.
- **Compute first.** Writing out fifteen terms made the claims believable and suggested the parent–child structure that drove the proofs.

What's really going on

The Calkin–Wilf tree

Arrange the positive fractions in a tree: put $\frac{1}{1}$ at the top, and give each fraction $\frac{a}{b}$ two children, $\frac{a}{a+b}$ on the left and $\frac{a+b}{b}$ on the right. Reading the tree row by row, left to right, gives exactly the list $f(1), f(2), f(3), \ldots$ from this question (Figure 1.8.1). Stage 3 says every fraction has a unique parent, and Stage 4 says every fraction is in the tree. In 2003 Moshe Newman found that you don't even need the tree: each fraction in the list can be computed from the one before by

$$x_{n+1} = \frac{1}{2\lfloor x_n \rfloor - x_n + 1}.$$

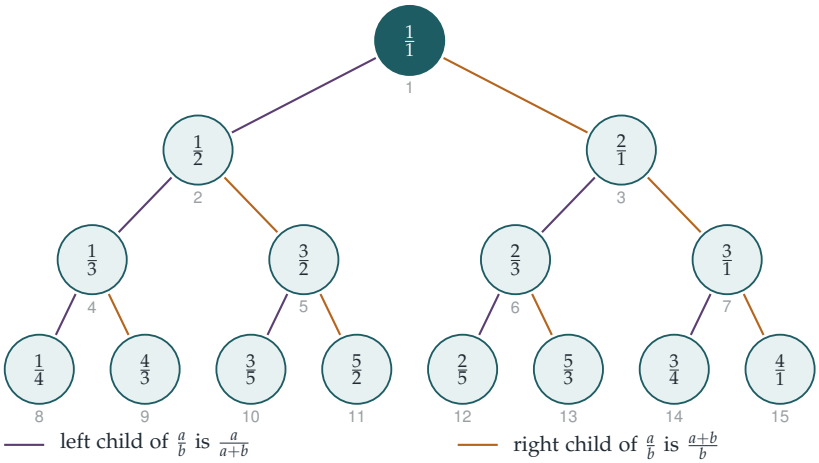


Figure 1.8.1 The first four rows of the Calkin–Wilf tree. Reading row by row, left to right, gives $f(1), f(2), f(3), \ldots$; the small grey numbers are the positions n. Every positive fraction appears exactly once, in lowest terms.

Different sizes of infinity

A set that can be listed like this is called *countable*. The fractions are countable, which is surprising, since they seem so much more numerous than the whole numbers. In your first term at university you will also meet Cantor's *diagonal argument*, which shows that the *real* numbers are not countable: any list of real numbers must miss some. So there really are different sizes of infinity, and the real numbers form a strictly bigger infinity than the rationals. This is one of the most startling results you will meet in a first-year course.

Stern's sequence

The sequence u_n is *Stern's diatomic sequence*, and it has many other lives. For example, u_{n+1} counts the number of ways to write n as a sum of powers of 2 with each power used at most twice. It is closely related to the *Stern–Brocot tree* and to continued fractions, which give the best rational approximations to irrational numbers.

The thinking skill

Building a bijection

To show two sets are the same size, match them up exactly: prove that every element of the target is hit (onto), and that nothing is hit twice (one-to-one).

Bijections are the foundation of counting, finite and infinite, and you will use them constantly at university. Here both halves were proved by chasing a structure back to its starting point.

Try this next

1. **Find the fraction.** For which n is $f(n) = \frac{3}{5}$? And $f(n) = \frac{5}{3}$? Use the parent rule to work backwards to $\frac{1}{1}$.

2. **Newman's formula.** Check that Newman's formula produces the first ten fractions of the list, starting from $x_1 = 1$.

3. **Cantor's diagonal argument.** Suppose you had a list of all real numbers between 0 and 1, written as decimals. Construct a decimal that differs from the nth number in the list in its nth digit. Why does this show the list is incomplete? (What care is needed with decimals like $0.0999\ldots = 0.1$?)

4. **Countable unions.** HARDER Show that the set of all pairs (m, n) of positive integers is countable by giving an explicit bijection with the positive integers. (Try $2^{m-1}(2n-1)$.)

Further reading

- **Neil Calkin & Herbert Wilf, "Recounting the rationals"** (*American Mathematical Monthly*, 2000). Four pages, entirely readable.
- **Aigner & Ziegler, *Proofs from THE BOOK***, Chapter 17, "Sets, functions, and the continuum hypothesis", which includes the Calkin–Wilf tree.
- **Ian Stewart & David Tall, *The Foundations of Mathematics*** (Oxford University Press). Countability, Cantor and the real numbers, as in a first-year course.

1.9 Tangents to a Cubic

The question

A curve has the equation

$$y^3 = x^3 + a^3 + b^3,$$

where a and b are positive constants. Show that the tangent to the curve at the point $(-a, b)$ is

$$b^2y - a^2x = a^3 + b^3.$$

In the case $a = 1$ and $b = 2$, show that the x-coordinates of the points where the tangent meets the curve satisfy

$$7x^3 - 3x^2 - 27x - 17 = 0.$$

Hence find positive integers p, q, r and s such that

$$p^3 = q^3 + r^3 + s^3.$$

The hook

This looks like a routine coordinate-geometry question: differentiate, find a tangent, solve a cubic. Then the last line asks for a cube that is the sum of three other cubes, and the geometry produces one as if by magic.

The trick, drawing a line through a known point on a curve to find a new point, goes back to Diophantus of Alexandria in the third century. In modern hands it became the theory of *elliptic curves*, which sits at the centre of present-day number theory, underlies the proof of Fermat's Last Theorem, and secures a large share of the world's internet traffic.

Before you start

Ask yourself The point $(-a, b)$ is on the curve. Why? And what does that have to do with sums of cubes?

Substitute: $(-a)^3 + a^3 + b^3 = b^3$, so yes. Points on $y^3 = x^3 + a^3 + b^3$ with *rational* coordinates are exactly what we want: if $x = q/s'$ and $y = p/s'$ then clearing denominators gives a relation between cubes. The plan is to start from one easy point and use the geometry to find a less obvious one.

Working it through

STAGE 1 The tangent

Differentiate $y^3 = x^3 + a^3 + b^3$ implicitly: $3y^2\dfrac{dy}{dx} = 3x^2$, so $\dfrac{dy}{dx} = \dfrac{x^2}{y^2}$. At $(-a, b)$ the gradient is a^2/b^2, so the tangent is

$$y - b = \frac{a^2}{b^2}(x + a) \quad\Longrightarrow\quad b^2y - b^3 = a^2x + a^3 \quad\Longrightarrow\quad b^2y - a^2x = a^3 + b^3. \qquad \blacksquare$$

STAGE 2 Where the tangent meets the curve again

With $a = 1$, $b = 2$ the curve is $y^3 = x^3 + 9$ and the tangent at $(-1, 2)$ is $4y - x = 9$, that is, $y = \dfrac{x+9}{4}$. Substitute into the curve:

$$\frac{(x+9)^3}{64} = x^3 + 9 \quad\Longrightarrow\quad x^3 + 27x^2 + 243x + 729 = 64x^3 + 576,$$

which rearranges to $63x^3 - 27x^2 - 243x - 153 = 0$, and dividing by 9 gives $7x^3 - 3x^2 - 27x - 17 = 0$. $\blacksquare$

STAGE 3 Solving the cubic

Ask yourself You already know one root of this cubic without doing any algebra. Which one, and why should it appear *twice*?

The tangent touches the curve at $x = -1$ (Figure 1.9.1), and "touching" means a repeated root, just as a tangent to a parabola meets it at a double root of a quadratic. So $(x+1)^2$ should be a factor. Check: $7(-1)^3 - 3(1) - 27(-1) - 17 = -7 - 3 + 27 - 17 = 0$, and the derivative $21x^2 - 6x - 27$ also vanishes at $x = -1$. Dividing,

$$7x^3 - 3x^2 - 27x - 17 = (x+1)^2(7x - 17).$$

Pause and try So the tangent meets the curve again where $x = \frac{17}{7}$. Find y, and turn the point into a relation between cubes.

$y = \dfrac{17/7 + 9}{4} = \dfrac{80/7}{4} = \dfrac{20}{7}$. The point $\left(\frac{17}{7}, \frac{20}{7}\right)$ lies on $y^3 = x^3 + 9$, so

$$\frac{20^3}{7^3} = \frac{17^3}{7^3} + 9 \quad\Longrightarrow\quad 20^3 = 17^3 + 9 \times 7^3.$$

Ask yourself We want 9×7^3 to be a sum of two cubes. Remember where the 9 came from.

$9 = a^3 + b^3 = 1^3 + 2^3$, so $9 \times 7^3 = 7^3 + 14^3$. Therefore

$$20^3 = 17^3 + 14^3 + 7^3.$$

Check: $8000 = 4913 + 2744 + 343$. ■

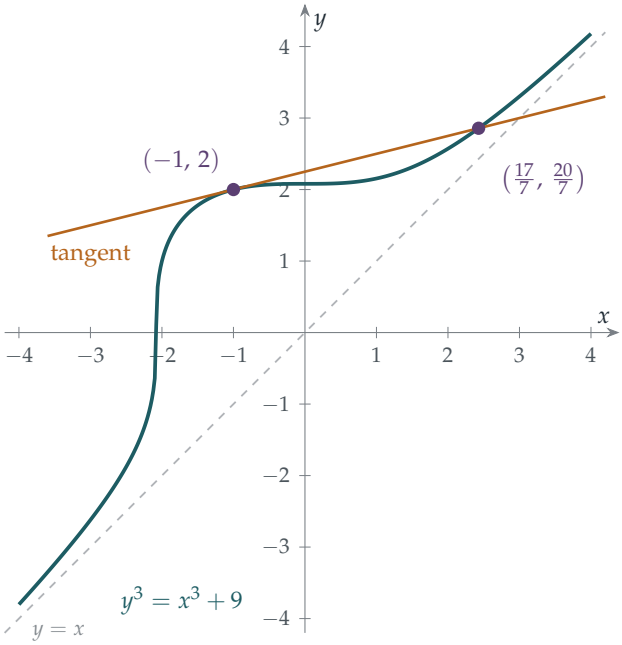


Figure 1.9.1 The curve $y^3 = x^3 + 9$ (teal) approaches the line $y = x$ far from the origin. The tangent at the rational point $(-1, 2)$ (amber) touches the curve there, which counts as a double intersection, and crosses it once more at another rational point, $\left(\frac{17}{7}, \frac{20}{7}\right)$.

Looking back

- **Tangency means a double root.** That turned a cubic with no obvious roots into one we could factorise immediately.
- **The new root is rational for free.** Once two of the three roots are rational (here both -1), the third must be too, because the sum of the roots, $\frac{3}{7}$, is rational. This is the heart of the method.
- **Geometry produced arithmetic.** We never searched for cubes; a tangent line found them.

What's really going on

The chord–tangent method

Take a cubic curve, such as $y^3 = x^3 + 9$ or the more standard form $y^2 = x^3 + Ax + B$. If you know two points on it with rational coordinates, the line through them meets the curve in exactly one more point, and that point also has rational coordinates (by the argument in *Looking back*). If you know only one point, use the tangent there instead. Either way, old rational points produce new ones.

Remarkably, this "chord–tangent" construction can be made into a way of *adding* points on the curve, and the rational points then form a *group*, an algebraic structure you will study in your first year. Louis Mordell proved in 1922 that this group is generated by finitely many points: all the rational points, however many there are, can be built from a finite set using chords and tangents. His proof is a version of Fermat's descent from Section 1.5.

Elliptic curves today

Curves like these are called *elliptic curves* (not because they are ellipses, but because of a historical link with the arc length of an ellipse). They are everywhere in modern mathematics.

- **Fermat's Last Theorem.** In the 1980s Gerhard Frey noticed that a solution of $x^n + y^n = z^n$ would produce a very strange elliptic curve. Andrew Wiles proved in 1994 that such strange curves can't exist, and so proved Fermat's Last Theorem.
- **The Birch and Swinnerton-Dyer conjecture.** How many generators does the group of rational points need? A conjecture made in the 1960s, using some of the earliest computer experiments in mathematics at Cambridge, predicts the answer in terms of a function built from counting points modulo primes. It is one of the seven Millennium Prize Problems.
- **Cryptography.** On an elliptic curve over a finite number system, adding a point P to itself n times is quick, even when n has hundreds of digits (double P repeatedly, much as you would calculate a high power by repeated squaring). But given P and the result nP, finding n seems to be extremely hard: this is the *elliptic-curve discrete logarithm problem*. That asymmetry protects much of today's online communication.

Sums of like powers

The smallest cube that is a sum of three positive cubes is $6^3 = 3^3 + 4^3 + 5^3$; the tangent method found $20^3 = 7^3 + 14^3 + 17^3$, which is not a multiple of it, and repeating the method keeps producing new identities (although some involve negative numbers). How many kth powers does it take to make another kth power? Euler conjectured in 1769 that you need at least k kth powers to make another kth power. For fourth and

fifth powers he was wrong: in 1966 Leon Lander and Thomas Parkin found

$$27^5 + 84^5 + 110^5 + 133^5 = 144^5$$

by computer search, and in 1988 Noam Elkies used elliptic curves to find counterexamples for fourth powers.

The thinking skill

Geometry solving arithmetic

To find whole-number or rational solutions of an equation, think of it as a curve and look for lines through known points.

The algebra behind it is simple: if a polynomial with rational coefficients has all but one of its roots rational, the last root is rational too. Translating between pictures and equations in this way is one of the most powerful habits a mathematician can have.

Try this next

1. **Rational points on a circle.** The line through $(-1, 0)$ with gradient t meets the circle $x^2 + y^2 = 1$ again. Show that the other point is $\left(\dfrac{1 - t^2}{1 + t^2}, \dfrac{2t}{1 + t^2}\right)$, and use this to produce infinitely many Pythagorean triples.

2. **A chord that misses.** The points $(1, 2)$ and $(2, 1)$ lie on $x^3 + y^3 = 9$. Show that the line through them meets the curve in no other point. Where has the "third point" gone? (Look up the *point at infinity*.)

3. **Repeat the trick.** Find the tangent to $y^3 = x^3 + 9$ at $\left(\frac{17}{7}, \frac{20}{7}\right)$ and the third point where it meets the curve. (A computer algebra system helps; notice how quickly the numbers grow.)

4. **Taxicab numbers.** $1729 = 1^3 + 12^3 = 9^3 + 10^3$. Find the third point where the line through $(1, 12)$ and $(9, 10)$ meets the curve $x^3 + y^3 = 1729$, and deduce a new identity of the form $p^3 = q^3 + r^3 + s^3$.

Further reading

- **Joseph Silverman & John Tate, *Rational Points on Elliptic Curves*** (Springer). Written for undergraduates; Chapter 1 needs nothing beyond A-level.
- **Avner Ash & Robert Gross, *Elliptic Tales*** (Princeton University Press). A popular account leading up to the Birch and Swinnerton-Dyer conjecture.

- **Simon Singh, *Fermat's Last Theorem***. How elliptic curves came to solve a 350-year-old problem.

What We Have Learned

At the start of this chapter we asked three questions about whole numbers, and promised a fourth theme running underneath them: how to prove things. It is time to see how far nine STEP questions have taken us.

1. How many primes are there, and how are they spread out?

In Section 1.1 we proved that there are infinitely many primes without finding a single new one. We built an infinite family of numbers, the Fermat numbers, that never share a factor, so each one needs a prime of its own. It is a proof about *structure*, not size.

In Section 1.2 we went further and showed that the product of all the primes up to n is less than 4^n. That was our first *quantitative* statement: it limits how densely the primes can be packed. Taking logarithms, it says $\sum_{p \le n} \ln p < 1.39\,n$, while the Prime Number Theorem says the true value is about n. Our STEP-level argument was already in the right ballpark.

What we have *not* done is find exactly how the primes are spread out. That is controlled by the Riemann zeta function, and making it precise is the Riemann Hypothesis, the most famous open problem in mathematics. You now know enough to understand what it is about.

2. How can you prove that a number is not a fraction?

We found three quite different ways.

- **Take the smallest one** (Section 1.3). If $\sqrt{2}$ were a fraction, there would be a smallest positive integer k making $k\sqrt{2}$ whole; but then $(\sqrt{2}-1)k$ would be a smaller one. The same argument handled cube roots.
- **Trap an integer between 0 and 1** (Section 1.4). If e or e^2 were a fraction, multiplying by a factorial would produce a whole number lying strictly between 0 and 1. This is the idea behind Hermite's and Lindemann's proofs that e and π are transcendental.
- **Find an impossible divisibility** (Section 1.5, exercise 3). Showing there is no rational point on $x^2 + y^2 = 3$ is the same as showing $\sqrt{3 - x^2}$ is never rational when x is.

And we learned how much remains unknown: nobody can yet prove that $e + \pi$ is irrational.

3. Which equations have whole-number solutions?

Sometimes the answer is *none*, and Section 1.5 showed Fermat's way of proving it: any solution would produce a smaller one, for ever. Working modulo a well-chosen prime told us which divisibility to exploit.

Sometimes the answer is *yes, and here is a machine for producing them*. In Section 1.6 the complex numbers turned sums of two squares into a multiplication rule. In Section 1.9 a tangent line to a cubic curve produced $20^3 = 17^3 + 14^3 + 7^3$.

And sometimes the right move is to *change the number system*. The Gaussian integers made sums of two squares easy. But Section 1.7 gave a warning: in a system that looks perfectly reasonable, numbers can factorise into "primes" in two different ways. That warning, ignored in 1847, led to the invention of ideals and modern algebra.

Section 1.8 added a twist of its own: the fractions, which seem far more numerous than the whole numbers, can be listed one by one. Infinity has more structure than it first appears.

4. A toolkit of proof

Here are the methods you have used in this chapter. Every one of them will appear again and again in a university mathematics course.

Method	Where you used it
Induction and strong induction	Fermat numbers (1.1); products of primes (1.2)
Well-ordering, minimal counterexample	$\sqrt{2}$ and $\sqrt[3]{2}$ (1.3); listing fractions (1.8)
Infinite descent	Impossible equations (1.5); coprime pairs (1.8)
Proof by contradiction	Throughout, especially 1.3 and 1.4
Working modulo n	1.5; 1.7
Bounding and squeezing	1.2; 1.4
Building a bijection	1.8
Constructing a counterexample	Non-unique factorisation (1.7)
Changing the setting	Complex numbers (1.6); geometry (1.9)

If you look back at the questions, you will notice that the STEP examiners often gave you the first step: "by considering the binomial expansion…", "by considering the complex numbers $2 + \mathrm{i}$ and $10 + 11\mathrm{i}$…". At university that help gradually disappears, and *finding* the right first step becomes the skill. The best preparation is exactly what you have been doing: when you finish a problem, ask *why* the hint worked, so that next time you can think of it yourself.

Where next

Several threads from this chapter continue elsewhere in the book.

- The approximation of e by fractions returns in Section 2.9, and the Riemann zeta function in Section 2.4, where $\zeta(2) = \pi^2/6$.

- Roots of unity, which appeared in Lamé's failed proof, are the subject of Chapter 5, and Gauss's construction of the regular 17-gon, promised in Section 1.1, is Section 6.2.
- The four-square version of the identity in Section 1.6 appears in Section 7.8, via quaternions.

If you want to go further before university

- **Harold Davenport, *The Higher Arithmetic***: the best single introduction to elementary number theory.
- **Aigner & Ziegler, *Proofs from THE BOOK***: Chapters 1–4, 8 and 17 connect directly with this chapter.
- **Joseph Silverman, *A Friendly Introduction to Number Theory***: covers almost everything here, and more, at a gentle pace.
- **Marcus du Sautoy, *The Music of the Primes*** and **Simon Singh, *Fermat's Last Theorem***: the human stories behind the mathematics.

Number theory has a habit of drawing people in. Its questions are simple enough to ask on the first day, and deep enough to occupy a lifetime. If some of these sections have left you wanting to know what happens next, that is exactly the point.

Answers to Try This Next

These are brief answers and hints, not full solutions: they give the result and the key step, so that you can check your own work. Try each exercise properly before looking.

1.1 A Prime Factory

1. Suppose $p_1, \dots, p_k$ are all the primes; then $p_1 \cdots p_k + 1$ has a prime factor not in the list, a contradiction. The Fermat-number proof is constructive: it exhibits infinitely many pairwise coprime numbers $F_0, F_1, \dots$, each of which supplies a new prime.

2. For $n \geq 2$, 2^n is a multiple of 4, so $2^{2^n} = 16^{2^{n-2}}$, and every power of 16 ends in 6. Hence F_n ends in 7.

3. (a) Induction: $a_{n+1} - 1 = a_n(a_n - 1) = a_n \cdot a_1 \cdots a_{n-1}$. (b) A common factor of a_m and a_n ($m < n$) divides $a_1 \cdots a_{n-1} = a_n - 1$ and a_n, so it is 1; each term has a prime factor shared with no other. (c) $\frac{1}{a_n} = \frac{1}{a_n - 1} - \frac{1}{a_{n+1}-1}$; the sum telescopes to $1 - \frac{1}{a_{n+1}-1}$.

4. (a) $p \mid F_n$ gives $2^{2^n} \equiv -1$; squaring gives $2^{2^{n+1}} \equiv 1$. (b) d divides every e with $2^e \equiv 1$, so $d \mid 2^{n+1}$; $d \nmid 2^n$ since $2^{2^n} \equiv -1 \not\equiv 1$ (as p is odd). So $d = 2^{n+1}$. (c) $2^{p-1} \equiv 1$, so $2^{n+1} \mid p - 1$. (d) $641 = 64 \times 10 + 1$. The primes of the form $64k + 1$ below 641 are 193, 257, 449 and 577: only four candidates to try, which is how Euler found 641.

5. The nth prime is at most the largest prime factor of any of $F_0, \dots, F_{n-1}$, so $p_n \leq F_{n-1} = 2^{2^{n-1}} + 1$. For $n = 5$ the bound is 65 537, while $p_5 = 11$: correct but extremely weak.

1.2 How Thinly Spread Are the Primes?

1. $P_{1,10} = 210$, $P_{1,20} = 9\,699\,690$, $P_{1,30} = 6\,469\,693\,230$. Compare $4^{30} \approx 1.15 \times 10^{18}$ and $e^{30} \approx 1.07 \times 10^{13}$: the product is far below 4^n and grows roughly like e^n (in fact $\ln P_{1,n} \sim n$, a form of the Prime Number Theorem).

2. $\binom{2m}{k+1} / \binom{2m}{k} = \frac{2m-k}{k+1}$, which exceeds 1 exactly when $k < m$, so the coefficients rise to the middle and then fall. The largest of $2m + 1$ terms adding to 4^m is at least their average.

3. The primes in $(\sqrt{n}, n]$ each exceed $\sqrt{n}$ and their product is less than 4^n, so $(\sqrt{n})^{\pi(n) - \pi(\sqrt{n})} < 4^n$; take logarithms. Since $2 \ln 4 \approx 2.77$ and $\pi(\sqrt{n}) \leq \sqrt{n}$, the bound follows. It has the right shape $n / \ln n$, with the constant 2.8 instead of the true 1.

4. Use the primes $2, 3, 5, 7, 13, 23, 43, 83$, each less than twice the one before. For any $n \leq 50$, the first prime in this list greater than n is less than twice the previous one, which is at most n, so it lies between n and $2n$.

1.3 The Smallest Counterexample

1. If $k\sqrt{3}$ is an integer for a smallest positive integer k, then $k' = (\sqrt{3}-1)k$ is a smaller positive integer with $k'\sqrt{3} = 3k - k\sqrt{3}$ an integer. The number $\sqrt{3}-1$ plays the role of $\sqrt{2}-1$.
2. With $0 < \sqrt{N} - m < 1$, $k' = (\sqrt{N} - m)k$ is a positive integer smaller than k, and $k'\sqrt{N} = kN - mk\sqrt{N}$ is an integer: a contradiction.
3. With $N = 4$, $m = 2 = \sqrt{4}$, so $\sqrt{N} - m = 0$ and $k' = 0$ is not a positive integer. The step "$0 < \sqrt{N} - m$" fails.
4. In the picture, the two overlapping squares of side b inside the square of side a cover a central square of side $2b - a$ twice and leave two uncovered squares of side $a - b$. Algebraically, $(2b-a)^2 - 2(a-b)^2 = 2b^2 - a^2 = 0$.

1.4 e Is Irrational, and So Is e^2

1. If $\frac{1}{e} = \frac{p}{q}$, multiply $\frac{1}{e} = \sum_{k\geq 0} \frac{(-1)^k}{k!}$ by $q!$. The terms with $k \leq q$ are integers, and the tail $\sum_{k>q}(-1)^k q!/k!$ is an alternating series of decreasing terms, so it lies strictly between 0 and $\frac{1}{q+1}$ in absolute value. So $q!/e$ is not an integer: contradiction.
2. If $\sin 1 = \frac{p}{q}$, multiply by $(2n+1)!$ with $2n+1 \geq q$. The left side is an integer, and so are the terms of the series up to $\frac{1}{(2n+1)!}$; the tail is alternating with first term of size $\frac{1}{(2n+2)(2n+3)} < 1$, so it is not an integer.
3. The method needs a series with rational terms whose tails shrink faster than the denominators grow, like $\frac{1}{k!}$; no such series is known for π. Niven's proof uses $\int_0^\pi \frac{x^n(\pi-x)^n}{n!} \sin x \, dx$, which would be a positive integer smaller than 1 for large n if π were rational.
4. The roots of $x^2 - (e+\pi)x + e\pi$ are e and π. If $e+\pi$ and $e\pi$ were both algebraic, π would be a root of a polynomial with algebraic coefficients, hence algebraic, contradicting Lindemann's theorem.

1.5 Infinite Descent

1. $3 \mid a$; writing $a = 3a_1$ gives $9a_1^3 + b^3 = 3c^3$, so $3 \mid b$, then $3a_1^3 + 9b_1^3 = c^3$, so $3 \mid c$, and (a_1, b_1, c_1) is a smaller solution: descent. For $p^4 + 2q^4 = 5r^4$: fourth powers are 0 or 1 modulo 5 (they end in 0, 1, 5 or 6), so $p^4 + 2q^4 \equiv 0$ forces $5 \mid p$ and $5 \mid q$; then $5^4 \mid 5r^4$, so $5 \mid r$, and descent again.
2. Squares are 0 or 1 modulo 4, so $x^2 + y^2$ is 0, 1 or 2, never 3.

3. A rational point $(\frac{a}{c}, \frac{b}{c})$ would give $a^2 + b^2 = 3c^2$ with $c \neq 0$, which part (iii) rules out.

4. For fixed x, y, the equation is the quadratic $z^2 - 3xyz + (x^2 + y^2) = 0$, whose two roots add to $3xy$. From $(1,1,1)$: $(1,1,2)$; replacing the 1 opposite $(1,2)$ gives $3 \cdot 1 \cdot 2 - 1 = 5$, so $(1,2,5)$; then $(1,5,13)$ and $(2,5,29)$.

1.6 Sums of Two Squares

1. Squares are 0 or 1 modulo 4, so a sum of two squares is never 3 modulo 4.

2. $65 = 1^2 + 8^2 = 4^2 + 7^2$ and $85 = 2^2 + 9^2 = 6^2 + 7^2$. With $65 = (2 + \mathrm{i})(2 - \mathrm{i})(3 + 2\mathrm{i})(3 - 2\mathrm{i})$: $(2 + \mathrm{i})(3 + 2\mathrm{i}) = 4 + 7\mathrm{i}$ and $(2 + \mathrm{i})(3 - 2\mathrm{i}) = 8 - \mathrm{i}$. Similarly $(2 + \mathrm{i})(4 \pm \mathrm{i})$ gives $7 + 6\mathrm{i}$ and $9 - 2\mathrm{i}$ for 85.

3. $1105 = 4^2 + 33^2 = 9^2 + 32^2 = 12^2 + 31^2 = 23^2 + 24^2$. Each of 5, 13, 17 splits as a Gaussian integer times its conjugate; choosing one factor from each pair gives 8 products, which pair up into conjugates (the same sum of squares), leaving 4.

4. Taking moduli squared, $9 = (a^2 + b^2)(c^2 + d^2)$. Since 3 is not a sum of two squares, one factor is 1.

1.7 When Factorisation Fails

1. $693 = 9 \times 77 = 21 \times 33$, all four factors being S-prime. Recipe: take four primes p, q, r, s of the form $4k + 3$; then $(pq)(rs) = (pr)(qs)$ gives two factorisations into S-primes.

2. The E-primes are the numbers $2 \pmod 4$ (they cannot be products of two even numbers). $60 = 2 \times 30 = 6 \times 10$.

3. $N(zw) = N(z)N(w)$ by expanding $(ac - 5bd)^2 + 5(ad + bc)^2$. $a^2 + 5b^2$ is never 2 or 3. Since $N(2) = 4$, $N(3) = 9$ and $N(1 \pm \sqrt{-5}) = 6$, a factorisation into non-units would need a factor of norm 2 or 3.

4. Let n be the smallest number with two different factorisations $p_1 \cdots p_r = q_1 \cdots q_s$. By Euclid's lemma p_1 divides some q_j, so $p_1 = q_j$; cancelling gives a smaller number with two different factorisations, a contradiction.

1.8 Counting the Rationals

1. $\frac{3}{5}$: parents $(3,5) \leftarrow (3,2) \leftarrow (1,2) \leftarrow (1,1)$, giving positions $1 \to 2 \to 5 \to 10$, so $f(10) = \frac{3}{5}$. $\frac{5}{3}$: $(5,3) \leftarrow (2,3) \leftarrow (2,1) \leftarrow (1,1)$, positions $1 \to 3 \to 6 \to 13$, so $f(13) = \frac{5}{3}$.

2. Newman's rule $x_{n+1} = \frac{1}{2\lfloor x_n \rfloor - x_n + 1}$ gives $1, \frac{1}{2}, 2, \frac{1}{3}, \frac{3}{2}, \frac{2}{3}, 3, \frac{1}{4}, \frac{4}{3}, \frac{3}{5}$, matching $f(1), \ldots, f(10)$.

3. Choose the nth digit to be 5 unless the nth number's nth digit is 5, in which case choose 4. The new decimal differs from every number in the list, and avoiding 0 and 9 means it has only one decimal expansion, so it really is a different number.

4. Every positive integer is uniquely a power of 2 times an odd number, $2^{m-1}(2n-1)$, so $(m,n) \mapsto 2^{m-1}(2n-1)$ is a bijection.

1.9 Tangents to a Cubic

1. Substituting $y = t(x+1)$ into $x^2 + y^2 = 1$ gives $(1+t^2)x^2 + 2t^2x + t^2 - 1 = 0$, with roots -1 and $\frac{1-t^2}{1+t^2}$. With $t = \frac{q}{p}$ the point is $(\frac{p^2-q^2}{p^2+q^2}, \frac{2pq}{p^2+q^2})$, giving the triples $(p^2-q^2, 2pq, p^2+q^2)$.
2. The line is $x + y = 3$, and $x^3 + (3-x)^3 = 9$ reduces to $9x^2 - 27x + 18 = 0$: the cubic terms cancel, leaving only $x = 1, 2$. The line is parallel to the asymptote $y = -x$, and the third intersection is at infinity.
3. The tangent has gradient $\frac{x^2}{y^2} = \frac{289}{400}$, and meets the curve again at $(-\frac{188479}{90391}, -\frac{36520}{90391})$. Clearing denominators, $188479^3 = 36520^3 + 90391^3 + 180782^3$.
4. The line meets $x^3 + y^3 = 1729$ again at $(-\frac{37}{3}, \frac{46}{3})$, so $46^3 = 37^3 + 27 \times 1729 = 37^3 + 3^3 + 36^3$ (using $27 = 3^3$ and $27 \times 1728 = 36^3$).

CHAPTER

2

Sequences, Series and Convergence

Infinity is where mathematics gets interesting, and where it is easiest to go wrong.

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2.3	A Divergent Series and Its Convergent Cousin	STEP III 2014
2.4	$\pi^2/6$	STEP III 2018
2.5	Viète's Product	STEP III 2022
2.6	The Arithmetic–Geometric Mean	STEP II 2024
2.7	Six Digits or Six Hundred	STEP II 2023
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Taming the Infinite

The infinite! No other question has ever moved so profoundly the spirit of man.

— David Hilbert

Two and a half thousand years ago, Zeno of Elea argued that motion is impossible. To walk across a room you must first walk half way, then half of what remains, then half of that, and so on for ever: infinitely many tasks, which surely cannot be completed in a finite time. Everyone knew Zeno was wrong, since people do cross rooms, but for a very long time nobody could say exactly *why* he was wrong. The answer is that the infinite sum $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$ has a finite value, 1. But what does it even *mean* for infinitely many numbers to add up to something?

That question sat at the heart of mathematics for two thousand years. Newton and Leibniz built calculus on top of it in the seventeenth century without ever quite answering it, and their followers produced wonderful results alongside some spectacular nonsense. Euler, the greatest calculator of all time, happily wrote $1 - 1 + 1 - 1 + \cdots = \frac{1}{2}$. Others "proved" contradictory things about the same infinite series. It took until the nineteenth century, and the work of Cauchy, Weierstrass and others, before mathematicians had definitions precise enough to tell the true results from the false ones. Those definitions are the subject of this chapter.

Why it matters

Almost every number you will ever compute with is the limit of something. Your calculator does not know $\sqrt{2}$ or π or e: it knows recipes that get closer and closer to them, and it stops when it is close enough. Understanding *how fast* a recipe gets close is the difference between an algorithm that runs in a millisecond and one that never finishes. The same ideas underlie the computer simulations used to forecast weather, design aircraft and price financial products. And on the purest side, the study of limits leads to some of the most beautiful formulas in mathematics: in this chapter you will prove that

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \frac{\pi^2}{6},$$

a result that made Euler famous and that still seems almost magical.

What you will meet at university

At almost every university, *Analysis* is one of the very first courses you take, and for many students it is the biggest shock. In school mathematics you learn to calculate

limits; in Analysis you learn to *prove* things about them, starting from precise definitions. You will define what it means for a sequence to converge, using the phrases "for every $\varepsilon > 0$" and "there exists N" that appear in Section 2.1. You will prove that every bounded increasing sequence converges, and discover that this depends on a property of the real numbers, *completeness*, that the rational numbers do not have. You will study tests for the convergence of series, and then use the same ideas to give rigorous meaning to continuity, differentiation and integration.

Later courses build on this foundation in many directions. *Numerical analysis* asks how quickly approximations converge and how to make them faster (Section 2.7, Section 2.6). *Dynamical systems* studies what happens when you apply a rule over and over, sometimes reaching a limit and sometimes producing chaos (Section 2.2, Section 2.8). *Complex analysis* explains why π appears in so many infinite sums and products (Section 2.4, Section 2.5). And *analytic number theory* uses limits to answer questions about whole numbers, as in Section 2.9.

The questions that run through this chapter

Three big questions

1. **What does it mean for an infinite process to have an answer?** How do we say precisely that a sequence "gets close to" a limit, or that an infinite sum "adds up to" a number?
2. **How can we prove that something converges without knowing what it converges to?** And once we know a limit exists, how do we find it?
3. **How fast?** Two processes can have the same limit but approach it at wildly different speeds. Why, and how do we measure it?

The first section (Section 2.1) is about language: the precise way mathematicians talk about "eventually". The next five sections are about famous limits: the golden ratio as a continued fraction, a divergent series with a convergent cousin, $\pi^2/6$, Viète's product for $2/\pi$, and Gauss's arithmetic–geometric mean. The last three are about speed and structure: linear versus quadratic convergence, an iteration that either explodes or wanders chaotically, and a family of integrals that close in on e.

Tools for convergence

You will meet a small number of tools again and again, each in a different setting.

§	Topic	Tool
2.1	"Eventually smaller"	Quantifiers: producing m, negating statements
2.2	The golden ratio	Monotone bounded subsequences
2.3	Harmonic and Kempner series	Comparison by blocks
2.4	$\pi^2/6$	Roots of a polynomial, then a squeeze
2.5	Viète's product	Telescoping, then one simple limit
2.6	The AGM	Two sequences squeezing together; an invariant
2.7	Approximating $\sqrt{2}$	Error recurrences: linear and quadratic
2.8	$x \mapsto x^2 - 2$	A substitution that makes the map simple
2.9	Fractions for e	Integrals that are small but not zero

How to read this chapter

As in Chapter 1, each section starts with a STEP question exactly as set. Try it before reading on. This chapter's questions often tell you which tool to use ("by considering…", "you may use the theorem…"); as you work through them, notice how often the same few tools reappear, because at university you will have to choose them yourself.

Several of the questions ask you to assume that limits exist, or that infinite sums and products converge. Where that happens, the section explains what would be needed to prove it. Being aware of the gap between "it obviously works" and "I can prove it works" is the first step towards thinking like an analyst.

2.1 What "Eventually" Means

The question

If $s_1, s_2, s_3, \ldots$ and $t_1, t_2, t_3, \ldots$ are sequences of positive numbers, we write

$$(s_n) \leq (t_n)$$

to mean

"there exists a positive integer m such that $s_n \leq t_n$ whenever $n \geq m$".

Determine whether each of the following statements is true or false. In the case of a true statement, you should give a proof which includes an explicit determination of an appropriate m; in the case of a false statement, you should give a counterexample.

(i) $(1000n) \leq (n^2)$.

(ii) If it is not the case that $(s_n) \leq (t_n)$, then it is the case that $(t_n) \leq (s_n)$.

(iii) If $(s_n) \leq (t_n)$ and $(t_n) \leq (u_n)$, then $(s_n) \leq (u_n)$.

(iv) $(n^2) \leq (2^n)$.

The hook

Almost every statement in a first university analysis course has the same shape: *there exists* something such that *for all* somethings, a condition holds. "The sequence tends to L" means: for every tolerance $\varepsilon > 0$ *there exists* a point N such that *for all* $n \geq N$, $|a_n - L| < \varepsilon$. Students who are fluent with this shape find analysis exciting. Students who aren't find it baffling.

This question is a gentle, beautifully designed introduction to that way of speaking. It defines one new idea, "eventually smaller than", and asks four true-or-false questions about it. One of the four is false, and the reason it is false is exactly the trap that catches most first-year students when they first try to *negate* a statement with quantifiers in it.

Before you start

Read the definition slowly. $(s_n) \leq (t_n)$ does not say that $s_n \leq t_n$ for every n. It says that *from some point onwards* it is true. The first few terms don't matter at all: you can change

a million terms at the start of either sequence without affecting whether $(s_n) \leq (t_n)$.

> **Ask yourself** The definition contains the phrase "there exists a positive integer m". To prove that $(s_n) \leq (t_n)$, what exactly do you have to produce? To prove that it is *false*, what do you have to show?

To prove it, you must *produce* a suitable m and then show that *every* $n \geq m$ works. To disprove it, you must show that *no* m works: whatever m someone proposes, there is some $n \geq m$ with $s_n > t_n$. Keep that second version in mind for part (ii).

Working it through

STAGE 1 A linear function against a quadratic

Part (i) asks whether $1000n \leq n^2$ eventually. Since $n > 0$, this is the same as $1000 \leq n$. So $m = 1000$ works: for every $n \geq 1000$, $n^2 = n \cdot n \geq 1000n$. The statement is **true**. ■

It doesn't matter that $1000n$ is far bigger than n^2 for the first 999 terms. "Eventually" ignores any finite number of terms.

STAGE 2 Is "eventually smaller" like ordinary "smaller"?

For ordinary numbers, if it is not the case that $a \leq b$, then $b < a$, so certainly $b \leq a$. Part (ii) asks whether the same is true for our new relation.

> **Ask yourself** What does "it is *not* the case that $(s_n) \leq (t_n)$" actually mean? Write it out in full, without the word "not" in front.

The statement "there exists m such that for all $n \geq m$, $s_n \leq t_n$" is false precisely when

$$\text{for every } m \text{ there is some } n \geq m \text{ with } s_n > t_n.$$

In other words, $s_n > t_n$ happens *infinitely often*. That is much weaker than "$s_n > t_n$ from some point on". The sequences could keep swapping places.

> **Pause and try** Find two sequences of positive numbers that swap places for ever, so that neither is eventually smaller than the other.

Take $s_n = 2 + (-1)^n$, so $s_n = 1, 3, 1, 3, \ldots$, and $t_n = 2$ for all n. For every m there is an even $n \geq m$ with $s_n = 3 > 2 = t_n$, so it is not the case that $(s_n) \leq (t_n)$. But equally, for every m there is an odd $n \geq m$ with $t_n = 2 > 1 = s_n$, so it is not the case that $(t_n) \leq (s_n)$ either. The statement is **false**. ■

STAGE 3 Chaining

Ask yourself In part (iii) you are given two values of m, one for each hypothesis. They might be different. Which one should you use for the conclusion?

Suppose $s_n \le t_n$ for all $n \ge m_1$ and $t_n \le u_n$ for all $n \ge m_2$. If $n \ge \max(m_1, m_2)$, both inequalities hold, so $s_n \le t_n \le u_n$. So $m = \max(m_1, m_2)$ works, and the statement is **true**. ■

Taking the larger of two thresholds is a move you will make hundreds of times in analysis. It is how you prove, for example, that the sum of two convergent sequences converges.

STAGE 4 A quadratic against an exponential

Try a few values first:

n	1	2	3	4	5	6	7	8
n^2	1	4	9	16	25	36	49	64
2^n	2	4	8	16	32	64	128	256

So $n^2 \le 2^n$ at $n = 1, 2$, fails at $n = 3$, and then seems to hold from $n = 4$ onwards (Figure 2.1.1). The natural guess is $m = 4$.

Ask yourself To prove $n^2 \le 2^n$ for all $n \ge 4$ by induction, you need to get from $n^2 \le 2^n$ to $(n+1)^2 \le 2^{n+1} = 2 \cdot 2^n$. What inequality between $(n+1)^2$ and n^2 would make that step work?

It would be enough to know that $(n+1)^2 \le 2n^2$, because then $(n+1)^2 \le 2n^2 \le 2 \cdot 2^n$. Now $2n^2 - (n+1)^2 = n^2 - 2n - 1 = (n-1)^2 - 2$, which is positive when $n \ge 3$. So:

- *Base case:* $4^2 = 16 = 2^4$.
- *Inductive step:* if $n \ge 4$ and $n^2 \le 2^n$, then $(n+1)^2 \le 2n^2 \le 2^{n+1}$.

Hence $n^2 \le 2^n$ for all $n \ge 4$, and the statement is **true** with $m = 4$. ■

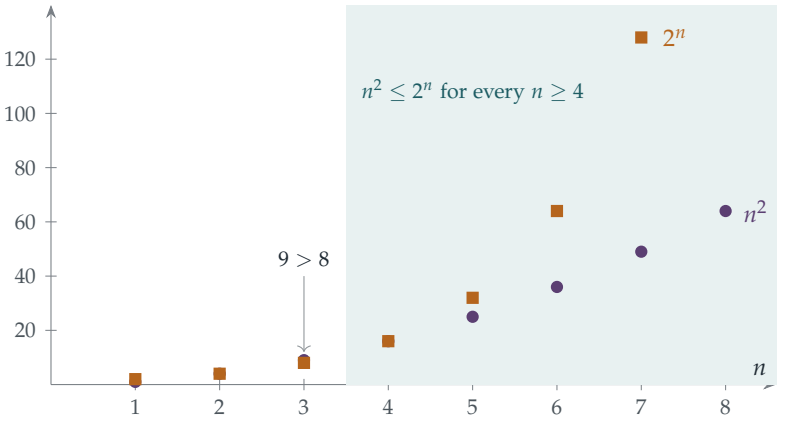


Figure 2.1.1 n^2 (plum circles) against 2^n (amber squares). The two sequences cross twice in the first few terms, but from $n = 4$ onwards 2^n stays ahead for ever. That is exactly what $(n^2) \le (2^n)$ means, with $m = 4$.

Looking back

- **Proving "there exists" means producing.** In parts (i), (iii) and (iv) the proof consisted of naming m and checking it.
- **Negating swaps the quantifiers.** "Not (there exists m, for all $n \ge m$, P)" becomes "for all m, there exists $n \ge m$, not P". Getting this right is what part (ii) tests.
- **A new relation need not behave like an old one.** "Eventually smaller than" is transitive, like $\le$, but unlike $\le$ it does not compare every pair of sequences.

What's really going on

The language of analysis

For about 150 years after Newton and Leibniz, mathematicians used limits and infinitesimals brilliantly but could not say exactly what they meant. Bishop Berkeley mocked infinitesimals in 1734 as "the ghosts of departed quantities", and he had a point. The repair came in the nineteenth century, when Augustin-Louis Cauchy and then Karl Weierstrass replaced vague talk of "becoming infinitely close" with statements built from "for every" and "there exists". The modern definition

$$a_n \to L \iff \text{for every } \varepsilon > 0 \text{ there exists } N \text{ such that } |a_n - L| < \varepsilon \text{ for all } n \ge N$$

has exactly the shape of this question's definition, with one extra quantifier at the front. It is the first thing you will meet in a university analysis course.

Orders of growth

Part (iv) is one small piece of a famous ladder. For any fixed $k > 0$ and $c > 1$,

$$(\ln n) \preceq (n^k), \qquad (n^k) \preceq (c^n), \qquad (c^n) \preceq (n!), \qquad (n!) \preceq (n^n),$$

and in fact each sequence is not just eventually smaller than the next but eventually smaller than *any constant multiple* of it. G. H. Hardy wrote a whole book, *Orders of Infinity* (1910), about this kind of comparison. Today it is the everyday language of computer science, where "this algorithm takes $O(n^2)$ steps" means that the number of steps is eventually at most a constant times n^2. The difference between n^2 and 2^n is the difference between a program that finishes and one that will not finish before the Sun burns out.

Orders that are not total

Part (ii) shows that "eventually smaller" is not a *total order*: some pairs of sequences cannot be compared at all. Relations like this, transitive but not comparing everything, are called *preorders* or *partial orders*, and they turn up everywhere: divisibility of integers, inclusion of sets, and "is a subgroup of" are all examples. Recognising which properties a relation has, and which it doesn't, is part of the training of a mathematician.

The thinking skill

Handling quantifiers

To prove "there exists m such that for all $n \geq m \ldots$", name m and then check every n. To disprove it, show that for every m some $n \geq m$ fails.

When a statement has several quantifiers, write it out in full before trying to prove or disprove it, and negate it by working from the left: each "for all" becomes "there exists", each "there exists" becomes "for all", and the condition at the end is negated.

Try this next

1 **The specimen warm-up.** (From the STEP III Specimen paper, issued before the first STEP in 1987.) Show by using the binomial expansion that $(1 + x)^n > nx$ whenever $x \geq 0$ and n is a positive integer. Deduce that if $y > 1$ then, given any number K, an integer N can be found such that $y^n > K$ for all $n \geq N$. Show further that if $y > 1$ then, given any K, an integer N can be found such that $y^n/n > K$ for all $n \geq N$. (*Hint for the last part:* use the next term of the binomial expansion.)

2 **Both ways round.** Prove that if $(s_n) \preceq (t_n)$ and $(t_n) \preceq (s_n)$, then $s_n = t_n$ for all sufficiently large n.

3 **Racing to infinity.** Find two *increasing* sequences of positive numbers, both tending to infinity, neither of which is eventually smaller than the other.

4 **Up the ladder.** Show that $(2^n) \leq (n!)$, giving an explicit m. Then show that $(n^{10}) \leq (2^n)$ and find the smallest m that works. (A spreadsheet helps for the second part; the answer is surprisingly large.)

Further reading

- **Lara Alcock, *How to Study for a Mathematics Degree*** (Oxford University Press). Chapter 3 is about exactly this kind of logical language and why new undergraduates find it hard.
- **Lara Alcock, *How to Think About Analysis*** (Oxford University Press). A friendly guide to the definitions of first-year analysis.
- **Daniel Velleman, *How to Prove It*** (Cambridge University Press). A systematic introduction to logic, quantifiers and proof.

2.2 The Golden Ratio as a Limit

The question

In this question, the following theorem may be used.

Let $u_1, u_2, \ldots$ be a sequence of (real) numbers. If the sequence is bounded above (that is, $u_n \leq b$ for all n, where b is some fixed number) and increasing (that is, $u_n \geq u_{n-1}$ for all n), then the sequence tends to a limit (that is, converges).

The sequence $u_1, u_2, \ldots$ is defined by $u_1 = 1$ and

$$u_{n+1} = 1 + \frac{1}{u_n} \qquad (n \geq 1). \tag{$*$}$$

(i) Show that, for $n \geq 3$,

$$u_{n+2} - u_n = \frac{u_n - u_{n-2}}{(1+u_n)(1+u_{n-2})}.$$

(ii) Prove, by induction or otherwise, that $1 \leq u_n \leq 2$ for all n.

(iii) Show that the sequence $u_1, u_3, u_5, \ldots$ tends to a limit, and that the sequence $u_2, u_4, u_6, \ldots$ tends to a limit. Find these limits and deduce that the sequence $u_1, u_2, u_3, \ldots$ tends to a limit.

Would this conclusion change if the sequence were defined by $(*)$ and $u_1 = 3$?

The hook

Write down the expression

$$1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cdots}}}$$

and ask what it equals. A quick answer: call it x; the part under the first fraction line is x again, so $x = 1 + 1/x$, which gives $x^2 = x + 1$ and $x = \frac{1+\sqrt{5}}{2}$, the golden ratio φ.

But is that argument valid? It assumes the infinite expression *means* something, that the process of building it up settles down to a single number. The same trick applied to $1 - 1 + 1 - 1 + \cdots$ ("call it S; then $S = 1 - S$, so $S = \frac{1}{2}$") produces nonsense. This question shows how to do the job properly, using the only tool a STEP candidate has

for proving convergence without knowing the limit in advance: *monotone sequences*.

Before you start

Compute some terms. With $u_1 = 1$:

$$1, \quad 2, \quad \frac{3}{2}, \quad \frac{5}{3}, \quad \frac{8}{5}, \quad \frac{13}{8}, \quad \frac{21}{13}, \quad \frac{34}{21}, \quad \ldots$$

Ask yourself Do you recognise the numerators and denominators? And look at the decimals: $1, 2, 1.5, 1.667, 1.6, 1.625, 1.615, \ldots$. Is the sequence increasing? Decreasing?

They are ratios of consecutive Fibonacci numbers. The sequence is neither increasing nor decreasing: it zig-zags, overshooting and undershooting. So the theorem we are given, which is about increasing sequences, cannot be applied to the whole sequence. But the odd-numbered terms $1, 1.5, 1.6, 1.615, \ldots$ seem to go up, and the even-numbered terms $2, 1.667, 1.625, \ldots$ seem to go down. That is the plan of the question.

Working it through

STAGE 1 Comparing terms two apart

Part (i) compares u_{n+2} with u_n, two steps apart. The natural first move is to use $(*)$ to write each of them in terms of the term before.

Ask yourself Using $(*)$ twice, write $u_{n+2} - u_n$ in terms of $u_{n+1} - u_{n-1}$, and then $u_{n+1} - u_{n-1}$ in terms of $u_n - u_{n-2}$.

From $(*)$,

$$u_{n+2} - u_n = \frac{1}{u_{n+1}} - \frac{1}{u_{n-1}} = -\frac{u_{n+1} - u_{n-1}}{u_{n+1}u_{n-1}},$$

and in the same way $u_{n+1} - u_{n-1} = -\dfrac{u_n - u_{n-2}}{u_n u_{n-2}}$. Putting these together, the two minus signs cancel:

$$u_{n+2} - u_n = \frac{u_n - u_{n-2}}{u_{n+1}u_n \cdot u_{n-1}u_{n-2}}.$$

Finally, $(*)$ says $u_{n+1}u_n = u_n + 1$ and $u_{n-1}u_{n-2} = u_{n-2} + 1$, which gives the required formula. (We need $n \geq 3$ so that u_{n-2} exists.) ■

STAGE 2 A trap

Pause and try Prove that $1 \leq u_n \leq 2$ for all n.

It is true for $u_1 = 1$. If $1 \le u_n \le 2$, then $\frac{1}{2} \le \frac{1}{u_n} \le 1$, so $\frac{3}{2} \le u_{n+1} \le 2$, which is certainly between 1 and 2. By induction, $1 \le u_n \le 2$ for all n. ■

In particular every u_n is positive, so the denominator $(1+u_n)(1+u_{n-2})$ in part (i) is positive.

STAGE 3 Two monotone subsequences

Ask yourself The formula in part (i) says that $u_{n+2} - u_n$ is a positive multiple of $u_n - u_{n-2}$. What does that tell you about the signs of $u_3 - u_1$, $u_5 - u_3$, $u_7 - u_5$, …?

They all have the same sign! Since $u_3 - u_1 = \frac{3}{2} - 1 > 0$, part (i) and induction give $u_5 > u_3$, $u_7 > u_5$, and so on: the odd terms are increasing. They are bounded above by 2, so by the theorem they tend to a limit; call it L.

Similarly $u_4 - u_2 = \frac{5}{3} - 2 < 0$, so the even terms are decreasing. They are bounded *below* by 1. The theorem is stated for increasing sequences, but $-u_2, -u_4, -u_6, \ldots$ is increasing and bounded above by -1, so it converges, and therefore so do the even terms; call their limit M.

STAGE 4 Finding the limits

Ask yourself To find L, you need an equation relating u_{n+2} to u_n directly. Can you get one from $(*)$?

Applying $(*)$ twice,

$$u_{n+2} = 1 + \frac{1}{u_{n+1}} = 1 + \frac{1}{1 + 1/u_n} = 1 + \frac{u_n}{u_n + 1} = \frac{2u_n + 1}{u_n + 1}.$$

Letting $n \to \infty$ through odd values, both u_n and u_{n+2} tend to L, so

$$L = \frac{2L+1}{L+1} \quad \Longrightarrow \quad L^2 + L = 2L + 1 \quad \Longrightarrow \quad L^2 - L - 1 = 0.$$

The roots are $\frac{1 \pm \sqrt{5}}{2}$, and since $L \ge 1$ we must have $L = \frac{1+\sqrt{5}}{2} = \varphi$. Exactly the same argument through even values gives $M = \varphi$.

Ask yourself The odd terms tend to φ and the even terms tend to φ. Why does this mean that the whole sequence tends to φ?

Intuitively, every term is either odd-numbered or even-numbered, and both kinds are getting close to φ. Precisely: given any tolerance $\varepsilon > 0$, the odd terms are within ε of φ from some point N_1 onwards, and the even terms from some point N_2 onwards. So *all* terms are within ε of φ from $\max(N_1, N_2)$ onwards, which is what convergence means. (This is the "take the larger threshold" move from Section 2.1.)

STAGE 5 Starting somewhere else

With $u_1 = 3$ the sequence is

$$3, \quad \tfrac{4}{3}, \quad \tfrac{7}{4}, \quad \tfrac{11}{7}, \quad \tfrac{18}{11}, \quad \tfrac{29}{18}, \quad \dots$$

Now $u_1 = 3$ is outside $[1,2]$, so the induction in part (ii) does not start. But $u_2 = \frac{4}{3}$ is inside, and the induction works from there: $1 \le u_n \le 2$ for all $n \ge 2$. Part (i) only ever uses terms from u_{n-2} onwards and only needs them to be positive, which they all are. This time $u_3 - u_1 = \frac{7}{4} - 3 < 0$ and $u_4 - u_2 = \frac{11}{7} - \frac{4}{3} > 0$, so the odd terms decrease and the even terms increase: the roles are swapped, but each subsequence is still monotone and bounded. The limits satisfy the same equation, and both are positive, so both are φ. The conclusion does **not** change. ■

Looking back

- **When a sequence isn't monotone, look for monotone pieces.** Splitting into odd and even terms turned a zig-zag into two well-behaved sequences.
- **Prove convergence first, find the limit second.** The equation $L = (2L+1)/(L+1)$ is only legitimate once we know L exists. That is exactly the step the quick argument in the hook skipped.
- **Check which root.** The limit equation had two solutions; the bounds from part (ii) told us which one was right.

What's really going on

Cobwebs and fixed points

Every sequence defined by $u_{n+1} = f(u_n)$ can be drawn as a *cobweb diagram* (Figure 2.2.1). Plot $y = f(x)$ and $y = x$. Start at u_1 on the x-axis, go vertically to the curve (height $u_2 = f(u_1)$), then horizontally to the line $y = x$ (which moves you to $x = u_2$), then vertically to the curve again, and so on. Here $f(x) = 1 + 1/x$ is decreasing, which is why the path spirals inwards, alternately overshooting and undershooting, towards the point where the curve crosses the line: the *fixed point* φ, where $f(\varphi) = \varphi$.

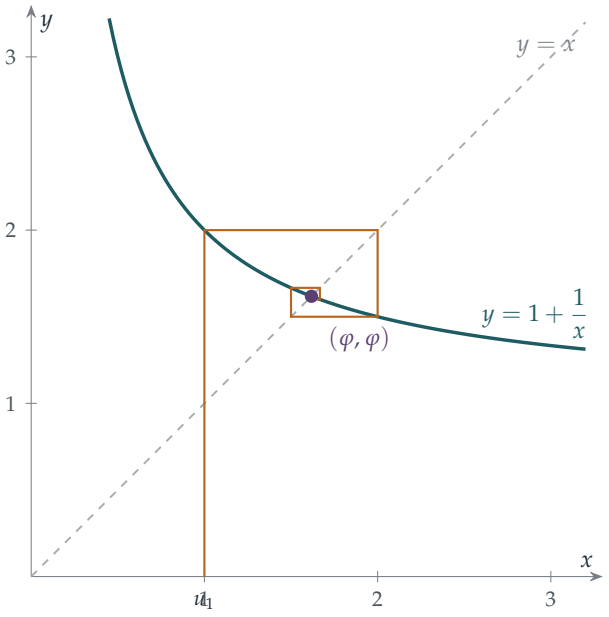


Figure 2.2.1 A cobweb diagram for $u_{n+1} = 1 + 1/u_n$ starting from $u_1 = 1$. The path goes up to the curve and across to the line $y = x$, over and over. Because the curve slopes downwards, the path spirals in towards the fixed point $\varphi \approx 1.618$, alternating sides: that is the zig-zag of odd and even terms.

How fast, and why

Where does the factor $(1 + u_n)(1 + u_{n-2})$ in part (i) come from? Near the fixed point, $u_{n+1} - \varphi = f(u_n) - f(\varphi) \approx f'(\varphi)(u_n - \varphi)$, and $f'(\varphi) = -1/\varphi^2 \approx -0.38$. So each step multiplies the error by about -0.38: the sign flips (zig-zag) and the size shrinks (convergence). Two steps multiply it by $1/\varphi^4 \approx 0.146$, which is what $1/\big((1 + u_n)(1 + u_{n-2})\big)$ tends to. At university this becomes the *contraction mapping theorem*: if $|f'| \le k < 1$ near a fixed point, iteration converges to it, with the error shrinking at least geometrically. It is one of the most useful theorems in mathematics, and it is how you prove that differential equations have solutions (Chapter 8). In Section 2.8 we will meet a map whose fixed points are *not* attracting, with very different results.

The most irrational number

The terms $u_n = F_{n+1}/F_n$ are the best possible fractional approximations to φ for their size, but "best" is not very good: φ is harder to approximate by fractions than any other number. The reason is visible in its continued fraction $[1; 1, 1, 1, \ldots]$: large entries in a continued fraction mean very good approximations (think of $\pi = [3; 7, 15, 1, 292, \ldots]$ and the excellent $\frac{355}{113}$), and φ has the smallest possible entries all the way. Adolf Hurwitz proved in 1891 that every irrational x has infinitely many fractions with $|x - p/q| < 1/(\sqrt{5}\,q^2)$, and that $\sqrt{5}$ cannot be improved, precisely because of φ. This "maximal irrationality" is the reason the golden angle appears in the spiral patterns of

sunflower seeds.

The thinking skill

Monotone and bounded implies convergent

To show a sequence converges without knowing its limit, find a monotone, bounded sequence, or split the sequence into monotone, bounded pieces.

Only once you know a limit exists may you find it by "letting $n \to \infty$" in the recurrence. At university this theorem is a consequence of the *completeness* of the real numbers, the property that distinguishes $\mathbb{R}$ from $\mathbb{Q}$, and you will see it proved from the axioms.

Try this next

1 **Fibonacci inside.** Prove that $u_n = F_{n+1}/F_n$, where $F_1 = F_2 = 1$ and $F_{n+1} = F_n + F_{n-1}$.

2 **The error shrinks.** Show that $u_{n+1} - \varphi = -\dfrac{u_n - \varphi}{\varphi\, u_n}$, and deduce that $|u_n - \varphi| \leq \varphi^{-(n-1)}$ for all n. (Use $\varphi^2 = \varphi + 1$, and $u_n \geq 1$.)

3 **Breaking the sequence.** Show that if $u_1 = \frac{1-\sqrt{5}}{2}$ then the sequence is constant. Find all the starting values u_1 for which some $u_n = 0$, so that the sequence cannot continue. (*Hint:* work backwards from $u_n = 0$.)

4 **A cousin for $\sqrt{2}$.** Let $v_1 = 1$ and $v_{n+1} = 1 + \dfrac{1}{1+v_n}$. Adapt the method of this question to show that $v_n \to \sqrt{2}$, and write $\sqrt{2}$ as a continued fraction.

Further reading

- **Mario Livio, *The Golden Ratio*** (Broadway). The history of φ, separating the genuine mathematics from the many myths.
- **C. D. Olds, *Continued Fractions*** (Mathematical Association of America). A short, very readable introduction aimed at strong school students.
- **J. C. Burkill, *A First Course in Mathematical Analysis*** (Cambridge University Press). Rigorous treatment of monotone sequences and completeness, with many exercises.

2.3 A Divergent Series and Its Convergent Cousin

The question

The numbers $f(r)$ satisfy $f(r) > f(r+1)$ for $r = 1, 2, \ldots$. Show that, for any non-negative integer n,

$$k^n(k-1)\,f(k^{n+1}) \leq \sum_{r=k^n}^{k^{n+1}-1} f(r) \leq k^n(k-1)\,f(k^n),$$

where k is an integer greater than 1.

(i) By taking $f(r) = 1/r$, show that

$$\frac{N+1}{2} \leq \sum_{r=1}^{2^{N+1}-1} \frac{1}{r} \leq N+1.$$

Deduce that the sum $\sum_{r=1}^{\infty} \frac{1}{r}$ does not converge.

(ii) By taking $f(r) = 1/r^3$, show that

$$\sum_{r=1}^{\infty} \frac{1}{r^3} \leq 1\tfrac{1}{3}.$$

(iii) Let $S(n)$ be the set of positive integers less than n which do not have a 2 in their decimal representation and let $\sigma(n)$ be the sum of the reciprocals of the numbers in $S(n)$, so for example $\sigma(5) = 1 + \frac{1}{3} + \frac{1}{4}$. Show that $S(1000)$ contains $9^3 - 1$ distinct numbers.

Show that $\sigma(n) < 80$ for all n.

The hook

The harmonic series $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$ is the most famous example of a sum whose terms shrink to zero but which nevertheless grows without limit. It grows absurdly slowly: you need more than 12,000 terms to pass 10, and about 1.5×10^{43} terms to pass 100. But it does pass every number eventually.

Now throw away every term whose denominator contains the digit 2: out go $\frac{1}{2}$, $\frac{1}{12}$, $\frac{1}{20}$ to $\frac{1}{29}$, $\frac{1}{32}$, and so on. You might expect that removing a tenth or so of the terms would make little difference to a sum that is infinite anyway. Instead, what is left adds up to

less than 80. This surprising fact was proved by A. J. Kempner in 1914, and the question walks you to it with one beautifully simple idea: chop the sum into blocks.

Before you start

Ask yourself The first inequality is about the block of terms from $r = k^n$ to $r = k^{n+1} - 1$. With $k = 2$ and $n = 3$, which terms are these, and how many of them are there?

They are $f(8), f(9), \dots, f(15)$: eight terms. In general the block runs from k^n to $k^{n+1} - 1$, so it has $k^{n+1} - k^n = k^n(k-1)$ terms. The inequality says the block sum lies between (number of terms) $\times$ (smallest possible term) and (number of terms) $\times$ (largest term).

Working it through

STAGE 1 Bounding a block

Since f is decreasing, every term in the block satisfies

$$f(k^{n+1}) < f(k^{n+1} - 1) \le f(r) \le f(k^n) \qquad (k^n \le r \le k^{n+1} - 1).$$

Adding up the $k^n(k-1)$ terms gives the inequality. ■

STAGE 2 The harmonic series

Ask yourself With $k = 2$ and $f(r) = 1/r$, what are the two bounds for the block from 2^n to $2^{n+1} - 1$?

The lower bound is $2^n \cdot \frac{1}{2^{n+1}} = \frac{1}{2}$ and the upper bound is $2^n \cdot \frac{1}{2^n} = 1$. So *every* block, however far out, contributes between $\frac{1}{2}$ and 1 (Figure 2.3.1). The blocks for $n = 0, 1, \dots, N$ fit together exactly to make the sum from $r = 1$ to $r = 2^{N+1} - 1$, so adding $N + 1$ blocks gives

$$\frac{N+1}{2} \le \sum_{r=1}^{2^{N+1}-1} \frac{1}{r} \le N + 1.$$

As $N \to \infty$ the left-hand side grows without bound, so the partial sums are unbounded and the series diverges. ■

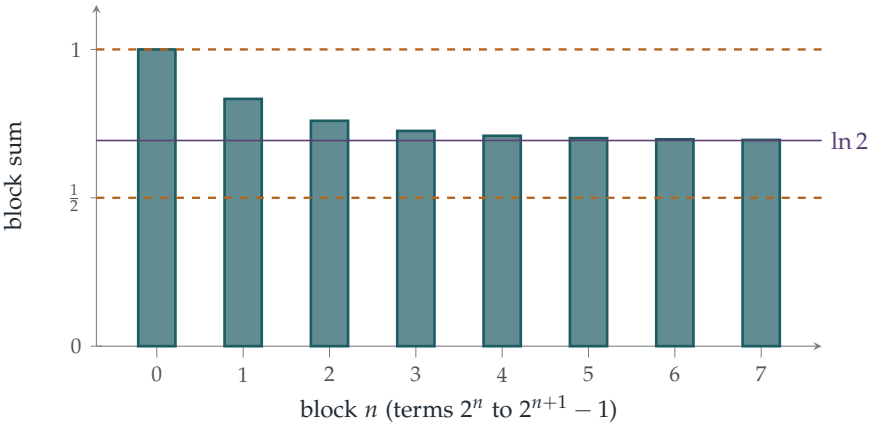


Figure 2.3.1 The harmonic series cut into blocks of $1, 2, 4, 8, \ldots$ terms. Every block sum lies between $\frac{1}{2}$ and 1 (amber, dashed), and in fact the block sums settle down towards $\ln 2 \approx 0.693$ (plum). Infinitely many blocks, each worth at least $\frac{1}{2}$, cannot have a finite total.

STAGE 3 The sum of reciprocal cubes

Now take $f(r) = 1/r^3$, still with $k = 2$. The upper bound for block n is

$$2^n \cdot \frac{1}{(2^n)^3} = \frac{1}{4^n}.$$

The blocks for $n = 0, 1, 2, \ldots$ cover every positive integer exactly once, so

$$\sum_{r=1}^{\infty} \frac{1}{r^3} \le \sum_{n=0}^{\infty} \frac{1}{4^n} = \frac{1}{1 - \frac{1}{4}} = \frac{4}{3} = 1\tfrac{1}{3}. \qquad \blacksquare$$

(Strictly, this shows every partial sum is at most $\frac{4}{3}$; an increasing sequence that is bounded above converges, so the infinite sum exists and is at most $\frac{4}{3}$. The true value is $1.2020\ldots$)

STAGE 4 Counting numbers without a 2

Ask yourself How many whole numbers from 0 to 999 can you write using only the digits $0, 1, 3, 4, 5, 6, 7, 8, 9$, if you allow leading zeros (so 7 is written 007)?

Each of the three digit positions has 9 choices, so there are 9^3 such strings. They correspond exactly to the numbers from 0 to 999 with no 2, and removing 0 (which is not positive) leaves $9^3 - 1 = 728$ numbers in $S(1000)$. $\blacksquare$

STAGE 5 Kempner's series

The block idea now works with blocks defined by the *number of digits*, which is the same as using $k = 10$.

Pause and try How many d-digit numbers have no 2 in them? And what is the smallest a d-digit number can be?

The first digit can be any of $1, 3, 4, \ldots, 9$ (eight choices: not 0, not 2), and each of the remaining $d-1$ digits has nine choices. So there are $8 \times 9^{d-1}$ such numbers, and each one is at least 10^{d-1}. So the d-digit numbers in $S(n)$ contribute at most

$$8 \times 9^{d-1} \times \frac{1}{10^{d-1}} = 8\left(\frac{9}{10}\right)^{d-1}$$

to $\sigma(n)$. Adding over all digit lengths $d = 1, 2, 3, \ldots$,

$$\sigma(n) \le \sum_{d=1}^{\infty} 8\left(\frac{9}{10}\right)^{d-1} = \frac{8}{1-\frac{9}{10}} = 80.$$

For the strict inequality, notice that most of the bounds are far from sharp (for instance $\frac{1}{3} < 1$, while we have bounded every one-digit term by 1), so $\sigma(n) < 80$ for all n. ■

Looking back

- **Group the terms.** Neither sum is easy term by term. Grouped into blocks of the right size, each was compared with a geometric series.
- **The right blocks depend on the problem.** Powers of 2 suited $1/r$ and $1/r^3$; powers of 10 suited a question about decimal digits.
- **Bounded and increasing.** Both convergence proofs rested on the same principle as Section 2.2: partial sums of positive terms increase, so a bound is all you need.

What's really going on

The condensation test

The first inequality in the question is the heart of *Cauchy's condensation test*: if f is positive and decreasing, then

$$\sum_{r=1}^{\infty} f(r) \text{ converges} \iff \sum_{n=0}^{\infty} 2^n f(2^n) \text{ converges.}$$

It turns hard series into easy ones. Applied to $f(r) = 1/r^p$, it gives $\sum 2^n \cdot 2^{-np} = \sum (2^{1-p})^n$, a geometric series that converges exactly when $p > 1$. Applied to $1/(r \ln r)$ it shows that even this diverges, and to $1/(r(\ln r)^2)$ that this converges. The argument goes back to Nicole Oresme, who proved the harmonic series diverges in about 1350 by exactly the blocks of Figure 2.3.1.

How slowly the harmonic series diverges

Figure 2.3.1 suggests that each block contributes about $\ln 2$, so the sum of the first N terms should be roughly $\log_2 N \times \ln 2 = \ln N$. In fact

$$1 + \frac{1}{2} + \cdots + \frac{1}{N} = \ln N + \gamma + (\text{something tending to } 0),$$

where $\gamma = 0.5772\ldots$ is the Euler–Mascheroni constant. Nobody knows whether γ is rational. You will prove the formula with integrals in first-year analysis, and the harmonic numbers turn up again in Section 9.11 and in Section 11.12.

Most numbers contain a 2

Why does Kempner's series converge? Because "numbers without a 2" are very rare among large numbers: of the numbers below 10^d, the proportion with no 2 is at most $(0.9)^d$, which tends to 0. Among 100-digit numbers, fewer than 0.003% avoid the digit 2. So we are not removing a tenth of the terms, but almost all of them. Our bound of 80 is generous: in 1979 Robert Baillie used an accelerated method to compute the sum to many decimal places, and it is about 19.257.

Sums of reciprocal powers

The sum in part (ii) is $\zeta(3)$, a value of the Riemann zeta function we met in Section 1.2. In Section 2.4 we will find $\zeta(2) = \pi^2/6$ exactly, and Euler found a formula for $\zeta(4), \zeta(6), \ldots$. But nobody has found a formula for $\zeta(3)$. In 1978 Roger Apéry stunned the mathematical world by proving that $\zeta(3)$ is irrational, using integrals not unlike the ones in Section 2.9. Whether $\zeta(5)$ is irrational is still unknown.

The thinking skill

Comparison by blocks

To decide whether a series of positive terms converges, group the terms into blocks on which they are roughly constant, and compare each block with a simple series.

Comparison tests are the bread and butter of first-year analysis. The skill is choosing the comparison: here, blocks of size 2^n or 10^d in which the terms change by at most a fixed factor.

Try this next

1 **Feel the slowness.** Using $\sum_{r \le N} 1/r \approx \ln N + 0.5772$, estimate how many terms of the harmonic series you need to pass 10, and to pass 100.

2 The test in action. Use the method of part (i) with $f(r) = 1/r^2$ to show that $\sum 1/r^2 \leq 2$. Then improve the bound by treating the first term separately.

3 Between convergence and divergence. Use the block inequality with $k = 2$ to show that $\sum_{r\geq 2} \dfrac{1}{r \ln r}$ diverges and $\sum_{r\geq 2} \dfrac{1}{r(\ln r)^2}$ converges.

4 Excluding more. HARDERShow that the sum of the reciprocals of the positive integers with no 2 *and* no 7 in their decimal representation is less than 35. (Can you see where 35 comes from?)

Further reading

- **Julian Havil, *Gamma: Exploring Euler's Constant*** (Princeton University Press). The harmonic series, γ and much more, written for a strong school student.
- **Mary Hart, *Guide to Analysis*** (Palgrave). A concise first-year text with a clear chapter on series and comparison tests.
- **Robert Baillie, "Sums of reciprocals of integers missing a given digit"**, *American Mathematical Monthly* 86 (1979). A short paper you can read with this section behind you.

2.4 $\pi^2/6$

The question

(i) Use De Moivre's theorem to show that, if $\sin\theta \neq 0$, then

$$\frac{(\cot\theta + \mathrm{i})^{2n+1} - (\cot\theta - \mathrm{i})^{2n+1}}{2\mathrm{i}} = \frac{\sin(2n+1)\theta}{\sin^{2n+1}\theta},$$

for any positive integer n.

Deduce that the solutions of the equation

$$\binom{2n+1}{1}x^n - \binom{2n+1}{3}x^{n-1} + \cdots + (-1)^n = 0$$

are

$$x = \cot^2\left(\frac{m\pi}{2n+1}\right)$$

where $m = 1, 2, \ldots, n$.

(ii) Hence show that

$$\sum_{m=1}^{n} \cot^2\left(\frac{m\pi}{2n+1}\right) = \frac{n(2n-1)}{3}.$$

(iii) Given that $0 < \sin\theta < \theta < \tan\theta$ for $0 < \theta < \frac{1}{2}\pi$, show that

$$\cot^2\theta < \frac{1}{\theta^2} < 1 + \cot^2\theta.$$

Hence show that

$$\sum_{m=1}^{\infty} \frac{1}{m^2} = \frac{\pi^2}{6}.$$

The hook

In 1650 Pietro Mengoli asked a simple question: what is

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \frac{1}{25} + \cdots ?$$

The terms shrink fast enough for the sum to converge (Section 2.3 showed it is at most 2), but nobody could find its value. The Bernoulli brothers tried and failed; Jacob Bernoulli publicly begged for help. The problem became known as the *Basel problem*,

after the Bernoullis' home city.

In 1734 the young Leonhard Euler solved it, and the answer made him famous: the sum is $\pi^2/6$. Why on earth should π, a number about circles, appear in a sum of reciprocal squares? This question gives a complete, rigorous proof using nothing beyond A-level: De Moivre's theorem, the sum of the roots of a polynomial, and a squeeze.

Before you start

The overall plan is worth seeing before the details.

- Part (i) builds a polynomial whose roots are the numbers $\cot^2\left(\frac{m\pi}{2n+1}\right)$.
- Part (ii) adds up those roots without knowing them individually, using the coefficients.
- Part (iii) uses the fact that $\cot\theta \approx 1/\theta$ for small θ to turn a sum of $\cot^2$ values into a sum of $1/m^2$ values, and squeezes.

> **Ask yourself** If a polynomial $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ has roots $r_1, \ldots, r_n$, what is $r_1 + \cdots + r_n$ in terms of the coefficients?

It is $-a_{n-1}/a_n$. That fact is the engine of part (ii).

Working it through

STAGE 1 De Moivre

> **Ask yourself** Write $\cot\theta + \mathrm{i}$ as a single fraction. What do you notice about the numerator?

$\cot\theta + \mathrm{i} = \dfrac{\cos\theta + \mathrm{i}\sin\theta}{\sin\theta}$, and similarly $\cot\theta - \mathrm{i} = \dfrac{\cos\theta - \mathrm{i}\sin\theta}{\sin\theta}$. By De Moivre's theorem,

$$(\cot\theta \pm \mathrm{i})^{2n+1} = \frac{\cos(2n+1)\theta \pm \mathrm{i}\sin(2n+1)\theta}{\sin^{2n+1}\theta}.$$

Subtracting, the cosines cancel and the sines double, and dividing by 2i gives

$$\frac{\sin(2n+1)\theta}{\sin^{2n+1}\theta}. \qquad \blacksquare$$

STAGE 2 The polynomial

Now expand the left-hand side with the binomial theorem instead. Write $c = \cot\theta$ and $N = 2n+1$.

Ask yourself In $(c+\mathrm{i})^N-(c-\mathrm{i})^N$, which terms of the two binomial expansions cancel, and which double?

The term containing i^j appears with a $+$ sign in the first expansion and with $(-1)^j$ in the second. So terms with even j cancel and terms with odd j double:

$$(c+\mathrm{i})^N-(c-\mathrm{i})^N=2\sum_{j\text{ odd}}\binom{N}{j}c^{N-j}\,\mathrm{i}^j.$$

With $j=2k+1$, $\mathrm{i}^j=\mathrm{i}\,(-1)^k$ and $c^{N-j}=c^{2n-2k}=(c^2)^{n-k}$. Dividing by 2i,

$$\frac{\sin(2n+1)\theta}{\sin^{2n+1}\theta}=\binom{2n+1}{1}x^n-\binom{2n+1}{3}x^{n-1}+\cdots+(-1)^n,$$

where $x=\cot^2\theta$.

Pause and try For which θ is the left-hand side zero? Use this to find n different roots of the polynomial.

The left-hand side is zero when $\sin(2n+1)\theta=0$ but $\sin\theta\neq 0$, for instance when $\theta=\frac{m\pi}{2n+1}$ for $m=1,2,\ldots,n$. These angles all lie strictly between 0 and $\frac{\pi}{2}$, where cot is positive and strictly decreasing, so the n values $\cot^2\left(\frac{m\pi}{2n+1}\right)$ are all different. A polynomial of degree n has at most n roots, so these are *all* of them. ■

STAGE 3 Adding the roots

The sum of the roots is $-(\text{coefficient of } x^{n-1})/(\text{coefficient of } x^n)$:

$$\sum_{m=1}^{n}\cot^2\left(\frac{m\pi}{2n+1}\right)=\frac{\binom{2n+1}{3}}{\binom{2n+1}{1}}=\frac{(2n+1)(2n)(2n-1)}{6\,(2n+1)}=\frac{n(2n-1)}{3}.$$ ■

Check with $n=1$: $\cot^2(\pi/3)=\frac{1}{3}$, and $\frac{1\times 1}{3}=\frac{1}{3}$.

STAGE 4 The squeeze

From $0<\sin\theta<\theta<\tan\theta$, take reciprocals (which reverses the inequalities) and square:

$$\cot^2\theta<\frac{1}{\theta^2}<\frac{1}{\sin^2\theta}=1+\cot^2\theta.$$

Ask yourself Put $\theta=\frac{m\pi}{2n+1}$ and add up over $m=1,\ldots,n$. What do the two outer sums become? What does the middle sum have to do with $\sum 1/m^2$?

The middle sum is $\displaystyle\sum_{m=1}^{n}\frac{(2n+1)^2}{m^2\pi^2}$. By part (ii) the left sum is $\frac{n(2n-1)}{3}$, and the right sum

is $n + \frac{n(2n-1)}{3} = \frac{2n(n+1)}{3}$. Multiplying through by $\frac{\pi^2}{(2n+1)^2}$:

$$\frac{\pi^2}{3} \cdot \frac{n(2n-1)}{(2n+1)^2} < \sum_{m=1}^{n} \frac{1}{m^2} < \frac{\pi^2}{3} \cdot \frac{2n(n+1)}{(2n+1)^2}.$$

As $n \to \infty$, both $\frac{n(2n-1)}{(2n+1)^2}$ and $\frac{2n(n+1)}{(2n+1)^2}$ tend to $\frac{2}{4} = \frac{1}{2}$ (divide top and bottom by n^2). So both bounds tend to $\frac{\pi^2}{6}$, and the partial sums, trapped between them, must tend to $\frac{\pi^2}{6}$ too (Figure 2.4.1). ■

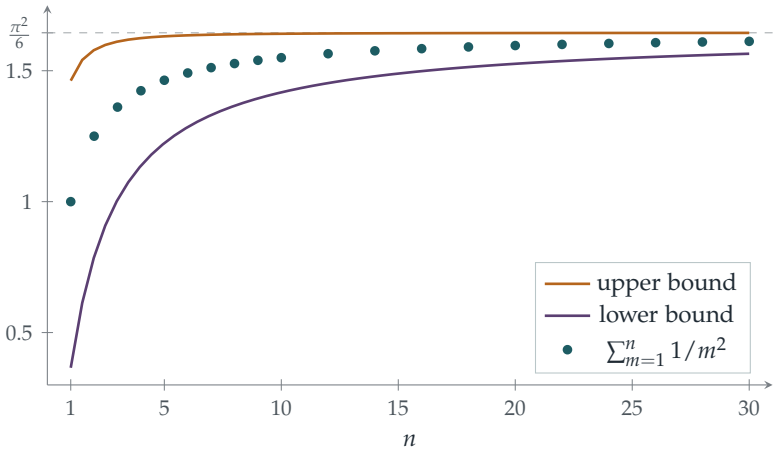


Figure 2.4.1 The squeeze in part (iii). The partial sums of $\sum 1/m^2$ (teal points) are trapped between the lower bound (plum) and the upper bound (amber), and both bounds tend to $\pi^2/6 \approx 1.645$.

Looking back

- **Information from coefficients.** We never found the roots $\cot^2(m\pi/(2n+1))$ in any simpler form, but we added them up exactly.
- **Approximate, then squeeze.** $\cot\theta$ is only approximately $1/\theta$, but the error was controlled by exact inequalities in both directions.
- **Finite to infinite.** Every step until the last concerned finite sums. The infinite sum appeared only when we let $n \to \infty$ in an inequality, which is the safe way to handle infinity.

What's really going on

Euler's bold argument

Euler's own 1734 argument was not rigorous by modern standards, but it was dazzling. A polynomial with roots $\pm a, \pm b, \pm c, \ldots$ and value 1 at 0 factorises as $(1 - x^2/a^2)(1 - x^2/b^2)\cdots$. The function $\frac{\sin x}{x}$ has value 1 at 0 and roots $\pm\pi, \pm 2\pi, \pm 3\pi, \ldots$, so, treating it as an "infinite polynomial", Euler wrote

$$\frac{\sin x}{x} = \left(1 - \frac{x^2}{\pi^2}\right)\left(1 - \frac{x^2}{4\pi^2}\right)\left(1 - \frac{x^2}{9\pi^2}\right)\cdots.$$

The coefficient of x^2 on the right is $-\frac{1}{\pi^2}\left(1 + \frac{1}{4} + \frac{1}{9} + \cdots\right)$, while from the Maclaurin series $\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \cdots$ it is $-\frac{1}{6}$. Comparing gives $\pi^2/6$. Our STEP proof is the finite, rigorous version of the same idea: a polynomial whose roots are tied to multiples of π, and its second coefficient.

Beyond reciprocal squares

Euler went on to find $\sum 1/m^4 = \pi^4/90$, $\sum 1/m^6 = \pi^6/945$, and in general $\sum 1/m^{2k}$ is a rational multiple of π^{2k}, involving the *Bernoulli numbers*. The odd powers are a different story: as we saw in Section 2.3, nobody knows a formula for $\sum 1/m^3$. These sums are the values $\zeta(2), \zeta(4), \ldots$ of the Riemann zeta function, whose behaviour for complex arguments controls the distribution of the primes (Section 1.2).

Where you will see it again

At university you will meet at least three more proofs of $\zeta(2) = \pi^2/6$: one from *Fourier series* (writing a function as a sum of sines and cosines), one from a double integral, and one from complex analysis. And the number turns up in probability: the chance that two randomly chosen whole numbers have no common factor is $6/\pi^2 \approx 0.61$. (A prime p divides both with probability $1/p^2$, so the chance that no prime divides both is $\prod_p (1 - 1/p^2)$, and Euler showed this product equals $1/\zeta(2)$.) A different appearance of π in a probability limit is the subject of Section 12.23.

The thinking skill

The squeeze, and symmetric functions of roots

To find a limit, trap the quantity between two simpler expressions with the same limit.

And when you need the sum (or product) of some awkward numbers, try to find a polynomial they are the roots of: the coefficients will give you symmetric

combinations of them for free.

Try this next

1 The 1992 version. (From STEP III, 1992.) Using the polynomial from part (i), show that

$$\sum_{k=1}^{n} \tan^2\left(\frac{k\pi}{2n+1}\right) = n(2n+1) \quad \text{and} \quad \sum_{k=1}^{n} \operatorname{cosec}^2\left(\frac{k\pi}{2n+1}\right) = \frac{2n(n+1)}{3}.$$

(*Hint for the first:* the numbers $\tan^2$ are the reciprocals of the roots. What polynomial has roots $1/r_i$?)

2 Odd squares. Deduce from $\sum 1/m^2 = \pi^2/6$ that $1 + \frac{1}{9} + \frac{1}{25} + \frac{1}{49} + \cdots = \frac{\pi^2}{8}$.

3 Fourth powers. HARDER Use $\sum r_i^2 = (\sum r_i)^2 - 2\sum_{i<j} r_i r_j$ and the coefficients of the polynomial to find $\sum_{m=1}^{n} \cot^4\left(\frac{m\pi}{2n+1}\right)$, and adapt the squeeze to prove $\sum 1/m^4 = \pi^4/90$.

4 How fast? Using the lower bound in part (iii), show that $\frac{\pi^2}{6} - \sum_{m=1}^{n} \frac{1}{m^2} < \frac{\pi^2}{4n}$. How large an n does this guarantee is enough for the partial sum to be within 0.001 of $\pi^2/6$? Compare with the true error, which is close to $1/n$.

Further reading

- **William Dunham, *Euler: The Master of Us All*** (Mathematical Association of America). Chapter 3 tells the story of the Basel problem with the mathematics in full.
- **Aigner & Ziegler, *Proofs from THE BOOK***, Chapter 9, "Three times $\pi^2/6$". Includes the proof in this section.
- **Paul Nahin, *Dr. Euler's Fabulous Formula*** (Princeton University Press). Complex numbers, Fourier series and sums like this one.

2.5 Viète's Product

The question

You may assume that all infinite sums and products in this question converge.

(i) Prove by induction that for all positive integers n,

$$\sinh x = 2^n \cosh\left(\frac{x}{2}\right)\cosh\left(\frac{x}{4}\right)\cdots\cosh\left(\frac{x}{2^n}\right)\sinh\left(\frac{x}{2^n}\right)$$

and deduce that, for $x \neq 0$,

$$\frac{\sinh x}{x}\cdot\frac{\frac{x}{2^n}}{\sinh\left(\frac{x}{2^n}\right)} = \cosh\left(\frac{x}{2}\right)\cosh\left(\frac{x}{4}\right)\cdots\cosh\left(\frac{x}{2^n}\right).$$

(ii) You are given that the Maclaurin series for $\sinh x$ is

$$\sinh x = \sum_{r=0}^{\infty}\frac{x^{2r+1}}{(2r+1)!}.$$

Use this result to show that, as y tends to 0, $\dfrac{y}{\sinh y}$ tends to 1.

Deduce that, for $x \neq 0$,

$$\frac{\sinh x}{x} = \cosh\left(\frac{x}{2}\right)\cosh\left(\frac{x}{4}\right)\cdots\cosh\left(\frac{x}{2^n}\right)\cdots.$$

(iii) Let $x = \ln 2$. Evaluate $\cosh\left(\frac{x}{2}\right)$ and show that

$$\cosh\left(\frac{x}{4}\right) = \frac{1+2^{\frac{1}{2}}}{2\times 2^{\frac{1}{4}}}.$$

Use part (ii) to show that

$$\frac{1}{\ln 2} = \frac{1+2^{\frac{1}{2}}}{2}\times\frac{1+2^{\frac{1}{4}}}{2}\times\frac{1+2^{\frac{1}{8}}}{2}\cdots.$$

(iv) Show that

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2}\times\frac{\sqrt{2+\sqrt{2}}}{2}\times\frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2}\cdots.$$

The hook

The last line of this question is one of the most beautiful formulas in mathematics, and one of the oldest of its kind. It was published in 1593 by the French lawyer and mathematician François Viète, and it was the first time anyone had written π *exactly* as the result of an infinite process. Before Viète, π could only be approximated; after him, it could be defined by a formula that, carried on for ever, gives every digit.

The formula looks mysterious: where could nested square roots of 2 come from? The question reveals that it comes from repeatedly using the double-angle formula, and it gets a second, less famous formula, for $\ln 2$, from the same machinery along the way.

Before you start

Ask yourself What is the double-angle formula for sinh? (If you are not sure, write $\sinh 2y$ and $\sinh y \cosh y$ in terms of exponentials and compare.)

$\sinh 2y = 2\sinh y \cosh y$, exactly like $\sin 2y = 2\sin y \cos y$. That one identity, used over and over, drives the whole question. The question works with hyperbolic functions until the last part, where we will switch to ordinary trigonometry.

Working it through

STAGE 1 Halving repeatedly

The case $n = 1$ is just the double-angle formula with $y = x/2$:

$$\sinh x = 2\cosh\left(\frac{x}{2}\right)\sinh\left(\frac{x}{2}\right).$$

Ask yourself Suppose the formula holds for n. The last factor is $\sinh(x/2^n)$. How can you rewrite it to get the formula for $n+1$?

Apply the double-angle formula to it: $\sinh(x/2^n) = 2\cosh(x/2^{n+1})\sinh(x/2^{n+1})$. Substituting,

$$\sinh x = 2^{n+1}\cosh\left(\frac{x}{2}\right)\cdots\cosh\left(\frac{x}{2^n}\right)\cosh\left(\frac{x}{2^{n+1}}\right)\sinh\left(\frac{x}{2^{n+1}}\right),$$

which is the formula for $n+1$. By induction it holds for all n. ■

For the deduction, divide both sides by x, and then multiply both sides by $\dfrac{x/2^n}{\sinh(x/2^n)}$. On the right, $2^n\sinh(x/2^n)\cdot\dfrac{x/2^n}{x\,\sinh(x/2^n)} = 1$, leaving just the product of cosh terms. ■

STAGE 2 The limit

From the series, for $y \neq 0$,

$$\frac{\sinh y}{y} = 1 + \frac{y^2}{3!} + \frac{y^4}{5!} + \cdots,$$

which tends to 1 as $y \to 0$, so its reciprocal $y/\sinh y$ also tends to 1. (If you want more care: every factorial in the tail $\frac{y^2}{3!} + \frac{y^4}{5!} + \cdots$ is at least 6, so for $0 < |y| < 1$ the tail is at most $\frac{1}{6}(y^2 + y^4 + \cdots) = \frac{y^2}{6(1-y^2)}$, which tends to 0.)

Now let $n \to \infty$ in part (i), with x fixed. Then $y = x/2^n \to 0$, so the left-hand side tends to $\dfrac{\sinh x}{x} \times 1$. The right-hand side is the partial product, which therefore tends to $\dfrac{\sinh x}{x}$. That is what the infinite product means. ■

STAGE 3 A formula for $\ln 2$

With $x = \ln 2$, $e^{x/2} = \sqrt{2}$, so

$$\cosh\left(\frac{x}{2}\right) = \frac{2^{1/2} + 2^{-1/2}}{2} = \frac{2+1}{2\sqrt{2}} = \frac{3}{2\sqrt{2}}.$$

Similarly $e^{x/4} = 2^{1/4}$, so $\cosh(x/4) = \dfrac{2^{1/4} + 2^{-1/4}}{2}$, and multiplying top and bottom by $2^{1/4}$ gives $\dfrac{2^{1/2}+1}{2 \times 2^{1/4}}$. ■

> **Ask yourself** What is the general pattern for $\cosh(x/2^k)$? What happens when you multiply all the denominators $2 \times 2^{1/2^k}$ together?

The same calculation gives, for every $k \geq 1$,

$$\cosh\left(\frac{x}{2^k}\right) = \frac{1 + 2^{1/2^{k-1}}}{2} \times 2^{-1/2^k}.$$

(For $k = 1$ this is $\frac{1+2}{2} \times 2^{-1/2} = \frac{3}{2\sqrt{2}}$, as before.) Multiplying over $k = 1, 2, 3, \ldots$, the powers of 2 combine to $2^{-(1/2+1/4+1/8+\cdots)} = 2^{-1}$, and the first fraction is $\frac{1+2}{2} = \frac{3}{2}$. So the infinite product is

$$\frac{3}{2} \times \frac{1}{2} \times \frac{1+2^{1/2}}{2} \times \frac{1+2^{1/4}}{2} \times \frac{1+2^{1/8}}{2} \cdots.$$

On the other side, $\sinh(\ln 2) = \frac{2 - \frac{1}{2}}{2} = \frac{3}{4}$, so $\dfrac{\sinh x}{x} = \dfrac{3}{4 \ln 2}$. The factor $\frac{3}{4}$ cancels from both sides, leaving

$$\frac{1}{\ln 2} = \frac{1+2^{1/2}}{2} \times \frac{1+2^{1/4}}{2} \times \frac{1+2^{1/8}}{2} \cdots. \quad ■$$

STAGE 4 Viète's formula

Ask yourself Part (iv) has π in it, so it must come from ordinary trigonometry, not hyperbolic functions. Does everything in parts (i) and (ii) still work with sin and cos in place of sinh and cosh?

It does. The induction used only the double-angle formula, and $\sin 2y = 2\sin y \cos y$. The limit used only $\frac{\sin y}{y} \to 1$ as $y \to 0$, which follows from $\sin y = y - \frac{y^3}{3!} + \cdots$ in the same way. So for $x \neq 0$ (and x not a multiple of π, so that no factor is zero along the way),

$$\frac{\sin x}{x} = \cos\left(\frac{x}{2}\right)\cos\left(\frac{x}{4}\right)\cos\left(\frac{x}{8}\right)\cdots.$$

Pause and try Choose x so that the left-hand side is $2/\pi$, and work out the first few factors. The half-angle formula $\cos\frac{\theta}{2} = \sqrt{\frac{1+\cos\theta}{2}}$ (for $0 \le \theta \le \pi$) will be useful.

Take $x = \frac{\pi}{2}$. Then $\frac{\sin x}{x} = \frac{1}{\pi/2} = \frac{2}{\pi}$, and the factors are $\cos\frac{\pi}{4}, \cos\frac{\pi}{8}, \cos\frac{\pi}{16}, \ldots$. The first is $\cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$. If a factor is $\cos\theta = \frac{c}{2}$, the half-angle formula makes the next one

$$\cos\frac{\theta}{2} = \sqrt{\frac{1+c/2}{2}} = \frac{\sqrt{2+c}}{2}.$$

So the numerators go $\sqrt{2}$, $\sqrt{2+\sqrt{2}}$, $\sqrt{2+\sqrt{2+\sqrt{2}}}$, …, each obtained by adding 2 to the previous one and taking the square root. That is Viète's formula. ■

Looking back

- **Telescoping a product.** The identity in part (i) is a product that collapses, just as some sums telescope.
- **Finite identity first, limit second.** As in Section 2.4, every step was exact for finite n, and the only limit needed was the very simple $\frac{\sin y}{y} \to 1$.
- **Hyperbolic and circular functions are twins.** The same proof worked for both, because both satisfy the same double-angle formula and the same small-angle limit. This is no coincidence: $\cosh(ix) = \cos x$, as you will see in Section 5.7.

What's really going on

Archimedes' polygons

Viète's formula has a geometric meaning that goes back to Archimedes. A regular polygon with 2^n sides inscribed in a circle of radius 1 has area $A_n = 2^{n-1}\sin\frac{2\pi}{2^n}$.

Doubling the number of sides multiplies the area by

$$\frac{A_{n+1}}{A_n} = \frac{2^n \sin(\pi/2^n)}{2^{n-1} \cdot 2 \sin(\pi/2^n)\cos(\pi/2^n)} = \frac{1}{\cos(\pi/2^n)}.$$

Starting from the square ($A_2 = 2$) and doubling for ever, the area tends to the area of the circle, π. So $\pi = 2 \div \left(\cos\frac{\pi}{4}\cos\frac{\pi}{8}\cos\frac{\pi}{16}\cdots\right)$, which is Viète's formula. Each factor in the product is the ratio of the areas of two successive polygons in Figure 2.5.1.

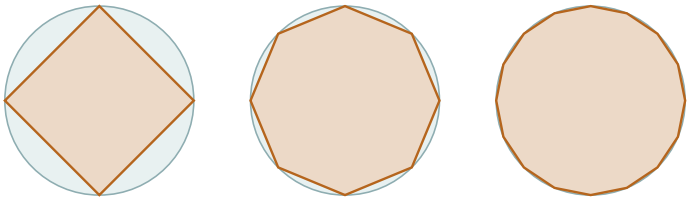


Figure 2.5.1 Regular polygons with 4, 8 and 16 sides inscribed in a circle of radius 1. Each doubling of the number of sides divides the area by one more factor of Viète's product, and the areas increase towards π.

Infinite products

Viète's product was the first infinite product in mathematics. Others followed: John Wallis found $\frac{\pi}{2} = \frac{2}{1}\cdot\frac{2}{3}\cdot\frac{4}{3}\cdot\frac{4}{5}\cdot\frac{6}{5}\cdot\frac{6}{7}\cdots$ in 1655, and Euler's product for $\frac{\sin x}{x}$ appeared in Section 2.4. At university you will learn when an infinite product converges. Roughly, $\prod(1+a_n)$ with small positive a_n behaves like $\sum a_n$, because $\ln(1+a_n) \approx a_n$. Here $1-\cos(\pi/2^n) \approx \frac{1}{2}(\pi/2^n)^2$, which shrinks by a factor of 4 each time, so the product converges quickly.

How quickly?

After n factors the partial product equals $\dfrac{1}{2^n \sin(\pi/2^{n+1})}$ exactly, and its error is about $\frac{\pi^2}{6\cdot 4^{n+1}}$ times $\frac{2}{\pi}$. Each new factor makes the answer about 4 times more accurate, gaining about 0.6 of a decimal digit. Ten factors give $2/\pi$ correct to six decimal places. That is good, but nothing like as fast as the methods of Section 2.6, which double the number of correct digits at each step.

The thinking skill

Iterating an identity

An identity that relates $f(x)$ to $f(x/2)$ can be applied again and again; the result, after a limit, is often an infinite sum or product.

Look for the "remainder" term after n steps (here $2^n \sinh(x/2^n)$) and find its limit separately. This pattern appears all over analysis, from infinite products to the construction of fractals.

Try this next

1 **The 1990 version.** (From STEP I, 1990.) Show directly, using only the double-angle formula for sin, that

$$\cos\left(\frac{\alpha}{2}\right)\cos\left(\frac{\alpha}{4}\right)\cdots\cos\left(\frac{\alpha}{2^n}\right) = \frac{\sin\alpha}{2^n \sin(\alpha/2^n)},$$

and deduce Viète's formula in the form $\dfrac{\pi}{2} = \dfrac{1}{\sqrt{\frac{1}{2}}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}\cdots}$.

2 **Another angle.** Use $x = \frac{\pi}{3}$ in the product for $\frac{\sin x}{x}$ to write $\frac{3\sqrt{3}}{2\pi}$ as an infinite product, and simplify the first two factors.

3 **How many factors?** Use a calculator to find how many factors of Viète's product are needed to give π correct to 4 decimal places. Explain why each extra factor gains about 0.6 of a digit.

4 **A formula for $\ln 3$.** Repeat part (iii) with $x = \ln 3$ to write $\frac{1}{\ln 3}$ as an infinite product. How many factors are needed to give $\frac{1}{\ln 3}$ to two decimal places? Why does this product converge more slowly than Viète's?

Further reading

- **Petr Beckmann, *A History of π*** (St Martin's Press). Archimedes, Viète, Wallis and the rest, told with gusto.
- **Jonathan Borwein & Peter Borwein, *Pi and the AGM*** (Wiley). The serious mathematics of computing π, including the method of Section 2.6. Chapter 1 is accessible with this chapter behind you.
- **Eli Maor, *Trigonometric Delights*** (Princeton University Press). Contains a chapter on Viète and his product.

2.6 The Arithmetic–Geometric Mean

The question

In this question, the following theorem may be used without proof.

Let $u_1, u_2, \ldots$ be a sequence of real numbers. If the sequence is

- *bounded above, so $u_n \leq b$ for all n, where b is some fixed number*
- *and increasing, so $u_n \leq u_{n+1}$ for all n*

then there is a number $L \leq b$ such that $u_n \to L$ as $n \to \infty$.

For positive real numbers x and y, define $\mathrm{a}(x,y) = \frac{1}{2}(x+y)$ and $\mathrm{g}(x,y) = \sqrt{xy}$. Let x_0 and y_0 be two positive real numbers with $y_0 < x_0$ and define, for $n \geq 0$,

$$x_{n+1} = \mathrm{a}(x_n, y_n), \qquad y_{n+1} = \mathrm{g}(x_n, y_n).$$

(i) By considering $\left(\sqrt{x_n} - \sqrt{y_n}\right)^2$, show that $y_{n+1} < x_{n+1}$, for $n \geq 0$. Show further that, for $n \geq 0$,

- $x_{n+1} < x_n$
- $y_n < y_{n+1}$.

Deduce that there is a value M such that $y_n \to M$ as $n \to \infty$.

Show that $0 < x_{n+1} - y_{n+1} < \frac{1}{2}(x_n - y_n)$ and hence that $x_n - y_n \to 0$ as $n \to \infty$.

Explain why x_n also tends to M as $n \to \infty$.

(ii) Let

$$\mathrm{I}(p,q) = \int_0^\infty \frac{1}{\sqrt{(p^2+x^2)(q^2+x^2)}}\,\mathrm{d}x,$$

where p and q are positive real numbers with $q < p$.

Show, using the substitution $t = \frac{1}{2}\left(x - \frac{pq}{x}\right)$ in the integral

$$\int_{-\infty}^\infty \frac{1}{\sqrt{\left(\frac{1}{4}(p+q)^2 + t^2\right)\left(pq + t^2\right)}}\,\mathrm{d}t,$$

that

$$\mathrm{I}(p,q) = \mathrm{I}\big(\mathrm{a}(p,q), \mathrm{g}(p,q)\big).$$

Hence evaluate $\mathrm{I}(x_0, y_0)$ in terms of M.

The hook

Take two positive numbers, say $\sqrt{2}$ and 1. Replace them by their arithmetic mean and their geometric mean. Do it again, and again. Here is what happens:

n	x_n (arithmetic)	y_n (geometric)
0	1.4142135623730950488	1
1	1.2071067811865475244	1.1892071150027210667
2	1.1981569480946342956	1.1981235214931201226
3	1.1981402347938772091	1.1981402346773072058
4	1.1981402347355922074	1.1981402347355922074

After four steps the two numbers agree to twenty decimal places. The common limit is the *arithmetic–geometric mean* $M(\sqrt{2}, 1) = 1.19814\ldots$

On 30 May 1799 the 21-year-old Carl Friedrich Gauss wrote in his diary that he had noticed $1/M(\sqrt{2}, 1)$ agreed, to eleven decimal places, with a constant that had appeared in a completely different problem: the arc length of a curve called the lemniscate. "This will surely open up an entirely new field of analysis," he wrote. It did. This question proves the key fact behind Gauss's discovery: the AGM evaluates an integral that cannot be done by any of the usual methods.

Before you start

Ask yourself Why is the arithmetic mean of two different positive numbers always bigger than their geometric mean? Can you prove it in one line?

That is the AM–GM inequality, and the hint in part (i) gives the one-line proof: $\left(\sqrt{x} - \sqrt{y}\right)^2 > 0$ when $x \neq y$, which expands to $x + y > 2\sqrt{xy}$. Everything in part (i) follows from this and from the fact that each mean lies between the two numbers it comes from.

Working it through

STAGE 1 The two sequences squeeze together

If $y_n < x_n$ then $\sqrt{x_n} \neq \sqrt{y_n}$, so $x_n - 2\sqrt{x_n y_n} + y_n > 0$, which says $x_{n+1} > y_{n+1}$. Since $y_0 < x_0$, induction gives $y_n < x_n$ for all n.

Ask yourself Why is the arithmetic mean of x_n and y_n less than x_n, and the geometric mean bigger than y_n?

Because $y_n < x_n$: $x_{n+1} = \frac{x_n + y_n}{2} < \frac{x_n + x_n}{2} = x_n$, and $y_{n+1} = \sqrt{x_n y_n} > \sqrt{y_n y_n} = y_n$. So

$$y_0 < y_1 < y_2 < \cdots < x_2 < x_1 < x_0.$$

The y_n are increasing and bounded above by x_0, so by the theorem they tend to a limit M.

For the gap,

$$x_{n+1} - y_{n+1} < x_{n+1} - y_n = \frac{x_n + y_n}{2} - y_n = \frac{1}{2}(x_n - y_n),$$

and it is positive by the first part. So the gap at least halves each time: $0 < x_n - y_n < (x_0 - y_0)/2^n$, which tends to 0.

Finally $x_n = y_n + (x_n - y_n)$ is the sum of a sequence tending to M and a sequence tending to 0, so $x_n \to M$ too. ■

STAGE 2 The substitution

The substitution looks unmotivated, but it is designed so that both brackets under the square root become perfect squares (or nearly).

Ask yourself With $t = \frac{1}{2}\left(x - \frac{pq}{x}\right)$, compute $t^2 + pq$. Then compute $t^2 + \frac{1}{4}(p+q)^2$. Do either of them factorise nicely?

First,

$$t^2 + pq = \frac{1}{4}\left(x^2 - 2pq + \frac{p^2q^2}{x^2}\right) + pq = \frac{1}{4}\left(x + \frac{pq}{x}\right)^2.$$

Second,

$$\begin{aligned} t^2 + \tfrac{1}{4}(p+q)^2 &= \frac{1}{4}\left(x^2 - 2pq + \frac{p^2q^2}{x^2} + p^2 + 2pq + q^2\right) \\ &= \frac{x^4 + (p^2+q^2)x^2 + p^2q^2}{4x^2} = \frac{(x^2+p^2)(x^2+q^2)}{4x^2}. \end{aligned}$$

And $\dfrac{\mathrm{d}t}{\mathrm{d}x} = \dfrac{1}{2}\left(1 + \dfrac{pq}{x^2}\right) = \dfrac{x^2+pq}{2x^2}$.

Pause and try Put these together. As x runs from 0 to ∞, what does t do? What does the t-integral become?

As x increases from 0 to ∞, t increases (its derivative is positive) from $-\infty$ to ∞. For $x > 0$,

$$\sqrt{\left(t^2 + \tfrac{1}{4}(p+q)^2\right)\left(t^2 + pq\right)} = \frac{\sqrt{(x^2+p^2)(x^2+q^2)}}{2x} \cdot \frac{x^2+pq}{2x},$$

so

$$\frac{\mathrm{d}t}{\sqrt{\cdots}} = \frac{x^2+pq}{2x^2} \cdot \frac{4x^2}{(x^2+pq)\sqrt{(x^2+p^2)(x^2+q^2)}}\,\mathrm{d}x = \frac{2\,\mathrm{d}x}{\sqrt{(x^2+p^2)(x^2+q^2)}}.$$

So the t-integral equals $2\,\mathrm{I}(p,q)$.

On the other hand, the t-integral has exactly the form of I with p and q replaced by $\frac{1}{2}(p+q) = \mathrm{a}(p,q)$ and $\sqrt{pq} = \mathrm{g}(p,q)$, except that it runs over the whole real line. The integrand is even in t, so the t-integral is $2\,\mathrm{I}\big(\mathrm{a}(p,q), \mathrm{g}(p,q)\big)$. Comparing,

$$\mathrm{I}(p,q) = \mathrm{I}\big(\mathrm{a}(p,q), \mathrm{g}(p,q)\big). \qquad \blacksquare$$

STAGE 3 Evaluating the integral

Ask yourself Apply this identity repeatedly, starting from (x_0, y_0). What is $\mathrm{I}(x_0, y_0)$ equal to after n steps? What happens as $n \to \infty$?

$\mathrm{I}(x_0,y_0) = \mathrm{I}(x_1,y_1) = \mathrm{I}(x_2,y_2) = \cdots = \mathrm{I}(x_n,y_n)$ for every n. As $n \to \infty$, both arguments tend to M, and

$$\mathrm{I}(M,M) = \int_0^\infty \frac{\mathrm{d}x}{M^2+x^2} = \left[\frac{1}{M}\arctan\frac{x}{M}\right]_0^\infty = \frac{\pi}{2M}.$$

So

$$\mathrm{I}(x_0,y_0) = \frac{\pi}{2M}. \qquad \blacksquare$$

The question lets us take the step "$\mathrm{I}(x_n,y_n) \to \mathrm{I}(M,M)$" on trust, but it can be justified neatly. Making p or q larger makes the integrand smaller, and $y_n < x_n$, so

$$\frac{\pi}{2x_n} = \mathrm{I}(x_n,x_n) < \mathrm{I}(x_n,y_n) < \mathrm{I}(y_n,y_n) = \frac{\pi}{2y_n}.$$

Both bounds tend to $\frac{\pi}{2M}$, and $\mathrm{I}(x_n,y_n)$ is the same number $\mathrm{I}(x_0,y_0)$ for every n, so it must equal $\frac{\pi}{2M}$. A squeeze again, as in Section 2.4.

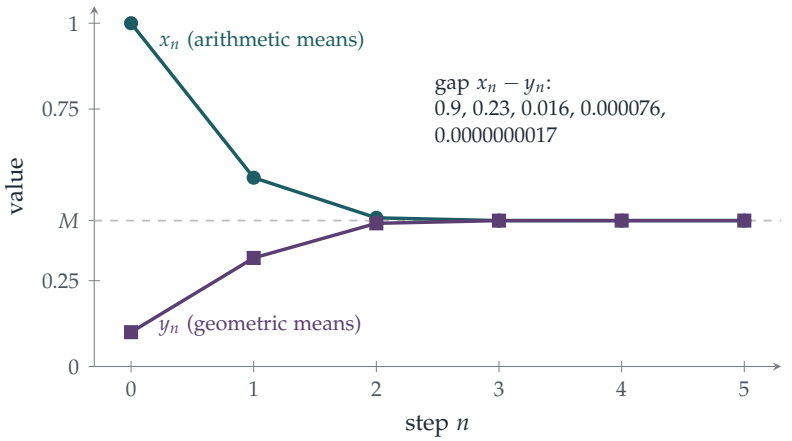


Figure 2.6.1 The arithmetic–geometric mean of $x_0 = 1$ and $y_0 = 0.1$. The arithmetic means x_n (teal) decrease, the geometric means y_n (plum) increase, and they close in on the common limit $M = 0.425040709\ldots$ astonishingly fast: once they are close, the number of correct digits roughly doubles at every step.

Looking back

- **Two monotone sequences squeezing together.** One goes up, one goes down, and the gap shrinks to zero. This is the cleanest way to prove two sequences have a common limit.
- **An invariant.** The quantity $\mathrm{I}(x_n, y_n)$ never changes from step to step. Finding something that stays fixed while everything else moves is one of the most powerful ideas in mathematics.
- **Clever substitutions are designed, not guessed.** The substitution was chosen so that the expressions under the square root factorised.

What's really going on

Faster than the question admits

Part (i) shows the gap at least halves at each step, which would give about one extra binary digit per step. The table in the hook shows something far more dramatic: the number of correct digits roughly *doubles* at each step $(2, 4, 10, 21, 43, \ldots)$. In fact

$$x_{n+1} - y_{n+1} = \frac{1}{2}\bigl(\sqrt{x_n} - \sqrt{y_n}\bigr)^2 = \frac{(x_n - y_n)^2}{2\bigl(\sqrt{x_n} + \sqrt{y_n}\bigr)^2},$$

so each gap is roughly a constant times the *square* of the previous one. This is called *quadratic convergence*, and we will study it properly in Section 2.7.

Elliptic integrals

Our integral $\mathrm{I}(p, q)$ is a disguised *complete elliptic integral*, a type of integral that arises in computing the arc length of an ellipse and that cannot be expressed using the functions you know. Its most famous application is the pendulum. The A-level formula $T = 2\pi\sqrt{L/g}$ for the period is only an approximation for small swings. The exact period of a pendulum released from angle θ_0 is

$$T = \frac{2\pi\sqrt{L/g}}{M\bigl(1, \cos\frac{\theta_0}{2}\bigr)},$$

an elliptic integral evaluated by the AGM. For $\theta_0 = 90^\circ$ the true period is about 18% longer than the simple formula predicts. Gauss's diary entry concerned the lemniscate, a figure-of-eight curve, whose total arc length involves $\int_0^1 \frac{\mathrm{d}t}{\sqrt{1-t^4}}$; that integral equals $\mathrm{I}(\sqrt{2}, 1)$ in disguise, so the AGM evaluates it.

Computing π

In 1976 Richard Brent and Eugene Salamin independently turned Gauss's ideas into an algorithm for computing π that doubles the number of correct digits at every step. Twenty-five steps give tens of millions of digits. Algorithms of this kind were used in record-breaking computations of π in the 1980s and 1990s, and quadratically convergent algorithms like it are now standard wherever high precision is needed. Compare Viète's product in Section 2.5, which gains about 0.6 of a digit per step.

The thinking skill

Look for an invariant

When a process changes many things at once, look for a quantity that does not change. Evaluate it at the end of the process, where everything is simple.

Here the process was the AGM step and the invariant was the integral I. At the end of the process the two arguments were equal and the integral was elementary. Invariants appear throughout mathematics, from conservation of energy in mechanics (Chapter 10) to the proof that some puzzles are impossible.

Try this next

1. **Quadratic convergence.** Prove the identity $x_{n+1} - y_{n+1} = \dfrac{(x_n - y_n)^2}{2(\sqrt{x_n} + \sqrt{y_n})^2}$, and deduce that $x_{n+1} - y_{n+1} \le \dfrac{(x_n - y_n)^2}{8y_0}$.

2. **By hand.** Compute $M(2,1)$ to six decimal places using a calculator. How many steps did you need?

3. **Scaling.** Show that $M(\lambda x, \lambda y) = \lambda M(x,y)$ for $\lambda > 0$, and that $\mathrm{I}(\lambda p, \lambda q) = \lambda^{-1}\mathrm{I}(p,q)$. Check that these two facts are consistent with $\mathrm{I}(x_0, y_0) = \pi/(2M)$.

4. **The pendulum.** Use the formula above to find, to three significant figures, how much longer the period of a pendulum released from 60° is than the small-angle prediction.

Further reading

- **David Cox, "The arithmetic–geometric mean of Gauss"**, *L'Enseignement Mathématique* 30 (1984). The classic account of the history and mathematics; the first sections need only calculus.
- **Jonathan Borwein & Peter Borwein, *Pi and the AGM*** (Wiley). The definitive book on the

AGM and fast computation of π.

- **Jonathan Borwein & Keith Devlin, *The Computer as Crucible*** (A. K. Peters). Experimental mathematics, including discoveries like Gauss's, made by spotting numerical coincidences.

2.7 Six Digits or Six Hundred

The question

(i) The sequence x_n for $n = 0, 1, 2, \ldots$ is defined by $x_0 = 1$ and by

$$x_{n+1} = \frac{x_n + 2}{x_n + 1}$$

for $n \geq 0$.

(a) Explain briefly why $x_n \geq 1$ for all n.

(b) Show that $x_{n+1}^2 - 2$ and $x_n^2 - 2$ have opposite sign, and that

$$|x_{n+1}^2 - 2| \leq \tfrac{1}{4}|x_n^2 - 2|.$$

(c) Show that

$$2 - 10^{-6} \leq x_{10}^2 \leq 2.$$

(ii) The sequence y_n for $n = 0, 1, 2, \ldots$ is defined by $y_0 = 1$ and by

$$y_{n+1} = \frac{y_n^2 + 2}{2y_n}$$

for $n \geq 0$.

(a) Show that, for $n \geq 0$,

$$y_{n+1} - \sqrt{2} = \frac{(y_n - \sqrt{2})^2}{2y_n}$$

and deduce that $y_n \geq 1$ for $n \geq 0$.

(b) Show that

$$y_n - \sqrt{2} \leq 2\left(\frac{\sqrt{2} - 1}{2}\right)^{2^n}$$

for $n \geq 1$.

(c) Using the fact that $\sqrt{2} - 1 < \frac{1}{2}$, or otherwise, show that

$$\sqrt{2} \leq y_{10} \leq \sqrt{2} + 10^{-600}.$$

The hook

Two simple recipes for approximating $\sqrt{2}$, both starting from 1, both using nothing but addition, multiplication and division. After ten steps, the first is correct to about 8 decimal places. The second is correct to more than 700.

The difference is not that the second recipe is cleverer about $\sqrt{2}$. It is a difference in the *shape* of the error: in the first, each step multiplies the error by a fixed fraction; in the second, each step *squares* it. The question makes this precise, and in doing so introduces one of the central ideas of numerical analysis, the subject that underlies every calculator, every weather forecast and every engineering simulation.

Before you start

Compute a few terms of each sequence as fractions.

$$x_n: \quad 1, \frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \frac{41}{29}, \frac{99}{70}, \dots \qquad y_n: \quad 1, \frac{3}{2}, \frac{17}{12}, \frac{577}{408}, \frac{665857}{470832}, \dots$$

Ask yourself Do you notice anything about the two lists? (Look at which fractions appear in both.)

Every y_n appears in the x list, but further and further along: $y_2 = x_3$, $y_3 = x_7$, and in general $y_n = x_{2^n-1}$. The second recipe is skipping ahead exponentially fast through the same sequence of approximations. Keep this in mind; we will explain it at the end.

Working it through

STAGE 1 The first recipe stays at least 1

We can write $x_{n+1} = 1 + \dfrac{1}{x_n + 1}$. If $x_n \geq 1$ then $x_n + 1 > 0$, so $x_{n+1} > 1$. Since $x_0 = 1$, induction gives $x_n \geq 1$ for all n. ■

STAGE 2 The error in the first recipe

Ask yourself The question measures the error by $x_n^2 - 2$ rather than $x_n - \sqrt{2}$. Write $x_{n+1}^2 - 2$ as a single fraction in terms of x_n. What do you notice about the numerator?

$$x_{n+1}^2 - 2 = \frac{(x_n+2)^2 - 2(x_n+1)^2}{(x_n+1)^2} = \frac{x_n^2 + 4x_n + 4 - 2x_n^2 - 4x_n - 2}{(x_n+1)^2} = \frac{2 - x_n^2}{(x_n+1)^2}.$$

The numerator is exactly $-(x_n^2 - 2)$. So $x_{n+1}^2 - 2$ and $x_n^2 - 2$ have opposite signs (the

approximations alternate either side of $\sqrt{2}$), and since $x_n \geq 1$ makes $(x_n+1)^2 \geq 4$,

$$|x_{n+1}^2 - 2| = \frac{|x_n^2 - 2|}{(x_n+1)^2} \leq \frac{1}{4}|x_n^2 - 2|. \qquad \blacksquare$$

STAGE 3 Ten steps

Applying this ten times, $|x_{10}^2 - 2| \leq 4^{-10}|x_0^2 - 2| = 4^{-10} = \frac{1}{1048576} < 10^{-6}$. For the sign: $x_0^2 - 2 = -1 < 0$, and the sign alternates, so $x_n^2 - 2 < 0$ for every even n. So $x_{10}^2 < 2$, and together these give $2 - 10^{-6} \leq x_{10}^2 \leq 2$. $\blacksquare$

The error in x_n^2 is multiplied by at most $\frac{1}{4}$ at each step. That is called *linear convergence*: a fixed number of extra correct digits per step (here at least $\log_{10} 4 \approx 0.6$; in fact closer to 0.77).

STAGE 4 The error in the second recipe

> **Ask yourself** Write $y_{n+1} - \sqrt{2}$ as a single fraction. The numerator should remind you of a perfect square.

$$y_{n+1} - \sqrt{2} = \frac{y_n^2 + 2 - 2\sqrt{2}\,y_n}{2y_n} = \frac{(y_n - \sqrt{2})^2}{2y_n}. \qquad \blacksquare$$

If $y_n > 0$, the right-hand side is non-negative, so $y_{n+1} \geq \sqrt{2} > 1$. Since $y_0 = 1$, induction gives $y_n \geq 1$ for all n. $\blacksquare$(In fact $y_n \geq \sqrt{2}$ for all $n \geq 1$: after the first step, the second recipe always approaches $\sqrt{2}$ from above.)

STAGE 5 Squaring the error

Write $e_n = y_n - \sqrt{2}$, so $e_n \geq 0$ for $n \geq 1$, and $e_{n+1} = \dfrac{e_n^2}{2y_n} \leq \dfrac{e_n^2}{2}$ because $y_n \geq 1$.

> **Pause and try** Prove by induction that $e_n \leq 2r^{2^n}$ for $n \geq 1$, where $r = \frac{\sqrt{2}-1}{2}$. Check the base case carefully.

Base case: $e_1 = \dfrac{(1-\sqrt{2})^2}{2 \cdot 1} = \dfrac{(\sqrt{2}-1)^2}{2}$, while $2r^2 = 2 \cdot \dfrac{(\sqrt{2}-1)^2}{4} = \dfrac{(\sqrt{2}-1)^2}{2}$. They are equal, so the base case holds (exactly).

Inductive step: if $e_n \leq 2r^{2^n}$ then

$$e_{n+1} \leq \frac{e_n^2}{2} \leq \frac{4r^{2^{n+1}}}{2} = 2r^{2^{n+1}}. \qquad \blacksquare$$

STAGE 6 Six hundred digits

Since $\sqrt{2}-1<\frac{1}{2}$, we have $r<\frac{1}{4}$, so

$$0 \le y_{10}-\sqrt{2} \le 2\left(\frac{1}{4}\right)^{2^{10}} = 2\cdot 2^{-2048} = 2^{-2047}.$$

Now $2^{10}=1024>10^3$, so $2^{-2047}<2^{-2040}=(2^{10})^{-204}<10^{-612}$, which is certainly less than 10^{-600}. ■

Looking back

- **Find the error recurrence.** Both parts worked by writing the new error as an exact function of the old error. Everything else followed.
- **Linear versus quadratic.** "New error $\le$ constant $\times$ old error" gives a fixed number of extra digits per step. "New error $\le$ constant $\times$ (old error)^2" doubles the number of correct digits per step (Figure 2.7.1).
- **Squared quantities control signs.** In part (ii) the error after one step is a square divided by something positive, so it can never be negative: the iteration approaches from above.

What's really going on

Newton's method

The second recipe is *Newton's method* applied to $f(y)=y^2-2$. Newton's method finds a root of f by repeatedly replacing a guess y by the point where the tangent at y crosses the axis:

$$y_{\text{new}} = y - \frac{f(y)}{f'(y)} = y - \frac{y^2-2}{2y} = \frac{y^2+2}{2y}.$$

Near a simple root α, a Taylor expansion shows that the new error is approximately $\frac{f''(\alpha)}{2f'(\alpha)}\times(\text{old error})^2$, which is why Newton's method converges quadratically. This particular case, "average y and $2/y$", was known to the Babylonians; a clay tablet from around 1700 BC, now called YBC 7289, gives $\sqrt{2}$ in base 60 with an error of less than one part in a million.

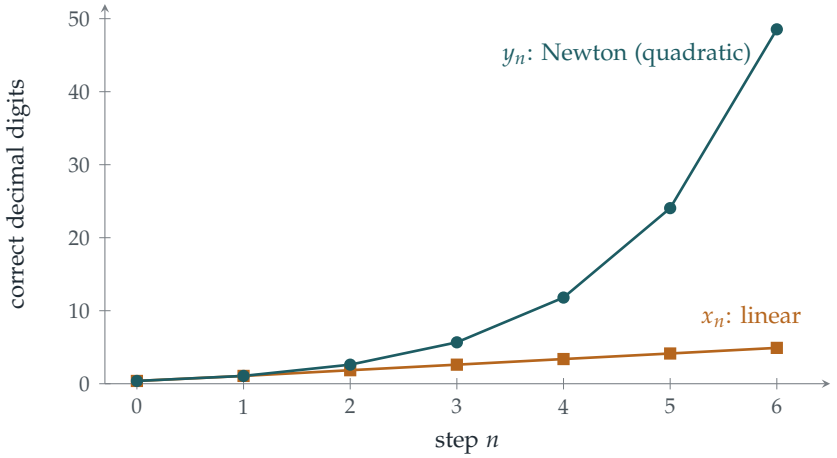


Figure 2.7.1 The number of correct decimal digits of $\sqrt{2}$ after n steps of each recipe. The linear recipe gains about 0.77 digits per step; Newton's method roughly doubles its number of correct digits at every step, reaching about 780 digits by $n = 10$.

Fixed points and their speed

Both recipes have the form $u_{n+1} = g(u_n)$ with $g(\sqrt{2}) = \sqrt{2}$, so $\sqrt{2}$ is a *fixed point*, as in the cobweb diagram of Section 2.2. Near a fixed point α, the error is multiplied by about $g'(\alpha)$ at each step. For the first recipe, $g(x) = \frac{x+2}{x+1}$ and $g'(\sqrt{2}) = -\frac{1}{(\sqrt{2}+1)^2} = -(3-2\sqrt{2}) \approx -0.17$: the minus sign explains the alternating signs and 0.17 explains the 0.77 digits per step ($\log_{10}(1/0.17) \approx 0.77$). For Newton's method $g'(\alpha) = 0$, so the linear term vanishes and the error is controlled by the next term, the square. Designing iterations with $g'(\alpha) = 0$ is the art of fast numerical methods.

Why the same fractions appear

The fractions $x_n = p_n/q_n$ satisfy $p_{n+1} = p_n + 2q_n$ and $q_{n+1} = p_n + q_n$, and $p_n^2 - 2q_n^2 = \pm 1$, alternating. These are the solutions of *Pell's equation* $p^2 - 2q^2 = \pm 1$, and they are the convergents of the continued fraction $\sqrt{2} = [1; 2, 2, 2, \ldots]$. They come from powers of $1 + \sqrt{2}$: $p_n + q_n\sqrt{2} = (1+\sqrt{2})^{n+1}$. Newton's step *squares* $p + q\sqrt{2}$ (try it: $(p+q\sqrt{2})^2 = p^2 + 2q^2 + 2pq\sqrt{2}$, and $\frac{p^2+2q^2}{2pq}$ is exactly Newton's formula applied to p/q). Squaring doubles the exponent, which is why $y_n = x_{2^n-1}$. The same pattern of "squaring" drives the iteration in Section 2.8.

The thinking skill

Analyse the error, not the answer

To understand how an approximation behaves, write down how the error at one step depends on the error at the step before.

This is the starting point of numerical analysis. The answer itself ($\sqrt{2}$) is the same for both recipes; what differs, and what matters in practice, is how the error evolves.

Try this next

1 **Pell's equation.** Show that if $y_n = p/q$ in lowest terms with $p^2 - 2q^2 = 1$, then $y_{n+1} = \dfrac{p^2 + 2q^2}{2pq}$ also satisfies this equation. Check that $577^2 - 2 \times 408^2 = 1$.

2 **Cube roots.** Write down Newton's method for $f(y) = y^3 - 2$, and show that the new error is $\dfrac{(y_n - \alpha)^2(2y_n + \alpha)}{3y_n^2}$, where $\alpha = \sqrt[3]{2}$. Deduce that if $y_n > \alpha$ then the new error is at most $\frac{1}{\alpha}$ times the square of the old error. (*Hint:* use $\alpha^3 = 2$.)

3 **Division without dividing.** Show that Newton's method for $f(y) = \frac{1}{y} - a$ gives $y_{n+1} = y_n(2 - ay_n)$, which needs no division. If $e_n = 1 - ay_n$, show that $e_{n+1} = e_n^2$. (Computer processors use iterations like this to divide.)

4 **When Newton fails.** Apply Newton's method to $f(y) = y^{1/3}$, whose only root is 0. Show that each step doubles the distance from the root, whatever you start with. Which hypothesis of the error formula fails?

Further reading

- **Endre Süli & David Mayers, *An Introduction to Numerical Analysis*** (Cambridge University Press). The standard undergraduate text; Chapter 1 covers fixed-point iteration and Newton's method.
- **Nick Higham (ed.), *The Princeton Companion to Applied Mathematics*** (Princeton University Press). Short articles on Newton's method and numerical analysis, written by experts for a wide audience.
- **David Fowler & Eleanor Robson, "Square root approximations in Old Babylonian mathematics: YBC 7289 in context"**, *Historia Mathematica* 25 (1998).

2.8 Squaring and Subtracting Two

The question

A sequence $x_1, x_2, \ldots$ of real numbers is defined by $x_{n+1} = x_n^2 - 2$ for $n \geq 1$ and $x_1 = a$.

(i) Show that if $a > 2$ then $x_n \geq 2 + 4^{n-1}(a-2)$.

(ii) Show also that $x_n \to \infty$ as $n \to \infty$ if and only if $|a| > 2$.

(iii) When $a > 2$, a second sequence $y_1, y_2, \ldots$ is defined by

$$y_n = \frac{A x_1 x_2 \cdots x_n}{x_{n+1}},$$

where A is a positive constant and $n \geq 1$.

Prove that, for a certain value of a, with $a > 2$, which you should find in terms of A,

$$y_n = \frac{\sqrt{x_{n+1}^2 - 4}}{x_{n+1}}$$

for all $n \geq 1$.

Determine whether, for this value of a, the second sequence converges.

The hook

"Square it and subtract two" is about the simplest rule you could use to generate a sequence. Yet it divides the real line into two worlds. Start outside $[-2, 2]$ and the numbers explode to infinity, faster than exponentially. Start inside and they stay trapped for ever, wandering in a way that is, in a precise sense, completely unpredictable: one of the simplest examples of *chaos*.

The same rule, carried out with very large whole numbers, is the engine of the Lucas–Lehmer test, an efficient test for whether a Mersenne number $2^p - 1$ is prime, which has found the current record prime and most of the other very largest known primes. This question explores the escaping side of the story and finds a hidden formula that makes the sequence completely solvable.

Before you start

Try some starting values. With $a = 3$: $3, 7, 47, 2207, \ldots$ With $a = 1$: $1, -1, -1, -1, \ldots$ With $a = 0$: $0, -2, 2, 2, 2, \ldots$ With $a = 0.3$: $0.3, -1.91, 1.648, 0.716, -1.487, 0.211, \ldots$

Ask yourself Why can a number in $[-2, 2]$ never escape? What is the largest and smallest that $x^2 - 2$ can be when $-2 \leq x \leq 2$?

If $-2 \leq x \leq 2$ then $0 \leq x^2 \leq 4$, so $-2 \leq x^2 - 2 \leq 2$. The interval $[-2, 2]$ is mapped into itself. That will be half of part (ii).

Working it through

STAGE 1 Escaping

Ask yourself Part (i) has a 4^{n-1} in it, suggesting that the distance above 2 at least quadruples at each step. Write $x_n = 2 + d$. What is x_{n+1} in terms of d?

$x_{n+1} = (2+d)^2 - 2 = 2 + 4d + d^2$. So if $d \geq 0$, the new distance above 2 is $4d + d^2 \geq 4d$. Now induct. When $n = 1$, $x_1 = a = 2 + 4^0(a-2)$. If $x_n \geq 2 + 4^{n-1}(a-2)$, then $d = x_n - 2 \geq 4^{n-1}(a-2) \geq 0$, so

$$x_{n+1} - 2 \geq 4d \geq 4^n(a-2). \qquad \blacksquare$$

STAGE 2 Escape if and only if $|a| > 2$

Ask yourself "If and only if" means two separate proofs. Which direction does part (i) almost give you, and what is missing?

If $|a| > 2$: when $a > 2$, part (i) gives $x_n \geq 2 + 4^{n-1}(a-2) \to \infty$. When $a < -2$, the first step fixes the sign: $x_2 = a^2 - 2 > 4 - 2 = 2$, and the sequence from x_2 onwards is the same sequence started at $x_2 > 2$, so it tends to infinity.

Only if: if $|a| \leq 2$, then $a \in [-2, 2]$, and by the observation above every term stays in $[-2, 2]$. A bounded sequence cannot tend to infinity. $\blacksquare$

STAGE 3 A hidden formula

Part (iii) looks forbidding. Before attempting it, it is worth finding a formula for x_n.

Ask yourself If $a > 2$, there is a number $t > 1$ with $a = t + \dfrac{1}{t}$. (Why?) What are x_2 and x_3 in terms of t?

The function $t + 1/t$ increases from 2 to ∞ as t goes from 1 to ∞, so every $a > 2$ can be

written this way. Then

$$x_2 = \left(t + \frac{1}{t}\right)^2 - 2 = t^2 + \frac{1}{t^2}, \qquad x_3 = \left(t^2 + \frac{1}{t^2}\right)^2 - 2 = t^4 + \frac{1}{t^4},$$

and by induction

$$x_n = t^{2^{n-1}} + t^{-2^{n-1}}.$$

Squaring and subtracting 2 simply doubles the exponent. Now the question becomes manageable.

STAGE 4 The product telescopes

Ask yourself What is $\left(t - \frac{1}{t}\right)\left(t + \frac{1}{t}\right)$? What happens if you multiply the result by $t^2 + \frac{1}{t^2}$?

$\left(t - \frac{1}{t}\right)\left(t + \frac{1}{t}\right) = t^2 - \frac{1}{t^2}$, and then $\left(t^2 - \frac{1}{t^2}\right)\left(t^2 + \frac{1}{t^2}\right) = t^4 - \frac{1}{t^4}$, and so on: each factor x_k doubles the exponent in a difference of two powers. So

$$\left(t - \frac{1}{t}\right) x_1 x_2 \cdots x_n = t^{2^n} - t^{-2^n}.$$

Pause and try Compute $x_{n+1}^2 - 4$ in terms of t, and compare with the product.

$$x_{n+1}^2 - 4 = \left(t^{2^n} + t^{-2^n}\right)^2 - 4 = \left(t^{2^n} - t^{-2^n}\right)^2.$$

Since $t > 1$, $t^{2^n} - t^{-2^n} > 0$, so $\sqrt{x_{n+1}^2 - 4} = t^{2^n} - t^{-2^n} = \left(t - \frac{1}{t}\right) x_1 x_2 \cdots x_n$. Therefore

$$\frac{\sqrt{x_{n+1}^2 - 4}}{x_{n+1}} = \frac{\left(t - \frac{1}{t}\right) x_1 \cdots x_n}{x_{n+1}},$$

which is y_n for every n exactly when $A = t - \frac{1}{t}$. Finally, $\left(t - \frac{1}{t}\right)^2 = \left(t + \frac{1}{t}\right)^2 - 4 = a^2 - 4$, so the condition is $A = \sqrt{a^2 - 4}$, that is,

$$a = \sqrt{A^2 + 4}.$$

(This is greater than 2, as required, because $A > 0$.) ■

STAGE 5 Does y_n converge?

With this value of a,

$$y_n = \frac{\sqrt{x_{n+1}^2 - 4}}{x_{n+1}} = \sqrt{1 - \frac{4}{x_{n+1}^2}}.$$

Since $x_{n+1} \to \infty$ by part (ii), $\frac{4}{x_{n+1}^2} \to 0$ and $y_n \to 1$. The sequence converges, to 1. ■

So the infinite product $x_1 x_2 x_3 \cdots$ grows at exactly the same rate as x_{n+1}/A: we have found the precise growth rate of a product of rapidly growing terms.

Looking back

- **An invariant region.** $[-2, 2]$ is mapped into itself, and that one fact settles half of part (ii).
- **A substitution that linearises.** Writing $a = t + 1/t$ turned "square and subtract 2" into "double the exponent". Messy nonlinear behaviour became simple.
- **Telescoping again.** As with the products in Section 2.5, a product of many factors collapsed because each factor fitted with the next.

What's really going on

Order and chaos

What happens inside $[-2, 2]$? There, $a = t + 1/t$ fails (it needs $|a| \geq 2$), but the trigonometric version works: write $a = 2\cos\theta$. Then $x_2 = 4\cos^2\theta - 2 = 2\cos 2\theta$, and in general

$$x_n = 2\cos\left(2^{n-1}\theta\right).$$

So inside the interval the map simply *doubles an angle*. That sounds harmless, but doubling an angle is chaotic. Write θ/π in binary: doubling shifts the binary point one place to the right, throwing away the leading digit. After 50 steps, x_n depends on the 51st binary digit of θ/π, so an error in the starting value of 10^{-15} (the precision of a typical computer) makes the answer after 50 steps completely unreliable. This sensitive dependence on initial conditions is the hallmark of chaos, illustrated in Figure 2.8.1. The map $x \mapsto x^2 - 2$ is equivalent, by a change of variable, to the famous logistic map $x \mapsto 4x(1-x)$, which we meet again in Section 8.7.

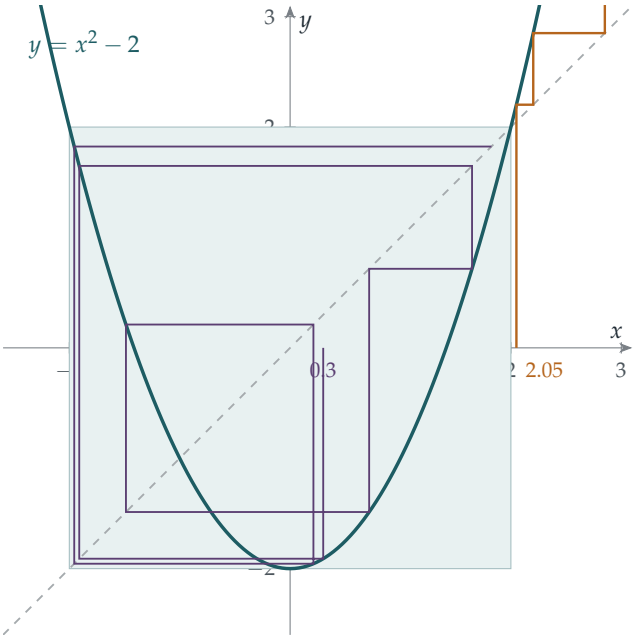


Figure 2.8.1 Cobweb diagrams for $x_{n+1} = x_n^2 - 2$. Starting at 0.3 (plum), the orbit stays in the shaded square $[-2, 2]^2$ for ever, wandering chaotically. Starting just outside, at 2.05 (amber), it escapes: $2.05, 2.20, 2.85, 6.13, \ldots$

Finding the largest primes

A *Mersenne number* is $M_p = 2^p - 1$. In 1876 Édouard Lucas found, and in 1930 Derrick Lehmer completed, a remarkable test for whether M_p is prime (for p an odd prime): start with $s_0 = 4$ and repeatedly square and subtract 2, reducing modulo M_p each time. Then

$$M_p \text{ is prime} \iff s_{p-2} \equiv 0 \pmod{M_p}.$$

For $M_7 = 127$: $s = 4, 14, 194 \equiv 67, 67^2 - 2 \equiv 42, 42^2 - 2 \equiv 111, 111^2 - 2 \equiv 0$, so 127 is prime. The connection with this question is the substitution: $4 = t + 1/t$ with $t = 2 + \sqrt{3}$, so $s_k = (2+\sqrt{3})^{2^k} + (2-\sqrt{3})^{2^k}$, and the proof of the test uses arithmetic with numbers of the form $a + b\sqrt{3}$, rather like the Gaussian integers of Section 1.6. Because each step needs only one squaring, the test is extremely fast, which is why most of the very largest known primes, including the current record holder, are Mersenne primes (not all of them are: other special forms, such as $a^{2^n} + 1$, have their own fast tests). As of the time of writing the record is $2^{136279841} - 1$, a number with more than 41 million digits, found in 2024.

The Mandelbrot set

Replace -2 by any constant c and allow complex numbers: $z \mapsto z^2 + c$. The set of c for which the orbit of 0 stays bounded is the *Mandelbrot set*. Our map has $c = -2$, the leftmost point of the Mandelbrot set, and the set of starting values whose orbits stay

bounded (its *Julia set*) is exactly the interval $[-2,2]$ that we found in part (ii). For almost every other c, that set is a fractal.

The thinking skill

Change variables to make a map simple

If an iteration looks complicated, look for a substitution that turns it into something simple, like doubling or adding a constant.

Here $x = t + 1/t$ and $x = 2\cos\theta$ turned $x \mapsto x^2 - 2$ into doubling. In the theory of dynamical systems this is called a *conjugacy*, and finding one is often the key to understanding a system completely.

Try this next

1 **Inside the interval.** Prove that if $a = 2\cos\theta$ then $x_n = 2\cos(2^{n-1}\theta)$. Find all values of a for which $x_3 = a$ but $x_2 \neq a$. (*Hint:* you want $4\theta \equiv \pm\theta$, modulo 2π.)

2 **Lucas–Lehmer by hand.** Use the test to show that $M_5 = 31$ is prime and that $M_{11} = 2047$ is not. (For M_{11} you will need nine steps; a calculator helps. Check your conclusion by factorising 2047.)

3 **An explicit example.** Take $A = \frac{3}{2}$. Find a and t, and check the formula $y_1 = \sqrt{x_2^2 - 4}/x_2$ numerically.

4 **A sum that telescopes.** HARDER With $a = t + 1/t$ and $t > 1$, show that

$$\frac{1}{x_1 x_2 \cdots x_n} = \frac{t - t^{-1}}{t^{2^n} - t^{-2^n}}.$$

Using the identity $\dfrac{1}{z-1} - \dfrac{1}{z^2-1} = \dfrac{z}{z^2-1}$ with $z = t^{2^n}$, deduce that

$$\frac{1}{x_1} + \frac{1}{x_1 x_2} + \frac{1}{x_1 x_2 x_3} + \cdots = \frac{1}{t}.$$

Check this numerically when $a = 3$.

Further reading

- **Ian Stewart, *Does God Play Dice?*** (Penguin). The classic popular account of chaos, including the logistic map.
- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press). The standard undergraduate introduction; Chapter 10 is about one-dimensional maps like this one.

- **Marcus du Sautoy, *The Music of the Primes*** (Fourth Estate). Includes the story of the hunt for Mersenne primes.

2.9 Fractions That Close In on e

The question

(i) Show that, for $n = 2, 3, 4, \ldots$,

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\left(t^n(1-t)^n\right) = nt^{n-2}(1-t)^{n-2}\Big[(n-1) - 2(2n-1)t(1-t)\Big].$$

(ii) The sequence $T_0, T_1, \ldots$ is defined by

$$T_n = \int_0^1 \frac{t^n(1-t)^n}{n!}\,\mathrm{e}^t\,\mathrm{d}t.$$

Show that, for $n \geq 2$,

$$T_n = T_{n-2} - 2(2n-1)T_{n-1}.$$

(iii) Evaluate T_0 and T_1 and deduce that, for $n \geq 0$, T_n can be written in the form

$$T_n = a_n + b_n\mathrm{e},$$

where a_n and b_n are integers (which you should not attempt to evaluate).

(iv) Show that $0 < T_n < \dfrac{\mathrm{e}}{n!}$ for $n \geq 0$. Given that b_n is non-zero for all n, deduce that $\dfrac{-a_n}{b_n}$ tends to e as n tends to infinity.

The hook

How would you find good fractions approximating e? You could truncate the series $1 + 1 + \frac{1}{2} + \frac{1}{6} + \cdots$, but the fractions you get are not especially good for the size of their denominators. This question does something much cleverer. It builds a family of integrals, each of which is a small number of the form "integer + integer $\times\, e$". If $a + be$ is tiny, then e is very close to $-a/b$, and the integrals hand us a sequence of excellent fractions: $\frac{19}{7}$, $\frac{193}{71}$, $\frac{2721}{1001}$, …

The same trick, integrals that are simultaneously small and of the form "integer combination of the number you care about", is behind some of the deepest irrationality proofs in mathematics, including Hermite's proof that e is transcendental and Apéry's proof that $\zeta(3)$ is irrational.

Before you start

Look at the integrand $\frac{t^n(1-t)^n}{n!}e^t$ on $[0,1]$.

> **Ask yourself** What is the largest value of $t(1-t)$ on $[0,1]$? What does that tell you about the size of T_n as n grows?

The maximum of $t(1-t)$ is $\frac{1}{4}$, at $t=\frac{1}{2}$. So the integrand is at most $\frac{e}{4^n n!}$, which shrinks extraordinarily fast (Figure 2.9.1). That smallness is half the story. The other half, that T_n has the special form $a_n + b_n e$, comes from integrating by parts.

Working it through

STAGE 1 Differentiating twice

Let $p(t) = t^n(1-t)^n$. By the product rule,

$$p'(t) = nt^{n-1}(1-t)^n - nt^n(1-t)^{n-1} = nt^{n-1}(1-t)^{n-1}(1-2t).$$

> **Ask yourself** Differentiate again. You will get two terms; take out the common factor $nt^{n-2}(1-t)^{n-2}$. To match the required form, you need to rewrite $(1-2t)^2$ in terms of $t(1-t)$. How?

Differentiating $n\,[t(1-t)]^{n-1}(1-2t)$ by the product rule (note that $\frac{\mathrm{d}}{\mathrm{d}t}[t(1-t)] = 1-2t$):

$$\begin{aligned} p''(t) &= n(n-1)[t(1-t)]^{n-2}(1-2t)^2 - 2n[t(1-t)]^{n-1} \\ &= nt^{n-2}(1-t)^{n-2}\Big[(n-1)(1-2t)^2 - 2t(1-t)\Big]. \end{aligned}$$

Now $(1-2t)^2 = 1-4t+4t^2 = 1-4t(1-t)$, so the bracket is $(n-1) - 4(n-1)t(1-t) - 2t(1-t) = (n-1) - 2(2n-1)t(1-t)$. ■

STAGE 2 Integrating by parts twice

> **Ask yourself** $p(t)$ is zero at both $t=0$ and $t=1$. What about $p'(t)$, for $n \geq 2$? Why does that matter when you integrate $p(t)e^t$ by parts?

For $n \geq 2$, $p'(t)$ contains the factor $t^{n-1}(1-t)^{n-1}$, so it also vanishes at both ends. Integrating by parts twice, every boundary term is zero:

$$\int_0^1 p\,e^t\,\mathrm{d}t = \Big[p\,e^t\Big]_0^1 - \int_0^1 p'\,e^t\,\mathrm{d}t = -\Big[p'\,e^t\Big]_0^1 + \int_0^1 p''\,e^t\,\mathrm{d}t = \int_0^1 p''\,e^t\,\mathrm{d}t.$$

From Stage 1, $p''(t) = n(n-1)\,t^{n-2}(1-t)^{n-2} - 2n(2n-1)\,t^{n-1}(1-t)^{n-1}$. Divide by $n!$:

$$\frac{p''(t)}{n!} = \frac{t^{n-2}(1-t)^{n-2}}{(n-2)!} - 2(2n-1)\frac{t^{n-1}(1-t)^{n-1}}{(n-1)!}.$$

Multiplying by e^t and integrating gives $T_n = T_{n-2} - 2(2n-1)T_{n-1}$. ■

STAGE 3 The integer structure

$T_0 = \int_0^1 e^t\,\mathrm{d}t = e - 1$. For $T_1 = \int_0^1 (t - t^2)e^t\,\mathrm{d}t$, integration by parts gives $\int_0^1 te^t\,\mathrm{d}t = 1$ and $\int_0^1 t^2 e^t\,\mathrm{d}t = e - 2$, so $T_1 = 1 - (e-2) = 3 - e$.

Ask yourself $T_0 = -1 + 1 \cdot e$ and $T_1 = 3 + (-1) \cdot e$. Why must every T_n have the form $a_n + b_n e$ with integer a_n, b_n?

By strong induction: if $T_{n-2} = a_{n-2} + b_{n-2}e$ and $T_{n-1} = a_{n-1} + b_{n-1}e$ with integer coefficients, then

$$T_n = \big(a_{n-2} - 2(2n-1)a_{n-1}\big) + \big(b_{n-2} - 2(2n-1)b_{n-1}\big)e,$$

and the new coefficients are integers too. ■

STAGE 4 Small, and therefore informative

For $0 < t < 1$, the integrand is positive, and $t^n(1-t)^n \le 1$ and $e^t < e$. So

$$0 < T_n < \int_0^1 \frac{e}{n!}\,\mathrm{d}t = \frac{e}{n!}.$$

Pause and try Divide $0 < a_n + b_n e < \dfrac{e}{n!}$ by $|b_n|$. What does that say about the distance between e and $-a_n/b_n$?

Dividing by $|b_n|$, which is a positive integer and so at least 1,

$$\left| e - \left(\frac{-a_n}{b_n} \right) \right| = \frac{|a_n + b_n e|}{|b_n|} < \frac{e}{n!\,|b_n|} \le \frac{e}{n!}.$$

The right-hand side tends to 0, so $-a_n/b_n \to e$. ■

Here are the first few, computed from the recurrence:

n	2	3	4	5	6
$-a_n/b_n$	$\dfrac{19}{7}$	$\dfrac{193}{71}$	$\dfrac{2721}{1001}$	$\dfrac{49171}{18089}$	$\dfrac{1084483}{398959}$
error	4×10^{-3}	3×10^{-5}	1×10^{-7}	3×10^{-10}	5×10^{-13}

The fraction $\frac{2721}{1001}$ is correct to 6 decimal places with only a four-digit denominator.

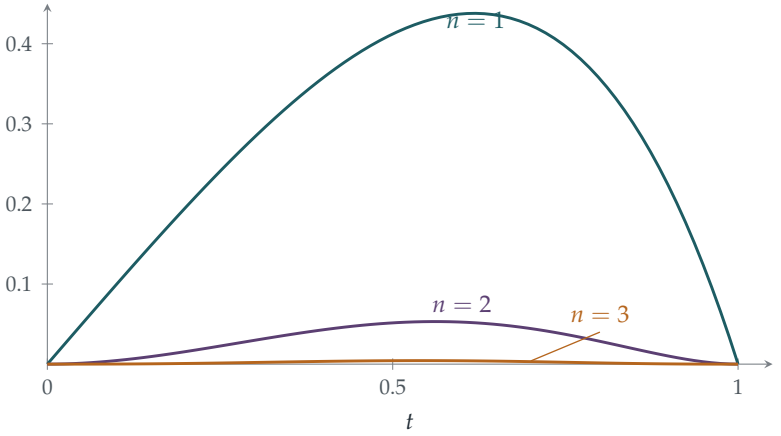


Figure 2.9.1 The integrands $\frac{t^n(1-t)^n}{n!}e^t$ for $n = 1, 2, 3$. Each is positive on $(0, 1)$, so each T_n is positive; and they shrink so quickly that by $n = 3$ the curve is barely visible. Positive but tiny is exactly what the argument needs.

Looking back

- **Integration by parts as a recurrence machine.** Because the polynomial and its derivative vanished at both ends, integrating by parts gave a clean recurrence with no boundary terms.
- **Structure and size together.** The recurrence showed T_n has the form $a_n + b_n e$; the bound showed it is small. Neither alone says anything about approximating e.
- **Don't compute what you don't need.** The question told us not to evaluate a_n and b_n, and the argument never needed them; it needed only that they were integers.

What's really going on

Another proof that e is irrational

The question stops just short of a famous theorem. Suppose $e = p/q$ with p, q positive integers. Then

$$q\,T_n = qa_n + b_n p$$

is an *integer*. But $0 < q\,T_n < \dfrac{qe}{n!}$, and for large n the right-hand side is less than 1. So qT_n would be an integer strictly between 0 and 1, which is impossible. Hence e is irrational. This is the "trap an integer between 0 and 1" idea from Section 1.4, delivered by integrals instead of series. Integrals of exactly this kind, with polynomials like $t^n(1-t)^n$, were used by Charles Hermite in 1873 to prove that e is *transcendental*: it is not a root of any polynomial with integer coefficients.

Continued fractions

The fractions in the table are not just good; they are among the *best possible*. The continued fraction of e, found by Euler, has a beautiful pattern:

$$e = [2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, \dots],$$

and its convergents are $2, 3, \frac{8}{3}, \frac{11}{4}, \frac{19}{7}, \frac{87}{32}, \frac{106}{39}, \frac{193}{71}, \dots, \frac{2721}{1001}, \dots$. Our $-a_n/b_n$ are exactly every third convergent. No fraction with a smaller denominator gets closer to e. Euler's pattern was itself the first proof that e is irrational, since only rational numbers have finite continued fractions.

Apéry's miracle

In 1978 Roger Apéry announced a proof that $\zeta(3) = 1 + \frac{1}{8} + \frac{1}{27} + \cdots$ is irrational, a question that had been open since Euler (see Section 2.3). His proof was so unexpected that the audience at first did not believe it. A year later Frits Beukers found a short version using integrals of the form

$$\int_0^1 \int_0^1 \frac{-\ln(xy)}{1 - xy}\, P_n(x) P_n(y)\, \mathrm{d}x\, \mathrm{d}y,$$

where P_n is built from $\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left(x^n(1-x)^n\right)$. These integrals are small, positive, and of the form "integer + integer $\times\, \zeta(3)$", divided by a controlled denominator: exactly the strategy of this question. Whether $\zeta(5)$, or $e + \pi$, is irrational is still unknown.

The thinking skill

Small but not zero

If a quantity is known to be an integer combination of some numbers, and also known to be small but not zero, it carries information: about approximations, and sometimes about irrationality.

The pattern is "structure from algebra, size from analysis". It recurs throughout number theory, and it is one of the places where analysis and algebra meet most fruitfully.

Try this next

1. **Run the recurrence.** Compute a_n and b_n for $n = 2, 3, 4$ from the recurrence and check the fractions in the table. Show that the signs of b_n alternate.

2. **A sharper bound.** Show that $0 < T_n < \dfrac{e}{4^n\, n!}$. How much does this improve the error estimate for $-a_n/b_n$?

3 **The irrationality proof.** Write out the argument in the box above carefully. Where exactly do you use that $T_n > 0$?

4 **A different weight.** HARDERRepeat the question with e^t replaced by e^{2t} (so $U_n = \int_0^1 \frac{t^n(1-t)^n}{n!} e^{2t}\,\mathrm{d}t$). Find a recurrence for U_n, show that U_n is a combination of 1 and e^2 with rational coefficients, and investigate whether the argument still proves that e^2 is irrational. (Compare Section 1.4.)

Further reading

- **Ivan Niven, *Irrational Numbers*** (Mathematical Association of America). The classic short book on irrationality proofs, including Hermite's method.
- **Alfred van der Poorten, "A proof that Euler missed"**, *The Mathematical Intelligencer* 1 (1979). A lively account of Apéry's proof and its reception.
- **Julian Havil, *The Irrationals*** (Princeton University Press). A history of irrational numbers, written for a general mathematical reader.

What We Have Learned

We began with Zeno, and with three questions about infinite processes: what it means for them to have an answer, how to prove that they do, and how fast they get there. Nine STEP questions later, here is how far we have come.

1. What does it mean for an infinite process to have an answer?

In Section 2.1 we learned the language. A statement about the long-run behaviour of a sequence has the shape "there exists a point m such that for all $n \geq m \ldots$". Proving it means producing m; disproving it means showing that every proposed m fails. The definition of a limit adds one more quantifier at the front, "for every tolerance $\varepsilon > 0$", and with that, Zeno's paradox dissolves: $\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n}$ is within any tolerance you like of 1 from some point on, and that is all that "the sum is 1" means.

The language also guards against nonsense. The quick "let x equal the infinite expression" argument gave the right answer for the golden ratio in Section 2.2, but only because we could separately prove that the limit existed. The same argument applied to $1 - 1 + 1 - \cdots$ gives an answer that means nothing.

2. How can we prove convergence without knowing the limit?

Most of the chapter's sections used one of a handful of tools.

- **Monotone and bounded.** An increasing sequence that is bounded above must converge (Section 2.2, Section 2.6). The same applies to partial sums of positive terms (Section 2.3). At university you will see that this is really a statement about the real numbers themselves: it is the property of *completeness*.
- **Comparison.** A series of positive terms converges if it is smaller than one that converges, and diverges if it is bigger than one that diverges. Grouping terms into blocks made the comparison easy, and showed that the harmonic series diverges while Kempner's thinned-out version converges (Section 2.3).
- **The squeeze.** Trap the quantity between two simpler ones with the same limit. This is how we proved $\sum 1/m^2 = \pi^2/6$ (Section 2.4) and evaluated the AGM integral rigorously (Section 2.6).
- **Exact formulas for finite stages.** In Section 2.5 and Section 2.8 we found exact expressions after n steps, by telescoping, and let $n \to \infty$ only at the very end.

Once a limit is known to exist, it can often be found by "letting $n \to \infty$" in a recurrence, as with $L = (2L + 1)/(L + 1)$ for the golden ratio. But the order matters: existence first, value second.

3. How fast?

Here the chapter held some surprises. Viète's product (Section 2.5) gains about 0.6 of a decimal digit per factor. The simple recipe for $\sqrt{2}$ in Section 2.7 gains about 0.8 digits per step. Newton's method and the AGM (Section 2.6) *double* the number of correct digits at each step: ten steps of Newton's method give more than 700 digits of $\sqrt{2}$. The difference is the shape of the error recurrence: "new error $\approx$ constant $\times$ old error" (linear) against "new error $\approx$ constant $\times$ (old error)^2" (quadratic). And in Section 2.8 we saw the opposite extreme: starting just above 2, the map $x \mapsto x^2 - 2$ pushes numbers away faster than exponentially, while inside $[-2, 2]$ it amplifies tiny errors until all predictability is lost.

Speed also matters for theory. The integrals in Section 2.9 produce fractions approaching e so quickly that they prove e is irrational: no fraction p/q could be approached that fast by other fractions.

4. A toolkit for analysis

Method	Where you used it
Producing and negating quantifiers	"Eventually" (2.1)
Monotone bounded sequences	Golden ratio (2.2); AGM (2.6)
Comparison and condensation	Harmonic and Kempner series (2.3)
Squeezing	$\pi^2/6$ (2.4); the AGM integral (2.6)
Symmetric functions of roots	Sum of $\cot^2$ values (2.4)
Telescoping sums and products	Viète (2.5); $x^2 - 2$ (2.8)
Invariants	The AGM integral (2.6)
Error recurrences	Linear and quadratic convergence (2.7)
Changing variables to simplify a map	$a = t + 1/t$, $a = 2\cos\theta$ (2.8)
Small but non-zero integer combinations	Approximating e (2.9)

Notice how many of these tools came with a "theorem you may use" attached. STEP often hands you the monotone convergence theorem, or tells you to assume that limits exist. At university you will prove those theorems from the axioms of the real numbers, and you will be expected to notice for yourself when an argument quietly assumes a limit exists. When you read a solution, it is a very good habit to ask: *where exactly was a limit taken, and was it justified?*

Where next

The ideas of this chapter run through the rest of the book.

- Fixed points and cobweb diagrams return in Section 8.7, where a chaotic map turns out to have an exact solution.
- The harmonic numbers $1 + \frac{1}{2} + \cdots + \frac{1}{n}$ appear in probability in Section 9.11 and in mechanics in Section 11.12.

- Another appearance of π in a limit, from probability, is in Section 12.23.
- The careful attitude to limits is the theme of Chapter 3, which opens with Section 3.1: arguments that look right but aren't.

If you want to go further before university

- **Lara Alcock, *How to Think About Analysis***: what first-year analysis is like, and how to cope with it.
- **David Bressoud, *A Radical Approach to Real Analysis***: analysis taught through its history, starting from the puzzles that forced mathematicians to be careful.
- **Tom Körner, *Calculus for the Ambitious*** (Cambridge University Press): written by a Cambridge analyst for exactly the readers of this book.
- **Julian Havil, *Gamma*** and **William Dunham, *Euler: The Master of Us All***: the stories behind many of the series in this chapter.

Zeno thought infinity made motion impossible. Mathematicians took two thousand years to answer him, and the answer turned out to be one of the most powerful ideas ever found: a way of making precise statements about processes that never end. You now know the first few words of that language. Analysis is where you will learn to speak it fluently.

Answers to Try This Next

These are brief answers and hints, not full solutions: they give the result and the key step, so that you can check your own work. Try each exercise properly before looking.

2.1 What "Eventually" Means

1. All terms of $(1+x)^n = 1 + nx + \binom{n}{2}x^2 + \cdots$ are non-negative, so $(1+x)^n \geq 1 + nx > nx$. With $y = 1 + x$, $y^n > n(y-1) > K$ once $n > K/(y-1)$. Using the next term, $y^n > \binom{n}{2}(y-1)^2$, so $y^n/n > \frac{1}{2}(n-1)(y-1)^2 > K$ once $n > 1 + 2K/(y-1)^2$.
2. If $s_n \leq t_n$ for $n \geq m_1$ and $t_n \leq s_n$ for $n \geq m_2$, then $s_n = t_n$ for $n \geq \max(m_1, m_2)$.
3. For example $s_n = n + 0.6$ for even n and $s_n = n$ for odd n, and $t_n = n + 0.6$ for odd n and $t_n = n$ for even n. Both increase, and each exceeds the other infinitely often.
4. $2^n \leq n!$ for $n \geq 4$ (induction: $2^{n+1} = 2 \cdot 2^n \leq (n+1)\,n!$). For $n^{10} \leq 2^n$, the inequality fails for every n from 2 to 58 and holds for all $n \geq 59$, so the smallest m is 59.

2.2 The Golden Ratio as a Limit

1. Induction: $u_1 = 1 = F_2/F_1$, and if $u_n = F_{n+1}/F_n$ then $u_{n+1} = 1 + F_n/F_{n+1} = F_{n+2}/F_{n+1}$.
2. Since $\varphi - 1 = 1/\varphi$, $u_{n+1} - \varphi = \frac{1}{u_n} - \frac{1}{\varphi} = -\frac{u_n - \varphi}{\varphi u_n}$. With $u_n \geq 1$, $|u_{n+1} - \varphi| \leq |u_n - \varphi|/\varphi$, and $|u_1 - \varphi| < 1$, so $|u_n - \varphi| \leq \varphi^{-(n-1)}$.
3. $\psi = \frac{1-\sqrt{5}}{2}$ satisfies $\psi^2 = \psi + 1$, so $1 + 1/\psi = \psi$. Working backwards, $u_{n-1} = \frac{1}{u_n - 1}$: from 0 we get $-1, -\frac{1}{2}, -\frac{2}{3}, -\frac{3}{5}, -\frac{5}{8}, \ldots$, that is, $u_1 = -F_{k-1}/F_k$ for some $k \geq 1$. These values approach ψ.
4. The fixed point satisfies $v = 1 + \frac{1}{1+v}$, i.e. $v^2 = 2$. The error is multiplied by $-\frac{1}{(1+v_n)(1+\sqrt{2})}$, of size at most $\frac{1}{2(1+\sqrt{2})}$, so $v_n \to \sqrt{2}$. And $\sqrt{2} = 1 + \cfrac{1}{2 + \cfrac{1}{2 + \cdots}}$.

2.3 A Divergent Series and Its Convergent Cousin

1. $\ln N + 0.5772 = 10$ gives $N \approx 12\,400$; passing 100 needs about $e^{99.4} \approx 1.5 \times 10^{43}$ terms.
2. With $k = 2$, block n is at most $2^n \cdot 4^{-n} = 2^{-n}$, so the sum is at most $\sum_{n \geq 0} 2^{-n} = 2$.

Adding $1+\frac{1}{4}+\frac{1}{9}$ exactly and bounding the blocks from $r=4$ by $\frac{1}{4}+\frac{1}{8}+\cdots=\frac{1}{2}$ gives about 1.86. (The true value is $\frac{\pi^2}{6}\approx 1.645$.)

3. For $1/(r\ln r)$ the block from 2^n to $2^{n+1}-1$ is at least $\frac{2^n}{2^{n+1}(n+1)\ln 2}=\frac{1}{2(n+1)\ln 2}$, and these sum like the harmonic series: divergence. For $1/(r(\ln r)^2)$ the block is at most $\frac{1}{n^2(\ln 2)^2}$, and $\sum\frac{1}{n^2}$ converges.

4. Allowed digits: 7 choices for the leading digit and 8 for each other digit, so there are at most $7\cdot 8^{d-1}$ allowed d-digit numbers, each at least 10^{d-1}. The sum is at most $\sum_{d\ge 1}7(0.8)^{d-1}=\frac{7}{1-0.8}=35$.

2.4 $\pi^2/6$

1. The numbers $\tan^2=1/\cot^2$ are the roots of the reversed polynomial $\sum_j(-1)^j\binom{2n+1}{2j+1}y^j$, whose roots add to $\binom{2n+1}{2n-1}=n(2n+1)$. Then $\operatorname{cosec}^2=1+\cot^2$ gives $n+\frac{n(2n-1)}{3}=\frac{2n(n+1)}{3}$.

2. The even terms add to $\frac{1}{4}\sum\frac{1}{m^2}=\frac{\pi^2}{24}$, so the odd terms add to $\frac{\pi^2}{6}-\frac{\pi^2}{24}=\frac{\pi^2}{8}$.

3. The roots have $e_1=\frac{n(2n-1)}{3}$ and $e_2=\binom{2n+1}{5}/\binom{2n+1}{1}$, giving $\sum\cot^4=e_1^2-2e_2=\frac{n(2n-1)(4n^2+10n-9)}{45}$. Squeezing with $\cot^4\theta<\theta^{-4}<(1+\cot^2\theta)^2$ and letting $n\to\infty$ gives $\frac{\pi^4}{90}$.

4. The lower bound gives $\sum_{m\le n}\frac{1}{m^2}>\frac{\pi^2 n(2n-1)}{3(2n+1)^2}$, so the error is less than $\frac{\pi^2(6n+1)}{6(2n+1)^2}<\frac{\pi^2}{4n}$. This guarantees accuracy 0.001 for $n\ge 2468$; in fact $n\approx 1000$ suffices.

2.5 Viète's Product

1. Repeatedly use $\sin\theta=2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$: the product telescopes. Let $n\to\infty$ (so $2^n\sin(\alpha/2^n)\to\alpha$) and take $\alpha=\frac{\pi}{2}$, using $\cos\frac{\pi}{4}=\sqrt{\frac{1}{2}}$ and $\cos\frac{\theta}{2}=\sqrt{\frac{1}{2}+\frac{1}{2}\cos\theta}$.

2. $\frac{\sin(\pi/3)}{\pi/3}=\frac{3\sqrt{3}}{2\pi}=\cos\frac{\pi}{6}\cos\frac{\pi}{12}\cos\frac{\pi}{24}\cdots$, with $\cos\frac{\pi}{6}=\frac{\sqrt{3}}{2}$ and $\cos\frac{\pi}{12}=\frac{\sqrt{2+\sqrt{3}}}{2}=\frac{\sqrt{6}+\sqrt{2}}{4}$.

3. 8 factors give 3.14157, which rounds to 3.1416. After n factors the error is about $\frac{\pi^3}{6\cdot 4^{n+1}}$, because the omitted factors are $\cos\frac{\pi}{2^{j+1}}\approx 1-\frac{\pi^2}{2\cdot 4^{j+1}}$: each factor divides the error by about 4, and $\log_{10}4\approx 0.6$.

4. With $x=\ln a$, the same argument gives

$$\frac{a-1}{\ln a}=\prod_{j\ge 1}\frac{1+a^{1/2^j}}{2},$$

so $\frac{1}{\ln 3}=\frac{1}{2}\prod_{j\ge 1}\frac{1+3^{1/2^j}}{2}$. Seven factors give 0.906, which rounds to 0.91 (the true value is 0.9102). Here the factors approach 1 like $1+\frac{\ln 3}{2^{j+1}}$, so each new factor

only halves the error (about 0.3 of a digit), whereas Viète's factors approach 1 like $1 - \frac{\pi^2}{2\cdot 4^{j+1}}$ and quarter it.

2.6 The Arithmetic–Geometric Mean

1. $x_{n+1} - y_{n+1} = \frac{1}{2}(\sqrt{x_n} - \sqrt{y_n})^2 = \frac{(x_n - y_n)^2}{2(\sqrt{x_n}+\sqrt{y_n})^2}$, and $(\sqrt{x_n} + \sqrt{y_n})^2 \geq 4y_n \geq 4y_0$.
2. $M(2,1) = 1.456791$; after three steps x_3 and y_3 already agree to seven decimal places.
3. Every step of the iteration scales by λ, so $M(\lambda x, \lambda y) = \lambda M(x,y)$. Substituting $x = \lambda u$ in the integral gives $\mathrm{I}(\lambda p, \lambda q) = \lambda^{-1}\mathrm{I}(p,q)$. Both sides of $\mathrm{I} = \frac{\pi}{2M}$ scale by λ^{-1}.
4. $M(1, \cos 30^\circ) \approx 0.9318$, so the period is longer by a factor 1.073: about 7.32%.

2.7 Six Digits or Six Hundred

1. $(p^2 + 2q^2)^2 - 2(2pq)^2 = (p^2 - 2q^2)^2 = 1$; and $577^2 - 2 \times 408^2 = 332\,929 - 332\,928 = 1$.
2. $y_{n+1} = \frac{2y_n^3+2}{3y_n^2}$, and $y_{n+1} - \alpha = \frac{2y^3 - 3\alpha y^2 + \alpha^3}{3y^2} = \frac{(y-\alpha)^2(2y+\alpha)}{3y^2}$. If $y > \alpha$, $\frac{2y+\alpha}{3y^2} < \frac{3y}{3y^2} = \frac{1}{y} < \frac{1}{\alpha}$.
3. $y - \frac{1/y - a}{-1/y^2} = y + y - ay^2 = y(2 - ay)$, and $1 - ay(2-ay) = (1-ay)^2$.
4. $y - \frac{y^{1/3}}{\frac{1}{3}y^{-2/3}} = y - 3y = -2y$. The error formula needs f′ to be non-zero and f″ bounded near the root; here f′ is infinite at 0.

2.8 Squaring and Subtracting Two

1. $(2\cos u)^2 - 2 = 2\cos 2u$, so induction gives $x_n = 2\cos(2^{n-1}\theta)$. $x_3 = a$ needs $4\theta \equiv \pm\theta$: 3θ or $5\theta \equiv 0 \pmod{2\pi}$. The cases with $x_2 = a$ (fixed points $a = 2$ and $a = -1$) are excluded, leaving $\theta = \frac{2\pi}{5}, \frac{4\pi}{5}$: $a = \frac{\sqrt{5}-1}{2}$ or $a = -\frac{\sqrt{5}+1}{2}$, the two points of a 2-cycle.
2. For M_5 the sequence is 4, 14, $194 \equiv 8$, $62 \equiv 0$: prime. For M_{11} it is 4, 14, 194, 788, 701, 119, 1877, 240, 282, $1736 \neq 0$: composite, and indeed $2047 = 23 \times 89$.
3. $A = t - \frac{1}{t} = \frac{3}{2}$ gives $t = 2$ and $a = \frac{5}{2}$. Then $x_2 = \frac{17}{4}$, $y_1 = \frac{Ax_1}{x_2} = \frac{15}{17}$, and $\frac{\sqrt{x_2^2 - 4}}{x_2} = \frac{15/4}{17/4} = \frac{15}{17}$.
4. $x_k = t^{2^{k-1}} + t^{-2^{k-1}}$, and multiplying $x_1 \cdots x_n$ by $t - t^{-1}$ telescopes to $t^{2^n} - t^{-2^n}$. The sum becomes $(t - t^{-1})\sum_{n\geq 1}\left(\frac{1}{t^{2^n}-1} - \frac{1}{t^{2^{n+1}}-1}\right) = \frac{t - t^{-1}}{t^2 - 1} = \frac{1}{t}$. For $a = 3$, $t = \frac{3+\sqrt{5}}{2}$ and $\frac{1}{t} \approx 0.3820$; the series $\frac{1}{3} + \frac{1}{21} + \frac{1}{987} + \cdots$ agrees.

2.9 Fractions That Close In on e

1. $T_0 = e - 1$, $T_1 = 3 - e$, $T_2 = 7e - 19$, $T_3 = 193 - 71e$, $T_4 = 1001e - 2721$, giving $\frac{19}{7}$, $\frac{193}{71}$, $\frac{2721}{1001}$. Since $b_n = b_{n-2} - 2(2n-1)b_{n-1}$ and (inductively) b_{n-1} has the opposite sign to b_{n-2}, b_n has the sign of b_{n-2}.

2. $t(1-t) \leq \frac{1}{4}$, so $T_n < \frac{e}{4^n n!}$; the error in $-a_n/b_n$ is $T_n/|b_n|$, now improved by a factor 4^n.

3. If $e = \frac{p}{q}$, then $qT_n = qa_n + pb_n$ is an integer. $T_n > 0$ makes it a *positive* integer, so at least 1; but $qT_n < \frac{qe}{n!} < 1$ for large n.

4. Integrating by parts twice (the boundary terms vanish), $U_n = \frac{1}{4}\big(U_{n-2} - 2(2n-1)U_{n-1}\big)$, with $U_0 = \frac{e^2-1}{2}$ and $U_1 = \frac{1}{2}$. So $2^{2n+1}U_n = A_n + B_n e^2$ with integers A_n, B_n, while $0 < U_n < \frac{e^2}{4^n n!}$. If $e^2 = \frac{p}{q}$, then $q2^{2n+1}U_n$ is a positive integer less than $\frac{2qe^2}{n!}$, impossible for large n. So the argument does prove that e^2 is irrational.

CHAPTER

3

Functions, Inequalities and Rigour

Knowing that something is true is not the same as knowing why, and the difference is where mathematics lives.

IN THIS CHAPTER

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3.4	Areas That Prove Inequalities	STEP III 2011
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Getting It Right

In mathematics you don't understand things. You just get used to them.

— John von Neumann

Here is a "proof" that $1 = 2$. Let $a = b$. Then $a^2 = ab$, so $a^2 - b^2 = ab - b^2$, which factorises as $(a+b)(a-b) = b(a-b)$. Cancelling $a - b$ gives $a + b = b$, and since $a = b$, that says $2b = b$, so $2 = 1$.

You probably spotted the problem: since $a = b$, "cancelling $a - b$" means dividing by zero. But notice how plausible every step looked, and how easily the error hid inside a routine move. Real mistakes in real mathematics are rarely so obvious, and they have fooled some of the greatest mathematicians who ever lived. This chapter is about the habits of mind that catch them: precise definitions, careful use of theorems, and a healthy suspicion of arguments that seem too easy.

Why rigour matters

At school, mathematics is mostly about getting answers. At university it is mostly about *knowing that your answers are right*, and being able to convince someone else. That shift is the biggest single change between school and university mathematics, and it can be unsettling. Why spend a page proving something that "obviously" works?

The answer is that "obvious" is unreliable. The history of analysis is full of results that seemed obvious and were false: that a continuous function must be differentiable except perhaps at a few isolated points (André-Marie Ampère published a "proof" in 1806; in 1872 Weierstrass found a continuous function differentiable *nowhere*), that a convergent sum of continuous functions is continuous (Cauchy believed it; Abel found a counterexample), that you can rearrange the terms of a convergent series without changing its sum (you can't, in general). Rigour is not pedantry. It is the only reliable way to tell the true from the plausible.

Inequalities: the working tools of analysis

The second theme of this chapter is *inequalities*. In school mathematics most results are equations: this equals that. In analysis, most results are inequalities: this is at most that. To prove a sequence converges you show the error is *less than* any tolerance; to prove an integral exists you show some sum is *bounded*; to estimate how good an approximation is, you bound the error. A professional analyst spends much more time proving inequalities than equations, and the classical inequalities you will meet here (AM–GM, Jensen, Young, Hölder, Cauchy–Schwarz) are the tools of the trade.

They are also remarkable in their own right. A single inequality about functions, with a well-chosen example, proves $\pi^2 \leq 10$ (Section 3.7). A fact about the shape of the logarithm settles whether π^e or e^π is bigger (Section 3.5). A chemist's question about measurement errors leads to some of the most important polynomials in mathematics (Section 3.6).

What you will meet at university

Rigour is the theme of first-year *Analysis*, and also of a course often called *Numbers and Sets* or *Foundations*, where you learn to write proofs precisely. You will meet the formal definitions of continuity and differentiability, and theorems with precise hypotheses (the intermediate value theorem, the mean value theorem, the extreme value theorem) whose hypotheses really matter. Later courses develop the inequalities in this chapter into whole subjects: *functional analysis* studies spaces of functions measured by integrals like those in Section 3.4 and Section 3.7; *convex analysis* and *optimisation* grow out of Section 3.3; *approximation theory* grows out of Section 3.6; and *special functions* like the one in Section 3.9 appear throughout applied mathematics and physics.

The questions that run through this chapter

Three big questions

1. **How do you know an argument is correct?** What are the common ways for plausible arguments to fail, and how do you check?
2. **How do you prove an inequality?** Where do the classical inequalities come from, and how do you choose which one to use?
3. **What is a function, really?** Can a function be defined by a property it has, rather than by a formula? And can we invent new functions when the old ones are not enough?

The chapter opens with error-spotting (Section 3.1), then works through the classical inequalities, each from a different angle: an unusual induction, convexity, areas, the shape of a graph, polynomial interpolation and test functions. It closes with two sections on functions defined by their properties: e^x and $\tanh x$ from their addition laws, and a brand-new function defined as an inverse.

Ideas in this chapter

§	Topic	Idea
3.1	Plausible wrong arguments	Checking calculations, logic and hidden hypotheses
3.2	AM–GM	Forward–backward induction
3.3	Jensen's inequality	Concave curves lie above their chords
3.4	Young's inequality	Comparing areas in a picture
3.5	π^e or e^π?	Reducing to one function and studying its graph
3.6	Slopes of quadratics	Expressing the unknown in terms of the known
3.7	$\pi^2 \leq 10$	Test functions in a general inequality
3.8	e^x and $\tanh x$	Functions defined by functional equations
3.9	The W function	Inventing a function as an inverse

How to read this chapter

As before, try each question before reading the solution. In this chapter, also try something else: after reading each solution, go back and find the one or two steps that a careless student would get wrong. Where would a hidden assumption slip in? Which hypothesis of which theorem needs checking? Reading in this way, critically and actively, is the single most useful habit you can build before university.

3.1 Spot the Flaw

The question

Criticise each step of the following arguments. You should correct the arguments where necessary and possible, and say (with justification) whether you think the conclusions are true even though the argument is incorrect.

(i) The function g defined by

$$g(x) = \frac{2x^3 + 3}{x^4 + 4}$$

satisfies $g'(x) = 0$ only for $x = 0$ or $x = \pm 1$. Hence the stationary values are given by $x = 0$, $g(x) = \frac{3}{4}$ and $x = \pm 1$, $g(x) = 1$. Since $\frac{3}{4} < 1$, there is a minimum at $x = 0$ and maxima at $x = \pm 1$. Thus we must have $\frac{3}{4} \leq g(x) \leq 1$ for all x.

(ii) $\displaystyle\int (1-x)^{-3}\,dx = -3(1-x)^{-4}$ and so $\displaystyle\int_{-1}^{3} (1-x)^{-3}\,dx = 0.$

The hook

Most exam questions ask you to produce an argument. This one hands you two arguments and asks you to take them apart. That is a very different skill, and it is the one you will use most at university: checking your own work, reading a proof in a book and deciding whether you believe it, marking a friend's attempt, refereeing a paper.

The arguments here are bad in an instructive way. Every step *looks* like something you have seen a hundred times. Some steps contain arithmetic slips, but the more serious errors are errors of *logic*: conclusions that do not follow even when the calculations are right. Those are the errors that survive into professional mathematics, and spotting them is what this chapter is about.

Before you start

Ask yourself When you are asked to "criticise each step", what should you check at each step? Make a list before you start.

A good checklist has at least three questions. *Is the calculation correct? Does this step follow from the previous ones? Are there hidden assumptions (that a function is continuous,*

that a limit exists, that a stationary point must be a maximum or minimum)? The question also asks whether the conclusions might be true anyway. A wrong argument can have a true conclusion, so that needs a separate check.

Working it through

STAGE 1 Is the derivative right?

Ask yourself Differentiate g with the quotient rule. Where is $g'(x) = 0$? Check $x = 1$ and $x = -1$ directly.

By the quotient rule,

$$g'(x) = \frac{6x^2(x^4+4) - (2x^3+3)\cdot 4x^3}{(x^4+4)^2} = \frac{-2x^6 - 12x^3 + 24x^2}{(x^4+4)^2} = \frac{-2x^2\left(x^4+6x-12\right)}{(x^4+4)^2}.$$

So $g'(x) = 0$ when $x = 0$ or $x^4 + 6x - 12 = 0$. At $x = 1$, $x^4 + 6x - 12 = -5$; at $x = -1$ it is -17. Neither is a stationary point. The first claim is **false**.

The quartic $x^4 + 6x - 12$ is -12 at $x = 0$ and tends to $+\infty$ in both directions, so it has (at least) one positive root and one negative root; it has exactly two real roots, because its second derivative $12x^2$ is non-negative, so it is convex and can cross zero at most twice. Numerically they are $x \approx 1.386$ and $x \approx -2.247$.

STAGE 2 Are the values right?

$g(0) = \frac{3}{4}$ is correct, and $g(1) = \frac{5}{5} = 1$. But $g(-1) = \frac{-2+3}{1+4} = \frac{1}{5}$, not 1. Even on its own terms, the argument has an arithmetic error: it seems to have assumed that g is even.

STAGE 3 Does comparing values tell you which is a maximum?

Ask yourself Suppose a function has exactly two stationary points, with values $\frac{3}{4}$ and 1. Does it follow that the first is a minimum and the second a maximum?

No. The *nature* of a stationary point is a local property: it depends on how the function behaves near that point, not on how its value compares with the value somewhere else. A function can have a local maximum lower than a local minimum (think of $x + 2\sin x$), or a stationary point that is neither. Here, near $x = 0$, the factor $x^4 + 6x - 12$ is close to -12, so

$$g'(x) = \frac{-2x^2(x^4+6x-12)}{(x^4+4)^2} > 0 \qquad \text{for small } x \neq 0.$$

The derivative is positive on both sides of 0: $x = 0$ is a stationary point of *inflection*, not a minimum.

STAGE 4 Do stationary values bound a function?

Pause and try Even with correct stationary points, why can't you conclude that every value of g lies between the smallest and largest stationary values? Find a value of x with $g(x) < \frac{3}{4}$.

Stationary points only describe *local* behaviour. The global behaviour also depends on what happens at the ends of the domain, here as $x \to \pm\infty$. Since $g(x) \to 0$ as $x \to \pm\infty$, and $g(x) < 0$ whenever $2x^3 + 3 < 0$, that is, $x < -\sqrt[3]{3/2} \approx -1.14$, the function takes negative values. For example $g(-2) = \frac{-13}{20}$. So the conclusion $\frac{3}{4} \le g(x) \le 1$ is **false** (Figure 3.1.1).

The correct statement: g increases to a global maximum of about 1.083 at $x \approx 1.386$, then decreases towards 0; to the left, it decreases from 0 to a global minimum of about -0.668 at $x \approx -2.247$, then increases through $x = 0$ (where it has a horizontal inflection) up to the maximum. So $-0.668 \lesssim g(x) \lesssim 1.083$ for all x.

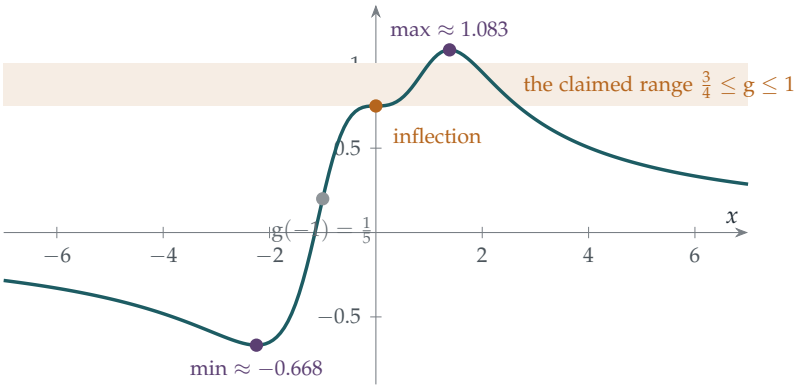


Figure 3.1.1 The true graph of $g(x) = \frac{2x^3+3}{x^4+4}$. The argument's claimed range (amber band) is wrong at every turn: $x = -1$ is not stationary and $g(-1) = \frac{1}{5}$; $x = 0$ is an inflection, not a minimum; and the function goes well outside the band, to a maximum above 1 and a minimum below 0.

STAGE 5 The integral

Ask yourself Differentiate $-3(1-x)^{-4}$. Do you get $(1-x)^{-3}$?

$\frac{d}{dx}\left[-3(1-x)^{-4}\right] = -3 \cdot (-4)(1-x)^{-5} \cdot (-1) = -12(1-x)^{-5}$. Wrong in both the power and the coefficient. The correct antiderivative is

$$\int (1-x)^{-3}\,dx = \tfrac{1}{2}(1-x)^{-2} + C,$$

as you can check by differentiating.

Ask yourself With the correct antiderivative, $\left[\frac{1}{2}(1-x)^{-2}\right]_{-1}^{3} = \frac{1}{2}\cdot\frac{1}{4} - \frac{1}{2}\cdot\frac{1}{4} = 0$. So the answer was right after all. Or was it?

It was not. The fundamental theorem of calculus, $\int_a^b F'(x)\,\mathrm{d}x = F(b) - F(a)$, requires the integrand to be continuous on the whole interval $[a, b]$. Here $(1-x)^{-3}$ is undefined at $x = 1$, which lies inside $[-1, 3]$, and near $x = 1$ it is enormous. The integral must be treated as an *improper* integral, split at the bad point:

$$\int_{-1}^{1} (1-x)^{-3}\,\mathrm{d}x = \lim_{\varepsilon\to 0^+}\left[\tfrac{1}{2}(1-x)^{-2}\right]_{-1}^{1-\varepsilon} = \lim_{\varepsilon\to 0^+}\left(\frac{1}{2\varepsilon^2} - \frac{1}{8}\right) = +\infty.$$

The integral from -1 to 1 is infinite (and the one from 1 to 3 is $-\infty$), so $\int_{-1}^{3}(1-x)^{-3}\,\mathrm{d}x$ **does not exist**. The conclusion "$= 0$" is false, not because the true value is something else, but because there is no value. ■

Looking back

- **Three kinds of error.** Calculation errors (the derivative, $g(-1)$, the antiderivative), logical errors (comparing values to classify stationary points; stationary values as global bounds), and hidden assumptions (continuity in the fundamental theorem of calculus).
- **Right answer, wrong reason.** Correcting the antiderivative in part (ii) reproduces the answer 0, but the argument is still invalid. A correct-looking answer is no evidence that the reasoning is sound.
- **Local versus global.** Derivatives tell you about the behaviour near a point. Global conclusions need more: what happens at the ends of the domain, or at points where the function misbehaves.

What's really going on

When great mathematicians got it wrong

Errors of exactly this kind were made by the founders of analysis. In his *Cours d'analyse* of 1821, Augustin-Louis Cauchy "proved" that a convergent sum of continuous functions is always continuous. In 1826 Niels Henrik Abel pointed out a counterexample: the Fourier series

$$\sin x - \tfrac{1}{2}\sin 2x + \tfrac{1}{3}\sin 3x - \cdots$$

converges for every x, and every term is continuous, but the sum jumps at $x = \pm\pi$ (it equals $x/2$ for $-\pi < x < \pi$). Working out exactly what had gone wrong in Cauchy's argument took decades, and led to the concept of *uniform convergence*, which every analysis student now learns. Hidden assumptions, often about swapping the order of two limits, are the most common source of error in all of analysis.

Theorems with hypotheses

Two of the errors in this question come from ignoring the hypotheses of a theorem.

- The *extreme value theorem* says that a continuous function on a closed, bounded interval $[a, b]$ attains a maximum and a minimum. To find them, check the endpoints, the stationary points, and any points where the function is not differentiable: the minimum of $|x|$ on $[-1, 1]$ is at 0, which is neither. If f is differentiable everywhere inside the interval, an interior maximum or minimum must be a stationary point. On the whole real line there are no endpoints to check, and the function might not have a maximum at all; you must examine the behaviour as $x \to \pm\infty$ instead.
- The *fundamental theorem of calculus* needs the integrand to be continuous on $[a, b]$. If it is not, the integral may still exist (try $\int_0^1 x^{-1/2}\,\mathrm{d}x$) or may not (as here), and you must check using limits.

At university, every theorem you meet will come with a precise list of hypotheses. Part of the training is to check them, every time, before using the theorem.

A value for a divergent integral

There is a curious footnote to part (ii). The interval $[-1, 3]$ is symmetric about the bad point $x = 1$, and $(1 - x)^{-3}$ is "odd" about that point. If you cut out a symmetric gap $(1 - \varepsilon, 1 + \varepsilon)$ and let $\varepsilon \to 0$, the two infinite pieces cancel and you do get 0. This is called the *Cauchy principal value*, and it is genuinely useful in physics and complex analysis. But it is a *different definition*, chosen deliberately, and an argument that arrives at it by accident has not proved anything.

The thinking skill

Reading arguments critically

Check every step for three things: is the calculation right, does the step follow logically, and what has been assumed without being said?

Then check the conclusion separately: an incorrect argument can have a true conclusion, and a "correct" answer can come from an invalid argument. Apply the same scrutiny to your own work; the arguments you are most likely to believe are your own.

Try this next

1 A substitution that lies. (From STEP I, 1987.) Explain why the substitution $x = 1/t$ does not show that

$$\int_{-1}^{1} \frac{\mathrm{d}x}{(1+x^2)^2} \quad \text{and} \quad \int_{-1}^{1} \frac{-t^2}{(1+t^2)^2}\,\mathrm{d}t$$

are equal. Evaluate both integrals correctly. (One is positive and one is negative, so they certainly differ.)

2 True or false? (From STEP I, 1998.) Justify your answers.

(a) $a^{\ln b} = b^{\ln a}$ for all $a, b > 0$.

(b) $\cos(\sin\theta) = \sin(\cos\theta)$ for all real θ.

(c) There is a polynomial P such that $|\mathrm{P}(\theta) - \cos\theta| \leq 10^{-6}$ for all real θ.

(d) $x^4 + 3 + x^{-4} \geq 5$ for all $x > 0$.

3 All horses are the same colour. Find the flaw in this "proof" by induction: any one horse is the same colour as itself; given $n+1$ horses, the first n are the same colour by the inductive hypothesis, and so are the last n; the two groups overlap, so all $n+1$ are the same colour.

4 An integral that does exist. Show that $\int_{-1}^{1} |x|^{-1/2}\,\mathrm{d}x$ exists, although the integrand is undefined at 0, and find its value.

Further reading

- **E. A. Maxwell, *Fallacies in Mathematics*** (Cambridge University Press). A classic collection of plausible wrong arguments, organised by topic, written for sixth-formers.
- **Bernard Gelbaum & John Olmsted, *Counterexamples in Analysis*** (Dover). Hundreds of examples showing why the hypotheses of theorems matter. Dip in once you have started an analysis course.
- **Imre Lakatos, *Proofs and Refutations*** (Cambridge University Press). A brilliant dialogue about how mathematicians find, and fix, errors in proofs.

3.2 AM–GM, Backwards

The question

Suppose that $a_i > 0$ for all $i > 0$. Show that

$$a_1 a_2 \leq \left(\frac{a_1 + a_2}{2}\right)^2.$$

Prove by induction that for all positive integers m

$$a_1 \cdots a_{2^m} \leq \left(\frac{a_1 + \cdots + a_{2^m}}{2^m}\right)^{2^m}. \qquad (*)$$

If $n < 2^m$, put $b_1 = a_1$, $b_2 = a_2$, …, $b_n = a_n$ and $b_{n+1} = \cdots = b_{2^m} = A$, where

$$A = \frac{a_1 + \cdots + a_n}{n}.$$

By applying $(*)$ to the b_i, show that

$$a_1 \cdots a_n A^{(2^m - n)} \leq A^{2^m}$$

(notice that $b_1 + \cdots + b_n = nA$). Deduce the (arithmetic mean)/(geometric mean) inequality

$$(a_1 \cdots a_n)^{1/n} \leq \frac{a_1 + \cdots + a_n}{n}.$$

The hook

The arithmetic mean–geometric mean inequality (AM–GM) is perhaps the single most useful inequality in mathematics. It says that for positive numbers, the ordinary average is always at least the "multiplicative average", with equality only when all the numbers are equal. It solves optimisation problems without calculus, it proves other inequalities, and it appears in every branch of analysis.

This question proves it by a strange and beautiful induction, due to Augustin-Louis Cauchy in 1821. Instead of going from n to $n + 1$, it leaps from 2 to 4 to 8 to 16, covering only the powers of 2, and then comes *back down* to fill in the numbers in between. It is a reminder that "induction" is a much more flexible idea than the textbook version suggests.

Before you start

The version of this paper in the online archive has a misprint: it reads "$b_1 = a_2$". It should of course be $b_1 = a_1$, as printed above; the idea is that the bs are the as, padded out with copies of their average.

> **Ask yourself** Why might it be easier to go from 2^m numbers to 2^{m+1} numbers than from n to $n+1$?

Because 2^{m+1} numbers split naturally into two groups of 2^m, and you can apply the inductive hypothesis to each half and then use the two-number case to combine the halves. Going from n to $n+1$ gives you nothing so symmetric to work with.

Working it through

STAGE 1 Two numbers

$$\left(\tfrac{a_1+a_2}{2}\right)^2 - a_1a_2 = \tfrac{a_1^2+2a_1a_2+a_2^2-4a_1a_2}{4} = \tfrac{(a_1-a_2)^2}{4} \geq 0. \qquad \blacksquare$$

STAGE 2 Doubling

The case $m = 1$ is Stage 1. Suppose $(*)$ holds for 2^m numbers, and take 2^{m+1} positive numbers.

> **Ask yourself** Split the 2^{m+1} numbers into a first half and a second half, and let P and Q be the arithmetic means of the two halves. What does the inductive hypothesis say about each half?

Applying $(*)$ to each half,

$$a_1 \cdots a_{2^m} \leq P^{2^m} \qquad \text{and} \qquad a_{2^m+1} \cdots a_{2^{m+1}} \leq Q^{2^m}.$$

Multiplying (everything is positive), the product of all 2^{m+1} numbers is at most $(PQ)^{2^m}$. Now the two-number case gives $PQ \leq \left(\frac{P+Q}{2}\right)^2$, and $\frac{P+Q}{2}$ is exactly the mean of all 2^{m+1} numbers. So

$$a_1 \cdots a_{2^{m+1}} \leq (PQ)^{2^m} \leq \left(\frac{P+Q}{2}\right)^{2^{m+1}} = \left(\frac{a_1 + \cdots + a_{2^{m+1}}}{2^{m+1}}\right)^{2^{m+1}},$$

which is $(*)$ for $m+1$. By induction, AM–GM holds whenever the number of terms is a power of 2. $\blacksquare$

STAGE 3 Filling the gaps

Now take any n, and choose m with $2^m > n$. We pad the list $a_1, \ldots, a_n$ with $2^m - n$ copies of its own average A.

Ask yourself What is the average of the padded list $b_1, \dots, b_{2^m}$?

The sum is $(a_1 + \cdots + a_n) + (2^m - n)A = nA + (2^m - n)A = 2^m A$, so the average is A itself. Adding copies of the average does not change the average. So $(*)$ applied to the bs gives

$$a_1 \cdots a_n \cdot A^{2^m - n} \le A^{2^m}.$$

Pause and try Finish the proof.

Divide both sides by $A^{2^m - n}$ (which is positive): $a_1 \cdots a_n \le A^n$. Taking nth roots,

$$(a_1 \cdots a_n)^{1/n} \le A = \frac{a_1 + \cdots + a_n}{n}. \qquad \blacksquare$$

Looking back

- **Induction along a different path.** Stage 2 is an induction on m, covering $n = 2, 4, 8, \dots$ Stage 3 is a way of stepping *down* from any power of 2 to any smaller n (Figure 3.2.1).
- **Padding with the average.** The trick of extending a list without changing its average is worth remembering; it is a way of turning a problem about any n into a problem about a convenient n.
- **When is there equality?** Tracing back through the proof, equality needs equality in every use of the two-number case, which happens only when all the numbers are equal.

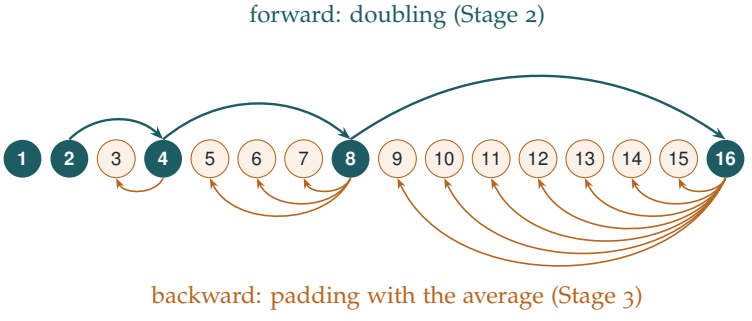


Figure 3.2.1 Cauchy's forward–backward induction. The teal arrows prove AM–GM for $2, 4, 8, 16, \dots$ numbers by doubling; the amber arrows fill in every other n by padding the list up to the next power of 2.

What's really going on

Cauchy's induction

Cauchy published this proof in his *Cours d'analyse* (1821), the book that began the rigorous treatment of analysis we discussed in Chapter 2. The technique is now called *forward–backward induction*: prove a statement for an infinite sequence of cases (here the powers of 2), and then show that the case n implies the case $n-1$. Together, these reach every n. Our Stage 3 jumps straight down from 2^m to n, but you can also do it one step at a time: padding $n-1$ numbers with one copy of their average shows that the case n implies the case $n-1$.

Many proofs of one inequality

AM–GM has dozens of proofs, and comparing them is a good way to see different styles of mathematics.

- **By convexity.** The logarithm is concave, so the average of the logarithms is at most the logarithm of the average. That is AM–GM after taking exponentials, and it is the subject of Section 3.3.
- **George Pólya's proof.** Use $e^{x-1} \geq x$ (the tangent line to e^{x-1} at $x=1$ lies below the curve). With A the mean, apply it to each a_i/A and multiply: $\prod \frac{a_i}{A} \leq e^{\sum(a_i/A-1)} = e^0 = 1$. Pólya said this proof came to him in a dream.
- **By smoothing.** If two of the numbers are unequal, replace them both by their average. The arithmetic mean is unchanged and, by the two-number case, the product increases. Repeating pushes all the numbers towards the average.

Optimisation without calculus

AM–GM turns many optimisation problems into one-liners. Of all boxes with a given surface area, which has the largest volume? Of all rectangles with a given perimeter, which has the largest area? (A square, by the two-number case: $xy \leq \left(\frac{x+y}{2}\right)^2$, with equality when $x=y$.) The equality condition tells you where the optimum is, and the inequality tells you it really is the optimum, not just a stationary point. Compare the warnings of Section 3.1.

The thinking skill

Flexible induction

Induction does not have to go $1, 2, 3, \ldots$ *You can induct along any path that reaches every case: by doubling, by strong induction, or forwards on a subsequence and then*

backwards to fill the gaps.

When the obvious step from n to $n+1$ is hard, ask which step would be easy, and whether you can reach every case using it.

Try this next

1 **One step back.** Show directly that if AM–GM holds for n numbers, it holds for $n-1$ numbers, by padding with one copy of the average.

2 **An increasing sequence.** Apply AM–GM to the $n+1$ numbers $1+\frac{1}{n}, \dots, 1+\frac{1}{n}, 1$ (with n copies of $1+\frac{1}{n}$) to show that $\left(1+\frac{1}{n}\right)^n < \left(1+\frac{1}{n+1}\right)^{n+1}$.

3 **The best box.** A closed rectangular box has surface area S. Use AM–GM on the areas of three different faces to show that its volume is at most $(S/6)^{3/2}$, with equality for a cube.

4 **Harmonic mean.** By applying AM–GM to the numbers $1/a_i$, show that the harmonic mean $\dfrac{n}{1/a_1 + \cdots + 1/a_n}$ is at most the geometric mean.

Further reading

- **J. Michael Steele, *The Cauchy–Schwarz Master Class*** (Cambridge University Press). A wonderful, problem-driven book on inequalities, starting from exactly this level. Chapter 2 is about AM–GM.
- **Hardy, Littlewood & Pólya, *Inequalities*** (Cambridge University Press). The classic reference, written by three of the great analysts of the twentieth century.
- **Arthur Engel, *Problem-Solving Strategies*** (Springer). Chapter 7 is a large collection of inequality problems from olympiads.

3.3 The Power of Convexity

The question

A function $f(x)$ is said to be concave on some interval if $f''(x) < 0$ in that interval. Show that $\sin x$ is concave for $0 < x < \pi$ and that $\ln x$ is concave for $x > 0$.

Let $f(x)$ be concave on a given interval and let $x_1, x_2, \ldots, x_n$ lie in the interval. *Jensen's inequality* states that

$$\frac{1}{n}\sum_{k=1}^{n} f(x_k) \leq f\left(\frac{1}{n}\sum_{k=1}^{n} x_k\right)$$

and that equality holds if and only if $x_1 = x_2 = \cdots = x_n$. You may use this result without proving it.

(i) Given that A, B and C are angles of a triangle, show that

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}.$$

(ii) By choosing a suitable function f, prove that

$$\sqrt[n]{t_1 t_2 \cdots t_n} \leq \frac{t_1 + t_2 + \cdots + t_n}{n}$$

for any positive integer n and for any positive numbers $t_1, t_2, \ldots, t_n$.

Hence:

(a) show that $x^4 + y^4 + z^4 + 16 \geq 8xyz$, where x, y and z are any positive numbers;

(b) find the minimum value of $x^5 + y^5 + z^5 - 5xyz$, where x, y and z are any positive numbers.

The hook

In Section 3.2 we proved AM–GM with a clever induction. Here it falls out in one line, as a special case of something much bigger. The bigger thing is *concavity*: the shape of a curve that bends downwards, like the top of a hill. A single geometric fact about such curves, that the curve lies above its chords, turns out to prove a startling range of inequalities, from a fact about triangles to AM–GM to inequalities in probability, information theory and economics.

At university this idea becomes *convexity*, and it is one of the great unifying themes of modern mathematics. Whole subjects, such as convex optimisation, are built on it.

Before you start

Ask yourself Draw a curve that bends downwards, such as $y = \sin x$ on $(0, \pi)$. Mark two points on it and join them with a straight line. Is the chord above or below the curve?

Below. For a concave curve, every chord lies below the curve between its endpoints. The average of two function values is the height of the midpoint of the chord; the function value at the average is the height of the curve above that midpoint. So "chord below curve" is exactly Jensen's inequality for $n = 2$:

$$\frac{f(x_1) + f(x_2)}{2} \leq f\left(\frac{x_1 + x_2}{2}\right).$$

For n points, the average of the points $(x_k, f(x_k))$ is their centre of mass, which lies inside the region under the curve, so it is below the curve too (Figure 3.3.1).

Working it through

STAGE 1 Checking concavity

$\frac{d^2}{dx^2} \sin x = -\sin x < 0$ for $0 < x < \pi$, and $\frac{d^2}{dx^2} \ln x = -\frac{1}{x^2} < 0$ for $x > 0$. ■

STAGE 2 Triangles

Ask yourself Which function, which points, and what is their average? Do the angles of a triangle lie in an interval where the function is concave?

Take $f(x) = \sin x$ and the three points A, B, C. Each is strictly between 0 and π, where sin is concave, and their average is $\frac{A+B+C}{3} = \frac{\pi}{3}$. Jensen gives

$$\frac{\sin A + \sin B + \sin C}{3} \leq \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2},$$

so $\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$, with equality exactly for the equilateral triangle. ■

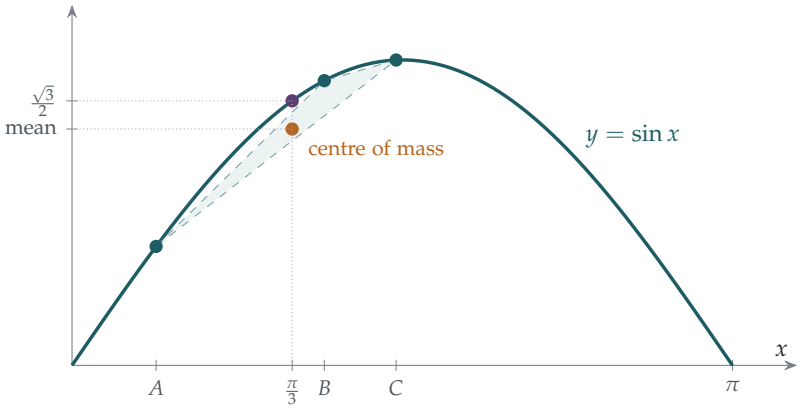


Figure 3.3.1 Jensen's inequality for $\sin x$ with three angles $A = 0.4$, $B = 1.2$, $C = \pi - 1.6$ of a triangle. The centre of mass of the three points on the curve (amber) lies inside the shaded triangle, and so below the curve. Its height is the average of $\sin A$, $\sin B$, $\sin C$, and the curve above it is at height $\sin \frac{\pi}{3}$.

STAGE 3 AM–GM in one line

Ask yourself AM–GM compares a product with a sum. Which concave function turns products into sums?

The logarithm. Apply Jensen to $\mathrm{f} = \ln$ and the positive numbers $t_1, \dots, t_n$:

$$\frac{\ln t_1 + \cdots + \ln t_n}{n} \le \ln\left(\frac{t_1 + \cdots + t_n}{n}\right).$$

The left-hand side is $\ln\left(\sqrt[n]{t_1 \cdots t_n}\right)$. Since $\ln$ is increasing, we can remove it from both sides, and that is AM–GM. Equality holds exactly when all the t_k are equal. ■

STAGE 4 Using AM–GM

Ask yourself For (a), the left-hand side is a sum of four terms and the right-hand side contains a product of three variables. How many terms should you apply AM–GM to, and why does the 16 help?

Apply AM–GM to the four positive numbers x^4, y^4, z^4, 16:

$$\frac{x^4 + y^4 + z^4 + 16}{4} \ge \sqrt[4]{16\, x^4 y^4 z^4} = 2xyz.$$

Multiplying by 4 gives $x^4 + y^4 + z^4 + 16 \ge 8xyz$. ■

Pause and try Now find the minimum of $x^5 + y^5 + z^5 - 5xyz$. (*Hint:* you want a fifth root of $x^5y^5z^5$, so you need five numbers. What should the other two

be?)

Apply AM–GM to $x^5, y^5, z^5, 1, 1$:

$$\frac{x^5+y^5+z^5+1+1}{5} \geq \sqrt[5]{x^5y^5z^5} = xyz,$$

so $x^5+y^5+z^5-5xyz \geq -2$. Equality holds when $x^5=y^5=z^5=1$, that is, $x=y=z=1$, where the expression is $3-5=-2$. So the minimum value is -2. ■

Notice how the equality condition does double duty: it shows the bound is actually attained, so -2 is the minimum and not just a lower bound.

Looking back

- **One idea, many inequalities.** A geometric fact about concave curves gave a result about triangles and AM–GM, with no calculation beyond checking a second derivative.
- **Choose the function and the points.** Using Jensen means deciding which function to apply it to and at which points. The logarithm, which turns products into sums, is a favourite.
- **Pad to fit.** In (b), adding two 1s made the AM–GM fit the problem, just as padding with the average did in Section 3.2.

What's really going on

Convex functions

At university the standard definition of a *convex* function (the upside-down version of concave) does not mention derivatives at all: f is convex if

$$\mathrm{f}\big(tx_1+(1-t)x_2\big) \leq t\,\mathrm{f}(x_1)+(1-t)\,\mathrm{f}(x_2) \qquad \text{for all } x_1, x_2 \text{ and } 0 \leq t \leq 1,$$

which says exactly that chords lie above the curve. This definition makes sense for functions that are not differentiable, such as $|x|$, and for functions of many variables. The Danish engineer and mathematician Johan Jensen made convex functions a subject in their own right in 1906, with the inequality that bears his name.

Convexity everywhere

Convexity appears throughout mathematics and its applications.

- **Probability.** Jensen's inequality says that for a concave f and any random variable X, $\mathrm{E}[\mathrm{f}(X)] \leq \mathrm{f}(\mathrm{E}[X])$. In economics, with f a concave "utility" function, this explains why people prefer a certain £100 to a fifty-fifty gamble on £0 or £200, and why insurance exists.

- **Information theory.** The proof that entropy is maximised by the uniform distribution, which underlies data compression, is Jensen's inequality for ln.
- **Optimisation.** A convex function has no local minima except the global one, so "walk downhill" always finds the best answer. This is why convex optimisation problems can be solved efficiently, and why so much effort goes into recasting problems in convex form.

In Section 3.4 we will see a different side of convexity: a geometric duality between a function and its inverse.

The thinking skill

> **Find the general principle**
>
> *When several results have similar proofs, look for the single principle behind them. Then prove the principle once and apply it many times.*
>
> AM–GM, the triangle inequality for sines, and many others are all instances of "a concave curve lies above its chords". Recognising the principle turns a collection of tricks into a method.

Try this next

1 **The definition.** (From STEP II, 2018.) Draw a sketch illustrating the chord definition of concavity in the box above. Explain why a function with $f''(x) < 0$ satisfies it. Then, by choosing $t = \frac{1}{3}$, $x_1 = u$ and $x_2 = \frac{v+w}{2}$ suitably, show that a concave function satisfies Jensen's inequality for three points.

2 **A product of sines.** (From STEP II, 2018.) By considering $\ln(\sin x)$, show that if A, B, C are the angles of a triangle then $\sin A \sin B \sin C \leq \frac{3\sqrt{3}}{8}$.

3 **Acute triangles.** Show that $\tan x$ is convex on $(0, \frac{\pi}{2})$, and deduce that in an acute-angled triangle, $\tan A + \tan B + \tan C \geq 3\sqrt{3}$.

4 **Power means.** Show that x^2 is convex, and deduce that $\left(\frac{a_1+\cdots+a_n}{n}\right)^2 \leq \frac{a_1^2+\cdots+a_n^2}{n}$. Interpret this as a statement about the variance of a set of numbers.

Further reading

- **J. Michael Steele, *The Cauchy–Schwarz Master Class*** (Cambridge University Press). Chapter 6 is about convexity and Jensen's inequality.
- **Claudi Alsina & Roger Nelsen, *When Less Is More: Visualizing Basic Inequalities*** (Mathematical Association of America). Inequalities proved by pictures, many of them

using convexity.

- **Stephen Boyd & Lieven Vandenberghe, *Convex Optimization*** (Cambridge University Press). A modern classic, free online; the first chapters show why convexity matters in practice.

3.4 Areas That Prove Inequalities

The question

The following result applies to any function f which is continuous, has positive gradient and satisfies $\mathrm{f}(0) = 0$:

$$ab \leq \int_0^a \mathrm{f}(x)\,\mathrm{d}x + \int_0^b \mathrm{f}^{-1}(y)\,\mathrm{d}y, \qquad (*)$$

where f^{-1} denotes the inverse function of f, and $a \geq 0$ and $b \geq 0$.

(i) By considering the graph of $y = \mathrm{f}(x)$, explain briefly why the inequality $(*)$ holds.

In the case $a > 0$ and $b > 0$, state a condition on a and b under which equality holds.

(ii) By taking $\mathrm{f}(x) = x^{p-1}$ in $(*)$, where $p > 1$, show that if $\dfrac{1}{p} + \dfrac{1}{q} = 1$ then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Verify that equality holds under the condition you stated above.

(iii) Show that, for $0 \leq a \leq \frac{1}{2}\pi$ and $0 \leq b \leq 1$,

$$ab \leq b \arcsin b + \sqrt{1 - b^2} - \cos a.$$

Deduce that, for $t \geq 1$,

$$\arcsin(t^{-1}) \geq t - \sqrt{t^2 - 1}.$$

The hook

Here is an inequality proved by a picture so simple that it fits on a postage stamp. Draw an increasing curve through the origin. The area under it, plus the area to its left, always covers a rectangle. From that one observation comes *Young's inequality*, $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$, which in turn is the key to Hölder's inequality and the Cauchy–Schwarz inequality, two of the most important tools in analysis.

The picture also contains a deeper idea. It pairs a function with its inverse in a perfectly symmetric way, and that pairing, generalised, is the *Legendre transform*, which connects

the two great formulations of classical mechanics and the two main descriptions of thermodynamics.

Before you start

Ask yourself $\int_0^a \mathrm{f}(x)\,\mathrm{d}x$ is the area between the curve and the x-axis. What area does $\int_0^b \mathrm{f}^{-1}(y)\,\mathrm{d}y$ represent on the same picture?

The graph of $x = \mathrm{f}^{-1}(y)$ is the same curve as $y = \mathrm{f}(x)$, viewed sideways. So $\int_0^b \mathrm{f}^{-1}(y)\,\mathrm{d}y$ is the area between the curve and the y-axis, from height 0 to height b. With that, part (i) is a matter of drawing.

Working it through

STAGE 1 The picture

Ask yourself Draw the rectangle with corners $(0,0)$ and (a,b). Shade the region under the curve from 0 to a in one colour and the region to the left of the curve from 0 to b in another. Do the two regions cover the rectangle? When is there no overlap and no excess?

The two regions together cover the whole rectangle $[0,a] \times [0,b]$, and usually some area outside it as well (Figure 3.4.1). Every point (x,y) of the rectangle is either below the curve (so in the first region, since $x \leq a$) or to the left of it (so in the second, since $y \leq b$). The total shaded area is therefore at least the area of the rectangle, ab. That is $(*)$.

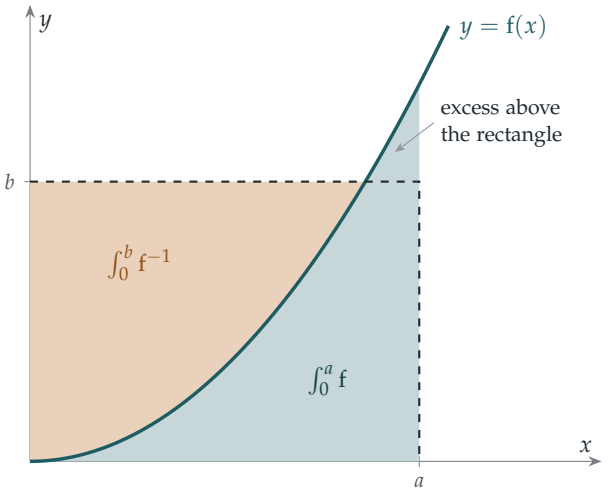


Figure 3.4.1 Young's inequality as a picture. The area under the curve from 0 to a (teal) and the area to the left of the curve from 0 to b (amber) together cover the dashed rectangle of area ab, with some area to spare. There is no excess exactly when the corner (a, b) lies on the curve.

The excess is the small curved region between the curve and the corner of the rectangle. It disappears exactly when the curve passes through the corner (a, b), that is, when

$$b = \mathrm{f}(a).$$

STAGE 2 Young's inequality

Ask yourself With $\mathrm{f}(x) = x^{p-1}$, what is $\mathrm{f}^{-1}(y)$? Compute both integrals, and use $\frac{1}{p} + \frac{1}{q} = 1$ to tidy the second.

Since $p > 1$, $\mathrm{f}(x) = x^{p-1}$ is continuous and increasing with $\mathrm{f}(0) = 0$, and $\mathrm{f}^{-1}(y) = y^{1/(p-1)}$. Then

$$\int_0^a x^{p-1}\,\mathrm{d}x = \frac{a^p}{p}, \qquad \int_0^b y^{1/(p-1)}\,\mathrm{d}y = \frac{b^{1+1/(p-1)}}{1+1/(p-1)}.$$

From $\frac{1}{p} + \frac{1}{q} = 1$ we get $q = \frac{p}{p-1}$, so $1 + \frac{1}{p-1} = \frac{p}{p-1} = q$, and the second integral is $\frac{b^q}{q}$. So $(*)$ becomes

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}. \qquad \blacksquare$$

Equality should hold when $b = \mathrm{f}(a) = a^{p-1}$. Check: then $ab = a^p$, and $b^q = a^{(p-1)q} = a^p$, so the right-hand side is $a^p\left(\frac{1}{p} + \frac{1}{q}\right) = a^p$. $\blacksquare$

With $p = q = 2$ this is $ab \le \frac{a^2+b^2}{2}$, the two-number AM–GM inequality of Section 3.2 in disguise.

STAGE 3 An inverse sine

Now take $f(x) = \sin x$ on $[0, \frac{\pi}{2}]$, which is continuous and increasing with $f(0) = 0$, and $f^{-1} = \arcsin$ on $[0, 1]$.

Ask yourself $\int_0^a \sin x \, dx$ is easy. How would you integrate $\arcsin y$?

$\int_0^a \sin x \, dx = 1 - \cos a$. For the inverse sine, integrate by parts with $\arcsin y \times 1$:

$$\int_0^b \arcsin y \, dy = \Big[y \arcsin y\Big]_0^b - \int_0^b \frac{y}{\sqrt{1-y^2}} \, dy = b \arcsin b + \sqrt{1-b^2} - 1.$$

Adding, $(*)$ gives $ab \le b \arcsin b + \sqrt{1-b^2} - \cos a$. ■

Pause and try Now deduce the inequality for $\arcsin(t^{-1})$. Which value of a should you choose? (Try the simplest possible one.)

Take $a = 0$: then $0 \le b \arcsin b + \sqrt{1-b^2} - 1$, that is, $b \arcsin b \ge 1 - \sqrt{1-b^2}$. (This is just the statement that $\int_0^b \arcsin y \, dy \ge 0$.) Put $b = 1/t$ with $t \ge 1$, so $0 < b \le 1$, and divide by b:

$$\arcsin\left(\frac{1}{t}\right) \ge t\left(1 - \sqrt{1 - \frac{1}{t^2}}\right) = t - \sqrt{t^2 - 1}. \qquad ■$$

Looking back

- **A picture as a proof.** Part (i) is a complete proof once you see the right picture. Translating a geometric fact into an inequality between integrals is a powerful technique.
- **Specialise a general result.** One general inequality, with different choices of f, gave Young's inequality and an inequality for arcsin.
- **The simplest choice can be the right one.** The final deduction used $a = 0$, where the inequality is almost trivial. Always try the easy cases.

What's really going on

From Young to Hölder

William Henry Young published his inequality in 1912. Its importance comes from what it leads to. Suppose $\sum a_i^p = 1$ and $\sum b_i^q = 1$. Applying Young to each pair and adding,

$$\sum a_i b_i \le \frac{\sum a_i^p}{p} + \frac{\sum b_i^q}{q} = \frac{1}{p} + \frac{1}{q} = 1.$$

Scaling the as and bs to make the sums 1 gives *Hölder's inequality*: for positive numbers,

$$\sum a_i b_i \le \Big(\sum a_i^p\Big)^{1/p}\Big(\sum b_i^q\Big)^{1/q}.$$

With $p = q = 2$ this is the *Cauchy–Schwarz inequality*, $\sum a_i b_i \le \sqrt{\sum a_i^2}\sqrt{\sum b_i^2}$, which says that the cosine of the angle between two vectors is at most 1. Hölder's inequality is the foundation of the theory of L^p spaces, spaces of functions in which "distance" is measured by $\big(\int |f|^p\big)^{1/p}$, and which you will meet in courses on functional analysis and probability.

The Legendre transform

The picture in Figure 3.4.1 treats f and f^{-1} completely symmetrically. Write $F(a) = \int_0^a \mathrm{f}$ and $G(b) = \int_0^b \mathrm{f}^{-1}$. The inequality says $F(a) + G(b) \ge ab$, with equality on the curve, so

$$G(b) = \max_a \big(ab - F(a)\big).$$

This operation, turning a convex function F into a new convex function G, is the *Legendre transform*, and applying it twice gets you back to F. In mechanics it converts the *Lagrangian*, a function of position and velocity, into the *Hamiltonian*, a function of position and momentum: two equivalent formulations of Newton's laws that you will meet in a second-year course. In thermodynamics it converts between energy, enthalpy and free energy. For $F(a) = a^p/p$ the transform is $G(b) = b^q/q$, which is why the two exponents in Young's inequality are linked by $\frac{1}{p} + \frac{1}{q} = 1$.

The thinking skill

Inequalities from pictures

An integral is an area. Comparing areas in a well-chosen picture can prove an inequality that is hard to see algebraically.

Look for pictures in which one region obviously contains another, or several regions obviously cover a third. Then translate each area into an integral.

Try this next

1 **The 1987 version.** (From STEP I, 1987.) Using the same picture with $y = \sin x$, and choosing s so that equality holds in $(*)$, show that $\displaystyle\int_0^t \arcsin y \,\mathrm{d}y = t\arcsin t - \big(1 - \cos(\arcsin t)\big)$ for $0 \le t \le 1$. Check the result by differentiating both sides with respect to t.

2 Cauchy–Schwarz. Use Young's inequality with $p = q = 2$ to prove $a_1b_1 + a_2b_2 + a_3b_3 \le \sqrt{a_1^2 + a_2^2 + a_3^2}\,\sqrt{b_1^2 + b_2^2 + b_3^2}$ for positive numbers. (*Hint:* first deal with the case where both square roots equal 1.)

3 An exponential version. Take $f(x) = e^x - 1$ in $(*)$ to show that, for $a, b \ge 0$,

$$ab \le e^a - 1 - a + (1 + b)\ln(1 + b) - b.$$

When is there equality?

4 How good is the bound? HARDER Use a calculator to compare $\arcsin(1/t)$ with $t - \sqrt{t^2 - 1}$ for $t = 1, 2, 10$. Show that for large t, $t - \sqrt{t^2 - 1} \approx \frac{1}{2t}$ while $\arcsin(1/t) \approx \frac{1}{t}$. Why is the bound off by a factor of 2? (Look at the picture with $a = 0$: how much of the excess is wasted?)

Further reading

- **J. Michael Steele, *The Cauchy–Schwarz Master Class*** (Cambridge University Press). Chapters 1 and 9 cover Cauchy–Schwarz and Hölder, with Young's inequality along the way.
- **Roger Nelsen, *Proofs Without Words*** (Mathematical Association of America). Includes Young's inequality and many other picture proofs.
- **V. I. Arnold, *Mathematical Methods of Classical Mechanics*** (Springer). Section 14 explains the Legendre transform with exactly this picture. Advanced, but the relevant section is short.

3.5 π^e or e^π?

The question

(i) Find the value of m for which the line $y = mx$ touches the curve $y = \ln x$.

If instead the line intersects the curve when $x = a$ and $x = b$, where $a < b$, show that $a^b = b^a$. Show by means of a sketch that $a < \mathrm{e} < b$.

(ii) The line $y = mx + c$, where $c > 0$, intersects the curve $y = \ln x$ when $x = p$ and $x = q$, where $p < q$. Show by means of a sketch, or otherwise, that $p^q > q^p$.

(iii) Show by means of a sketch that the straight line through the points $(p, \ln p)$ and $(q, \ln q)$, where $\mathrm{e} \leq p < q$, intersects the y-axis at a positive value of y. Which is greater, π^e or e^π?

(iv) Show, using a sketch or otherwise, that if $0 < p < q$ and $\dfrac{\ln q - \ln p}{q - p} = \mathrm{e}^{-1}$, then $q^p > p^q$.

The hook

Which is bigger, π^e or e^π? Your calculator will tell you in a second: $e^\pi \approx 23.14$ and $\pi^e \approx 22.46$. But a calculator only tells you *that*; it does not tell you *why*, and it cannot help with infinitely many similar questions at once. Is 3^π bigger than π^3? Are there any whole numbers with $n^m = m^n$ apart from the obvious $n = m$?

All these questions turn out to be about a single curve, $y = \ln x$, and the straight lines that cross it. This question shows how a sketch, used carefully, can settle comparisons that look hopeless by algebra, and how the number e emerges as the natural dividing point.

Before you start

Ask yourself Taking logarithms, $a^b = b^a$ is the same as $b \ln a = a \ln b$. Rearrange this so that a appears only on one side and b only on the other.

$b \ln a = a \ln b$ is the same as

$$\frac{\ln a}{a} = \frac{\ln b}{b}.$$

So every comparison of a^b with b^a is a comparison of the values of the single function $h(x) = \frac{\ln x}{x}$ at a and b. And $\frac{\ln a}{a}$ is the gradient of the line from the origin to the point $(a, \ln a)$ on the curve $y = \ln x$. That is why the question is about lines.

Working it through

STAGE 1 The tangent through the origin

Ask yourself If $y = mx$ touches $y = \ln x$ at $x = x_0$, what two equations must hold?

The line and curve must meet, $mx_0 = \ln x_0$, and have the same gradient, $m = 1/x_0$. Substituting the second into the first, $1 = \ln x_0$, so $x_0 = e$ and $m = 1/e$. The tangent from the origin touches the curve at $(e, 1)$.

If instead $y = mx$ crosses the curve at $x = a$ and $x = b$, then $\ln a = ma$ and $\ln b = mb$, so $\frac{\ln a}{a} = m = \frac{\ln b}{b}$, which gives $b \ln a = a \ln b$ and hence $a^b = b^a$. ■

For the sketch: a line through the origin that is slightly less steep than the tangent cuts the curve twice, once on each side of the point of tangency, so $a < e < b$. Lines steeper than $1/e$ miss the curve altogether, because $\ln x$ is concave and lies entirely below its tangent $y = x/e$.

STAGE 2 Lines that cut the y-axis above the origin

Ask yourself Now the line is $y = mx + c$ with $c > 0$. What are $\frac{\ln p}{p}$ and $\frac{\ln q}{q}$ in terms of m and c?

From $\ln p = mp + c$ and $\ln q = mq + c$,

$$\frac{\ln p}{p} = m + \frac{c}{p}, \qquad \frac{\ln q}{q} = m + \frac{c}{q}.$$

Since $c > 0$ and $p < q$, $\frac{c}{p} > \frac{c}{q}$, so $\frac{\ln p}{p} > \frac{\ln q}{q}$. Multiplying by $pq > 0$ gives $q \ln p > p \ln q$, so $p^q > q^p$. ■

Geometrically: from the origin, the point $(p, \ln p)$ is seen at a steeper angle than $(q, \ln q)$ when the line through them crosses the y-axis above the origin.

STAGE 3 Chords to the right of e

Ask yourself The chord from $(p, \ln p)$ to $(q, \ln q)$, with $e \le p < q$, has gradient less than the gradient of the curve at p. Why? And why does that make its y-intercept positive?

Because $\ln x$ is concave, its gradient $1/x$ decreases, so the chord from p to q is less steep

than the tangent at p: its gradient s satisfies $s < \frac{1}{p}$. The chord's y-intercept is

$$c = \ln p - sp > \ln p - \frac{1}{p} \cdot p = \ln p - 1 \geq 0,$$

using $p \geq e$. So the chord crosses the y-axis above the origin. ■

Pause and try Use this, and Stage 2, to decide which is greater, π^e or e^π.

Take $p = e$ and $q = \pi$. The chord between them has a positive intercept, so by Stage 2, $p^q > q^p$: that is,

$$e^\pi > \pi^e.$$

STAGE 4 Chords parallel to the tangent

Now suppose the chord from p to q has gradient exactly $\frac{1}{e}$, the same as the tangent $y = \frac{x}{e}$.

Ask yourself The chord is parallel to the tangent at $(e, 1)$. Since $\ln x$ is concave, the curve lies below that tangent. Does the chord cut the y-axis above or below the origin?

The chord is the line $y = \frac{x}{e} + c$ for some c. Its endpoints lie on the curve, which lies on or below the tangent $y = \frac{x}{e}$, and touches it only at $x = e$. The endpoints cannot both be at $x = e$, so at least one lies strictly below the tangent, and therefore the whole parallel chord lies below the tangent: $c < 0$. Now repeat Stage 2 with $c < 0$: the inequality reverses, $\frac{\ln p}{p} < \frac{\ln q}{q}$, and so $q^p > p^q$. ■

Looking back

- **Reduce to one function.** Comparing a^b with b^a became comparing $h(a)$ with $h(b)$, where $h(x) = \frac{\ln x}{x}$.
- **A sketch with reasons.** Each "by means of a sketch" was backed up by a precise fact: concavity of ln (chords below the curve, tangents above it) and the sign of an intercept.
- **The special role of e.** The tangent from the origin touches at $x = e$, and that is the dividing point for all of these comparisons.

What's really going on

The graph of $\frac{\ln x}{x}$

Everything in this question is visible in the graph of $h(x) = \frac{\ln x}{x}$ (Figure 3.5.1). Its derivative is $h'(x) = \frac{1-\ln x}{x^2}$, so h increases on $(0, e)$, reaches its maximum $\frac{1}{e}$ at $x = e$, and then decreases towards 0. A horizontal line at height m between 0 and $\frac{1}{e}$ cuts the graph twice, at points $a < e < b$ with $h(a) = h(b)$, which is exactly $a^b = b^a$. To the right of e, h is decreasing, so for $e \le p < q$ we have $h(p) > h(q)$, that is, $p^q > q^p$. In particular $e^\pi > \pi^e$, and more generally $e^x > x^e$ for every $x \neq e$: no other number beats e at this game.

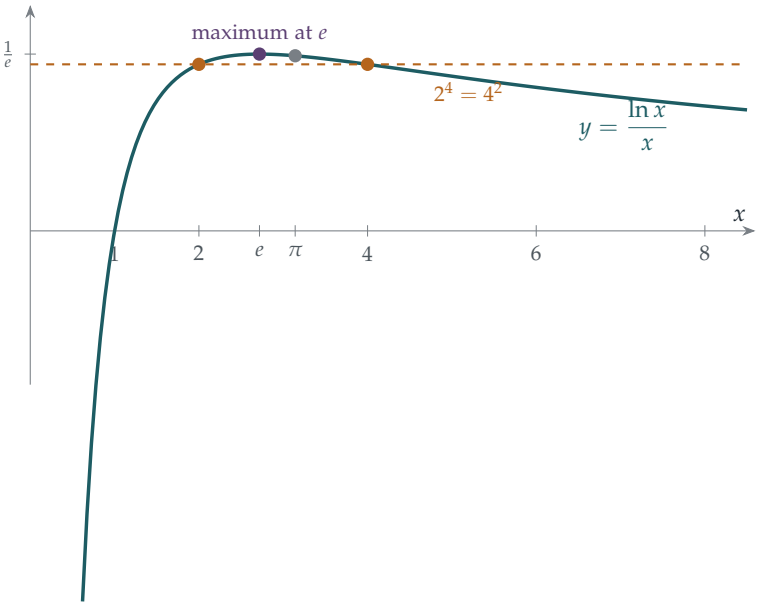


Figure 3.5.1 The graph of $h(x) = \frac{\ln x}{x}$. It rises to a maximum of $\frac{1}{e}$ at $x = e$ and then falls. Because $h(e) > h(\pi)$, we get $e^\pi > \pi^e$. The horizontal line at height $\frac{\ln 2}{2}$ (amber) meets the graph at $x = 2$ and $x = 4$: the only pair of distinct whole numbers with $n^m = m^n$.

All solutions of $x^y = y^x$

Which pairs of distinct positive reals satisfy $x^y = y^x$? A neat trick finds them all. Put $y = tx$ with $t > 1$; then $x^{tx} = (tx)^x$ gives $x^t = tx$, so $x^{t-1} = t$ and

$$x = t^{1/(t-1)}, \qquad y = t^{t/(t-1)}.$$

Writing $t = 1 + \frac{1}{n}$ gives $x = \left(1 + \frac{1}{n}\right)^n$ and $y = \left(1 + \frac{1}{n}\right)^{n+1}$. With $n = 1$ this is $(2, 4)$; with $n = 2$ it is $\left(\frac{9}{4}, \frac{27}{8}\right)$, and indeed $\left(\frac{9}{4}\right)^{27/8} = \left(\frac{27}{8}\right)^{9/4}$. As $n \to \infty$ both tend to e, the

point where the two branches meet. The problem was discussed in letters between Daniel Bernoulli and Christian Goldbach in 1728, and later by Euler.

The most economical base

The function $x^{1/x} = e^{h(x)}$ is also maximised at $x = e$. This answers a curious question: if the "cost" of a number system is the base times the number of digits needed, the most economical base is e, and among whole numbers, 3 (narrowly beating 2). A few experimental computers, such as the Soviet Setun of 1958, have used base 3.

The thinking skill

Rearrange to separate the variables, then study one function

An inequality or equation involving two quantities in a tangled way often becomes a statement about one function evaluated at two points.

Once you have that function, its graph (where it increases, decreases and peaks) answers every question at once. Here a^b versus b^a became $h(a)$ versus $h(b)$.

Try this next

1 **Whole numbers.** (From STEP I, 1988.) Use the graph of $h(x) = \frac{\ln x}{x}$ to find all pairs of distinct positive integers m and n with $n^m = m^n$.

2 **More comparisons.** (From STEP II, 2025.) Use the graph to show that $3^\pi > \pi^3$, and that $\left(\frac{9}{4}\right)^{\sqrt{5}} > \sqrt{5}^{9/4}$. Given that $1 < x < 2$, decide which is larger, x^{x+2} or $(x+2)^x$. (*Hint:* $h(2) = h(4)$.)

3 **Using $e^2 < 8$.** (From STEP II, 2025.) Show that $9^{\sqrt{2}} > \sqrt{2}^9$ and $3^{2\sqrt{2}} > (2\sqrt{2})^3$ are equivalent. Given that $e^2 < 8$, decide which of $9^{\sqrt{2}}$ and $\sqrt{2}^9$ is larger.

4 **A one-line proof.** Use $e^x > 1 + x$ for $x \neq 0$, with $x = \frac{\pi}{e} - 1$, to prove $e^\pi > \pi^e$ directly.

Further reading

- **Eli Maor, *e: The Story of a Number*** (Princeton University Press). The history of e, including the equation $x^y = y^x$.
- **Paul Nahin, *An Imaginary Tale*** (Princeton University Press). Not only about $\sqrt{-1}$: full of curious calculations of exactly this kind.
- **Michael Spivak, *Calculus*** (Cambridge University Press). Chapter 11 covers the use of derivatives to understand the shape of graphs with complete rigour.

3.6 A Bound on Slopes

The question

The function f is defined by

$$f(x) = ax^2 + bx + c.$$

Show that

$$f'(x) = f(1)\left(x + \tfrac{1}{2}\right) + f(-1)\left(x - \tfrac{1}{2}\right) - 2f(0)x.$$

If a, b and c are real and such that $|f(x)| \leq 1$ for $|x| \leq 1$, show that $|f'(x)| \leq 4$ for $|x| \leq 1$.

Find particular values of a, b and c such that, for the corresponding function f of the above form, $|f(x)| \leq 1$ for all x with $|x| \leq 1$ and $|f'(x)| = 4$ for some x satisfying $|x| \leq 1$.

The hook

If you know a function stays between -1 and 1 on an interval, can you say anything about how steep it is? For a general function, no: $\sin(1000x)$ stays between -1 and 1 but has slopes as large as 1000. But a quadratic cannot wiggle like that. This question shows that a quadratic trapped in the box $|x| \leq 1$, $|y| \leq 1$ can never have slope bigger than 4 inside the box, and that 4 is the best possible.

The problem has a surprising origin. In 1887 the chemist Dmitri Mendeleev, famous for the periodic table, was studying how the density of a solution depends on its concentration, and fitting quadratics to his data. He needed to know how badly errors in the data could affect the slope of his curves, and he proved exactly the result in this question. Two years later the young Andrei Markov solved the general problem for polynomials of any degree, and the answer led straight to some of the most important polynomials in mathematics.

Before you start

? **Ask yourself** The formula for $f'(x)$ uses only three values of f: at -1, 0 and 1. Why should three values be enough to determine the derivative of a quadratic?

A quadratic has three coefficients, so it is determined by its values at any three points. So its derivative, which is another function of the coefficients, must be expressible in terms of those three values. The formula tells you how. And it is exactly what you need, because the hypothesis $|f(x)| \le 1$ tells you about the *values* of f, while the conclusion is about its *derivative*.

Working it through

STAGE 1 The formula

$f(1) = a+b+c$, $f(-1) = a-b+c$ and $f(0) = c$. So the right-hand side is

$$(a+b+c)\left(x+\tfrac{1}{2}\right) + (a-b+c)\left(x-\tfrac{1}{2}\right) - 2cx.$$

The coefficient of x is $(a+b+c)+(a-b+c)-2c = 2a$, and the constant term is $\frac{1}{2}(a+b+c) - \frac{1}{2}(a-b+c) = b$. So the right-hand side is $2ax+b = f'(x)$. ■

STAGE 2 The bound

Ask yourself Take absolute values in the formula and use the triangle inequality, together with $|f(1)|, |f(-1)|, |f(0)| \le 1$. What bound on $|f'(x)|$ do you get, as a function of x?

By the triangle inequality $|u+v+w| \le |u|+|v|+|w|$,

$$|f'(x)| \le |f(1)|\,\left|x+\tfrac{1}{2}\right| + |f(-1)|\,\left|x-\tfrac{1}{2}\right| + 2|f(0)|\,|x| \le \left|x+\tfrac{1}{2}\right| + \left|x-\tfrac{1}{2}\right| + 2|x|.$$

Pause and try Show that $\left|x+\frac{1}{2}\right| + \left|x-\frac{1}{2}\right| + 2|x| \le 4$ for $|x| \le 1$. (Split into cases according to the signs.)

If $|x| \le \frac{1}{2}$, then $x+\frac{1}{2} \ge 0$ and $x-\frac{1}{2} \le 0$, so $\left|x+\frac{1}{2}\right| + \left|x-\frac{1}{2}\right| = (x+\frac{1}{2}) - (x-\frac{1}{2}) = 1$, and the total is $1+2|x| \le 2$.

If $\frac{1}{2} \le |x| \le 1$, both brackets have the same sign as x, so $\left|x+\frac{1}{2}\right| + \left|x-\frac{1}{2}\right| = 2|x|$, and the total is $4|x| \le 4$.

Either way $|f'(x)| \le 4$. ■

The bound is only achieved at $x = \pm 1$; in the middle half of the interval the slope is at most 2.

STAGE 3 An extreme example

Ask yourself For the bound to be attained at $x = 1$, every inequality in Stage 2 must be an equality. What values must $f(1)$, $f(-1)$ and $f(0)$ take, and with

? what signs?

At $x = 1$, the formula is $f'(1) = \frac{3}{2}f(1) + \frac{1}{2}f(-1) - 2f(0)$. To make this equal to 4 we need $f(1) = 1$, $f(-1) = 1$ and $f(0) = -1$. The quadratic through $(-1, 1)$, $(0, -1)$ and $(1, 1)$ is

$$f(x) = 2x^2 - 1,$$

so $a = 2$, $b = 0$, $c = -1$. On $|x| \le 1$, $x^2 \in [0, 1]$, so $2x^2 - 1 \in [-1, 1]$, and $f'(1) = 4$. ■

Looking back

- **Express what you want in terms of what you know.** The formula rewrote the derivative in terms of values of f, which is where the hypothesis gave information.
- **The triangle inequality.** Bounding a sum by the sum of the absolute values is one of the most common moves in analysis.
- **Find the extremal case from the equality conditions.** Asking when every inequality is tight produced the example $2x^2 - 1$ directly.

What's really going on

Markov's inequality and Chebyshev polynomials

Andrei Markov proved in 1889 that if p is a polynomial of degree n with $|p(x)| \le 1$ on $[-1, 1]$, then

$$|p'(x)| \le n^2 \qquad \text{for } |x| \le 1.$$

For $n = 2$ this is our bound of 4. The extreme cases are the *Chebyshev polynomials* T_n, defined by

$$T_n(\cos\theta) = \cos n\theta.$$

For example $\cos 2\theta = 2\cos^2\theta - 1$ gives $T_2(x) = 2x^2 - 1$, our extreme quadratic, and $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ gives $T_3(x) = 4x^3 - 3x$. Because $T_n(x) = \cos n\theta$, these polynomials oscillate between -1 and 1 as often as a polynomial of degree n possibly can, and they are as steep as possible at the ends (Figure 3.6.1). Markov's brother Vladimir extended the result to higher derivatives in 1892.

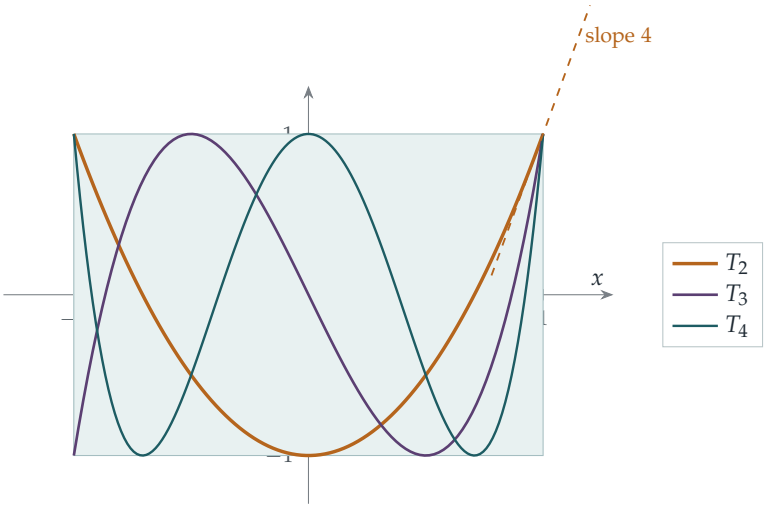


Figure 3.6.1 The Chebyshev polynomials $T_2 = 2x^2 - 1$, $T_3 = 4x^3 - 3x$ and $T_4 = 8x^4 - 8x^2 + 1$ stay inside the box $|x| \le 1$, $|y| \le 1$, touching its top and bottom as often as possible. They are the steepest polynomials of their degree at the ends: $T_n'(1) = n^2$. The dashed line is the tangent to T_2 at $x = 1$, with slope 4.

Interpolation

The formula in the question is a piece of *Lagrange interpolation*: given the values of a function at a few points, write down the unique polynomial of lowest degree through them, and use it to approximate the function or its derivative. Here the three points are $-1, 0, 1$. Formulas like this are used everywhere in numerical analysis, for example to estimate derivatives from measured data. Markov's inequality is the warning that comes with them: a small error in the values can be magnified by a factor of up to n^2 in the derivative, which is exactly the issue Mendeleev was worried about.

Chebyshev polynomials in practice

Chebyshev polynomials are the workhorses of approximation theory. Interpolating a function at the roots of T_n, rather than at equally spaced points, keeps the error under control; with equally spaced points, high-degree interpolation can go wildly wrong near the ends of the interval (Runge's phenomenon). Software that evaluates functions to full machine precision, or solves differential equations very accurately, often works with Chebyshev expansions internally.

The thinking skill

Express the unknown in terms of the known

If the hypothesis tells you about one kind of quantity and the conclusion is about another, look for an identity linking them, then bound term by term.

And when you have a bound, ask whether it is best possible. Following the equality cases backwards usually leads you to the extreme example, and the extreme example is often interesting in its own right.

Try this next

1 **The general idea.** Show that if f is a quadratic then $f(x) = f(-1)\dfrac{x(x-1)}{2} + f(0)(1-x^2) + f(1)\dfrac{x(x+1)}{2}$, and differentiate to recover the formula in the question.

2 **Uniqueness.** HARDER Show that $\pm(2x^2-1)$ are the only quadratics with $|f(x)| \le 1$ on $[-1,1]$ and $|f'(x)| = 4$ somewhere in $[-1,1]$.

3 **Chebyshev polynomials.** Use $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ to check that $T_3(x) = 4x^3 - 3x$, and verify that $T_3'(1) = 9$. Show that $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$, and find T_4.

4 **Slopes of T_n.** By differentiating $T_n(\cos\theta) = \cos n\theta$ with respect to θ, show that $T_n'(\cos\theta) = \dfrac{n\sin n\theta}{\sin\theta}$, and deduce that $T_n'(1) = n^2$ by letting $\theta \to 0$.

Further reading

- **Lloyd N. Trefethen, *Approximation Theory and Approximation Practice*** (SIAM). A beautiful modern book about Chebyshev polynomials and interpolation, readable by a strong school student, with computer experiments throughout.
- **Peter Borwein & Tamás Erdélyi, *Polynomials and Polynomial Inequalities*** (Springer). For the full story of Markov's inequality and its relatives. Advanced.
- **George M. Phillips, *Interpolation and Approximation by Polynomials*** (Springer). An undergraduate text covering Lagrange interpolation, Chebyshev polynomials and Markov-type inequalities.

3.7 $\pi^2 \leq 10$

The question

This question concerns the inequality

$$\int_0^{\pi} \big(\mathrm{f}(x)\big)^2 \,\mathrm{d}x \leq \int_0^{\pi} \big(\mathrm{f}'(x)\big)^2 \,\mathrm{d}x. \qquad (*)$$

(i) Show that $(*)$ is satisfied in the case $\mathrm{f}(x) = \sin nx$, where n is a positive integer.

Show by means of counterexamples that $(*)$ is not necessarily satisfied if either $\mathrm{f}(0) \neq 0$ or $\mathrm{f}(\pi) \neq 0$.

(ii) You may now assume that $(*)$ is satisfied for any (differentiable) function f for which $\mathrm{f}(0) = \mathrm{f}(\pi) = 0$.

By setting $\mathrm{f}(x) = ax^2 + bx + c$, where a, b and c are suitably chosen, show that $\pi^2 \leq 10$.

By setting $\mathrm{f}(x) = p \sin \frac{1}{2}x + q \cos \frac{1}{2}x + r$, where p, q and r are suitably chosen, obtain another inequality for π.

Which of these inequalities leads to a better estimate for π^2?

The hook

Every school student knows that π is a little more than 3. How would you *prove* an upper bound for it? Archimedes did it with polygons. This question does it in a completely unexpected way: by taking a general inequality about functions, plugging in a simple parabola, and reading off $\pi^2 \leq 10$. A different choice of function gives the famous approximation $\pi \leq \frac{22}{7}$.

The inequality $(*)$ is *Wirtinger's inequality*. It says that a function pinned to zero at both ends cannot be "big" without also being "steep". Behind it lies one of the central ideas of mathematical physics: the vibrations of a string, the energy levels of a quantum particle, and the theory of eigenvalues.

Before you start

Ask yourself Why might a function that is zero at both ends of $[0, \pi]$ have to be steep if it is large in the middle? Why does the condition $f(0) = f(\pi) = 0$ matter?

To get from 0 up to some height H and back down to 0 within a distance π, a function must climb at an average rate of at least $H/(\pi/2)$. So its derivative cannot be small everywhere. Without the condition at the ends there is no such constraint: a constant function can be as large as you like with zero derivative. The inequality $(*)$ makes this precise, measuring "size" and "steepness" by integrals of squares.

Working it through

STAGE 1 Sine waves

Ask yourself What is $\int_0^\pi \sin^2 nx\,dx$? And $\int_0^\pi \cos^2 nx\,dx$? (Use $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$.)

Both are $\frac{\pi}{2}$, since $\int_0^\pi \cos 2nx\,dx = 0$. With $f(x) = \sin nx$ we have $f'(x) = n\cos nx$, so

$$\int_0^\pi f^2 = \frac{\pi}{2} \le \frac{n^2\pi}{2} = \int_0^\pi (f')^2,$$

since $n \ge 1$. ■Notice that $n = 1$ gives equality, so the inequality is as tight as it could be.

STAGE 2 The end conditions matter

Ask yourself Find a simple function with $f(0) \neq 0$ for which $(*)$ fails. Then find one with $f(0) = 0$ but $f(\pi) \neq 0$.

The constant function $f(x) = 1$ has $f(0) = 1 \neq 0$, and $\int_0^\pi 1\,dx = \pi > 0 = \int_0^\pi 0\,dx$.

For the second, $f(x) = x$ has $f(0) = 0$ but $f(\pi) = \pi$, and

$$\int_0^\pi x^2\,dx = \frac{\pi^3}{3} \approx 10.3 > \pi = \int_0^\pi 1^2\,dx.$$

So $(*)$ fails in both cases. ■

STAGE 3 A parabola

Ask yourself Which quadratic vanishes at $x = 0$ and $x = \pi$? (Up to a constant multiple, there is only one. Why doesn't the multiple matter?)

The quadratic must be a multiple of $x(\pi - x)$, and multiplying f by a constant k multiplies both sides of $(*)$ by k^2, so we may take $f(x) = x(\pi - x)$, that is, $a = -1$, $b = \pi$, $c = 0$. Then $f'(x) = \pi - 2x$, and with the substitution $u = x - \frac{\pi}{2}$ (which makes both integrands even),

$$\int_0^\pi x^2(\pi - x)^2\,dx = \int_{-\pi/2}^{\pi/2} \left(\frac{\pi^2}{4} - u^2\right)^2 du = \frac{\pi^5}{30},$$
$$\int_0^\pi (\pi - 2x)^2\,dx = \int_{-\pi/2}^{\pi/2} 4u^2\,du = \frac{\pi^3}{3}.$$

(For the first: $\int_{-h}^{h}(h^2 - u^2)^2\,du = 2\big(h^5 - \frac{2}{3}h^5 + \frac{1}{5}h^5\big) = \frac{16}{15}h^5$ with $h = \frac{\pi}{2}$.) So $(*)$ says $\frac{\pi^5}{30} \le \frac{\pi^3}{3}$, that is,

$$\pi^2 \le 10. \qquad \blacksquare$$

STAGE 4 A trigonometric choice

Ask yourself For $f(x) = p\sin\frac{1}{2}x + q\cos\frac{1}{2}x + r$, what conditions do $f(0) = 0$ and $f(\pi) = 0$ impose on p, q, r?

$f(0) = q + r = 0$ and $f(\pi) = p + r = 0$. So $p = q = -r$, and scaling, we take $f(x) = \sin\frac{x}{2} + \cos\frac{x}{2} - 1$.

Pause and try Compute $\int_0^\pi f^2$ and $\int_0^\pi (f')^2$ for this function. What inequality for π results?

Write $s = \sin\frac{x}{2}$ and $c = \cos\frac{x}{2}$. Then $(s + c)^2 = 1 + 2sc = 1 + \sin x$, so

$$f^2 = (s + c)^2 - 2(s + c) + 1 = 2 + \sin x - 2\sin\tfrac{x}{2} - 2\cos\tfrac{x}{2}.$$

Using $\int_0^\pi \sin x = 2$, $\int_0^\pi \sin\frac{x}{2} = 2$ and $\int_0^\pi \cos\frac{x}{2} = 2$,

$$\int_0^\pi f^2\,dx = 2\pi + 2 - 4 - 4 = 2\pi - 6.$$

And $f' = \frac{1}{2}(c - s)$, so $(f')^2 = \frac{1}{4}(1 - \sin x)$ and $\int_0^\pi (f')^2\,dx = \frac{1}{4}(\pi - 2)$. Now $(*)$ gives

$$2\pi - 6 \le \frac{\pi - 2}{4} \quad\Longrightarrow\quad 8\pi - 24 \le \pi - 2 \quad\Longrightarrow\quad \pi \le \frac{22}{7}. \qquad \blacksquare$$

STAGE 5 Which is better?

From $\pi \le \frac{22}{7}$ we get $\pi^2 \le \frac{484}{49} \approx 9.878$, which is better than $\pi^2 \le 10$. (The true value is $\pi^2 \approx 9.870$.) Notice that this proves $\frac{22}{7}$ is an *upper* bound for π; a quite different integral, in Section 4.3, shows that it is not equal to π.

Looking back

- **A general inequality, a specific function.** Choosing a test function turned a statement about all functions into a statement about a number.
- **Scaling.** Both sides of $(*)$ scale by the same factor, so we could fix the free constant in each family.
- **Better test functions give better bounds.** The closer the test function is to the extreme case $\sin x$, the closer the inequality is to equality, and the better the estimate (Figure 3.7.1).

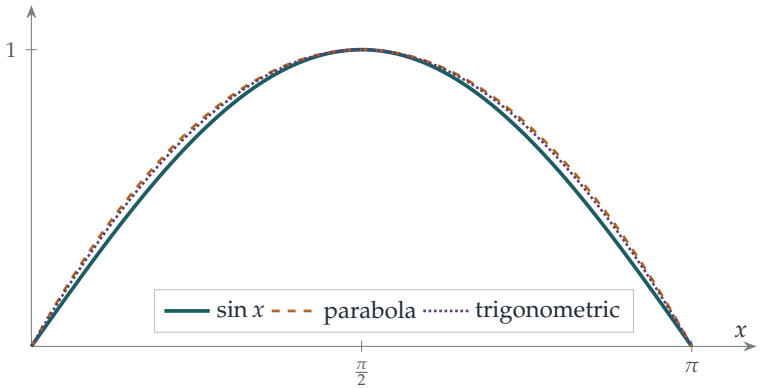


Figure 3.7.1 The three functions in this section, each scaled to have maximum 1: the extreme case $\sin x$ (solid), which gives equality in $(*)$; the parabola $x(\pi - x)$ (dashed), which gives $\pi^2 \leq 10$; and $\sin \frac{x}{2} + \cos \frac{x}{2} - 1$ (dotted), which gives $\pi \leq \frac{22}{7}$. Both test functions are very close to $\sin x$, which is why the bounds are so good.

What's really going on

A proof with Fourier series

Why is $(*)$ true? At university you will learn that any reasonable function on $[0, \pi]$ that vanishes at both ends can be written as a *Fourier sine series*,

$$f(x) = b_1 \sin x + b_2 \sin 2x + b_3 \sin 3x + \cdots .$$

Because $\int_0^\pi \sin mx \sin nx \, dx = 0$ for $m \neq n$, all the cross terms vanish when you square and integrate, giving

$$\int_0^\pi f^2 = \frac{\pi}{2}(b_1^2 + b_2^2 + b_3^2 + \cdots), \qquad \int_0^\pi (f')^2 = \frac{\pi}{2}(1^2 b_1^2 + 2^2 b_2^2 + 3^2 b_3^2 + \cdots).$$

Each term on the right is at least the corresponding term on the left, which proves $(*)$, and shows that equality holds only when f is a multiple of $\sin x$. Part (i) of the question is the single-term case.

Vibrating strings and quantum particles

The ratio $\int (f')^2 / \int f^2$ is called a *Rayleigh quotient*, and $(*)$ says its smallest possible value, for functions pinned at 0 and π, is 1. This number has physical meaning. For a string fixed at both ends, $\int (f')^2$ measures the stretching energy and $\int f^2$ the size of the displacement, and the minimum value of the ratio determines the lowest note the string can play. In quantum mechanics, the same minimum is the lowest energy of a particle trapped in a box: it cannot be zero, because a wave pinned at both ends must bend. Lord Rayleigh used exactly this trick, guessing a test function and computing the ratio, to estimate the frequencies of vibrating systems; because the extreme case gives the minimum, any guess gives an upper bound on the true lowest frequency, and good guesses give very accurate ones.

The calculus of variations

Finding the function that minimises a ratio of integrals is a problem in the *calculus of variations*, where the unknown is a whole function rather than a number. The minimiser here, $\sin x$, satisfies the differential equation $f'' + f = 0$, and this is no coincidence: minimising an integral always leads to a differential equation for the minimiser. We return to this idea in Section 4.7.

The thinking skill

Test functions

An inequality that holds for every function in a class can be applied to a cleverly chosen function to give information about a specific number.

The art is in the choice: a test function close to the extreme case gives a nearly sharp result. This idea runs through analysis, physics and engineering, from estimating eigenvalues to the finite element method.

Try this next

1 **A quartic.** Use $f(x) = x^2(\pi - x)^2$ in $(*)$. Show that it gives only $\pi^2 \leq 12$. Why is this test function worse than the parabola? (Compare its shape with $\sin x$.)

2 **Two sine waves.** Check $(*)$ directly for $f(x) = \sin x + \sin 2x$, computing both sides.

3 **A different interval.** Show by a substitution that if $g(0) = g(L) = 0$ then $\displaystyle\int_0^L g^2 \leq \frac{L^2}{\pi^2}\int_0^L (g')^2$. What does this say about strings of different lengths?

4 **Rayleigh's method.** HARDER Take $f(x) = x(\pi - x) + \lambda x^2(\pi - x)^2$ and choose λ to make the bound on π^2 as good as possible. (The algebra is heavy; a computer algebra system helps.)

Further reading

- **T. W. Körner, *Fourier Analysis*** (Cambridge University Press). A wonderful, discursive book on Fourier series and their applications, written by a Cambridge analyst.
- **Hardy, Littlewood & Pólya, *Inequalities*** (Cambridge University Press). Chapter 7 discusses Wirtinger's inequality and its relatives.
- **Richard Feynman, *The Feynman Lectures on Physics*,** Volume II, Chapter 19 ("The principle of least action"). Feynman's lecture on minimising integrals, with physical intuition throughout. Free online.

3.8 Functions from Their Addition Laws

The question

The definition of the derivative f' of a (differentiable) function f is

$$\mathrm{f}'(x) = \lim_{h\to 0} \frac{\mathrm{f}(x+h) - \mathrm{f}(x)}{h}. \qquad (*)$$

(i) The function f has derivative f' and satisfies

$$\mathrm{f}(x+y) = \mathrm{f}(x)\mathrm{f}(y)$$

for all x and y, and $\mathrm{f}'(0) = k$ where $k \neq 0$. Show that $\mathrm{f}(0) = 1$.

Using $(*)$, show that $\mathrm{f}'(x) = k\mathrm{f}(x)$ and find $\mathrm{f}(x)$ in terms of x and k.

(ii) The function g has derivative g' and satisfies

$$\mathrm{g}(x+y) = \frac{\mathrm{g}(x) + \mathrm{g}(y)}{1 + \mathrm{g}(x)\mathrm{g}(y)}$$

for all x and y, $|\mathrm{g}(x)| < 1$ for all x, and $\mathrm{g}'(0) = k$ where $k \neq 0$.

Find $\mathrm{g}'(x)$ in terms of $\mathrm{g}(x)$ and k, and hence find $\mathrm{g}(x)$ in terms of x and k.

The hook

How do you know what e^x is? At school it is introduced through compound interest, or as the function that is its own derivative, or as $\lim(1 + x/n)^n$. This question offers another answer: e^{kx} is the *only* differentiable function that turns addition into multiplication, $\mathrm{f}(x+y) = \mathrm{f}(x)\mathrm{f}(y)$. The rule alone, plus one number $\mathrm{f}'(0)$, pins the function down completely.

The second part does the same for a stranger rule, $\mathrm{g}(x+y) = \frac{\mathrm{g}(x)+\mathrm{g}(y)}{1+\mathrm{g}(x)\mathrm{g}(y)}$. If that formula looks familiar, it should: it is Einstein's rule for adding velocities in special relativity, with the speed of light set to 1. The function it forces is tanh, and that explains why nothing can be accelerated past the speed of light.

Before you start

An equation like $\mathrm{f}(x+y) = \mathrm{f}(x)\mathrm{f}(y)$, where the unknown is a function, is called a *functional equation*.

Ask yourself The equation holds for *all* x and y. What is the most useful thing to try first?

Special values. Setting $y = 0$ (or $x = y$, or $y = -x$) often gives a lot of information cheaply. That is how both parts start.

Working it through

STAGE 1 The value at zero

Put $x = y = 0$: $\mathrm{f}(0) = \mathrm{f}(0)^2$, so $\mathrm{f}(0) = 0$ or 1.

Ask yourself What goes wrong if $\mathrm{f}(0) = 0$?

If $\mathrm{f}(0) = 0$, then putting $y = 0$ gives $\mathrm{f}(x) = \mathrm{f}(x)\mathrm{f}(0) = 0$ for every x. Then f is identically zero and $\mathrm{f}'(0) = 0$, contradicting $\mathrm{f}'(0) = k \neq 0$. So $\mathrm{f}(0) = 1$. ■

STAGE 2 The derivative everywhere from the derivative at zero

Ask yourself In $(*)$, write $\mathrm{f}(x+h) = \mathrm{f}(x)\mathrm{f}(h)$. What can you take outside the limit?

$$\begin{aligned}\mathrm{f}'(x) &= \lim_{h\to 0} \frac{\mathrm{f}(x)\mathrm{f}(h) - \mathrm{f}(x)}{h} = \mathrm{f}(x) \lim_{h\to 0} \frac{\mathrm{f}(h) - 1}{h} \\ &= \mathrm{f}(x) \lim_{h\to 0} \frac{\mathrm{f}(0+h) - \mathrm{f}(0)}{h} = \mathrm{f}(x)\,\mathrm{f}'(0) = k\,\mathrm{f}(x). \qquad \blacksquare\end{aligned}$$

The functional equation lets the behaviour near 0 be transported to every other point.

STAGE 3 Solving $\mathrm{f}' = k\mathrm{f}$

Consider $\mathrm{u}(x) = \mathrm{f}(x)e^{-kx}$. Then $\mathrm{u}'(x) = \big(\mathrm{f}'(x) - k\mathrm{f}(x)\big)e^{-kx} = 0$, so u is constant, and $\mathrm{u}(0) = \mathrm{f}(0) = 1$. Therefore

$$\mathrm{f}(x) = e^{kx}. \qquad \blacksquare$$

(Separating the variables, $\int \frac{\mathrm{df}}{\mathrm{f}} = \int k\,\mathrm{d}x$, gives the same answer, but needs $\mathrm{f}(x) \neq 0$; the trick with u avoids that worry.)

STAGE 4 The second rule

Ask yourself Put $y = 0$ in the rule for g. What does that tell you about $\mathrm{g}(0)$? Where does $|\mathrm{g}(x)| < 1$ come in?

With $y = 0$: $\mathrm{g}(x)\big(1 + \mathrm{g}(x)\mathrm{g}(0)\big) = \mathrm{g}(x) + \mathrm{g}(0)$, which simplifies to $\mathrm{g}(0)\big(\mathrm{g}(x)^2 - 1\big) = 0$. Since $|\mathrm{g}(x)| < 1$, the bracket is never zero, so $\mathrm{g}(0) = 0$.

Pause and try Now compute $\mathrm{g}(x+h)-\mathrm{g}(x)$ using the rule, simplify, divide by h and let $h\to 0$.

$$\mathrm{g}(x+h)-\mathrm{g}(x)=\frac{\mathrm{g}(x)+\mathrm{g}(h)-\mathrm{g}(x)-\mathrm{g}(x)^2\mathrm{g}(h)}{1+\mathrm{g}(x)\mathrm{g}(h)}=\frac{\mathrm{g}(h)\bigl(1-\mathrm{g}(x)^2\bigr)}{1+\mathrm{g}(x)\mathrm{g}(h)}.$$

Divide by h. As $h\to 0$, $\frac{\mathrm{g}(h)}{h}=\frac{\mathrm{g}(h)-\mathrm{g}(0)}{h}\to \mathrm{g}'(0)=k$, and $\mathrm{g}(h)\to \mathrm{g}(0)=0$ (a differentiable function is continuous), so the denominator tends to 1. Therefore

$$\mathrm{g}'(x)=k\bigl(1-\mathrm{g}(x)^2\bigr). \qquad \blacksquare$$

STAGE 5 Solving it

Ask yourself Separate the variables. Which function has derivative $\frac{1}{1-g^2}$ for $|g|<1$?

Since $|\mathrm{g}|<1$, we can divide: $\dfrac{\mathrm{g}'}{1-\mathrm{g}^2}=k$. The function $\operatorname{artanh} g=\frac{1}{2}\ln\frac{1+g}{1-g}$ has derivative $\frac{1}{1-g^2}$, so $\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{artanh}\mathrm{g}(x)=k$ and $\operatorname{artanh}\mathrm{g}(x)=kx+C$. From $\mathrm{g}(0)=0$, $C=0$, so

$$\mathrm{g}(x)=\tanh kx. \qquad \blacksquare$$

Check: the addition formula $\tanh(A+B)=\dfrac{\tanh A+\tanh B}{1+\tanh A\tanh B}$ is exactly the given rule.

Looking back

- **Special values first.** Putting $y=0$ found $\mathrm{f}(0)$ and $\mathrm{g}(0)$.
- **From one point to everywhere.** The addition law moved the derivative at 0 to every x, turning a functional equation into a differential equation.
- **Uniqueness.** The argument does not just check that e^{kx} and $\tanh kx$ work; it shows they are the *only* possibilities. That is what makes a functional equation a definition.

What's really going on

Velocities in relativity

In everyday life, if a train moves at speed u and you walk forwards inside it at speed v, your speed relative to the ground is $u+v$. Einstein showed in 1905 that this is only an approximation. The correct rule, with c the speed of light, is

$$u\oplus v=\frac{u+v}{1+uv/c^2}.$$

Measuring speeds as fractions of c, this is exactly the rule for g. Our result says that if we write each velocity as $v = \tanh\phi$, then velocities combine by *adding* the ϕs. The quantity $\phi = \operatorname{artanh} v$ is called the *rapidity*. Each boost from a rocket engine adds a fixed amount of rapidity, but since $\tanh\phi < 1$ for every ϕ, the velocity creeps closer and closer to the speed of light without ever reaching it (Figure 3.8.1). Combining $0.9c$ with $0.9c$ gives not $1.8c$ but $\frac{1.8}{1.81}c \approx 0.9945c$.

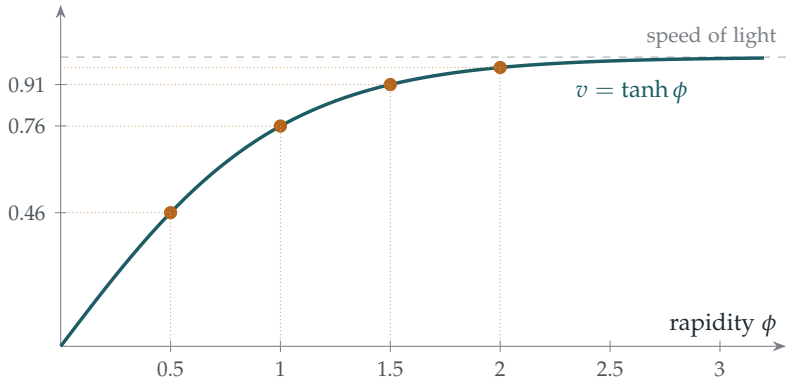


Figure 3.8.1 Velocity (as a fraction of the speed of light) against rapidity. Equal steps in rapidity, which is what equal boosts from an engine provide, give smaller and smaller increases in velocity: 0.46, 0.76, 0.91, 0.96, ..., never reaching 1.

Cauchy's functional equation

The simplest functional equation is $f(x+y) = f(x) + f(y)$, studied by Cauchy in 1821. If f is differentiable, the method of this question shows $f(x) = kx$. In fact continuity is enough. But without any regularity assumption at all, there are other, utterly wild solutions, whose graphs are scattered densely over the whole plane. Georg Hamel constructed them in 1905; the construction needs the *axiom of choice*, and it is impossible to write any of them down explicitly. This is one reason the question insists that f is differentiable: the hypothesis is not a formality.

Homomorphisms

$f(x+y) = f(x)f(y)$ says that f turns the addition of real numbers into the multiplication of positive real numbers, respecting the structure of both. In the language of first-year algebra, f is a *group homomorphism* from $(\mathbb{R}, +)$ to $(\mathbb{R}_{>0}, \times)$, and the logarithm is its inverse. Many of the most important functions in mathematics are characterised in this way, by the structure they preserve. In Section 7.6 we meet the geometric meaning of tanh: it measures "hyperbolic angle", and its addition law comes from composing hyperbolic rotations.

The thinking skill

Define a function by its properties

Instead of a formula, describe a function by a rule it obeys, and prove that only one function obeys it.

This is how mathematicians often think about exp, sin, cos and many other functions: not as formulas, but as the unique solutions of functional or differential equations. It explains *why* they matter, not just how to calculate them.

Try this next

1 **A tangent rule.** Suppose g is differentiable, $g'(0) = k \neq 0$, and $g(x+y) = \dfrac{g(x)+g(y)}{1-g(x)g(y)}$ whenever both sides are defined. Show that $g'(x) = k(1+g(x)^2)$ and deduce that $g(x) = \tan kx$ near $x = 0$.

2 **Cauchy's equation.** If f is differentiable and $f(x+y) = f(x)+f(y)$ for all x, y, show that $f(x) = kx$.

3 **Without derivatives.** HARDERSuppose only that $f(x+y) = f(x)f(y)$ and $f(1) = a > 0$. Show that $f(n) = a^n$ for positive integers n, then that $f(1/n) = a^{1/n}$, and then that $f(q) = a^q$ for every positive rational q. (Why is continuity needed to go further?)

4 **Rocket science.** A rocket can increase its speed by $0.5c$ relative to its current frame each time it fires. Starting from rest, what is its speed (relative to the start) after 2, 3 and 4 firings? Use rapidities.

Further reading

- **Christopher Small, *Functional Equations and How to Solve Them*** (Springer). An accessible introduction aimed at problem-solvers.
- **Edwin Taylor & John Wheeler, *Spacetime Physics*** (W. H. Freeman). The classic introduction to special relativity, using rapidity throughout.
- **Tom Körner, *Calculus for the Ambitious*** (Cambridge University Press). Defines the exponential and trigonometric functions from their properties, exactly in the spirit of this section.

3.9 Inventing a New Function

The question

(i) Sketch the curve $y = xe^x$, giving the coordinates of any stationary points.

(ii) The function f is defined by $f(x) = xe^x$ for $x \geq a$, where a is the minimum possible value such that f has an inverse function. What is the value of a?

Let g be the inverse of f. Sketch the curve $y = g(x)$.

(iii) For each of the following equations, find a real root in terms of a value of the function g, or demonstrate that the equation has no real root. If the equation has two real roots, determine whether the root you have found is greater than or less than the other root.

(a) $e^{-x} = 5x$ **(b)** $2x \ln x + 1 = 0$ **(c)** $3x \ln x + 1 = 0$ **(d)** $x = 3 \ln x$

(iv) Given that the equation $x^x = 10$ has a unique positive root, find this root in terms of a value of the function g.

The hook

Can you solve $e^{-x} = 5x$? Not with any function on your calculator. Rearranging gets nowhere: the x is stuck both inside and outside the exponential. You could find a numerical answer by trial and improvement, but there is no "formula".

Or is there? This question takes a bold step: if a family of equations has no solution in terms of known functions, *invent a new function* that solves them. That is exactly what mathematicians did with $\sqrt{x}$ (the inverse of x^2), with $\ln x$ (the inverse of e^x) and with $\arcsin x$. The new function here is the inverse of xe^x, now called the *Lambert W function*. Once you have it, a surprising range of equations, from $x^x = 10$ to a law of physics, can be solved exactly.

Before you start

Ask yourself What does it actually mean to "solve" an equation? Is $x = \sqrt{2}$ a solution of $x^2 = 2$, or just a name for the solution?

Both. $\sqrt{2}$ is a name for "the positive number whose square is 2"; we accept it as an answer because the function $\sqrt{\ }$ is well understood, easy to compute and has known

properties. Whether an answer counts as a "formula" depends on which functions we have agreed to name. This question adds one more to the list.

Working it through

STAGE 1 **The curve $y = xe^x$**

$\frac{dy}{dx} = e^x + xe^x = (1+x)e^x$, which is zero only at $x = -1$, where $y = -e^{-1}$. The derivative is negative for $x < -1$ and positive for $x > -1$, so this is a minimum.

> **Ask yourself** What happens to xe^x as $x \to +\infty$ and as $x \to -\infty$? Where does the curve cross the axes?

As $x \to \infty$, $xe^x \to \infty$. As $x \to -\infty$, $xe^x \to 0$ from below, because the exponential decay beats the linear growth. The curve passes through the origin. So it comes in from just below the negative x-axis, dips to its minimum $(-1, -\frac{1}{e})$, then rises through $(0,0)$ and shoots upwards (Figure 3.9.1).

STAGE 2 **The inverse**

To have an inverse, f must be one-to-one, so it must be restricted to a region where it is strictly increasing or strictly decreasing. The largest such interval containing the right-hand part is $[-1,\infty)$, so $a = -1$. On $[-1,\infty)$, f increases from $-\frac{1}{e}$ to ∞, so the inverse g is defined on $[-\frac{1}{e},\infty)$, increases from $g(-\frac{1}{e}) = -1$, passes through $g(0) = 0$, and grows slowly (for large x, even more slowly than $\ln x$). Its graph is the reflection of the right-hand part of $y = xe^x$ in the line $y = x$, with a vertical tangent at $(-\frac{1}{e}, -1)$.

The defining property of g is

$$g(z)\, e^{g(z)} = z, \qquad g(z) \geq -1.$$

So the strategy for every part of (iii) is the same: rearrange the equation into the form (something) $e^{(\text{the same something})} = (\text{a number})$.

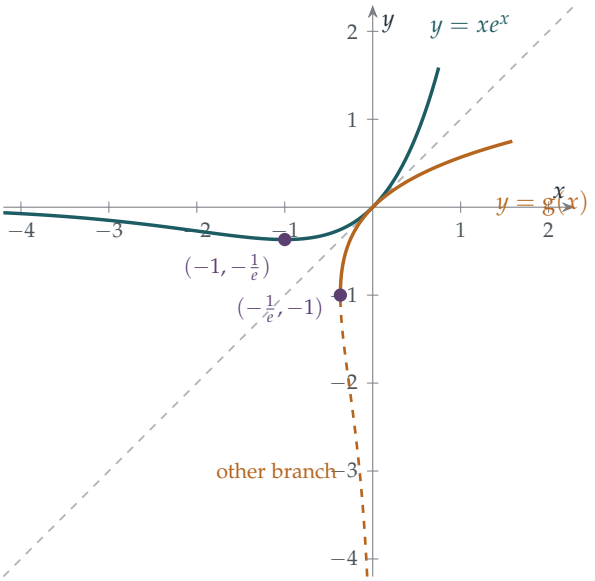


Figure 3.9.1 The curve $y = xe^x$ (teal) has a minimum at $(-1, -\frac{1}{e})$. Restricting to $x \geq -1$ and reflecting in $y = x$ gives the inverse g (amber, solid), the Lambert W function. The dashed amber curve is the reflection of the left-hand part of $y = xe^x$: a second branch, needed for the second roots in part (iii).

STAGE 3 Four equations

(a) $e^{-x} = 5x$. Multiply by e^x and divide by 5: $xe^x = \frac{1}{5}$. So $x = g(\frac{1}{5}) \approx 0.169$. Since xe^x is negative for $x < 0$, any root is positive, and xe^x is increasing for $x > 0$, so this is the only root.

Ask yourself For (b) and (c), the unknown appears as $x \ln x$. What substitution turns $x \ln x$ into something of the form ue^u?

Put $x = e^u$, so $\ln x = u$ and $x \ln x = ue^u$.

(b) $2x \ln x + 1 = 0$ becomes $ue^u = -\frac{1}{2}$. But $ue^u \geq -\frac{1}{e} \approx -0.368$ for all real u, and $-\frac{1}{2} < -\frac{1}{e}$. So there is **no real root**.

(c) $3x \ln x + 1 = 0$ becomes $ue^u = -\frac{1}{3}$, and $-\frac{1}{e} < -\frac{1}{3} < 0$. A horizontal line at height $-\frac{1}{3}$ cuts $y = ue^u$ *twice*, once on each side of the minimum at $u = -1$. The root on the right is $u = g(-\frac{1}{3})$, giving

$$x = e^{g(-1/3)} \approx 0.538.$$

The other root has $u < -1 < g(-\frac{1}{3})$, and $x = e^u$ is increasing in u, so the root we have found is the **greater** of the two. (The other is about 0.220.)

Pause and try Now try (d), $x = 3 \ln x$. (*Hint:* exponentiate and aim for (something) $e^{\text{something}}$ with the something being a multiple of x.)

(d) $x = 3\ln x$ means $e^{x/3} = x$, so $xe^{-x/3} = 1$, and multiplying by $-\frac{1}{3}$,

$$\left(-\frac{x}{3}\right) e^{-x/3} = -\frac{1}{3}.$$

So $-\frac{x}{3}$ is one of the two solutions of $ue^u = -\frac{1}{3}$ found in (c). Taking the one given by g, $x = -3\,\mathrm{g}\left(-\frac{1}{3}\right) \approx 1.857$. The other root comes from the other solution $u < \mathrm{g}(-\frac{1}{3})$, which gives a larger $x = -3u$. So this time the root we have found is the **smaller** of the two. (The other is about 4.536.) ■

STAGE 4 $x^x = 10$

Taking logarithms, $x\ln x = \ln 10$. With $x = e^u$ this is $ue^u = \ln 10$, so $u = \mathrm{g}(\ln 10)$ and

$$x = e^{\mathrm{g}(\ln 10)} \approx 2.506.$$

Since $\mathrm{g}(z)e^{\mathrm{g}(z)} = z$, we can also write this as $x = \dfrac{\ln 10}{\mathrm{g}(\ln 10)}$. ■

Looking back

- **Name the missing function.** When an equation cannot be solved with known functions, defining a new function as an inverse is a legitimate, and often illuminating, move.
- **Rearrange to a standard form.** Every part of (iii) and (iv) came down to $ue^u =$ constant, using a substitution such as $x = e^u$ or $u = -\frac{x}{3}$.
- **Count roots with the graph.** Whether $ue^u = c$ has zero, one or two solutions depends only on where c sits relative to the minimum value $-\frac{1}{e}$.

What's really going on

A function with a long history and a recent name

Johann Heinrich Lambert studied equations of this type in 1758, and Euler took them further in 1783. But the function had no standard name until the 1990s, when the developers of the computer algebra system Maple needed one. In 1996 Robert Corless, Gaston Gonnet, David Hare, David Jeffrey and Donald Knuth published a paper, "On the Lambert W function", that collected its properties and applications. It has been cited thousands of times, and W is now built into most mathematical software. The function g in the question is the *principal branch* W_0; the dashed curve in Figure 3.9.1, which gave the second roots in (c) and (d), is the other real branch W_{-1}.

Where W appears

- **Wien's displacement law.** The wavelength at which a hot body radiates most strongly is inversely proportional to its temperature. The constant comes from solving

$5(1-e^{-x})=x$, whose positive solution is $x=5+W(-5e^{-5})\approx 4.965$. That is why the Sun, at about 5800 K, looks yellow-white, and why a cooler red dwarf looks red.

- **Power towers.** The infinite tower $x^{x^{x^{\cdot^{\cdot^{\cdot}}}}}$ converges for $e^{-e}\le x\le e^{1/e}$, and its value is $\dfrac{W(-\ln x)}{-\ln x}$.
- **Population and epidemic models.** The final size of an epidemic in the simplest model, and the time to reach a given population in some growth models, are both given by W.

Elementary functions

Is W really new, or could it be written using exp, ln, roots and the trigonometric functions? It is genuinely new: a theorem going back to Joseph Liouville in the 1830s shows that W is not an *elementary function*. The same theory shows that some innocent-looking integrals, such as $\int e^{-x^2}\,dx$, cannot be done in elementary terms, which is why the normal distribution needs its own special function.

The thinking skill

Inverse functions as solutions

An equation $f(x)=c$ is solved by $x=f^{-1}(c)$, provided f is restricted to make it one-to-one. The restriction determines which solution you get.

Many equations that look different are the same equation in disguise; finding the substitution that reveals the common form is the key step.

Try this next

1 **Two, four, and one more.** The equation $x^2=2^x$ has the obvious solutions $x=2$ and $x=4$ (compare Section 3.5). Show that it also has a negative solution, and that this solution is
$$x=-\frac{2}{\ln 2}\,g\!\left(\frac{\ln 2}{2}\right)\approx -0.767.$$

2 **A derivative.** By differentiating $g(x)e^{g(x)}=x$ implicitly, show that, for $x\neq 0$,
$$g'(x)=\frac{g(x)}{x\bigl(1+g(x)\bigr)}.$$
What happens at $x=-\frac{1}{e}$?

3 **An integral.** Use the substitution $x=we^{w}$ to show that $\displaystyle\int_0^e g(x)\,dx=e-1$.

4 The tower. Assuming the tower $T = \sqrt{2}^{\sqrt{2}^{\sqrt{2}^{\cdots}}}$ converges, show that $T = \sqrt{2}^{T}$, and deduce that $T = 2$ or $T = 4$. Which is it? (Compute a few terms of the sequence $t_1 = \sqrt{2}$, $t_{n+1} = \sqrt{2}^{t_n}$, and think about the cobweb diagrams of Section 2.2.)

Further reading

- **Robert Corless et al., "On the Lambert W function"**, *Advances in Computational Mathematics* 5 (1996). The paper that named the function. The first sections are readable with this section behind you.
- **Brian Hayes, "Why W?"**, *American Scientist* 93 (2005). A lively essay on the function's history and uses. Free online.
- **Brian Conrad, "Impossibility theorems for elementary integration"**. Lecture notes, available online, explaining Liouville's theory. Advanced.

What We Have Learned

We began with a "proof" that $1 = 2$ and three questions: how to know an argument is correct, how to prove inequalities, and what a function really is. Here is what the chapter's nine questions have taught us.

1. How do you know an argument is correct?

Section 3.1 gave us a checklist. At every step, ask three things: *is the calculation right? does the step follow logically? what has been assumed without being said?* The errors there were typical. Comparing the values at two stationary points says nothing about which is a maximum. Stationary values do not bound a function on an infinite domain. The fundamental theorem of calculus needs a continuous integrand. And an argument can arrive at a "correct-looking" answer by accident.

The same vigilance ran through the rest of the chapter. Section 3.7 showed by counterexample that dropping a hypothesis ($f(0) = f(\pi) = 0$) breaks an inequality. In Section 3.8, the assumption that the function is differentiable was essential: without it, Cauchy's functional equation has wild solutions that nobody can write down. And in Section 3.6 and Section 3.3, the equality conditions told us whether a bound was best possible, which is the kind of question a careful mathematician always asks.

2. How do you prove an inequality?

We met the classical inequalities from several directions, and saw how they are related.

- **Squares are non-negative.** The two-number AM–GM is just $(a - b)^2 \geq 0$. Cauchy's forward–backward induction built the general case from it (Section 3.2).
- **Convexity.** A concave curve lies above its chords. That one fact gives Jensen's inequality, and with the logarithm it gives AM–GM in a line; with $\sin x$ it gives a sharp inequality for triangles (Section 3.3).
- **Areas.** The area under a curve plus the area beside it covers a rectangle. That gives Young's inequality, which leads to Hölder and Cauchy–Schwarz (Section 3.4).
- **Shape of a graph.** Rewriting a^b versus b^a as a comparison of $\frac{\ln x}{x}$ at two points settled π^e versus e^π (Section 3.5).
- **Interpolation and the triangle inequality.** Writing the derivative in terms of values, then bounding term by term, gave Markov's inequality for quadratics (Section 3.6).
- **Test functions.** Applying a general inequality to a cleverly chosen function gave $\pi^2 \leq 10$ and $\pi \leq \frac{22}{7}$ (Section 3.7).

Choosing among these is a skill that comes with practice. A rough guide: products and sums suggest AM–GM; averages of function values suggest Jensen; a function and its inverse suggest Young; a comparison of two expressions of the same form suggests finding the single function behind them.

3. What is a function, really?

The last two sections answered this in two ways. In Section 3.8, e^{kx} and $\tanh kx$ were not given by formulas at all; they were the *only* differentiable functions obeying certain addition laws. That is a deeper kind of understanding than a formula: it explains why these functions keep appearing, in compound interest and in Einstein's velocity addition. In Section 3.9 we went further and *invented* a function, the inverse of xe^x, because the functions we had were not enough to solve a natural family of equations. Every function on your calculator was once new in the same way.

4. A toolkit for rigour and inequalities

Method	Where you used it
Checking every step and every hypothesis	Plausible wrong arguments (3.1)
Local versus global behaviour	Stationary points and limits at infinity (3.1)
Forward–backward induction	AM–GM (3.2)
Padding with extra terms	AM–GM (3.2, 3.3)
Jensen's inequality	Triangles; AM–GM (3.3)
Comparing areas	Young's inequality (3.4)
Separating variables into one function	π^e versus e^π (3.5)
Triangle inequality, equality cases	Markov's inequality (3.6)
Test functions	$\pi^2 \leq 10$ (3.7)
Functional equations	e^x and $\tanh x$ (3.8)
Inverse functions and substitutions	The W function (3.9)

Where next

- The calculus of variations, glimpsed in Section 3.7, is the subject of Section 4.7.
- The geometry behind tanh and its addition law appears in Section 7.6.
- Chapter 4 takes the careful attitude of this chapter to integration itself, starting from what an integral actually is.

If you want to go further before university

- **J. Michael Steele, *The Cauchy–Schwarz Master Class***: the best introduction to inequalities, built entirely from problems.

- **Daniel Velleman, *How to Prove It***: a systematic guide to writing correct proofs.
- **Michael Spivak, *Calculus***: rigorous calculus from the ground up; many Cambridge students read it the summer before they arrive.
- **E. A. Maxwell, *Fallacies in Mathematics***: for practice in spotting what has gone wrong.

Von Neumann joked that in mathematics you don't understand things, you just get used to them. There is some truth in it: the precise definitions and careful arguments of this chapter can feel strange at first. But getting used to them is exactly what turns a student who can calculate into a mathematician who can be sure. The proof that $1 = 2$ will never fool you again; the aim of a mathematics degree is to make every subtler error just as easy to see.

Answers to Try This Next

These are brief answers and hints, not full solutions: they give the result and the key step, so that you can check your own work. Try each exercise properly before looking.

3.1 Spot the Flaw

1. As x runs from -1 to 1 it passes through 0, where $t = 1/x$ is undefined: the substitution is not continuous on the interval. Correctly, $\int_{-1}^{1} \frac{\mathrm{d}x}{(1+x^2)^2} = \frac{1}{2} + \frac{\pi}{4}$ (use $x = \tan\theta$), while $\int_{-1}^{1} \frac{-t^2}{(1+t^2)^2}\,\mathrm{d}t = \frac{1}{2} - \frac{\pi}{4}$.

2. (a) True: both sides have logarithm $\ln a \ln b$. (b) False: at $\theta = 0$, $\cos 0 = 1 \neq \sin 1$. (In fact they are never equal, since $|\sin\theta \pm \cos\theta| \leq \sqrt{2} < \frac{\pi}{2}$.) (c) False: a non-constant polynomial is unbounded, and a constant cannot be within 10^{-6} of both 1 and -1. (d) True: $x^4 + x^{-4} \geq 2$ by AM–GM, with equality at $x = 1$.

3. The step from $n = 1$ to $n = 2$ fails: the "first 1" and "last 1" of two horses do not overlap.

4. $\int_{\varepsilon}^{1} x^{-1/2}\,\mathrm{d}x = 2 - 2\sqrt{\varepsilon} \to 2$, so by symmetry the integral exists (as a limit) and equals 4.

3.2 AM–GM, Backwards

1. If $a_1, \ldots, a_{n-1}$ have mean A and geometric mean G, add A as an nth number. The mean is still A, so $(G^{n-1}A)^{1/n} \leq A$, which gives $G \leq A$.

2. The $n+1$ numbers have mean $1 + \frac{1}{n+1}$ and geometric mean $(1 + \frac{1}{n})^{n/(n+1)}$, and they are not all equal, so the inequality is strict; raise both sides to the power $n+1$.

3. The faces have areas ab, bc, ca, with $S = 2(ab + bc + ca)$. AM–GM gives $\frac{S}{6} \geq (abc)^{2/3} = V^{2/3}$, with equality when $ab = bc = ca$, a cube.

4. AM–GM for $1/a_i$: $\frac{1}{n}\sum \frac{1}{a_i} \geq \frac{1}{G}$. Take reciprocals.

3.3 The Power of Convexity

1. If $\mathrm{f}'' < 0$ the gradient decreases, so the curve lies above each chord. With $t = \frac{1}{3}$: $\mathrm{f}\big(\frac{u+v+w}{3}\big) \geq \frac{1}{3}\mathrm{f}(u) + \frac{2}{3}\mathrm{f}\big(\frac{v+w}{2}\big) \geq \frac{1}{3}\big(\mathrm{f}(u) + \mathrm{f}(v) + \mathrm{f}(w)\big)$.

2. $(\ln\sin x)'' = -\operatorname{cosec}^2 x < 0$, so $\ln\sin$ is concave on $(0, \pi)$. Jensen gives $\frac{1}{3}\ln(\sin A \sin B \sin C) \leq \ln\sin\frac{\pi}{3}$, so the product is at most $\big(\frac{\sqrt{3}}{2}\big)^3$.

3. $(\tan x)'' = 2\sec^2 x \tan x > 0$ on $(0, \frac{\pi}{2})$; Jensen gives $\tan A + \tan B + \tan C \geq 3\tan\frac{\pi}{3} = 3\sqrt{3}$.

4. $(x^2)'' = 2 > 0$; Jensen gives the inequality, which says that $\frac{1}{n}\sum a_i^2 - \bar{a}^2$, the variance, is non-negative.

3.4 Areas That Prove Inequalities

1. With $\mathrm{f} = \sin$ on $[0, \frac{\pi}{2}]$, equality holds when $b = \mathrm{f}(a)$; take $a = \arcsin t$, $b = t$: $t\arcsin t = (1 - \cos(\arcsin t)) + \int_0^t \arcsin y\,\mathrm{d}y$. Differentiating the right-hand side of the result gives $\arcsin t + \frac{t}{\sqrt{1-t^2}} - \frac{t}{\sqrt{1-t^2}} = \arcsin t$.

2. Young with $p = q = 2$ gives $ab \leq \frac{1}{2}(a^2 + b^2)$. If $\sum a_i^2 = \sum b_i^2 = 1$, adding gives $\sum a_i b_i \leq 1$. In general, apply this to $a_i / \sqrt{\sum a_j^2}$ and $b_i / \sqrt{\sum b_j^2}$.

3. $\int_0^a (e^x - 1)\,\mathrm{d}x = e^a - 1 - a$ and $\int_0^b \ln(1+y)\,\mathrm{d}y = (1+b)\ln(1+b) - b$. Equality holds when $b = e^a - 1$.

4. $t = 1$: 1.571 against 1; $t = 2$: 0.524 against 0.268; $t = 10$: 0.1002 against 0.0501. For large t, $t - \sqrt{t^2-1} = \frac{1}{t+\sqrt{t^2-1}} \approx \frac{1}{2t}$. The bound comes from $a = 0$, where $(*)$ just says $0 \leq \int_0^b \arcsin y\,\mathrm{d}y$; for small b this area is about $\frac{b^2}{2}$, half of $b\arcsin b \approx b^2$, and discarding it loses the factor 2.

3.5 π^e or e^π?

1. $n^m = m^n$ means $h(n) = h(m)$. h increases on $(1, e)$ and decreases afterwards, so one of the two is less than e: it must be 2 (as $h(1) = 0$ is not taken again), and then the other is 4. The only pair is $\{2, 4\}$.

2. $e < 3 < \pi$ and h decreases there, so $h(3) > h(\pi)$, i.e. $3^\pi > \pi^3$. Both $\frac{9}{4}$ and $\sqrt{5}$ are less than e, with $\frac{9}{4} > \sqrt{5}$, so $h(\frac{9}{4}) > h(\sqrt{5})$, giving the second. For $1 < x < 2$: $h(x) < h(2) = h(4) < h(x+2)$ since $3 < x + 2 < 4$, so $x^{x+2} < (x+2)^x$.

3. $9^{\sqrt{2}} = 3^{2\sqrt{2}}$ and $\sqrt{2}^9 = 16\sqrt{2} = (2\sqrt{2})^3$. Since $e^2 < 8$, $e < 2\sqrt{2} < 3$, where h decreases, so $h(2\sqrt{2}) > h(3)$, which gives $(2\sqrt{2})^3 > 3^{2\sqrt{2}}$: $\sqrt{2}^9$ is larger.

4. $e^{\pi/e - 1} > \frac{\pi}{e}$, so $e^{\pi/e} > \pi$; raise to the power e.

3.6 A Bound on Slopes

1. Both sides are quadratics agreeing at $x = -1, 0, 1$, so they are equal (Lagrange interpolation). Differentiating gives $\mathrm{f}(-1)(x - \frac{1}{2}) - 2\mathrm{f}(0)x + \mathrm{f}(1)(x + \frac{1}{2})$.

2. $|\mathrm{f}'(x)| \leq |x - \frac{1}{2}| + 2|x| + |x + \frac{1}{2}| \leq 4$ on $[-1, 1]$, with equality only at $x = \pm 1$, and only if $\mathrm{f}(1)$, $\mathrm{f}(-1)$, $-\mathrm{f}(0)$ are all equal to 1 or all equal to -1. A quadratic is determined by these three values, giving $\pm(2x^2 - 1)$.

3. $T_3(\cos\theta) = \cos 3\theta$ gives $4x^3 - 3x$, and $T_3'(1) = 12 - 3 = 9$. From $\cos(n+1)\theta + \cos(n-1)\theta = 2\cos\theta\cos n\theta$, $T_{n+1} = 2xT_n - T_{n-1}$, and $T_4 = 8x^4 - 8x^2 + 1$.

4. $-\sin\theta\, T_n'(\cos\theta) = -n\sin n\theta$. As $\theta \to 0$, $\frac{n\sin n\theta}{\sin\theta} \to n^2$.

3.7 $\pi^2 \le 10$

1. $\int_0^\pi \mathrm{f}^2 = \frac{\pi^9}{630}$ and $\int_0^\pi (\mathrm{f}')^2 = \frac{2\pi^7}{105}$, so $(*)$ gives $\pi^2 \le 12$. The quartic is too flat near the ends and too peaked in the middle compared with $\sin x$, so its ratio of slope to size is further from the optimum.
2. $\int_0^\pi (\sin x + \sin 2x)^2 = \frac{\pi}{2} + \frac{\pi}{2} = \pi$ (the cross term integrates to 0), and $\int_0^\pi (\cos x + 2\cos 2x)^2 = \frac{\pi}{2} + 2\pi = \frac{5\pi}{2}$.
3. Put $\mathrm{g}(x) = \mathrm{f}(\pi x/L)$. The left side scales by $\frac{L}{\pi}$ and $\int(\mathrm{g}')^2$ by $\frac{\pi}{L}$. For strings, the lowest frequency is proportional to $\frac{1}{L}$: halving the length doubles the pitch.
4. The bound $\pi^2 \le \int(\mathrm{f}')^2 / \int \mathrm{f}^2$ (in suitable units) is minimised at $\lambda \approx 1.133/\pi^2$, giving $\pi^2 \le 9.86975$, against the true value 9.86960.

3.8 Functions from Their Addition Laws

1. Putting $x = y = 0$ gives $\mathrm{g}(0) = 0$. Differentiating with respect to y at $y = 0$ gives $\mathrm{g}'(x) = k(1 + \mathrm{g}(x)^2)$, so $\frac{\mathrm{d}}{\mathrm{d}x}\arctan \mathrm{g}(x) = k$ and $\mathrm{g}(x) = \tan kx$.
2. Differentiating with respect to y gives $\mathrm{f}'(x+y) = \mathrm{f}'(y)$, so f' is a constant k; and $\mathrm{f}(0) = 0$ (put $x = y = 0$).
3. $\mathrm{f}(x) = \mathrm{f}(x/2)^2 \ge 0$, and f is never 0 (or it would be 0 everywhere). Induction gives $\mathrm{f}(n) = a^n$; $\mathrm{f}(1/n)^n = \mathrm{f}(1) = a$ with $\mathrm{f}(1/n) > 0$ gives $a^{1/n}$; then $\mathrm{f}(m/n) = \mathrm{f}(1/n)^m$. Without continuity the values at irrationals are unconstrained: there are strange solutions, built using a "Hamel basis", that are not a^x.
4. Each firing adds rapidity $\phi = \operatorname{artanh}\frac{1}{2}$, so after k firings the speed is $\tanh k\phi = \frac{3^k-1}{3^k+1}$: $0.8c$, $\frac{13}{14}c \approx 0.929c$ and $\frac{40}{41}c \approx 0.976c$.

3.9 Inventing a New Function

1. At $x = -1$, $x^2 - 2^x = \frac{1}{2} > 0$; at $x = 0$ it is -1: so there is a negative root. Writing $x = -2u/\ln 2$ turns $x^2 = 2^x$ (with $x < 0$, taking $x = -2^{x/2}$) into $ue^u = \frac{\ln 2}{2}$, so $u = \mathrm{g}\left(\frac{\ln 2}{2}\right)$ and $x \approx -0.7667$.
2. $\mathrm{g}' e^{\mathrm{g}}(1 + \mathrm{g}) = 1$ and $e^{\mathrm{g}} = x/\mathrm{g}$. At $x = -\frac{1}{e}$, $\mathrm{g} = -1$ and the slope is infinite (the graph is vertical there).
3. With $x = we^w$, $\mathrm{d}x = (1+w)e^w\,\mathrm{d}w$ and w runs from 0 to 1: $\int_0^1 w(1+w)e^w\,\mathrm{d}w = \left[(w^2 - w + 1)e^w\right]_0^1 = e - 1$.
4. $T = \sqrt{2}^{\,T}$ means $\frac{\ln T}{T} = \frac{\ln 2}{2}$, so $T = 2$ or 4. The terms $1.414, 1.633, 1.761, \ldots$ increase and stay below 2 (if $t_n < 2$ then $t_{n+1} < \sqrt{2}^{\,2} = 2$), so $T = 2$; the fixed point 4 is repelling.

CHAPTER

4

Calculus Done Properly

Integration is where analysis, geometry and number theory meet, and where some of the most beautiful formulas in mathematics live.

IN THIS CHAPTER

4.1	What Is an Integral?	STEP III 1989
4.2	Building Trigonometry from an Integral	STEP III 2017
4.3	22/7 Is Bigger Than π	STEP I 1997
4.4	Wallis, Beta and the Factorial of a Half	STEP II 1995
4.5	The Sophomore's Dream	STEP III 2009
4.6	Integrals With No Formula	STEP III 2016
4.7	The Calculus of Variations	STEP III 2014
4.8	Slicing a Spherical Loaf	STEP I 2001
4.9	Holditch's Theorem	STEP III 2011
A	Answers to the exercises	p. 303

More Than Antiderivatives

I turn away with fright and horror from the lamentable evil of functions which do not have derivatives.

— Charles Hermite, in a letter of 1893

Ask a school student what integration is, and you will probably hear "the opposite of differentiation". That is a wonderful fact, the *fundamental theorem of calculus*, and it makes most of the integrals you meet at school routine. But it is a theorem about integrals, not a definition of them, and it hides most of what makes integration interesting.

Integration is older than differentiation by about two thousand years. Archimedes found the area of a parabolic segment and the surface area of a sphere by summing thin slices, long before anyone thought of derivatives. And integration reaches places differentiation cannot. Every reasonable function can be differentiated by mechanical rules, but most functions *cannot* be integrated in a formula, which raises a natural question: when can they, and what do we do when they can't? Integrals define new functions, prove inequalities between numbers, compute areas and volumes that have no other description, and give exact values of infinite sums and products. This chapter is about all of these.

What an integral really is

The chapter begins at the beginning, with the definition (Section 4.1). An integral is a limit of sums of thin rectangles; computing even $\int_0^a x^5\,dx$ from that definition, without antiderivatives, is a substantial piece of work, and it makes you appreciate the fundamental theorem. With a precise definition comes a precise question: which functions are integrable? The answer, and the problems it raised, led in the early twentieth century to Lebesgue's theory of integration, the foundation of modern probability.

Integrals that prove things

Some of the most surprising results in this chapter use integrals to prove facts that seem to have nothing to do with areas. A single integral of a positive function equals $\frac{22}{7} - \pi$, which proves $\pi < \frac{22}{7}$ in one line (Section 4.3). Powers of $\sin t$ lead to Wallis's infinite product for π and to the value $\left(\frac{1}{2}\right)! = \frac{\sqrt{\pi}}{2}$ (Section 4.4). The integral of x^x turns into the sum $\sum (-1)^{n+1} n^{-n}$ (Section 4.5). And a single integral, $\int_0^x \frac{du}{1+u^2}$, can be used to build trigonometry from scratch (Section 4.2).

What you will meet at university

The first-year *Analysis* course proves the fundamental theorem of calculus properly, from the definition of the integral. Later courses build in several directions. *Measure theory* and the *Lebesgue integral* make integration work smoothly with limits and underpin probability. *Vector calculus* integrates over curves, surfaces and volumes, and Green's and Stokes's theorems relate them (Section 4.9 is a taste). *Complex analysis* evaluates real integrals by detours through the complex plane. The *calculus of variations* asks which function makes an integral as small as possible (Section 4.7), and leads to the principle of least action in physics. And *differential algebra* explains which integrals have formulas at all (Section 4.6).

The questions that run through this chapter

Three big questions

1. **What is an integral, and which functions have one?** How does the definition as a limit of sums connect with antiderivatives?
2. **What can integrals prove?** How can an area tell us about π, about factorials, about infinite sums?
3. **What can we do when there is no formula?** How do we know an integral has no formula, and what can we compute anyway?

Ideas in this chapter

§	Topic	Idea
4.1	The definition of the integral	Lower and upper sums; telescoping
4.2	Trigonometry from $\int \frac{du}{1+u^2}$	Substitutions that preserve an integral
4.3	$\frac{22}{7} > \pi$	An inequality as an integral of something positive
4.4	Wallis and $\left(\frac{1}{2}\right)!$	Reduction formulae, an invariant, a squeeze
4.5	$\int_0^1 x^x \, dx$	Series expansion and term-by-term integration
4.6	Integrals with no formula	Impossibility via polynomial algebra
4.7	The calculus of variations	Completing the square in an integral
4.8	Archimedes' hat-box theorem	Simplifying an integrand reveals geometry
4.9	Holditch's theorem	Area from a boundary integral; elimination

How to read this chapter

Several sections involve heavy but routine calculation: polynomial division, integration by parts, expanding products. Do these yourself, with pencil and paper, before reading the text. Then look for the idea that made the calculation work: an invariant, a symmetry, a cancellation. The calculation is what an examiner checks; the idea is what you will remember and use again.

4.1 What Is an Integral?

The question

(i) Prove that

$$\sum_{r=1}^{n} r(r+1)(r+2)(r+3)(r+4) = \tfrac{1}{6}n(n+1)(n+2)(n+3)(n+4)(n+5)$$

and deduce that

$$\sum_{r=1}^{n} r^5 < \tfrac{1}{6}n(n+1)(n+2)(n+3)(n+4)(n+5).$$

(ii) Prove that, if $n > 1$,

$$\sum_{r=0}^{n-1} r^5 > \tfrac{1}{6}(n-5)(n-4)(n-3)(n-2)(n-1)n.$$

(iii) Let f be an increasing function. If the limits

$$\lim_{n\to\infty} \sum_{r=0}^{n-1} \frac{a}{n} \mathrm{f}\Big(\frac{ra}{n}\Big) \qquad \text{and} \qquad \lim_{n\to\infty} \sum_{r=1}^{n} \frac{a}{n} \mathrm{f}\Big(\frac{ra}{n}\Big)$$

both exist and are equal, the definite integral $\int_0^a \mathrm{f}(x)\,\mathrm{d}x$ is defined to be their common value. Using this definition, prove that

$$\int_0^a x^5\,\mathrm{d}x = \tfrac{1}{6}a^6.$$

The hook

You have been integrating for years. You know that $\int_0^a x^5\,\mathrm{d}x = \frac{1}{6}a^6$ because "integration is the reverse of differentiation". But that is a theorem, not a definition. What *is* an integral?

The honest answer is: an area, defined as a limit of sums of thin rectangles. This question asks you to compute an integral from that definition alone, with no antiderivatives, no fundamental theorem, nothing but adding up rectangles and taking a limit. It is harder than you might expect, which is exactly why the fundamental theorem of calculus,

which makes it easy, is such a remarkable result.

Before you start

Ask yourself The two sums in part (iii) are sums of areas of rectangles of width $\frac{a}{n}$. For an increasing function f, which sum is an underestimate of the area under the curve, and which an overestimate?

The first sum uses the height at the *left* end of each strip, $x = \frac{ra}{n}$ for $r = 0, \dots, n-1$. For an increasing function that is the lowest point on the strip, so the rectangles fit under the curve: a *lower sum*. The second uses the right ends, $r = 1, \dots, n$, giving rectangles that poke above the curve: an *upper sum* (Figure 4.1.1). The definition says the integral exists if both sums approach the same limit as the strips get thinner.

For $f(x) = x^5$ both sums involve $\sum r^5$. Parts (i) and (ii) bound that sum above and below.

Working it through

STAGE 1 A sum that telescopes

Ask yourself Write $P(r) = r(r+1)(r+2)(r+3)(r+4)(r+5)$, a product of six consecutive integers. What is $P(r) - P(r-1)$?

$P(r)$ and $P(r-1) = (r-1)r(r+1)(r+2)(r+3)(r+4)$ share the five factors $r(r+1)(r+2)(r+3)(r+4)$, so

$$\begin{aligned} P(r) - P(r-1) &= r(r+1)(r+2)(r+3)(r+4)\big[(r+5) - (r-1)\big] \\ &= 6\,r(r+1)(r+2)(r+3)(r+4). \end{aligned}$$

Summing from $r = 1$ to n, the right-hand sides telescope to $P(n) - P(0) = P(n)$. So

$$\sum_{r=1}^{n} r(r+1)(r+2)(r+3)(r+4) = \tfrac{1}{6}P(n) = \tfrac{1}{6}n(n+1)(n+2)(n+3)(n+4)(n+5).$$

■

Since $r^5 < r(r+1)(r+2)(r+3)(r+4)$ for every $r \geq 1$, the inequality follows. ■

STAGE 2 A lower bound

Pause and try Adapt Stage 1 to a product of five consecutive integers going *down*: $r(r-1)(r-2)(r-3)(r-4)$. Find its sum from $r = 0$ to $n-1$, and compare it with r^5.

Let $Q(r) = (r-5)(r-4)(r-3)(r-2)(r-1)r$. Then $Q(r+1) - Q(r) = (r-4)(r-3)(r-2)(r-1)r\big[(r+1)-(r-5)\big] = 6\,(r-4)(r-3)(r-2)(r-1)r$. Summing from $r=0$ to $n-1$ telescopes to $Q(n) - Q(0) = Q(n)$, so

$$\sum_{r=0}^{n-1} (r-4)(r-3)(r-2)(r-1)r = \tfrac{1}{6}(n-5)(n-4)(n-3)(n-2)(n-1)n.$$

Now compare term by term. For $r = 0$ both sides are 0. For $1 \le r \le 4$ the product contains a zero factor, while $r^5 > 0$. For $r \ge 5$ every factor is positive and at most r, and some are strictly smaller, so the product is less than r^5. Since $n > 1$, the sum includes $r = 1$, where the inequality is strict, so

$$\sum_{r=0}^{n-1} r^5 > \tfrac{1}{6}(n-5)(n-4)(n-3)(n-2)(n-1)n. \qquad \blacksquare$$

STAGE 3 The integral

Ask yourself Write the lower and upper sums for $f(x) = x^5$ in terms of $\sum r^5$. What power of n appears in the denominator?

With $f(x) = x^5$,

$$L_n = \sum_{r=0}^{n-1} \frac{a}{n}\left(\frac{ra}{n}\right)^5 = \frac{a^6}{n^6}\sum_{r=0}^{n-1} r^5, \qquad U_n = \frac{a^6}{n^6}\sum_{r=1}^{n} r^5.$$

By parts (i) and (ii), and since $L_n < U_n$,

$$\frac{a^6}{6}\cdot\frac{(n-5)(n-4)\cdots n}{n^6} \;<\; L_n \;<\; U_n \;<\; \frac{a^6}{6}\cdot\frac{n(n+1)\cdots(n+5)}{n^6}.$$

Each of the two fractions is a product of six factors like $\frac{n-3}{n} = 1 - \frac{3}{n}$ or $\frac{n+3}{n} = 1 + \frac{3}{n}$, each tending to 1. So both outer bounds tend to $\frac{a^6}{6}$, and by the squeeze, so do L_n and U_n. Both limits exist and are equal, so by the definition

$$\int_0^a x^5\,dx = \tfrac{1}{6}a^6. \qquad \blacksquare$$

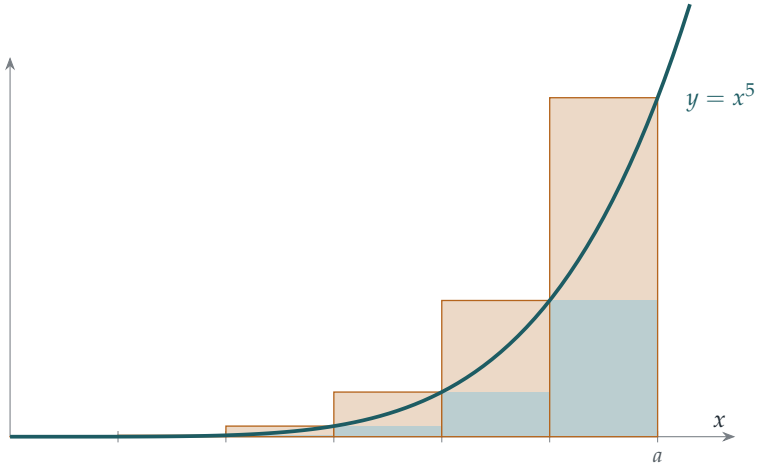


Figure 4.1.1 Lower and upper sums for x^5 on $[0, a]$ with $n = 6$ strips. The lower sum is the teal area (rectangles under the curve); the upper sum adds the amber pieces. The difference between them is exactly the area of the last, tallest rectangle, $\frac{a}{n}\,\mathrm{f}(a)$, which tends to 0 as n grows.

Looking back

- **Telescoping again.** The sums of products of consecutive integers collapsed, exactly like the sums and products in Chapter 2.
- **Bound, then squeeze.** We never found $\sum r^5$ exactly (it is $\frac{1}{12}n^2(n+1)^2(2n^2+2n-1)$); bounds on both sides were enough.
- **The definition does the work.** No antiderivative appeared anywhere. Everything followed from the definition of the integral as a limit of sums.

What's really going on

Riemann's definition

The idea of finding areas by summing thin strips goes back to Archimedes, who found the area under a parabola by a version of exactly this argument around 250 BC. Newton and Leibniz turned integration into the reverse of differentiation, and for a century and a half that was how most people thought about it. In 1854 Bernhard Riemann gave the definition that university courses still begin with: divide $[a, b]$ into small pieces, form sums of (width) $\times$ (height of f somewhere in the piece), and say f is integrable if all such sums approach one limit as the pieces shrink. The version in this question, with equal strips and left and right endpoints, is a special case that works for increasing functions; Jean-Gaston Darboux's "upper and lower sums" make the general case clean.

Not every function has an integral

With a precise definition, you can ask which functions are integrable. Every continuous function on a closed interval is, and every increasing function. But consider Dirichlet's function: $\mathrm{f}(x) = 1$ if x is rational and 0 if irrational. Every strip, however thin, contains both rationals and irrationals, so every upper sum is 1 and every lower sum is 0. They never agree, and $\int_0^1 \mathrm{f}$ does not exist in Riemann's sense. In 1902 Henri Lebesgue invented a more powerful integral that handles this function (its Lebesgue integral is 0) and behaves much better under limits. The Lebesgue integral is the foundation of modern probability theory, and you will meet it in your second or third year.

Discrete calculus

The telescoping in Stage 1 is calculus in disguise. The "rising factorial" $r(r+1)\cdots(r+k-1)$ behaves under summation exactly as x^k behaves under integration: summing it raises the number of factors by one and divides by the new number of factors, just as $\int x^k \,\mathrm{d}x = \frac{x^{k+1}}{k+1}$. This parallel between sums and integrals, differences and derivatives, is called the *calculus of finite differences*, and it is a powerful tool for evaluating sums.

The thinking skill

Work from the definition

When asked to prove something "using the definition", use nothing else: no theorems that depend on what you are proving, no familiar shortcuts.

It is surprisingly hard, and it shows you exactly what the shortcuts are worth. At university, the first weeks of every course work this way: definitions first, then theorems that let you stop working from definitions.

Try this next

1 **A limit as an integral.** (From STEP III, 1999.) Justify by a sketch that

$$\lim_{n\to\infty} \frac{1}{n} \sum_{m=1}^{n} \mathrm{f}\Big(1 + \frac{m}{n}\Big) = \int_1^2 \mathrm{f}(x)\,\mathrm{d}x.$$

Deduce that

$$\lim_{n\to\infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} \right) = \ln 2,$$

and evaluate $\lim_{n\to\infty} \left(\frac{n}{n^2+1} + \frac{n}{n^2+4} + \cdots + \frac{n}{n^2+n^2} \right)$.

2 **Archimedes' parabola.** Using $\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$ and the definition in part (iii), prove that $\int_0^a x^2\,\mathrm{d}x = \frac{1}{3}a^3$.

3 Rising factorials. Show that $\displaystyle\sum_{r=1}^{n} r(r+1)\cdots(r+k-1) = \frac{n(n+1)\cdots(n+k)}{k+1}$ for every positive integer k.

4 The exact sum. Use the identity $r(r+1)(r+2)(r+3)(r+4) = r^5 + 10r^4 + 35r^3 + 50r^2 + 24r$ and the known formulas for $\sum r^k$ with $k \le 4$ to find $\sum_{r=1}^{n} r^5$ exactly.

Further reading

- **David Bressoud, *A Radical Approach to Lebesgue's Theory of Integration*** (Cambridge University Press). The story from Riemann's integral to Lebesgue's, told through the problems that forced each step.
- **Michael Spivak, *Calculus*** (Cambridge University Press). Chapter 13 defines the integral by upper and lower sums, exactly as here.
- **Graham, Knuth & Patashnik, *Concrete Mathematics*** (Addison-Wesley). Chapter 2 develops the calculus of finite differences, with rising and falling factorials.

4.2 Building Trigonometry from an Integral

The question

In this question, you are not permitted to use any properties of trigonometric functions or inverse trigonometric functions.

The function T is defined for $x > 0$ by

$$\mathrm{T}(x) = \int_0^x \frac{1}{1+u^2}\,\mathrm{d}u,$$

and $\mathrm{T}_\infty = \displaystyle\int_0^\infty \frac{1}{1+u^2}\,\mathrm{d}u$ (which has a finite value).

(i) By making an appropriate substitution in the integral for $\mathrm{T}(x)$, show that

$$\mathrm{T}(x) = \mathrm{T}_\infty - \mathrm{T}(x^{-1}).$$

(ii) Let $v = \dfrac{u+a}{1-au}$, where a is a constant. Verify that, for $u \neq a^{-1}$,

$$\frac{\mathrm{d}v}{\mathrm{d}u} = \frac{1+v^2}{1+u^2}.$$

Hence show that, for $a > 0$ and $x < \dfrac{1}{a}$,

$$\mathrm{T}(x) = \mathrm{T}\left(\frac{x+a}{1-ax}\right) - \mathrm{T}(a).$$

Deduce that

$$\mathrm{T}(x^{-1}) = 2\mathrm{T}_\infty - \mathrm{T}\left(\frac{x+a}{1-ax}\right) - \mathrm{T}(a^{-1})$$

and hence that, for $b > 0$ and $y > \dfrac{1}{b}$,

$$\mathrm{T}(y) = 2\mathrm{T}_\infty - \mathrm{T}\left(\frac{y+b}{by-1}\right) - \mathrm{T}(b).$$

(iii) Use the above results to show that $\mathrm{T}(\sqrt{3}) = \frac{2}{3}\mathrm{T}_\infty$ and $\mathrm{T}(\sqrt{2}-1) = \frac{1}{4}\mathrm{T}_\infty$.

The hook

What is π? At school it is the ratio of a circle's circumference to its diameter, and sin and cos are defined by triangles. That is fine for geometry, but it leaves awkward questions. What exactly is "length" along a curve? Why does $\sin x$ have the Maclaurin series it does? How do we know the addition formulas hold for all angles, not just ones we can draw?

This question shows a completely different way in. Forget triangles and circles. Start from a single integral, $\mathrm{T}(x) = \int_0^x \frac{\mathrm{d}u}{1+u^2}$, which needs nothing but the definition of the integral from Section 4.1. Define π to be $2\mathrm{T}_\infty$. Then, using only substitutions, prove that $\mathrm{T}(\sqrt{3}) = \frac{\pi}{3}$ and $\mathrm{T}(\sqrt{2}-1) = \frac{\pi}{8}$, facts you know as $\arctan\sqrt{3} = 60^\circ$ and $\tan 22\frac{1}{2}^\circ = \sqrt{2}-1$. Trigonometry rebuilt from nothing.

Before you start

You know, secretly, that $\mathrm{T}(x) = \arctan x$ and $\mathrm{T}_\infty = \frac{\pi}{2}$. The question forbids you to use this, and that is the point: everything must come from the integral. But your secret knowledge is a useful guide.

> **Ask yourself** Which trigonometric identities do parts (i) and (ii) correspond to? (Think of $\arctan x + \arctan\frac{1}{x}$, and the formula for $\tan(A+B)$.)

Part (i) is $\arctan x + \arctan\frac{1}{x} = \frac{\pi}{2}$. Part (ii) is the addition formula $\tan(A+B) = \frac{\tan A+\tan B}{1-\tan A\tan B}$ in disguise: with $u = \tan A$ and $a = \tan B$, $v = \tan(A+B)$. The question proves these identities from the integral, without any geometry.

Working it through

STAGE 1 The reflection $u \mapsto 1/u$

Substitute $u = 1/w$ in $\mathrm{T}(x)$. Then $\mathrm{d}u = -\mathrm{d}w/w^2$, and $\dfrac{1}{1+u^2} = \dfrac{w^2}{w^2+1}$, so

$$\mathrm{T}(x) = \int_\infty^{1/x} \frac{w^2}{w^2+1}\cdot\left(-\frac{1}{w^2}\right)\mathrm{d}w = \int_{1/x}^\infty \frac{\mathrm{d}w}{1+w^2} = \mathrm{T}_\infty - \mathrm{T}(x^{-1}). \qquad \blacksquare$$

In other words, the area under $\frac{1}{1+u^2}$ from 0 to x equals the area from $\frac{1}{x}$ to ∞ (Figure 4.2.1).

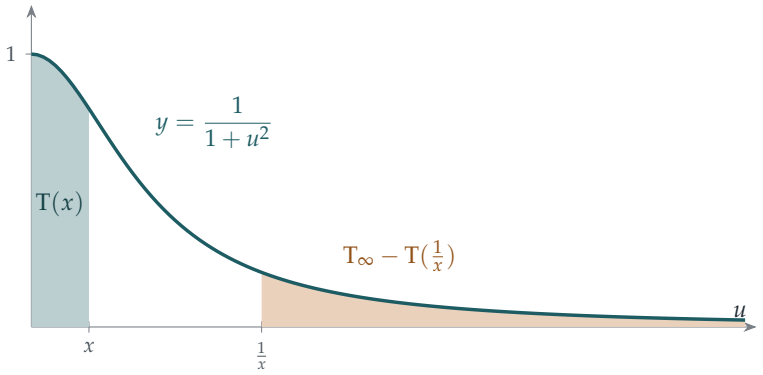


Figure 4.2.1 The substitution $u = 1/w$ shows that the two shaded areas under $\frac{1}{1+u^2}$ are equal: the area from 0 to x (teal) and the infinitely long area from $\frac{1}{x}$ onwards (amber). Here $x = \frac{1}{2}$.

STAGE 2 The addition substitution

Ask yourself Differentiate $v = \dfrac{u+a}{1-au}$ by the quotient rule. Then compute $1+v^2$ as a single fraction and compare.

$$\frac{\mathrm{d}v}{\mathrm{d}u} = \frac{(1-au)+a(u+a)}{(1-au)^2} = \frac{1+a^2}{(1-au)^2},$$
$$1+v^2 = \frac{(1-au)^2+(u+a)^2}{(1-au)^2} = \frac{(1+a^2)(1+u^2)}{(1-au)^2}.$$

Dividing, $\dfrac{1+v^2}{1+u^2} = \dfrac{1+a^2}{(1-au)^2} = \dfrac{\mathrm{d}v}{\mathrm{d}u}$. ■

Ask yourself This says $\dfrac{\mathrm{d}u}{1+u^2} = \dfrac{\mathrm{d}v}{1+v^2}$. So the substitution $u \mapsto v$ doesn't change the integrand at all. What does it do to the limits of $\mathrm{T}(x)$?

As u runs from 0 to x (with $x < \frac{1}{a}$, so $1-au > 0$ throughout and v is continuous and increasing), v runs from a to $\frac{x+a}{1-ax}$. So

$$\mathrm{T}(x) = \int_0^x \frac{\mathrm{d}u}{1+u^2} = \int_a^{(x+a)/(1-ax)} \frac{\mathrm{d}v}{1+v^2} = \mathrm{T}\left(\frac{x+a}{1-ax}\right) - \mathrm{T}(a). \quad ■$$

STAGE 3 Two more identities

Combining with part (i), twice:

$$\mathrm{T}(x^{-1}) = \mathrm{T}_\infty - \mathrm{T}(x) = \mathrm{T}_\infty - \mathrm{T}\left(\frac{x+a}{1-ax}\right) + \mathrm{T}(a) = 2\mathrm{T}_\infty - \mathrm{T}\left(\frac{x+a}{1-ax}\right) - \mathrm{T}(a^{-1}). \quad ■$$

Now rename: put $y = 1/x$ and $b = 1/a$. The condition $x < 1/a$ becomes $1/y < b$, that is $y > 1/b$, and

$$\frac{x+a}{1-ax} = \frac{1/y + 1/b}{1 - 1/(by)} = \frac{b+y}{by-1}.$$

So $\mathrm{T}(y) = 2\mathrm{T}_\infty - \mathrm{T}\left(\dfrac{y+b}{by-1}\right) - \mathrm{T}(b)$. ■

STAGE 4 Special values

Pause and try Choose y and b in the last identity so that the argument $\frac{y+b}{by-1}$ equals y itself. What do you get?

Take $y = b = \sqrt{3}$: then $y > \frac{1}{b}$, and $\dfrac{\sqrt{3}+\sqrt{3}}{3-1} = \sqrt{3}$. The identity becomes $\mathrm{T}(\sqrt{3}) = 2\mathrm{T}_\infty - 2\mathrm{T}(\sqrt{3})$, so

$$\mathrm{T}(\sqrt{3}) = \tfrac{2}{3}\mathrm{T}_\infty. \quad ■$$

Ask yourself For $\sqrt{2}-1$, try the addition identity with $x = a = \sqrt{2}-1$. What is $\frac{2x}{1-x^2}$? And what is $\mathrm{T}(1)$?

With $x = a = \sqrt{2}-1$ (and $x < \frac{1}{a}$ since $x^2 = 3 - 2\sqrt{2} < 1$), the addition identity says $2\mathrm{T}(x) = \mathrm{T}\left(\frac{2x}{1-x^2}\right)$. Now $1 - x^2 = 2\sqrt{2} - 2 = 2(\sqrt{2}-1) = 2x$, so $\frac{2x}{1-x^2} = 1$, and $2\mathrm{T}(\sqrt{2}-1) = \mathrm{T}(1)$. Finally, part (i) with $x = 1$ gives $\mathrm{T}(1) = \mathrm{T}_\infty - \mathrm{T}(1)$, so $\mathrm{T}(1) = \frac{1}{2}\mathrm{T}_\infty$. Therefore

$$\mathrm{T}(\sqrt{2}-1) = \tfrac{1}{4}\mathrm{T}_\infty. \quad ■$$

Looking back

- **Substitutions that preserve the integrand.** Both key steps used substitutions under which $\frac{\mathrm{d}u}{1+u^2}$ is unchanged; only the limits moved. Such "symmetries" of an integral are powerful.
- **Fixed points.** The value $\mathrm{T}(\sqrt{3})$ came from choosing numbers that made an expression equal to itself, so an identity became an equation.
- **Definitions can come from analysis.** We never used a triangle, yet we recovered $\frac{\pi}{3}$ and $\frac{\pi}{8}$ (with π defined as $2\mathrm{T}_\infty$).

What's really going on

Building the elementary functions

In a rigorous analysis course, the functions exp, sin, cos and ln cannot simply be assumed; they have to be *constructed*, and their properties proved. There are three standard approaches. You can define them by power series ($\sin x = x - \frac{x^3}{3!} + \cdots$). You can define them as solutions of differential equations, as in the 1986 specimen question in the exercises ($f' = f$, $f(0) = 1$ defines exp). Or you can define inverse functions by integrals, as here: $\ln x = \int_1^x \frac{du}{u}$ and $\arctan x = \int_0^x \frac{du}{1+u^2}$, and then define $\pi = 2\int_0^\infty \frac{du}{1+u^2}$. Each approach has its strengths, and proving that they all give the same functions is a satisfying exercise. Compare the functional-equation approach of Section 3.8.

Beyond the circle: elliptic integrals

The integral approach generalises in a way the triangle approach cannot. In 1718 Giulio Fagnano studied the arc length of the *lemniscate*, a figure-of-eight curve, which leads to $\int_0^x \frac{du}{\sqrt{1-u^4}}$, and found a substitution that doubles the arc, just as our substitution doubled T. Euler, reading Fagnano's work in 1751, found a full addition formula, and this led eventually, through Abel and Jacobi, to the theory of *elliptic functions*: inverse functions of integrals like this one, which are to the lemniscate what sin and cos are to the circle. We met the same integral in Gauss's work on the AGM (Section 2.6). Elliptic functions are now central to number theory; the elliptic curves of Section 1.9 are their geometric shadow.

Computing π

The addition identity is also the basis of the most famous formulas for computing π. Repeatedly applying it shows, for instance, that $T(1) = 4T(\frac{1}{5}) - T(\frac{1}{239})$, which is Machin's formula of 1706; combined with the series $T(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$, it gave 100 digits of π and stayed the method of choice for two and a half centuries. That is the subject of Section 5.1.

The thinking skill

Symmetries of an integral

Look for substitutions that leave the integrand unchanged. They turn into identities for the function the integral defines.

Here $u \mapsto 1/u$ and $u \mapsto \frac{u+a}{1-au}$ preserve $\frac{\mathrm{d}u}{1+u^2}$, and they give the reflection and addition formulas. The same idea, applied to other integrals, underlies the theory of special functions.

Try this next

1 **The exponential and the cosine.** (From the STEP II Specimen paper.) *Without using standard properties of exponential or trigonometric functions:* a function f satisfies $\mathrm{f}' = \mathrm{f}$ with $\mathrm{f}(0) = 1$. Show that $\frac{\mathrm{d}}{\mathrm{d}x}\big[\mathrm{f}(a-x)\mathrm{f}(x)\big] = 0$ and deduce that $\mathrm{f}(x)\mathrm{f}(y) = \mathrm{f}(x+y)$. Functions c and s satisfy $\mathrm{c}' = -\mathrm{s}$, $\mathrm{s}' = \mathrm{c}$, $\mathrm{s}(0) = 0$, $\mathrm{c}(0) = 1$. Show that $\mathrm{c}(x+y) = \mathrm{c}(x)\mathrm{c}(y) - \mathrm{s}(x)\mathrm{s}(y)$. (*Hint:* fix y and consider $\mathrm{c}(x+y) - \mathrm{c}(x)\mathrm{c}(y) + \mathrm{s}(x)\mathrm{s}(y)$ and a similar combination for s; show the sum of their squares is constant.)

2 **The inverse sine.** (From STEP III, 2007.) Let $\mathrm{S}(x) = \int_0^x \frac{\mathrm{d}u}{\sqrt{1-u^2}}$. Using $y = \frac{u}{\sqrt{1+u^2}}$, show that $\mathrm{T}(x) = \mathrm{S}\Big(\frac{x}{\sqrt{1+x^2}}\Big)$ and deduce that $\mathrm{S}\Big(\frac{1}{\sqrt{2}}\Big) = \frac{1}{2}\mathrm{T}_\infty$.

3 **An early Machin-type formula.** Use the addition identity to show that $\mathrm{T}(\frac{1}{2}) + \mathrm{T}(\frac{1}{3}) = \mathrm{T}(1) = \frac{1}{2}\mathrm{T}_\infty$.

4 T_∞ **is finite.** Show that $\mathrm{T}(x) \le 1 + \int_1^x \frac{\mathrm{d}u}{u^2} < 2$ for all $x \ge 1$, explain why $\mathrm{T}(x) < 2$ for $0 < x < 1$ as well, and deduce from the monotone convergence theorem (Section 2.2) that T_∞ exists.

Further reading

- **Tom Körner,** ***Calculus for the Ambitious*** (Cambridge University Press). Builds the elementary functions rigorously, discussing the different approaches.
- **Walter Rudin,** ***Principles of Mathematical Analysis*** (McGraw-Hill). The famous "baby Rudin"; the prologue to Chapter 8 constructs exp, sin and cos from power series in a few elegant pages. Advanced.
- **Victor Prasolov & Yuri Solovyev,** ***Elliptic Functions and Elliptic Integrals*** (American Mathematical Society). For where the lemniscate leads. Advanced.

4.3 22/7 Is Bigger Than π

The question

Find constants a_0, a_1, a_2, a_3, a_4, a_5, a_6 and b such that

$$x^4(1-x)^4 = (a_6x^6 + a_5x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0)(x^2+1) + b.$$

Hence, or otherwise, prove that

$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,\mathrm{d}x = \frac{22}{7} - \pi.$$

Evaluate $\displaystyle\int_0^1 x^4(1-x)^4\,\mathrm{d}x$ and deduce that

$$\frac{22}{7} > \pi > \frac{22}{7} - \frac{1}{630}.$$

The hook

Generations of students have been told that π is "about $\frac{22}{7}$". It is a very good approximation: $\frac{22}{7} = 3.142857\ldots$ and $\pi = 3.141592\ldots$. But is $\frac{22}{7}$ too big or too small? And can you *prove* it, without simply looking up the digits of π, which someone else had to compute?

This question gives one of the most elegant proofs in all of mathematics. A single integral of a positive function, which therefore must be positive, turns out to equal exactly $\frac{22}{7} - \pi$. Positive integrand, positive integral, so $\frac{22}{7} > \pi$. The whole proof fits in a few lines, and it is so neat that it appears as a puzzle on mathematics T-shirts.

Before you start

Ask yourself Why does π come out of $\int_0^1 \frac{\text{polynomial}}{1+x^2}\,\mathrm{d}x$? Which simple integral of this form gives π on its own?

$\int_0^1 \frac{\mathrm{d}x}{1+x^2} = \arctan 1 = \frac{\pi}{4}$. When you divide a polynomial by $1+x^2$, you get a polynomial quotient (which integrates to a rational number) plus a constant remainder over $1+x^2$ (which integrates to a rational multiple of π). The first part of the question asks you to do exactly that division.

Working it through

STAGE 1 Polynomial division

First expand: $(1-x)^4 = 1 - 4x + 6x^2 - 4x^3 + x^4$, so

$$x^4(1-x)^4 = x^8 - 4x^7 + 6x^6 - 4x^5 + x^4.$$

Ask yourself Divide by x^2+1, either by long division or by comparing coefficients. What is the remainder?

Comparing coefficients in $(a_6x^6 + \cdots + a_0)(x^2+1) + b$, from the top: $a_6 = 1$; $a_5 = -4$; $a_4 + a_6 = 6$, so $a_4 = 5$; $a_3 + a_5 = -4$, so $a_3 = 0$; $a_2 + a_4 = 1$, so $a_2 = -4$; $a_1 + a_3 = 0$, so $a_1 = 0$; $a_0 + a_2 = 0$, so $a_0 = 4$; and finally $a_1 = 0$ (the coefficient of x) and $a_0 + b = 0$, so $b = -4$. That is,

$$x^4(1-x)^4 = (x^6 - 4x^5 + 5x^4 - 4x^2 + 4)(x^2+1) - 4.$$

STAGE 2 The integral

Dividing by $1 + x^2$,

$$\frac{x^4(1-x)^4}{1+x^2} = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2},$$

so

$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx = \frac{1}{7} - \frac{4}{6} + \frac{5}{5} - \frac{4}{3} + 4 - 4\cdot\frac{\pi}{4} = \frac{1}{7} - \frac{2}{3} + 1 - \frac{4}{3} + 4 - \pi.$$

Now $-\frac{2}{3} - \frac{4}{3} = -2$, so the rational part is $\frac{1}{7} + 3 = \frac{22}{7}$, and the integral is $\frac{22}{7} - \pi$. ■

STAGE 3 Bounding the integral

Ask yourself The integrand is positive on $(0,1)$, so $\frac{22}{7} > \pi$ at once. To get a lower bound for π we need an *upper* bound for the integral. How big can $\frac{1}{1+x^2}$ be on $[0,1]$?

On $[0,1]$, $1 \le 1 + x^2 \le 2$, so

$$\tfrac{1}{2}x^4(1-x)^4 \le \frac{x^4(1-x)^4}{1+x^2} \le x^4(1-x)^4,$$

with strict inequality except at the endpoints.

Pause and try Evaluate $\int_0^1 x^4(1-x)^4\,dx$, and finish the proof.

Integrating the expansion term by term,

$$\int_0^1 x^4(1-x)^4\,dx = \frac{1}{9} - \frac{4}{8} + \frac{6}{7} - \frac{4}{6} + \frac{1}{5} = \frac{1}{630}.$$

(Over the common denominator 2520: $280 - 1260 + 2160 - 1680 + 504 = 4$, and $\frac{4}{2520} = \frac{1}{630}$.) So

$$0 < \frac{22}{7} - \pi < \frac{1}{630},$$

which is $\frac{22}{7} > \pi > \frac{22}{7} - \frac{1}{630}$. ■

Numerically, $\frac{22}{7} - \frac{1}{630} = 3.14127\ldots$, so we have proved $3.1412 < \pi < 3.1429$. Using the left-hand inequality as well, $\frac{1}{1260} < \frac{22}{7} - \pi$, which improves the upper bound to $\pi < 3.14207$.

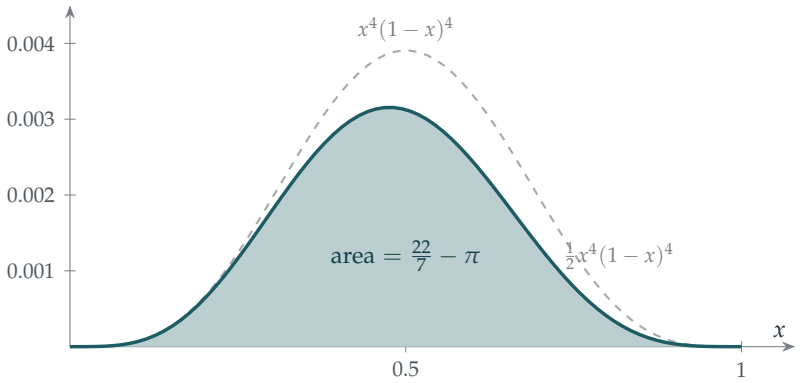


Figure 4.3.1 The integrand $\frac{x^4(1-x)^4}{1+x^2}$ (teal) is positive but tiny: its maximum is about 0.003. The shaded area is exactly $\frac{22}{7} - \pi \approx 0.00126$. The integrand lies between $\frac{1}{2}x^4(1-x)^4$ (dotted) and $x^4(1-x)^4$ (dashed), whose areas are $\frac{1}{1260}$ and $\frac{1}{630}$.

Looking back

- **Positivity proves inequalities.** The integral of a positive function is positive. Once the integral was identified as $\frac{22}{7} - \pi$, the inequality was immediate.
- **Small integrands, sharp bounds.** The factor $x^4(1-x)^4$ is tiny on $[0,1]$ (at most $\frac{1}{256}$), which makes the integral small and the bound sharp.
- **Rational plus a multiple of π.** As in Section 2.9, an integral of the form "rational number + rational multiple of a constant" that is small gives a good rational approximation to the constant.

What's really going on

A proof with a history

Archimedes proved $\frac{223}{71} < \pi < \frac{22}{7}$ around 250 BC by computing the perimeters of 96-sided polygons inside and outside a circle (compare the polygons of Section 2.5). The integral proof is much more recent. It was published by D. P. Dalzell in 1944, and became famous when it was set as a problem in the 1968 Putnam competition for American undergraduates. Its charm is that it needs no numerical work at all: the inequality $\frac{22}{7} > \pi$ drops out of the fact that an integral of a positive function is positive.

Better approximations from the same idea

The method generalises. With $x^8(1-x)^8$ in place of $x^4(1-x)^4$, the same division gives

$$\int_0^1 \frac{x^8(1-x)^8}{1+x^2}\,\mathrm{d}x = 4\pi - \frac{188684}{15015},$$

which proves $\pi > \frac{47171}{15015} = 3.1415917\ldots$, correct to six significant figures. In 2009 Stephen Lucas found families of integrals of this kind with non-negative integrands that prove inequalities for other famous approximations, including $\frac{355}{113}$, which is correct to seven significant figures.

Approximations and irrationality

Why does π have such good fractional approximations as $\frac{22}{7}$ and $\frac{355}{113}$? Every irrational number has infinitely many fractions $\frac{p}{q}$ within $\frac{1}{q^2}$ of it, and π happens to have some exceptionally good ones early on, because its continued fraction $[3;7,15,1,292,\ldots]$ contains the large entry 292 (see Section 2.2). Integrals like the one in this question, made smaller and smaller, are the starting point for proofs that numbers *cannot* be approximated too well by fractions, and so are irrational: Ivan Niven's famous one-page proof that π is irrational (1947) uses integrals of $\frac{x^n(1-x)^n}{n!}\sin(\pi x)$, the trigonometric cousins of those in Section 2.9.

The thinking skill

Encode the inequality as an integral of something positive

To prove $A > B$, find a positive function whose integral is $A - B$.

This turns an inequality into an identity plus an obvious fact. It is one of the most useful tricks in analysis, and the same idea, with sums of squares instead of integrals, proved AM–GM in Section 3.2.

Try this next

1 **Check the numbers.** Compute $\frac{22}{7} - \pi$ to four significant figures and confirm that it lies between $\frac{1}{1260}$ and $\frac{1}{630}$.

2 **A Beta integral.** Use the formula $\int_0^1 x^n(1-x)^n \, dx = \frac{(n!)^2}{(2n+1)!}$ (proved in Section 4.4) to check that $\int_0^1 x^4(1-x)^4 \, dx = \frac{1}{630}$.

3 **A lower approximation.** Show that

$$\int_0^1 \frac{x(1-x)^2}{1+x^2} \, dx = \frac{\pi}{2} - \frac{3}{2} \quad \text{and} \quad \int_0^1 \frac{x^2(1-x)^4}{1+x^2} \, dx = \pi - \frac{47}{15},$$

and deduce that $\pi > 3$ and $\pi > \frac{47}{15}$. Why are these less accurate than $\frac{22}{7}$?

4 **Going further.** HARDER Verify the identity for $\int_0^1 \frac{x^8(1-x)^8}{1+x^2} \, dx$ given above. (A computer algebra system will do the division; the point is to see that the remainder is 16, giving 4π.)

Further reading

- **Petr Beckmann, *A History of* π** (St Martin's Press). Archimedes' polygon method in detail, and much else.
- **Stephen Lucas, "Approximations to π derived from integrals with nonnegative integrands"**, *American Mathematical Monthly* 116 (2009). Short and readable.
- **Ivan Niven, *Irrational Numbers*** (Mathematical Association of America). Includes Niven's proof that π is irrational.

4.4 Wallis, Beta and the Factorial of a Half

The question

Let

$$u_n = \int_0^{\frac{1}{2}\pi} \sin^n t \, \mathrm{d}t$$

for each integer $n \geq 0$. By integrating

$$\int_0^{\frac{1}{2}\pi} \sin t \sin^{n-1} t \, \mathrm{d}t$$

by parts, or otherwise, obtain a formula connecting u_n and u_{n-2} when $n \geq 2$ and deduce that

$$nu_n u_{n-1} = (n-1)u_{n-1}u_{n-2}$$

for all $n \geq 2$. Deduce that

$$nu_n u_{n-1} = \tfrac{1}{2}\pi.$$

Sketch graphs of $\sin^n t$ and $\sin^{n-1} t$, for $0 \leq t \leq \frac{1}{2}\pi$, on the same diagram and explain why $0 < u_n < u_{n-1}$. By using the result of the previous paragraph show that

$$nu_n^2 < \tfrac{1}{2}\pi < nu_{n-1}^2$$

for all $n \geq 1$. Hence show that

$$\left(\frac{n}{n+1}\right)\tfrac{1}{2}\pi < nu_n^2 < \tfrac{1}{2}\pi$$

and deduce that $nu_n^2 \to \frac{1}{2}\pi$ as $n \to \infty$.

The hook

What is $\left(\frac{1}{2}\right)!$? The question sounds like nonsense: $n!$ means $1 \times 2 \times \cdots \times n$, and you cannot multiply "half a number" of terms. Yet there is a natural answer, and it is

$$\left(\tfrac{1}{2}\right)! = \frac{\sqrt{\pi}}{2}.$$

Where does π come from? From integrals of powers of $\sin t$, which this question studies. Along the way it proves one of the oldest and strangest formulas for π, John Wallis's infinite product of 1655,

$$\frac{\pi}{2} = \frac{2}{1}\cdot\frac{2}{3}\cdot\frac{4}{3}\cdot\frac{4}{5}\cdot\frac{6}{5}\cdot\frac{6}{7}\cdots,$$

and it finds the hidden constant in Stirling's formula for $n!$, which every probabilist and statistician uses.

Before you start

Ask yourself What are u_0 and u_1? And roughly what does the graph of $\sin^n t$ look like on $[0, \frac{\pi}{2}]$ when n is large?

$u_0 = \int_0^{\pi/2} 1\,\mathrm{d}t = \frac{\pi}{2}$ and $u_1 = \int_0^{\pi/2} \sin t\,\mathrm{d}t = 1$. For large n, $\sin^n t$ is tiny except near $t = \frac{\pi}{2}$, where $\sin t$ is close to 1: the graph becomes a narrow spike against the right-hand edge (Figure 4.4.1). So u_n shrinks as n grows. The question finds exactly how fast: like $\sqrt{\pi/(2n)}$.

Working it through

STAGE 1 A reduction formula

Ask yourself Integrate $\int \sin t \cdot \sin^{n-1} t\,\mathrm{d}t$ by parts, integrating $\sin t$ and differentiating $\sin^{n-1} t$. The new integrand contains $\cos^2 t$; how can you get back to powers of $\sin t$?

With $\mathrm{d}v = \sin t\,\mathrm{d}t$ (so $v = -\cos t$) and $w = \sin^{n-1} t$ (so $w' = (n-1)\sin^{n-2} t \cos t$),

$$u_n = \Big[-\cos t \sin^{n-1} t\Big]_0^{\pi/2} + (n-1)\int_0^{\pi/2} \sin^{n-2} t \cos^2 t\,\mathrm{d}t.$$

The boundary term is zero for $n \geq 2$. Writing $\cos^2 t = 1 - \sin^2 t$, the integral is $u_{n-2} - u_n$, so $u_n = (n-1)(u_{n-2} - u_n)$, that is,

$$nu_n = (n-1)u_{n-2}. \qquad \blacksquare$$

Multiplying both sides by u_{n-1} gives $nu_nu_{n-1} = (n-1)u_{n-1}u_{n-2}$. $\blacksquare$

STAGE 2 An invariant

Ask yourself The relation says the quantity nu_nu_{n-1} does not change when n goes down by one. What is it equal to at $n = 1$?

By repeated use, $nu_nu_{n-1} = (n-1)u_{n-1}u_{n-2} = \cdots = 1 \cdot u_1u_0 = 1 \cdot 1 \cdot \frac{\pi}{2}$. So $nu_nu_{n-1} = \frac{1}{2}\pi$ for all $n \geq 1$. $\blacksquare$

STAGE 3 A squeeze

For $0 < t < \frac{\pi}{2}$ we have $0 < \sin t < 1$, so $\sin^n t < \sin^{n-1} t$: the graph of $\sin^n t$ lies below that of $\sin^{n-1} t$ except at the endpoints. Integrating, $0 < u_n < u_{n-1}$.

Pause and try Use $u_n < u_{n-1}$ and $nu_nu_{n-1} = \frac{1}{2}\pi$ to trap nu_n^2.

Replacing u_{n-1} by the smaller u_n gives $nu_n^2 < nu_nu_{n-1} = \frac{1}{2}\pi$. Replacing u_n by the larger u_{n-1} gives $\frac{1}{2}\pi = nu_nu_{n-1} < nu_{n-1}^2$. So $nu_n^2 < \frac{1}{2}\pi < nu_{n-1}^2$. ■

Applying the right-hand inequality with $n+1$ in place of n: $\frac{1}{2}\pi < (n+1)u_n^2$, so $nu_n^2 > \frac{n}{n+1} \cdot \frac{1}{2}\pi$. Together,

$$\left(\frac{n}{n+1}\right)\frac{1}{2}\pi < nu_n^2 < \frac{1}{2}\pi. \qquad \blacksquare$$

Since $\frac{n}{n+1} \to 1$, the squeeze gives $nu_n^2 \to \frac{1}{2}\pi$. ■

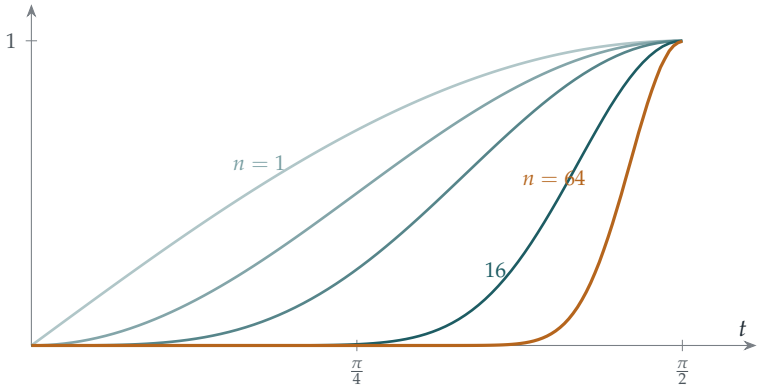


Figure 4.4.1 The graphs of $\sin^n t$ for $n = 1, 2, 4, 16, 64$. As n grows the area u_n under the curve shrinks, concentrating near $t = \frac{\pi}{2}$. The question shows that $u_n \approx \sqrt{\pi/(2n)}$; for $n = 64$ that is about 0.157.

Looking back

- **Reduction formulae.** Integration by parts turned u_n into u_{n-2}, so all the u_n are determined by u_0 and u_1.
- **Look for an invariant.** The product nu_nu_{n-1} stayed fixed, just as the integral did under the AGM step in Section 2.6.
- **Monotonicity plus an invariant gives a squeeze.** Knowing $u_n < u_{n-1}$ and the exact value of nu_nu_{n-1} trapped nu_n^2 between two quantities with the same limit.

What's really going on

Wallis's product

Running the reduction formula down to u_0 or u_1 gives explicit values:

$$u_{2m} = \frac{1\cdot 3\cdot 5\cdots(2m-1)}{2\cdot 4\cdot 6\cdots 2m}\cdot\frac{\pi}{2}, \qquad u_{2m+1} = \frac{2\cdot 4\cdot 6\cdots 2m}{3\cdot 5\cdot 7\cdots(2m+1)}.$$

The squeeze shows $u_{2m}/u_{2m+1} \to 1$. Writing out that ratio and rearranging gives Wallis's product,

$$\frac{\pi}{2} = \lim_{m\to\infty} \frac{2\cdot 2}{1\cdot 3}\cdot\frac{4\cdot 4}{3\cdot 5}\cdots\frac{2m\cdot 2m}{(2m-1)(2m+1)}.$$

John Wallis found it in 1655 by a daring process of interpolation, guessing what $\int_0^1 (1-x^2)^{1/2}\,dx$ "should" be by fitting it into a table of integrals with whole-number exponents. It was one of the first infinite products in mathematics, after Viète's (Section 2.5), and the first to use only rational numbers.

The factorial of a half

Wallis's interpolation idea was made precise by Euler in 1729. He found a function, now called the *Gamma function*,

$$\Gamma(s) = \int_0^\infty t^{s-1}e^{-t}\,dt,$$

which satisfies $\Gamma(s+1) = s\,\Gamma(s)$ (integrate by parts) and $\Gamma(n+1) = n!$ for whole numbers n. It extends the factorial smoothly to all positive real numbers. The closely related *Beta function*

$$\mathrm{B}(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1}\,dx = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$$

includes the integrals of Section 4.3 and Section 2.9, and, after the substitution $x = \sin^2 t$, our u_n. In particular, $\int_0^1 x^{1/2}(1-x)^{1/2}\,dx = \frac{\pi}{8}$ (exercise 2), which gives $\Gamma(\frac{3}{2})^2/\Gamma(3) = \frac{\pi}{8}$, that is, $\left(\frac{1}{2}\right)! = \Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$. Equivalently, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, which is the same fact as the famous Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$, the reason π appears in the formula for the normal distribution.

Stirling's formula

Stirling's formula says $n! \approx \sqrt{2\pi n}\,(n/e)^n$. The $(n/e)^n$ part is relatively easy to obtain, but the constant $\sqrt{2\pi}$ is harder, and the standard way to find it is exactly this question: the explicit values of u_{2m} involve $(2m)!/(m!)^2$, and the limit $nu_n^2 \to \frac{\pi}{2}$ pins down the constant. A completely different route to the same constant, through probability, is in Section 12.23.

The thinking skill

Recurrences from integrals

Integration by parts often relates an integral with parameter n to the same integral with a smaller parameter. That recurrence can then be solved, or analysed for its behaviour as $n \to \infty$.

Watch for quantities that the recurrence leaves unchanged, and for monotonicity in n; together they often give limits without any explicit formula.

Try this next

1 **Wallis's product.** Using the explicit values of u_{2m} and u_{2m+1}, and $u_{2m+1} < u_{2m} < u_{2m-1}$, prove Wallis's product.

2 **A Beta integral.** (From STEP II, 2013.) Let $I_n = \int_0^1 x^n (1-x)^n \,\mathrm{d}x$. Show by integration by parts and symmetry that $I_n = \frac{n}{2(2n+1)} I_{n-1}$, and deduce that $I_n = \frac{(n!)^2}{(2n+1)!}$. Use $x = \sin^2\theta$ to show that $I_{1/2} = \frac{\pi}{8}$. What does the formula $I_n = \frac{(n!)^2}{(2n+1)!}$ suggest $\left(\frac{1}{2}\right)!$ should be?

3 **The Beta function.** (From STEP III, 2025.) For $p, q > 0$ let $\mathrm{b}(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1}\,\mathrm{d}x$. Show that $\mathrm{b}(p,q) = \mathrm{b}(q,p)$, that $\mathrm{b}(p+1,q) = \mathrm{b}(p,q) - \mathrm{b}(p,q+1)$, and, with the substitution $x = \frac{t}{1+t}$, that $\mathrm{b}(p,q) = \int_0^\infty \frac{t^{p-1}}{(1+t)^{p+q}}\,\mathrm{d}t$. Hence evaluate $\displaystyle\int_0^\infty \frac{t^{3/2}}{(1+t)^6}\,\mathrm{d}t$. (The answer is $\frac{3\pi}{256}$.)

4 **Central binomial coefficients.** HARDER Show that $u_{2m} = \binom{2m}{m}\frac{\pi}{2^{2m+1}}$, and use $n u_n^2 \to \frac{\pi}{2}$ to deduce that $\binom{2m}{m} \approx \dfrac{4^m}{\sqrt{\pi m}}$ for large m. Check this for $m = 10$.

Further reading

- **Julian Havil, *Gamma*** (Princeton University Press). Includes the Gamma function and its history.
- **Philip J. Davis, "Leonhard Euler's integral: a historical profile of the Gamma function"**, *American Mathematical Monthly* 66 (1959). A classic, prize-winning article, readable with this section behind you.
- **Emil Artin, *The Gamma Function*** (Dover). A short, beautiful monograph for when you have done some analysis.

4.5 The Sophomore's Dream

The question

Let m be a positive integer and let n be a non-negative integer.

(i) Use the result $\lim_{t\to\infty} e^{-mt}t^n = 0$ to show that

$$\lim_{x\to 0} x^m(\ln x)^n = 0.$$

By writing x^x as $e^{x\ln x}$ show that

$$\lim_{x\to 0} x^x = 1.$$

(ii) Let $I_n = \displaystyle\int_0^1 x^m(\ln x)^n\,\mathrm{d}x$. Show that

$$I_{n+1} = -\frac{n+1}{m+1}I_n$$

and hence evaluate I_n.

(iii) Show that

$$\int_0^1 x^x\,\mathrm{d}x = 1 - \left(\tfrac{1}{2}\right)^2 + \left(\tfrac{1}{3}\right)^3 - \left(\tfrac{1}{4}\right)^4 + \cdots.$$

The hook

Every student learns, sooner or later, that $(a+b)^2 \neq a^2+b^2$. The false version is jokingly called the "freshman's dream". This question proves something that looks just as much like a beginner's mistake:

$$\int_0^1 x^x\,\mathrm{d}x = \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^n} = 1 - \frac{1}{2^2} + \frac{1}{3^3} - \frac{1}{4^4} + \cdots,$$

and its companion $\int_0^1 x^{-x}\,\mathrm{d}x = \sum_{n=1}^{\infty}\frac{1}{n^n}$. An integral of x^x turns into a sum of n^{-n}: it seems too good to be true, which is why it is called the *sophomore's dream*. But it is true, and Johann Bernoulli found it in 1697.

The proof uses three ideas you will meet constantly at university: limits that compare exponentials with powers, reduction formulae, and expanding a function as a power series and integrating term by term.

Before you start

Ask yourself The function x^x is not defined at $x = 0$ by any formula you know (0^0 is ambiguous). How does it behave as x approaches 0 from above? Try $x = 0.1, 0.01, 0.001$ on a calculator.

$0.1^{0.1} \approx 0.794$, $0.01^{0.01} \approx 0.955$, $0.001^{0.001} \approx 0.993$. It seems to approach 1. Part (i) proves this, which tells us the integrand is well behaved near 0 and the integral makes sense. The graph of x^x on $(0, 1]$ starts near 1, dips to a minimum and returns to 1 at $x = 1$ (Figure 4.5.1).

Working it through

STAGE 1 Powers beat logarithms

Ask yourself As $x \to 0^+$, $\ln x \to -\infty$. What substitution turns $x \to 0^+$ into $t \to \infty$, and turns $x^m (\ln x)^n$ into something like $e^{-mt} t^n$?

Put $x = e^{-t}$. As $x \to 0^+$, $t \to \infty$, and $\ln x = -t$, so

$$x^m (\ln x)^n = e^{-mt}(-t)^n = (-1)^n e^{-mt} t^n \to 0.$$

In words: as $x \to 0$, any positive power of x goes to zero faster than any power of $\ln x$ blows up. ■

With $m = n = 1$, $x \ln x \to 0$, and since the exponential function is continuous, $x^x = e^{x \ln x} \to e^0 = 1$. ■

STAGE 2 A reduction formula

Ask yourself Integrate $I_{n+1} = \int_0^1 x^m (\ln x)^{n+1}\, dx$ by parts. Which factor should you integrate? What happens to the boundary term at $x = 0$?

Integrate x^m and differentiate $(\ln x)^{n+1}$:

$$I_{n+1} = \left[\frac{x^{m+1}}{m+1} (\ln x)^{n+1} \right]_0^1 - \int_0^1 \frac{x^{m+1}}{m+1} \cdot \frac{(n+1)(\ln x)^n}{x}\, dx.$$

At $x = 1$ the boundary term is 0 because $\ln 1 = 0$; as $x \to 0$ it tends to 0 by part (i) (with $m+1$ and $n+1$). The remaining integral is $\frac{n+1}{m+1} I_n$, so

$$I_{n+1} = -\frac{n+1}{m+1} I_n. \qquad ■$$

Starting from $I_0 = \int_0^1 x^m\, dx = \frac{1}{m+1}$ and applying the formula n times,

$$I_n = (-1)^n \frac{n!}{(m+1)^n} \cdot \frac{1}{m+1} = \frac{(-1)^n\, n!}{(m+1)^{n+1}}. \qquad ■$$

STAGE 3 The series

Ask yourself Expand $x^x = e^{x\ln x}$ using the series for e^y. What does each term look like? Which value of m makes the integrals in part (ii) appear?

Since $e^y = \sum_{n=0}^{\infty} \frac{y^n}{n!}$ for every y,

$$x^x = e^{x\ln x} = \sum_{n=0}^{\infty} \frac{x^n (\ln x)^n}{n!}.$$

Pause and try Integrate term by term from 0 to 1, using part (ii) with $m = n$ (and treating $n = 0$ separately). What do you get?

The $n = 0$ term is 1, which integrates to 1. For $n \geq 1$, part (ii) with $m = n$ gives $\int_0^1 x^n (\ln x)^n \, dx = \frac{(-1)^n \, n!}{(n+1)^{n+1}}$, so the nth term contributes $\frac{(-1)^n}{(n+1)^{n+1}}$. Adding up,

$$\int_0^1 x^x \, dx = 1 - \frac{1}{2^2} + \frac{1}{3^3} - \frac{1}{4^4} + \cdots . \qquad \blacksquare$$

(The $n = 0$ term fits the same pattern, $\frac{1}{1^1}$.) The terms shrink extremely fast: seven terms give the value $0.783430\ldots$ to six decimal places.

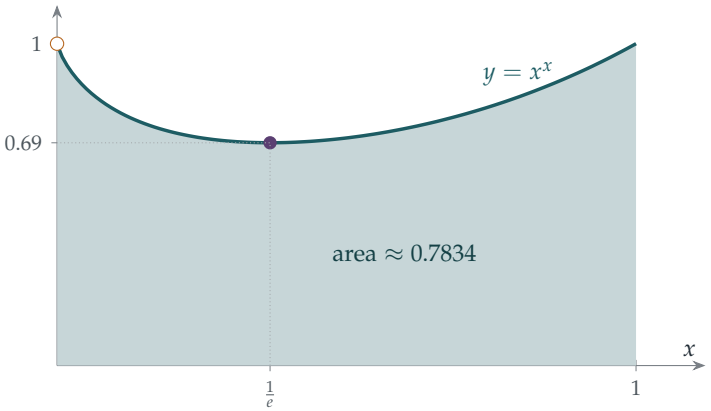


Figure 4.5.1 The graph of x^x on $(0, 1]$. It tends to 1 as $x \to 0$ (open circle), dips to a minimum of $e^{-1/e} \approx 0.692$ at $x = \frac{1}{e}$, and returns to 1 at $x = 1$. The shaded area is $\sum (-1)^{n+1} n^{-n} \approx 0.7834$.

Looking back

- **Substitute to compare growth rates.** Putting $x = e^{-t}$ turned a limit at 0 into a familiar limit at ∞.
- **Boundary terms need checking.** The reduction formula depended on the boundary term vanishing at $x = 0$, which is where part (i) was used.

- **Series, then integrate.** Expanding the integrand as a power series and integrating each term turned an integral with no elementary antiderivative into a rapidly convergent sum.

What's really going on

Swapping sums and integrals

Stage 3 quietly swapped an infinite sum with an integral: $\int \sum = \sum \int$. That is not always allowed. At university you will meet examples where it fails, and theorems that say when it is safe. The simplest says that it is safe when the series converges *uniformly*: the tail of the series can be made small at every x simultaneously. Here, $|x \ln x| \le \frac{1}{e}$ on $(0, 1]$, so the nth term is at most $\frac{e^{-n}}{n!}$ in size, whatever x is. The sum of these bounds converges, so the tails are uniformly small and the swap is justified. This argument is called the *Weierstrass M-test*, and it is one of the workhorses of analysis.

Why the "dream" works

The identity is special. It depends on the exact match between the powers in $\frac{(x \ln x)^n}{n!}$ and the integral $\int_0^1 x^n (\ln x)^n \, dx = \frac{(-1)^n n!}{(n+1)^{n+1}}$, where the $n!$ cancels perfectly. Underneath, this integral is a Gamma function in disguise: the substitution $x = e^{-t/(n+1)}$ turns it into $\frac{(-1)^n}{(n+1)^{n+1}} \int_0^\infty t^n e^{-t} \, dt = \frac{(-1)^n \, \Gamma(n+1)}{(n+1)^{n+1}}$, connecting with Section 4.4.

The name

The name "sophomore's dream" was coined by Jonathan Borwein and his co-authors in their 2004 book *Experimentation in Mathematics*, as a companion to the "freshman's dream" $(a + b)^n = a^n + b^n$ (which, curiously, is true in some number systems: in arithmetic modulo a prime p, $(a + b)^p = a^p + b^p$, because all the middle binomial coefficients are divisible by p). Borwein was a pioneer of *experimental mathematics*, in which computers are used to discover identities like this one, which are then proved by hand.

The thinking skill

Expand and integrate term by term

If an integrand has a power series whose terms you can integrate, integrate each term and add up the results.

It is a powerful way to evaluate integrals that have no elementary antiderivative. Always ask whether swapping the sum and the integral is justified; usually it is, but you should know why.

Try this next

1 **The companion.** Show that $\displaystyle\int_0^1 x^{-x}\,dx = \sum_{n=1}^{\infty} \frac{1}{n^n} \approx 1.2913$.

2 **The minimum.** Show that x^x has its minimum on $(0,\infty)$ at $x = \frac{1}{e}$, with value $e^{-1/e}$.

3 **How many terms?** For an alternating series with decreasing terms, the error after stopping is less than the first omitted term. How many terms of the series for $\int_0^1 x^x\,dx$ are needed for six decimal places?

4 **A cousin.** Use the same method to show that $\displaystyle\int_0^1 x^2 \ln x\,dx = -\frac{1}{9}$, and that $\displaystyle\int_0^1 \frac{\ln x}{1-x}\,dx = -\sum_{n=1}^{\infty} \frac{1}{n^2}$. (By Section 2.4, this is $-\frac{\pi^2}{6}$.)

Further reading

- **Jonathan Borwein, David Bailey & Roland Girgensohn, *Experimentation in Mathematics*** (A. K. Peters). The source of the name, and many more identities discovered by computer.
- **William Dunham, *The Calculus Gallery*** (Princeton University Press). Masterpieces of analysis from Newton to Lebesgue, including Bernoulli's work.
- **Mary Hart, *Guide to Analysis*** (Palgrave). Clear treatment of uniform convergence and when sums and integrals can be swapped.

4.6 Integrals With No Formula

The question

(i) Given that

$$\int \frac{x^3-2}{(x+1)^2}\,e^x\,\mathrm{d}x = \frac{\mathrm{P}(x)}{\mathrm{Q}(x)}\,e^x + \text{constant},$$

where $\mathrm{P}(x)$ and $\mathrm{Q}(x)$ are polynomials, show that $\mathrm{Q}(x)$ has a factor of $x+1$. Show also that the degree of $\mathrm{P}(x)$ is exactly one more than the degree of $\mathrm{Q}(x)$, and find $\mathrm{P}(x)$ in the case $\mathrm{Q}(x) = x+1$.

(ii) Show that there are no polynomials $\mathrm{P}(x)$ and $\mathrm{Q}(x)$ such that

$$\int \frac{1}{x+1}\,e^x\,\mathrm{d}x = \frac{\mathrm{P}(x)}{\mathrm{Q}(x)}\,e^x + \text{constant}.$$

You need consider only the case when $\mathrm{P}(x)$ and $\mathrm{Q}(x)$ have no common factors.

The hook

Differentiation is mechanical: there are rules, and any function built from the usual pieces can be differentiated by applying them. Integration is not. Every student has met integrals that resist every technique, and wondered whether they simply weren't clever enough. Sometimes the answer is: no one is clever enough, because *there is no formula*.

This question proves such a thing in a special case. The integral $\int \frac{e^x}{x+1}\,\mathrm{d}x$ certainly exists, as an area. But it cannot be written in the form (rational function) $\times\, e^x$, which is the form you would expect from integration by parts. Proving that something is impossible, rather than finding something, is a different and very characteristic kind of mathematics.

Before you start

Ask yourself If $\int \mathrm{f}(x)e^x\,\mathrm{d}x = \frac{\mathrm{P}}{\mathrm{Q}}e^x + C$, what equation must P and Q satisfy? (Differentiate the right-hand side.)

Differentiating, $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\dfrac{\mathrm{P}}{\mathrm{Q}}e^x\right) = \dfrac{\mathrm{P}'\mathrm{Q} - \mathrm{P}\mathrm{Q}' + \mathrm{P}\mathrm{Q}}{\mathrm{Q}^2}\,e^x$, and this must equal $\mathrm{f}(x)e^x$. So the

whole question is about the polynomial identity

$$P'Q - PQ' + PQ = fQ^2.$$

Questions about integrals have become questions about polynomials: their factors and their degrees.

Working it through

STAGE 1 The factor $x+1$

With $f(x) = \dfrac{x^3-2}{(x+1)^2}$, multiply the identity through by $(x+1)^2$:

$$(x+1)^2(P'Q - PQ' + PQ) = (x^3-2)\,Q^2. \qquad (\dagger)$$

Ask yourself Put $x=-1$ in $(\dagger)$. What must happen?

The left-hand side is 0 at $x=-1$. The right-hand side is $(-1-2)\,Q(-1)^2 = -3\,Q(-1)^2$. So $Q(-1)=0$, and by the factor theorem, $x+1$ is a factor of Q. ■

STAGE 2 Degrees

Let P have degree p and Q have degree q.

Ask yourself In $P'Q - PQ' + PQ$, which term has the highest degree? Can it be cancelled by the others?

PQ has degree $p+q$, while $P'Q$ and PQ' have degree at most $p+q-1$. So the left-hand side of $(\dagger)$ has degree exactly $2+p+q$, and the right-hand side has degree $3+2q$. Equating, $p = q+1$. ■

STAGE 3 Finding P

With $Q = x+1$, we have $Q' = 1$ and P has degree 2. Dividing $(\dagger)$ by $(x+1)^2$:

$$P'(x+1) - P + P(x+1) = x^3 - 2, \qquad \text{that is,} \qquad (x+1)P' + xP = x^3 - 2.$$

Try $P = \alpha x^2 + \beta x + \gamma$. The left-hand side is

$$(x+1)(2\alpha x + \beta) + \alpha x^3 + \beta x^2 + \gamma x = \alpha x^3 + (2\alpha+\beta)x^2 + (2\alpha+\beta+\gamma)x + \beta.$$

Matching with $x^3 - 2$: $\alpha = 1$, $2\alpha + \beta = 0$ so $\beta = -2$, and then $\gamma = 0$; the constant term $\beta = -2$ is consistent. So

$$P(x) = x^2 - 2x, \qquad \int \frac{x^3-2}{(x+1)^2}\,e^x\,dx = \frac{x^2-2x}{x+1}\,e^x + C. \qquad ■$$

(Check by differentiating, if you like; it is a good exercise in the quotient rule.)

STAGE 4 An impossibility

Now suppose $\int \frac{e^x}{x+1}\,dx = \frac{P}{Q}e^x + C$ with P and Q having no common factor. The identity becomes

$$(x+1)\big(P'Q - PQ' + PQ\big) = Q^2.$$

As in Stage 1, putting $x = -1$ shows $Q(-1) = 0$. Write $Q = (x+1)^k R$, where $k \geq 1$ and $R(-1) \neq 0$ (take out as many factors of $x+1$ as possible).

Pause and try Substitute $Q = (x+1)^k R$, cancel as many factors of $(x+1)$ as you can, and then put $x = -1$. What do you learn about $P(-1)$?

Then $Q' = k(x+1)^{k-1}R + (x+1)^k R'$, and

$$P'Q - PQ' + PQ = (x+1)^{k-1}\Big[(x+1)\big(P'R - PR' + PR\big) - kPR\Big].$$

So the identity reads $(x+1)^k\big[(x+1)(P'R - PR' + PR) - kPR\big] = (x+1)^{2k}R^2$. Dividing by $(x+1)^k$:

$$(x+1)\big(P'R - PR' + PR\big) - kPR = (x+1)^k R^2.$$

At $x = -1$, since $k \geq 1$, this gives $-kP(-1)R(-1) = 0$. As $k \neq 0$ and $R(-1) \neq 0$, we must have $P(-1) = 0$. But then $x+1$ divides both P and Q, contradicting the assumption that they have no common factor. So no such P and Q exist. ■

Looking back

- **Translate into algebra.** "Can this be integrated in this form?" became "does this polynomial identity have a solution?"
- **Evaluate at a special point.** Putting $x = -1$ extracted information about factors, three times.
- **Proof by contradiction, using lowest terms.** As with $\sqrt{2}$ in Section 1.3, assuming the fraction is in lowest terms and finding a common factor gave the contradiction.

What's really going on

Poles and logarithms

There is a simple picture behind part (ii). The integrand $\frac{e^x}{x+1}$ has a *simple pole* at $x = -1$: near there it behaves like $\frac{e^{-1}}{x+1}$. Integrating $\frac{1}{x+1}$ gives $\ln|x+1|$, so the antiderivative should behave like $e^{-1}\ln|x+1|$ near -1, growing to $-\infty$ logarithmically (Figure 4.6.1). But a rational function times e^x can only blow up like a power $(x+1)^{-k}$, and its

derivative then has a pole of order $k+1 \geq 2$, never a simple pole. Logarithmic behaviour cannot be produced by rational functions; the algebra in Stage 4 is a rigorous version of this observation. In part (i), by contrast, the integrand has a *double* pole, which is exactly what the derivative of something with a simple pole looks like, so a rational answer was possible.

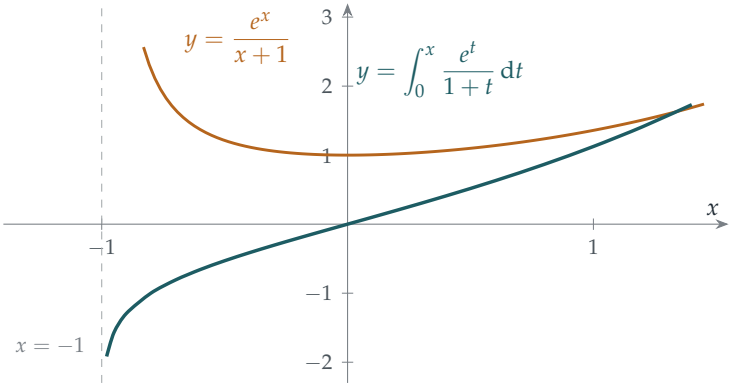


Figure 4.6.1 The integrand $\frac{e^x}{x+1}$ (amber) has a simple pole at $x = -1$. Its antiderivative (teal, computed numerically) exists perfectly well, but it tends to $-\infty$ *logarithmically* as $x \to -1^+$, like $e^{-1}\ln(x+1)$. No rational function times e^x can behave like that.

Liouville and the Risch algorithm

In the 1830s Joseph Liouville developed a theory of which integrals can be expressed in terms of *elementary functions* (those built from polynomials, exponentials, logarithms, roots and trigonometric functions by the usual operations). One of his results says that $\int \mathrm{f}(x)e^{\mathrm{g}(x)}\,\mathrm{d}x$, with f and g rational and g not constant, is elementary only if it equals $\mathrm{R}(x)e^{\mathrm{g}(x)}$ (plus a constant) for some rational R. (The condition on g matters: with $\mathrm{g} = 0$ and $\mathrm{f}(x) = \frac{1}{x}$ the integral is $\ln|x|$, which is elementary but not rational.) So this question, which rules out the rational form, actually proves that $\int \frac{e^x}{x+1}\,\mathrm{d}x$ is not elementary at all. Versions of the same theorem show that $\int e^{-x^2}\,\mathrm{d}x$ and $\int \frac{\sin x}{x}\,\mathrm{d}x$ are not elementary. In 1969 Robert Risch turned Liouville's ideas into an algorithm that decides, for a large class of functions, whether an elementary integral exists and finds it if it does. Versions of it run inside computer algebra systems.

Special functions

When an integral has no elementary form, mathematicians give it a name and study it as a new function, just as we did with the Lambert W function in Section 3.9. The integral here is essentially the *exponential integral* $\mathrm{Ei}(x)$, which appears in heat flow and in the *logarithmic integral* $\mathrm{li}(x) = \mathrm{Ei}(\ln x)$, the best simple approximation to the number of primes up to x (Section 1.2). For $x < 0$ it is $\int_{-\infty}^{x} \frac{e^t}{t}\,\mathrm{d}t$. For $x > 0$ that integral runs

through the infinite spike at $t = 0$, so $\mathrm{Ei}(x)$ is defined by cutting out a small interval $(-\varepsilon, \varepsilon)$ around the spike and letting $\varepsilon \to 0$: the *Cauchy principal value*.

The thinking skill

Proving impossibility

To show something cannot be done, assume it can, translate the assumption into precise algebra, and derive a contradiction.

The hard part is the translation: finding a precise statement of what "can be done" means. Here it meant "there are polynomials P, Q with no common factor satisfying this identity". Many of the deepest results in mathematics are impossibility proofs of this kind.

Try this next

1 **Two that work.** Show that $\displaystyle\int \frac{xe^x}{(x+1)^2}\,\mathrm{d}x = \frac{e^x}{x+1} + C$ and $\displaystyle\int \frac{(x^2+1)e^x}{(x+1)^2}\,\mathrm{d}x = \frac{x-1}{x+1}\,e^x + C$.

2 **The Gaussian.** Show that there is no polynomial P with $\displaystyle\int e^{-x^2}\,\mathrm{d}x = \mathrm{P}(x)e^{-x^2} + C$. (*Hint:* differentiate and compare degrees.)

3 **Another pole.** Adapt the argument of part (ii) to show that $\displaystyle\int \frac{e^x}{x}\,\mathrm{d}x$ is not of the form $\frac{\mathrm{P}(x)}{\mathrm{Q}(x)}e^x + C$.

4 **When is it possible?** Suppose f is a polynomial and $\displaystyle\int \frac{\mathrm{f}(x)}{(x+1)^2}\,e^x\,\mathrm{d}x = \frac{\mathrm{P}(x)}{x+1}\,e^x + C$ for some polynomial P. Show that $(x+1)\mathrm{P}' + x\mathrm{P} = \mathrm{f}$, and by evaluating this and its derivative at $x = -1$, show that $\mathrm{f}(-1) + \mathrm{f}'(-1) = 0$. Check that this condition holds for $\mathrm{f}(x) = x^3 - 2$ and for the examples in exercise 1. HARDERCan you show that the condition is also sufficient?

Further reading

- **Brian Conrad, "Impossibility theorems for elementary integration"**. Freely available lecture notes explaining Liouville's theorem with examples.
- **Elena Anne Marchisotto & Gholam-Ali Zakeri, "An invitation to integration in finite terms"**, *The College Mathematics Journal* 25 (1994). A gentle introduction to Liouville's theory.

- **Manuel Bronstein, *Symbolic Integration I*** (Springer). How computer algebra systems actually integrate. Advanced.

4.7 The Calculus of Variations

The question

(i) Let

$$I = \int_0^1 \left((y')^2 - y^2\right) \mathrm{d}x \qquad \text{and} \qquad I_1 = \int_0^1 (y' + y\tan x)^2 \, \mathrm{d}x,$$

where y is a given function of x satisfying $y = 0$ at $x = 1$. Show that $I - I_1 = 0$ and deduce that $I \geq 0$. Show further that $I = 0$ only if $y = 0$ for all x $(0 \leq x \leq 1)$.

(ii) Let

$$J = \int_0^1 \left((y')^2 - a^2y^2\right) \mathrm{d}x,$$

where a is a given positive constant and y is a given function of x, not identically zero, satisfying $y = 0$ at $x = 1$. By considering an integral of the form

$$\int_0^1 (y' + ay\tan bx)^2 \, \mathrm{d}x,$$

where b is suitably chosen, show that $J \geq 0$. You should state the range of values of a, in the form $a < k$, for which your proof is valid.

In the case $a = k$, find a function y (not everywhere zero) such that $J = 0$.

The hook

Most optimisation problems at school ask for the best *number*: the value of x that maximises an area, say. The *calculus of variations* asks for the best *function*. Which curve between two points does a bead slide down fastest? What shape does a hanging chain take? Which path does light follow through a lens? In each case the quantity to be optimised is an integral, and the unknown is a whole function.

This question studies the simplest interesting example. The integral $J = \int_0^1 \left((y')^2 - a^2y^2\right) \mathrm{d}x$ balances two competing terms: steepness pushes it up, size pushes it down. For small a, steepness always wins and $J \geq 0$ for every function. But there is a critical value of a at which the balance tips, and the question finds it exactly: $a = \frac{\pi}{2}$. Where π comes from will be the story of the section.

Before you start

Ask yourself Why is it not obvious that $I \geq 0$? Try $y = 1 - x$, which satisfies $y(1) = 0$.

I is a difference of two positive quantities, so it could have either sign. For $y = 1 - x$: $\int_0^1 1\,\mathrm{d}x = 1$ and $\int_0^1 (1-x)^2\,\mathrm{d}x = \frac{1}{3}$, so $I = \frac{2}{3} > 0$. But a cleverer function might make $\int y^2$ larger relative to $\int (y')^2$. The trick in the question is to write I as an integral of a *square*, which is obviously non-negative. This is "completing the square", for integrals.

Working it through

STAGE 1 Completing the square

Expanding the square in I_1:

$$I_1 = \int_0^1 \left((y')^2 + 2yy' \tan x + y^2 \tan^2 x\right) \mathrm{d}x,$$

so

$$I - I_1 = -\int_0^1 \left(y^2 + y^2 \tan^2 x + 2yy' \tan x\right) \mathrm{d}x = -\int_0^1 \left(y^2 \sec^2 x + 2yy' \tan x\right) \mathrm{d}x.$$

Ask yourself The integrand $y^2 \sec^2 x + 2yy' \tan x$ is an exact derivative. Of what?

By the product rule, $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(y^2 \tan x\right) = 2yy' \tan x + y^2 \sec^2 x$. So

$$I - I_1 = -\Big[y^2 \tan x\Big]_0^1 = -\left(y(1)^2 \tan 1 - y(0)^2 \tan 0\right) = 0,$$

because $y(1) = 0$ and $\tan 0 = 0$. So $I = I_1 \geq 0$, being the integral of a square. ■

STAGE 2 When is $I = 0$?

Ask yourself If the integral of a continuous non-negative function is zero, what can you say about the function? So what equation must y satisfy if $I = 0$?

A continuous non-negative function with zero integral must be identically zero (if it were positive at one point, it would be positive on a small interval around it, contributing a positive amount). So $I = 0$ forces $y' + y \tan x = 0$ on $[0, 1]$. Multiplying by $\sec x$, this says $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\cos x}\right) = \frac{y' \cos x + y \sin x}{\cos^2 x} = 0$, so $y = C \cos x$. Then $y(1) = C \cos 1 = 0$, and since $\cos 1 \neq 0$, $C = 0$. So $I = 0$ only if $y \equiv 0$. ■

STAGE 3 The general case

Ask yourself Repeat Stage 1 with $\int_0^1 (y' + ay\tan bx)^2\,\mathrm{d}x$. What must b be for the leftover terms to form an exact derivative?

Call this integral K. Then

$$\begin{aligned} J - K &= -\int_0^1 \left(a^2y^2 + a^2y^2\tan^2 bx + 2ayy'\tan bx\right)\mathrm{d}x \\ &= -\int_0^1 \left(a^2y^2\sec^2 bx + 2ayy'\tan bx\right)\mathrm{d}x. \end{aligned}$$

On the other hand $\frac{\mathrm{d}}{\mathrm{d}x}\left(ay^2\tan bx\right) = 2ayy'\tan bx + ab\,y^2\sec^2 bx$. These match exactly when $ab = a^2$, that is, $b = a$. Then

$$J - K = -\Big[ay^2\tan ax\Big]_0^1 = -a\,y(1)^2\tan a = 0,$$

so $J = K \geq 0$.

Pause and try Where can this argument break down? (Look at $\tan ax$ on the interval $0 \leq x \leq 1$.)

The argument needs $\tan ax$ to be defined and continuous for all x in $[0,1]$; otherwise K might not exist and the fundamental theorem of calculus cannot be applied. As x runs over $[0,1]$, ax runs over $[0,a]$, and tan is continuous there exactly when $a < \frac{\pi}{2}$. So the proof is valid for

$$a < k = \frac{\pi}{2}. \qquad \blacksquare$$

STAGE 4 The critical case

Ask yourself When $a = \frac{\pi}{2}$, the proof fails. In Stage 2, the function making $I = 0$ would have satisfied $y' + y\tan x = 0$. What is the analogous equation now, and what are its solutions?

The analogue is $y' + ay\tan ax = 0$, whose solutions are $y = C\cos ax$. With $a = \frac{\pi}{2}$, $y = \cos\frac{\pi x}{2}$ satisfies $y(1) = \cos\frac{\pi}{2} = 0$. Check directly:

$$\begin{aligned} J &= \frac{\pi^2}{4}\int_0^1 \left(\sin^2\frac{\pi x}{2} - \cos^2\frac{\pi x}{2}\right)\mathrm{d}x = -\frac{\pi^2}{4}\int_0^1 \cos \pi x\,\mathrm{d}x \\ &= -\frac{\pi^2}{4}\left[\frac{\sin \pi x}{\pi}\right]_0^1 = 0. \qquad \blacksquare \end{aligned}$$

So at the critical value, a non-zero function makes J exactly zero.

Looking back

- **Completing the square in an integral.** Adding and subtracting an exact derivative turned J into the integral of a square plus a boundary term that vanished.

- **Check where the tools fail.** The method worked only while $\tan ax$ stayed finite on the interval. The point where it failed, $a = \frac{\pi}{2}$, was exactly the critical value.
- **The extreme case solves a differential equation.** The function giving $J = 0$ came from the first-order equation $y' + ay\tan ax = 0$, and it also satisfies $y'' + a^2 y = 0$.

What's really going on

The brachistochrone and the birth of a subject

In 1696 Johann Bernoulli challenged the mathematicians of Europe: find the curve down which a bead slides from one point to another in the least time. Solutions came from Newton (anonymously, overnight; Bernoulli said he recognised "the lion by his claw"), Leibniz, l'Hôpital and Jakob Bernoulli. The answer is a cycloid. In the 1750s Euler and the young Joseph-Louis Lagrange turned these special tricks into a general method: if y makes $\int F(x, y, y')\,\mathrm{d}x$ stationary, then y satisfies the *Euler–Lagrange equation*

$$\frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial F}{\partial y'} = \frac{\partial F}{\partial y}.$$

For our J, $F = (y')^2 - a^2y^2$, and the Euler–Lagrange equation is $y'' + a^2y = 0$: simple harmonic motion.

Why $\pi/2$?

For $a < \frac{\pi}{2}$, J has a minimum value of 0, achieved only by $y \equiv 0$. At $a = \frac{\pi}{2}$ the minimum is still 0 but a second function achieves it. For $a > \frac{\pi}{2}$, the same function $y = \cos\frac{\pi x}{2}$ gives

$$J = \frac{\pi^2}{8} - \frac{a^2}{2} < 0,$$

and by scaling y up, J can be made as negative as you like: there is no minimum at all (Figure 4.7.1). The critical value is the first a for which $y'' + a^2y = 0$ has a non-zero solution satisfying the boundary conditions ($y(1) = 0$, and a "natural" condition $y'(0) = 0$ at the free end). That is an *eigenvalue problem*, and $a^2 = \frac{\pi^2}{4}$ is its smallest eigenvalue. Compare Section 3.7, where pinning the function at both ends of an interval of length π gave critical value 1; here the interval has length 1 and only one end is pinned, which is like half of an interval of length 2, so the critical a is $\frac{\pi}{2}$.

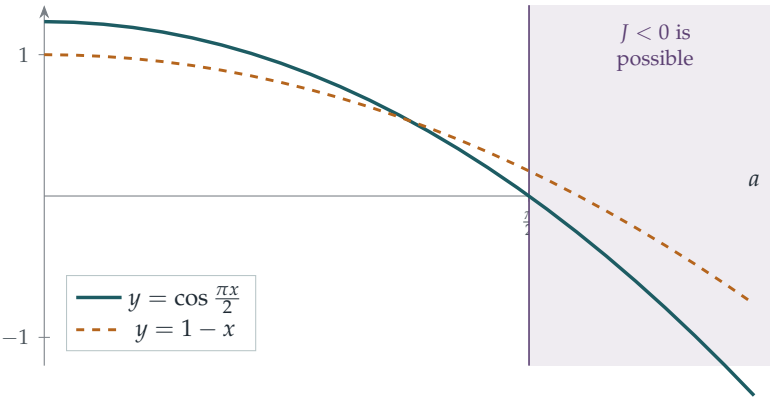


Figure 4.7.1 The value of J against a for two test functions with $y(1) = 0$. For $y = 1 - x$, $J = 1 - \frac{a^2}{3}$ becomes negative only when $a > \sqrt{3}$. For $y = \cos \frac{\pi x}{2}$, $J = \frac{\pi^2}{8} - \frac{a^2}{2}$ becomes negative as soon as $a > \frac{\pi}{2}$, and no function does better: for $a < \frac{\pi}{2}$, $J \geq 0$ for every y.

Physics and the principle of least action

The calculus of variations turned out to describe nature itself. In the eighteenth and nineteenth centuries it emerged that the laws of motion can be stated as: a mechanical system moves so as to make an integral called the *action* stationary. Light, similarly, travels along the path of least time (Fermat's principle), which explains refraction. In the twentieth century the same idea became the foundation of quantum mechanics, through Richard Feynman's "sum over paths". The quadratic integrals in this question are the simplest examples, and the eigenvalue problem they lead to reappears as the energy levels of a quantum oscillator in Section 8.6.

The thinking skill

Complete the square, then mind the boundary

To show a quadratic expression is non-negative, write it as a square plus something you can control. For integrals, the "something" is often an exact derivative, which integrates to boundary terms.

And always ask where the method stops working. The point of failure here was not an accident; it was the most important number in the problem.

Try this next

1 **Beyond the critical value.** Show that if $a > \frac{\pi}{2}$ then $y = \cos \frac{\pi x}{2}$ gives $J < 0$.

2 The Euler–Lagrange equation. HARDERSuppose y minimises J among functions with $y(1) = 0$. For any function η with $\eta(1) = 0$ and any number ε, $J(y + \varepsilon\eta) \geq J(y)$. Expand $J(y + \varepsilon\eta)$ in powers of ε and deduce that $\int_0^1 (y'\eta' - a^2 y\eta)\,\mathrm{d}x = 0$. Integrate by parts to show that $y'' + a^2 y = 0$ and $y'(0) = 0$ (assuming y is smooth enough).

3 A longer interval. Show by a substitution that if $y(L) = 0$, then $\int_0^L \left((y')^2 - a^2 y^2\right)\mathrm{d}x \geq 0$ for all y exactly when $a \leq \frac{\pi}{2L}$, and that a non-zero y gives equality only when $a = \frac{\pi}{2L}$.

4 The cycloid. Look up the brachistochrone problem and find the parametric equations of the cycloid. Show that a bead released from rest anywhere on an upside-down cycloid reaches the bottom in the same time (the *tautochrone* property, found by Huygens in 1659).

Further reading

- **Paul Nahin, *When Least Is Best*** (Princeton University Press). The history and mathematics of optimisation, including the brachistochrone, written for students.
- **Richard Feynman, *The Feynman Lectures on Physics*,** Volume II, Chapter 19 (“The principle of least action”). Free online.
- **Bruce van Brunt, *The Calculus of Variations*** (Springer). An undergraduate textbook; the first chapters need only A-level calculus.

4.8 Slicing a Spherical Loaf

The question

A spherical loaf of bread is cut into parallel slices of equal thickness. Show that, after any number of the slices have been eaten, the area of crust remaining is proportional to the number of slices remaining.

A European ruling decrees that a parallel-sliced spherical loaf can only be referred to as 'crusty' if the ratio of volume V (in cubic metres) of bread remaining to area A (in square metres) of crust remaining after any number of slices have been eaten satisfies $V/A < 1$. Show that the radius of a crusty parallel-sliced spherical loaf must be less than $2\frac{2}{3}$ metres.

[*The area A and volume V formed by rotating a curve in the x–y plane round the x-axis from $x = -a$ to $x = -a + t$ are given by*

$$A = 2\pi \int_{-a}^{-a+t} y \left(1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2\right)^{\frac{1}{2}} \mathrm{d}x, \qquad V = \pi \int_{-a}^{-a+t} y^2 \, \mathrm{d}x.]$$

The hook

Slice a spherical loaf into slices of equal thickness. The slices near the middle are fat discs with a thin band of crust round the edge. The slices near the ends are small caps that seem to be almost all crust. Surely the end slices have more crust?

They don't. Every slice has exactly the same area of crust, however near the middle or the end it is cut. The thin central band is wide in circumference but nearly vertical; the end cap is small in circumference but slopes steeply, so its crust is stretched over a larger width. The two effects cancel exactly. This is a theorem of Archimedes, and it was so dear to him that he asked for a sphere inside a cylinder to be carved on his tomb. The second part of the question is a lighter-hearted piece of regulation, but it rests on the same calculation.

Before you start

Ask yourself The sphere of radius a is obtained by rotating which curve about the x-axis? What is $\frac{dy}{dx}$ for that curve?

Rotating the semicircle $y = \sqrt{a^2 - x^2}$, $-a \le x \le a$, gives the sphere. Then $\frac{dy}{dx} = \frac{-x}{\sqrt{a^2-x^2}} = -\frac{x}{y}$. Keep that in mind: the slope is steep near the ends ($x \to \pm a$) and flat in the middle, which is exactly the effect described in the hook.

Working it through

STAGE 1 The crust formula simplifies

Ask yourself Compute $y\sqrt{1 + (y')^2}$ for the semicircle. Something remarkable happens.

With $y' = -x/y$,

$$1 + (y')^2 = 1 + \frac{x^2}{y^2} = \frac{y^2 + x^2}{y^2} = \frac{a^2}{y^2},$$

so $y\sqrt{1 + (y')^2} = y \cdot \frac{a}{y} = a$, a constant! The crust area between any two planes $x = x_1$ and $x = x_2$ is therefore

$$2\pi \int_{x_1}^{x_2} a \, dx = 2\pi a(x_2 - x_1).$$

It depends only on the distance between the planes, not on where they are.

STAGE 2 Crust is proportional to slices

If the loaf is cut into slices of thickness h, each slice has crust area $2\pi ah$, the same for all of them. Eat some slices, from one end, from both ends, or from anywhere, and the crust remaining is $2\pi ah$ times the number of slices remaining. ■

STAGE 3 The crusty condition

The hint in the question describes the bread that is left when slices are eaten from one end, so follow it: suppose the slices are eaten from the end $x = a$, so that what remains runs from $x = -a$ to $x = -a + t$. A ruling about "any number of slices" should hold however finely the loaf is cut, so we treat t as able to take any value between 0 and $2a$. (Stage 4 looks at slices eaten from both ends.)

Ask yourself Compute V using the given formula. A substitution $s = x + a$ might make the algebra cleaner.

With $s = x + a$, $y^2 = a^2 - x^2 = a^2 - (s-a)^2 = 2as - s^2$, so

$$V = \pi \int_0^t (2as - s^2)\,\mathrm{d}s = \pi\left(at^2 - \frac{t^3}{3}\right), \qquad A = 2\pi at,$$

and

$$\frac{V}{A} = \frac{at^2 - t^3/3}{2at} = \frac{t}{2} - \frac{t^2}{6a}.$$

Pause and try The ruling requires $V/A < 1$ *after any number of slices have been eaten*, that is, for every t with $0 < t \le 2a$. What is the largest value of $\frac{t}{2} - \frac{t^2}{6a}$ on this range?

Differentiating with respect to t: $\frac{1}{2} - \frac{t}{3a} = 0$ at $t = \frac{3a}{2}$, which is inside the range $(0, 2a]$, and this is a maximum (the expression is a downward parabola in t). The maximum value is

$$\frac{3a}{4} - \frac{9a^2/4}{6a} = \frac{3a}{4} - \frac{3a}{8} = \frac{3a}{8}.$$

A crusty loaf needs $V/A < 1$ for every t, and in particular at $t = \frac{3a}{2}$, so $\frac{3a}{8} < 1$, that is, $a < \frac{8}{3} = 2\frac{2}{3}$ metres. ■

STAGE 4 Beyond the question: is $2\frac{2}{3}$ enough?

Ask yourself The question asks only for a condition the radius *must* satisfy. Is $a < \frac{8}{3}$ also enough? What if the slices are eaten from both ends?

It is not enough. Eating from both ends leaves a band in the middle, and a middle band holds more bread for its crust than a cap at the end. For the band $-\frac{w}{2} \le x \le \frac{w}{2}$,

$$V = \pi \int_{-w/2}^{w/2} (a^2 - x^2)\,\mathrm{d}x = \pi\left(a^2 w - \frac{w^3}{12}\right), \qquad A = 2\pi a w, \qquad \frac{V}{A} = \frac{a}{2} - \frac{w^2}{24a},$$

which is close to $\frac{a}{2}$ for a thin band. For example, a loaf of radius 2.4 metres, comfortably under $2\frac{2}{3}$, cut into four slices with the two end slices eaten, leaves the band with $w = a$ and $\frac{V}{A} = \frac{11a}{24} = 1.1$. In fact $\frac{V}{A} < \frac{a}{2}$ for any collection of slices, so if slices may be eaten from anywhere, a loaf is crusty exactly when $a \le 2$ metres. The question's $2\frac{2}{3}$ is a correct necessary condition, but the ruling is stricter than the question lets on.

Looking back

- **Simplify before integrating.** The integrand for the crust looked complicated but turned out to be constant. Always simplify an integrand before attacking it.
- **Read "for any" carefully.** The condition had to hold after *any* number of slices, so we needed the maximum of V/A, not its value for the whole loaf (which is $\frac{a}{3}$).
- **A surprising answer from a simple calculation.** The equal-crust property sounds implausible, but the calculation is two lines long.

What's really going on

Archimedes' hat-box theorem

Archimedes proved, in *On the Sphere and Cylinder*, that the area of the band of a sphere between two parallel planes equals the area of the band they cut from the cylinder that wraps around the sphere (Figure 4.8.1). Our calculation is this theorem: the cylinder of radius a has crust area $2\pi a$ per unit length, and so does the sphere. Taking the whole sphere, its surface area equals that of the cylinder's curved surface, $2\pi a \times 2a = 4\pi a^2$. Archimedes regarded this, together with the fact that the sphere has two-thirds the volume of the cylinder, as his finest work. Cicero, visiting Syracuse in 75 BC, found the overgrown tomb by looking for the sphere and cylinder carved on it.

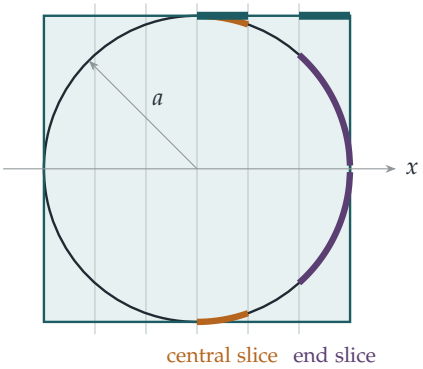


Figure 4.8.1 Cross-section of a spherical loaf of radius a inside its wrapping cylinder, cut into slices of equal thickness. The central slice (amber) has a short, nearly horizontal arc of crust far from the axis; the end slice (plum) has a long, steep arc close to the axis. When rotated about the x-axis, both give exactly the same crust area as the corresponding band of the cylinder (teal): $2\pi a$ times the thickness.

Maps of the Earth

The hat-box theorem gives a way to draw a map of the Earth that preserves area. Project each point of the globe horizontally outwards onto the wrapping cylinder, then unroll the cylinder. Bands of the sphere map to bands of the cylinder of the same area, and the same is true for any region. This is *Lambert's cylindrical equal-area projection* (1772). It distorts shapes badly near the poles, but areas are honest, unlike the familiar Mercator projection, which makes Greenland look as big as Africa (it is about fourteen times smaller). A famous theorem of Gauss, the *Theorema Egregium* (1827), shows that no map of a sphere can preserve both areas and shapes: every flat map of the Earth must distort something.

Random points on a sphere

Here is a practical consequence. To choose a point uniformly at random on a sphere, you might try choosing a latitude and longitude uniformly, but that crowds points near the poles. The hat-box theorem gives the right recipe: choose the height x uniformly between $-a$ and a, and the longitude uniformly between 0 and 2π. Because area is proportional to height, this is exactly uniform. Computer simulations in physics and graphics use this method. (Remarkably, it works only in three dimensions: the corresponding statement for spheres in four or more dimensions is false.)

The thinking skill

Look for hidden simplicity

When a calculation produces something unexpectedly simple, such as a constant, stop and ask why. There is usually a geometric reason, and finding it is where the real understanding lies.

Here the integrand $y\sqrt{1+(y')^2}$ was constant because steepness and distance from the axis exactly compensate each other on a sphere: that is Archimedes' theorem.

Try this next

1. **The whole loaf.** Show that for the whole uncut loaf, $V/A = \frac{a}{3}$. Why is the condition in the question harder to satisfy than $\frac{a}{3} < 1$?

2. **The tropics.** What fraction of the Earth's surface lies between latitudes 23.4° south and 23.4° north? (Treat the Earth as a sphere. Since area is proportional to height, you only need $\sin 23.4^\circ$.)

3. **Other shapes.** Show that for a loaf in the shape of a paraboloid (rotate $y = \sqrt{x}$ about the x-axis), the crust per slice is *not* constant. For which curves $y(x)$ is $y\sqrt{1+(y')^2}$ constant? (Solve the differential equation.)

4. **Volumes.** Use the volume formula to show that a sphere has two-thirds the volume of its wrapping cylinder, as Archimedes found.

Further reading

- **Sherman Stein, *Archimedes: What Did He Do Besides Cry Eureka?*** (Mathematical Association of America). Archimedes' mathematics explained for a modern reader.
- **Timothy Gowers, *Mathematics: A Very Short Introduction*** (Oxford University Press). The chapter on geometry introduces the geometry of curved surfaces.

- **John Snyder, *Flattening the Earth*** (University of Chicago Press). Two thousand years of map projections.

4.9 Holditch's Theorem

The question

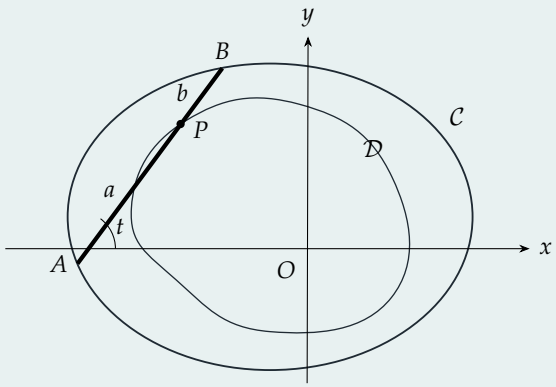
A movable point P has cartesian coordinates (x, y), where x and y are functions of t. The polar coordinates of P with respect to the origin O are r and θ. Starting with the expression

$$\tfrac{1}{2}\int r^2\,\mathrm{d}\theta$$

for the area swept out by OP, obtain the equivalent expression

$$\tfrac{1}{2}\int\left(x\frac{\mathrm{d}y}{\mathrm{d}t}-y\frac{\mathrm{d}x}{\mathrm{d}t}\right)\mathrm{d}t. \qquad (*)$$

The ends of a thin straight rod AB lie on a closed convex curve $\mathcal{C}$. The point P on the rod is a fixed distance a from A and a fixed distance b from B. The angle between AB and the positive x direction is t. As A and B move anticlockwise round $\mathcal{C}$, the angle t increases from 0 to 2π and P traces a closed convex curve $\mathcal{D}$ inside $\mathcal{C}$, with the origin O lying inside $\mathcal{D}$, as shown in the diagram.



Let (x, y) be the coordinates of P. Write down the coordinates of A and B in terms of a, b, x, y and t.

The areas swept out by OA, OB and OP are denoted by $[A]$, $[B]$ and $[P]$, respectively. Show, using $(*)$, that

$$[A] = [P] + \pi a^2 - af$$

where

$$f = \tfrac{1}{2}\int_0^{2\pi}\left(\left(x+\frac{\mathrm{d}y}{\mathrm{d}t}\right)\cos t+\left(y-\frac{\mathrm{d}x}{\mathrm{d}t}\right)\sin t\right)\mathrm{d}t.$$

Obtain a corresponding expression for $[B]$ involving b. Hence show that the area between the curves $\mathcal{C}$ and $\mathcal{D}$ is πab.

The hook

Take any oval, any shape at all as long as it is convex, and slide a rod of fixed length round inside it, keeping both ends on the curve. Paint a dot on the rod, dividing it into lengths a and b. As the rod goes once round, the dot traces out a smaller oval. What is the area of the ring between the two curves?

The astonishing answer is *πab, whatever the shape of the original curve*. A long thin oval, an egg, a rounded triangle: the area of the ring is always the same, and it depends only on where the dot is on the rod. The result was found in 1858 by the Reverend Hamnet Holditch, a fellow of Gonville and Caius College, Cambridge, and it is one of the most surprising theorems in elementary geometry. This question proves it with a single formula for area.

Before you start

Ask yourself Check the theorem in the simplest case. If $\mathcal{C}$ is a circle of radius R, what curve does P trace, and what is the area of the ring?

By symmetry, P traces a circle of some radius r with the same centre. The rod AB is a chord of the big circle through P, and the *intersecting chords theorem* (or "power of a point") says that for any chord through P, $PA \times PB = R^2 - r^2$. So $ab = R^2 - r^2$, and the ring has area $\pi R^2 - \pi r^2 = \pi ab$. The theorem says this remains true for every convex curve, not just circles.

Working it through

STAGE 1 Area from coordinates

Ask yourself With $x = r\cos\theta$ and $y = r\sin\theta$, all functions of t, compute $x\dot{y} - y\dot{x}$, where dots denote $\frac{\mathrm{d}}{\mathrm{d}t}$.

$\dot{x} = \dot{r}\cos\theta - r\sin\theta\,\dot{\theta}$ and $\dot{y} = \dot{r}\sin\theta + r\cos\theta\,\dot{\theta}$. Then

$$x\dot{y} - y\dot{x} = r\cos\theta\,(\dot{r}\sin\theta + r\cos\theta\,\dot{\theta}) - r\sin\theta\,(\dot{r}\cos\theta - r\sin\theta\,\dot{\theta}) = r^2\dot{\theta}.$$

So $\frac{1}{2}\int r^2\,\mathrm{d}\theta = \frac{1}{2}\int r^2\dot{\theta}\,\mathrm{d}t = \frac{1}{2}\int (x\dot{y} - y\dot{x})\,\mathrm{d}t$. ■

This formula needs no polar coordinates, only the cartesian position as a function of a parameter. When a point goes once anticlockwise round a closed curve enclosing O, it

gives the area inside the curve.

STAGE 2 The ends of the rod

The rod points in the direction $(\cos t, \sin t)$ from A to B, with P a distance a from A and b from B. So

$$A = (x - a\cos t,\ y - a\sin t), \qquad B = (x + b\cos t,\ y + b\sin t).$$

STAGE 3 The area swept by OA

Ask yourself Apply $(*)$ to A. Expand $x_A\dot{y}_A - y_A\dot{x}_A$ and sort the terms by powers of a.

With $x_A = x - a\cos t$, $y_A = y - a\sin t$, we have $\dot{x}_A = \dot{x} + a\sin t$ and $\dot{y}_A = \dot{y} - a\cos t$. Expanding,

$$\begin{aligned} x_A\dot{y}_A - y_A\dot{x}_A &= (x - a\cos t)(\dot{y} - a\cos t) - (y - a\sin t)(\dot{x} + a\sin t) \\ &= (x\dot{y} - y\dot{x}) + a^2(\cos^2 t + \sin^2 t) - a\big[(x + \dot{y})\cos t + (y - \dot{x})\sin t\big]. \end{aligned}$$

Integrating from 0 to 2π and halving: the first term gives $[P]$, the second gives $\frac{1}{2}a^2 \cdot 2\pi = \pi a^2$, and the third gives $-af$. So

$$[A] = [P] + \pi a^2 - af. \qquad \blacksquare$$

Pause and try Now find $[B]$. (*Hint:* B is obtained from the formula for A by replacing a with what?)

$B = (x + b\cos t, y + b\sin t)$ is the formula for A with a replaced by $-b$. So

$$[B] = [P] + \pi b^2 + bf.$$

STAGE 4 Eliminating f

Ask yourself Both A and B go once round the same curve $\mathcal{C}$. What does that tell you about $[A]$ and $[B]$?

Each of A and B travels once anticlockwise round $\mathcal{C}$ as t goes from 0 to 2π, so $[A] = [B] = \text{area}(\mathcal{C})$. Equating the two expressions,

$$\pi a^2 - af = \pi b^2 + bf \quad\Longrightarrow\quad f(a+b) = \pi(a^2 - b^2) \quad\Longrightarrow\quad f = \pi(a - b).$$

Since P goes once round $\mathcal{D}$, $[P] = \text{area}(\mathcal{D})$, and

$$\text{area}(\mathcal{C}) - \text{area}(\mathcal{D}) = [A] - [P] = \pi a^2 - a\pi(a - b) = \pi ab. \qquad \blacksquare$$

The mysterious quantity f, which depends on the shape of the curve, was never computed: it was eliminated.

Looking back

- **A formula suited to the problem.** The area formula $(*)$ needed only coordinates as functions of a parameter, which is exactly how the rod's motion is described.
- **Two expressions, one unknown.** We could not compute f, but we had two equations involving it, so we eliminated it, like solving simultaneous equations.
- **Test the special case first.** The circle gave the answer πab before we started, and told us what to aim for.

What's really going on

Green's theorem and the planimeter

The formula $(*)$ is a special case of *Green's theorem*, which relates an integral round a closed curve to an integral over the region inside it. In its area form, it says that the area enclosed by a curve is $\frac{1}{2}\oint(x\,\mathrm{d}y - y\,\mathrm{d}x)$, an integral that can be computed by travelling round the boundary alone. Engineers exploited this with the *planimeter*, a mechanical instrument invented in the nineteenth century (Jakob Amsler's 1854 design was the most successful): you trace the boundary of a shape on a map or diagram with a pointer, and a wheel records the enclosed area. Planimeters were used to measure land areas, to compute the work done in steam engines from indicator diagrams, and in many other applications, until digital methods replaced them. Holditch's theorem is closely related: the rod is like the arm of a planimeter.

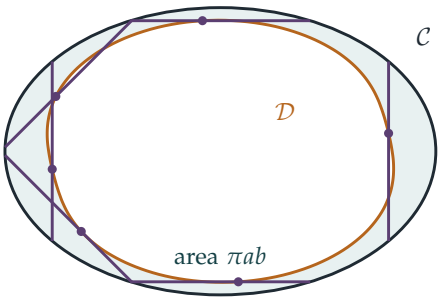


Figure 4.9.1 Holditch's theorem for an ellipse with semi-axes 3 and 2, and a rod of length 2.5 with $a = 1$, $b = 1.5$. Some positions of the rod are shown (plum), with the point P marked. The curve $\mathcal{D}$ traced by P (amber) was computed numerically, and the shaded ring between the curves has area $4.7124\ldots = 1.5\pi$, exactly as the theorem predicts.

The fine print

Holditch stated his theorem without much care for conditions, and it needs some. The rod must be able to go all the way round with both ends on the curve, and the traced

curve $\mathcal{D}$ must be a simple closed curve traced once; if the rod is too long compared with the width of $\mathcal{C}$, $\mathcal{D}$ can cross itself, and the "area" given by the formula counts some regions twice or with a negative sign. The question avoids these difficulties by assuming $\mathcal{D}$ is a convex curve containing O. Finding exactly when the theorem holds is a good example of how a beautiful idea must be made precise, a theme of Chapter 3.

The thinking skill

Eliminate what you cannot compute

If a quantity is too complicated to calculate, look for two different equations that contain it, and eliminate it.

Here f depended on every detail of the curve, but it appeared in the expressions for both $[A]$ and $[B]$, and the fact that $[A] = [B]$ removed it. The answer, πab, could only be independent of the curve because the curve-dependent part cancelled.

Try this next

1 **The area of an ellipse.** Use $(*)$ with $x = p\cos t$, $y = q\sin t$ to show that an ellipse with semi-axes p and q has area πpq.

2 **The circle case.** Prove the intersecting chords theorem: if two chords of a circle meet at P, the products of the lengths of their two segments are equal. Deduce Holditch's theorem for a circle.

3 **The midpoint.** Show that $f = \pi(a - b)$ means that when P is the midpoint of the rod, $f = 0$. What is the ring's area then, in terms of the rod's length L?

4 **A triangle.** HARDERUse $(*)$ to find the area of the triangle with vertices $(0,0)$, (x_1, y_1), (x_2, y_2), traversing its boundary, and show it equals $\frac{1}{2}|x_1y_2 - x_2y_1|$. This is the start of the "shoelace formula" for the area of any polygon.

Further reading

- **Mark Levi, *The Mathematical Mechanic*** (Princeton University Press). Uses physical reasoning, with rods, wheels and springs, to prove mathematical results, in the spirit of this section.
- **H. M. Schey, *Div, Grad, Curl, and All That*** (Norton). An informal introduction to vector calculus, including Green's theorem.
- **Hamnet Holditch, "Geometrical theorem"**, *Quarterly Journal of Pure and Applied Mathematics* 2 (1858). The original note, a single short paragraph with none of the fine

print.

What We Have Learned

We began by asking what an integral really is, what integrals can prove, and what to do when an integral has no formula. Here is how the chapter's nine questions answered them.

1. What is an integral, and which functions have one?

In Section 4.1 we defined the integral as the common limit of lower and upper sums of thin rectangles, and computed $\int_0^a x^5\,\mathrm{d}x$ from that definition alone. It took some work: telescoping sums of consecutive products, bounds on both sides, and a squeeze. That effort is the measure of how much the fundamental theorem of calculus gives you for free.

The definition also told us that not every function is integrable. Dirichlet's function, 1 on the rationals and 0 on the irrationals, has upper sums all equal to 1 and lower sums all equal to 0. Problems like this one led Lebesgue to a more powerful theory of integration, which you will meet later at university.

2. What can integrals prove?

Far more than areas.

- **Inequalities between numbers.** An integral of a positive function equals $\frac{22}{7} - \pi$, so $\frac{22}{7} > \pi$ (Section 4.3). The idea, "write $A - B$ as the integral of something positive", is as useful as it is simple.
- **Infinite products and special values.** Reduction formulae for $\int \sin^n t\,\mathrm{d}t$, an invariant and a squeeze gave Wallis's product, the constant in Stirling's formula and $\left(\frac{1}{2}\right)! = \frac{\sqrt{\pi}}{2}$ (Section 4.4).
- **Infinite sums.** Expanding x^x as a power series and integrating term by term gave the sophomore's dream (Section 4.5).
- **The foundations of trigonometry.** Defining $\mathrm{T}(x) = \int_0^x \frac{\mathrm{d}u}{1+u^2}$ and $\pi = 2\mathrm{T}(\infty)$, substitutions alone produced the addition formula and special values like $\mathrm{T}(\sqrt{3}) = \frac{\pi}{3}$ (Section 4.2).
- **Geometry.** An integrand that simplified to a constant revealed Archimedes' hat-box theorem (Section 4.8), and an area formula written as an integral round the boundary proved Holditch's astonishing theorem (Section 4.9).
- **Optimal functions.** Completing the square inside an integral showed exactly when $\int \left((y')^2 - a^2y^2\right)$ must be non-negative, and the critical value $\frac{\pi}{2}$ turned out to be an

eigenvalue (Section 4.7).

3. What can we do when there is no formula?

In Section 4.6 we proved that $\int \frac{e^x}{x+1}\,\mathrm{d}x$ cannot be written as a rational function times e^x, and Liouville's theory shows it has no elementary formula at all. The reason, seen through the lens of poles, is that the antiderivative must behave like a logarithm near $x = -1$, which rational functions cannot do. Yet the integral exists perfectly well, and we can compute it: numerically, by series (as for $\int x^x$ in Section 4.5), or by giving it a name and studying it as a new function, as we did with the W function in Section 3.9.

4. A toolkit for integration

Method	Where you used it
Working from the definition	$\int_0^a x^5$ from rectangles (4.1)
Telescoping sums	Sums of consecutive products (4.1)
Substitutions that preserve an integral	Trigonometry from $\int \frac{\mathrm{d}u}{1+u^2}$ (4.2)
Polynomial division	$\frac{22}{7} - \pi$ (4.3)
Positivity and bounding the integrand	$\pi < \frac{22}{7}$ (4.3)
Reduction formulae	$\int \sin^n t$ (4.4); $\int x^m (\ln x)^n$ (4.5)
Invariants and squeezes	$nu_n u_{n-1} = \frac{\pi}{2}$ (4.4)
Term-by-term integration	The sophomore's dream (4.5)
Factor and degree arguments	No rational antiderivative (4.6)
Completing the square	The calculus of variations (4.7)
Simplifying before integrating	The spherical loaf (4.8)
Area as a boundary integral	Holditch's theorem (4.9)

Where next

- The Gamma and Beta functions of Section 4.4, and the constant $\sqrt{2\pi}$ in Stirling's formula, reappear in probability in Section 12.23.
- The eigenvalue problem of Section 4.7 becomes the quantum harmonic oscillator in Section 8.6.
- Machin's formula, built from the addition identity of Section 4.2, is the subject of Section 5.1.
- Chapter 5 turns to complex numbers, which explain why so many of this chapter's integrals involve π.

If you want to go further before university

- **Paul Nahin, *Inside Interesting Integrals*** (Springer): a feast of clever integrals

and the tricks behind them.

- **William Dunham, *The Calculus Gallery***: masterpieces of analysis from Newton to Lebesgue.
- **Michael Spivak, *Calculus***: the definition of the integral and the fundamental theorem, done properly.
- **Sherman Stein, *Archimedes***: for the ancient roots of integration.

Hermite recoiled from functions without derivatives, but integration has never had that problem: it is generous, accepting functions that differentiation cannot handle, and it keeps producing surprises. If a single theme runs through this chapter, it is that integrals are not just a technique for finding areas. They are a language for expressing and proving facts about numbers, functions and shapes, and you will be speaking it for the rest of your mathematical life.

Answers to Try This Next

These are brief answers and hints, not full solutions: they give the result and the key step, so that you can check your own work. Try each exercise properly before looking.

4.1 What Is an Integral?

1. The sum is the total area of n rectangles of width $\frac{1}{n}$ under $y = \mathrm{f}(x)$ on $[1,2]$. With $\mathrm{f}(x) = \frac{1}{x}$, $\frac{1}{n}\sum \frac{1}{1+m/n} = \sum \frac{1}{n+m} \to \ln 2$. The second sum is $\frac{1}{n}\sum \frac{1}{1+(m/n)^2} \to \int_0^1 \frac{\mathrm{d}x}{1+x^2} = \frac{\pi}{4}$.
2. The upper sum with n strips is $\frac{a}{n}\sum_{r=1}^{n}\left(\frac{ra}{n}\right)^2 = \frac{a^3(n+1)(2n+1)}{6n^2}$, and the lower sum is the same with $n-1$ in place of $n+1$ in the first factor; both tend to $\frac{a^3}{3}$.
3. The terms telescope: $r(r+1)\cdots(r+k) - (r-1)r\cdots(r+k-1) = (k+1)\,r(r+1)\cdots(r+k-1)$.
4. $\sum_{r=1}^{n} r^5 = \frac{n^2(n+1)^2(2n^2+2n-1)}{12}$.

4.2 Building Trigonometry from an Integral

1. The derivative is $-\mathrm{f}(a-x)\mathrm{f}(x) + \mathrm{f}(a-x)\mathrm{f}(x) = 0$, so $\mathrm{f}(a-x)\mathrm{f}(x) = \mathrm{f}(a)$; put $a = x+y$. For the cosine, with y fixed let $p(x) = \mathrm{c}(x+y) - \mathrm{c}(x)\mathrm{c}(y) + \mathrm{s}(x)\mathrm{s}(y)$ and $q(x) = \mathrm{s}(x+y) - \mathrm{s}(x)\mathrm{c}(y) - \mathrm{c}(x)\mathrm{s}(y)$. Then $p' = -q$ and $q' = p$, so p^2+q^2 is constant, and it is 0 at $x = 0$.
2. $\frac{\mathrm{d}y}{\mathrm{d}u} = (1+u^2)^{-3/2}$ and $\sqrt{1-y^2} = (1+u^2)^{-1/2}$, so $\frac{\mathrm{d}y}{\sqrt{1-y^2}} = \frac{\mathrm{d}u}{1+u^2}$. Then $\mathrm{S}(\frac{1}{\sqrt{2}}) = \mathrm{T}(1)$, and part (i) with $x = 1$ gives $\mathrm{T}(1) = \frac{1}{2}\mathrm{T}_\infty$.
3. With $a = \frac{1}{2}$ and $x = \frac{1}{3}$, $\frac{x+a}{1-ax} = 1$, so $\mathrm{T}(\frac{1}{3}) = \mathrm{T}(1) - \mathrm{T}(\frac{1}{2})$.
4. $\frac{1}{1+u^2} \le 1$ on $[0,1]$ and $\le \frac{1}{u^2}$ on $[1,x]$, so for $x \ge 1$, $\mathrm{T}(x) \le 1 + (1 - \frac{1}{x}) < 2$; for $0 < x < 1$, $\mathrm{T}(x) < \mathrm{T}(1) \le 1$. (The bound $1 + \int_1^x u^{-2}\,\mathrm{d}u$ is not valid for $x < 1$: at $x = \frac{1}{2}$ it equals 0.) T is increasing and bounded, so it has a limit.

4.3 $22/7$ Is Bigger Than π

1. $\frac{22}{7} - \pi \approx 0.001264$, between $\frac{1}{1260} \approx 0.000794$ and $\frac{1}{630} \approx 0.001587$.
2. $\frac{(4!)^2}{9!} = \frac{576}{362\,880} = \frac{1}{630}$.
3. $\frac{x(1-x)^2}{1+x^2} = x - 2 + \frac{2}{1+x^2}$, which integrates to $\frac{1}{2} - 2 + \frac{\pi}{2}$. Similarly $\frac{x^2(1-x)^4}{1+x^2} =$

$x^4 - 4x^3 + 5x^2 - 4 + \frac{4}{1+x^2}$, which integrates to $\frac{1}{5} - 1 + \frac{5}{3} - 4 + \pi = \pi - \frac{47}{15}$. Both integrals are positive. The integrands are much larger than $\frac{x^4(1-x)^4}{1+x^2}$ (whose maximum is about 0.003), so the errors $\frac{\pi}{2} - \frac{3}{2} \approx 0.07$ and $\pi - \frac{47}{15} \approx 0.008$ are larger.

4. Polynomial division leaves the remainder $\frac{16}{1+x^2}$, which integrates to 4π; the integral is $4\pi - \frac{188684}{15015} \approx 3.6 \times 10^{-6}$, so $\pi \approx 3.14159174$.

4.4 Wallis, Beta and the Factorial of a Half

1. $u_{2m} = \frac{1\cdot3\cdots(2m-1)}{2\cdot4\cdots2m} \cdot \frac{\pi}{2}$ and $u_{2m+1} = \frac{2\cdot4\cdots2m}{3\cdot5\cdots(2m+1)}$. Then $1 < \frac{u_{2m}}{u_{2m+1}} < \frac{u_{2m-1}}{u_{2m+1}} = \frac{2m+1}{2m}$, so the ratio tends to 1, which rearranges to $\frac{\pi}{2} = \prod_{k\geq1} \frac{(2k)^2}{(2k-1)(2k+1)}$.

2. Integration by parts gives $I_n = \frac{n}{n+1} \int_0^1 x^{n+1}(1-x)^{n-1}\,\mathrm{d}x$, and writing $x^{n+1}(1-x)^{n-1} = x^n(1-x)^{n-1} - x^n(1-x)^n$ together with symmetry leads to $I_n = \frac{n}{2(2n+1)} I_{n-1}$; induction gives $\frac{(n!)^2}{(2n+1)!}$. With $x = \sin^2\theta$, $I_{1/2} = \int_0^{\pi/2} 2\sin^2\theta\cos^2\theta\,\mathrm{d}\theta = \frac{\pi}{8}$. The formula suggests $\left(\left(\frac{1}{2}\right)!\right)^2/2! = \frac{\pi}{8}$, so $\left(\frac{1}{2}\right)! = \frac{\sqrt{\pi}}{2}$ (which is $\Gamma\left(\frac{3}{2}\right)$).

3. Symmetry from $x \mapsto 1-x$; the recurrence from $x^p = x^{p-1} - x^{p-1}(1-x)$; and with $x = \frac{t}{1+t}$, $1-x = \frac{1}{1+t}$ and $\mathrm{d}x = \frac{\mathrm{d}t}{(1+t)^2}$. The integral is $\mathrm{b}\left(\frac{5}{2}, \frac{7}{2}\right)$; using the recurrence and symmetry to reduce to $\mathrm{b}\left(\frac{1}{2}, \frac{1}{2}\right) = \pi$ gives $\frac{3\pi}{256}$.

4. From part (i)'s explicit value, $u_{2m} = \frac{(2m)!}{(2^m m!)^2} \cdot \frac{\pi}{2} = \binom{2m}{m} \frac{\pi}{2^{2m+1}}$. Then $2mu_{2m}^2 \to \frac{\pi}{2}$ gives $\binom{2m}{m}^2 \approx \frac{4^{2m}}{\pi m}$. For $m = 10$: $\binom{20}{10} = 184\,756$, against $\frac{4^{10}}{\sqrt{10\pi}} \approx 187\,079$ (about 1.3% high).

4.5 The Sophomore's Dream

1. $x^{-x} = e^{-x\ln x} = \sum_{n\geq0} \frac{(-x\ln x)^n}{n!}$, and $\int_0^1 \frac{x^n(-\ln x)^n}{n!}\,\mathrm{d}x = \frac{1}{(n+1)^{n+1}}$ (substitute $x = e^{-u}$). So the integral is $\sum_{n\geq1} n^{-n} \approx 1.2913$.

2. $\frac{\mathrm{d}}{\mathrm{d}x} x^x = x^x(\ln x + 1)$, zero at $x = \frac{1}{e}$; the value is $(e^{-1})^{e^{-1}} = e^{-1/e} \approx 0.692$.

3. The terms are $\frac{1}{n^n}$. Stopping after 7 terms gives error less than $\frac{1}{8^8} \approx 6 \times 10^{-8}$, enough; after 6 terms the bound $\frac{1}{7^7} \approx 1.2 \times 10^{-6}$ is not. So 7 terms.

4. By parts: $\int_0^1 x^2 \ln x\,\mathrm{d}x = \left[\frac{x^3 \ln x}{3}\right]_0^1 - \int_0^1 \frac{x^2}{3}\,\mathrm{d}x = -\frac{1}{9}$. In general $\int_0^1 x^n \ln x\,\mathrm{d}x = -\frac{1}{(n+1)^2}$, and $\frac{\ln x}{1-x} = \sum_{n\geq0} x^n \ln x$.

4.6 Integrals With No Formula

1. Differentiate: $\frac{\mathrm{d}}{\mathrm{d}x} \frac{e^x}{x+1} = \frac{xe^x}{(x+1)^2}$, and $\frac{\mathrm{d}}{\mathrm{d}x} \frac{(x-1)e^x}{x+1} = e^x\left(\frac{x-1}{x+1} + \frac{2}{(x+1)^2}\right) = \frac{(x^2+1)e^x}{(x+1)^2}$.

2. $\frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{P}e^{-x^2}\right) = (\mathrm{P}' - 2x\mathrm{P})e^{-x^2}$, and $\mathrm{P}' - 2x\mathrm{P}$ has degree $\deg \mathrm{P} + 1 \geq 1$ (for $\mathrm{P} \neq 0$), so it cannot equal 1.

3. If $\mathrm{R} = \mathrm{P}/\mathrm{Q}$ (in lowest terms) had a pole of order $k \geq 1$ at 0, then $\mathrm{R}' + \mathrm{R}$ would have a pole of order $k+1 \geq 2$ there, but $\frac{1}{x}$ has a pole of order 1. If R has no pole at 0, neither has $\mathrm{R}' + \mathrm{R}$. Either way $(\mathrm{R}e^x)' \neq \frac{e^x}{x}$.

4. Differentiating $\frac{\mathrm{P}e^x}{x+1}$ gives $e^x \frac{(x+1)\mathrm{P}' + x\mathrm{P}}{(x+1)^2}$. At $x = -1$: $\mathrm{f}(-1) = -\mathrm{P}(-1)$, and $\mathrm{f}' = (x+1)\mathrm{P}'' + (x+1)\mathrm{P}' + \mathrm{P}$ gives $\mathrm{f}'(-1) = \mathrm{P}(-1)$. For $x^3 - 2$: $-3 + 3 = 0$; for x: $-1 + 1 = 0$; for $x^2 + 1$: $2 - 2 = 0$. The condition is also sufficient: the map $\mathrm{P} \mapsto (x+1)\mathrm{P}' + x\mathrm{P}$ from polynomials of degree less than d to degree at most d is injective, so its image has dimension d, and it lies inside the d-dimensional space of f with $\mathrm{f}(-1) + \mathrm{f}'(-1) = 0$; so it is that whole space.

4.7 The Calculus of Variations

1. With $y = \cos\frac{\pi x}{2}$, $J = \frac{\pi^2}{4}\int_0^1 \sin^2 - a^2\int_0^1 \cos^2 = \frac{1}{2}\left(\frac{\pi^2}{4} - a^2\right) < 0$.

2. $J(y+\varepsilon\eta) = J(y) + 2\varepsilon\int_0^1 (y'\eta' - a^2 y\eta) + \varepsilon^2 J(\eta)$; a minimum at $\varepsilon = 0$ forces the linear term to vanish. Integrating by parts, $-y'(0)\eta(0) - \int_0^1 (y'' + a^2 y)\eta = 0$ for every η with $\eta(1) = 0$, which forces $y'' + a^2 y = 0$ and $y'(0) = 0$.

3. With $x = Lu$, $\int_0^L \left((y')^2 - a^2 y^2\right)\mathrm{d}x = \frac{1}{L}\int_0^1 \left((Y')^2 - (aL)^2 Y^2\right)\mathrm{d}u$, so the condition is $aL \leq \frac{\pi}{2}$. (At $aL = \frac{\pi}{2}$ the integral is still never negative, as in the main text, but $y = \cos\frac{\pi x}{2L}$ makes it zero; for $aL > \frac{\pi}{2}$ that same function makes it negative.)

4. The cycloid is $x = R(\theta - \sin\theta)$, $y = R(1 - \cos\theta)$. For the tautochrone, see Section 11.4: in terms of arc length the motion is exactly simple harmonic.

4.8 Slicing a Spherical Loaf

1. $V/A = \frac{\frac{4}{3}\pi a^3}{4\pi a^2} = \frac{a}{3}$. But a remaining piece of height t has $A = 2\pi a t$ and $V = \frac{1}{3}\pi t^2(3a - t)$, so $V/A = \frac{t(3a-t)}{6a}$, which is largest, $\frac{3a}{8}$, at $t = \frac{3}{2}a$: bigger than $\frac{a}{3}$.

2. The band has height $2R\sin 23.4^\circ$ out of the sphere's $2R$, so the fraction is $\sin 23.4^\circ \approx 0.40$.

3. For $y = \sqrt{x}$, $y\sqrt{1 + (y')^2} = \sqrt{x + \frac{1}{4}}$, not constant. $y\sqrt{1 + (y')^2} = c$ gives $(y')^2 = \frac{c^2}{y^2} - 1$, whose solutions are $y = c$ (a cylinder) and the circles $(x - x_0)^2 + y^2 = c^2$ (spheres).

4. The sphere has volume $\frac{4}{3}\pi a^3$ and the cylinder $\pi a^2 \cdot 2a = 2\pi a^3$; the ratio is $\frac{2}{3}$.

4.9 Holditch's Theorem

1. $\frac{1}{2}\int_0^{2\pi} (p\cos t \cdot q\cos t + q\sin t \cdot p\sin t)\,\mathrm{d}t = \frac{1}{2}\cdot 2\pi pq = \pi pq$.

2. The triangles formed by joining the ends of the two chords are similar (equal angles subtended by the same arcs), so $PA \cdot PB = PC \cdot PD$. For a circle of radius R with P at distance r from the centre, the diameter through P gives $(R - r)(R + r)$, so $ab = R^2 - r^2$ and the ring has area $\pi R^2 - \pi r^2 = \pi ab$.

3. If $a = b$ then $f = \pi(a - b) = 0$, and the ring's area is $\pi a^2 = \frac{\pi L^2}{4}$: the area of the circle with the rod as diameter.

4. Along the two sides through O, $x\,\mathrm{d}y - y\,\mathrm{d}x = 0$. On the third side, parametrised by $(x_1, y_1) + s(x_2 - x_1, y_2 - y_1)$ for $0 \leq s \leq 1$, it is $(x_1 y_2 - x_2 y_1)\,\mathrm{d}s$. So the area is $\frac{1}{2}|x_1 y_2 - x_2 y_1|$.

CHAPTER

5

Complex Numbers

The shortest path between two truths in the real domain passes through the complex domain.

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Imaginary Numbers, Real Answers

The shortest path between two truths in the real domain passes through the complex domain.

— Jacques Hadamard

In 1545 Gerolamo Cardano published a formula for solving cubic equations, and found something disturbing. For some cubics with three perfectly good real roots, the formula required taking square roots of negative numbers along the way. The square roots cancelled out at the end, leaving real answers, but in the middle of the calculation there were quantities that, by every standard of the time, did not exist. Cardano called the manipulation "as subtle as it is useless". For two centuries, mathematicians used these "imaginary" numbers uneasily, as a trick that happened to work.

It took until the early nineteenth century, when Caspar Wessel, Jean-Robert Argand and Gauss pictured complex numbers as points in a plane, for them to become respectable. Once they did, they turned out to be not a trick but a revelation. Complex numbers explain why real phenomena behave as they do, and they keep providing the shortest route to results about real numbers, exactly as Hadamard said.

Why complex numbers matter

Every polynomial equation has a solution in the complex numbers: the *fundamental theorem of algebra*. So the complex numbers are, in a precise sense, where algebra is complete. They are also the natural language of rotation and oscillation. Multiplying by a complex number of modulus 1 rotates the plane, so anything involving angles, waves or periodic behaviour is simpler with complex numbers. Electrical engineers use them to analyse alternating currents; physicists cannot write down quantum mechanics without them; signal processing, from mobile phones to MRI scanners, runs on the complex Fourier transform. And in pure mathematics, functions of a complex variable turn out to be astonishingly rigid and beautiful: knowing a complex function on a tiny region determines it everywhere.

What you will meet at university

You will use complex numbers from the first week, but the big step comes with *complex analysis*, usually in the second year. Its central objects are *holomorphic* functions, complex functions that can be differentiated, and the theory is full of surprises. Every holomorphic function is automatically infinitely differentiable and equal to its Taylor series. Integrals round closed curves can be evaluated by looking only at the points where the function blows up (Cauchy's residue theorem), which lets you compute real integrals that resist every other method. The theory leads on to conformal maps

(Section 5.7), Riemann surfaces, and the Riemann zeta function that controls the primes (Section 1.2). In parallel, the algebra of roots of unity (Section 5.2, Section 5.3) leads to Galois theory, and complex geometry leads to the non-Euclidean geometry of Section 5.8.

The questions that run through this chapter

Three big questions

1. **How do complex numbers help with real problems?** Why should a detour through the complex plane produce facts about π, binomial coefficients or cosines?
2. **What is special about the roots of unity?** Why do the solutions of $z^n = 1$ keep appearing, in sums, products, polygons and angles?
3. **What does a complex function look like?** How can we picture functions whose inputs and outputs are both points of a plane, and what geometry do they reveal?

The chapter moves from arguments and multiplication (Machin's formula), through the algebra of roots of unity (filters, factorisations, polygon distances and irrational angles), to geometry with complex numbers (the Fermat point) and finally to complex functions as maps (the exponential, and a hidden distance in the unit disc).

Ideas in this chapter

§	Topic	Idea
5.1	Machin's formula	Arguments add when complex numbers multiply
5.2	Every third binomial coefficient	Averaging over roots of unity filters by residue
5.3	Products of cosines	Factorise over the roots, then substitute
5.4	Chords of a regular polygon	Distances are moduli; $z^n - 1 = \prod(z - \omega^k)$
5.5	$\cos^{-1}\frac{1}{3}$ is irrational	De Moivre, parity and a minimal counterexample
5.6	The Fermat point	Rotation is multiplication; good coordinates
5.7	e^w as a map	Picturing complex functions; many-to-one maps
5.8	A hidden distance	An inequality as a triangle inequality in hyperbolic geometry

How to read this chapter

Draw pictures. Almost every result in this chapter has a geometric meaning, and a quick sketch of the complex plane (where the roots of unity are, which way a multiplication rotates, where the grid lines go) often shows you what to do before any algebra. Then do the algebra carefully, because the pictures can mislead about signs and multiples of 2π.

5.1 Machin's Formula

The question

(i) If $z = x + \mathrm{i}y$, with x, y real, show that

$$|x|\cos\alpha + |y|\sin\alpha \leq |z| \leq |x| + |y|$$

for all real α.

(ii) By considering $(5-\mathrm{i})^4(1+\mathrm{i})$, show that

$$\frac{\pi}{4} = 4\tan^{-1}\left(\frac{1}{5}\right) - \tan^{-1}\left(\frac{1}{239}\right).$$

Prove similarly that

$$\frac{\pi}{4} = 3\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{1}{20}\right) + \tan^{-1}\left(\frac{1}{1985}\right).$$

The hook

In 1706 the London mathematician John Machin computed π to 100 decimal places, by hand. Nobody had come close before. His secret was the formula in part (ii), which expresses $\frac{\pi}{4}$ in terms of angles whose tangents are tiny fractions. Tiny fractions make the series $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$ converge very fast, and Machin's formula, or variants of it, remained the standard way to compute π for two and a half centuries, until computers and the methods of Section 2.6 took over.

The formula looks like a miracle of trigonometric identities. The question shows it is really a fact about multiplying complex numbers: when you multiply complex numbers, their arguments add. One multiplication of whole-number complex numbers proves the whole thing.

Before you start

Ask yourself What is the argument of $5-\mathrm{i}$? Of $1+\mathrm{i}$? What happens to arguments when complex numbers are multiplied?

$\arg(5-\mathrm{i}) = -\tan^{-1}\frac{1}{5}$, a small negative angle, and $\arg(1+\mathrm{i}) = \frac{\pi}{4}$. When you multiply

complex numbers, the moduli multiply and the arguments add (up to multiples of 2π). So

$$\arg\big((5-\mathrm{i})^4(1+\mathrm{i})\big) = \frac{\pi}{4} - 4\tan^{-1}\frac{1}{5} \quad \text{(possibly plus a multiple of } 2\pi\text{)}.$$

If we can compute the product and find its argument another way, we get an identity.

Working it through

STAGE 1 Bounds on the modulus

Ask yourself For the upper bound, square both sides. For the lower bound, think of $|x|\cos\alpha + |y|\sin\alpha$ as a dot product of two vectors.

Upper bound: $(|x|+|y|)^2 = x^2+y^2+2|x||y| \geq x^2+y^2 = |z|^2$, and both sides are non-negative, so $|z| \leq |x|+|y|$.

Lower bound: $|x|\cos\alpha + |y|\sin\alpha$ is the dot product of the vector $(|x|,|y|)$, of length $|z|$, with the unit vector $(\cos\alpha, \sin\alpha)$. A dot product is at most the product of the lengths, so it is at most $|z|$. (Algebraically: $|z|^2 - (|x|\cos\alpha + |y|\sin\alpha)^2 = (|x|\sin\alpha - |y|\cos\alpha)^2 \geq 0$.) ■

STAGE 2 The key multiplication

Ask yourself Compute $(5-\mathrm{i})^2$, then square it, then multiply by $1+\mathrm{i}$.

$(5-\mathrm{i})^2 = 25 - 10\mathrm{i} - 1 = 24 - 10\mathrm{i}$. Then $(24-10\mathrm{i})^2 = 576 - 480\mathrm{i} - 100 = 476 - 480\mathrm{i}$. Finally

$$(476-480\mathrm{i})(1+\mathrm{i}) = 476 + 476\mathrm{i} - 480\mathrm{i} + 480 = 956 - 4\mathrm{i}.$$

Pause and try Now compare arguments. What is the argument of $956-4\mathrm{i}$? Why are you sure there is no extra multiple of 2π?

$956 - 4\mathrm{i} = 4(239-\mathrm{i})$, which has argument $-\tan^{-1}\frac{1}{239}$. So

$$\frac{\pi}{4} - 4\tan^{-1}\frac{1}{5} = -\tan^{-1}\frac{1}{239} + 2k\pi$$

for some integer k. But $0 < \tan^{-1}\frac{1}{5} < \frac{1}{5}$, so the left-hand side lies between $\frac{\pi}{4} - \frac{4}{5}$ and $\frac{\pi}{4}$, that is, between about -0.02 and 0.79; and the right-hand side without the $2k\pi$ lies between -0.005 and 0. Since 2π is much bigger than the gap, $k = 0$. Rearranging,

$$\frac{\pi}{4} = 4\tan^{-1}\frac{1}{5} - \tan^{-1}\frac{1}{239}. \qquad \blacksquare$$

Figure 5.1.1 shows what happened: starting at $45°$, each multiplication by $5-\mathrm{i}$ rotates clockwise by about $11.3°$, and after four of them we have overshot the real axis by just $0.24°$.

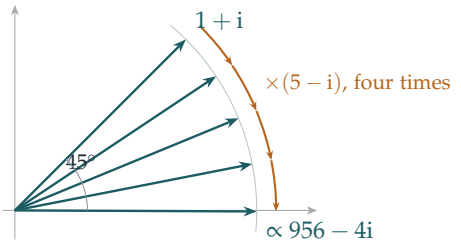


Figure 5.1.1 Multiplying $1+\mathrm{i}$ by $5-\mathrm{i}$ four times. Each multiplication rotates clockwise by $\tan^{-1}\frac{1}{5} \approx 11.31°$ (and stretches, which is ignored here). Four rotations take $45°$ to $-0.24°$, the argument of $956-4\mathrm{i}$, which is $-\tan^{-1}\frac{1}{239}$.

STAGE 3 The second formula

Ask yourself By analogy, which product should you compute this time?

Consider $(4+\mathrm{i})^3(20+\mathrm{i})(1985+\mathrm{i})$, whose argument is $3\tan^{-1}\frac{1}{4}+\tan^{-1}\frac{1}{20}+\tan^{-1}\frac{1}{1985}$. Now $(4+\mathrm{i})^2 = 15+8\mathrm{i}$, so $(4+\mathrm{i})^3 = (15+8\mathrm{i})(4+\mathrm{i}) = 52+47\mathrm{i}$. Then $(52+47\mathrm{i})(20+\mathrm{i}) = 1040+52\mathrm{i}+940\mathrm{i}-47 = 993+992\mathrm{i}$, and

$$(993+992\mathrm{i})(1985+\mathrm{i}) = (1971105-992)+(993+1969120)\mathrm{i} = 1970113+1970113\mathrm{i}.$$

Real part equals imaginary part, so the argument is $\frac{\pi}{4}$. All three angles are small and positive (their sum is less than $3\cdot\frac{1}{4}+\frac{1}{20}+\frac{1}{1985} < 1 < 2\pi$), so there is no multiple of 2π to worry about, and the formula follows. ■

Looking back

- **Arguments add under multiplication.** That single fact turned a product of Gaussian integers into an identity between angles.
- **Mind the multiples of 2π.** Arguments are only defined up to 2π, so each identity needed a size estimate to pin down the multiple.
- **Whole numbers do the work.** All the arithmetic was with integers; the identity for π came out exactly, with no approximation.

What's really going on

Computing π by hand

Why is Machin's formula so good for computation? The series $\tan^{-1}x = x-\frac{x^3}{3}+\frac{x^5}{5}-\cdots$ converges for $|x|\le 1$, but painfully slowly at $x=1$: you need about a thousand terms of $1-\frac{1}{3}+\frac{1}{5}-\cdots$ for three decimal places of $\frac{\pi}{4}$. With $x=\frac{1}{5}$, each term is about 25 times smaller than the last, giving roughly 1.4 decimal digits per term; with $x=\frac{1}{239}$, about 4.8 digits per term. Machin's 100 digits needed only about 70 terms of

the first series and 20 of the second. In 1873 William Shanks used a Machin formula to compute 707 digits, spending years on the arithmetic; in 1944 it was discovered that he had made an error at the 528th place, and everything after it was wrong.

Gaussian integers again

The numbers $a + b\mathrm{i}$ with a, b whole numbers are the *Gaussian integers* of Section 1.6. The reason $5 - \mathrm{i}$ works so well is a factorisation: $239^2 + 1 = 57122 = 2 \times 13^4$, which in the Gaussian integers reads $(239 + \mathrm{i})(239 - \mathrm{i}) = (1 + \mathrm{i})(1 - \mathrm{i})\,(3 + 2\mathrm{i})^4(3 - 2\mathrm{i})^4$, and $5 - \mathrm{i} = (1 - \mathrm{i})(3 + 2\mathrm{i})$ is built from the same Gaussian primes. Finding Machin-type formulas is really a problem about factorising numbers of the form $n^2 + 1$. In 1899 Carl Størmer used exactly this idea to prove that there are only four formulas of the form $\frac{\pi}{4} = a\tan^{-1}\frac{1}{m} + b\tan^{-1}\frac{1}{n}$ with whole numbers a, b, m, n: Machin's, and three found by Euler, Hermann and Hutton.

Rotations

Multiplying by a complex number of modulus 1 is a rotation of the plane. That is why complex numbers are the natural language for anything involving rotation and angles, from the geometry later in this chapter to alternating currents in electrical engineering and the wave functions of quantum mechanics. In Section 4.2 the addition formula for $\tan^{-1}$ came from a clever substitution; here it comes from the multiplication of complex numbers, and the two are the same fact seen from different sides.

The thinking skill

Translate angles into complex multiplication

An identity between sums of angles is an identity between arguments of a product. Compute the product exactly and read off its argument.

Arithmetic with complex numbers is mechanical and exact, while manipulating inverse trigonometric functions is error-prone. Translating from one to the other is often the quickest route to a proof.

Try this next

1. **A cousin.** (From STEP II, 2000.) By considering $(5 - \mathrm{i})^2(1 + \mathrm{i})$, prove that $\tan^{-1}\frac{7}{17} + 2\tan^{-1}\frac{1}{5} = \frac{\pi}{4}$.
2. **Euler's formula.** Show that $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3} = \frac{\pi}{4}$ by considering $(2 + \mathrm{i})(3 + \mathrm{i})$.
3. **Hutton's formula.** Show that $\frac{\pi}{4} = 2\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{7}$.

4 **How many terms?** Estimate how many terms of each series in Machin's formula you would need to compute π to 20 decimal places. Then do it for the first few digits with a calculator.

Further reading

- **Jonathan Borwein & Peter Borwein, *Pi and the AGM*** (Wiley). Includes the history of Machin-type formulas.
- **Jörg Arndt & Christoph Haenel, *Pi Unleashed*** (Springer). A lively account of methods for computing π, old and new.
- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). The best book for seeing complex numbers geometrically. Chapter 1 is about multiplication as rotation and scaling.

5.2 The Roots-of-Unity Filter

The question

Let $\omega = e^{2\pi i/3}$. Show that $1 + \omega + \omega^2 = 0$ and calculate the modulus and argument of $1 + \omega^2$.

Let n be a positive integer. By evaluating $(1 + \omega^r)^n$ in two ways, taking $r = 1, 2$ and 3, or otherwise, prove that

$$\binom{n}{0} + \binom{n}{3} + \binom{n}{6} + \cdots + \binom{n}{k} = \frac{1}{3}\left(2^n + 2\cos\left(\frac{n\pi}{3}\right)\right),$$

where k is the largest multiple of 3 less than or equal to n. Without using a calculator, evaluate

$$\binom{25}{0} + \binom{25}{3} + \cdots + \binom{25}{24}$$

and

$$\binom{24}{2} + \binom{24}{5} + \cdots + \binom{24}{23}.$$

[$2^{25} = 33554432$.]

The hook

The binomial coefficients in a row of Pascal's triangle add up to 2^n: put $x = 1$ in $(1 + x)^n$. Put $x = -1$ instead and you get the alternating sum, which is 0; adding the two shows that the coefficients in even positions add up to 2^{n-1}, exactly half. So the even-numbered and odd-numbered entries split the total evenly.

What about every *third* coefficient? There is no real number x that can pick out positions $0, 3, 6, \ldots$. But there is a complex one. The cube roots of unity act as a filter: averaged over 1, ω and ω^2, the powers ω^k give 1 when k is a multiple of 3 and 0 otherwise. This idea, the *roots-of-unity filter*, is a small version of the *discrete Fourier transform*, one of the most important algorithms ever devised.

Before you start

Ask yourself ω is a root of $z^3 - 1 = 0$. Factorise $z^3 - 1$. Why does this give $1 + \omega + \omega^2 = 0$?

$z^3 - 1 = (z-1)(z^2 + z + 1)$. Since $\omega^3 = 1$ but $\omega \neq 1$, ω must be a root of the second factor: $\omega^2 + \omega + 1 = 0$. Geometrically, 1, ω and ω^2 are the vertices of an equilateral triangle centred at the origin, and their sum is the centre of mass of the triangle, which is 0.

Working it through

STAGE 1 Moduli and arguments

From $1 + \omega + \omega^2 = 0$, $1 + \omega^2 = -\omega$. Since $\omega = e^{2\pi\mathrm{i}/3}$, $-\omega = e^{\mathrm{i}\pi} e^{2\pi\mathrm{i}/3} = e^{5\pi\mathrm{i}/3} = e^{-\pi\mathrm{i}/3}$. So $1 + \omega^2$ has modulus 1 and argument $-\frac{\pi}{3}$. Similarly $1 + \omega = -\omega^2 = e^{\pi\mathrm{i}/3}$, with modulus 1 and argument $\frac{\pi}{3}$.

STAGE 2 The filter

Ask yourself What is $1 + \omega^k + \omega^{2k}$ when k is a multiple of 3? When it is not?

If $3 \mid k$, then $\omega^k = 1$ and the sum is 3. If not, ω^k is ω or ω^2, and in either case $1 + \omega^k + \omega^{2k} = 1 + \omega + \omega^2 = 0$. So the expression $\frac{1}{3}(1 + \omega^k + \omega^{2k})$ is 1 when $3 \mid k$ and 0 otherwise: it is a detector for multiples of 3.

Pause and try Add up $(1 + \omega^r)^n$ for $r = 1, 2, 3$, expanding each by the binomial theorem. What do you get?

Since $\omega^3 = 1$,

$$(1+\omega)^n + (1+\omega^2)^n + (1+1)^n = \sum_{k=0}^{n} \binom{n}{k} (\omega^k + \omega^{2k} + 1) = 3 \sum_{3 \mid k} \binom{n}{k}.$$

On the other hand, by Stage 1,

$$(1+\omega)^n + (1+\omega^2)^n + 2^n = e^{n\pi\mathrm{i}/3} + e^{-n\pi\mathrm{i}/3} + 2^n = 2\cos\frac{n\pi}{3} + 2^n.$$

Dividing by 3 gives the formula. ■

STAGE 3 Two calculations

For $n = 25$: $\frac{25\pi}{3} = 8\pi + \frac{\pi}{3}$, so $\cos\frac{25\pi}{3} = \frac{1}{2}$, and

$$\binom{25}{0} + \binom{25}{3} + \cdots + \binom{25}{24} = \frac{33554432 + 1}{3} = 11184811.$$

Ask yourself For the second sum we need positions $k \equiv 2 \pmod 3$. How can you modify the filter to detect $k \equiv 2$ instead of $k \equiv 0$?

Shift the detector: $\frac{1}{3}\left(1+\omega^{k-2}+\omega^{2(k-2)}\right)$ is 1 exactly when $k \equiv 2 \pmod 3$. Since $\omega^{-2}=\omega$ and $\omega^{-4}=\omega^2$, this gives

$$\sum_{k\equiv 2}\binom{n}{k}=\frac{1}{3}\left(2^n+\omega(1+\omega)^n+\omega^2(1+\omega^2)^n\right).$$

With $n=24$, $(1+\omega)^{24}=e^{8\pi i}=1$ and likewise $(1+\omega^2)^{24}=1$, so the bracket is $2^{24}+\omega+\omega^2=2^{24}-1$. With $2^{24}=16777216$,

$$\binom{24}{2}+\binom{24}{5}+\cdots+\binom{24}{23}=\frac{16777215}{3}=5592405. \qquad \blacksquare$$

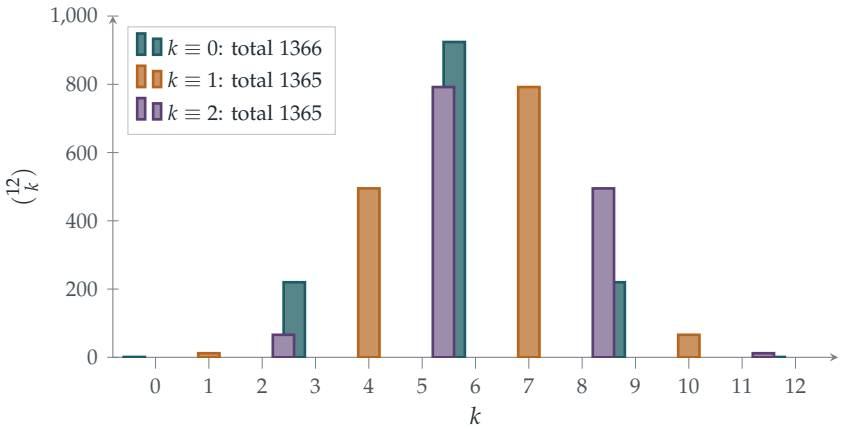


Figure 5.2.1 Row 12 of Pascal's triangle, coloured by the remainder of k on division by 3. The three classes have totals 1366, 1365 and 1365: almost exactly a third each of $2^{12}=4096$. The roots-of-unity filter predicts the small differences from $\frac{4096}{3}$ exactly: $+\frac{2}{3}$, $-\frac{1}{3}$ and $-\frac{1}{3}$.

Looking back

- **Averaging over roots of unity filters by residue.** $\frac{1}{3}\sum_r \omega^{rk}$ is 1 or 0 according as 3 divides k. Summing a polynomial's values at the roots of unity picks out every third coefficient.
- **Shifting the filter.** Multiplying by ω^{-rj} before averaging picks out the coefficients with $k \equiv j$.
- **Complex numbers for real answers.** The answer is a whole number, and the question was about whole numbers, but the route went through the complex plane.

What's really going on

The discrete Fourier transform

Replace 3 by any N and ω by $e^{2\pi i/N}$. For a list of numbers $a_0, a_1, \ldots, a_{N-1}$, the numbers

$$\hat{a}_r = \sum_{k=0}^{N-1} a_k \, \omega^{rk}$$

form its *discrete Fourier transform*, and the filter identity $\sum_r \omega^{rk} = N$ or 0 is what lets you recover the a_k from the $\hat{a}_r$. The transform splits a signal into its component frequencies. In 1965 James Cooley and John Tukey published a way to compute it in about $N \log_2 N$ steps instead of N^2, the *fast Fourier transform*; Gauss had found essentially the same idea in 1805 but never published it. The FFT is used in audio and image compression (MP3 and JPEG), in mobile phone signals, in medical scanners and in multiplying enormous numbers, and it is often listed among the most important algorithms of the twentieth century.

Generating functions

The filter is one instance of a bigger idea: a sequence $a_0, a_1, a_2, \ldots$ can be packaged as a polynomial or power series $A(x) = \sum a_k x^k$, its *generating function*, and operations on the sequence become operations on the function. Evaluating at 1 sums the sequence; evaluating at roots of unity picks out residue classes; differentiating weights the terms by k. This turns many counting problems into algebra, and it is a central technique in Chapter 9. The same filter, applied to $(1+x)^n$ with fifth roots of unity, connects to the regular pentagon and the golden ratio; we pursue the algebra of roots of unity in Section 6.1.

The thinking skill

Use symmetry to filter

To pick out the terms of a sum that satisfy a condition, find a function that is 1 *on the terms you want and* 0 *on the rest, built from a symmetric average.*

Averaging over the roots of unity is the classic example. The same principle, "average over a group of symmetries to isolate the part you want", appears throughout algebra, from the roots-of-unity filter to the representation theory of groups.

Try this next

1 **Even positions.** Use the filter with square roots of unity (1 and -1) to show that $\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots = 2^{n-1}$.

2 **The other residue.** Find a formula for $\binom{n}{1} + \binom{n}{4} + \binom{n}{7} + \cdots$, and check that the three formulas add up to 2^n.

3 **Every fourth.** Using i, show that $\binom{n}{0} + \binom{n}{4} + \binom{n}{8} + \cdots = \frac{1}{4}\left(2^n + 2^{n/2+1}\cos\frac{n\pi}{4}\right)$.

4 **Divisibility.** How many subsets of $\{1, 2, \ldots, 12\}$ have a number of elements divisible by 3? How many have a sum divisible by 3? (*Hint for the second:* consider $\prod_{j=1}^{12}(1 + x^j)$ at the cube roots of unity.)

Further reading

- **Herbert Wilf, *generatingfunctionology*** (A. K. Peters; free online). A delightful introduction to generating functions; the roots-of-unity filter appears early on.
- **Arthur Benjamin & Jennifer Quinn, *Proofs That Really Count*** (Mathematical Association of America). Binomial identities proved by counting.
- **T. W. Körner, *Fourier Analysis*** (Cambridge University Press). Includes the discrete Fourier transform and the FFT.

5.3 Factorising $z^{2n}+1$

The question

Show that $(z-\mathrm{e}^{\mathrm{i}\theta})(z-\mathrm{e}^{-\mathrm{i}\theta}) = z^2 - 2z\cos\theta + 1$.

Write down the $(2n)$th roots of -1 in the form $\mathrm{e}^{\mathrm{i}\theta}$, where $-\pi < \theta \leq \pi$, and deduce that

$$z^{2n}+1 = \prod_{k=1}^{n}\left(z^2 - 2z\cos\left(\tfrac{(2k-1)\pi}{2n}\right)+1\right).$$

Here, n is a positive integer, and the $\prod$ notation denotes the product.

(i) By substituting $z = \mathrm{i}$ show that, when n is even,

$$\cos\left(\tfrac{\pi}{2n}\right)\cos\left(\tfrac{3\pi}{2n}\right)\cos\left(\tfrac{5\pi}{2n}\right)\cdots\cos\left(\tfrac{(2n-1)\pi}{2n}\right) = (-1)^{\frac{1}{2}n}2^{1-n}.$$

(ii) Show that, when n is odd,

$$\cos^2\left(\tfrac{\pi}{2n}\right)\cos^2\left(\tfrac{3\pi}{2n}\right)\cos^2\left(\tfrac{5\pi}{2n}\right)\cdots\cos^2\left(\tfrac{(n-2)\pi}{2n}\right) = n2^{1-n}.$$

You may use without proof the fact that $1+z^{2n} = (1+z^2)(1-z^2+z^4-\cdots+z^{2n-2})$ when n is odd.

The hook

What is $\cos\frac{\pi}{8}\cos\frac{3\pi}{8}\cos\frac{5\pi}{8}\cos\frac{7\pi}{8}$? Each factor is an ugly number involving nested square roots, but the product is exactly $\frac{1}{8}$. Products of cosines at equally spaced angles keep collapsing to simple numbers like this, and the question explains why. The trick is to recognise the angles as arguments of roots of a polynomial, factorise the polynomial, and then substitute a cleverly chosen value.

Over the complex numbers, every polynomial factorises completely into linear factors. Over the real numbers, the non-real roots come in conjugate pairs, which combine into real quadratic factors. Both facts are on display here, and the second is how $z^{2n}+1$, which has no real roots at all, can be written as a product of real quadratics.

Before you start

Ask yourself Where are the solutions of $z^{2n} = -1$ in the complex plane? Draw them for $n = 2$ and $n = 4$.

They all have modulus 1, and their arguments θ satisfy $2n\theta = \pi + 2m\pi$, so θ is an odd multiple of $\frac{\pi}{2n}$. They form a regular $2n$-gon on the unit circle, rotated so that no vertex lies on the real axis (Figure 5.3.1). They pair up into conjugates $e^{\pm i\theta}$, mirror images in the real axis.

Working it through

STAGE 1 Conjugate pairs give real quadratics

$$(z - e^{i\theta})(z - e^{-i\theta}) = z^2 - (e^{i\theta} + e^{-i\theta})z + 1 = z^2 - 2z\cos\theta + 1.$$ ■

STAGE 2 The factorisation

The $2n$th roots of -1 with $-\pi < \theta \leq \pi$ are $e^{\pm i\theta_k}$ with

$$\theta_k = \frac{(2k-1)\pi}{2n}, \qquad k = 1, 2, \ldots, n.$$

These are $2n$ distinct numbers, all roots of the degree-$2n$ polynomial $z^{2n} + 1$, so they are all its roots and

$$z^{2n} + 1 = \prod_{k=1}^{n} (z - e^{i\theta_k})(z - e^{-i\theta_k}) = \prod_{k=1}^{n} (z^2 - 2z\cos\theta_k + 1).$$ ■

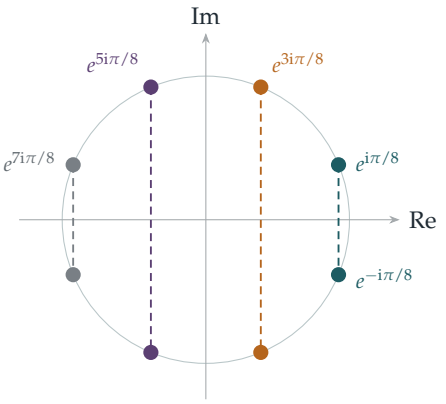


Figure 5.3.1 The eight solutions of $z^8 = -1$ (the case $n = 4$). They pair into four conjugate pairs, joined by dashed lines; each pair gives one real quadratic factor $z^2 - 2z\cos\theta_k + 1$ of $z^8 + 1$.

STAGE 3 Substituting $z = \mathrm{i}$

Ask yourself What is $\mathrm{i}^2 - 2\mathrm{i}\cos\theta + 1$? And what is $\mathrm{i}^{2n}+1$ when n is even?

Each factor becomes

$$\mathrm{i}^2 - 2\mathrm{i}\cos\theta_k + 1 = -2\mathrm{i}\cos\theta_k,$$

so the right-hand side is $(-2\mathrm{i})^n \prod_k \cos\theta_k$. The left-hand side is $\mathrm{i}^{2n}+1 = (-1)^n + 1$, which is 2 when n is even. And for even n,

$$(-2\mathrm{i})^n = 2^n(-\mathrm{i})^n = 2^n\big((-\mathrm{i})^2\big)^{n/2} = 2^n(-1)^{n/2}.$$

So

$$\prod_{k=1}^{n}\cos\theta_k = \frac{2}{2^n(-1)^{n/2}} = (-1)^{\frac{1}{2}n}2^{1-n}. \qquad \blacksquare$$

For $n=4$ this is the product in the hook: $(-1)^2 2^{-3} = \frac{1}{8}$.

STAGE 4 When n is odd

Ask yourself For odd n, one of the angles θ_k equals $\frac{\pi}{2}$. Which one, and why does that make the substitution $z = \mathrm{i}$ useless as it stands?

When $k = \frac{n+1}{2}$, $\theta_k = \frac{n\pi}{2n} = \frac{\pi}{2}$, so that factor is z^2+1, which vanishes at $z=\mathrm{i}$; both sides are 0, and we learn nothing. The way round is to divide out that factor first, which is what the given identity does.

Pause and try Divide both sides of the factorisation by $1+z^2$, substitute $z=\mathrm{i}$, and simplify. You will need to pair up the remaining cosines.

Dividing by the factor z^2+1 (the term $k=\frac{n+1}{2}$),

$$1 - z^2 + z^4 - \cdots + z^{2n-2} = \prod_{k\neq\frac{n+1}{2}}\big(z^2 - 2z\cos\theta_k + 1\big).$$

At $z=\mathrm{i}$, every term on the left is $(-z^2)^j = 1^j = 1$, and there are n terms, so the left-hand side is n. The right-hand side is $(-2\mathrm{i})^{n-1}\prod_{k\neq\frac{n+1}{2}}\cos\theta_k$. Since $n-1$ is even, $(-2\mathrm{i})^{n-1} = 2^{n-1}(-1)^{(n-1)/2}$.

Now pair the angles: $\theta_{n+1-k} = \pi - \theta_k$, so $\cos\theta_{n+1-k} = -\cos\theta_k$. The $n-1$ remaining factors form $\frac{n-1}{2}$ such pairs, with $k = 1, \ldots, \frac{n-1}{2}$, and each pair contributes $-\cos^2\theta_k$. So

$$n = 2^{n-1}(-1)^{(n-1)/2}\cdot(-1)^{(n-1)/2}\prod_{k=1}^{(n-1)/2}\cos^2\theta_k = 2^{n-1}\prod_{k=1}^{(n-1)/2}\cos^2\theta_k,$$

giving $\prod\cos^2\theta_k = n2^{1-n}$. The last factor, with $k=\frac{n-1}{2}$, is $\cos^2\frac{(n-2)\pi}{2n}$, as in the question. $\blacksquare$

Check with $n=3$: $\cos^2\frac{\pi}{6} = \frac{3}{4} = 3\cdot 2^{-2}$.

Looking back

- **Know the roots, know the polynomial.** A polynomial is determined (up to a constant) by its roots, so listing the roots of $z^{2n}+1$ gave its factorisation.
- **Substitute a clever value.** Putting $z=\mathrm{i}$ turned every quadratic factor into a multiple of a cosine.
- **When a substitution gives $0=0$, divide first.** For odd n one factor vanished at $z=\mathrm{i}$, so we removed it before substituting.

What's really going on

The fundamental theorem of algebra

That $z^{2n}+1$ can be factorised completely into linear factors over $\mathbb{C}$ is a special case of the *fundamental theorem of algebra*: every non-constant polynomial with complex coefficients has a complex root, and so factorises completely into linear factors. Gauss gave the first widely accepted proof in his doctoral thesis of 1799, and returned to the theorem several times with new proofs. A consequence is that every real polynomial factorises into real linear and quadratic factors, because non-real roots come in conjugate pairs, exactly as in Stage 1. The theorem is "algebraic" in name but analytic in substance: every proof uses some property of the real numbers, such as continuity, and you will see one of the slickest proofs in a first course on complex analysis.

Products of sines and cosines

The method gives a whole family of identities. Using z^n-1 in place of $z^{2n}+1$ leads to

$$\prod_{k=1}^{n-1}\sin\frac{k\pi}{n}=\frac{n}{2^{n-1}},$$

which is proved, in its geometric form, in Section 5.4. Identities like these appear in the analysis of vibrating systems (the frequencies of n masses joined by springs involve $\sin\frac{k\pi}{2(n+1)}$) and in counting problems (the number of spanning trees of some graphs, and the number of domino tilings of a rectangle, are given by products of cosines). Kasteleyn's 1961 formula for domino tilings of an $m\times n$ rectangle is a double product of expressions in $\cos^2\frac{j\pi}{m+1}$ and $\cos^2\frac{k\pi}{n+1}$.

The thinking skill

From roots to products

To evaluate a product of trigonometric values at equally spaced angles, recognise them as coming from the roots of a simple polynomial, factorise, and substitute a convenient value.

The choice of value is the art: here $z = \mathrm{i}$ turned each quadratic factor into a cosine. Other choices (such as $z = 1$ or $z = -1$) give other identities.

Try this next

1 **A heptagonal product.** Use part (ii) with $n = 7$ to show that $\cos\frac{\pi}{14}\cos\frac{3\pi}{14}\cos\frac{5\pi}{14} = \frac{\sqrt{7}}{8}$. Then show that $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7} = \frac{1}{8}$. (*Hint:* relate the angles using $\cos\theta = \sin(\frac{\pi}{2}-\theta)$, or use the identity $\sin 2x = 2\sin x\cos x$ repeatedly.)

2 **Another substitution.** Substitute $z = 1$ in the factorisation to show that

$$\prod_{k=1}^{n} \sin\frac{(2k-1)\pi}{4n} = 2^{\frac{1}{2}-n}.$$

3 **Factorise z^6+1.** Write z^6+1 as a product of three real quadratics, and check your answer by multiplying out.

4 **Sines.** By factorising $z^{2n}-1$ similarly and dividing by z^2-1, show that

$$\prod_{k=1}^{n-1} \sin\frac{k\pi}{2n} = \frac{\sqrt{n}}{2^{n-1}}.$$

Further reading

- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). Chapter 2 on roots of unity and polynomials, with beautiful pictures.
- **Benjamin Fine & Gerhard Rosenberger, *The Fundamental Theorem of Algebra*** (Springer). Several proofs of the theorem, from analysis, algebra and topology, written for undergraduates.
- **Paul Nahin, *An Imaginary Tale*** (Princeton University Press). A history of $\sqrt{-1}$, including roots of unity and their uses.

5.4 Distances in a Regular Polygon

The question

Let $\omega = \mathrm{e}^{2\pi \mathrm{i}/n}$, where n is a positive integer. Show that, for any complex number z,

$$(z-1)(z-\omega)\cdots(z-\omega^{n-1}) = z^n - 1.$$

The points $X_0, X_1, \ldots, X_{n-1}$ lie on a circle with centre O and radius 1, and are the vertices of a regular polygon.

(i) The point P is equidistant from X_0 and X_1. Show that, if n is even,

$$|PX_0| \times |PX_1| \times \cdots \times |PX_{n-1}| = |OP|^n + 1,$$

where $|PX_k|$ denotes the distance from P to X_k.

Give the corresponding result when n is odd. (There are two cases to consider.)

(ii) Show that

$$|X_0X_1| \times |X_0X_2| \times \cdots \times |X_0X_{n-1}| = n.$$

The hook

Draw a regular heptagon in a circle of radius 1. From one vertex, draw the six chords to the other vertices and measure them. Multiply the six lengths together. The answer is exactly 7. For a regular 100-gon, the product of the 99 chords from one vertex is exactly 100.

Why should anything so clean be true? The chord lengths themselves are messy ($2\sin\frac{\pi}{7}$, $2\sin\frac{2\pi}{7}$, and so on). The question shows that the result is a geometric shadow of a simple algebraic identity, the factorisation of $z^n - 1$ over the roots of unity, and that distances in the plane are moduli of complex numbers.

Before you start

Ask yourself Put the polygon's vertices at the nth roots of unity, $X_k = \omega^k$. If P is the point z, what is $|PX_k|$ in terms of z? So what is the product of all the distances?

$|PX_k| = |z - \omega^k|$, and the product of all of them is $|(z-1)(z-\omega)\cdots(z-\omega^{n-1})|$, which the first part of the question says is $|z^n - 1|$. A product of n distances has become a single modulus. Everything else is choosing z cleverly.

Working it through

STAGE 1 The factorisation

Each ω^k satisfies $(\omega^k)^n = (\omega^n)^k = 1$, so $1, \omega, \ldots, \omega^{n-1}$ are roots of $z^n - 1$. They are distinct (their arguments $\frac{2\pi k}{n}$ are different, between 0 and 2π), so they are all n roots of this degree-n polynomial with leading coefficient 1, and $z^n - 1 = \prod_{k=0}^{n-1}(z - \omega^k)$. ■

STAGE 2 A point on the perpendicular bisector

Rotate the picture so that $X_k = \omega^k$. The points equidistant from $X_0 = 1$ and $X_1 = \omega$ form the perpendicular bisector of X_0X_1, which is the line through O at angle $\frac{\pi}{n}$.

Ask yourself So P is $z = te^{i\pi/n}$ for some real t (positive or negative), with $|OP| = |t|$. What is z^n?

$z^n = t^n e^{i\pi} = -t^n$. So

$$|PX_0| \times \cdots \times |PX_{n-1}| = |z^n - 1| = |-t^n - 1| = |t^n + 1|.$$

If n is even, $t^n = |t|^n \geq 0$, so this is $|OP|^n + 1$. ■

Pause and try What happens when n is odd? Consider $t > 0$ and $t < 0$ separately, and say what each case means geometrically.

If n is odd, t^n has the same sign as t.

- If $t \geq 0$, so P is on the same side of O as the midpoint of X_0X_1, the product is $|OP|^n + 1$.
- If $t < 0$, so P is on the opposite side of O (towards the vertex opposite the side X_0X_1), then $t^n = -|OP|^n$ and the product is $\left|1 - |OP|^n\right|$.

For odd n, the line through O perpendicular to X_0X_1 passes through a vertex on the far side; in the second case, the product vanishes when P is that vertex ($|OP| = 1$), as it must, since one of the distances is then zero.

STAGE 3 Chords from a vertex

Ask yourself Now $P = X_0$, and one of the distances, $|X_0X_0|$, is zero, so $|z^n - 1| = 0$ is no use. What polynomial has the roots $\omega, \omega^2, \ldots, \omega^{n-1}$ but not 1?

Dividing $z^n - 1$ by $z - 1$,

$$(z-\omega)(z-\omega^2)\cdots(z-\omega^{n-1}) = \frac{z^n-1}{z-1} = 1 + z + z^2 + \cdots + z^{n-1}.$$

The right-hand side is a polynomial, so this holds for every z, including $z = 1$, where it equals n. Taking moduli at $z = 1$,

$$|X_0X_1| \times |X_0X_2| \times \cdots \times |X_0X_{n-1}| = |1-\omega|\,|1-\omega^2|\cdots|1-\omega^{n-1}| = n. \qquad \blacksquare$$

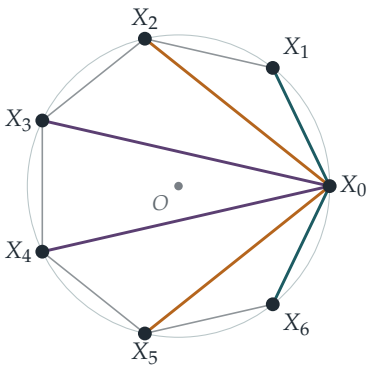


Figure 5.4.1 A regular heptagon inscribed in a circle of radius 1. The six chords from X_0 have lengths $2\sin\frac{k\pi}{7}$, about 0.868 (teal), 1.564 (amber) and 1.950 (plum), each occurring twice. Their product is exactly 7.

Looking back

- **Distances are moduli.** $|PX_k| = |z - \omega^k|$, so a product of distances is the modulus of a product, and the product had a closed form.
- **Use symmetry to choose coordinates.** Rotating so that the vertices are the roots of unity, and using the symmetry line through O, made z^n easy to compute.
- **Cancel the zero before substituting.** As in Section 5.3, dividing out the factor that vanished at the point of interest rescued the substitution.

What's really going on

Cotes's theorem

Part (i) is a theorem of Roger Cotes, the brilliant young Cambridge mathematician who edited the second edition of Newton's *Principia* and died in 1716 at the age of thirty-three (Newton said "if he had lived we might have known something"). Cotes found that if P lies on a ray from O through a vertex of a regular n-gon, then the product of the distances from P to the vertices is $\big||OP|^n - 1\big|$, and if P lies on a

ray through the midpoint of a side, it is $|OP|^n + 1$. He used this, without complex numbers, to factorise $x^n \pm 1$ into real quadratics, which he needed for integrating rational functions. The theorem was published after his death, in 1722.

A product of sines

Since $|1 - \omega^k| = |e^{-\mathrm{i}\pi k/n} - e^{\mathrm{i}\pi k/n}| = 2\sin\frac{k\pi}{n}$, part (ii) says

$$\prod_{k=1}^{n-1} \sin\frac{k\pi}{n} = \frac{n}{2^{n-1}}.$$

This is the companion of the cosine products in Section 5.3. A related product gives the *discriminant* of $z^n - 1$: the product of the squared distances between all pairs of distinct nth roots of unity is n^n. Discriminants measure how far apart the roots of a polynomial are, and they are central to the algebra of polynomials and of number fields, which we take up in Chapter 6.

Electrostatics and equal spacing

There is also a physical reading. Place n equal electric charges on a circle; the configuration that minimises the potential energy (with the logarithmic potential of two-dimensional electrostatics) is the regular polygon, and the energy involves exactly $\sum \ln|X_jX_k|$, the logarithm of a product of distances. Points that are "as spread out as possible" are important in numerical analysis (Section 3.6, where the best interpolation points come from a regular polygon via $x = \cos\theta$) and in the theory of random matrices.

The thinking skill

Turn geometry into algebra with complex numbers

Points in the plane are complex numbers, distances are moduli, and regular polygons are roots of unity. A product of distances becomes the modulus of a polynomial.

The translation lets algebraic identities, like the factorisation of $z^n - 1$, prove geometric facts that would be very hard to see directly.

Try this next

1. **Check small cases.** Verify part (ii) directly for a square ($n = 4$) and an equilateral triangle ($n = 3$).
2. **Through a vertex.** Show that if P lies on the ray from O through X_0, then $|PX_0| \times \cdots \times |PX_{n-1}| = \big||OP|^n - 1\big|$.

3 The average distance. Using part (ii) and the AM–GM inequality of Section 3.2, show that the average length of the $n-1$ chords from X_0 is at least $n^{1/(n-1)}$. Find the actual average, $\frac{1}{n-1}\sum_k 2\sin\frac{k\pi}{n}$, for $n=6$, and compare.

4 All the chords. HARDERShow that the product of all $\binom{n}{2}$ chord lengths of a regular n-gon inscribed in a unit circle is $n^{n/2}$.

Further reading

- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). Geometry with complex numbers, beautifully illustrated.
- **Ross Honsberger, *Mathematical Gems*** (Mathematical Association of America). Several elegant problems about regular polygons.
- **Ronald Gowing, *Roger Cotes: Natural Philosopher*** (Cambridge University Press). A biography of Cotes and his mathematics.

5.5 Rational Angles That Aren't

The question

(i) Use De Moivre's theorem to prove that for any positive integer $k > 1$,

$$\sin(k\theta) = \sin\theta\cos^{k-1}\theta\left(k - \binom{k}{3}(\sec^2\theta - 1) + \binom{k}{5}(\sec^2\theta - 1)^2 - \cdots\right)$$

and find a similar expression for $\cos(k\theta)$.

(ii) Let $\theta = \cos^{-1}\left(\frac{1}{a}\right)$, where θ is measured in degrees, and a is an odd integer greater than 1.

Suppose that there is a positive integer k such that $\sin(k\theta) = 0$ and $\sin(m\theta) \neq 0$ for all integers m with $0 < m < k$.

Show that it would be necessary to have k even and $\cos(\frac{1}{2}k\theta) = 0$.

Deduce that θ is irrational.

(iii) Show that if $\phi = \cot^{-1}\left(\frac{1}{b}\right)$, where ϕ is measured in degrees, and b is an even integer greater than 1, then ϕ is irrational.

The hook

The angles you meet in school geometry are usually whole numbers of degrees: $30°$, $45°$, $60°$. Their cosines are numbers like $\frac{\sqrt{3}}{2}$ and $\frac{1}{2}$. What about going the other way? If the cosine is a simple fraction, such as $\frac{1}{3}$, is the angle a nice number of degrees?

The angle $\cos^{-1}\frac{1}{3} \approx 70.528779\ldots°$ appears all over science. It is the angle between two faces of a regular tetrahedron, and its supplement, $109.47°$, is the angle between the bonds in a methane molecule. This question proves that it is not a rational number of degrees: not $70\frac{1}{2}$, not $\frac{1411}{20}$, not any fraction at all. Its decimal expansion never repeats. This seemingly obscure fact solved one of the famous problems that David Hilbert posed in 1900.

Before you start

Ask yourself Suppose θ were rational, say $\theta = \frac{p}{q}$ degrees. Find a positive integer k with $\sin(k\theta) = 0$.

Take $k = 180q$: then $k\theta = 180p$ degrees, a whole number of half-turns, so $\sin(k\theta) = 0$. So rational angles always have some multiple that lands on a multiple of 180°. To prove θ is irrational, it is enough to show that *no* multiple $k\theta$ has zero sine. Part (ii) does this by taking the smallest such k, if there is one, and deriving a contradiction: the well-ordering principle of Section 1.3 again.

Working it through

STAGE 1 Multiple-angle formulas

Ask yourself Write $\cos k\theta + \mathrm{i}\sin k\theta = (\cos\theta + \mathrm{i}\sin\theta)^k$, and take out a factor of $\cos^k\theta$. What is left inside the bracket?

$(\cos\theta + \mathrm{i}\sin\theta)^k = \cos^k\theta\,(1 + \mathrm{i}\tan\theta)^k = \cos^k\theta \sum_{j=0}^{k} \binom{k}{j} \mathrm{i}^j \tan^j\theta$. The imaginary part comes from odd $j = 2m+1$, where $\mathrm{i}^j = \mathrm{i}(-1)^m$:

$$\begin{aligned}\sin k\theta &= \cos^k\theta \sum_m (-1)^m \binom{k}{2m+1} \tan^{2m+1}\theta \\ &= \sin\theta\cos^{k-1}\theta \sum_m (-1)^m \binom{k}{2m+1} \tan^{2m}\theta.\end{aligned}$$

Since $\tan^2\theta = \sec^2\theta - 1$, this is the required formula. ■ The real part, from even $j = 2m$, gives

$$\cos k\theta = \cos^k\theta \left(1 - \binom{k}{2}(\sec^2\theta - 1) + \binom{k}{4}(\sec^2\theta - 1)^2 - \cdots\right).$$

STAGE 2 Parity

Now $\sec\theta = a$, an odd integer, so $\sec^2\theta - 1 = a^2 - 1$ is an *even* integer. And θ is strictly between 0° and 90°, so $\sin\theta \neq 0$ and $\cos\theta \neq 0$.

Ask yourself If $\sin(k\theta) = 0$, the bracket $k - \binom{k}{3}(a^2-1) + \cdots$ must be zero. Every term after the first is a multiple of $a^2 - 1$. What does that tell you about k?

The bracket is k plus a sum of multiples of the even number $a^2 - 1$. If k were odd, the bracket would be odd, and an odd integer cannot be 0. So k must be even.

Then $\sin(k\theta) = 2\sin(\frac{1}{2}k\theta)\cos(\frac{1}{2}k\theta) = 0$. But $\frac{1}{2}k$ is a positive integer smaller than k, so by the minimality of k, $\sin(\frac{1}{2}k\theta) \neq 0$. Hence $\cos(\frac{1}{2}k\theta) = 0$. ■

STAGE 3 The contradiction

Pause and try Apply the cosine formula with $m = \frac{1}{2}k$. Why can $\cos(m\theta)$ never be zero?

By Stage 1, $\cos(m\theta) = a^{-m}\left(1 - \binom{m}{2}(a^2-1) + \binom{m}{4}(a^2-1)^2 - \cdots\right)$. The bracket is 1 plus multiples of the even number $a^2 - 1$, so it is odd, and in particular not zero. So $\cos(m\theta) \neq 0$, contradicting Stage 2.

Therefore there is no positive integer k with $\sin(k\theta) = 0$. (If there were any, there would be a smallest one, and we have just shown that the smallest one cannot exist.) By the observation in *Before you start*, θ cannot be a rational number of degrees. ■

STAGE 4 The cotangent version

Ask yourself If $\cot\phi = \frac{1}{b}$, what is $\sec^2\phi - 1$? Is it even?

$\tan\phi = b$, so $\sec^2\phi - 1 = \tan^2\phi = b^2$, which is even (indeed a multiple of 4) since b is even. Also $0^\circ < \phi < 90^\circ$, so $\sin\phi$ and $\cos\phi$ are non-zero. Exactly the same argument now works: if k is the smallest positive integer with $\sin(k\phi) = 0$, the bracket $k - \binom{k}{3}b^2 + \cdots$ must vanish, forcing k even; then $\cos(\frac{1}{2}k\phi) = 0$; but the cosine bracket $1 - \binom{m}{2}b^2 + \cdots$ is odd. Contradiction, so ϕ is irrational. ■

For example, $\cot^{-1}\frac{1}{2} = \tan^{-1}2 \approx 63.43^\circ$ is irrational.

Looking back

- **Rational angles have periodic multiples.** That reformulation turned irrationality into a statement about $\sin(k\theta)$ for all k.
- **Parity arguments.** Working modulo 2 showed that certain integers were odd and therefore not zero. Simple, but decisive.
- **The smallest counterexample.** Assuming a smallest k and showing a smaller one must exist (here $\frac{1}{2}k$) is the well-ordering argument of Chapter 1.

What's really going on

Irrational rotations

If θ is a rational number of degrees, repeatedly rotating by θ eventually returns you to where you started. If θ is irrational, you never return, and something remarkable happens: the points $k\theta$ (taken modulo 360°) spread out *evenly* around the circle, coming arbitrarily close to every angle. This is the *equidistribution theorem*, proved independently by Hermann Weyl, Wacław Sierpiński and Piers Bohl around 1910. Figure 5.5.1 shows the first 40 multiples of $\cos^{-1}\frac{1}{3}$: they are already filling the circle, and none ever repeats.

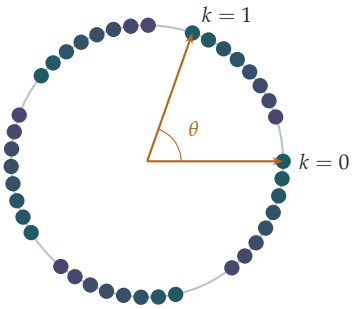


Figure 5.5.1 The points at angles $k\theta$, for $k = 0, 1, \ldots, 39$, where $\theta = \cos^{-1}\frac{1}{3} \approx 70.53^\circ$ (shading from teal to plum as k increases). Because θ is irrational, no point is ever repeated, and the points spread evenly around the whole circle.

Hilbert's third problem

In 1900 David Hilbert listed twenty-three problems for the new century. The third asked: if two polyhedra have the same volume, can one always be cut into finitely many polyhedral pieces and reassembled into the other? (For polygons in the plane the answer is yes: this is the Wallace–Bolyai–Gerwien theorem.) Within a year, Hilbert's student Max Dehn showed the answer is *no*: a regular tetrahedron cannot be cut up and reassembled into a cube of the same volume. Dehn's proof attaches to each polyhedron a quantity built from its edge lengths and dihedral angles, now called the *Dehn invariant*, which is unchanged by cutting and reassembling; for a cube it is zero, and for a regular tetrahedron it is non-zero precisely because the dihedral angle $\cos^{-1}\frac{1}{3}$ is not a rational multiple of 180°. That is exactly what this question proves.

Niven's theorem

How far does this go? In 1956 Ivan Niven proved that if θ is a rational number of degrees and $\cos\theta$ is rational, then $\cos\theta$ is one of $0, \pm\frac{1}{2}, \pm 1$. So the only "nice" angles with rational cosines are 0°, 60°, 90°, 120°, 180° and their relatives. The proof uses the

fact that $2\cos\theta$ is a root of a monic polynomial with integer coefficients, an *algebraic integer*, the idea mentioned at the end of Section 1.3.

The thinking skill

Parity as a proof of non-vanishing

To show an integer expression is never zero, show it is always odd: reduce it modulo 2.

Combined with a minimal counterexample, this simple idea proved a deep fact. Reducing modulo a small number to rule out an equation is one of the most common moves in number theory, as in Section 1.5.

Try this next

1. **The tetrahedron.** Show that the angle between two faces of a regular tetrahedron is $\cos^{-1}\frac{1}{3}$, and that the angle between two bonds in a methane molecule (from the centre of a regular tetrahedron to two vertices) is $\cos^{-1}\left(-\frac{1}{3}\right)$.

2. **The 3–4–5 triangle.** Show that the smallest angle of a 3–4–5 triangle is an irrational number of degrees. (*Hint:* it is $\cot^{-1}\frac{4}{3}$; adapt the argument, or consider powers of the Gaussian integer $4+3\mathrm{i}$ and show none is real.)

3. **Why odd a?** What goes wrong in the argument when $a=2$? Is $\cos^{-1}\frac{1}{2}$ rational?

4. **Niven for small cases.** Find all whole numbers of degrees θ between 0° and 90° for which $\cos\theta$ is rational, and check that your list agrees with Niven's theorem.

Further reading

- **Ivan Niven, *Irrational Numbers*** (Mathematical Association of America). Includes Niven's theorem on rational values of trigonometric functions.
- **Martin Aigner & Günter Ziegler, *Proofs from THE BOOK***, chapter on Hilbert's third problem. A beautiful account of Dehn's solution.
- **Benjamin Yandell, *The Honors Class: Hilbert's Problems and Their Solvers*** (A. K. Peters). The stories behind all twenty-three problems.

5.6 The Fermat Point

The question

Given that $\alpha = e^{i\pi/3}$, prove that $1 + \alpha^2 = \alpha$.

A triangle in the Argand plane has vertices A, B, and C represented by the complex numbers p, $q\alpha^2$ and $-r\alpha$ respectively, where p, q and r are positive real numbers. Sketch the triangle ABC.

Three equilateral triangles ABL, BCM and CAN (each lettered clockwise) are erected on sides AB, BC and CA respectively. Show that the complex number representing N is $(1-\alpha)p - \alpha^2 r$ and find similar expressions for the complex numbers representing L and M.

Show that lines LC, MA and NB all meet at the origin, and that these three line segments have the common length $p + q + r$.

The hook

Three villages want to build a road network connecting all of them, as short as possible. Where should the junction go? In about 1640 Pierre de Fermat posed the problem of finding the point P inside a triangle that minimises $PA + PB + PC$, and Evangelista Torricelli solved it: it is the point from which the three sides are seen at angles of $120°$.

This question uses complex numbers to prove a beautiful construction of that point. Build an equilateral triangle outwards on each side of the triangle, and join each new apex to the opposite vertex of the original. The three lines meet at a single point, the *Fermat point*, and all three have the same length, which is the minimum total road length. With complex numbers and one identity about $e^{i\pi/3}$, the whole thing is a few lines of algebra.

Before you start

Ask yourself Where are A, B and C? What are their arguments? What angles do OA, OB and OC make with each other?

$A = p$ lies on the positive real axis (argument $0°$). $B = q\alpha^2 = qe^{2\pi i/3}$ has argument $120°$. $C = -r\alpha = re^{i\pi}e^{i\pi/3}$ has argument $240°$. So the origin is a point inside the triangle from which the three vertices are seen at $120°$ to one another, at distances p, q and r. The origin *is* the Fermat point, by construction; the question shows that the

equilateral-triangle construction finds it.

Working it through

STAGE 1 The key identity

$\alpha^3 = e^{i\pi} = -1$, so α is a root of $z^3+1 = (z+1)(z^2-z+1)$. Since $\alpha \neq -1$, $\alpha^2 - \alpha + 1 = 0$, that is, $1+\alpha^2 = \alpha$. ■

Two consequences will be useful: $1-\alpha = -\alpha^2$, and $\alpha - \alpha^2 = 1$. Also $1-\alpha = \frac{1}{2} - \frac{\sqrt{3}}{2}i = e^{-i\pi/3}$, the complex conjugate of α.

STAGE 2 Equilateral triangles as rotations

Ask yourself Triangle CAN is equilateral, lettered clockwise. If you stand at C, how do you get from A to N? Express $N-C$ in terms of $A-C$.

Going clockwise round C, A, N, the point N is obtained by rotating A about C through $60°$ clockwise. Multiplying by $e^{-i\pi/3} = 1-\alpha$ rotates by $60°$ clockwise, so

$$\begin{aligned} N = C + (1-\alpha)(A-C) &= -r\alpha + (1-\alpha)(p+r\alpha) \\ &= (1-\alpha)p + r(\alpha - \alpha^2 - \alpha) = (1-\alpha)p - \alpha^2 r. \end{aligned}$$ ■

(Rotating clockwise rather than anticlockwise puts N on the far side of CA from the origin, outside the triangle.)

Pause and try Find L and M in the same way, and simplify using the identities from Stage 1.

Similarly $L = A + (1-\alpha)(B-A)$ and $M = B + (1-\alpha)(C-B)$:

$$\begin{aligned} L &= p + (1-\alpha)(q\alpha^2 - p) = \alpha p + (\alpha^2 - \alpha^3)q = \alpha p + (\alpha^2+1)q = \alpha(p+q), \\ M &= q\alpha^2 + (1-\alpha)(-r\alpha - q\alpha^2) = q\alpha^3 + r(\alpha^2 - \alpha) = -q - r. \end{aligned}$$

And using $1-\alpha = -\alpha^2$, $N = -\alpha^2 p - \alpha^2 r = -\alpha^2(p+r)$.

STAGE 3 Three lines through the origin

Ask yourself Compare $L = \alpha(p+q)$ with $C = -r\alpha$. What do you notice? Do the same for M and A, and for N and B.

L is a positive multiple of α and C is a negative multiple of α. So L and C lie on opposite sides of O on the same line through O, and the segment LC passes through the origin, with length $|L-C| = |\alpha|(p+q+r) = p+q+r$.

Likewise $M = -(q+r)$ and $A = p$ lie on the real axis on opposite sides of O, so MA passes through O with length $p+q+r$. And $N = -\alpha^2(p+r)$, $B = q\alpha^2$ lie on the line

through O in the direction α^2, on opposite sides, so NB passes through O with length $p+q+r$. ■

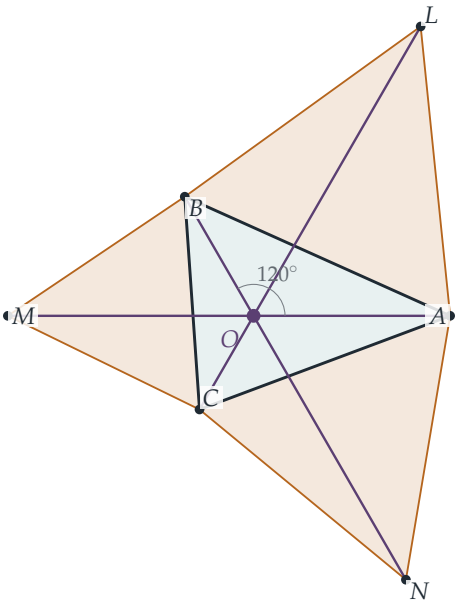


Figure 5.6.1 The Fermat point, with $p = 2$, $q = 1.4$, $r = 1.1$. Equilateral triangles (amber) are built outwards on the sides of ABC. The lines LC, MA and NB (plum) all pass through the origin O, where the sides of ABC subtend $120°$, and each has length $p+q+r = 4.5$.

Looking back

- **Rotation is multiplication.** Building an equilateral triangle meant rotating one vertex about another by $60°$, which is multiplication by $e^{\pm i\pi/3}$.
- **Clever coordinates.** Placing the Fermat point at the origin, with the vertices along the directions 1, α^2, $-\alpha$, made every expression collapse.
- **One identity, used repeatedly.** Every simplification came from $\alpha^2 - \alpha + 1 = 0$ and $\alpha^3 = -1$.

What's really going on

Why the Fermat point minimises distance

Why should $PA + PB + PC$ be smallest at the point where the sides subtend $120°$? Here is a lovely argument. Rotate the whole picture about B by $60°$, sending A to L' (the apex of an equilateral triangle on AB) and P to P'. Triangle BPP' is equilateral,

so $PB = PP'$, and $PA = P'L'$. Then $PA + PB + PC = L'P' + P'P + PC$, the length of a path from L' to C. That is at least $|L'C|$, with equality exactly when the path is straight, which happens when the angles at P are $120°$. So the minimum total length is $|LC| = p + q + r$, as found in the question. (This works provided every angle of the triangle is less than $120°$; otherwise the minimum is at the obtuse vertex.)

Soap films and Steiner trees

Dip a frame with three pegs (representing A, B, C) into soapy water: the soap film, minimising its surface tension energy, forms three walls meeting at $120°$ at the Fermat point. For more points the shortest connecting network is a *Steiner tree*, with extra junctions where three edges meet at $120°$. Finding Steiner trees is computationally hard in general (it is an NP-hard problem), but they matter in practice for designing networks of roads, pipelines and the wiring on computer chips.

Napoleon's theorem

The same configuration contains another gem: the *centres* of the three equilateral triangles on the sides of any triangle form an equilateral triangle. This is traditionally credited to Napoleon Bonaparte (who was a keen amateur mathematician), though there is no evidence he discovered it. It too has a short proof with complex numbers, using $1 + \alpha^2 = \alpha$.

The thinking skill

Choose coordinates that fit the problem

Put the most important point at the origin and line things up with the symmetry of the problem. A good choice can make the algebra almost trivial.

Here the origin was the Fermat point, and the three vertices lay along the cube roots of -1 (up to sign). With any other coordinates the same calculation would be much messier.

Try this next

1. **Check the lettering.** Show that if the equilateral triangles were lettered anticlockwise (built inwards), the three lines would generally not pass through O.

2. **Napoleon's theorem.** Show that the centre of the equilateral triangle ABL is $\frac{1}{3}(A + B + L)$. Using the expressions above, show that the three centres form an equilateral triangle.

3 **The minimum.** With $p = 2$, $q = 1.4$, $r = 1.1$, compute $PA + PB + PC$ for P at each vertex and at the centroid $\frac{1}{3}(A + B + C)$, and check that all exceed 4.5.

4 **An equilateral test.** Show that z_1, z_2, z_3 form an equilateral triangle (in either orientation) if and only if $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$.

Further reading

- **Richard Courant & Herbert Robbins, *What Is Mathematics?*** (Oxford University Press). Chapter 7 discusses the Fermat point, Steiner trees and soap films.
- **Liang-shin Hahn, *Complex Numbers and Geometry*** (Mathematical Association of America). Many geometric theorems proved with complex numbers, at exactly this level.
- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). Chapter 1 includes Napoleon's theorem and other geometric results.

5.7 The Complex Exponential as a Map

The question

The complex numbers $w = u + \mathrm{i}v$ and $z = x + \mathrm{i}y$ are related by the equation

$$z = (\cos v + \mathrm{i}\sin v)\mathrm{e}^u.$$

Find all w which correspond to $z = \mathrm{i}\,\mathrm{e}$.

Find the loci in the x–y plane corresponding to the lines $u =$ constant in the u–v plane. Find also the loci corresponding to the lines $v =$ constant. Illustrate your answers with clearly labelled sketches.

Identify two subsets W_1 and W_2 of the u–v plane each of which is in one-to-one correspondence with the first quadrant $\{(x, y) : x > 0, y > 0\}$ of the x–y plane. Identify also two subsets W_3 and W_4 each of which is in one-to-one correspondence with the set $\{z : 0 < |z| < 1\}$.

[**NB** 'one-to-one' means here that to each value of w there is only one corresponding value of z, and vice-versa.]

The hook

You know the exponential function as a curve, $y = e^x$. But a function of a complex variable cannot be drawn as a curve: its input is a point of a plane and so is its output. The way to picture it is as a *map*, a transformation that moves every point of one plane to a point of another, stretching and bending as it goes.

The formula in the question is $z = e^u(\cos v + \mathrm{i}\sin v) = e^{u+\mathrm{i}v} = e^w$. Seen as a map, the complex exponential does something remarkable: it rolls up a horizontal strip of the plane into a whole plane (minus the origin), wrapping vertical lines round circles and laying horizontal lines out as rays. And because it wraps round, it is not one-to-one: infinitely many points go to each point. This is why the complex logarithm is "multi-valued", one of the first surprises of complex analysis.

Before you start

Ask yourself What are the modulus and argument of $z = e^u(\cos v + \mathrm{i}\sin v)$ in terms of u and v?

$|z| = e^u$ and $\arg z = v$ (up to multiples of 2π). So the real part u of w controls the distance from the origin, exponentially, and the imaginary part v controls the direction. That one observation answers almost every part of the question.

Working it through

STAGE 1 Solving $e^w = \mathrm{i}e$

Ask yourself What are the modulus and argument of $\mathrm{i}e$?

$|\mathrm{i}e| = e$ and $\arg(\mathrm{i}e) = \frac{\pi}{2}$. So we need $e^u = e$, giving $u = 1$, and $v = \frac{\pi}{2} + 2k\pi$ for any integer k. The solutions are

$$w = 1 + \mathrm{i}\left(\tfrac{\pi}{2} + 2k\pi\right), \qquad k \in \mathbb{Z}.$$

There are infinitely many, spaced 2π apart vertically.

STAGE 2 The images of grid lines

On a vertical line $u = c$, $|z| = e^c$ is fixed while v varies: the image is the *circle* of radius e^c centred at the origin, traced round infinitely often as v runs over all real numbers.

On a horizontal line $v = d$, $\arg z = d$ is fixed while $|z| = e^u$ takes every positive value: the image is the open *ray* (half-line) from the origin at angle d, not including the origin itself. Lines $v = d$ and $v = d + 2\pi$ give the same ray.

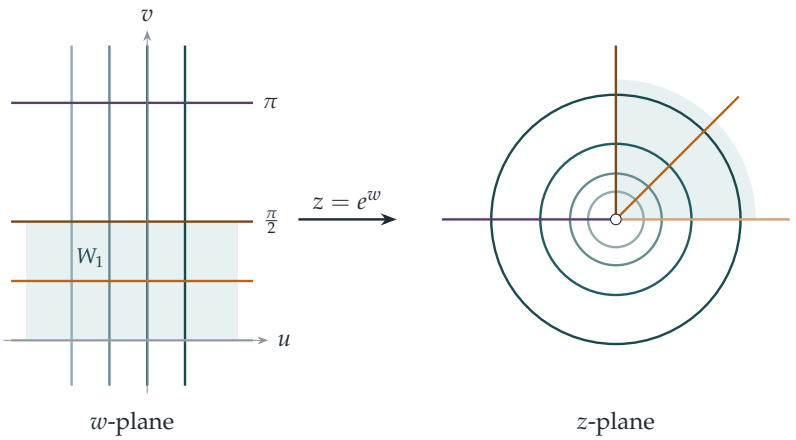


Figure 5.7.1 The complex exponential as a map. Vertical lines $u = c$ in the w-plane (teal) go to circles of radius e^c; horizontal lines $v = d$ (amber, plum) go to rays at angle d. The shaded strip $W_1 = \{0 < v < \frac{\pi}{2}\}$ goes exactly onto the first quadrant. Notice that the grid lines meet at right angles in both pictures: the map preserves angles. The origin (open circle) is never reached.

STAGE 3 One-to-one pieces

Ask yourself Which points w map into the first quadrant $x > 0$, $y > 0$? Why is that set too big to be in one-to-one correspondence with the quadrant?

z is in the open first quadrant exactly when $\arg z$ is strictly between 0 and $\frac{\pi}{2}$ (modulo 2π), with any modulus. So w must satisfy $0 < v < \frac{\pi}{2}$ *or* $2\pi < v < \frac{5\pi}{2}$ or $\ldots$: infinitely many horizontal strips, each of which maps onto the whole quadrant. Any one strip is in one-to-one correspondence with it, since within a strip u and v are determined uniquely by $|z|$ and $\arg z$. So we can take

$$W_1 = \{u + iv : 0 < v < \tfrac{\pi}{2}\}, \qquad W_2 = \{u + iv : 2\pi < v < \tfrac{5\pi}{2}\}.$$

Pause and try Now find two sets in the w-plane in one-to-one correspondence with the punctured disc $0 < |z| < 1$.

$|z| < 1$ means $e^u < 1$, that is $u < 0$; and every argument must be covered exactly once, so v should range over an interval of length exactly 2π, including one end but not the other. For example,

$$W_3 = \{u + iv : u < 0,\ 0 \le v < 2\pi\}, \qquad W_4 = \{u + iv : u < 0,\ -\pi < v \le \pi\}.$$

These are half-strips with one edge included and one excluded. (If both edges were included, the points on the positive real axis would be hit twice; if neither, they would be missed.) ■

Looking back

- **Modulus and argument separate the variables.** $|e^w| = e^{\operatorname{Re} w}$ and $\arg e^w = \operatorname{Im} w$, so the real and imaginary parts of w control distance and direction independently.
- **Picture a complex function as a map.** Tracking where grid lines go gives a clear picture of a function that cannot be drawn as a graph.
- **Periodicity destroys injectivity.** $e^{w+2\pi\mathrm{i}} = e^w$, so the map is many-to-one, and must be restricted to a strip of height 2π to have an inverse.

What's really going on

The complex logarithm

On the real line, ln is the inverse of exp. In the complex plane, $e^w = z$ has infinitely many solutions w for each $z \neq 0$, differing by multiples of $2\pi\mathrm{i}$, so "$\log z$" is not a single number. To get a genuine function you must choose a *branch*, which means choosing a strip like W_3 or W_4: the *principal branch* takes $-\pi < \arg z \leq \pi$. Every branch has a jump somewhere, along a ray from the origin called a *branch cut*. Bernhard Riemann's resolution, in his thesis of 1851, was to stop insisting that the logarithm lives on a plane: it lives on an infinite spiral staircase, a *Riemann surface*, on which it is single-valued and smooth. Riemann surfaces are now central to geometry, number theory and string theory.

Angles are preserved

Look again at Figure 5.7.1. The vertical and horizontal grid lines meet at right angles, and so do the circles and rays they become. This is not a coincidence: wherever the derivative of a complex function is non-zero, the function preserves the angles between curves. Such maps are called *conformal*. Locally, a conformal map is just a rotation and a scaling, which is exactly what multiplication by the complex number $f'(z)$ does. Conformal maps are used to solve problems in fluid flow, electrostatics and heat conduction, by transforming a complicated region into a simple one where the answer is known.

Mercator's map

Here is a surprise connecting this section with Section 4.8. Gerardus Mercator's map of 1569 was designed so that a course of constant compass bearing is a straight line, which requires preserving angles. If you first project the globe stereographically onto a plane (from the north pole), then take the complex logarithm, the result is exactly Mercator's projection: meridians become vertical lines and lines of latitude become horizontal lines, just as rays and circles become horizontal and vertical lines under log. The exponential map in this section, run backwards, is the mathematics of the navigator's chart.

The thinking skill

Visualise functions as transformations

When a function cannot be drawn as a graph, draw what it does: follow a grid of simple curves and see where they go.

For complex functions, the images of horizontal and vertical lines, or of circles and rays, usually tell the whole story. This way of thinking, about what a map does to shapes, runs through all of geometry and analysis.

Try this next

1 **Solving equations.** Find all complex w with $e^w = -1$, and all w with $e^w = 1 + \mathrm{i}$.

2 **The image of a square.** Find and sketch the image under $z = e^w$ of the square with corners 0, 1, $1 + \mathrm{i}$ and i.

3 **Right angles.** Show directly that the circle $|z| = e^c$ and the ray $\arg z = d$ meet at right angles, and explain why this is consistent with the lines $u = c$ and $v = d$ being perpendicular.

4 **Squaring.** Let c and d be non-zero real numbers. Find the images of the lines $x = c$ and $y = d$ under the map $z \mapsto z^2$. Show that they are parabolas, and that they meet at right angles wherever they meet. (What happens if $c = 0$ or $d = 0$?)

Further reading

- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). Chapters 2 and 4 treat the exponential, the logarithm and conformal maps with superb pictures.
- **Hilary Priestley, *Introduction to Complex Analysis*** (Oxford University Press). A clear first undergraduate text.
- **Ian Stewart & David Tall, *Complex Analysis*** (Cambridge University Press). Another excellent undergraduate introduction, with an emphasis on geometry.

5.8 A Hidden Distance in the Disc

The question

Let z and w be complex numbers. Use a diagram to show that $|z-w| \leq |z|+|w|$.

For any complex numbers z and w, E is defined by

$$E = zw^* + z^*w + 2|zw|.$$

(i) Show that $|z-w|^2 = \left(|z|+|w|\right)^2 - E$, and deduce that E is real and non-negative.

(ii) Show that $|1-zw^*|^2 = \left(1+|zw|\right)^2 - E$.

Hence show that, if both $|z|>1$ and $|w|>1$, then

$$\frac{|z-w|}{|1-zw^*|} \leq \frac{|z|+|w|}{1+|zw|}.$$

Does this inequality also hold if both $|z|<1$ and $|w|<1$?

The hook

At first sight this is an exercise in manipulating moduli and conjugates, ending in a strange-looking inequality. Why would anyone care about $\frac{|z-w|}{|1-zw^*|}$?

Because it is a distance. Not the ordinary distance between z and w, but the distance in a different geometry, one in which the unit disc is an entire infinite universe, straight lines are circular arcs, and the angles of a triangle add up to less than 180°. This is *hyperbolic geometry*, discovered in the nineteenth century after two thousand years of attempts to prove Euclid's parallel postulate, and the inequality in this question turns out to be its triangle inequality, in disguise. Even more surprisingly, the right-hand side is the formula for adding velocities in special relativity, which we met in Section 3.8.

Before you start

Ask yourself How does $|z-w| \leq |z|+|w|$ follow from a diagram? (Think of z and $-w$ as vectors.)

$z-w = z+(-w)$. Draw z as an arrow from the origin, then $-w$ as an arrow from the

tip of z. The arrow from the origin to the final point is $z - w$, and the three arrows form a triangle with sides $|z|$, $|w|$ and $|z - w|$. One side of a triangle is at most the sum of the other two.

Also recall that $|a|^2 = aa^*$ for any complex number a. That converts every modulus in the question into algebra.

Working it through

STAGE 1 Expanding the moduli

Ask yourself Expand $|z - w|^2 = (z - w)(z^* - w^*)$. Compare with $(|z| + |w|)^2$.

$|z - w|^2 = zz^* - zw^* - z^*w + ww^* = |z|^2 + |w|^2 - zw^* - z^*w$. Meanwhile $(|z| + |w|)^2 - E = |z|^2 + 2|z||w| + |w|^2 - zw^* - z^*w - 2|zw|$, and since $|z||w| = |zw|$ this is the same. ■

$E = (zw^*) + (zw^*)^* + 2|zw| = 2\,\mathrm{Re}(zw^*) + 2|zw|$, which is real. And $E = (|z| + |w|)^2 - |z - w|^2 \geq 0$ by the triangle inequality. ■

STAGE 2 The second identity

$|1 - zw^*|^2 = (1 - zw^*)(1 - z^*w) = 1 - zw^* - z^*w + |z|^2|w|^2$, while $(1 + |zw|)^2 - E = 1 + 2|zw| + |zw|^2 - zw^* - z^*w - 2|zw|$, which is the same. ■

STAGE 3 The inequality

Ask yourself Write $A = (|z| + |w|)^2$ and $B = (1 + |zw|)^2$. The inequality, squared, says $\frac{A-E}{B-E} \leq \frac{A}{B}$. When is that true?

Both sides of the required inequality are non-negative, so it is equivalent to its square, $\frac{A-E}{B-E} \leq \frac{A}{B}$. Assuming $B - E = |1 - zw^*|^2 > 0$, multiply out:

$$(A - E)B \leq A(B - E) \iff -EB \leq -EA \iff E(B - A) \geq 0.$$

Since $E \geq 0$, this holds whenever $B \geq A$, that is, $1 + |zw| \geq |z| + |w|$.

Pause and try Show that $1 + |z||w| - |z| - |w|$ factorises. When is it non-negative?

$1 + |z||w| - |z| - |w| = (1 - |z|)(1 - |w|)$. If $|z| > 1$ and $|w| > 1$, both factors are negative and the product is positive. So $B \geq A$ and the inequality holds. ■

If instead $|z| < 1$ and $|w| < 1$, both factors are positive, the product is again positive, and the same argument works: **yes**, the inequality holds in this case too. (In both cases $|zw^*| \neq 1$, so $1 - zw^* \neq 0$ and the left-hand side is defined.) It is the mixed case, one

inside the unit circle and one outside, where it can fail.

Looking back

- $|a|^2 = aa^*$. Every identity came from expanding products of conjugates. This is the standard way to handle moduli algebraically.
- **Reduce to the sign of one quantity.** After squaring and cross-multiplying, the whole inequality depended on the sign of $(1 - |z|)(1 - |w|)$.
- **Check other cases.** The final question asked about $|z|, |w| < 1$; the same factorisation answered it immediately.

What's really going on

Non-Euclidean geometry

For two thousand years mathematicians tried to prove Euclid's fifth postulate (through a point not on a line, there is exactly one parallel to the line) from his other axioms. In the 1820s and 1830s Nikolai Lobachevsky, János Bolyai and (privately) Gauss realised that it cannot be done: there is a perfectly consistent geometry in which there are infinitely many parallels. In 1882 Henri Poincaré found a beautiful model of it inside the unit disc. Points are points of the open disc; "straight lines" are diameters and circular arcs that meet the boundary circle at right angles; and the distance between z and w is

$$d(z, w) = 2 \tanh^{-1} \rho(z, w), \qquad \text{where} \qquad \rho(z, w) = \frac{|z - w|}{|1 - zw^*|}.$$

As a point approaches the boundary circle, its distance from the centre tends to infinity: the disc is a whole infinite plane, seen through a distorting lens.

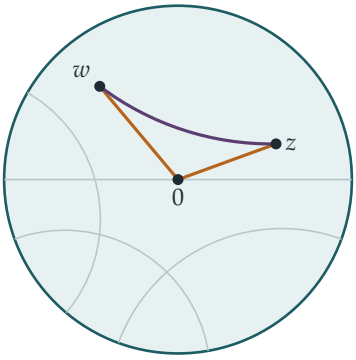


Figure 5.8.1 The Poincaré disc. Hyperbolic "straight lines" are diameters or arcs of circles meeting the boundary at right angles (grey). The hyperbolic triangle with vertices 0, z and w has two sides along diameters (amber) and one along an arc (plum). The inequality in the question says that the plum side is no longer, in hyperbolic distance, than the other two put together.

The triangle inequality in disguise

Here is what the question proved, for $|z|, |w| < 1$. Write $|z| = \tanh a$ and $|w| = \tanh b$, so that the hyperbolic distances from 0 to z and w are $2a$ and $2b$. Then by the addition formula for tanh from Section 3.8,

$$\frac{|z| + |w|}{1 + |zw|} = \frac{\tanh a + \tanh b}{1 + \tanh a \tanh b} = \tanh(a + b).$$

So the inequality says $\tanh\left(\frac{1}{2}d(z, w)\right) \leq \tanh(a + b)$, that is,

$$d(z, w) \leq 2a + 2b = d(z, 0) + d(0, w).$$

This is the triangle inequality for hyperbolic distance (Figure 5.8.1). The relativistic velocity-addition formula and hyperbolic geometry are secretly the same thing: the set of possible velocities in special relativity, with the right notion of distance, *is* a hyperbolic space. We return to this in Section 7.6.

Symmetries of the disc

Why is ρ the right quantity? The maps $z \mapsto \dfrac{z - a}{1 - a^* z}$, for $|a| < 1$, send the unit disc to itself, preserve the hyperbolic distance, and move a to the centre. They are the hyperbolic analogues of translations, and $\rho(z, w)$ is exactly |image of z| when w is moved to the centre. These *Möbius transformations* are a central topic in complex analysis. M. C. Escher's *Circle Limit* woodcuts, inspired by a diagram of the mathematician H. S. M. Coxeter, show tilings of the Poincaré disc by congruent fish and angels, which shrink towards the edge because hyperbolic distances blow up there.

The thinking skill

Ask what an expression means

When an algebraic expression keeps appearing, ask whether it has a meaning: a distance, an angle, a probability, a symmetry.

Here a strange ratio of moduli turned out to be a distance in another geometry, and a strange inequality turned out to be its triangle inequality. Recognising the meaning explains why the result is true and where it leads.

Try this next

1. **A numerical check.** Take $z = \frac{1}{2}$ and $w = -\frac{1}{2}$. Compute both sides of the inequality. Then compute the hyperbolic distances $d(z, 0)$, $d(0, w)$ and $d(z, w)$ and check the triangle inequality is an equality here. Why?

2. **The mixed case.** Find z, w with $|z| < 1 < |w|$ for which the inequality fails.

3. **An isometry.** Let $f(z) = \dfrac{z - a}{1 - a^* z}$ with $|a| < 1$. Show that $|f(z)| = 1$ whenever $|z| = 1$, and that $f(a) = 0$. Show that $\rho(f(z), f(w)) = \rho(z, w)$.

4. **Infinity is near.** Show that the hyperbolic distance from 0 to r (with $0 < r < 1$) is $\ln \frac{1+r}{1-r}$, and that it tends to infinity as $r \to 1$. How close to the boundary must a point be to be hyperbolic distance 10 from the centre?

Further reading

- **Caroline Series, "Hyperbolic geometry" lecture notes** (University of Warwick, freely available). A clear first course, starting from the Poincaré disc.
- **Jeremy Gray, *Ideas of Space*** (Oxford University Press). The history of non-Euclidean geometry from Euclid to Einstein.
- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). Chapter 6 covers non-Euclidean geometry and Möbius transformations, with beautiful pictures.

What We Have Learned

We began with Cardano's unsettling discovery that real answers can require imaginary steps, and with three questions: how complex numbers help with real problems, what is special about roots of unity, and what complex functions look like. Here is what the chapter's eight questions showed.

1. How do complex numbers help with real problems?

Again and again, a fact about real numbers had a short proof through the complex plane.

- **Angles become products.** Machin's formula for π is the statement that $(5-\mathrm{i})^4(1+\mathrm{i})$ is a real multiple of $239-\mathrm{i}$, checked with whole-number arithmetic (Section 5.1).
- **Counting becomes averaging.** Sums of every third binomial coefficient came from evaluating $(1+x)^n$ at the cube roots of unity (Section 5.2).
- **Trigonometric products become polynomial values.** Products of cosines at equally spaced angles came from factorising $z^{2n}+1$ and putting $z=\mathrm{i}$ (Section 5.3).
- **Geometry becomes algebra.** Distances became moduli, so the product of chords from a vertex of a regular n-gon is $|1+1+\cdots+1|=n$ (Section 5.4); rotations became multiplications, so the Fermat point construction collapsed to the identity $1+\alpha^2=\alpha$ (Section 5.6).
- **Irrationality from De Moivre.** Expanding $(\cos\theta+\mathrm{i}\sin\theta)^k$ and arguing about parity showed that $\cos^{-1}\frac{1}{3}$ is not a rational number of degrees, the key fact in Dehn's solution of Hilbert's third problem (Section 5.5).

2. What is special about the roots of unity?

The nth roots of unity are the vertices of a regular n-gon, they are the roots of the simplest possible polynomial z^n-1, and they form a group under multiplication. Those three facts together explain their ubiquity. Because they are evenly spread round a circle, they sum to zero, which gives the filter of Section 5.2 and, in general, the discrete Fourier transform. Because they are the roots of z^n-1, products over them are values of simple polynomials (Section 5.3, Section 5.4). And because they are closed under multiplication, powers of a root of unity cycle round, which is exactly what fails for an angle like $\cos^{-1}\frac{1}{3}$ (Section 5.5). In Chapter 6 roots of unity become the heart of algebra, leading towards Galois theory.

3. What does a complex function look like?

In Section 5.7 we pictured e^w as a map that rolls a strip into a plane, sending vertical lines to circles and horizontal lines to rays, and wrapping round infinitely often, so that the logarithm is many-valued. We saw that it preserves angles, the first example of a *conformal* map, and that it underlies Mercator's chart. In Section 5.8 a strange inequality turned out to be the triangle inequality of hyperbolic geometry, where the unit disc is an infinite plane, straight lines are circular arcs, and distances combine by the relativistic velocity-addition formula.

4. A toolkit for complex numbers

Method	Where you used it
Arguments add under multiplication	Machin's formula (5.1)
Checking multiples of 2π	Machin's formula (5.1)
Averaging over roots of unity	Every third coefficient (5.2)
Factorising over roots, then substituting	Cosine products (5.3); chords (5.4)
Dividing out a vanishing factor	5.3; 5.4
De Moivre's theorem	Multiple-angle formulas (5.5)
Parity and minimal counterexamples	Irrational angles (5.5)
Rotation as multiplication	Equilateral triangles (5.6)
Mapping grids	The exponential map (5.7)
$\lvert a\rvert^2 = aa^*$	The Poincaré disc inequality (5.8)

Where next

- Roots of unity drive Chapter 6, including the cyclotomic polynomials of Section 6.1 and Gauss's construction of the regular 17-gon in Section 6.2.
- The connection between hyperbolic geometry, tanh and relativity is developed in Section 7.6.
- Gaussian integers, which appeared in Machin's formula, were the subject of Section 1.6.

If you want to go further before university

- **Tristan Needham, *Visual Complex Analysis***: the most beautiful book on complex numbers ever written, full of pictures. Start with Chapter 1.
- **Paul Nahin, *An Imaginary Tale***: the history of $\sqrt{-1}$, from Cardano to modern physics.
- **Liang-shin Hahn, *Complex Numbers and Geometry***: Euclidean geometry with complex numbers, at exactly the level of this chapter.
- **Hilary Priestley, *Introduction to Complex Analysis***: for a first look at the univer-

sity course.

Cardano thought his imaginary quantities were as useless as they were subtle. Five centuries later, it is hard to find a corner of mathematics or physics where they do not appear. The lesson of this chapter is Hadamard's: when a problem about real numbers is hard, try stepping off the real line. The complex plane is bigger, and in it, many things that look complicated turn out to be simple.

Answers to Try This Next

These are brief answers and hints, not full solutions: they give the result and the key step, so that you can check your own work. Try each exercise properly before looking.

5.1 Machin's Formula

1. $(5-\mathrm{i})^2(1+\mathrm{i}) = (24-10\mathrm{i})(1+\mathrm{i}) = 34+14\mathrm{i}$, whose argument is $\tan^{-1}\frac{7}{17}$. Its argument is also $-2\tan^{-1}\frac{1}{5}+\frac{\pi}{4}$.
2. $(2+\mathrm{i})(3+\mathrm{i}) = 5+5\mathrm{i}$, with argument $\frac{\pi}{4}$.
3. $(3+\mathrm{i})^2(7+\mathrm{i}) = (8+6\mathrm{i})(7+\mathrm{i}) = 50+50\mathrm{i}$.
4. The series alternate, so the error is less than the first omitted term. For $16\tan^{-1}\frac{1}{5}$ you need the omitted term $\frac{16}{(2N+1)5^{2N+1}}$ below 5×10^{-21}, which needs $N=14$ terms; for $4\tan^{-1}\frac{1}{239}$, 4 terms suffice. With a calculator, $16\left(\frac{1}{5}-\frac{1}{3\cdot5^3}+\frac{1}{5\cdot5^5}\right)-\frac{4}{239}\approx 3.14162$.

5.2 The Roots-of-Unity Filter

1. $(1+1)^n+(1-1)^n = 2\left(\binom{n}{0}+\binom{n}{2}+\cdots\right)$, so the sum is 2^{n-1} for $n\geq 1$.
2. Multiply by ω^{-1} and ω^{-2} before averaging: $\frac{1}{3}\left(2^n+\omega^{-1}(1+\omega)^n+\omega^{-2}(1+\omega^2)^n\right)$. Since $1+\omega = \mathrm{e}^{\mathrm{i}\pi/3}$, this is $\frac{1}{3}\left(2^n+2\cos\frac{(n-2)\pi}{3}\right)$. The third residue gives $\frac{1}{3}\left(2^n+2\cos\frac{(n+2)\pi}{3}\right)$, and the three cosines $\cos x+\cos(x-\frac{2\pi}{3})+\cos(x+\frac{2\pi}{3})$ sum to 0.
3. Average $(1+\mathrm{i}^j)^n$ over $j=0,1,2,3$: the sum is $\frac{1}{4}\left(2^n+(1+\mathrm{i})^n+0+(1-\mathrm{i})^n\right)$, and $(1\pm\mathrm{i})^n = 2^{n/2}\mathrm{e}^{\pm\mathrm{i}n\pi/4}$.
4. By size: $\frac{1}{3}(2^{12}+2\cos 4\pi) = 1366$. By sum: $\prod(1+\omega^j)$ has four factors each of 2, $1+\omega$ and $1+\omega^2$, and $(1+\omega)(1+\omega^2)=1$, so it equals 16 (and so does the product at ω^2). The count is $\frac{1}{3}(4096+16+16) = 1376$.

5.3 Factorising $z^{2n}+1$

1. Part (ii) with $n=7$ gives $\cos^2\frac{\pi}{14}\cos^2\frac{3\pi}{14}\cos^2\frac{5\pi}{14} = \frac{7}{64}$, and all three cosines are positive. For the second, let $c = \cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}$. Then $8c\sin\frac{\pi}{7} = \sin\frac{8\pi}{7} = -\sin\frac{\pi}{7}$, so $c=-\frac{1}{8}$; and $\cos\frac{4\pi}{7} = -\cos\frac{3\pi}{7}$.
2. At $z=1$, $2 = \prod\left(2-2\cos\frac{(2k-1)\pi}{2n}\right) = \prod 4\sin^2\frac{(2k-1)\pi}{4n}$, so the product of the sines (all positive) is $\sqrt{2/4^n} = 2^{\frac{1}{2}-n}$.

3. $z^6+1=(z^2-\sqrt{3}z+1)(z^2+1)(z^2+\sqrt{3}z+1)$; the outer two multiply to z^4-z^2+1.
4. The roots of $z^{2n}=1$ are ±1 and the conjugate pairs $e^{\pm ik\pi/n}$ for $1\le k\le n-1$, so $\frac{z^{2n}-1}{z^2-1}=\prod_{k=1}^{n-1}\left(z^2-2z\cos\frac{k\pi}{n}+1\right)$. Let $z\to1$: the left side tends to n, and the right side is $\prod 4\sin^2\frac{k\pi}{2n}$.

5.4 Distances in a Regular Polygon

1. Square: $\sqrt{2}\times2\times\sqrt{2}=4$. Triangle: $\sqrt{3}\times\sqrt{3}=3$.
2. Put $X_k=\omega^k$ and $P=r$ on the positive real axis. Then $\prod|r-\omega^k|=|r^n-1|$.
3. By AM–GM the average is at least the geometric mean, $n^{1/(n-1)}$. For $n=6$ the chords are 1, $\sqrt{3}$, 2, $\sqrt{3}$, 1, with average $\frac{4+2\sqrt{3}}{5}\approx1.493$, against $6^{1/5}\approx1.431$.
4. By part (ii), the product of the $n-1$ chords from each vertex is n. Multiplying over all n vertices gives n^n, and counts every chord twice, so the product of the $\binom{n}{2}$ chords is $n^{n/2}$.

5.5 Rational Angles That Aren't

1. Take vertices $(1,1,1)$, $(1,-1,-1)$, $(-1,1,-1)$, $(-1,-1,1)$ and centre O. Two bond vectors have dot product -1 and lengths $\sqrt{3}$, so the bond angle is $\cos^{-1}(-\frac{1}{3})$. The outward normal to a face is the negative of the vector to the opposite vertex, so the angle between outward normals is also $\cos^{-1}(-\frac{1}{3})$, and the angle between the faces is its supplement, $\cos^{-1}\frac{1}{3}$.
2. Let $(4+3i)^k=a_k+ib_k$. Since $4+3i$ and its conjugate are the roots of $t^2-8t+25$, $b_{k+1}=8b_k-25b_{k-1}\equiv3b_k\pmod5$, and $b_1=3$, so $b_k\equiv3^k\not\equiv0\pmod5$. No power is real. If the angle were a rational number of degrees, some multiple of it would be a multiple of 180°, and that power would be real.
3. With $a=2$, $a^2-1=3$ is odd, so the parity argument fails. Indeed $\cos^{-1}\frac{1}{2}=60^\circ$ is rational; with $k=3$ the sine bracket is $3-\binom{3}{3}\cdot3=0$.
4. 0°, 60° and 90°, with cosines 1, $\frac{1}{2}$ and 0. Niven's theorem says that the only rational values of cos at rational multiples of 180° are 0, $\pm\frac{1}{2}$ and ±1.

5.6 The Fermat Point

1. Inwards, the apexes are $A+(1-\bar{\alpha})(B-A)$ and so on. In the example with $p=2$, $q=1.4$, $r=1.1$, the line through C and the new L misses O (the test $\operatorname{Im}(\bar{C}L')=0$ fails), and so do the other two.
2. The centre of an equilateral triangle is its centroid. With $L=\alpha(p+q)$, $M=-(q+r)$ and $N=-\alpha^2(p+r)$, the centres G_1, G_2, G_3 of ABL, BCM, CAN satisfy $G_3-G_1=\alpha(G_2-G_1)$ (expand using $\alpha^2=\alpha-1$), which says that $G_1G_2G_3$ is equilateral. In the example its side is $\frac{p+q+r}{\sqrt{3}}\approx2.598$.

3. $PA + PB + PC$ is about 5.68 at A, 5.13 at B, 4.89 at C and 4.54 at the centroid; all exceed 4.5.

4. Equilateral (in one orientation or the other) means $z_1 + \omega z_2 + \omega^2 z_3 = 0$ or $z_1 + \omega^2 z_2 + \omega z_3 = 0$, where $\omega = \mathrm{e}^{2\pi\mathrm{i}/3}$. Using $\omega + \omega^2 = -1$, the product of the two left sides is $z_1^2 + z_2^2 + z_3^2 - z_1 z_2 - z_2 z_3 - z_3 z_1$.

5.7 The Complex Exponential as a Map

1. $\mathrm{e}^w = -1$ when $w = (2k+1)\pi\mathrm{i}$; $\mathrm{e}^w = 1 + \mathrm{i}$ when $w = \frac{1}{2}\ln 2 + \mathrm{i}\left(\frac{\pi}{4} + 2k\pi\right)$, for integers k.

2. The region $1 \le |z| \le \mathrm{e}$, $0 \le \arg z \le 1$ (one radian): a piece of an annulus. The bottom edge goes to $[1, \mathrm{e}]$ on the real axis, the right edge to an arc of radius e, the top edge to a radial segment at angle 1, and the left edge to an arc of the unit circle.

3. The ray is along the radius; the circle's tangent is perpendicular to the radius. This matches the lines $u = c$ and $v = d$ because e^w preserves angles (its derivative is never 0).

4. $(x + \mathrm{i}y)^2 = x^2 - y^2 + 2\mathrm{i}xy$. The line $x = c$ goes to $u = c^2 - \frac{v^2}{4c^2}$ and $y = d$ to $u = \frac{v^2}{4d^2} - d^2$: parabolas opening left and right, both with focus at 0. Equating them gives $v^2 = 4c^2d^2$, so they meet twice, at $u = c^2 - d^2$, $v = \pm 2cd$. At $v = 2cd$ (the image of $c + \mathrm{i}d$, where the lines cross) the slopes $\frac{\mathrm{d}u}{\mathrm{d}v}$ are $-\frac{d}{c}$ and $\frac{c}{d}$; at $v = -2cd$ they are $\frac{d}{c}$ and $-\frac{c}{d}$. In both cases the product is -1. (The second meeting point is the image of $c - \mathrm{i}d$ on the first line and of $-c + \mathrm{i}d$ on the second: squaring sends z and $-z$ to the same place.) If $c = 0$ the image of $x = 0$ is the ray $u \le 0$, $v = 0$, and if $d = 0$ the image of $y = 0$ is the ray $u \ge 0$, $v = 0$: these are not parabolas. A ray still meets the other image at right angles when the lines cross away from 0, but if $c = d = 0$ both images lie along the real axis: at 0 the map doubles angles (its derivative $2z$ is zero there), so the right angle between the axes becomes a straight angle.

5.8 A Hidden Distance in the Disc

1. Both sides equal $\frac{1}{5/4} = \frac{4}{5}$. Then $d(z,0) = d(0,w) = 2\tanh^{-1}\frac{1}{2} = \ln 3$ and $d(z,w) = 2\tanh^{-1}\frac{4}{5} = \ln 9$, so equality holds: 0 lies on the hyperbolic line segment (a diameter) from z to w.

2. From parts (i) and (ii), when $E > 0$ the inequality is equivalent to $|z| + |w| \le 1 + |zw|$, that is, $(1 - |z|)(1 - |w|) \ge 0$. In the mixed case this fails, for instance with $z = \frac{1}{2}$, $w = 3$: the left side is $\frac{5/2}{1/2} = 5$ and the right side is $\frac{7/2}{5/2} = \frac{7}{5}$.

3. If $|z| = 1$ then $|1 - a^*z| = |\bar{z} - a^*| = |z - a|$. Directly, $f(z) - f(w) = \frac{(1-|a|^2)(z-w)}{(1-a^*z)(1-a^*w)}$ and $1 - f(z)f(w)^* = \frac{(1-|a|^2)(1-zw^*)}{(1-a^*z)(1-aw^*)}$; dividing and taking moduli gives $\rho(f(z), f(w)) = \rho(z, w)$.

4. $d(0,r) = 2\tanh^{-1} r = \ln \frac{1+r}{1-r}$, which tends to ∞ as $r \to 1$. Distance 10 needs $r = \tanh 5 \approx 0.99991$: within about 9×10^{-5} of the edge.

CHAPTER

6

Algebra and Structure

Behind every clever trick in algebra there is a structure, and understanding the structure is worth more than any number of tricks.

IN THIS CHAPTER

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6.2	Gauss's 17-gon	STEP II 1991
6.3	A Polynomial for $\sqrt{2}+\sqrt{3}+\sqrt{5}$	STEP II 2023
6.4	Solving the Quartic	STEP III 2017
6.5	Differentiation from Axioms	STEP II 1992
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A	Answers to the exercises	p. 423

From Equations to Structures

Mathematics is the art of giving the same name to different things.

— Henri Poincaré

For most of its history, algebra meant solving equations. The Babylonians solved quadratics four thousand years ago; Italian mathematicians of the sixteenth century solved cubics and quartics; and then, for nearly three hundred years, the quintic resisted every attempt. The resolution, when it came in the 1820s and 1830s, changed algebra completely. Niels Henrik Abel proved that no general formula for the quintic exists, and Évariste Galois explained exactly which equations can be solved, by studying not the equations themselves but the *symmetries of their roots*. To do so he had to invent a new kind of mathematical object: the group.

That shift, from computing answers to studying structures, is the subject of this chapter. Instead of asking "what is the solution?", modern algebra asks "what kind of object is this, and what operations does it support?" The answers turn out to explain why the old tricks worked, and to settle questions that tricks alone never could.

Why structure matters

Here are three ancient problems that resisted all attempts for two thousand years: with ruler and compasses, double the volume of a cube; trisect an arbitrary angle; construct a regular 7-sided polygon. None of them can be done, and the proofs, found in the nineteenth century, are not clever constructions but arguments about structure: about the *degree* of the field of numbers that ruler and compasses can reach (Section 6.3). The same theory explains why the regular 17-gon *can* be constructed (Section 6.2), and why the quartic can be solved but the quintic cannot (Section 6.4).

Structures also travel. The axioms for a group are satisfied by the symmetries of a crystal, the permutations of the roots of a polynomial, the rotations of three-dimensional space, the moves of Rubik's cube and the points of an elliptic curve. A theorem proved once from the axioms holds in all of them. That economy, giving "the same name to different things" as Poincaré put it, is what makes abstract algebra so powerful.

What you will meet at university

In the first year you will meet *groups* (sets with an operation satisfying three axioms) and, usually, *rings* and *fields* (sets with addition and multiplication). You will prove Lagrange's theorem, that the size of a subgroup divides the size of the group, and meet permutations, symmetry groups and matrix groups. In the second and third years these lead to *Galois theory*, which connects fields and groups and answers the

classical questions about equations and constructions; to *representation theory*, which studies groups through matrices; and to *algebraic number theory*, where the ideas of Section 1.7 are put on a firm footing. The last three sections of this chapter are about groups. Groups are no longer on the STEP syllabus (they were removed in 2019), but the questions are self-contained and each section gives the definitions it needs.

The questions that run through this chapter

Three big questions

1. **What can be built from what?** Which numbers can be reached from the rationals by square roots, or by solving equations, and how do we prove something cannot be reached?
2. **Why do the formulas for equations work?** What is the hidden reason that quadratics, cubics and quartics have formulas, and what changes at degree five?
3. **What is gained by abstraction?** What happens when we describe objects only by the rules they obey?

Ideas in this chapter

§	Topic	Idea
6.1	Cyclotomic polynomials	Classify roots of unity by order
6.2	The regular 17-gon	Gauss periods: sums and products give quadratics
6.3	Minimal polynomials	Eliminate radicals; degrees multiply
6.4	Solving the quartic	Symmetric functions; the resolvent cubic
6.5	Differentiation from axioms	An operation determined by its rules
6.6	Unions of subgroups	Arguing from the group axioms
6.7	The quaternion group	Representing an operation by matrices
6.8	A group of singular matrices	Reading definitions literally

How to read this chapter

Many of these questions can be done by brute force: expand everything, compare coefficients, check every case. Do that first if you need to. Then look back and ask what made the brute force work, and whether there is a structural reason behind it. The structural reason is usually shorter, always more illuminating, and it is what you will need when the brute force becomes impossible.

6.1 Polynomials Whose Roots Are Roots of Unity

The question

For any given positive integer n, a number a (which may be complex) is said to be a *primitive nth root of unity* if $a^n = 1$ and there is no integer m such that $0 < m < n$ and $a^m = 1$. Write down the two primitive 4th roots of unity.

Let $C_n(x)$ be the polynomial such that the roots of the equation $C_n(x) = 0$ are the primitive nth roots of unity, the coefficient of the highest power of x is one and the equation has no repeated roots. Show that $C_4(x) = x^2 + 1$.

(i) Find $C_1(x)$, $C_2(x)$, $C_3(x)$, $C_5(x)$ and $C_6(x)$, giving your answers as unfactorised polynomials.

(ii) Find the value of n for which $C_n(x) = x^4 + 1$.

(iii) Given that p is prime, find an expression for $C_p(x)$, giving your answer as an unfactorised polynomial.

(iv) Prove that there are no positive integers q, r and s such that $C_q(x) \equiv C_r(x)\,C_s(x)$.

The hook

The nth roots of unity, the solutions of $z^n = 1$, sit at the vertices of a regular n-gon. But they are not all alike. Among the 12th roots of unity, for instance, 1 is already a first root of unity, -1 is a square root of unity, i is a fourth root, and only four of the twelve, the *primitive* ones, need all twelve steps to get back to 1.

Sorting the roots of unity by the number of steps they need, and building a polynomial from each class, produces the *cyclotomic polynomials* ("circle-dividing", from the Greek). They are the basic building blocks of $z^n - 1$, in the same way that primes are the building blocks of whole numbers, and they lie behind Gauss's construction of the regular 17-gon (Section 6.2), the roots-of-unity filter (Section 5.2) and a good deal of modern number theory.

Before you start

Ask yourself Every nth root of unity satisfies $z^n = 1$. But the smallest positive power that gives 1 might be less than n. For each of the four 4th roots of unity, 1, i, -1, $-\mathrm{i}$, what is that smallest power?

$1^1 = 1$, $(-1)^2 = 1$, and $\mathrm{i}^4 = (-\mathrm{i})^4 = 1$ with no smaller power working. So the smallest powers are 1, 4, 2, 4. This smallest power is called the *order* of the root of unity, and the primitive nth roots of unity are exactly those of order n. The key fact for the whole question is that *every root of unity has exactly one order*.

Working it through

STAGE 1 The first few

The primitive 4th roots are i and $-\mathrm{i}$, so $\mathrm{C}_4(x) = (x-\mathrm{i})(x+\mathrm{i}) = x^2+1$. ■

Ask yourself Which roots of unity have order 1, 2, 3, 5 and 6? (Draw the 6th roots of unity and sort them by order.)

Order 1: just 1, so $\mathrm{C}_1(x) = x-1$. Order 2: just -1, so $\mathrm{C}_2(x) = x+1$.

Order 3: the cube roots of unity other than 1, that is, the roots of $\frac{x^3-1}{x-1}$. So $\mathrm{C}_3(x) = x^2+x+1$.

Order 5: the fifth roots other than 1, which are the roots of $\frac{x^5-1}{x-1}$, so $\mathrm{C}_5(x) = x^4+x^3+x^2+x+1$.

Order 6: of the sixth roots $e^{k\pi\mathrm{i}/3}$, $k = 0,\ldots,5$, those with $k = 0, 3$ have orders 1, 2, and $k = 2, 4$ have order 3. That leaves $e^{\pm\pi\mathrm{i}/3}$, so

$$\mathrm{C}_6(x) = (x-e^{\mathrm{i}\pi/3})(x-e^{-\mathrm{i}\pi/3}) = x^2 - 2x\cos\tfrac{\pi}{3} + 1 = x^2 - x + 1. \quad ■$$

STAGE 2 Which n gives x^4+1?

The roots of $x^4+1=0$ satisfy $x^4 = -1$, so $x^8 = 1$ but $x^4 \neq 1$. Their order divides 8 but not 4, so it is 8. Conversely, a root of order 8 satisfies $x^8 = 1$, $x^4 \neq 1$, so $x^4 = -1$. There are four primitive 8th roots and four roots of x^4+1, so $n = 8$. ■

STAGE 3 Primes

If p is prime, the order of a pth root of unity divides p, so it is 1 or p. Every pth root except 1 is therefore primitive, and

$$\mathrm{C}_p(x) = \frac{x^p-1}{x-1} = x^{p-1} + x^{p-2} + \cdots + x + 1. \quad ■$$

STAGE 4 **No cyclotomic polynomial is a product of two others**

Pause and try Suppose $C_q = C_r C_s$. Take a root of C_r. What is its order? What does the identity say about its order? What does that force?

Every root of C_r (there is at least one, since $e^{2\pi i/r}$ is primitive) is a root of $C_r C_s = C_q$, so it has order q. But it also has order r, and a root of unity has only one order, so $r = q$. Then $C_q = C_q C_s$, and since C_q is not the zero polynomial, $C_s = 1$. That is impossible, because C_s has at least one root, $e^{2\pi i/s}$. So no such q, r, s exist. ■

Looking back

- **Sort by order.** Every root of unity has a unique order, and grouping by order splits the roots of $z^n - 1$ into disjoint classes.
- **Define a polynomial by its roots.** Each C_n was pinned down by saying exactly which numbers are its roots.
- **Uniqueness gives impossibility.** Part (iv) needed only one idea: a root cannot have two different orders.

What's really going on

Factorising $x^n - 1$

Every nth root of unity has some order d that divides n, and every dth root of unity of order d is an nth root of unity when $d \mid n$. So the roots of $x^n - 1$ split into classes by order, giving

$$x^n - 1 = \prod_{d \mid n} C_d(x).$$

For example $x^6 - 1 = C_1 C_2 C_3 C_6 = (x-1)(x+1)(x^2+x+1)(x^2-x+1)$. Figure 6.1.1 shows the twelve 12th roots of unity sorted in this way. The number of primitive nth roots, which is the degree of C_n, is the number of k between 1 and n with no common factor with n: *Euler's totient function* $\varphi(n)$. Comparing degrees gives the identity $\sum_{d \mid n} \varphi(d) = n$.

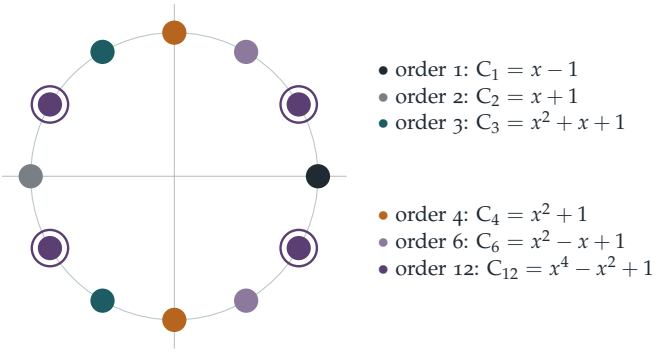


Figure 6.1.1 The twelve 12th roots of unity, coloured by order. The four primitive 12th roots (ringed) are the roots of C_{12}. Together the classes give $x^{12} - 1 = C_1C_2C_3C_4C_6C_{12}$, with degrees $1 + 1 + 2 + 2 + 2 + 4 = 12$.

Integer coefficients, and a surprise

It is not obvious from the definition, but every cyclotomic polynomial has *integer* coefficients. (Use the factorisation of $x^n - 1$ and induction: C_n is $x^n - 1$ divided by a product of earlier cyclotomic polynomials, and long division of integer polynomials by monic integer polynomials keeps the coefficients integers.) For a long time all known examples had coefficients 0, 1 and -1. The first exception is C_{105}, which contains the term $-2x^7$, found by Migotti in 1883; $105 = 3 \times 5 \times 7$ is the smallest number with three distinct odd prime factors. In fact the coefficients of cyclotomic polynomials can be made as large as you like.

Irreducibility

Part (iv) shows that no cyclotomic polynomial is a product of two others. Something much stronger is true: C_n cannot be factorised at all into polynomials with rational coefficients of lower degree. Gauss proved this for prime n (it follows from *Eisenstein's criterion*, which you will meet in a first algebra course), and Kronecker and Dedekind for general n. This *irreducibility* is what makes the cyclotomic polynomials the "primes" among polynomials related to roots of unity, and it is the starting point for the Galois theory of roots of unity.

The thinking skill

Classify, then build

Sort the objects in a problem into classes with a single, well-defined invariant, and build structure class by class.

Here the invariant was the order of a root of unity. Because each root has exactly one order, the classes do not overlap, and $x^n - 1$ splits as a product over them. The same pattern (a unique invariant giving a disjoint decomposition) underlies prime factorisation and the classification of finite abelian groups.

Try this next

1 **Check the factorisation.** Verify that $(x-1)(x+1)(x^2+x+1)(x^2-x+1) = x^6 - 1$, and that $x^{12} - 1$ factorises as in Figure 6.1.1.

2 **Twice a prime.** Show that if p is an odd prime then $C_{2p}(x) = C_p(-x)$. Check this for $p = 3$.

3 **Totients.** Show that $\varphi(p^k) = p^k - p^{k-1}$ for a prime p, and that $C_{p^k}(x) = C_p\left(x^{p^{k-1}}\right)$. Use this to find C_9.

4 **A longer one.** HARDERUsing $x^{15} - 1 = C_1 C_3 C_5 C_{15}$, show that $C_{15}(x) = x^8 - x^7 + x^5 - x^4 + x^3 - x + 1$.

Further reading

- **Ian Stewart, *Galois Theory*** (CRC Press). The standard undergraduate introduction; cyclotomic polynomials and the regular polygon appear early.
- **Harold Davenport, *The Higher Arithmetic*** (Cambridge University Press). Includes roots of unity and their arithmetic.
- **Marcus du Sautoy, *Finding Moonshine*** (Fourth Estate). A popular account of symmetry and group theory, including the story of Galois.

6.2 Gauss's 17-gon

The question

Let $y = \cos\phi + \cos 2\phi$, where $\phi = \dfrac{2\pi}{5}$. Verify by direct substitution that y satisfies the quadratic equation $2y^2 = 3y + 2$ and deduce that the value of y is $-\frac{1}{2}$.

Let $\theta = \dfrac{2\pi}{17}$. Show that

$$\sum_{k=0}^{16} \cos k\theta = 0.$$

If $z = \cos\theta + \cos 2\theta + \cos 4\theta + \cos 8\theta$, show that the value of z is $-(1 - \sqrt{17})/4$.

The hook

On 30 March 1796, an eighteen-year-old student at Göttingen made the first entry in a mathematical diary he would keep for the rest of his life. He had discovered how to construct a regular 17-sided polygon with ruler and compasses. The Greeks had known how to construct regular polygons with 3, 4, 5 and 15 sides, and doubles of these, and for two thousand years nobody had added a single new one. Carl Friedrich Gauss later said that this discovery decided him to become a mathematician rather than a philologist, and he is said to have asked for a 17-gon to be carved on his tombstone.

The heart of the construction is the second half of this question: the sum of four particular cosines equals $\frac{\sqrt{17}-1}{4}$, a number involving only a square root. Lengths built from square roots can be constructed with ruler and compasses, and with three more steps of the same kind Gauss could construct $\cos\frac{2\pi}{17}$ itself.

Before you start

Ask yourself Why does the first part, about $\cos\frac{2\pi}{5}$, appear in a question about the 17-gon?

It is a warm-up, using the same idea on a smaller polygon. For the pentagon, the sum $\cos\phi + \cos 2\phi$ satisfies a quadratic with rational coefficients, so it can be found by solving a quadratic. For the 17-gon, the sum z of four cosines will turn out to satisfy a quadratic too. The trick in both cases is to multiply cosines together and use

$\cos A \cos B = \frac{1}{2}\big(\cos(A+B) + \cos(A-B)\big)$ to turn products back into sums of cosines of the same angles.

Working it through

STAGE 1 The pentagon

Ask yourself Expand $y^2 = (\cos\phi + \cos 2\phi)^2$, turn each product into a sum, and use $5\phi = 2\pi$ to reduce angles like 3ϕ and 4ϕ.

$y^2 = \cos^2\phi + 2\cos\phi\cos 2\phi + \cos^2 2\phi = \frac{1}{2}(1+\cos 2\phi) + (\cos 3\phi + \cos\phi) + \frac{1}{2}(1+\cos 4\phi)$.

Since $5\phi = 2\pi$, $\cos 3\phi = \cos(2\pi - 2\phi) = \cos 2\phi$ and $\cos 4\phi = \cos\phi$. So

$$y^2 = 1 + \tfrac{1}{2}(\cos\phi + \cos 2\phi) + (\cos\phi + \cos 2\phi) = 1 + \tfrac{3}{2}y,$$

that is, $2y^2 = 3y + 2$. ■The roots of $2y^2 - 3y - 2 = (2y+1)(y-2) = 0$ are $y = 2$ and $y = -\frac{1}{2}$. But $y = 2$ would need both cosines to equal 1, which they do not, so $y = -\frac{1}{2}$. ■

STAGE 2 The seventeenth roots of unity sum to zero

$\sum_{k=0}^{16} \cos k\theta$ is the real part of $\sum_{k=0}^{16} \omega^k$, where $\omega = e^{i\theta}$ is a primitive 17th root of unity. That geometric series is $\frac{\omega^{17}-1}{\omega-1} = 0$. ■

So $\sum_{k=1}^{16} \cos k\theta = -1$. Since $\cos(17-k)\theta = \cos k\theta$, the terms pair up, and

$$\sum_{k=1}^{8} \cos k\theta = -\tfrac{1}{2}.$$

STAGE 3 Splitting the cosines into two groups

Ask yourself Let $z = \cos\theta + \cos 2\theta + \cos 4\theta + \cos 8\theta$, and let z' be the sum of the other four cosines, $\cos 3\theta + \cos 5\theta + \cos 6\theta + \cos 7\theta$. What is $z + z'$?

By Stage 2, $z + z' = -\frac{1}{2}$. If we can also find zz', then z and z' are the roots of a quadratic.

Pause and try Compute zz'. Expand it as 16 products $\cos a\theta \cos b\theta$, turn each into a sum of two cosines, and count how often each of $\cos\theta, \cos 2\theta, \ldots, \cos 8\theta$ appears. (Reduce angles using $\cos k\theta = \cos(17-k)\theta$ and $\cos(-k\theta) = \cos k\theta$.)

Each product gives $\cos a\theta \cos b\theta = \frac{1}{2}\big(\cos(a+b)\theta + \cos(a-b)\theta\big)$, so zz' is $\frac{1}{2}$ times a sum of 32 cosines. Working through the 16 pairs with $a \in \{1,2,4,8\}$ and $b \in \{3,5,6,7\}$: for

example $a = 1$, $b = 3$ gives $\cos 4\theta$ and $\cos 2\theta$; $a = 8$, $b = 7$ gives $\cos 15\theta = \cos 2\theta$ and $\cos\theta$. Carrying this out for all sixteen pairs, each of $\cos\theta, \dots, \cos 8\theta$ appears exactly four times. Therefore

$$zz' = \tfrac{1}{2} \times 4 \times \sum_{k=1}^{8} \cos k\theta = 2 \times \left(-\tfrac{1}{2}\right) = -1.$$

STAGE 4 The quadratic

So z and z' are the roots of $t^2 + \frac{1}{2}t - 1 = 0$, that is, $4t^2 + 2t - 4 = 0$, with roots $t = \frac{-1 \pm \sqrt{17}}{4}$. Which is z? Numerically $\cos\theta \approx 0.932$, $\cos 2\theta \approx 0.739$, $\cos 4\theta \approx 0.092$ and $\cos 8\theta \approx -0.983$, so $z \approx 0.78 > 0$. Hence

$$z = \frac{-1 + \sqrt{17}}{4} = -\frac{1 - \sqrt{17}}{4}. \qquad \blacksquare$$

Looking back

- **Products to sums.** The product formula for cosines turned y^2 and zz' into sums of the same cosines, so the algebra closed up.
- **Sum and product give a quadratic.** Knowing $z + z'$ and zz' is the same as knowing the quadratic with roots z and z'.
- **The right split matters.** Splitting the eight cosines into $\{1, 2, 4, 8\}$ and $\{3, 5, 6, 7\}$ made every index appear equally often in zz'. A random split would not work.

What's really going on

Gauss periods

Why $\{1, 2, 4, 8\}$? Modulo 17, the powers $3^0, 3^1, 3^2, \dots$ of 3 run through all the non-zero residues:

$$1, 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6.$$

Taking every other one, $3^0, 3^2, 3^4, \dots$, gives 1, 9, 13, 15, 16, 8, 4, 2, which are ± 1, ± 2, ± 4, ± 8 modulo 17: exactly the angles in z. The other half gives ± 3, ± 5, ± 6, ± 7, the angles in z'. Gauss called the sums of ω^k over such classes *periods*. Because $17 - 1 = 16 = 2^4$, the residues can be halved four times, and at each stage the periods satisfy quadratic equations whose coefficients involve the periods of the previous stage. Four quadratics, so four square roots, lead from the rational numbers down to $\cos\frac{2\pi}{17}$:

$$\cos\tfrac{2\pi}{17} = \tfrac{1}{16}\Big(-1 + \sqrt{17} + \sqrt{34 - 2\sqrt{17}} + 2\sqrt{17 + 3\sqrt{17} - \sqrt{34 - 2\sqrt{17}} - 2\sqrt{34 + 2\sqrt{17}}}\Big).$$

Every step involves only square roots, and square roots can be constructed with ruler and compasses (using similar triangles or a semicircle), so the 17-gon is constructible.

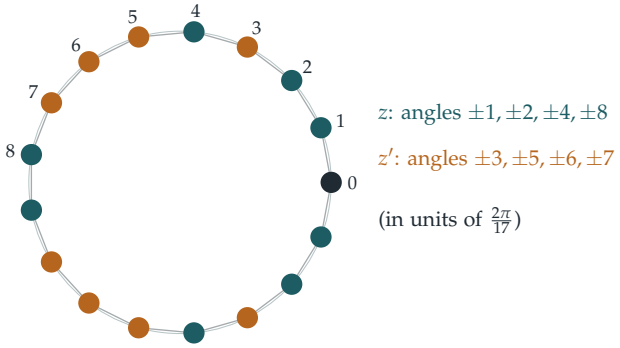


Figure 6.2.1 The vertices of the regular 17-gon, split into Gauss's two periods. The teal vertices are at angles $\pm1, \pm2, \pm4, \pm8$ (in units of $\frac{2\pi}{17}$), and the sum of their x-coordinates is $2z = \frac{-1+\sqrt{17}}{2}$; the amber vertices give $2z' = \frac{-1-\sqrt{17}}{2}$.

Which polygons can be constructed?

Gauss claimed, and Pierre Wantzel proved in 1837, that a regular n-gon can be constructed with ruler and compasses exactly when n is a power of 2 times a product of distinct *Fermat primes*, primes of the form $2^{2^m}+1$. The known Fermat primes are 3, 5, 17, 257 and 65537, the Fermat numbers of Section 1.1. (Fermat guessed that all such numbers are prime, but Euler showed that $2^{32}+1$ is divisible by 641, and no further Fermat primes have ever been found.) So the 257-gon and the 65537-gon are constructible; in the nineteenth century Johann Hermes spent about ten years working out the construction of the 65537-gon, and his manuscript is kept at Göttingen. The 7-gon and 11-gon are not constructible, because 7 and 11 are not Fermat primes ($7-1=6$ and $11-1=10$ are not powers of 2), and neither is the 9-gon, because $9=3^2$ repeats the Fermat prime 3. Wantzel's proof uses the theory of field degrees from Section 6.3.

The thinking skill

Find a quadratic by computing sum and product

To find a quantity, pair it with a natural partner, compute their sum and their product, and solve the resulting quadratic.

The art is in choosing the partner so that both sum and product can be computed. Here the partner came from the symmetry of the residues modulo 17, and repeating the idea four times gave Gauss his construction.

Try this next

1 **The heptagon.** (From STEP II, 1995.) Let $\alpha = e^{2\pi i/7}$. Show that $\eta = \alpha + \alpha^2 + \alpha^4$ and $\eta' = \alpha^3 + \alpha^5 + \alpha^6$ satisfy $\eta + \eta' = -1$ and $\eta\eta' = 2$. Deduce that $\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{8\pi}{7} = -\frac{1}{2}$, and that $\sin\frac{2\pi}{7} + \sin\frac{4\pi}{7} + \sin\frac{8\pi}{7} = \frac{\sqrt{7}}{2}$, explaining how you decide the sign.

2 **The pentagon.** Show that $\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$, and use it to describe a ruler-and-compasses construction of a regular pentagon.

3 **The other period.** Check numerically that $z' = \cos 3\theta + \cos 5\theta + \cos 6\theta + \cos 7\theta = \frac{-1-\sqrt{17}}{4}$.

4 **Generators.** Show that the powers of 3 modulo 17 give every non-zero residue, but the powers of 2 do not. Which residues do the powers of 2 give?

Further reading

- **Ian Stewart, *Galois Theory*** (CRC Press). Chapter on ruler-and-compass constructions and the regular 17-gon.
- **Richard Courant & Herbert Robbins, *What Is Mathematics?*** (Oxford University Press). Chapter 3 is a beautiful elementary account of constructions, including the 17-gon.
- **Daniel Kehlmann, *Measuring the World*** (Quercus). A comic novel about Gauss and Alexander von Humboldt, for light relief.

6.3 A Polynomial for $\sqrt{2}+\sqrt{3}+\sqrt{5}$

The question

(i) Show that, if $(x-\sqrt{2})^2 = 3$, then $x^4 - 10x^2 + 1 = 0$.

Deduce that, if $\mathrm{f}(x) = x^4 - 10x^2 + 1$, then $\mathrm{f}(\sqrt{2}+\sqrt{3}) = 0$.

(ii) Find a polynomial g of degree 8 with integer coefficients such that $\mathrm{g}(\sqrt{2}+\sqrt{3}+\sqrt{5}) = 0$. Write your answer in a form without brackets.

(iii) Let a, b and c be the three roots of $t^3 - 3t + 1 = 0$.

Find a polynomial h of degree 6 with integer coefficients such that $\mathrm{h}(a+\sqrt{2}) = 0$, $\mathrm{h}(b+\sqrt{2}) = 0$ and $\mathrm{h}(c+\sqrt{2}) = 0$. Write your answer in a form without brackets.

(iv) Find a polynomial k with integer coefficients such that $\mathrm{k}(\sqrt[3]{2}+\sqrt[3]{3}) = 0$. Write your answer in a form without brackets.

The hook

$\sqrt{2}$ is a root of $x^2 - 2$, a polynomial with integer coefficients. So is $\sqrt{3}$, of $x^2 - 3$. Is $\sqrt{2}+\sqrt{3}$ also the root of some polynomial with integer coefficients? What about $\sqrt{2}+\sqrt{3}+\sqrt{5}$, or $\sqrt[3]{2}+\sqrt[3]{3}$?

Numbers that are roots of polynomials with integer coefficients are called *algebraic*, and this question shows, by hands-on elimination, that sums of algebraic numbers are algebraic. The degrees that come out, 4, 8, 6, 9, are not accidents. They measure how "complicated" each number is, and that measurement, the *degree of a field extension*, is what finally settled the ancient Greek problems of doubling the cube and trisecting the angle.

Before you start

? **Ask yourself** To find a polynomial with $\sqrt{2}$ as a root, you square $x = \sqrt{2}$ to get rid of the square root. How might you get rid of two different square roots, one at a time?

Isolate one square root and square, which removes it; then isolate the other and square

again. Each squaring may double the degree. That is the whole strategy of the question: eliminate the radicals one at a time.

Working it through

STAGE 1 Two square roots

If $(x-\sqrt{2})^2 = 3$, then $x^2 - 2\sqrt{2}\,x + 2 = 3$, so $x^2 - 1 = 2\sqrt{2}\,x$. Squaring removes the remaining square root: $(x^2-1)^2 = 8x^2$, that is,

$$x^4 - 10x^2 + 1 = 0. \qquad \blacksquare$$

And $x = \sqrt{2}+\sqrt{3}$ satisfies $(x-\sqrt{2})^2 = (\sqrt{3})^2 = 3$, so $\mathrm{f}(\sqrt{2}+\sqrt{3}) = 0$. $\blacksquare$

STAGE 2 Three square roots

Ask yourself Let $x = \sqrt{2}+\sqrt{3}+\sqrt{5}$, so $x-\sqrt{5} = \sqrt{2}+\sqrt{3}$ is a root of f. Write $\mathrm{f}(x-\sqrt{5}) = 0$ in the form $P(x) + \sqrt{5}\,Q(x) = 0$. How do you then eliminate $\sqrt{5}$?

Expand using $(x-\sqrt{5})^2 = x^2 + 5 - 2\sqrt{5}x$ and $(x-\sqrt{5})^4 = \left(x^2+5-2\sqrt{5}x\right)^2 = x^4 + 30x^2 + 25 - 4\sqrt{5}\,x(x^2+5)$:

$$\mathrm{f}(x-\sqrt{5}) = \left(x^4+30x^2+25\right) - 4\sqrt{5}\,x(x^2+5) - 10\left(x^2+5\right) + 20\sqrt{5}\,x + 1.$$

Collecting, $P(x) = x^4 + 20x^2 - 24$ and $Q(x) = -4x^3 - 20x + 20x = -4x^3$. So $x^4 + 20x^2 - 24 = 4\sqrt{5}\,x^3$. Squaring,

$$\left(x^4+20x^2-24\right)^2 = 80x^6,$$

and expanding gives

$$\mathrm{g}(x) = x^8 - 40x^6 + 352x^4 - 960x^2 + 576. \qquad \blacksquare$$

(Check: $(x^4+20x^2-24)^2 = x^8 + 40x^6 + 352x^4 - 960x^2 + 576$, using $400 - 48 = 352$, and subtracting $80x^6$ gives g.)

STAGE 3 A cubic and a square root

Ask yourself If $x = a + \sqrt{2}$, then $x - \sqrt{2}$ is a root of $t^3 - 3t + 1$. Substitute, separate the terms with and without $\sqrt{2}$, and square.

$(x-\sqrt{2})^3 = x^3 - 3\sqrt{2}\,x^2 + 6x - 2\sqrt{2}$, so

$$(x-\sqrt{2})^3 - 3(x-\sqrt{2}) + 1 = \left(x^3+3x+1\right) - \sqrt{2}\left(3x^2-1\right) = 0.$$

Hence $(x^3+3x+1)^2 = 2(3x^2-1)^2$. Expanding, $x^6+6x^4+2x^3+9x^2+6x+1 = 18x^4-12x^2+2$, so

$$\mathrm{h}(x) = x^6-12x^4+2x^3+21x^2+6x-1. \qquad \blacksquare$$

The same argument works for $b+\sqrt{2}$ and $c+\sqrt{2}$, since only the fact that a is a root of t^3-3t+1 was used. (The other three roots of h are $a-\sqrt{2}$, $b-\sqrt{2}$ and $c-\sqrt{2}$, which came in when we squared.)

STAGE 4 Two cube roots

Pause and try Let $x = \sqrt[3]{2}+\sqrt[3]{3}$. Cube both sides. Can you write the result in terms of x and a single cube root?

Using $(u+v)^3 = u^3+v^3+3uv(u+v)$ with $u=\sqrt[3]{2}$, $v=\sqrt[3]{3}$:

$$x^3 = 2+3+3\sqrt[3]{6}\,x, \qquad \text{so} \qquad x^3-5 = 3\sqrt[3]{6}\,x.$$

Cubing again removes the last cube root: $(x^3-5)^3 = 27\cdot 6\,x^3 = 162x^3$. Expanding $(x^3-5)^3 = x^9-15x^6+75x^3-125$,

$$\mathrm{k}(x) = x^9-15x^6-87x^3-125. \qquad \blacksquare$$

Looking back

- **Eliminate one radical at a time.** Isolate a radical, raise to the appropriate power, and repeat.
- **Separate rational and irrational parts.** Writing an expression as $P+\sqrt{5}\,Q$ and then using $P^2 = 5Q^2$ is the key move.
- **Squaring adds extra roots.** Each squaring brings in "conjugate" roots, such as $a-\sqrt{2}$ alongside $a+\sqrt{2}$. The final polynomial has all of them as roots.

What's really going on

Fields and their degrees

The set $\mathbb{Q}(\sqrt{2})$ of all numbers $p+q\sqrt{2}$ with p, q rational is closed under addition, subtraction, multiplication and division (by non-zero numbers): it is a *field*. It is a two-dimensional "vector space" over $\mathbb{Q}$, with basis $1, \sqrt{2}$, so we say it has *degree* 2 over $\mathbb{Q}$. Adjoining $\sqrt{3}$ gives $\mathbb{Q}(\sqrt{2},\sqrt{3})$, with basis $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$ and degree 4; adjoining $\sqrt{5}$ as well gives degree 8 (Figure 6.3.1). The *tower law* says degrees multiply up a tower of fields. The degree of the field generated by a number equals the degree of its *minimal polynomial*, the lowest-degree polynomial with rational coefficients that it

satisfies. One can show that the polynomials f and g in this question are minimal, so $\sqrt{2}+\sqrt{3}$ has degree 4 and $\sqrt{2}+\sqrt{3}+\sqrt{5}$ has degree 8: in fact the single number $\sqrt{2}+\sqrt{3}+\sqrt{5}$ generates the whole degree-8 field.

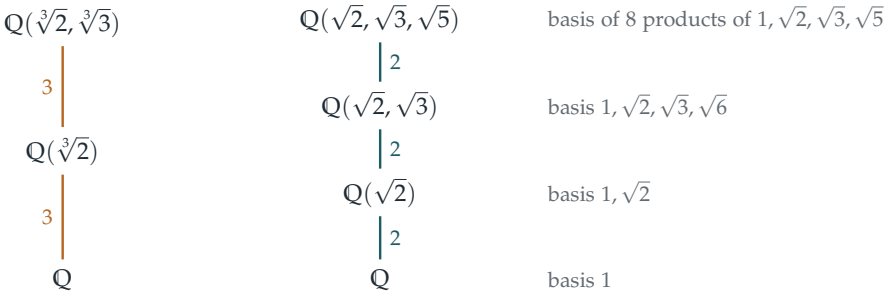


Figure 6.3.1 Towers of fields. Right: each square root doubles the degree, giving $2\times2\times2=8$, the degree of g. Left: each cube root triples it, giving $3\times3=9$, the degree of k. Degrees multiply up a tower; this *tower law* is the key to the impossibility proofs for ruler-and-compass constructions.

Doubling the cube

Ruler-and-compass constructions can only produce lengths obtained from rational numbers by repeatedly solving linear and quadratic equations. Each such step gives a field extension of degree 1 or 2, so by the tower law every constructible length lies in a field of degree a power of 2, and has degree a power of 2 itself. But $\sqrt[3]{2}$ has minimal polynomial x^3-2, of degree 3. Since 3 is not a power of 2, $\sqrt[3]{2}$ is not constructible, and the Delian problem of doubling the volume of a cube with ruler and compasses, posed around 430 BC, is impossible. Pierre Wantzel proved this in 1837, together with the impossibility of trisecting a general angle and the classification of constructible polygons (Section 6.2). Two thousand years of attempts were ended by an argument about degrees.

Algebraic numbers form a field

The method of this question shows, in examples, that sums of algebraic numbers are algebraic. In general this is proved with linear algebra: if α and β have degrees m and n, then all the numbers $\alpha^i\beta^j$ (with $i<m$, $j<n$) span a space of dimension at most mn, and the $mn+1$ powers $1,\alpha+\beta,(\alpha+\beta)^2,\ldots$ must be linearly dependent, which gives a polynomial of degree at most mn. Our degrees $4=2\times2$, $6=3\times2$ and $9=3\times3$ fit this pattern exactly. Numbers that are *not* algebraic are called *transcendental*; e and π are examples (Hermite, 1873, and Lindemann, 1882), and Lindemann's theorem that π is transcendental proved that squaring the circle is impossible too.

The thinking skill

Eliminate, then measure

To find a polynomial satisfied by an expression in radicals, remove the radicals one at a time by isolating and raising to a power.

Then ask what the degree tells you. The degree of the minimal polynomial is an invariant of the number, and comparing degrees is one of the most powerful ways to prove that something is impossible.

Try this next

1 **A primitive element.** Let $y = \sqrt{2}+\sqrt{3}$. Show that $y^3 = 11\sqrt{2}+9\sqrt{3}$, and deduce that $\sqrt{2} = \frac{1}{2}(y^3 - 9y)$. Hence every number $p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6}$ can be written as a polynomial in y.

2 **Mixed roots.** Find a polynomial of degree 6 with integer coefficients that has $\sqrt{2} + \sqrt[3]{3}$ as a root.

3 **Irreducibility.** Show that $x^4 - 10x^2 + 1$ has no rational roots and cannot be written as a product of two quadratics with integer coefficients. (*Hint:* try $(x^2 + ax + b)(x^2 - ax + c)$.) Deduce that $\sqrt{2}+\sqrt{3}$ has degree 4.

4 **Trisection.** Show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, and deduce that $\cos 20^\circ$ is a root of $8x^3 - 6x - 1 = 0$. Assuming this cubic has no rational roots (check this), explain why a 60° angle cannot be trisected with ruler and compasses.

Further reading

- **Ian Stewart, *Galois Theory*** (CRC Press). Field extensions, the tower law and the classical construction problems, written for undergraduates.
- **Richard Courant & Herbert Robbins, *What Is Mathematics?*** (Oxford University Press). Chapter 3 proves the impossibility of the Greek problems with no prerequisites.
- **Robin Hartshorne, *Geometry: Euclid and Beyond*** (Springer). Ruler-and-compass constructions treated with modern algebra.

6.4 Solving the Quartic

The question

Let α, β, γ and δ be the roots of the quartic equation

$$x^4 + px^3 + qx^2 + rx + s = 0.$$

You are given that, for any such equation, $\alpha\beta + \gamma\delta$, $\alpha\gamma + \beta\delta$ and $\alpha\delta + \beta\gamma$ satisfy a cubic equation of the form

$$y^3 + Ay^2 + (pr - 4s)y + (4qs - p^2s - r^2) = 0.$$

Determine A.

Now consider the quartic equation given by $p = 0$, $q = 3$, $r = -6$ and $s = 10$.

(i) Find the value of $\alpha\beta + \gamma\delta$, given that it is the largest root of the corresponding cubic equation.

(ii) Hence, using the values of q and s, find the value of $(\alpha + \beta)(\gamma + \delta)$ and the value of $\alpha\beta$ given that $\alpha\beta > \gamma\delta$.

(iii) Using these results, and the values of p and r, solve the quartic equation.

The hook

Every student learns the formula for solving quadratic equations. Formulas for cubics and quartics exist too; they were found in sixteenth-century Italy amid secret contests and bitter disputes over priority. Gerolamo Cardano published both in his *Ars Magna* of 1545, the cubic from Niccolò Tartaglia (who had sworn him to secrecy) and the quartic from his own student Lodovico Ferrari. Then, for nearly three hundred years, mathematicians searched in vain for a formula for the quintic.

This question shows the trick that solves the quartic, and it is a trick with a deep meaning. Out of the four roots, form three new quantities by pairing the roots up in the three possible ways. Those three quantities satisfy a cubic, which you can solve, and from them the original roots follow. The question *why* this works, why four roots can be reduced to three quantities but five roots cannot be similarly reduced, is what led Galois to invent group theory.

Before you start

Ask yourself How many ways are there to split four objects $\alpha, \beta, \gamma, \delta$ into two pairs?

Three: $\{\alpha\beta, \gamma\delta\}$, $\{\alpha\gamma, \beta\delta\}$ and $\{\alpha\delta, \beta\gamma\}$ (Figure 6.4.1). The three quantities in the question, $\alpha\beta + \gamma\delta$ and so on, correspond to these three pairings. Any rearrangement of the four roots just shuffles the three pairings among themselves, so any symmetric expression in the three quantities is a symmetric expression in the four roots, and can be written in terms of p, q, r, s. That is why they satisfy a cubic whose coefficients involve only p, q, r, s.

Working it through

STAGE 1 The missing coefficient

Ask yourself In a cubic $y^3 + Ay^2 + By + C = 0$, what is the sum of the roots in terms of A? And what is $(\alpha\beta + \gamma\delta) + (\alpha\gamma + \beta\delta) + (\alpha\delta + \beta\gamma)$ in terms of the quartic's coefficients?

The sum of the roots of the cubic is $-A$. The sum of the three quantities is the sum of all six products of pairs of $\alpha, \beta, \gamma, \delta$, which for the quartic $x^4 + px^3 + qx^2 + rx + s$ is q. So $A = -q$.

STAGE 2 The specific cubic

With $p = 0$, $q = 3$, $r = -6$, $s = 10$, the cubic is

$$y^3 - 3y^2 + (0 - 40)y + (120 - 0 - 36) = y^3 - 3y^2 - 40y + 84 = 0.$$

Ask yourself Try small divisors of 84.

$y = 2$ gives $8 - 12 - 80 + 84 = 0$. Dividing, $y^3 - 3y^2 - 40y + 84 = (y - 2)(y^2 - y - 42) = (y - 2)(y - 7)(y + 6)$. The largest root is 7, so $\alpha\beta + \gamma\delta = 7$. ■

STAGE 3 Pairs of roots

$(\alpha + \beta)(\gamma + \delta) = \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta$, which is the sum of all six pair-products minus $\alpha\beta + \gamma\delta$, that is, $q - 7 = 3 - 7 = -4$.

Also $\alpha\beta \cdot \gamma\delta = \alpha\beta\gamma\delta = s = 10$. So $\alpha\beta$ and $\gamma\delta$ have sum 7 and product 10: they are the roots of $t^2 - 7t + 10 = (t - 2)(t - 5)$. With $\alpha\beta > \gamma\delta$, $\alpha\beta = 5$ and $\gamma\delta = 2$. ■

STAGE 4 Solving the quartic

Pause and try Find $\alpha+\beta$ and $\gamma+\delta$. You know their sum (from p) and their product (from Stage 3); to decide which is which, use r.

$\alpha+\beta+\gamma+\delta=-p=0$ and $(\alpha+\beta)(\gamma+\delta)=-4$, so $\{\alpha+\beta,\gamma+\delta\}=\{2,-2\}$. The coefficient r gives the sum of the four triple products: $\alpha\beta\gamma+\alpha\beta\delta+\alpha\gamma\delta+\beta\gamma\delta=-r=6$. This can be written as $\alpha\beta(\gamma+\delta)+\gamma\delta(\alpha+\beta)=5(\gamma+\delta)+2(\alpha+\beta)$. If $\alpha+\beta=2$ this is $-10+4=-6$; if $\alpha+\beta=-2$ it is $10-4=6$. So $\alpha+\beta=-2$ and $\gamma+\delta=2$.

Now α, β are the roots of $t^2+2t+5=0$, giving $-1\pm2\mathrm{i}$; and γ, δ are the roots of $t^2-2t+2=0$, giving $1\pm\mathrm{i}$. The four roots are

$$x=-1\pm2\mathrm{i}, \qquad x=1\pm\mathrm{i}. \qquad \blacksquare$$

Check: $(x^2+2x+5)(x^2-2x+2)=x^4+3x^2-6x+10$.

Looking back

- **Symmetric functions.** Every step used the relations between roots and coefficients: sums and products of roots.
- **Reduce the degree.** A quartic problem became a cubic (the *resolvent cubic*), and then two quadratics.
- **Use every piece of information.** p and q gave the sums and products; s split $\alpha\beta$ from $\gamma\delta$; r decided which pair went with which sum.

What's really going on

Why the trick works: permutations of roots

In 1770 Joseph-Louis Lagrange asked why the methods for cubics and quartics work, and found the answer in the *permutations* of the roots. There are 24 ways to rearrange four roots. Under all of them, the three quantities $y_1=\alpha\beta+\gamma\delta$, $y_2=\alpha\gamma+\beta\delta$, $y_3=\alpha\delta+\beta\gamma$ are merely shuffled among themselves: each rearrangement of the four roots induces a rearrangement of the three pairings (Figure 6.4.1). So they satisfy a polynomial of degree 3 with coefficients symmetric in the roots. In modern language, there is a group homomorphism from the symmetric group S_4 (all 24 permutations) onto S_3 (the 6 permutations of the pairings), and that is why a quartic can be reduced to a cubic.

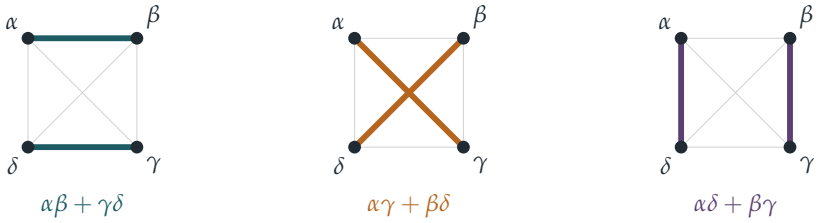


Figure 6.4.1 The three ways to split four roots into two pairs, and the corresponding roots of the resolvent cubic. Any rearrangement of $\alpha, \beta, \gamma, \delta$ permutes these three pictures among themselves, which is why the three quantities satisfy a cubic with coefficients expressible in terms of the quartic's.

The quintic

For five roots, Lagrange looked for a similar quantity taking only a few values under permutations of the roots, and could not find one that helped. In 1799 Paolo Ruffini, and in 1824 Niels Henrik Abel (completing the proof), showed that there is no general formula for the roots of a quintic using only arithmetic and radicals. A few years later Évariste Galois explained exactly which polynomials can be solved by radicals: those whose *group of symmetries of the roots* can be broken down in a particular way into simpler pieces. The symmetric group S_4 can be (via S_3 and smaller groups), but S_5 cannot, because it contains a "simple" piece of size 60 that cannot be broken down further. Galois wrote up his theory in a hurry the night before he was killed in a duel in 1832, at the age of twenty. It is now one of the jewels of a mathematics degree, usually met in the third year.

Solving cubics

The cubic itself can be solved by a similar trick: write a depressed cubic $x^3 - 3px + q$ as a combination of two perfect cubes, which reduces it to a quadratic. That is the method of the 2011 question in the exercises, and it is equivalent to Cardano's formula. When the cubic has three real roots, the formula still requires complex numbers along the way, the phenomenon that forced mathematicians to take complex numbers seriously (see the introduction to Chapter 5).

The thinking skill

Symmetry reduces complexity

If some quantities built from the roots are only permuted among themselves when the roots are permuted, they satisfy a polynomial of lower degree whose coefficients you can compute.

This is the seed of Galois theory: understand an equation by understanding the symmetries of its roots.

Try this next

1 **Check the cubic.** Verify that $y_1y_2 + y_2y_3 + y_3y_1 = pr - 4s$, where $y_1 = \alpha\beta + \gamma\delta$ etc. (Expand and use $\sum\alpha = -p$, $\sum\alpha\beta\gamma = -r$, $\alpha\beta\gamma\delta = s$.)

2 **The cubic as a sum of cubes.** (From STEP III, 2011.) Show that if $p \neq 0$ and $q^2 \neq 4p^3$ then $x^3 - 3px + q = a(x-\alpha)^3 + b(x-\beta)^3$, where α, β are the roots of $pt^2 - qt + p^2 = 0$. Hence show that one root of $x^3 - 24x + 48 = 0$ is $x = \dfrac{2(2-2^{1/3})}{1-2^{1/3}}$, and check this numerically.

3 **Another quartic.** Use the method of this section to solve $x^4 - 4x^3 + 7x^2 - 6x + 2 = 0$. (One root of the resolvent cubic is a small integer.)

4 **Counting permutations.** Of the 24 permutations of $\alpha, \beta, \gamma, \delta$, find all those that send the pairing $\{\alpha\beta, \gamma\delta\}$ to itself. Show that there are 8 of them. (They form a subgroup of S_4.)

Further reading

- **Mario Livio, *The Equation That Couldn't Be Solved*** (Simon & Schuster). The story of the quintic and the birth of group theory, for a general reader.
- **Ian Stewart, *Why Beauty Is Truth: A History of Symmetry*** (Basic Books). From Babylonian equations to Galois and beyond.
- **Ian Stewart, *Galois Theory*** (CRC Press). The undergraduate text, with a historical introduction covering exactly this material.

6.5 Differentiation from Axioms

The question

$\Diamond$ is an operation which takes polynomials in x to polynomials in x; that is, given a polynomial $\mathrm{h}(x)$ there is another polynomial called $\Diamond\mathrm{h}(x)$. It is given that, if $\mathrm{f}(x)$ and $\mathrm{g}(x)$ are any two polynomials in x, the following are always true:

(i) $\Diamond\big(\mathrm{f}(x)\mathrm{g}(x)\big) = \mathrm{g}(x)\Diamond\mathrm{f}(x) + \mathrm{f}(x)\Diamond\mathrm{g}(x)$,

(ii) $\Diamond\big(\mathrm{f}(x) + \mathrm{g}(x)\big) = \Diamond\mathrm{f}(x) + \Diamond\mathrm{g}(x)$,

(iii) $\Diamond x = 1$,

(iv) if λ is a constant then $\Diamond\big(\lambda\mathrm{f}(x)\big) = \lambda\Diamond\mathrm{f}(x)$.

Show that, if $\mathrm{f}(x)$ is a constant (i.e., a polynomial of degree zero), then $\Diamond\mathrm{f}(x) = 0$.

Calculate $\Diamond x^2$ and $\Diamond x^3$. Prove that $\Diamond\mathrm{h}(x) = \dfrac{\mathrm{d}}{\mathrm{d}x}\big(\mathrm{h}(x)\big)$ for any polynomial $\mathrm{h}(x)$.

The hook

What is differentiation? At school it is defined by a limit, $\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$, and then you learn rules: the derivative of a sum is the sum of the derivatives, the product rule, and so on. This question turns that upside down. Forget limits entirely. Suppose all you know about an operation is that it obeys the sum rule, the product rule and the rule for constants, and that it sends x to 1. Then, the question shows, it *must* be differentiation, at least on polynomials.

This is the algebraist's view of calculus: an operation is characterised by the rules it obeys, not by a formula. Operations obeying the product rule are called *derivations*, and they turn up throughout algebra, geometry and physics. Surprisingly, the same idea is at work inside the software that trains modern machine-learning systems.

Before you start

Ask yourself Rule (i) is the product rule. How could the product rule tell you anything about a constant? (Think of a constant as a product.)

The number 1 equals 1×1. Applying the product rule to 1×1 relates $\Diamond 1$ to itself, and that will force it to be zero. In general, the trick with axioms like these is to apply them

to cleverly chosen simple inputs.

Working it through

STAGE 1 Constants

Ask yourself Apply rule (i) with $f = g = 1$. Then use rule (iv) for any other constant.

$\Diamond 1 = \Diamond(1 \cdot 1) = 1 \cdot \Diamond 1 + 1 \cdot \Diamond 1 = 2\Diamond 1$, so $\Diamond 1 = 0$. For any constant λ, rule (iv) gives $\Diamond \lambda = \Diamond(\lambda \cdot 1) = \lambda \Diamond 1 = 0$. ■

STAGE 2 Powers

$\Diamond x^2 = \Diamond(x \cdot x) = x\Diamond x + x\Diamond x = 2x$, using rule (iii). Then

$$\Diamond x^3 = \Diamond(x^2 \cdot x) = x\Diamond x^2 + x^2 \Diamond x = x \cdot 2x + x^2 = 3x^2.$$

Pause and try Prove by induction that $\Diamond x^n = nx^{n-1}$ for every positive integer n.

The case $n = 1$ is rule (iii). If $\Diamond x^n = nx^{n-1}$, then

$$\Diamond x^{n+1} = \Diamond(x^n \cdot x) = x\Diamond x^n + x^n \Diamond x = x \cdot nx^{n-1} + x^n = (n+1)x^n. \quad ■$$

STAGE 3 All polynomials

A general polynomial is $h(x) = a_0 + a_1 x + \cdots + a_m x^m$. By rule (ii) (applied repeatedly) and rule (iv),

$$\begin{aligned} \Diamond h(x) &= \Diamond a_0 + a_1 \Diamond x + a_2 \Diamond x^2 + \cdots + a_m \Diamond x^m \\ &= a_1 + 2a_2 x + \cdots + m a_m x^{m-1} = h'(x). \end{aligned} \quad ■$$

Looking back

- **Axioms determine the operation.** Four rules, with no mention of limits, pinned down $\Diamond$ completely on polynomials.
- **Generators.** Every polynomial is built from constants and x using addition and multiplication. Knowing what $\Diamond$ does to the building blocks, and how it interacts with the building operations, determines it everywhere.
- **Induction along the construction.** The proof followed the way polynomials are built: powers by repeated multiplication, then sums.

What's really going on

Derivations

An operation D on an algebraic system that is additive and satisfies the product rule $D(fg) = fD(g) + gD(f)$ is called a *derivation*. The question shows that on polynomials, a derivation that kills constants is completely determined by $D(x)$: in fact $D(h) = h'(x)\,D(x)$, so the only freedom is the choice of the polynomial $D(x)$. Rule (iii) fixes $D(x) = 1$. Derivations appear across mathematics: in geometry, a *vector field* on a surface can be defined as a derivation on the functions on that surface (it differentiates functions in the direction of the field); in algebra, derivations of a structure measure its "infinitesimal symmetries"; in physics, the commutator $[A, B] = AB - BA$ of quantum mechanics is a derivation in each argument.

Calculus without limits

The algebraic derivative works in settings where limits make no sense at all. Over a finite field, such as arithmetic modulo a prime p, there are no limits, but the formal derivative of a polynomial is defined by the same rules and still detects repeated roots: a polynomial f has a repeated root exactly when f and f' have a common factor. That is how one shows, for example, that $x^n - 1$ has no repeated roots when $p \nmid n$, a fact needed to define the cyclotomic polynomials of Section 6.1 over finite fields. The same idea is used in coding theory and cryptography.

Automatic differentiation

Here is a modern application. Invent a number ε with $\varepsilon^2 = 0$ but $\varepsilon \neq 0$, and do arithmetic with *dual numbers* $a + b\varepsilon$. Then $(a + b\varepsilon)(c + d\varepsilon) = ac + (ad + bc)\varepsilon$: the ε-part follows exactly the product rule. So for any polynomial,

$$f(x + \varepsilon) = f(x) + f'(x)\,\varepsilon,$$

and evaluating f at the dual number $x + \varepsilon$ computes the value and the derivative at the same time, exactly, with no limits and no rounding error from small differences (Figure 6.5.1). This is called *forward-mode automatic differentiation*. Its cousin, reverse mode, is the algorithm (*backpropagation*) used to train neural networks with billions of parameters: every time such a system learns, it is applying the product and chain rules, as derivations, step by step.

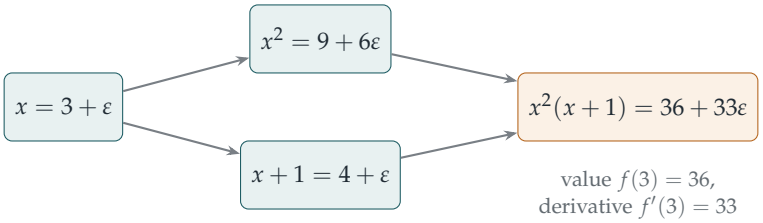


Figure 6.5.1 Automatic differentiation with dual numbers ($\varepsilon^2 = 0$) for $f(x) = x^2(x+1)$ at $x = 3$. Each step follows the rules of the question: $(3+\varepsilon)^2 = 9 + 6\varepsilon$, and $(9+6\varepsilon)(4+\varepsilon) = 36 + (9+24)\varepsilon$. The coefficient of ε is $f'(3) = 3 \cdot 9 + 2 \cdot 3 = 33$, computed exactly.

The thinking skill

Characterise by properties

Describe an operation by the rules it obeys, then show that the rules determine it uniquely.

This is the axiomatic method, central to university algebra. Its power is that anything you prove from the axioms applies to every operation that satisfies them, whether it is differentiation of polynomials, of functions on a surface, or something not yet invented.

Try this next

1. **A different derivation.** Suppose $\Diamond$ satisfies rules (i), (ii) and (iv), but $\Diamond x = x$ instead of 1. Show that $\Diamond \mathrm{h}(x) = x\,\mathrm{h}'(x)$. What does $\Diamond$ do to x^n?

2. **The quotient rule.** Extend $\Diamond$ to rational functions f/g, assuming rule (i) still holds. By applying rule (i) to $\mathrm{g} \cdot \frac{\mathrm{f}}{\mathrm{g}} = \mathrm{f}$, derive the quotient rule.

3. **Repeated roots.** Show that if $(x-a)^2$ divides a polynomial f, then $x - a$ divides f'. Use this to show that $x^n - 1$ has no repeated roots.

4. **Dual numbers.** Use dual numbers to compute $f(2)$ and $f'(2)$ for $f(x) = (x^2+1)^3$, and check with the chain rule.

Further reading

- **Peter Cameron, *Introduction to Algebra*** (Oxford University Press). Rings, polynomials and fields at first-year level, with the formal derivative.
- **Andreas Griewank & Andrea Walther, *Evaluating Derivatives*** (SIAM). The standard

reference on automatic differentiation. Advanced.

- **Michael Nielsen, *Neural Networks and Deep Learning*** (free online). Chapter 2 explains backpropagation clearly.

6.6 Unions of Subgroups

The question

Give a careful argument to show that, if G_1 and G_2 are subgroups of a finite group G such that every element of G is either in G_1 or in G_2, then either $G_1 = G$ or $G_2 = G$.

Give an example of a group H which has three subgroups H_1, H_2 and H_3 such that every element of H is either in H_1, H_2 or H_3 and $H_1 \neq H$, $H_2 \neq H$, $H_3 \neq H$.

The hook

This is one of the shortest questions in the book, and one of the most characteristic of university mathematics. It is about *groups*, the mathematical language of symmetry, and it asks something that sounds like a puzzle: can a group be covered by two smaller groups sitting inside it? The answer is no, and the proof is three lines long. But three smaller groups can do it, which is a surprise.

Group theory was on the STEP syllabus until 2019 and is now a staple of the first year at university. This question needs only the definitions, which are given below, and it is a perfect first taste of how group theorists think: a short, clever argument that uses nothing but the axioms.

Before you start

A *group* is a set G with an operation (written as multiplication, ab) such that:

- the operation is *associative*: $(ab)c = a(bc)$;
- there is an *identity* element e with $ea = ae = a$ for every a;
- every a has an *inverse* a^{-1} with $aa^{-1} = a^{-1}a = e$.

Examples: the integers (positive, negative and zero) under addition; the non-zero real numbers under multiplication; the rotations of a square under composition. A *subgroup* is a subset K of G that is a group in its own right under the same operation: it contains e, and if $a, b \in K$ then $ab \in K$ and $a^{-1} \in K$.

? **Ask yourself** If K is a subgroup and $a \in K$, but $b \notin K$, can ab be in K?

No. If $ab \in K$, then $b = a^{-1}(ab)$ would be a product of two elements of K, and so in K.

This tiny observation is the whole proof.

Working it through

STAGE 1 Two subgroups cannot cover

Suppose, for a contradiction, that $G = G_1 \cup G_2$ but neither G_1 nor G_2 is all of G.

Ask yourself Since $G_1 \neq G$, there is an element not in G_1. Where must it be? Similarly for G_2. Now multiply the two elements together.

Choose $a \in G$ with $a \notin G_1$; since every element is in G_1 or G_2, $a \in G_2$. Choose $b \in G$ with $b \notin G_2$; then $b \in G_1$. Consider the product ab. It lies in G_1 or in G_2.

- If $ab \in G_1$: since $b \in G_1$, $a = (ab)b^{-1} \in G_1$. Contradiction.
- If $ab \in G_2$: since $a \in G_2$, $b = a^{-1}(ab) \in G_2$. Contradiction.

Either way we reach a contradiction, so $G_1 = G$ or $G_2 = G$. ■

Notice that we never used the fact that G is finite. The result holds for every group.

STAGE 2 Three subgroups can

Pause and try Find a group that is the union of three subgroups, none of them the whole group. (*Hint:* try a small group in which every element is its own inverse.)

Take the *Klein four-group* $V = \{e, a, b, c\}$, with the rules $a^2 = b^2 = c^2 = e$ and $ab = ba = c$, $bc = cb = a$, $ca = ac = b$. (Concretely: the four symmetries of a non-square rectangle, namely doing nothing, rotating by $180°$, and reflecting in either axis. Or the pairs (x, y) of bits 0, 1 under addition modulo 2.) Its three subgroups

$$H_1 = \{e, a\}, \qquad H_2 = \{e, b\}, \qquad H_3 = \{e, c\}$$

each have two elements, none is the whole group, and together they contain every element of V (Figure 6.6.1). ■

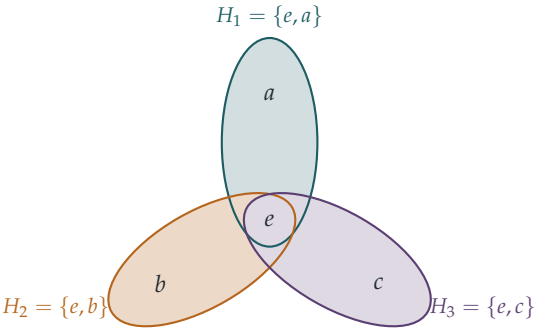


Figure 6.6.1 The Klein four-group $\{e, a, b, c\}$ is the union of its three subgroups of order 2, which overlap only in the identity e. By the first part of the question, no group can be the union of just two proper subgroups.

Looking back

- **Pick witnesses and combine them.** The proof chose an element outside each subgroup and multiplied them. This "take one from each and combine" move is extremely common in algebra.
- **Closure is powerful.** The only property of subgroups used was that they are closed under multiplication and inverses.
- **Check the hypotheses you need.** Finiteness was given but not used. Noticing that a hypothesis is unnecessary is part of understanding a proof.

What's really going on

The language of symmetry

Groups arose in the nineteenth century from several directions at once: Galois's study of permutations of the roots of polynomials (Section 6.4), Cauchy's work on permutations, and the study of symmetries in geometry and crystallography. The abstract definition above, in which a group is anything satisfying the three axioms, was settled by the end of the century. Its power is generality: a theorem proved from the axioms, like the one in this question, holds simultaneously for groups of numbers, of matrices, of symmetries and of permutations. Today groups are everywhere: in particle physics (the standard model is built on symmetry groups), in chemistry (molecular symmetry determines spectra), in cryptography (elliptic curve groups, Section 1.9) and in the solution of Rubik's cube.

How many subgroups does it take?

The question suggests a natural problem: what is the smallest number of proper subgroups whose union is the whole group? It is never 2, by the question. In 1926 Gaetano Scorza showed that a group is the union of three proper subgroups exactly when it has the Klein four-group as a "quotient". In 1994 J. H. E. Cohn studied the general problem and conjectured several results, and in 1997 M. J. Tomkinson proved that there is no group for which this smallest number is seven, although every number from 3 to 6 occurs. Questions like this, simple to state and surprisingly subtle, are typical of group theory.

Vector spaces

A similar argument shows that a vector space over the real numbers (such as the plane, or three-dimensional space) can never be the union of finitely many proper subspaces: a plane is not the union of finitely many lines through the origin. But over a finite field it can be: the plane over the field with two elements, which is just the Klein four-group again, is the union of its three lines.

The thinking skill

Argue from the axioms

In abstract algebra, you often know nothing about the objects except the axioms they satisfy. The proof must use only those axioms.

This forces arguments to be short and clean, like the one here, and it makes them universally applicable. It takes some getting used to, and it is one of the most distinctive experiences of a first-year algebra course.

Try this next

1. **Check the group.** Write out the multiplication table of the Klein four-group and check the group axioms. Show that it is not the same as the group $\{1, \mathrm{i}, -1, -\mathrm{i}\}$ under multiplication. (*Hint:* which elements are their own inverses?)

2. **Cyclic groups.** A group is *cyclic* if every element is a power of a single element g. Show that a cyclic group is never the union of proper subgroups, however many. (*Hint:* which subgroup contains g?)

3. **The whole numbers.** Show directly that $(\mathbb{Z}, +)$ is not the union of the subgroups $2\mathbb{Z}$ (even numbers) and $3\mathbb{Z}$ (multiples of 3). Is it the union of any two proper subgroups?

4 Lines in the plane. HARDER Show that the plane $\mathbb{R}^2$ is not the union of finitely many lines through the origin. (*Hint:* each line contains at most one point of the parabola $y = x^2$ other than the origin.)

Further reading

- **M. A. Armstrong, *Groups and Symmetry*** (Springer). A beautifully illustrated first course, driven by geometric examples.
- **Nathan Carter, *Visual Group Theory*** (Mathematical Association of America). Groups through pictures, ideal before university.
- **Peter Neumann, Gabrielle Stoy & Edward Thompson, *Groups and Geometry*** (Oxford University Press). A classic Oxford first-year text.

6.7 The Quaternion Group

The question

The set S consists of ordered pairs of complex numbers (z_1, z_2) and a binary operation $\circ$ on S is defined by

$$(z_1, z_2) \circ (w_1, w_2) = \left(z_1 w_1 - z_2 w_2^*,\ z_1 w_2 + z_2 w_1^*\right).$$

Show that the operation $\circ$ is associative and determine whether it is commutative. Evaluate $(z, 0) \circ (w, 0)$, $(z, 0) \circ (0, w)$, $(0, z) \circ (w, 0)$ and $(0, z) \circ (0, w)$.

The set S_1 is the subset of S consisting of $A, B, \ldots, H$, where $A = (1, 0)$, $B = (0, 1)$, $C = (\mathrm{i}, 0)$, $D = (0, \mathrm{i})$, $E = (-1, 0)$, $F = (0, -1)$, $G = (-\mathrm{i}, 0)$ and $H = (0, -\mathrm{i})$. Show that S_1 is closed under $\circ$ and that it has an identity element. Determine the inverse and order of each element of S_1. Show that S_1 is a group under $\circ$. [You are not required to compute the multiplication table in full.]

Show that $\{A, B, E, F\}$ is a subgroup of S_1 and determine whether it is isomorphic to the group generated by the 2×2 matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ under matrix multiplication.

The hook

On 16 October 1843, walking along the Royal Canal in Dublin with his wife, William Rowan Hamilton had a flash of inspiration that he had been chasing for years. He had been trying to multiply triples of numbers, the way complex numbers multiply pairs, and failing. The answer, he suddenly saw, needed *four* numbers, and a multiplication in which the order matters: $ij \neq ji$. He carved the rules into the stone of Brougham Bridge:

$$\mathrm{i}^2 = \mathrm{j}^2 = \mathrm{k}^2 = \mathrm{ijk} = -1.$$

These are the *quaternions*. The question builds them from pairs of complex numbers, and finds inside them a small group of eight elements with a twisted multiplication. Quaternions are now used to describe rotations in three dimensions in every computer game, flight simulator and spacecraft guidance system.

Before you start

For the group definitions, see Section 6.6. Two more words: the *order* of an element g of a group is the smallest positive n with $g^n = e$; and two groups are *isomorphic* if there is a one-to-one correspondence between them that respects the operations, so that they

are "the same group with different labels".

Ask yourself Think of the pair (z_1, z_2) as standing for a "number" $z_1 + z_2\mathrm{j}$, where j is a new symbol with $\mathrm{j}^2 = -1$ and $\mathrm{j}w = w^*\mathrm{j}$ for complex w. Multiply $(z_1 + z_2\mathrm{j})(w_1 + w_2\mathrm{j})$ using these rules. Do you get the formula for $\circ$?

$(z_1 + z_2\mathrm{j})(w_1 + w_2\mathrm{j}) = z_1w_1 + z_1w_2\mathrm{j} + z_2\mathrm{j}w_1 + z_2\mathrm{j}w_2\mathrm{j} = z_1w_1 + z_1w_2\mathrm{j} + z_2w_1^*\mathrm{j} + z_2w_2^*\mathrm{j}^2$, which is $(z_1w_1 - z_2w_2^*) + (z_1w_2 + z_2w_1^*)\mathrm{j}$. Yes: the operation $\circ$ is quaternion multiplication in disguise.

Working it through

STAGE 1 Associativity via matrices

Checking $(a \circ b) \circ c = a \circ (b \circ c)$ directly is a long calculation. There is a slicker way.

Ask yourself Associate to (z_1, z_2) the complex matrix $M = \begin{pmatrix} z_1 & z_2 \\ -z_2^* & z_1^* \end{pmatrix}$. What is the top row of the matrix product $M(z_1, z_2)\, M(w_1, w_2)$?

The top row of the product is $\left(z_1w_1 - z_2w_2^*,\ z_1w_2 + z_2w_1^*\right)$, exactly $(z_1, z_2) \circ (w_1, w_2)$, and the bottom row is $\left(-(z_1w_2 + z_2w_1^*)^*,\ (z_1w_1 - z_2w_2^*)^*\right)$, so the product has the same special form. So $M(a \circ b) = M(a)M(b)$. Matrix multiplication is associative, and M is one-to-one (you can read off z_1, z_2 from the top row), so $\circ$ is associative. ■

STAGE 2 Not commutative

$(0,1) \circ (\mathrm{i}, 0) = (0 \cdot \mathrm{i} - 1 \cdot 0,\ 0 \cdot 0 + 1 \cdot \mathrm{i}^*) = (0, -\mathrm{i})$, but $(\mathrm{i}, 0) \circ (0, 1) = (0, \mathrm{i})$. So $\circ$ is not commutative.

The four special products follow directly from the definition:

$$(z,0) \circ (w,0) = (zw, 0), \qquad (z,0) \circ (0,w) = (0, zw),$$
$$(0,z) \circ (w,0) = (0, zw^*), \qquad (0,z) \circ (0,w) = (-zw^*, 0).$$

STAGE 3 The group of eight

In quaternion notation, $A = 1$, $C = \mathrm{i}$, $B = \mathrm{j}$, $D = \mathrm{ij} = \mathrm{k}$, and E, G, F, H are their negatives. Every element of S_1 has the form $(u, 0)$ or $(0, u)$ with $u \in \{1, \mathrm{i}, -1, -\mathrm{i}\}$.

Ask yourself Use the four special products to show that S_1 is closed. What is the identity?

By the four special products, the product of two elements of the form $(u, 0)$ or $(0, u)$ is again of one of these forms, with first or second entry uv, uv^* or $-uv^*$. Since

$\{1, \mathrm{i}, -1, -\mathrm{i}\}$ is closed under multiplication, conjugation and negation, the product is in S_1. The identity is $A = (1,0)$: $(1,0) \circ (w_1, w_2) = (w_1, w_2)$ and $(w_1, w_2) \circ (1,0) = (w_1, w_2)$.

Pause and try Find the order and inverse of each element.

A has order 1. $E \circ E = (-1,0) \circ (-1,0) = (1,0)$, so E has order 2. For $C = (\mathrm{i}, 0)$: $C \circ C = (-1,0) = E$, so C has order 4 and $C^{-1} = C^3 = E \circ C = G$. For $B = (0,1)$: $B \circ B = (-1 \cdot 1^*, 0) = (-1,0) = E$, so B has order 4 and $B^{-1} = F$. Similarly $D \circ D = (-\mathrm{i} \cdot \mathrm{i}^*, 0) = (-\mathrm{i} \cdot (-\mathrm{i}), 0) = (-1,0) = E$, so D has order 4 with inverse H. And G, F, H are the inverses of C, B, D, also of order 4. In summary:

element	A	B	C	D	E	F	G	H
inverse	A	F	G	H	E	B	C	D
order	1	4	4	4	2	4	4	4

With closure, associativity (Stage 1), an identity and inverses, S_1 is a group. ■It is called the *quaternion group* Q_8.

STAGE 4 A subgroup

$\{A, B, E, F\} = \{1, \mathrm{j}, -1, -\mathrm{j}\}$ consists of the powers of B: $B^1 = B$, $B^2 = E$, $B^3 = F$, $B^4 = A$. So it is closed and contains inverses: a subgroup, *cyclic* of order 4. The matrix $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ satisfies $J^2 = -I$, $J^3 = -J$, $J^4 = I$, so it generates a cyclic group of order 4 too. Matching $B^k \leftrightarrow J^k$ is an isomorphism. ■(Indeed J is exactly $M(0,1)$, the matrix of B.)

Looking back

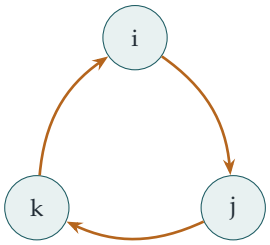
- **Represent by matrices.** Associativity, painful to check directly, was inherited from matrix multiplication.
- **Find the structure behind a formula.** The strange operation $\circ$ was quaternion multiplication, and seeing that made everything easier.
- **Orders and inverses describe a group.** The table of orders already shows Q_8 is unusual: one element of order 2 and six of order 4.

What's really going on

Beyond the complex numbers

The real numbers form a one-dimensional number system in which you can add, multiply and divide; the complex numbers form a two-dimensional one. Hamilton's

quaternions are four-dimensional, and you can still divide, but multiplication is no longer commutative (Figure 6.7.1). In 1878 Ferdinand Frobenius proved that these three, $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{H}$ (for Hamilton), are the only finite-dimensional "division algebras" over the reals with associative multiplication. If you give up associativity but keep a size that multiplies, $|xy| = |x|\,|y|$, as it does for real and complex numbers and quaternions, there is exactly one more: the eight-dimensional *octonions*, discovered by John Graves and Arthur Cayley in the 1840s. Adolf Hurwitz proved in 1898 that these four are the only such systems. (Without the condition on size there are other, stranger non-associative division algebras, but a deep theorem of 1958, due to Raoul Bott, John Milnor and Michel Kervaire, shows that every finite-dimensional real division algebra has dimension 1, 2, 4 or 8.)



With the arrows: $\mathrm{ij} = \mathrm{k}$, $\mathrm{jk} = \mathrm{i}$, $\mathrm{ki} = \mathrm{j}$

Against them: $\mathrm{ji} = -\mathrm{k}$, $\mathrm{kj} = -\mathrm{i}$, $\mathrm{ik} = -\mathrm{j}$

and $\mathrm{i}^2 = \mathrm{j}^2 = \mathrm{k}^2 = -1$

Figure 6.7.1 Hamilton's multiplication rules. Multiplying two of i, j, k in the direction of the arrows gives the third; multiplying against the arrows gives minus the third. The eight elements $\pm 1, \pm\mathrm{i}, \pm\mathrm{j}, \pm\mathrm{k}$ form the quaternion group Q_8 of this question.

Rotations in three dimensions

The quaternions of modulus 1 describe rotations of three-dimensional space: the rotation through angle θ about a unit axis $\mathbf{u}$ corresponds to $q = \cos\frac{\theta}{2} + \sin\frac{\theta}{2}(u_1\mathrm{i} + u_2\mathrm{j} + u_3\mathrm{k})$, acting on a vector $\mathbf{v}$ (written as $v_1\mathrm{i} + v_2\mathrm{j} + v_3\mathrm{k}$) by $\mathbf{v} \mapsto q\mathbf{v}q^{-1}$. Composing rotations is multiplying quaternions. This is more efficient and numerically stable than using matrices or angles, and it avoids "gimbal lock", a loss of one degree of freedom that the Apollo astronauts had to work around with their gyroscope-based navigation. The half-angle $\frac{\theta}{2}$ is also a clue to something deep: rotating by 360° gives $q = -1$, not 1, a fact that physicists see in the behaviour of electrons (spin).

Four squares

The modulus of a quaternion is multiplicative, $|pq| = |p||q|$, just like that of a complex number. Writing it out gives Euler's *four-square identity*: a product of two sums of four squares is itself a sum of four squares. That is the key step in Lagrange's theorem that every positive integer is a sum of four squares, which we prove in Section 7.8, the four-dimensional cousin of the two-squares result in Section 1.6.

The thinking skill

Represent abstract objects concretely

If an abstract operation is hard to reason about, find a concrete model, such as matrices, in which it becomes a familiar operation.

Properties like associativity then come for free. This idea, studying groups by representing them as matrices, grows into the large and beautiful subject of *representation theory*.

Try this next

1 **Hamilton's rules.** Using the pairs, check that $C \circ B = D$ and $B \circ C = H$, that is, ij $=$ k and ji $= -$k.

2 **Only one element of order** 2**.** Show that $E = -1$ is the only element of order 2 in Q_8. Deduce that Q_8 is not isomorphic to the group of symmetries of a square (which has five elements of order 2).

3 **Four squares.** Show that $\det M(z_1, z_2) = |z_1|^2 + |z_2|^2$. Using $\det(M_1 M_2) = \det M_1 \det M_2$, deduce that a product of two sums of four squares is a sum of four squares.

4 **Subgroups.** Find all the subgroups of Q_8. Show that every subgroup contains E unless it is $\{A\}$, and that Q_8 is the union of its three cyclic subgroups of order 4 (compare Section 6.6).

Further reading

- **John Conway & Derek Smith, *On Quaternions and Octonions*** (A. K. Peters). The geometry and arithmetic of quaternions and octonions. Advanced but fascinating.
- **John Baez, "The octonions"**, *Bulletin of the American Mathematical Society* 39 (2002); also freely available online. A wonderful survey; the introduction is readable now.
- **Andrew Hanson, *Visualizing Quaternions*** (Morgan Kaufmann). Quaternions for rotations, with applications to graphics.

6.8 A Group of Singular Matrices

The question

(i) Let S be the set of matrices of the form

$$\begin{pmatrix} a & a \\ a & a \end{pmatrix},$$

where a is any real non-zero number. Show that S is closed under matrix multiplication and, further, that S is a group under matrix multiplication.

(ii) Let G be a set of $n \times n$ matrices which is a group under matrix multiplication, with identity element $\mathbf{E}$. By considering equations of the form $\mathbf{BC} = \mathbf{D}$ for suitable elements $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ of G, show that if a given element $\mathbf{A}$ of G is a singular matrix (i.e. $\det \mathbf{A} = 0$), then all elements of G are singular. Give, with justification, an example of such a group of singular matrices in the case $n = 3$.

The hook

Every student learns that a matrix with zero determinant has no inverse. So a group of matrices, in which every element has an inverse, surely cannot contain any singular matrices?

It can. The catch is in the word "inverse". In a group, the inverse of $\mathbf{A}$ is the element that combines with $\mathbf{A}$ to give the *group's* identity, and nothing says the group's identity has to be the identity matrix $\mathbf{I}$. This question builds a group in which every matrix is singular, and the identity element is a matrix with determinant zero. It is a lesson in reading definitions literally, and in the difference between an abstract structure and the familiar example you first met it in.

Before you start

For the definition of a group, see Section 6.6.

? **Ask yourself** In part (i), the identity matrix $\mathbf{I}$ is not of the form $\begin{pmatrix} a & a \\ a & a \end{pmatrix}$. So what could the identity element of S be?

It must be some matrix $\mathbf{E}$ in S with $\mathbf{EX} = \mathbf{XE} = \mathbf{X}$ for every $\mathbf{X}$ in S. That is a condition

only on how $\mathbf{E}$ acts on matrices of this special form, not on all matrices. Find it by solving the equation.

Working it through

STAGE 1 Closure

Write $\mathbf{M}_a$ for the matrix with all four entries equal to a. Then

$$\mathbf{M}_a\mathbf{M}_b = \begin{pmatrix} a & a \\ a & a \end{pmatrix}\begin{pmatrix} b & b \\ b & b \end{pmatrix} = \begin{pmatrix} 2ab & 2ab \\ 2ab & 2ab \end{pmatrix} = \mathbf{M}_{2ab},$$

and $2ab \neq 0$ when $a, b \neq 0$. So S is closed. ■

STAGE 2 The group axioms

> **Ask yourself** Matrix multiplication is associative. For the identity, solve $\mathbf{M}_e\mathbf{M}_a = \mathbf{M}_a$. For inverses, solve $\mathbf{M}_a\mathbf{M}_b = \mathbf{M}_e$.

Associativity is inherited from matrix multiplication. *Identity:* $\mathbf{M}_e\mathbf{M}_a = \mathbf{M}_{2ea} = \mathbf{M}_a$ for all a exactly when $e = \frac{1}{2}$; by the symmetry of the formula, $\mathbf{M}_a\mathbf{M}_{1/2} = \mathbf{M}_a$ too. So

$$\mathbf{E} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

is the identity of S. *Inverses:* $\mathbf{M}_a\mathbf{M}_b = \mathbf{M}_{2ab} = \mathbf{M}_{1/2}$ when $b = \frac{1}{4a}$, which is non-zero, so $\mathbf{M}_{1/(4a)}$ is the inverse of $\mathbf{M}_a$ in S. So S is a group. ■Every element has determinant $a^2 - a^2 = 0$.

STAGE 3 Singularity spreads

> **Pause and try** Suppose $\mathbf{A} \in G$ is singular, and let $\mathbf{B}$ be any element of G. Write $\mathbf{B}$ as a product involving $\mathbf{A}$, and take determinants.

Let $\mathbf{A}^{-1}$ denote the inverse of $\mathbf{A}$ *in the group* G (not necessarily a matrix inverse). Then $\mathbf{C} = \mathbf{A}^{-1}\mathbf{B}$ is in G, and

$$\mathbf{AC} = \mathbf{AA}^{-1}\mathbf{B} = \mathbf{EB} = \mathbf{B}.$$

Taking determinants, $\det\mathbf{B} = \det\mathbf{A}\,\det\mathbf{C} = 0$. So every element of G is singular. ■In particular, if one element of a matrix group is invertible as a matrix, all are; a matrix group is either entirely invertible or entirely singular.

STAGE 4 A 3×3 example

Ask yourself Can you generalise part (i) to 3×3 matrices? Or find something even simpler, using a matrix that "ignores" two of the three coordinates?

Take

$$G = \left\{ \begin{pmatrix} a & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} : a \neq 0 \right\}.$$

The product of the matrices with corner entries a and b is the one with corner entry ab, so G is closed; multiplication is associative; the identity is the matrix with corner entry 1; the inverse of the one with corner a is the one with corner $\frac{1}{a}$. So G is a group, and every element has determinant 0. ■(Another example: the 3×3 matrices with all nine entries equal to $a \neq 0$, whose identity has all entries $\frac{1}{3}$.)

Looking back

- **Read definitions literally.** A group's identity is whatever element acts as an identity *within the group*; it need not be the familiar identity.
- **Solve for the structure.** The identity and inverses in S were found by solving equations, not by guessing.
- **Determinants are multiplicative.** That one property showed that singularity spreads from one element to all of them.

What's really going on

Idempotents and hidden invertibility

The identity $\mathbf{E}$ of any matrix group satisfies $\mathbf{E}^2 = \mathbf{E}$: it is *idempotent*. Geometrically, an idempotent matrix is a *projection*: it maps every vector onto a subspace and leaves the vectors in that subspace alone. In part (i), $\mathbf{E}$ projects the plane onto the line $y = x$, and every matrix $\mathbf{M}_a$ first projects onto that line and then stretches along it by the factor $2a$ (Figure 6.8.1). On the line, the group acts by invertible stretches; it simply ignores the direction perpendicular to it. So S is really the group of non-zero real numbers under multiplication (via $\mathbf{M}_a \mapsto 2a$), living on a line inside the plane. This is true in general: every group of matrices is, after a change of basis, a group of invertible matrices acting on the image of its identity element, padded with zeros.

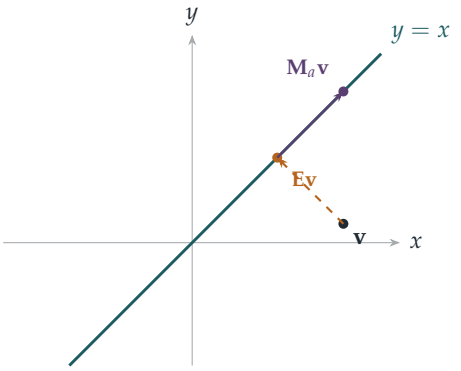


Figure 6.8.1 What the matrices in part (i) do. The identity element $\mathbf{E}$ projects every vector $\mathbf{v}$ onto the line $y = x$ (amber, dashed). Each $\mathbf{M}_a$ does the same projection and then stretches along the line by the factor $2a$ (plum; here $2a = \frac{16}{9}$). On the line itself, the group acts by ordinary invertible scaling.

Abstract and concrete

The first groups people studied were concrete: permutations, symmetries, invertible matrices. The abstract definition (a set with an operation satisfying three axioms) was distilled from them. This question shows the value of the abstract view: S and the group of non-zero reals under multiplication are *isomorphic*, the same group in different clothing, even though one consists of singular matrices and the other of numbers. Recognising when two different-looking structures are "the same" is one of the central skills of algebra, and the notion of isomorphism is how mathematicians make that precise.

The thinking skill

Question hidden assumptions about familiar objects

When a familiar object appears in an abstract definition, check which of its familiar properties the definition actually requires.

"The identity of a group of matrices is $\mathbf{I}$" sounds obvious but is not part of the definition, and dropping it led to a genuinely new example. Questioning assumptions like this is how many new mathematical structures are found.

Try this next

1 **An isomorphism.** Show that the map $\mathbf{M}_a \mapsto 2a$ is a one-to-one correspondence from S to the non-zero reals that turns matrix multiplication into multiplication of numbers.

2 Not a group. Show that the set of matrices $\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix}$ with $a \neq 0$ is closed under multiplication but is not a group. (*Hint:* look for an identity element. Is there a left identity? A right identity?)

3 Idempotents. Find all 2×2 real matrices $\mathbf{E}$ with $\mathbf{E}^2 = \mathbf{E}$. Describe them geometrically.

4 All entries equal. Show that the $n \times n$ matrices with all entries equal to $a \neq 0$ form a group under multiplication, and find its identity element.

Further reading

- **M. A. Armstrong, *Groups and Symmetry*** (Springer). Matrix groups and isomorphisms, with many geometric examples.
- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge). Projections and idempotent matrices, clearly explained.
- **Joseph Gallian, *Contemporary Abstract Algebra*** (CRC Press). A friendly first course with many examples of unusual groups.

What We Have Learned

We began with the shift in algebra from solving equations to studying structures, and with three questions: what can be built from what, why the formulas for equations work, and what is gained by abstraction. Here is what the chapter's eight questions showed.

1. What can be built from what?

In Section 6.3 we eliminated radicals one at a time to find polynomials satisfied by $\sqrt{2}+\sqrt{3}$, $\sqrt{2}+\sqrt{3}+\sqrt{5}$ and $\sqrt[3]{2}+\sqrt[3]{3}$, of degrees 4, 8 and 9. Those degrees are invariants: the degree of the field a number generates. Because degrees multiply up a tower of fields, numbers reachable by square roots have degrees that are powers of 2. That single fact proves that the cube cannot be doubled ($\sqrt[3]{2}$ has degree 3) and that a 60° angle cannot be trisected.

In the other direction, Section 6.2 showed how something *can* be built. Splitting the 17th roots of unity into Gauss's periods, the sum $\cos\frac{2\pi}{17}+\cos\frac{4\pi}{17}+\cos\frac{8\pi}{17}+\cos\frac{16\pi}{17}$ satisfied a quadratic, with root $\frac{-1+\sqrt{17}}{4}$; three more quadratics lead to $\cos\frac{2\pi}{17}$, and so to the construction of the regular 17-gon. Underneath both sections were the cyclotomic polynomials of Section 6.1, the building blocks of x^n-1, obtained by sorting the roots of unity by their order.

2. Why do the formulas for equations work?

Section 6.4 showed the secret of the quartic: from four roots, form the three quantities $\alpha\beta+\gamma\delta$, $\alpha\gamma+\beta\delta$, $\alpha\delta+\beta\gamma$, one for each way of pairing the roots. Any rearrangement of the roots merely shuffles these three, so they satisfy a cubic whose coefficients are known. Solving the cubic, then two quadratics, gave the roots. Lagrange saw that this works because of how permutations of four objects act on the three pairings; Galois saw that it fails for five roots because the group of permutations of five objects contains a piece that cannot be broken down. The formulas are shadows of group structure.

3. What is gained by abstraction?

The last four sections described objects by the rules they obey.

- In Section 6.5, four rules (additivity, the product rule, constants, $\Diamond x = 1$) forced an operation to be differentiation, with no limits in sight. The same rules power automatic differentiation in modern software.
- In Section 6.6, a three-line argument from the group axioms showed that no group is the union of two proper subgroups, for every group at once.

- In Section 6.7, a strange operation on pairs of complex numbers turned out to be Hamilton's quaternion multiplication, and representing it by matrices made associativity automatic.
- In Section 6.8, reading the definition of a group literally produced a group of singular matrices whose identity is not the identity matrix, and which is "really" the group of non-zero real numbers in disguise.

In each case, the abstract view explained something that calculation alone would have obscured.

4. A toolkit for algebra

Method	Where you used it
Classifying by a unique invariant	Roots of unity by order (6.1)
Sum and product give a quadratic	Gauss periods (6.2); the quartic (6.4)
Product-to-sum formulas	The 17-gon (6.2)
Eliminating radicals	Minimal polynomials (6.3)
Degrees and the tower law	Impossibility of constructions (6.3)
Symmetric functions of roots	The resolvent cubic (6.4)
Determining an operation by its axioms	Derivations (6.5)
Arguing from group axioms	Unions of subgroups (6.6)
Matrix representations	Quaternions (6.7)
Isomorphism	Singular matrix groups (6.8)

Where next

- Quaternions lead to Lagrange's four-square theorem in Section 7.8.
- Matrices as algebraic objects, with their own rules, are the subject of Chapter 7.
- The arithmetic of number systems like the Gaussian integers, where factorisation can fail, was the theme of Section 1.7.

If you want to go further before university

- **Mario Livio, *The Equation That Couldn't Be Solved***: the quintic, Galois and the birth of group theory.
- **Nathan Carter, *Visual Group Theory***: groups through pictures, the ideal first look.
- **M. A. Armstrong, *Groups and Symmetry***: a first university course, built on geometry.
- **Ian Stewart, *Galois Theory***: for when you are ready for the full story.

Galois died at twenty, after a duel, leaving a few dozen pages that took the mathematical

world decades to understand. What he saw was that an equation's deepest secrets are not in its coefficients but in the symmetries of its solutions. That insight, giving the same name to different things and studying the structure they share, now runs through all of mathematics. You have seen it at work in polynomials, polygons, quaternions and matrices; at university you will see it almost everywhere.

Answers to Try This Next

These are brief answers and hints, not full solutions: they give the result and the key step, so that you can check your own work. Try each exercise properly before looking.

6.1 Polynomials Whose Roots Are Roots of Unity

1. $(x-1)(x+1) = x^2-1$ and $(x^2+x+1)(x^2-x+1) = x^4+x^2+1$, whose product is x^6-1. Similarly $x^{12}-1 = (x^6-1)(x^6+1)$ and $x^6+1 = (x^2+1)(x^4-x^2+1) = \mathrm{C}_4\mathrm{C}_{12}$.

2. The divisors of $2p$ are $1, 2, p, 2p$, so $\mathrm{C}_{2p} = \frac{x^{2p}-1}{(x-1)(x+1)\mathrm{C}_p(x)} = \frac{x^p+1}{x+1}$. And $\mathrm{C}_p(-x) = \frac{(-x)^p-1}{-x-1} = \frac{x^p+1}{x+1}$ since p is odd. For $p=3$: $\mathrm{C}_6 = x^2-x+1 = \mathrm{C}_3(-x)$.

3. The numbers up to p^k that share a factor with p^k are the p^{k-1} multiples of p. The primitive p^kth roots are the roots of $x^{p^k}-1$ that are not roots of $x^{p^{k-1}}-1$, so $\mathrm{C}_{p^k} = \frac{x^{p^k}-1}{x^{p^{k-1}}-1} = \mathrm{C}_p\big(x^{p^{k-1}}\big)$. So $\mathrm{C}_9 = x^6+x^3+1$.

4. $\mathrm{C}_{15} = \frac{(x^{15}-1)(x-1)}{(x^3-1)(x^5-1)}$. Check by multiplying out that $(x-1)(x^2+x+1)(x^4+x^3+x^2+x+1)$ times the stated octic is $x^{15}-1$.

6.2 Gauss's 17-gon

1. $\eta+\eta'$ is the sum of all six non-real seventh roots, -1. Expanding $\eta\eta'$ gives nine terms α^{a+b} with exponents $4,6,0,5,0,1,0,2,3 \pmod 7$: three 1s and each non-real root once, so $\eta\eta' = 3-1 = 2$. So η, η' are $\frac{-1\pm\mathrm{i}\sqrt{7}}{2}$. The real part of η gives the cosine sum. Its imaginary part is $\sin\frac{2\pi}{7}+\sin\frac{4\pi}{7}-\sin\frac{\pi}{7}$, which is positive since $\sin\frac{4\pi}{7} = \sin\frac{3\pi}{7} > \sin\frac{\pi}{7}$; so it is $+\frac{\sqrt{7}}{2}$.

2. With $\zeta = \mathrm{e}^{2\pi\mathrm{i}/5}$, the periods $\zeta+\zeta^4 = 2\cos\frac{2\pi}{5}$ and $\zeta^2+\zeta^3$ have sum -1 and product -1, so they are roots of $t^2+t-1=0$, and the positive one gives $\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$. To construct: in a unit circle, a right triangle with legs 1 and $\frac{1}{2}$ has hypotenuse $\frac{\sqrt{5}}{2}$; subtract $\frac{1}{2}$ and halve to get $\cos\frac{2\pi}{5}$. The perpendicular to the axis at that distance meets the circle at a vertex adjacent to $(1,0)$; step this chord round the circle.

3. Both are -1.28078 to five decimal places.

4. The powers of 3 are $3,9,10,13,5,15,11,16,14,8,7,4,12,2,6,1$: all sixteen. The powers of 2 repeat after eight steps ($2^8 = 256 \equiv 1$) and give $\{1,2,4,8,9,13,15,16\}$, the non-zero squares modulo 17.

6.3 A Polynomial for $\sqrt{2}+\sqrt{3}+\sqrt{5}$

1. $y^3 = 2\sqrt{2}+6\sqrt{3}+9\sqrt{2}+3\sqrt{3} = 11\sqrt{2}+9\sqrt{3}$, and $9y = 9\sqrt{2}+9\sqrt{3}$, so $y^3-9y = 2\sqrt{2}$. Then $\sqrt{3} = y-\sqrt{2}$ and $\sqrt{6} = \sqrt{2}\sqrt{3}$ are polynomials in y too.
2. Put $x-\sqrt{2} = \sqrt[3]{3}$ and cube: $x^3+6x-3 = \sqrt{2}(3x^2+2)$. Squaring gives $x^6-6x^4-6x^3+12x^2-36x+1=0$.
3. The only possible rational roots are ±1, and neither works. Since there is no x^3 term, a factorisation into monic quadratics has the form $(x^2+ax+b)(x^2-ax+c)$, with $a(c-b)=0$, $b+c-a^2=-10$ and $bc=1$. If $a=0$ then $b+c=-10$ and $bc=1$, impossible in integers; if $b=c=\pm1$ then $a^2=12$ or 8, also impossible. So the quartic is irreducible and $\sqrt{2}+\sqrt{3}$ has degree 4.
4. From De Moivre, $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$; with $\theta = 20^\circ$, $\frac{1}{2} = 4c^3-3c$. The candidates $\pm1, \pm\frac{1}{2}, \pm\frac{1}{4}, \pm\frac{1}{8}$ all fail. So $\cos 20^\circ$ has degree 3, but every constructible length has degree a power of 2, so 20° cannot be constructed from 60°.

6.4 Solving the Quartic

1. $y_1y_2+y_2y_3+y_3y_1$ is the sum of the twelve terms $\alpha^2\beta\gamma$ (one squared root times two others). And $\left(\sum\alpha\right)\left(\sum\alpha\beta\gamma\right)$ is those twelve terms plus $4\alpha\beta\gamma\delta$. So the sum is $(-p)(-r)-4s$.
2. Matching coefficients gives $a+b=1$, $a\alpha+b\beta=0$ and $a\alpha^2+b\beta^2=-p$, solvable when $\alpha\neq\beta$, which is what the two conditions guarantee: $p\neq0$ makes the equation for t a genuine quadratic (if $p=0$ it collapses to $-qt=0$, with the single root 0), and $q^2\neq4p^3$ makes its discriminant q^2-4p^3 non-zero. For $x^3-24x+48$: $p=8$, $q=48$, and $8t^2-48t+64=0$ gives $\alpha=2$, $\beta=4$. Then $\left(\frac{x-2}{x-4}\right)^3 = -\frac{b}{a} = \frac{\alpha}{\beta} = \frac{1}{2}$, so $x-2 = 2^{-1/3}(x-4)$, which rearranges to the given root, $x\approx-5.6946$.
3. The resolvent cubic is $y^3-7y^2+16y-12 = (y-3)(y-2)^2$. The root $y=3$ leads to $(x^2-2x+1)(x^2-2x+2)$, so the roots are 1 (twice) and $1\pm\mathrm{i}$.
4. Swap within either pair or not (4 choices), and swap the two pairs or not (2 choices): the identity, $(\alpha\beta)$, $(\gamma\delta)$, $(\alpha\beta)(\gamma\delta)$, $(\alpha\gamma)(\beta\delta)$, $(\alpha\delta)(\beta\gamma)$, $(\alpha\gamma\beta\delta)$ and $(\alpha\delta\beta\gamma)$. This is a copy of the symmetry group of a square.

6.5 Differentiation from Axioms

1. Constants still give 0, and by rule (i) and induction $\Diamond x^n = x^{n-1}\Diamond x + x\Diamond x^{n-1} = nx^n$. By linearity $\Diamond \mathrm{h} = x\mathrm{h}'$.
2. $\Diamond\mathrm{f} = \frac{\mathrm{f}}{\mathrm{g}}\Diamond\mathrm{g} + \mathrm{g}\Diamond\frac{\mathrm{f}}{\mathrm{g}}$, so $\Diamond\frac{\mathrm{f}}{\mathrm{g}} = \frac{\mathrm{g}\Diamond\mathrm{f}-\mathrm{f}\Diamond\mathrm{g}}{\mathrm{g}^2}$.
3. If $\mathrm{f} = (x-a)^2\mathrm{g}$ then $\mathrm{f}' = (x-a)\big(2\mathrm{g}+(x-a)\mathrm{g}'\big)$. A repeated root of x^n-1 would be a root of nx^{n-1}, so it would be 0; but 0 is not a root of x^n-1.
4. $(2+\varepsilon)^2+1 = 5+4\varepsilon$, and $(5+4\varepsilon)^3 = 125+300\varepsilon$. So $f(2)=125$ and $f'(2)=300$; the chain rule gives $3(x^2+1)^2\cdot2x = 3\cdot25\cdot4 = 300$.

6.6 Unions of Subgroups

1. With $V = \{e, a, b, c\}$: $a^2 = b^2 = c^2 = e$ and the product of any two of a, b, c is the third. Every element is its own inverse. In $\{1, \mathrm{i}, -1, -\mathrm{i}\}$, i has inverse $-\mathrm{i}$, so the groups are different.
2. If g lay in a proper subgroup H, then so would every power of g, so H would be the whole group.
3. 1 is neither even nor a multiple of 3. More generally, $\mathbb{Z}$ is cyclic (generated by 1), so by the previous exercise it is not the union of any number of proper subgroups.
4. The line $y = mx$ meets $y = x^2$ only at $x = 0$ and $x = m$, and the line $x = 0$ only at the origin. So k lines contain at most k of the infinitely many points (t, t^2) with $t \neq 0$.

6.7 The Quaternion Group

1. $C \circ B = (\mathrm{i}, 0) \circ (0, 1) = (0, \mathrm{i}) = D$, and $B \circ C = (0, 1) \circ (\mathrm{i}, 0) = (0, \mathrm{i}^*) = (0, -\mathrm{i}) = H$.
2. A has order 1 and E order 2; each of the other six squares to E, so has order 4. The square's symmetry group has five elements of order 2 (four reflections and the half-turn), so the groups are not isomorphic.
3. $\det M = z_1 z_1^* + z_2 z_2^* = |z_1|^2 + |z_2|^2$, which is a sum of four real squares. Since $M(z)M(w)$ is the matrix of $z \circ w$, multiplying determinants writes $(a_1^2 + \cdots + a_4^2)(b_1^2 + \cdots + b_4^2)$ as $|u_1|^2 + |u_2|^2$, again four squares.
4. The subgroups are $\{A\}$, $\{A, E\}$, $\{A, E, C, G\}$, $\{A, E, B, F\}$, $\{A, E, D, H\}$ and Q_8. Every element other than A has E as a power, so any subgroup other than $\{A\}$ contains E. The three subgroups of order 4 together contain all eight elements.

6.8 A Group of Singular Matrices

1. $\mathbf{M}_a \mathbf{M}_b = \mathbf{M}_{2ab}$, and $2(2ab) = (2a)(2b)$. The map is one-to-one and onto the non-zero reals.
2. $\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} \begin{pmatrix} c & d \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} ac & ad \\ 0 & 0 \end{pmatrix}$ with $ac \neq 0$, so the set is closed. Every $\begin{pmatrix} 1 & e \\ 0 & 0 \end{pmatrix}$ is a left identity. But $\mathbf{XE} = \mathbf{X}$ with $\mathbf{E} = \begin{pmatrix} e & f \\ 0 & 0 \end{pmatrix}$ needs $e = 1$ and $f = b/a$, which depends on $\mathbf{X}$: there is no right identity, so no identity, and no group.
3. $\mathbf{0}$, $\mathbf{I}$, and the matrices $\begin{pmatrix} a & b \\ c & 1-a \end{pmatrix}$ with $bc = a(1-a)$ (trace 1, determinant 0). These are the projections of the plane onto one line along another.
4. The matrix $\mathbf{J}_a$ with all entries a satisfies $\mathbf{J}_a \mathbf{J}_b = \mathbf{J}_{nab}$. The identity is $\mathbf{J}_{1/n}$ and the inverse of $\mathbf{J}_a$ is $\mathbf{J}_{1/(n^2 a)}$.

CHAPTER

7

Linear Algebra and Geometry

A matrix is not a table of numbers. It is a transformation of space, and it has a life of its own.

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Transformations of Space

We think basis-free, we write basis-free, but when the chips are down we close the office door and compute with matrices like fury.

— Irving Kaplansky, on his and Paul Halmos's approach to linear algebra

At school, a matrix usually appears as a tool for solving simultaneous equations, or as a recipe for transforming points: multiply the matrix by a column vector and read off the new coordinates. Both are useful, but they hide what makes matrices one of the most important ideas in mathematics. A matrix is a *linear transformation*: a way of stretching, rotating, reflecting and shearing space, and matrices can be added, multiplied, raised to powers and differentiated, just like numbers. Except that they are not like numbers. Their multiplication depends on the order, they can multiply to zero, and they have hidden invariants (trace, determinant, eigenvalues) that capture their essential behaviour regardless of how they are written down.

Why linear algebra matters

Linear algebra is probably the most widely used branch of mathematics. Every computer graphics engine, from video games to film animation, is a machine for multiplying matrices: every rotation of a character, every camera angle, every perspective projection is a matrix. The search engines and recommendation systems of the internet are built on eigenvectors. Engineers find the vibration frequencies of bridges and buildings from eigenvalues; statisticians find the main patterns in data with them; quantum mechanics is written entirely in the language of linear algebra. And modern machine learning, at its core, is very large amounts of matrix multiplication.

There is a reason for this ubiquity. Linear problems are the ones we can solve completely, and almost every non-linear problem looks linear if you zoom in close enough. That is the idea of calculus, and it means that linear algebra is the language in which the local behaviour of almost everything is described.

What you will meet at university

Linear algebra is one of the two pillars of the first year, alongside analysis. You will meet *vector spaces* (sets where you can add and scale, of which the plane and three-dimensional space are just the first examples) and *linear maps* between them, and learn that matrices are simply linear maps written in coordinates. You will study determinants, eigenvalues and eigenvectors, and the *spectral theorem*, which says that symmetric matrices can be diagonalised by a rotation. Later courses use this everywhere: in differential equations (Chapter 8), in geometry (where rotations and reflections form

groups), in the theory of *Lie groups*, which describe continuous symmetries, and in functional analysis, which is linear algebra in infinitely many dimensions.

The questions that run through this chapter

Three big questions

1. **How are matrices like numbers, and how are they different?** Which rules of arithmetic survive, and which fail?
2. **What does a matrix really do?** How can we see the geometry of a transformation, and which numbers capture it independently of coordinates?
3. **What stays the same?** Which quantities are preserved by transformations, or by motion, and what do they tell us?

The chapter starts with the rules of matrix algebra (Section 7.1), then turns to the geometry of transformations: eigenvectors, determinants, matrices of finite order, rotations in three dimensions and the hyperbolic rotations of special relativity. It continues with conserved quantities of evolving matrices, the four-square identity from determinants, and ends with a counting argument, Euler's formula, that classifies the Platonic solids.

Ideas in this chapter

§	Topic	Idea
7.1	Rules that fail	Non-commutativity; zero divisors
7.2	Eigenvectors	Directions a transformation does not rotate
7.3	The determinant	Characterised by multiplicativity; signed area
7.4	Matrices of finite order	Trace, determinant and Cayley–Hamilton
7.5	Rotations in 3D	Rodrigues' formula from the cross product
7.6	Lorentz boosts	Hyperbolic angles add
7.7	Conserved quantities	The trace of a commutator is zero
7.8	Four squares	Multiplicative determinants
7.9	Platonic solids	Double counting and Euler's formula

How to read this chapter

For every matrix in this chapter, draw what it does. Take the unit square, or the unit circle, or a few vectors, and sketch their images. A two-minute sketch will often show you the answer (an eigenvector, a counterexample, an invariant line) before you start calculating, and it will help you check the calculation afterwards.

7.1 Rules That Fail for Matrices

The question

In this question, $\mathbf{A}$, $\mathbf{B}$ *and* $\mathbf{X}$ *are non-zero* 2×2 *real matrices.*

Are the following assertions true or false? You must provide a proof or a counterexample in each case.

(i) If $\mathbf{AB} = \mathbf{0}$ then $\mathbf{BA} = \mathbf{0}$.

(ii) $(\mathbf{A} - \mathbf{B})(\mathbf{A} + \mathbf{B}) = \mathbf{A}^2 - \mathbf{B}^2$.

(iii) The equation $\mathbf{AX} = \mathbf{0}$ has a non-zero solution $\mathbf{X}$ if and only if $\det \mathbf{A} = 0$.

(iv) For any $\mathbf{A}$ and $\mathbf{B}$ there are at most two matrices $\mathbf{X}$ such that $\mathbf{X}^2 + \mathbf{AX} + \mathbf{B} = \mathbf{0}$.

The hook

Matrices can be added and multiplied, so it is tempting to treat them like numbers. This question tests that temptation with four "rules" that are true for numbers, and asks which survive for matrices. Most of them fail, and the way they fail is instructive. Matrix multiplication is not commutative; there are non-zero matrices whose product is zero; and a quadratic equation for matrices can have infinitely many solutions.

These failures are not defects. They are the reason matrices are useful: they describe transformations of space, and doing two transformations in different orders genuinely gives different results. In 1925 Werner Heisenberg discovered that the quantities of quantum mechanics behave like matrices rather than numbers, and that the failure of $\mathbf{AB} = \mathbf{BA}$ is the mathematical form of the uncertainty principle.

Before you start

Ask yourself For real numbers, why is it true that $ab = 0$ implies $a = 0$ or $b = 0$? Which property of numbers does the argument use, and does it hold for matrices?

If $ab = 0$ and $a \neq 0$, divide by a to get $b = 0$. The argument needs every non-zero number to have an inverse. Non-zero matrices need not have inverses (those with determinant zero do not), so the argument breaks down. Keep the *geometric* picture in mind too: a matrix is a transformation of the plane, and a product $\mathbf{AB}$ means "do $\mathbf{B}$, then $\mathbf{A}$".

Working it through

STAGE 1 Can $\mathbf{AB} = \mathbf{0}$ but $\mathbf{BA} \neq \mathbf{0}$?

Ask yourself Think geometrically. You want $\mathbf{B}$ to send everything into a line that $\mathbf{A}$ then squashes to zero. But in the other order, $\mathbf{A}$ should not squash everything that $\mathbf{B}$ then kills.

Let $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, which projects onto the x-axis (killing the y-axis), and $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, which sends (x, y) to $(0, x)$, into the y-axis. Then

$$\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \mathbf{BA} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \neq \mathbf{0}.$$

The assertion is **false** (Figure 7.1.1). ■

STAGE 2 The difference of two squares

Expanding carefully, keeping the order of every product:

$$(\mathbf{A} - \mathbf{B})(\mathbf{A} + \mathbf{B}) = \mathbf{A}^2 + \mathbf{AB} - \mathbf{BA} - \mathbf{B}^2.$$

This equals $\mathbf{A}^2 - \mathbf{B}^2$ exactly when $\mathbf{AB} = \mathbf{BA}$. With the matrices of Stage 1, $\mathbf{AB} - \mathbf{BA} = -\mathbf{BA} \neq \mathbf{0}$, so the assertion is **false**. ■

STAGE 3 Non-zero solutions of $\mathbf{AX} = \mathbf{0}$

Ask yourself If $\det \mathbf{A} \neq 0$, what can you do to $\mathbf{AX} = \mathbf{0}$? If $\det \mathbf{A} = 0$, can you find a non-zero column vector $\mathbf{v}$ with $\mathbf{Av} = \mathbf{0}$?

If $\det \mathbf{A} \neq 0$, then $\mathbf{A}^{-1}$ exists and $\mathbf{X} = \mathbf{A}^{-1}\mathbf{AX} = \mathbf{0}$: no non-zero solution.

If $\det \mathbf{A} = 0$, write $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $ad = bc$. The vectors $\begin{pmatrix} b \\ -a \end{pmatrix}$ and $\begin{pmatrix} d \\ -c \end{pmatrix}$ satisfy $\mathbf{A}\begin{pmatrix} b \\ -a \end{pmatrix} = \begin{pmatrix} ab - ba \\ cb - da \end{pmatrix} = \mathbf{0}$, and similarly for the other. Since $\mathbf{A} \neq \mathbf{0}$, at least one of them is non-zero; call it $\mathbf{v}$. Then the matrix $\mathbf{X}$ whose two columns are both $\mathbf{v}$ is non-zero and $\mathbf{AX} = \mathbf{0}$. So the assertion is **true**. ■

STAGE 4 How many square roots?

Pause and try Find non-zero $\mathbf{A}$ and $\mathbf{B}$ for which $\mathbf{X}^2 + \mathbf{AX} + \mathbf{B} = \mathbf{0}$ has infinitely many solutions. (*Hint:* make the left-hand side a perfect square, $(\mathbf{X} - \mathbf{I})^2$, and think about matrices whose square is zero.)

Take $\mathbf{A} = -2\mathbf{I}$ and $\mathbf{B} = \mathbf{I}$, both non-zero. The equation becomes $(\mathbf{X} - \mathbf{I})^2 = \mathbf{0}$. Any matrix $\mathbf{N}$ with $\mathbf{N}^2 = \mathbf{0}$ gives a solution $\mathbf{X} = \mathbf{I} + \mathbf{N}$, and there are infinitely many such $\mathbf{N}$:

$$\mathbf{N} = \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix}, \qquad \mathbf{N}^2 = \mathbf{0} \text{ for every real } t.$$

So $\mathbf{X} = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ is a solution for every t, and the assertion is **false**. ■

Looking back

- **Order matters.** Every expansion must keep products in order; $\mathbf{AB}$ and $\mathbf{BA}$ are usually different.
- **Zero divisors.** Non-zero matrices can multiply to zero, which breaks the argument behind "a quadratic has at most two roots".
- **Think of transformations.** Each counterexample was easy to find by picturing what the matrices do to the plane.

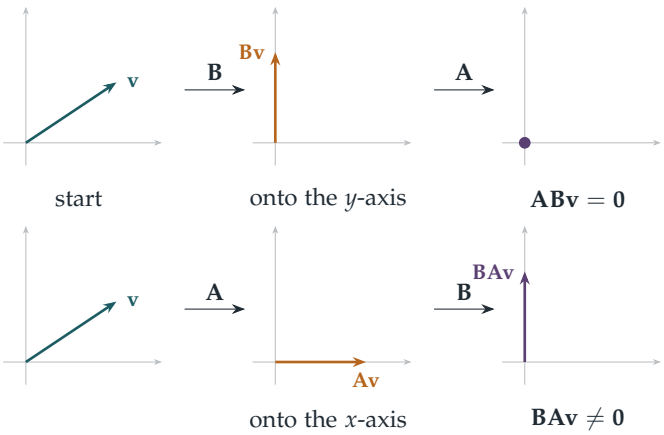


Figure 7.1.1 Why $\mathbf{AB} = \mathbf{0}$ but $\mathbf{BA} \neq \mathbf{0}$. Top: $\mathbf{B}$ sends every vector onto the y-axis, which $\mathbf{A}$ (projection onto the x-axis) then crushes to zero. Bottom: in the other order, $\mathbf{A}$ keeps the x-component, and $\mathbf{B}$ moves it onto the y-axis, where it survives.

What's really going on

Matrices as an algebra

Arthur Cayley introduced matrix algebra in 1858, treating matrices as single objects that can be added and multiplied, rather than just as tables of coefficients. He noticed at once that multiplication is not commutative, and he proved the *Cayley–Hamilton*

theorem: every square matrix satisfies its own characteristic equation. For a 2×2 matrix with trace t and determinant d, this says $\mathbf{M}^2 - t\mathbf{M} + d\mathbf{I} = \mathbf{0}$, a fact we use repeatedly in this chapter (Section 7.4). The 2×2 real matrices form a *ring*: you can add, subtract and multiply, but multiplication is not commutative, and there are *zero divisors*, non-zero elements whose product is zero.

Infinitely many square roots

Stage 4 can be made even more striking. The equation $\mathbf{X}^2 = \mathbf{I}$ has the obvious solutions $\pm\mathbf{I}$, but also every reflection of the plane in a line through the origin, $\mathbf{X} = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$, since reflecting twice does nothing. That is a whole circle of square roots of $\mathbf{I}$. A polynomial of degree n with number coefficients has at most n roots because numbers have no zero divisors; matrices do, and the count fails.

Non-commutativity in physics

In classical mechanics, position x and momentum p are numbers and $xp = px$. In quantum mechanics they become operators (infinite matrices) with $xp - px = \mathrm{i}\hbar$, where $\hbar$ is Planck's constant divided by 2π. This single non-commutation relation implies Heisenberg's uncertainty principle: position and momentum cannot both be known exactly. When Heisenberg discovered the rules of quantum mechanics in 1925 he did not know what matrices were; Max Born recognised that the strange multiplication rule Heisenberg had invented was matrix multiplication.

The thinking skill

Test familiar rules in a new setting

When you meet a new kind of object, check which familiar rules still hold, and find out exactly which property each rule depends on.

The rule "$ab = 0$ implies $a = 0$ or $b = 0$" depends on division; "$(a-b)(a+b) = a^2 - b^2$" depends on commutativity. Knowing what each rule rests on tells you when you can still use it.

Try this next

1 **When do they commute?** Find all 2×2 matrices that commute with $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

2 **Square roots of zero.** Find all 2×2 real matrices $\mathbf{N}$ with $\mathbf{N}^2 = \mathbf{0}$. (*Hint:* use the Cayley–Hamilton theorem to show $\operatorname{tr}\mathbf{N} = \det\mathbf{N} = 0$.)

3 **Reflections.** Show that every reflection matrix $\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$ squares to $\mathbf{I}$, and find the line it reflects in.

4 **Cayley–Hamilton.** Prove that every 2×2 matrix satisfies $\mathbf{M}^2 - (\operatorname{tr}\mathbf{M})\mathbf{M} + (\det\mathbf{M})\mathbf{I} = \mathbf{0}$ by direct calculation.

Further reading

- **Gilbert Strang,** ***Introduction to Linear Algebra*** (Wellesley-Cambridge). The most popular first linear algebra text, with an emphasis on understanding.
- **Sheldon Axler,** ***Linear Algebra Done Right*** (Springer; free online). A more abstract approach, centred on linear maps rather than matrices.
- **Jim Baggott,** ***The Quantum Story*** (Oxford University Press). The history of quantum mechanics, including Heisenberg's matrix mechanics.

7.2 Stretching Without Turning

The question

The transformation T from $\begin{pmatrix} x \\ y \end{pmatrix}$ to $\begin{pmatrix} X \\ Y \end{pmatrix}$ is given by

$$\begin{pmatrix} X \\ Y \end{pmatrix} = \frac{2}{5}\begin{pmatrix} 9 & -2 \\ -2 & 6 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}.$$

Show that T leaves the vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ unchanged in direction but multiplied by a scalar, and that $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ is similarly transformed.

The circle C whose equation is $x^2 + y^2 = 1$ transforms under T to a curve E. Show that E has equation

$$8X^2 + 12XY + 17Y^2 = 80,$$

and state the area of the region bounded by E. Show also that the greatest value of X on E is $2\sqrt{17/5}$.

Find the equation of the tangent to E at the point which corresponds to the point $\frac{1}{5}(3,4)$ on C.

The hook

A matrix transformation usually moves every vector to a new direction. But many matrices have a few special directions they do not turn at all, only stretching or shrinking along them. These are their *eigenvectors* (from the German *eigen*, "own" or "characteristic"), and the stretch factors are their *eigenvalues*. When there are enough of them, as there always are for a symmetric matrix like the one in this question, the transformation becomes transparent: in the right coordinates it is just a stretch along two axes.

This question finds the eigenvectors of a particular matrix and uses them to understand what it does to a circle. Eigenvectors are one of the most useful ideas in all of applied mathematics. They describe the natural vibrations of a bridge, the principal axes of a spinning object, the energy levels of an atom, the most important directions in a cloud of data, and the ranking of web pages by Google's original PageRank algorithm.

Before you start

Ask yourself What does it mean for T to leave a vector "unchanged in direction but multiplied by a scalar"? Write it as an equation.

It means $T\mathbf{v} = \lambda\mathbf{v}$ for some number λ: $\mathbf{v}$ is an eigenvector with eigenvalue λ. Notice that the two vectors in the question, $(1,2)$ and $(2,-1)$, are perpendicular. That is not a coincidence: the matrix is *symmetric*, and symmetric matrices always have perpendicular eigenvectors.

Working it through

STAGE 1 The eigenvectors

$$T\begin{pmatrix}1\\2\end{pmatrix} = \frac{2}{5}\begin{pmatrix}9-4\\-2+12\end{pmatrix} = \frac{2}{5}\begin{pmatrix}5\\10\end{pmatrix} = 2\begin{pmatrix}1\\2\end{pmatrix}, \qquad T\begin{pmatrix}2\\-1\end{pmatrix} = \frac{2}{5}\begin{pmatrix}20\\-10\end{pmatrix} = 4\begin{pmatrix}2\\-1\end{pmatrix}.$$

So $(1,2)$ is stretched by 2 and $(2,-1)$ by 4. ■

STAGE 2 The image of the circle

Ask yourself To find the equation of E, express x and y in terms of X and Y (invert the matrix), and substitute into $x^2+y^2=1$.

The matrix $\frac{2}{5}\begin{pmatrix}9&-2\\-2&6\end{pmatrix}$ has determinant $\frac{4}{25}(54-4)=8$, and its inverse is $\frac{1}{20}\begin{pmatrix}6&2\\2&9\end{pmatrix}$. So $x = \frac{6X+2Y}{20}$ and $y = \frac{2X+9Y}{20}$, and $x^2+y^2=1$ becomes

$$(6X+2Y)^2 + (2X+9Y)^2 = 400 \quad\Longrightarrow\quad 40X^2+60XY+85Y^2 = 400,$$

which, divided by 5, is $8X^2+12XY+17Y^2=80$. ■

Ask yourself What shape is E, and what is its area? Use the eigenvectors.

The unit circle has radius 1 in every direction. In the direction of $(2,-1)$ it is stretched by 4, and in the perpendicular direction $(1,2)$ by 2. So E is an ellipse with semi-axes 4 and 2, along the eigenvector directions (Figure 7.2.1), and its area is $\pi\times4\times2 = 8\pi$. (Equivalently, any linear map multiplies areas by $|\det T| = 8$, and the circle has area π.)

STAGE 3 The greatest value of X

Ask yourself On C, $X = \frac{2}{5}(9x - 2y)$. How large can $9x - 2y$ be when $x^2 + y^2 = 1$?

$9x - 2y$ is the dot product of $(9, -2)$ with the unit vector (x, y), so its largest value is $|(9, -2)| = \sqrt{85}$, attained when (x, y) points along $(9, -2)$. Hence the greatest value of X is $\frac{2}{5}\sqrt{85} = 2\sqrt{\frac{85}{25}} = 2\sqrt{\frac{17}{5}}$. ■

STAGE 4 A tangent

Pause and try The point $\frac{1}{5}(3, 4)$ on C has tangent line $3x + 4y = 5$. A linear map sends tangent lines to tangent lines. Use this to find the tangent to E.

Substituting $x = \frac{6X+2Y}{20}$, $y = \frac{2X+9Y}{20}$ into $3x + 4y = 5$:

$$\frac{3(6X + 2Y) + 4(2X + 9Y)}{20} = 5 \quad \Longrightarrow \quad 26X + 42Y = 100,$$

that is, $13X + 21Y = 50$. ■

(Check: the point itself is $T\left(\frac{3}{5}, \frac{4}{5}\right) = \left(\frac{38}{25}, \frac{36}{25}\right)$, and $13 \cdot \frac{38}{25} + 21 \cdot \frac{36}{25} = \frac{494+756}{25} = 50$.)

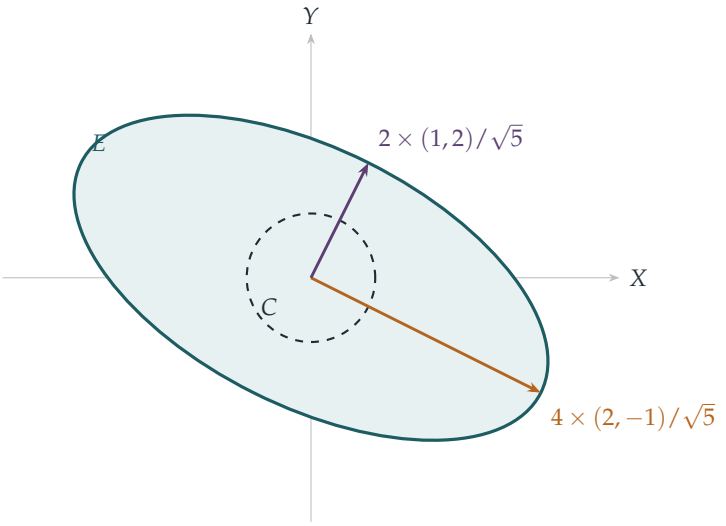


Figure 7.2.1 The unit circle C (dashed) and its image E under T. Along the eigenvector $(2, -1)$ the circle is stretched by a factor of 4; along the perpendicular eigenvector $(1, 2)$, by a factor of 2. So E is an ellipse with semi-axes 4 and 2 along those directions, and area 8π.

Looking back

► **Eigenvectors are the natural axes.** In the coordinates given by the eigenvectors, T is

just a stretch by 4 and by 2, and its effect on a circle is obvious.

- **Determinants measure area.** The area of E is $|\det T|$ times the area of C.
- **Linear maps preserve tangency.** Transforming the circle's tangent gave the ellipse's tangent with almost no calculation.

What's really going on

The spectral theorem

Every real symmetric matrix has a full set of perpendicular eigenvectors, with real eigenvalues. This is the *spectral theorem*, one of the central results of a first linear algebra course. It says that every symmetric linear map is, in suitable perpendicular coordinates, just a stretch along each axis. Equivalently, every quadratic form like $8X^2 + 12XY + 17Y^2$ can be rotated into the form $\lambda_1 u^2 + \lambda_2 v^2$, which is why every central conic is an ellipse or hyperbola with perpendicular axes (the *principal axis theorem*). The idea goes back to Cauchy in the 1820s, who used it to find the principal axes of quadric surfaces and of the stress in a solid body.

Symmetry is what makes this work. A matrix that is not symmetric need not be so obliging: a rotation through 90° turns every direction, so it has no real eigenvectors at all, and the shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has only one eigenvector direction, so it is not a stretch along any pair of axes.

Eigenvectors everywhere

Here are a few of the places eigenvectors appear.

- **Vibrations.** A system of masses and springs vibrates in *normal modes*, patterns in which every part moves in step at a single frequency. These are eigenvectors of a matrix, and the frequencies come from the eigenvalues (Section 11.2).
- **Data.** The directions in which a cloud of data points is most spread out are the eigenvectors of its covariance matrix. Finding them is *principal component analysis*, one of the basic tools of statistics and machine learning (Section 13.7).
- **Quantum mechanics.** The possible results of measuring a quantity are the eigenvalues of a matrix (or operator), and the energy levels of an atom are eigenvalues.
- **Web search.** Google's PageRank, the original idea behind the search engine, ranks pages by the entries of the eigenvector, with eigenvalue 1, of a huge matrix describing the links between pages.

The thinking skill

Choose coordinates adapted to the transformation

To understand a linear transformation, find the directions it does not rotate. In those coordinates it becomes simple.

This is the same idea as choosing the Fermat point as origin in Section 5.6, or changing variables to simplify a map in Section 2.8: the right coordinates make the structure obvious.

Try this next

1 **Finding eigenvalues.** Show that the eigenvalues of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ are the roots of $\lambda^2 - (a+d)\lambda + (ad - bc) = 0$, and check that the matrix in the question gives $\lambda = 2$ and 4.

2 **Perpendicular eigenvectors.** Show that if $\mathbf{M}$ is symmetric and $\mathbf{Mu} = \lambda\mathbf{u}$, $\mathbf{Mv} = \mu\mathbf{v}$ with $\lambda \neq \mu$, then $\mathbf{u} \cdot \mathbf{v} = 0$. (*Hint:* compute $\mathbf{v} \cdot \mathbf{Mu}$ in two ways.)

3 **A rotation.** Show that the rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ has no real eigenvectors unless θ is a multiple of π. What are its complex eigenvalues?

4 **Powers.** Using the eigenvectors, find $T^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. (*Hint:* write $(1, 0)$ as a combination of $(1, 2)$ and $(2, -1)$.)

Further reading

- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge). Chapter 6 is an excellent introduction to eigenvalues and their applications.
- **Kurt Bryan & Tanya Leise, "The $25,000,000,000 eigenvector: the linear algebra behind Google"**, *SIAM Review* 48 (2006). Readable and freely available.
- **3Blue1Brown, *Essence of Linear Algebra*** (video series, free online). Beautiful animations of exactly the geometric ideas in this section.

7.3 What Makes a Determinant

The question

The domain of the function f is the set of all 2×2 matrices and its range is the set of real numbers. Thus, if $\mathbf{M}$ is a 2×2 matrix, then $\mathrm{f}(\mathbf{M}) \in \mathbb{R}$.

The function f has the property that $\mathrm{f}(\mathbf{MN}) = \mathrm{f}(\mathbf{M})\mathrm{f}(\mathbf{N})$ for any 2×2 matrices $\mathbf{M}$ and $\mathbf{N}$.

(i) You are given that there is a matrix $\mathbf{M}$ such that $\mathrm{f}(\mathbf{M}) \neq 0$. Let $\mathbf{I}$ be the 2×2 identity matrix. By considering $\mathrm{f}(\mathbf{MI})$, show that $\mathrm{f}(\mathbf{I}) = 1$.

(ii) Let $\mathbf{J} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. You are given that $\mathrm{f}(\mathbf{J}) \neq 1$. By considering $\mathbf{J}^2$, evaluate $\mathrm{f}(\mathbf{J})$.

Using $\mathbf{J}$, show that, for any real numbers a, b, c and d,

$$\mathrm{f}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = -\mathrm{f}\begin{pmatrix} c & d \\ a & b \end{pmatrix} = \mathrm{f}\begin{pmatrix} d & c \\ b & a \end{pmatrix}.$$

(iii) Let $\mathbf{K} = \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$ where $k \in \mathbb{R}$. Use $\mathbf{K}$ to show that, if the second row of the matrix $\mathbf{A}$ is a multiple of the first row, then $\mathrm{f}(\mathbf{A}) = 0$.

(iv) Let $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. By considering the matrices $\mathbf{P}^2$, $\mathbf{P}^{-1}$, and $\mathbf{K}^{-1}\mathbf{PK}$ for suitable values of k, evaluate $\mathrm{f}(\mathbf{P})$.

The hook

The determinant of a 2×2 matrix is $ad - bc$. Where does that formula come from, and why is it so important? One answer is that it is the factor by which the matrix scales areas. Another, which this question explores, is that the determinant is essentially *the only* function on matrices that turns matrix multiplication into ordinary multiplication: $\det(\mathbf{MN}) = \det \mathbf{M} \det \mathbf{N}$.

The question gives you an unknown function f with that one property (plus two mild conditions to rule out trivial cases), and asks you to deduce, step by step, how it must behave. Like the derivations of Section 6.5, it is an exercise in pinning an object down by its properties alone. By the end, f has been forced to change sign when rows are swapped, vanish on singular matrices and equal 1 on shears: exactly the behaviour of

the determinant.

Before you start

Ask yourself What does each special matrix in the question do? Work out $\mathbf{JM}$, $\mathbf{MJ}$, $\mathbf{KM}$ and $\mathbf{PM}$ for a general $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$\mathbf{JM}$ swaps the rows of $\mathbf{M}$; $\mathbf{MJ}$ swaps its columns. $\mathbf{KM}$ multiplies the second row by k. $\mathbf{PM}$ adds the second row to the first. These are the *elementary row operations* of Gaussian elimination. Since f turns products into products, knowing f on these special matrices tells you how f responds to each row operation.

Working it through

STAGE 1 The identity

$\mathrm{f}(\mathbf{M}) = \mathrm{f}(\mathbf{MI}) = \mathrm{f}(\mathbf{M})\,\mathrm{f}(\mathbf{I})$. Since $\mathrm{f}(\mathbf{M}) \neq 0$, divide to get $\mathrm{f}(\mathbf{I}) = 1$. ■

STAGE 2 Swapping

$\mathbf{J}^2 = \mathbf{I}$, so $\mathrm{f}(\mathbf{J})^2 = \mathrm{f}(\mathbf{I}) = 1$, giving $\mathrm{f}(\mathbf{J}) = \pm 1$. Since $\mathrm{f}(\mathbf{J}) \neq 1$, $\mathrm{f}(\mathbf{J}) = -1$.

Now $\begin{pmatrix} c & d \\ a & b \end{pmatrix} = \mathbf{JM}$, so its value is $\mathrm{f}(\mathbf{J})\mathrm{f}(\mathbf{M}) = -\mathrm{f}(\mathbf{M})$. And $\begin{pmatrix} d & c \\ b & a \end{pmatrix} = \mathbf{JMJ}$ (swap rows, then columns), so its value is $\mathrm{f}(\mathbf{J})^2\mathrm{f}(\mathbf{M}) = \mathrm{f}(\mathbf{M})$. ■

STAGE 3 Dependent rows

Ask yourself First consider a matrix $\mathbf{B}$ with two *equal* rows. What is $\mathbf{JB}$? What does that say about $\mathrm{f}(\mathbf{B})$?

If both rows of $\mathbf{B}$ are (a, b), swapping them changes nothing: $\mathbf{JB} = \mathbf{B}$. So $\mathrm{f}(\mathbf{B}) = \mathrm{f}(\mathbf{J})\mathrm{f}(\mathbf{B}) = -\mathrm{f}(\mathbf{B})$, which forces $\mathrm{f}(\mathbf{B}) = 0$.

Now if $\mathbf{A}$ has rows (a, b) and $k(a, b)$, then $\mathbf{A} = \mathbf{KB}$, where $\mathbf{B}$ has both rows equal to (a, b). So $\mathrm{f}(\mathbf{A}) = \mathrm{f}(\mathbf{K})\mathrm{f}(\mathbf{B}) = 0$. ■(This includes $k = 0$, where the second row is zero.)

STAGE 4 The shear

Pause and try Compute $\mathbf{P}^2$ and $\mathbf{K}^{-1}\mathbf{PK}$. Choose k to make them equal. What does that give for $\mathrm{f}(\mathbf{P})$?

$\mathbf{P}^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ and, for $k \neq 0$,

$$\mathbf{K}^{-1}\mathbf{P}\mathbf{K} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{k} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix} = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}.$$

With $k = 2$, $\mathbf{K}^{-1}\mathbf{P}\mathbf{K} = \mathbf{P}^2$. Taking f of both sides, and using $\mathrm{f}(\mathbf{K}^{-1})\mathrm{f}(\mathbf{K}) = \mathrm{f}(\mathbf{I}) = 1$,

$$\mathrm{f}(\mathbf{P}) = \mathrm{f}(\mathbf{P})^2,$$

so $\mathrm{f}(\mathbf{P}) = 0$ or 1. But $\mathbf{P}$ is invertible, and $\mathrm{f}(\mathbf{P})\mathrm{f}(\mathbf{P}^{-1}) = \mathrm{f}(\mathbf{I}) = 1$, so $\mathrm{f}(\mathbf{P}) \neq 0$. Therefore $\mathrm{f}(\mathbf{P}) = 1$. ■

Looking back

- **Special elements reveal the function.** Applying the multiplicative property to $\mathbf{I}$, $\mathbf{J}$, $\mathbf{K}$ and $\mathbf{P}$ gave the value of f on each, and its effect on each row operation.
- **An element equal to its own transform.** A matrix with equal rows is unchanged by a row swap, which changes the sign of f; so f must be zero there.
- **Conjugation preserves** f. $\mathrm{f}(\mathbf{K}^{-1}\mathbf{P}\mathbf{K}) = \mathrm{f}(\mathbf{P})$, a fact used again and again in algebra.

What's really going on

Every multiplicative function is a function of the determinant

The question stops at $\mathrm{f}(\mathbf{P}) = 1$, but the argument can be completed. Every invertible matrix is a product of elementary matrices (row swaps, row scalings and shears), and the question shows how f behaves on each. It follows that $\mathrm{f}(\mathbf{M}) = \phi(\det \mathbf{M})$ for some function ϕ of a single real number with $\phi(xy) = \phi(x)\phi(y)$. The determinant itself is the case $\phi(x) = x$, but $\phi(x) = x^3$, or $\phi(x) = |x|^{1/2}\operatorname{sign}(x)$, work too. In other words, *the determinant is the only fundamentally new multiplicative function on matrices*. In the language of group theory, the determinant captures everything about the group of invertible matrices that can be seen by a commutative group.

The determinant as area

Geometrically, a 2×2 matrix sends the unit square to a parallelogram, and $\det \mathbf{M}$ is its *signed area*: positive if the orientation is preserved, negative if it is reversed (Figure 7.3.1). The multiplicative property then has an obvious meaning: doing two transformations in turn multiplies their area scale factors. A row swap reverses orientation, which is why $\det \mathbf{J} = -1$; a shear slides the square into a parallelogram of the same area, which is why $\det \mathbf{P} = 1$; and a matrix with dependent rows squashes the square flat, which is why its determinant is zero. Every step of the question has a picture.

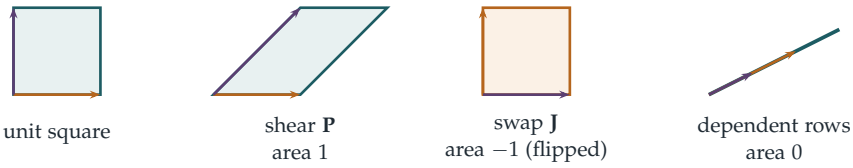


Figure 7.3.1 The determinant as signed area. The images of the two basis vectors (amber, plum) span a parallelogram. A shear preserves area; a swap of the basis vectors reverses orientation; dependent rows (or columns) collapse the square onto a line. Doing two transformations in turn multiplies the signed areas, which is the property $\mathrm{f}(\mathbf{MN}) = \mathrm{f}(\mathbf{M})\mathrm{f}(\mathbf{N})$.

Determinants in higher dimensions

For $n \times n$ matrices the determinant is the signed volume scale factor, and it is characterised by three properties: it is linear in each row, it changes sign when two rows are swapped, and it equals 1 on the identity. From these alone you can derive the familiar expansion formulas, and Gaussian elimination gives an efficient way to compute it. The determinant appears in the change-of-variables formula for multiple integrals (the *Jacobian*), in the formula for the inverse of a matrix, and, as we saw in Section 5.4, in the discriminant of a polynomial.

The thinking skill

Probe an unknown function with special inputs

When a function is known only through a property, apply the property to the simplest special inputs you can think of, and see what it forces.

Here the identity, a swap, a scaling and a shear were enough to determine almost everything. Choosing inputs that are their own squares, inverses or conjugates is particularly effective.

Try this next

1 **Other examples.** Show that $\mathrm{f}(\mathbf{M}) = (\det \mathbf{M})^3$ satisfies all the conditions of the question, and that $\mathrm{f}(\mathbf{M}) = |\det \mathbf{M}|$ satisfies the multiplicative property but not $\mathrm{f}(\mathbf{J}) \neq 1$.

2 **Column operations.** Show that f is also zero on any matrix whose second column is a multiple of the first.

3 **Scalings.** Let $\mathbf{K}_k = \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$. Show that $\mathrm{f}\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix} = \mathrm{f}(\mathbf{K}_k)$, and that $\mathrm{f}(\mathbf{K}_k)\mathrm{f}(\mathbf{K}_l) = \mathrm{f}(\mathbf{K}_{kl})$.

4 Reduction to the determinant. HARDER Show that every invertible 2×2 matrix can be written as a product of matrices of the forms $\mathbf{J}$, $\mathbf{K}_k$ and $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$. Deduce that $\mathrm{f}(\mathbf{M})$ depends only on $\det \mathbf{M}$.

Further reading

- **Sheldon Axler, *Linear Algebra Done Right*** (Springer; free online). Defines the determinant from its properties, in the spirit of this question.
- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge). Chapter 5 develops determinants from three rules.
- **3Blue1Brown, "The determinant"** (video, free online). The area interpretation, beautifully animated.

7.4 Matrices That Return to the Start

The question

A 2×2 matrix $\mathbf{M}$ is real if it can be written as $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where a, b, c and d are real. In this case, the *trace* of matrix $\mathbf{M}$ is defined to be $\mathrm{tr}(\mathbf{M}) = a + d$ and $\det(\mathbf{M})$ is the determinant of matrix $\mathbf{M}$. In this question, $\mathbf{M}$ is a real 2×2 matrix.

(i) Prove that
$$\mathrm{tr}(\mathbf{M}^2) = \mathrm{tr}(\mathbf{M})^2 - 2\det(\mathbf{M}).$$

(ii) Prove that
$$\mathbf{M}^2 = \mathbf{I} \text{ but } \mathbf{M} \neq \pm\mathbf{I} \iff \mathrm{tr}(\mathbf{M}) = 0 \text{ and } \det(\mathbf{M}) = -1,$$
and that
$$\mathbf{M}^2 = -\mathbf{I} \iff \mathrm{tr}(\mathbf{M}) = 0 \text{ and } \det(\mathbf{M}) = 1.$$

(iii) Use part (ii) to prove that
$$\mathbf{M}^4 = \mathbf{I} \iff \mathbf{M}^2 = \pm\mathbf{I}.$$
Find a necessary and sufficient condition on $\det(\mathbf{M})$ and $\mathrm{tr}(\mathbf{M})$ so that $\mathbf{M}^4 = -\mathbf{I}$.

(iv) Give an example of a matrix $\mathbf{M}$ for which $\mathbf{M}^8 = \mathbf{I}$, but which does not represent a rotation or reflection. [Note that the matrices $\pm\mathbf{I}$ are both rotations.]

The hook

Some transformations, repeated enough times, bring everything back to where it started. A rotation by $90°$ does it in four steps, a reflection in two. Which 2×2 matrices have this property, $\mathbf{M}^n = \mathbf{I}$ for some n, and how can you tell?

The surprise of this question is that two numbers, the *trace* and the *determinant*, are enough to decide. You do not need to know the four entries, only $a + d$ and $ad - bc$. And the answer has a striking consequence for the physical world: it explains why crystals can have 2-, 3-, 4- and 6-fold symmetry but never 5-fold, a "law" of crystallography that held for two centuries until a discovery in 1982 won a Nobel Prize.

Before you start

Ask yourself Every 2×2 matrix satisfies $\mathbf{M}^2 - (\operatorname{tr}\mathbf{M})\mathbf{M} + (\det\mathbf{M})\mathbf{I} = \mathbf{0}$ (the Cayley–Hamilton theorem; check it by multiplying out). How could that help with questions about $\mathbf{M}^2$?

It expresses $\mathbf{M}^2$ in terms of $\mathbf{M}$ and $\mathbf{I}$: $\mathbf{M}^2 = t\mathbf{M} - d\mathbf{I}$, where $t = \operatorname{tr}\mathbf{M}$ and $d = \det\mathbf{M}$. So a condition on $\mathbf{M}^2$ becomes a condition linking $\mathbf{M}$, t and d. That is the key to the whole question.

Working it through

STAGE 1 The trace of the square

$\mathbf{M}^2 = \begin{pmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{pmatrix}$, so $\operatorname{tr}(\mathbf{M}^2) = a^2 + 2bc + d^2 = (a+d)^2 - 2(ad-bc)$. ■

STAGE 2 Square roots of $\pm\mathbf{I}$

Ask yourself Suppose $\mathbf{M}^2 = \mathbf{I}$. Substitute into $\mathbf{M}^2 = t\mathbf{M} - d\mathbf{I}$. If $t \neq 0$, what does that say about $\mathbf{M}$?

If $\mathbf{M}^2 = \mathbf{I}$, then $\mathbf{I} = t\mathbf{M} - d\mathbf{I}$, so $t\mathbf{M} = (1+d)\mathbf{I}$. If $t \neq 0$, then $\mathbf{M}$ is a multiple of $\mathbf{I}$, say $c\mathbf{I}$, and $c^2 = 1$, so $\mathbf{M} = \pm\mathbf{I}$, which is excluded. Hence $t = 0$, and then $(1+d)\mathbf{I} = \mathbf{0}$, so $d = -1$.

Conversely, if $t = 0$ and $d = -1$, then $\mathbf{M}^2 = t\mathbf{M} - d\mathbf{I} = \mathbf{I}$, and $\mathbf{M} \neq \pm\mathbf{I}$ because $\operatorname{tr}(\pm\mathbf{I}) = \pm 2 \neq 0$. ■

For $\mathbf{M}^2 = -\mathbf{I}$: substituting gives $t\mathbf{M} = (d-1)\mathbf{I}$. If $t \neq 0$ then $\mathbf{M} = c\mathbf{I}$ with $c^2 = -1$, impossible for real c. So $t = 0$ and $d = 1$. Conversely, $t = 0$ and $d = 1$ give $\mathbf{M}^2 = -\mathbf{I}$. ■

STAGE 3 Fourth roots

Pause and try Let $\mathbf{N} = \mathbf{M}^2$, so $\mathbf{M}^4 = \mathbf{I}$ means $\mathbf{N}^2 = \mathbf{I}$. By part (ii), either $\mathbf{N} = \pm\mathbf{I}$ or $\operatorname{tr}\mathbf{N} = 0$ and $\det\mathbf{N} = -1$. Why is the second case impossible?

$\det\mathbf{N} = \det(\mathbf{M}^2) = (\det\mathbf{M})^2 \geq 0$, so $\det\mathbf{N} \neq -1$. Hence $\mathbf{N} = \mathbf{M}^2 = \pm\mathbf{I}$. The converse is clear. ■

For $\mathbf{M}^4 = -\mathbf{I}$: by part (ii) applied to $\mathbf{N} = \mathbf{M}^2$, this holds exactly when $\operatorname{tr}(\mathbf{M}^2) = 0$ and $\det(\mathbf{M}^2) = 1$, that is, $t^2 - 2d = 0$ and $d^2 = 1$. Since $t^2 = 2d \geq 0$, we need $d = 1$, and then $t^2 = 2$. So

$$\mathbf{M}^4 = -\mathbf{I} \iff \det\mathbf{M} = 1 \text{ and } \operatorname{tr}\mathbf{M} = \pm\sqrt{2}. \quad ■$$

STAGE 4 An eighth root of $\mathbf{I}$ that is not a rotation

Any matrix with $\mathbf{M}^4 = -\mathbf{I}$ has $\mathbf{M}^8 = \mathbf{I}$. A rotation by 45° has trace $\sqrt{2}$ and determinant 1; we want another matrix with the same trace and determinant that is not a rotation. Take

$$\mathbf{M} = \begin{pmatrix} \sqrt{2} & -1 \\ 1 & 0 \end{pmatrix}, \qquad \operatorname{tr}\mathbf{M} = \sqrt{2}, \quad \det\mathbf{M} = 1.$$

By Stage 3, $\mathbf{M}^4 = -\mathbf{I}$, so $\mathbf{M}^8 = \mathbf{I}$. But $\mathbf{M}$ is not a rotation or reflection, because those preserve lengths, and $\mathbf{M}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{2} \\ 1 \end{pmatrix}$ has length $\sqrt{3} \neq 1$. ■

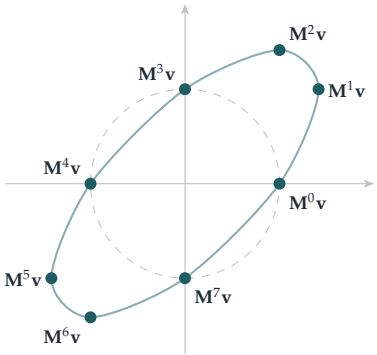


Figure 7.4.1 The orbit of $\mathbf{v} = (1, 0)$ under $\mathbf{M} = \begin{pmatrix} \sqrt{2} & -1 \\ 1 & 0 \end{pmatrix}$. After eight steps it returns to the start. The points do not lie on a circle (dashed), so $\mathbf{M}$ is not a rotation, but they do lie on an ellipse: $\mathbf{M}$ is a 45° rotation viewed in skewed coordinates.

Looking back

- **Cayley–Hamilton turns powers into linear combinations.** Every power of a 2×2 matrix is a combination of $\mathbf{M}$ and $\mathbf{I}$, with coefficients depending only on trace and determinant.
- **Determinants of squares are non-negative.** That simple fact ruled out a whole case in part (iii).
- **Trace and determinant are invariants.** Two matrices with the same trace and determinant (and not multiples of $\mathbf{I}$) behave alike under powers, whether or not they look alike.

What's really going on

Similar matrices

Why is $\mathbf{M}$ in Stage 4 "a rotation in disguise"? Two matrices $\mathbf{M}$ and $\mathbf{R}$ are *similar* if $\mathbf{M} = \mathbf{S}\mathbf{R}\mathbf{S}^{-1}$ for some invertible $\mathbf{S}$: they represent the same transformation in different coordinates. Similar matrices have the same trace and determinant, and (for 2×2 matrices that are not multiples of $\mathbf{I}$) the converse holds too. So any real matrix with trace $\sqrt{2}$ and determinant 1 is similar to the rotation by 45°, and in the skewed coordinates given by $\mathbf{S}$ it simply rotates (Figure 7.4.1). In general, a real 2×2 matrix with $\mathbf{M}^n = \mathbf{I}$ is always similar to a rotation or a reflection.

Why crystals have no five-fold symmetry

A crystal is a regular, repeating arrangement of atoms. A rotational symmetry of the crystal sends the lattice of atoms to itself, so in coordinates based on the lattice it is given by a matrix with *integer* entries. Its trace is therefore an integer. But a rotation by angle θ has trace $2\cos\theta$ in any coordinates, and $2\cos\theta$ is an integer only when it is $-2, -1, 0, 1$ or 2, that is, for rotations of order 2, 3, 4 and 6 (and 1). Five-fold symmetry would need $2\cos 72^\circ = \frac{\sqrt{5}-1}{2}$, which is not an integer. This is the *crystallographic restriction theorem*. In 1982 Dan Shechtman observed an alloy with apparent five-fold symmetry and was at first ridiculed; it turned out to be a *quasicrystal*, ordered but not periodic, like the Penrose tilings, and so not covered by the theorem. He received the Nobel Prize in Chemistry in 2011.

The thinking skill

Look for invariants that capture the essential behaviour

When a question is about an object with many parameters, ask whether a few combinations of them, invariant under a change of viewpoint, are enough.

Here four entries reduced to two invariants, the trace and the determinant, because the question ("does $\mathbf{M}^n = \mathbf{I}$?") does not depend on the coordinates. Finding the right invariants is often the key to classification problems.

Try this next

1 **Rotations of order three.** (From STEP II, 2021.) The matrix $\mathbf{R}$ represents an anticlockwise rotation through angle ϕ, and $\mathbf{R} + \mathbf{I}$ also represents a rotation. Find the possible values of ϕ and deduce that $\mathbf{R}^3 = \mathbf{I}$. Then show that if $\mathbf{S}$ is a real matrix with $\mathbf{S}^3 = \mathbf{I}$ but $\mathbf{S} \neq \mathbf{I}$, then $\det \mathbf{S} = 1$ and $\operatorname{tr} \mathbf{S} = -1$.

2 **Order six.** Find a condition on the trace and determinant for $\mathbf{M}^6 = \mathbf{I}$ but $\mathbf{M}^2 \neq \mathbf{I}$ and $\mathbf{M}^3 \neq \mathbf{I}$.

3 **The crystallographic restriction.** Show that if $\mathbf{M}$ has integer entries, $\det \mathbf{M} = 1$ and $\mathbf{M}^n = \mathbf{I}$ for some n, then $\operatorname{tr} \mathbf{M} \in \{-2, -1, 0, 1, 2\}$. Find an integer matrix of order 6.

4 **Cayley–Hamilton for powers.** Use $\mathbf{M}^2 = t\mathbf{M} - d\mathbf{I}$ to express $\mathbf{M}^3$ and $\mathbf{M}^4$ in the form $p\mathbf{M} + q\mathbf{I}$, and use this to rederive the condition for $\mathbf{M}^4 = -\mathbf{I}$.

Further reading

- **M. A. Armstrong, *Groups and Symmetry*** (Springer). Includes the crystallographic restriction and the classification of wallpaper patterns.
- **Marjorie Senechal, *Quasicrystals and Geometry*** (Cambridge University Press). The mathematics of quasicrystals and aperiodic order.
- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge). Similar matrices and the Cayley–Hamilton theorem.

7.5 Rotating in Three Dimensions

The question

Let $\mathbf{n}$ be a vector of unit length in three dimensions. For each vector $\mathbf{r}$, $\mathrm{f}(\mathbf{r})$ is defined by $\mathrm{f}(\mathbf{r}) = \mathbf{n} \times \mathbf{r}$.

(i) Given that

$$\mathbf{n} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \quad \text{and} \quad \mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix},$$

show that the x-component of $\mathrm{f}(\mathrm{f}(\mathbf{r}))$ is $-x(b^2 + c^2) + aby + acz$. Show further that

$$\mathrm{f}(\mathrm{f}(\mathbf{r})) = (\mathbf{n}.\mathbf{r})\mathbf{n} - \mathbf{r}.$$

Explain, by means of a diagram, how $\mathrm{f}(\mathrm{f}(\mathbf{r}))$ is related to $\mathbf{n}$ and $\mathbf{r}$.

(ii) Let R be the point with position vector $\mathbf{r}$ and P be the point with position vector $\mathrm{g}(\mathbf{r})$, where g is defined by

$$\mathrm{g}(\mathbf{s}) = \mathbf{s} + \sin\theta\,\mathrm{f}(\mathbf{s}) + (1 - \cos\theta)\,\mathrm{f}(\mathrm{f}(\mathbf{s})).$$

By considering $\mathrm{g}(\mathbf{n})$ and $\mathrm{g}(\mathbf{r})$ when $\mathbf{r}$ is perpendicular to $\mathbf{n}$, state, with justification, the geometric transformation which maps R onto P.

(iii) Let R be the point with position vector $\mathbf{r}$ and Q be the point with position vector $\mathrm{h}(\mathbf{r})$, where h is defined by

$$\mathrm{h}(\mathbf{s}) = -\mathbf{s} - 2\,\mathrm{f}(\mathrm{f}(\mathbf{s})).$$

State, with justification, the geometric transformation which maps R onto Q.

The hook

In two dimensions, rotating a point is easy: multiply by $e^{\mathrm{i}\theta}$. In three dimensions it is much harder. There is no single complex number to multiply by, rotations about different axes do not commute, and writing down the matrix for "rotate by 37° about the axis through $(1, 2, 2)$" looks daunting.

This question builds that matrix, for any axis and any angle, out of the cross product. The result is *Rodrigues' rotation formula*, published by Olinde Rodrigues in 1840. It is used every day in robotics, computer graphics and spacecraft navigation, and it reveals

a lovely fact: crossing with $\mathbf{n}$ is "rotation by 90° about $\mathbf{n}$", and every rotation about $\mathbf{n}$ is built from it, just as $e^{i\theta} = \cos\theta + i\sin\theta$ is built from multiplication by i.

Before you start

Ask yourself What does $\mathbf{n} \times \mathbf{r}$ look like geometrically, when $\mathbf{r}$ is perpendicular to $\mathbf{n}$? And when $\mathbf{r}$ is parallel to $\mathbf{n}$?

If $\mathbf{r} \parallel \mathbf{n}$, then $\mathbf{n} \times \mathbf{r} = \mathbf{0}$. If $\mathbf{r} \perp \mathbf{n}$, then $\mathbf{n} \times \mathbf{r}$ has the same length as $\mathbf{r}$ (since $|\mathbf{n}| = 1$ and the angle is 90°), is perpendicular to both, and by the right-hand rule is $\mathbf{r}$ rotated by 90° anticlockwise about $\mathbf{n}$ (looking down $\mathbf{n}$). So f kills the component of $\mathbf{r}$ along $\mathbf{n}$ and rotates the perpendicular component by 90°. That picture explains the whole question.

Working it through

STAGE 1 Crossing twice

$\mathrm{f}(\mathbf{r}) = \mathbf{n} \times \mathbf{r} = (bz - cy,\ cx - az,\ ay - bx)$. Crossing again with $\mathbf{n} = (a, b, c)$, the x-component of $\mathrm{f}(\mathrm{f}(\mathbf{r}))$ is

$$b(ay - bx) - c(cx - az) = -x(b^2 + c^2) + aby + acz. \qquad \blacksquare$$

Since $a^2 + b^2 + c^2 = 1$, this is $-x(1 - a^2) + aby + acz = a(ax + by + cz) - x$, which is the x-component of $(\mathbf{n}.\mathbf{r})\mathbf{n} - \mathbf{r}$. The other components follow by symmetry, so $\mathrm{f}(\mathrm{f}(\mathbf{r})) = (\mathbf{n}.\mathbf{r})\mathbf{n} - \mathbf{r}$. $\blacksquare$

Ask yourself Write $\mathbf{r} = \mathbf{r}_\parallel + \mathbf{r}_\perp$, where $\mathbf{r}_\parallel = (\mathbf{n}.\mathbf{r})\mathbf{n}$ is the component along $\mathbf{n}$. What is $\mathrm{f}(\mathrm{f}(\mathbf{r}))$ in terms of these?

$\mathrm{f}(\mathrm{f}(\mathbf{r})) = \mathbf{r}_\parallel - \mathbf{r} = -\mathbf{r}_\perp$. Applying f twice kills the parallel part and rotates the perpendicular part by 180° about $\mathbf{n}$, which simply reverses it (Figure 7.5.1). This is the analogue of $i^2 = -1$.

STAGE 2 The rotation

Ask yourself Compute $\mathrm{g}(\mathbf{n})$. Then compute $\mathrm{g}(\mathbf{r})$ for $\mathbf{r} \perp \mathbf{n}$, using $\mathrm{f}(\mathrm{f}(\mathbf{r})) = -\mathbf{r}$ for such $\mathbf{r}$.

$\mathrm{f}(\mathbf{n}) = \mathbf{n} \times \mathbf{n} = \mathbf{0}$, so $\mathrm{g}(\mathbf{n}) = \mathbf{n}$: the axis is fixed. If $\mathbf{r} \perp \mathbf{n}$, then $\mathrm{f}(\mathrm{f}(\mathbf{r})) = -\mathbf{r}$ and

$$\mathrm{g}(\mathbf{r}) = \mathbf{r} + \sin\theta\,(\mathbf{n} \times \mathbf{r}) - (1 - \cos\theta)\mathbf{r} = \cos\theta\,\mathbf{r} + \sin\theta\,(\mathbf{n} \times \mathbf{r}).$$

Since $\mathbf{n} \times \mathbf{r}$ is $\mathbf{r}$ turned through 90° about $\mathbf{n}$, with the same length, this is $\mathbf{r}$ rotated through θ about $\mathbf{n}$, exactly like $(\cos\theta + i\sin\theta)z$ in the plane. Since g is linear and every vector is a sum of a multiple of $\mathbf{n}$ and a vector perpendicular to $\mathbf{n}$, g is the **rotation**

through angle θ about the axis through the origin in the direction $\mathbf{n}$ (anticlockwise when viewed from the tip of $\mathbf{n}$). ■

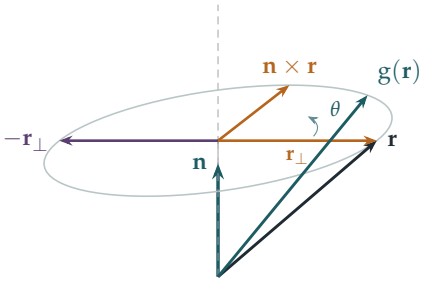


Figure 7.5.1 Rodrigues' formula. The vector $\mathbf{r}$ splits into a part along the axis $\mathbf{n}$ and a part $\mathbf{r}_\perp$ perpendicular to it, which sweeps out a circle. Crossing with $\mathbf{n}$ turns $\mathbf{r}_\perp$ through 90° (amber); crossing twice gives $-\mathbf{r}_\perp$ (plum). The rotated vector $g(\mathbf{r})$ is $\mathbf{r}_\parallel + \cos\theta\,\mathbf{r}_\perp + \sin\theta\,(\mathbf{n}\times\mathbf{r})$.

STAGE 3 The reflection

Pause and try Simplify $h(\mathbf{s}) = -\mathbf{s} - 2f(f(\mathbf{s}))$ using part (i). What does it do to vectors along $\mathbf{n}$, and to vectors perpendicular to $\mathbf{n}$?

$h(\mathbf{s}) = -\mathbf{s} - 2\big((\mathbf{n}.\mathbf{s})\mathbf{n} - \mathbf{s}\big) = \mathbf{s} - 2(\mathbf{n}.\mathbf{s})\mathbf{n}$. Vectors perpendicular to $\mathbf{n}$ are unchanged, and $h(\mathbf{n}) = -\mathbf{n}$. So h is the **reflection in the plane through the origin perpendicular to $\mathbf{n}$**. ■

Looking back

- **Decompose into parallel and perpendicular parts.** Every step became clear once $\mathbf{r}$ was split into its components along and across $\mathbf{n}$.
- **Linearity reduces the work.** To identify a linear map, it is enough to know what it does to a few well-chosen vectors.
- **An analogue of complex numbers.** On vectors perpendicular to $\mathbf{n}$, f behaves like multiplication by i: $f^2 = -1$, and $g = \cos\theta + \sin\theta\, f$.

What's really going on

Euler's rotation theorem

Every rotation of three-dimensional space that fixes the origin is a rotation about some axis through the origin. Euler proved this in 1775. In matrix terms, every 3×3 rotation matrix has an eigenvector with eigenvalue 1, the axis. Rodrigues' formula gives the

converse: for every axis $\mathbf{n}$ and angle θ, the rotation matrix is

$$\mathbf{R} = \mathbf{I} + \sin\theta\,\mathbf{K} + (1-\cos\theta)\,\mathbf{K}^2, \qquad \mathbf{K} = \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix},$$

where $\mathbf{K}$ is the matrix of $\mathbf{r} \mapsto \mathbf{n} \times \mathbf{r}$. Since $\mathbf{K}^3 = -\mathbf{K}$, the exponential series $e^{\theta\mathbf{K}} = \mathbf{I} + \theta\mathbf{K} + \frac{\theta^2}{2!}\mathbf{K}^2 + \cdots$ collapses to exactly this formula, the three-dimensional version of $e^{i\theta} = \cos\theta + i\sin\theta$. This connection between rotations and exponentials of "infinitesimal rotations" $\mathbf{K}$ is the starting point of the theory of *Lie groups*.

Rotations do not commute

Hold a book flat, rotate it 90° about a vertical axis and then 90° about a horizontal axis; then start again and do the two rotations in the other order. The book ends up in different positions. Composition of rotations in three dimensions is not commutative, unlike in two dimensions. This is one reason the quaternions of Section 6.7 are useful: the rotation by θ about $\mathbf{n}$ corresponds to the quaternion $\cos\frac{\theta}{2} + \sin\frac{\theta}{2}(a\mathrm{i} + b\mathrm{j} + c\mathrm{k})$, and composing rotations is multiplying quaternions, with the non-commutativity built in. Rodrigues himself, in his 1840 paper, found the composition formula for rotations three years before Hamilton invented quaternions.

The thinking skill

Split into invariant parts

To understand a transformation, split space into pieces it treats simply: here, the axis (left alone) and the plane perpendicular to it (rotated).

This is the same idea as eigenvectors in Section 7.2: find subspaces the transformation preserves, and study it on each piece.

Try this next

1 **Rotations in the plane.** (From STEP III, 2017.) Show that rotation by θ about the point a in the complex plane sends z to $e^{i\theta}z + a(1-e^{i\theta})$. Show that rotating by θ about a and then by ϕ about b is a rotation by $\theta+\phi$ about some point c, provided $\theta+\phi$ is not a multiple of 2π. What is it when $\theta+\phi = 2\pi$?

2 **A specific rotation.** Find the matrix of the rotation by 120° about the axis in the direction $(1,1,1)$. Show that it permutes the coordinate axes cyclically.

3 **Powers of K.** Show that $\mathbf{K}^2\mathbf{r} = (\mathbf{n}.\mathbf{r})\mathbf{n} - \mathbf{r}$ and $\mathbf{K}^3 = -\mathbf{K}$, and deduce Rodrigues' formula from the exponential series.

4 **Two reflections make a rotation.** Show that reflecting in two planes through the origin, at angle α to each other, is a rotation by 2α about their line of intersection.

Further reading

- **Simon Altmann, *Rotations, Quaternions, and Double Groups*** (Dover). The definitive account of three-dimensional rotations, including Rodrigues' contribution.
- **Simon Altmann, "Hamilton, Rodrigues, and the quaternion scandal"**, *Mathematics Magazine* 62 (1989). A short, lively history.
- **John Stillwell, *Naive Lie Theory*** (Springer). Rotations, quaternions and Lie groups for undergraduates.

7.6 Hyperbolic Rotations and Relativity

The question

The transformation T from $\begin{pmatrix} x \\ y \end{pmatrix}$ to $\begin{pmatrix} x' \\ y' \end{pmatrix}$ in two-dimensional space is given by

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cosh u & \sinh u \\ \sinh u & \cosh u \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix},$$

where u is a positive real constant. Show that the curve with equation $x^2 - y^2 = 1$ is transformed into itself. Find the equations of two straight lines through the origin which transform into themselves.

A line, not necessarily through the origin, which has gradient $\tanh v$ transforms under T into a line with gradient $\tanh v'$. Show that $v' = v + u$.

The lines ℓ_1 and ℓ_2 with gradients $\tanh v_1$ and $\tanh v_2$ transform under T into lines with gradients $\tanh v_1'$ and $\tanh v_2'$ respectively. Find the relation satisfied by v_1 and v_2 that is necessary and sufficient for ℓ_1 and ℓ_2 to intersect at the same angle as their transforms.

In the case when ℓ_1 and ℓ_2 meet at the origin, illustrate in a diagram the relation between ℓ_1, ℓ_2 and their transforms.

The hook

A rotation matrix $\begin{pmatrix} \cos u & -\sin u \\ \sin u & \cos u \end{pmatrix}$ moves points round the circle $x^2 + y^2 = 1$, and rotating by u then by w is rotating by $u + w$. Replace cos and sin by cosh and sinh, adjust a sign, and you get the matrix in this question. It moves points along the hyperbola $x^2 - y^2 = 1$ instead, and it too adds its "angles". It is a *hyperbolic rotation*.

This is not just a curiosity. With x standing for time and y for distance (in units where the speed of light is 1), the matrix is a *Lorentz boost*: the transformation, discovered in Einstein's special relativity, between the measurements of two observers moving at constant velocity relative to each other. The "hyperbolic angle" u is the *rapidity*, and the fact that rapidities add explains the strange velocity-addition law we met in Section 3.8.

Before you start

Ask yourself What identity do $\cosh u$ and $\sinh u$ satisfy, analogous to $\cos^2 u + \sin^2 u = 1$? And what are the addition formulas for $\cosh(u+v)$ and $\sinh(u+v)$?

$\cosh^2 u - \sinh^2 u = 1$, and

$$\cosh(u+v) = \cosh u \cosh v + \sinh u \sinh v,$$
$$\sinh(u+v) = \sinh u \cosh v + \cosh u \sinh v.$$

You can check these from the definitions $\cosh u = \frac{1}{2}(e^u + e^{-u})$ and $\sinh u = \frac{1}{2}(e^u - e^{-u})$. The point $(\cosh v, \sinh v)$ lies on the hyperbola $x^2 - y^2 = 1$, just as $(\cos v, \sin v)$ lies on the circle.

Working it through

STAGE 1 The hyperbola is invariant

$$\begin{aligned} x'^2 - y'^2 &= (x\cosh u + y \sinh u)^2 - (x \sinh u + y \cosh u)^2 \\ &= (x^2 - y^2)(\cosh^2 u - \sinh^2 u) = x^2 - y^2, \end{aligned}$$

since the cross terms $2xy \cosh u \sinh u$ cancel. So T preserves $x^2 - y^2$, and in particular maps the hyperbola $x^2 - y^2 = 1$ to itself. ■

Ask yourself Which lines through the origin are mapped to themselves? (Look for eigenvectors.)

$T\begin{pmatrix}1\\1\end{pmatrix} = (\cosh u + \sinh u)\begin{pmatrix}1\\1\end{pmatrix} = e^u\begin{pmatrix}1\\1\end{pmatrix}$ and $T\begin{pmatrix}1\\-1\end{pmatrix} = e^{-u}\begin{pmatrix}1\\-1\end{pmatrix}$. So the lines $y = x$ and $y = -x$ are each mapped to themselves: one is stretched by e^u, the other shrunk by e^{-u}. These are the asymptotes of the hyperbola. ■

STAGE 2 Adding hyperbolic angles

A line with gradient $\tanh v$ has direction vector $(\cosh v, \sinh v)$. Its image has direction

$$T\begin{pmatrix}\cosh v\\ \sinh v\end{pmatrix} = \begin{pmatrix}\cosh u \cosh v + \sinh u \sinh v\\ \sinh u \cosh v + \cosh u \sinh v\end{pmatrix} = \begin{pmatrix}\cosh(u+v)\\ \sinh(u+v)\end{pmatrix},$$

which has gradient $\tanh(u+v)$. A linear map sends parallel lines to parallel lines, so this holds whether or not the line passes through the origin. Hence $v' = v + u$. ■

STAGE 3 When are angles preserved?

Ask yourself The angle ψ between two lines with gradients m_1, m_2 satisfies $\tan\psi = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$. Simplify this when $m_i = \tanh v_i$. (*Hint:* put everything over $\cosh v_1 \cosh v_2$.)

Using $\tanh v_1 - \tanh v_2 = \dfrac{\sinh(v_1 - v_2)}{\cosh v_1 \cosh v_2}$ and $1 + \tanh v_1 \tanh v_2 = \dfrac{\cosh(v_1 + v_2)}{\cosh v_1 \cosh v_2}$,

$$\tan\psi = \left|\frac{\sinh(v_1 - v_2)}{\cosh(v_1 + v_2)}\right|.$$

After the transformation, $v_1 - v_2$ is unchanged and $v_1 + v_2$ becomes $v_1 + v_2 + 2u$.

Pause and try When is $\cosh(v_1 + v_2) = \cosh(v_1 + v_2 + 2u)$?

Assuming the lines are not parallel ($v_1 \neq v_2$), the angles are equal exactly when $\cosh(v_1 + v_2) = \cosh(v_1 + v_2 + 2u)$. Since cosh is even and strictly increasing on $[0, \infty)$, this means $v_1 + v_2 = \pm(v_1 + v_2 + 2u)$. The plus sign gives $u = 0$, which is excluded, so

$$v_1 + v_2 = -u. \qquad \blacksquare$$

STAGE 4 The picture

When $v_1 + v_2 = -u$, the transformed angles are $v_1' = v_1 + u = -v_2$ and $v_2' = v_2 + u = -v_1$. So the image of ℓ_1 has gradient $-\tanh v_2$: it is the reflection of ℓ_2 in the x-axis, and the image of ℓ_2 is the reflection of ℓ_1. For lines through the origin, the transformation swaps the two lines and reflects them, which explains why the angle between them is unchanged (Figure 7.6.1).

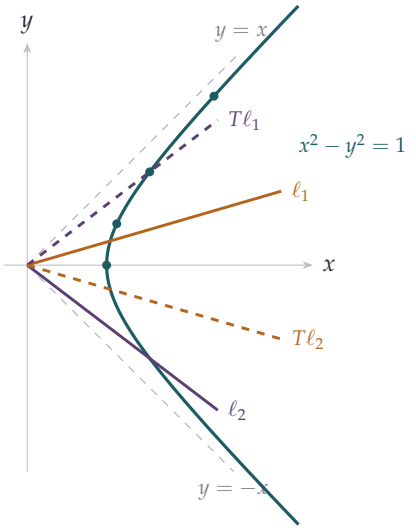


Figure 7.6.1 A hyperbolic rotation with $u = 0.7$ applied to lines ℓ_1, ℓ_2 through the origin with $v_1 = 0.3$, $v_2 = -1.0$, so that $v_1 + v_2 = -u$. The image of ℓ_1 (dashed plum) is the mirror image of ℓ_2, and the image of ℓ_2 (dashed amber) is the mirror image of ℓ_1, so the angle between the lines is preserved. The teal dots on the hyperbola are equally spaced in hyperbolic angle ($v = 0, 0.5, 1, 1.5$); T moves each one to the next point 0.7 further along.

Looking back

- **An invariant quadratic form.** T preserves $x^2 - y^2$, just as a rotation preserves $x^2 + y^2$.
- **Eigenvectors are the asymptotes.** The lines left invariant are $y = \pm x$, stretched and shrunk by $e^{\pm u}$.
- **Hyperbolic angles add.** Parametrising directions by $\tanh v$ turned the action of T into simple addition, $v \mapsto v + u$.

What's really going on

Special relativity

In Einstein's special relativity (1905), different observers moving at constant velocity relative to each other disagree about time and distance, but they all agree on the quantity $t^2 - x^2$ (with the speed of light $c = 1$), the *spacetime interval*. The transformations preserving it are the *Lorentz transformations*, and in one space dimension they are exactly the matrices in this question, with x as time and y as distance. An observer moving with velocity $v = \tanh u$ uses coordinates related to ours by this matrix, with $\cosh u = \frac{1}{\sqrt{1-v^2}}$, the famous factor γ of time dilation. Hermann Minkowski saw in

1908 that this makes spacetime a four-dimensional geometry with a "distance" that has a minus sign in it: "Henceforth space by itself, and time by itself, are doomed to fade away into mere shadows."

What the question means physically

- **The invariant lines** $y = \pm x$ are the paths of light rays. They are the same for every observer: the speed of light is the same for everyone, the founding principle of relativity.
- **The hyperbola** $x^2 - y^2 = 1$ is the set of events one unit of "proper time" from the origin, as measured by observers moving at all possible constant velocities.
- **Adding rapidities.** A line of gradient $\tanh v$ through the origin is the path of an object moving at velocity $\tanh v$. A boost by rapidity u changes its velocity to $\tanh(v+u) = \frac{\tanh v + \tanh u}{1 + \tanh v \tanh u}$, which is Einstein's velocity-addition formula from Section 3.8. Velocities do not add; rapidities do.

And in Section 5.8 we saw the same addition formula as the distance rule in the Poincaré disc: the set of possible velocities in relativity, measured with rapidity, is a hyperbolic space.

The thinking skill

Find the right parametrisation

Describe your objects by a parameter in which the transformation becomes simple.

Describing directions by their gradient, T acts by a messy formula; describing them by v with gradient $\tanh v$, it simply adds u. The same idea made rotations simple with angles, and made the map $x \mapsto x^2 - 2$ simple in Section 2.8.

Try this next

1. **Composing boosts.** Show that $T(u)T(w) = T(u+w)$, where $T(u)$ is the matrix in the question. What is $T(u)^{-1}$?
2. **Time dilation.** A spaceship travels at 0.8 times the speed of light. Find its rapidity u and $\cosh u$. If 10 years pass on Earth, how much time passes on the ship? (Use $t_{\text{ship}} = t_{\text{Earth}} / \cosh u$.)
3. **Hyperbolic area.** Show that the area enclosed by the hyperbola $x^2 - y^2 = 1$, the x-axis and the line from the origin to $(\cosh v, \sinh v)$ is $\frac{v}{2}$, just as the corresponding area for a circle is $\frac{\theta}{2}$. Why must T preserve such areas?

4 Squeeze. Change coordinates to $p = x + y$, $q = x - y$. Show that T becomes $p \mapsto e^u p$, $q \mapsto e^{-u} q$, and that the hyperbola becomes $pq = 1$. Describe the transformation geometrically in these coordinates.

Further reading

- **Edwin Taylor & John Wheeler, *Spacetime Physics*** (W. H. Freeman). Special relativity with rapidity and spacetime diagrams, for students with A-level mathematics.
- **Tevian Dray, *The Geometry of Special Relativity*** (CRC Press). Relativity as hyperbolic trigonometry, in exactly the spirit of this section.
- **Albert Einstein, *Relativity: The Special and the General Theory*** (various editions). Einstein's own popular account, still very readable.

7.7 Quantities That Never Change

The question

In this question, $\mathbf{M}$ and $\mathbf{N}$ are non-singular 2×2 matrices.

The *trace* of the matrix $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is defined as $\mathrm{tr}(\mathbf{M}) = a + d$.

(i) Prove that, for any two matrices $\mathbf{M}$ and $\mathbf{N}$, $\mathrm{tr}(\mathbf{MN}) = \mathrm{tr}(\mathbf{NM})$ and derive an expression for $\mathrm{tr}(\mathbf{M} + \mathbf{N})$ in terms of $\mathrm{tr}(\mathbf{M})$ and $\mathrm{tr}(\mathbf{N})$.

The entries in matrix $\mathbf{M}$ are functions of t and $\dfrac{\mathrm{d}\mathbf{M}}{\mathrm{d}t}$ denotes the matrix whose entries are the derivatives of the corresponding entries in $\mathbf{M}$.

(ii) Show that

$$\frac{1}{\det \mathbf{M}} \frac{\mathrm{d}}{\mathrm{d}t}(\det \mathbf{M}) = \mathrm{tr}\left(\mathbf{M}^{-1} \frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t}\right).$$

(iii) In this part, matrix $\mathbf{M}$ satisfies the differential equation

$$\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = \mathbf{MN} - \mathbf{NM},$$

where the entries in matrix $\mathbf{N}$ are also functions of t.

Show that $\det \mathbf{M}$, $\mathrm{tr}(\mathbf{M})$ and $\mathrm{tr}(\mathbf{M}^2)$ are independent of t.

In the case $\mathbf{N} = \begin{pmatrix} t & t \\ 0 & t \end{pmatrix}$, and given that $\mathbf{M} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ when $t = 0$, find $\mathbf{M}$ as a function of t.

(iv) In this part, matrix $\mathbf{M}$ satisfies the differential equation

$$\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = \mathbf{MN},$$

where the entries in matrix $\mathbf{N}$ are again functions of t.

The trace of $\mathbf{M}$ is non-zero and independent of t. Is it necessarily true that $\mathrm{tr}(\mathbf{N}) = 0$?

The hook

Some of the deepest facts in physics are *conservation laws*: energy, momentum and angular momentum stay the same however complicated the motion. When a system

has enough conserved quantities, its motion can be solved exactly; when it has too few, it can be chaotic.

This question shows a remarkable way of manufacturing conserved quantities. If a matrix changes in time according to $\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t} = \mathbf{MN} - \mathbf{NM}$, then however wildly its entries vary, its trace and determinant, and hence its eigenvalues, never change. Such a pair $(\mathbf{M}, \mathbf{N})$ is called a *Lax pair*, after Peter Lax, who introduced the idea in 1968 to explain the astonishing behaviour of *solitons*, solitary waves that pass through each other unchanged.

Before you start

Ask yourself The expression $\mathbf{MN} - \mathbf{NM}$ is called the *commutator* of $\mathbf{M}$ and $\mathbf{N}$. When is it zero? What is its trace, given part (i)?

It is zero exactly when $\mathbf{M}$ and $\mathbf{N}$ commute. And by part (i) its trace is $\operatorname{tr}(\mathbf{MN}) - \operatorname{tr}(\mathbf{NM}) = 0$, always. That single fact is the beginning of every conservation law in the question.

Working it through

STAGE 1 The trace

With $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{N} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$, the diagonal entries of $\mathbf{MN}$ are $ap + br$ and $cq + ds$, so $\operatorname{tr}(\mathbf{MN}) = ap + br + cq + ds$. Swapping the roles of $\mathbf{M}$ and $\mathbf{N}$ gives $pa + qc + rb + sd$, the same. Clearly $\operatorname{tr}(\mathbf{M} + \mathbf{N}) = \operatorname{tr}\mathbf{M} + \operatorname{tr}\mathbf{N}$. ■

STAGE 2 The derivative of a determinant

Ask yourself Differentiate $\det\mathbf{M} = ad - bc$ by the product rule. Then compute $\mathbf{M}^{-1}\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}t}$ and its trace, using $\mathbf{M}^{-1} = \frac{1}{\det\mathbf{M}}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

Writing dots for derivatives, $\frac{\mathrm{d}}{\mathrm{d}t}\det\mathbf{M} = \dot{a}d + a\dot{d} - \dot{b}c - b\dot{c}$. And

$$\mathbf{M}^{-1}\dot{\mathbf{M}} = \frac{1}{\det\mathbf{M}}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}\begin{pmatrix} \dot{a} & \dot{b} \\ \dot{c} & \dot{d} \end{pmatrix}$$

has diagonal entries $\frac{d\dot{a} - b\dot{c}}{\det\mathbf{M}}$ and $\frac{-c\dot{b} + a\dot{d}}{\det\mathbf{M}}$. Their sum is $\frac{1}{\det\mathbf{M}}\frac{\mathrm{d}}{\mathrm{d}t}\det\mathbf{M}$. ■

STAGE 3 Three conserved quantities

Ask yourself Take the trace of $\dot{\mathbf{M}} = \mathbf{MN} - \mathbf{NM}$. For the determinant, use part (ii): what is $\mathbf{M}^{-1}(\mathbf{MN} - \mathbf{NM})$?

Trace: $\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{tr}\mathbf{M} = \operatorname{tr}\dot{\mathbf{M}} = \operatorname{tr}(\mathbf{MN}) - \operatorname{tr}(\mathbf{NM}) = 0$.

Determinant: by part (ii),

$$\frac{1}{\det\mathbf{M}}\frac{\mathrm{d}}{\mathrm{d}t}\det\mathbf{M} = \operatorname{tr}\big(\mathbf{M}^{-1}(\mathbf{MN} - \mathbf{NM})\big) = \operatorname{tr}\mathbf{N} - \operatorname{tr}(\mathbf{M}^{-1}\mathbf{NM}) = \operatorname{tr}\mathbf{N} - \operatorname{tr}\mathbf{N} = 0,$$

using part (i) in the form $\operatorname{tr}\big(\mathbf{M}^{-1}(\mathbf{NM})\big) = \operatorname{tr}\big((\mathbf{NM})\mathbf{M}^{-1}\big) = \operatorname{tr}\mathbf{N}$.

Trace of the square: $\operatorname{tr}(\mathbf{M}^2) = (\operatorname{tr}\mathbf{M})^2 - 2\det\mathbf{M}$ (see Section 7.4), which is constant because both terms are. ∎

STAGE 4 An explicit solution

Pause and try With $\mathbf{N} = t\mathbf{I} + t\mathbf{E}$, where $\mathbf{E} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, compute $\mathbf{MN} - \mathbf{NM}$ and write down the differential equations for a, b, c, d. Solve them in the order c, then a and d, then b.

The part $t\mathbf{I}$ commutes with everything, so $\mathbf{MN} - \mathbf{NM} = t(\mathbf{ME} - \mathbf{EM})$. Now $\mathbf{ME} = \begin{pmatrix} 0 & a \\ 0 & c \end{pmatrix}$ and $\mathbf{EM} = \begin{pmatrix} c & d \\ 0 & 0 \end{pmatrix}$, so

$$\dot{a} = -tc, \qquad \dot{b} = t(a - d), \qquad \dot{c} = 0, \qquad \dot{d} = tc.$$

So $c = C$ is constant; then $a = A - \frac{1}{2}Ct^2$ and $d = D + \frac{1}{2}Ct^2$; then $\dot{b} = t(A - D - Ct^2)$, so $b = B + \frac{1}{2}(A - D)t^2 - \frac{1}{4}Ct^4$. Therefore

$$\mathbf{M} = \begin{pmatrix} A - \frac{1}{2}Ct^2 & B + \frac{1}{2}(A - D)t^2 - \frac{1}{4}Ct^4 \\ C & D + \frac{1}{2}Ct^2 \end{pmatrix}. \qquad ∎$$

Check: the trace is $A + D$, and the determinant is $(A - \frac{1}{2}Ct^2)(D + \frac{1}{2}Ct^2) - C\big(B + \frac{1}{2}(A - D)t^2 - \frac{1}{4}Ct^4\big) = AD - BC$, constant as promised (Figure 7.7.1).

STAGE 5 A different equation

Ask yourself If $\dot{\mathbf{M}} = \mathbf{MN}$, then $\mathbf{N} = \mathbf{M}^{-1}\dot{\mathbf{M}}$, and by part (ii) $\operatorname{tr}\mathbf{N}$ is the logarithmic derivative of $\det\mathbf{M}$. Can the trace of $\mathbf{M}$ stay constant while its determinant changes?

Certainly. Take $\mathbf{M} = \begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}$ for $|t| < 1$ (where $\det\mathbf{M} = 1 - t^2 \neq 0$). Its trace is constantly 2. Define $\mathbf{N} = \mathbf{M}^{-1}\dot{\mathbf{M}}$, so that $\dot{\mathbf{M}} = \mathbf{MN}$:

$$\mathbf{N} = \frac{1}{1 - t^2}\begin{pmatrix} 1 & -t \\ -t & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{1}{1 - t^2}\begin{pmatrix} -t & 1 \\ 1 & -t \end{pmatrix},$$

which is non-singular, with $\operatorname{tr}\mathbf{N} = \frac{-2t}{1-t^2} \neq 0$ for $t \neq 0$. So **no**, $\operatorname{tr}\mathbf{N}$ need not be zero. ■

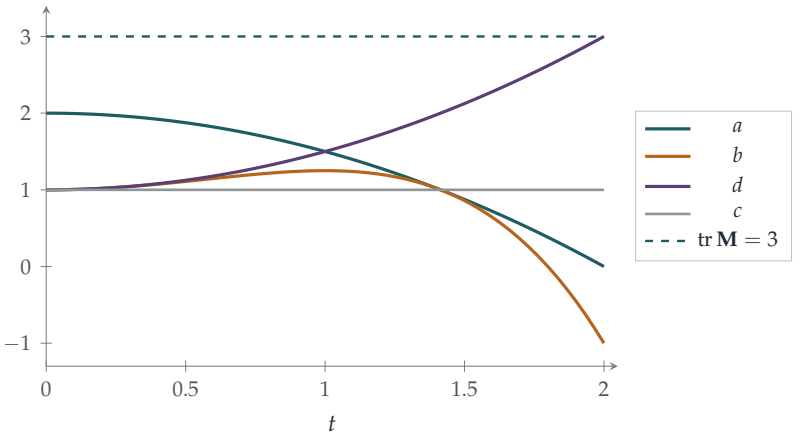


Figure 7.7.1 The entries of $\mathbf{M}(t)$ from Stage 4, with $A = 2$, $B = 1$, $C = 1$, $D = 1$. All four entries except c change, but the trace stays at 3 and the determinant stays at 1 throughout. So the eigenvalues, $\frac{3\pm\sqrt{5}}{2}$, never change: the flow is *isospectral*.

Looking back

- **The trace of a commutator is zero.** From $\operatorname{tr}(\mathbf{MN}) = \operatorname{tr}(\mathbf{NM})$, every conservation law followed.
- **Logarithmic derivatives.** Part (ii) turned the derivative of a determinant into a trace, which is linear and easy to handle.
- **Check conservation in the explicit solution.** The formula for $\mathbf{M}(t)$ was checked by confirming that its trace and determinant are constant.

What's really going on

Solitons and Lax pairs

In 1834 the Scottish engineer John Scott Russell followed, on horseback, a solitary wave travelling along the Union Canal near Edinburgh for over a mile without losing its shape. Such waves are described by the *Korteweg–de Vries (KdV) equation*, and in the 1960s computer experiments by Norman Zabusky and Martin Kruskal showed that two of these "solitons" can collide and emerge unchanged, as if they were particles. Explaining this needed infinitely many conserved quantities. Peter Lax showed in 1968 that the KdV equation can be written as $\dot{L} = LP - PL$ for a pair of differential operators L and P; then, exactly as in this question, the eigenvalues of L are conserved, and they provide the infinitely many conservation laws. Systems with a Lax pair are

called *integrable*, and they are among the few nonlinear systems that can be solved exactly.

Isospectral flows

Since the trace and determinant of $\mathbf{M}$ determine its characteristic polynomial $\lambda^2 - (\operatorname{tr}\mathbf{M})\lambda + \det\mathbf{M}$, the eigenvalues of $\mathbf{M}$ do not change with time. The flow $\dot{\mathbf{M}} = [\mathbf{M}, \mathbf{N}]$ moves $\mathbf{M}$ around inside the set of matrices with fixed eigenvalues. In fact, the solution is always of the form $\mathbf{M}(t) = \mathbf{S}(t)^{-1}\mathbf{M}(0)\mathbf{S}(t)$ for some invertible $\mathbf{S}(t)$: the matrix stays similar to its starting value. The same idea runs through physics: in quantum mechanics, an observable $\mathbf{A}$ evolves by $\frac{d\mathbf{A}}{dt} = \frac{i}{\hbar}[\mathbf{H}, \mathbf{A}]$, where $\mathbf{H}$ is the energy, so anything that commutes with the energy is conserved.

Jacobi's formula

Part (ii) holds for invertible matrices of any size and is called *Jacobi's formula*. It is safest written without the logarithm, as $\frac{d}{dt}\det\mathbf{M} = \det\mathbf{M}\,\operatorname{tr}\!\left(\mathbf{M}^{-1}\dot{\mathbf{M}}\right)$, because $\det\mathbf{M}$ may be negative; the logarithmic form $\frac{d}{dt}\ln\det\mathbf{M} = \operatorname{tr}\!\left(\mathbf{M}^{-1}\dot{\mathbf{M}}\right)$ needs $\det\mathbf{M} > 0$ (or $\ln|\det\mathbf{M}|$ in general). A consequence: if $\dot{\mathbf{M}} = \mathbf{A}\mathbf{M}$ for a constant matrix $\mathbf{A}$, then $\det\mathbf{M}(t) = e^{t\operatorname{tr}\mathbf{A}}\det\mathbf{M}(0)$. In the study of differential equations this tells you how volumes in phase space grow or shrink; for the equations of classical mechanics the trace is zero and volume is preserved, a result known as *Liouville's theorem*.

The thinking skill

Find what does not change

When something evolves, look for quantities built from it that stay constant. They constrain the motion and often reveal its structure.

Here the trace and determinant were conserved because the rate of change was a commutator. In mechanics, conserved quantities come from symmetries (Noether's theorem), and finding them is often the first step in solving a problem.

Try this next

1. **Check the solution.** Verify by differentiating that the matrix $\mathbf{M}(t)$ in Stage 4 satisfies $\dot{\mathbf{M}} = \mathbf{M}\mathbf{N} - \mathbf{N}\mathbf{M}$.

2. **Similarity.** Show that if $\mathbf{M}(t) = \mathbf{S}^{-1}\mathbf{M}_0\mathbf{S}$, where $\mathbf{M}_0$ is constant and $\mathbf{S}(t)$ satisfies $\dot{\mathbf{S}} = \mathbf{S}\mathbf{N}$, then $\dot{\mathbf{M}} = \mathbf{M}\mathbf{N} - \mathbf{N}\mathbf{M}$. (*Hint:* $\frac{d}{dt}\mathbf{S}^{-1} = -\mathbf{S}^{-1}\dot{\mathbf{S}}\mathbf{S}^{-1}$.)

3 **Volume growth.** Let $\mathbf{M}(t) = \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{-t} \end{pmatrix}$. Find $\mathbf{N}$ with $\dot{\mathbf{M}} = \mathbf{M}\mathbf{N}$, and check that $\frac{\mathrm{d}}{\mathrm{d}t} \ln \det \mathbf{M} = \operatorname{tr} \mathbf{N}$.

4 **Higher powers.** Show directly that $\frac{\mathrm{d}}{\mathrm{d}t} \operatorname{tr}(\mathbf{M}^k) = 0$ for every positive integer k when $\dot{\mathbf{M}} = \mathbf{M}\mathbf{N} - \mathbf{N}\mathbf{M}$.

Further reading

- **P. G. Drazin & R. S. Johnson, *Solitons: An Introduction*** (Cambridge University Press). The standard undergraduate introduction, including Lax pairs.
- **Thierry Dauxois & Michel Peyrard, *Physics of Solitons*** (Cambridge University Press). Includes the story of Scott Russell's wave.
- **Dwight Neuenschwander, *Emmy Noether's Wonderful Theorem*** (Johns Hopkins University Press). How symmetries give conservation laws.

7.8 Four Squares

The question

Let a, b, c, d, p, q, r and s be real numbers. By considering the determinant of the matrix product

$$\begin{pmatrix} z_1 & z_2 \\ -z_2^* & z_1^* \end{pmatrix} \begin{pmatrix} z_3 & z_4 \\ -z_4^* & z_3^* \end{pmatrix},$$

where z_1, z_2, z_3 and z_4 are suitably chosen complex numbers, find expressions L_1, L_2, L_3 and L_4, each of which is linear in a, b, c and d and also linear in p, q, r and s, such that

$$(a^2 + b^2 + c^2 + d^2)(p^2 + q^2 + r^2 + s^2) = L_1^2 + L_2^2 + L_3^2 + L_4^2.$$

The hook

7 is not a sum of two squares, or of three. But it is a sum of four: $7 = 4 + 1 + 1 + 1$. In 1770 Joseph-Louis Lagrange proved that *every* positive whole number is a sum of four squares (allowing 0^2). The theorem had been conjectured by Bachet in 1621 and claimed by Fermat, who as usual left no proof; Euler worked on it for forty years without quite finishing.

The key tool, found by Euler in 1748, is in this question: a product of two sums of four squares is itself a sum of four squares. It means you only need to prove Lagrange's theorem for prime numbers, and the rest follows by multiplying. The question derives the identity in a few lines from determinants of complex matrices, which are the quaternions of Section 6.7 in disguise.

Before you start

Ask yourself Recall the two-square version: $(a^2 + b^2)(p^2 + q^2)$ is a sum of two squares because $|z|^2|w|^2 = |zw|^2$ for complex numbers $z = a + b\mathrm{i}$, $w = p + q\mathrm{i}$. How could matrices extend this to four squares?

The idea is to find a kind of "number" made from four real numbers whose "size squared" is $a^2 + b^2 + c^2 + d^2$ and which is multiplicative. Matrices are multiplicative under determinants: $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$. So we want a matrix built from a, b, c, d whose determinant is $a^2 + b^2 + c^2 + d^2$, and such that products of such matrices are again of the same form.

Working it through

STAGE 1 The determinant is a sum of four squares

Ask yourself What is the determinant of $\begin{pmatrix} z_1 & z_2 \\ -z_2^* & z_1^* \end{pmatrix}$? If $z_1 = a + b\mathrm{i}$ and $z_2 = c + d\mathrm{i}$, what is it in terms of a, b, c, d?

$\det = z_1 z_1^* + z_2 z_2^* = |z_1|^2 + |z_2|^2 = a^2 + b^2 + c^2 + d^2$. Similarly with $z_3 = p + q\mathrm{i}$ and $z_4 = r + s\mathrm{i}$, the second matrix has determinant $p^2 + q^2 + r^2 + s^2$.

STAGE 2 The product has the same form

Multiplying,

$$\begin{pmatrix} z_1 & z_2 \\ -z_2^* & z_1^* \end{pmatrix} \begin{pmatrix} z_3 & z_4 \\ -z_4^* & z_3^* \end{pmatrix} = \begin{pmatrix} z_1 z_3 - z_2 z_4^* & z_1 z_4 + z_2 z_3^* \\ -z_2^* z_3 - z_1^* z_4^* & -z_2^* z_4 + z_1^* z_3^* \end{pmatrix}.$$

Writing $w_1 = z_1 z_3 - z_2 z_4^*$ and $w_2 = z_1 z_4 + z_2 z_3^*$, the bottom row is $(-w_2^*, w_1^*)$, so the product is $\begin{pmatrix} w_1 & w_2 \\ -w_2^* & w_1^* \end{pmatrix}$, of the same form. Its determinant is $|w_1|^2 + |w_2|^2$, and since determinants multiply,

$$(a^2 + b^2 + c^2 + d^2)(p^2 + q^2 + r^2 + s^2) = |w_1|^2 + |w_2|^2.$$

STAGE 3 Reading off the four squares

Pause and try Expand $w_1 = (a + b\mathrm{i})(p + q\mathrm{i}) - (c + d\mathrm{i})(r - s\mathrm{i})$ and $w_2 = (a + b\mathrm{i})(r + s\mathrm{i}) + (c + d\mathrm{i})(p - q\mathrm{i})$ into real and imaginary parts.

$$w_1 = (ap - bq - cr - ds) + \mathrm{i}(aq + bp + cs - dr),$$
$$w_2 = (ar - bs + cp + dq) + \mathrm{i}(as + br - cq + dp).$$

So

$$L_1 = ap - bq - cr - ds, \qquad L_2 = aq + bp + cs - dr,$$
$$L_3 = ar - bs + cp + dq, \qquad L_4 = as + br - cq + dp,$$

each linear in a, b, c, d and in p, q, r, s, with $(a^2 + b^2 + c^2 + d^2)(p^2 + q^2 + r^2 + s^2) = L_1^2 + L_2^2 + L_3^2 + L_4^2$. ■

For example, $7 = 2^2 + 1^2 + 1^2 + 1^2$ and $3 = 1^2 + 1^2 + 1^2 + 0^2$. With $(a, b, c, d) = (2, 1, 1, 1)$ and $(p, q, r, s) = (1, 1, 1, 0)$ the formulas give $(L_1, L_2, L_3, L_4) = (0, 2, 4, 1)$, and indeed $0^2 + 2^2 + 4^2 + 1^2 = 21 = 7 \times 3$.

Looking back

- **Multiplicativity does the work.** The identity came from $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$, exactly as the two-square identity comes from $|zw| = |z||w|$.
- **Closure under multiplication.** The special form $\begin{pmatrix} z_1 & z_2 \\ -z_2^* & z_1^* \end{pmatrix}$ is preserved by products, which is what makes the result another sum of four squares.
- **Bookkeeping matters.** The final step is routine but error-prone; signs and conjugates must be handled carefully.

What's really going on

Lagrange's four-square theorem

Here is the shape of Lagrange's proof. By Euler's identity, it is enough to show every prime p is a sum of four squares. A counting argument shows that some multiple mp, with $m < p$, is a sum of four squares. A clever "descent" then shows that if mp is a sum of four squares with $1 < m < p$, so is $m'p$ for some smaller m'; repeating, we reach $m = 1$. (Compare Fermat's descent in Section 1.5.) In 1834 Jacobi went much further and found the exact number of ways to write n as a sum of four squares: 8 times the sum of the divisors of n that are not multiples of 4. For example, 1 can be written in 8 ways (± 1 in any of four positions).

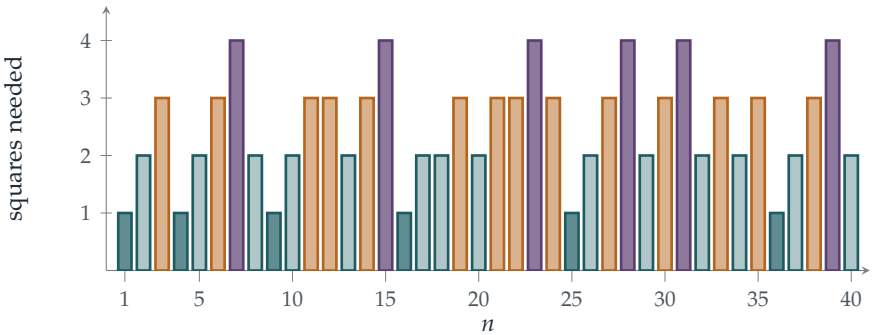


Figure 7.8.1 The least number of squares needed to make each n from 1 to 40. Every number needs at most four, by Lagrange's theorem. The numbers that need four (plum) are $7, 15, 23, 28, 31, 39, \ldots$: exactly those of the form $4^k(8m+7)$, by a theorem of Legendre and Gauss.

Two, three, four

The pattern across dimensions is subtle. Sums of two squares come from complex numbers, and a number is a sum of two squares exactly when its primes of the form $4k+3$ appear to even powers (Section 1.6). Sums of four squares come from quaternions,

and every number is one. But sums of three squares have no multiplication identity at all (there is no three-dimensional number system with a multiplicative size; that is why Hamilton failed with triples, Section 6.7), and the theorem is harder: Legendre and Gauss showed that n is a sum of three squares unless $n = 4^k(8m + 7)$ (Figure 7.8.1). Beyond four, the eight-dimensional octonions give an eight-square identity, found by Degen and Graves, and Hurwitz proved in 1898 that identities of this kind exist only for 1, 2, 4 and 8 squares.

Waring's problem

Lagrange's theorem is the first case of a question asked by Edward Waring in 1770: is there, for each k, a number $g(k)$ such that every positive integer is a sum of at most $g(k)$ kth powers? David Hilbert proved in 1909 that there is. We now know $g(2) = 4$, $g(3) = 9$ (for example $23 = 2 \cdot 2^3 + 7 \cdot 1^3$ needs nine cubes) and $g(4) = 19$.

The thinking skill

Find a multiplicative structure

To show a property is preserved under multiplication, find an algebraic structure (complex numbers, quaternions, matrices) in which the property is the "size" of an object and size is multiplicative.

Once such a structure is found, a product identity comes for free, and a problem about all numbers reduces to a problem about primes.

Try this next

1. **A worked example.** Use the formulas for $L_1, \ldots, L_4$ with $(a, b, c, d) = (p, q, r, s) = (2, 1, 1, 1)$ to write $49 = 7 \times 7$ as a sum of four squares, and check your answer. Is it the same as the obvious $49 = 7^2 + 0^2 + 0^2 + 0^2$?

2. **Two squares inside four.** Show that if $c = d = r = s = 0$, the identity reduces to the two-square identity of Section 1.6.

3. **Numbers needing four squares.** Show that no number of the form $8m + 7$ is a sum of three squares, by considering squares modulo 8.

4. **Jacobi's formula.** Count the representations of 2 as a sum of four squares (with order and signs mattering), and check that the answer is $8 \times (1 + 2) = 24$.

Further reading

► **Harold Davenport, *The Higher Arithmetic*** (Cambridge University Press). Chapter V

proves Lagrange's four-square theorem.

- **John Conway & Derek Smith, *On Quaternions and Octonions*** (A. K. Peters). The arithmetic of quaternions, including a proof of the four-square theorem via quaternion integers.
- **Aigner & Ziegler, *Proofs from THE BOOK*,** chapter "Representing numbers as sums of two squares". The two-square companion to this section.

7.9 Only Five Platonic Solids

The question

A polyhedron is a solid bounded by F plane faces, which meet in E edges and V vertices. You may assume *Euler's formula*, that $V - E + F = 2$.

In a regular polyhedron the faces are equal regular m-sided polygons, n of which meet at each vertex. Show that

$$F = \frac{4n}{h},$$

where $h = 4 - (n-2)(m-2)$.

By considering the possible values of h, or otherwise, prove that there are only five regular polyhedra, and find V, E and F for each.

The hook

There are infinitely many regular polygons: triangles, squares, pentagons, and so on for every number of sides. You might expect infinitely many regular polyhedra too. There are exactly five: the tetrahedron, cube, octahedron, dodecahedron and icosahedron. Plato associated four of them with the elements (fire, earth, air, water) and the fifth with the cosmos; Euclid's *Elements*, the most influential mathematics book ever written, ends with a proof that there are no others; Kepler built a model of the solar system from them.

This question proves the classification in a few lines, from a single remarkable formula found by Euler in 1750: for any convex polyhedron, $V - E + F = 2$. A formula about counting turns into a theorem about shapes, and the number 2 turns out to be a fingerprint of the sphere.

Before you start

Ask yourself Check Euler's formula on a cube: how many vertices, edges and faces does it have? And on a square-based pyramid?

The cube has $V = 8$, $E = 12$, $F = 6$, and $8 - 12 + 6 = 2$. The square-based pyramid has 5 vertices, 8 edges and 5 faces, and $5 - 8 + 5 = 2$. The formula holds for any polyhedron without holes, however irregular.

Ask yourself In a regular polyhedron, each face has m edges and each edge borders two faces. What equation does that give between E and F? Similarly, between E and V?

Counting edge–face incidences in two ways: each of the F faces has m edges, and each edge is counted twice (once for each of its two faces), so $mF = 2E$. Similarly each vertex has n edges and each edge has two ends, so $nV = 2E$. This "count the same thing in two ways" trick is called *double counting*.

Working it through

STAGE 1 A formula for F

From $mF = 2E$ and $nV = 2E$, we have $E = \frac{mF}{2}$ and $V = \frac{mF}{n}$. Substituting into Euler's formula,

$$\frac{mF}{n} - \frac{mF}{2} + F = 2 \quad \Longrightarrow \quad F\,\frac{2m - mn + 2n}{2n} = 2 \quad \Longrightarrow \quad F = \frac{4n}{2m + 2n - mn}.$$

And $4 - (n-2)(m-2) = 4 - (mn - 2m - 2n + 4) = 2m + 2n - mn = h$. So $F = \frac{4n}{h}$. ■

STAGE 2 Which values of m and n are possible?

Ask yourself What is the smallest possible value of m (sides of a face) and of n (faces at a vertex)? And since $F > 0$, what sign must h have?

A polygon needs at least 3 sides, so $m \geq 3$, and at least 3 faces must meet at each vertex of a solid, so $n \geq 3$. Since $F > 0$, we need $h > 0$, that is, $(n-2)(m-2) < 4$. With both factors at least 1, the only possibilities for $(m-2, n-2)$ are $(1,1)$, $(1,2)$, $(2,1)$, $(1,3)$ and $(3,1)$.

Pause and try For each of the five possibilities, compute h, then F, then $E = \frac{mF}{2}$ and $V = \frac{mF}{n}$. Identify each solid.

m	n	h	F	E	V	solid
3	3	3	4	6	4	tetrahedron
3	4	2	8	12	6	octahedron
4	3	2	6	12	8	cube
3	5	1	20	30	12	icosahedron
5	3	1	12	30	20	dodecahedron

These are the only possibilities, so there are at most five regular polyhedra; and all five exist (Euclid shows how to construct each of them). ■

Looking back

- **Double counting.** Counting edge–face and edge–vertex incidences in two ways gave $mF = 2E$ and $nV = 2E$.
- **An inequality from positivity.** The simple requirement $F > 0$ cut infinitely many candidates down to five.
- **Existence is separate.** The argument shows there are *at most* five; that each actually exists has to be checked by constructing it.

What's really going on

Euler's formula and topology

Euler stated $V - E + F = 2$ in 1750, but his proof had gaps, and the formula has a fascinating history of attempted proofs, counterexamples and repairs. A "picture frame" polyhedron, with a square hole through the middle, has $V - E + F = 0$; a polyhedron made of two cubes joined at a vertex has $V - E + F = 3$. Imre Lakatos turned this history into a famous book, *Proofs and Refutations*, about how mathematical concepts are sharpened by counterexamples. The resolution is that $V - E + F$ depends only on the *shape* of the surface in the loose sense of topology: it is 2 for any surface that can be deformed into a sphere, 0 for a torus (a doughnut), and $2 - 2g$ for a surface with g holes. This number, the *Euler characteristic*, was one of the first ideas of topology, the study of properties unchanged by stretching and bending.

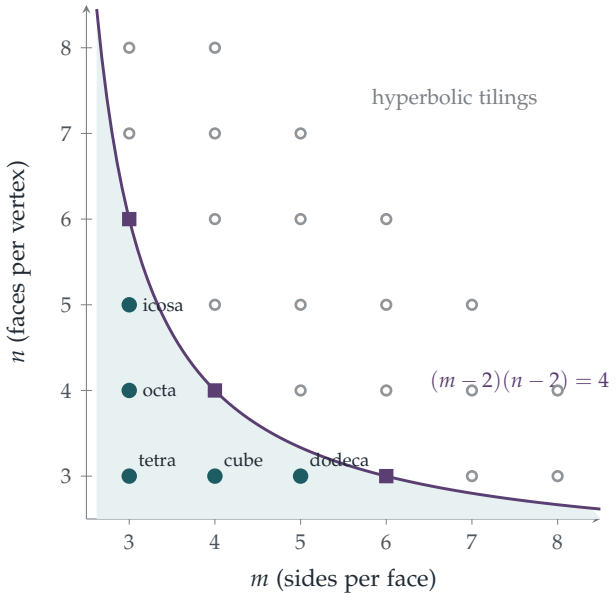


Figure 7.9.1 The possible pairs (m, n) with $m, n \geq 3$. Below the curve $(m-2)(n-2) = 4$ (shaded) are the five Platonic solids (teal). On the curve are three tilings of the flat plane (plum squares): by triangles ($m = 3$, $n = 6$), squares $(4, 4)$ and hexagons $(6, 3)$. Every point above the curve (open circles) gives a regular tiling of the hyperbolic plane.

Beyond the sphere

What happens when $h = 0$ or $h < 0$? When $(m-2)(n-2) = 4$, there are three solutions, $(3, 6)$, $(4, 4)$ and $(6, 3)$: six triangles, four squares or three hexagons meet at a point with angles adding up to exactly $360°$. These are the three regular tilings of the flat plane, the patterns of bathroom floors and honeycombs. And when $(m-2)(n-2) > 4$, the angles at a vertex add up to *more* than $360°$, which is impossible in the flat plane but perfectly possible in the hyperbolic plane of Section 5.8. There are infinitely many regular tilings of the hyperbolic plane, one for every point above the curve in Figure 7.9.1; Escher's *Circle Limit* woodcuts are drawings of some of them. So the classification of Platonic solids is really the first case of a trichotomy (sphere, plane, hyperbolic plane) that runs through all of geometry.

Symmetry groups

The symmetries of the Platonic solids form groups: 12 rotations for the tetrahedron, 24 for the cube and octahedron (which are dual: swap faces and vertices, $m \leftrightarrow n$), and 60 for the dodecahedron and icosahedron. The icosahedral group of order 60 is the "simple" group that, as we mentioned in Section 6.4, prevents the quintic from being solved by radicals. Felix Klein wrote a whole book, *Lectures on the Icosahedron* (1884), connecting the icosahedron to the solution of quintic equations.

The thinking skill

Double counting

Count the same collection in two different ways and equate the results.

Here the collection was the set of (edge, face) pairs with the edge on the face. Double counting is one of the most powerful tools in combinatorics, and it reappears throughout Chapter 9.

Try this next

1 **Duality.** Show that swapping m and n swaps V and F and leaves E unchanged. Which pairs of Platonic solids are dual to each other? Which is self-dual?

2 **A football.** A traditional football is made of 12 pentagons and 20 hexagons, with three faces meeting at each vertex. Use double counting and Euler's formula to check that $V - E + F = 2$. Show more generally that any such polyhedron made of pentagons and hexagons, three at each vertex, has exactly 12 pentagons.

3 **A torus.** Make a "picture frame" polyhedron (a square block with a square hole) and count V, E and F. Check that $V - E + F = 0$.

4 **Angles at a vertex.** The interior angle of a regular m-gon is $180^\circ(1 - \frac{2}{m})$. Show that n of them fit round a vertex of a convex solid (total less than 360°) exactly when $(m-2)(n-2) < 4$, giving a second proof of the classification.

Further reading

- **David Richeson, *Euler's Gem*** (Princeton University Press). The history of $V - E + F = 2$ and how it led to topology. Excellent for a sixth-former.
- **Imre Lakatos, *Proofs and Refutations*** (Cambridge University Press). A dialogue about Euler's formula and the nature of proof.
- **H. S. M. Coxeter, *Regular Polytopes*** (Dover). The classic book on regular polyhedra and their higher-dimensional cousins.

What We Have Learned

We began by saying that a matrix is a transformation of space with a life of its own, and asked three questions: how matrices are like numbers and how they differ, what a matrix really does, and what stays the same. Here is what the chapter's nine questions showed.

1. How are matrices like numbers, and how are they different?

Matrices can be added and multiplied, but Section 7.1 showed which familiar rules fail. Multiplication is not commutative, so $(\mathbf{A}-\mathbf{B})(\mathbf{A}+\mathbf{B})$ need not be $\mathbf{A}^2-\mathbf{B}^2$; there are zero divisors, so $\mathbf{AB}=\mathbf{0}$ does not force $\mathbf{BA}=\mathbf{0}$; and a quadratic matrix equation can have infinitely many roots. Each failure had a geometric explanation, in terms of what the matrices do to the plane. What survives, crucially, is that determinants multiply, $\det(\mathbf{AB})=\det\mathbf{A}\det\mathbf{B}$. Section 7.3 showed that this property essentially *characterises* the determinant, and Section 7.8 used it to prove Euler's four-square identity, the key step in Lagrange's theorem that every whole number is a sum of four squares.

2. What does a matrix really do?

Several sections answered this by finding the right point of view.

- **Eigenvectors** (Section 7.2) are the directions a matrix does not turn. In their coordinates a symmetric matrix is just a stretch, and the image of a circle is an ellipse whose axes and area can be read off.
- **The determinant** (Section 7.3) is the signed area scale factor: shears preserve area, swaps reverse orientation, singular matrices squash the plane flat.
- **Trace and determinant** (Section 7.4) decide, via the Cayley–Hamilton theorem, whether a matrix returns to the identity after a few steps. A matrix with $\mathbf{M}^8=\mathbf{I}$ need not be a rotation; it is a rotation seen in skewed coordinates. And because traces of integer matrices are integers, crystals cannot have five-fold symmetry.
- **Rotations in three dimensions** (Section 7.5) are built from the cross product by Rodrigues' formula, the three-dimensional analogue of $e^{\mathrm{i}\theta}=\cos\theta+\mathrm{i}\sin\theta$.
- **Hyperbolic rotations** (Section 7.6) move points along a hyperbola and add their hyperbolic angles. They are the Lorentz boosts of special relativity, and their invariant lines are the paths of light.

3. What stays the same?

Invariants were everywhere. Rotations preserve $x^2 + y^2$, Lorentz boosts preserve $x^2 - y^2$, and similar matrices share their trace and determinant. In Section 7.7, a matrix evolving by $\dot{\mathbf{M}} = \mathbf{MN} - \mathbf{NM}$ kept its trace and determinant, and so its eigenvalues, constant, however much its entries changed: the mechanism behind the conservation laws of solitons. And in Section 7.9, the invariant was a number, $V - E + F = 2$, which holds for every convex polyhedron and pins the regular ones down to just five. That number, the Euler characteristic, was one of the first invariants of topology.

4. A toolkit for linear algebra

Method	Where you used it
Geometric counterexamples	Rules that fail (7.1)
Eigenvectors as natural axes	The ellipse (7.2); boosts (7.6)
Probing with special matrices	The determinant (7.3)
Cayley–Hamilton	Finite order (7.4)
Splitting into parallel and perpendicular parts	Rodrigues' formula (7.5)
Choosing the right parameter	Rapidity (7.6)
$\operatorname{tr}(\mathbf{MN}) = \operatorname{tr}(\mathbf{NM})$	Conserved quantities (7.7)
Multiplicativity of determinants	Four squares (7.8)
Double counting	Platonic solids (7.9)

Where next

- Eigenvalues and eigenvectors govern the vibrations of systems of springs and masses in Section 11.2.
- The eigenvectors of a covariance matrix appear in statistics in Section 13.7.
- Differential equations and dynamical systems are the subject of Chapter 8.

If you want to go further before university

- **3Blue1Brown, *Essence of Linear Algebra***: a free video series that shows what matrices do. Watch it first.
- **Gilbert Strang, *Introduction to Linear Algebra***: the classic first course, with a strong emphasis on understanding and applications.
- **Sheldon Axler, *Linear Algebra Done Right***: a more abstract approach, centred on linear maps; free online.
- **David Richeson, *Euler's Gem***: Euler's formula and the birth of topology.

Kaplansky joked that he and Halmos thought without coordinates but computed with matrices "like fury" behind closed doors. That captures what this chapter has tried

to show: matrices are concrete and computable, but behind every calculation there is geometry (a stretch, a rotation, a projection, a boost) and an invariant that does not depend on how you write it down. Learning to move between the two views is one of the most valuable skills a mathematician can have.

Answers to Try This Next

These are brief answers and hints, not full solutions: they give the result and the key step, so that you can check your own work. Try each exercise properly before looking.

7.1 Rules That Fail for Matrices

1. Comparing $\mathbf{XP}$ and $\mathbf{PX}$ with $\mathbf{X} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ forces $c = 0$ and $a = d$: the matrices $\begin{pmatrix} a & b \\ 0 & a \end{pmatrix}$.

2. Cayley–Hamilton gives $\mathbf{N}^2 = t\mathbf{N} - d\mathbf{I}$, so $t\mathbf{N} = d\mathbf{I}$. If $t \neq 0$ then $\mathbf{N}$ is a multiple of $\mathbf{I}$, and $\mathbf{N}^2 = \mathbf{0}$ forces $\mathbf{N} = \mathbf{0}$, contradicting $t \neq 0$. So $t = 0$, and then $d = 0$. The answer: $\begin{pmatrix} a & b \\ c & -a \end{pmatrix}$ with $a^2 + bc = 0$.

3. The diagonal entries of the square are $\cos^2\theta + \sin^2\theta = 1$ and the off-diagonal ones are 0. The fixed vector is $(\cos\frac{\theta}{2}, \sin\frac{\theta}{2})$, so it reflects in the line through the origin at angle $\frac{\theta}{2}$.

4. With $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\mathbf{M}^2 = \begin{pmatrix} a^2 + bc & b(a+d) \\ c(a+d) & d^2 + bc \end{pmatrix}$; subtract $(a+d)\mathbf{M}$ and add $(ad - bc)\mathbf{I}$ to get $\mathbf{0}$.

7.2 Stretching Without Turning

1. $\mathbf{Mu} = \lambda\mathbf{u}$ with $\mathbf{u} \neq \mathbf{0}$ needs $\det(\mathbf{M} - \lambda\mathbf{I}) = 0$, which is the quadratic. Here the trace is 6 and the determinant is $\frac{4}{25}(54 - 4) = 8$, so $\lambda^2 - 6\lambda + 8 = 0$: $\lambda = 2$ (for $(1, 2)$) and 4 (for $(2, -1)$).

2. $\mathbf{v} \cdot \mathbf{Mu} = \lambda\,\mathbf{v} \cdot \mathbf{u}$, and since $\mathbf{M}$ is symmetric, $\mathbf{v} \cdot \mathbf{Mu} = \mathbf{Mv} \cdot \mathbf{u} = \mu\,\mathbf{v} \cdot \mathbf{u}$. So $(\lambda - \mu)\,\mathbf{u} \cdot \mathbf{v} = 0$.

3. The characteristic equation $\lambda^2 - 2\lambda\cos\theta + 1 = 0$ has discriminant $-4\sin^2\theta$, negative unless θ is a multiple of π. The eigenvalues are $e^{\pm i\theta}$.

4. $(1, 0) = \frac{1}{5}(1, 2) + \frac{2}{5}(2, -1)$, so $T^n(1, 0) = \frac{2^n}{5}(1, 2) + \frac{2 \cdot 4^n}{5}(2, -1) = \frac{1}{5}(2^n + 4^{n+1},\ 2^{n+1} - 2 \cdot 4^n)$. For $n = 1$ this is $(\frac{18}{5}, -\frac{4}{5})$, as it should be.

7.3 What Makes a Determinant

1. $(\det\mathbf{MN})^3 = (\det\mathbf{M})^3(\det\mathbf{N})^3$, and $(\det\mathbf{J})^3 = -1 \neq 1$. The function $|\det\mathbf{M}|$ is multiplicative, but $|\det\mathbf{J}| = 1$.

2. $\begin{pmatrix} a & \lambda a \\ c & \lambda c \end{pmatrix} = \begin{pmatrix} a & 0 \\ c & 0 \end{pmatrix}\begin{pmatrix} 1 & \lambda \\ 0 & 0 \end{pmatrix}$, and the second factor has second row 0 times its first, so f of it is 0.

3. $\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{J}\mathbf{K}_k\mathbf{J}$, so its value is $\mathrm{f}(\mathbf{J})^2\mathrm{f}(\mathbf{K}_k) = \mathrm{f}(\mathbf{K}_k)$. And $\mathbf{K}_k\mathbf{K}_l = \mathbf{K}_{kl}$.

4. Row reduction writes any invertible matrix as a product of elementary matrices: row swaps ($\mathbf{J}$), row scalings ($\mathbf{K}_k$ or $\mathbf{J}\mathbf{K}_k\mathbf{J}$), and shears ($\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ or its conjugate by $\mathbf{J}$). Shears have $\mathrm{f} = 1$ (as in part (iv), using $\mathbf{K}^{-1}\mathbf{P}\mathbf{K}$). Also $\mathbf{J}$ is a product of shears and $\mathbf{K}_{-1}$, so $\mathrm{f}(\mathbf{K}_{-1}) = \mathrm{f}(\mathbf{J}) = -1$. Hence $\mathrm{f}(\mathbf{M}) = \mathrm{g}(\det\mathbf{M})$ where $\mathrm{g}(k) = \mathrm{f}(\mathbf{K}_k)$; for singular $\mathbf{M}$, $\mathrm{f}(\mathbf{M}) = 0$ by part (iii).

7.4 Matrices That Return to the Start

1. $|1 + \mathrm{e}^{\mathrm{i}\phi}|^2 = 2 + 2\cos\phi = 1$ gives $\phi = \pm\frac{2\pi}{3}$, so $\mathbf{R}^3 = \mathbf{I}$. For $\mathbf{S}$: its eigenvalues are cube roots of 1. If both were 1, then, since $x^3 - 1$ has distinct roots, $\mathbf{S}$ would be diagonalisable and equal to $\mathbf{I}$. Non-real eigenvalues of a real matrix come in conjugate pairs, so they are $\mathrm{e}^{\pm 2\pi\mathrm{i}/3}$: $\det\mathbf{S} = 1$ and $\operatorname{tr}\mathbf{S} = -1$.

2. The eigenvalues must be the primitive sixth roots $\mathrm{e}^{\pm\mathrm{i}\pi/3}$ (any other choice of sixth roots, paired with conjugates, gives $\mathbf{M}^2 = \mathbf{I}$ or $\mathbf{M}^3 = \mathbf{I}$). So $\operatorname{tr}\mathbf{M} = 1$ and $\det\mathbf{M} = 1$.

3. The eigenvalues are roots of unity with product 1, so they are $\mathrm{e}^{\pm\mathrm{i}\theta}$ (or both ± 1), and the trace $2\cos\theta$ is an integer in $[-2, 2]$. Example of order 6: $\begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$.

4. $\mathbf{M}^3 = (t^2 - d)\mathbf{M} - td\mathbf{I}$ and $\mathbf{M}^4 = (t^3 - 2td)\mathbf{M} - d(t^2 - d)\mathbf{I}$. For $\mathbf{M}^4 = -\mathbf{I}$ ($\mathbf{M}$ cannot be a multiple of $\mathbf{I}$), $t(t^2 - 2d) = 0$ and $d(t^2 - d) = 1$. $t = 0$ gives $-d^2 = 1$, impossible; so $t^2 = 2d$, $d^2 = 1$, and $d = 1$, $t^2 = 2$.

7.5 Rotating in Three Dimensions

1. $z \mapsto a + \mathrm{e}^{\mathrm{i}\theta}(z - a)$. The composite is $z \mapsto \mathrm{e}^{\mathrm{i}(\theta+\phi)}z + k$ with $k = \mathrm{e}^{\mathrm{i}\phi}a(1 - \mathrm{e}^{\mathrm{i}\theta}) + b(1 - \mathrm{e}^{\mathrm{i}\phi})$; if $\mathrm{e}^{\mathrm{i}(\theta+\phi)} \neq 1$ it has the fixed point $c = \frac{k}{1-\mathrm{e}^{\mathrm{i}(\theta+\phi)}}$. If $\theta + \phi = 2\pi$ it is the translation by $(b - a)(1 - \mathrm{e}^{-\mathrm{i}\theta})$.

2. $\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$: it fixes $(1, 1, 1)$, has trace $0 = 1 + 2\cos 120^\circ$, and sends $\mathbf{i} \to \mathbf{j} \to \mathbf{k} \to \mathbf{i}$.

3. $\mathbf{K}^2\mathbf{r} = \mathbf{n} \times (\mathbf{n} \times \mathbf{r}) = (\mathbf{n}.\mathbf{r})\mathbf{n} - \mathbf{r}$, and then $\mathbf{K}^3\mathbf{r} = \mathbf{n} \times \big((\mathbf{n}.\mathbf{r})\mathbf{n} - \mathbf{r}\big) = -\mathbf{K}\mathbf{r}$. In the series, the odd powers give $(\theta - \frac{\theta^3}{3!} + \cdots)\mathbf{K} = \sin\theta\,\mathbf{K}$ and the even powers (from $\mathbf{K}^2$ on) give $(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \cdots)\mathbf{K}^2 = (1 - \cos\theta)\mathbf{K}^2$.

4. The composite fixes every point of the line of intersection, and has determinant

$(-1)^2 = 1$. In each plane perpendicular to that line it is the product of two reflections in lines at angle α, which is a rotation by 2α.

7.6 Hyperbolic Rotations and Relativity

1. The entries of $T(u)T(w)$ are $\cosh u \cosh w + \sinh u \sinh w = \cosh(u+w)$ and $\sinh u \cosh w + \cosh u \sinh w = \sinh(u+w)$. So $T(u)^{-1} = T(-u)$.
2. $\tanh u = 0.8$ gives $u = \frac{1}{2}\ln\frac{1.8}{0.2} = \ln 3 \approx 1.099$ and $\cosh u = \frac{1}{2}(3+\frac{1}{3}) = \frac{5}{3}$. On the ship, $10 \times \frac{3}{5} = 6$ years pass.
3. The area is $\frac{1}{2}\oint(x\,\mathrm{d}y - y\,\mathrm{d}x)$. The straight sides contribute nothing, and along the hyperbola, with $x = \cosh s$, $y = \sinh s$, the integrand is $(\cosh^2 s - \sinh^2 s)\,\mathrm{d}s = \mathrm{d}s$, giving $\frac{v}{2}$. T has determinant $\cosh^2 u - \sinh^2 u = 1$, so it preserves area; it maps the hyperbola to itself and moves the point with parameter v to $v+u$.
4. $x' + y' = (\cosh u + \sinh u)(x+y) = \mathrm{e}^u p$ and $x' - y' = \mathrm{e}^{-u} q$; also $x^2 - y^2 = pq$. In these coordinates T is a squeeze: a stretch by e^u along the asymptote $y = x$ and a compression by e^{-u} along $y = -x$, which slides points along each hyperbola $pq =$ constant.

7.7 Quantities That Never Change

1. $\dot a = -Ct = -tc$, $\dot d = Ct = tc$, $\dot c = 0$, and $\dot b = (A-D)t - Ct^3 = t(a-d)$, matching $\mathbf{MN} - \mathbf{NM} = t\begin{pmatrix} -c & a-d \\ 0 & c \end{pmatrix}$.
2. $\dot{\mathbf{M}} = -\mathbf{S}^{-1}\dot{\mathbf{S}}\mathbf{S}^{-1}\mathbf{M}_0\mathbf{S} + \mathbf{S}^{-1}\mathbf{M}_0\dot{\mathbf{S}} = -\mathbf{NM} + \mathbf{MN}$, using $\dot{\mathbf{S}} = \mathbf{SN}$.
3. $\mathbf{N} = \mathbf{M}^{-1}\dot{\mathbf{M}} = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$, and $\ln\det\mathbf{M} = 2t - t = t$, whose derivative is $1 = \operatorname{tr}\mathbf{N}$.
4. $\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{tr}(\mathbf{M}^k) = \sum_j \operatorname{tr}(\mathbf{M}^j\dot{\mathbf{M}}\mathbf{M}^{k-1-j}) = k\operatorname{tr}(\mathbf{M}^{k-1}\dot{\mathbf{M}})$ by cycling, and $\operatorname{tr}(\mathbf{M}^{k-1}(\mathbf{MN} - \mathbf{NM})) = \operatorname{tr}(\mathbf{M}^k\mathbf{N}) - \operatorname{tr}(\mathbf{M}^{k-1}\mathbf{NM}) = 0$, again by cycling.

7.8 Four Squares

1. $(L_1, L_2, L_3, L_4) = (1, 4, 4, 4)$, and $1 + 16 + 16 + 16 = 49$. This is a different representation from $7^2 + 0^2 + 0^2 + 0^2$.
2. With $c = d = r = s = 0$: $L_1 = ap - bq$, $L_2 = aq + bp$, $L_3 = L_4 = 0$, which is $(a^2+b^2)(p^2+q^2) = (ap-bq)^2 + (aq+bp)^2$.
3. Squares are 0, 1 or 4 modulo 8. Sums of three of these give $0, 1, 2, 3, 4, 5, 6$ modulo 8, never 7.
4. $2 = (\pm1)^2 + (\pm1)^2 + 0^2 + 0^2$: choose the two non-zero places (6 ways) and their signs (4 ways), giving 24. Since 2 is not divisible by 4, Jacobi's formula gives $8\sigma(2) = 8 \times 3 = 24$.

7.9 Only Five Platonic Solids

1. With $h = 2m + 2n - mn$, $V = \frac{4m}{h}$, $E = \frac{2mn}{h}$ and $F = \frac{4n}{h}$; swapping m and n swaps V and F. The cube and octahedron are dual, as are the dodecahedron and icosahedron; the tetrahedron is self-dual.

2. $F = 32$, $E = \frac{1}{2}(12 \cdot 5 + 20 \cdot 6) = 90$ and $V = \frac{1}{3}(180) = 60$, and $60 - 90 + 32 = 2$. In general, with P pentagons and H hexagons, $E = \frac{1}{2}(5P + 6H)$ and $V = \frac{1}{3}(5P + 6H)$, so $V - E + F = -\frac{1}{6}(5P + 6H) + P + H = \frac{P}{6}$. This equals 2, so $P = 12$.

3. Divide the top and bottom (which have holes) into four trapezia each. Then $V = 16$, $F = 4 + 4 + 4 + 4 = 16$ and $E = 12 + 12 + 8 = 32$, so $V - E + F = 0$.

4. $180°n(1 - \frac{2}{m}) < 360°$ is $n(m - 2) < 2m$, that is, $mn - 2m - 2n < 0$, or $(m - 2)(n - 2) < 4$. With $m, n \geq 3$ the solutions are $(3, 3)$, $(3, 4)$, $(4, 3)$, $(3, 5)$ and $(5, 3)$.

CHAPTER

8

Differential Equations and Dynamical Systems

Most differential equations cannot be solved. The art is to understand them anyway.

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Rules for Change

Nature laughs at the difficulties of integration.

— Attributed to Pierre-Simon Laplace

At school, a differential equation is something to be solved. You separate the variables, or find an integrating factor, or guess a solution of the form $e^{\lambda x}$, and at the end you have a formula. The equations in textbooks are chosen because this works. Unfortunately, almost every differential equation that comes from the real world (the motion of three planets, the weather, a population of predators and prey, a swinging pendulum with a large amplitude) has no solution that can be written down with the functions we know.

This chapter is about what to do instead. Rather than asking "what is the formula?", it asks questions that can be answered without one. Is there a solution at all, and is there only one? Does it stay bounded? Does it settle down to a steady state, or to a repeating cycle, or does it wander for ever? What does it do in the long run? This change of viewpoint, from solving to understanding, was begun by Henri Poincaré in the 1880s, and it created the subject now called *dynamical systems*.

Why differential equations matter

Differential equations are the language of science. Newton's second law is a differential equation; so are Maxwell's equations of electromagnetism, the Schrödinger equation of quantum mechanics, and the equations of fluid flow used to forecast the weather and design aircraft. Epidemiologists model the spread of a disease with them, ecologists the populations of fish stocks, economists the growth of an economy, and engineers the response of every control system from a thermostat to an autopilot. Whenever a quantity changes at a rate that depends on its current state, there is a differential equation, and usually one that nobody can solve exactly.

Alongside these continuous models are discrete ones, where time moves in steps: the population this year determines the population next year. These *iterated maps* are simpler to write down, but the last two sections show that they can behave just as wildly. One of the great discoveries of the 1970s was that very simple rules can produce *chaos*: behaviour that is completely determined, yet in practice unpredictable.

What you will meet at university

A first-year course on differential equations covers the standard methods of solution, but it also introduces the questions of this chapter. You will meet existence and uniqueness theorems, which tell you when a solution exists and when it is determined by its starting value. You will learn to sketch *phase portraits*, pictures of all solutions at once,

and to classify equilibria as stable or unstable by linearising. Later courses go much further: *nonlinear dynamics* studies limit cycles, bifurcations and chaos; *mathematical biology* applies them to populations and epidemics; and the theory of partial differential equations, Fourier series and orthogonal functions (glimpsed in Section 8.6) underpins quantum mechanics and much of applied mathematics.

The questions that run through this chapter

Three big questions

1. **Does the present determine the future?** When does a differential equation, with a starting value, have exactly one solution?
2. **What can we know without solving?** How can we bound a solution, find where it settles, or describe its shape, when no formula exists?
3. **What happens in the long run?** Do solutions settle to a steady state, approach a cycle, or behave chaotically, and how can we tell?

The chapter begins with uniqueness and what happens when it fails (Section 8.1), then develops tools for understanding equations without solving them: energy-like quantities that cannot increase, equilibria and their stability, and coordinates matched to the symmetry of the problem. Two sections apply these ideas to physical problems: dogs chasing each other in a square, and the quantum harmonic oscillator. The chapter ends with discrete maps, where a formula reveals chaos, and where a single cycle of length three forces cycles of every length.

Ideas in this chapter

§	Topic	Idea
8.1	Uniqueness	Envelopes and singular solutions; gluing
8.2	Energy methods	A quantity that never increases
8.3	Population models	Equilibria, stability and a tipping point
8.4	Limit cycles	Polar coordinates and the Bernoulli substitution
8.5	Pursuit curves	Symmetry reduces four bodies to one
8.6	Orthogonal functions	The Wronskian and orthogonality
8.7	Chaos	Change of variable: the logistic map is angle doubling in disguise
8.8	Periodic orbits	Covering intervals; Sharkovskii's theorem

How to read this chapter

Before solving anything, sketch. For a differential equation $y' = F(y)$, mark where F is positive, negative and zero, and draw arrows on the y-axis; for a map $x \mapsto f(x)$, draw its graph with the line $y = x$ and trace a few sequences by the cobweb method. These pictures take a minute, and they will usually tell you what the answer should look like

before the algebra begins.

8.1 When Solutions Aren't Unique

The question

Find the two solutions of the differential equation

$$\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = 4y$$

which pass through the point (a, b^2), where $b \neq 0$.

Find two distinct points $(a_1, 1)$ and $(a_2, 1)$ such that one of the solutions through each of them also passes through the origin. Show that the graphs of these two solutions coincide and sketch their common graph, together with the other solutions through $(a_1, 1)$ and $(a_2, 1)$.

Now sketch sufficient members of the family of solutions (for varying a and b) to indicate the general behaviour. Use your sketch to identify a common tangent, and comment briefly on its relevance to the differential equation.

The hook

A differential equation is a rule that tells you how a quantity changes: Newton's laws tell a planet how to move, given where it is now. The classical picture of physics is that this is enough: know the state of the world now, and the laws determine its entire future. Pierre-Simon Laplace imagined an intellect that, knowing every position and velocity, could predict everything. That picture rests on a mathematical assumption: that a differential equation, with a starting condition, has *exactly one* solution.

This question shows that the assumption can fail. The innocent-looking equation $(y')^2 = 4y$ has solutions that run along the x-axis and then, at a moment of their own choosing, peel away. Through some points pass infinitely many solutions. Knowing the present does not determine the future. Finding out exactly when uniqueness holds is one of the first theorems of a university course on differential equations.

Before you start

Ask yourself Take the square root of both sides: $y' = \pm 2\sqrt{y}$. Try $y = (x-c)^2$. Does it satisfy the equation? Is there an even simpler solution?

If $y = (x-c)^2$, then $y' = 2(x-c)$ and $(y')^2 = 4(x-c)^2 = 4y$. So every parabola $y = (x-c)^2$ is a solution, for every x. And $y = 0$ is also a solution: $(0)^2 = 4 \times 0$. Keep that last one in mind; it is the key to the whole question.

Working it through

STAGE 1 Solutions through a point

Ask yourself Which parabolas $y = (x-c)^2$ pass through (a, b^2)?

We need $(a-c)^2 = b^2$, so $a - c = \pm b$, that is, $c = a - b$ or $c = a + b$. Since $b \neq 0$ these are different, giving the two solutions

$$y = (x-a+b)^2 \qquad \text{and} \qquad y = (x-a-b)^2. \qquad \blacksquare$$

(We should also check that we have not missed any: on a region where $y > 0$, $y' = \pm 2\sqrt{y}$ with a fixed sign, and $\frac{\mathrm{d}}{\mathrm{d}x}\sqrt{y} = \frac{y'}{2\sqrt{y}} = \pm 1$, so $\sqrt{y} = \pm(x-c)$, giving these parabolas. So near the given point, and for as long as y stays positive, these are the only two solutions. Once a solution reaches the x-axis, though, it can be glued to the zero solution, as we shall see in Stage 3, and that gives infinitely many more solutions defined for all x. The "two solutions" of the question are the two parabolas.)

STAGE 2 Two points and a common solution

A solution through the origin must be $y = x^2$ (the parabola with $c = 0$). It passes through height 1 at $x = 1$ and $x = -1$. So take $(a_1, 1) = (1, 1)$ and $(a_2, 1) = (-1, 1)$.

Through $(1, 1)$ (so $a = 1$, $b = \pm 1$), the solutions are $y = x^2$ and $y = (x-2)^2$. Through $(-1, 1)$ they are $y = (x+2)^2$ and $y = x^2$. The solution through the origin is the same curve $y = x^2$ in both cases. $\blacksquare$

STAGE 3 The whole family, and a common tangent

Pause and try Sketch several of the parabolas $y = (x-c)^2$ for different c. Which line touches every one of them? Is that line itself a solution?

Every parabola $y = (x-c)^2$ touches the x-axis at its vertex $(c, 0)$, so the x-axis is a common tangent to the whole family: their *envelope* (Figure 8.1.1). And the x-axis, $y = 0$, is itself a solution of the differential equation. It is not one of the parabolas; it is

a separate, *singular* solution.

STAGE 4 Why it matters

Ask yourself Start at the origin. Which solutions pass through it? Can you build a solution that follows the x-axis for a while and then leaves it?

Through the origin pass $y = x^2$ and $y = 0$. But much more is true. Because each parabola meets the x-axis tangentially (both have gradient 0 there), you can glue pieces together smoothly: for any $c_1 \le 0 \le c_2$,

$$y = \begin{cases} (x - c_1)^2 & x < c_1, \\ 0 & c_1 \le x \le c_2, \\ (x - c_2)^2 & x > c_2 \end{cases}$$

is a differentiable function satisfying $(y')^2 = 4y$ everywhere, and it passes through the origin. There are infinitely many solutions through every point on the x-axis: a solution can sit at $y = 0$ for as long as it likes, and then leave. The common tangent is exactly where uniqueness breaks down. ■

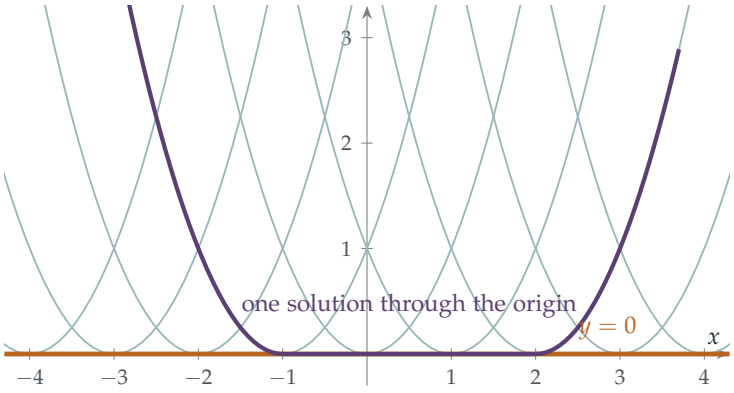


Figure 8.1.1 The solutions $y = (x - c)^2$ of $(y')^2 = 4y$ (teal) all touch the x-axis, which is their envelope and is itself a solution (amber). Gluing pieces gives further solutions, such as the plum one, which follows a parabola down to the axis, runs along it, and leaves again at a time of its choosing. Through any point of the x-axis pass infinitely many solutions.

Looking back

- **Look for simple solutions first.** Trying $y = (x - c)^2$ and $y = 0$ found the whole picture.
- **Envelopes give singular solutions.** The common tangent to a family of solutions can itself be a solution that is not in the family.
- **Gluing.** Where two solutions touch tangentially, they can be joined into a new differentiable solution. This is how non-uniqueness spreads.

What's really going on

When are solutions unique?

The *Picard–Lindelöf theorem* (proved by Émile Picard and Ernst Lindelöf around 1890–1894) says that the equation $y' = F(x, y)$ with $y(x_0) = y_0$ has exactly one solution near x_0, provided F is *Lipschitz* in y: there is a constant L with $|F(x, y_1) - F(x, y_2)| \le L|y_1 - y_2|$. Roughly, F must not change infinitely fast as y changes. Our equation is $y' = \pm 2\sqrt{y}$, and $\sqrt{y}$ has infinite slope at $y = 0$: it is not Lipschitz there, and that is exactly where uniqueness fails. (Giuseppe Peano showed in 1886 that continuity of F alone guarantees that a solution *exists*, but not that it is unique.) The proof of Picard–Lindelöf uses an iteration that converges by the contraction mapping principle of Section 2.2.

The leaking bucket

Here is a physical example. A bucket with a hole in the bottom empties according to Torricelli's law: the height h of water satisfies $h' = -k\sqrt{h}$. Run time forwards and everything is fine: the bucket empties and stays empty. But run time backwards: if you find an empty bucket, the equation cannot tell you when it emptied. The solution $h = 0$ for all time and the solutions that were full until some moment and then drained are all consistent with "empty now". The non-uniqueness is the mathematical shadow of the fact that information about the past has been lost.

Norton's dome

In 2003 the philosopher John Norton described a frictionless dome shaped so that a ball placed exactly at the top obeys Newton's laws in the form $r'' = \sqrt{r}$. One solution is that the ball sits at the top for ever. But there are others, in which it sits there for any length of time and then spontaneously rolls off in any direction, without any cause. The example is still debated: is classical mechanics deterministic after all? Mathematically, it is non-uniqueness of exactly the kind in this question.

The thinking skill

Don't assume uniqueness

Finding a solution does not show it is the only one. Ask whether others exist, especially where the equation involves functions that are not smooth.

At university, existence and uniqueness theorems tell you exactly what conditions are needed. Here a square root, with its infinite slope at zero, was enough to break uniqueness.

Try this next

1 **Two rates.** (From STEP II, 2018.) Using $v = \sqrt{y}$, solve $\dfrac{\mathrm{d}y}{\mathrm{d}t} = \alpha y^{1/2} - \beta y$ (with $\alpha, \beta > 0$) and find the non-constant solution y_1 with $y_1(0) = 0$. Do the same for $\dfrac{\mathrm{d}y}{\mathrm{d}t} = \alpha y^{2/3} - \beta y$, using $v = y^{1/3}$. Why does $y = 0$ also satisfy both equations with $y(0) = 0$?

2 **The leaking bucket.** Solve $h' = -k\sqrt{h}$ with $h(0) = h_0 > 0$. When is the bucket empty? Write down two different solutions with $h(T) = 0$ for some T and $h(t) = 0$ for all $t \geq T$.

3 **Uniqueness restored.** Show that $y' = 2y$ with $y(0) = 0$ has only the solution $y = 0$. (*Hint:* consider $\frac{\mathrm{d}}{\mathrm{d}x}(ye^{-2x})$.) What property does $2y$ have that $2\sqrt{y}$ lacks?

4 **Cubes.** Find all solutions of $y' = 3y^{2/3}$ through the origin.

Further reading

- **Morris Hirsch, Stephen Smale & Robert Devaney, *Differential Equations, Dynamical Systems, and an Introduction to Chaos*** (Academic Press). Existence and uniqueness, and much of the rest of this chapter.
- **George Simmons, *Differential Equations with Applications and Historical Notes*** (CRC Press). A classic, full of history; discusses singular solutions and envelopes.
- **John Norton, "The dome: an unexpectedly simple failure of determinism"**, *Philosophy of Science* 75 (2008). Short and readable.

8.2 Bounding Without Solving

The question

(i) Let $y(x)$ be a solution of the differential equation $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + y^3 = 0$ with $y = 1$ and $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$ at $x = 0$, and let

$$\mathrm{E}(x) = \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + \tfrac{1}{2}y^4.$$

Show by differentiation that E is constant and deduce that $|y(x)| \leq 1$ for all x.

(ii) Let $v(x)$ be a solution of the differential equation $\dfrac{\mathrm{d}^2v}{\mathrm{d}x^2} + x\dfrac{\mathrm{d}v}{\mathrm{d}x} + \sinh v = 0$ with $v = \ln 3$ and $\dfrac{\mathrm{d}v}{\mathrm{d}x} = 0$ at $x = 0$, and let

$$\mathrm{E}(x) = \left(\frac{\mathrm{d}v}{\mathrm{d}x}\right)^2 + 2\cosh v.$$

Show that $\dfrac{\mathrm{dE}}{\mathrm{d}x} \leq 0$ for $x \geq 0$ and deduce that $\cosh v(x) \leq \frac{5}{3}$ for $x \geq 0$.

(iii) Let $w(x)$ be a solution of the differential equation

$$\frac{\mathrm{d}^2w}{\mathrm{d}x^2} + (5\cosh x - 4\sinh x - 3)\frac{\mathrm{d}w}{\mathrm{d}x} + (w\cosh w + 2\sinh w) = 0$$

with $\dfrac{\mathrm{d}w}{\mathrm{d}x} = \dfrac{1}{\sqrt{2}}$ and $w = 0$ at $x = 0$. Show that $\cosh w(x) \leq \frac{5}{4}$ for $x \geq 0$.

The hook

Most differential equations cannot be solved. There is no formula for the solution of $y'' + y^3 = 0$ in terms of familiar functions, let alone for the monster in part (iii). Yet we can still say a great deal about the solutions: that they stay bounded, that they oscillate, that they settle down. The trick is to find a quantity, built from the solution and its derivative, that never increases. Then the solution is trapped.

In physics, that quantity is usually *energy*. A ball rolling in a bowl cannot climb higher than its starting height, because energy is conserved, or, with friction, decreases. This

question applies the same reasoning to three equations, the last of them chosen so that you have to invent the "energy" yourself. The method, due to Aleksandr Lyapunov in 1892, is one of the most important tools in the study of differential equations.

Before you start

Ask yourself For part (i), think of y as the position of a particle and x as time. Then $y'' = -y^3$ is Newton's law with force $-y^3$. What are the kinetic and potential energies?

The kinetic energy is $\tfrac{1}{2}(y')^2$ (taking mass 1), and the force $-y^3$ comes from the potential $V(y) = \tfrac{1}{4}y^4$, since force is $-V'(y)$. The total energy is $\tfrac{1}{2}(y')^2 + \tfrac{1}{4}y^4$, which is half of E. Conservation of energy is exactly the statement that E is constant.

Working it through

STAGE 1 Conservation

By the chain rule,

$$\frac{\mathrm{dE}}{\mathrm{d}x} = 2y'y'' + 2y^3y' = 2y'\left(y'' + y^3\right) = 0.$$

So E is constant, equal to its value at $x = 0$: $\mathrm{E} = 0 + \tfrac{1}{2} = \tfrac{1}{2}$. Since $(y')^2 \geq 0$, we get $\tfrac{1}{2}y^4 \leq \tfrac{1}{2}$, so $y^4 \leq 1$ and $|y| \leq 1$. ■

The solution oscillates between -1 and 1, tracing the closed curve $(y')^2 + \tfrac{1}{2}y^4 = \tfrac{1}{2}$ in the (y, y') plane (Figure 8.2.1).

STAGE 2 Dissipation

Ask yourself Differentiate $\mathrm{E} = (v')^2 + 2\cosh v$. Use the differential equation to replace v''. What sign does the result have?

$$\frac{\mathrm{dE}}{\mathrm{d}x} = 2v'v'' + 2\sinh v\, v' = 2v'\left(v'' + \sinh v\right) = 2v'\left(-xv'\right) = -2x(v')^2 \leq 0 \quad \text{for } x \geq 0.$$

The term xv' acts like friction, and it removes energy. So $\mathrm{E}(x) \leq \mathrm{E}(0)$ for $x \geq 0$. At $x = 0$, $\cosh(\ln 3) = \frac{3 + \frac{1}{3}}{2} = \frac{5}{3}$, so $\mathrm{E}(0) = \frac{10}{3}$. Hence

$$2\cosh v \leq (v')^2 + 2\cosh v = \mathrm{E}(x) \leq \tfrac{10}{3}, \qquad \text{so} \qquad \cosh v(x) \leq \tfrac{5}{3}. \quad ■$$

STAGE 3 Inventing the energy

Part (iii) gives no hint. We need a function $\mathrm{E} = (w')^2 + F(w)$ that does not increase.

Ask yourself If $\mathrm{E} = (w')^2 + F(w)$, then $\mathrm{E}' = 2w'w'' + F'(w)w'$. What should $F'(w)$ be, so that the equation can be used to replace w'' and leave only a multiple of $(w')^2$?

We want $\mathrm{E}' = 2w'\big(w'' + w\cosh w + 2\sinh w\big)$, so we need $F'(w) = 2(w\cosh w + 2\sinh w)$. Since $\frac{\mathrm{d}}{\mathrm{d}w}(w\sinh w) = \sinh w + w\cosh w$, we can take

$$F(w) = 2w\sinh w + 2\cosh w, \qquad \mathrm{E} = (w')^2 + 2w\sinh w + 2\cosh w.$$

Then, using the equation,

$$\frac{\mathrm{dE}}{\mathrm{d}x} = -2(5\cosh x - 4\sinh x - 3)(w')^2.$$

Pause and try Show that $5\cosh x - 4\sinh x - 3 \geq 0$ for all x. (Write it in terms of e^x and look for a perfect square.) Then finish the argument.

$5\cosh x - 4\sinh x - 3 = \frac{5}{2}(e^x + e^{-x}) - 2(e^x - e^{-x}) - 3 = \frac{1}{2}e^x - 3 + \frac{9}{2}e^{-x} = \frac{1}{2}\big(e^{x/2} - 3e^{-x/2}\big)^2 \geq 0$. So E never increases. At $x = 0$, $\mathrm{E}(0) = \frac{1}{2} + 0 + 2 = \frac{5}{2}$. Since $w\sinh w \geq 0$ for every w (both factors have the same sign),

$$2\cosh w \leq (w')^2 + 2w\sinh w + 2\cosh w = \mathrm{E}(x) \leq \tfrac{5}{2}, \qquad \text{so} \qquad \cosh w(x) \leq \tfrac{5}{4}. \qquad \blacksquare$$

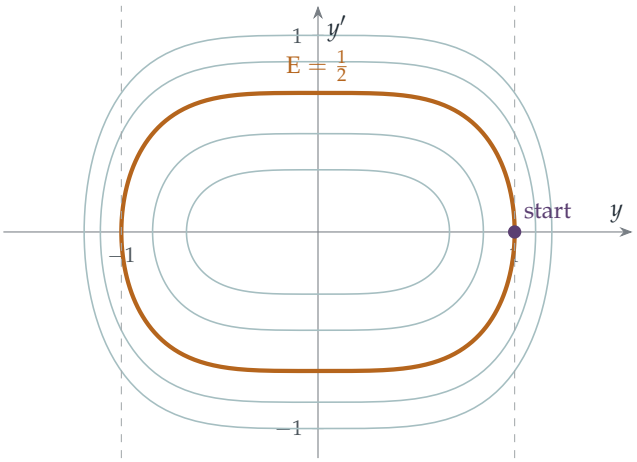


Figure 8.2.1 Level curves of the energy $\mathrm{E} = (y')^2 + \frac{1}{2}y^4$ in the (y, y') plane. Because E is constant, the solution of part (i), starting at $y = 1$, $y' = 0$, stays on the curve $\mathrm{E} = \frac{1}{2}$ (amber) for ever, and so never leaves the strip $|y| \leq 1$ (dashed). We know this without ever finding a formula for y.

Looking back

► **Multiply by y'.** In each part, E' was $2y'$ times the left-hand side of the equation, plus

whatever was left over. That leftover term decided whether E was conserved or decreasing.

- **Energy bounds position.** Since the kinetic part $(y')^2$ is non-negative, a bound on E is a bound on the potential part, and hence on y.
- **Spot hidden squares.** The damping coefficient in part (iii) looked as if it might change sign, but it was a perfect square in disguise.

What's really going on

Lyapunov functions

In his 1892 doctoral thesis, Aleksandr Lyapunov studied the *stability* of motion: if a system is disturbed slightly from equilibrium, does it return, or wander away? His "second method" was to find a function, now called a *Lyapunov function*, that is smallest at the equilibrium and never increases along solutions. Then solutions that start near the equilibrium stay near it. The beauty of the method is that it needs no formula for the solutions, only the differential equation itself, which is exactly what this question demonstrates. Lyapunov functions are now used to prove the stability of everything from aircraft control systems to power grids to ecosystems.

Energy in physics and engineering

Part (i) is a nonlinear oscillator: like a spring, but with a restoring force proportional to the cube of the displacement. Unlike a simple harmonic oscillator, its period depends on its amplitude (larger oscillations are faster), but energy conservation still confines it. Part (ii) adds damping that grows with time, and energy drains away. In engineering, energy methods are used to show that structures will not shake themselves apart and that control systems will not oscillate out of control. In the theory of partial differential equations, "energy estimates" of this kind are the main tool for proving that solutions exist and behave well; questions about the equations of fluid flow, including one of the Millennium Prize Problems (the Navier–Stokes problem), come down to whether such bounds can be found.

The thinking skill

Find a quantity that cannot increase

If you cannot solve an equation, look for a quantity built from the solution whose derivative has a definite sign. It will bound the solution.

The search is guided by physics ("what is the energy?") and by algebra ("what must I add so that the derivative collapses?"). Part (iii) shows the algebraic route

when there is no physical intuition to lean on.

Try this next

1. **Simple harmonic motion.** For $y'' + y = 0$ with $y(0) = A$, $y'(0) = 0$, find the conserved energy and deduce $|y| \le A$. Check against the exact solution.

2. **The pendulum.** The angle θ of a pendulum satisfies $\theta'' + \sin\theta = 0$ (in suitable units). Show that $(\theta')^2 - 2\cos\theta$ is constant. If the pendulum starts at rest at angle θ_0, where $|\theta_0| \le \pi$, show that $|\theta|$ never exceeds $|\theta_0|$. What goes wrong if $|\theta_0| > \pi$?

3. **Damping.** For $y'' + 2ky' + y = 0$ with $k > 0$, show that $(y')^2 + y^2$ decreases. Does it necessarily tend to zero?

4. **An escape.** For $y'' - y^3 = 0$ with $y(0) = 1$, $y'(0) = 0$, find a conserved quantity and show that y is not bounded by it. What goes wrong?

Further reading

- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press). Chapters 6 and 7 cover phase planes, conservative systems and Lyapunov functions.
- **D. W. Jordan & P. Smith, *Nonlinear Ordinary Differential Equations*** (Oxford University Press). A standard undergraduate text with many examples of energy methods.
- **Keith Devlin, *The Millennium Problems*** (Basic Books). Includes an accessible account of the Navier–Stokes problem.

8.3 Harvesting a Population

The question

If there are x micrograms of bacteria in a nutrient medium, the population of bacteria will grow at the rate $(2K - x)x$ micrograms per hour. Show that, if $x = K$ when $t = 0$, the population at time t is given by

$$x(t) = K + K\frac{1 - e^{-2Kt}}{1 + e^{-2Kt}}.$$

Sketch, for $t \geq 0$, the graph of x against t. What happens to $x(t)$ as $t \to \infty$?

Now suppose that the situation is as described in the first paragraph, except that we remove the bacteria from the nutrient medium at a rate L micrograms per hour where $K^2 > L$. We set $\alpha = \sqrt{K^2 - L}$. Write down the new differential equation for x. By considering a new variable $y = x - K + \alpha$, or otherwise, show that, if $x(0) = K$ then $x(t) \to K + \alpha$ as $t \to \infty$.

The hook

A population with unlimited food grows exponentially. Real populations run out of room: growth slows as the population approaches what the environment can support, its *carrying capacity*. The simplest model of this, the *logistic equation*, is in the first half of the question, and it describes everything from bacteria in a dish to the spread of a new technology.

The second half asks a question with enormous practical stakes: what happens if we harvest the population at a steady rate? Below a certain threshold, the population adjusts to a new, lower equilibrium. But push the harvest past that threshold, even slightly, and the equilibrium vanishes and the population crashes to zero. This is the mathematics behind the collapse of fisheries, and the reason fishing quotas exist.

Before you start

Ask yourself Before solving anything, sketch the growth rate $(2K - x)x$ as a function of x. Where is it zero? Where positive? What does that tell you about how x will behave?

The rate is zero at $x = 0$ and $x = 2K$, positive in between and negative beyond $2K$. So a population starting between 0 and $2K$ grows towards $2K$, and one starting above $2K$

shrinks towards it. The carrying capacity is $2K$, and the growth is fastest at $x = K$, where the rate is K^2. You can predict the long-term behaviour without solving the equation.

Working it through

STAGE 1 Solving the logistic equation

Ask yourself Separate the variables: $\displaystyle\int \frac{\mathrm{d}x}{(2K-x)x} = \int \mathrm{d}t$. Split the left-hand side into partial fractions.

$\dfrac{1}{(2K-x)x} = \dfrac{1}{2K}\left(\dfrac{1}{x} + \dfrac{1}{2K-x}\right)$, so

$$\frac{1}{2K}\ln\frac{x}{2K-x} = t + C.$$

At $t = 0$, $x = K$, so $\ln 1 = 0$ and $C = 0$. Then $\frac{x}{2K-x} = e^{2Kt}$, giving

$$x = \frac{2Ke^{2Kt}}{1+e^{2Kt}} = \frac{2K}{1+e^{-2Kt}} = K + K\frac{1-e^{-2Kt}}{1+e^{-2Kt}}. \qquad \blacksquare$$

The fraction is $\tanh(Kt)$, so $x = K + K\tanh Kt$: an S-shaped curve rising from K towards $2K$. As $t \to \infty$, $e^{-2Kt} \to 0$ and $x \to 2K$.

STAGE 2 Harvesting

Removing bacteria at rate L gives

$$\frac{\mathrm{d}x}{\mathrm{d}t} = (2K-x)x - L.$$

Ask yourself Complete the square in $(2K-x)x - L$. Where are its zeros?

$(2K-x)x - L = K^2 - L - (x-K)^2 = \alpha^2 - (x-K)^2 = \big(\alpha - (x-K)\big)\big(\alpha + (x-K)\big)$. The equilibria are $x = K \pm \alpha$.

Pause and try Now substitute $y = x - K + \alpha$. What equation does y satisfy? Compare it with the original logistic equation.

With $y = x - K + \alpha$, we have $x - K = y - \alpha$, so $\alpha + (x-K) = y$ and $\alpha - (x-K) = 2\alpha - y$. Hence

$$\frac{\mathrm{d}y}{\mathrm{d}t} = (2\alpha - y)y,$$

the logistic equation again, with K replaced by α. Initially $y(0) = K - K + \alpha = \alpha$, exactly as $x(0) = K$ in the first part. So by Stage 1 (with α for K), $y(t) = \alpha + \alpha \tanh \alpha t \to 2\alpha$, and therefore

$$x(t) = K + \alpha \tanh \alpha t \to K + \alpha. \qquad \blacksquare$$

Looking back

- **Qualitative first.** The sign of the growth rate predicted the long-term behaviour before any solving.
- **Completing the square reveals the equilibria.** The harvested equation had equilibria at $K \pm \alpha$.
- **A substitution reduces a new problem to an old one.** Shifting the variable turned the harvested equation into the logistic equation already solved.

What's really going on

Tipping points

The harvested equation has two equilibria, $K + \alpha$ and $K - \alpha$. The upper one is *stable*: a population slightly above or below it returns to it. The lower one is *unstable*: a population just below $K - \alpha$ declines to extinction. As the harvest rate L increases, $\alpha = \sqrt{K^2 - L}$ shrinks, and the two equilibria approach each other; at $L = K^2$ they merge, and for $L > K^2$ there is no equilibrium at all, and every population collapses (Figure 8.3.1). This is a *saddle-node bifurcation*, the simplest kind of *tipping point*. Its sting is that nothing dramatic warns of it: as L approaches K^2, the equilibrium population declines only gently, and then suddenly there is no equilibrium. Reducing the harvest afterwards may not help, because the population may already be below the unstable threshold.

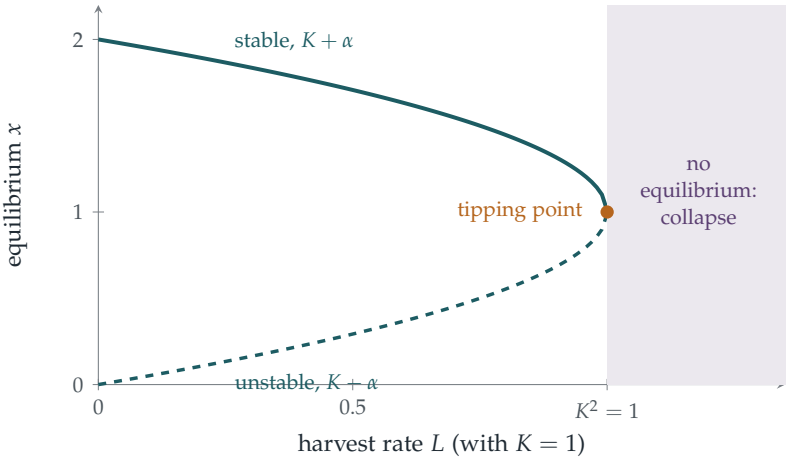


Figure 8.3.1 Equilibrium populations against the harvest rate L, with $K = 1$. The stable equilibrium (solid) and the unstable threshold (dashed) approach each other as L increases, and meet at $L = K^2$. Beyond that, no equilibrium exists and the population collapses whatever its size.

Maximum sustainable yield

What is the largest harvest the population can sustain? It is $L = K^2$, the peak of the growth rate $(2K - x)x$, achieved when the population is held at K, half the carrying capacity. Fisheries science adopted this "maximum sustainable yield" as a management target in the mid-twentieth century. But harvesting exactly at the maximum puts the system exactly at the tipping point, where the slightest overestimate of K, or a bad year, causes collapse. The Grand Banks cod fishery off Newfoundland, one of the richest in the world for five hundred years, collapsed in the early 1990s, and in 1992 the Canadian government imposed a moratorium that put tens of thousands of people out of work. The stocks have still not fully recovered.

Logistic growth elsewhere

The logistic equation was proposed by Pierre-François Verhulst in 1838 to model human populations. The same S-shaped curve describes the spread of an epidemic in its early stages, the adoption of new technologies, autocatalytic chemical reactions and the growth of tumours. Its discrete-time version, the logistic *map*, behaves very differently: it can be chaotic, as we will see in Section 8.7.

The thinking skill

Equilibria and their stability

For an equation $x' = f(x)$, the equilibria are the zeros of f, and the sign of f between them tells you which way solutions move.

An equilibrium where f changes from positive to negative is stable; one where it changes from negative to positive is unstable. Drawing the graph of f often answers the important questions without solving anything.

Try this next

1. **Starting low.** With harvesting and $K^2 > L$, show that if $x(0) < K - \alpha$ then the population reaches zero in finite time. (*Hint:* what does y do when $y(0) < 0$?)

2. **Over-harvesting.** If $L > K^2$, write $\beta = \sqrt{L - K^2}$. Solve $x' = -(x - K)^2 - \beta^2$ with $x(0) = 2K$ and find when the population reaches zero.

3. **Proportional harvesting.** Suppose instead that a fixed *fraction* of the population is removed: $x' = (2K - x)x - hx$. Find the equilibria and show that the population survives for all $h < 2K$. Why is this policy safer?

4. **Half-time.** In the unharvested case, find the time at which the population reaches $\frac{3}{2}K$.

Further reading

- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press). Chapter 3 is about bifurcations, with this harvesting model as an exercise.
- **Nicholas Britton, *Essential Mathematical Biology*** (Springer). Population models for undergraduates.
- **Mark Kurlansky, *Cod: A Biography of the Fish That Changed the World*** (Penguin). The history of the cod fishery and its collapse.

8.4 Spiralling onto a Circle

The question

(i) Show that under the changes of variable $x = r\cos\theta$ and $y = r\sin\theta$, where r is a function of θ with $r > 0$, the differential equation

$$(y + x)\frac{\mathrm{d}y}{\mathrm{d}x} = y - x$$

becomes

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} + r = 0.$$

Sketch a solution in the x-y plane.

(ii) Show that the solutions of

$$\left(y + x - x(x^2 + y^2)\right)\frac{\mathrm{d}y}{\mathrm{d}x} = y - x - y(x^2 + y^2)$$

can be written in the form

$$r^2 = \frac{1}{1 + A\mathrm{e}^{2\theta}}$$

and sketch the different forms of solution that arise according to the value of A.

The hook

Many systems in nature settle into a regular rhythm that does not depend on how they started: a heart beating, a cricket chirping, a violin string bowed steadily, the population cycles of predators and prey. Disturb them a little and they return to the same rhythm. In the language of differential equations, their solutions approach a closed loop, called a *limit cycle*.

This question constructs one. The equation looks hopeless in x and y, but in polar coordinates it collapses to something you can solve, and the solutions turn out to spiral onto the unit circle, from inside and from outside. The key move, choosing coordinates that match the symmetry of the problem, is one we have used throughout the book.

Before you start

Ask yourself Both equations contain the combinations $x\,dx + y\,dy$ and $x\,dy - y\,dx$ in disguise. What are these in polar coordinates?

Since $r^2 = x^2 + y^2$, differentiating gives $x\,dx + y\,dy = r\,dr$. And $x\,dy - y\,dx = r^2\,d\theta$ (this is the area formula from Section 4.9). These two identities make the change of variables almost painless: rewrite the equation so that only these combinations appear.

Working it through

STAGE 1 Part (i) in polar coordinates

Write the equation as $(y + x)\,dy - (y - x)\,dx = 0$, and rearrange:

$$(x\,dx + y\,dy) + (x\,dy - y\,dx) = 0.$$

Ask yourself Check that rearrangement by expanding. Now use the two identities above.

$(y + x)\,dy - (y - x)\,dx = y\,dy + x\,dy - y\,dx + x\,dx$, which is the expression above. In polar form it is $r\,dr + r^2\,d\theta = 0$, and dividing by $r^2\,d\theta$ (with $r > 0$),

$$\frac{dr}{d\theta} + r = 0. \qquad \blacksquare$$

The solutions are $r = Ce^{-\theta}$: *logarithmic spirals*, winding in towards the origin as θ increases and out to infinity as θ decreases.

STAGE 2 Part (ii)

Pause and try The new equation is the old one plus extra terms involving $x^2 + y^2 = r^2$. Rewrite it in the same way and find $\frac{dr}{d\theta}$.

The equation is $\left(y + x - xr^2\right)dy - \left(y - x - yr^2\right)dx = 0$, that is,

$$\left[(y + x)\,dy - (y - x)\,dx\right] - r^2\left(x\,dy - y\,dx\right) = 0.$$

The first bracket is $r\,dr + r^2\,d\theta$ as before, and the second is $r^2\,d\theta$. So $r\,dr + r^2(1 - r^2)\,d\theta = 0$, giving

$$\frac{dr}{d\theta} = r^3 - r.$$

Ask yourself This is not linear in r, but it becomes linear in $u = r^{-2}$. What equation does u satisfy?

With $u = r^{-2}$, $\frac{du}{d\theta} = -2r^{-3}\frac{dr}{d\theta} = -2r^{-3}(r^3 - r) = -2 + 2u$. So $\frac{du}{d\theta} - 2u = -2$, whose solutions are $u = 1 + Ae^{2\theta}$. Hence

$$r^2 = \frac{1}{1 + Ae^{2\theta}}. \qquad \blacksquare$$

STAGE 3 The three kinds of solution

- $A = 0$: $r = 1$. The unit circle is a solution.
- $A > 0$: $r < 1$ always. As $\theta \to -\infty$, $r \to 1$; as $\theta \to +\infty$, $r \to 0$. The curve spirals between the circle and the origin.
- $A < 0$: $r > 1$, and the formula only makes sense while $1 + Ae^{2\theta} > 0$, that is, $\theta < \frac{1}{2}\ln\left(-\frac{1}{A}\right)$. As $\theta \to -\infty$, $r \to 1$; as θ approaches that bound, $r \to \infty$. The curve spirals between the circle and infinity.

So every solution curve approaches the unit circle as $\theta \to -\infty$, from inside or outside (Figure 8.4.1).

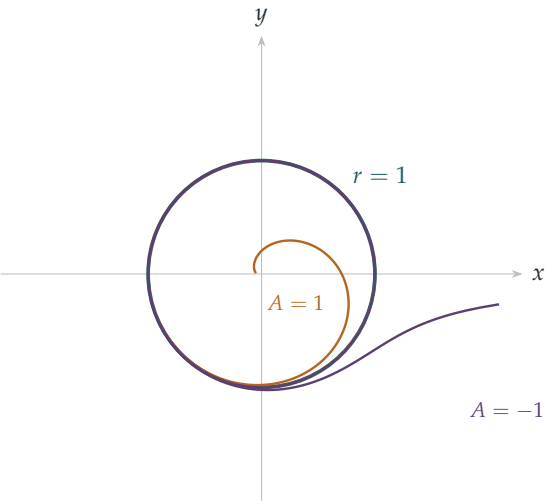


Figure 8.4.1 Solutions of part (ii). The unit circle (teal) is a solution. The solution with $A = 1$ (amber) lies inside it, spiralling from the circle (as $\theta \to -\infty$) towards the origin; the solution with $A = -1$ (plum) lies outside, spiralling from the circle out to infinity. Seen in the other direction (decreasing θ), every solution winds onto the circle: a limit cycle.

Looking back

- **Use the symmetry.** The equations treated all directions alike, and polar coordinates made that visible.
- **Recognise standard combinations.** $x\,dx + y\,dy = r\,dr$ and $x\,dy - y\,dx = r^2\,d\theta$ did most of the work.
- **A substitution linearises.** The nonlinear equation $r' = r^3 - r$ became linear in $u = r^{-2}$ (a *Bernoulli equation*).

What's really going on

Limit cycles

Henri Poincaré introduced limit cycles in the 1880s while studying the qualitative behaviour of differential equations in the plane. Interpret the question's equation as describing the path of a point moving in the plane. The unit circle is an isolated closed orbit, and every nearby orbit winds towards it (or away from it, depending on the direction of time). In 1926 the Dutch engineer Balthasar van der Pol found a limit cycle in an electrical circuit with a valve, and proposed that similar equations describe the beating of the heart. Limit cycles model any system with a self-sustained, stable rhythm: circadian clocks, nerve impulses, the cycles of predator and prey populations.

No chaos in the plane

The *Poincaré–Bendixson theorem* says that for a smooth system in the plane, an orbit that stays in a bounded region without approaching an equilibrium must approach a closed orbit. So in two dimensions, long-term behaviour is limited to equilibria and cycles: there is no room for chaos, because orbits cannot cross. Chaos needs at least three dimensions, as Edward Lorenz discovered in 1963 with his famous three-variable model of convection (the "butterfly"). In discrete time, as we will see in Section 8.7, chaos is possible even in one dimension.

An unsolved problem

In 1900 David Hilbert asked, as part of his sixteenth problem: what is the largest number of limit cycles that a system $x' = P(x, y)$, $y' = Q(x, y)$ can have, when P and Q are polynomials of degree n? It is still unknown even for $n = 2$, where systems with four limit cycles are known and it is not known whether more are possible. It is one of the very few of Hilbert's problems that remain completely open.

The thinking skill

Change coordinates to match the symmetry

If a problem has rotational symmetry, use polar coordinates; if it has another symmetry, find coordinates that respect it.

The right coordinates turned an intractable equation into a separable one, and made the geometry of the solutions (circles and spirals) obvious.

Try this next

1 **The spiral.** For the solution $r = e^{-\theta}$ of part (i), show that the curve crosses every ray from the origin at the same angle, 45°. (This is why logarithmic spirals are called *equiangular*.)

2 **Polar identities.** Prove that $x\,\mathrm{d}x + y\,\mathrm{d}y = r\,\mathrm{d}r$ and $x\,\mathrm{d}y - y\,\mathrm{d}x = r^2\,\mathrm{d}\theta$.

3 **A system in time.** Consider the system $\dot{r} = r(1 - r^2)$, $\dot{\theta} = 1$. Show that every solution with $r(0) > 0$ approaches the unit circle as $t \to \infty$. Write the system in terms of x and y.

4 **Two cycles.** Find a system of the form $\dot{r} = f(r)$, $\dot{\theta} = 1$ with two limit cycles, at $r = 1$ and $r = 2$, one attracting and one repelling.

Further reading

- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press). Chapter 7 is about limit cycles and the Poincaré–Bendixson theorem.
- **Steven Strogatz, *Sync*** (Hyperion). A popular book about rhythms and synchronisation in nature.
- **James Gleick, *Chaos: Making a New Science*** (Penguin). The story of Lorenz and the discovery of chaos.

8.5 Greyhounds Chasing Greyhounds

The question

Four greyhounds A, B, C and D are held at positions such that $ABCD$ is a large square. At a given instant, the dogs are released and A runs directly towards B at constant speed v, B runs directly towards C at constant speed v, and so on. Show that A's path is given in polar coordinates (referred to an origin at the centre of the field and a suitable initial line) by $r = \lambda e^{-\theta}$, where λ is a constant.

Generalise this result to the case of n dogs held at the vertices of a regular n-gon ($n \geq 3$).

The hook

Four dogs stand at the corners of a square field. Each chases the dog to its left. Where do they meet, how long does it take, and what paths do they follow? The paths are beautiful spirals that wind round the centre infinitely many times, and yet each dog runs only a finite distance, exactly the length of a side of the square.

This is a *pursuit problem*, a type first studied by Pierre Bouguer in 1732 for a ship chasing a pirate. The trick that makes it solvable is symmetry: whatever happens to one dog happens to all of them, so the four dogs always form a square. Once you see that, the differential equation almost writes itself.

Before you start

Ask yourself By symmetry, what shape do the four dogs form at every moment? Where is its centre?

The situation is unchanged by a quarter-turn about the centre of the square (which moves $A \to B \to C \to D \to A$), so the dogs always form a square with the same centre O. The square rotates and shrinks, but it stays a square. In particular, A and B are always the same distance r from O, and OB is always a quarter-turn ahead of OA.

Working it through

STAGE 1 The direction of motion

Ask yourself A runs towards B. In the triangle OAB, what angle does AB make with AO (the direction towards the centre)?

Triangle OAB has $OA = OB$ and a right angle at O, so its other two angles are $45°$. So A's velocity, of size v, points at $45°$ to the inward radial direction, and is turned towards B, in the direction of increasing θ (if we label the angles so that B is anticlockwise from A).

STAGE 2 The differential equation

Ask yourself Split A's velocity into radial and transverse parts. In polar coordinates the radial velocity is $\dot{r}$ and the transverse velocity is $r\dot{\theta}$.

The radial component (inwards) is $v\cos 45°$ and the transverse component is $v\sin 45°$, so

$$\dot{r} = -\frac{v}{\sqrt{2}}, \qquad r\dot{\theta} = \frac{v}{\sqrt{2}}.$$

Dividing, $\dfrac{\mathrm{d}r}{\mathrm{d}\theta} = \dfrac{\dot{r}}{\dot{\theta}} = -r$, so $r = \lambda e^{-\theta}$, with λ the initial distance from the centre if $\theta = 0$ initially. ■

STAGE 3 n dogs

Pause and try For a regular n-gon, what is the angle at O in triangle OAB? What angle does AB make with AO? Repeat Stage 2.

The angle at O is $\frac{2\pi}{n}$, so the two equal base angles are $\frac{1}{2}\left(\pi - \frac{2\pi}{n}\right) = \frac{\pi}{2} - \frac{\pi}{n}$. The velocity makes this angle with the inward radius, so

$$\dot{r} = -v\cos\left(\frac{\pi}{2} - \frac{\pi}{n}\right) = -v\sin\frac{\pi}{n}, \qquad r\dot{\theta} = v\sin\left(\frac{\pi}{2} - \frac{\pi}{n}\right) = v\cos\frac{\pi}{n}.$$

Hence $\dfrac{\mathrm{d}r}{\mathrm{d}\theta} = -r\tan\dfrac{\pi}{n}$, and

$$r = \lambda e^{-\theta\tan(\pi/n)}. \qquad ■$$

For $n = 4$, $\tan\frac{\pi}{4} = 1$, recovering the square case.

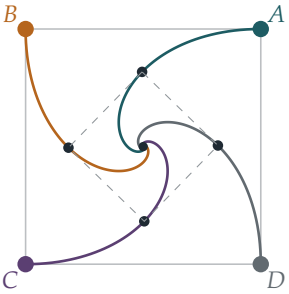


Figure 8.5.1 Four dogs chasing each other from the corners of a square. Each follows a logarithmic spiral $r = \lambda e^{-\theta}$ (relative to its own starting direction), winding infinitely often round the centre. At every moment the dogs form a square (dashed, shown at one intermediate time), rotating and shrinking until they meet at the centre.

Looking back

- **Symmetry reduces many bodies to one.** Four interacting dogs became a single particle obeying a simple rule, because the configuration stays a square.
- **Polar coordinates for motion about a centre.** Radial and transverse components of velocity, $\dot{r}$ and $r\dot{\theta}$, gave the equation directly.
- **Divide to eliminate time.** The ratio $\dot{r}/\dot{\theta}$ gave the shape of the path without needing to know when each point is reached.

What's really going on

Infinitely many turns, finite distance

The radial speed $\dot{r} = -v\sin\frac{\pi}{n}$ is constant, so the dogs reach the centre after time $T = \frac{r_0}{v\sin(\pi/n)}$, having each run a distance $vT = \frac{r_0}{\sin(\pi/n)}$. For a regular n-gon of side s, $r_0 = \frac{s}{2\sin(\pi/n)}$, so each dog runs

$$\frac{s}{2\sin^2(\pi/n)}.$$

For four dogs this is exactly s: each dog runs precisely the length of one side. (Here is a slick way to see it: B's velocity is always perpendicular to AB, so it does not change the distance AB; only A's motion, at speed v straight along AB, closes the gap, and the gap is initially s.) For three dogs the distance is $\frac{2}{3}s$, and for six it is $2s$. Yet in every case the spiral winds round the centre infinitely many times, because $\theta \to \infty$ as $r \to 0$. A curve of finite length with infinitely many turns is no paradox: the turns get smaller fast enough.

The logarithmic spiral

The curve $r = \lambda e^{-k\theta}$ is the *logarithmic* or *equiangular* spiral: it crosses every ray from the centre at the same angle. It appears in nature, in the shell of the nautilus and the arms of some galaxies, because growth that preserves shape produces it. It fascinated Jacob Bernoulli so much that he asked for it to be carved on his tombstone with the motto *Eadem mutata resurgo* ("Changed, I shall arise the same"), referring to the fact that it looks the same at every scale. The stonemason carved an Archimedean spiral instead.

Pursuit in practice

"Pure pursuit", always heading straight at the target, is the simplest guidance law for a missile or a predator. It is not the best: heading for where the target *will* be ("proportional navigation") is more efficient, and is what both modern missiles and some predators, such as dragonflies and falcons, actually do. The mathematics of pursuit and evasion became a subject of its own in the twentieth century under the name *differential games*.

The thinking skill

Exploit symmetry to reduce a many-body problem

If a system is symmetric and the rules respect the symmetry, the symmetry is preserved for all time, and you can follow just one part.

This idea solves many problems in mechanics and dynamics; it is a simple case of the deep connection between symmetry and conservation.

Try this next

1. **The slick argument.** Explain carefully why the distance AB decreases at rate exactly v in the four-dog problem, and deduce that the dogs meet after time s/v.

2. **Three dogs.** Three dogs start at the corners of an equilateral triangle of side s. Find the equation of each path, the time until they meet, and the distance each runs.

3. **How many turns?** For four dogs starting 100 m from the centre, how close to the centre are they after one full turn? After how many full turns are they within 1 mm?

4. **Two dogs.** What happens with $n = 2$ (two dogs facing each other)? Does the formula still make sense?

Further reading

- **Paul Nahin,** ***Chases and Escapes: The Mathematics of Pursuit and Evasion*** (Princeton University Press). Everything about pursuit curves, from Bouguer to missiles.
- **Eli Maor,** ***e: The Story of a Number*** (Princeton University Press). Includes the logarithmic spiral and Bernoulli's tombstone.
- **D'Arcy Thompson,** ***On Growth and Form*** (Cambridge University Press). The classic book on mathematics in nature, including spiral growth.

8.6 The Quantum Oscillator

The question

For $n = 0, 1, 2, \ldots$, the functions y_n satisfy the differential equation

$$\frac{\mathrm{d}^2 y_n}{\mathrm{d}x^2} - \omega^2 x^2 y_n = -(2n+1)\omega y_n,$$

where ω is a positive constant, and $y_n \to 0$ and $\mathrm{d}y_n/\mathrm{d}x \to 0$ as $x \to +\infty$ and as $x \to -\infty$. Verify that these conditions are satisfied, for $n = 0$ and $n = 1$, by

$$y_0(x) = \mathrm{e}^{-\lambda x^2} \qquad \text{and} \qquad y_1(x) = x\mathrm{e}^{-\lambda x^2}$$

for some constant λ, to be determined.

Show that

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(y_m \frac{\mathrm{d}y_n}{\mathrm{d}x} - y_n \frac{\mathrm{d}y_m}{\mathrm{d}x}\right) = 2(m-n)\omega y_m y_n,$$

and deduce that, if $m \neq n$,

$$\int_{-\infty}^{\infty} y_m(x) y_n(x)\,\mathrm{d}x = 0.$$

The hook

A mass on a spring oscillates with any energy you like: pull it back a little or a lot. But in 1900 Max Planck found that he could only explain the colours of light emitted by hot objects if the vibrating atoms could have only certain energies, in steps of a fixed size. Energy, at the atomic scale, comes in *quanta*. Twenty-five years later, quantum mechanics explained why.

This question contains the explanation. The differential equation is, after a change of units, the Schrödinger equation for a quantum particle in a spring-like potential: the *quantum harmonic oscillator*. It has well-behaved solutions, ones that die away at infinity, only for special values of the energy, labelled by $n = 0, 1, 2, \ldots$, in equally spaced steps. And the solutions for different n are *orthogonal*: the integral of their product is zero, just like $\sin mx$ and $\sin nx$ in a Fourier series. The quantum oscillator is the starting point for the physics of molecules, solids and light.

Before you start

Ask yourself Why should a function like $e^{-\lambda x^2}$ be a sensible guess? What happens to $e^{-\lambda x^2}$ and its derivative as $x \to \pm\infty$?

For large $|x|$ the equation is dominated by $y'' \approx \omega^2 x^2 y$, and a Gaussian $e^{-\lambda x^2}$ has $y'' \approx 4\lambda^2 x^2 y$ for large x, which matches if $4\lambda^2 = \omega^2$. With $\lambda > 0$, a Gaussian and all its derivatives decay to 0 faster than any power of x. So the boundary conditions will be automatic; the work is in matching the equation.

Working it through

STAGE 1 The ground state

Ask yourself Differentiate $y_0 = e^{-\lambda x^2}$ twice, substitute into the equation with $n = 0$, and compare coefficients of x^2 and of the constant term.

$y_0' = -2\lambda x e^{-\lambda x^2}$ and $y_0'' = (4\lambda^2 x^2 - 2\lambda)e^{-\lambda x^2}$. The equation with $n = 0$ requires

$$4\lambda^2 x^2 - 2\lambda - \omega^2 x^2 = -\omega$$

for all x. Matching coefficients: $4\lambda^2 = \omega^2$ and $2\lambda = \omega$. Both hold with $\lambda = \frac{\omega}{2}$ (taking $\lambda > 0$, so that $y_0 \to 0$). And $y_0, y_0' \to 0$ as $x \to \pm\infty$. ■

STAGE 2 The first excited state

$y_1 = xe^{-\lambda x^2}$ has $y_1' = (1 - 2\lambda x^2)e^{-\lambda x^2}$ and $y_1'' = (4\lambda^2 x^3 - 6\lambda x)e^{-\lambda x^2}$. The equation with $n = 1$ requires

$$4\lambda^2 x^3 - 6\lambda x - \omega^2 x^3 = -3\omega x,$$

which holds with the same $\lambda = \frac{\omega}{2}$. Again y_1 and y_1' decay at $\pm\infty$. ■

STAGE 3 An identity

Ask yourself Differentiate $y_m y_n' - y_n y_m'$ by the product rule. Two of the four terms cancel. Then use the differential equation to replace y_n'' and y_m''.

$\frac{\mathrm{d}}{\mathrm{d}x}(y_m y_n' - y_n y_m') = y_m' y_n' + y_m y_n'' - y_n' y_m' - y_n y_m'' = y_m y_n'' - y_n y_m''$. From the equation, $y_n'' = (\omega^2 x^2 - (2n+1)\omega)y_n$, and similarly for m. So

$$y_m y_n'' - y_n y_m'' = y_m y_n \left[\omega^2 x^2 - (2n+1)\omega - \omega^2 x^2 + (2m+1)\omega\right] = 2(m-n)\omega y_m y_n.$$ ■

STAGE 4 Orthogonality

Pause and try Integrate the identity from $-\infty$ to ∞. What is the left-hand side?

The left-hand side integrates to $\left[y_m y_n' - y_n y_m'\right]_{-\infty}^{\infty} = 0 - 0 = 0$, because each y and y' tends to 0 at both ends. So $2(m-n)\omega \int_{-\infty}^{\infty} y_m y_n \,\mathrm{d}x = 0$, and since $m \neq n$ and $\omega > 0$,

$$\int_{-\infty}^{\infty} y_m(x) y_n(x)\,\mathrm{d}x = 0. \qquad \blacksquare$$

For y_0 and y_1 this is easy to see directly: $y_0 y_1 = xe^{-\omega x^2}$ is an odd function. But the identity proves it for every pair, without knowing the functions.

Looking back

- **Match coefficients.** Substituting a guessed form and requiring the equation to hold for all x determined λ.
- **The Wronskian trick.** The combination $y_m y_n' - y_n y_m'$ has a derivative in which the second derivatives can be replaced using the equation, leaving something simple.
- **Boundary conditions kill boundary terms.** Decay at infinity made the integrated left-hand side vanish, which gave orthogonality.

What's really going on

Schrödinger's equation

In 1926 Erwin Schrödinger proposed that a quantum particle is described by a *wave function* $\psi(x)$, which for a particle of energy E in a potential $V(x)$ satisfies

$$-\frac{\hbar^2}{2m}\psi'' + V(x)\psi = E\psi.$$

For a spring, $V(x) = \tfrac{1}{2}m\Omega^2 x^2$, and after rescaling x this is exactly the equation in the question, with $E = \left(n + \tfrac{1}{2}\right)\hbar\Omega$. The physical requirement that ψ dies away at infinity (so that the particle is somewhere) can only be met for these energies: that is *quantisation*, falling out of the mathematics. Notice that the lowest energy is not zero but $\tfrac{1}{2}\hbar\Omega$: a quantum oscillator can never be completely at rest. This *zero-point energy* is real and measurable; it is why liquid helium does not freeze at ordinary pressure, even at absolute zero.

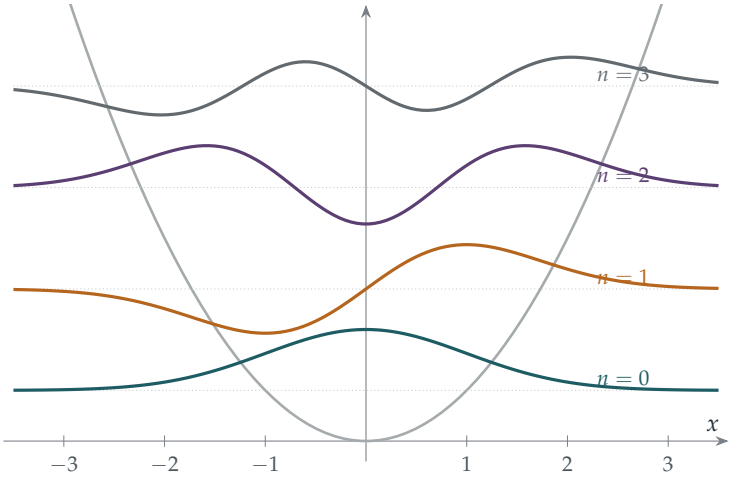


Figure 8.6.1 The quantum harmonic oscillator (with $\omega = 1$). The parabola is the potential energy. Each wave function y_n is drawn at the height of its energy level, $n + \frac{1}{2}$; the levels are equally spaced, and the lowest is above zero. The function y_n crosses zero n times, and any two of them are orthogonal.

Hermite functions and Sturm–Liouville theory

The general solution is $y_n = H_n(\sqrt{\omega}x)e^{-\omega x^2/2}$, where H_n is a polynomial of degree n, the *Hermite polynomial* (Figure 8.6.1). Orthogonality is not special to this equation. Whenever an equation can be written as $(py')' + qy = -\lambda wy$ with suitable boundary conditions, solutions belonging to different values of λ are orthogonal, by exactly the argument of Stage 4. This is *Sturm–Liouville theory* (1830s), and it covers the sine functions of Fourier series (Section 3.7), the vibrations of a drum, the Legendre polynomials of electrostatics and the wave functions of the hydrogen atom. The eigenvalue problem of Section 4.7 is another instance: the critical values found there are the eigenvalues λ, and the orthogonal functions are the natural "modes" of the system.

The thinking skill

Orthogonality from an identity

To show two solutions of related equations are orthogonal, find an exact derivative whose integral vanishes by the boundary conditions, and which equals a multiple of their product.

This is far more powerful than computing the integral directly: it works for every pair at once, even when you cannot write the solutions down.

Try this next

1 **The next state.** Show that $y_2 = (4\lambda x^2 - 1)e^{-\lambda x^2}$, with $\lambda = \frac{\omega}{2}$, satisfies the equation with $n = 2$. Check directly that $\int_{-\infty}^{\infty} y_0 y_2 \, dx = 0$, using $\int_{-\infty}^{\infty} e^{-\omega x^2} \, dx = \sqrt{\pi/\omega}$ and $\int_{-\infty}^{\infty} x^2 e^{-\omega x^2} \, dx = \frac{1}{2\omega}\sqrt{\pi/\omega}$.

2 **No other energies.** Try $y = e^{-\lambda x^2}$ with n replaced by a general number ν. Show that it works only if $\nu = 0$. (In fact, decaying solutions exist only when ν is a non-negative integer.)

3 **Sine waves.** Use the same method to show that $\int_0^{\pi} \sin mx \sin nx \, dx = 0$ for $m \neq n$, starting from $y'' = -n^2 y$ with $y(0) = y(\pi) = 0$.

4 **A ladder.** Show that if y_n solves the equation, then $y_{n+1} = \omega x y_n - y_n'$ solves it with $n + 1$ in place of n. Use this to generate y_1 and y_2 from y_0. (This is the "raising operator" of quantum mechanics.)

Further reading

- **David Griffiths, *Introduction to Quantum Mechanics*** (Cambridge University Press). The standard undergraduate physics text; Chapter 2 treats the harmonic oscillator.
- **Richard Feynman, *QED: The Strange Theory of Light and Matter*** (Princeton University Press). A popular account of quantum mechanics by one of its masters.
- **Mary Boas, *Mathematical Methods in the Physical Sciences*** (Wiley). Covers Hermite polynomials and Sturm–Liouville theory for students of mathematics and physics.

8.7 Chaos You Can Solve

The question

(i) The sequence of numbers $u_0, u_1, \ldots$ is given by $u_0 = u$ and, for $n \geq 0$,

$$u_{n+1} = 4u_n(1 - u_n). \qquad (*)$$

In the case $u = \sin^2\theta$ for some given angle θ, write down and simplify expressions for u_1 and u_2 in terms of θ. Conjecture an expression for u_n and prove your conjecture.

(ii) The sequence of numbers $v_0, v_1, \ldots$ is given by $v_0 = v$ and, for $n \geq 0$,

$$v_{n+1} = -pv_n^2 + qv_n + r,$$

where p, q and r are given numbers, with $p \neq 0$. Show that a substitution of the form $v_n = \alpha u_n + \beta$, where α and β are suitably chosen, results in the sequence $(*)$ provided that

$$4pr = 8 + 2q - q^2.$$

Hence obtain the sequence satisfying $v_0 = 1$ and, for $n \geq 0$, $v_{n+1} = -v_n^2 + 2v_n + 2$.

The hook

The rule $u_{n+1} = ku_n(1 - u_n)$ is the *logistic map*, the simplest model of a population with limited resources: u_n is the population in year n as a fraction of the maximum the environment can support, k is the breeding rate, and the factor $1 - u_n$ represents overcrowding. In 1976 the biologist Robert May published a short article in *Nature* called "Simple mathematical models with very complicated dynamics". He showed that, as k increases towards 4, this one-line rule goes from settling down, to oscillating, to behaving in a way that looks completely random. It became one of the founding examples of *chaos*.

Chaotic systems are usually impossible to solve: there is no formula for u_n, and the only way to find u_{100} is to compute all the terms before it. This question finds the exception. At $k = 4$, the most chaotic value of all, there *is* a formula, and it explains exactly why the sequence is unpredictable.

Before you start

Ask yourself Where have you seen the expression $4s(1-s)$ before, with $s = \sin^2\theta$? Which identity connects $\sin^2\theta$, $\cos^2\theta$ and $\sin 2\theta$?

If $s = \sin^2\theta$ then $1 - s = \cos^2\theta$, so $4s(1-s) = 4\sin^2\theta\cos^2\theta = (2\sin\theta\cos\theta)^2 = \sin^2 2\theta$. The map $s \mapsto 4s(1-s)$ is the double-angle formula in disguise.

Working it through

STAGE 1 Guessing the pattern

With $u_0 = \sin^2\theta$, the identity above gives

$$u_1 = 4\sin^2\theta\cos^2\theta = \sin^2 2\theta, \qquad u_2 = 4\sin^2 2\theta\cos^2 2\theta = \sin^2 4\theta.$$

Ask yourself What happens to the angle at each step? What is your conjecture for u_n?

The angle doubles each time, so we conjecture $u_n = \sin^2(2^n\theta)$.

STAGE 2 Proving it

Ask yourself Which proof method fits a statement about every term of a sequence defined by a rule from one term to the next?

Induction. The statement holds for $n = 0$, since $u_0 = \sin^2\theta = \sin^2(2^0\theta)$. Suppose $u_n = \sin^2(2^n\theta)$ for some n. Then

$$u_{n+1} = 4\sin^2(2^n\theta)\bigl(1 - \sin^2(2^n\theta)\bigr) = \bigl(2\sin(2^n\theta)\cos(2^n\theta)\bigr)^2 = \sin^2(2^{n+1}\theta),$$

which is the statement for $n+1$. So $u_n = \sin^2(2^n\theta)$ for all $n \geq 0$. ■

Notice that every starting value u in $[0,1]$ can be written as $\sin^2\theta$ (take $\theta = \arcsin\sqrt{u}$), so this solves the logistic map with $k = 4$ for every start in $[0,1]$.

STAGE 3 Changing variables

Ask yourself Substitute $v_n = \alpha u_n + \beta$ into the rule for v_{n+1}. You need the result to be $u_{n+1} = 4u_n - 4u_n^2$. How many conditions does that give, and how many unknowns do you have?

Substituting,

$$\alpha u_{n+1} + \beta = -p(\alpha u_n + \beta)^2 + q(\alpha u_n + \beta) + r,$$

so

$$u_{n+1} = -p\alpha\, u_n^2 + (q - 2p\beta)\, u_n + \frac{-p\beta^2 + q\beta + r - \beta}{\alpha}.$$

For this to be $4u_n - 4u_n^2$ we need three things:

$$p\alpha = 4, \qquad q - 2p\beta = 4, \qquad -p\beta^2 + (q-1)\beta + r = 0.$$

The first two fix $\alpha = 4/p$ and $\beta = (q-4)/(2p)$. There are no unknowns left, so the third equation is a genuine condition on p, q and r. Substituting β and multiplying by $4p$,

$$-(q-4)^2 + 2(q-1)(q-4) + 4pr = 0,$$

and expanding, $-(q^2 - 8q + 16) + (2q^2 - 10q + 8) + 4pr = 0$, that is $q^2 - 2q - 8 + 4pr = 0$, or

$$4pr = 8 + 2q - q^2. \qquad \blacksquare$$



Pause and try Three equations but only two unknowns: this is why a condition appears. Can you see, before calculating, why a general quadratic map should not always be equivalent to $(*)$? (Think about what a linear change of variable can and cannot change.)

STAGE 4 The example

For $v_{n+1} = -v_n^2 + 2v_n + 2$ we have $p = 1$, $q = 2$, $r = 2$, and $4pr = 8 = 8 + 4 - 4$, so the condition holds. Then $\alpha = 4$ and $\beta = -1$, so $v_n = 4u_n - 1$, with u_n satisfying $(*)$. Since $v_0 = 1$, $u_0 = \frac{1}{2} = \sin^2 \frac{\pi}{4}$, so

$$u_n = \sin^2\!\left(\frac{2^n \pi}{4}\right), \qquad v_n = 4\sin^2\!\left(\frac{2^n \pi}{4}\right) - 1 = 1 - 2\cos\!\left(\frac{2^n \pi}{2}\right).$$

This gives $v_0 = 1$, $v_1 = 3$ and $v_n = -1$ for all $n \geq 2$. (Check directly: $-1 + 2 + 2 = 3$, then $-9 + 6 + 2 = -1$, then $-1 - 2 + 2 = -1$.) $\blacksquare$

The particular starting value was chosen so that the angle lands on a multiple of π and the sequence stops. That hides the interesting part, which comes next.

Looking back

- **Recognise the disguise.** The map $s \mapsto 4s(1-s)$ became angle doubling once we wrote $s = \sin^2\theta$.
- **Conjecture, then prove by induction.** Two terms suggested the pattern; induction proved it.
- **Count equations and unknowns.** A linear substitution has two free parameters, a quadratic has three coefficients, so one condition must be left over.

What's really going on

Why the sequence is unpredictable

The formula $u_n = \sin^2(2^n\theta)$ looks like the end of the story: we can compute any term directly. But look at what it says. Since $\sin^2$ has period π, only the value of $2^n\theta$ modulo π matters. Write $\theta = \pi x$ with $0 \le x < 1$, and write x in binary:

$$x = 0.b_1b_2b_3\ldots \quad \text{(base 2)}.$$

Multiplying by 2 shifts the binary point one place to the right, and taking the result modulo 1 throws away the digit that crosses the point. So u_n depends only on the binary digits of x from b_{n+1} onwards. Each step of the logistic map reads one more digit of the starting value and discards one from the front.

This is *sensitive dependence on initial conditions*, the defining feature of chaos. Two starting values that agree to 20 binary places (about six decimal places) typically produce sequences that agree for about 20 steps and then have nothing to do with each other (Figure 8.7.1). To predict u_{100}, you need to know u_0 to 100 binary places. Any error in the starting angle, however small, doubles at every step. (An error in the population u itself does not grow by exactly the same factor each time, since the map stretches some parts of $[0,1]$ more than others, but on average it grows at the same rate.) The formula does not help: it simply makes the mechanism visible.

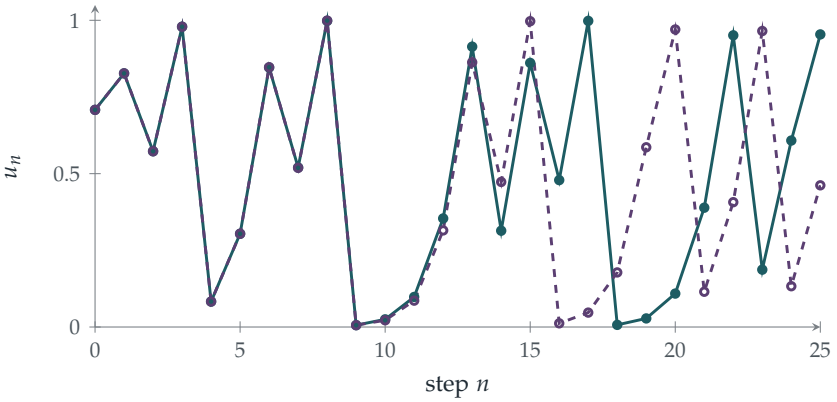


Figure 8.7.1 Two sequences of the logistic map with $k = 4$, starting from $\sin^2(1)$ (teal) and $\sin^2(1.00001)$ (plum, dashed). They are indistinguishable for about ten steps, visibly different by step 12, and unrelated after that: by step 12 the difference in angle, 10^{-5}, has been doubled twelve times, to about $2^{12} \times 10^{-5} \approx 0.04$.

Periodic points everywhere

The formula also finds every cycle of the map. The sequence returns to its starting value after N steps exactly when $\sin^2(2^N\theta) = \sin^2\theta$, that is, when $2^N\theta \equiv \pm\theta \pmod{\pi}$. So $\theta = m\pi/(2^N - 1)$ or $\theta = m\pi/(2^N + 1)$ gives a point of period dividing N. For

example $\theta = \pi/5$ satisfies $4\theta \equiv -\theta$, so $\sin^2\frac{\pi}{5} \to \sin^2\frac{2\pi}{5} \to \sin^2\frac{4\pi}{5} = \sin^2\frac{\pi}{5}$ is a 2-cycle. There are periodic points of every period, and they are dense in $[0,1]$: in every interval, however small, there is a starting value whose sequence repeats. Yet each one is unstable, because any error doubles. Dense periodic points that repel everything nearby are the second hallmark of chaos.

The road to chaos

For smaller k there is no formula, and the behaviour depends dramatically on k (Figure 8.7.2). For $1 < k < 3$ every sequence starting in $(0,1)$ settles down to the fixed point $1 - 1/k$. At $k = 3$ the fixed point becomes unstable and a 2-cycle takes over (this is the subject of STEP II 2002 Q5; see the exercises). At $k = 1 + \sqrt{6} \approx 3.449$ the 2-cycle gives way to a 4-cycle, then an 8-cycle, with the doublings coming faster and faster, until at $k \approx 3.5699$ the cycles have infinite length and chaos begins. Inside the chaotic region are windows of order: the clearest, near $k = 3.83$, contains a 3-cycle, which is the subject of Section 8.8.

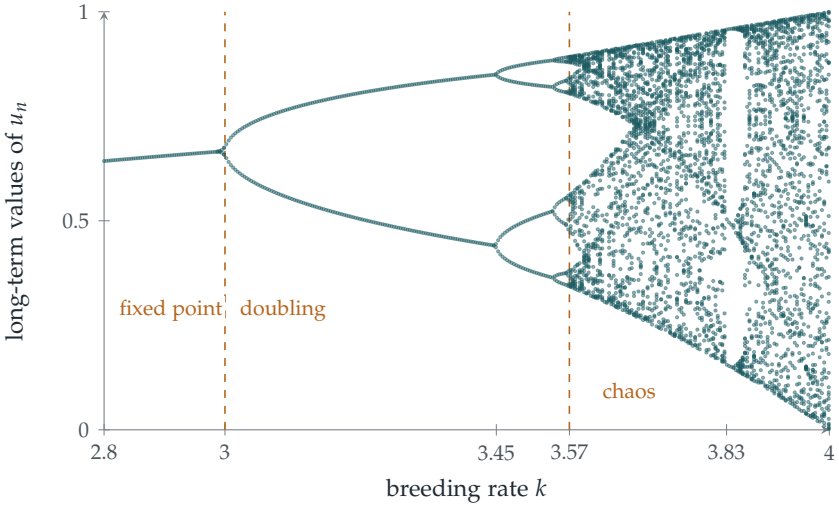


Figure 8.7.2 The bifurcation diagram of the logistic map. For each k, the map was iterated 300 times from $u_0 = 0.31416$ and the next 40 values plotted. A single fixed point splits into a 2-cycle at $k = 3$, then 4, 8, ..., until chaos sets in near $k = 3.57$. The pale vertical gap near $k = 3.83$ is the window containing a stable 3-cycle. At $k = 4$ the values fill all of $[0,1]$.

Feigenbaum's constant

In 1975 Mitchell Feigenbaum, working at Los Alamos with a programmable pocket calculator, measured the values of k at which the period doubles: $k_1 = 3$, $k_2 \approx 3.4495$,

$k_3 \approx 3.5441$, $k_4 \approx 3.5644$, ... He noticed that the gaps shrink by a constant factor,

$$\frac{k_n - k_{n-1}}{k_{n+1} - k_n} \to 4.6692\ldots,$$

and then, astonishingly, that the same number 4.6692... appears for $x \mapsto k\sin(\pi x)$ and for every other smooth map with a single hump that is rounded like a parabola at its top (a *quadratic maximum*). The constant is *universal* for that whole class; maps whose hump is flatter at the top, such as one shaped like $-x^4$, have universal constants of their own. It has since been measured in experiments on dripping taps, heated fluids and electronic circuits. Explaining why required ideas borrowed from the physics of phase transitions, and was only made rigorous by Oscar Lanford in 1982, with the help of a computer.

The same trick, three times

The substitution in part (ii) is an example of *conjugacy*: two maps are conjugate if a change of variable turns one into the other, and then they have exactly the same dynamics. The question shows that every quadratic map satisfying the condition is conjugate to the logistic map with $k = 4$, which in turn is angle doubling in disguise, through $u = \sin^2\theta$. (Strictly, this last link is a *semiconjugacy*: θ and $\pi - \theta$ give the same u, so the change of variable is not one-to-one. The logistic map is genuinely conjugate to the *tent map*, $t \mapsto 2t$ for $t \le \frac{1}{2}$ and $t \mapsto 2 - 2t$ for $t \ge \frac{1}{2}$, through $u = \sin^2\frac{\pi t}{2}$, which is one-to-one on $[0,1]$.) In Section 2.8 you met $x \mapsto x^2 - 2$, solved by $x = 2\cos\theta$ with the angle doubling at each step; it satisfies the condition too (try it). So these are all the same map in disguise. In Section 8.8, a third member of the family appears: a map built from straight lines, the *tent map*.

The thinking skill

Change variables to see the structure

A good substitution can turn a complicated rule into a simple one. Look for a change of variable that turns the problem into one you already understand.

Here $u = \sin^2\theta$ turned a quadratic map into angle doubling, and a linear substitution turned a whole family of quadratics into one. At university, conjugacy is a central tool: in dynamical systems, in linear algebra (diagonalising a matrix is choosing coordinates in which it is simple), and in group theory.

Try this next

1 Cycles of the logistic map. (From STEP II, 2002.) The sequence x_n satisfies $x_{n+1} = kx_n(1 - x_n)$.

(a) Prove that, if $0 < k < 4$ and $0 < x_0 < 1$, then $0 < x_n < 1$ for all n.

(b) Given that $x_0 = x_1 = x_2 = \cdots = a$, with $a \neq 0$ and $a \neq 1$, find k in terms of a.

(c) Given instead that $x_0 = x_2 = x_4 = \cdots = a$, with $a \neq 0, 1$, show that $ab^3 - b^2 + (1 - a) = 0$, where $b = k(1 - a)$. Given, in addition, that $x_1 \neq a$, find the possible values of k in terms of a. Hence show that a 2-cycle exists only when $k > 3$.

2 Squaring minus two. Show that $v_{n+1} = v_n^2 - 2$ satisfies the condition of part (ii) (write it as $-pv_n^2 + qv_n + r$ with $p = -1$). Find α and β, and check that your answer agrees with $v_n = 2\cos(2^n\phi)$.

3 A 3-cycle. Use the formula $u_n = \sin^2(2^n\theta)$ to find a 3-cycle of $u_{n+1} = 4u_n(1 - u_n)$. (*Hint:* try $\theta = \pi/7$, or $\theta = \pi/9$.) How many points have period exactly 3?

4 Losing digits. A computer stores numbers to about 53 binary places. If you iterate $u_{n+1} = 4u_n(1 - u_n)$ on a computer, roughly how many steps before the computed values bear no relation to the true sequence? Is the computed sequence then useless?

Further reading

- **Robert May, "Simple mathematical models with very complicated dynamics"**, *Nature* 261 (1976). A classic, still very readable.
- **James Gleick, *Chaos: Making a New Science*** (Penguin). The story of May, Feigenbaum and the discovery of chaos.
- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press), Chapter 10. One-dimensional maps, the logistic map and Feigenbaum's constant, at first-year undergraduate level.
- **Ian Stewart, *Does God Play Dice?*** (Penguin). A popular account of chaos, with a good chapter on the logistic map.

8.8 Period Three Implies Chaos

The question

Let $f(x) = 7 - 2|x|$. A sequence $u_0, u_1, u_2, \ldots$ is defined by $u_0 = a$ and $u_n = f(u_{n-1})$ for $n > 0$.

(i) (a) Sketch, on the same axes, the graphs with equations $y = f(x)$ and $y = f(f(x))$.

(b) Find all solutions of the equation $f(f(x)) = x$.

(c) Find the values of a for which the sequence $u_0, u_1, u_2, \ldots$ has period 2.

(d) Show that, if $a = \frac{28}{5}$, then the sequence $u_2, u_3, u_4, \ldots$ has period 2, but neither u_0 nor u_1 is equal to either of u_2 or u_3.

(ii) (a) Sketch, on the same axes, the graphs with equations $y = f(x)$ and $y = f(f(f(x)))$.

(b) Consider the sequence $u_0, u_1, u_2, \ldots$ in the cases $a = 1$ and $a = -\frac{7}{9}$. Hence find all the solutions of the equation $f(f(f(x))) = x$.

(c) Find a value of a such that the sequence $u_3, u_4, u_5, \ldots$ has period 3, but where none of u_0, u_1 or u_2 is equal to any of u_3, u_4 or u_5.

The hook

In 1975 Tien-Yien Li and James Yorke published a paper in the *American Mathematical Monthly* with one of the most famous titles in mathematics: "Period three implies chaos". They proved that if a continuous function from an interval to itself has a point of period 3 (a starting value that returns after exactly three steps), then it has points of *every* period: 1, 2, 4, 5, 17, 1000, all of them. Their paper also introduced the word "chaos" into mathematics.

This question, set in 2025, is a hands-on tour of that theorem. The function $f(x) = 7 - 2|x|$ is built from two straight lines, so everything can be computed exactly. You find its 2-cycles and its 3-cycles, and in doing so see why the existence of one forces the existence of the rest.

Before you start

Ask yourself What does the graph of $y = 7 - 2|x|$ look like? Where does it cross the line $y = x$? Where does it send the interval $[-7, 7]$?

The graph is an upside-down V with its peak at $(0, 7)$ and gradients ± 2. It meets $y = x$ where $x = 7 - 2x$ (with $x \geq 0$), giving $x = \frac{7}{3}$, and where $x = 7 + 2x$ (with $x < 0$), giving $x = -7$. These are the *fixed points*: start there and the sequence never moves. Also $f(-7) = f(7) = -7$ and $f(0) = 7$, so f maps $[-7, 7]$ onto itself, stretching it to twice its length and folding it in half.

Working it through

STAGE 1 The graph of $f \circ f$

Ask yourself As x goes from -7 to 7, $f(x)$ goes from -7 up to 7 and back down to -7. What does $f(f(x))$ do?

As x runs from -7 to 0, $f(x)$ runs over the whole of $[-7, 7]$, so $f(f(x))$ traces out a complete copy of the V: up from -7 to 7 and down again. The same happens as x runs from 0 to 7. So on $[-7, 7]$ the graph of $f \circ f$ is a W-shape turned upside down: four straight pieces of gradient ± 4, each running between heights -7 and 7, with peaks at $x = \pm\frac{7}{2}$ (where $f(x) = 0$) and a dip to -7 at $x = 0$ (Figure 8.8.1). Outside $[-7, 7]$ both graphs fall away steeply.

STAGE 2 Solving $f(f(x)) = x$

Ask yourself Each of the four straight pieces crosses the diagonal $y = x$ exactly once. Two of the crossings you already know. Why? How can you find the other two without expanding nested absolute values?

The fixed points $\frac{7}{3}$ and -7 certainly satisfy $f(f(x)) = x$. Any other solution p belongs to a 2-cycle $p \to q \to p$ with $q = f(p) \neq p$. Try the signs. If p and q are both positive, then $q = 7 - 2p$ and $p = 7 - 2q$, which forces $p = q = \frac{7}{3}$; if both are negative, the same argument gives -7. So one is positive and one negative, say $p > 0 > q$:

$$q = 7 - 2p, \qquad p = 7 + 2q.$$

Then $p = 7 + 14 - 4p$, so $p = \frac{21}{5}$ and $q = -\frac{7}{5}$, which indeed has $q < 0$.

Outside $[-7, 7]$ there are no solutions: for $x > 7$ we have $f(f(x)) = 21 - 4x < x$, and for $x < -7$ we have $f(f(x)) = 21 + 4x < x$. So the four solutions are

$$x = -7, \quad -\tfrac{7}{5}, \quad \tfrac{7}{3}, \quad \tfrac{21}{5}. \qquad \blacksquare$$

The sequence has period 2 exactly when $a = \frac{21}{5}$ or $a = -\frac{7}{5}$. (At the fixed points the sequence is constant, with period 1.)

STAGE 3 Arriving at a cycle late

With $a = \frac{28}{5}$: $u_1 = 7 - \frac{56}{5} = -\frac{21}{5}$, $u_2 = 7 - \frac{42}{5} = -\frac{7}{5}$, $u_3 = 7 - \frac{14}{5} = \frac{21}{5}$, $u_4 = -\frac{7}{5}$, and so on. From u_2 onwards the sequence is the 2-cycle, but $u_0 = \frac{28}{5}$ and $u_1 = -\frac{21}{5}$ are not in it. ■

Ask yourself How could you have found $\frac{28}{5}$ yourself? Work backwards: which numbers does f send to $-\frac{7}{5}$?

$f(x) = -\frac{7}{5}$ means $|x| = \frac{21}{5}$, so $x = \pm\frac{21}{5}$. One of these is in the cycle; the other, $-\frac{21}{5}$, is a way in. Then $f(x) = -\frac{21}{5}$ means $|x| = \frac{28}{5}$. Because f is two-to-one, every point of $(-7, 7)$ has two preimages, and so every cycle has a tree of starting values that feed into it.

STAGE 4 Three-cycles

Pause and try Compute the sequences starting from $a = 1$ and $a = -\frac{7}{9}$.

From $a = 1$: $1 \to 5 \to -3 \to 1$. From $a = -\frac{7}{9}$: $-\frac{7}{9} \to \frac{49}{9} \to -\frac{35}{9} \to -\frac{7}{9}$. Both are 3-cycles.

Ask yourself How many solutions can $f(f(f(x))) = x$ have? Think about the shape of its graph, as in Stage 1.

Repeating the argument of Stage 1, each time we apply f again the number of straight pieces doubles. So on $[-7, 7]$ the graph of $f \circ f \circ f$ is a zigzag of eight pieces, of gradient ± 8, each running the full height from -7 to 7 (Figure 8.8.2). Each piece crosses the diagonal exactly once, so there are exactly eight solutions in $[-7, 7]$, and (as before) none outside. We have found eight: the two fixed points and the six points of the two 3-cycles. So the solutions are

$$x = -7,\ \tfrac{7}{3},\ 1,\ 5,\ -3,\ -\tfrac{7}{9},\ \tfrac{49}{9},\ -\tfrac{35}{9}.$$ ■

STAGE 5 A late arrival at a 3-cycle

Work backwards from the cycle $1 \to 5 \to -3$. We want u_3 in the cycle but u_2 not. The preimages of 1 satisfy $|x| = 3$, so $x = \pm 3$; -3 is in the cycle, so take $u_2 = 3$. The preimages of 3 satisfy $|x| = 2$; take $u_1 = 2$. The preimages of 2 satisfy $|x| = \frac{5}{2}$; take $u_0 = \frac{5}{2}$. Then

$$\tfrac{5}{2} \to 2 \to 3 \to 1 \to 5 \to -3 \to 1 \to \cdots,$$

and none of $\frac{5}{2}$, 2, 3 is in the cycle. So $a = \frac{5}{2}$ works. ■

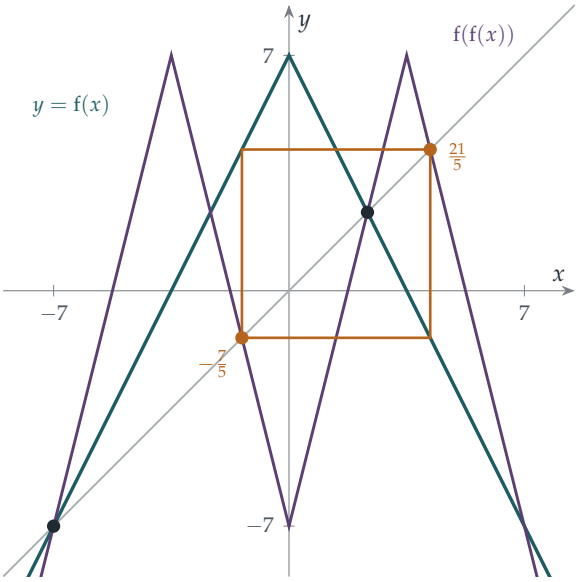


Figure 8.8.1 The graphs of $y = \mathrm{f}(x)$ (teal), $y = \mathrm{f}(\mathrm{f}(x))$ (plum) and $y = x$ (grey). The plum graph meets the diagonal four times: at the fixed points -7 and $\frac{7}{3}$ (black), and at the 2-cycle $\frac{21}{5} \leftrightarrow -\frac{7}{5}$ (amber). The amber square is the *cobweb* of the 2-cycle: go up to the graph of f, across to the diagonal, and repeat.

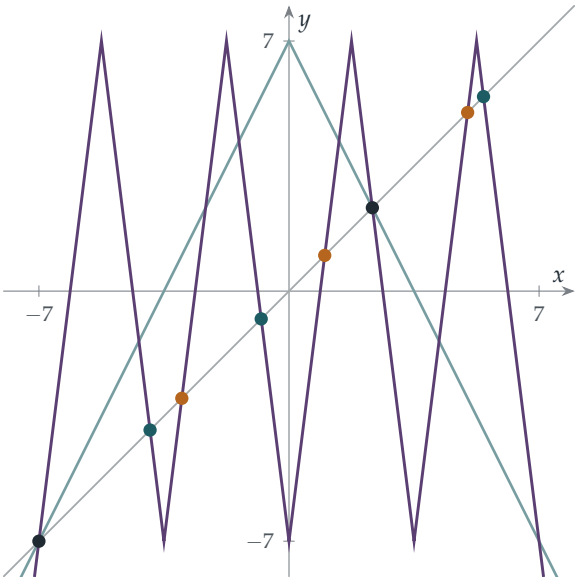


Figure 8.8.2 The graph of $y = \mathrm{f}(\mathrm{f}(\mathrm{f}(x)))$ (plum) on $[-7, 7]$ is a zigzag of eight pieces with gradient ± 8, each crossing the diagonal once. The eight crossings are the two fixed points (black), the 3-cycle $1 \to 5 \to -3$ (amber) and the 3-cycle $-\frac{7}{9} \to \frac{49}{9} \to -\frac{35}{9}$ (teal).

Looking back

- **Count by shape.** Instead of expanding nested absolute values, we counted the straight pieces of the graph. That told us how many solutions to look for, and so when we had found them all.
- **Use the structure of cycles.** Solutions of $f(f(x)) = x$ come as fixed points or in pairs; solutions of $f(f(f(x))) = x$ as fixed points or in threes.
- **Work backwards.** Preimages built the "late arrivals" in (i)(d) and (ii)(c).

What's really going on

Cycles of every length

The counting in Stage 4 works for any number of steps. Write f^n for f applied n times. On $[-7,7]$ the graph of f^n has 2^n straight pieces, each running the full height, so $f^n(x) = x$ has exactly 2^n solutions. Some of them have a shorter period d dividing n, but there are at most 2^d of those for each d, and

$$2^n > 2^{n/2} + 2^{n/3} + \cdots + 2^1 \quad \text{(summing over the divisors } d < n \text{ of } n)$$

because all the divisors less than n are at most $n/2$, and $2^1 + 2^2 + \cdots + 2^{\lfloor n/2 \rfloor} < 2^{\lfloor n/2 \rfloor + 1} \leq 2^n$. So for every n there are points whose period is exactly n. This function has cycles of every length.

Why period three forces everything

Li and Yorke's theorem says this is not special to our function: *any* continuous g from an interval to itself with a 3-cycle has cycles of every length. The proof uses nothing more than the intermediate value theorem, and our 3-cycle shows how it goes.

? **Ask yourself** Let $I_1 = [-3,1]$ and $I_2 = [1,5]$, the two intervals between the points of the cycle $1 \to 5 \to -3$. Where does f send each of them?

On I_2, f is decreasing from $f(1) = 5$ to $f(5) = -3$, so $f(I_2) = [-3,5] = I_1 \cup I_2$. On I_1, f goes from $f(-3) = 1$ up to $f(0) = 7$ and down to $f(1) = 5$, so $f(I_1) = [1,7]$, which contains I_2. Notice that only the three points of the cycle were used: any continuous function with a 3-cycle $a \to b \to c$, $a < b < c$, covers its intervals in the same way. We can summarise it as a diagram of which interval covers which (Figure 8.8.3):

$$I_1 \to I_2, \qquad I_2 \to I_1, \qquad I_2 \to I_2.$$

The theorem then follows from two facts, both consequences of the intermediate value theorem.

- If $g(I) \supseteq I$ then g has a fixed point in I. (Apply the intermediate value theorem to $g(x) - x$.)

- If $g(I) \supseteq J$ then there is a smaller closed interval $I' \subseteq I$ with $g(I') = J$ exactly.

To find a point of period n, follow the loop $I_2 \to I_2 \to \cdots \to I_2 \to I_1 \to I_2$ in the diagram: $n-1$ steps inside I_2 and one step out to I_1 and back. Using the second fact repeatedly, we can shrink I_2 to an interval of starting points whose orbits follow this itinerary, and g^n maps that interval onto all of I_2. The first fact gives a point x with $g^n(x) = x$ that visits I_1 only at step $n-1$. Its period cannot be smaller than n, because then the visit to I_1 would be repeated earlier. (There is one subtlety, since I_1 and I_2 share the point 1; checking that it causes no trouble is an exercise.)

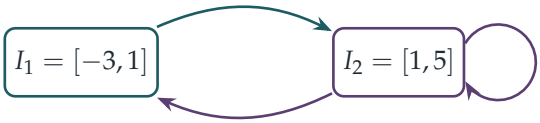


Figure 8.8.3 The covering diagram of a 3-cycle. An arrow from one interval to another means the image of the first contains the second. Every loop in the diagram gives a cycle of the function: the loop $I_2 \to I_2$ gives a fixed point, $I_1 \to I_2 \to I_1$ a 2-cycle, and going round the loop at I_2 $n-1$ times before visiting I_1 gives a cycle of length n.

Sharkovskii's theorem

Li and Yorke were not the first. In 1964 the Ukrainian mathematician Oleksandr Sharkovskii had proved a far more complete result, published in Russian in the *Ukrainian Mathematical Journal* and unknown in the West for a decade. Order the positive integers as

$$3,\ 5,\ 7,\ 9,\ \ldots,\ \ 2\cdot 3,\ 2\cdot 5,\ \ldots,\ \ 2^2\cdot 3,\ 2^2\cdot 5,\ \ldots,\ \ \ldots,\ \ 2^3,\ 2^2,\ 2,\ 1:$$

first the odd numbers from 3 upwards, then twice the odd numbers, then four times the odd numbers, and so on, ending with the powers of 2 in decreasing order. Sharkovskii's theorem says that if a continuous function on an interval has a cycle of length m, then it has cycles of every length that comes after m in this list. Since 3 comes first, period 3 implies every period. And the theorem is sharp: for every m there is a function with a cycle of length m and no cycles of lengths earlier in the list. The powers of 2 at the end are exactly the period-doubling cascade of the logistic map in Section 8.7, and the 3-cycle appears there at $k = 1 + \sqrt{8}$.

The tent map and the logistic map

Substitute $x = 14y - 7$, which maps $[0,1]$ onto $[-7,7]$. A short calculation shows that f becomes the *tent map*

$$T(y) = \begin{cases} 2y & 0 \le y \le \frac{1}{2}, \\ 2 - 2y & \frac{1}{2} \le y \le 1, \end{cases}$$

and the further substitution $u = \sin^2(\pi y/2)$ turns the tent map into the logistic map $u \mapsto 4u(1-u)$. So the function in this question is the chaotic logistic map of Section 8.7, straightened out. Its eight solutions of $f^3(x) = x$ correspond to the eight solutions of $\sin^2(8\theta) = \sin^2\theta$ in $[0, \frac{\pi}{2}]$.

The thinking skill

Existence by the intermediate value theorem

To show that a solution exists, you do not always need to find it. Show that a continuous function changes sign, or that an interval is mapped over itself.

In this question every solution could be computed. But Li and Yorke's theorem applies to functions we cannot compute with at all, and it proves the existence of infinitely many cycles using only the intermediate value theorem and a diagram of which interval covers which. At university this style of argument, existence without construction, appears in analysis (the intermediate value theorem itself), in topology (fixed point theorems) and throughout dynamical systems.

Try this next

1. **A 4-cycle.** Use the covering diagram to predict where a 4-cycle of f should be (three steps in I_2, one in I_1). Then find it exactly. (*Hint:* the cycle has three positive points and one negative point; write down the four linear equations.)

2. **The subtlety.** In the proof that period 3 implies period n, explain why the point x found cannot be the point 1 (or any point of the original 3-cycle) when $n \geq 4$.

3. **Period 5 without period 3.** Sketch the graph of a continuous function on $[1,5]$ with $g(1) = 3$, $g(3) = 4$, $g(4) = 2$, $g(2) = 5$, $g(5) = 1$, joining these points with straight lines. Draw its covering diagram for the intervals $[1,2]$, $[2,3]$, $[3,4]$, $[4,5]$. Can you find a loop of length 3? (Sharkovskii's theorem says there is no 3-cycle.)

4. **The tent map.** Check that $x = 14y - 7$ turns f into the tent map. Then show that the tent map doubles y written in binary, flipping all the digits whenever the leading digit is 1, and use this to find its 3-cycles.

Further reading

- **Tien-Yien Li & James Yorke, "Period three implies chaos"**, *American Mathematical Monthly* 82 (1975). Short, and readable with the intermediate value theorem.
- **Robert Devaney, *A First Course in Chaotic Dynamical Systems*** (CRC Press). Written for undergraduates; covers the tent map, the logistic map and Sharkovskii's theorem

with full proofs.

- **Keith Burns & Boris Hasselblatt, "The Sharkovsky theorem: a natural direct proof"**, *American Mathematical Monthly* 118 (2011). A modern proof using covering diagrams.

What We Have Learned

We began by saying that most differential equations cannot be solved, and that the art is to understand them anyway. We asked three questions: does the present determine the future, what can we know without solving, and what happens in the long run? Here is what the chapter's eight questions showed.

1. Does the present determine the future?

Usually, but not always. In Section 8.1, the equation $(y')^2 = 4y$ had infinitely many solutions through every point of the x-axis: a solution could rest at $y = 0$ for as long as it liked and then leave. The culprit was the square root, whose infinite slope at zero breaks the Lipschitz condition of the Picard–Lindelöf theorem. The same mechanism explains why a leaking bucket cannot tell you when it emptied, and it lies behind Norton's dome, a puzzle about determinism in classical mechanics.

Even when the future is determined, it may not be predictable. The logistic map of Section 8.7 has an exact formula, $u_n = \sin^2(2^n\theta)$, and yet every step doubles any error in the starting angle θ, and errors in the starting value grow, on average, just as fast. Determinism in principle is not predictability in practice.

2. What can we know without solving?

A great deal.

- **Energy.** In Section 8.2, multiplying the equation by y' produced a quantity that could not increase, and that bounded the solution of an equation nobody can solve. Such quantities are called *Lyapunov functions*, and they are the main tool for proving stability.
- **Equilibria and stability.** In Section 8.3, the sign of the growth rate predicted the fate of a fish population, and showed that harvesting beyond a threshold causes sudden collapse.
- **The right coordinates.** In Section 8.4, polar coordinates turned a coupled system into a single equation for r, showing every solution spiralling onto a circle. In Section 8.5, symmetry reduced four chasing dogs to one, and dividing to eliminate time gave the shape of their paths.
- **Structure from an identity.** In Section 8.6, the Wronskian showed that the solutions of an equation are orthogonal, without integrating a single one of them.
- **Counting and covering.** In Section 8.8, counting the pieces of a graph found every cycle, and the intermediate value theorem proved the existence of cycles of every

length.

3. What happens in the long run?

The chapter met every kind of long-term behaviour. Solutions settled to stable equilibria (Section 8.3), approached a repeating cycle (Section 8.4), spiralled into a point in finite time (Section 8.5), or behaved chaotically (Section 8.7). For the logistic map, a single parameter moves the system through all of these: a fixed point, a cascade of period doublings governed by Feigenbaum's universal constant, and finally chaos, with windows of order such as the 3-cycle. And Section 8.8 showed that once a 3-cycle appears, cycles of every length must exist, the first step in Sharkovskii's ordering of the integers.

4. A toolkit for differential equations and dynamics

Method	Where you used it
Looking for singular solutions	Non-uniqueness (8.1)
A quantity that cannot increase	Bounding (8.2)
Sign of the growth rate	Harvesting (8.3)
Coordinates that match the symmetry	Limit cycle (8.4); pursuit (8.5)
Substitution to linearise	Bernoulli equation (8.4)
The Wronskian	Orthogonality (8.6)
Conjugacy	Logistic map (8.7)
Induction	$u_n = \sin^2(2^n\theta)$ (8.7)
Counting pieces of a graph	Cycles of f^n (8.8)
Intermediate value theorem	Period three (8.8)

Where next

- Energy and conservation laws return in mechanics, starting with Section 10.1.
- Linear differential equations with several unknowns, solved by eigenvectors, govern vibrating systems in Section 11.2; the nonlinear pendulum, which cannot be solved in elementary functions, is in Section 11.3.
- Iterating a map to find its fixed points decides whether a family name dies out in Section 12.15, and long-run behaviour of random sequences is the subject of Section 12.16.

If you want to go further before university

- **Steven Strogatz, *Nonlinear Dynamics and Chaos***: the best first book on dynamical systems, full of applications and pictures. Strogatz's recorded lectures based on it are free online.

- **James Gleick, *Chaos: Making a New Science***: the story of how chaos was discovered.
- **George Simmons, *Differential Equations with Applications and Historical Notes***: a classic first course, rich in history.
- **Robert Devaney, *A First Course in Chaotic Dynamical Systems***: iterated maps, with proofs, at first-year level.

Laplace imagined that, knowing the present exactly, one could compute the whole future. This chapter has shown where that dream stops. Uniqueness can fail; and where it holds, chaos can amplify the smallest uncertainty until prediction is impossible. But it has also shown that a great deal can be understood without a formula: bounds, equilibria, cycles and their stability, and the structure of chaos itself. Moving from "solve the equation" to "understand the solutions" is one of the most important shifts in thinking that a university course in mathematics asks of you.

Answers to Try This Next

These are brief answers and hints, not full solutions: they give the result and the key step, so that you can check your own work. Try each exercise properly before looking.

8.1 When Solutions Aren't Unique

1. With $y = v^2$: $2vv' = \alpha v - \beta v^2$, so $v' = \frac{1}{2}(\alpha - \beta v)$ and $y_1 = \frac{\alpha^2}{\beta^2}\left(1 - \mathrm{e}^{-\beta t/2}\right)^2$. With $y = v^3$: $v' = \frac{1}{3}(\alpha - \beta v)$ and $y_1 = \frac{\alpha^3}{\beta^3}\left(1 - \mathrm{e}^{-\beta t/3}\right)^3$. The constant $y = 0$ makes both sides zero; dividing by v in the derivation threw it away.

2. $2\sqrt{h} = 2\sqrt{h_0} - kt$, so $h = \left(\sqrt{h_0} - \frac{1}{2}kt\right)^2$, and the bucket is empty at $T = \frac{2\sqrt{h_0}}{k}$. Both $h \equiv 0$ and $h = \frac{k^2}{4}(T - t)^2$ for $t \le T$, $h = 0$ for $t \ge T$, satisfy the equation and are zero from T on: looking at an empty bucket, you cannot tell when it emptied.

3. $\frac{\mathrm{d}}{\mathrm{d}x}(y\mathrm{e}^{-2x}) = (y' - 2y)\mathrm{e}^{-2x} = 0$, so $y = C\mathrm{e}^{2x}$ and $y(0) = 0$ forces $C = 0$. The function $2y$ is Lipschitz (it changes by at most $2|y_1 - y_2|$), while $2\sqrt{y}$ has infinite slope at 0.

4. For any $b \le 0 \le a$ (allowing $b = -\infty$ or $a = \infty$): $y = (x - b)^3$ for $x < b$, $y = 0$ for $b \le x \le a$, and $y = (x - a)^3$ for $x > a$. The case $a = b = 0$ is $y = x^3$.

8.2 Bounding Without Solving

1. $\frac{\mathrm{d}}{\mathrm{d}x}\left((y')^2 + y^2\right) = 2y'(y'' + y) = 0$, so $(y')^2 + y^2 = A^2$ and $|y| \le A$. The exact solution is $A\cos x$.

2. $\frac{\mathrm{d}}{\mathrm{d}t}\left((\theta')^2 - 2\cos\theta\right) = 2\theta'(\theta'' + \sin\theta) = 0$. Starting at rest, $(\theta')^2 = 2(\cos\theta - \cos\theta_0) \ge 0$, so $\cos\theta \ge \cos\theta_0$; for $|\theta_0| < \pi$, continuity keeps θ in $[-|\theta_0|, |\theta_0|]$, and at $|\theta_0| = \pi$ the pendulum balances upside down and stays there. If $|\theta_0| > \pi$, say $\theta_0 = \frac{3\pi}{2}$, then $\theta''(0) = -\sin\frac{3\pi}{2} = 1 > 0$, so θ starts by increasing: the pendulum is really at the angle $\theta_0 - 2\pi = -\frac{\pi}{2}$, and swings between $\pm\frac{\pi}{2}$ measured from there.

3. $\frac{\mathrm{d}}{\mathrm{d}t}\left((y')^2 + y^2\right) = 2y'(y'' + y) = -4k(y')^2 \le 0$. It does tend to zero, but the monotonicity alone does not show this: you need, for example, the explicit solutions, whose exponents $-k \pm \sqrt{k^2 - 1}$ have negative real part.

4. $(y')^2 - \frac{1}{2}y^4$ is constant, equal to $-\frac{1}{2}$. This gives $(y')^2 = \frac{1}{2}(y^4 - 1)$, which bounds nothing: the "potential" $-\frac{1}{2}y^4$ has no well to trap y, and in fact y blows up in finite time.

8.3 Harvesting a Population

1. The equation is $x' = \alpha^2 - (x-K)^2$, which becomes $y' = y(2\alpha - y)$. With $w = \frac{1}{y}$, $w' = 1 - 2\alpha w$, so $w = \frac{1}{2\alpha} + Ce^{-2\alpha t}$. If $y(0) < 0$ then $w(0) < 0 < \frac{1}{2\alpha}$, so w passes through 0 in finite time and $y \to -\infty$; x reaches 0 before that.
2. With $z = x - K$, $z' = -(z^2 + \beta^2)$, so $\tan^{-1}\frac{z}{\beta} = \tan^{-1}\frac{K}{\beta} - \beta t$. The population reaches zero ($z = -K$) at $t = \frac{2}{\beta}\tan^{-1}\frac{K}{\beta}$.
3. $x' = x(2K - h - x)$ has equilibria 0 and $2K - h$. For $h < 2K$ this is the logistic equation with a positive carrying capacity, and every positive solution tends to $2K - h$. The catch falls as the population falls, so it can never drive the population to zero.
4. $x = K + K\tanh Kt$, and $\tanh Kt = \frac{1}{2}$ gives $t = \frac{1}{K}\tanh^{-1}\frac{1}{2} = \frac{\ln 3}{2K}$.

8.4 Spiralling onto a Circle

1. The angle ψ between the curve and the radius satisfies $\tan\psi = \frac{r}{\mathrm{d}r/\mathrm{d}\theta} = -1$, the same at every point: the curve meets every ray at 45°.
2. $x\,\mathrm{d}x + y\,\mathrm{d}y = \frac{1}{2}\,\mathrm{d}(x^2+y^2) = r\,\mathrm{d}r$. And $x\,\mathrm{d}y - y\,\mathrm{d}x = r\cos\theta(\sin\theta\,\mathrm{d}r + r\cos\theta\,\mathrm{d}\theta) - r\sin\theta(\cos\theta\,\mathrm{d}r - r\sin\theta\,\mathrm{d}\theta) = r^2\,\mathrm{d}\theta$.
3. With $s = r^2$, $\dot{s} = 2s(1-s)$, a logistic equation, so $s \to 1$ for any $s(0) > 0$. In Cartesian form: $\dot{x} = x(1 - x^2 - y^2) - y$, $\dot{y} = y(1 - x^2 - y^2) + x$.
4. $\dot{r} = r(r-1)(r-2)$, $\dot{\theta} = 1$. Between 0 and 1, $\dot{r} > 0$; between 1 and 2, $\dot{r} < 0$; beyond 2, $\dot{r} > 0$. So $r = 1$ attracts and $r = 2$ repels.

8.5 Greyhounds Chasing Greyhounds

1. By symmetry the dogs always form a square. A runs along AB at speed v, closing the gap at rate v; B runs along BC, which is perpendicular to AB, and so does not change the distance to first order. So AB shrinks at rate exactly v, from s to 0, in time $\frac{s}{v}$.
2. $r = \lambda e^{-\sqrt{3}\theta}$. The gap closes at rate $v - v\cos 120° = \frac{3}{2}v$, so they meet after time $\frac{2s}{3v}$, each having run $\frac{2s}{3}$ (which agrees with $\frac{s}{2\sin^2(\pi/n)}$ for $n = 3$).
3. Each turn multiplies r by $e^{-2\pi} \approx \frac{1}{535}$: after one turn, about 19 cm. After two turns, about 0.35 mm, so two full turns suffice.
4. The two dogs run straight at each other and meet in the middle after time $\frac{s}{2v}$. The formula breaks down: $\tan\frac{\pi}{2}$ is infinite, matching the fact that there is no turning at all.

8.6 The Quantum Oscillator

1. This is a direct (if tedious) differentiation; alternatively, it follows from Exercise 4. With $2\lambda = \omega$, $\int y_0 y_2 = \int (2\omega x^2 - 1)\mathrm{e}^{-\omega x^2}\,\mathrm{d}x = 2\omega \cdot \frac{1}{2\omega}\sqrt{\frac{\pi}{\omega}} - \sqrt{\frac{\pi}{\omega}} = 0$.

2. $y'' = (4\lambda^2 x^2 - 2\lambda)y$, so the equation needs $4\lambda^2 = \omega^2$ and $2\lambda = (2\nu + 1)\omega$. With $\lambda = \frac{\omega}{2} > 0$ (for decay), $\nu = 0$.

3. $\frac{\mathrm{d}}{\mathrm{d}x}(y_m y_n' - y_n y_m') = y_m y_n'' - y_n y_m'' = (m^2 - n^2) y_m y_n$. Integrate from 0 to π: the left side gives 0 because y_m and y_n vanish at both ends, and $m^2 \neq n^2$.

4. Write $c = (2n + 1)\omega$, so $y_n'' = (\omega^2 x^2 - c)y_n$. Differentiating $z = \omega x y_n - y_n'$ twice and using this (and its derivative) gives $z'' - \omega^2 x^2 z = -(c + 2\omega)z$, and $c + 2\omega = (2n + 3)\omega$. From $y_0 = \mathrm{e}^{-\omega x^2/2}$: $2\omega x \mathrm{e}^{-\omega x^2/2}$, then $2\omega(2\omega x^2 - 1)\mathrm{e}^{-\omega x^2/2}$, which are multiples of y_1 and y_2.

8.7 Chaos You Can Solve

1. (a) By induction: if $0 < x < 1$ then $0 < x(1 - x) \leq \frac{1}{4}$, so $0 < kx(1 - x) < 1$. (b) $a = ka(1 - a)$ gives $k = \frac{1}{1-a}$. (c) With $x_1 = ka(1 - a) = ab$, the condition $x_2 = a$ is $kab(1 - ab) = a$; using $k = \frac{b}{1-a}$ this becomes $ab^3 - b^2 + 1 - a = 0$. The root $b = 1$ is the fixed point ($x_1 = a$), so factor it out: $ab^2 + (a - 1)b + (a - 1) = 0$; solving for b gives $k = \frac{b}{1-a}$. Equivalently, the two points of a 2-cycle are the roots of $k^2 x^2 - k(k + 1)x + (k + 1) = 0$, whose discriminant $k^2(k + 1)(k - 3)$ is positive only when $k > 3$.

2. $p = -1$, $q = 0$, $r = -2$: $4pr = 8 = 8 + 2q - q^2$. Matching coefficients gives $\alpha = \frac{4}{p} = -4$ and $\beta = \frac{q-4}{2p} = 2$, so $v_n = 2 - 4\sin^2(2^n\theta) = 2\cos(2^{n+1}\theta)$, which is $2\cos(2^n\phi)$ with $\phi = 2\theta$.

3. $u_3 = u_0$ needs $\sin^2 8\theta = \sin^2\theta$, so $8\theta = \pm\theta + k\pi$. With $\theta = \frac{\pi}{7}$: the cycle $\sin^2\frac{\pi}{7} \to \sin^2\frac{2\pi}{7} \to \sin^2\frac{4\pi}{7} \to \sin^2\frac{8\pi}{7} = \sin^2\frac{\pi}{7}$. With $\theta = \frac{\pi}{9}$: $\sin^2\frac{\pi}{9}$, $\sin^2\frac{2\pi}{9}$, $\sin^2\frac{4\pi}{9}$. The map $u \mapsto u_3$ has $2^3 = 8$ fixed points; two of them (0 and $\frac{3}{4}$) are fixed points of the map itself, leaving 6 points of period exactly 3: two 3-cycles.

4. Each step doubles θ, which loses one binary digit of accuracy, so after roughly 50 steps the computed values bear no relation to the true ones. They are not useless: the computed sequence stays close to a true orbit from a slightly different starting point (the *shadowing* property), so its statistics are still right.

8.8 Period Three Implies Chaos

1. With $a, b, c > 0$ and $d < 0$: $b = 7 - 2a$, $c = 7 - 2b$, $d = 7 - 2c$, $a = 7 + 2d$. Solving gives $17a = 49$, so the cycle is $\frac{49}{17} \to \frac{21}{17} \to \frac{77}{17} \to -\frac{35}{17} \to \frac{49}{17}$, with three points in $I_2 = [1, 5]$ and one in $I_1 = [-3, 1]$.

2. The point found has itinerary $I_2, I_2, \ldots, I_2, I_1$ (with $n - 1 \geq 3$ steps in I_2). The cycle $1 \to 5 \to -3$ cannot follow this: -3 is not in I_2, so neither 1 (whose second image is -3) nor 5 (whose first image is -3) nor -3 itself can be the point.

3. The covering diagram is $[1,2] \to [3,4], [4,5]$; $[2,3] \to [4,5]$; $[3,4] \to [2,3], [3,4]$; $[4,5] \to [1,2]$. The only loop of length 3 is $[3,4] \to [3,4] \to [3,4]$, which gives the fixed point, not a 3-cycle.

4. With $x = 14y - 7$, $\frac{f(x)+7}{14} = 1 - |2y - 1|$, the tent map. Doubling shifts the binary digits left; if the digit shifted out was 1, the result is $2 - 2y$, which flips the remaining digits. The 3-cycles are $\frac{2}{7} \to \frac{4}{7} \to \frac{6}{7}$ and $\frac{2}{9} \to \frac{4}{9} \to \frac{8}{9}$ (that is, $x = -3, 1, 5$ and $x = -\frac{35}{9}, -\frac{7}{9}, \frac{49}{9}$); in binary $\frac{2}{7} = 0.\overline{010}$ and $\frac{2}{9} = 0.\overline{001110}$.

CHAPTER

9

Counting and Chance

Counting is the oldest mathematics there is. Doing it cleverly is still one of the hardest.

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The Art of Counting

Combinatorics is the slums of topology.

— J. H. C. Whitehead, in the 1950s. It is not a view anyone holds today.

Everyone can count. But ask how many ways there are to make £1 from coins, or to seat ten people so that nobody sits in their own place, or to colour a map with four colours, and simple counting quickly breaks down. There are too many cases to list, and the cases overlap. *Combinatorics*, the mathematics of counting and arrangement, is the art of organising such problems so that they can be solved without listing.

For a long time, combinatorics was regarded as a collection of clever tricks rather than a serious subject, as Whitehead's remark suggests. That changed in the second half of the twentieth century, driven partly by computer science, where counting the steps an algorithm takes is a combinatorial problem, and partly by the discovery of deep connections with algebra, geometry and probability. Today it is one of the most active areas of mathematics. In 2022 June Huh was awarded a Fields Medal, the highest honour in mathematics, partly for proving a 1968 conjecture about the chromatic polynomials of Section 9.3, using ideas from algebraic geometry.

Why counting matters

Probability, in its simplest form, is counting: when all outcomes are equally likely, the probability of an event is the number of favourable outcomes divided by the total. So every question about cards, dice and lotteries is a counting problem, and the second half of this chapter is drawn from the probability papers for exactly this reason. But counting also matters well beyond games. Computer scientists count the operations an algorithm needs, and the ways it might fail. Chemists count the possible structures of molecules. Biologists count the ways a strand of DNA can fold. Cryptographers count the keys an attacker must try, and, as Section 9.9 shows, how many are needed before two of something coincide.

What you will meet at university

Most universities offer a first- or second-year course in combinatorics or discrete mathematics, and its themes are the themes of this chapter. You will learn the basic counting principles (the product rule, bijections, inclusion–exclusion, the pigeonhole principle), and how to set up and solve recurrence relations. You will meet *generating functions*, which turn sequences into functions and counting into algebra. You will study graphs, from colourings to Ramsey theory, and partially ordered sets such as the lattice of subsets. And probability courses will use the same tools, together with linearity of expectation, to analyse random structures. Further on, combinatorics

connects with algebra (symmetric functions, representation theory), with number theory (partitions, additive combinatorics) and with computer science (the analysis of algorithms, complexity theory).

The questions that run through this chapter

Three big questions

1. **How do you count without listing?** What structures (recurrences, bijections, generating functions) let you count huge collections exactly?
2. **When must something happen?** When does size alone force a pattern, a coincidence or a match?
3. **How do counting and probability help each other?** When is a probability best found by counting, and when is a count best found by probability?

The chapter has two halves. The first seven sections come from the pure papers. They begin with recurrences (lines cutting the plane, colouring fence posts) and the pigeonhole principle (Ramsey's theorem), then develop generating functions and use them to count coins, design strange dice and split numbers into teams with equal power sums. The last six sections are classic counting problems from the probability papers: derangements and the appearance of $1/e$, the birthday problem, the ballot problem and its reflection principle, the coupon collector, random subsets, and the impossibility of loading dice so that every total is equally likely.

Ideas in this chapter

§	Topic	Idea
9.1	Regions of the plane	What changes when you add one more
9.2	Ramsey theory	The pigeonhole principle
9.3	Graph colouring	Coupled recurrences; chromatic polynomials
9.4	Generating functions	A sequence becomes a function
9.5	Partitions	Counting via coefficients
9.6	Sicherman dice	Factorising polynomials
9.7	Thue–Morse	Repeated roots give balanced sums
9.8	Derangements	Inclusion–exclusion; $1/e$
9.9	Coincidences	Complements; the square-root law
9.10	The ballot problem	The reflection principle
9.11	Coupon collector	Linearity of expectation; $n \ln n$
9.12	Random subsets	One element at a time
9.13	Loaded dice	Impossibility by AM–GM

How to read this chapter

Always start with small cases. Draw three lines, list the six orders of three cards, count the ways to make 5p. Small cases give you data to guess from, and they check your formulas at the end. But do not trust a pattern until you have a reason for it: Section 9.1 includes a famous sequence, $1, 2, 4, 8, 16, \ldots$, whose next term is 31.

9.1 Lines Cutting the Plane

The question

A plane contains n distinct given lines, no two of which are parallel, and no three of which intersect at a point. By first considering the cases $n = 1, 2, 3$ and 4, provide and justify, by induction or otherwise, a formula for the number of line segments (including the infinite segments).

Prove also that the plane is divided into $\frac{1}{2}(n^2 + n + 2)$ regions (including those extending to infinity).

The hook

How many pieces can you cut a pizza into with n straight cuts, if the pieces need not be the same size? One cut gives 2, two cuts give 4, three cuts give 7 (not 8: try it), four give 11. The sequence $1, 2, 4, 7, 11, 16, \ldots$ is known as the *lazy caterer's sequence*, and this question asks for its formula.

The question is a first lesson in a technique that runs through all of combinatorics: do not try to count everything at once. Instead, ask what changes when you add one more object. And the answer turns out to be a special case of Euler's formula $V - E + F = 2$, which you met with the Platonic solids in Section 7.9.

Before you start

Pause and try Draw the cases $n = 1, 2, 3, 4$ carefully, keeping the lines in general position (no two parallel, no three through a point). Count the segments and the regions in each case. Can you guess the patterns?

You should find:

n	0	1	2	3	4
segments	0	1	4	9	16
regions	1	2	4	7	11

The segments look like n^2. The regions go up by $1, 2, 3, 4$: the differences are the natural numbers.

Working it through

STAGE 1 Counting segments

Ask yourself Concentrate on one line. How many of the other lines cross it, and into how many pieces do they cut it?

Each line is crossed by each of the other $n-1$ lines (none is parallel to it), and at $n-1$ different points (no three lines meet at a point). Those $n-1$ points cut the line into n segments: two infinite ones at the ends, and $n-2$ finite ones between. With n lines, there are

$$n \times n = n^2 \text{ segments.} \qquad \blacksquare$$

This is a direct count, with no induction needed: we counted segments line by line.

STAGE 2 What happens when we add a line?

Ask yourself Suppose $n-1$ lines are drawn, and you add the nth. It is cut into n segments by the others. What does each segment do to the region it passes through?

As the new line is drawn, each of its n segments runs through one of the existing regions and splits it into two (Figure 9.1.1). Different segments lie in different regions, because to pass from one region to another the line must cross an existing line. So adding the nth line increases the number of regions by exactly n:

$$R(n) = R(n-1) + n.$$

STAGE 3 Solving the recurrence

Starting from $R(0) = 1$ (no lines, one region),

$$R(n) = 1 + (1 + 2 + \cdots + n) = 1 + \frac{n(n+1)}{2} = \frac{n^2 + n + 2}{2}.$$

To present this as an induction: the formula holds for $n = 1$, since $\frac{1}{2}(1+1+2) = 2$; and if $R(n-1) = \frac{1}{2}\big((n-1)^2 + (n-1) + 2\big) = \frac{1}{2}(n^2 - n + 2)$, then $R(n) = \frac{1}{2}(n^2 - n + 2) + n = \frac{1}{2}(n^2 + n + 2)$. $\blacksquare$

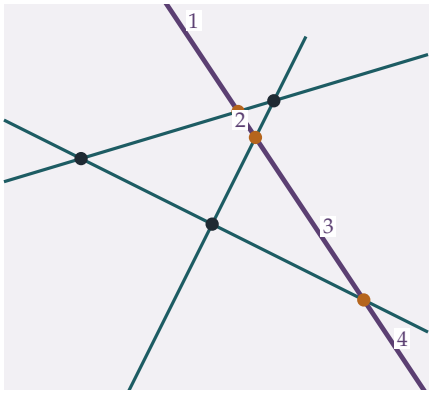


Figure 9.1.1 Three lines (teal) divide the plane into 7 regions. The fourth line (plum) is crossed by the other three at the amber points, which cut it into four segments, numbered 1 to 4. Each segment splits one region into two, so the fourth line adds 4 regions, giving 11.

Looking back

- **Count one piece at a time.** Segments were counted line by line; regions were counted by what one new line adds.
- **Turn a geometric step into a recurrence.** The observation "each new segment splits one region" became $R(n) = R(n-1) + n$.
- **General position matters.** Parallel lines, or three lines through a point, would give fewer regions. Find where each assumption was used.

What's really going on

Euler's formula in disguise

Here is a second proof, which explains the answer rather than just deriving it. Think of the arrangement of lines as a map: the crossing points are *vertices*, the segments are *edges* and the regions are *faces*. There are $V = \binom{n}{2}$ vertices (one for each pair of lines) and $E = n^2$ edges. The infinite segments have only one end, so add a single extra vertex "at infinity" where they all meet; now the picture is a map drawn on a sphere (imagine the plane wrapped round a ball, with infinity at the top). Euler's formula $V - E + F = 2$ from Section 7.9 gives

$$\left(\binom{n}{2} + 1\right) - n^2 + F = 2, \qquad \text{so} \qquad F = n^2 - \frac{n(n-1)}{2} + 1 = \frac{n^2 + n + 2}{2}.$$

The same number, from a completely different direction. The regions also split neatly into types: $2n$ of them are unbounded (one between each pair of neighbouring infinite rays), and the remaining $\frac{1}{2}(n^2 + n + 2) - 2n = \binom{n-1}{2}$ are bounded.

Binomial coefficients and higher dimensions

The formula can be written as

$$R(n) = \binom{n}{0} + \binom{n}{1} + \binom{n}{2},$$

and this is not a coincidence. Turn the picture so that no line is horizontal. A region that is bounded below has a lowest point, which must be a crossing, and each crossing is the lowest point of exactly one region (the one directly above it): that gives $\binom{n}{2}$ regions, one for each pair of lines. Every other region stretches down for ever, and so meets a horizontal line drawn far below all the crossings; the n lines cut that horizontal line into $n+1$ intervals, one in each such region. In total, $\binom{n}{2} + n + 1$. In three dimensions, n planes in general position cut space into

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} = \frac{n^3 + 5n + 6}{6}$$

pieces (the *cake numbers* $1, 2, 4, 8, 15, 26, \ldots$), and the pattern continues in every dimension. The proof is the same induction: a new plane is cut by the others into regions arranged like lines in a plane, and each of those regions splits a piece of space in two.

Steiner and the arrangement of hyperplanes

The formula for regions cut by lines and planes was found by Jakob Steiner in 1826. Today it is the starting point of the theory of *hyperplane arrangements*, which counts the regions formed by arbitrary collections of lines, planes and their higher-dimensional analogues, including non-general ones. A remarkable 1975 theorem of Thomas Zaslavsky counts the regions of any arrangement using a polynomial that encodes which hyperplanes meet. The same ideas appear in machine learning: a neural network built from "ReLU" units cuts its input space into regions by hyperplanes, and the number of regions measures how complicated a function it can represent.

The thinking skill

Ask what changes when you add one more

To count a configuration of n objects, compare it with the configuration of $n-1$. The change is often much easier to count than the whole.

This is induction used as a tool for discovery, not just for proof. It is one of the most common moves in combinatorics, and it turns a geometric question into a recurrence relation that can be solved.

Try this next

1 **Circles.** Show that n circles, each pair meeting in two points and no three through a point, divide the plane into $n^2 - n + 2$ regions.

2 **Parallel lines.** If the n lines include k that are all parallel to each other (and otherwise general position), how many regions are there?

3 **Cake numbers.** Prove by induction that n planes in general position divide space into $\frac{1}{6}(n^3 + 5n + 6)$ regions.

4 **A pattern that breaks.** Put n points on a circle and join every pair with a chord, with no three chords through one point inside the circle. For $n = 1, 2, 3, 4, 5$ you get $1, 2, 4, 8, 16$ regions. Count the regions for $n = 6$. (*Hint:* use Euler's formula; each set of four points gives one crossing.) Show that the number is $1 + \binom{n}{2} + \binom{n}{4}$.

Further reading

- **Ronald Graham, Donald Knuth & Oren Patashnik, *Concrete Mathematics*** (Addison-Wesley), Chapter 1. Starts with exactly this problem, and the "pizza" and "zig" variants.
- **Miklós Bóna, *A Walk Through Combinatorics*** (World Scientific). A friendly first course, with chapters on induction, graphs and Euler's formula.
- **Richard Guy, "The strong law of small numbers"**, *American Mathematical Monthly* 95 (1988). Many patterns that look obvious and turn out to be false, including the circle problem in the exercises.

9.2 Ramsey's Theorem in Miniature

The question

Six points A, B, C, D, E and F lie in three dimensional space and are in general positions, that is, no three are collinear and no four lie on a plane. All possible line segments joining pairs of points are drawn and coloured either gold or silver. Prove that there is a triangle whose edges are entirely of one colour. [**Hint**: consider segments radiating from A.]

Give a sketch showing that the result is false for five points in general positions.

The hook

At any party of six people, there are either three who all know each other or three who are all strangers to each other. This is guaranteed, whoever the six people are. With five people it is not guaranteed. This little fact is the first case of *Ramsey's theorem*, and it captures a principle that runs through modern combinatorics: *complete disorder is impossible*. Any sufficiently large structure, however it is coloured or arranged, must contain a highly ordered piece.

The proof takes four lines. But the obvious next question (how many people guarantee four mutual friends or four mutual strangers? Five?) leads to some of the hardest open problems in mathematics.

Before you start

Ask yourself The six points matter only through which pairs are joined by which colour: the geometry is a distraction. How many segments are there, and how many leave each point?

There are $\binom{6}{2} = 15$ segments, and 5 leave each point. (The general-position condition just makes sure every three points form a genuine triangle. The problem is really about the *complete graph* on six vertices, K_6, with its edges coloured in two colours.)

Working it through

STAGE 1 **Five segments, two colours**

Ask yourself Five segments leave A, each gold or silver. What can you say for certain about their colours?

At least three of them have the same colour. If at most two were gold and at most two silver, there would be at most four segments. This is the *pigeonhole principle*: five objects in two boxes put at least three in one box. Say (swapping the names of the colours if necessary) that AB, AC and AD are all gold.

STAGE 2 Look at the far ends

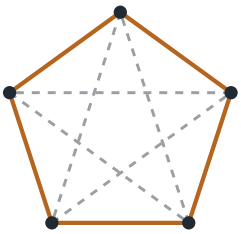
Ask yourself Now look at the triangle BCD. What happens if any one of its sides is gold? What if none of them is?

If any side of triangle BCD is gold, say BC, then ABC has all three sides gold: AB, AC and BC. If no side of BCD is gold, then BC, CD and DB are all silver, and BCD is a silver triangle. Either way there is a one-colour triangle (Figure 9.2.1, right). ■

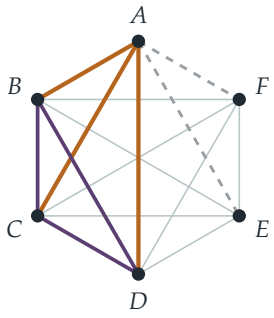
STAGE 3 Five points are not enough

Ask yourself With five points, each point has only four segments. How might you colour them so that each point has exactly two of each colour?

Place the five points as a regular pentagon (lifting them slightly off a plane if you want general position). Colour the five sides of the pentagon gold and the five diagonals silver (Figure 9.2.1, left). The gold edges form a five-cycle, which contains no triangle; the silver edges form a pentagram, which is also a five-cycle and contains no triangle. So there is no one-colour triangle. ■



five points: no one-colour triangle



six points: look at BCD

Figure 9.2.1 Left: five points with the pentagon gold (solid) and the pentagram silver (dashed) have no one-colour triangle. Right: with six points, three of the five segments from A share a colour, say gold to B, C, D. The triangle BCD (plum) then decides: one gold side makes a gold triangle with A, and no gold side makes BCD a silver triangle.

Looking back

- **Strip away the irrelevant.** The points in space became a coloured graph; only the colours of the edges mattered.
- **Pigeonhole.** Five edges and two colours forced three of one colour.
- **Case analysis on the right object.** Once three gold edges were found, the triangle at their far ends settled everything.
- **Sharpness.** The pentagon shows the number six cannot be lowered. A good theorem comes with an example showing it is best possible.

What's really going on

Ramsey numbers

The *Ramsey number* $R(s,t)$ is the smallest N such that every colouring of the edges of K_N in gold and silver contains either a gold K_s (a set of s points with all edges between them gold) or a silver K_t. The question has shown that $R(3,3) = 6$: six points always work, and the pentagon shows five do not.

Frank Ramsey proved in 1928 that these numbers always exist: for every s and t there is some N that works. The proof is an induction that generalises the argument above, and shows $R(s,t) \le R(s-1,t) + R(s,t-1)$. But exact values are extraordinarily hard to find:

$$R(3,3) = 6, \qquad R(4,4) = 18, \qquad 43 \le R(5,5) \le 46.$$

The value of $R(4,4)$ was found in 1955. Despite enormous computer searches, $R(5,5)$ is still unknown; the upper bound of 46 was proved only in 2024. A computer cannot simply check every colouring of K_{43}: there are 2^{903} of them, far more than the number of atoms in the universe.

Erdős and the aliens

Paul Erdős, who did more than anyone to develop Ramsey theory, liked to tell this story. Suppose aliens invade the Earth and threaten to destroy it unless we tell them the value of $R(5,5)$. We should marshal all our computers and mathematicians, and we might find it. But if they asked for $R(6,6)$, we should try to destroy the aliens. Erdős also proved, in 1947, that $R(k,k) > 2^{k/2}$, using a startling method: he showed that a *randomly* coloured graph on fewer than $2^{k/2}$ points has a positive probability of containing no one-colour K_k, so a good colouring must exist, even though nobody knows how to construct one. This *probabilistic method* is now one of the most powerful tools in combinatorics.

Order from chaos

Ramsey's theorem has many relatives, all expressing the same idea: large enough structures cannot avoid order. *Van der Waerden's theorem* says that if the whole numbers are

coloured with finitely many colours, there are arbitrarily long arithmetic progressions of one colour. The *Erdős–Szekeres theorem* says that any sequence of $n^2 + 1$ distinct numbers contains an increasing or a decreasing subsequence of length $n + 1$. And Ramsey himself was interested in a question of logic: he proved his theorem as a lemma in a paper on deciding the truth of logical statements.

The thinking skill

The pigeonhole principle

If more than kn objects are placed in n boxes, some box contains more than k objects.

The principle is obvious, but using it well is an art: the skill is in choosing what the objects and boxes should be. Here they were the five edges at one point and the two colours. At university, pigeonhole arguments appear in number theory (approximating irrationals by fractions), in analysis (the Bolzano–Weierstrass theorem) and throughout combinatorics.

Try this next

1 **At least two.** Show that in any two-colouring of K_6 there are in fact at least *two* one-colour triangles. (*Hint:* count the "bichromatic angles": pairs of edges of different colours at a common vertex. A point with g gold and $5 - g$ silver edges is at the apex of $g(5 - g) \leq 6$ such angles. Each triangle that is not one-colour contains exactly two.)

2 **Three colours.** If the edges of K_{17} are coloured in three colours, show that there is a one-colour triangle. (*Hint:* one point has at least six edges of one colour.)

3 **Three or four.** Show that $R(3,4) \leq 10$: in any gold/silver colouring of K_{10} there is a gold triangle or a silver K_4. (*Hint:* a point has either at least 4 gold edges or at least 6 silver edges.) In fact $R(3,4) = 9$.

4 **Monotone subsequences.** Prove the Erdős–Szekeres theorem for $n = 2$: any sequence of 5 distinct numbers contains an increasing or decreasing subsequence of length 3. Find a sequence of 4 with neither.

Further reading

- **Miklós Bóna, *A Walk Through Combinatorics*** (World Scientific), the chapters on the pigeonhole principle and Ramsey theory.
- **Ronald Graham & Joel Spencer, "Ramsey theory"**, *Scientific American* (July 1990). A popular introduction by two leading researchers.

- **Noga Alon & Joel Spencer, *The Probabilistic Method*** (Wiley). The standard book on Erdős's method; the first chapter is accessible.
- **Paul Hoffman, *The Man Who Loved Only Numbers*** (Fourth Estate). A biography of Erdős.

9.3 Colouring Fence Posts

The question

I have n fence posts placed in a line and, as part of my spouse's birthday celebrations, I wish to paint them using three different colours red, white and blue in such a way that no adjacent fence posts have the same colours. (This allows the possibility of using fewer than three colours as well as exactly three.) Let r_n be the number of ways (possibly zero) that I can paint them if I paint the first and the last post red and let s_n be the number of ways that I can paint them if I paint the first post red but the last post either of the other two colours. Explain why $r_{n+1} = s_n$ and find $r_n + s_n$. Hence find the value of $r_{n+1} + r_n$ for all $n \geq 1$.

Prove, by induction, that

$$r_n = \frac{2^{n-1} + 2(-1)^{n-1}}{3}.$$

Find the number of ways of painting n fence posts (where $n \geq 3$) placed in a circle using three different colours in such a way that no adjacent fence posts have the same colours.

The hook

Painting posts in a line is easy: 3 choices for the first, then 2 for each of the rest (anything but the colour of the previous post), giving $3 \times 2^{n-1}$. Bend the line into a circle and it becomes surprisingly awkward, because the last post now has to differ from the first as well as from its neighbour, and whether that rules out one colour or none depends on everything that came before.

The question gets round this with a classic trick: split the count into cases according to the colour of the last post, and find how the cases feed into each other. The answer, $2^n + 2(-1)^n$, is the first example of a *chromatic polynomial*, part of the mathematics behind the four colour theorem.

Before you start

Pause and try Count the ways to paint 3 posts in a circle, and 4. Does $3 \times 2^{n-1}$ overcount, and by how much?

Three posts in a circle are all adjacent to each other, so they need three different colours: $3! = 6$ ways. Four posts in a circle: 18 ways (you can check this at the end). Compare $3 \times 2^2 = 12$ and $3 \times 2^3 = 24$ for the line: the circle loses the colourings in which the last post matches the first.

Working it through

STAGE 1 Two kinds of colouring

Ask yourself Take a colouring of $n+1$ posts with the first and last red. Remove the last post. What can you say about the colouring of the first n posts that remains?

Post n is next to post $n+1$, which is red, so post n is not red. Removing post $n+1$ leaves a colouring of n posts with the first red and the last not red: one of the s_n colourings. Conversely, any of the s_n colourings can be extended in exactly one way, by painting post $n+1$ red. So the two sets correspond exactly, and $r_{n+1} = s_n$.

Ask yourself What is $r_n + s_n$?

Every colouring with the first post red has its last post either red or not, so $r_n + s_n$ counts all colourings of n posts with the first red. After the first post, each post has 2 choices, so $r_n + s_n = 2^{n-1}$. Hence

$$r_{n+1} + r_n = s_n + r_n = 2^{n-1}. \qquad \blacksquare$$

STAGE 2 The formula by induction

For $n = 1$ there is one post, painted red, so $r_1 = 1$, and the formula gives $\frac{1}{3}(1+2) = 1$. Suppose the formula holds for r_n. Then

$$r_{n+1} = 2^{n-1} - r_n = 2^{n-1} - \frac{2^{n-1} + 2(-1)^{n-1}}{3} = \frac{2 \cdot 2^{n-1} - 2(-1)^{n-1}}{3} = \frac{2^n + 2(-1)^n}{3},$$

which is the formula for r_{n+1}. $\blacksquare$

Pause and try Where did the formula come from? The recurrence $r_{n+1} + r_n = 2^{n-1}$ is like a differential equation: a "particular solution" $c \cdot 2^n$ plus a

"complementary function" $A(-1)^n$ that solves $r_{n+1} + r_n = 0$. Find c and A.

STAGE 3 The circle

Ask yourself In a circle of n posts, fix the colour of post 1, say red. Now cut the circle between post n and post 1, and straighten it into a line. Which of the line colourings correspond to valid circle colourings?

Straightened out, the circle becomes a line of n posts starting with red, in which the last post must also differ from post 1: the last post is not red. These are exactly the s_n colourings. Post 1 can be any of 3 colours, so the number of circle colourings is

$$3s_n = 3(2^{n-1} - r_n) = 3 \cdot 2^{n-1} - 2^{n-1} - 2(-1)^{n-1} = 2^n + 2(-1)^n. \qquad \blacksquare$$

Check: $n = 3$ gives $8 - 2 = 6$, and $n = 4$ gives $16 + 2 = 18$.

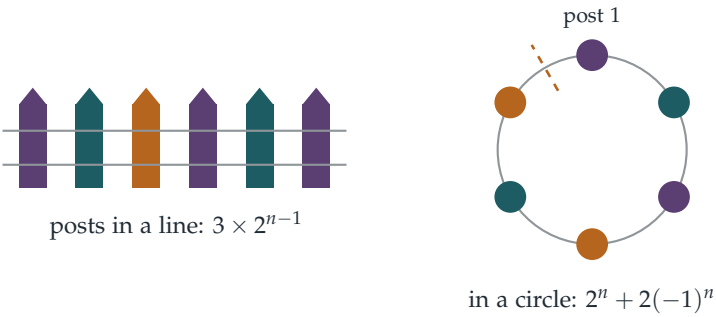


Figure 9.3.1 Left: in a line, each post after the first just has to differ from its predecessor. Right: in a circle, the last post must also differ from post 1. Cutting the circle at the dashed line turns a circle colouring into a line colouring whose first and last posts differ: one of the s_n colourings counted in the question.

Looking back

- **Split into cases that feed each other.** Recording whether the last post was red gave two sequences, r_n and s_n, linked by simple recurrences.
- **Bijections.** $r_{n+1} = s_n$ was proved by matching colourings one-to-one, not by calculation.
- **Cut the circle.** A cyclic problem became a linear one with a condition on the ends.

What's really going on

Chromatic polynomials

The posts and their adjacencies form a *graph*: a path for the line, a cycle for the circle. A *proper colouring* gives adjacent vertices different colours. The number of proper colourings with k colours is a polynomial in k, the *chromatic polynomial* of the graph, introduced by George Birkhoff in 1912. Repeating the question's argument with k colours gives

$$\text{path: } k(k-1)^{n-1}, \qquad \text{cycle: } (k-1)^n + (-1)^n(k-1).$$

With $k = 3$ the cycle formula is $2^n + 2(-1)^n$. Birkhoff hoped to prove the four colour theorem (every map can be coloured with four colours so that neighbouring countries differ) by showing that the chromatic polynomial of every map's graph is positive at $k = 4$. That approach never succeeded; the theorem was finally proved in 1976 by Kenneth Appel and Wolfgang Haken, with extensive computer help. Chromatic polynomials remain an active field, with surprising connections to statistical physics, where they appear as the *Potts model* of magnetism.

The same count by matrices

There is another way to see the answer, using linear algebra from Chapter 7. Let $\mathbf{M}$ be the 3×3 matrix with 0 on the diagonal and 1 elsewhere: entry (i, j) is 1 when colour j may follow colour i. Then entry (i, j) of $\mathbf{M}^{n-1}$ counts line colourings that start with colour i and end with colour j, so r_n is a diagonal entry of $\mathbf{M}^{n-1}$. A circle colouring is a sequence that returns to its start after n steps, so the number of circle colourings is the *trace* of $\mathbf{M}^n$. Since $\mathbf{M} = \mathbf{J} - \mathbf{I}$, where every entry of $\mathbf{J}$ is 1, the eigenvalues of $\mathbf{M}$ are $3 - 1 = 2$ (eigenvector $(1, 1, 1)$) and -1 (twice), and the trace of $\mathbf{M}^n$ is the sum of the nth powers of the eigenvalues:

$$\operatorname{tr} \mathbf{M}^n = 2^n + (-1)^n + (-1)^n = 2^n + 2(-1)^n.$$

This *transfer matrix* method is used throughout combinatorics and statistical physics to count arrangements built up one step at a time.

The thinking skill

Set up coupled recurrences

When one sequence will not satisfy a simple recurrence on its own, record a little extra information (here, the colour of the last post) and find recurrences linking the cases.

This is the idea of a *state*: keep track of exactly what you need to know to continue, and nothing more. It is the basis of dynamic programming in computer science, of Markov chains in probability (Section 12.16), and of the transfer matrix method.

Try this next

1 **k colours.** Repeat the question with k colours. Show that $r_{n+1} = s_n$ still holds and that $r_{n+1} + r_n = (k-1)^{n-1}$, find r_n, and deduce the chromatic polynomial of the cycle, $(k-1)^n + (-1)^n(k-1)$.

2 **No two reds together.** How many ways are there to paint n posts in a line red or white, with no two adjacent posts red? (You should find the Fibonacci numbers.)

3 **A wheel.** Add a central post adjacent to every post of the circle (a "wheel"). How many ways are there to colour it with 4 colours? (*Hint:* the centre uses one colour, and the circle must use the other three.)

4 **Necklaces.** If two colourings of the circle count as the same when one is a rotation of the other, how many different colourings are there for $n = 4$? For $n = 5$? (This is the start of *Burnside's lemma*.)

Further reading

- **Robin Wilson, *Introduction to Graph Theory*** (Pearson). The standard first course; covers colourings and the four colour problem.
- **Robin Wilson, *Four Colours Suffice*** (Penguin). The story of the four colour theorem, for a general reader.
- **Herbert Wilf, *generatingfunctionology*** (free online). Recurrences and how to solve them, including the transfer matrix method.

9.4 Sequences as Functions

The question

Given an infinite sequence of numbers $u_0, u_1, u_2, \ldots$, we define the *generating function*, f, for the sequence by

$$\mathrm{f}(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots .$$

Issues of convergence can be ignored in this question.

(i) Using the binomial series, show that the sequence given by $u_n = n$ has generating function $x(1-x)^{-2}$, and find the sequence that has generating function $x(1-x)^{-3}$.

Hence, or otherwise, find the generating function for the sequence $u_n = n^2$. You should simplify your answer.

(ii) (a) The sequence $u_0, u_1, u_2, \ldots$ is determined by $u_n = k u_{n-1}$ $(n \geq 1)$, where k is independent of n, and $u_0 = a$. By summing the identity $u_n x^n \equiv k u_{n-1} x^n$, or otherwise, show that the generating function, f, satisfies

$$\mathrm{f}(x) = a + kx\,\mathrm{f}(x).$$

Write down an expression for $\mathrm{f}(x)$.

(b) The sequence $u_0, u_1, u_2, \ldots$ is determined by $u_n = u_{n-1} + u_{n-2}$ $(n \geq 2)$ and $u_0 = 0$, $u_1 = 1$. Obtain the generating function.

The hook

Work out $1/89$ on a calculator: $0.011235955\ldots$. The digits begin $0, 1, 1, 2, 3, 5$, the Fibonacci numbers. Is this a coincidence? It is not, and this question explains why.

The idea is one of the most powerful in combinatorics. Take an infinite sequence of numbers and package it as a single function, by using the terms as the coefficients of a power series. Then facts about the sequence become facts about the function, and you can use algebra and calculus on them. Herbert Wilf put it memorably: "A generating function is a clothesline on which we hang up a sequence of numbers for display."

Before you start

Ask yourself What sequence has generating function $\dfrac{1}{1-x}$? What about $\dfrac{1}{1-2x}$? What happens to the sequence if you multiply its generating function by x?

The geometric series $\frac{1}{1-x} = 1 + x + x^2 + \cdots$ is the generating function of $1, 1, 1, \ldots$, and $\frac{1}{1-2x} = 1 + 2x + 4x^2 + \cdots$ that of the powers of 2. Multiplying by x shifts every term one place along: $x \cdot \frac{1}{1-x} = x + x^2 + \cdots$ belongs to $0, 1, 1, 1, \ldots$.

Working it through

STAGE 1 The natural numbers and the triangle numbers

Ask yourself Expand $(1-x)^{-2}$ by the binomial series. What is the coefficient of x^n?

The binomial series gives

$$(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \cdots = \sum_{n \geq 0} (n+1)x^n,$$

so $x(1-x)^{-2} = x + 2x^2 + 3x^3 + \cdots = \sum nx^n$: the generating function of $u_n = n$.

Similarly, the coefficient of x^n in $(1-x)^{-3}$ is $\frac{(-3)(-4)\cdots(-3-n+1)}{n!}(-1)^n = \frac{(n+1)(n+2)}{2}$, so

$$x(1-x)^{-3} = \sum_{n \geq 1} \frac{n(n+1)}{2} x^n = x + 3x^2 + 6x^3 + 10x^4 + \cdots,$$

the generating function of the triangle numbers $u_n = \frac{1}{2}n(n+1)$.

Ask yourself How can you write n^2 in terms of n and $\frac{1}{2}n(n+1)$?

Since $n^2 = 2 \cdot \frac{1}{2}n(n+1) - n$, and generating functions add just as sequences do,

$$\sum n^2 x^n = \frac{2x}{(1-x)^3} - \frac{x}{(1-x)^2} = \frac{2x - x(1-x)}{(1-x)^3} = \frac{x(1+x)}{(1-x)^3}. \qquad \blacksquare$$

STAGE 2 A recurrence becomes an equation

Ask yourself Multiply $u_n = ku_{n-1}$ by x^n and add up over all $n \geq 1$. What is each side, in terms of f?

The left side is $\sum_{n\geq 1} u_n x^n = f(x) - u_0 = f(x) - a$. The right side is $kx \sum_{n\geq 1} u_{n-1} x^{n-1} = kx\, f(x)$. So $f(x) = a + kx\, f(x)$, and

$$f(x) = \frac{a}{1-kx}. \qquad \blacksquare$$

Of course, $u_n = ak^n$; expanding $\frac{a}{1-kx}$ as a geometric series confirms it.

STAGE 3 The Fibonacci numbers

Do the same with $u_n = u_{n-1} + u_{n-2}$, for $n \geq 2$:

$$\sum_{n\geq 2} u_n x^n = x \sum_{n\geq 2} u_{n-1} x^{n-1} + x^2 \sum_{n\geq 2} u_{n-2} x^{n-2},$$

that is,

$$f(x) - u_0 - u_1 x = x\big(f(x) - u_0\big) + x^2 f(x).$$

With $u_0 = 0$ and $u_1 = 1$, this says $f(x) - x = x\, f(x) + x^2 f(x)$, so

$$f(x) = \frac{x}{1 - x - x^2}. \qquad \blacksquare$$

STAGE 4 Explaining $1/89$

Pause and try Ignoring convergence (which is fine here: the series converges for $|x| < 0.618$), put $x = \frac{1}{10}$ in f. What do you get?

$$\frac{1}{10}F_1 + \frac{1}{100}F_2 + \frac{1}{1000}F_3 + \cdots = f\Big(\frac{1}{10}\Big) = \frac{1/10}{1 - 1/10 - 1/100} = \frac{10}{89}.$$

Dividing by 10, $\frac{1}{89} = \sum F_n/10^{n+1}$. The Fibonacci numbers are placed in successive decimal places, overlapping when they get bigger than 9 (Figure 9.4.1). The early digits of $1/89$ show the Fibonacci numbers; later ones are blurred by carrying.

```
0.  1
0.    1
0.      2
0.        3
0.          5
0.            8
0.            1 3
0.              2 1
0.                3 4
0.                  5 5
 ⋮
0. 0 1 1 2 3 5 9 5 5 0 5
```

Figure 9.4.1 Adding $F_n/10^{n+1}$ for $n = 1, 2, 3, \ldots$. Each Fibonacci number ends one decimal place further to the right. Once they have two digits they overlap, and carrying changes the digits from the seventh place on: $\frac{1}{89} = 0.01123595505\ldots$.

Looking back

- **Turn a sequence into a function.** The whole infinite sequence became one rational function.
- **Recurrences become equations.** Multiplying by x^n and summing turned the rule $u_n = u_{n-1} + u_{n-2}$ into an equation for f that could be solved by algebra.
- **Build new from old.** Adding, shifting and differentiating generating functions produced new ones without starting from scratch.

What's really going on

From generating function to formula

Generating functions do not just encode a sequence: they can solve for it. The denominator factorises as $1 - x - x^2 = (1 - \varphi x)(1 - \psi x)$, where $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$ are the roots of $t^2 = t + 1$. Partial fractions give

$$\frac{x}{1 - x - x^2} = \frac{1}{\sqrt{5}}\Big(\frac{1}{1 - \varphi x} - \frac{1}{1 - \psi x}\Big),$$

and expanding each fraction as a geometric series,

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

This is *Binet's formula*. Since $|\psi| < 1$, the second term is tiny, so F_n is the nearest integer to $\varphi^n/\sqrt{5}$, and the ratio of successive Fibonacci numbers tends to the golden ratio (the subject of Section 2.2). The same method solves any linear recurrence with constant coefficients; the exercises include an example from STEP I 2004.

Why generating functions are so useful

The real power appears when we multiply. The coefficient of x^n in a product $\mathrm{f}(x)\mathrm{g}(x)$ is $\sum_k u_k v_{n-k}$, which adds up over all ways of splitting n into two parts. That is exactly what we need to count things built from two pieces. The next two sections use it to count the ways of making change (Section 9.5) and to find strangely numbered dice with the same totals as ordinary ones (Section 9.6). In probability, the same idea gives the *probability generating function*, which turns sums of independent random variables into products.

De Moivre, Euler and Laplace

Abraham de Moivre used generating functions in the 1720s to solve linear recurrences, essentially by the method above. Leonhard Euler used them in 1748 to study partitions of whole numbers, and Pierre-Simon Laplace made them the foundation of his *Théorie analytique des probabilités* (1812). In the twentieth century they became central to

combinatorics, and analytic methods (studying a generating function as a function of a complex variable) now give precise estimates for counting problems far too complicated to solve exactly.

The thinking skill

Change the representation

A problem about a sequence may become easy when the sequence is turned into a function, a matrix, or a picture.

Generating functions translate recurrences into algebra and counting into multiplication. At university you will meet many such translations: Fourier series turn functions into sequences, Laplace transforms turn differential equations into algebra, and matrices turn geometry into arithmetic.

Try this next

1. **Two digits at a time.** (From STEP II, 2022.) Using $F_r \le 2^{r-n}F_n$ for $r \ge n$, show that $\sum_{r\ge 7} F_r/10^r < 2\times 10^{-6}$, and hence find the first six digits after the decimal point of $1/89$ with justification. Then find a fraction r/s, with r and s positive integers less than 10000, whose decimal expansion starts $0.000101020305081321345 5\ldots$.

2. **Solving a recurrence.** (From STEP I, 2004.) Let $a_0 = 2$, $a_1 = 7$ and $a_n = 7a_{n-1} - 10a_{n-2}$. Show that the generating function is $\frac{1}{1-2x} + \frac{1}{1-5x}$, and hence that $a_n = 2^n + 5^n$. If $b_0 = 5$, $b_1 = 10$, $b_2 = 40$, $b_3 = 100$ and $b_n = pb_{n-1} + qb_{n-2}$, find p, q and b_n.

3. **Counting numbers.** Show that $\frac{1}{9801} = 0.00\,01\,02\,03\,04\ldots$, with the two-digit blocks counting upwards. Where does the pattern first go wrong, and why?

4. **Sums of squares.** Explain why $\frac{1}{1-x}f(x)$ is the generating function of the partial sums $u_0 + u_1 + \cdots + u_n$. Use this and the answer to part (i) to find a formula for $1^2 + 2^2 + \cdots + n^2$.

Further reading

- **Herbert Wilf, *generatingfunctionology*** (A K Peters; free online). The classic introduction. Chapter 1 is perfect after this section.
- **Ronald Graham, Donald Knuth & Oren Patashnik, *Concrete Mathematics*** (Addison-Wesley), Chapter 7.
- **Philippe Flajolet & Robert Sedgewick, *Analytic Combinatorics*** (Cambridge University

Press; free online). Far beyond first year, but the opening pages show where the subject leads.

9.5 Making Change

The question

Show that you can make up 10 pence in eleven ways using 10p, 5p, 2p and 1p coins.

In how many ways can you make up 20 pence using 20p, 10p, 5p, 2p and 1p coins?

[You are reminded that no credit will be given for unexplained answers.]

The hook

This looks like the easiest question in the book: just list the possibilities. But listing gets out of hand very quickly. There are 41 ways to make 20p, 451 ways to make 50p, and 4563 ways to make £1 from the coins in a British pocket. The question is really asking for a *system*, a way to organise the count so that nothing is missed and nothing counted twice.

Behind the system is a beautiful piece of mathematics. Each coin corresponds to a geometric series, the number of ways to make n pence is a coefficient in their product, and the same idea, applied to all whole numbers at once, gives Euler's theory of *partitions*, one of the deepest topics in number theory.

Before you start

Ask yourself How many ways are there to make n pence using only 2p and 1p coins? Why does the answer depend only on how many 2p coins you use?

Once the number of 2p coins is chosen, the rest must be 1p coins, so there is exactly one way for each possible number of 2p coins: $0, 1, \ldots, \lfloor n/2 \rfloor$. That gives $\lfloor n/2 \rfloor + 1$ ways. Call this $a(n)$. For example, $a(10) = 6$ and $a(5) = 3$.

Working it through

STAGE 1 **Ten pence**

Ask yourself Organise the count by the largest coins. How many 10p coins can you use? Given that, how many 5p coins?

- One 10p coin: 1 way.
- No 10p, two 5p: 1 way.
- No 10p, one 5p: the other 5p is made from 2p and 1p, in $a(5) = 3$ ways.
- No 10p or 5p: 10p from 2p and 1p, in $a(10) = 6$ ways.

The total is $1 + 1 + 3 + 6 = 11$. ■

STAGE 2 Building up layer by layer

Ask yourself Let $b(n)$ be the number of ways to make n pence from 5p, 2p and 1p coins. How can you express $b(n)$ using the function a?

Choose the number k of 5p coins; the remaining $n - 5k$ pence are made from 2p and 1p coins. So

$$b(n) = a(n) + a(n-5) + a(n-10) + \cdots.$$

This gives

n	0	5	10	15	20
$a(n)$	1	3	6	8	11
$b(n) = a(n) + b(n-5)$	1	4	10	18	29

Similarly, let $c(n)$ count ways using 10p coins as well; then $c(n) = b(n) + b(n-10) + \cdots$, so $c(10) = 10 + 1 = 11$ (agreeing with Stage 1) and $c(20) = b(20) + b(10) + b(0) = 29 + 10 + 1 = 40$.

STAGE 3 Twenty pence

Finally the 20p coin: either we use it (one way) or we do not ($c(20)$ ways). So there are

$$40 + 1 = 41 \text{ ways.}$$ ■

Pause and try Notice the structure: each new coin adds a layer, and each layer is a sum of the previous layer at spacings of the new coin's value. Can you see how a computer would carry out this calculation for £1, or £100?

Looking back

- **Organise by the largest part.** Fixing the number of the largest coins first made the count systematic.

- **Build a table.** Each new coin added a row computed from the row before. This is *dynamic programming*, the standard way computers count such things.
- **Check against a known case.** The table reproduced the 11 ways for 10p.

What's really going on

Coins as geometric series

Here is the idea that turns this into algebra. The number of 5p coins used can be $0, 1, 2, \ldots$, contributing $0, 5, 10, \ldots$ pence. Record this as the series

$$1 + x^5 + x^{10} + x^{15} + \cdots = \frac{1}{1 - x^5}.$$

Do the same for each coin and multiply:

$$\frac{1}{(1-x)(1-x^2)(1-x^5)(1-x^{10})(1-x^{20})} = \sum_n w(n) x^n.$$

When the brackets are multiplied out, each term x^n arises from choosing one term from each bracket, that is, a number of each coin, with total value n. So $w(n)$, the coefficient of x^n, is exactly the number of ways to make n pence. The table in Stage 2 was simply multiplying out this product one bracket at a time. This is the generating function idea of Section 9.4, and multiplication is where it shows its power: products of generating functions count combinations of choices.

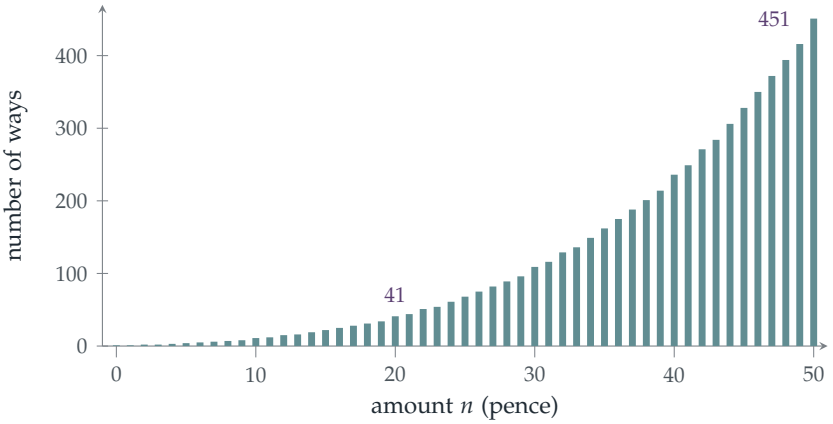


Figure 9.5.1 The number of ways to make n pence from 1p, 2p, 5p, 10p, 20p and 50p coins. The jumps at multiples of 5 and 10 show where a new coin first becomes usable. With six coins the count grows roughly like n^5: in general, like a polynomial of degree one less than the number of coins.

Partitions

What if every whole number is allowed as a coin? Then the number of ways to write n as a sum of positive integers, ignoring order, is the *partition number* $p(n)$. For example

$p(4) = 5$, since $4 = 3+1 = 2+2 = 2+1+1 = 1+1+1+1$. Euler saw in 1748 that

$$\sum_{n\geq 0} p(n)x^n = \frac{1}{(1-x)(1-x^2)(1-x^3)(1-x^4)\cdots},$$

and used it to discover remarkable identities. One of them: the number of ways to write n as a sum of *distinct* parts equals the number of ways to write it as a sum of *odd* parts. (The proof is a single line of algebra with generating functions: try it in the exercises.)

Hardy, Ramanujan and $p(200)$

The partition numbers grow fast: $p(10) = 42$, $p(100) = 190\,569\,292$. In 1918 G. H. Hardy and Srinivasa Ramanujan found an astonishing formula for $p(n)$, an infinite series whose first term is

$$p(n) \approx \frac{1}{4n\sqrt{3}}\, e^{\pi\sqrt{2n/3}}.$$

To test it, Percy MacMahon computed $p(200) = 3\,972\,999\,029\,388$ by hand, using Euler's identities. The first few terms of Hardy and Ramanujan's series gave the right answer to within a fraction of one. Ramanujan also discovered that $p(5n+4)$ is always divisible by 5, $p(7n+5)$ by 7, and $p(11n+6)$ by 11. Explaining why there are no more such patterns took until the twenty-first century.

The thinking skill

Count via coefficients

When an object is built from independent choices, each choice is a factor, and the number of objects of total size n is the coefficient of x^n in the product.

This turns counting into algebra. At university it is the basis of enumerative combinatorics, and it is how probability generating functions handle sums of independent random variables.

Try this next

1 **Two kinds of six.** (From STEP II, 2012.) Show that the coefficient of x^{24} in $(1-x^6)^{-2}(1-x^3)^{-1}$ is 15, and find the coefficient of x^n in general. Then show that the coefficient of x^{24} in $(1-x^6)^{-2}(1-x^3)^{-1}(1-x)^{-1}$ is 55, and find the coefficients of x^{25} and x^{66}. Describe a coin-counting problem that these coefficients answer.

2 **Distinct and odd.** Show that $(1+x)(1+x^2)(1+x^3)\cdots$ is the generating function for partitions into distinct parts. Use $1+x^k = \frac{1-x^{2k}}{1-x^k}$ to show that it equals $\frac{1}{(1-x)(1-x^3)(1-x^5)\cdots}$, and deduce Euler's theorem. Check it for $n = 6$.

3 **A formula.** Show that the number of ways to make n pence using only 1p, 2p and 5p coins is the nearest integer to $\frac{(n+4)^2}{20}$. (Check it first for small n.)

4 **Order matters.** In how many ways can you make n pence by putting 1p and 2p coins into a slot one at a time, if the order matters? (You have met this sequence before.)

Further reading

- **George Andrews & Kimmo Eriksson, *Integer Partitions*** (Cambridge University Press). A short, readable introduction for undergraduates.
- **George Pólya, "On picture-writing"**, *American Mathematical Monthly* 63 (1956). Pólya's classic explanation of generating functions using the problem of changing a dollar (there are 292 ways).
- **Robert Kanigel, *The Man Who Knew Infinity*** (Abacus). The life of Ramanujan, and his partnership with Hardy.

9.6 Strange Dice

The question

I have two dice whose faces are all painted different colours. I number the faces of one of them $1, 2, 2, 3, 3, 6$ and the other $1, 3, 3, 4, 5, 6$. I can now throw a total of 3 in two different ways using the two number 2's on the first die once each. Show that there are seven different ways of throwing a total of 6.

I now renumber the dice (again only using integers in the range 1 to 6) with the results shown in the following table.

Total shown by the two dice	2	3	4	5	6	7	8	9	10	11	12
Different ways of obtaining the total	0	2	1	1	4	3	8	6	5	6	0

Find how I have numbered the dice explaining your reasoning.

[You will only get high marks if the examiner can follow your argument.]

The hook

Is there any other way to number two six-sided dice, with positive whole numbers, so that every total from 2 to 12 comes up exactly as often as with ordinary dice? In 1978 Martin Gardner announced in *Scientific American* that there is exactly one: a die numbered $1, 2, 2, 3, 3, 4$ and a die numbered $1, 3, 4, 5, 6, 8$. They had been discovered by Colonel George Sicherman, and are now sold as *Sicherman dice*. A game of Monopoly played with them would be indistinguishable from an ordinary one.

This question is a warm-up: deduce how two dice are numbered from the table of totals. The systematic way to do it, and the proof that Sicherman's dice are the only strange pair, both come from turning each die into a polynomial.

Before you start

Ask yourself If the first die has p_i faces showing i and the second has q_j faces showing j, how many ways are there to throw a total of t?

The number of ways is $\sum_i p_i q_{t-i}$: choose the number i on the first die, then the second must show $t - i$. For the first pair of dice, a total of 6 comes from $1 + 5$ (in 1×1 way), $2 + 4$ (2×1), $3 + 3$ (2×2), and nothing else, since the first die has no 4 or 5 and the second has no 0. That is $1 + 2 + 4 = 7$ ways. ■

Working it through

STAGE 1 The extremes

Call the dice A and B, and write a_i, b_i for the number of faces of each showing i.

Ask yourself There is no way to throw 2, nor 12. What does that say about the 1s and the 6s? There *are* two ways to throw 3. Which die has the 1s?

No total of 2 means $a_1 b_1 = 0$: at most one die has any 1s. A total of 3 can only be $1+2$ or $2+1$, and there are two ways, so some die does have a 1; call it A, so $b_1 = 0$. Then

$$\text{total } 3: \quad a_1 b_2 = 2, \qquad \text{so } (a_1, b_2) = (2,1) \text{ or } (1,2).$$

Similarly, no total of 12 means $a_6 b_6 = 0$.

STAGE 2 Working upwards

Ask yourself Write down the equation for a total of 4, remembering $b_1 = 0$. What does it say in each of the two cases?

A total of 4 is $1+3$ or $2+2$ (and $3+1$ is impossible as $b_1 = 0$), so $a_1 b_3 + a_2 b_2 = 1$.

- **Case (a)**, $a_1 = 2$, $b_2 = 1$: $2b_3 + a_2 = 1$, so $b_3 = 0$ and $a_2 = 1$.
- **Case (b)**, $a_1 = 1$, $b_2 = 2$: $b_3 + 2a_2 = 1$, so $a_2 = 0$ and $b_3 = 1$.

Total 5: $a_1 b_4 + a_2 b_3 + a_3 b_2 = 1$.

- Case (a): $2b_4 + 0 + a_3 = 1$, so $b_4 = 0$, $a_3 = 1$. So far A has $1,1,2,3$ and B has 2; B's other five faces are 5s and 6s.
- Case (b): $b_4 + 0 + 2a_3 = 1$, so $a_3 = 0$, $b_4 = 1$. So far A has 1 and B has $2,2,3,4$; A's other five faces are from $4,5,6$, and B's other two from $5,6$.

STAGE 3 Case (a) fails

Ask yourself In case (a), write down the equation for a total of 7, and use $b_5 + b_6 = 5$.

Total 7: $a_1 b_6 + a_2 b_5 + a_3 b_4 + a_4 b_3 + a_5 b_2 = 2b_6 + b_5 + 0 + 0 + a_5 = 3$. With $b_6 = 5 - b_5$ this says $10 - b_5 + a_5 = 3$, so $b_5 = 7 + a_5 \geq 7$. But B only has six faces. Case (a) is impossible.

STAGE 4 Case (b) succeeds

Ask yourself In case (b), use the totals 6 and 11, and the fact that $a_6 b_6 = 0$.

Total 6: $a_1b_5 + a_4b_2 = b_5 + 2a_4 = 4$ (the other products involve $a_2 = a_3 = 0$ or $b_1 = 0$). Since $b_5 \le 2$ and $b_5 = 4 - 2a_4$ is even, either $b_5 = 0, a_4 = 2$ or $b_5 = 2, a_4 = 1$.

- If $b_5 = 0$ then B is $2,2,3,4,6,6$, so $a_6 = 0$ and A is $1,4,4,5,5,5$. But then total 7 has $a_1b_6 + a_4b_3 + a_5b_2 = 2 + 2 + 6 = 10$ ways, not 3.
- If $b_5 = 2$ then B is $2,2,3,4,5,5$, with $b_6 = 0$, and A is $1,4$ and four faces from 5 and 6. Total 11: $a_6b_5 = 2a_6 = 6$, so $a_6 = 3$ and $a_5 = 1$.

The only possibility is

$$A:\ 1,4,5,6,6,6, \qquad B:\ 2,2,3,4,5,5.$$

Finally we must check that these dice give the whole table, since we have used only some of the totals. Multiplying out (or using the polynomials below) confirms every entry. ■

Looking back

- **Start at the extremes.** The smallest and largest totals involve only one or two products, so they give the strongest information.
- **Split into cases and kill them.** Each total cut the possibilities down, until only one case survived.
- **Check sufficiency.** The deduction showed what the dice *must* be; a final check showed that they work.

What's really going on

Dice as polynomials

Write each die as a polynomial, with one term x^i for each face showing i. An ordinary die is $x + x^2 + x^3 + x^4 + x^5 + x^6$, and the dice in the question are

$$A(x) = x + x^4 + x^5 + 3x^6, \qquad B(x) = 2x^2 + x^3 + x^4 + 2x^5.$$

Multiplying, each pair of faces contributes $x^i x^j = x^{i+j}$, so the coefficient of x^t in $A(x)B(x)$ is the number of ways to throw a total of t. This is the generating function idea of Section 9.4 once more: the table of totals is just the product $A(x)B(x)$, and our case-by-case deduction was a way of factorising it.

Sicherman's dice

For two ordinary dice the product is $(x + x^2 + \cdots + x^6)^2$. Any pair of dice with the same totals must have polynomials $P(x)$, $Q(x)$ with $P(x)Q(x) = (x + \cdots + x^6)^2$. Now factorise:

$$x + x^2 + \cdots + x^6 = x(1 + x)(1 + x + x^2)(1 - x + x^2),$$

and these factors cannot be broken down further (into polynomials with integer coefficients). So each of P and Q must be built from the factors x, $1+x$, $1+x+x^2$, $1-x+x^2$, each appearing twice in total. Three conditions narrow the choice:

- each die has 6 faces, so $P(1) = Q(1) = 6$; the factors take the values 1, 2, 3, 1 at $x = 1$, so each die gets one $1+x$ and one $1+x+x^2$;
- no face shows 0, so $P(0) = Q(0) = 0$: each die gets one factor x;
- the only freedom left is how the two copies of $1-x+x^2$ are shared. One each gives ordinary dice. Both on one die gives

$$x(1+x)(1+x+x^2) = x+2x^2+2x^3+x^4,$$
$$x(1+x)(1+x+x^2)(1-x+x^2)^2 = x+x^3+x^4+x^5+x^6+x^8.$$

These are Sicherman's dice, $1,2,2,3,3,4$ and $1,3,4,5,6,8$ (Figure 9.6.1), and the argument proves they are the only alternative. The key fact is that polynomials, like integers, factorise uniquely into irreducibles: number theory and combinatorics meet.

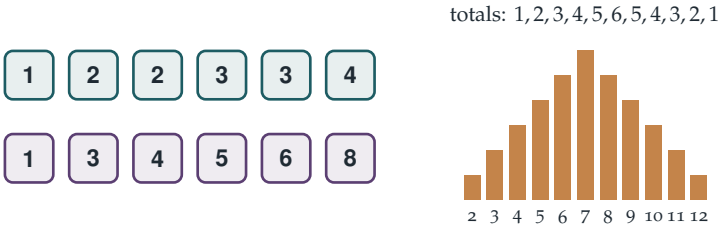


Figure 9.6.1 Sicherman's dice. Thrown together, they give every total from 2 to 12 with exactly the same frequency as two ordinary dice (right): one way to throw 2, two ways to throw 3, and so on up to six ways to throw 7.

Probability generating functions

In probability the same idea appears as the *probability generating function*: a random variable X taking values $0,1,2,\ldots$ has $G(t) = E(t^X) = \sum P(X=k)t^k$, and for independent X and Y, $E(t^{X+Y}) = E(t^X)E(t^Y)$. Sums of independent random variables correspond to products of generating functions. STEP III 2003 Q14 (in the exercises) finds Sicherman's dice this way, and Section 9.13 uses a related argument to show that no loading of two dice can make all totals equally likely.

The thinking skill

Encode a counting problem as a polynomial

When objects combine by adding their values, represent each one as a polynomial in x. Combining becomes multiplying, and questions about the combinations become questions about factorisation.

Here a question about dice turned into the unique factorisation of polynomials. At university, such encodings are everywhere: in combinatorics, in probability, and in algebra, where polynomials are studied in their own right as a ring with arithmetic like that of the integers.

Try this next

1. **Tetrahedral dice.** (From STEP III, 2003.) Standard four-faced dice are labelled $1, 2, 3, 4$. Find all the ways in which two four-faced dice can be labelled with positive integers so that every total has the same probability as for two standard four-faced dice. (*Hint:* $x + x^2 + x^3 + x^4 = x(1 + x)(1 + x^2)$.)

2. **Checking the table.** Multiply out $(x + x^4 + x^5 + 3x^6)(2x^2 + x^3 + x^4 + 2x^5)$ and check it against the table in the question.

3. **Three dice.** Are there three six-sided dice, not all standard, labelled with positive integers, whose totals have the same distribution as three standard dice? (*Hint:* the factors of $(x + \cdots + x^6)^3$ now include three copies of $1 - x + x^2$.)

4. **A coin and an octahedron.** An eight-sided die labelled $1, \ldots, 8$ and a two-sided "die" (a coin) labelled $1, 2$: can they be relabelled differently with the same totals?

Further reading

- **Martin Gardner, *Penrose Tiles to Trapdoor Ciphers*** (Mathematical Association of America). Includes the column on Sicherman dice.
- **Joseph Gallian & David Rusin, "Cyclotomic polynomials and nonstandard dice"**, *Discrete Mathematics* 27 (1979). The general theory, for dice with any number of faces.
- **Herbert Wilf, *generatingfunctionology*** (free online), Chapter 1.

9.7 Equal Sums of Powers

The question

Let $p_0(x) = (1-x)(1-x^2)(1-x^4)$. Show that $(1-x)^3$ is a factor of $p_0(x)$. If $p_1(x) = xp_0'(x)$ show, by considering factors of the polynomials involved, that $p_0'(1) = 0$ and $p_1'(1) = 0$.

By writing $p_0(x)$ in the form

$$p_0(x) = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7,$$

deduce that

$$1 + 2 + 4 + 7 = 3 + 5 + 6$$
$$1^2 + 2^2 + 4^2 + 7^2 = 3^2 + 5^2 + 6^2.$$

Show that we can write the integers $1, 2, \ldots, 15$ in some order as $a_1, a_2, \ldots, a_{15}$ in such a way that

$$a_1^r + a_2^r + \cdots + a_8^r = a_9^r + a_{10}^r + \cdots + a_{15}^r$$

for $r = 1, 2, 3$.

The hook

Split the numbers 1 to 16 into two teams of eight:

$$\{1, 4, 6, 7, 10, 11, 13, 16\} \quad \text{and} \quad \{2, 3, 5, 8, 9, 12, 14, 15\}.$$

Each team sums to 68. The sums of their squares are both 748. The sums of their cubes are both 9248. It is hard to believe this could be anything but a fluke, yet it is the third step of a pattern that goes on for ever: split 1 to 32 the right way and the fourth powers agree as well.

The split was found by Eugène Prouhet in 1851, and the rule behind it, the *Thue–Morse sequence*, turns up in an astonishing range of places: in number theory, in the theory of computation, in chess, and in the fairest way to take turns. This question uncovers it through a polynomial and its derivatives.

Before you start

Ask yourself If a polynomial $p(x) = \sum c_k x^k$ has $p(1) = 0$, what does that say about the coefficients? What about $p'(1) = 0$?

$p(1) = \sum c_k$, so $p(1) = 0$ says the coefficients add to zero. And $p'(x) = \sum k c_k x^{k-1}$, so $p'(1) = \sum k c_k$. If the coefficients are all ± 1, these say that the positions of the $+1$s and the -1s are equal in number, and have equal sums.

Working it through

STAGE 1 A triple root

Ask yourself Factorise $1 - x^2$ and $1 - x^4$. How many factors of $1 - x$ appear?

Since $1 - x^2 = (1 - x)(1 + x)$ and $1 - x^4 = (1 - x)(1 + x)(1 + x^2)$,

$$p_0(x) = (1 - x)^3 (1 + x)^2 (1 + x^2),$$

so $(1 - x)^3$ is a factor. ■

STAGE 2 Derivatives at 1

Ask yourself If $(1 - x)^3$ divides p_0, what power of $(1 - x)$ divides p_0'? And then $p_1 = x p_0'$ and p_1'?

Write $p_0(x) = (1 - x)^3 q(x)$. By the product rule, $p_0'(x) = -3(1 - x)^2 q(x) + (1 - x)^3 q'(x)$, which has $(1 - x)^2$ as a factor. So $p_0'(1) = 0$. Then $p_1(x) = x p_0'(x)$ also has $(1 - x)^2$ as a factor, and the same argument shows p_1' has $(1 - x)$ as a factor, so $p_1'(1) = 0$. ■

STAGE 3 Reading off the coefficients

Multiplying out, $(1 - x)(1 - x^2) = 1 - x - x^2 + x^3$, and then

$$p_0(x) = 1 - x - x^2 + x^3 - x^4 + x^5 + x^6 - x^7.$$

Ask yourself What are $p_0'(1)$ and $p_1'(1)$ in terms of the coefficients c_k?

Since $p_1(x) = x p_0'(x) = \sum k c_k x^k$, we have $p_1'(1) = \sum k^2 c_k$. The coefficient is $+1$ at $k = 0, 3, 5, 6$ and -1 at $k = 1, 2, 4, 7$. So $p_0'(1) = 0$ and $p_1'(1) = 0$ say

$$0 + 3 + 5 + 6 = 1 + 2 + 4 + 7 \quad (= 14),$$
$$0^2 + 3^2 + 5^2 + 6^2 = 1^2 + 2^2 + 4^2 + 7^2 \quad (= 70),$$

which, dropping the 0, are the identities required. ■

STAGE 4 One more factor

Pause and try To make the cubes balance too, we need a fourth factor of $(1-x)$. Which polynomial should replace p_0?

Take $P(x) = (1-x)(1-x^2)(1-x^4)(1-x^8)$, which has $(1-x)^4$ as a factor. Its coefficients are $p_0(x) - x^8 p_0(x)$: the pattern of signs for 0 to 7, followed by the same pattern reversed in sign for 8 to 15:

k	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
c_k	+	−	−	+	−	+	+	−	−	+	+	−	+	−	−	+

The argument of Stage 2, applied one more time, shows that $\sum c_k k^r = 0$ for $r = 0, 1, 2, 3$. Take the + positions $\{0, 3, 5, 6, 9, 10, 12, 15\}$ and the − positions $\{1, 2, 4, 7, 8, 11, 13, 14\}$. For $r = 1, 2, 3$, the term from 0 is zero, so it can be dropped. So the order

$$a_1, \ldots, a_8 = 1, 2, 4, 7, 8, 11, 13, 14, \qquad a_9, \ldots, a_{15} = 3, 5, 6, 9, 10, 12, 15$$

works: both halves sum to 60, their squares to 620, and their cubes to 7200. ■

Looking back

- **Derivatives detect repeated roots.** A factor $(1-x)^m$ forces the first $m-1$ derivatives to vanish at $x = 1$.
- **Multiplying by x before differentiating** turned $\sum k c_k$ into $\sum k^2 c_k$, and so on to higher powers.
- **Doubling.** Each extra factor $(1 - x^{2^m})$ doubled the length and added one more power that balances.

What's really going on

The Thue–Morse sequence

The signs of the coefficients of $(1-x)(1-x^2)(1-x^4)(1-x^8)\cdots$ form the *Thue–Morse sequence*. Writing 0 for + and 1 for −:

$$0\,1\,1\,0\,1\,0\,0\,1\,1\,0\,0\,1\,0\,1\,1\,0 \ \ldots$$

There are three equivalent descriptions.

- **Doubling:** start with 0, and repeatedly append the complement (swap 0s and 1s) of what you have: $0 \to 01 \to 0110 \to 01101001 \to \cdots$.

- **Binary digits:** the kth term is 0 if k has an even number of 1s when written in binary, and 1 if odd. (This is why the sign of x^k is $(-1)^{\text{number of 1s in } k}$: to make x^k from the product, you choose the factors x^{2^m} that make up k in binary, and each brings a minus sign.)
- **Polynomial:** the coefficients of $\prod_m (1 - x^{2^m})$, as in the question.

Splitting $0, 1, \ldots, 2^N - 1$ by the Thue–Morse sequence gives two sets whose rth powers have equal sums for every $r < N$ (Figure 9.7.1). This is the *Prouhet–Tarry–Escott problem*: find two different sets of integers with equal sums of rth powers for $r = 1, \ldots, k$. Prouhet's solution uses 2^{k+1} numbers; it is a famous open problem to find out how few numbers are really needed.



$1 + 4 + 6 + 7 + 10 + 11 + 13 + 16 = 68$

$2 + 3 + 5 + 8 + 9 + 12 + 14 + 15 = 68$

squares: $748 = 748$ cubes: $9248 = 9248$ fourth powers: $122\,692 \neq 121\,156$

Figure 9.7.1 The numbers 1 to 16 split by the Thue–Morse sequence. The two sets have equal sums, equal sums of squares and equal sums of cubes, but not equal sums of fourth powers; for that you need 1 to 32. Reading the colours left to right gives the pattern teal–plum–plum–teal, plum–teal–teal–plum, ...

The fairest way to take turns

Two children take turns choosing sweets from a pile, best first. If they alternate $ABABAB\ldots$, then A always gets the better sweet of each pair, a systematic advantage. The order $ABBA$ is fairer, and $ABBABAAB$ fairer still: this is the Thue–Morse sequence, and the balanced power sums are exactly the reason. If the sweets have values given by a smooth function of their position, expanding that function as a polynomial shows that the Thue–Morse order cancels its leading terms. The same idea has been proposed for penalty shoot-outs in football (the "$ABBA$" order trialled by football's rule-makers from 2017) and is used for the order of serving in tennis tie-breaks.

Thue, Morse and Euwe

The sequence has been discovered many times. Prouhet used it in 1851. Axel Thue rediscovered it in 1912 while studying sequences without repetitive patterns: it never contains the same block three times in a row (it is "cube-free"). Marston Morse found it again in 1921 in the study of geodesics on surfaces. And the chess world champion Max Euwe used it in 1929 to show that the rule then in force for drawing a game by repetition did not prevent infinitely long games. Today it is a central example in the theory of *automatic sequences*, sequences computed by simple machines reading the binary digits of n.

The thinking skill

Encode a combinatorial property in a polynomial

Balanced sums of powers are about where the $+$ and $-$ signs fall. A polynomial with those signs as coefficients turns the property into a statement about roots.

The question never asks you to search: a repeated root at $x = 1$ does all the work. Using algebra to produce combinatorial objects, rather than finding them by trial, is a recurring idea in university combinatorics.

Try this next

1 **Shifting.** (From STEP II, 2025.) Suppose $\sum a_k = \sum b_k$ and $\sum a_k^2 = \sum b_k^2$ for two lists of the same length m. Show that $\sum a_k^3 + \sum (c + b_k)^3 = \sum b_k^3 + \sum (c + a_k)^3$ for every c. Use this, starting from a balanced split of $\{1, \dots, 8\}$, to find a split of $\{1, \dots, 16\}$ in which the cubes balance as well.

2 **Squares of squares.** (Also from STEP II, 2025.) The sets $A = \{1, 3, 4, 5, 9, 11\}$ and $B = \{2, 6, 7, 8, 10\}$ have equal sums and equal sums of squares. Split $\{n^2, (n+1)^2, \dots, (n+11)^2\}$ into two sets with equal sums, for every positive integer n. (*Hint:* add 0 to the second set.)

3 **Translation invariance.** Show that if two sets of numbers have equal sums of rth powers for $r = 0, 1, \dots, k$, then so do the sets obtained by adding the same number c to every element. Why does this mean that the question's sets $\{1, \dots, 15\}$ could equally be replaced by any 16 consecutive integers?

4 **Fair sweets.** Sweets have values $1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{8}$. Compare the totals A gets with the orders $ABABABAB$ and $ABBABAAB$.

Further reading

- **Jean-Paul Allouche & Jeffrey Shallit, "The ubiquitous Prouhet–Thue–Morse sequence"**, in *Sequences and Their Applications* (Springer, 1999). A survey of the sequence's many appearances; free online.
- **Matt Parker, *Things to Make and Do in the Fourth Dimension*** (Penguin). Includes equal power sums and other number curiosities.
- **Joshua Cooper & Aaron Dutle, "Greedy Galois games"**, *American Mathematical Monthly* 120 (2013). Fair turn-taking and Thue–Morse.

9.8 Nobody Gets Their Own Shirt

The question

Starting with the result $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$, prove that

$$\mathrm{P}(A \cup B \cup C) = \mathrm{P}(A) + \mathrm{P}(B) + \mathrm{P}(C) - \mathrm{P}(A \cap B) - \mathrm{P}(B \cap C) - \mathrm{P}(C \cap A) + \mathrm{P}(A \cap B \cap C).$$

Write down, without proof, the corresponding result for four events A, B, C and D.

A pack of n cards, numbered $1, 2, \ldots, n$, is shuffled and laid out in a row. The result of the shuffle is that each card is equally likely to be in any position in the row. Let E_i be the event that the card bearing the number i is in the ith position in the row. Write down the following probabilities:

(i) $\mathrm{P}(E_i)$;

(ii) $\mathrm{P}(E_i \cap E_j)$, where $i \neq j$;

(iii) $\mathrm{P}(E_i \cap E_j \cap E_k)$, where $i \neq j$, $j \neq k$ and $k \neq i$.

Hence show that the probability that at least one card is in the same position as the number it bears is

$$1 - \frac{1}{2!} + \frac{1}{3!} - \cdots + (-1)^{n+1}\frac{1}{n!}.$$

Find the probability that exactly one card is in the same position as the number it bears.

The hook

A football team of eleven players changes in a hurry and grabs shirts at random. What is the chance that nobody ends up in their own shirt? Now suppose it is a squad of fifty, or a crowd of a million. Does the chance go up, because there are more people to get it wrong? Or down, because there are more people who might get it right by luck? Remarkably, it hardly changes at all: from five people onwards, the chance is within half a per cent of $1/e \approx 0.368$.

This is the oldest problem in combinatorial probability, posed by Pierre de Montmort in 1708 as the *problème des rencontres*. The question solves it with the *inclusion–exclusion principle*, a tool for counting things that overlap, and the answer is one of the most

unexpected appearances of e in mathematics.

Before you start

Pause and try Try $n = 3$: list all six orders of cards $1, 2, 3$ and count how many have no card in its own place. Then try $n = 4$, if you are patient (there are 24 orders).

For $n = 3$ only 231 and 312 have no card in its place, so the probability is $\frac{2}{6} = \frac{1}{3}$. For $n = 4$ there are 9 such orders out of 24, giving $\frac{3}{8} = 0.375$. An arrangement with no fixed card is called a *derangement*.

Working it through

STAGE 1 Inclusion–exclusion for three events

Ask yourself How can you write $A \cup B \cup C$ as a union of two events, so that the two-event formula applies?

Treat $A \cup B$ as a single event and apply the formula to it and C:

$$\mathrm{P}(A \cup B \cup C) = \mathrm{P}(A \cup B) + \mathrm{P}(C) - \mathrm{P}\big((A \cup B) \cap C\big).$$

Now $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$, and applying the formula once more, $\mathrm{P}\big((A \cap C) \cup (B \cap C)\big) = \mathrm{P}(A \cap C) + \mathrm{P}(B \cap C) - \mathrm{P}(A \cap B \cap C)$. Expanding $\mathrm{P}(A \cup B)$ too and collecting terms gives the result. ■

For four events the pattern continues: add the four single probabilities, subtract the six pairwise intersections, add the four triple intersections, and subtract $\mathrm{P}(A \cap B \cap C \cap D)$.

STAGE 2 The probabilities

Ask yourself If card i is in position i, how many ways are there to arrange the rest? And if cards i and j are both fixed?

There are $n!$ equally likely orders. If card i is fixed in position i, the other $n - 1$ cards can be in any order: $(n-1)!$ ways. So

$$\mathrm{P}(E_i) = \frac{(n-1)!}{n!} = \frac{1}{n}, \qquad \mathrm{P}(E_i \cap E_j) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)},$$
$$\mathrm{P}(E_i \cap E_j \cap E_k) = \frac{(n-3)!}{n!} = \frac{1}{n(n-1)(n-2)}.$$

In general, the probability that k specified cards are all in their own positions is $\frac{(n-k)!}{n!}$.

STAGE 3 Adding it up

Ask yourself In the inclusion–exclusion formula for $\mathrm{P}(E_1 \cup \cdots \cup E_n)$, how many terms are there involving intersections of k events, and what is their total?

There are $\binom{n}{k}$ ways to choose k of the events, and each intersection has probability $\frac{(n-k)!}{n!}$. So the k-fold terms add up to

$$\binom{n}{k}\frac{(n-k)!}{n!} = \frac{n!}{k!\,(n-k)!}\cdot\frac{(n-k)!}{n!} = \frac{1}{k!}.$$

Everything except the $k!$ cancels. With alternating signs,

$$\mathrm{P}(\text{at least one card in its place}) = 1 - \frac{1}{2!} + \frac{1}{3!} - \cdots + (-1)^{n+1}\frac{1}{n!}. \qquad \blacksquare$$

So the probability of a derangement is

$$d_n = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n\frac{1}{n!} = \sum_{k=0}^{n}\frac{(-1)^k}{k!}.$$

STAGE 4 Exactly one

Ask yourself For exactly one card to be in place, which card is it, and what must the others do?

Choose which card is in its place (n ways). The probability that card i is in place is $\frac{1}{n}$, and given that, the other $n-1$ cards must form a derangement among themselves, which has probability d_{n-1}. So

$$\mathrm{P}(\text{exactly one}) = n\cdot\frac{1}{n}\cdot d_{n-1} = \sum_{k=0}^{n-1}\frac{(-1)^k}{k!}. \qquad \blacksquare$$

Looking back

- **Overcount, then correct.** Adding the $\mathrm{P}(E_i)$ counts arrangements with two fixed cards twice; subtracting the pairs corrects this but goes too far for triples; and so on.
- **Symmetry makes the terms equal.** Every k-fold intersection has the same probability, so each layer of the formula is one number times a binomial coefficient.
- **Reduce to a smaller case.** "Exactly one" became a derangement of $n-1$ cards.

What's really going on

Where e comes from

The series $\sum(-1)^k/k!$ is the exponential series $e^x = \sum x^k/k!$ at $x = -1$. So as $n \to \infty$,

$$d_n \to e^{-1} \approx 0.3679,$$

and the probability of exactly one match tends to the same limit. The convergence is extraordinarily fast (Figure 9.8.1): the error in stopping at $1/n!$ is less than $1/(n+1)!$, so already for $n = 7$ the answer agrees with $1/e$ to four decimal places. In fact the number of derangements of n objects, $n!\,d_n$, is exactly the nearest integer to $n!/e$.

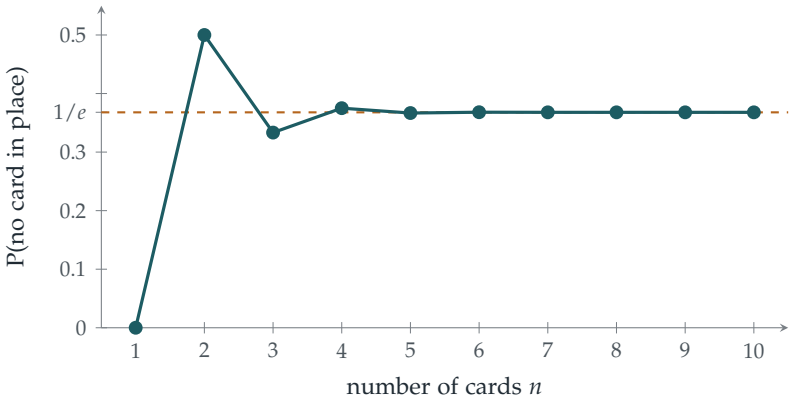


Figure 9.8.1 The probability that a random shuffle of n cards leaves no card in its own place. It alternates above and below $1/e$, and from $n = 6$ onwards it is indistinguishable from $1/e$ on this scale.

The number of matches is Poisson

More is true. Let N be the number of cards in their own place. Then $\mathrm{P}(N = k) \to \frac{e^{-1}}{k!}$ for each k: the number of matches has, in the limit, the *Poisson distribution* with mean 1. You can see why the mean is 1 without any calculation: N is the sum of the indicators of $E_1, \ldots, E_n$, each with expectation $\frac{1}{n}$, and expectation adds even though the events are not independent. STEP III 2004 Q12 (in the exercises) shows that the variance is 1 too. The Poisson distribution appears whenever there are many unlikely events that are nearly independent; derangements are a classic example, and the subject of Section 12.11 in Chapter 12.

Montmort's game of *treize*

Pierre Rémond de Montmort, a French nobleman, studied games of chance in his *Essay d'analyse sur les jeux de hazard* (1708). In the game of *treize* (thirteen), a player deals thirteen cards, calling out "one, two, three, ..." and wins if the card dealt ever matches the number called. Montmort found the probability of winning, and in correspondence with Nicolaus Bernoulli the general formula was established by 1713. Euler rediscovered it in 1753. The inclusion–exclusion principle was later formalised

by Abraham de Moivre and, in its general form, by James Joseph Sylvester in the nineteenth century.

The thinking skill

Inclusion–exclusion

To count a union of overlapping sets, add the sizes of the sets, subtract the pairwise overlaps, add the triple overlaps, and so on.

It is most powerful when symmetry makes all the overlaps of each size equal, as here. At university it appears in combinatorics, in probability and in number theory, where it counts integers not divisible by any of a list of primes, and leads to Euler's totient function and the Möbius function.

Try this next

1 **Shirts.** (From STEP III, 2004.) A team of m players put on m numbered shirts at random. Let $C_i = 1$ if player i wears shirt i, and 0 otherwise. Find $\mathrm{E}(C_1)$, $\mathrm{Var}(C_1)$ and $\mathrm{Cov}(C_1, C_2)$, and show that $N = C_1 + \cdots + C_m$ has $\mathrm{E}(N) = \mathrm{Var}(N) = 1$. For $m = 4$, compare $\mathrm{P}(N = 0)$ with the Poisson approximation e^{-1}.

2 **A recurrence.** Let D_n be the number of derangements of n objects. By considering where card 1 goes, and whether the card in that position goes to position 1, show that $D_n = (n-1)(D_{n-1} + D_{n-2})$. Check it against $D_3 = 2$, $D_4 = 9$.

3 **Exactly k.** Show that the probability that exactly k cards are in their own places is $\frac{1}{k!} d_{n-k}$, and that it tends to $\frac{e^{-1}}{k!}$.

4 **Coprime integers.** Use inclusion–exclusion to count the integers from 1 to 30 divisible by none of 2, 3 and 5. How does your answer relate to $30\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{5}\right)$?

Further reading

- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press). A first-year text that covers inclusion–exclusion and the matching problem.
- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter IV. A classic, with many variations on the matching problem.
- **Miklós Bóna, *A Walk Through Combinatorics*** (World Scientific), on inclusion–exclusion.

9.9 The Birthday Problem

The question

Given that $0 < r < n$ and r is much smaller than n, show that $\dfrac{n-r}{n} \approx e^{-r/n}$.

There are k guests at a party. Assuming that there are exactly 365 days in the year, and that the birthday of any guest is equally likely to fall on any of these days, show that the probability that there are at least two guests with the same birthday is approximately $1 - e^{-k(k-1)/730}$.

Using the approximation $\frac{253}{365} \approx \ln 2$, find the smallest value of k such that the probability that at least two guests share the same birthday is at least $\frac{1}{2}$.

How many guests must there be at the party for the probability that at least one guest has the same birthday as the host to be at least $\frac{1}{2}$?

The hook

How many people do you need in a room for it to be more likely than not that two of them share a birthday? Most people guess somewhere around 180, half of 365. The answer is 23. In a class of thirty there is a 70% chance of a shared birthday; at a party of fifty, 97%.

The surprise comes from confusing two different questions: "does someone share *my* birthday?" (which does need about 253 people) and "do *any two people* share a birthday?". With 23 people there are $\binom{23}{2} = 253$ pairs, and each pair is a chance for a match. This question makes that intuition precise, and the same mathematics decides how secure the cryptography protecting your bank account is.

Before you start

Ask yourself Why is it easier to calculate the probability that *no* two guests share a birthday? Line the guests up: what is the chance that the second avoids the first's birthday? That the third avoids both?

"At least two share" covers many overlapping cases (two share, three share, two different pairs, ...). Its complement, "all different", is a single clean event. The second guest avoids the first's birthday with probability $\frac{364}{365}$; given that, the third avoids both with probability $\frac{363}{365}$; and so on.

Working it through

STAGE 1 A useful approximation

Ask yourself What is the tangent line to $y = e^{-x}$ at $x = 0$?

Since $e^{-x} = 1 - x + \frac{x^2}{2} - \cdots$, for small x we have $e^{-x} \approx 1 - x$. With $x = r/n$,

$$\frac{n-r}{n} = 1 - \frac{r}{n} \approx e^{-r/n}. \qquad \blacksquare$$

The error is about $\frac{r^2}{2n^2}$, small when r is much smaller than n.

STAGE 2 The probability of a shared birthday

The probability that all k birthdays are different is

$$\frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{365-(k-1)}{365} = \prod_{r=0}^{k-1} \frac{365-r}{365}.$$

Using the approximation on each factor, and the fact that exponentials multiply by adding their exponents,

$$\prod_{r=0}^{k-1} e^{-r/365} = \exp\Big(-\frac{0+1+\cdots+(k-1)}{365}\Big) = \exp\Big(-\frac{k(k-1)}{730}\Big).$$

So the probability of at least one shared birthday is approximately $1 - e^{-k(k-1)/730}$. $\blacksquare$

Ask yourself Look at the exponent: $\frac{k(k-1)}{730} = \binom{k}{2}\Big/365$. What does that say in words?

It is the number of pairs of guests, times the probability $\frac{1}{365}$ that a given pair shares a birthday: the expected number of shared pairs. The approximation says the chance of *no* match is $e^{-(\text{expected number of matches})}$, a hint of the Poisson distribution again, as in Section 9.8.

STAGE 3 Twenty-three

We need $1 - e^{-k(k-1)/730} \geq \frac{1}{2}$, that is $e^{-k(k-1)/730} \leq \frac{1}{2}$, or

$$\frac{k(k-1)}{730} \geq \ln 2 \approx \frac{253}{365}, \qquad \text{that is} \qquad k(k-1) \geq 506.$$

Since $22 \times 21 = 462$ and $23 \times 22 = 506$, the smallest value is $k = 23$. $\blacksquare$

(The exact probability for 23 guests is 0.507, and for 22 guests 0.476, so the approximation gives the right answer.)

STAGE 4 Sharing with the host

Ask yourself For each guest, what is the probability of *not* sharing the host's birthday? Are the guests independent of each other in this respect?

Each guest independently misses the host's birthday with probability $\frac{364}{365} \approx e^{-1/365}$. So the probability that none of k guests shares it is $\left(\frac{364}{365}\right)^k \approx e^{-k/365}$, and we need

$$e^{-k/365} \leq \tfrac{1}{2}, \qquad \text{that is} \qquad k \geq 365 \ln 2 \approx 253.$$

So there must be 253 guests. ■

The contrast is the whole point: 253 guests to match one particular person, but only 23 for some pair to match, because 23 people make 253 pairs.

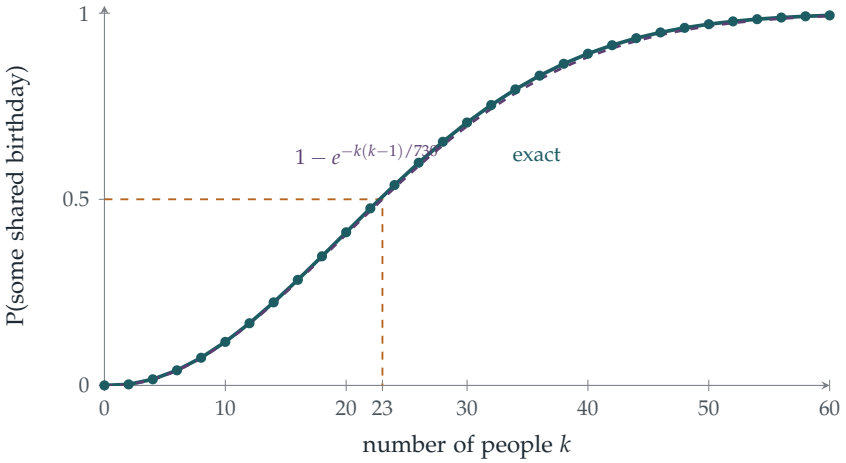


Figure 9.9.1 The probability that at least two of k people share a birthday: exact values (teal) and the approximation $1 - e^{-k(k-1)/730}$ (plum, dashed). The two are very close, and both pass $\frac{1}{2}$ between $k = 22$ and $k = 23$.

Looking back

- **Use the complement.** "All different" was a single product; "at least one match" was a mess.
- **Linearise, then exponentiate.** Replacing each factor $1 - x$ by e^{-x} turned a product into a sum.
- **Count pairs.** The exponent was the number of pairs times the chance that a pair matches. The square in $k(k-1)$ is the whole explanation of the paradox.

What's really going on

The square-root law

With n equally likely days instead of 365, the same argument gives

$$\mathrm{P}(\text{some match among } k) \approx 1 - e^{-k^2/(2n)},$$

so matches become likely once k is around $\sqrt{n}$, not n. More precisely, the probability reaches $\frac{1}{2}$ at $k \approx 1.18\sqrt{n}$, and the expected number of people you need to meet before the first repeat is about $\sqrt{\pi n/2}$ (for $n = 365$, about 24.6). This *square-root law* is the reason coincidences are so common: in any large collection of events, some pair is likely to match in some way, even though any particular match is unlikely.

Birthday attacks

Computers check that a file has not been tampered with by computing a *hash*: a fingerprint of, say, $n = 2^{128}$ possible values. A forger who wants two documents with the same hash (a harmless contract and a fraudulent one) does not need to try 2^{128} documents. By the birthday argument, generating about $\sqrt{2^{128}} = 2^{64}$ variants of each will probably produce a match. This *birthday attack* is why modern hash functions use 256 bits or more: half the bits are lost to the square-root law. Similar calculations decide how long random identifiers must be so that two computers never accidentally choose the same one.

Who first asked?

The birthday problem is usually credited to Richard von Mises, who discussed it in 1939, though it circulated earlier. Harold Davenport is also said to have known it earlier, and not to have published it because it seemed too simple. As Warren Weaver told the story, he liked to demonstrate it at dinners of army officers during the Second World War: he bet that two of the twenty-two people present shared a birthday, and lost, until a waitress revealed that she shared her birthday with one of the officers.

The thinking skill

Calculate the complement; approximate carefully

When "at least one" is complicated, find the probability of "none". When a product is awkward, approximate each factor by an exponential and add the exponents.

Both moves are standard at university. The second, $1 - x \approx e^{-x}$, is the start of the theory of Poisson approximation and of the asymptotic methods used to estimate quantities too complicated to calculate exactly.

Try this next

1 The first repeat. (From STEP II, 2014.) A random number generator produces integers from 1 to n, independently and equally likely. Let X be the position of the first repeat. Find $\mathrm{P}(X = r)$ for $2 \le r \le n+1$, and deduce that

$$\frac{1}{n} + \left(1 - \frac{1}{n}\right)\frac{2}{n} + \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\frac{3}{n} + \cdots = 1.$$

Show that $\mathrm{E}(Y) = \sum_{k=1}^{N} \mathrm{P}(Y \ge k)$ for any random variable taking values $1, \ldots, N$, and hence that

$$\left(1 - \frac{1^2}{n}\right) + \left(1 - \frac{1}{n}\right)\left(1 - \frac{2^2}{n}\right) + \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\left(1 - \frac{3^2}{n}\right) + \cdots = 0.$$

2 Three in common. Roughly how many people are needed for a better-than-even chance that some *three* share a birthday? (*Hint:* the expected number of triples sharing a birthday is $\binom{k}{3}/365^2$. Set this to about $\ln 2$.) The true answer is 88.

3 Near misses. How many people are needed for an even chance that two have birthdays on the same day or on adjacent days?

4 Uneven birthdays. Real birthdays are not uniform. Show that if the days have probabilities $p_1, \ldots, p_{365}$, the chance that two people share a birthday is $\sum p_i^2$, and that this is smallest when all p_i are equal. (So non-uniformity only makes matches more likely.)

Further reading

- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter II. The birthday problem and its approximations.
- **Persi Diaconis & Frederick Mosteller, "Methods for studying coincidences"**, *Journal of the American Statistical Association* 84 (1989). Why coincidences are common; very readable.
- **David Hand, *The Improbability Principle*** (Bantam). The square-root law and other reasons why unlikely things happen all the time.

9.10 Always Able to Give Change

The question

I am selling raffle tickets for £1 per ticket. In the queue for tickets, there are m people each with a single £1 coin and n people each with a single £2 coin. Each person in the queue wants to buy a single raffle ticket and each arrangement of people in the queue is equally likely to occur. Initially, I have no coins and a large supply of tickets. I stop selling tickets if I cannot give the required change.

(i) In the case $n = 1$ and $m \geq 1$, find the probability that I am able to sell one ticket to each person in the queue.

(ii) By considering the first three people in the queue, show that the probability that I am able to sell one ticket to each person in the queue in the case $n = 2$ and $m \geq 2$ is $\dfrac{m-1}{m+1}$.

(iii) Show that the probability that I am able to sell one ticket to each person in the queue in the case $n = 3$ and $m \geq 3$ is $\dfrac{m-2}{m+1}$.

The hook

The pattern is irresistible: $\frac{m}{m+1}$, $\frac{m-1}{m+1}$, $\frac{m-2}{m+1}$. Surely, with n people holding £2 coins, the answer is $\frac{m-n+1}{m+1}$. It is, and the slickest proof is one of the most elegant ideas in combinatorics: the *reflection principle*. It turns a hard count (queues that never run out of change) into an easy one by reflecting part of a picture in a line.

The same problem has many disguises: an election count in which the winner is never behind (Bertrand's ballot problem of 1887), a gambler who never goes broke, brackets that match correctly. When $m = n$ the count gives the *Catalan numbers*, perhaps the most frequently occurring sequence in combinatorics after the binomial coefficients.

Before you start

Ask yourself Each £1 customer gives you a coin; each £2 customer takes one away (as change). When do you get stuck?

Keep track of the number of £1 coins you hold. It goes up by 1 for each £1 customer and down by 1 for each £2 customer. You are stuck exactly when a £2 customer arrives

and you hold no coins: the count would go to -1. So you succeed if and only if, at every point in the queue, the number of £1 customers so far is at least the number of £2 customers so far.

Working it through

STAGE 1 One £2 coin

With $n = 1$, you fail only if the £2 customer is first. The £2 customer is equally likely to be in any of the $m+1$ positions, so

$$\mathrm{P}(\text{success}) = 1 - \frac{1}{m+1} = \frac{m}{m+1}. \qquad \blacksquare$$

STAGE 2 Two £2 coins

Ask yourself The first person must have a £1 coin. What can happen with the second person, and in each case, what situation are you left in?

The first person has a £1 coin with probability $\frac{m}{m+2}$; otherwise you fail at once. Then you hold one coin, and the remaining queue has $m-1$ £1 people and two £2 people.

- If the second person has a £1 coin (probability $\frac{m-1}{m+1}$), you hold two coins, enough to change both £2 coins whenever they come. Success is certain.
- If the second person has a £2 coin (probability $\frac{2}{m+1}$), you are back to no coins, with $m-1$ £1 people and one £2 person left. By part (i) with $m-1$ in place of m, you succeed with probability $\frac{m-1}{m}$.

So

$$\mathrm{P}(\text{success}) = \frac{m}{m+2}\left(\frac{m-1}{m+1} + \frac{2}{m+1}\cdot\frac{m-1}{m}\right) = \frac{m}{m+2}\cdot\frac{m-1}{m+1}\cdot\frac{m+2}{m} = \frac{m-1}{m+1}. \qquad \blacksquare$$

Pause and try Try the same approach for $n = 3$. It works, but the cases multiply. Is there a way to count the successful queues all at once?

STAGE 3 Queues as paths

Ask yourself Draw the number of coins you hold against the number of people served. What does a queue look like? What does a successful queue look like?

A queue is a path made of m up-steps (£1) and n down-steps (£2), starting at height 0 and ending at height $m-n$. There are $\binom{m+n}{n}$ such paths, all equally likely. A queue is

successful exactly when its path never goes below 0. So we need to count the *bad* paths, those that touch height -1.

STAGE 4 The reflection

Ask yourself Take a bad path. Find the first moment it touches -1, and reflect the part of the path *before* that moment in the line $y = -1$. Where does the new path start, and where does it end?

The reflected path starts at -2 (the mirror image of 0 in the line $y = -1$) and ends where the original did, at $m - n$ (Figure 9.10.1). Conversely, every path from -2 to $m - n$ must cross $y = -1$ somewhere (it starts below and ends above, since $m \geq n$); reflecting it before its first visit gives back a bad path from 0. So bad paths from 0 correspond exactly to *all* paths from -2 to $m - n$.

A path from -2 to $m - n$ in $m + n$ steps rises by $m - n + 2$, so it has $m + 1$ up-steps and $n - 1$ down-steps: there are $\binom{m+n}{n-1}$ of them. Hence

$$\mathrm{P}(\text{failure}) = \frac{\binom{m+n}{n-1}}{\binom{m+n}{n}} = \frac{n!\,m!}{(n-1)!\,(m+1)!} = \frac{n}{m+1},$$

and

$$\mathrm{P}(\text{success}) = 1 - \frac{n}{m+1} = \frac{m-n+1}{m+1}.$$

With $n = 3$ this is $\frac{m-2}{m+1}$. ■

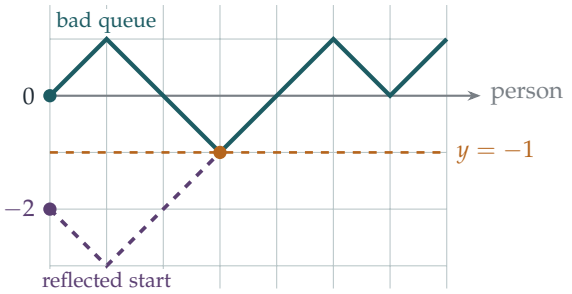


Figure 9.10.1 A queue with $m = 4$ £1 coins and $n = 3$ £2 coins, drawn as a path (teal). It first hits -1 at the third person (amber), when you cannot give change. Reflecting the path up to that point in the line $y = -1$ gives a path from -2 (plum, dashed) to the same end point. This matching between bad queues and paths starting at -2 is the reflection principle.

Looking back

- **Find the right state.** The only thing that mattered was the number of coins held, which turned the queue into a walk.
- **Condition on the first steps.** Parts (i) and (ii) were done by asking what the first few

people do, reducing to smaller cases.

- **Count the complement by a bijection.** The bad paths were hard to count directly, but reflection matched them with a set that was easy to count.

What's really going on

Catalan numbers

When $m = n$ (exactly enough £1 coins), the number of successful queues is

$$\binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1}\binom{2n}{n},$$

the nth *Catalan number* C_n: $1, 1, 2, 5, 14, 42, 132, \ldots$. Richard Stanley's book *Catalan Numbers* lists more than two hundred different things that they count. Among them: the number of ways to arrange n pairs of brackets correctly ("(())()" is allowed, ")()(()" is not, by exactly the coin argument with "(" as £1 and ")" as £2), the number of ways to cut a polygon with $n+2$ sides into triangles by non-crossing diagonals, and the number of ways to bracket a product of $n+1$ factors.

Random walks

Queues are random walks, and the reflection principle is one of the basic tools for studying them. It shows, for example, that a fair random walk starting at 0 is certain to return to 0 eventually, and that the probability that it stays strictly positive for the first $2n$ steps is $\frac{1}{2}\binom{2n}{n}2^{-2n} \approx \frac{1}{2\sqrt{\pi n}}$. In continuous form, the same reflection argument applied to *Brownian motion* gives the distribution of its maximum. This is the basis of the pricing of certain financial options, whose payoff depends on whether a share price ever crosses a barrier.

Bertrand's ballot problem

In 1887 Joseph Bertrand asked: in an election where candidate A receives a votes and B receives $b < a$, what is the probability that A is strictly ahead throughout the count? The answer, $\frac{a-b}{a+b}$, was proved in the same year by Désiré André, whose argument is usually remembered as the reflection principle (although historians have pointed out that André's actual proof used a slightly different bijection). Our raffle problem is a variant in which ties are allowed: success means the £1 count never falls *below* the £2 count. One of the exercises deduces Bertrand's answer from ours.

The thinking skill

Bijective counting

To count a set, find a one-to-one correspondence with a set you can count.

Reflection matched bad queues with unrestricted paths from a different start. Bijections are the most satisfying proofs in combinatorics, because they explain *why* two numbers are equal rather than just showing that they are. At university you will meet bijective proofs throughout combinatorics, and the reflection principle again in the study of random walks and Brownian motion.

Try this next

1 **Three by cases.** Prove part (iii) by conditioning on the first people in the queue, as in part (ii), to see how much work the reflection principle saves.

2 **Bertrand's ballot.** In an election, A gets a votes and B gets $b < a$. Show that the probability that A is strictly ahead throughout the count is $\frac{a-b}{a+b}$. (*Hint:* the first vote must be for A; after that, A must never fall behind B by more than…)

3 **Brackets.** List the 5 correct arrangements of three pairs of brackets, and the 5 ways to cut a pentagon into triangles. Can you find a bijection between them?

4 **A £5 note.** A queue contains m people with £1 coins and n with £5 notes, for a ticket costing £1. Explain why the reflection principle no longer gives a simple answer. (This generalisation is much harder.)

Further reading

- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter III. The classic treatment of random walks and the reflection principle.
- **Richard Stanley, *Catalan Numbers*** (Cambridge University Press). Over two hundred interpretations, with exercises.
- **Marc Renault, "Lost (and found) in translation: André's actual method and its application to the generalized ballot problem"**, *American Mathematical Monthly* 115 (2008). The history of the reflection principle.

9.11 Collecting the Set

The question

The makers of Cruncho ('The Cereal Which Cares') are giving away a series of cards depicting n great mathematicians. Each packet of Cruncho contains one picture chosen at random. Show that when I have collected r different cards the expected number of packets I must open to find a new card is $n/(n-r)$ where $0 \leq r \leq n-1$.

Show by means of a diagram, or otherwise, that

$$\frac{1}{r+1} \leq \int_r^{r+1} \frac{1}{x}\,\mathrm{d}x \leq \frac{1}{r}$$

and deduce that

$$\sum_{r=2}^{n} \frac{1}{r} \leq \ln n \leq \sum_{r=1}^{n-1} \frac{1}{r}$$

for all $n \geq 2$.

My children will give me no peace until we have the complete set of cards, but I am the only person in our household prepared to eat Cruncho and my spouse will only buy the stuff if I eat it. If n is large, roughly how many packets must I expect to consume before we have the set?

The hook

A football sticker album has 600 stickers to collect, sold one at a time at random. How many stickers should you expect to buy to fill the album? The first few hundred go quickly, since almost every sticker is new. But as the album fills, new stickers get rarer, and the very last one takes on average 600 purchases on its own. The total comes to about 4200, seven times the size of the album.

This is the *coupon collector's problem*. Its answer combines two ideas: the waiting time for a success, and the harmonic series $1 + \frac{1}{2} + \frac{1}{3} + \cdots$, which grows like $\ln n$. You met the harmonic series diverging in Section 2.3; here its slow growth is exactly what you pay for completing a collection.

Before you start

Ask yourself If each packet independently gives a new card with probability p, what is the expected number of packets you open up to and including the first new card?

The number of packets X has a *geometric distribution*: $\mathrm{P}(X = k) = (1-p)^{k-1}p$. Its expectation is $\frac{1}{p}$. (One quick way to see this: $\mathrm{E}(X) = 1 + (1-p)\mathrm{E}(X)$, since you open one packet, and with probability $1-p$ you are back where you started.) If a new card turns up a third of the time, you expect to wait three packets.

Working it through

STAGE 1 Waiting for a new card

With r different cards already, a packet gives a new card with probability $\frac{n-r}{n}$, independently of previous packets. So the expected number of packets until the next new card is

$$\frac{1}{(n-r)/n} = \frac{n}{n-r}. \qquad \blacksquare$$

STAGE 2 Comparing a sum with an integral

Ask yourself On the interval $r \le x \le r+1$, between which two values does $\frac{1}{x}$ lie? What does that say about the area under the curve?

Since $\frac{1}{x}$ is decreasing, $\frac{1}{r+1} \le \frac{1}{x} \le \frac{1}{r}$ on $[r, r+1]$. Integrating over an interval of length 1 gives

$$\frac{1}{r+1} \le \int_r^{r+1} \frac{\mathrm{d}x}{x} \le \frac{1}{r}. \qquad \blacksquare$$

In pictures (Figure 9.11.1): the area under the curve is squeezed between a rectangle of height $\frac{1}{r+1}$ and one of height $\frac{1}{r}$. Adding these from $r = 1$ to $r = n-1$, the integrals join up to $\int_1^n \frac{\mathrm{d}x}{x} = \ln n$, and so

$$\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \le \ln n \le 1 + \frac{1}{2} + \cdots + \frac{1}{n-1}. \qquad \blacksquare$$

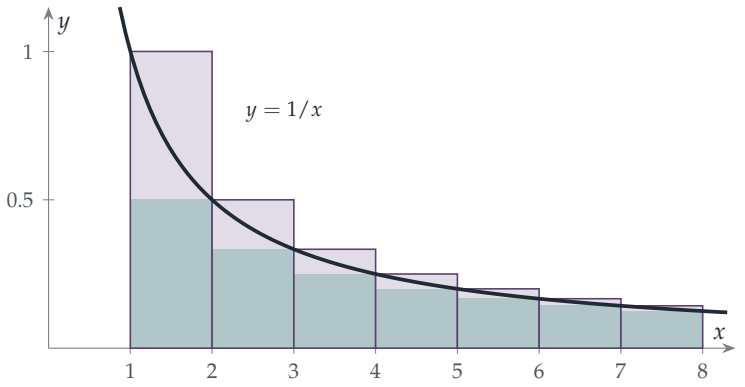


Figure 9.11.1 On each interval $[r, r+1]$ the area under $y = 1/x$ lies between the teal rectangle (height $\frac{1}{r+1}$) and the larger plum one (height $\frac{1}{r}$). Adding up, $\ln n$ lies between $\frac{1}{2} + \cdots + \frac{1}{n}$ and $1 + \frac{1}{2} + \cdots + \frac{1}{n-1}$. The gaps between the curve and the plum rectangles add up to a finite total: Euler's constant $\gamma \approx 0.577$.

STAGE 3 Completing the set

Ask yourself The total number of packets is the sum of the waiting times for the first card, the second new card, and so on. What is its expectation?

Let X_r be the number of packets needed to go from r different cards to $r+1$. The total is $X_0 + X_1 + \cdots + X_{n-1}$, and since expectation adds,

$$\mathrm{E}(\text{total}) = \sum_{r=0}^{n-1} \frac{n}{n-r} = n\Big(\frac{1}{n} + \frac{1}{n-1} + \cdots + 1\Big) = nH_n,$$

where $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ is the nth *harmonic number*. From Stage 2, $\ln n + \frac{1}{n} \le H_n \le \ln n + 1$, so H_n is about $\ln n$ when n is large, and I must expect to eat roughly

$$n \ln n \text{ packets.} \qquad \blacksquare$$

Looking back

- **Break a long wait into stages.** The total time split into n waiting times, each of a simple geometric kind.
- **Expectation is additive.** We never needed the distribution of the total, only the expectations of the pieces.
- **Compare sums with integrals.** The diagram turned a sum we cannot evaluate into a logarithm, with error at most 1.

What's really going on

How good is the estimate?

The comparison can be sharpened. The shaded gaps between the curve and the upper rectangles in Figure 9.11.1 have a finite total area, so $H_n - \ln n$ tends to a limit,

$$\gamma = \lim_{n\to\infty} (H_n - \ln n) = 0.5772156649\ldots,$$

Euler's constant. (Nobody knows whether γ is rational.) So the expected number of packets is $n \ln n + \gamma n + \frac{1}{2} + \cdots$. For the 600-sticker album, $600 \times (\ln 600 + 0.577) \approx 4185$, which is the figure quoted in the hook.

Where does the time go? The stages are very unequal. Collecting the first half of the set takes only about $n \ln 2 \approx 0.69n$ packets, while the last card alone takes n on average. Most of the effort goes into the last few cards, which is why collectors swap duplicates.

Concentration

The expected value is only the start: how much does the total vary? The variance of the total is $\sum nr/(n-r)^2$, which is about $\frac{\pi^2}{6}n^2$ (the sum $\sum 1/k^2 = \pi^2/6$ from Section 2.4 appears). So the standard deviation is about $1.28n$, much smaller than the mean $n \ln n$ when n is large: the total is *concentrated* around its mean. Paul Erdős and Alfréd Rényi showed in 1961 that, more precisely, the total minus $n \ln n$, divided by n, has a limiting distribution, the *Gumbel distribution*, which also describes the largest of many random quantities (Section 12.25).

Coupons everywhere

The problem goes back to Abraham de Moivre (1712) and Laplace. It now appears in many disguises. How many random tests are needed to check every component of a system? How long until a random walk on a network has visited every node? How many times must a computer hash table be probed? In each case the answer involves $n \ln n$. In biology, the same calculation estimates how many sequencing "reads" are needed to cover a whole genome.

The thinking skill

Linearity of expectation

The expectation of a sum is the sum of the expectations, whether or not the pieces are independent. Split a complicated random quantity into simple pieces.

This is one of the most useful facts in probability. Here it reduced the coupon collector's problem to n geometric waits; in Section 9.8 it gave the expected number of matches without any calculation. At university, the same idea powers the analysis of algorithms and the probabilistic method in combinatorics.

Try this next

1 **Half a set.** Show that the expected number of packets to collect $\frac{n}{2}$ different cards (for even n) is $n(H_n - H_{n/2})$, and that this is about $n \ln 2$.

2 **Distinct cards.** After opening k packets, show that the expected number of different cards you have is $n\big(1 - (1 - \frac{1}{n})^k\big)$. (*Hint:* for each card, find the probability that you have it, and add.) Roughly how many packets give you 95% of the set?

3 **Two collectors.** Two children collect the same set, sharing packets so that a duplicate for one may be new for the other. Show that the expected number of packets to complete both sets is about $n(\ln n + \ln \ln n)$. (This is harder; first convince yourself that it is much less than $2n \ln n$.)

4 **A sharper bound.** Use the trapezium rule picture instead of rectangles to show that $H_n - \ln n$ is decreasing, and that $H_n - \ln n - \frac{1}{2n}$ is increasing. Deduce $0.5 < \gamma < 0.7$ from small values of n.

Further reading

- **Geoffrey Grimmett & David Stirzaker, *One Thousand Exercises in Probability*** (Oxford University Press). Many variations on the coupon collector.
- **Julian Havil, *Gamma: Exploring Euler's Constant*** (Princeton University Press). Everything about γ and the harmonic series, at an accessible level.
- **Michael Mitzenmacher & Eli Upfal, *Probability and Computing*** (Cambridge University Press). The coupon collector and its uses in computer science.

9.12 Random Subsets

The question

The set S is the set of all integers from 1 to n. The set T is the set of all distinct subsets of S, including the empty set $\varnothing$ and S itself. Show that T contains exactly 2^n sets.

The sets $A_1, A_2, \ldots, A_m$, which are not necessarily distinct, are chosen randomly and independently from T, and for each k ($1 \leq k \leq m$), the set A_k is equally likely to be any of the sets in T.

(i) Write down the value of $\mathrm{P}(1 \in A_1)$.

(ii) By considering each integer separately, show that $\mathrm{P}(A_1 \cap A_2 = \varnothing) = \left(\frac{3}{4}\right)^n$. Find $\mathrm{P}(A_1 \cap A_2 \cap A_3 = \varnothing)$ and $\mathrm{P}(A_1 \cap A_2 \cap \cdots \cap A_m = \varnothing)$.

(iii) Find $\mathrm{P}(A_1 \subseteq A_2)$, $\mathrm{P}(A_1 \subseteq A_2 \subseteq A_3)$ and $\mathrm{P}(A_1 \subseteq A_2 \subseteq \cdots \subseteq A_m)$.

The hook

Choose two subsets of $\{1, 2, \ldots, 100\}$ completely at random. How likely is it that they have no element in common? Your intuition may say "quite likely: most pairs of sets are different". In fact the probability is $\left(\frac{3}{4}\right)^{100}$, about 3×10^{-13}. Random sets are big (about half of everything) and so they almost always overlap.

The question is short, and its one idea is powerful: do not think about the sets as a whole; think about one element at a time. A random subset is just n independent coin tosses, one for each element, and every question about random subsets becomes a question about those tosses.

Before you start

Ask yourself How can you describe a subset of $\{1, \ldots, n\}$ by a sequence of n answers to yes/no questions?

For each element i, answer the question "is i in the set?". The subset determines the answers, and the answers determine the subset. There are 2 choices for each element, so 2^n subsets in all. Choosing a subset uniformly at random is the same as answering each question by tossing a fair coin, independently.

Working it through

STAGE 1 One element

By the coin-tossing description, $\mathrm{P}(1 \in A_1) = \frac{1}{2}$. Indeed, exactly half of the 2^n subsets contain 1: pair each subset that does not contain 1 with the one obtained by adding 1.

STAGE 2 Disjoint sets

Ask yourself For $A_1 \cap A_2 = \varnothing$, what must be true of each element i? What is the probability of that, for one element? Why can you multiply over the elements?

A_1 and A_2 are disjoint exactly when no element is in both. For a single element i, there are four equally likely possibilities: in neither set, in A_1 only, in A_2 only, in both (Figure 9.12.1, left). Only the last is forbidden, so the probability that i is not in both is $\frac{3}{4}$. Different elements are decided by different, independent coin tosses, so

$$\mathrm{P}(A_1 \cap A_2 = \varnothing) = \left(\frac{3}{4}\right)^n. \qquad \blacksquare$$

For three sets, element i is in all three with probability $\frac{1}{8}$, so $\mathrm{P}(A_1 \cap A_2 \cap A_3 = \varnothing) = \left(\frac{7}{8}\right)^n$. In general,

$$\mathrm{P}(A_1 \cap \cdots \cap A_m = \varnothing) = \left(1 - \frac{1}{2^m}\right)^n. \qquad \blacksquare$$

STAGE 3 Subsets

Ask yourself For $A_1 \subseteq A_2$, which of the four possibilities for an element is forbidden?

$A_1 \subseteq A_2$ fails exactly when some element is in A_1 but not in A_2. Again one pattern out of four is forbidden, so

$$\mathrm{P}(A_1 \subseteq A_2) = \left(\frac{3}{4}\right)^n,$$

the same as for disjointness. That is not a coincidence: $A_1 \subseteq A_2$ exactly when A_1 is disjoint from the complement of A_2, and the complement of a random set is another random set.

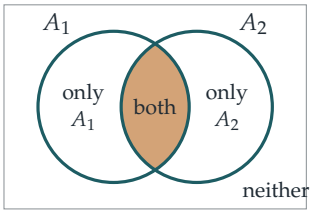
Ask yourself For $A_1 \subseteq A_2 \subseteq A_3$, which of the 8 patterns of membership (in or out of each set) are allowed for an element?

Going along the chain A_1, A_2, A_3, once an element is in, it must stay in. So the allowed patterns are "out, out, out", "out, out, in", "out, in, in" and "in, in, in": 4 out of 8 (Figure 9.12.1, right). Hence $\mathrm{P}(A_1 \subseteq A_2 \subseteq A_3) = \left(\frac{4}{8}\right)^n = \left(\frac{1}{2}\right)^n$.

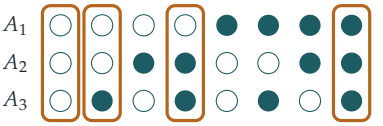
For a chain of m sets, an allowed pattern is determined by the first set the element

enters (one of $A_1, \dots, A_m$, or never), so $m+1$ of the 2^m patterns are allowed:

$$\mathrm{P}(A_1 \subseteq A_2 \subseteq \cdots \subseteq A_m) = \left(\frac{m+1}{2^m}\right)^n. \qquad \blacksquare$$



each element: 4 equally likely places



allowed for $A_1 \subseteq A_2 \subseteq A_3$: 4 of 8

Figure 9.12.1 Left: an element of S can be in one of four places relative to A_1 and A_2, each with probability $\frac{1}{4}$. Disjointness forbids "both" (amber); $A_1 \subseteq A_2$ forbids "only A_1". Right: the eight patterns of membership of A_1, A_2, A_3 (filled circle means "in"). The chain $A_1 \subseteq A_2 \subseteq A_3$ allows only the four boxed patterns, in which an element, once in, stays in.

Looking back

- **Decompose by elements.** Each set-level condition (disjoint, subset, chain) was a condition on each element separately.
- **Independence across elements.** Because the elements are decided by independent coin tosses, the probabilities multiply.
- **Count patterns.** Each answer was (allowed patterns ÷ all patterns)^n.

What's really going on

Counting without probability

Multiplying by the number of choices turns each probability into a count. The number of ordered pairs (A_1, A_2) with $A_1 \subseteq A_2$ is $4^n \times \left(\frac{3}{4}\right)^n = 3^n$. You can also count these pairs directly: choose the size k of A_2, then A_2 itself in $\binom{n}{k}$ ways, then A_1 as any of its 2^k subsets. So

$$\sum_{k=0}^{n} \binom{n}{k} 2^k = 3^n,$$

which is the binomial theorem for $(1+2)^n$. Counting the same thing in two ways, once by elements and once by sizes, proves an identity: this is the method of *double counting*, which you met with the Platonic solids in Section 7.9. Similarly, the number of chains $A_1 \subseteq \cdots \subseteq A_m$ is $(m+1)^n$.

The lattice of subsets

The subsets of $\{1, \dots, n\}$, ordered by inclusion, form a structure called the *Boolean lattice*, which can be pictured as an n-dimensional cube: each subset is a corner, labelled by its sequence of yes/no answers, and moving along an edge adds or removes one element. Chains are paths going upwards. Deep results describe how large families of subsets can be, given conditions on them. *Sperner's theorem* (1928) says that the largest family in which no set contains another has $\binom{n}{\lfloor n/2 \rfloor}$ sets; the *Erdős–Ko–Rado theorem* (1961) says how many k-element sets can be chosen so that every two overlap. Both are standard in a first course on combinatorics, and the coin-tossing view of subsets gives some of the slickest proofs.

Boole's algebra

George Boole's *The Laws of Thought* (1854) treated logical statements as sets and "and", "or" and "not" as intersection, union and complement. Encoding a set as a string of yes/no answers is exactly how a computer represents it: a subset of $\{1, \dots, 64\}$ is a single 64-bit number, and intersection is the "bitwise AND" instruction. The method of this question, one element (one bit) at a time, is how such operations are analysed.

The thinking skill

Think one element at a time

When objects are built from independent choices for each element, a condition that applies to every element separately has probability equal to a product.

The method turned a question about 2^n subsets into a question about 4 or 8 patterns. At university it reappears in combinatorics (generating functions for sets and multisets), in probability (independent coordinates) and in computer science (bitwise reasoning).

Try this next

1. **Covering.** Find $\mathrm{P}(A_1 \cup A_2 = S)$ and $\mathrm{P}(A_1 \cup A_2 \cup \cdots \cup A_m = S)$.

2. **Same size.** Show that $\mathrm{P}(|A_1| = |A_2|) = \binom{2n}{n}/4^n$. (*Hint:* $|A_1| = |A_2|$ exactly when $|A_1| + |S \setminus A_2| = n$.) Show that this is about $1/\sqrt{\pi n}$ for large n.

3. **Partitions into three.** Show that the number of ordered pairs of disjoint subsets (A, B) of S is 3^n, and find the number of unordered pairs $\{A, B\}$ of disjoint subsets.

4. **Sperner for small n.** For $n = 3$ and $n = 4$, find the largest family of subsets of $\{1, \dots, n\}$ in which no set contains another.

Further reading

- **Miklós Bóna, *A Walk Through Combinatorics*** (World Scientific), the chapters on subsets and partially ordered sets.
- **Béla Bollobás, *Combinatorics: Set Systems, Hypergraphs, Families of Vectors and Combinatorial Probability*** (Cambridge University Press). A short, elegant book on families of subsets, from Sperner's theorem onwards.
- **Peter Cameron, *Combinatorics: Topics, Techniques, Algorithms*** (Cambridge University Press). A first course with a strong emphasis on counting in two ways.

9.13 Loaded Dice

The question

(i) Prove that if $x > 0$ then $x + x^{-1} \geq 2$.

I have a pair of six-faced dice, each with faces numbered from 1 to 6. The probability of throwing i with the first die is q_i and the probability of throwing j with the second die is r_j $(1 \leq i, j \leq 6)$. The two dice are thrown independently and the sum noted. By considering the probabilities of throwing 2, 12 and 7, show the sums $2, 3, \ldots, 12$ are not equally likely.

(ii) The first die described above is thrown twice and the two numbers on the die noted. Is it possible to find values of q_j so that the probability that the numbers are the same is less than $1/36$?

The hook

With two fair dice, a total of 7 is six times as likely as a total of 2. Board-game designers sometimes wish it were otherwise: could you load the dice (weight them so that some faces come up more often) so that every total from 2 to 12 has the same chance, $\frac{1}{11}$? You can choose the twelve probabilities freely, so there seems to be plenty of room.

The answer is no, and the proof is a gem: it uses only three of the eleven totals and the inequality $x + \frac{1}{x} \geq 2$. It is a model of how to prove that something is *impossible*, which is often harder, and more interesting, than showing that something can be done.

Before you start

Ask yourself How would you prove $x + \frac{1}{x} \geq 2$ for $x > 0$? When is there equality?

Multiply by $x > 0$: the inequality becomes $x^2 + 1 \geq 2x$, that is $(x-1)^2 \geq 0$, which is true. Equality holds only when $x = 1$. (Alternatively, $x + \frac{1}{x} - 2 = \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2$.) This is the AM–GM inequality in its simplest form: the average of x and $\frac{1}{x}$ is at least their geometric mean, 1.

Working it through

STAGE 1 The extreme totals

Ask yourself In how many ways can you throw a total of 2? A total of 12? Write their probabilities in terms of the q_i and r_j.

A total of 2 needs 1 on both dice, and 12 needs 6 on both, so $\mathrm{P}(2) = q_1 r_1$ and $\mathrm{P}(12) = q_6 r_6$.

Suppose, for contradiction, that all eleven totals are equally likely, each with probability $\frac{1}{11}$. Then

$$q_1 r_1 = \frac{1}{11}, \qquad q_6 r_6 = \frac{1}{11}.$$

In particular, all four of q_1, r_1, q_6, r_6 are positive.

STAGE 2 The middle total

Ask yourself A total of 7 can be made in six ways. Which two of them involve only the faces 1 and 6? Write their probabilities using the equations above.

The total 7 includes the outcomes (1, 6) and (6, 1), so

$$\mathrm{P}(7) \geq q_1 r_6 + q_6 r_1.$$

Now use the equations of Stage 1 to bring in a ratio. Since $q_1 = \frac{1}{11 r_1}$ and $q_6 = \frac{1}{11 r_6}$,

$$q_1 r_6 + q_6 r_1 = \frac{1}{11}\left(\frac{r_6}{r_1} + \frac{r_1}{r_6}\right) \geq \frac{1}{11} \times 2 = \frac{2}{11},$$

using $x + \frac{1}{x} \geq 2$ with $x = r_6/r_1 > 0$. So $\mathrm{P}(7) \geq \frac{2}{11}$, contradicting $\mathrm{P}(7) = \frac{1}{11}$. The totals cannot all be equally likely. ■

Pause and try Look at what made the proof work. Making 2 and 12 as likely as 7 forces the 1s and 6s to be fairly likely, and then the "mixed" outcomes (1, 6) and (6, 1) alone make 7 too likely. Figure 9.13.1 shows the cells involved.

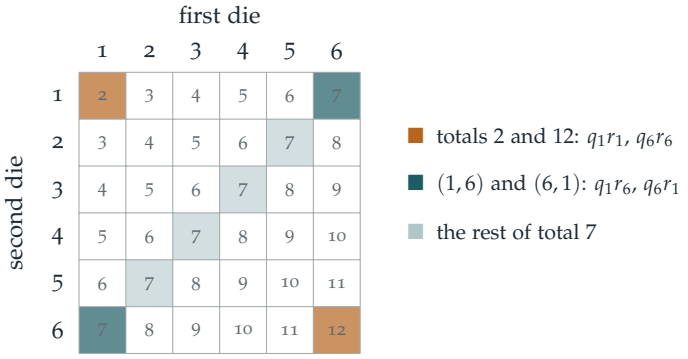


Figure 9.13.1 The 36 outcomes of two dice, labelled by their totals. The proof uses only the four corner cells. The product of the two amber probabilities equals the product of the two dark teal ones, $q_1r_1q_6r_6 = q_1r_6 \cdot q_6r_1$, and by AM–GM the sum of the teal pair is at least twice the geometric mean of the amber pair.

STAGE 3 Two throws of one die

Ask yourself If one die is thrown twice, what is the probability that the two numbers are the same, in terms of the q_j? What is it for a fair die?

The two throws agree with probability $q_1^2 + q_2^2 + \cdots + q_6^2$. For a fair die this is $6 \times \frac{1}{36} = \frac{1}{6}$, which is already much bigger than $\frac{1}{36}$.

Ask yourself Can loading make $\sum q_j^2$ smaller than $\frac{1}{6}$? You know that $\sum q_j = 1$.

Write $q_j = \frac{1}{6} + e_j$, where the deviations e_j add to zero. Then

$$\sum q_j^2 = \sum\Big(\frac{1}{36} + \frac{e_j}{3} + e_j^2\Big) = \frac{1}{6} + \frac{1}{3}\sum e_j + \sum e_j^2 = \frac{1}{6} + \sum e_j^2 \geq \frac{1}{6}.$$

So the probability of a repeat is always at least $\frac{1}{6}$, with equality only for a fair die. It certainly cannot be less than $\frac{1}{36}$. ■

Looking back

- **Proof by contradiction.** We assumed all totals equally likely and derived an impossibility.
- **Use only what you need.** Of the eleven equations, three were enough. Choosing the right ones was the key step.
- **Inequalities prove impossibility.** AM–GM in part (i), and "a sum of squares is non-negative" in part (ii), ruled out whole families of possibilities at once.

What's really going on

A proof with polynomials

Here is a completely different proof, using the generating functions of Section 9.6. Let $Q(x) = q_1x + \cdots + q_6x^6$ and $R(x) = r_1x + \cdots + r_6x^6$. If all totals had probability $\frac{1}{11}$, then

$$Q(x)R(x) = \frac{x^2 + x^3 + \cdots + x^{12}}{11} = \frac{x^2}{11}(1 + x + \cdots + x^{10}).$$

Now $Q(x)/x$ is a polynomial of degree 5 with real coefficients, and every real polynomial of odd degree has a real root (its graph goes from $-\infty$ to $+\infty$, or the other way). So the left side has a real non-zero root. But $1 + x + \cdots + x^{10} = \frac{x^{11}-1}{x-1}$, whose roots are the 11th roots of unity other than 1, none of which is real. Contradiction. The same argument shows that no loading of two n-sided dice can make the totals equally likely whenever $n - 1$ is odd, that is, for n even. (For odd n the polynomial argument needs more work, but the AM–GM proof above works for every $n \geq 2$.)

Collision probability

The quantity $\sum q_j^2$ in part (ii), the chance that two independent draws agree, is called the *collision probability* of a distribution. Part (ii) shows that it is smallest for the uniform distribution. It measures how concentrated a distribution is: in ecology, $\sum q_j^2$ is Simpson's index of diversity, where q_j is the proportion of species j in a habitat; in economics, it is the Herfindahl index of market concentration; in cryptography, it measures how guessable a password distribution is. And it explains the remark at the end of Section 9.9: unequal birthday probabilities can only make shared birthdays more likely.

Impossibility proofs

Some of the most famous results in mathematics say that something cannot be done: no fraction squares to 2; no ruler-and-compass construction trisects every angle; no formula in radicals solves every quintic; no algorithm decides every mathematical statement. Each changed its subject, because to prove that *no* method works, you must understand the whole space of methods. The loaded dice problem is a miniature: to rule out every loading, we needed an invariant property (the AM–GM inequality) that every loading must obey.

The thinking skill

Prove impossibility by contradiction and inequality

To show that no choice of parameters works, assume one does, pick out the few conditions that clash, and use an inequality to show they cannot all hold.

At university, contradiction combined with inequalities is the standard way to prove that objects do not exist, in analysis, in combinatorics and in optimisation, where inequalities like AM–GM and Cauchy–Schwarz set limits on what can be achieved.

Try this next

1 **Cauchy–Schwarz.** Use the Cauchy–Schwarz inequality, $\left(\sum a_j b_j\right)^2 \le \sum a_j^2 \sum b_j^2$, with $a_j = q_j$ and $b_j = 1$, to give a one-line proof of part (ii).

2 **Two different dice.** If two *different* loaded dice are each thrown once, can the probability that they show the same number be less than $\frac{1}{6}$? Can it be 0?

3 **Uniform totals with other faces.** Show that two dice *can* give equally likely totals if their faces are numbered differently: try one die numbered $0, 1, 2, 3, 4, 5$ and the other $0, 6, 12, 18, 24, 30$. What totals arise? Why does this not contradict the question?

4 **Three dice.** Can three loaded six-sided dice, with faces 1 to 6, make all totals from 3 to 18 equally likely? (*Hint:* use the totals 3, 8, 13 and 18, and AM–GM for six numbers. Why does the polynomial proof fail this time?)

Further reading

- **J. Michael Steele, *The Cauchy–Schwarz Master Class*** (Cambridge University Press). A beautiful introduction to inequalities, starting from AM–GM.
- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press). Generating functions and their use in proofs like the one above.
- **Ian Stewart, *Galois Theory*** (CRC Press), the opening chapters. The most famous impossibility proof of all, for the quintic.

What We Have Learned

We began by saying that counting is the oldest mathematics there is, and that doing it cleverly is still one of the hardest. We asked three questions: how do you count without listing, when must something happen, and how do counting and probability help each other? Here is what the chapter's thirteen questions showed.

1. How do you count without listing?

The chapter built up a toolkit.

- **Recurrences.** Adding one line to the plane (Section 9.1) or one post to a fence (Section 9.3) changed the count in a simple way, giving a recurrence that could be solved. Recording a little extra information, the colour of the last post, made the recurrences close up.
- **Generating functions.** Packaging a sequence as a power series turned recurrences into algebra (Section 9.4), explained why $1/89 = 0.01123\ldots$, and turned counting combinations into multiplying polynomials: coins in Section 9.5, dice in Section 9.6, and balanced power sums in Section 9.7, where a repeated root at $x = 1$ did all the work.
- **Bijections.** The reflection principle (Section 9.10) counted bad queues by matching them with unrestricted paths, and complements of random sets turned subsets into disjoint pairs (Section 9.12).
- **Inclusion–exclusion.** Overcounting and correcting found the number of derangements (Section 9.8).
- **Double counting.** Counting regions with Euler's formula (Section 9.1) and pairs of subsets by sizes (Section 9.12) gave the same answers by different routes, and proved identities on the way.

2. When must something happen?

Several sections showed that size alone forces structure. Among six people there must be three mutual friends or three mutual strangers (Section 9.2), by the pigeonhole principle, though five people are not enough. Twenty-three people make a shared birthday more likely than not (Section 9.9), because the number of pairs grows like the square of the number of people. Random subsets of a large set almost always overlap (Section 9.12). And some things *cannot* happen, however you try: no loading of two dice makes every total equally likely (Section 9.13), and no numbering of dice other than Sicherman's reproduces the ordinary totals (Section 9.6).

3. How do counting and probability help each other?

In the second half of the chapter, probabilities were found by counting: derangements by inclusion–exclusion, the birthday problem by a product, the ballot problem by reflection. But the traffic went the other way too. Linearity of expectation gave the coupon collector's nH_n (Section 9.11) and the expected number of matches in a shuffle without any counting at all. Thinking of a random subset as n coin tosses (Section 9.12) turned counts like 3^n and $(m+1)^n$ into one-line calculations. And Erdős's probabilistic method (Section 9.2) proved that good colourings exist by showing that a random colouring works with positive probability. Two constants appeared from nowhere: e, in the limiting chance of a derangement, and Euler's γ, in the cost of completing a collection.

4. A toolkit for counting

Method	Where you used it
What changes when you add one	Lines (9.1); fence posts (9.3)
Pigeonhole principle	Ramsey (9.2)
Coupled recurrences	Fence posts (9.3)
Generating functions	9.4, 9.5, 9.6, 9.7, 9.13
Case analysis from the extremes	Strange dice (9.6); loaded dice (9.13)
Inclusion–exclusion	Derangements (9.8)
Complement and exponential approximation	Birthdays (9.9)
Reflection principle	Ballot problem (9.10)
Linearity of expectation	Coupons (9.11); matches (9.8)
One element at a time	Random subsets (9.12)
Contradiction with AM–GM	Loaded dice (9.13)

Where next

- The Poisson limit of derangements returns in Section 12.11, and generating functions of random variables drive the study of family trees in Section 12.15.
- Coupled recurrences become Markov chains in Section 12.16.
- The Gumbel distribution of the coupon collector describes the largest of many random quantities in Section 12.25.

If you want to go further before university

- **Miklós Bóna, *A Walk Through Combinatorics***: a friendly first course covering almost every idea in this chapter.
- **Herbert Wilf, *generatingfunctionology***: the classic on generating functions; free online.

- **Ronald Graham, Donald Knuth & Oren Patashnik, *Concrete Mathematics***: sums, recurrences and generating functions, with wit and many exercises.
- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction***: the counting side of first-year probability.

Whitehead called combinatorics "the slums of topology". This chapter has tried to show why nobody would say that now. Behind every clever count there was an idea that reaches far beyond it: a generating function, a bijection, a recurrence, an inequality. Those ideas connect counting with algebra, analysis and probability, and learning to look for them, rather than for one more trick, is what turns counting into mathematics.

Answers to Try This Next

These are brief answers and hints, not full solutions: they give the result and the key step, so that you can check your own work. Try each exercise properly before looking.

9.1 Lines Cutting the Plane

1. One circle gives 2 regions. The nth circle meets the others in $2(n-1)$ points, which cut it into $2(n-1)$ arcs, each splitting a region in two. So $R_n = R_{n-1} + 2(n-1)$, giving $n^2 - n + 2$.

2. Each line adds 1 plus the number of crossings on it, so the total is $1 + n +$ (number of crossings). The k parallel lines lose $\binom{k}{2}$ crossings: $1 + n + \binom{n}{2} - \binom{k}{2}$.

3. The nth plane is cut by the others in $n-1$ lines in general position, which divide it into $\frac{1}{2}(n^2 - n + 2)$ pieces, each splitting a region of space. Adding these to $\frac{1}{6}\big((n-1)^3 + 5(n-1) + 6\big)$ gives $\frac{1}{6}(n^3 + 5n + 6)$.

4. For $n = 6$ there are 31 regions, not 32. Inside the circle the graph has $V = n + \binom{n}{4}$ vertices (each crossing comes from four points) and $E = n + \binom{n}{2} + 2\binom{n}{4}$ edges (the arcs of the circle, plus the chords, each crossing adding two pieces). Euler's formula $V - E + F = 2$ counts the outside too, so the number of regions is $F - 1 = 1 + \binom{n}{2} + \binom{n}{4}$.

9.2 Ramsey's Theorem in Miniature

1. There are at most $6 \times 6 = 36$ bichromatic angles. Each triangle that is not one-colour contains exactly two, so there are at most 18 such triangles. Since $\binom{6}{3} = 20$, at least 2 triangles are one-colour.

2. A point has 16 edges, so at least 6 of one colour, say red, going to points $P_1, \ldots, P_6$. If any edge among the P_i is red, there is a red triangle. If not, the P_i form a two-coloured K_6, which has a one-colour triangle.

3. A point v has 9 edges, so either at least 4 are gold or at least 6 are silver. With 4 gold neighbours: a gold edge among them gives a gold triangle with v, and otherwise they form a silver K_4. With 6 silver neighbours: among them there is a one-colour triangle; if gold we are done, and if silver it forms a silver K_4 with v.

4. Label each term with (i, d): the lengths of the longest increasing and decreasing subsequences ending there. For two terms, the later one has a larger i or a larger d, so the labels are all different. If no subsequence of length 3 existed, there would be only 4 possible labels for 5 terms. Example with neither: $2, 1, 4, 3$.

9.3 Colouring Fence Posts

1. $r_{n+1} = s_n$ as before (post n is not red, so post $n+1$ can be red), and $r_n + s_n = (k-1)^{n-1}$, since each post after the first has $k-1$ choices. So $r_{n+1} + r_n = (k-1)^{n-1}$, and $r_n = \frac{(k-1)^{n-1} + (k-1)(-1)^{n-1}}{k}$. For the circle: k choices of colour for post 1, then a line of $n+1$ posts ending in the same colour, which gives $kr_{n+1} = (k-1)^n + (-1)^n(k-1)$.

2. $2, 3, 5, 8, \ldots$: with a_n ways, a line ending in white continues any line of $n-1$, and one ending in red continues a line of $n-2$ ending in white, so $a_n = a_{n-1} + a_{n-2}$. The answer is F_{n+2}.

3. 4 choices for the centre, then a 3-colouring of the n-cycle: $4\big(2^n + 2(-1)^n\big)$.

4. For $n = 4$: 18 colourings. A quarter-turn fixes none, and a half-turn fixes the 6 with opposite posts alike, so there are $\frac{1}{4}(18 + 0 + 6 + 0) = 6$ necklaces. For $n = 5$: 30 colourings, no rotation fixes any, so 6.

9.4 Sequences as Functions

1. The tail is at most $F_7 \sum_{r \geq 7} 2^{r-7} 10^{-r} = \frac{13 \times 10^{-7}}{0.8} < 2 \times 10^{-6}$. Since $\sum F_r 10^{-r} = \frac{10}{89}$ and the first six terms give 0.112358, $\frac{10}{89}$ lies between 0.112358 and 0.112360, so $\frac{1}{89} = 0.011235\ldots$. With two-digit blocks, $\sum F_r 100^{-r-1} = \frac{1}{9899}$.

2. Summing the recurrence gives $f(x)(1 - 7x + 10x^2) = 2 - 7x$, and $\frac{2-7x}{(1-2x)(1-5x)}$ splits as stated. For b_n: $10p + 5q = 40$ and $40p + 10q = 100$ give $p = 1$, $q = 6$; the roots of $t^2 = t + 6$ are 3 and -2, and $b_n = 4 \cdot 3^n + (-2)^n$.

3. $\sum n x^n = \frac{x}{(1-x)^2}$; at $x = \frac{1}{100}$ this is $\frac{100}{9801}$, so $\frac{1}{9801} = \sum n \cdot 100^{-n-1}$. The pattern goes wrong at the block 98: the block for 100 carries 1 into the 99, which carries into the 98, so the expansion runs $\ldots 96\,97\,99\,00\,01\ldots$.

4. Multiplying by $\frac{1}{1-x} = 1 + x + x^2 + \cdots$ makes the coefficient of x^n equal to $u_0 + \cdots + u_n$. The squares have generating function $\frac{x(1+x)}{(1-x)^3}$, so the partial sums have $\frac{x+x^2}{(1-x)^4}$, and the coefficient of x^n is $\binom{n+2}{3} + \binom{n+1}{3} = \frac{n(n+1)(2n+1)}{6}$.

9.5 Making Change

1. The coefficient of x^{24} counts $6a + 6b + 3c = 24$: for each $s = a + b$ from 0 to 4 there are $s + 1$ choices, giving 15. In general the coefficient of x^n is 0 unless $n = 3m$, and then it is $\frac{1}{2}(k+1)(k+2)$ with $k = \lfloor m/2 \rfloor$. With the extra factor the coefficients are partial sums: 55, 55 and 650 for x^{24}, x^{25} and x^{66}. They count the ways of paying with 1p and 3p coins and two different kinds of 6p coin.

2. Each part k is used once or not at all. The product telescopes: $\prod \frac{1-x^{2k}}{1-x^k}$ cancels every factor with an even exponent from the denominator, leaving $\prod \frac{1}{1-x^{\text{odd}}}$, the generating function for partitions into odd parts. For $n = 6$: 6, $5+1$, $4+2$, $3+2+1$, and $5+1$, $3+3$, $3+1+1+1$, $1+1+1+1+1+1$: four of each.

3. The values for $n = 0, 1, 2, \ldots$ are $1, 1, 2, 2, 3, 4, 5, 6, 7, 8, 10, \ldots$, and $\frac{(n+4)^2}{20}$ is always within $\frac{1}{2}$ of them (checked up to $n = 60$; a proof uses the partial fractions of $\frac{1}{(1-x)(1-x^2)(1-x^5)}$).
4. The last coin is 1p or 2p, so $c_n = c_{n-1} + c_{n-2}$: the Fibonacci numbers, $c_n = F_{n+1}$.

9.6 Strange Dice

1. Each die is x times a product of the factors $1 + x$, $1 + x$, $1 + x^2$, $1 + x^2$ with value 4 at $x = 1$, so it takes two of them. Apart from the standard pair, the only option is $x(1 + x)^2$ and $x(1 + x^2)^2$: faces $1, 2, 2, 3$ and $1, 3, 3, 5$.
2. The product is $2x^3 + x^4 + x^5 + 4x^6 + 3x^7 + 8x^8 + 6x^9 + 5x^{10} + 6x^{11}$, matching the table.
3. Yes: the Sicherman pair $1, 2, 2, 3, 3, 4$ and $1, 3, 4, 5, 6, 8$ together with a standard die.
4. $(x + x^2)(x + \cdots + x^8) = x^2(1 + x)^2(1 + x^2)(1 + x^4)$. The coin needs a factor with value 2: besides the standard choice $x(1 + x)$, it can be $x(1 + x^2)$ (faces $1, 3$, with the octahedron $1, 2, 2, 3, 5, 6, 6, 7$) or $x(1 + x^4)$ (faces $1, 5$, with $1, 2, 2, 3, 3, 4, 4, 5$).

9.7 Equal Sums of Powers

1. $\sum(c + b_k)^3 - \sum(c + a_k)^3 = 3c^2\big(\sum b_k - \sum a_k\big) + 3c\big(\sum b_k^2 - \sum a_k^2\big) + \sum b_k^3 - \sum a_k^3$, and the first two brackets vanish. Starting from $\{1, 4, 6, 7\}$, $\{2, 3, 5, 8\}$ with $c = 8$: $\{1, 4, 6, 7, 10, 11, 13, 16\}$ and $\{2, 3, 5, 8, 9, 12, 14, 15\}$.
2. A and $B \cup \{0\}$ both have 6 elements, sum 33 and sum of squares 253. Then $\sum_{a \in A}(n + a)^2 = \sum_{b \in B \cup \{0\}}(n + b)^2$, which splits $\{n^2, \ldots, (n + 11)^2\}$.
3. Expand $(x + c)^r$ by the binomial theorem: it involves only powers up to r, whose sums agree. The question's sets, with 0 added to the second, are $\{0, \ldots, 15\}$, and translating gives any 16 consecutive integers.
4. With $ABABABAB$, A gets about 1.676 and B about 1.042. With $ABBABAAB$, A gets about 1.560 and B about 1.158: still unequal, but much closer.

9.8 Nobody Gets Their Own Shirt

1. $\mathrm{E}(C_1) = \frac{1}{m}$, $\mathrm{Var}(C_1) = \frac{1}{m} - \frac{1}{m^2}$, and $\mathrm{Cov}(C_1, C_2) = \frac{1}{m(m-1)} - \frac{1}{m^2} = \frac{1}{m^2(m-1)}$. Then $\mathrm{E}(N) = 1$ and $\mathrm{Var}(N) = m\big(\frac{1}{m} - \frac{1}{m^2}\big) + m(m-1)\frac{1}{m^2(m-1)} = 1$. For $m = 4$: $\mathrm{P}(N = 0) = \frac{9}{24} = 0.375$, against $\mathrm{e}^{-1} \approx 0.368$.
2. Card 1 goes to some position i ($n - 1$ choices). If card i goes to position 1, the rest is a derangement of $n - 2$; if not, treat position 1 as card i's forbidden place, giving a derangement of $n - 1$. Check: $D_3 = 2(1 + 0) = 2$ and $D_4 = 3(2 + 1) = 9$.
3. Choose the k fixed cards and derange the rest: $\binom{n}{k}\frac{D_{n-k}}{n!} = \frac{1}{k!} \cdot \frac{D_{n-k}}{(n-k)!} = \frac{d_{n-k}}{k!}$, which tends to $\frac{\mathrm{e}^{-1}}{k!}$.

4. $30-(15+10+6)+(5+3+2)-1=8$, which equals $30\cdot\frac{1}{2}\cdot\frac{2}{3}\cdot\frac{4}{5}$: the inclusion–exclusion sum factorises (this is Euler's $\varphi(30)$).

9.9 The Birthday Problem

1. $\mathrm{P}(X=r)=\bigl(1-\frac{1}{n}\bigr)\cdots\bigl(1-\frac{r-2}{n}\bigr)\frac{r-1}{n}$, and these sum to 1. For the expectation, $\mathrm{P}(Y=r)$ is counted once in each of $\mathrm{P}(Y\ge 1),\dots,\mathrm{P}(Y\ge r)$. Write $Q_k=\mathrm{P}(X>k)=\prod_{j=1}^{k-1}(1-\frac{j}{n})$. Then $\mathrm{E}(X)=\sum(k+1)\frac{k}{n}Q_k$ and also $\mathrm{E}(X)=1+\sum Q_k$; using $\sum\frac{k}{n}Q_k=1$ and subtracting gives $\sum Q_k\bigl(1-\frac{k^2}{n}\bigr)=0$.

2. $\binom{k}{3}\approx 365^2\ln 2$ gives $k\approx 82$; the approximation ignores the dependence between triples, and the true answer is 88.

3. 14 people. (Each pair is "close" with probability $\frac{3}{365}$, and $\binom{k}{2}\frac{3}{365}\approx\ln 2$ gives $k\approx 14$; an exact count gives a probability of about 0.537 for 14 and 0.483 for 13.)

4. Two people share day i with probability p_i^2; add. By Cauchy–Schwarz, $1=\bigl(\sum p_i\bigr)^2\le 365\sum p_i^2$, with equality only when all p_i are equal.

9.10 Always Able to Give Change

1. The first person must have a £1 coin. Conditioning on the next few people splits into many cases (which coins you hold after each £2 customer), each finishing with the answers to parts (i) and (ii); collecting them gives $\frac{m-2}{m+1}$. The reflection principle does it in one line.

2. The first vote must be for A. Among the counts that start with A, the bad ones (those that reach a tie) correspond, by reflecting the part before the first tie, to the counts that start with B. So the number of good counts is $\binom{a+b-1}{a-1}-\binom{a+b-1}{a}$, and dividing by $\binom{a+b}{a}$ gives $\frac{a-b}{a+b}$.

3. $((()))$, $(()())$, $(())()$, $()(())$, $()()()$. A bijection: the first bracket pair splits a sequence into the part inside it and the part after it; correspondingly the triangle on a fixed edge splits the polygon into two smaller polygons. Recurse on both.

4. Each £5 customer needs four £1 coins in change, so the path goes up 1 or down 4. A path can jump past the barrier instead of touching it, and reflection no longer matches the bad paths with a simple set.

9.11 Collecting the Set

1. The expected waits are $\frac{n}{n}+\frac{n}{n-1}+\cdots+\frac{n}{n/2+1}=n(H_n-H_{n/2})\approx n(\ln n-\ln\frac{n}{2})=n\ln 2$.

2. Each card is missing with probability $(1-\frac{1}{n})^k$; add over the n cards. For 95%, $(1-\frac{1}{n})^k\approx \mathrm{e}^{-k/n}=0.05$, so $k\approx n\ln 20\approx 3n$.

3. After t packets a given card has been seen fewer than twice with probability about $\mathrm{e}^{-t/n}(1+\frac{t}{n})$. Both children finish roughly when the expected number of such

cards, $ne^{-t/n}(1+\frac{t}{n})$, falls to about 1; that gives $\frac{t}{n} \approx \ln n + \ln\ln n$, far less than $2\ln n$.

4. $\frac{1}{n+1} < \ln\frac{n+1}{n}$ (area under $\frac{1}{x}$ on $[n, n+1]$), so $H_n - \ln n$ decreases. By convexity the trapezium overestimates the area: $\ln\frac{n+1}{n} < \frac{1}{2}\left(\frac{1}{n} + \frac{1}{n+1}\right)$, so $H_n - \ln n - \frac{1}{2n}$ increases. With $n = 1$, $\gamma > \frac{1}{2}$; with $n = 4$, $\gamma < H_4 - \ln 4 \approx 0.697$.

9.12 Random Subsets

1. Each element must be in at least one set: $\left(\frac{3}{4}\right)^n$, and $\left(1 - 2^{-m}\right)^n$ in general.
2. $|S \setminus A_2|$ has the same distribution as $|A_2|$, and is independent of A_1, so the probability is that of $|A_1| + |S \setminus A_2| = n$. The pair $(A_1, S \setminus A_2)$ corresponds to a random subset of a $2n$-element set, of size n with probability $\binom{2n}{n}/4^n$. Stirling (or Wallis, Section 4.4) gives $\approx \frac{1}{\sqrt{\pi n}}$.
3. Each element goes in A, in B or in neither: 3^n. The only ordered pair with $A = B$ is $(\varnothing, \varnothing)$, so there are $\frac{3^n-1}{2} + 1 = \frac{3^n+1}{2}$ unordered pairs.
4. For $n = 3$, 3 sets (all the singletons, or all the pairs). For $n = 4$, 6 sets (all the pairs). In general $\binom{n}{\lfloor n/2 \rfloor}$ (Sperner's theorem).

9.13 Loaded Dice

1. $1 = \left(\sum q_j\right)^2 \le 6\sum q_j^2$, so $\mathrm{P}(\text{same}) = \sum q_j^2 \ge \frac{1}{6} > \frac{1}{36}$.
2. Yes, and yes: the probability is $\sum q_j r_j$, which is 0 if, say, one die always shows 1 and the other always 2.
3. The totals are $0, 1, \ldots, 35$, each exactly once: uniform. The question requires both dice to be numbered 1 to 6, and here they are not.
4. No. Write $\mathrm{P}(3) = q_1 r_1 s_1$ and $\mathrm{P}(18) = q_6 r_6 s_6$, both $\frac{1}{16}$. $\mathrm{P}(8) + \mathrm{P}(13)$ includes the six terms $q_1 r_1 s_6, \ldots, q_1 r_6 s_6$ (two 1s and a 6, or two 6s and a 1), whose product is $(\mathrm{P}(3)\mathrm{P}(18))^3$. AM–GM gives $\mathrm{P}(8) + \mathrm{P}(13) \ge 6 \cdot \frac{1}{16} > \frac{2}{16}$. The polynomial proof as given fails because $1 + x + \cdots + x^{15}$ *does* have a real root, -1. (It can be rescued: three real quintics need three real roots, and -1 is only a simple root.)

CHAPTER

10

Conservation Laws, Collisions and Gravity

Some things never change. Knowing which ones is half of physics.

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What Stays the Same

> *There is a fact, or if you wish, a law, governing all natural phenomena that are known to date. There is no known exception to this law.*
>
> — Richard Feynman, on the conservation of energy, *The Feynman Lectures on Physics* (1963)

Mechanics at school is usually a matter of forces: draw a diagram, resolve, apply $F = ma$. That works for a block on a slope. It does not work for a collision lasting a thousandth of a second, during which the forces are huge and unknown, or for a rocket that changes its mass as it goes, or for three stars pulling on each other at once. In all of these, the way forward is to stop asking about forces and ask instead what *cannot change*.

Momentum is conserved whenever no outside force acts. Energy is conserved in elastic collisions, and its loss can be measured in the others. Mass, angular momentum and the motion of the centre of mass follow similar rules. These laws do not tell you everything, but they are always true, they apply to systems far too complicated to analyse in detail, and they often rule out most of the possibilities before any calculation begins.

Why conservation laws matter

The conservation laws are the most reliable tools in physics. In the radioactive process called beta decay, the energies of the emerging particles did not add up; rather than abandon conservation of energy, Wolfgang Pauli predicted in 1930 an invisible particle carrying the missing energy, and the neutrino was detected twenty-six years later. Engineers design crash barriers, airbags and rockets by momentum and energy budgets. Astronomers weigh stars and find planets by watching centres of mass. And in 1918 Emmy Noether explained why the laws hold: each conservation law is the shadow of a symmetry of nature. Momentum is conserved because the laws of physics are the same everywhere; energy, because they are the same at all times.

Gravity is the second theme of this chapter. It is the force Newton used to unify the heavens and the Earth, and it is a natural partner of the conservation laws: orbits, slingshots and collapsing clouds are all understood through energy and momentum, together with one remarkable fact about spheres.

What you will meet at university

A first-year dynamics course starts here. You will meet systems of particles, the centre of mass and the reduced mass, collisions in different frames, variable-mass systems such as rockets, and the motion of bodies under gravity: Kepler's laws, orbits, escape velocity.

Later courses reformulate mechanics in the languages of Lagrange and Hamilton, where Noether's theorem becomes a precise statement and conservation laws can be read off from the symmetries of a single function. Beyond that lie celestial mechanics and the three-body problem, kinetic theory and statistical mechanics (where the piston of Section 10.4 becomes the theory of gases), and Einstein's theory of gravity.

The questions that run through this chapter

Three big questions

1. **What can conservation laws decide on their own?** When do momentum and energy determine an outcome, and when do they only rule things out?
2. **How do small steps add up?** What happens when a simple event, a collision or a firing, is repeated many times, or infinitely often?
3. **How do we tame many bodies?** Which choices of coordinates, and which symmetries, turn a problem about several interacting bodies into one we can solve?

The chapter begins with collisions: what Newton's cradle can and cannot do (Section 10.1), and how a chain of balls multiplies speed (Section 10.2). It then follows repeated processes: guns fired from a truck and the rocket equation, a molecule speeding up against a piston, and a puck that bounces infinitely often in a finite time. The last three sections turn to bodies that attract or repel: the centre of mass and reduced mass, the gravity of a spherical shell, and Lagrange's rotating triangle of stars.

Ideas in this chapter

§	Topic	Idea
10.1	Newton's cradle	Conservation laws as constraints
10.2	Stacked balls	Iterating one collision; slingshots
10.3	Rockets	Variable mass; the rocket equation
10.4	Kinetic theory	An adiabatic invariant
10.5	Inelastic collisions	Infinitely many bounces in finite time
10.6	The two-body problem	Centre of mass and reduced mass
10.7	Newtonian gravity	The shell theorem; free-fall collapse
10.8	The three-body problem	Lagrange's equilateral solution

How to read this chapter

Before calculating, list what is conserved, and in which frame. Then ask whether the problem looks simpler from somewhere else: from the centre of mass, from the moving piston, from the bouncing ball. Many of the questions in this chapter become almost trivial in the right frame, and most of the work in mechanics is finding it.

10.1 Newton's Cradle

The question

The identical uniform smooth spherical marbles $A_1, A_2, \ldots, A_n$, where $n \geq 3$, each of mass m, lie in that order in a smooth straight trough, with each marble touching the next. The marble A_{n+1}, which is similar to A_n but has mass λm, is placed in the trough so that it touches A_n. Another marble A_0, identical to A_n, slides along the trough with speed u and hits A_1. It is given that kinetic energy is conserved throughout.

(i) Show that if $\lambda < 1$, there is a possible subsequent motion in which only A_n and A_{n+1} move (and A_0 is reduced to rest), but that if $\lambda > 1$, such a motion is not possible.

(ii) If $\lambda > 1$, show that a subsequent motion in which only A_{n-1}, A_n and A_{n+1} move is not possible.

(iii) If $\lambda > 1$, find a possible subsequent motion in which only two marbles move.

The hook

Pull back one ball of a Newton's cradle and let it go: one ball flies out at the other end. Pull back two, and two fly out. Why? Why not one ball at twice the speed, or three balls at two-thirds of the speed? Both would carry the right momentum.

The answer begins with the two great conservation laws of mechanics. Momentum and energy together are much more restrictive than either alone, and they rule out most of the outcomes you might imagine. This question uses them as constraints: not to calculate what happens, but to decide what *can* happen. That way of thinking, with conservation laws as rules that every possible outcome must obey, runs through the whole of physics.

Before you start

Ask yourself A marble of mass m moving at speed u hits a stationary marble of mass M, and kinetic energy is conserved. What are the velocities afterwards? What if $M = m$?

Let the velocities afterwards be p and q. Momentum gives $mp + Mq = mu$ and energy gives $mp^2 + Mq^2 = mu^2$. Writing these as $m(u - p) = Mq$ and $m(u - p)(u + p) = Mq^2$

and dividing, $u + p = q$. Then

$$p = \frac{m - M}{m + M}\,u, \qquad q = \frac{2m}{m + M}\,u.$$

When $M = m$, the first marble stops and the second moves off at speed u: the two simply exchange velocities. That is why the marbles of equal mass pass the motion along the line.

Working it through

STAGE 1 Only the last two move

Ask yourself Suppose only A_n (velocity p) and A_{n+1} (velocity q) move afterwards. Write down the conservation laws. Which marble's motion constrains the sign of p?

Momentum and energy give

$$mp + \lambda mq = mu, \qquad \tfrac{1}{2}mp^2 + \tfrac{1}{2}\lambda mq^2 = \tfrac{1}{2}mu^2.$$

These are the equations for a marble of mass m hitting one of mass λm, so (apart from the solution $p = u$, $q = 0$, which would need A_n to pass through A_{n+1})

$$p = \frac{1 - \lambda}{1 + \lambda}\,u, \qquad q = \frac{2}{1 + \lambda}\,u.$$

If $\lambda < 1$, both are positive and $p < q$: A_n follows A_{n+1} but more slowly, so they separate, and everything is consistent. If $\lambda > 1$, then $p < 0$: A_n moves backwards. But A_{n-1} is at rest directly behind it, touching it, and A_n cannot pass through it. So this motion is impossible. ■

STAGE 2 Three marbles

Ask yourself Now suppose only A_{n-1}, A_n and A_{n+1} move, with velocities a, b, c. What inequalities must a, b, c satisfy, given which marbles are at rest and which are behind which?

The marble A_{n-2} is at rest behind A_{n-1} (it exists, since $n \geq 3$), so $a \geq 0$. And no marble can overtake the one in front, so $a \leq b \leq c$. Altogether $0 \leq a \leq b \leq c$. The conservation laws are

$$a + b + \lambda c = u, \qquad a^2 + b^2 + \lambda c^2 = u^2.$$

Ask yourself Square the momentum equation and compare it with the energy equation. What is left?

Squaring, $u^2 = a^2 + b^2 + \lambda^2 c^2 + 2ab + 2\lambda ac + 2\lambda bc$. Subtracting the energy equation,

$$0 = \lambda(\lambda - 1)c^2 + 2ab + 2\lambda c(a + b).$$

With $\lambda > 1$ and $a, b, c \geq 0$, every term on the right is non-negative, so every term is zero. In particular $c = 0$, and then $0 \leq a \leq b \leq c$ forces $a = b = 0$. But then the total momentum is 0, not mu. Contradiction: this motion is impossible. ■

Pause and try Notice the structure of the argument. Squaring the momentum equation produces cross terms like $2ab$, and the energy equation has none. The difference measures how "spread out" the motion is. Can you see why the same argument rules out *any* outcome in which only $A_k, \ldots, A_{n+1}$ move with non-negative velocities, when $\lambda > 1$?

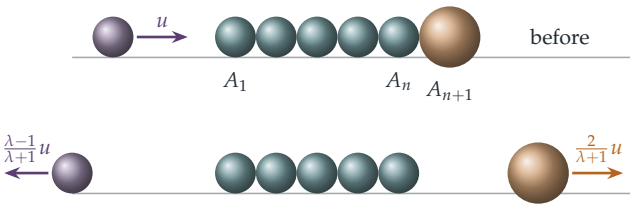
STAGE 3 Two marbles that can move

Ask yourself If the marbles moving forwards must all be at the far end, and that fails, perhaps one marble moves *backwards*. Which marble is free to move backwards?

Only A_0 has nothing behind it. Try: A_0 moves with velocity p (possibly negative), A_{n+1} with velocity q, and all the others are at rest. The conservation laws are again those of a collision between masses m and λm, so

$$p = -\frac{\lambda - 1}{\lambda + 1}\,u, \qquad q = \frac{2}{\lambda + 1}\,u.$$

A_0 bounces back, A_{n+1} moves forwards, and the marbles in between stay at rest, all consistent. It is exactly as if A_0 had hit A_{n+1} directly, with the line of equal marbles in between acting as a perfect messenger. ■



after: the possible motion of part (iii)

Figure 10.1.1 A row of equal marbles (teal) ending in a heavier one (amber, $\lambda > 1$). After the impact, A_0 bounces back, the heavy marble moves off, and every marble in between is left at rest, exactly as if A_0 had hit A_{n+1} directly.

Looking back

- **Conservation laws as constraints.** We never followed the collisions in detail. We asked which outcomes are allowed by momentum, energy and geometry.

- **Physical inequalities matter.** "Marbles cannot pass through each other" gave the inequalities $0 \le a \le b \le c$, and these, not the conservation laws, ruled out parts (i) and (ii).
- **Square and subtract.** Comparing the square of the momentum with the energy gives a sum of cross terms that must vanish.

What's really going on

Why one ball comes out

In a Newton's cradle of equal balls, suppose k balls leave together at speed w. Momentum gives $kw = u$ and energy gives $kw^2 = u^2$. Dividing, $w = u$ and $k = 1$. So among outcomes where the departing balls move together, only "one ball at the original speed" obeys both laws. Two balls at half speed carry the right momentum but only half the energy; one ball at twice the speed would have the right momentum only if it were half a ball.

But this does not settle the matter. With n balls there are n unknown velocities and only two conservation laws, so there are infinitely many outcomes that obey both (try finding one with three balls moving at different speeds). Which one happens depends on how the balls deform during contact. In the early 1980s Frank Herrmann and his colleagues showed that the familiar behaviour depends on the shock passing along the line *without spreading out*, which happens for hard steel balls with tiny gaps between them. With soft balls, or balls pressed tightly together, the cradle behaves less cleanly, and careful measurements show that the other balls always move a little.

Conservation laws and symmetry

Where do the conservation laws come from? In 1918 Emmy Noether proved a theorem that is one of the foundations of modern physics: every continuous symmetry of a physical system gives a conserved quantity. If the laws of physics are the same here as one metre to the left (symmetry under translation in space), momentum is conserved. If they are the same today as tomorrow (symmetry under translation in time), energy is conserved. If they are the same in every direction (rotational symmetry), angular momentum is conserved. You will meet Noether's theorem in a second-year course on Lagrangian mechanics; this chapter uses its consequences throughout.

Huygens, Wren and Wallis

In 1668 the Royal Society invited solutions to the problem of colliding bodies. John Wallis treated inelastic collisions, and Christopher Wren and Christiaan Huygens independently solved the elastic case. Huygens's key insight was to change frame: view the collision from a boat moving with the centre of mass, where two equal balls simply bounce back with their speeds reversed, and then transform back to the shore. He also noticed that a certain quantity, the sum of the masses times the squares of the

speeds, was the same before and after, though it was another century before kinetic energy was recognised as a conserved quantity in its own right.

The thinking skill

Use conserved quantities to rule things out

When a process is too complicated to follow, list the quantities that cannot change, and see which outcomes they allow.

This is how physicists reason about collisions in particle accelerators, where the detailed interactions are unknown but momentum, energy and charge must balance. Combined with an inequality (here, that marbles cannot pass through each other), it can prove that an outcome is impossible.

Try this next

1 **The same problem again.** (From STEP I, 2006.) Particles $A_1, \dots, A_n$ lie at rest in a trough; A_{n-1} has mass m and A_n has mass λm with $\lambda > 1$. A particle A_0 of mass m hits A_1 with speed u. Show that it is impossible for exactly one particle to be moving after all collisions, or for only A_{n-1} and A_n, or only A_{n-2}, A_{n-1}, A_n. Given that exactly two particles move, find their speeds.

2 **Following the collisions.** Model the impact in part (iii) as a sequence of separate two-marble collisions, starting with A_0 hitting A_1. Follow the sequence through and show that it ends in the motion found in part (iii).

3 **Two balls in.** In a cradle of five equal balls, two balls are released together and hit the other three with speed u. Show that "two balls leave with speed u" is consistent with both conservation laws, and find another outcome that is.

4 **Light end.** If $\lambda < 1$, follow the sequence of two-marble collisions. Do any marbles other than A_n and A_{n+1} end up moving?

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 5. Conservation of momentum and energy, with many problems on collisions.
- **Frank Herrmann & Peter Schmälzle, "Simple explanation of a well-known collision experiment"**, *American Journal of Physics* 49 (1981). Why Newton's cradle works.
- **Dwight Neuenschwander, *Emmy Noether's Wonderful Theorem*** (Johns Hopkins University Press). An accessible account of the link between symmetry and conservation.

10.2 The Stacked-Ball Effect

The question

(i) In an experiment, a particle A of mass m is at rest on a smooth horizontal table. A particle B of mass bm, where $b > 1$, is projected along the table directly towards A with speed u. The collision is perfectly elastic.

Find an expression for the speed of A after the collision in terms of b and u, and show that, irrespective of the relative masses of the particles, A cannot be made to move at twice the initial speed of B.

(ii) In a second experiment, a particle B_1 is projected along the table directly towards A with speed u. This time, particles $B_2, B_3, \ldots, B_n$ are at rest in order on the line between B_1 and A. The mass of B_i $(i = 1, 2, \ldots, n)$ is $\lambda^{n+1-i}m$, where $\lambda > 1$. All collisions are perfectly elastic. Show that, by choosing n sufficiently large, there is no upper limit on the speed at which A can be made to move.

In the case $\lambda = 4$, determine the least value of n for which A moves at more than $20u$. You may use the approximation $\log_{10} 2 \approx 0.30103$.

The hook

Hold a tennis ball on top of a basketball and drop them together. The basketball bounces off the floor, and the tennis ball shoots up to several times the height you dropped it from, sometimes high enough to hit the ceiling. Try it outdoors.

Part (i) of the question shows that a single collision has a speed limit: however heavy the ball that hits it, a stationary ball cannot be sent off at more than twice the speed of the incoming one. Part (ii) shows how to break the limit: pass the motion down a chain of ever-lighter balls, and the speed multiplies at every step. The same idea sends space probes to the outer planets, and has been suggested as part of the mechanism of a supernova.

Before you start

Ask yourself From Section 10.1: if a mass M moving at speed u hits a stationary mass m elastically, what is the speed of m afterwards?

The result there (with the roles of the masses swapped) gives the speed of the struck particle as $\frac{2M}{M+m}u$. Another way to remember it: in an elastic collision the *relative* velocity is reversed. If the heavy ball barely slows down, the light ball must leave at about u more than the heavy ball's speed, that is about $2u$.

Working it through

STAGE 1 The speed limit

With $M = bm$, the speed of A after the collision is

$$v_A = \frac{2bm}{bm+m}\,u = \frac{2b}{b+1}\,u.$$

Ask yourself How does $\frac{2b}{b+1}$ compare with 2? What happens as $b \to \infty$?

Since $\frac{2b}{b+1} = 2 - \frac{2}{b+1} < 2$ for every b, the speed of A is always less than $2u$, however heavy B is. It approaches $2u$ as $b \to \infty$: an infinitely heavy ball would bounce A off as if from a moving wall. ■

STAGE 2 Passing it along a chain

Ask yourself In part (ii), each particle is λ times as heavy as the next one. What does part (i) say about each collision in the chain?

B_1 (mass $\lambda^n m$) hits B_2 (mass $\lambda^{n-1}m$), which is λ times lighter. By part (i) with $b = \lambda$, B_2 moves off at $\frac{2\lambda}{\lambda+1}u$. Then B_2 hits B_3, which is again λ times lighter, and B_3 moves off at $\left(\frac{2\lambda}{\lambda+1}\right)^2 u$. After n such collisions, the last one being B_n hitting A,

$$v_A = \left(\frac{2\lambda}{\lambda+1}\right)^n u.$$

(Later collisions between the Bs can only push A further; they cannot slow it, since nothing ever hits A from in front.)

Ask yourself Is $\frac{2\lambda}{\lambda+1}$ bigger or smaller than 1?

Since $\lambda > 1$, $2\lambda > \lambda + 1$, so the ratio $r = \frac{2\lambda}{\lambda+1}$ exceeds 1, and $r^n \to \infty$ as $n \to \infty$. So by taking n large enough, A can be made to move as fast as we like. ■

STAGE 3 **Twenty times as fast**

With $\lambda = 4$, $r = \frac{8}{5} = 1.6$, and we need $1.6^n > 20$. Taking logarithms to base 10,

$$\log_{10} 1.6 = \log_{10} 16 - 1 = 4\log_{10} 2 - 1 \approx 0.20412, \qquad \log_{10} 20 = 1 + \log_{10} 2 \approx 1.30103,$$

so we need $n > \frac{1.30103}{0.20412} \approx 6.37$. The least value is $n = 7$. (Indeed $1.6^6 \approx 16.8$ and $1.6^7 \approx 26.8$.) ■

Looking back

- **Iterate a single result.** Part (i) was used n times in part (ii), turning a bounded gain into an unbounded one.
- **A limit that is never reached.** The bound $2u$ is approached but never attained, so the right statement is a strict inequality.
- **Logarithms turn powers into products.** The question of how many steps are needed became a division.

What's really going on

The basketball and the tennis ball

Let the two balls fall together from height H, reaching the floor at speed $U = \sqrt{2gH}$. Suppose everything is perfectly elastic and the basketball is much heavier. It bounces off the floor with speed U upwards, and immediately meets the tennis ball, still moving down at speed U. In the basketball's frame, the tennis ball approaches at $2U$ and, bouncing off a very heavy object, leaves at $2U$. Back in the ground frame, that is $2U + U = 3U$ upwards (Figure 10.2.1). Since height is proportional to the square of the launch speed, the tennis ball rises to $9H$.

Real balls are not perfectly elastic. With coefficient of restitution e and mass ratio M/m, the tennis ball leaves at speed $\frac{Me(2+e)-m}{M+m}U$. STEP I 2014 Q10, in the exercises, uses a measured height to work backwards to the mass ratio.

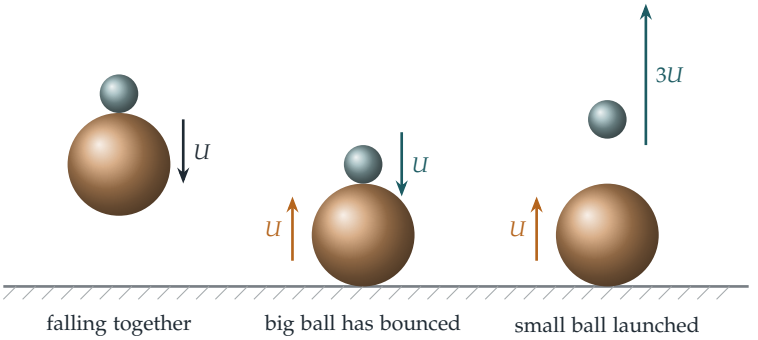


Figure 10.2.1 The stacked-ball effect, for perfectly elastic balls with the lower one much heavier. The big ball bounces up at speed U and meets the small ball coming down at U. Relative to the big ball, the small one arrives at $2U$ and leaves at $2U$, so it leaves the ground frame at $3U$, reaching nine times the original height.

Gravitational slingshots

A space probe flying past a planet does not touch it, but gravity acts like a very soft elastic collision with a very heavy object. In the planet's frame, the probe swings round and leaves with the same speed it arrived with. If the planet is moving towards the probe at speed V, then in the Sun's frame the probe can gain up to $2V$, exactly as the tennis ball does. The two *Voyager* probes, launched in 1977, used a rare alignment of the outer planets to pass from Jupiter to Saturn (and, for *Voyager 2*, on to Uranus and Neptune), gaining speed at each encounter. Without these gravity assists, the journey to Neptune would have taken about thirty years instead of twelve. The energy comes from the planet, which slows down by an immeasurably small amount.

Superballs and supernovae

In 1968 W. R. Mellen described in the *American Journal of Physics* a "superball rebound projectile" made by stacking several balls of decreasing size, and noted that the top ball can be launched at many times the impact speed. Astrophysicists have pointed out a loose analogy with a core-collapse supernova: the infalling outer layers of a star rebound off the stiff, dense core, and a shock wave carries momentum outwards into ever less dense material, accelerating it to enormous speeds. Demonstrations of the stacked-ball effect are used to explain the idea to students, though real supernovae involve much more physics, including neutrinos, than bouncing balls.

The thinking skill

Iterate a single result

When a process is a repetition of one simple step, solve the step once and compose it.

A gain factor less than 2 per step becomes unbounded after many steps, because the factors multiply. At university this is the idea behind transfer matrices, multi-stage amplifiers and multi-stage rockets (Section 10.3): the gain of a chain is the product of the gains of its links.

Try this next

1 **The real thing.** (From STEP I, 2014.) A ball of mass M and radius R is released from rest with its centre at height $H + R$, with a ball of mass m and radius r on top. The coefficient of restitution for every impact is e. Show that, bouncing alone, the large ball would rise until its centre is at height $R + He^2$. Given $R = 0.2$, $r = 0.05$, $H = 1.8$, $e = \frac{2}{3}$ and that the small ball's centre reaches height 4.5, show that $M/m = 9$.

2 **An inelastic chain.** (From STEP I, 2017.) Particles $P_1, P_2, \ldots$ of masses $\lambda m, \lambda^2 m, \ldots$ lie at $x = 1, 2, \ldots$, and a particle of mass m hits P_1 at speed u; the coefficient of restitution is e. Show that P_1 moves off at $\frac{1+e}{1+\lambda}u$. Find the speed of P_n after its first collision, and show that if $e > \lambda$ each particle is involved in exactly two collisions. What happens if $\lambda e = 1$?

3 **Three balls.** Three perfectly elastic balls are stacked, each much lighter than the one below. Show that the top ball leaves at $7U$ and rises to $49H$. With k balls, show that the top one leaves at $(2^k - 1)U$.

4 **Where does the energy come from?** In part (ii), show that the total kinetic energy is conserved even though A's speed is unbounded. What happens to the heavy particles?

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 5, which includes the stacked-ball problem.
- **W. R. Mellen, "Superball rebound projectiles"**, *American Journal of Physics* 36 (1968). Short and fun.
- **Gary Flandro, "Fast reconnaissance missions to the outer solar system utilizing energy derived from the gravitational field of Jupiter"**, *Astronautica Acta* 12 (1966). The paper that proposed the Voyager "grand tour".

10.3 Rockets and Guns

The question

A railway truck, initially at rest, can move forwards without friction on a long straight horizontal track. On the truck, n guns are mounted parallel to the track and facing backwards, where $n > 1$. Each of the guns is loaded with a single projectile of mass m. The mass of the truck and guns (but not including the projectiles) is M.

When a gun is fired, the projectile leaves its muzzle horizontally with a speed $v - V$ relative to the ground, where V is the speed of the truck immediately before the gun is fired.

(i) All n guns are fired simultaneously. Find the speed, u, with which the truck moves, and show that the kinetic energy, K, which is gained by the system (truck, guns and projectiles) is given by

$$K = \tfrac{1}{2}nmv^2\left(1 + \frac{nm}{M}\right).$$

(ii) Instead, the guns are fired one at a time. Let u_r be the speed of the truck when r guns have been fired, so that $u_0 = 0$. Show that, for $1 \leq r \leq n$,

$$u_r - u_{r-1} = \frac{mv}{M + (n-r)m} \qquad (*)$$

and hence that $u_n < u$.

(iii) Let K_r be the total kinetic energy of the system when r guns have been fired (one at a time), so that $K_0 = 0$. Using $(*)$, show that, for $1 \leq r \leq n$,

$$K_r - K_{r-1} = \tfrac{1}{2}mv^2 + \tfrac{1}{2}mv(u_r - u_{r-1})$$

and hence show that

$$K_n = \tfrac{1}{2}nmv^2 + \tfrac{1}{2}mvu_n.$$

Deduce that $K_n < K$.

The hook

A rocket in space has nothing to push against. It moves forward by throwing mass backwards, and conservation of momentum does the rest. A truck firing guns off its back is a rocket broken into pieces: each projectile is a lump of exhaust.

So should the guns fire all at once, or one after another? This question gives a definite answer, but look carefully at what it compares. Firing all at once gives the truck more speed, but it also uses more energy, so the guns are not doing the same job in the two cases. Change what is held fixed, and the answer reverses. Sorting this out leads to the most important equation in spaceflight, Tsiolkovsky's *rocket equation*, and to the reason why real rockets have several stages.

Before you start

Ask yourself Before a gun fires, the truck moves at V. Afterwards the projectile's velocity is $V - v$ (that is, speed $v - V$ backwards). What is the projectile's velocity *relative to the truck before firing*?

It is $(V - v) - V = -v$. So each gun throws its projectile backwards at speed v relative to the truck's velocity just before firing. That is the fixed rule of this question; keep it in mind, because it is what decides the answer.

Working it through

STAGE 1 All at once

The system starts at rest, with zero momentum. After firing, the truck (mass M) moves forward at u, and the n projectiles move backwards at speed v (since $V = 0$ before firing). So

$$Mu - nmv = 0, \qquad u = \frac{nmv}{M}.$$

The kinetic energy gained is

$$K = \tfrac{1}{2}Mu^2 + \tfrac{1}{2}nmv^2 = \tfrac{1}{2}\frac{n^2m^2v^2}{M} + \tfrac{1}{2}nmv^2 = \tfrac{1}{2}nmv^2\Big(1 + \frac{nm}{M}\Big). \qquad \blacksquare$$

STAGE 2 One at a time

Ask yourself Just before the rth gun fires, what is the mass of the truck with its remaining projectiles, and what is its speed? Write down conservation of momentum for the rth firing.

Before the rth firing, the truck carries $n - r + 1$ projectiles, so it has mass $M + (n - r + 1)m$ and speed u_{r-1}. Afterwards, the truck with $n - r$ projectiles has speed u_r, and the fired projectile has velocity $u_{r-1} - v$. Momentum is conserved:

$$\big(M + (n - r + 1)m\big)u_{r-1} = \big(M + (n - r)m\big)u_r + m(u_{r-1} - v).$$

The mu_{r-1} terms rearrange to give $\big(M + (n - r)m\big)(u_r - u_{r-1}) = mv$, which is $(*)$. $\blacksquare$

Adding up the increments,

$$u_n = \sum_{r=1}^{n} \frac{mv}{M + (n-r)m} = \frac{mv}{M} + \frac{mv}{M+m} + \cdots + \frac{mv}{M+(n-1)m}.$$

Every term after the first is smaller than $\frac{mv}{M}$, and $n > 1$, so $u_n < \frac{nmv}{M} = u$. ■

Pause and try Why does firing one at a time do worse here? When the first gun fires, it has to accelerate the truck *and* the $n-1$ unfired projectiles, so it produces less of a speed increase than it would on the bare truck.

STAGE 3 The energy bill

Ask yourself Write down the change in kinetic energy at the rth firing. How can $(*)$ help you simplify the term involving $u_r^2 - u_{r-1}^2$?

The kinetic energy changes by

$$K_r - K_{r-1} = \tfrac{1}{2}\big(M + (n-r)m\big)u_r^2 + \tfrac{1}{2}m(u_{r-1} - v)^2 - \tfrac{1}{2}\big(M + (n-r+1)m\big)u_{r-1}^2.$$

The last two terms combine to $\frac{1}{2}m(v^2 - 2vu_{r-1}) - \frac{1}{2}\big(M + (n-r)m\big)u_{r-1}^2$. Then, factorising a difference of squares and using $(*)$,

$$\tfrac{1}{2}\big(M + (n-r)m\big)(u_r^2 - u_{r-1}^2) = \tfrac{1}{2}mv(u_r + u_{r-1}).$$

So

$$K_r - K_{r-1} = \tfrac{1}{2}mv(u_r + u_{r-1}) + \tfrac{1}{2}mv^2 - mvu_{r-1} = \tfrac{1}{2}mv^2 + \tfrac{1}{2}mv(u_r - u_{r-1}).$$ ■

Summing from $r = 1$ to n, the increments $u_r - u_{r-1}$ telescope to u_n:

$$K_n = \tfrac{1}{2}nmv^2 + \tfrac{1}{2}mvu_n.$$ ■

Finally, $K = \frac{1}{2}nmv^2 + \frac{1}{2}nmv^2 \cdot \frac{nm}{M} = \frac{1}{2}nmv^2 + \frac{1}{2}mvu$, so $K_n < K$ because $u_n < u$. ■

Looking back

- **Momentum for each step.** Each firing was a small two-body problem: the truck with its load, and one projectile.
- **Telescoping sums.** The increments $(*)$ and the energy increments added up neatly.
- **Read the conclusion carefully.** Firing all at once gives more speed, but costs more energy. The question compares two processes that do different amounts of work.

What's really going on

What should be held fixed?

A real gun, or a rocket engine, throws its projectile at a fixed speed relative to *itself*, measured just after the projectile leaves. In this question the reference is the truck's speed *before* firing. For a small projectile the difference is tiny, but when all n projectiles are fired at once it matters a great deal: in part (i), a heavy load of projectiles is thrown back at speed v as if the truck did not recoil at all, which needs a lot of energy.

If instead every projectile leaves at speed v relative to the truck *after* firing, the conclusion reverses. All at once gives $u = \frac{nmv}{M+nm}$; one at a time gives $u_n = \sum_{k=1}^{n} \frac{mv}{M+km}$, which is larger, because each term is at least $\frac{mv}{M+nm}$. Now firing one at a time wins: each projectile pushes off from a truck that is already moving, and each later gun has less mass to accelerate. (Check this in the exercises.)

The rocket equation

Now let the projectiles become very small and very numerous, with total mass $\mu = nm$ fixed: this is a rocket burning fuel continuously. The sum for u_n becomes an integral (both conventions give the same limit):

$$u_n = \sum_{k=0}^{n-1} \frac{mv}{M+km} \;\to\; \int_0^{\mu} \frac{v\,\mathrm{d}s}{M+s} = v \ln \frac{M+\mu}{M}.$$

This is *Tsiolkovsky's rocket equation*: a rocket whose exhaust leaves at speed v relative to the rocket gains speed

$$\Delta u = v \ln \frac{m_{\text{initial}}}{m_{\text{final}}}.$$

Figure 10.3.1 compares the three answers. The logarithm is the problem: to reach the orbital speed of about 7.8 km/s with an exhaust speed of about 4.5 km/s (good chemical fuel), the mass ratio must be $e^{7.8/4.5} \approx 5.7$, and once air resistance and gravity are included it is nearer 10 or more. Almost all of a rocket is fuel.

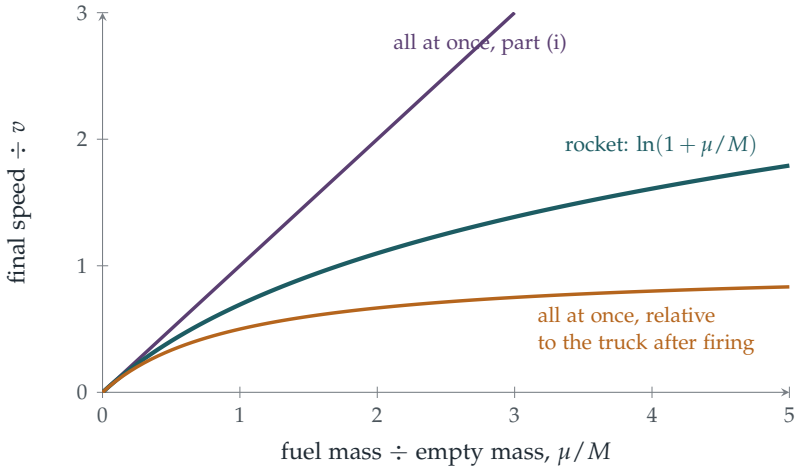


Figure 10.3.1 Final speed against the ratio of fuel to empty mass. Firing everything at once gives μ/M (plum) with the question's convention, but only $\frac{\mu/M}{1+\mu/M}$ (amber) if the exhaust speed is measured relative to the recoiling truck. The rocket, firing continuously, lies in between at $\ln(1+\mu/M)$ (teal). The picture is the inequality $\frac{x}{1+x} \leq \ln(1+x) \leq x$.

Why rockets have stages

The logarithm also explains staging. A rocket that drops its empty fuel tanks part way up no longer has to accelerate their dead mass. Since the gains of successive stages *add* (like the guns), while each stage's gain depends on the *ratio* of its masses, splitting a rocket into stages lets it reach speeds that a single stage of the same total mass cannot. The Saturn V that took astronauts to the Moon had three stages; its first stage burned about 2000 tonnes of fuel in under three minutes.

Tsiolkovsky

Konstantin Tsiolkovsky, a deaf schoolteacher in the provincial Russian town of Kaluga, published the rocket equation in 1903, in a paper entitled "Exploration of outer space by means of reaction devices". He worked out multi-stage rockets, liquid hydrogen and oxygen fuel, and space stations, decades before any of them could be built. The equation had appeared earlier in less prominent places (William Moore derived a version in 1810), but Tsiolkovsky was the first to see what it meant for spaceflight. "The Earth is the cradle of humanity," he wrote, "but one cannot live in the cradle forever."

The thinking skill

From a discrete process to a limit

A continuous process can often be understood as the limit of many small discrete steps. Solve the discrete problem exactly, then let the step size shrink: sums become integrals.

Here n guns became a rocket and a sum became a logarithm. At university you will see the same move whenever a physical law is derived: a small element is analysed, and the results are added up into an integral or a differential equation.

Try this next

1 **The other convention.** Redo parts (i) and (ii) supposing each projectile leaves at speed v relative to the truck *after* firing. Show that now firing one at a time gives the larger final speed.

2 **A comet in a dust cloud.** (From STEP III, 2009.) A comet of mass M enters a stationary dust cloud at speed V; after travelling a distance x its mass is $M(1+bx)$. If no other force acts, show that $(1+bx)v = V$ and find x in terms of t. If a constant force Mf acts in the direction of motion, use force = rate of change of momentum to show that $v = \frac{ft+V}{1+bx}$. Show that the comet can move at constant speed if $bV^2 = f$, and that otherwise its speed tends to $\sqrt{f/b}$.

3 **Two stages.** A rocket has exhaust speed v, empty mass M and fuel mass $9M$. Find its final speed. Now split it into two stages, each with empty mass $\frac{1}{2}M$ and fuel $4.5M$, the first stage being dropped when empty. Compare the final speeds.

4 **The inequality.** Prove that $\frac{x}{1+x} \le \ln(1+x) \le x$ for $x \ge 0$, by comparing $\frac{1}{1+s}$ with 1 and with $\frac{1}{1+x}$ on $[0, x]$.

Further reading

- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), Chapter 4. Momentum, variable mass and the rocket equation.
- **George Sutton & Oscar Biblarz, *Rocket Propulsion Elements*** (Wiley). The standard engineering text; the opening chapters are readable after this section.
- **Hermann Oberth, *Ways to Spaceflight*** (NASA translation, free online). A pioneer's own account.

10.4 A Molecule and a Piston

The question

A bicycle pump consists of a cylinder and a piston. The piston is pushed in with steady speed u. A particle of air moves to and fro between the piston and the end of the cylinder, colliding perfectly elastically with the piston and the end of the cylinder, and always moving parallel with the axis of the cylinder. Initially, the particle is moving towards the piston at speed v. Show that the speed, v_n, of the particle just after the nth collision with the piston is given by $v_n = v + 2nu$.

Let d_n be the distance between the piston and the end of the cylinder at the nth collision, and let t_n be the time between the nth and $(n+1)$th collisions. Express $d_n - d_{n+1}$ in terms of u and t_n, and show that

$$d_{n+1} = \frac{v + (2n-1)u}{v + (2n+1)u}\, d_n.$$

Express d_n in terms of d_1, u, v and n.

In the case $v = u$, show that $ut_n = \dfrac{d_1}{n(n+1)}$.

The hook

Pump up a bicycle tyre and feel the pump: it gets hot. Squeezing a gas warms it, and this is the principle behind diesel engines (which ignite their fuel by compression alone), refrigerators and the formation of stars. But why does compression heat a gas? Temperature is the average kinetic energy of the molecules, so the question is: why do the molecules speed up?

This question answers it with a single molecule. Each time it bounces off the advancing piston, it picks up a little extra speed, just as a tennis ball does off a moving racket. Following the bounces exactly reveals a hidden law: the molecule's speed times the length of the cylinder stays almost constant. That is an *adiabatic invariant*, and it is the microscopic origin of the gas laws you meet in physics.

Before you start

Ask yourself A ball moving at speed w hits a wall that is moving towards it at speed u, and bounces elastically. What is its speed afterwards? (Think in the wall's frame.)

In the wall's frame, the ball approaches at $w+u$ and, bouncing elastically off a very heavy object, leaves at $w+u$. Back in the ground frame, the wall is moving at u in the same direction as the rebounding ball, so the ball's speed is $w+2u$. Each bounce off the advancing piston adds $2u$. (Bouncing off the stationary end of the cylinder changes nothing.)

Working it through

STAGE 1 The speed after n bounces

The particle starts at speed v. Each collision with the piston adds $2u$, and collisions with the end do not change the speed. So after n collisions with the piston,

$$v_n = v + 2nu. \qquad \blacksquare$$

STAGE 2 A recurrence for the length

Ask yourself Between the nth and $(n+1)$th collisions, the particle goes from the piston to the end and back. How far does it travel? How far does the piston move?

In time t_n the piston moves ut_n, so $d_n - d_{n+1} = ut_n$. Meanwhile the particle, at speed v_n, travels from the piston to the end (distance d_n) and back to the piston (distance d_{n+1}): so $d_n + d_{n+1} = v_n t_n$. Dividing one equation by the other eliminates t_n:

$$\frac{d_n + d_{n+1}}{d_n - d_{n+1}} = \frac{v_n}{u}, \qquad \text{so} \qquad d_{n+1}(v_n + u) = d_n(v_n - u).$$

With $v_n = v + 2nu$,

$$d_{n+1} = \frac{v + (2n-1)u}{v + (2n+1)u}\, d_n. \qquad \blacksquare$$

STAGE 3 Telescoping

Ask yourself Write d_n as d_1 times a product of $n-1$ such fractions. What cancels?

$$d_n = d_1 \cdot \frac{v+u}{v+3u} \cdot \frac{v+3u}{v+5u} \cdots \frac{v+(2n-3)u}{v+(2n-1)u} = \frac{v+u}{v+(2n-1)u}\, d_1. \qquad \blacksquare$$

Each numerator cancels the previous denominator, leaving only the first numerator and the last denominator.

STAGE 4 **The case $v = u$**

Then $d_n = \frac{2u}{2nu} d_1 = \frac{d_1}{n}$, and

$$ut_n = d_n - d_{n+1} = d_1\Big(\frac{1}{n} - \frac{1}{n+1}\Big) = \frac{d_1}{n(n+1)}. \qquad \blacksquare$$

Adding these up, $u(t_1 + t_2 + \cdots) = d_1\big(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \cdots\big) = d_1$: the piston reaches the end of the cylinder at time d_1/u after the first collision, as it must. Before it gets there, the particle bounces infinitely often (Figure 10.4.1).

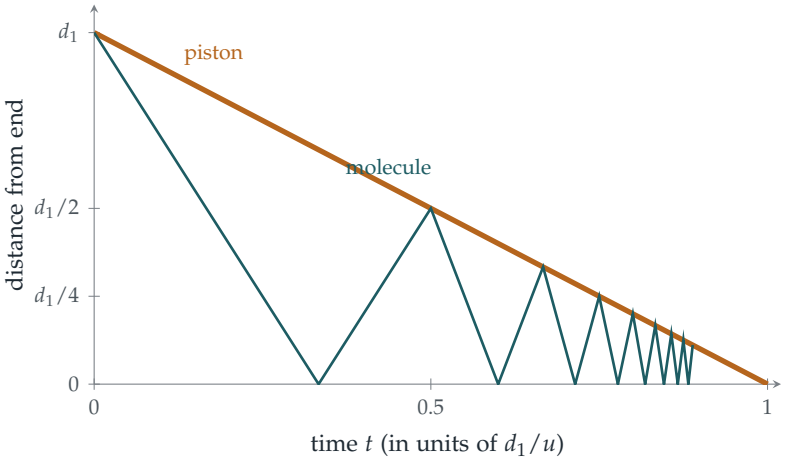


Figure 10.4.1 Space–time diagram for the case $v = u$: distance from the end of the cylinder against time. The piston (amber) advances steadily; the molecule (teal) bounces between it and the end. Its zigzags get steeper (faster) and shorter. The nth collision with the piston happens at distance d_1/n, so the molecule's speed is roughly proportional to $1/(\text{length})$.

Looking back

- **Change frame for a collision.** In the piston's frame the bounce is trivial.
- **Two equations, eliminate the unknown.** The distances travelled by the particle and the piston in the same time gave a recurrence without t_n.
- **Telescoping.** The product collapsed, just as the sum of $\frac{1}{n(n+1)}$ did.

What's really going on

An adiabatic invariant

Multiply the speed by the length:

$$v_n d_n = (v + 2nu) \cdot \frac{v+u}{v+(2n-1)u}\, d_1 \approx (v+u) d_1$$

when n is large, or when u is much smaller than v. So

$$\text{speed} \times \text{length} \approx \text{constant}.$$

If the piston moves slowly compared with the molecule, the product vd hardly changes, however far the piston moves. Quantities like this, which stay nearly constant when a system is changed slowly, are called *adiabatic invariants*. In general, the adiabatic invariant of a back-and-forth motion is $\oint p\,\mathrm{d}q$, the momentum integrated round one cycle; here that is $mv \times 2d$.

From one molecule to a gas

Since $v \propto 1/d$, the kinetic energy $\frac{1}{2}mv^2$ is proportional to $1/d^2$. For a gas of many molecules moving along one direction, temperature is proportional to the average kinetic energy, so $T \propto 1/d^2$: halving the length quadruples the temperature. In three dimensions the energy is shared between three directions of motion, and the same argument (which Feynman gives in his *Lectures*) leads to the law of adiabatic compression of a monatomic gas,

$$TV^{2/3} = \text{constant}, \qquad \text{or equivalently} \qquad pV^{5/3} = \text{constant}.$$

The exponent $\frac{5}{3}$ is the famous ratio γ of specific heats for a monatomic gas. Compressing air in a bicycle pump to a fifth of its volume would, if done quickly and with no heat escaping, raise its absolute temperature by a factor of $5^{0.4} \approx 1.9$ (air molecules have two atoms, giving $\gamma = 1.4$ instead): from 20°C to about 280°C. The pump does not get that hot only because the compression is slow and heat leaks away.

Fermi's cosmic rays

In 1949 Enrico Fermi proposed that cosmic rays, particles arriving from space with enormous energies, are accelerated exactly as the molecule in this question is: by bouncing repeatedly off moving magnetic fields in interstellar gas clouds, gaining a little energy each time. "Fermi acceleration" is now thought to operate at the shock waves of supernova remnants, where particles cross the shock back and forth like a molecule between a piston and a wall. Adiabatic invariants also played a role in the birth of quantum mechanics: Paul Ehrenfest argued in 1916 that the quantities that are quantised must be adiabatic invariants, since a slow change cannot make a system jump between quantum states.

The thinking skill

Build a recurrence from a physical process

When a process repeats, describe one cycle exactly and write down how the state changes from one cycle to the next.

Here the state was the pair (speed, length) at each collision. The recurrence could be solved exactly, and its solution revealed an approximately conserved quantity. At university, the same approach (one cycle, then a recurrence or a "return map") is used for orbits, oscillators and the iterated maps of Section 8.7.

Try this next

1 **The invariant exactly.** Show that $v_n d_n - (v+u)d_1 = \frac{u(v+u)d_1}{v+(2n-1)u}$, and deduce that $v_n d_n \to (v+u)d_1$ as $n \to \infty$.

2 **Pulling the piston out.** Repeat the question with the piston withdrawn at speed $u < v$. Show that the particle slows down, and find $v_n d_n$. (Expanding gases cool.)

3 **Energy.** In the case $v = u$, the molecule's speed grows without limit as the piston approaches the end. Find the average force on the piston during the nth cycle (change of momentum divided by t_n) and show that it grows like n^3. What does this say about the work needed to push the piston all the way in?

4 **A bouncing ball.** A ball bounces elastically on a floor under gravity, and the floor is raised slowly. Guess an adiabatic invariant for this motion. (*Hint:* the analogue of vd is the average speed times the height of the bounce, or equivalently the energy times the time between bounces.)

Further reading

- **Richard Feynman, Robert Leighton & Matthew Sands, *The Feynman Lectures on Physics*, Vol. I** (free online), Chapter 39, "The kinetic theory of gases". The same piston argument, extended to three dimensions.
- **Stephen Blundell & Katherine Blundell, *Concepts in Thermal Physics*** (Oxford University Press). Adiabatic processes and kinetic theory at first-year level.
- **Lev Landau & Evgeny Lifshitz, *Mechanics*** (Butterworth-Heinemann), §49. A concise treatment of adiabatic invariants, for later.

10.5 Infinitely Many Bounces in Finite Time

The question

A particle moves on a smooth triangular horizontal surface AOB with angle $AOB = 30°$. The surface is bounded by two vertical walls OA and OB and the coefficient of restitution between the particle and the walls is e, where $e < 1$. The particle, which is initially at point P on the surface and moving with velocity u_1, strikes the wall OA at M_1, with angle $PM_1A = \theta$, and rebounds, with velocity v_1, to strike the wall OB at N_1, with angle $M_1N_1B = \theta$. Find e and $\dfrac{v_1}{u_1}$ in terms of θ.

The motion continues, with the particle striking side OA at $M_2, M_3, \ldots$ and striking side OB at $N_2, N_3, \ldots$. Show that, if $\theta < 60°$, the particle reaches O in a finite time.

The hook

Zeno of Elea argued that you can never reach a wall: first you must cover half the distance, then half of what remains, and so on for ever. The resolution, that infinitely many steps can take a finite total time, is an infinite series that converges. This question is Zeno's paradox made real: a puck sliding into the corner of a wedge bounces off the walls infinitely many times, yet arrives at the corner after a finite time.

Nature does this too. A ball bouncing on the floor with a coefficient of restitution less than 1 bounces infinitely often in a finite time and then comes to rest. In computer simulations of granular materials (sand, grain, snow) this "inelastic collapse" is a real difficulty: the computer is asked to handle infinitely many collisions in a finite time.

Before you start

Ask yourself When a particle hits a smooth wall with coefficient of restitution e, what happens to the component of velocity along the wall? And the component perpendicular to it?

A smooth wall exerts no force along itself, so the component of velocity parallel to the wall is unchanged. The perpendicular component is reversed and multiplied by e. So if the particle arrives at angle α to the wall and leaves at angle β, then

$$\tan\beta = e\tan\alpha,$$

and since the parallel component is unchanged, (speed in)$\cos\alpha$ = (speed out)$\cos\beta$.

Working it through

STAGE 1 The angle of departure

Ask yourself The particle arrives at M_1 at angle θ to the wall OA (measured on the side away from O). At what angle φ to OA does it leave? Use the triangle OM_1N_1 to find φ in terms of θ.

The particle leaves M_1 heading towards O's side, at angle φ to the wall, so the angle OM_1N_1 is φ. In triangle OM_1N_1, the angle at O is 30°. At N_1, the angle M_1N_1B is θ and O, N_1, B lie on a line, so the angle ON_1M_1 is $180^\circ - \theta$. The angles of the triangle add to 180°:

$$30^\circ + \varphi + (180^\circ - \theta) = 180^\circ, \qquad \text{so} \qquad \varphi = \theta - 30^\circ.$$

(In particular, $\theta > 30^\circ$.) The restitution law $\tan\varphi = e\tan\theta$ then gives

$$e = \frac{\tan(\theta - 30^\circ)}{\tan\theta},$$

and conservation of the parallel component, $u_1\cos\theta = v_1\cos\varphi$, gives

$$\frac{v_1}{u_1} = \frac{\cos\theta}{\cos(\theta - 30^\circ)}. \qquad \blacksquare$$

STAGE 2 The pattern repeats

Ask yourself At N_1, the particle arrives at angle θ to the wall OB, measured on the side away from O. Compare this with its arrival at M_1. What happens next?

The situation at N_1 is the mirror image of the situation at M_1: the particle meets a wall at angle θ, on the side away from O. So it leaves at angle $\theta - 30^\circ$ and meets OA again at angle θ, and so on for ever. Every bounce is a scaled copy of the first. Each triangle OM_kN_k, ON_kM_{k+1}, ... is similar to the one before (Figure 10.5.1), and the speed is multiplied by the same factor at every bounce.

STAGE 3 Two ratios

Ask yourself By what factor ρ does the distance from O shrink from one bounce to the next? By what factor k does the speed shrink? So by what factor does the time for each leg shrink?

The sine rule in triangle OM_1N_1 gives

$$\rho = \frac{ON_1}{OM_1} = \frac{\sin\varphi}{\sin(180^\circ - \theta)} = \frac{\sin(\theta - 30^\circ)}{\sin\theta}.$$

All lengths in the next triangle are ρ times those in this one, so each leg of the path is ρ times as long as the previous leg. The speed is multiplied by $k = \frac{\cos\theta}{\cos(\theta-30^\circ)}$ at each bounce. Time is length divided by speed, so each leg takes

$$\frac{\rho}{k} = \frac{\sin(\theta-30^\circ)\cos(\theta-30^\circ)}{\sin\theta\cos\theta} = \frac{\sin(2\theta-60^\circ)}{\sin 2\theta}$$

times as long as the previous one.

STAGE 4 A geometric series

The total time is $T_1(1+r+r^2+\cdots)$, where T_1 is the time for the first leg and $r = \frac{\sin(2\theta-60^\circ)}{\sin 2\theta}$. This is finite if $0 < r < 1$.

Ask yourself When is $\sin(2\theta-60^\circ) < \sin 2\theta$? Use the identity $\sin X - \sin Y = 2\cos\frac{1}{2}(X+Y)\sin\frac{1}{2}(X-Y)$.

$$\sin 2\theta - \sin(2\theta-60^\circ) = 2\cos(2\theta-30^\circ)\sin 30^\circ = \cos(2\theta-30^\circ),$$

which is positive when $2\theta - 30^\circ < 90^\circ$, that is, when $\theta < 60^\circ$. And $r > 0$ since $\theta > 30^\circ$. So for $30^\circ < \theta < 60^\circ$ the series converges: the particle makes infinitely many bounces, its distance from O shrinks by the factor $\rho < 1$ each time, and it reaches O after the finite time $\frac{T_1}{1-r}$. ■

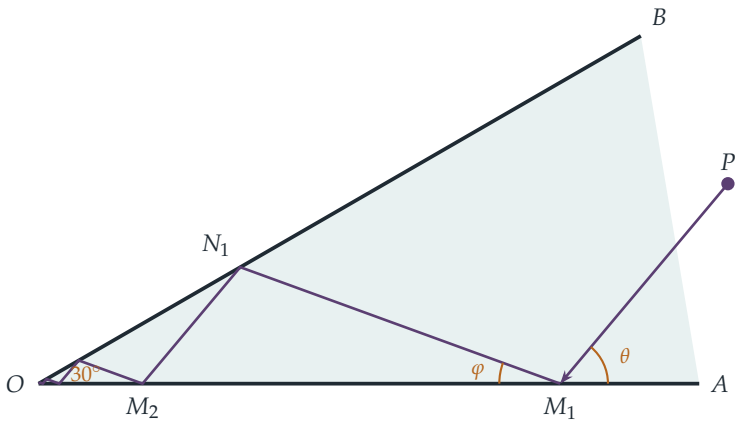


Figure 10.5.1 The path in the wedge with $\theta = 50^\circ$, so $\varphi = 20^\circ$ and $e = \tan 20^\circ / \tan 50^\circ \approx 0.31$. Each bounce is a copy of the previous one, scaled by $\rho \approx 0.45$, and the time for each leg is about 0.65 times that of the leg before. The particle reaches O after infinitely many bounces, in a finite time.

Looking back

- **Self-similarity.** Once one bounce was understood, every later bounce was a scaled copy, so the whole infinite process reduced to two ratios.

- **Combine ratios correctly.** Distance shrinks by ρ and speed by k, so time shrinks by ρ/k.
- **Infinitely many events, finite total.** A geometric series with ratio less than 1 converges.

What's really going on

When $\theta \geq 60°$

At $\theta = 60°$, the ratio is exactly 1: every leg takes the same time, and the particle never reaches O, although it gets ever closer. (The speed and the leg lengths both shrink by the same factor.) For $\theta > 60°$, the legs take longer and longer. In every case the particle approaches O and makes infinitely many bounces; only for $\theta < 60°$ does it get there in finite time. The value $e = \tan 30° / \tan 60° = \frac{1}{3}$ is the dividing line.

Bouncing balls and inelastic collapse

The simplest example of the same phenomenon: a ball dropped onto a floor rises to height e^2H, then e^4H, and so on. The time between bounces is proportional to the speed at each bounce, which shrinks by the factor e, so the total time is a geometric series: with $T_0 = \sqrt{2H/g}$ the time of the first fall, the ball stops bouncing after

$$T_0 + 2eT_0 + 2e^2T_0 + \cdots = T_0 \frac{1+e}{1-e}.$$

Infinitely many bounces, and then rest. In 1992 Sean McNamara and William Young showed that the same thing happens in simulations of many inelastic particles in a line: a group of them can undergo infinitely many collisions in a finite time, an "inelastic collapse". Simulations of sand and granular flows must include special rules to avoid it, and the phenomenon helps explain why granular gases spontaneously form clusters.

Zeno and convergence

Zeno's paradoxes, from around 450 BC, troubled philosophers for two thousand years. The mathematical resolution, that an infinite sum can have a finite value, was made precise only in the nineteenth century, when Cauchy, Weierstrass and others defined convergence by means of limits (the subject of Chapter 2). The surprising feature of this question is that the process is *physically* infinite: there really are infinitely many collisions, in a finite time. Mathematicians studying billiards in polygons and the motion of colliding particles call such behaviour a *Zeno phenomenon*, and it is one of the subtleties in proving that systems of colliding hard spheres have well-defined motions for all time.

The thinking skill

Sum an infinite process

When a process repeats with a constant scale factor, the total of any additive quantity (time, distance, energy) is a geometric series. Decide convergence from the ratio.

Here the key was to find the right ratio: not the ratio of distances, nor of speeds, but of times. At university, the same reasoning governs the convergence of iterative methods, the theory of fractals, and the analysis of algorithms that repeatedly halve a problem.

Try this next

1. **The total distance.** Show that the total distance travelled from M_1 to O is finite for every θ between 30° and 90°, and find it in terms of OM_1 and θ.

2. **Other wedges.** Repeat the question for a wedge of angle α instead of 30°. Show that $\varphi = \theta - \alpha$, and find the condition on θ for the particle to reach O in a finite time.

3. **Elastic walls.** If $e = 1$, can the particle ever reach O? (*Hint:* reflect the wedge repeatedly in its walls, so that the path becomes a straight line. How many copies of a 30° wedge fit round O?)

4. **The bouncing ball.** A ball is dropped from height 1 m with $e = 0.8$. How long does it bounce for, and how far does it travel in total? (Take $g = 9.8\ \mathrm{m\,s^{-2}}$.)

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 5. Includes several problems on bouncing balls and infinite sequences of collisions.
- **Sean McNamara & William Young, "Inelastic collapse and clumping in a one-dimensional granular medium"**, *Physics of Fluids A* 4 (1992).
- **Serge Tabachnikov, *Geometry and Billiards*** (American Mathematical Society; free online). The mathematics of billiards, including the unfolding trick in the exercises.

10.6 Two Bodies Become One

The question

Particles P, of mass 2, and Q, of mass 1, move along a line. Their distances from a fixed point are x_1 and x_2, respectively where $x_2 > x_1$. Each particle is subject to a repulsive force from the other of magnitude $\dfrac{2}{z^3}$, where $z = x_2 - x_1$.

Initially, $x_1 = 0$, $x_2 = 1$, Q is at rest and P moves towards Q with speed 1. Show that z obeys the equation $\dfrac{\mathrm{d}^2 z}{\mathrm{d}t^2} = \dfrac{3}{z^3}$.

By first writing $\dfrac{\mathrm{d}^2 z}{\mathrm{d}t^2} = v\dfrac{\mathrm{d}v}{\mathrm{d}z}$, where $v = \dfrac{\mathrm{d}z}{\mathrm{d}t}$, show that $z = \sqrt{4t^2 - 2t + 1}$.

By considering the equation satisfied by $2x_1 + x_2$, find x_1 and x_2 in terms of t.

The hook

Two bodies attract or repel each other: the Earth and the Moon, two stars, a proton and an electron. Each body's motion depends on where the other one is, so the equations are coupled, and it looks as if both motions must be found at once. Newton's great simplification was to change variables. Instead of the two positions, use the *centre of mass*, which moves as if no forces acted at all, and the *separation*, which moves like a single particle with a modified mass. One hard problem becomes two easy ones.

This question carries out that separation for two particles on a line. Along the way there is a small miracle: the separation turns out to be the square root of a quadratic in t, for a reason that has nothing to do with the question as set and everything to do with motion in a plane.

Before you start

Ask yourself Write down Newton's second law for each particle. In which direction does each force act?

The force is repulsive, so it pushes P (on the left) to the left and Q (on the right) to the right:

$$2\ddot{x}_1 = -\frac{2}{z^3}, \qquad 1 \cdot \ddot{x}_2 = +\frac{2}{z^3}.$$

These are coupled, because each involves $z = x_2 - x_1$.

Working it through

STAGE 1 The separation

Ask yourself Which combination of the two equations involves only z?

Subtract: $\ddot{z} = \ddot{x}_2 - \ddot{x}_1 = \frac{2}{z^3} + \frac{1}{z^3} = \frac{3}{z^3}$. ■

The separation behaves like a single particle of mass $\frac{2}{3}$ pushed by the force $\frac{2}{z^3}$. The number $\frac{2}{3} = \frac{2\times 1}{2+1}$ is the *reduced mass* $\frac{m_1 m_2}{m_1+m_2}$.

STAGE 2 Energy for the separation

Ask yourself With $\ddot{z} = v\frac{\mathrm{d}v}{\mathrm{d}z}$, separate the variables and integrate. What are the initial values of z and $v = \dot{z}$?

From $v\frac{\mathrm{d}v}{\mathrm{d}z} = \frac{3}{z^3}$, integrating gives $\frac{1}{2}v^2 = -\frac{3}{2z^2} + C$. Initially $z = 1$ and $\dot{z} = \dot{x}_2 - \dot{x}_1 = 0 - 1 = -1$, so $\frac{1}{2} = -\frac{3}{2} + C$, giving $C = 2$:

$$v^2 = 4 - \frac{3}{z^2}.$$

(This is conservation of energy for the separation: kinetic energy plus potential energy $\frac{3}{2z^2}$ per unit reduced mass is constant.)

STAGE 3 A surprisingly simple solution

Ask yourself The answer to show is $z^2 = 4t^2 - 2t + 1$. So look at $w = z^2$ instead of z. What is $\dot{w}^2$ in terms of w?

With $w = z^2$, $\dot{w} = 2z\dot{z}$, so

$$\dot{w}^2 = 4z^2\dot{z}^2 = 4z^2\left(4 - \frac{3}{z^2}\right) = 16w - 12.$$

Differentiating, $2\dot{w}\ddot{w} = 16\dot{w}$, so (whenever $\dot{w} \neq 0$) $\ddot{w} = 8$. Hence w is a quadratic in t: $w = 4t^2 + \alpha t + \beta$. Initially $w = 1$ and $\dot{w} = 2z\dot{z} = -2$, so $\beta = 1$ and $\alpha = -2$:

$$z = \sqrt{4t^2 - 2t + 1}. \quad ■$$

(You can check directly: $\dot{w}^2 = (8t-2)^2 = 64t^2 - 32t + 4 = 16(4t^2 - 2t + 1) - 12$.) The particles come closest at $t = \frac{1}{4}$, when $z = \frac{\sqrt{3}}{2}$, and then separate for ever.

STAGE 4 The centre of mass

Ask yourself What equation does $2x_1 + x_2$ satisfy? Why is it the natural combination to look at?

Adding the two equations of motion, $2\ddot{x}_1 + \ddot{x}_2 = -\frac{2}{z^3} + \frac{2}{z^3} = 0$. The forces are equal and opposite (Newton's third law), so the total momentum $2\dot{x}_1 + \dot{x}_2$ is constant. It starts at $2 \times 1 + 0 = 2$, and $2x_1 + x_2$ starts at 1, so

$$2x_1 + x_2 = 1 + 2t.$$

The centre of mass, at $\frac{2x_1+x_2}{3}$, moves steadily at speed $\frac{2}{3}$. Combining with $x_2 - x_1 = z$,

$$x_1 = \frac{1 + 2t - \sqrt{4t^2 - 2t + 1}}{3}, \qquad x_2 = \frac{1 + 2t + 2\sqrt{4t^2 - 2t + 1}}{3}. \qquad \blacksquare$$

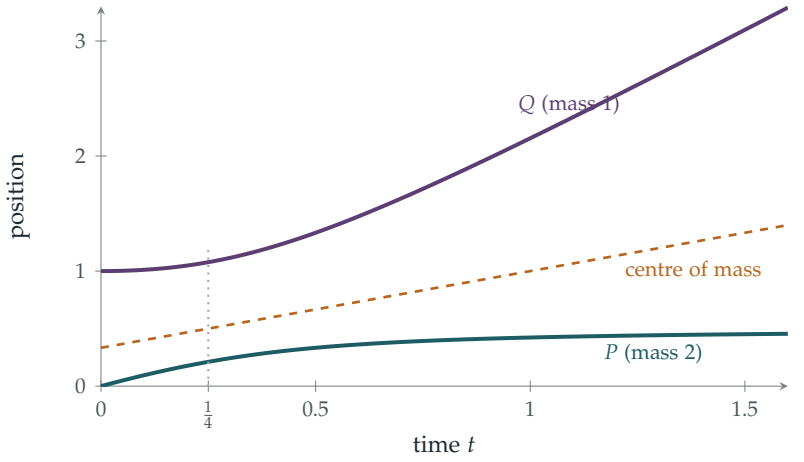


Figure 10.6.1 Positions of P (teal) and Q (plum) against time. P moves towards Q, is slowed and turned back by the repulsion, and approaches a constant position, while Q is pushed away. Their closest approach is at $t = \frac{1}{4}$. The centre of mass (amber, dashed) moves at constant speed throughout.

Looking back

- **Decouple by changing variables.** The difference of the equations involved only z; the weighted sum involved only the centre of mass.
- **Energy integral.** Writing $\ddot{z} = v\frac{\mathrm{d}v}{\mathrm{d}z}$ turned a second-order equation into a first-order one: conservation of energy.
- **Change the unknown.** The equation for z was awkward; the equation for z^2 was trivial.

What's really going on

Why z^2 is quadratic

For any force law of the form $\ddot{z} = k/z^3$, the same calculation gives $\frac{d^2}{dt^2}(z^2) = 2\dot{z}^2 + 2z\ddot{z} = 2\dot{z}^2 + \frac{2k}{z^2} =$ constant, since $\frac{1}{2}\dot{z}^2 + \frac{k}{2z^2}$ is the conserved energy. So z^2 is always a quadratic in t. Why?

Here is the hidden reason. Think of a particle moving freely, in a straight line at constant speed, in a *plane*. Its distance r from the origin satisfies $r^2 =$ (a quadratic in t), since $r^2 = (x_0 + at)^2 + (y_0 + bt)^2$. And in polar coordinates its radial equation of motion is $\ddot{r} - r\dot{\theta}^2 = 0$; with the angular momentum $h = r^2\dot{\theta}$ constant, that becomes

$$\ddot{r} = \frac{h^2}{r^3}.$$

The inverse-cube repulsion is exactly the *centrifugal* effect of the hidden angular motion. In our problem, $z^2 = 4t^2 - 2t + 1 = (2t - \frac{1}{2})^2 + \frac{3}{4}$ is the squared distance from the origin of a point moving at speed 2 along a straight line that passes the origin at distance $\frac{\sqrt{3}}{2}$. The one-dimensional repelling particles are the shadow of free motion in two dimensions.

The two-body problem

The method of this question works for any force between two bodies that depends only on their separation and acts along the line between them, in three dimensions as well as one. The centre of mass moves uniformly, and the separation vector $\mathbf{r}$ obeys $\mu\ddot{\mathbf{r}} = \mathbf{F}(\mathbf{r})$, with $\mu = \frac{m_1 m_2}{m_1 + m_2}$. For gravity this is the *Kepler problem*, solved by Newton in the *Principia*: the separation traces an ellipse, parabola or hyperbola. The Earth and the Sun both orbit their common centre of mass, which lies inside the Sun; the Earth and the Moon orbit a point about 4700 km from the Earth's centre, inside the Earth but well away from its centre, which is why the Earth wobbles monthly. Astronomers find planets round other stars by detecting exactly this wobble in the star.

Integrable systems

In 1969 Francesco Calogero studied many particles on a line repelling each other with inverse-square potentials (that is, inverse-cube forces), and found that, remarkably, the system can be solved exactly for any number of particles. Jürgen Moser explained the reason in 1975, using a matrix whose eigenvalues stay constant as it evolves: the *Lax pairs* of Section 7.7. The *Calogero–Moser system* is now a central example of an *integrable system*, a system with enough conserved quantities to be solved completely, and it connects mechanics with random matrices, quantum field theory and the theory of solitons.

The thinking skill

Choose coordinates that decouple

When two quantities are coupled, look for combinations that are not. Sums and differences, weighted by the right constants, often separate a problem into independent pieces.

Centre of mass and relative coordinates are the first example. At university the same idea appears as diagonalising a matrix (normal modes, Section 11.2), as separation of variables for partial differential equations, and in quantum mechanics.

Try this next

1 **Two masses on a string.** (From STEP III, 2023.) Particles A (mass m) and B (mass M) are joined by a light string of length r on a smooth table. Initially A is at the origin and B at $(r, 0)$; B is at rest and A is given velocity u in the y direction. Writing θ for the angle of $\overrightarrow{AB}$ clockwise from the x-axis, show that $\theta = ut/r$, and find $my + MY$ in terms of t. Show that $y = \frac{1}{m+M}\left(mut + Mr\sin\frac{ut}{r}\right)$, and that if $M > m$ the y-velocity of A is sometimes negative.

2 **Attraction.** Repeat the question with an attractive force $\frac{2}{z^3}$. Show that z^2 is still a quadratic in t, and find when the particles collide.

3 **The hidden plane.** For the motion in the question, find the straight line in the plane whose distance from the origin is $z(t)$, and the angle $\theta(t)$ swept out. Check that $z^2\dot{\theta}$ is constant and that $(z^2\dot{\theta})^2 = 3$.

4 **Kepler's third law.** Two stars of masses m_1 and m_2 move in circles about their centre of mass, a distance a apart, under gravity. Show that the period T satisfies $T^2 = \frac{4\pi^2 a^3}{G(m_1+m_2)}$.

Further reading

- **John Taylor, *Classical Mechanics*** (University Science Books), Chapter 8. The two-body problem, reduced mass and orbits.
- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), Chapter 10, on central forces.
- **Francesco Calogero, *Classical Many-Body Problems Amenable to Exact Treatments*** (Springer). For the ambitious: the world of exactly solvable systems.

10.7 Gravity Inside a Shell

The question

The gravitational force between two point particles of masses m and m' is mutually attractive and has magnitude

$$\frac{Gmm'}{r^2},$$

where G is a constant and r is the distance between them.

A particle of unit mass lies on the axis of a thin uniform circular ring of radius r and mass m, at a distance x from its centre. Explain why the net force on the particle is directed towards the centre of the ring and show that its magnitude is

$$\frac{Gmx}{(x^2+r^2)^{3/2}}.$$

The particle now lies inside a thin hollow spherical shell of uniform density, mass M and radius a, at a distance b from its centre. Show that the particle experiences no gravitational force due to the shell.

The hook

Imagine a hollow planet: a thin spherical shell of rock with nothing inside. Standing inside it, near the wall, you might expect to be pulled towards the nearby rock. In fact you would float: the gravitational pull of the shell is exactly zero everywhere inside it. The near side of the shell is closer, but the far side is bigger, and the two effects cancel perfectly.

This is *Newton's shell theorem*. It took Newton years to prove, and it was essential to his theory of gravity: together with its partner (outside a sphere, the pull is as if all the mass were at the centre) it justifies treating the Earth and the Sun as points. In the exercises and the What's Really Going On section, the theorem is used to compute how long a cloud of gas takes to collapse into a star.

Before you start

Ask yourself Consider two small, diametrically opposite pieces of the ring. Draw the forces they exert on the particle. Which components cancel, and which add?

The two pieces are at the same distance from the particle. Their pulls have equal components along the axis, which add, and equal and opposite components perpendicular to the axis, which cancel. Pairing up the whole ring in this way, the net force is along the axis, towards the centre of the ring.

Working it through

STAGE 1 The ring

Every piece of the ring is at distance $\sqrt{x^2+r^2}$ from the particle, and the line to it makes angle ψ with the axis, where $\cos\psi = \frac{x}{\sqrt{x^2+r^2}}$. A piece of mass δm pulls with force $\frac{G\,\delta m}{x^2+r^2}$, of which the component along the axis is $\frac{G\,\delta m}{x^2+r^2}\cdot\frac{x}{\sqrt{x^2+r^2}}$. Every piece has the same distance and the same angle, so the components simply add up to

$$\frac{Gmx}{(x^2+r^2)^{3/2}}. \qquad \blacksquare$$

STAGE 2 Slicing the shell into rings

Ask yourself Put the centre of the shell at O and the particle on the axis at distance b from O. Slice the shell into thin rings perpendicular to the axis, labelled by the angle ϕ that a point of the ring makes with the axis at O. What are the radius, the position and the mass of the ring?

The ring at angle ϕ has radius $a\sin\phi$ and its centre is at distance $a\cos\phi$ from O along the axis, so at distance $a\cos\phi - b$ from the particle. Its width is $a\,\delta\phi$, so its area is $2\pi a\sin\phi\cdot a\,\delta\phi$, a fraction $\frac{1}{2}\sin\phi\,\delta\phi$ of the shell's area $4\pi a^2$. Its mass is $\frac{1}{2}M\sin\phi\,\delta\phi$. By Stage 1, it pulls the particle along the axis, towards the $\phi = 0$ end, with force

$$\frac{G\cdot\frac{1}{2}M\sin\phi\,\delta\phi\,(a\cos\phi - b)}{\left((a\cos\phi-b)^2 + a^2\sin^2\phi\right)^{3/2}} = \frac{GM}{2}\cdot\frac{(a\cos\phi-b)\sin\phi\,\delta\phi}{(a^2-2ab\cos\phi+b^2)^{3/2}}.$$

(The sign takes care of itself: rings beyond the particle, with $a\cos\phi > b$, pull one way, and rings behind it pull the other.)

STAGE 3 The right variable

Ask yourself The denominator involves $s^2 = a^2 - 2ab\cos\phi + b^2$. What does s represent geometrically? What is $\mathrm{d}s$ in terms of $\mathrm{d}\phi$?

By the cosine rule, s is the distance from the particle to the ring. Differentiating, $2s\,\mathrm{d}s = 2ab\sin\phi\,\mathrm{d}\phi$, and solving $s^2 = a^2 - 2ab\cos\phi + b^2$ for $a\cos\phi$ gives $a\cos\phi - b = \frac{a^2-b^2-s^2}{2b}$. So the total force is

$$\frac{GM}{2}\int \frac{a^2-b^2-s^2}{2b\,s^3}\cdot\frac{s\,\mathrm{d}s}{ab} = \frac{GM}{4ab^2}\int_{a-b}^{a+b}\Big(\frac{a^2-b^2}{s^2}-1\Big)\mathrm{d}s,$$

where s runs from $a-b$ (the nearest point, $\phi=0$) to $a+b$ (the farthest, $\phi=\pi$).

STAGE 4 **It vanishes**

$$\int_{a-b}^{a+b}\Big(\frac{a^2-b^2}{s^2}-1\Big)\mathrm{d}s = (a^2-b^2)\Big(\frac{1}{a-b}-\frac{1}{a+b}\Big) - 2b = (a^2-b^2)\cdot\frac{2b}{a^2-b^2} - 2b = 0.$$

The shell exerts no force on the particle. ■

Pause and try What changes if the particle is *outside* the shell, with $b > a$? The limits of s become $b-a$ and $b+a$. Redo the last line and show that the force is $\frac{GM}{b^2}$ towards the centre, as if all the mass were concentrated there.

Looking back

- **Symmetry first.** Pairing opposite pieces showed the direction of the force before any calculation.
- **Build up from simpler shapes.** Points made a ring; rings made a shell.
- **Choose the variable that matches the physics.** Changing from the angle ϕ to the distance s turned an awkward integral into a trivial one.

What's really going on

Newton's cone argument

Newton's own proof (*Principia*, Book I, Proposition 70) uses no integration at all. Draw a narrow double cone with its vertex at the particle, cutting the shell in two small patches (Figure 10.7.1). The areas of the patches are proportional to the squares of their distances from the particle, r_1^2 and r_2^2 (the patches make equal angles with the cone, since a chord meets a circle at equal angles at its two ends). So their masses are proportional to r_1^2 and r_2^2, and their pulls, proportional to mass divided by distance squared, are *equal* and opposite. Every double cone through the particle cancels, so the

total force is zero. The inverse-square law is exactly what makes the cancellation work: with any other power of distance, the shell would push or pull.

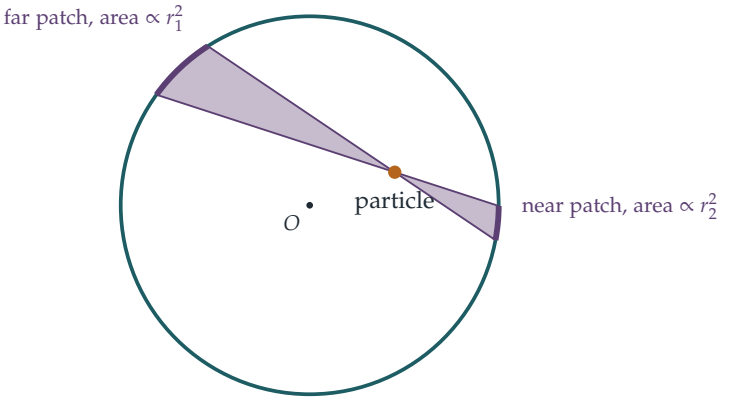


Figure 10.7.1 Newton's argument. A thin double cone from the particle cuts the shell in two patches. The far patch is bigger, by the ratio of the squared distances, but its pull is weaker by exactly the same ratio. The two pulls cancel, and so does every other pair: inside the shell, gravity vanishes.

Collapsing clouds

STEP III 1987 Q14 uses the theorem the other way round. Outside a uniform sphere, the pull on a particle at the surface is GMm/r^2, as if the whole mass were at the centre. Now take a uniform spherical cloud of dust, at rest, of radius R and mass M, and let it collapse under its own gravity. A particle on the surface feels only the mass inside it, which stays M as everything falls inwards, so its inward speed $v = -\dot{r}$ satisfies

$$v\frac{\mathrm{d}v}{\mathrm{d}r} = -\frac{GM}{r^2}, \qquad \text{so} \qquad v^2 = 2GM\left(\frac{1}{r} - \frac{1}{R}\right).$$

Substituting $r = R\cos^2\alpha$ (one of the exercises) gives the time to reach radius r as $\sqrt{\frac{R^3}{2GM}}\left(\alpha + \frac{1}{2}\sin 2\alpha\right)$. The whole cloud collapses to a point when $\alpha = \frac{\pi}{2}$, after the *free-fall time*

$$t_{\mathrm{ff}} = \frac{\pi}{2}\sqrt{\frac{R^3}{2GM}} = \sqrt{\frac{3\pi}{32G\rho}},$$

where ρ is the density. Remarkably, every shell of the cloud reaches the centre at the same moment, since the time depends only on the density (Figure 10.7.2). For a typical interstellar cloud, with about 10^{-17} kg per cubic metre, t_{ff} is around 700 000 years: this sets the timescale on which stars form.

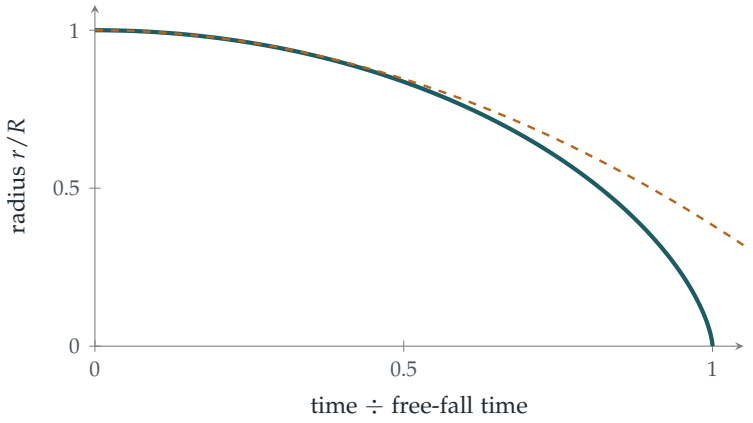


Figure 10.7.2 Radius of a collapsing uniform dust cloud against time (teal). It starts slowly, like a dropped stone (the dashed amber parabola is the constant-acceleration approximation $r = R - \frac{1}{2}\frac{GM}{R^2}t^2$), then accelerates sharply and reaches zero at $t = t_{\text{ff}}$. The curve is half of a cycloid.

Newton's delay and Gauss's law

Newton had the inverse-square law by the 1660s, but did not publish his theory of gravity until 1687. One reason, according to some historians, was that he could not justify treating the Earth as a point mass until he had proved the shell theorem; once he had, the rest of the *Principia* followed. Today the theorem is a short consequence of *Gauss's law*, which says that the total gravitational field flowing out through any closed surface is proportional to the mass inside. Take a sphere inside the shell, with the same centre. It encloses no mass, so the total flow through it is zero. That alone does not make the field zero (a uniform field has zero total flow through any closed surface, flowing in on one side and out on the other), but here symmetry helps: the field must point straight towards or away from the centre, with the same strength everywhere on the sphere, so zero total flow means zero field. The same law governs electric fields, which is why there is no electric field inside a charged metal shell, and why your car protects you from lightning.

The thinking skill

Integration as summation

To find the effect of a continuous body, cut it into pieces whose effects you know, and add them up. Use symmetry to choose the pieces.

Points made a ring, rings made a shell, shells make a solid sphere. At university this is how every field is computed from its sources (gravitational, electric, magnetic), and it leads to the vector calculus of Gauss's and Stokes's theorems.

Try this next

1 **Outside the shell.** Complete the Pause box: show that outside the shell the force is $\frac{GM}{b^2}$. Deduce that a solid uniform sphere of mass M pulls an outside particle as if its mass were at the centre.

2 **The collapse time.** (From STEP III, 1987.) Using $v^2 = 2GM(\frac{1}{r} - \frac{1}{R})$ and the substitution $r = R\cos^2\alpha$, show that the time to reach radius r is $\sqrt{\frac{R^3}{2GM}}\left(\alpha + \frac{1}{2}\sin 2\alpha\right)$.

3 **A tunnel through the Earth.** Inside a uniform solid sphere, show that the force at distance b from the centre is $\frac{GMb}{a^3}$, proportional to b. Deduce that a ball dropped down a tunnel through the centre of the Earth oscillates with period $2\pi\sqrt{a^3/(GM)}$, about 84 minutes, the same as the period of a satellite skimming the surface.

4 **Another force law.** If gravity followed an inverse-cube law, would the force inside a shell still be zero? (Repeat Stage 3 with $\frac{1}{s^3}$ in place of $\frac{1}{s^2}$.)

Further reading

- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), the chapters on gravitation, including the shell theorem.
- **Subrahmanyan Chandrasekhar, *Newton's Principia for the Common Reader*** (Oxford University Press). Newton's proofs, including Propositions 70 and 71, explained by a Nobel laureate.
- **Andrew Pontzen, *The Universe in a Box*** (Riverhead). How gravity, collapse and simulation explain the structure of the universe.

10.8 Three Stars in a Triangle

The question

The force of attraction between two stars of masses m_1 and m_2 a distance r apart is $\gamma m_1 m_2 / r^2$. The Starmakers of Kryton place three stars of equal mass m at the corners of an equilateral triangle of side a. Show that it is possible for each star to revolve round the centre of mass of the system with angular velocity $(3\gamma m/a^3)^{1/2}$.

Find a corresponding result if the Starmakers place a fourth star, of mass λm, at the centre of mass of the system.

The hook

Two bodies attracting each other by gravity can be solved completely, as Newton showed and as Section 10.6 explained: each moves on a conic about their common centre of mass, an ellipse if they are bound together, or a parabola or hyperbola if they have enough energy to escape each other. Add a third body and everything changes. The *three-body problem* has no general solution in terms of familiar functions, and in 1890 Henri Poincaré discovered that its solutions can be chaotic, the first appearance of chaos in mathematics.

Yet some special solutions are perfectly simple. In 1772 Joseph-Louis Lagrange found that three bodies at the corners of an equilateral triangle can rotate rigidly, the triangle turning like a wheel. This question finds that solution for equal masses. Lagrange's triangle is not just a curiosity: the Sun, Jupiter and thousands of "Trojan" asteroids form exactly this configuration, and space telescopes are parked at similar points today.

Before you start

Ask yourself For a star to move in a circle of radius R with angular velocity ω, what must the total force on it be?

It must have magnitude $m\omega^2 R$ and point towards the centre of the circle. So we need to check that the total gravitational force on each star points at the centre of the triangle, and find the ω that makes its size right.

Working it through

STAGE 1 The geometry

Ask yourself How far is each corner of an equilateral triangle of side a from its centre?

The centre lies on each median, two-thirds of the way from the corner. A median has length $a\sin 60^\circ = \frac{\sqrt{3}}{2}a$, so the distance from each corner to the centre is

$$R = \frac{2}{3}\cdot\frac{\sqrt{3}}{2}a = \frac{a}{\sqrt{3}}.$$

By symmetry the centre of the triangle is the centre of mass of three equal stars.

STAGE 2 The force on one star

Ask yourself Each of the other two stars pulls with force $\frac{\gamma m^2}{a^2}$. What is the angle between the two pulls? What is their resultant, and in which direction does it point?

The two pulls point along the two sides of the triangle that meet at the star, so the angle between them is 60°. By symmetry their resultant lies along the bisector of that angle, which passes through the centre. Each pull makes 30° with the bisector, so the resultant has magnitude

$$2\cdot\frac{\gamma m^2}{a^2}\cos 30^\circ = \frac{\sqrt{3}\,\gamma m^2}{a^2},$$

pointing at the centre (Figure 10.8.1).

STAGE 3 The angular velocity

Setting the resultant equal to $m\omega^2 R$:

$$m\omega^2\frac{a}{\sqrt{3}} = \frac{\sqrt{3}\,\gamma m^2}{a^2}, \qquad \text{so} \qquad \omega^2 = \frac{3\gamma m}{a^3}.$$

With this angular velocity, the same calculation applies to each star (by symmetry), so all three move round the centre in circles, keeping the triangle rigid. ■

STAGE 4 A fourth star at the centre

Ask yourself A star of mass λm at the centre is at distance $\frac{a}{\sqrt{3}}$ from each corner. What extra force does it exert on each corner star? Does the centre of mass move? Does the central star feel any net force?

The central star does not change the centre of mass, and by symmetry the three pulls on it cancel, so it stays at rest. It pulls each corner star towards the centre with force $\frac{\gamma\lambda m^2}{(a/\sqrt{3})^2} = \frac{3\gamma\lambda m^2}{a^2}$. The total inward force on a corner star is now $(\sqrt{3}+3\lambda)\frac{\gamma m^2}{a^2}$, so

$$m\omega^2\frac{a}{\sqrt{3}} = (\sqrt{3}+3\lambda)\frac{\gamma m^2}{a^2}, \qquad \omega^2 = \frac{(3+3\sqrt{3}\,\lambda)\gamma m}{a^3} = \frac{3\gamma m}{a^3}(1+\sqrt{3}\,\lambda). \qquad \blacksquare$$

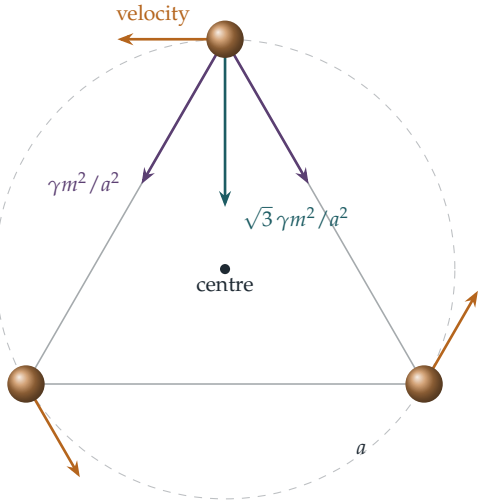


Figure 10.8.1 Three equal stars at the corners of an equilateral triangle. The two pulls on the top star (plum) combine into a single force towards the centre (teal). If each star moves on the dashed circle with the right speed (amber), the whole triangle rotates rigidly. This is Lagrange's equilateral solution of the three-body problem.

Looking back

- **Guess a symmetric solution.** We did not solve the three-body problem; we guessed a highly symmetric motion and checked that Newton's laws allow it.
- **Symmetry fixes directions.** The resultant force had to lie along the bisector, so only its size needed calculating.
- **Check every body.** By symmetry, verifying one star verified all three; the central star needed a separate (easy) check.

What's really going on

Kepler's law in disguise

Write the answer as $\omega^2 a^3 = \gamma(3m)$. For two bodies in circular orbit a distance a apart, Kepler's third law (an exercise in Section 10.6) says $\omega^2 a^3 = \gamma(m_1+m_2)$, the total mass.

The three stars obey the same law, with the total mass $3m$. In fact Lagrange showed that the equilateral solution exists for *any* three masses, not just equal ones: the triangle rotates about the centre of mass with $\omega^2 a^3 = \gamma(m_1 + m_2 + m_3)$. Euler had found a second family a few years earlier, with the three bodies on a rotating straight line. These five configurations (three collinear, two triangular, up to labelling) are the only ones in which three bodies keep the same shape.

Lagrange points

Take m_1 to be the Sun, m_2 a planet, and m_3 a tiny asteroid. The equilateral solution says the asteroid can sit at the third corner of an equilateral triangle with the Sun and the planet, ahead of or behind the planet in its orbit. These are the *Lagrange points* L_4 and L_5; Euler's collinear solutions give L_1, L_2 and L_3. Jupiter has over ten thousand known "Trojan" asteroids clustered around its L_4 and L_5 points. The James Webb Space Telescope orbits near the Sun–Earth L_2 point, 1.5 million km beyond the Earth, where it can keep the Sun, Earth and Moon all behind its sunshield.

Stability

Would the Starmakers' triangle last? For equal masses, no: the slightest disturbance grows, and the stars fly apart or collide. Edward Routh showed in 1875 that Lagrange's triangle is stable only if one mass dominates, precisely when

$$27(m_1 m_2 + m_2 m_3 + m_3 m_1) < (m_1 + m_2 + m_3)^2.$$

For the Sun and Jupiter this holds comfortably, which is why the Trojans survive. For three equal masses, $27 \times 3m^2 > 9m^2$, so the configuration is unstable.

Poincaré's mistake and the figure eight

In 1887 King Oscar II of Sweden offered a prize for a solution of the many-body problem. Henri Poincaré won it in 1889 with a long memoir on the three-body problem, but while the memoir was being printed an editor found a gap. In correcting it, Poincaré discovered that some solutions behave with "a complexity so astonishing that I cannot even attempt to draw it": the first glimpse of chaos. The printed copies were recalled and destroyed, at Poincaré's expense. New special solutions are still being found: in 1993 Cristopher Moore discovered numerically, and in 2000 Alain Chenciner and Richard Montgomery proved, that three equal masses can chase each other round a figure-eight curve. Thousands more periodic orbits have since been found by computer.

The thinking skill

Guess a symmetric solution and verify it

When a problem is too hard to solve in general, look for solutions with as much symmetry as possible. A symmetric guess reduces many unknowns to one or two, and checking it is often easy.

This is a powerful strategy in mathematics and physics: symmetric equilibria, steady states, travelling waves and rigidly rotating configurations are often the first solutions found, and the starting points for studying the rest. The next question is always whether the symmetric solution is stable.

Try this next

1. **Four stars in a square.** Four stars of mass m are placed at the corners of a square of side a. Show that they can rotate rigidly about the centre with $\omega^2 = \frac{(4+\sqrt{2})\gamma m}{2a^3}$.

2. **Two equal stars and a planet.** Two stars of mass M orbit their centre of mass, a distance a apart. A planet of negligible mass sits at the centre. Is it in equilibrium? Is it stable to small displacements along the line joining the stars? Perpendicular to it?

3. **Unequal masses.** Three stars of masses m_1, m_2, m_3 are at the corners of an equilateral triangle of side a, rotating about their centre of mass G. Show that the gravitational force on m_1 is $\frac{\gamma m_1(m_1+m_2+m_3)}{a^3}\,\overrightarrow{P_1G}$, where P_1 is its position. (*Hint:* write the forces as vectors and use the definition of the centre of mass.) Deduce Lagrange's result.

4. **n stars.** n equal stars sit at the corners of a regular n-gon inscribed in a circle of radius R. Find ω^2 for rigid rotation as a sum over k of $\frac{1}{\sin(k\pi/n)}$.

Further reading

- **June Barrow-Green, *Poincaré and the Three Body Problem*** (American Mathematical Society). The story of the prize and Poincaré's discovery of chaos.
- **Richard Montgomery, "A new solution to the three-body problem"**, *Notices of the American Mathematical Society* 48 (2001). The figure eight, explained by one of its discoverers; free online.
- **Florin Diacu & Philip Holmes, *Celestial Encounters*** (Princeton University Press). A popular history of the three-body problem and chaos.

What We Have Learned

We began by saying that some things never change, and that knowing which ones is half of physics. We asked three questions: what conservation laws can decide on their own, how small steps add up, and how to tame many bodies. Here is what the chapter's eight questions showed.

1. What can conservation laws decide on their own?

For two bodies colliding elastically in a line, momentum and energy determine the outcome completely, and that single result was used again and again. For more bodies, they do not. In Section 10.1, a row of marbles had many outcomes consistent with both laws, and it took a further physical fact (marbles cannot pass through each other) to rule some out. The square of the momentum minus the energy gave a sum of cross terms that had to vanish, an inequality argument that proved certain motions impossible. And the familiar behaviour of Newton's cradle turned out to depend on details beyond the conservation laws: how the shock travels through the balls.

Conservation laws also exposed hidden assumptions. In Section 10.3, firing guns all at once gave more speed than firing them one at a time, but only because it used more energy; holding the right quantity fixed reversed the conclusion.

2. How do small steps add up?

Several sections repeated one simple step. In Section 10.2, a collision that can at most double a speed became, through a chain of lighter and lighter balls, unlimited gain: the principle behind the basketball and tennis ball and behind gravitational slingshots. In Section 10.3, many small firings became an integral, and Tsiolkovsky's logarithm, which explains why rockets are mostly fuel and why they have stages. In Section 10.4, each bounce off a moving piston added $2u$ to a molecule's speed, and an exact recurrence revealed that speed times length stays almost constant: an adiabatic invariant, and the reason compressed gases heat up. And in Section 10.5, infinitely many bounces took a finite time, because the times formed a convergent geometric series.

3. How do we tame many bodies?

By choosing coordinates and exploiting symmetry. In Section 10.6, the centre of mass moved uniformly and the separation behaved like a single particle with the reduced mass; a hidden plane even explained why the separation squared was a quadratic in time. In Section 10.7, symmetry and integration showed that a spherical shell exerts no gravity inside and acts as a point mass outside, which let a collapsing cloud of dust be treated as a single equation, with a free-fall time that sets the pace of star formation. And

in Section 10.8, a symmetric guess solved a special case of the unsolvable three-body problem: Lagrange's rotating triangle, realised in the Trojan asteroids.

4. A toolkit for mechanics

Method	Where you used it
Conservation laws as constraints	Newton's cradle (10.1)
Change of frame	Collisions (10.2); piston (10.4)
Iterating one result	Stacked balls (10.2)
Momentum for variable mass	Rockets (10.3)
Sum to integral	Rocket equation (10.3); shell (10.7)
Recurrence and telescoping	Piston (10.4)
Geometric series	Wedge (10.5)
Centre of mass and reduced mass	Two bodies (10.6)
Energy integral $v\,\mathrm{d}v/\mathrm{d}z$	Two bodies (10.6); collapse (10.7)
Symmetric guess	Three stars (10.8)

Where next

- Chapter 11 turns from conservation to oscillation and rotation: resonance, normal modes, the real pendulum, and motion in accelerating and rotating frames.
- The energy method of Section 8.2 and the limit cycles of Section 8.4 are the differential-equation side of the same ideas.
- The chaos Poincaré found in the three-body problem is the subject of Section 8.7.

If you want to go further before university

- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics***: the classic first-year text; demanding but rewarding.
- **David Morin, *Introduction to Classical Mechanics***: hundreds of problems, many of them as surprising as the ones in this chapter.
- **Richard Feynman, *The Character of Physical Law***: lectures for a general audience on conservation laws, symmetry and gravity.
- **Dwight Neuenschwander, *Emmy Noether's Wonderful Theorem***: why symmetry implies conservation.

Feynman said that there is no known exception to the conservation of energy. This chapter has tried to show how much they can do: rule out impossible outcomes, reveal hidden invariants, and turn hopelessly complicated systems into ones we can understand. But it has also shown their limits. They constrain, but they do not always decide; and where they stop, the details of forces, symmetries and frames must take over.

Answers to Try This Next

These are brief answers and hints, not full solutions: they give the result and the key step, so that you can check your own work. Try each exercise properly before looking.

10.1 Newton's Cradle

1. After all collisions, a moving particle cannot be followed by a slower one. So if exactly one moves, it is A_n, and $\lambda m v = mu$, $\lambda m v^2 = mu^2$ are inconsistent for $\lambda \neq 1$. If only A_{n-1} and A_n move, the equations are those of a collision between m and λm, giving A_{n-1} the velocity $\frac{1-\lambda}{1+\lambda}u < 0$; but then it would hit A_{n-2}. Three moving particles are ruled out exactly as in part (ii). The two that move are A_0, backwards at $\frac{\lambda-1}{\lambda+1}u$, and A_n, forwards at $\frac{2u}{\lambda+1}$.
2. Equal marbles exchange velocities, so the speed u passes along to A_n. A_n then hits A_{n+1}: A_{n+1} leaves at $\frac{2u}{1+\lambda}$ and A_n rebounds at $\frac{\lambda-1}{\lambda+1}u$. This backward velocity passes back along the row to A_0, which leaves backwards at that speed.
3. "Two out at u" carries momentum $2mu$ and energy mu^2, the same as before. Another solution, with no further collisions: the last three balls move off at $\frac{u}{3}$, $\frac{u}{3}$ and $\frac{4u}{3}$ (check: total $2u$, squares total $2u^2$). Real cradles choose the first because the balls are not quite touching, so the collisions happen one pair at a time.
4. The speed passes along to A_n, which hits the lighter A_{n+1}: A_{n+1} leaves at $\frac{2u}{1+\lambda}$ and A_n follows more slowly at $\frac{1-\lambda}{1+\lambda}u$. Nothing else moves.

10.2 The Stacked-Ball Effect

1. The ball reaches the floor at $U = \sqrt{2gH}$ and rebounds at eU, rising e^2H. With the small ball on top: the large ball, moving up at eU, meets the small one moving down at U. Momentum and restitution give the small ball the upward speed $w = U\frac{Me(2+e)-m}{M+m}$. Its centre starts at $2R + r = 0.45$ and rises to 4.5, so $\frac{w^2}{2g} = 4.05$ and $\frac{w}{U} = \sqrt{\frac{4.05}{1.8}} = 1.5$. With $e = \frac{2}{3}$, $\frac{16}{9}M - m = 1.5(M + m)$, giving $M = 9m$.
2. Momentum and restitution give P_1 the speed $\frac{1+e}{1+\lambda}u$; each collision repeats this with the same mass ratio, so P_n moves at $\left(\frac{1+e}{1+\lambda}\right)^n u$. A particle with speed s is left with $\frac{1-\lambda e}{1+\lambda}s$ after hitting the next one, which is itself slowed to $\frac{(1+e)(1-\lambda e)}{(1+\lambda)^2}s$ by its own next collision. When $\lambda e < 1$, the first catches the second again only if $\frac{1-\lambda e}{1+\lambda} > \frac{(1+e)(1-\lambda e)}{(1+\lambda)^2}$, that is, $\lambda > e$; so if $e > \lambda$ each particle is hit once and hits once. If $\lambda e = 1$, each particle stops dead after hitting the next.
3. The bottom ball rebounds at U and meets the middle one (moving down at U):

relative speed $2U$, so the middle ball leaves at $3U$. The top ball meets it with relative speed $4U$ and leaves at $7U$, rising $49H$. In general $u_{j+1} = 2u_j + U$, so $u_j + U = 2^j U$.

4. Elastic collisions conserve kinetic energy. A's energy is $\frac{1}{2}m\left(\frac{2\lambda}{1+\lambda}\right)^{2n}u^2$, while the initial energy is $\frac{1}{2}\lambda^n m u^2$, and $\frac{4\lambda}{(1+\lambda)^2} < 1$, so A never takes more than was there. The heavy particles keep moving forwards, each with the fraction $\frac{\lambda-1}{\lambda+1}$ of its speed, and keep the rest of the energy.

10.3 Rockets and Guns

1. All at once: $Mu = nm(v-u)$, so $u = \frac{nmv}{M+nm}$. One at a time: the rth firing gives $\big(M+(n-r+1)m\big)(u_r - u_{r-1}) = mv$, so $u_n = \sum_{j=1}^{n} \frac{mv}{M+jm}$, and every term except the last exceeds $\frac{mv}{M+nm}$. So $u_n > u$.

2. Momentum: $M(1+bx)v = MV$. Then $(1+bx)\frac{\mathrm{d}x}{\mathrm{d}t} = V$, so $x + \frac{1}{2}bx^2 = Vt$ and $x = \frac{\sqrt{1+2bVt}-1}{b}$. With the force, $\frac{\mathrm{d}}{\mathrm{d}t}\big(M(1+bx)v\big) = Mf$ gives $(1+bx)v = ft + V$. Constant speed c needs $x = ct$ and $(1+bct)c = ft + V$, so $c = V$ and $bV^2 = f$. Otherwise $x + \frac{1}{2}bx^2 = \frac{1}{2}ft^2 + Vt$, so $x \sim t\sqrt{f/b}$ and $v \to \sqrt{f/b}$.

3. One stage: $v \ln 10 \approx 2.30v$. Two stages: $v \ln \frac{10}{5.5} + v \ln \frac{5}{0.5} \approx 0.60v + 2.30v = 2.90v$.

4. For $0 \le s \le x$, $\frac{1}{1+x} \le \frac{1}{1+s} \le 1$; integrate over $[0, x]$.

10.4 A Molecule and a Piston

1. The recurrence telescopes to $d_n = \frac{(v+u)d_1}{v+(2n-1)u}$, so $v_n d_n = \frac{(v+2nu)(v+u)d_1}{v+(2n-1)u}$, which exceeds $(v+u)d_1$ by $\frac{u(v+u)d_1}{v+(2n-1)u} \to 0$.

2. Replace u by $-u$: each bounce off the retreating piston costs $2u$, so $v_n = v - 2nu$ (while the particle can still catch the piston), $d_n = \frac{(v-u)d_1}{v-(2n-1)u}$, and $v_n d_n = \frac{(v-2nu)(v-u)d_1}{v-(2n-1)u}$, again close to $(v-u)d_1$.

3. With $v = u$, $v_n = (2n+1)u$. At the nth collision the momentum changes by $m(v_{n-1} + v_n) = 4nmu$, and $t_n = \frac{d_1}{un(n+1)}$, so the average force is $\frac{4mu^2n^2(n+1)}{d_1}$, of order n^3. The work in each cycle is about force $\times\, ut_n$, roughly $4nmu^2$, and these add up to infinity: pushing the piston all the way in would need infinite work, just as the molecule's energy $\frac{1}{2}m(2n+1)^2u^2$ grows without limit.

4. The invariant is $\oint p\,\mathrm{d}q = 2\int_0^h m\sqrt{2g(h-z)}\,\mathrm{d}z \propto h^{3/2}$, where h is the height of the bounce above the floor. So as the floor rises slowly, $h^{3/2}$ (equivalently v^3, or energy times period) stays nearly constant.

10.5 Infinitely Many Bounces in Finite Time

1. By the sine rule $M_1N_1 = \frac{OM_1 \sin 30^\circ}{\sin\theta}$, and successive legs shrink by $\rho = \frac{\sin(\theta-30^\circ)}{\sin\theta} < 1$. The total is $\frac{M_1N_1}{1-\rho} = \frac{OM_1}{2\left(\sin\theta - \sin(\theta-30^\circ)\right)} = \frac{OM_1}{4\sin 15^\circ \cos(\theta-15^\circ)}$.

2. The angles of triangle OM_1N_1 give $\varphi = \theta - \alpha$. The time ratio is $\frac{\rho}{k} = \frac{\sin(\theta-\alpha)\cos(\theta-\alpha)}{\sin\theta\cos\theta} = \frac{\sin(2\theta-2\alpha)}{\sin 2\theta}$, which is less than 1 exactly when $\alpha < \theta < 45^\circ + \frac{1}{2}\alpha$.

3. Reflecting the wedge in its walls unfolds the path into a straight line. Twelve copies of the wedge fit round O, and a straight line not through O crosses at most six of them. So the particle bounces finitely often and escapes, unless it is aimed straight at O.

4. It first falls for $t_0 = \sqrt{2/9.8} \approx 0.452\,\text{s}$. Each bounce lasts $2e^k t_0$, so the total time is $t_0\left(1 + \frac{2e}{1-e}\right) = 9t_0 \approx 4.07\,\text{s}$. The distance is $1 + 2\sum_{k\ge1} e^{2k} = 1 + \frac{2(0.64)}{0.36} \approx 4.56\,\text{m}$.

10.6 Two Bodies Become One

1. The string stays taut and the relative velocity is perpendicular to it, with constant size u, so $\theta = \frac{ut}{r}$. Momentum gives $my + MY = mut$. With $A = G - \frac{M}{m+M}\overrightarrow{AB}$ and $\overrightarrow{AB} = r(\cos\theta, -\sin\theta)$, $y = \frac{1}{m+M}\left(mut + Mr\sin\frac{ut}{r}\right)$. Then $\dot y = \frac{u}{m+M}\left(m + M\cos\frac{ut}{r}\right)$, which is negative when $\cos\frac{ut}{r} < -\frac{m}{M}$, possible if $M > m$.

2. Now $\ddot z = -\frac{3}{z^3}$, so $\dot z^2 = \frac{3}{z^2} - 2$ (using $z = 1, \dot z = -1$). With $w = z^2$, $\ddot w = 2\dot z^2 + 2z\ddot z = -4$, so $z^2 = 1 - 2t - 2t^2$. They collide when $t = \frac{\sqrt3 - 1}{2} \approx 0.366$.

3. $z^2 = 4t^2 - 2t + 1 = |\mathbf{p} + \mathbf{q}t|^2$ with $\mathbf{p} = (1,0)$ and $\mathbf{q} = (-1, \sqrt3)$: the line through $(1,0)$ in direction $(-1,\sqrt3)$, with $\theta = \tan^{-1}\frac{\sqrt3 t}{1-t}$. Then $z^2\dot\theta = |\mathbf{p}\times\mathbf{q}| = \sqrt3$. The repulsion is exactly the "centrifugal" term $\frac{h^2}{z^3}$ of free motion in a plane with angular momentum $h = \sqrt3$.

4. The stars are at distances $\frac{m_2 a}{m_1+m_2}$ and $\frac{m_1 a}{m_1+m_2}$ from the centre of mass. For the first, $\frac{Gm_1m_2}{a^2} = m_1\omega^2\frac{m_2 a}{m_1+m_2}$, so $\omega^2 = \frac{G(m_1+m_2)}{a^3}$ and $T = \frac{2\pi}{\omega}$.

10.7 Gravity Inside a Shell

1. Outside ($b > a$), the same integral as Stage 3 runs over s from $b-a$ to $b+a$ and gives $\frac{GM}{b^2}$. A solid sphere is a union of concentric shells, each acting as if its mass were at the centre.

2. With $r = R\cos^2\alpha$, $\frac{1}{r} - \frac{1}{R} = \frac{\tan^2\alpha}{R}$, so the speed is $\sqrt{2GM/R}\tan\alpha$. Then $\mathrm{d}t = \frac{2R\sin\alpha\cos\alpha\,\mathrm{d}\alpha}{\sqrt{2GM/R}\tan\alpha} = \sqrt{\frac{R^3}{2GM}}\,2\cos^2\alpha\,\mathrm{d}\alpha$; integrate.

3. Only the mass inside radius b, which is $M\frac{b^3}{a^3}$, pulls (by the shell theorem), giving $\frac{GMb}{a^3}$. This is simple harmonic motion with $\omega^2 = \frac{GM}{a^3}$, period $2\pi\sqrt{a^3/GM} \approx 84$

minutes with $a = 6.37 \times 10^6$ m and $GM = 3.99 \times 10^{14}\,\mathrm{m^3\,s^{-2}}$. A satellite at radius a has $\omega^2 a = \frac{GM}{a^2}$: the same ω.

4. No. With $\frac{1}{s^3}$ the integrand no longer simplifies to a constant, and the near side of the shell wins: the particle is pulled towards the nearest part of the shell. Only the inverse-square law gives exact cancellation (the area of a patch of the shell grows like s^2, exactly offsetting the force).

10.8 Three Stars in a Triangle

1. Each star is $\frac{a}{\sqrt{2}}$ from the centre. The two neighbours pull with total component $\frac{2\gamma m^2}{a^2}\cos 45^\circ = \frac{\sqrt{2}\gamma m^2}{a^2}$ towards the centre, and the opposite star with $\frac{\gamma m^2}{2a^2}$. Setting $\frac{\gamma m^2}{a^2}\left(\sqrt{2} + \frac{1}{2}\right) = m\omega^2 \frac{a}{\sqrt{2}}$ gives $\omega^2 = \frac{(4+\sqrt{2})\gamma m}{2a^3}$.

2. At the centre the pulls cancel, so the planet is in equilibrium. Along the line of the stars it is unstable: moving a distance x towards one star gives a net outward force of about $\frac{32\gamma M x}{a^3}$. Perpendicular to the orbital plane it is stable, with restoring force about $\frac{16\gamma M z}{a^3}$. (In the orbital plane, perpendicular to the line, gravity restores, but the rotation couples this motion to the unstable direction.)

3. The force on m_1 is $\frac{\gamma m_1}{a^3}\big(m_2(P_2 - P_1) + m_3(P_3 - P_1)\big)$. Since $m_1P_1 + m_2P_2 + m_3P_3 = (m_1 + m_2 + m_3)G$, this is $\frac{\gamma m_1(m_1+m_2+m_3)}{a^3}(G - P_1)$. Every star is pulled towards G in proportion to its distance from G, with the same constant, so all can rotate rigidly with $\omega^2 = \frac{\gamma(m_1+m_2+m_3)}{a^3}$.

4. The star k steps away is at distance $2R\sin\frac{k\pi}{n}$, and its pull has component $\sin\frac{k\pi}{n}$ towards the centre. So $m\omega^2 R = \frac{\gamma m^2}{4R^2}\sum_{k=1}^{n-1}\frac{1}{\sin(k\pi/n)}$, giving $\omega^2 = \frac{\gamma m}{4R^3}\sum_{k=1}^{n-1}\frac{1}{\sin(k\pi/n)}$. For $n = 3$, $R = \frac{a}{\sqrt{3}}$, this gives $\frac{3\gamma m}{a^3}$.

CHAPTER

11

Oscillations, Rotation and Moving Frames

Things that swing, things that spin, and things seen from somewhere that moves.

IN THIS CHAPTER

11.1	Resonance	STEP II 1991
11.2	Normal Modes	STEP III 2014
11.3	The Real Pendulum	STEP III 1998
11.4	The Tautochrone	STEP III 1994
11.5	Gravity in an Accelerating Train	STEP II 2021
11.6	Backspin	STEP III 1997
11.7	Simpson's Rule in a Rod	STEP II 1989
11.8	Tilted Equilibria	STEP III 2025
11.9	The Safety Parabola	STEP II 2014
11.10	A Ball That Comes Back Towards You	STEP II 2019
11.11	How Fielders Catch	STEP II 1998
11.12	The Ant on the Rubber Rope	STEP II 2018
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Motion, Seen Clearly

The career of a young theoretical physicist consists of treating the harmonic oscillator in ever-increasing levels of abstraction.

— Attributed to Sidney Coleman

The previous chapter used conservation laws to understand collisions and gravity. This one turns to the other great themes of a first course in dynamics: things that oscillate, things that rotate, and the art of describing motion from the most useful point of view.

Oscillation is everywhere. Pendulums and springs are the textbook examples, but the same mathematics describes the vibrations of molecules, the swaying of buildings, the circuits in a radio and, in quantum mechanics, the behaviour of light itself. The reason is that almost anything near a stable equilibrium behaves like a spring: pull it a little way and the restoring force is proportional to the displacement. So the simple harmonic oscillator is the first approximation to almost every physical system, and much of physics consists of understanding what happens when the approximation is not quite right.

Why this matters

Engineers must know the natural frequencies of every structure they build, to keep them away from the frequencies of wind, traffic and earthquakes. Chemists identify molecules by their vibrations. Timekeeping, from Huygens's pendulum clocks to quartz watches and atomic clocks, is the art of making a perfect oscillator. Rotation governs everything from the spin of a snooker ball to the stability of satellites. And the question of which frames of reference are "natural" led, through the equivalence of gravity and acceleration, to Einstein's general relativity.

The chapter ends with six questions grouped as *surprising trajectories*: projectiles, ants, fielders and snowploughs whose behaviour is not what intuition predicts. They are short, elegant problems, and each has a proper mathematical idea at its centre: envelopes, monotonicity, feedback, divergence, linear differential equations and the difference between a parabola and a catenary.

What you will meet at university

A first-year dynamics course covers simple, damped and forced harmonic motion, resonance, and coupled oscillators and their normal modes (where linear algebra and mechanics meet). It treats rigid bodies, moments of inertia and rolling, and motion in accelerating and rotating frames, with the centrifugal and Coriolis forces. Later courses develop all of this into Lagrangian and Hamiltonian mechanics, where the

choice of coordinates becomes a central theme, and into the theory of waves, which are oscillations spread out in space. Nonlinear oscillations, like the large-amplitude pendulum, lead to elliptic functions and to the dynamical systems of Chapter 8.

The questions that run through this chapter

Three big questions

1. **When is motion simple?** When does a system oscillate with a single, fixed frequency, and what happens when it does not?
2. **What is the right point of view?** Which coordinates, which variables and which frame of reference make a problem easy?
3. **Can intuition be trusted?** Where does everyday intuition about motion go wrong, and how does mathematics correct it?

The chapter begins with oscillations: resonance (Section 11.1), normal modes, the real pendulum and the tautochrone. It then turns to frames and rigid bodies: an accelerating train, a spinning marble, a rod that behaves like three particles, and a solid that can rest at three different tilts. The last six sections are the surprising trajectories.

Ideas in this chapter

§	Topic	Idea
11.1	Forced oscillations	Resonance and beats
11.2	Coupled oscillators	Normal modes are eigenvectors
11.3	The real pendulum	Elliptic integrals; the AGM
11.4	The tautochrone	Simple harmonic in arc length
11.5	Moving frames	Fictitious forces; equivalence principle
11.6	Rolling and sliding	A conserved angular momentum
11.7	Moments of inertia	Dynamical equivalence; Simpson's rule
11.8	Stability	Normals through the centre of mass
11.9	Projectiles	The envelope of trajectories
11.10	Projectiles	Distance need not increase
11.11	Catching	A feedback rule
11.12	Stretching ropes	Fractions and the harmonic series
11.13	Snowploughs	Time as a function of distance
11.14	Suspension bridges	Parabola, not catenary

Two sections, Section 11.6 and Section 11.7, use rotational dynamics and moments of inertia, which were removed from the STEP specification in 2019. They are included because they are excellent questions; each gives the facts it needs.

How to read this chapter

Before you start any calculation, ask which variable the problem wants. Arc length for the tautochrone, $x \pm y$ for coupled springs, the fraction of the rope for the ant, time as a function of distance for the snowplough: in each case, the right choice turns a difficult problem into a familiar one.

11.1 Resonance

The question

A particle is attached to one end B of a light elastic string of unstretched length a. Initially the other end A is at rest and the particle hangs at rest at a distance $a + c$ vertically below A. At time $t = 0$, the end A is forced to oscillate vertically, its downwards displacement at time t being $b \sin pt$. Let $x(t)$ be the downwards displacement of the particle at time t from its initial equilibrium position. Show that, while the string remains taut, $x(t)$ satisfies

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -n^2(x - b\sin pt),$$

where $n^2 = g/c$, and that if $0 < p < n$, $x(t)$ is given by

$$x(t) = \frac{bn}{n^2 - p^2}(n\sin pt - p\sin nt).$$

Write down a necessary and sufficient condition that the string remains taut throughout the subsequent motion, and show that it is satisfied if $pb < (n - p)c$.

The hook

Push a child on a swing at just the right moments and small pushes build into large swings. Push at the wrong rhythm and nothing much happens. Every oscillating system has a *natural frequency*, and driving it at that frequency produces a response out of all proportion to the effort. This is *resonance*. It is why a singer can shatter a wine glass, why soldiers break step when crossing a bridge, and why a radio can pick out one station from thousands.

This question shakes the top of an elastic string and asks when the string stays taut. The answer contains the factor $\frac{1}{n-p}$, which blows up as the driving frequency p approaches the natural frequency n: a gentle shake near resonance is enough to make the string go slack.

Before you start

Ask yourself The particle hangs in equilibrium with extension c. If the string has modulus λ, what is the relation between λ, a, c, m and g? What is the extension when the top has moved down by $b\sin pt$ and the particle by x?

In equilibrium, tension equals weight: $\frac{\lambda c}{a} = mg$. When the top is displaced down by $b\sin pt$ and the particle by x, the string's length increases by $x - b\sin pt$, so the extension is $c + x - b\sin pt$.

Working it through

STAGE 1 The equation of motion

Taking downwards as positive, the forces on the particle are its weight mg and the tension $\frac{\lambda}{a}(c + x - b\sin pt)$ upwards:

$$m\ddot{x} = mg - \frac{\lambda}{a}(c + x - b\sin pt) = -\frac{\lambda}{a}(x - b\sin pt),$$

using $\frac{\lambda c}{a} = mg$. Since $\frac{\lambda}{ma} = \frac{g}{c} = n^2$, this is $\ddot{x} = -n^2(x - b\sin pt)$. ■

The particle is pulled towards the moving point $b\sin pt$ with a spring force. Without the driving, it would oscillate with angular frequency n, its natural frequency.

STAGE 2 Solving

Ask yourself The equation is $\ddot{x} + n^2x = n^2b\sin pt$. Try a particular solution of the form $x = C\sin pt$. Then add the general solution of $\ddot{x} + n^2x = 0$. Which initial conditions does the particle satisfy?

Substituting $x = C\sin pt$ gives $(-p^2 + n^2)C = n^2b$, so $C = \frac{n^2b}{n^2 - p^2}$ (we need $p \neq n$). The complementary function is $D\cos nt + E\sin nt$. The particle starts at rest in equilibrium: $x(0) = 0$ forces $D = 0$, and $\dot{x}(0) = 0$ gives $Cp + En = 0$, so $E = -\frac{nbp}{n^2 - p^2}$. Altogether,

$$x(t) = \frac{bn}{n^2 - p^2}(n\sin pt - p\sin nt). \quad ■$$

STAGE 3 Staying taut

Ask yourself The string is taut exactly when its extension is positive. Write the extension $c + x - b\sin pt$ in terms of t.

Subtracting $b \sin pt$ from $x(t)$,

$$\begin{aligned} x - b\sin pt &= \frac{b}{n^2 - p^2}\left(n^2 \sin pt - np \sin nt - (n^2 - p^2)\sin pt\right) \\ &= \frac{bp}{n^2 - p^2}(p \sin pt - n \sin nt). \end{aligned}$$

So the string remains taut throughout if and only if

$$c + \frac{bp}{n^2 - p^2}(p \sin pt - n \sin nt) \geq 0 \qquad \text{for all } t \geq 0.$$

Ask yourself How negative can $p \sin pt - n \sin nt$ possibly be?

Since $|\sin| \leq 1$, we have $p \sin pt - n \sin nt \geq -(p + n)$. So the extension is at least

$$c - \frac{bp(n + p)}{n^2 - p^2} = c - \frac{bp}{n - p},$$

which is positive if $pb < (n - p)c$. ■

Pause and try What happens to this condition as p approaches n? How small must the shaking amplitude b be when $p = 0.9n$? When $p = 0.99n$?

Looking back

- **Measure from equilibrium.** Using the equilibrium extension c removed gravity from the equation.
- **Particular plus complementary.** The forced response at frequency p plus the free oscillation at frequency n, with constants fixed by the initial conditions.
- **Bound, do not solve.** We did not find the minimum extension exactly; a crude bound on the sines was enough for a sufficient condition.

What's really going on

The resonance limit

Let $p \to n$ in the solution. Both the numerator and the denominator tend to zero, and a limit calculation (or l'Hôpital's rule, differentiating with respect to p) gives

$$x(t) \to \frac{b}{2}(\sin nt - nt \cos nt).$$

The term $nt \cos nt$ grows without limit: driving exactly at the natural frequency makes the amplitude increase linearly in time, each push adding to the motion in step with

it. Just below resonance, the two frequencies drift in and out of step, producing *beats*: the amplitude swells and dies away with the slow frequency $n - p$ (Figure 11.1.1). The same beats are heard when two guitar strings are almost in tune.

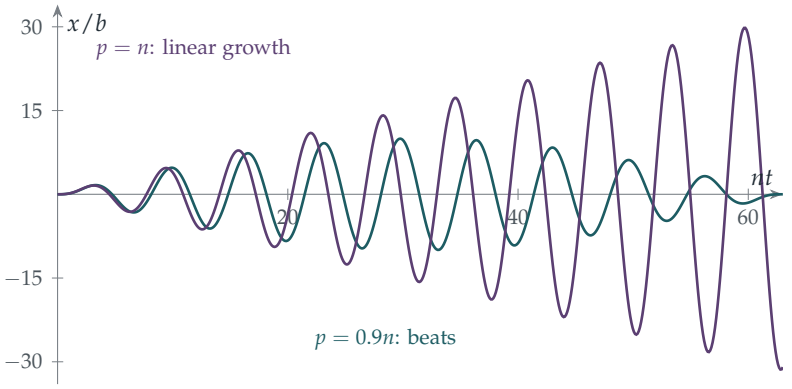


Figure 11.1.1 The response of the particle (ignoring slackness), in units of the shaking amplitude b. With $p = 0.9n$ (teal) the response swells to about ten times b and dies away again, with a beat period of $\frac{2\pi}{n-p}$. At exact resonance, $p = n$ (plum), the amplitude grows without limit.

Damping and the resonance curve

Real systems lose energy, through friction or air resistance, and that limits the growth. With damping, the steady response to a driving force at frequency p has an amplitude that peaks near the natural frequency; the sharper the peak, the less damping there is. Engineers describe this by the *quality factor* Q: roughly, the number of oscillations a system makes before losing most of its energy. A wine glass has Q in the thousands, which is why a sustained note at exactly its natural frequency can break it. A car's suspension is deliberately given a low Q, so that bumps do not set it bouncing.

Bridges that move

In 1831 a column of soldiers marching in step over the Broughton Suspension Bridge near Manchester set it vibrating until it collapsed; since then, troops have been ordered to break step on bridges. The London Millennium Bridge, opened in June 2000, began to sway sideways as soon as crowds crossed it: pedestrians unconsciously adjusted their steps to the sway, feeding it energy in step, and the bridge closed for two years while dampers were fitted. The famous collapse of the Tacoma Narrows Bridge in 1940 is often described as resonance, but it was a different phenomenon, *aeroelastic flutter*, in which the wind's force depends on the bridge's own motion. Its analysis needs the ideas of self-excited oscillation and limit cycles (Section 8.4).

The thinking skill

Superpose a forced response and a free response

For a linear equation driven by a periodic force, the solution is the response at the driving frequency plus a free oscillation at the natural frequency, chosen to fit the starting conditions.

Linearity makes this work: solutions can be added. At university, the same principle, applied to sums of many driving frequencies, leads to Fourier analysis, transfer functions in engineering, and Green's functions in physics.

Try this next

1 **A falling lift.** (From STEP II, 1996.) A particle hangs from the ceiling of a lift by an elastic string, extended by a. The lift starts to descend with constant acceleration $f < \frac{g}{2}$. Show that the extension y of the string satisfies $\ddot{y} + \frac{g}{a}y = g - f$, that the string never becomes slack, and that the oscillation has amplitude $\frac{af}{g}$. After time T the lift stops accelerating. Show that the amplitude becomes $\frac{2af}{g}\left|\sin\frac{1}{2}\omega T\right|$, where $\omega^2 = \frac{g}{a}$.

2 **At resonance.** Solve $\ddot{x} + n^2x = n^2 b\sin nt$ with $x(0) = \dot{x}(0) = 0$ directly, by trying a particular solution $x = Ct\cos nt$. Check that it agrees with the limit above.

3 **Beats.** Show that $n\sin pt - p\sin nt$ can be written as a sum of two terms, one of which is approximately $2n\cos\frac{(n+p)t}{2}\sin\frac{(p-n)t}{2}$ when p is close to n. Hence explain the beat period $\frac{2\pi}{n-p}$.

4 **Damped resonance.** For $\ddot{x} + 2k\dot{x} + n^2x = F\cos pt$ with small $k > 0$, find the steady-state amplitude, and show that it is greatest when $p^2 = n^2 - 2k^2$.

Further reading

- **A. P. French, *Vibrations and Waves*** (MIT Introductory Physics Series). Chapter 4 on forced oscillations and resonance is the classic account.
- **Steven Strogatz, *Sync*** (Penguin). How things fall into step, from fireflies to the Millennium Bridge.
- **Henry Petroski, *To Engineer Is Human*** (Vintage). Engineering failures, including the bridges in the history box.

11.2 Normal Modes

The question

Two particles X and Y, of equal mass m, lie on a smooth horizontal table and are connected by a light elastic spring of natural length a and modulus of elasticity λ. Two more springs, identical to the first, connect X to a point P on the table and Y to a point Q on the table. The distance between P and Q is $3a$.

Initially, the particles are held so that $XP = a$, $YQ = \frac{1}{2}a$, and $PXYQ$ is a straight line. The particles are then released.

At time t, the particle X is a distance $a + x$ from P and the particle Y is a distance $a + y$ from Q. Show that

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{\lambda}{a}(2x + y)$$

and find a similar expression involving $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2}$. Deduce that

$$x - y = A\cos\omega t + B\sin\omega t$$

where A and B are constants to be determined and $ma\omega^2 = \lambda$. Find a similar expression for $x + y$.

Show that Y will never return to its initial position.

The hook

One mass on a spring bounces up and down in a simple, regular way. Join two masses with springs and the motion looks complicated: each mass jerks about irregularly, energy sloshing from one to the other. But hidden inside the complication are two perfectly simple motions, the *normal modes*, in which the masses move together or exactly opposite, each mode oscillating at its own frequency. Every motion of the system is a combination of the two.

This question finds the normal modes of two masses between three springs, and then proves something striking: started from rest with one mass displaced, the system *never* returns to its starting position. The reason is that the ratio of the two frequencies is $\sqrt{3}$, and $\sqrt{3}$ is irrational.

Before you start

Ask yourself With X at distance $a+x$ from P and Y at distance $a+y$ from Q, how long is the middle spring XY? Which springs are stretched and which compressed when x and y are positive?

The three lengths add up to $3a$, so $XY = 3a-(a+x)-(a+y) = a-x-y$. When $x, y > 0$, the outer springs are stretched (by x and y) and the middle one is compressed (by $x+y$).

Working it through

STAGE 1 The equations of motion

Measure x in the direction from P towards Q. The spring PX has extension x, so its tension $\frac{\lambda x}{a}$ pulls X back towards P. The spring XY has extension $-(x+y)$, so its tension $-\frac{\lambda(x+y)}{a}$ pulls X towards Y (a negative tension is a push). So

$$m\ddot{x} = -\frac{\lambda x}{a} - \frac{\lambda(x+y)}{a} = -\frac{\lambda}{a}(2x+y). \qquad \blacksquare$$

By symmetry (swap the roles of X and Y, P and Q),

$$m\ddot{y} = -\frac{\lambda}{a}(2y+x).$$

STAGE 2 Decoupling

Ask yourself Each equation involves both x and y. Which combinations of the two equations involve only one combination of x and y?

Subtracting and adding,

$$m(\ddot{x}-\ddot{y}) = -\frac{\lambda}{a}(x-y), \qquad m(\ddot{x}+\ddot{y}) = -\frac{3\lambda}{a}(x+y).$$

Each is the equation of simple harmonic motion: $x-y$ oscillates with $\omega^2 = \frac{\lambda}{ma}$, and $x+y$ with $3\omega^2$.

STAGE 3 The initial conditions

Initially $XP = a$ and $YQ = \frac{1}{2}a$, so $x = 0$ and $y = -\frac{1}{2}a$, and both particles are at rest. So $x-y = \frac{1}{2}a$ and $x+y = -\frac{1}{2}a$ at $t = 0$, with zero rates of change, giving

$$x-y = \tfrac{1}{2}a\cos\omega t, \qquad x+y = -\tfrac{1}{2}a\cos\sqrt{3}\,\omega t.$$

So $A = \frac{1}{2}a$ and $B = 0$. $\blacksquare$

STAGE 4 **Never returning**

Ask yourself Write y in terms of t. For Y to be back at its initial position, $y = -\frac{1}{2}a$. What must both cosines be?

Subtracting, $y = \frac{1}{2}\big((x+y)-(x-y)\big) = -\frac{1}{4}a\big(\cos\sqrt{3}\,\omega t + \cos\omega t\big)$. For $y = -\frac{1}{2}a$ we need $\cos\sqrt{3}\,\omega t + \cos\omega t = 2$, and since each cosine is at most 1, both must equal 1:

$$\omega t = 2\pi k, \qquad \sqrt{3}\,\omega t = 2\pi j$$

for integers k and j. For $t > 0$, $k \geq 1$, and dividing, $\sqrt{3} = \frac{j}{k}$, which is impossible because $\sqrt{3}$ is irrational. So Y never returns to its initial position. ■

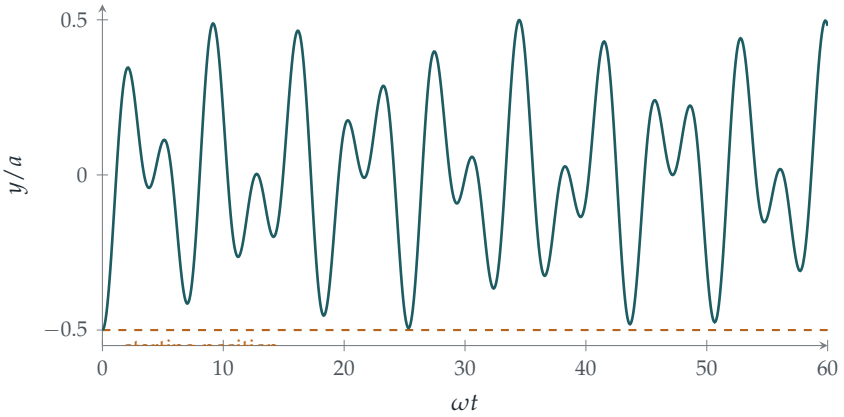


Figure 11.2.1 The displacement y of the particle Y. It is the sum of two oscillations whose frequencies are in the ratio $\sqrt{3}:1$, so the pattern never repeats. Y comes close to its starting position (dashed) again and again, for instance near $\omega t \approx 25$ (where $\omega t \approx 8\pi$ and $\sqrt{3}\,\omega t \approx 14\pi$), but never reaches it exactly.

Looking back

- **Find the combinations that decouple.** $x - y$ and $x + y$ each satisfied an equation of their own.
- **Symmetry suggests them.** The system is symmetric under swapping the two sides, so symmetric and antisymmetric combinations are natural.
- **Irrationality has physical consequences.** The motion is not periodic because a frequency ratio is irrational.

What's really going on

Normal modes are eigenvectors

Write the equations as a matrix equation:

$$m\begin{pmatrix}\ddot{x}\\ \ddot{y}\end{pmatrix} = -\frac{\lambda}{a}\begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}.$$

The combinations that decouple are the eigenvectors of the matrix: $(1,-1)$ with eigenvalue 1, and $(1,1)$ with eigenvalue 3 (Section 7.2). In the first mode, $x = -y$: both particles move the same way along the line (remember that x and y are measured from opposite ends), and the middle spring keeps its length. In the second, $x = y$: they move in opposite directions, squeezing and stretching the middle spring, which makes the frequency higher. The squares of the frequencies are the eigenvalues, multiplied by $\frac{\lambda}{ma}$.

This works for any number of masses and springs. A system of N coupled oscillators has N normal modes, given by the eigenvectors of a symmetric matrix, and the spectral theorem guarantees that they always exist and are perpendicular. Molecules vibrate in normal modes, which is how chemists identify them by the infrared light they absorb. Buildings and bridges have normal modes, which engineers must keep away from the frequencies of wind and earthquakes. And as the number of masses tends to infinity, the normal modes become the harmonics of a vibrating string, the starting point of Fourier analysis.

Coming close without arriving

Although Y never returns exactly, it comes as close as you like. The reason is a theorem of Kronecker: if α is irrational, the multiples $\alpha, 2\alpha, 3\alpha, \ldots$, taken modulo 1, come arbitrarily close to every number in $[0,1]$. So we can find times when ωt is a multiple of 2π and $\sqrt{3}\,\omega t$ is within any tolerance of one. The motion is *quasi-periodic*: it winds round a torus (one angle for each mode) without ever closing up. The same distinction, between rational and irrational frequency ratios, is at the heart of the KAM theorem on the stability of the solar system.

The thinking skill

Change variables to decouple equations

When equations are coupled, look for combinations of the variables that evolve independently. In linear problems, these are the eigenvectors.

This is the same idea as the centre of mass and relative coordinates of Section 10.6. At university it becomes diagonalisation: choose a basis in which the matrix is diagonal, and a system of coupled equations falls apart into independent ones.

Try this next

1 **Energy.** Show that the total energy $\frac{1}{2}m(\dot{x}^2+\dot{y}^2)+\frac{\lambda}{2a}\left(x^2+y^2+(x+y)^2\right)$ is constant, and that it splits into a sum of two terms, one for each normal mode, each of which is separately constant.

2 **Pure modes.** How should the particles be started so that the motion is periodic, with period $\frac{2\pi}{\omega}$? With period $\frac{2\pi}{\sqrt{3}\,\omega}$?

3 **A stiffer middle spring.** If the middle spring has modulus μ instead of λ, find the two normal-mode frequencies. For which values of μ/λ is every motion periodic?

4 **Three masses.** Three equal masses are connected in a line by four identical springs between two fixed points. Find the matrix and its eigenvalues $(2-\sqrt{2},\,2,\,2+\sqrt{2})$, and describe the three normal modes.

Further reading

- **A. P. French, *Vibrations and Waves*** (MIT Introductory Physics Series), Chapter 5, on coupled oscillators and normal modes.
- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge), the chapter on eigenvalues, which includes springs and masses as a main example.
- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press), Chapter 8, on quasi-periodic motion on a torus.

11.3 The Real Pendulum

The question

Consider a simple pendulum of length l and angular displacement θ, which is **not** assumed to be small. Show that

$$\frac{1}{2}l\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2 = g(\cos\theta - \cos\gamma),$$

where γ is the maximum value of θ. Show also that the period P is given by

$$P = 2\sqrt{\frac{l}{g}}\int_0^{\gamma}\left(\sin^2(\gamma/2) - \sin^2(\theta/2)\right)^{-\frac{1}{2}}\,\mathrm{d}\theta.$$

By using the substitution $\sin(\theta/2) = \sin(\gamma/2)\sin\phi$, and then finding an approximate expression for the integrand using the binomial expansion, show that for small values of γ the period is approximately

$$2\pi\sqrt{\frac{l}{g}}\left(1 + \frac{\gamma^2}{16}\right).$$

The hook

Every physics student learns that a pendulum's period is $2\pi\sqrt{l/g}$, independent of the size of the swing. Galileo believed this, supposedly after timing a swinging lamp in Pisa cathedral against his pulse, and it is the reason pendulum clocks work. But it is not quite true. The formula comes from the approximation $\sin\theta \approx \theta$, and a pendulum swinging through a wide arc takes longer. At 90° the period is 18% longer; a pendulum released from almost upside down can take as long as you like.

This question finds the exact period as an integral, and then the first correction for moderate swings. The integral cannot be evaluated in elementary functions: it is an *elliptic integral*, and the story of how such integrals were tamed runs from Gauss's arithmetic–geometric mean to the theory of elliptic curves.

Before you start

Ask yourself The bob moves on a circle of radius l. What are its speed and its height (relative to the lowest point) in terms of θ and $\dot\theta$?

The speed is $l\dot\theta$, and the height above the lowest point is $l(1-\cos\theta)$. Since the string does no work (its tension is always perpendicular to the motion), energy is conserved.

Working it through

STAGE 1 Energy

At the extreme angle γ the bob is momentarily at rest. Conservation of energy gives

$$\tfrac{1}{2}ml^2\dot\theta^2 + mgl(1-\cos\theta) = mgl(1-\cos\gamma),$$

and dividing by ml, $\tfrac{1}{2}l\dot\theta^2 = g(\cos\theta - \cos\gamma)$. ■

STAGE 2 The period as an integral

Ask yourself How can you write $\cos\theta - \cos\gamma$ in terms of half-angles? And how much of the period is spent going from $\theta = 0$ to $\theta = \gamma$?

Using $\cos\theta = 1 - 2\sin^2\frac{\theta}{2}$, we have $\cos\theta - \cos\gamma = 2\left(\sin^2\frac{\gamma}{2} - \sin^2\frac{\theta}{2}\right)$, so

$$\dot\theta = 2\sqrt{\frac{g}{l}}\left(\sin^2\frac{\gamma}{2} - \sin^2\frac{\theta}{2}\right)^{1/2}.$$

By symmetry the swing from 0 to γ takes a quarter of the period, so

$$P = 4\int_0^\gamma \frac{\mathrm{d}\theta}{\dot\theta} = 2\sqrt{\frac{l}{g}}\int_0^\gamma \left(\sin^2\frac{\gamma}{2} - \sin^2\frac{\theta}{2}\right)^{-1/2}\mathrm{d}\theta. \qquad ■$$

STAGE 3 Removing the singularity

Ask yourself The integrand becomes infinite at $\theta = \gamma$. Write $k = \sin\frac{\gamma}{2}$ and substitute $\sin\frac{\theta}{2} = k\sin\phi$. What range of ϕ corresponds to $0 \le \theta \le \gamma$? What happens to the integrand?

As θ goes from 0 to γ, ϕ goes from 0 to $\frac{\pi}{2}$. Differentiating, $\frac{1}{2}\cos\frac{\theta}{2}\,\mathrm{d}\theta = k\cos\phi\,\mathrm{d}\phi$, so

$$\mathrm{d}\theta = \frac{2k\cos\phi\,\mathrm{d}\phi}{\cos\frac{\theta}{2}} = \frac{2k\cos\phi\,\mathrm{d}\phi}{\sqrt{1-k^2\sin^2\phi}}.$$

Meanwhile $\sin^2\frac{\gamma}{2} - \sin^2\frac{\theta}{2} = k^2(1-\sin^2\phi) = k^2\cos^2\phi$. The awkward factors cancel:

$$P = 2\sqrt{\frac{l}{g}}\int_0^{\pi/2} \frac{1}{k\cos\phi}\cdot\frac{2k\cos\phi}{\sqrt{1-k^2\sin^2\phi}}\,\mathrm{d}\phi = 4\sqrt{\frac{l}{g}}\int_0^{\pi/2}\frac{\mathrm{d}\phi}{\sqrt{1-k^2\sin^2\phi}}.$$

The new integrand is perfectly well behaved, since $k < 1$.

STAGE 4 **The first correction**

Ask yourself For small k, expand $(1-k^2\sin^2\phi)^{-1/2}$ by the binomial theorem. What is $\int_0^{\pi/2}\sin^2\phi\,\mathrm{d}\phi$?

To first order, $(1-k^2\sin^2\phi)^{-1/2} \approx 1 + \frac{1}{2}k^2\sin^2\phi$, and $\int_0^{\pi/2}\sin^2\phi\,\mathrm{d}\phi = \frac{\pi}{4}$. So

$$P \approx 4\sqrt{\frac{l}{g}}\left(\frac{\pi}{2} + \frac{k^2}{2}\cdot\frac{\pi}{4}\right) = 2\pi\sqrt{\frac{l}{g}}\left(1+\frac{k^2}{4}\right).$$

For small γ, $k = \sin\frac{\gamma}{2} \approx \frac{\gamma}{2}$, so $k^2 \approx \frac{\gamma^2}{4}$ and

$$P \approx 2\pi\sqrt{\frac{l}{g}}\left(1+\frac{\gamma^2}{16}\right). \qquad \blacksquare$$

For a swing of 20° ($\gamma \approx 0.35$) the correction is less than 1%; at 60° it is about 7% (Figure 11.3.1).

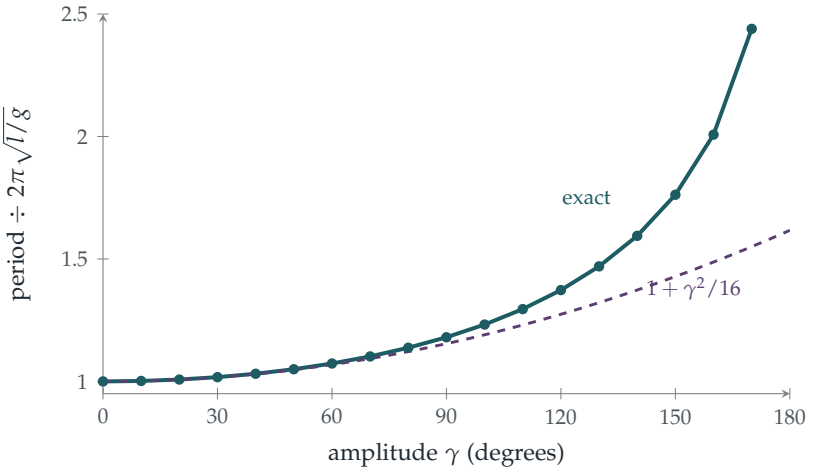


Figure 11.3.1 The exact period of a pendulum (teal), relative to the small-angle value, against its amplitude, compared with the approximation $1+\gamma^2/16$ (plum, dashed). The approximation is excellent up to about 60°. As $\gamma \to 180°$ the true period tends to infinity: a pendulum balanced upside down takes for ever to fall.

Looking back

- **Energy gives a first integral.** A second-order equation became a first-order one, and the period became an integral.
- **A substitution tames a singularity.** The integrand blew up at the turning point; the substitution $\sin\frac{\theta}{2} = k\sin\phi$ removed the problem.
- **Exact first, then approximate.** We found an exact formula and expanded it, which also tells us how good the approximation is.

What's really going on

Elliptic integrals

The integral

$$K(k) = \int_0^{\pi/2} \frac{\mathrm{d}\phi}{\sqrt{1 - k^2\sin^2\phi}}$$

is the *complete elliptic integral of the first kind*, so called because a close relative gives the perimeter of an ellipse. The exact period is $P = 4\sqrt{l/g}\,K\bigl(\sin\frac{\gamma}{2}\bigr)$. Expanding the integrand fully and integrating term by term using the Wallis integrals $\int_0^{\pi/2}\sin^{2n}\phi\,\mathrm{d}\phi$ of Section 4.4, one finds

$$P = 2\pi\sqrt{\frac{l}{g}}\Bigl(1 + \bigl(\tfrac{1}{2}\bigr)^2k^2 + \bigl(\tfrac{1\cdot3}{2\cdot4}\bigr)^2k^4 + \bigl(\tfrac{1\cdot3\cdot5}{2\cdot4\cdot6}\bigr)^2k^6 + \cdots\Bigr).$$

No finite combination of familiar functions gives $K(k)$; it is a genuinely new function.

Gauss's remarkable formula

There is a far better way to compute K. In Section 2.6 you met the *arithmetic–geometric mean*: start with a and b, and repeatedly replace them by their arithmetic and geometric means. Gauss discovered in 1799 that

$$K(k) = \frac{\pi}{2\,\mathrm{AGM}\bigl(1, \sqrt{1-k^2}\bigr)},$$

so the period of a pendulum with amplitude γ is exactly

$$P = \frac{2\pi\sqrt{l/g}}{\mathrm{AGM}\bigl(1, \cos\frac{\gamma}{2}\bigr)}.$$

The AGM converges so fast that, for swings of up to about 160°, three or four steps give the period to ten significant figures, and the values in Figure 11.3.1 were computed this way. Nearer 180° the two starting values, 1 and $\cos\frac{\gamma}{2}$, are far apart and it takes a step or two more, but five steps suffice even at 179°. Gauss wrote in his diary that this discovery "opens an entirely new field of analysis", and he was right: inverting

elliptic integrals led Abel and Jacobi to *elliptic functions*, and those led to *elliptic curves*, which are now central to number theory (they were the key to the proof of Fermat's Last Theorem) and to the cryptography that secures the internet.

Galileo's lamp and Huygens's clock

Galileo's claim that a pendulum's period does not depend on its amplitude, made around 1602, was a good approximation for the small swings he studied. Christiaan Huygens realised that it was not exact, and in 1659 found the curve along which the period really is independent of amplitude: the cycloid, the subject of Section 11.4. His pendulum clocks, made from 1656 onwards, improved timekeeping from errors of about a quarter of an hour a day to about fifteen seconds. Marine chronometers, which solved the problem of finding longitude at sea, later abandoned the pendulum altogether, but pendulum clocks remained the most accurate timekeepers in the world until the 1920s.

The thinking skill

From an exact integral to a series approximation

When a quantity is given by an integral that cannot be evaluated, transform it into a well-behaved form, then expand the integrand and integrate term by term.

This is how physicists find corrections to idealised results, and it is the start of *perturbation theory*. At university you will learn when term-by-term integration is justified (uniform convergence), and how to estimate the error made by stopping.

Try this next

1 **A tethered particle.** (From STEP III, 2015.) A particle of mass m on a smooth horizontal rail is attached to a peg at distance a from the rail by an elastic string of natural length a and modulus λ. Starting with the string just taut, it is given speed v along the rail. Show that $\dot{x}^2 = v^2 - k^2\left(\sqrt{x^2+a^2}-a\right)^2$, where $k^2 = \frac{\lambda}{ma}$, find the greatest displacement, and show that for $v \ll ka$ the period is approximately $\sqrt{\frac{32a}{kv}}\int_0^1 \frac{\mathrm{d}u}{\sqrt{1-u^4}}$. Why does the period now *depend* on v, even for small oscillations?

2 **The next term.** Carry the binomial expansion one term further and show that $P \approx 2\pi\sqrt{\frac{l}{g}}\left(1+\frac{\gamma^2}{16}+\frac{11\gamma^4}{3072}\right)$.

3 **The AGM by hand.** Use three steps of the AGM to estimate the period of a pendulum with amplitude 90°, and compare with the value 1.1803 in the figure.

4 Clocks. A pendulum clock is regulated to keep perfect time with an amplitude of 4°. If its amplitude increases to 5°, how many seconds a day does it lose?

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 4, which treats the large-amplitude pendulum.
- **Jonathan Borwein & Peter Borwein, *Pi and the AGM*** (Wiley). Gauss's discovery and its uses, including the fastest ways to compute π.
- **Dava Sobel, *Longitude*** (Fourth Estate). The story of timekeeping at sea, beyond the pendulum.

11.4 The Tautochrone

The question

A smooth, axially symmetric bowl has its vertical cross-sections determined by $s = 2\sqrt{ky}$, where s is the arc-length measured from its lowest point V, and y is the height above V. A particle is released from rest at a point on the surface at a height h above V. Explain why

$$\left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2 + 2gy$$

is constant.

Show that the time for the particle to reach V is

$$\pi\sqrt{\frac{k}{2g}}.$$

Two elastic particles of mass m and αm, where $\alpha < 1$, are released simultaneously from opposite sides of the bowl at heights $\alpha^2 h$ and h respectively. If the coefficient of restitution between the particles is α, describe the subsequent motion.

The hook

Release two marbles at the same moment inside a bowl, one from near the rim and one from just above the bottom. Which reaches the bottom first? In an ordinary bowl, the one starting lower usually wins. But there is a bowl in which they always arrive together, wherever they start. It is the *tautochrone* (Greek for "same time"), and its cross-section is a *cycloid*, the curve traced by a point on the rim of a rolling wheel.

Huygens discovered this in 1659 and used it to design a pendulum clock whose period would not depend on its swing, correcting the flaw found in Section 11.3. This question proves the tautochrone property in a few lines, by choosing the right variable: distance along the curve, rather than horizontal position.

Before you start

Ask yourself The bowl is smooth. Which forces act on the particle, and which of them does work?

Gravity and the normal reaction. The normal reaction is perpendicular to the surface, and so to the motion, so it does no work. Only gravity does work, and energy is conserved.

Working it through

STAGE 1 Energy along the curve

The speed of the particle along the surface is $\left|\frac{\mathrm{d}s}{\mathrm{d}t}\right|$, since s is arc length. Conservation of energy gives $\frac{1}{2}m\dot{s}^2 + mgy = \text{constant}$, so $\dot{s}^2 + 2gy$ is constant. ■

STAGE 2 Simple harmonic motion in s

Ask yourself Use $s = 2\sqrt{ky}$ to replace y by a function of s. What kind of motion does the resulting equation describe?

Since $y = \frac{s^2}{4k}$, the energy equation becomes

$$\dot{s}^2 + \frac{g}{2k}s^2 = \text{constant} = \frac{g}{2k}s_0^2,$$

where $s_0 = 2\sqrt{kh}$ is the starting arc length (the particle starts at rest). Differentiating, $2\dot{s}\ddot{s} + \frac{g}{k}s\dot{s} = 0$, so

$$\ddot{s} = -\frac{g}{2k}\,s.$$

This is simple harmonic motion in the variable s, with $\omega^2 = \frac{g}{2k}$, whatever the amplitude: $s = s_0 \cos \omega t$.

STAGE 3 The time to the bottom

The particle reaches V (where $s = 0$) after a quarter of a period:

$$\frac{\pi}{2\omega} = \frac{\pi}{2}\sqrt{\frac{2k}{g}} = \pi\sqrt{\frac{k}{2g}},$$

which does not depend on h. ■

Pause and try Why does this work? The "restoring force" along the curve is g times the slope, $g\frac{\mathrm{d}y}{\mathrm{d}s} = \frac{g}{2k}s$: exactly proportional to the distance from the

bottom, measured along the curve. That is the defining property of simple harmonic motion. A particle starting higher has further to go, but is pushed proportionally harder.

STAGE 4 Two particles

Ask yourself When do the particles reach V, and how fast are they moving? Who has more momentum?

Both reach V at the same time $T = \pi\sqrt{k/(2g)}$, from opposite sides. The speed at V is ω times the amplitude in s, which is proportional to $\sqrt{\text{height}}$. Let the heavier particle, released from h, have speed U at V; the lighter one, released from $\alpha^2 h$, has speed αU. Their momenta are $\alpha m \cdot U$ and $m \cdot \alpha U$ in opposite directions: equal and opposite. The total momentum at the collision is zero.

Ask yourself With total momentum zero and coefficient of restitution α, what are the velocities just after the collision?

The momentum stays zero, so after the collision the particles again move in opposite directions with momenta of equal size, and their relative speed is α times the approach speed $\alpha U + U$. If the lighter particle (mass αm) now moves at speed w, the heavier (mass m) moves at speed αw, and $w + \alpha w = \alpha(1+\alpha)U$, so $w = \alpha U$. Each particle bounces back with its speed multiplied by α: the particle of mass αm leaves at αU, and the one of mass m at $\alpha^2 U$.

STAGE 5 The whole motion

Each particle now climbs back up its own side, to a height reduced by the factor α^2 (since height is proportional to speed squared): the particle of mass αm to $\alpha^2 h$ and the particle of mass m to $\alpha^4 h$. By the tautochrone property, both return to V at exactly the same moment, a time $2T$ later, with momenta again equal and opposite. The same collision happens again, with all speeds scaled by α. So:

- the particles always collide at V, at times $T, 3T, 5T, \ldots$;
- after the nth collision they rise to heights $\alpha^{2n} h$ and $\alpha^{2n+2} h$;
- the motion dies away geometrically but never stops: there are infinitely many collisions, and they continue for ever. ■

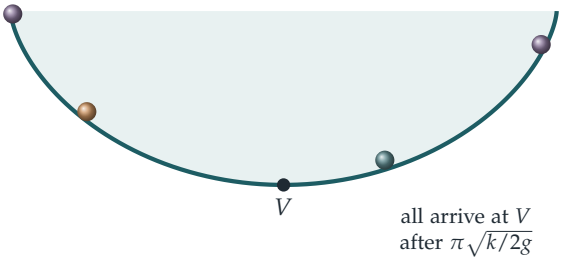


Figure 11.4.1 The tautochrone: an upside-down cycloid, traced by a point on the rim of a wheel rolling along a line. Marbles released from rest anywhere on it reach the lowest point V at the same moment. For the cycloid generated by a wheel of radius R, arc length and height satisfy $s^2 = 8Ry$, so $k = 2R$.

Looking back

- **Choose the natural variable.** In terms of arc length s, the motion is exactly simple harmonic. In terms of x or y it is a mess.
- **Energy gives the equation.** Differentiating the energy equation produced the equation of motion directly.
- **Symmetry in time.** Because every journey to the bottom takes the same time, the two particles were synchronised for ever, and the collisions repeated exactly.

What's really going on

Why the cycloid?

A wheel of radius R rolling under a line traces the cycloid $x = R(t + \sin t)$, $y = R(1 - \cos t)$. Then $\frac{dx}{dt} = R(1 + \cos t)$ and $\frac{dy}{dt} = R\sin t$, so the speed along the curve is $R\sqrt{2 + 2\cos t} = 2R\cos\frac{t}{2}$. Integrating, the arc length from the bottom is $s = 4R\sin\frac{t}{2}$, while $y = 2R\sin^2\frac{t}{2}$. Hence $s^2 = 8Ry$: the cycloid is exactly the curve with $s = 2\sqrt{ky}$, with $k = 2R$. Conversely, the requirement that the restoring force be proportional to s forces the curve to be a cycloid.

The brachistochrone

In 1696 Johann Bernoulli challenged the mathematicians of Europe: along what curve does a bead slide from one point to a lower point (not directly below it) in the *shortest* time? The answer, the *brachistochrone*, is again a cycloid. Newton is said to have solved the problem in a single evening after receiving it; he published his solution anonymously, but Bernoulli recognised the author, "as the lion by its claw". The problem is the founding example of the *calculus of variations*, which you met in Section 4.7: finding the curve that minimises a quantity defined by an integral.

Huygens's cycloidal clock

Huygens knew that a simple pendulum keeps imperfect time because its period depends on its amplitude. In his *Horologium Oscillatorium* (1673) he showed how to make the bob move along a cycloid: hang the string between two curved "cheeks" shaped as cycloids, and the string wraps round them in such a way that the bob traces a cycloid itself (the *involute* of a cycloid is another cycloid). In practice the cheeks caused more error through friction than they removed, and clockmakers instead kept the swings small. But Huygens's analysis invented the theory of evolutes and involutes, and his book was one of the great works of seventeenth-century mechanics.

The thinking skill

Find the variable in which the motion is simple

Motion along a curve is often simplest in terms of the distance along the curve. Look for the variable that makes the equation linear.

At university this becomes the idea of *generalised coordinates* in Lagrangian mechanics: any variables that describe the configuration can be used, and choosing them well can make a problem solvable.

Try this next

1. **Different heights.** A particle is released from height h in the tautochrone bowl. Show that its speed at V is $\sqrt{2gh}$, as it must be by energy conservation, and check that this agrees with ωs_0.

2. **An ordinary bowl.** In a hemispherical bowl of radius R, compare the time taken to reach the bottom from a small height with the time from the rim. Which is longer? (Use the result of Section 11.3.)

3. **Equal masses.** Repeat the second part of the question with two particles of equal mass m, released from heights h_1 and h_2, and coefficient of restitution e. Do they still always collide at V?

4. **Cycloid arc length.** Show that one arch of a cycloid has length $8R$, and that the area under it is $3\pi R^2$, three times the area of the rolling circle.

Further reading

- **George Simmons, *Differential Equations with Applications and Historical Notes*** (CRC Press). Includes the tautochrone and the brachistochrone, with their history.

- **E. A. Whitman, "Some historical notes on the cycloid"**, *American Mathematical Monthly* 50 (1943). Why the cycloid was called "the Helen of geometers".
- **Paul Nahin, *When Least Is Best*** (Princeton University Press). The brachistochrone and other optimisation problems, told with history.

11.5 Gravity in an Accelerating Train

The question

A train moves westwards on a straight horizontal track with constant acceleration a, where $a > 0$. Axes are chosen as follows: the origin is fixed in the train; the x-axis is in the direction of the track with the positive x-axis pointing to the East; and the positive y-axis points vertically upwards.

A smooth wire is fixed in the train. It lies in the x-y plane and is bent in the shape given by $ky = x^2$, where k is a positive constant. A small bead is threaded onto the wire. Initially, the bead is held at the origin. It is then released.

(i) Explain why the bead cannot remain stationary relative to the train at the origin.

(ii) Show that, in the subsequent motion, the coordinates (x, y) of the bead satisfy

$$\dot{x}(\ddot{x} - a) + \dot{y}(\ddot{y} + g) = 0$$

and deduce that $\frac{1}{2}(\dot{x}^2 + \dot{y}^2) - ax + gy$ is constant during the motion.

(iii) Find an expression for the maximum vertical displacement, b, of the bead from its initial position in terms of a, k and g.

(iv) Find the value of x for which the speed of the bead relative to the train is greatest and give this maximum speed in terms of a, k and g.

The hook

When a train pulls out of a station, you feel pressed back into your seat; a cup of coffee on the table tilts its surface; a ball on the floor rolls towards the back. From inside the train it looks exactly as if gravity had tilted, pointing slightly backwards as well as down. Einstein called the realisation that this is more than an analogy "the happiest thought of my life": gravity and acceleration are, locally, indistinguishable.

This question puts a bead on a curved wire in an accelerating train. Seen from the train, the bead behaves exactly as if it were in a gravitational field tilted by the angle whose tangent is a/g. The bottom of the wire is no longer the lowest point, and the bead oscillates about a new equilibrium.

Before you start

Ask yourself The train accelerates westwards (in the negative x direction) at a. For an observer in the train, what "fictitious" force appears to act on a mass m?

In the train's frame, Newton's second law holds if we add a fictitious force $-m \times$ (acceleration of the frame). The frame accelerates at $-a$ in the x direction, so the fictitious force is $+ma$, pointing east. Together with gravity, the bead feels an effective gravity of $(a, -g)$ per unit mass: tilted towards the east.

Working it through

STAGE 1 No rest at the origin

At the origin the wire $ky = x^2$ is horizontal. The normal reaction is vertical there, so it cannot cancel the eastward fictitious force ma. The net force along the wire is $ma \neq 0$, so the bead cannot stay at rest relative to the train. ■

STAGE 2 An energy equation

Ask yourself Write Newton's law in the train's frame, including the normal reaction $\mathbf{N}$. Then take the dot product with the velocity $(\dot{x}, \dot{y})$. Why does $\mathbf{N}$ disappear?

In the train's frame,

$$m(\ddot{x}, \ddot{y}) = \mathbf{N} + (ma, -mg).$$

The wire is smooth, so $\mathbf{N}$ is perpendicular to the wire, and hence to the velocity $(\dot{x}, \dot{y})$, which is along the wire. Taking the dot product with the velocity kills $\mathbf{N}$:

$$m\dot{x}\ddot{x} + m\dot{y}\ddot{y} = ma\dot{x} - mg\dot{y}, \qquad \text{so} \qquad \dot{x}(\ddot{x} - a) + \dot{y}(\ddot{y} + g) = 0. \quad ■$$

The left side is the derivative of $\frac{1}{2}(\dot{x}^2 + \dot{y}^2) - ax + gy$, which is therefore constant. ■

This is conservation of energy in the train's frame, with a potential energy $-ax + gy$ per unit mass that includes the fictitious force.

STAGE 3 How high?

Ask yourself The constant is 0, since the bead starts at rest at the origin. At the highest point the bead is momentarily at rest. What equation does that give?

At rest, $\frac{1}{2}(\dot{x}^2 + \dot{y}^2) = 0$, so $ax = gy$. With $y = \frac{x^2}{k}$, this gives $ax = \frac{gx^2}{k}$, so (apart from the

starting point $x = 0$) $x = \frac{ak}{g}$, and

$$b = \frac{a^2 k}{g^2}. \qquad \blacksquare$$

The bead swings east and up the wire to the point $\left(\frac{ak}{g}, \frac{a^2 k}{g^2}\right)$, then back to the origin, and so on.

STAGE 4 The fastest point

Ask yourself Write the speed squared as a function of x alone. Where is it greatest?

The speed squared is $2(ax - gy) = 2\left(ax - \frac{gx^2}{k}\right)$, a quadratic in x that is greatest at $x = \frac{ak}{2g}$, halfway between the turning points. There the speed squared is $2\left(\frac{a^2 k}{2g} - \frac{a^2 k}{4g}\right) = \frac{a^2 k}{2g}$, so the maximum speed is

$$a\sqrt{\frac{k}{2g}}. \qquad \blacksquare$$

Ask yourself Find the gradient of the wire at $x = \frac{ak}{2g}$. How does it relate to the direction of the effective gravity $(a, -g)$?

The gradient is $\frac{2x}{k} = \frac{a}{g}$. So at this point the wire is perpendicular to the effective gravity: it is the "lowest" point of the wire in the tilted field, and the bead oscillates about it (Figure 11.5.1).

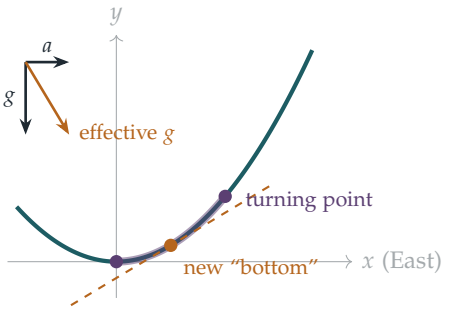


Figure 11.5.1 The wire $ky = x^2$ (teal, with $k = 1$ and $a/g = 0.6$) in the frame of the train. Effective gravity (amber arrow) is tilted east by the angle with tangent a/g. The bead oscillates (plum) between the origin and $x = ak/g$, about the point where the wire is perpendicular to the effective gravity, which acts as the new lowest point. The dashed line is a "level" line in the tilted field.

Looking back

- **Work in the convenient frame, and add the fictitious force.** In the train's frame the wire is fixed, and the only price is an extra constant force.
- **Dot with the velocity.** This removes constraint forces and gives an energy equation.
- **Interpret the answer.** The turning point and the fastest point have a geometric meaning: effective gravity is tilted.

What's really going on

Fictitious forces

Newton's laws hold in *inertial* frames, those that are not accelerating. In an accelerating frame they can be rescued by adding fictitious forces: $-m\mathbf{A}$ for a frame with acceleration $\mathbf{A}$, and, in a rotating frame, the centrifugal and Coriolis forces. They are called fictitious because no object exerts them; they are artefacts of the frame. But they are perfectly real to anyone inside it. The centrifugal force flattens the Earth at the poles, and the Coriolis force sets the direction in which hurricanes spin (anticlockwise in the northern hemisphere) and deflects long-range artillery shells.

The equivalence principle

A constant fictitious force $-m\mathbf{A}$ is proportional to mass, exactly like gravity. No experiment in a closed box can tell the difference between the box resting on a planet and the box accelerating in empty space: this is the *equivalence principle*. Einstein took it as the starting point of general relativity. Since light travelling across an accelerating box appears to bend, light must bend in a gravitational field too; and since clocks at the front and back of an accelerating rocket run at different rates, clocks at different heights in a gravitational field must too. Both predictions have been confirmed, and the second must be corrected for in every GPS satellite, whose clocks run faster than clocks on the ground by about 45 microseconds a day from this effect alone.

Einstein's happiest thought

In 1907, working at the patent office in Bern, Einstein was writing a review article on special relativity when, as he later recalled, "the happiest thought of my life" came to him: a person falling freely from a roof would not feel their own weight. Gravity could be "transformed away" by choosing a falling frame. It took him eight more years, and a great deal of new mathematics (the differential geometry of curved spaces, developed by Gauss and Riemann), to turn the thought into the general theory of relativity of 1915.

The thinking skill

Find a conserved quantity in a moving frame

In a frame with constant acceleration, a constant fictitious force acts. It can be included in a potential energy, and energy is conserved in that frame.

The trick of dotting the equation of motion with the velocity to eliminate constraint forces works in any frame. At university it becomes a general principle: a system whose equations do not depend explicitly on time has a conserved energy.

Try this next

1 **A cannon on a train.** (From STEP I, 2000.) A railway carriage moves north with acceleration a, and a child fires a toy cannon southwards at angle θ and speed V. Show that the range on the carriage floor is greatest when $\tan 2\theta = -\frac{g}{a}$, that this angle is approximately $\frac{\pi}{4} + \frac{a}{2g}$ when a is small, and that the maximum range is then approximately $\frac{V^2}{g} + \frac{V^2 a}{g^2}$.

2 **A headwind.** (From STEP II, 2006.) A projectile is fired north at speed u and angle θ, against a wind that exerts a constant southward force f per unit mass. Find where it lands. Show that if θ exceeds a certain angle α the wind blows it back towards O before it lands, and find α. Describe the motion when $f = g$ and $\theta = 45^\circ$.

3 **Coffee in a train.** Show that the surface of a cup of coffee in the accelerating train is a plane tilted at angle $\arctan(a/g)$ to the horizontal.

4 **Small oscillations.** Find the period of small oscillations of the bead about $x = \frac{ak}{2g}$. (*Hint:* near the new bottom the wire is approximately a circle of radius equal to its radius of curvature.)

Further reading

- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), the chapter on non-inertial frames and fictitious forces.
- **Albert Einstein, *Relativity: The Special and the General Theory*** (various editions). Einstein's own popular account, including the accelerating box.
- **Pedro Ferreira, *The Perfect Theory*** (Little, Brown). A biography of general relativity.

11.6 Backspin

The question

By pressing a finger down on it, a uniform spherical marble of radius a is made to slide along a horizontal table top with an initial linear velocity v_0 and an initial *backward* angular velocity ω_0 about the horizontal axis perpendicular to v_0. The frictional force between the marble and the table is constant (independent of speed).

For what value of $v_0/(a\omega_0)$ does the marble

(i) slide to a complete stop,

(ii) come to a stop and then roll back towards its initial position with linear speed $v_0/7$?

The hook

Press down on the back of a marble with your finger so that it squirts forwards while spinning backwards. It slides away from you, slows, stops, and then, as if it had changed its mind, rolls back. Snooker players use the same effect in the "screw" or "draw" shot, hitting the cue ball low so that it spins backwards, stops on hitting the object ball, and comes back.

The question asks for the exact amount of backspin that makes the marble stop dead, and the amount that makes it return at one-seventh of its starting speed. The calculation tracks two quantities, linear and angular velocity, both changed by the same frictional force. Behind it there is a beautiful shortcut: one combination of the two is not changed by friction at all.

A note on the syllabus

Rotational dynamics was removed from the STEP specification in 2019. All you need here is: a solid sphere of mass m and radius a has moment of inertia $I = \frac{2}{5}ma^2$ about a diameter, and a torque τ changes its angular velocity according to $I\dot{\omega} = \tau$, the rotational version of $m\dot{v} = F$.

Before you start

Ask yourself The marble moves forwards at v and spins backwards at ω. How fast is the bottom point of the marble moving over the table? Which way does friction act?

The bottom point moves at v due to the marble's motion, plus $a\omega$ due to the backward spin (a backward spin carries the bottom point forwards). So it slides forwards at $v + a\omega$, and friction acts backwards, opposing the sliding. The marble rolls without slipping only when $v + a\omega = 0$.

Working it through

STAGE 1 What friction does

Ask yourself Let the friction force be F, backwards. What does it do to v? Its torque about the centre is Fa: does it increase or decrease the backspin?

Friction decelerates the marble: $m\dot{v} = -F$, so $v = v_0 - \frac{F}{m}t$. Acting backwards at the bottom of the marble, it also exerts a torque Fa that opposes the backspin: $I\dot{\omega} = -Fa$, so with $I = \frac{2}{5}ma^2$,

$$\omega = \omega_0 - \frac{5F}{2ma}t.$$

Both v and ω decrease steadily while the marble slides.

STAGE 2 Stopping dead

Ask yourself For the marble to stop completely, v and ω must reach zero at the same moment. When does each reach zero?

$v = 0$ at $t = \frac{mv_0}{F}$, and $\omega = 0$ at $t = \frac{2ma\omega_0}{5F}$. These are equal when $v_0 = \frac{2}{5}a\omega_0$, that is,

$$\frac{v_0}{a\omega_0} = \frac{2}{5}. \quad \blacksquare$$

At that moment the marble has no linear or angular velocity, and friction (which only acts while there is sliding) switches off: it stays put.

STAGE 3 Coming back

Ask yourself If the backspin is larger, v reaches zero while ω is still positive. The bottom point is still sliding forwards, so friction still acts backwards. What happens next, and when does sliding stop?

Friction continues to push the marble backwards, so v becomes negative, while ω keeps decreasing. Sliding stops when $v + a\omega = 0$:

$$v_0 + a\omega_0 - \left(\frac{F}{m} + \frac{5F}{2m}\right)t = 0, \qquad t^* = \frac{2m(v_0 + a\omega_0)}{7F}.$$

At that moment,

$$v = v_0 - \frac{2(v_0 + a\omega_0)}{7} = \frac{5v_0 - 2a\omega_0}{7}.$$

From then on the marble rolls without slipping, friction stops acting, and v stays constant. For it to roll back at speed $\frac{v_0}{7}$ we need $5v_0 - 2a\omega_0 = -v_0$, so

$$\frac{v_0}{a\omega_0} = \frac{1}{3}. \qquad \blacksquare$$

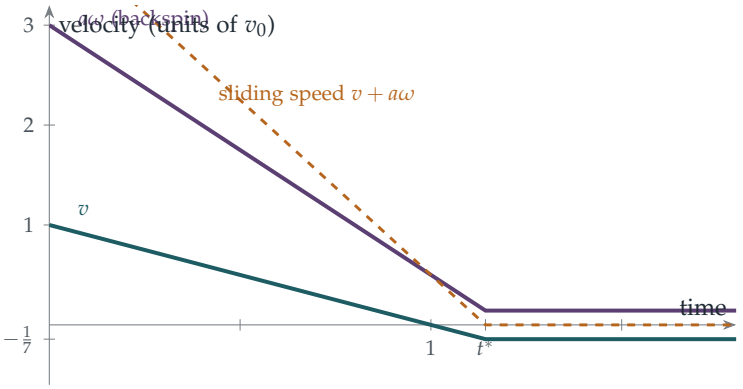


Figure 11.6.1 The case $v_0/(a\omega_0) = \frac{1}{3}$, with time in units of mv_0/F. Friction reduces the forward velocity v (teal) and the backspin $a\omega$ (plum) at constant rates, in the ratio 2 : 5. The marble stops moving forwards at $t = 1$, but is still sliding, so it is pushed backwards until the sliding speed (amber) reaches zero at $t^* = \frac{8}{7}$. It then rolls back at $\frac{v_0}{7}$.

Looking back

- **Track two quantities.** Friction changed both v and ω, at rates in a fixed ratio.
- **Know when a force switches off.** Sliding friction acts only while there is sliding; once the marble rolls, it stops.
- **The sliding speed is the key variable.** It decreased linearly, and its zero marked the change from sliding to rolling.

What's really going on

A conserved quantity

There is a quicker way to the final speed. Consider the angular momentum of the marble about the point of contact with the table. Friction acts *at* that point, so it has no moment about it, and the normal reaction and gravity both act along the vertical through it. So this angular momentum is conserved while the marble slides. It is mva (from the motion of the centre) minus $I\omega$ (the backspin counts negatively):

$$L = mav - \tfrac{2}{5}ma^2\omega.$$

When rolling at speed V, with $a\omega = -V$, it is $maV + \frac{2}{5}maV = \frac{7}{5}maV$. Equating with the initial value,

$$\tfrac{7}{5}maV = mav_0 - \tfrac{2}{5}ma^2\omega_0, \qquad V = \frac{5v_0 - 2a\omega_0}{7},$$

the same answer with no reference to the size of the friction force, or even to whether it is constant. The final speed is determined by conservation alone; friction only decides how long it takes to get there.

Snooker, bowling and tennis

The same analysis explains many sports. A snooker ball struck centrally slides at first and then starts to roll, finishing at $\frac{5}{7}$ of its initial speed (set $\omega_0 = 0$ in the formula). A ten-pin bowling ball thrown with sidespin skids down the lane and then "hooks" when it starts to roll. A tennis ball hit with topspin dips sharply, though there the main effect is aerodynamic: a spinning ball deflects the air around it, the *Magnus effect*.

Coriolis and billiards

Gaspard-Gustave Coriolis, now remembered for the fictitious force in rotating frames (Section 11.5), wrote in 1835 a whole book on the mathematics of billiards, *Théorie mathématique des effets du jeu de billard*. He showed, among other things, that a ball struck with spin follows a parabola while it slides before rolling straight, and analysed exactly the screw shot of this question. The book is still cited by physicists who study the game.

The thinking skill

Track two quantities changed by one force

When a single force affects several quantities, write down how it changes each, and look for a combination it does not change at all.

Here friction changed v and ω but not the angular momentum about the contact point. At university, finding such combinations is a recurring theme: choosing

the point about which to take moments so that unknown forces drop out is the rotational version of choosing a frame.

Try this next

1 **Topspin.** If the marble starts with forward spin ω_0 instead of backspin, find its final rolling speed. For what value of $v_0/(a\omega_0)$ does it speed up?

2 **A snooker ball.** A ball struck centrally starts sliding at v_0 with no spin. Show that it ends up rolling at $\frac{5}{7}v_0$, and find the fraction of kinetic energy lost. (The kinetic energy of rolling is $\frac{1}{2}mV^2 + \frac{1}{2}I\omega^2$.)

3 **A hoop.** Repeat the question for a thin hoop rolling in its own plane, with $I = ma^2$. What ratio $v_0/(a\omega_0)$ now makes it return at $\frac{v_0}{7}$?

4 **The draw shot.** Where must a snooker cue strike the ball (at what height above the centre, in terms of a) for the ball to roll without sliding from the start? (*Hint:* take angular momentum about the contact point during the impulse.)

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 8, on angular momentum and rolling with slipping.
- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), the chapters on rigid body motion.
- **David Alciatore, *The Illustrated Principles of Pool and Billiards*** (Sterling). The physics of the game, with many diagrams.

11.7 Simpson's Rule in a Rod

The question

A body of mass m and centre of mass O is said to be *dynamically equivalent* to a system of particles of total mass m and centre of mass O if the moment of inertia of the system of particles is the same as the moment of inertia of the body, about any axis through O. Show that this implies that the moment of inertia of the system of particles is the same as that of the body about *any* axis.

Show that a uniform rod of length $2a$ and mass m is dynamically equivalent to a suitable system of three particles, one at each end of the rod, and one at the midpoint.

Use this result to deduce that a uniform rectangular lamina of mass M is dynamically equivalent to a system consisting of particles each of mass $\frac{1}{36}M$ at the corners, particles each of mass $\frac{1}{9}M$ at the midpoint of each side, and a particle of mass $\frac{4}{9}M$ at the centre. Hence find the moment of inertia of a square lamina, of side $2a$ and mass M, about one of its diagonals.

The mass per unit length of a thin rod of mass m is proportional to the distance from one end of the rod, and a dynamically equivalent system consists of one particle at each end of the rod and one at the midpoint. Write down a set of equations which determines these masses, and show that, in fact, only two particles are required.

The hook

How does a uniform rod respond to being pushed and spun? It turns out that, for every purpose in mechanics, a rod behaves exactly like three point masses: one-sixth of its mass at each end and two-thirds at the middle. Nothing you can do to the rod (push it, spin it about any axis) distinguishes it from the three particles.

The numbers $\frac{1}{6}$, $\frac{2}{3}$, $\frac{1}{6}$ may look familiar. They are the weights in *Simpson's rule*, $\int_{-a}^{a} f \approx 2a\left(\frac{1}{6}f(-a) + \frac{2}{3}f(0) + \frac{1}{6}f(a)\right)$, the numerical integration method you may have met at A level. This is no coincidence: a dynamically equivalent system and a numerical integration rule are solving the same problem.

A note on the syllabus

Moments of inertia were removed from the STEP specification in 2019. The

moment of inertia of a body about an axis is $\sum m_i d_i^2$, the sum of each mass times the square of its distance from the axis (an integral for a continuous body). A uniform rod of mass m and length $2a$ has moment of inertia $\frac{1}{3}ma^2$ about a perpendicular axis through its centre. The *parallel axis theorem* says that the moment of inertia about any axis is the moment about the parallel axis through the centre of mass plus md^2, where d is the distance between the axes.

Before you start

Ask yourself Two systems have the same mass, the same centre of mass, and the same moment of inertia about every axis through it. Why does the parallel axis theorem then give the same moment of inertia about every other axis too?

For any axis, both moments of inertia equal (moment about the parallel axis through O) $+md^2$. The first terms agree by assumption, and m and d are the same for both systems. ■

Working it through

STAGE 1 The rod

Ask yourself Put masses p at each end and q at the middle. By symmetry the centre of mass is right. What do the total mass and the moment of inertia about a perpendicular axis through the centre require?

We need $2p + q = m$, and the moment of inertia about a perpendicular axis through the middle must be $\frac{1}{3}ma^2$: $2pa^2 = \frac{1}{3}ma^2$. So $p = \frac{1}{6}m$ and $q = \frac{2}{3}m$.

Ask yourself Does this work for *every* axis through the centre, not just the perpendicular one?

For an axis through the centre at angle ψ to the rod, a point at distance r along the rod is at distance $r\sin\psi$ from the axis. So every moment of inertia (rod or particles) is $\sin^2\psi$ times the value for the perpendicular axis, and the two agree for all ψ. (Axes not in a plane containing the rod work the same way: only the distance of each point from the axis matters, and for points on a line through O this is proportional to r.) So the rod is dynamically equivalent to masses $\frac{1}{6}m$, $\frac{2}{3}m$, $\frac{1}{6}m$. ■

STAGE 2 The rectangle

Ask yourself Cut the lamina into thin strips parallel to one side. Replace each strip by its three equivalent particles. What do you get, and what can you do next?

Each thin strip is a rod, equivalent to $\frac{1}{6}$, $\frac{2}{3}$, $\frac{1}{6}$ of its mass on the left edge, the centre line and the right edge. Adding over all the strips, the lamina is equivalent to three uniform rods (along the left edge, the middle and the right edge) with masses $\frac{1}{6}M$, $\frac{2}{3}M$, $\frac{1}{6}M$. Now replace each of these rods by three particles in the same ratio. The mass at each point is a product of two fractions (Figure 11.7.1):

$$\text{corners: } \tfrac{1}{6}\cdot\tfrac{1}{6}=\tfrac{1}{36}, \qquad \text{edge midpoints: } \tfrac{1}{6}\cdot\tfrac{2}{3}=\tfrac{1}{9}, \qquad \text{centre: } \tfrac{2}{3}\cdot\tfrac{2}{3}=\tfrac{4}{9},$$

all times M. ■

(Strictly, each replacement must preserve the moment of inertia about every axis through the centre of the whole lamina, which it does: each replacement keeps the mass, the centre of mass and the moments of inertia about axes through the centre of mass of the piece, and the parallel axis theorem does the rest.)

STAGE 3 The square about its diagonal

For a square of side $2a$, the diagonal passes through two corners and the centre, which contribute nothing. The other two corners are at distance $\sqrt{2}a$ from it, and the four edge midpoints at distance $\frac{a}{\sqrt{2}}$. So

$$I = 2\cdot\tfrac{1}{36}M\cdot 2a^2 + 4\cdot\tfrac{1}{9}M\cdot\tfrac{a^2}{2} = \tfrac{1}{9}Ma^2+\tfrac{2}{9}Ma^2 = \tfrac{1}{3}Ma^2. \qquad ■$$

This is the same as the moment of inertia about an axis through the centre parallel to a side. (That is no accident: for a square, all axes through the centre in its plane give the same answer, by a symmetry argument in the exercises.)

STAGE 4 A non-uniform rod

Ask yourself Let the rod have length L and density $\frac{2mx}{L^2}$ at distance x from one end. Put masses p_0, p_1, p_2 at $x=0$, $\frac{L}{2}$, L. Matching the total mass, the centre of mass and the moment of inertia about the end gives three equations. What are they?

The rod has total mass m, first moment $\int_0^L x\cdot\frac{2mx}{L^2}\,\mathrm{d}x = \frac{2}{3}mL$ and second moment $\int_0^L x^2\cdot\frac{2mx}{L^2}\,\mathrm{d}x = \frac{1}{2}mL^2$ about the light end. Matching these:

$$p_0+p_1+p_2 = m, \qquad \tfrac{1}{2}Lp_1+Lp_2 = \tfrac{2}{3}mL, \qquad \tfrac{1}{4}L^2p_1+L^2p_2 = \tfrac{1}{2}mL^2.$$

Subtracting the third (divided by L^2) from the second (divided by L) gives $\frac{1}{4}p_1 = \frac{1}{6}m$, so $p_1 = \frac{2}{3}m$, then $p_2 = \frac{1}{3}m$ and $p_0 = 0$. Only two particles are needed: $\frac{2}{3}m$ at the midpoint and $\frac{1}{3}m$ at the heavy end. ■

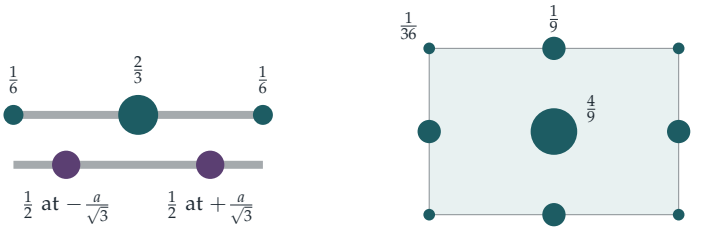


Figure 11.7.1 Left, top: a uniform rod is dynamically equivalent to masses $\frac{1}{6}$, $\frac{2}{3}$, $\frac{1}{6}$ (as fractions of its mass) at its ends and centre, the weights of Simpson's rule. Left, bottom: it is also equivalent to two halves at distance $\frac{a}{\sqrt{3}}$ from the centre, the nodes of two-point Gaussian integration. Right: a rectangular lamina is equivalent to nine masses, with areas of the dots proportional to the masses.

Looking back

- **Match moments.** Dynamical equivalence means matching the zeroth, first and second moments of the mass distribution.
- **Build up dimension by dimension.** A rectangle is a rod of rods, so its weights are products.
- **Count equations and unknowns.** Three moments to match, three unknown masses; and sometimes a solution happens to have a zero.

What's really going on

Simpson's rule

Matching moments of a mass distribution is the same problem as designing a numerical integration rule. A rule $\int_{-a}^{a} f(x)\,\mathrm{d}x \approx w_0 f(-a) + w_1 f(0) + w_2 f(a)$ that is exact for $f = 1, x, x^2$ must satisfy

$$w_0 + w_1 + w_2 = 2a, \qquad -aw_0 + aw_2 = 0, \qquad a^2 w_0 + a^2 w_2 = \tfrac{2}{3}a^3,$$

exactly the conditions for the rod (with $f = 1, x, x^2$ playing the roles of mass, first moment and moment of inertia). The solution is $w = 2a(\frac{1}{6}, \frac{2}{3}, \frac{1}{6})$: Simpson's rule. A bonus comes from symmetry: the rule is also exact for $f = x^3$, since both sides vanish, so Simpson's rule integrates cubics exactly, and its error for smooth functions is proportional to the fourth derivative. A rod therefore matches its three particles not just in mass, centre and moment of inertia, but in the third moment too.

Gaussian quadrature and the inertia tensor

If the positions as well as the masses are free, two particles suffice: half the mass at each of the points $\pm\frac{a}{\sqrt{3}}$. Check: $2\cdot\frac{1}{2}m\cdot\frac{a^2}{3}=\frac{1}{3}ma^2$. This is the two-point *Gauss–Legendre rule*, $\int_{-a}^{a} f \approx a\big(f(-\frac{a}{\sqrt{3}})+f(\frac{a}{\sqrt{3}})\big)$, which is also exact for cubics. Carl Friedrich Gauss showed in 1814 that n cleverly placed points can integrate polynomials of degree $2n-1$ exactly, and Gaussian quadrature is now a standard method of numerical analysis.

In mechanics, the moments of inertia of a body about all axes through its centre are encoded in a symmetric 3×3 matrix, the *inertia tensor*. Two bodies with the same mass, centre of mass and inertia tensor respond identically to all forces, which is why engineers can replace a complicated part by a few point masses in a simulation. The eigenvectors of the inertia tensor are the *principal axes*, the axes about which a body can spin steadily.

The thinking skill

Replace a system by an equivalent one

If two systems agree in every quantity that matters for the problem, replace the complicated one by the simple one.

Here only three moments mattered, so a continuous rod could be replaced by three particles. At university, this idea appears as Gaussian quadrature, as multipole expansions in electromagnetism and gravitation (far away, only a few moments of a charge or mass distribution matter), and as the method of moments in statistics.

Try this next

1. **Simpson's rule.** Use Simpson's rule with a single interval to estimate $\int_0^1 \frac{\mathrm{d}x}{1+x^2}$, and compare with $\frac{\pi}{4}$. Check that the rule is exact for $\int_{-1}^{1} x^3\,\mathrm{d}x$ and $\int_{-1}^{1} x^2\,\mathrm{d}x$ but not for $\int_{-1}^{1} x^4\,\mathrm{d}x$.

2. **All axes alike.** Show that the moment of inertia of a square lamina about any axis through its centre in its plane is $\frac{1}{3}Ma^2$. (*Hint:* the nine equivalent particles are symmetric under rotation by 90°; what does that say about a quadratic expression in $\cos\psi$ and $\sin\psi$?)

3. **A triangle.** Show that a uniform triangular lamina of mass M is dynamically equivalent to three particles of mass $\frac{1}{3}M$ at the midpoints of its sides.

4. **A solid cube.** Find a dynamically equivalent system of 27 particles for a uniform solid cube, and use it to find the moment of inertia of a cube of side $2a$ about an

edge.

Further reading

- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), the chapter on rigid bodies and moments of inertia.
- **Endre Süli & David Mayers, *An Introduction to Numerical Analysis*** (Cambridge University Press), the chapters on numerical integration, including Simpson's rule and Gaussian quadrature.
- **Philip Davis & Philip Rabinowitz, *Methods of Numerical Integration*** (Dover). The classic reference.

11.8 Tilted Equilibria

The question

In this question, $n \geq 2$.

(i) A solid, of uniform density, is formed by rotating through 360° about the y-axis the region bounded by the part of the curve $r^{n-1}y = r^n - x^n$ with $0 \leq x \leq r$, and the x- and y-axes. Show that the y-coordinate of the centre of mass of this solid is $\dfrac{nr}{2(n+1)}$.

(ii) Show that the normal to the curve $r^{n-1}y = r^n - x^n$ at the point $\left(rp, r(1-p^n)\right)$, where $0 < p \leq 1$, meets the y-axis at $(0, Y)$, where $Y = r\left(1 - p^n - \dfrac{1}{np^{n-2}}\right)$.

In the case $n = 4$, show that the greatest value of Y is $\frac{1}{4}r$.

(iii) A solid is formed by rotating through 360° about the y-axis the region bounded by the curves $r^3y = r^4 - x^4$ and $ry = -(r^2 - x^2)$, both for $0 \leq x \leq r$. A and B are the points $(0, -r)$ and $(0, r)$, respectively, on the surface of the solid.

Show that the solid can rest in equilibrium on a horizontal surface with the vector $\overrightarrow{AB}$ at three different, non-zero, angles to the upward vertical. You should not attempt to find these angles.

The hook

Put a rugby ball on a table and it settles on its side; stand it on its end and, with great care, it balances there too, but the slightest nudge topples it. Every rigid body resting on a table has a set of equilibrium positions, some stable and some unstable, and which ones exist depends only on its shape and its centre of mass. How many can a body have? How few?

This question builds a solid, shaped like a spinning top with a domed upper half and a pointed lower half, and shows that besides resting upright or upside down it can rest at three different tilts. The key idea is a single geometric condition: a body rests on a point of its surface exactly when the normal to the surface at that point passes through the centre of mass. Behind it lies a recent discovery: a convex shape, the Gömböc, with only one stable and one unstable position.

Before you start

Ask yourself A solid rests on a table, touching at a single point X of its curved surface. The forces on it are its weight (acting at the centre of mass G) and the reaction at X. When are these in equilibrium?

The table is tangent to the surface at X, and the reaction acts at X, perpendicular to the table: along the normal to the surface. The weight acts vertically through G. Two forces balance only if they act along the same line, so the vertical through X, which is the normal at X, must pass through G. So: *the body can rest touching at X exactly when the normal at X passes through G.*

Working it through

STAGE 1 The centre of mass

Ask yourself Slice the solid into thin cylindrical shells of radius x, thickness δx and height y. What is the volume of a shell, and where is its centre of mass?

With $y = \frac{r^n - x^n}{r^{n-1}}$, a shell has volume $2\pi x y\,\delta x$ and its centre of mass at height $\frac{y}{2}$. So

$$V = \int_0^r 2\pi x y \,\mathrm{d}x = \frac{2\pi}{r^{n-1}}\left(\frac{r^{n+2}}{2} - \frac{r^{n+2}}{n+2}\right) = \frac{n\pi r^3}{n+2},$$

and the moment about the base is

$$\begin{aligned}\int_0^r 2\pi x \cdot \frac{y^2}{2}\,\mathrm{d}x &= \frac{\pi}{r^{2n-2}}\int_0^r x(r^n - x^n)^2\,\mathrm{d}x \\ &= \pi r^4\left(\frac{1}{2} - \frac{2}{n+2} + \frac{1}{2n+2}\right) = \frac{n^2\pi r^4}{2(n+1)(n+2)}.\end{aligned}$$

(The bracket simplifies because $(n+1)(n+2) - 4(n+1) + (n+2) = n^2$.) Dividing,

$$\bar{y} = \frac{nr}{2(n+1)}. \qquad \blacksquare$$

For $n = 2$ (a paraboloid) this is $\frac{r}{3}$, the familiar result; as $n \to \infty$ the solid becomes a cylinder and $\bar{y} \to \frac{r}{2}$.

STAGE 2 Where normals meet the axis

The gradient of the curve at $x = rp$ is $-\frac{nx^{n-1}}{r^{n-1}} = -np^{n-1}$, so the normal has gradient $\frac{1}{np^{n-1}}$. Moving along the normal from $\big(rp, r(1-p^n)\big)$ to the y-axis, x decreases by rp and y by $\frac{rp}{np^{n-1}} = \frac{r}{np^{n-2}}$. So

$$Y = r\left(1 - p^n - \frac{1}{np^{n-2}}\right). \qquad \blacksquare$$

Ask yourself For $n = 4$, $Y = r\left(1 - p^4 - \frac{1}{4p^2}\right)$. Where is it greatest? What happens as $p \to 0$, and at $p = 1$?

$\frac{dY}{dp} = r\left(-4p^3 + \frac{1}{2p^3}\right) = 0$ when $p^6 = \frac{1}{8}$, that is $p^2 = \frac{1}{2}$. Then $Y = r\left(1 - \frac{1}{4} - \frac{1}{2}\right) = \frac{1}{4}r$. ■

As $p \to 0$, $Y \to -\infty$ (near the top the dome is very flat, and its normals are almost vertical, meeting the axis far below); at $p = 1$, $Y = -\frac{1}{4}r$. So as p runs from 0 to 1, Y rises from $-\infty$ to $\frac{1}{4}r$ and falls back to $-\frac{1}{4}r$.

STAGE 3 The spinning top

Ask yourself The solid in part (iii) is a dome ($n = 4$) on top of an upside-down paraboloid ($n = 2$). Where is its centre of mass?

From Stage 1, the dome has volume $\frac{2\pi r^3}{3}$ and centre of mass at $\frac{2r}{5}$; the paraboloid has volume $\frac{\pi r^3}{2}$ and, being upside down, centre of mass at $-\frac{r}{3}$. So

$$\bar{y} = \frac{\frac{2}{3}\cdot\frac{2}{5} - \frac{1}{2}\cdot\frac{1}{3}}{\frac{2}{3} + \frac{1}{2}}\, r = \frac{\frac{4}{15} - \frac{1}{6}}{\frac{7}{6}}\, r = \frac{3r}{35}.$$

G is at $\left(0, \frac{3r}{35}\right)$, slightly above the widest point.

STAGE 4 Counting equilibria

Ask yourself A tilted equilibrium needs a point (not A or B) whose normal passes through G. On each part of the surface, how many points have normals meeting the axis at height $\frac{3r}{35}$?

The dome. Its normals meet the axis at heights Y which rise from $-\infty$ to $\frac{1}{4}r$ and fall to $-\frac{1}{4}r$. Since $-\frac{1}{4}r < \frac{3r}{35} < \frac{1}{4}r$, by the intermediate value theorem the value $\frac{3r}{35}$ is taken once on the rising part and once on the falling part: two points.

The pointed base. By part (ii) with $n = 2$, reflected in the x-axis, the normal at the point with parameter p meets the axis at $-r\left(1 - p^2 - \frac{1}{2}\right) = r\left(p^2 - \frac{1}{2}\right)$, which increases from $-\frac{1}{2}r$ to $\frac{1}{2}r$. It equals $\frac{3r}{35}$ exactly once, at $p^2 = \frac{41}{70}$.

Each of these three points gives an equilibrium, and the three normals make different angles with the axis, so $\overrightarrow{AB}$ makes three different non-zero angles with the vertical (Figure 11.8.1). ■

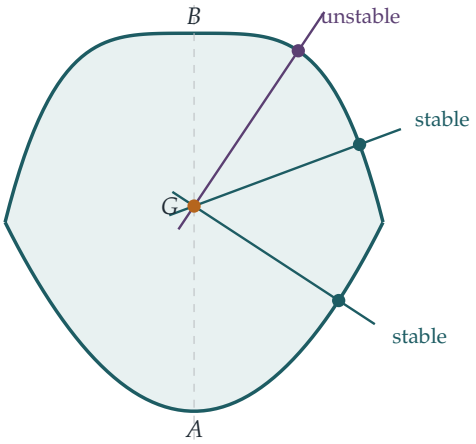


Figure 11.8.1 Cross-section of the solid of part (iii), with its centre of mass G (amber). At each of the three marked points the normal to the surface passes through G, so the solid can rest on that point. Two of these equilibria are stable (the surface is locally nearest to G there) and one is unstable (locally furthest).

Looking back

- **Turn a physical condition into geometry.** Equilibrium became "the normal passes through G".
- **Existence without solving.** We never found the angles. The intermediate value theorem, with the shape of the function $Y(p)$, showed how many there are.
- **Reuse a general result.** Part (ii) for general n handled both pieces of the surface in part (iii).

What's really going on

Equilibria as critical points

When the solid rests on the point X, the height of G above the table is the distance from G to the tangent plane at X, which for a convex body is $|GX|$ when GX is along the normal. In general, the equilibria are the *critical points* of the distance from G to the surface: a point X where $|GX|$ is stationary is exactly a point whose normal passes through G. Stable equilibria are local minima of this distance (G as low as possible), and unstable ones are local maxima or saddle points.

Going round the cross-section from A to B, the distance from G is: large at A (a maximum: balanced on its point, unstable), smaller at the tilted point on the base (a minimum, stable), larger at the rim, smaller at the lower tilted point of the dome (stable), larger at the upper one (unstable), and smaller at B (stable: resting on the top of the dome). Minima and maxima must alternate, as the figure shows. (In fact there is one more equilibrium that the question does not count: the solid can balance, unstably,

on its sharp rim, where the surface has no single normal.)

The Gömböc

How few equilibria can a convex body have? Any body has at least one stable and one unstable position: the nearest and furthest points of the surface from G. A two-dimensional convex shape of uniform density always has at least four (two minima and two maxima of distance, alternating). In 1995 Vladimir Arnold asked whether a three-dimensional convex body of uniform density could have just two: one stable, one unstable. In 2006 Gábor Domokos and Péter Várkonyi constructed one, the *Gömböc*: put it down in any position, and it rolls and rocks until it reaches its single stable position. Domokos later noticed that some tortoises have shells close to this shape, which helps them right themselves when turned over.

Archimedes and floating bodies

The study of the stability of bodies goes back to Archimedes, who in *On Floating Bodies* analysed when a floating paraboloid (a model of a ship's hull) is stable upright and when it capsizes, in terms of its shape and density. His answer, worked out with geometry alone, involves exactly the kind of centre-of-mass and normal calculations used here. Naval architects still use his ideas, in the form of the *metacentre*: a ship is stable if the point where the vertical through the centre of buoyancy meets the ship's axis lies above the centre of mass.

The thinking skill

Potential energy landscapes

Equilibria are the stationary points of potential energy, and stable equilibria are its minima. Count stationary points instead of solving for them.

Here the potential energy was proportional to the height of G, the distance from G to the surface. At university this viewpoint runs through mechanics, chemistry (molecules settle into minima of energy) and topology, where *Morse theory* shows that the numbers of minima, maxima and saddles of any function on a surface are constrained by the surface's shape.

Try this next

1 **A self-righting toy.** (From STEP II, 1996.) A toy is made of a solid cone of height λa fixed to a solid hemisphere of radius a along their common circular face. Show that if $\lambda < \sqrt{3}$ the toy returns upright when placed with its hemisphere on a table, but that if $\lambda > \sqrt{3}$ it may overbalance. (*Hint:* where must G be relative to the centre of

the hemisphere?) Show that if it is placed with the cone touching the table, it stays there whatever λ.

2 **A sphere.** For a uniform solid sphere, every normal passes through the centre. What does that say about its equilibria? What if the sphere's centre of mass is displaced slightly from its centre?

3 **An ellipse.** Show that an elliptical disc rolling in its own plane on a line has exactly four equilibria, and identify which are stable.

4 **Two paraboloids.** Replace the dome in part (iii) by a paraboloid $ry = r^2 - x^2$, so that the solid is symmetric about the x-axis. Show that G is at the origin, find the tilted equilibria, and explain why they come in pairs with equal tilts.

Further reading

- **Péter Várkonyi & Gábor Domokos, "Static equilibria of rigid bodies: dice, pebbles, and the Poincaré–Hopf theorem"**, *Journal of Nonlinear Science* 16 (2006). The Gömböc paper.
- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), on equilibrium and stability.
- **Chris Rorres, "Completing Book II of Archimedes's *On Floating Bodies*"**, *Mathematical Intelligencer* 26 (2004). Archimedes' stability analysis, with modern pictures.

11.9 The Safety Parabola

The question

A particle is projected from a point O on horizontal ground with initial speed u and at an angle of θ above the ground. The motion takes place in the x-y plane, where the x-axis is horizontal, the y-axis is vertical and the origin is O. Obtain the Cartesian equation of the particle's trajectory in terms of u, g and λ, where $\lambda = \tan\theta$.

Now consider the trajectories for different values of θ with u fixed. Show that for a given value of x, the coordinate y can take all values up to a maximum value, Y, which you should determine as a function of x, u and g.

Sketch a graph of Y against x and indicate on your graph the set of points that can be reached by a particle projected from O with speed u.

Hence find the furthest distance from O that can be achieved by such a projectile.

The hook

A fountain sprays water at the same speed in every direction. The jets form a glittering dome with a sharp outer edge: inside, water passes through every point; outside, none reaches. That edge is a parabola, called the *parabola of safety*, because a target outside it cannot be hit by a cannon of the given muzzle speed, whatever the angle.

The surprising thing is that the boundary is not itself one of the trajectories. It is the *envelope* of the whole family, a curve that touches every trajectory once, exactly like the line $y = 0$ touching the family of parabolas in Section 8.1. This question finds it with nothing more than the discriminant of a quadratic.

Before you start

Ask yourself Write x and y in terms of t, and eliminate t. How can you write $\sec^2\theta$ in terms of $\lambda = \tan\theta$?

With $x = ut\cos\theta$ and $y = ut\sin\theta - \frac{1}{2}gt^2$, substitute $t = \frac{x}{u\cos\theta}$:

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta} = \lambda x - \frac{gx^2}{2u^2}(1+\lambda^2),$$

using $\sec^2\theta = 1 + \tan^2\theta$.

Working it through

STAGE 1 Fix x, vary the angle

Ask yourself For a fixed $x > 0$, think of y as a function of λ. What kind of function is it? Which values does it take as λ ranges over all real numbers?

For fixed x, y is a quadratic in λ with negative leading coefficient $-\frac{gx^2}{2u^2}$: a downward parabola in λ. It has a maximum, and as $\lambda \to \pm\infty$, $y \to -\infty$. Since it is continuous, it takes every value up to its maximum. The maximum is where $\frac{dy}{d\lambda} = x - \frac{gx^2}{u^2}\lambda = 0$, that is $\lambda = \frac{u^2}{gx}$, and there

$$Y = \frac{u^2}{gx}\,x - \frac{gx^2}{2u^2}\left(1 + \frac{u^4}{g^2x^2}\right) = \frac{u^2}{2g} - \frac{gx^2}{2u^2}. \qquad \blacksquare$$

(Equivalently: the point (x, y) can be reached exactly when the quadratic $\frac{gx^2}{2u^2}\lambda^2 - x\lambda + \frac{gx^2}{2u^2} + y = 0$ has a real root λ, and the discriminant condition gives $y \le Y$.)

STAGE 2 The region

The curve $Y = \frac{u^2}{2g} - \frac{gx^2}{2u^2}$ is a downward parabola with its vertex at $\left(0, \frac{u^2}{2g}\right)$, the height reached by firing straight up, crossing the ground at $x = \pm\frac{u^2}{g}$, the maximum range. Every point on or below it (and above the ground) can be reached, and no point above it can (Figure 11.9.1).

Ask yourself Through a point strictly below the curve, how many trajectories pass? And through a point on it?

A point below the curve corresponds to a quadratic in λ with two real roots: two angles of projection, one high and one low. A point on the curve corresponds to a repeated root: exactly one trajectory, which touches the curve there.

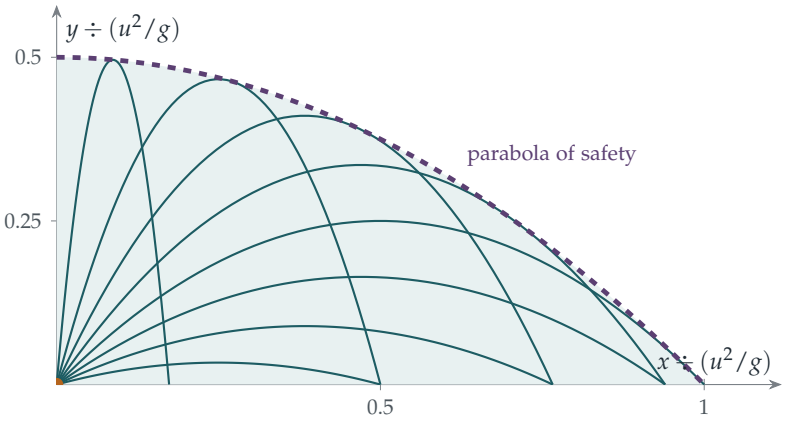


Figure 11.9.1 Trajectories at a fixed speed u and angles $15°, 25°, \ldots, 85°$ (teal). Each touches the dashed parabola of safety $y = \frac{u^2}{2g} - \frac{gx^2}{2u^2}$ exactly once, and the region beneath it (shaded) is exactly the set of points that can be reached. The origin O (amber) is the focus of the parabola of safety.

STAGE 3 The furthest point

Ask yourself The furthest reachable point from O (on or above the ground) lies on the boundary. Write the squared distance $x^2 + Y^2$ as a function of x^2. Is it increasing or decreasing?

Let $X = x^2$. Then

$$x^2 + Y^2 = X + \left(\frac{u^2}{2g} - \frac{gX}{2u^2}\right)^2, \qquad \frac{\mathrm{d}}{\mathrm{d}X}(x^2 + Y^2) = 1 - \frac{g}{u^2}\left(\frac{u^2}{2g} - \frac{gX}{2u^2}\right) = \frac{1}{2} + \frac{g^2 X}{2u^4} > 0.$$

So the distance increases as we move out along the boundary, and the furthest point is where the boundary meets the ground, at $x = \frac{u^2}{g}$. The furthest distance is

$$\frac{u^2}{g},$$

achieved by the ordinary maximum-range shot at 45°. ■

Looking back

- **Turn the question around.** Instead of asking where a given trajectory goes, ask which trajectories pass through a given point.
- **A quadratic in the parameter.** For each point, the equation for $\lambda = \tan\theta$ is a quadratic, and its discriminant decides whether the point can be reached.
- **Two, one or none.** Two angles reach a point inside, one reaches a point on the boundary, none reaches a point outside.

What's really going on

Envelopes

A one-parameter family of curves $\mathrm{F}(x, y, \lambda) = 0$ often has an *envelope*: a curve touching every member of the family. It is found by solving $\mathrm{F} = 0$ and $\frac{\partial \mathrm{F}}{\partial \lambda} = 0$ together, which is exactly what we did by maximising y over λ. Envelopes appear throughout science. The bright curve of light on the surface of a cup of coffee is the envelope of reflected rays, a *caustic*. A supersonic aircraft's sonic boom is carried by the envelope of the sound waves it emits, the *Mach cone*. And in Section 8.1, the envelope of a family of solutions of a differential equation was itself a solution, a "singular" one.

The focus

The parabola of safety has a remarkable property: its focus is the launch point O. (A parabola $y = c - \frac{x^2}{4c}$ has its focus at the origin; here $c = \frac{u^2}{2g}$.) Every trajectory, too, is a parabola, and the foci of all the trajectories lie on a circle of radius $\frac{u^2}{2g}$ about O. There is an energy explanation: the speed of the projectile at height y is $\sqrt{u^2 - 2gy}$, which is zero at the height $\frac{u^2}{2g}$ of the directrix, and parabolas whose directrix is the same line and which pass through O have their foci on that circle.

Torricelli's parabola

Galileo showed around 1609 that projectiles move along parabolas, and that the maximum range on level ground is at 45°. His student Evangelista Torricelli, better known for inventing the barometer, found the parabola of safety in 1644, in his *De motu gravium*. For gunners it answered a practical question: which targets are within range at all? Torricelli also showed that the trajectories through a given point come in pairs with angles symmetric about the direction of the point and the vertical, the high and low shots of the question.

The thinking skill

Envelopes of families of curves

To find the boundary of the region covered by a family of curves, fix a point and ask for which parameter values a curve passes through it. The boundary is where the number of solutions changes.

The same idea, counting solutions as a point moves, is the basis of the study of caustics in optics, of bifurcations in dynamical systems, and of catastrophe theory, which classifies how the number of solutions of an equation can change.

Try this next

1 **An exploding firework.** (From STEP II, 2023.) An explosion at the origin sends particles in all directions with speed u. Show that at time t they lie on a sphere of radius ut whose centre falls freely from the origin. Show that a particle projected horizontally with speed u from the top of the paraboloid of safety moves along its surface for ever, and find when it must be launched to collide with the particle fired at angle α.

2 **Firing up a slope.** Find the maximum range of a projectile with speed u up a plane inclined at angle β. (*Hint:* where does the line $y = x\tan\beta$ meet the parabola of safety?) Show that the best angle bisects the angle between the slope and the vertical.

3 **Two angles.** Show that the two angles that reach a point P are symmetric about the bisector of the angle between OP and the vertical.

4 **The foci.** Show that the trajectory with angle θ has its focus at $\left(\frac{u^2}{2g}\sin 2\theta, -\frac{u^2}{2g}\cos 2\theta\right)$, and hence that the foci lie on a circle about O.

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 3, which discusses the envelope of trajectories.
- **J. W. Bruce, P. J. Giblin & C. G. Gibson, "On caustics of plane curves"**, *American Mathematical Monthly* 88 (1981). Envelopes and caustics in geometry.
- **E. H. Lockwood, *A Book of Curves*** (Cambridge University Press). Envelopes, caustics and many classical curves, with constructions.

11.10 A Ball That Comes Back Towards You

The question

A particle P is projected from a point O on horizontal ground with speed u and angle of projection α, where $0 < \alpha < \frac{1}{2}\pi$.

(i) Show that if $\sin\alpha < \frac{2\sqrt{2}}{3}$, then the distance OP is increasing throughout the flight.

Show also that if $\sin\alpha > \frac{2\sqrt{2}}{3}$, then OP will be decreasing at some time before the particle lands.

(ii) At the same time as P is projected, a particle Q is projected horizontally from O with speed v along the ground in the opposite direction from the trajectory of P. The ground is smooth. Show that if

$$2\sqrt{2}\,v > (\sin\alpha - 2\sqrt{2}\cos\alpha)u,$$

then QP is increasing throughout the flight of P.

The hook

Throw a ball forwards and it moves away from you until it lands. Obviously. Throw it straight up and it comes back. Somewhere in between, something must change: there must be angles at which the ball moves away, then comes *closer*, then moves away again before landing. This question finds the exact threshold, $\sin\alpha = \frac{2\sqrt{2}}{3}$, or about $70.5°$.

The result is a nice example of a common situation in mathematics: a quantity everyone assumes is monotonic, which is not. It also shows a standard technique, used constantly at university: to find out whether a distance is increasing, differentiate its *square*, which avoids square roots.

Before you start

Ask yourself Why is it easier to work with OP^2 than with OP? Is "OP is increasing" the same as "OP^2 is increasing"?

$OP^2 = x^2 + y^2$ is a polynomial in t, while OP involves a square root. Since $OP \geq 0$, and squaring is increasing on non-negative numbers, OP increases exactly when OP^2 does.

Working it through

STAGE 1 The squared distance

With $x = ut\cos\alpha$ and $y = ut\sin\alpha - \frac{1}{2}gt^2$,

$$OP^2 = u^2t^2\cos^2\alpha + u^2t^2\sin^2\alpha - ugt^3\sin\alpha + \tfrac{1}{4}g^2t^4 = u^2t^2 - ugt^3\sin\alpha + \tfrac{1}{4}g^2t^4.$$

Ask yourself Differentiate with respect to t. After taking out a factor of t, what is left? When is it positive for all t?

$$\frac{\mathrm{d}}{\mathrm{d}t}OP^2 = 2u^2t - 3ugt^2\sin\alpha + g^2t^3 = t\left(g^2t^2 - 3ug\sin\alpha\, t + 2u^2\right).$$

For $t > 0$, the sign is that of the quadratic $g^2t^2 - 3ug\sin\alpha\, t + 2u^2$. This quadratic has positive leading coefficient, so it is positive for every t if its discriminant is negative:

$$9u^2g^2\sin^2\alpha - 8u^2g^2 < 0 \quad\Longleftrightarrow\quad \sin^2\alpha < \frac{8}{9} \quad\Longleftrightarrow\quad \sin\alpha < \frac{2\sqrt{2}}{3}.$$

In that case OP is increasing throughout the flight. ■

STAGE 2 Steeper throws

Ask yourself If $\sin\alpha > \frac{2\sqrt{2}}{3}$, the quadratic has two real roots. Are they positive? Is the smaller one before the particle lands?

The roots have sum $\frac{3u\sin\alpha}{g} > 0$ and product $\frac{2u^2}{g^2} > 0$, so both are positive, and between them the quadratic, and so $\frac{\mathrm{d}}{\mathrm{d}t}OP^2$, is negative. The particle lands at $T = \frac{2u\sin\alpha}{g}$. The smaller root is

$$t_1 = \frac{u}{2g}\left(3\sin\alpha - \sqrt{9\sin^2\alpha - 8}\right) < \frac{u}{2g}\cdot 3\sin\alpha < \frac{2u\sin\alpha}{g} = T.$$

So just after t_1 (and before landing) OP is decreasing. ■

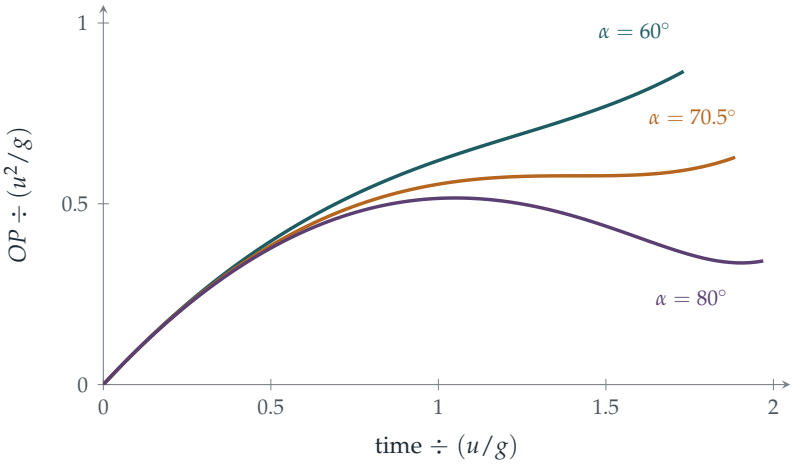


Figure 11.10.1 Distance from the thrower against time, until landing, for three angles. At 60° (teal) it always increases. At the critical angle $\arcsin \frac{2\sqrt{2}}{3} \approx 70.5°$ (amber) it pauses for an instant. At 80° (plum) the ball moves away, comes back towards the thrower for a while, and moves away again just before landing.

STAGE 3 A moving observer

Ask yourself Q moves away from O along the ground in the opposite direction, at speed v. Measured from Q, the horizontal separation is $x + vt$. What does that do to the calculation?

The separation QP has horizontal component $(u\cos\alpha + v)t$ and vertical component $ut\sin\alpha - \frac{1}{2}gt^2$. So

$$QP^2 = \big((u\cos\alpha + v)^2 + u^2\sin^2\alpha\big)t^2 - ugt^3\sin\alpha + \tfrac{1}{4}g^2t^4,$$

the same as before with u^2 replaced by $w^2 = (u\cos\alpha + v)^2 + u^2\sin^2\alpha$ in the first term. The derivative is $t\big(g^2t^2 - 3ug\sin\alpha\, t + 2w^2\big)$, and the condition for it to be positive for all $t > 0$ is

$$9u^2\sin^2\alpha < 8w^2 \quad\Longleftrightarrow\quad u^2\sin^2\alpha < 8(u\cos\alpha + v)^2,$$

after cancelling $8u^2\sin^2\alpha$ from both sides. Taking square roots (both $u\sin\alpha$ and $u\cos\alpha + v$ are positive), this is $u\sin\alpha < 2\sqrt{2}(u\cos\alpha + v)$, which rearranges to

$$2\sqrt{2}\,v > (\sin\alpha - 2\sqrt{2}\cos\alpha)u. \qquad \blacksquare$$

(This is sufficient for QP to increase throughout the flight, since the derivative is then positive for all $t > 0$.)

Looking back

- **Differentiate the square.** Squared distances are polynomials, and their derivatives have the same sign as the derivative of the distance.

- **A quadratic that is always positive.** The condition "increasing for all time" became a discriminant condition.
- **A new problem as an old one.** The moving observer only changed one coefficient, so the earlier analysis applied directly.

What's really going on

A geometric view

OP decreases exactly when the velocity of P points "inwards", making an obtuse angle with the vector $\overrightarrow{OP}$, that is, when $\overrightarrow{OP} \cdot \mathbf{v} < 0$. For a steep throw, near the top of the flight the particle is high above and slightly in front of the thrower, moving almost horizontally away, but starting to fall; once the downward part of its velocity outweighs the outward part, relative to the direction of $\overrightarrow{OP}$, it comes closer. For shallow throws the particle is always far enough in front that its velocity never points back towards O. The critical case is when the velocity is perpendicular to $\overrightarrow{OP}$ at exactly one instant, the moment when the distance pauses in Figure 11.10.1.

Monotonicity is not automatic

Many quantities that "obviously" increase do not. The distance of the Moon from the Earth oscillates; the distance between two cars on a winding road can decrease even when one is always ahead; the population of a predator can fall while its prey multiplies. Proving that a quantity is monotonic usually means showing that a derivative has a fixed sign, and the technique here (reduce to a polynomial, then check its roots or discriminant) is the standard way to do it. In analysis, monotonicity is often the key to proving convergence: a bounded monotonic sequence converges, as you saw in Chapter 2.

The thinking skill

Check monotonicity through a derivative

To decide whether a quantity always increases, differentiate it (or a convenient function of it, such as its square) and ask whether the derivative can change sign.

At university, sign arguments of this kind are used to prove inequalities, to locate maxima and minima, and to study the stability of solutions of differential equations, where a quantity that always decreases (as in Section 8.2) proves that a system settles down.

Try this next

1. **How close?** For $\alpha = 80^\circ$, find the times at which OP has its local maximum and minimum, and the distances there, in units of u^2/g.

2. **On a slope.** Repeat part (i) for a ball thrown up a plane inclined at angle β to the horizontal, landing on the plane. Does the critical angle (measured from the plane) change?

3. **With air resistance.** Suppose the ball also experiences a resistance proportional to its velocity. Without calculating, would you expect the critical angle to be larger or smaller than 70.5°? Why?

4. **Throwing from a height.** If the ball is thrown from the top of a cliff and lands far below, show that OP eventually increases for any angle, and find whether there are angles for which it decreases at some stage.

Further reading

- **James Walker, *The Flying Circus of Physics*** (Wiley). Hundreds of surprising everyday physics puzzles.
- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 3, on projectiles, with several counter-intuitive problems.
- **J. M. Steele, *The Cauchy–Schwarz Master Class*** (Cambridge University Press). The art of proving that one thing is always bigger than another.

11.11 How Fielders Catch

The question

A fielder, who is perfectly placed to catch a ball struck by the batsman in a game of cricket, watches the ball in flight. Assuming that the ball is struck at the fielder's eye level and is caught just in front of her eye, show that $\frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta)$ is constant, where θ is the angle between the horizontal and the fielder's line of sight.

In order to catch the next ball, which is also struck towards her but at a different velocity, the fielder runs at constant speed v towards the batsman. Assuming that the ground is horizontal, show that the fielder should choose v so that $\frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta)$ remains constant.

The hook

A ball is struck high into the outfield. A good fielder sets off at once and arrives just as it comes down. How? Nobody can solve the equations of projectile motion in their head in a fraction of a second, and in any case the fielder does not know the ball's speed or angle. Yet fielders, and dogs catching frisbees, do it routinely.

In 1968 the physicist Seville Chapman proposed an answer, and this question proves it. If the fielder watches the angle of elevation θ of the ball, and moves so that $\tan\theta$ increases at a *constant rate*, she will arrive exactly where the ball lands. She needs no knowledge of physics, only the ability to notice whether the ball seems to be rising faster or slower than steadily. A hard prediction problem is replaced by a simple feedback rule.

Before you start

Ask yourself Put the batsman at the origin and the fielder at distance D. If the ball is at (x, y), what is $\tan\theta$?

The ball is at height y above the fielder's eye, and at horizontal distance $D - x$ in front of her (it is coming towards her). So $\tan\theta = \frac{y}{D-x}$.

Working it through

STAGE 1 The perfectly placed fielder

Ask yourself Let the ball have horizontal speed U and initial vertical speed W, and let it land (at eye level) at time T. Write x, y and D in terms of U, W, T, g and t. Can you factorise y?

We have $x = Ut$ and $y = Wt - \frac{1}{2}gt^2$, and the ball returns to eye level at $T = \frac{2W}{g}$, where the fielder is: $D = UT$. Since $W = \frac{1}{2}gT$,

$$y = \tfrac{1}{2}gt(T - t), \qquad D - x = U(T - t).$$

The factor $T - t$ cancels:

$$\tan\theta = \frac{\frac{1}{2}gt(T-t)}{U(T-t)} = \frac{g}{2U}\,t.$$

So $\frac{\mathrm{d}}{\mathrm{d}t}\tan\theta = \frac{g}{2U}$, a constant. ■

STAGE 2 The running fielder

Ask yourself Now the fielder starts at distance D_0 and runs towards the batsman at speed v, so her position is $D_0 - vt$. She catches the ball at time T. What is D_0? Repeat the calculation.

To catch the ball at time T, she must be where it lands: $D_0 - vT = UT$, so $D_0 = (U+v)T$. Her horizontal distance from the ball is $D_0 - vt - Ut = (U+v)(T-t)$, and again the factor cancels:

$$\tan\theta = \frac{\frac{1}{2}gt(T-t)}{(U+v)(T-t)} = \frac{g}{2(U+v)}\,t.$$

So if she is running at the right speed, $\tan\theta$ increases at a constant rate.

Ask yourself The question asks for the converse: if she runs so that $\frac{\mathrm{d}}{\mathrm{d}t}\tan\theta$ is constant, will she catch the ball?

Suppose she runs at speed v and $\tan\theta = \frac{\frac{1}{2}gt(T-t)}{D_0-(U+v)t}$ increases at a constant rate c, so $\tan\theta = ct$ (it is 0 at $t = 0$). Then

$$\tfrac{1}{2}g(T-t) = c\big(D_0 - (U+v)t\big) \quad \text{for all } t.$$

Comparing the coefficients of t and the constant terms: $\frac{1}{2}g = c(U+v)$ and $\frac{1}{2}gT = cD_0$. Dividing, $D_0 = (U+v)T$: exactly the condition to be at the landing point at time T. So she should choose v to keep $\tan\theta$ rising steadily, and then she catches the ball. ■

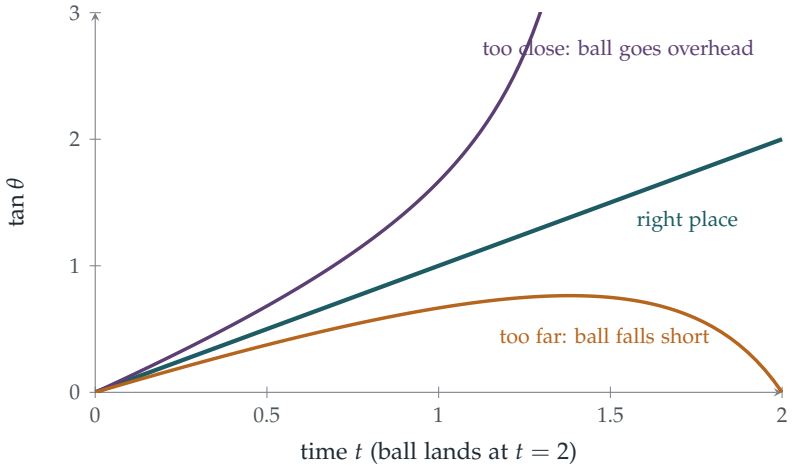


Figure 11.11.1 What a stationary fielder sees, for a ball that lands at distance 2 (with $g = 2$ and horizontal speed 1). In the right place, $\tan\theta$ rises in a straight line (teal). Standing too close, $\tan\theta$ accelerates upwards and the ball goes over her head (plum); too far back, it rises ever more slowly and then falls (amber). The rule: move so that the image of the ball rises steadily.

Looking back

- **Look for a common factor.** Both the height and the horizontal distance contained $T - t$, and cancelling it made $\tan\theta$ linear.
- **Prove the converse.** Showing that the rule *works* (it leads to the catch) needed the reverse argument, comparing coefficients.
- **An invariant that is easy to perceive.** A steady rise in $\tan\theta$ can be judged by eye; the landing point cannot.

What's really going on

Optical acceleration cancellation

Chapman's rule is now called *optical acceleration cancellation*: the fielder moves so as to keep the "optical acceleration" $\frac{d^2}{dt^2}\tan\theta$ at zero. If the ball's image accelerates upwards, the ball will go over her head, so she backs away; if it decelerates, the ball will fall short, so she runs in (Figure 11.11.1). The strategy does not tell her where the ball will land, but it gets her there. It is a feedback control law, like the ones that steer guided missiles: missiles use *proportional navigation*, keeping the line-of-sight angle to the target constant, which is the sideways version of the same idea.

What fielders really do

Do real fielders use Chapman's rule? Experiments in the 1990s, with cameras mounted on the heads of baseball outfielders, gave mixed results. Michael McBeath and colleagues argued in *Science* in 1995 that, for balls hit to one side, fielders run along curved paths that keep the ball's image moving in a straight line across their field of view (a "linear optical trajectory"), which also leads them to the catch. Later work has suggested that both strategies, and a mixture of them, are used, and that fielders often do not run at constant speed. A 2004 study found that dogs catching frisbees follow paths consistent with the same optical rules. Whatever the details, the remarkable conclusion stands: catching does not require prediction, only the right feedback.

Chapman's paper

Seville Chapman's "Catching a baseball", published in the *American Journal of Physics* in 1968, is three pages long and contains essentially the calculation in this question. He noted that the strategy works only without air resistance, and that a real baseball's trajectory is distorted by drag and spin, which would complicate things. Nevertheless, the idea that the brain uses simple invariants of the visual field to solve complicated control problems has become important in psychology and robotics.

The thinking skill

Find an invariant that is easy to observe

When a quantity is hard to predict, look for a related quantity that can be observed directly and that behaves simply exactly when things are on track.

Here the landing point was hard to predict, but the rate of change of $\tan\theta$ was easy to monitor. At university this is the idea of feedback control, and of choosing an observable that encodes the information you need.

Try this next

1. **Running backwards.** If the ball will land behind the fielder's starting position, show that she should run away from the batsman, and that the same rule applies.

2. **Not θ itself.** Show that, for a perfectly placed fielder, $\frac{\mathrm{d}\theta}{\mathrm{d}t}$ is not constant. Why is $\tan\theta$ the right quantity to watch?

3. **Eye level.** If the ball is struck from a height h below the fielder's eye level, is $\tan\theta$ still linear in t for a well-placed fielder? (Assume she catches it at eye level.)

4 **Air resistance.** With linear air resistance, the horizontal velocity decays exponentially. Is $\tan\theta$ still linear in t for a well-placed fielder?

Further reading

- **Seville Chapman, "Catching a baseball"**, *American Journal of Physics* 36 (1968).
- **Michael McBeath, Dennis Shaffer & Mary Kaiser, "How baseball outfielders determine where to run to catch fly balls"**, *Science* 268 (1995).
- **Dennis Shaffer et al., "How dogs navigate to catch Frisbees"**, *Psychological Science* 15 (2004).

11.12 The Ant on the Rubber Rope

The question

A uniform elastic string lies on a smooth horizontal table. One end of the string is attached to a fixed peg, and the other end is pulled at constant speed u. At time $t = 0$, the string is taut and its length is a. Obtain an expression for the speed, at time t, of the point on the string which is a distance x from the peg at time t.

An ant walks along the string starting at $t = 0$ at the peg. The ant walks at constant speed v along the string (so that its speed relative to the peg is the sum of v and the speed of the point on the string beneath the ant). At time t, the ant is a distance x from the peg. Write down a first order differential equation for x, and verify that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{x}{a+ut}\right) = \frac{v}{a+ut}.$$

Show that the time T taken for the ant to reach the end of the string is given by

$$uT = a(e^k - 1),$$

where $k = u/v$.

On reaching the end of the string, the ant turns round and walks back to the peg. Find in terms of T and k the time taken for the journey back.

The hook

An ant sets off along a rubber rope one kilometre long, walking at one centimetre per second. Every second, the rope is stretched uniformly by another kilometre. The far end races away at a thousand metres per second, a hundred thousand times faster than the ant. Does the ant ever reach the end?

Astonishingly, yes. The stretching carries the ant along with it, and although the ant's progress as a *fraction* of the rope shrinks, the fractions add up like the harmonic series, which diverges (Section 2.3). The ant arrives, though only after a time that dwarfs the age of the universe. This question solves the continuous version exactly, and finds that the time grows exponentially with the ratio of the speeds.

Before you start

Ask yourself The string is stretched uniformly: every piece is stretched by the same factor. At time t its length is $a+ut$. If the point at distance x from the peg is a fraction $\frac{x}{a+ut}$ of the way along, how fast does it move?

Each point on the string keeps the same fraction f of the length, so its distance from the peg is $f(a+ut)$, and its speed is $fu = \frac{ux}{a+ut}$. Points near the peg barely move; the far end moves at u.

Working it through

STAGE 1 The ant's equation

The ant moves at v relative to the string, plus the speed of the string beneath it:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v + \frac{ux}{a+ut}.$$

Ask yourself Differentiate $\frac{x}{a+ut}$ by the quotient rule and use the equation for $\dot{x}$.

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{x}{a+ut}\right) = \frac{\dot{x}}{a+ut} - \frac{ux}{(a+ut)^2} = \frac{1}{a+ut}\left(v + \frac{ux}{a+ut}\right) - \frac{ux}{(a+ut)^2} = \frac{v}{a+ut}. \qquad \blacksquare$$

The fraction of the string the ant has covered increases at the rate $\frac{v}{a+ut}$: its own speed divided by the current length. The stretching drops out completely, because it does not change anyone's fraction.

STAGE 2 Reaching the end

Ask yourself Integrate from 0 to t. When does the fraction reach 1?

The ant starts at the peg, so

$$\frac{x}{a+ut} = \int_0^t \frac{v\,\mathrm{d}s}{a+us} = \frac{v}{u}\ln\frac{a+ut}{a}.$$

The ant reaches the end when the fraction is 1: $\ln\frac{a+uT}{a} = \frac{u}{v} = k$, so $a+uT = ae^k$, that is

$$uT = a(e^k - 1). \qquad \blacksquare$$

The logarithm grows without limit, so the ant always arrives, however fast the rope is stretched.

STAGE 3 The way back

Ask yourself Walking back, the ant's speed relative to the string is $-v$. What is the rate of change of its fraction now? It starts at fraction 1 at time T.

The same calculation with $-v$ gives $\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{x}{a+ut}\right) = -\frac{v}{a+ut}$, so

$$\frac{x}{a+ut} = 1 - \frac{v}{u}\ln\frac{a+ut}{a+uT}.$$

The fraction reaches 0 when $a+ut = (a+uT)e^k$. Since $a+uT = ae^k$, the return time is

$$t - T = \frac{(a+uT)(e^k-1)}{u} = \frac{ae^k \cdot uT/a}{u} = Te^k. \qquad \blacksquare$$

The journey back takes e^k times as long as the journey out: by the time the ant turns round, the string is e^k times longer than at the start.

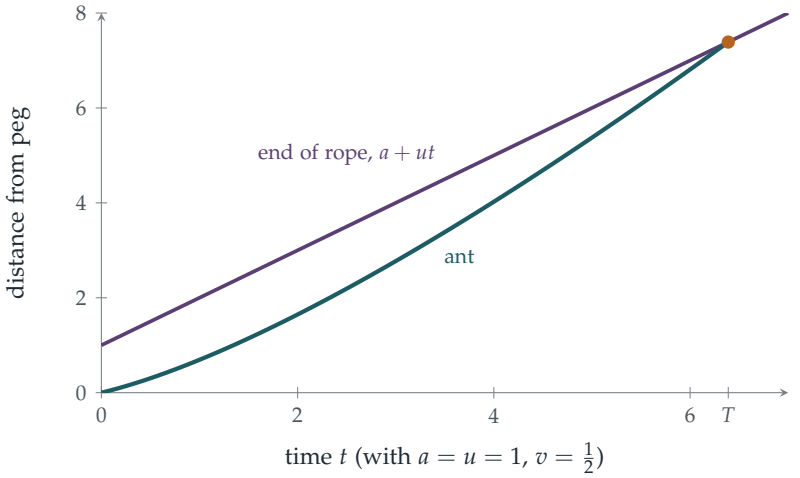


Figure 11.12.1 The ant (teal) walking along a rope whose end (plum) moves at twice the ant's walking speed ($k = 2$). The ant is carried along by the stretching, and its distance from the peg grows faster and faster. It reaches the end at $T = e^2 - 1 \approx 6.39$.

Looking back

- **Use the right variable.** The fraction of the rope covered was the natural quantity: the stretching does not change it.
- **Integrating factor in disguise.** Multiplying by $\frac{1}{a+ut}$ turned a linear differential equation into an exact derivative.
- **Divergence beats speed.** The fraction grew like a logarithm, which is unbounded, so the ant always arrives.

What's really going on

The harmonic series version

The puzzle is usually told with discrete stretches. A rope is 1 km long; the ant walks 1 cm in the first second; then the rope is stretched by 1 km; then the ant walks another centimetre, and so on. In the nth second, the ant covers 1 cm of a rope n km long: a fraction $\frac{1}{100\,000\,n}$. Stretching does not change its fraction, so after N seconds it has covered

$$\frac{1}{100\,000}\left(1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{N}\right)\approx\frac{\ln N+0.577}{100\,000}$$

of the rope. It arrives when $H_N \approx 100\,000$, after about $e^{100\,000}$ seconds, a number with over 43 000 digits. (For comparison, the universe is about 4×10^{17} seconds old.) Our continuous answer agrees: $k=\frac{u}{v}=100\,000$, and $T\approx\frac{a}{u}e^{k}$.

Light in an expanding universe

The ant on the rope is a model of light crossing an expanding universe. Distant galaxies recede from us, carried by the expansion of space itself, and light from them travels towards us at speed c *through* the expanding space, as the ant walks along the stretching rope. Whether light from a distant galaxy ever reaches us depends on how the expansion rate changes with time. In a universe expanding at a steady rate, like the rope, light always arrives eventually. In a universe whose expansion accelerates, as ours appears to, there are galaxies whose light will never reach us: there is a *cosmic event horizon*. The analogue for the ant is a rope stretched exponentially, so that the end's speed grows in proportion to the length (try it in the exercises).

Gardner's ant

The puzzle seems to have been invented by the French mathematician Denys Wilquin in 1972, as a worm on an elastic band, and was popularised by Martin Gardner in his *Scientific American* column and in his book *Time Travel and Other Mathematical Bewilderments*. It is one of the best-known examples of a result that contradicts intuition and is nevertheless easy to prove, and it is a favourite illustration of the divergence of the harmonic series.

The thinking skill

Find the quantity the complication does not affect

When part of a problem changes everything (here, the stretching), look for a quantity it leaves alone.

Stretching changes every distance but no fraction. Working with the fraction removed the stretching from the problem. At university, this idea appears as *comoving coordinates* in cosmology and fluid mechanics, and more generally as the search for invariants under a transformation.

Try this next

1. **Halfway.** When is the ant halfway along the rope? Is that before or after time $\frac{T}{2}$?
2. **Exponential stretching.** Suppose instead the rope's length is $ae^{\lambda t}$. Show that the ant reaches the end only if $v > \lambda a$. (This is the cosmic event horizon.)
3. **An accelerating rope.** If the length is $a + \frac{1}{2}bt^2$, does the ant always arrive?
4. **The discrete puzzle.** In the discrete version, how many seconds does the ant need to cover 1% of the rope? Use $H_N \approx \ln N + 0.577$.

Further reading

- **Martin Gardner,** ***Time Travel and Other Mathematical Bewilderments*** (W. H. Freeman). Includes the ant on the rubber rope.
- **Julian Havil,** ***Gamma: Exploring Euler's Constant*** (Princeton University Press). The harmonic series, and the ant.
- **Barbara Ryden,** ***Introduction to Cosmology*** (Cambridge University Press). Expanding space, horizons and comoving coordinates, at undergraduate level.

11.13 The Snowplough Problem

The question

Two identical snowploughs plough the same stretch of road in the same direction. The first starts at time $t = 0$ when the depth of snow is h metres, and the second starts from the same point T seconds later. Snow falls so that the depth of snow increases at a constant rate of k m s^{-1}. It may be assumed that each snowplough moves at a speed equal to $k/(\alpha z)$ m s^{-1} with α a constant, where z is the depth of snow it is ploughing, and that it clears all the snow.

(i) Show that the time taken for the first snowplough to travel x metres is

$$(e^{\alpha x} - 1)\frac{h}{k} \text{ seconds.}$$

(ii) Show that at time $t > T$, the second snowplough has moved y metres, where t satisfies

$$\frac{1}{\alpha}\frac{\mathrm{d}t}{\mathrm{d}y} = t - (e^{\alpha y} - 1)\frac{h}{k}.$$

Verify that the required solution of this equation is

$$t = (e^{\alpha y} - 1)\frac{h}{k} + \left(T - \frac{\alpha h y}{k}\right)e^{\alpha y}$$

and deduce that the snowploughs collide when they have moved a distance $kT/(\alpha h)$ metres.

The hook

Snow is falling steadily. A snowplough sets off down a road, slowing down as the snow it meets gets deeper. A second plough follows a little later along the cleared road. It meets only the snow that has fallen since the first plough passed, which is shallower, so it travels faster, and eventually it catches up. When?

This is a version of a famous problem published by Ralph Agnew in 1942, which became one of the best-known puzzles in differential equations. Its charm is that the second plough's speed depends on *when the first plough passed each point*, so the solution to part (i) feeds into the equation for part (ii). And the equation is cleverest when written "upside down", with time as a function of distance.

Before you start

Ask yourself The first plough meets snow of depth $h + kt$ at time t. What is its speed? Is it easier to find x as a function of t, or t as a function of x?

Its speed is $\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{k}{\alpha(h+kt)}$. This separates neatly. Turning it over, $\frac{\mathrm{d}t}{\mathrm{d}x} = \frac{\alpha(h+kt)}{k}$, which is a simple linear equation for t as a function of x. Either way works for the first plough; for the second, the upside-down form is essential.

Working it through

STAGE 1 The first plough

From $\frac{\mathrm{d}t}{\mathrm{d}x} = \alpha\left(\frac{h}{k} + t\right)$, the quantity $\frac{h}{k} + t$ grows exponentially in x: $\frac{h}{k} + t = Ce^{\alpha x}$. At $x = 0$, $t = 0$, so $C = \frac{h}{k}$ and

$$t_1(x) = \frac{h}{k}\left(e^{\alpha x} - 1\right). \qquad \blacksquare$$

The first plough slows down exponentially: equal distances take exponentially increasing times.

STAGE 2 The second plough's equation

Ask yourself The second plough reaches position y at time t. How deep is the snow there? (What was the depth just after the first plough passed?)

The first plough cleared all the snow at position y at time $t_1(y)$. Since then, snow has accumulated at rate k, so the depth at time t is $z = k\big(t - t_1(y)\big)$. The second plough's speed is therefore

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{k}{\alpha k\big(t - t_1(y)\big)} = \frac{1}{\alpha\big(t - t_1(y)\big)},$$

and turning it over,

$$\frac{1}{\alpha}\frac{\mathrm{d}t}{\mathrm{d}y} = t - \frac{h}{k}\left(e^{\alpha y} - 1\right). \qquad \blacksquare$$

STAGE 3 Solving

Ask yourself This is linear in t: $\frac{\mathrm{d}t}{\mathrm{d}y} - \alpha t = -\frac{\alpha h}{k}(e^{\alpha y} - 1)$. What integrating factor works? What is the starting condition?

Multiplying by $e^{-\alpha y}$,

$$\frac{\mathrm{d}}{\mathrm{d}y}\left(te^{-\alpha y}\right) = -\frac{\alpha h}{k}\left(1 - e^{-\alpha y}\right), \qquad te^{-\alpha y} = -\frac{\alpha h}{k}y - \frac{h}{k}e^{-\alpha y} + C.$$

The second plough starts at $y = 0$ at $t = T$, so $T = -\frac{h}{k} + C$, giving $C = T + \frac{h}{k}$. Multiplying back by $e^{\alpha y}$ and rearranging,

$$t_2(y) = \frac{h}{k}\left(e^{\alpha y} - 1\right) + \left(T - \frac{\alpha h y}{k}\right)e^{\alpha y}. \qquad \blacksquare$$

(To verify, as the question asks: differentiate and substitute; the terms combine to $\alpha\big(t - t_1(y)\big)$.)

STAGE 4 The collision

Ask yourself The ploughs collide when both reach the same point at the same time. Compare $t_2(y)$ with $t_1(y)$.

The difference is $t_2(y) - t_1(y) = \left(T - \frac{\alpha h y}{k}\right)e^{\alpha y}$, the time gap between the two ploughs at position y. It shrinks to zero when

$$y = \frac{kT}{\alpha h}. \qquad \blacksquare$$

At that point the second plough reaches the first: the snow in front of it has zero depth, and in the model its speed becomes infinite (Figure 11.13.1).

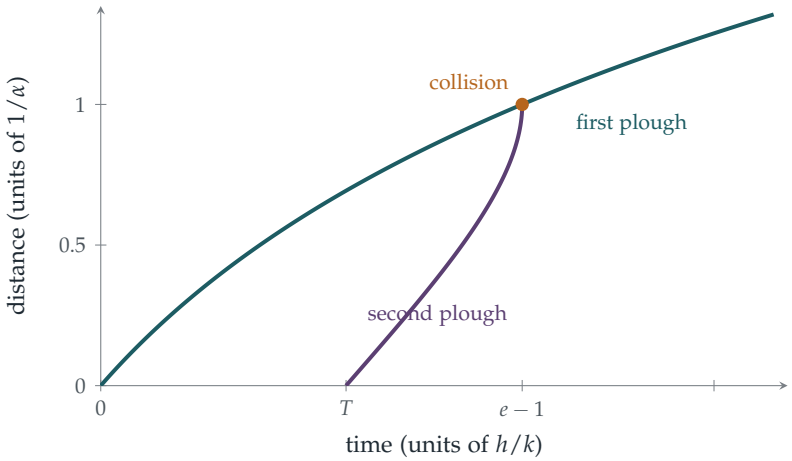


Figure 11.13.1 Distance against time for the two ploughs, with $T = \frac{h}{k}$ (so the second starts once the snow depth has doubled). The first plough (teal) slows as the snow deepens. The second (plum) starts later on a partly cleared road and speeds up as it closes the gap, meeting the first at distance $\frac{kT}{\alpha h} = \frac{1}{\alpha}$, at time $(e-1)\frac{h}{k}$.

Looking back

- **Turn the equation upside down.** With t as a function of position, both equations were linear.

- **One solution feeds the next.** The first plough's timetable $t_1(y)$ appeared in the second plough's equation.
- **Compare, do not solve twice.** The collision condition came from the difference $t_2 - t_1$, which had a simple factor.

What's really going on

Agnew's snowplough

In Ralph Palmer Agnew's version, from his 1942 textbook *Differential Equations*, it starts snowing heavily and steadily one morning; a snowplough sets out at noon and covers 2 miles in the first hour and 1 mile in the second; when did it start snowing? The plough's speed is inversely proportional to the depth, which is proportional to the time since the snow started. The answer (an exercise below) involves the golden ratio: it started snowing about 37 minutes before noon. Versions with two ploughs, like this question, followed.

Information that travels

The second plough's motion depends on the history recorded in the road: the time at which each point was last cleared. This is a simple example of a system with *memory*, in which the state at one place depends on what happened there earlier. Traffic flow, the motion of glaciers and the spreading of a forest fire all have this character, and their mathematics, the theory of first-order partial differential equations and their *characteristics*, is a standard second-year topic. The idea of writing time as a function of position ("when does the front arrive here?") is exactly the method of *arrival times* used for such problems.

The thinking skill

Choose the independent variable

A differential equation may be much simpler with the roles of the variables reversed. If the rate depends on the time at which something happened at a place, write time as a function of place.

Both equations here became linear when written for $t(y)$. At university, the freedom to choose which variable is independent (and to change it when convenient) is used constantly, from the hodograph method in fluid mechanics to the use of arc length as a parameter in geometry.

Try this next

1. **Agnew's problem.** Snow starts falling at a steady rate at an unknown time before noon. A snowplough whose speed is inversely proportional to the depth starts at noon and goes 2 miles in the first hour and 1 mile in the second. Show that if it started snowing t_0 hours before noon, then $\left(1+\frac{2}{t_0}\right)^2 = \left(1+\frac{1}{t_0}\right)^3$, deduce that $\frac{1}{t_0}$ is the golden ratio, and find when it started snowing.

2. **Verification.** Verify by differentiation that $t_2(y)$ satisfies the differential equation of part (ii).

3. **The time gap.** Show that if $kT > h$, the time gap $t_2(y) - t_1(y)$ first increases and then decreases to zero, and find where it is greatest. What happens if $kT \le h$? Interpret this.

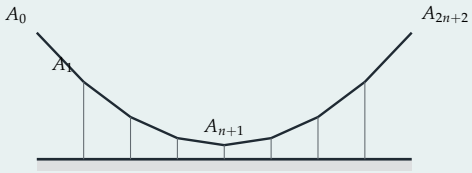
4. **Three ploughs.** A third plough starts at time $2T$. Write down the equation for its motion. Does it catch the second plough before or after the second catches the first?

Further reading

- **Ralph Palmer Agnew, *Differential Equations*** (McGraw-Hill, 1942). The original snowplough problem.
- **George Simmons, *Differential Equations with Applications and Historical Notes*** (CRC Press), which discusses the snowplough problem among many others.
- **Richard Haberman, *Mathematical Models*** (SIAM). Traffic flow and characteristics, at an accessible level.

11.14 Parabola, Not Catenary

The question



The above diagram represents a suspension bridge. A heavy uniform horizontal roadway is attached by vertical struts to a light flexible chain at points $A_1 = (x_1, y_1)$, $A_2 = (x_2, y_2), \ldots, A_{2n+1} = (x_{2n+1}, y_{2n+1})$, where the coordinates are referred to horizontal and vertically upward axes Ox, Oy. The chain is fixed to external supports at points

$$A_0 = (x_0, y_0) \quad \text{and} \quad A_{2n+2} = (x_{2n+2}, y_{2n+2})$$

at the same height. The weight of the chain and struts may be neglected. Each strut carries the same weight w. The horizontal spacing h between A_i and A_{i+1} (for $0 \leq i \leq 2n+1$) is constant. Write down equations satisfied by the tensions T_i in the portion $A_{i-1}A_i$ of the chain for $1 \leq i \leq n+1$. Hence or otherwise show that

$$\frac{h}{y_n - y_{n+1}} = \frac{3h}{y_{n-1} - y_n} = \cdots = \frac{(2n+1)h}{y_0 - y_1}.$$

Verify that the points $A_0, A_1, \ldots, A_{2n+1}, A_{2n+2}$ lie on a parabola.

The hook

Hold a chain by its two ends and it hangs in a curve called the *catenary* (from the Latin for chain). Galileo thought it was a parabola; it is not, as the teenage Huygens showed in 1646. But hang a heavy roadway from the chain by evenly spaced struts, as in a suspension bridge, and the chain *does* form a parabola. The difference is in how the weight is spread: along the curved chain itself for the catenary, evenly along the horizontal for the bridge.

This question proves the parabola with nothing more than resolving forces at each joint. The key is that the horizontal component of tension is the same all the way along, so each segment's slope is proportional to the weight it supports. The result is a beautiful pattern: the drops between successive joints go up in the odd numbers $1, 3, 5, 7, \ldots$,

whose partial sums are the perfect squares.

Before you start

Ask yourself At each joint A_i, three forces act: the tensions in the two neighbouring segments, and the weight w hanging from the strut. Which component of tension can the weight not affect?

The weight is vertical, so at each joint the horizontal components of the two tensions must balance. So the horizontal component of tension, H say, is the same in every segment of the chain.

Working it through

STAGE 1 The equations at each joint

Let segment $A_{i-1}A_i$ make angle ϕ_i below the horizontal (going from A_{i-1} towards the middle), so its tension T_i has horizontal component $T_i\cos\phi_i$ and vertical component $T_i\sin\phi_i$. Resolving at A_i for $1 \le i \le n$:

$$T_i\cos\phi_i = T_{i+1}\cos\phi_{i+1} = H, \qquad T_i\sin\phi_i - T_{i+1}\sin\phi_{i+1} = w.$$

(The segment on the outer side pulls up and out with the larger vertical component; the inner segment pulls in and slightly down; the difference supports the strut.)

Ask yourself What happens at the middle joint A_{n+1}? Use the symmetry of the bridge.

By symmetry, the two segments meeting at A_{n+1} are mirror images, so each supports half of its strut: $T_{n+1}\sin\phi_{n+1} = \frac{1}{2}w$.

STAGE 2 Slopes from weights

Working outwards from the middle, the vertical components are

$$T_{n+1}\sin\phi_{n+1} = \tfrac{1}{2}w, \quad T_n\sin\phi_n = \tfrac{3}{2}w, \quad \ldots, \quad T_1\sin\phi_1 = \tfrac{2n+1}{2}w.$$

Dividing each by H, the slopes are $\tan\phi_i = \frac{(2(n+1-i)+1)w}{2H}$. The slope of segment $A_{i-1}A_i$ is also $\frac{y_{i-1}-y_i}{h}$, so

$$\frac{y_n - y_{n+1}}{h} = \frac{w}{2H}, \qquad \frac{y_{n-1}-y_n}{h} = \frac{3w}{2H}, \qquad \ldots, \qquad \frac{y_0 - y_1}{h} = \frac{(2n+1)w}{2H}.$$

Inverting and multiplying by the odd numbers, each of $\frac{h}{y_n-y_{n+1}}, \frac{3h}{y_{n-1}-y_n}, \ldots, \frac{(2n+1)h}{y_0-y_1}$ equals $\frac{2H}{w}$. ■

STAGE 3 The parabola

Ask yourself Let j count joints outwards from the middle, so that A_{n+1-j} is at horizontal distance jh from A_{n+1}. How far is it above A_{n+1}?

Adding the drops, with $c = \frac{wh}{2H}$:

$$y_{n+1-j} - y_{n+1} = c\big(1 + 3 + 5 + \cdots + (2j-1)\big) = cj^2.$$

With x measured from the middle, $x = jh$, so the height above the lowest joint is $\frac{c}{h^2}x^2 = \frac{w}{2Hh}x^2$. By symmetry the same holds on the other side, and the supports ($j = n+1$) fit the same formula. So all the points lie on the parabola

$$y = y_{n+1} + \frac{w}{2Hh}(x - x_{n+1})^2. \qquad \blacksquare$$

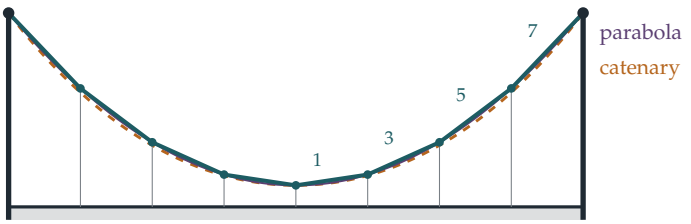


Figure 11.14.1 A chain carrying equal loads at equal horizontal spacings (teal). Its joints lie on a parabola (plum), and the drops between successive joints, from the middle outwards, are in the ratio 1 : 3 : 5 : 7. A free chain under its own weight, with the same ends and lowest point, would hang in a catenary (amber, dashed), which is flatter at the bottom and steeper near the supports.

Looking back

- **Find what is constant.** With only vertical loads, the horizontal tension is the same everywhere.
- **Accumulate from the middle.** Each segment supports all the weight between it and the middle, so the vertical components grow in steps of w.
- **Odd numbers sum to squares.** $1 + 3 + \cdots + (2j-1) = j^2$ turned a list of slopes into a parabola.

What's really going on

Parabola or catenary?

The continuous version of the argument explains both curves. For a chain $y(x)$ under constant horizontal tension H, the vertical component of tension is Hy', and over a short stretch it changes by the load carried there: $H\,\mathrm{d}y' = \mathrm{d}W$.

- If the load is spread evenly *horizontally* (a heavy roadway, weight ρ per unit length of road), then $Hy'' = \rho$, so y is a quadratic: a parabola.
- If the load is the chain's own weight, spread evenly *along the chain* (ρ per unit length of chain), then $Hy'' = \rho\sqrt{1 + y'^2}$, whose solution is the catenary $y = c\cosh\frac{x}{c}$, with $c = H/\rho$.

Near the bottom, where the chain is nearly horizontal, the two are almost the same; they differ near the supports, where the chain is steep and a given horizontal distance contains more chain. Real suspension bridges, whose deck is much heavier than their cables, are very close to parabolas.

Upside-down chains and arches

A chain is in pure tension; turn its shape upside down and you have an arch in pure compression, with no tendency to bend. Robert Hooke published this in 1675 as an anagram, which decoded reads: "As hangs the flexible line, so but inverted will stand the rigid arch." Christopher Wren and Hooke used the idea for the dome of St Paul's Cathedral, and Antoni Gaudí designed the Sagrada Família in Barcelona by hanging weighted strings from the ceiling and photographing them upside down. The Gateway Arch in St Louis is a (weighted) inverted catenary.

The catenary challenge

In 1690 Jakob Bernoulli challenged mathematicians to find the shape of a hanging chain. Within a year, solutions were published by his brother Johann, by Leibniz and by Huygens. Johann Bernoulli later recalled with glee that his brother, who had set the problem, had failed to solve it. The catenary was one of the first triumphs of the new calculus, and it led to the hyperbolic functions cosh and sinh, studied systematically by Vincenzo Riccati and Johann Heinrich Lambert in the eighteenth century.

The thinking skill

Resolve forces at every joint

For a structure made of joints and straight members, balance forces at each joint. Constant components and accumulating loads often reveal the shape.

At university this becomes the analysis of trusses and frameworks in engineering, and, in the continuous limit, the differential equations of cables, beams and membranes. The step from a discrete chain to a continuous one is the same move from sums to integrals seen in Section 10.3.

Try this next

1 **An even number of struts.** Repeat the question with $2n$ struts, so that the lowest point of the chain is a horizontal segment in the middle. Do the joints still lie on a parabola?

2 **The tension.** Find the greatest tension in the chain, in terms of H, w and n. Where is it?

3 **The catenary.** Verify that $y = c\cosh\frac{x}{c}$ satisfies $y'' = \frac{1}{c}\sqrt{1+y'^2}$. Show that the length of the catenary from $x = 0$ to $x = X$ is $c\sinh\frac{X}{c}$.

4 **Galileo's error.** A chain of length $2L$ hangs between two supports at distance $2d$ apart. For L only slightly larger than d, show that the sag of the catenary is approximately the same as that of a parabola. Why did Galileo's experiments not reveal the difference?

Further reading

- **E. H. Lockwood, *A Book of Curves*** (Cambridge University Press). The catenary, the parabola and their relatives.
- **Jacques Heyman, *The Stone Skeleton*** (Cambridge University Press). How masonry arches and domes stand up, from Hooke's chain onwards.
- **David Billington, *The Tower and the Bridge*** (Princeton University Press). The art and engineering of bridges.

What We Have Learned

We began with things that swing, things that spin, and things seen from somewhere that moves. We asked three questions: when is motion simple, what is the right point of view, and can intuition be trusted? Here is what the chapter's fourteen questions showed.

1. When is motion simple?

Simple harmonic motion, with a single frequency independent of amplitude, is the ideal. Driving it at its natural frequency produced resonance (Section 11.1), with beats just off resonance and unlimited growth exactly on it. Two coupled oscillators turned out to be two independent simple oscillators in disguise (Section 11.2), but with frequencies in the irrational ratio $\sqrt{3}$, so the combined motion never repeats. The real pendulum is not simple at all (Section 11.3): its period grows with amplitude, through an elliptic integral that Gauss's arithmetic–geometric mean computes with astonishing speed. And the tautochrone (Section 11.4) showed how to make motion exactly simple again, by bending the track into a cycloid.

2. What is the right point of view?

Almost every question was solved by choosing the right variable or the right frame.

- **Coordinates.** Arc length made the tautochrone simple; the combinations $x \pm y$ decoupled the springs; the fraction of the rope removed the stretching from the ant's problem (Section 11.12); writing time as a function of distance made the snowploughs' equations linear (Section 11.13).
- **Frames.** In the accelerating train (Section 11.5), a fictitious force tilted gravity, and the bead oscillated about a new lowest point. Einstein turned this observation into general relativity.
- **Points to take moments about.** For the spinning marble (Section 11.6), angular momentum about the contact point was conserved, because friction acts there, giving the final speed without knowing the friction.
- **Equivalent systems.** A rod behaves exactly like three particles (Section 11.7), with the weights of Simpson's rule; and the equilibria of a solid are where its normals pass through the centre of mass (Section 11.8).

3. Can intuition be trusted?

Often not. The boundary of the region a cannon can reach is not a trajectory but their envelope (Section 11.9). A ball thrown steeply enough comes back towards you before it lands (Section 11.10). A fielder can catch a ball without predicting where it will land, by keeping its image rising steadily (Section 11.11). An ant walking slowly along a rope stretched a hundred thousand times faster still reaches the end (Section 11.12). A hanging chain is not a parabola, but a chain carrying a bridge is (Section 11.14). In each case, a short calculation replaced a confident but wrong intuition with a precise statement, and usually with an explanation.

4. A toolkit for oscillations and frames

Method	Where you used it
Particular plus complementary solution	Resonance (11.1)
Decoupling by eigenvectors	Normal modes (11.2)
Energy integral and substitution	Pendulum (11.3)
The natural variable	Tautochrone (11.4); ant (11.12)
Fictitious forces	Train (11.5)
Angular momentum about a point	Backspin (11.6)
Matching moments	Simpson's rod (11.7)
Counting stationary points	Tilted equilibria (11.8)
Discriminants and envelopes	Safety parabola (11.9)
Sign of a derivative	Coming back (11.10)
Comparing coefficients	Fielders (11.11)
Integrating factors	Ant (11.12); snowplough (11.13)
Resolving at joints	Bridge (11.14)

Where next

- The next two chapters turn to probability and statistics, starting with Section 12.1.
- Normal modes are eigenvectors (Section 7.2), and the quantum harmonic oscillator of Section 8.6 is the oscillator of this chapter in quantum mechanics.
- Limit cycles (Section 8.4) are oscillations that sustain themselves, and chaos (Section 8.7) is what nonlinear oscillators can do when driven.

If you want to go further before university

- **A. P. French, *Vibrations and Waves***: the classic introduction to oscillations, resonance and normal modes.
- **David Morin, *Introduction to Classical Mechanics***: rotation, frames and many surprising problems.

- **Steven Strogatz, *Sync***: how oscillators fall into step, from fireflies to bridges.
- **George Simmons, *Differential Equations with Applications and Historical Notes***: the tautochrone, the brachistochrone and the snowplough, with their history.

Coleman joked that a physicist's career consists of treating the harmonic oscillator in ever more abstract ways. This chapter has tried to show why: the oscillator is the simplest motion there is, and almost everything looks like one if you choose the right point of view. Learning to find that point of view, a variable, a frame or a combination, is the most useful skill that mechanics teaches.

Answers to Try This Next

These are brief answers and hints, not full solutions: they give the result and the key step, so that you can check your own work. Try each exercise properly before looking.

11.1 Resonance

1. In the lift's frame gravity is effectively $g - f$, and the string's stiffness is $\frac{mg}{a}$ (it stretches by a under mg). So $\ddot{y} + \frac{g}{a}y = g - f$, with $y = a$, $\dot{y} = 0$ at the start. The new equilibrium is $a - \frac{af}{g}$, so $y = a - \frac{af}{g}(1 - \cos\omega t)$: amplitude $\frac{af}{g}$, and the least extension $a(1 - \frac{2f}{g})$ is positive. When the lift stops accelerating, the equilibrium returns to a, and the particle is at displacement $-\frac{af}{g}(1 - \cos\omega T)$ with velocity $-\frac{af}{g}\omega\sin\omega T$. The new amplitude is $\frac{af}{g}\sqrt{(1-\cos\omega T)^2 + \sin^2\omega T} = \frac{2af}{g}\left|\sin\frac{1}{2}\omega T\right|$.

2. $x = Ct\cos nt$ gives $\ddot{x} + n^2x = -2Cn\sin nt$, so $C = -\frac{nb}{2}$. Adding $\frac{b}{2}\sin nt$ to meet the initial conditions: $x = \frac{b}{2}(\sin nt - nt\cos nt)$. L'Hôpital's rule in p, applied to the formula in the question, gives the same as $p \to n$.

3. $n\sin pt - p\sin nt = n(\sin pt - \sin nt) + (n-p)\sin nt = 2n\cos\frac{(n+p)t}{2}\sin\frac{(p-n)t}{2} + (n-p)\sin nt$. The second term is small. The first is a fast oscillation inside the slowly varying envelope $\left|\sin\frac{(n-p)t}{2}\right|$, which repeats every $\frac{2\pi}{n-p}$.

4. The amplitude is $\frac{F}{\sqrt{(n^2-p^2)^2+4k^2p^2}}$. Minimising the denominator over p^2: $-2(n^2 - p^2) + 4k^2 = 0$, so $p^2 = n^2 - 2k^2$.

11.2 Normal Modes

1. Differentiating, the energy changes at rate $\dot{x}\left(m\ddot{x} + \frac{\lambda}{a}(2x+y)\right) + \dot{y}\left(m\ddot{y} + \frac{\lambda}{a}(x+2y)\right) = 0$. With $u = x - y$ and $v = x + y$, the energy is $\left(\frac{m}{4}\dot{u}^2 + \frac{\lambda}{4a}u^2\right) + \left(\frac{m}{4}\dot{v}^2 + \frac{3\lambda}{4a}v^2\right)$, and each bracket is the energy of a simple harmonic oscillator, so each is constant.

2. Period $\frac{2\pi}{\omega}$: start with $x = -y$ and $\dot{x} = -\dot{y}$ (so that $v \equiv 0$). Period $\frac{2\pi}{\sqrt{3}\omega}$: start with $x = y$ and $\dot{x} = \dot{y}$.

3. The force on X is $-\frac{\lambda}{a}x - \frac{\mu}{a}(x+y)$. The mode $x - y$ still has $\omega^2 = \frac{\lambda}{ma}$, and $x + y$ now has $\frac{\lambda+2\mu}{ma}$. Every motion is periodic exactly when the ratio of frequencies $\sqrt{1 + 2\mu/\lambda}$ is rational.

4. The matrix is $\frac{k}{m}\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$, with eigenvectors $(1,\sqrt{2},1)$, $(1,0,-1)$ and $(1,-\sqrt{2},1)$ for $2-\sqrt{2}$, 2 and $2+\sqrt{2}$. The modes: all three swing together, the middle one most; the outer two swing oppositely with the middle at rest; and the middle swings against the outer two.

11.3 The Real Pendulum

1. Energy: $\frac{1}{2}m\dot{x}^2 + \frac{\lambda}{2a}\big(\sqrt{x^2+a^2}-a\big)^2 = \frac{1}{2}mv^2$. The greatest displacement has $\sqrt{x^2+a^2} = a + \frac{v}{k}$, so $x_{\max} = \sqrt{\frac{2av}{k} + \frac{v^2}{k^2}}$. For small v the extension is about $\frac{x^2}{2a}$, so $\dot{x}^2 \approx v^2 - \frac{k^2x^4}{4a^2}$. With $x = Xu$ and $X = \sqrt{2av/k}$, the period $4\int_0^X \frac{\mathrm{d}x}{\dot{x}}$ becomes $\frac{4X}{v}\int_0^1 \frac{\mathrm{d}u}{\sqrt{1-u^4}}$. The restoring force is proportional to x^3, not x (at $x=0$ the string is exactly at its natural length), so there is no simple harmonic limit.

2. $(1-m\sin^2\phi)^{-1/2} \approx 1 + \frac{m}{2}\sin^2\phi + \frac{3m^2}{8}\sin^4\phi$ with $m = \sin^2\frac{\gamma}{2} \approx \frac{\gamma^2}{4} - \frac{\gamma^4}{48}$. Averaging over ϕ gives $1 + \frac{m}{4} + \frac{9m^2}{64}$, and the γ^4 terms are $-\frac{1}{192} + \frac{9}{1024} = \frac{11}{3072}$.

3. Starting from 1 and $\cos 45^\circ = 0.70711$: $(0.85355, 0.84090)$, then $(0.84722, 0.84720)$, then 0.847213. The period factor is $\frac{1}{0.847213} = 1.18034$.

4. The period factor changes by $\frac{1}{16}(\gamma_5^2 - \gamma_4^2) \approx 1.71\times10^{-4}$ (in radians), so the clock loses about 15 seconds a day.

11.4 The Tautochrone

1. Since $y = \frac{s^2}{4k}$, $\ddot{s} = -g\frac{\mathrm{d}y}{\mathrm{d}s} = -\frac{g}{2k}s$: simple harmonic motion with $\omega^2 = \frac{g}{2k}$. From $s_0 = 2\sqrt{kh}$, the speed at V is $\omega s_0 = \sqrt{2gh}$.

2. The bowl is a pendulum of length R. From a small height the time to the bottom is $\frac{\pi}{2}\sqrt{R/g}$. From the rim ($\gamma = 90^\circ$) it is 1.180 times longer, about $1.85\sqrt{R/g}$.

3. Yes. Both reach V after a quarter period and collide there. Whatever velocities they leave with, each performs simple harmonic motion in s with the same period, so each returns to V after exactly half a period, and they collide at V again. (A particle left at rest at V simply stays there.)

4. With $x = R(\theta - \sin\theta)$ and $y = R(1-\cos\theta)$, $\mathrm{d}s = 2R\sin\frac{\theta}{2}\,\mathrm{d}\theta$, so one arch has length $\int_0^{2\pi} 2R\sin\frac{\theta}{2}\,\mathrm{d}\theta = 8R$. The area is $\int y\,\mathrm{d}x = R^2\int_0^{2\pi}(1-\cos\theta)^2\,\mathrm{d}\theta = 3\pi R^2$.

11.5 Gravity in an Accelerating Train

1. In the carriage's frame there is a backward (southward) force a per unit mass, in the direction of firing. With flight time $\frac{2V\sin\theta}{g}$, the range is $\frac{V^2}{g}\big(\sin 2\theta + \frac{a}{g}(1-\cos 2\theta)\big)$. Its derivative vanishes when $\tan 2\theta = -\frac{g}{a}$, so $2\theta \approx \frac{\pi}{2} + \frac{a}{g}$. The maximum is $\frac{V^2}{g}\big(\sqrt{1+a^2/g^2} + \frac{a}{g}\big) \approx \frac{V^2}{g} + \frac{V^2 a}{g^2}$.

2. $x = ut\cos\theta - \frac{1}{2}ft^2$, landing at $t = \frac{2u\sin\theta}{g}$ and $x = \frac{2u^2\sin\theta}{g^2}(g\cos\theta - f\sin\theta)$. The horizontal velocity $u\cos\theta - ft$ turns negative before landing if $\tan\theta > \frac{g}{2f}$, so $\alpha = \tan^{-1}\frac{g}{2f}$. With $f = g$ and $\theta = 45°$, the acceleration is exactly opposite to the initial velocity: the projectile goes straight up the line $y = x$, stops, and comes straight back to O.

3. The effective gravity is g downwards plus a backwards. The free surface is perpendicular to it, so it is tilted at $\tan^{-1}\frac{a}{g}$.

4. The effective gravity has size $\sqrt{g^2 + a^2}$, and at $x = \frac{ak}{2g}$ the slope is $\frac{a}{g}$ and the radius of curvature is $\rho = \frac{k}{2}\left(1 + \frac{a^2}{g^2}\right)^{3/2}$. The period is $2\pi\sqrt{\rho/\sqrt{g^2+a^2}} = 2\pi\sqrt{\frac{k(g^2+a^2)}{2g^3}}$, which reduces to $2\pi\sqrt{\frac{k}{2g}}$ when $a = 0$.

11.6 Backspin

1. Friction acts at the contact point, so angular momentum about that point is conserved: $maV + \frac{2}{5}ma^2\frac{V}{a} = mav_0 + \frac{2}{5}ma^2\omega_0$. So $V = \frac{5v_0 + 2a\omega_0}{7}$, which exceeds v_0 when $\frac{v_0}{a\omega_0} < 1$.

2. $V = \frac{5}{7}v_0$. The final energy is $\frac{1}{2}m\left(\frac{5}{7}v_0\right)^2\left(1 + \frac{2}{5}\right) = \frac{5}{7}\cdot\frac{1}{2}mv_0^2$, so $\frac{2}{7}$ of the energy is lost.

3. The same argument with $I = ma^2$ gives $V = \frac{1}{2}(v_0 - a\omega_0)$. Setting $V = -\frac{v_0}{7}$ gives $\frac{v_0}{a\omega_0} = \frac{7}{9}$.

4. An impulse J at height h above the centre gives $mv = J$ and $\frac{2}{5}ma^2\omega = Jh$. Rolling needs $v = a\omega$, so $h = \frac{2}{5}a$.

11.7 Simpson's Rule in a Rod

1. $\frac{1}{6}\left(1 + 4\cdot\frac{4}{5} + \frac{1}{2}\right) = \frac{47}{60} \approx 0.7833$, against $\frac{\pi}{4} \approx 0.7854$. For $\int_{-1}^{1}$, the rule $\frac{1}{3}\big(\mathrm{f}(-1) + 4\mathrm{f}(0) + \mathrm{f}(1)\big)$ gives 0 for x^3 and $\frac{2}{3}$ for x^2 (both exact), but $\frac{2}{3}$ for x^4, whose integral is $\frac{2}{5}$.

2. The moment of inertia about the axis at angle ψ is a quadratic form $A\cos^2\psi + B\sin^2\psi + 2C\sin\psi\cos\psi$. Invariance under $\psi \mapsto \psi + 90°$ forces $A = B$ and $C = 0$, so it is the same for every axis: $\frac{1}{3}Ma^2$, the value about the axis parallel to a side.

3. Placing $\frac{1}{3}$ of the area at each midpoint integrates every quadratic exactly over a triangle (check 1, x, x^2, xy on a right triangle). So the three particles have the same mass, centre of mass and second moments as the lamina.

4. Take products of the rod weights $\frac{1}{6}$, $\frac{2}{3}$, $\frac{1}{6}$ at $-a$, 0, a in each direction: $\frac{M}{216}$ at the corners, $\frac{M}{54}$ at the edge midpoints, $\frac{2M}{27}$ at the face centres and $\frac{8M}{27}$ at the centre. About an axis through the centre parallel to an edge, $I = M\left(\frac{a^2}{3} + \frac{a^2}{3}\right) = \frac{2}{3}Ma^2$. By the parallel axis theorem, about an edge it is $\frac{2}{3}Ma^2 + M(\sqrt{2}a)^2 = \frac{8}{3}Ma^2$.

11.8 Tilted Equilibria

1. Measured from the common face towards the cone, G is at $\frac{-\frac{2}{3}\cdot\frac{3a}{8}+\frac{\lambda}{3}\cdot\frac{\lambda a}{4}}{\frac{2}{3}+\frac{\lambda}{3}} = \frac{a(\lambda^2-3)}{4(2+\lambda)}$. On the hemisphere the normal at the contact point always passes through the centre of the face, so the toy rights itself when G is below that centre, that is, $\lambda < \sqrt{3}$. On its side, the toy touches the table along a generator of the cone and at the rim, and it stays there if the vertical through G falls between the apex and the rim. This needs the height of G to be at least $-\frac{a}{\lambda}$, which reduces to $\lambda^3+\lambda+8>0$: always true.

2. Every point of a uniform sphere is an equilibrium (neutral). If the centre of mass is slightly off-centre, there are just two: with G lowest (stable) and with G highest (unstable).

3. Equilibria occur where the normal passes through the centre: at the four ends of the axes. Resting on an end of the minor axis is stable (G as low as possible); on an end of the major axis, unstable.

4. By symmetry G is at the origin. For $n=2$, $Y = r\left(\frac{1}{2}-p^2\right)$, and a tilted equilibrium needs the normal to pass through G, so $Y=0$ and $p=\frac{1}{\sqrt{2}}$. There the slope is $-\sqrt{2}$, so $\overrightarrow{AB}$ is tilted at $\tan^{-1}\sqrt{2} \approx 54.7°$, or $125.3°$ when resting on the other paraboloid. The two are reflections of each other in the x-axis, which is why the tilts pair up.

11.9 The Safety Parabola

1. Each particle is at $ut\,\hat{\mathbf{n}} - \frac{1}{2}gt^2\mathbf{j}$, a sphere of radius ut centred at $-\frac{1}{2}gt^2\mathbf{j}$. Fired horizontally from $(0, \frac{u^2}{2g})$, a particle follows $y = \frac{u^2}{2g} - \frac{gx^2}{2u^2}$: exactly the parabola of safety. The particle fired at angle α touches it at $x = \frac{u^2}{g\tan\alpha}$ at time $\frac{u}{g\sin\alpha}$; the horizontal particle takes $\frac{x}{u}$ to get there, so it must be launched at time $\frac{u(1-\cos\alpha)}{g\sin\alpha} = \frac{u}{g}\tan\frac{\alpha}{2}$ (this needs $\alpha \geq 45°$, so that the touching point is above the ground).

2. The line meets the parabola of safety where $R\sin\beta = \frac{u^2}{2g} - \frac{gR^2\cos^2\beta}{2u^2}$, giving $R = \frac{u^2}{g(1+\sin\beta)}$. At the maximum the two angles of Exercise 3 coincide, so the best angle is the bisector, $45° + \frac{1}{2}\beta$.

3. With $\lambda = \tan\theta$, the two values satisfy $\lambda_1+\lambda_2 = \frac{2u^2}{gx}$ and $\lambda_1\lambda_2 = 1 + \frac{2u^2y}{gx^2}$. So $\tan(\theta_1+\theta_2) = -\frac{x}{y}$ and $\theta_1+\theta_2 = 90° + \varphi$, where φ is the elevation of OP. The bisector of OP and the vertical is at $\frac{1}{2}(90°+\varphi)$.

4. The vertex is at $\left(\frac{u^2}{2g}\sin 2\theta, \frac{u^2}{2g}\sin^2\theta\right)$ and the focal length is $\frac{u^2\cos^2\theta}{2g}$, so the focus is at height $\frac{u^2}{2g}(\sin^2\theta - \cos^2\theta)$. Its distance from O is $\frac{u^2}{2g}$, whatever θ.

11.10 A Ball That Comes Back Towards You

1. In units $u = g = 1$, $\frac{\mathrm{d}}{\mathrm{d}t}OP^2 = t(t^2 - 3t\sin\alpha + 2)$, with roots $t \approx 1.050$ (local maximum, $OP \approx 0.516\frac{u^2}{g}$) and $t \approx 1.904$ (local minimum, $OP \approx 0.336\frac{u^2}{g}$). The ball lands at $t \approx 1.970$.
2. Resolving along and perpendicular to the plane, $OP^2 = u^2t^2 - ugt^3\sin(\alpha' + \beta) + \frac{1}{4}g^2t^4$, where α' is measured from the plane. This is the level-ground formula with $\alpha = \alpha' + \beta$. So the condition $\sin(\alpha' + \beta) > \frac{2\sqrt{2}}{3}$ concerns the angle from the *horizontal*; measured from the plane, the critical angle drops to $70.5° - \beta$ (provided the ball has not already landed).
3. Smaller. Drag kills the horizontal motion and makes the descent steeper, so the ball drops back more nearly vertically, which makes it easier for OP to shrink.
4. The formula for OP^2 does not involve the landing point, so OP decreases for a time exactly when $\sin\alpha > \frac{2\sqrt{2}}{3}$, as before (and the longer flight means the decreasing phase always happens before landing). Eventually the fall dominates and OP increases.

11.11 How Fielders Catch

1. If she stands at D_0 and runs at velocity v away from the batsman, the ball is at distance $D_0 + vt - Ut$ in front of her. To catch it she needs $D_0 + vT = UT$, and then the distance is $(U - v)(T - t)$, so $\tan\theta = \frac{gt}{2(U-v)}$, again linear. If the ball lands behind her, $UT > D_0$ and $v > 0$: she runs backwards.
2. $\theta = \tan^{-1}\frac{gt}{2U}$, so $\dot\theta = \frac{g/2U}{1+(gt/2U)^2}$, which decreases. $\tan\theta$ is linear because the height $\frac{1}{2}gt(T - t)$ and the distance $U(T - t)$ share the factor $T - t$.
3. Yes. The height relative to her eye is a quadratic that vanishes at the catch, so it factorises as $(T - t)(c + dt)$; dividing by $U(T - t)$ leaves a linear function (which starts negative: she first looks down).
4. No. With drag, the horizontal distance is $\frac{U}{k}(\mathrm{e}^{-kt} - \mathrm{e}^{-kT})$, and the height is not this times a linear function of t, so $\tan\theta$ is no longer linear.

11.12 The Ant on the Rubber Rope

1. $\frac{x}{a+ut} = \frac{1}{k}\ln\frac{a+ut}{a}$, which is $\frac{1}{2}$ at $t = \frac{a}{u}(\mathrm{e}^{k/2} - 1)$. Since $\frac{T}{2} = \frac{a}{2u}(\mathrm{e}^{k/2} - 1)(\mathrm{e}^{k/2} + 1)$ and $\frac{1}{2}(\mathrm{e}^{k/2} + 1) > 1$, this is before $\frac{T}{2}$.
2. $\frac{\mathrm{d}}{\mathrm{d}t}\frac{x}{L} = \frac{v}{L} = \frac{v}{a}\mathrm{e}^{-\lambda t}$, so $\frac{x}{L} = \frac{v}{a\lambda}(1 - \mathrm{e}^{-\lambda t})$, which reaches 1 only if $v > \lambda a$.
3. $\frac{x}{L} = v\int_0^t \frac{\mathrm{d}s}{a+\frac{1}{2}bs^2}$, which tends to $\frac{v\pi}{\sqrt{2ab}}$. So the ant arrives only if $v > \frac{\sqrt{2ab}}{\pi}$.
4. It needs $\frac{H_N}{100\,000} = \frac{1}{100}$, so $H_N = 1000$ and $N \approx \mathrm{e}^{999.4} \approx 10^{434}$ seconds.

11.13 The Snowplough Problem

1. The depth at time t after noon is proportional to $t + t_0$, so the distance covered is $c \ln \frac{t+t_0}{t_0}$. The conditions $x(1) = 2$ and $x(2) = 3$ give $\left(1 + \frac{1}{t_0}\right)^3 = \left(1 + \frac{2}{t_0}\right)^2$. With $u = \frac{1}{t_0}$ this reduces to $u^2 - u - 1 = 0$, so $u = \frac{1+\sqrt{5}}{2}$ and $t_0 \approx 0.618$ hours: it started snowing at about 11:23.

2. $\frac{\mathrm{d}t}{\mathrm{d}y} = \frac{\alpha h}{k} \mathrm{e}^{\alpha y} - \frac{\alpha h}{k} \mathrm{e}^{\alpha y} + \alpha\left(T - \frac{\alpha h y}{k}\right)\mathrm{e}^{\alpha y}$, and $\alpha\left(t - (\mathrm{e}^{\alpha y} - 1)\frac{h}{k}\right)$ is the same.

3. The gap is $\left(T - \frac{\alpha h y}{k}\right)\mathrm{e}^{\alpha y}$, whose derivative vanishes at $y = \frac{kT-h}{\alpha h}$. If $kT > h$, the gap grows until then and shrinks to 0 at $y = \frac{kT}{\alpha h}$. If $kT \le h$, the derivative $\alpha \mathrm{e}^{\alpha y}\left(T - \frac{h}{k} - \frac{\alpha h y}{k}\right)$ is negative for all $y > 0$, so the gap shrinks from the start. The second plough faces snow of depth $k\times$ (gap); while this is deeper than the snow the first plough met at the same place, it falls further behind.

4. $\frac{1}{\alpha}\frac{\mathrm{d}t_3}{\mathrm{d}y} = t_3 - t_2(y)$. Writing $t_3 = t_2 + g$ gives $g = \mathrm{e}^{\alpha y}\left(T - \alpha T y + \frac{\alpha^2 h y^2}{2k}\right)$. If $kT \ge 2h$, this vanishes at $y = \frac{k}{\alpha h}\left(T - \sqrt{T^2 - 2hT/k}\right)$, before the second plough catches the first; if $kT < 2h$, the second catches the first first.

11.14 Parabola, Not Catenary

1. Yes. The horizontal tension H is still constant, the middle segment has slope 0, and the slopes then go up in steps of $\frac{w}{H}$. With x measured from the middle of the horizontal segment, the kth joint is at $x = (k - \frac{1}{2})h$ and height $\frac{wh}{2H}k(k-1)$ above the lowest joints, which is $\frac{w}{2Hh}\left(x^2 - \frac{h^2}{4}\right)$: a parabola.

2. The tension is $\sqrt{H^2 + V^2}$, greatest where the vertical component V is greatest: in the end segments, where V supports half the load, $\frac{(2n+1)w}{2}$. So $T_{\max} = \sqrt{H^2 + \frac{1}{4}(2n+1)^2 w^2}$.

3. $y' = \sinh\frac{x}{c}$ and $y'' = \frac{1}{c}\cosh\frac{x}{c} = \frac{1}{c}\sqrt{1 + \sinh^2\frac{x}{c}}$. The length is $\int_0^X \cosh\frac{x}{c}\,\mathrm{d}x = c\sinh\frac{X}{c}$.

4. For a shallow curve with sag s, both the catenary and the parabola have length about $2d + \frac{4s^2}{3d}$, so both give $s \approx \sqrt{\frac{3}{2}d(L-d)}$; they differ only at order s^4. For a nearly taut chain, the difference is far smaller than anything Galileo could have measured.

CHAPTER

12

Probability

Uncertainty can be reasoned about exactly. Intuition, it turns out, cannot.

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Reasoning About Chance

Probability theory is nothing but common sense reduced to calculation.

— Pierre-Simon Laplace, *Théorie analytique des probabilités* (1812)

Laplace's famous remark is half true. Probability is common sense made precise; but, as this chapter shows again and again, common sense left to itself goes wrong. People think a positive test for a rare disease means you probably have it; that the second passenger to board a plane is as badly off as the last; that a sequence of coin tosses cannot beat another sequence of the same length; that the average of many measurements always settles down. All of these are false, and in each case a short calculation reveals the truth and, better still, the reason for it.

Probability began in the seventeenth century with questions about gambling, in the letters between Blaise Pascal and Pierre de Fermat. It became a branch of rigorous mathematics only in 1933, when Andrey Kolmogorov set out its axioms. In between, and since, it has grown into one of the most widely used parts of mathematics.

Why probability matters

Every field that deals with uncertainty uses probability: medicine (diagnostic tests and clinical trials), law (the weight of evidence), finance (risk and insurance), engineering (reliability and floods), computer science (randomised algorithms, cryptography, machine learning), physics (quantum mechanics and statistical mechanics) and biology (genetics and epidemics). Modern artificial intelligence is, in large part, applied probability: a language model predicts the probability of the next word. And the ability to reason correctly about risk, chance and evidence is one of the most valuable skills any educated person can have.

What you will meet at university

First-year probability covers the axioms, conditional probability and Bayes' theorem, independence, discrete and continuous random variables, expectation and variance, generating functions, and the two great limit theorems: the law of large numbers and the central limit theorem. Later courses study random processes (Markov chains, Poisson processes, random walks, branching processes), measure-theoretic probability (which puts limits and continuous distributions on a firm footing) and martingales, and connect probability with statistics, analysis, combinatorics and physics.

The questions that run through this chapter

Three big questions

1. **What exactly is the question?** What is the random experiment, what is being conditioned on, and what does "at random" mean?
2. **What is the clever way in?** When can symmetry, indicators, conditioning or a well-chosen state replace a long calculation?
3. **What happens in the long run?** How do random processes behave over time, and what do sums, averages and maxima of many random quantities look like?

The chapter has five parts. *Thinking clearly* (Section 12.1 to Section 12.4) deals with the traps: conditioning, base rates, the meaning of "random" and the meaning of independence. *Symmetry and clever arguments* (Section 12.5 to Section 12.9) collects problems with beautiful short solutions. *Expectation, games and stopping* (Section 12.10 to Section 12.14) turns to averages and decisions: when to stop, whom to shoot at, how to randomise. *Random processes* (Section 12.15 to Section 12.19) follows randomness through time and space. And *Limits and continuous probability* (Section 12.20 to Section 12.25) ends with geometric probability and the great limit theorems.

Ideas in this chapter

§	Topic	Idea
12.1	Conditioning	"A specific one" versus "at least one"
12.2	Bayes' theorem	Base rates and natural frequencies
12.3	Bertrand's paradox	"Random" needs a definition
12.4	Independence	Pairwise is not mutual
12.5	The airline seat	Symmetry between two seats
12.6	One more coin	Complementary symmetric events
12.7	Tournaments	Counting matches and pairs
12.8	Penney's game	First-step analysis; non-transitivity
12.9	Fair coins	Extracting fairness from bias
12.10	Random loops	A memoryless sum of indicators
12.11	Cups and e	Factorials and the exponential series
12.12	Optimal stopping	The $\frac{1}{e}$ rule
12.13	The truel	Strategic weakness
12.14	Game theory	Randomised strategies; minimax
12.15	Branching processes	Extinction as a fixed point
12.16	Markov chains	Stationary distributions; reversal
12.17	Banach's matchboxes	The negative binomial
12.18	The exponential	Memorylessness and independence
12.19	Spatial randomness	Nearest neighbours; Rayleigh
12.20	Buffon's needle	Geometric probability and π
12.21	Random triangles	Probability as area
12.22	The Cauchy distribution	A distribution with no mean
12.23	Stirling's formula	Factorials from the normal curve
12.24	Central limit theorem	Moment generating functions
12.25	Extreme values	The Gumbel distribution

One section, Section 12.24, uses moment generating functions, which were removed from the STEP specification in 2019; it explains what it needs.

How to read this chapter

When a probability question seems confusing, write down the sample space for a small case and count. When an answer comes out suspiciously simple, look for the symmetry that explains it. And when your intuition and the calculation disagree, trust the calculation, then find out why your intuition was wrong: that is where the learning happens.

12.1 Precise Conditioning

The question

A bag contains eleven small discs, which are identical except that six of the discs are blank and five of the discs are numbered, using the numbers 1, 2, 3, 4 and 5. The bag is shaken, and four discs are taken one at a time without replacement.

Calculate the probability that:

(i) all four discs taken are numbered;

(ii) all four discs taken are numbered, given that the disc numbered "3" is taken first;

(iii) exactly two numbered discs are taken, given that the disc numbered "3" is taken first;

(iv) exactly two numbered discs are taken, given that the disc numbered "3" is taken;

(v) exactly two numbered discs are taken, given that a numbered disc is taken first;

(vi) exactly two numbered discs are taken, given that a numbered disc is taken.

The hook

"Given that" is the most dangerous phrase in probability. The last four parts of this question sound almost identical: the disc numbered 3 is taken first; the disc numbered 3 is taken; a numbered disc is taken first; a numbered disc is taken. Three of them give the answer $\frac{1}{2}$. The fourth does not.

The difference is exactly the one behind the famous "boy or girl" puzzle (a family has two children, at least one is a boy; what is the chance both are boys?), which has tripped up generations of students and a fair number of professional mathematicians. The lesson is that a condition is a precise statement about which outcomes remain possible, and the way information was obtained matters as much as the information itself.

Before you start

Ask yourself Four discs are taken from eleven. Does the order of drawing matter for the event "all four are numbered"? How many equally likely sets of four discs are there?

Every set of four discs is equally likely to be the set drawn, so we can ignore order: there are $\binom{11}{4} = 330$ equally likely sets.

Working it through

STAGE 1 No condition

There are $\binom{5}{4} = 5$ sets of four numbered discs, so

$$\mathrm{P}(\text{all four numbered}) = \frac{5}{330} = \frac{1}{66}.$$

STAGE 2 The 3 comes first

Ask yourself Once the disc numbered 3 has been drawn first, what are the remaining three draws?

The other three discs form a random set of three from the remaining ten (four numbered, six blank), all $\binom{10}{3} = 120$ sets equally likely. So

$$\text{(ii)}\quad \frac{\binom{4}{3}}{120} = \frac{4}{120} = \frac{1}{30}, \qquad \text{(iii)}\quad \frac{\binom{4}{1}\binom{6}{2}}{120} = \frac{4 \times 15}{120} = \frac{1}{2},$$

since in (iii) exactly one of the other three must be numbered.

STAGE 3 The 3 is taken (at some point)

Ask yourself Does it matter *when* the disc numbered 3 was drawn? Given that 3 is among the four, what can you say about the other three?

By symmetry, given that the set of four contains the 3, the other three discs are a random set of three from the remaining ten, exactly as in Stage 2. So (iv) is also $\frac{1}{2}$.

STAGE 4 A numbered disc comes first

Given that the first disc is numbered (whichever number it is), the other three are again a random set of three from the remaining ten discs, of which four are numbered and six blank. So (v) is $\frac{1}{2}$ too.

STAGE 5 A numbered disc is taken

Ask yourself Now the condition is "at least one of the four is numbered". Is that the same as any of the previous conditions? How many of the 330 sets satisfy it?

This condition is different: it does not single out any particular disc. Only $\binom{6}{4} = 15$ sets have no numbered disc, so 315 satisfy the condition. Of these, $\binom{5}{2}\binom{6}{2} = 10 \times 15 = 150$ contain exactly two numbered discs. So

$$\text{(vi)} \quad \frac{150}{315} = \frac{10}{21}. \qquad \blacksquare$$

Pause and try Why is (vi) different from (v)? In (v) we know *which* disc is numbered (the first), and the other three are unconstrained. In (vi) we only know that *some* disc is numbered. Sets with three or four numbered discs satisfy "some disc is numbered" in more ways, so they are not under-represented the way they are in (v).

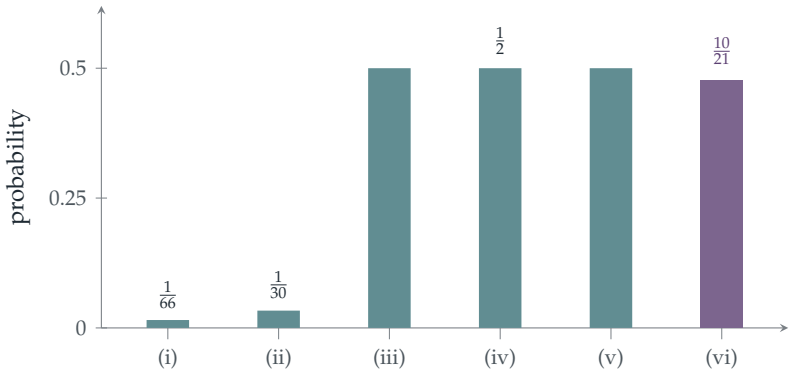


Figure 12.1.1 The six answers. Parts (iii), (iv) and (v) all condition on a *specific* disc being numbered, and give $\frac{1}{2}$. Part (vi) conditions only on *some* disc being numbered, and gives $\frac{10}{21}$.

Looking back

- **A condition restricts the sample space.** $\mathrm{P}(A \mid B)$ is the proportion of outcomes in B that are also in A.
- **Symmetry saves work.** Whether the 3 was drawn first or at some point made no difference, because all positions are alike.
- **"A specific one" is not "at least one".** Conditions that sound alike can select quite different sets of outcomes.

What's really going on

The boy-or-girl puzzle

Martin Gardner posed this in 1959: Mr Smith has two children, at least one of whom is a boy; what is the probability that both are boys? Listing the equally likely families BB, BG, GB, GG and removing GG leaves three, of which one is BB: the answer is $\frac{1}{3}$. But if we know that the *elder* child is a boy, the answer is $\frac{1}{2}$. These are parts (vi) and (v) of our question in miniature. Gardner later admitted that the question as he first posed it was ambiguous: it depends on how we came to know that there is a boy. If we met one of the children at random and he was a boy, the answer is $\frac{1}{2}$; if we asked "do you have at least one son?" and were told yes, it is $\frac{1}{3}$. Probability depends on the process that produced the information, not just the information.

Monty Hall

The same lesson underlies the most argued-about puzzle in probability. On a game show, a car is behind one of three doors. You pick door 1. The host, who knows where the car is, always opens one of the other two doors to reveal a goat, and always offers you the chance to switch; when the car is behind your door, so that he has a choice, he picks between the other two at random. He opens door 3, and offers you door 2. Should you switch? Yes: switching wins with probability $\frac{2}{3}$. The host's choice of door is not random; it depends on where the car is, and that carries information. The precise rules matter. If the host always opens door 3 whenever he can, then seeing door 3 open makes doors 1 and 2 equally likely, and switching wins with probability only $\frac{1}{2}$; yet a contestant who decides in advance always to switch still wins $\frac{2}{3}$ of the time, because switching wins exactly when the first choice was wrong. When Marilyn vos Savant gave this answer in a magazine column in 1990, she received thousands of letters telling her she was wrong, some from mathematicians. A computer simulation, or a careful list of the equally likely cases, settles it.

Bayes and Laplace

Conditional probability was put on a systematic footing by Thomas Bayes, whose essay was published after his death in 1763, and by Pierre-Simon Laplace, who in 1774 stated the rule now called Bayes' theorem in its general form. Andrey Kolmogorov's axioms of 1933, which every university course now starts from, define $\mathrm{P}(A \mid B) = \mathrm{P}(A \cap B)/\mathrm{P}(B)$, and the rest of probability theory is built on that definition. The next section, Section 12.2, shows what Bayes' theorem does in practice.

The thinking skill

Be precise about the condition

Before computing a conditional probability, write down exactly which outcomes satisfy the condition. Ask how the information was obtained.

At university, conditional probability and conditional expectation are among the most important tools in probability and statistics, and the most frequent source of errors. Writing out the sample space, at least for a small case, is the best defence.

Try this next

1 **Generalise.** With b blank and n numbered discs, and k drawn, show that the answers to (iv) and (v) are always equal, and find when (vi) equals them.

2 **Boys born on Tuesday.** A family has two children, and at least one is a boy born on a Tuesday. Show that the probability that both are boys is $\frac{13}{27}$. (List the 14×14 equally likely possibilities, or count cleverly.)

3 **Monty Hall.** Verify the $\frac{2}{3}$ by listing outcomes. What if the host does not know where the car is, opens one of the other doors at random, and it happens to show a goat?

4 **Two aces.** A bridge hand of 13 cards contains at least one ace. What is the probability that it contains at least two? Compare with the probability given that it contains the ace of spades.

Further reading

- **Jason Rosenhouse, *The Monty Hall Problem*** (Oxford University Press). The puzzle, its history and its many variations.
- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press), the chapters on conditional probability.
- **Joseph Blitzstein & Jessica Hwang, *Introduction to Probability*** (CRC Press; free online). Excellent on conditioning, with many paradoxes.

12.2 The Base-Rate Fallacy

The question

The diagnostic test AL has a probability 0.9 of giving a positive result when applied to a person suffering from the rare disease mathematitis. It also has a probability 1/11 of giving a false positive result when applied to a non-sufferer. It is known that only 1% of the population suffer from the disease. Given that the test AL is positive when applied to Frankie, who is chosen at random from the population, what is the probability that Frankie is a sufferer?

In an attempt to identify sufferers more accurately, a second diagnostic test STEP is given to those for whom the test AL gave a positive result. The probability of STEP giving a positive result on a sufferer is 0.9, and the probability that it gives a false positive result on a non-sufferer is p. Half of those for whom AL was positive and on whom STEP then also gives a positive result are sufferers. Find p.

The hook

A test for a disease is 90% accurate on people who have it and gives few false alarms. Frankie tests positive. How worried should Frankie be? Most people, including, in many studies, most doctors, say "about 90%". The true answer in this question is $\frac{1}{11}$, about 9%. Ten times smaller.

The mistake is to ignore the *base rate*: the disease is rare, so even a small false-positive rate among the many healthy people produces far more false alarms than there are true cases. This *base-rate fallacy* matters in medicine, in law (where it is called the prosecutor's fallacy), in security screening and in machine learning. Bayes' theorem is the cure.

Before you start

Ask yourself Imagine 1000 people. How many have the disease? How many of them test positive? How many of the healthy people test positive?

Of 1000 people, 10 have the disease and 9 of them test positive. Of the 990 healthy people, $\frac{1}{11}$ test positive: 90 false positives. So 99 people test positive, of whom only 9 are ill.

Working it through

STAGE 1 Bayes' theorem

Ask yourself Write D for "Frankie has the disease" and $+$ for "AL is positive". You know $\mathrm{P}(+ \mid D)$ and want $\mathrm{P}(D \mid +)$. How are they related?

By the definition of conditional probability, $\mathrm{P}(D \mid +) = \frac{\mathrm{P}(D \cap +)}{\mathrm{P}(+)}$, and $\mathrm{P}(+)$ splits according to whether Frankie has the disease:

$$\begin{aligned}\mathrm{P}(D \mid +) &= \frac{\mathrm{P}(+ \mid D)\mathrm{P}(D)}{\mathrm{P}(+ \mid D)\mathrm{P}(D) + \mathrm{P}(+ \mid \bar{D})\mathrm{P}(\bar{D})} \\ &= \frac{0.9 \times 0.01}{0.9 \times 0.01 + \frac{1}{11} \times 0.99} = \frac{0.009}{0.009 + 0.09} = \frac{1}{11}.\end{aligned}$$

This is the $\frac{9}{99}$ of the thousand people.

STAGE 2 A second test

Ask yourself The second test is given only to those who tested positive on AL. What is the proportion of sufferers in that group? Use it as the new "base rate".

Among those positive on AL, a fraction $\frac{1}{11}$ are sufferers and $\frac{10}{11}$ are not. Applying Bayes' theorem again, with these as the prior probabilities,

$$\mathrm{P}(D \mid \text{both positive}) = \frac{0.9 \times \frac{1}{11}}{0.9 \times \frac{1}{11} + p \times \frac{10}{11}} = \frac{0.9}{0.9 + 10p}.$$

Setting this equal to $\frac{1}{2}$ gives $10p = 0.9$, so $p = 0.09$.

(Assuming, as the question implicitly does, that the two tests' errors are independent given whether Frankie is ill.)

Pause and try With $p = 0.09$, STEP's false-positive rate is about the same as AL's, yet a second positive result raises the probability from $\frac{1}{11}$ to $\frac{1}{2}$. Why does the same quality of test have a much bigger effect the second time?

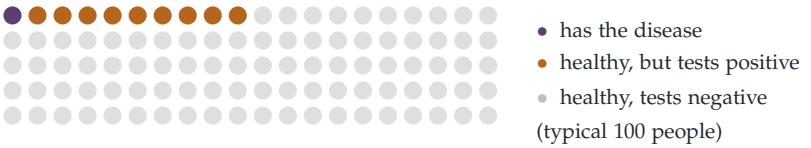


Figure 12.2.1 A typical hundred people. One has the disease (plum), and will test positive nine times in ten. About 9 of the 99 healthy people test positive by mistake (amber). Among those who test positive, the healthy outnumber the sick roughly ten to one.

Looking back

- **Natural frequencies.** Imagining 1000 people made the answer obvious and easy to check.
- **Prior times likelihood.** The posterior probability depends on both the test's accuracy and the base rate.
- **Updating in stages.** Yesterday's posterior is today's prior: the second test started from $\frac{1}{11}$, not $\frac{1}{100}$.

What's really going on

Odds and evidence

Bayes' theorem is clearest in terms of *odds*. The odds of D are $\mathrm{P}(D) : \mathrm{P}(\bar{D})$. A piece of evidence E multiplies the odds by the *likelihood ratio* $\frac{\mathrm{P}(E|D)}{\mathrm{P}(E|\bar{D})}$:

$$\text{posterior odds} = \text{prior odds} \times \text{likelihood ratio}.$$

Here the prior odds are $1 : 99$, and AL's likelihood ratio is $0.9 \div \frac{1}{11} = 9.9$, giving posterior odds $9.9 : 99 = 1 : 10$, that is, probability $\frac{1}{11}$. A second independent test with likelihood ratio 10 (as STEP with $p = 0.09$ has) brings the odds to $1 : 1$. Taking logarithms, evidence *adds*: each independent test contributes the logarithm of its likelihood ratio. Alan Turing used exactly this, measuring evidence in "bans" and "decibans", to break Enigma messages at Bletchley Park.

The prosecutor's fallacy

In 1999 the solicitor Sally Clark was convicted of murdering her two baby sons, partly on the strength of evidence that the chance of two cot deaths in one family was "1 in 73 million". Quite apart from the error in that number (it assumed the two deaths were independent, ignoring genetic and environmental factors), the reasoning confused $\mathrm{P}(\text{evidence} \mid \text{innocent})$ with $\mathrm{P}(\text{innocent} \mid \text{evidence})$. Double murder by a mother is also extremely rare, and the relevant comparison is between the two explanations. The Royal Statistical Society wrote publicly to point out the error, and her conviction was overturned in 2003. The mathematics is exactly that of this question.

Gigerenzer and natural frequencies

In the 1990s the psychologist Gerd Gigerenzer showed that doctors who were given test statistics as conditional probabilities mostly got questions like this one badly wrong, but that when the same information was presented as natural frequencies ("of 1000 women, 10 have breast cancer; of these, 8 test positive…") most got them right. His work has changed how medical risks are communicated to patients in many countries.

The thinking skill

Reverse a conditional with Bayes' theorem

$\mathrm{P}(A \mid B)$ *and* $\mathrm{P}(B \mid A)$ *are different numbers. To get from one to the other you need the base rate* $\mathrm{P}(A)$.

Bayesian reasoning, updating beliefs in the light of evidence, is the foundation of modern statistics and machine learning, from spam filters to medical diagnosis. At university you will meet it as Bayesian inference, with prior and posterior distributions instead of single probabilities.

Try this next

1 **A body in the fridge.** (From STEP II, 1987.) Two friends, who tell the truth independently with probabilities p and q, both tell me there is a body in my fridge. If I had thought this equally likely as not, explain why my new estimate should be $\frac{pq}{1-p-q+2pq}$. There is indeed a body. It could be my enemy A, my enemy B, or both, which I judged equally likely; both friends now say A is in the fridge. What is my new estimate that both are there?

2 **How rare?** For a test with sensitivity 0.9 and false-positive rate $\frac{1}{11}$, how common must a disease be for a positive result to mean a better than even chance of having it?

3 **Repeat the test.** If AL itself is repeated, and the results are independent given the person's true state, what is the probability of disease after two positive AL results? After three?

4 **Screening.** A screening programme tests a million people for a condition with prevalence 1 in 10 000, using a test with sensitivity 99% and false-positive rate 1%. How many people will be told they may have the condition, and how many of them actually do?

Further reading

- **Gerd Gigerenzer, *Reckoning with Risk*** (Penguin). Natural frequencies, medical tests and the courtroom.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican). Includes medical screening and the Sally Clark case.
- **Sharon Bertsch McGrayne, *The Theory That Would Not Die*** (Yale University Press). The history of Bayes' theorem, including Turing's use of it.

12.3 What Does "Random" Mean?

The question

(i) The point A lies on the circumference of a circle of radius a and centre O. The point B is chosen at random on the circumference, so that the angle AOB has a uniform distribution on $[0, 2\pi]$. Find the expected length of the chord AB.

(ii) The point C is chosen at random in the interior of a circle of radius a and centre O, so that the probability that it lies in any given region is proportional to the area of the region. The random variable R is defined as the distance between C and O. Find the probability density function of R.

Obtain a formula in terms of a, R and t for the length of a chord through C that makes an acute angle of t with OC.

Show that as C varies (with t fixed), the expected length $L(t)$ of such chords is given by

$$L(t) = \frac{4a(1-\cos^3 t)}{3\sin^2 t}.$$

Show further that

$$L(t) = \frac{4a}{3}\left(\cos t + \tfrac{1}{2}\sec^2\left(\tfrac{1}{2}t\right)\right).$$

(iii) The random variable T is uniformly distributed on $[0, \frac{1}{2}\pi]$. Find the expected value of $L(T)$.

The hook

"Choose a chord of a circle at random. What is its expected length?" The question sounds as if it has an answer. It does not, until you say what "at random" means. Pick two random points on the circle and join them, and the expected length is $\frac{4a}{\pi} \approx 1.27a$. Pick a random point inside the circle and a random direction through it, and the answer is $\frac{16a}{3\pi} \approx 1.70a$. Both are perfectly reasonable ways to choose "a random chord".

This is *Bertrand's paradox*, published by Joseph Bertrand in 1889 as a warning. Probability is only well defined once the random experiment is specified. The paradox helped to push mathematicians towards the rigorous foundations of probability built in the twentieth century, in which every probability lives on a precisely defined space of outcomes.

Before you start

Ask yourself If the angle AOB is θ, what is the length of the chord AB? (Drop a perpendicular from O.)

The perpendicular from O bisects the chord and the angle, so $AB = 2a\sin\frac{\theta}{2}$.

Working it through

STAGE 1 Two points on the circle

$$\mathrm{E}(AB) = \frac{1}{2\pi}\int_0^{2\pi} 2a\sin\frac{\theta}{2}\,\mathrm{d}\theta = \frac{a}{\pi}\Big[-2\cos\frac{\theta}{2}\Big]_0^{2\pi} = \frac{a}{\pi}\cdot 4 = \frac{4a}{\pi}. \qquad \blacksquare$$

STAGE 2 A random point inside

Ask yourself What is $\mathrm{P}(R \le r)$ for $0 \le r \le a$? Differentiate to get the density.

$\mathrm{P}(R \le r)$ is the area of the disc of radius r divided by the area of the whole disc, $\frac{r^2}{a^2}$. So R has density $\frac{2r}{a^2}$ on $[0, a]$. (Points are more likely to be far from the centre, because there is more area there.)

Ask yourself A chord through C makes angle t with OC. How far is the chord from O?

The distance from O to the line is $R\sin t$, so the chord has length $2\sqrt{a^2 - R^2\sin^2 t}$.

STAGE 3 Averaging over C

With $s = \sin t$,

$$L(t) = \int_0^a 2\sqrt{a^2 - r^2s^2}\cdot\frac{2r}{a^2}\,\mathrm{d}r = \frac{4}{a^2}\Big[-\frac{(a^2 - r^2s^2)^{3/2}}{3s^2}\Big]_0^a$$
$$= \frac{4}{a^2}\cdot\frac{a^3 - a^3(1-s^2)^{3/2}}{3s^2} = \frac{4a(1-\cos^3 t)}{3\sin^2 t}. \qquad \blacksquare$$

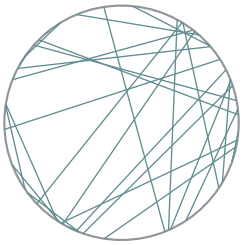
Ask yourself Factorise $1 - \cos^3 t$ and $\sin^2 t = 1 - \cos^2 t$. What cancels?

With $c = \cos t$, $\frac{1-c^3}{1-c^2} = \frac{(1-c)(1+c+c^2)}{(1-c)(1+c)} = \frac{1+c+c^2}{1+c} = c + \frac{1}{1+c}$, and $1 + \cos t = 2\cos^2\frac{t}{2}$. So

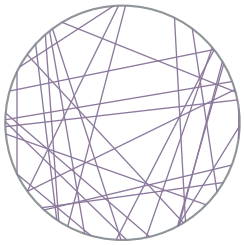
$$L(t) = \frac{4a}{3}\Big(\cos t + \tfrac{1}{2}\sec^2\tfrac{t}{2}\Big). \qquad \blacksquare$$

STAGE 4 **A random direction too**

$$\begin{aligned}\mathrm{E}\big(L(T)\big) &= \frac{2}{\pi}\int_0^{\pi/2} \frac{4a}{3}\Big(\cos t + \tfrac{1}{2}\sec^2\tfrac{t}{2}\Big)\,\mathrm{d}t \\ &= \frac{8a}{3\pi}\Big[\sin t + \tan\tfrac{t}{2}\Big]_0^{\pi/2} = \frac{8a}{3\pi}(1+1) = \frac{16a}{3\pi}. \qquad \blacksquare\end{aligned}$$



two random endpoints
mean length $\frac{4a}{\pi} \approx 1.27a$



random midpoint in the disc
mean length $\frac{4a}{3} \approx 1.33a$

Figure 12.3.1 Twenty-eight "random chords" chosen in two different ways. Left: two independent uniform points on the circle. Right: a uniformly random point in the disc as the chord's midpoint. The right-hand method produces fewer short chords near the edge. Different methods give different distributions, and different answers.

Looking back

- **Specify the experiment.** "A random chord" became a precise procedure in each part, and each procedure gave a different answer.
- **Densities from areas.** $\mathrm{P}(R \le r)$ came directly from geometry; differentiating gave the density.
- **Average in stages.** First over the position of C (with t fixed), then over t.

What's really going on

Bertrand's three answers

Bertrand's own question was: what is the probability that a random chord is longer than the side of the inscribed equilateral triangle, $\sqrt{3}a$? He gave three methods and three answers.

- Random endpoints: fix one endpoint; the chord is long if the other lies in the middle third of the circle. Probability $\frac{1}{3}$.
- Random radius and a random point on it as the midpoint: the chord is long if its distance from O is less than $\frac{a}{2}$. Probability $\frac{1}{2}$.

- Random midpoint anywhere in the disc: the chord is long if the midpoint lies within $\frac{a}{2}$ of O. Probability $\frac{1}{4}$.

Each method is a different probability distribution on the set of chords, and there is no contradiction: they answer different questions. The lesson is that "uniformly at random" only makes sense relative to a chosen parametrisation, and for continuous sets there is usually no single natural one.

Is there a right answer?

In 1973 the physicist Edwin Jaynes argued that the second method is the "right" one, on the grounds that the answer should not change if the circle is moved or rescaled slightly: a random chord should be like a straw thrown onto a circle drawn on the floor. Only the distribution with uniform distance from the centre has this invariance, and experiments throwing straws confirm $\frac{1}{2}$. The same idea, that the right distribution is the one that respects the symmetries of the situation, underlies *integral geometry*, the mathematics of random lines, which also explains Buffon's needle in Section 12.20.

From paradox to axioms

Bertrand published his chord paradox in *Calcul des probabilités* (1889), partly to mock what he saw as careless uses of "geometric probability". Henri Poincaré discussed it too. In 1900 David Hilbert listed the axiomatisation of probability among his famous problems for the new century, and Andrey Kolmogorov solved it in 1933: probability is a measure on a space of outcomes, and every question about "random" objects must first say which space and which measure. Bertrand's paradox is the standard example of why this is necessary.

The thinking skill

Define the random experiment

Before calculating a probability, describe exactly how the random object is generated. Different descriptions can give different answers.

At university, every probability calculation starts from a sample space and a probability measure. In statistics and physics, choosing the right model of randomness, often by symmetry, is a large part of the work.

Try this next

1 **Bertrand's probabilities.** Verify the three probabilities $\frac{1}{3}$, $\frac{1}{2}$, $\frac{1}{4}$ above.

2 **The other mean.** Show that if the chord's midpoint is chosen uniformly in the disc (the right-hand picture), the expected length is $\frac{4a}{3}$. (*Hint:* this is $L(\frac{\pi}{2})$. Why?)

3 **Random distance.** If the distance of the chord from O is uniform on $[0, a]$, show that the expected length is $\frac{\pi a}{2}$.

4 **Three methods, three distributions.** For each of the three methods, find the probability density of the chord's length ℓ on $[0, 2a]$.

Further reading

- **Edwin Jaynes, "The well-posed problem"**, *Foundations of Physics* 3 (1973). The invariance argument, readable and provocative.
- **Gábor Székely, *Paradoxes in Probability Theory and Mathematical Statistics*** (Reidel). Bertrand's paradox and many others.
- **Daniel Klain & Gian-Carlo Rota, *Introduction to Geometric Probability*** (Cambridge University Press). Integral geometry, for later.

12.4 Independent in Pairs, Not All Together

The question

(i) In a game, each member of a team of n players rolls a fair six-sided die. The total score of the team is the number of pairs of players rolling the same number. For example, if 7 players roll $3, 3, 3, 3, 6, 6, 2$ the total score is 7, as six different pairs of players both score 3 and one pair of players both score 6.

Let X_{ij}, for $1 \leq i < j \leq n$, be the random variable that takes the value 1 if players i and j roll the same number and the value 0 otherwise. Show that X_{12} is independent of X_{23}. Hence find the mean and variance of the team's total score.

(ii) Show that, if Y_i, for $1 \leq i \leq m$, are random variables with mean zero, then

$$\mathrm{Var}(Y_1 + Y_2 + \cdots + Y_m) = \sum_{i=1}^{m} \mathrm{E}(Y_i^2) + 2 \sum_{i=1}^{m-1} \sum_{j=i+1}^{m} \mathrm{E}(Y_i Y_j).$$

(iii) In a different game, each member of a team of n players rolls a fair six-sided die. The total score of the team is the number of pairs of players rolling the same even number minus the number of pairs of players rolling the same odd number. For example, if 7 players roll $3, 3, 3, 3, 6, 6, 2$ the total score is -5.

Let Z_{ij}, for $1 \leq i < j \leq n$, be the random variable that takes the value 1 if players i and j roll the same even number, the value -1 if players i and j roll the same odd number and the value 0 otherwise. Show that Z_{12} is not independent of Z_{23}. Find the mean of the team's total score and show that the variance of the team's total score is $\frac{1}{36} n(n^2 - 1)$.

The hook

Players 1 and 2 roll the same number. Does that tell you anything about whether players 2 and 3 roll the same number? Surprisingly, no: the events are independent. The same goes for players 1 and 3. But if 1 matches 2 and 2 matches 3, then certainly 1 matches 3. Three events, each pair independent, but not independent all together.

Pairwise independence is weaker than full independence, and the difference matters. For variances, pairwise independence is all you need, which this question exploits. For many other purposes it is not enough. Computer scientists use pairwise independent

random numbers deliberately, because they are much cheaper to generate than fully independent ones and are good enough for many algorithms.

Before you start

Ask yourself What is $\mathrm{P}(X_{12} = 1)$? What is $\mathrm{P}(X_{12} = 1 \text{ and } X_{23} = 1)$?

Whatever player 1 rolls, player 2 matches it with probability $\frac{1}{6}$. Both $X_{12} = 1$ and $X_{23} = 1$ means all three roll the same number: probability $\frac{6}{216} = \frac{1}{36}$.

Working it through

STAGE 1 Pairwise independence

Since $\mathrm{P}(X_{12} = 1, X_{23} = 1) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6} = \mathrm{P}(X_{12} = 1)\mathrm{P}(X_{23} = 1)$, and each variable takes only two values, X_{12} and X_{23} are independent. (For two-valued variables, one cell of the joint table determines the rest; exercise 1 makes this precise.) Pairs with no player in common, such as X_{12} and X_{34}, are clearly independent. So all the X_{ij} are pairwise independent.

STAGE 2 Mean and variance of the score

The score is $S = \sum_{i<j} X_{ij}$, a sum of $\binom{n}{2}$ variables, each with mean $\frac{1}{6}$ and variance $\frac{1}{6} \cdot \frac{5}{6} = \frac{5}{36}$. Expectation always adds, so $\mathrm{E}(S) = \frac{1}{6}\binom{n}{2}$.

Ask yourself When does the variance of a sum equal the sum of the variances?

The variance of a sum is the sum of the variances plus twice the sum of the covariances, and independent pairs have zero covariance. Pairwise independence is enough:

$$\mathrm{E}(S) = \frac{n(n-1)}{12}, \qquad \mathrm{Var}(S) = \frac{5}{36}\binom{n}{2} = \frac{5n(n-1)}{72}.$$

STAGE 3 The variance formula

If the Y_i have mean zero, then $\mathrm{Var}(\sum Y_i) = \mathrm{E}\big((\sum Y_i)^2\big)$, and expanding the square gives every Y_i^2 once and every product Y_iY_j (with $i < j$) twice. Taking expectations gives the formula. ■

STAGE 4 The odd-and-even game

Ask yourself What is $\mathrm{E}(Z_{12})$? What values can $Z_{12}Z_{23}$ take, and when is it non-zero?

$Z_{12} = 1$ with probability $\frac{3}{36} = \frac{1}{12}$ (both roll 2, both 4 or both 6), and -1 with probability $\frac{1}{12}$; so $\mathrm{E}(Z_{12}) = 0$ and $\mathrm{E}(Z_{12}^2) = \frac{1}{6}$.

$Z_{12}Z_{23} \neq 0$ only when all three players roll the same number, and then $Z_{12} = Z_{23}$ (both $+1$ or both -1), so the product is 1. Hence $\mathrm{E}(Z_{12}Z_{23}) = \frac{1}{36}$, which is not $\mathrm{E}(Z_{12})\mathrm{E}(Z_{23}) = 0$: they are not independent. ■

(Even more directly, $\mathrm{P}(Z_{12} = 1, Z_{23} = -1) = 0$, while each event alone has probability $\frac{1}{12}$.)

STAGE 5 The variance

The total score $T = \sum Z_{ij}$ has mean 0. By part (ii), $\mathrm{Var}(T)$ is $\binom{n}{2} \cdot \frac{1}{6}$ plus twice the sum of $\mathrm{E}(Z_{ij}Z_{kl})$ over pairs of pairs. Pairs of pairs with no common player contribute 0 (they are independent, each with mean zero). Pairs of pairs sharing exactly one player each contribute $\frac{1}{36}$, and there are $n\binom{n-1}{2}$ of them (choose the shared player, then two others). So

$$\begin{aligned}\mathrm{Var}(T) &= \frac{n(n-1)}{12} + 2 \cdot \frac{1}{36} \cdot \frac{n(n-1)(n-2)}{2} \\ &= \frac{n(n-1)}{36}\big(3 + (n-2)\big) = \frac{n(n^2-1)}{36}.\end{aligned}$$ ■

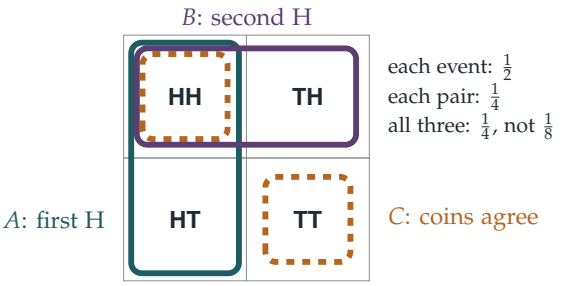


Figure 12.4.1 Bernstein's example: two fair coins. The events "first coin is heads" (teal), "second coin is heads" (plum) and "the coins agree" (amber, dashed) each have probability $\frac{1}{2}$, and each pair meets in a single square of probability $\frac{1}{4}$, so they are pairwise independent. But all three meet in HH, which has probability $\frac{1}{4}$, not $\frac{1}{8}$: any two of them determine the third.

Looking back

- **Indicators.** Writing the score as a sum of indicator variables reduced everything to pairs of players.
- **Variance needs only pairs.** The variance of a sum depends only on the pairwise covariances, so pairwise independence is enough to add variances.
- **Check independence, do not assume it.** Two games that look alike had different dependence structures.

What's really going on

Independence of several events

Events $A_1, \dots, A_n$ are *independent* if, for every subcollection, the probability of the intersection is the product of the probabilities. For three events this means four conditions: three for the pairs, and $\mathrm{P}(A \cap B \cap C) = \mathrm{P}(A)\mathrm{P}(B)\mathrm{P}(C)$. Sergei Bernstein gave the example in Figure 12.4.1 in the 1920s to show that the pairwise conditions do not imply the triple one. (Conversely, the triple condition alone does not imply the pairwise ones either.) The matching events X_{12}, X_{23}, X_{13} in the question are exactly like Bernstein's coins.

Uncorrelated is not independent

Zero covariance, $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$, is weaker still than independence. STEP III 2000 Q14 (in the exercises) shows that for variables taking only two values the two notions coincide, but with three values they do not. A simple example: let X be uniform on $\{-1, 0, 1\}$ and $Y = X^2$. Then $\mathrm{E}(XY) = \mathrm{E}(X^3) = 0 = \mathrm{E}(X)\mathrm{E}(Y)$, yet Y is completely determined by X. Covariance measures only linear association.

Cheap randomness

Random numbers are expensive for a computer to generate. In 1979 Larry Carter and Mark Wegman showed that for many algorithms, such as *hashing*, which stores data at pseudo-random locations, pairwise independent random choices are just as good as fully independent ones. Pairwise independent values can be generated very cheaply: for example, $a + bk \pmod{p}$ for $k = 0, 1, \dots, p-1$, with a and b random, gives p values any two of which are independent. The variance calculation of this question is why it works: Chebyshev's inequality needs only variances, and variances need only pairs.

The thinking skill

Decompose into indicators

To find the mean and variance of a count, write it as a sum of indicator variables, one for each thing that might be counted. The mean needs single indicators; the variance needs pairs.

This is one of the most useful techniques in probability, and it appeared already in Section 9.8. At university it powers the analysis of random graphs, of algorithms and of statistical tests.

Try this next

1 **Two values.** (From STEP III, 2000.) X takes the values $x_1 \neq x_2$ and Y the values $y_1 \neq y_2$, with $\mathrm{P}(X = x_1, Y = y_1) = a$, $\mathrm{P}(X = x_1) = q$, $\mathrm{P}(Y = y_1) = p$. Show that if $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$ then $(a - pq)(x_1 - x_2)(y_1 - y_2) = 0$, and deduce that X and Y are independent. Give a joint distribution for A and B, each uniform on $\{-1, 0, 1\}$, with $\mathrm{E}(AB) = \mathrm{E}(A)\mathrm{E}(B)$ but not independent.

2 **A bigger team.** For $n = 7$ in part (i), find the mean and standard deviation of the score, and compare the example score 7 with them.

3 **Birthdays again.** Use indicators to find the mean and variance of the number of pairs sharing a birthday among k people (Section 9.9). Are the indicators pairwise independent?

4 **Pairwise but not triple.** Find three events, each of probability $\frac{1}{2}$, with $\mathrm{P}(A \cap B \cap C) = \frac{1}{8}$ but which are not pairwise independent.

Further reading

- **Geoffrey Grimmett & David Stirzaker, *Probability and Random Processes*** (Oxford University Press), the opening chapters on independence.
- **Michael Mitzenmacher & Eli Upfal, *Probability and Computing*** (Cambridge University Press). Pairwise independence and its uses in algorithms.
- **Joseph Blitzstein & Jessica Hwang, *Introduction to Probability*** (CRC Press; free online), on indicators and covariance.

12.5 The Last Passenger's Seat

The question

A train has n seats, where $n \geq 2$. For a particular journey, all n seats have been sold, and each of the n passengers has been allocated a seat. The passengers arrive one at a time and are labelled $T_1, \ldots, T_n$ according to the order in which they arrive: T_1 arrives first and T_n arrives last. The seat allocated to T_r $(r = 1, \ldots, n)$ is labelled S_r.

Passenger T_1 ignores their allocation and decides to choose a seat at random (each of the n seats being equally likely). However, for each $r \geq 2$, passenger T_r sits in S_r if it is available or, if S_r is not available, chooses from the available seats at random.

(i) Let P_n be the probability that, in a train with n seats, T_n sits in S_n. Write down the value of P_2 and find the value of P_3.

(ii) Explain why, for $k = 2, 3, \ldots, n-1$,

$$\mathrm{P}(T_n \text{ sits in } S_n \mid T_1 \text{ sits in } S_k) = P_{n-k+1},$$

and deduce that, for $n \geq 3$,

$$P_n = \frac{1}{n}\Big(1 + \sum_{r=2}^{n-1} P_r\Big).$$

(iii) Give the value of P_n in its simplest form and prove your result by induction.

(iv) Let Q_n be the probability that, in a train with n seats, T_{n-1} sits in S_{n-1}. Determine Q_n for $n \geq 2$.

The hook

A hundred passengers board a full plane. The first has lost their boarding card and sits anywhere. Each later passenger takes their own seat if it is free, and otherwise a random free seat. What is the probability that the last passenger gets their own seat? The chaos might cascade through the whole plane, so you might expect a small probability; or perhaps most passengers are unaffected and it should be close to 1.

The answer is exactly $\frac{1}{2}$, for any number of passengers. This question proves it with a recurrence and induction. But there is also a beautiful one-line argument by symmetry, which explains why the answer is $\frac{1}{2}$ and lets you write down the answer for any

passenger at once.

Before you start

Pause and try Work out P_2 and P_3 by listing what can happen.

With two seats, T_1 takes S_1 (then T_2 gets S_2) or S_2, each with probability $\frac{1}{2}$: $P_2 = \frac{1}{2}$. With three seats: if T_1 takes S_1 (probability $\frac{1}{3}$), everyone is fine; if T_1 takes S_3, T_3 loses; if T_1 takes S_2, then T_2 chooses between S_1 and S_3 at random, and T_3 is fine only if T_2 chooses S_1. So $P_3 = \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{2}$.

Working it through

STAGE 1 The recurrence

Ask yourself Suppose T_1 sits in S_k, with $2 \le k \le n-1$. What happens to passengers $T_2, \dots, T_{k-1}$? When T_k arrives, which seats are free, and how does the situation compare with a smaller train?

Passengers $T_2, \dots, T_{k-1}$ find their seats free and sit in them. When T_k arrives, the free seats are S_1 and $S_{k+1}, \dots, S_n$: there are $n-k+1$ of them. T_k chooses one at random, and from then on each passenger takes their own seat if free and a random one if not. This is exactly the original problem with $n-k+1$ seats, T_k playing the role of the first passenger and S_1 playing the role of T_k's "own" seat (the one that, if taken, sorts everything out). So the conditional probability that T_n gets S_n is P_{n-k+1}.

Now condition on T_1's seat, each with probability $\frac{1}{n}$: S_1 gives success for sure, S_n gives certain failure, and S_k gives P_{n-k+1}. So

$$P_n = \frac{1}{n}\Big(1 + 0 + \sum_{k=2}^{n-1} P_{n-k+1}\Big) = \frac{1}{n}\Big(1 + \sum_{r=2}^{n-1} P_r\Big). \qquad \blacksquare$$

STAGE 2 The answer

Ask yourself $P_2 = P_3 = \frac{1}{2}$. Guess P_n and prove it by strong induction.

Claim: $P_n = \frac{1}{2}$ for all $n \ge 2$. It holds for $n = 2$. If $P_r = \frac{1}{2}$ for $2 \le r \le n-1$, then

$$P_n = \frac{1}{n}\Big(1 + \frac{n-2}{2}\Big) = \frac{1}{n} \cdot \frac{n}{2} = \frac{1}{2}. \qquad \blacksquare$$

STAGE 3 The penultimate passenger

Ask yourself Try to find a slicker argument. At the end, which seat is left for T_n? Could it be S_2?

The last seat left for T_n is either S_1 or S_n. It cannot be S_k for $2 \le k \le n-1$, because T_k would have taken it. And every random choice made before then treats S_1 and S_n symmetrically: whenever a displaced passenger chooses, S_1 and S_n are equally likely to be picked. The first of the two to be chosen decides the outcome, so each is equally likely to be the one left: $P_n = \frac{1}{2}$.

The same argument works for any passenger T_k, $k \ge 2$. T_k loses their seat only if some earlier passenger took it. The displacement continues until someone picks S_1; among the seats $S_1, S_k, S_{k+1}, \ldots, S_n$, which are all treated alike by the choosers before T_k arrives, the first to be chosen is equally likely to be any of these $n-k+2$ seats, and T_k loses only if it is S_k. So

$$\mathrm{P}(T_k \text{ sits in } S_k) = \frac{n-k+1}{n-k+2}.$$

For $k = n-1$ (with $n \ge 3$) this gives $Q_n = \frac{2}{3}$; and $Q_2 = \frac{1}{2}$, since then $T_{n-1} = T_1$, who sits in S_1 with probability $\frac{1}{2}$. ■

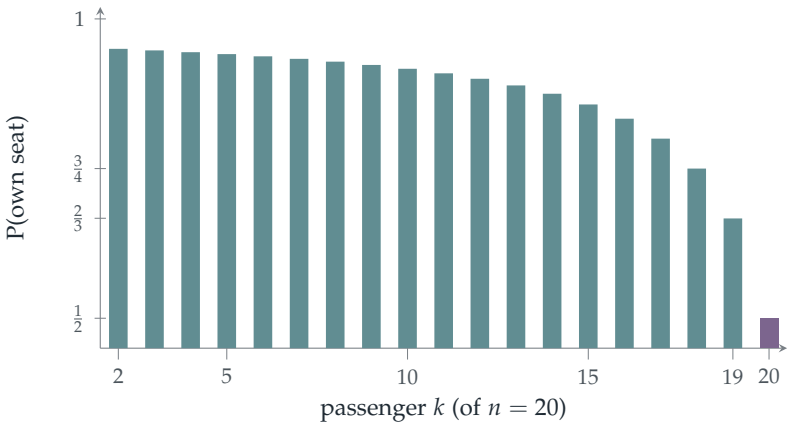


Figure 12.5.1 The probability that passenger k gets their own seat, in a train of $n = 20$. For $2 \le k \le n$ it is $\frac{n-k+1}{n-k+2}$: from $\frac{19}{20}$ for T_2 down to $\frac{2}{3}$ for T_{19} and $\frac{1}{2}$ for the last passenger (plum).

Looking back

- **Reduce to a smaller copy.** After T_1 sat in S_k, the rest of the problem was the same problem with fewer seats.
- **Strong induction.** The recurrence involved all smaller cases, so the induction assumed the result for all of them.
- **Find the symmetry.** The seats S_1 and S_n are interchangeable throughout, which explains the $\frac{1}{2}$ without any algebra.

What's really going on

Symmetry arguments

The second proof is an example of a powerful idea in probability: if two outcomes are related by a symmetry of the random process, they have the same probability. The process here does not distinguish between S_1 and S_n until one of them is chosen, so they must be equally likely to be the survivor. Symmetry arguments often replace long calculations with a single observation. The next few sections are full of them: an extra coin toss in Section 12.6, a knockout draw in Section 12.7.

How many passengers are displaced?

The number of displaced passengers (those who do not get their own seat) is also nicely behaved. T_1 is displaced unless they choose S_1, and T_k (for $k \geq 2$) is displaced with probability $\frac{1}{n-k+2}$. Adding, the expected number of displaced passengers is

$$\left(1 - \frac{1}{n}\right) + \left(\frac{1}{n} + \frac{1}{n-1} + \cdots + \frac{1}{2}\right) = H_n - \frac{1}{n},$$

roughly $\ln n$: even in a plane of 300 the chaos affects only about six passengers on average. The harmonic numbers, again (Section 9.11).

The thinking skill

Look for a symmetry

If the random process cannot tell two outcomes apart, they are equally likely. Find the moment at which they first become distinguishable.

At university, symmetry arguments appear throughout probability, from the reflection principle (Section 9.10) to exchangeability, the property that a sequence of random variables has the same distribution in any order, which is the basis of much of modern statistics.

Try this next

1. **Two forgetful passengers.** If the first *two* passengers both choose at random, what is the probability that the last gets their own seat?
2. **Displaced passengers.** Verify the expected number of displaced passengers for $n = 3$ by listing the possibilities.
3. **An indirect route.** Show, using the recurrence, that $\sum_{r=2}^{n-1} P_r = \frac{n-2}{2}$ is consistent with the symmetry argument.

4 **The last seat.** Show that, conditional on the last passenger not getting their own seat, the last passenger gets S_1 with probability 1. Why?

Further reading

- **Peter Winkler, *Mathematical Puzzles: A Connoisseur's Collection*** (A K Peters). Includes the airline seat puzzle and many others solved by symmetry.
- **Frederick Mosteller, *Fifty Challenging Problems in Probability*** (Dover). A classic collection.
- **Joseph Blitzstein & Jessica Hwang, *Introduction to Probability*** (CRC Press; free online).

12.6 One More Coin

The question

(i) Alice tosses a fair coin twice and Bob tosses a fair coin three times. Calculate the probability that Bob gets more heads than Alice.

(ii) Alice tosses a fair coin three times and Bob tosses a fair coin four times. Calculate the probability that Bob gets more heads than Alice.

(iii) Let p_1 be the probability that Bob gets the same number of heads as Alice, and let p_2 be the probability that Bob gets more heads than Alice, when Alice and Bob each toss a fair coin n times.

Alice tosses a fair coin n times and Bob tosses a fair coin $n+1$ times. Express the probability that Bob gets more heads than Alice in terms of p_1 and p_2, and hence obtain a generalisation of the results of parts (i) and (ii).

The hook

Bob tosses one more coin than Alice. What is the probability that he gets more heads? He has an advantage, but heads are not guaranteed, and ties go against him. Work out the first two cases by brute force and the answer is $\frac{1}{2}$ both times. Is that a coincidence?

It is not, and there is a proof that takes one sentence, once you notice that Bob must get either more heads than Alice or more tails, but not both. This question is a model of how a calculation can suggest a pattern, and how a clever symmetry can then explain it.

Before you start

Ask yourself Alice tosses twice: what is the distribution of her number of heads? And Bob, tossing three times?

Alice gets $0, 1, 2$ heads with probabilities $\frac{1}{4}, \frac{2}{4}, \frac{1}{4}$; Bob gets $0, 1, 2, 3$ with probabilities $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$.

Working it through

STAGE 1 Brute force

For part (i), add up over Alice's number of heads:

$$\begin{aligned} \mathrm{P}(B > A) &= \tfrac{1}{4}\,\mathrm{P}(B \geq 1) + \tfrac{2}{4}\,\mathrm{P}(B \geq 2) + \tfrac{1}{4}\,\mathrm{P}(B = 3) \\ &= \tfrac{1}{4} \cdot \tfrac{7}{8} + \tfrac{1}{2} \cdot \tfrac{4}{8} + \tfrac{1}{4} \cdot \tfrac{1}{8} = \frac{7+8+1}{32} = \frac{1}{2}. \end{aligned}$$

Part (ii) is the same calculation with more terms; counting the $8 \times 16 = 128$ equally likely outcomes, 64 have Bob ahead (Figure 12.6.1), so again the answer is $\frac{1}{2}$.

STAGE 2 Splitting Bob's tosses

Ask yourself Think of Bob's $n+1$ tosses as n tosses followed by one extra toss. After his first n tosses, he is ahead of Alice, level with her, or behind. What happens in each case?

After n tosses each, Bob is ahead with probability p_2, level with probability p_1, and behind with probability p_2 (by symmetry between Alice and Bob). If he is ahead, he stays ahead. If he is level, his extra toss puts him ahead with probability $\frac{1}{2}$. If he is behind, one extra head can at best draw level. So

$$\mathrm{P}(\text{Bob more heads}) = p_2 + \tfrac{1}{2}p_1.$$

Since the three cases cover everything, $2p_2 + p_1 = 1$, and so $p_2 + \frac{1}{2}p_1 = \frac{1}{2}$. With one extra coin, Bob gets more heads with probability exactly $\frac{1}{2}$, for every n. ■

Pause and try Here is the one-sentence proof. Bob has $n+1$ tosses, so his heads and tails add to $n+1$, one more than Alice's. Show that Bob has more heads than Alice if and only if he does *not* have more tails than Alice. Since heads and tails are symmetric, these two events have equal probability, so each is $\frac{1}{2}$.

Indeed, writing $H_A + T_A = n$ and $H_B + T_B = n+1$: if $H_B > H_A$ then $H_B \geq H_A + 1$, so $T_B = n + 1 - H_B \leq n - H_A = T_A$; and conversely if $T_B \leq T_A$ then $H_B \geq H_A + 1$. So "more heads" and "more tails" are complementary, and swapping heads and tails (which does not change the probabilities, as the coins are fair) swaps them.

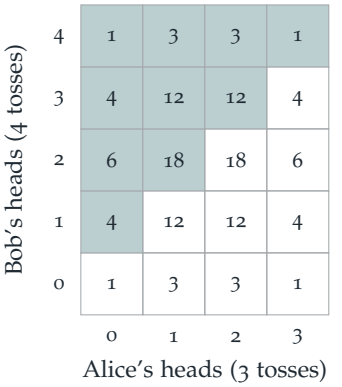


Figure 12.6.1 Part (ii): the number of the 128 equally likely outcomes with each combination of heads. Bob is ahead in the shaded cells, which contain exactly half of the outcomes.

Looking back

- **Compute small cases.** Two calculations suggested the answer $\frac{1}{2}$.
- **Condition on a natural split.** Separating Bob's extra toss reduced the problem to an even contest plus one coin.
- **Find the complementary pair.** "More heads" and "more tails" are complements and are swapped by a symmetry.

What's really going on

Symmetry and complements

The slick proof combines two facts: two events that are swapped by a symmetry have equal probabilities, and two complementary events have probabilities adding to 1. If both hold, each has probability $\frac{1}{2}$. The same reasoning appears throughout probability. In the airline problem (Section 12.5), S_1 and S_n were swapped by a symmetry and exactly one of them is left at the end. In a game of Hex, the board's symmetry and the fact that exactly one player must win show that the first player has the advantage, though nobody knows a winning strategy on large boards.

How likely is a tie?

For equal numbers of tosses the tie probability p_1 matters. Alice and Bob tie when Alice's heads equal Bob's heads, which (replacing Bob's heads by his tails, by symmetry) has the same probability as Alice's heads plus Bob's heads equalling n: a total of n heads in $2n$ tosses. So

$$p_1 = \binom{2n}{n}\frac{1}{4^n} \approx \frac{1}{\sqrt{\pi n}},$$

using Stirling's approximation (Section 12.23). Ties become rare, but slowly: with 100 tosses each, the chance of a tie is still about 5.6%.

The thinking skill

Calculate, conjecture, then explain

When a calculation gives a suspiciously simple answer, look for a reason. A simple answer usually has a simple explanation.

At university this habit separates understanding from computation. The first proof here confirms the answer; the second explains it, and generalises immediately (try exercise 2).

Try this next

1 **Two extra coins.** If Bob tosses $n+2$ coins and Alice n, prove $\mathrm{P}(\text{Bob more heads}) = \frac{1}{2} + \frac{1}{2}\mathrm{P}(H' = H_A)$, where H' is the number of heads in Bob's first $n+1$ tosses.

2 **Biased coins.** If both use a coin with probability p of heads, is the answer still $\frac{1}{2}$ with one extra toss? Which step of the slick proof fails?

3 **Ties.** Show that $p_1 = \binom{2n}{n}/4^n$ by the trick in the text, and evaluate it for $n = 1, 2, 3$.

4 **Dice.** Alice rolls n dice and Bob rolls $n+1$. Is the probability that Bob gets more sixes equal to $\frac{1}{2}$?

Further reading

- **Peter Winkler, *Mathematical Mind-Benders*** (A K Peters). Many probability puzzles with symmetric solutions.
- **Gabor Székely, *Paradoxes in Probability Theory and Mathematical Statistics*** (Reidel).
- **Frederick Mosteller, *Fifty Challenging Problems in Probability*** (Dover).

12.7 Who Meets Whom in a Knockout

The question

A tennis tournament is arranged for 2^n players. It is organised as a knockout tournament so that only the winners in any given round proceed to the next round. Opponents in each round except the final are drawn at random, and in any match either player has a probability of $\frac{1}{2}$ of winning. Two players are chosen at random before the draw for the first round. Find the probabilities that they play each other:

(i) in the first round,

(ii) in the final,

(iii) in the tournament.

The hook

At Wimbledon, 128 players enter the draw. Pick two at random: what is the chance that they meet at some point? You might try to add up the probabilities of meeting in each round, each a product of chances of surviving and being drawn together. That works, but there is a much better way. Count the matches: every match is between a pair of players, and there are exactly 127 matches. By symmetry, every pair of players is equally likely to be any of them. So the chance is 127 divided by the number of pairs.

This question is a perfect example of solving a probability problem by counting the right thing. The answer, $\frac{1}{2^{n-1}}$, comes out in one line.

Before you start

Ask yourself How many matches are played in a knockout tournament with 2^n players? (Each match eliminates one player.)

Each match eliminates exactly one player, and all but one player is eliminated, so there are $2^n - 1$ matches.

Working it through

STAGE 1 The first round

Call the two chosen players X and Y. In the first round, X's opponent is equally likely to be any of the other $2^n - 1$ players, so

$$\mathrm{P}(\text{meet in round } 1) = \frac{1}{2^n - 1}.$$

STAGE 2 The key symmetry

Ask yourself The draw in each round is random and every match is a fair coin toss. Does the process treat any pair of players differently from any other?

No. Relabelling the players does not change any probability, since draws are random and every match is 50–50. So any particular match (say, the final) is equally likely to be between any of the $\binom{2^n}{2}$ pairs.

STAGE 3 The final

The final is one specific match, so the probability that it is between X and Y is

$$\frac{1}{\binom{2^n}{2}} = \frac{2}{2^n(2^n - 1)}.$$

(Check: both must reach the final, which needs them in different halves and each to win $n - 1$ matches.)

STAGE 4 At any stage

Ask yourself X and Y can meet at most once. How does that let you add probabilities over matches?

Let I_m be 1 if match m is between X and Y. Each of the $2^n - 1$ matches has $\mathrm{P}(I_m = 1) = \binom{2^n}{2}^{-1}$, and the events are mutually exclusive (they cannot meet twice), so

$$\mathrm{P}(\text{meet}) = \frac{2^n - 1}{\binom{2^n}{2}} = \frac{2(2^n - 1)}{2^n(2^n - 1)} = \frac{1}{2^{n-1}}. \qquad \blacksquare$$

For Wimbledon, with $n = 7$, the probability that two given players meet is $\frac{1}{64}$. As a check, the first round contains 2^{n-1} of the $2^n - 1$ matches, giving $\frac{2^{n-1}}{\binom{2^n}{2}} = \frac{1}{2^n - 1}$, agreeing with Stage 1.

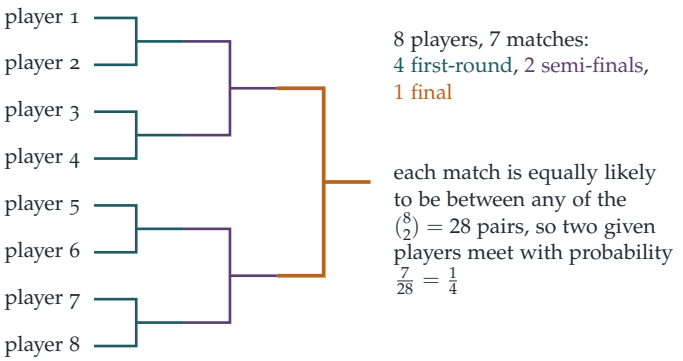


Figure 12.7.1 A knockout tournament for $2^3 = 8$ players (drawn as a fixed bracket, though the question redraws each round at random). There are 7 matches. By symmetry each is equally likely to involve any of the 28 pairs, so two chosen players meet with probability $\frac{7}{28} = \frac{1}{4} = \frac{1}{2^{n-1}}$.

Looking back

- **Count the matches.** The number of matches, $2^n - 1$, was the key fact.
- **Symmetry among pairs.** Every pair of players is equally likely to contest any particular match.
- **Linearity over matches.** Since two players meet at most once, the probability of meeting is the expected number of meetings, a sum over matches.

What's really going on

Expected counts

The argument in Stage 4 is really an expectation calculation: the number of times X and Y meet is a sum of indicators, one per match, and its expectation (which equals the probability of meeting, since it is 0 or 1) is the sum of the indicators' expectations. This is linearity of expectation, used in Section 9.11 and Section 12.4. Its great advantage is that it ignores all the complicated dependence between the rounds.

Seeded tournaments

Real tournaments are *seeded*: the strongest players are placed so that they cannot meet until the late rounds. And real matches are not 50–50. Both destroy the symmetry, and calculating the chance that two given players meet becomes a genuine computation. Statisticians who model sport use exactly such calculations, with each match's probability estimated from players' ratings, to forecast tournaments. The symmetric case is the baseline against which the real effects are measured.

The thinking skill

Count the right thing

A probability can often be found as (number of favourable objects) ÷ (number of equally likely objects), once the right objects are chosen. Here they were matches and pairs.

Choosing what to count is the art of combinatorial probability. At university the same idea appears as *double counting* (Section 7.9) and in the probabilistic method, where the expected number of some structure proves that it exists.

Try this next

1. **Round by round.** Show directly that the probability that X and Y meet in round k is $\frac{2^{n-k}}{\binom{2^n}{2}}$, and check that these add up to $\frac{1}{2^{n-1}}$.
2. **A fixed bracket.** If instead the bracket is fixed in advance (as in the figure) but all players are placed in it at random, does the answer change?
3. **A strong player.** Suppose player X wins every match. What is now the probability that X meets Y?
4. **A league.** In a league of $2n$ teams where every pair meets once, how many matches are there? What is the analogous "probability of meeting"?

Further reading

- **Frederick Mosteller, *Fifty Challenging Problems in Probability*** (Dover), which includes tournament problems.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican), on modelling and forecasting.
- **Joseph Blitzstein & Jessica Hwang, *Introduction to Probability*** (CRC Press; free online), on indicators and symmetry.

12.8 Penney's Game

The question

In a game for two players, a fair coin is tossed repeatedly. Each player is assigned a sequence of heads and tails and the player whose sequence appears first wins. Four players, A, B, C and D take turns to play the game. Each time they play, A is assigned the sequence TTH (i.e. Tail then Tail then Head), B is assigned THH, C is assigned HHT and D is assigned HTT.

(i) A and B play the game. Let p_{HH}, p_{HT}, p_{TH} and p_{TT} be the probabilities of A winning the game given that the first two tosses of the coin show HH, HT, TH and TT, respectively. Explain why $p_{\mathrm{TT}} = 1$, and why $p_{\mathrm{HT}} = \frac{1}{2}p_{\mathrm{TH}} + \frac{1}{2}p_{\mathrm{TT}}$. Show that $p_{\mathrm{HH}} = p_{\mathrm{HT}} = \frac{2}{3}$ and that $p_{\mathrm{TH}} = \frac{1}{3}$. Deduce that the probability that A wins the game is $\frac{2}{3}$.

(ii) B and C play the game. Find the probability that B wins.

(iii) Show that if C plays D, then C is more likely to win than D, but that if D plays A, then D is more likely to win than A.

The hook

You and a friend each choose a sequence of three coin tosses, and a coin is tossed until one of the sequences appears. Every sequence of three tosses is equally likely to occur at any given moment, so surely every choice is equally good? Not at all. Whatever sequence your friend chooses, you can choose one that beats it, sometimes with odds of 7 to 1.

This is *Penney's game*, invented by Walter Penney in 1969. The question shows that A beats B, B beats C, C beats D, and D beats A: a cycle, like rock–paper–scissors. "Beats" is not transitive. Whoever chooses second can always win in the long run.

Before you start

Ask yourself Suppose the tosses so far end in TT and nobody has won yet. Can THH ever appear before TTH?

From TT, every further T keeps the last two tosses as TT, and the first H completes TTH. For THH to appear we would need a TH, which can only arise from a T followed by an

H, and that H already completes TTH. So once TT has occurred, A wins.

Working it through

STAGE 1 States

Ask yourself Neither sequence can be completed in two tosses. After two tosses, what is the next toss's effect from each state?

Since the game depends only on the last two tosses, we can track the state (last two tosses):

- From TT: A wins (shown above), so $p_{\mathrm{TT}} = 1$.
- From HT: the next toss gives TH or TT, so $p_{\mathrm{HT}} = \frac{1}{2}p_{\mathrm{TH}} + \frac{1}{2}p_{\mathrm{TT}}$.
- From TH: an H completes THH (B wins); a T gives HT. So $p_{\mathrm{TH}} = \frac{1}{2}\cdot 0 + \frac{1}{2}p_{\mathrm{HT}}$.
- From HH: an H gives HH again; a T gives HT. So $p_{\mathrm{HH}} = \frac{1}{2}p_{\mathrm{HH}} + \frac{1}{2}p_{\mathrm{HT}}$, giving $p_{\mathrm{HH}} = p_{\mathrm{HT}}$.

STAGE 2 Solving

Substituting $p_{\mathrm{TH}} = \frac{1}{2}p_{\mathrm{HT}}$ into the HT equation: $p_{\mathrm{HT}} = \frac{1}{4}p_{\mathrm{HT}} + \frac{1}{2}$, so $p_{\mathrm{HT}} = \frac{2}{3}$, $p_{\mathrm{HH}} = \frac{2}{3}$ and $p_{\mathrm{TH}} = \frac{1}{3}$. Each of the four starting states has probability $\frac{1}{4}$, so

$$\mathrm{P}(A \text{ wins}) = \tfrac{1}{4}\left(\tfrac{2}{3} + \tfrac{2}{3} + \tfrac{1}{3} + 1\right) = \tfrac{2}{3}. \qquad \blacksquare$$

STAGE 3 B against C

Ask yourself C wants HHT and B wants THH. When does C win? (Look at what must come just before the first HH.)

If the first two tosses are HH, then the first T after that completes HHT, before THH can appear: C wins. Otherwise, there is a T somewhere before the first occurrence of HH, and so the first HH is immediately preceded by a T: THH appears at that moment, before HHT (which needs a T after HH). So C wins only if the game starts HH:

$$\mathrm{P}(B \text{ wins}) = \tfrac{3}{4}. \qquad \blacksquare$$

STAGE 4 The cycle

The same method (or the shortcut in the What's Really Going On section) shows that C (HHT) beats D (HTT) with probability $\frac{2}{3}$, and D (HTT) beats A (TTH) with probability $\frac{3}{4}$; the last is the mirror image of Stage 3 with heads and tails swapped. So

$$A \text{ beats } B, \qquad B \text{ beats } C, \qquad C \text{ beats } D, \qquad D \text{ beats } A. \qquad \blacksquare$$

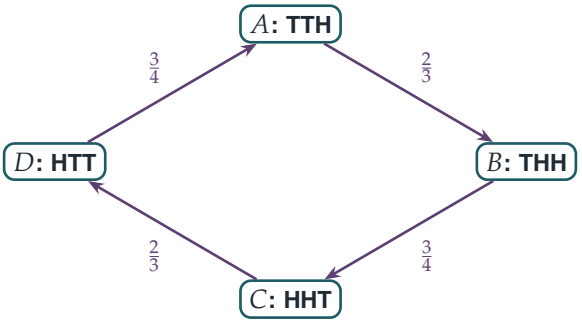


Figure 12.8.1 Penney's cycle. An arrow from X to Y means that X's sequence beats Y's, with the probability shown. Every sequence is beaten by another, so whoever chooses second can always pick a winner.

Looking back

- **Choose the right state.** Only the last two tosses mattered, so four states and four equations captured the whole game.
- **First-step analysis.** Each equation conditioned on the next toss: the probability from a state is the average over where the next toss leads.
- **Look for a decisive event.** In Stage 3, "the first HH is preceded by a T" settled the game without equations.

What's really going on

Why some sequences are better

All sequences of three tosses are equally likely at any given position, and the expected waiting time for each is easy to compute. But the waiting times differ: you wait on average 8 tosses for TTH but 14 for TTT, because a sequence that overlaps itself (like TTT, where a near miss TTH ruins two tosses of progress) is slow to appear. And in a race between two sequences, what matters is how each overlaps the *other*. THH beats HHT because every occurrence of HHT (except at the very start) is preceded by a T, which means THH has just occurred. The second player's trick is always the same: take the first two symbols of your opponent's sequence, and put in front of them the opposite of the opponent's middle symbol.

Conway's formula

John Conway found a remarkable formula for the odds. For two sequences X and Y of length n, define the *correlation* XY as a binary number with n digits: reading from the left, the ith digit is 1 if the part of X starting at its ith symbol equals the beginning of Y of the same length. Then the odds that Y appears before X are

$$(XX - XY) : (YY - YX).$$

For $X = \text{HHT}$ and $Y = \text{THH}$: $XX = 100_2 = 4$ (only the whole of HHT matches itself); $XY = 001_2 = 1$ (only the final T of HHT matches the start of THH); $YY = 100_2 = 4$; and $YX = 011_2 = 3$ (both HH and H at the end of THH match the start of HHT). So the odds that THH comes first are $(4-1) : (4-3) = 3 : 1$, the $\frac{3}{4}$ of Stage 3. Conway's formula works for sequences of any length and any alphabet, and its proof uses the theory of *martingales*: imagine a casino in which gamblers bet on the sequence appearing, and use the fact that a fair game has zero expected profit.

Penney and Gardner

Walter Penney described the game in a short note in the *Journal of Recreational Mathematics* in 1969. Martin Gardner made it famous in his *Scientific American* column in 1974, including Conway's formula, which Conway had found but not published. Non-transitive relations of this kind (Efron's dice, where die A beats B, B beats C, C beats D and D beats A, are another example) are a recurring source of surprise, and they matter in voting theory, where majority preferences can form cycles: Condorcet's paradox.

The thinking skill

First-step analysis

To find a probability or expectation for a process that moves between states, condition on the first step. This gives one linear equation per state.

This is the basic technique for Markov chains, which you will meet in Section 12.16 and throughout a university course in probability. The same method finds the probability of reaching one state before another, the expected time to absorption, and the probability of ruin in gambling problems.

Try this next

1 **Four players together.** (From STEP II, 2015.) Players choose HHT, THH, TTH and HTT. Show that if only HHT and THH play, HHT wins with probability $\frac{1}{4}$. If all four play together, show that each wins with probability $\frac{1}{4}$. If only THH and TTH play, find the probability that TTH wins.

2 **Waiting times.** Show that the expected number of tosses until HH first appears is 6, and until HT is 4. (Use first-step analysis.)

3 **The best reply.** For each of the eight sequences of length three, find the sequence that beats it with the greatest probability. Which sequence is beaten most easily?

4 Conway's formula. Use Conway's formula to check that HHT beats HTT with probability $\frac{2}{3}$.

Further reading

- **Martin Gardner, *Time Travel and Other Mathematical Bewilderments*** (W. H. Freeman). Includes Penney's game and Conway's formula.
- **Ronald Graham, Donald Knuth & Oren Patashnik, *Concrete Mathematics*** (Addison-Wesley), Section 8.4. Penney's game with generating functions.
- **David Williams, *Probability with Martingales*** (Cambridge University Press). The martingale proof, for later.

12.9 A Fair Decision from a Biased Coin

The question

(i) Two people adopt the following procedure for deciding where to go for a cup of tea: either to a hotel or to a tea shop. Each person has a coin which has a probability p of showing heads and q of showing tails (where $p+q=1$). In each round of the procedure, both people toss their coins once. If both coins show heads, then both people go to the hotel; if both coins show tails, then both people go to the tea shop; otherwise, they continue to the next round. This process is repeated until a decision is made.

Show that the probability that they make a decision on the nth round is

$$(q^2+p^2)(2qp)^{n-1}.$$

Show also that the probability that they make a decision on or before the nth round is at least

$$1-\frac{1}{2^n}$$

whatever the value of p.

(ii) Three people adopt the following procedure for deciding where to go for a cup of tea: either to a hotel or to a tea shop. Each person has a coin which has a probability p of showing heads and q of showing tails (where $p+q=1$). In the first round of the procedure, all three people toss their coins once. If all three coins show heads, then all three people go to the hotel; if all three coins show tails, then all three people go to the tea shop; otherwise, they continue to the next round.

In the next round the two people whose coins showed the same face toss again, but the third person just turns over his or her coin. If all three coins show heads, then all three people go to the hotel; if all three coins show tails, then all three people go to the tea shop; otherwise, they go to the third round.

Show that the probability that they make a decision on or before the second round is at least $\frac{7}{16}$, whatever the value of p.

The hook

You need to make a fair decision, but the only coin available is bent, and you do not know how biased it is. Can you still make a fair choice? John von Neumann's answer,

from 1951: toss it twice. If it comes up HT, choose the hotel; if TH, the tea shop; if HH or TT, toss twice more. Since HT and TH both have probability pq, the decision is perfectly fair, whatever p is.

The procedure in this question is the opposite: it decides on HH and TT, which is quick but unfair (hotel with probability $\frac{p^2}{p^2+q^2}$). The question shows that it never takes long, whatever the coin. Comparing the two procedures shows a basic trade-off in generating randomness: speed against fairness.

Before you start

Ask yourself In one round, what is the probability of a decision? Of no decision? What is the largest value $2pq$ can take?

A decision is made if the coins agree, with probability $p^2 + q^2$; otherwise, with probability $2pq$, they continue. Since $pq = p(1-p) \leq \frac{1}{4}$ (with equality at $p = \frac{1}{2}$), we have $2pq \leq \frac{1}{2}$.

Working it through

STAGE 1 Decision on the nth round

Rounds are independent. A decision on the nth round needs $n-1$ rounds of disagreement followed by agreement:

$$\mathrm{P}(\text{decision on round } n) = (2pq)^{n-1}(p^2+q^2). \qquad \blacksquare$$

STAGE 2 Decision by round n

Ask yourself What is the probability of *no* decision in the first n rounds?

No decision in n rounds has probability $(2pq)^n \leq \left(\frac{1}{2}\right)^n$, so a decision by round n has probability at least $1 - \frac{1}{2^n}$, whatever p. $\blacksquare$

STAGE 3 Three people

Ask yourself After a first round with two heads and one tail, the third person turns the tail over to a head. What must happen in round two for a decision? And after two tails and one head?

Round one decides with probability $p^3 + q^3$. Otherwise two coins agree and one differs. If the majority showed heads (probability $3p^2q$), the odd coin is turned to heads, and the round-two decision needs the other two to toss heads: probability p^2 (all tails is

now impossible). Similarly with a tails majority (probability $3pq^2$), a decision needs two tails: q^2. So

$$\mathrm{P}(\text{decision by round 2}) = p^3 + q^3 + 3p^2q \cdot p^2 + 3pq^2 \cdot q^2.$$

Ask yourself Write everything in terms of $s = pq$. (Use $p + q = 1$.)

We have $p^3 + q^3 = (p+q)^3 - 3pq(p+q) = 1 - 3s$, and $3p^4q + 3pq^4 = 3s(p^3 + q^3) = 3s(1 - 3s)$. So

$$\mathrm{P} = 1 - 3s + 3s - 9s^2 = 1 - 9s^2 \geq 1 - \frac{9}{16} = \frac{7}{16},$$

since $s \leq \frac{1}{4}$. ■

Looking back

- **Complements.** "At least once by round n" was easiest as $1-$ ("never in n rounds").
- **Bound the worst case.** The unknown p was handled by the inequality $pq \leq \frac{1}{4}$.
- **Symmetric functions.** Expressions symmetric in p and q simplify when written in terms of $p + q = 1$ and pq.

What's really going on

Von Neumann's trick

The procedure in the question is quick but biased. Von Neumann's procedure is fair: toss twice, decide on HT or TH, repeat on HH or TT. Each attempt succeeds with probability $2pq$, so it takes on average $\frac{1}{2pq}$ attempts, which is slow when the coin is very biased (Figure 12.9.1). The question's procedure decides with probability $p^2 + q^2 \geq \frac{1}{2}$ per round, but its answer favours the hotel when $p > \frac{1}{2}$.

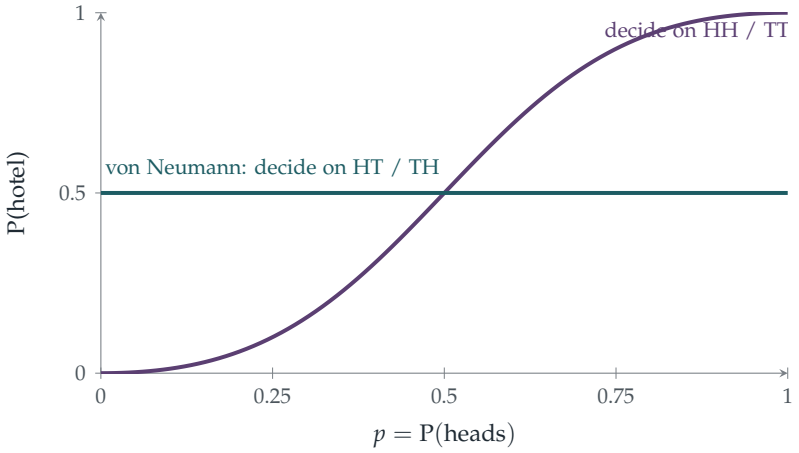


Figure 12.9.1 The probability of choosing the hotel, against the bias of the coin. Deciding when the coins agree (plum) is fast but favours the more likely face. Von Neumann's procedure (teal) is exactly fair for every p, at the cost of taking longer when the coin is very biased.

Extracting randomness

Von Neumann's trick is the first *randomness extractor*: a procedure that turns imperfect randomness into perfect randomness. It wastes a lot (each pair of tosses yields at most one fair bit, and often none), and better methods recycle the discarded HH and TT pairs to extract almost all the randomness the coin contains, which is measured by its *entropy* $-p \log_2 p - q \log_2 q$ bits per toss. Extractors are used in computers' hardware random number generators, which harvest unpredictable but biased physical noise and must turn it into unbiased bits for cryptography.

Von Neumann and random numbers

John von Neumann described the trick in a 1951 paper, "Various techniques used in connection with random digits", written when he and his colleagues were running some of the first computer simulations (the "Monte Carlo method") for the design of nuclear weapons. In the same paper he made a famous remark about generating random numbers by arithmetic: "Anyone who considers arithmetical methods of producing random digits is, of course, in a state of sin."

The thinking skill

Use symmetry to cancel an unknown

When a parameter is unknown, look for events whose probabilities are equal whatever its value, and base decisions on those.

HT and TH have the same probability pq, so comparing them removes p altogether. At university, the same idea appears in statistics as *pivotal quantities* and in the design of experiments, where randomisation and symmetry cancel out unknown effects.

Try this next

1 **Expected rounds.** Find the expected number of rounds for the procedure in part (i), and for von Neumann's procedure. For which p is von Neumann's procedure slower?

2 **Fair three-way choice.** Using a biased coin, devise a procedure that chooses one of three options, each with probability exactly $\frac{1}{3}$.

3 **A biased die from a fair coin.** Using a fair coin, devise a procedure that simulates a coin with probability of heads exactly $\frac{1}{3}$. What is the expected number of tosses?

4 **Three people again.** In part (ii), what is the probability that they go to the hotel, as a function of p? Is the procedure fair when $p = \frac{1}{2}$?

Further reading

- **John von Neumann, "Various techniques used in connection with random digits"**, *National Bureau of Standards Applied Mathematics Series* 12 (1951). Short and famous.
- **Donald Knuth, *The Art of Computer Programming*, Vol. 2** (Addison-Wesley), Chapter 3, on random numbers.
- **Michael Mitzenmacher & Eli Upfal, *Probability and Computing*** (Cambridge University Press), on extracting randomness and entropy.

12.10 Tying Loose Ends

The question

A box contains n pieces of string, each of which has two ends. I select two string ends at random and tie them together. This creates either a ring (if the two ends are from the same string) or a longer piece of string. I repeat the process of tying together string ends chosen at random until there are none left.

Find the expected number of rings created at the first step and hence obtain an expression for the expected number of rings created by the end of the process. Find also an expression for the variance of the number of rings created.

Given that $\ln 20 \approx 3$ and that $1 + \frac{1}{2} + \cdots + \frac{1}{n} \approx \ln n$ for large n, determine approximately the expected number of rings created in the case $n = 40\,000$.

The hook

Put a hundred pieces of string in a bag, so that only their ends poke out. Pick two ends at random and tie them; repeat until no loose ends are left. You end up with a tangle of loops. How many? A hundred strings could in principle make a hundred little rings, or one enormous loop. Remarkably, the expected number is only about 3.3; with forty thousand strings, it is still only about 6.

The key observation makes the problem almost trivial: after each knot, the box simply contains one fewer piece of string, whatever happened. The process "forgets" its history, and the number of rings becomes a sum of independent coin tosses with changing probabilities.

Before you start

Ask yourself After tying two ends together, how many pieces of string are left (counting a ring as no longer a "piece")? Does it depend on whether a ring was made?

If the two ends belong to the same string, it becomes a ring and $n - 1$ pieces remain. If they belong to different strings, those two become one longer piece, and again $n - 1$ pieces remain. Either way, we are left with $n - 1$ pieces of string, each with two ends.

Working it through

STAGE 1 The first knot

There are $2n$ ends. Choose one; its partner is equally likely to be any of the other $2n-1$ ends, and exactly one of these belongs to the same string. So the first knot makes a ring with probability $\frac{1}{2n-1}$, and the expected number of rings created at the first step is $\frac{1}{2n-1}$.

STAGE 2 All the knots

Ask yourself By the observation above, what is the situation before the second knot? The third? Let I_k indicate a ring at the knot made when k pieces remain. What are the $\mathrm{E}(I_k)$?

Before each knot the box contains k pieces for $k = n, n-1, \ldots, 1$, and the knot makes a ring with probability $\frac{1}{2k-1}$. (The last knot, with one piece left, always makes a ring.) The total number of rings is $R = I_n + I_{n-1} + \cdots + I_1$, so

$$\mathrm{E}(R) = \sum_{k=1}^{n} \frac{1}{2k-1} = 1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2n-1}. \qquad \blacksquare$$

STAGE 3 The variance

Ask yourself Are the indicators I_k independent? (Does the probability of a ring at one step depend on what happened earlier?)

At each step, the probability of a ring is $\frac{1}{2k-1}$ whatever happened before, because the configuration is always "k pieces of string". So the I_k are independent, and variances add:

$$\mathrm{Var}(R) = \sum_{k=1}^{n} \frac{1}{2k-1}\left(1 - \frac{1}{2k-1}\right) = \sum_{k=1}^{n} \frac{2k-2}{(2k-1)^2}. \qquad \blacksquare$$

STAGE 4 Forty thousand strings

Ask yourself Write the sum of the reciprocals of the odd numbers up to $2n-1$ using harmonic numbers.

The sum of reciprocals of all numbers up to $2n$, minus those of the even numbers, is

$$\sum_{k=1}^{n} \frac{1}{2k-1} = H_{2n} - \tfrac{1}{2}H_n \approx \ln(2n) - \tfrac{1}{2}\ln n = \tfrac{1}{2}\ln(4n).$$

For $n = 40\,000$, $4n = 160\,000 = 20^4$, so $\mathrm{E}(R) \approx \frac{1}{2} \cdot 4 \ln 20 \approx 6$. ■(The exact value is 6.28; the difference comes from Euler's constant, as $H_n \approx \ln n + 0.577$.)

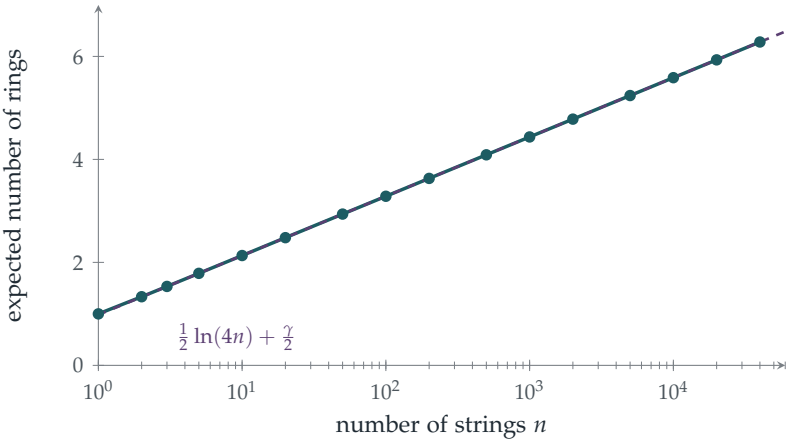


Figure 12.10.1 The expected number of rings against the number of strings, on a logarithmic scale. The exact values (teal) lie on the straight line $\frac{1}{2}\ln(4n) + \frac{\gamma}{2}$ (plum, dashed), where $\gamma \approx 0.577$ is Euler's constant. Multiplying the number of strings by 10 adds only about 1.15 rings.

Looking back

- **Find what the process forgets.** After each knot the box was "k pieces of string", regardless of history.
- **Indicators again.** The number of rings was a sum of indicators, one per knot.
- **Independence gives the variance.** Because each step's probability did not depend on the past, the indicators were independent.

What's really going on

Logarithmic growth

The expected number of rings grows like $\frac{1}{2}\ln n$, and the variance is close to the mean (both are sums of small probabilities). By the central limit theorem (Section 12.24), the number of rings is approximately normal for large n, with mean and variance about $\frac{1}{2}\ln(4n)$. With 40 000 strings, you will usually get between 2 and 11 rings.

Cycles of random permutations

The same logarithm appears in the number of cycles of a random permutation of n objects: on average $H_n \approx \ln n$ (the strings problem is a close relative, with $\frac{1}{2}\ln n$). Random permutations have few cycles: one giant cycle usually contains most of the elements, and the fraction of elements in the largest cycle averages about 0.62, the Golomb–Dickman constant. Tangling strings at random behaves the same way: most of the string ends up in one or two long loops. These facts are part of the study of *random*

combinatorial structures, which has connections to number theory (the prime factors of a random integer behave like the cycles of a random permutation).

Random knots and polymers

The string-tying problem is a toy model for physical systems. Long polymer molecules in a solution join end to end to form rings and chains, and the statistics of loop formation follow similar rules. In knot theory, mathematicians ask the harder question of how likely a random closed loop is to be knotted; for long random loops, the answer tends to 1, a result proved by De Witt Sumners and Stuart Whittington in 1988.

The thinking skill

Find the memoryless structure

If, after each step, the system is in a state that depends only on the number of steps taken, then the steps behave independently, and expectations and variances simply add.

This is the idea behind the coupon collector (Section 9.11), where each stage is a fresh geometric wait, and it reappears in the theory of Markov chains, where only the current state matters.

Try this next

1. **Small cases.** For $n = 2$, list the possibilities and check that the expected number of rings is $\frac{4}{3}$.
2. **One big loop.** Show that the probability that the process ends with a single loop containing all the strings is $\prod_{k=2}^{n}\left(1 - \frac{1}{2k-1}\right)$, and that this is approximately $\sqrt{\frac{\pi}{4n}}$ for large n. (*Hint:* relate the product to $\binom{2n}{n}$.)
3. **Cycles of permutations.** Show that the expected number of cycles of a random permutation of n objects is H_n, by building the permutation one element at a time (the "Chinese restaurant" construction).
4. **The variance.** Show that $\mathrm{Var}(R) = \mathrm{E}(R) - \sum_{k=1}^{n} \frac{1}{(2k-1)^2}$, and that the subtracted sum tends to $\frac{\pi^2}{8}$.

Further reading

- **Peter Winkler, *Mathematical Puzzles: A Connoisseur's Collection*** (A K Peters), which includes the string-tying puzzle.

- **Richard Arratia, Andrew Barbour & Simon Tavaré, *Logarithmic Combinatorial Structures*** (European Mathematical Society). Cycles, primes and random structures, for later.
- **Colin Adams, *The Knot Book*** (American Mathematical Society). An introduction to knot theory, including random knots.

12.11 Stamps, Cups and e

The question

I have a Penny Black stamp which I want to sell to my friend Jim, but we cannot agree a price. So I put the stamp under one of two cups, jumble them up, and let Jim guess which one it is under. If he guesses correctly, I add a third cup, jumble them up, and let Jim guess correctly, adding another cup each time. The price he pays for the stamp is £N, where N is the number of cups present when Jim fails to guess correctly. Find $\mathrm{P}(N = k)$. Show that $\mathrm{E}(N) = e$ and calculate $\mathrm{Var}(N)$.

The hook

A strange way to sell a stamp: the buyer keeps guessing which cup hides it, with one more cup each round, and pays one pound per cup on the table when he first guesses wrong. On average he pays £e, about £2.72.

The number e appears in probability with uncanny frequency: the chance that nobody gets their own shirt (Section 9.8), the secretary problem (Section 12.12), the Poisson distribution. This question is perhaps the simplest of all its appearances, and it turns out to have exactly the same distribution as another famous puzzle: how many random numbers between 0 and 1 must you add before the total exceeds 1? On average, e.

Before you start

Ask yourself For $N = 4$, what must happen? With what probability?

Jim must guess correctly with 2 cups (probability $\frac{1}{2}$) and with 3 cups ($\frac{1}{3}$), then fail with 4 cups ($\frac{3}{4}$). So $\mathrm{P}(N = 4) = \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{4} = \frac{3}{4!} = \frac{1}{8}$.

Working it through

STAGE 1 The distribution

For $N = k$ (with $k \geq 2$), Jim succeeds with $2, 3, \ldots, k-1$ cups and fails with k:

$$\mathrm{P}(N = k) = \frac{1}{2} \cdot \frac{1}{3} \cdots \frac{1}{k-1} \cdot \frac{k-1}{k} = \frac{k-1}{k!}.$$

(Check: $\frac{1}{2!}+\frac{2}{3!}+\frac{3}{4!}+\cdots=\sum_{k\geq 2}\left(\frac{1}{(k-1)!}-\frac{1}{k!}\right)=1$, a telescoping sum.)

STAGE 2 The expectation

Ask yourself Simplify $k\cdot\frac{k-1}{k!}$. What series do you get?

$$\mathrm{E}(N)=\sum_{k\geq 2}\frac{k(k-1)}{k!}=\sum_{k\geq 2}\frac{1}{(k-2)!}=\frac{1}{0!}+\frac{1}{1!}+\cdots=e. \qquad \blacksquare$$

STAGE 3 The variance

Ask yourself Find $\mathrm{E}(N^2)$ by writing $k^2=k(k-1)+k$, or by simplifying $\frac{k^2(k-1)}{k!}$ directly.

$$\mathrm{E}(N^2)=\sum_{k\geq 2}\frac{k^2(k-1)}{k!}=\sum_{k\geq 2}\frac{k}{(k-2)!}=\sum_{j\geq 0}\frac{j+2}{j!}=\sum_{j\geq 1}\frac{1}{(j-1)!}+2\sum_{j\geq 0}\frac{1}{j!}=e+2e=3e.$$

So $\mathrm{Var}(N)=3e-e^2=e(3-e)\approx 0.766$. $\blacksquare$

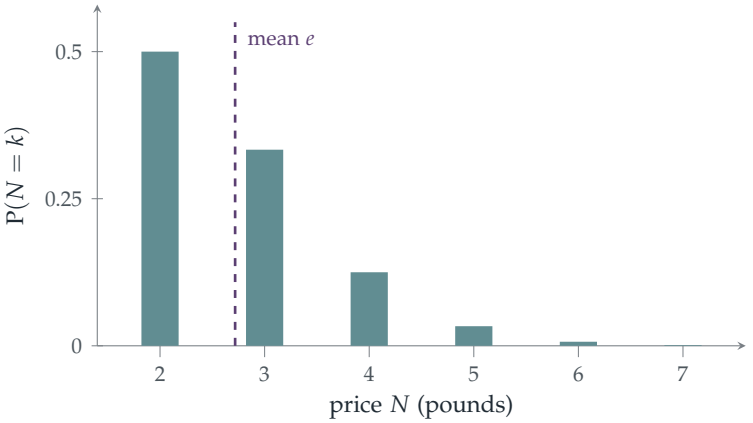


Figure 12.11.1 The distribution of the price, $\mathrm{P}(N=k)=\frac{k-1}{k!}$. Half the time Jim pays £2; the long but thin tail brings the average up to $£e\approx$ £2.72.

Looking back

- **Products of probabilities.** A sequence of independent successes followed by a failure gave a product that collapsed to $\frac{k-1}{k!}$.
- **Cancel factorials.** Multiplying by k or k^2 and cancelling turned the sums into the exponential series.
- $\mathrm{E}(N^2)$ **via a shift.** Re-indexing with $j=k-2$ split the sum into two copies of e.

What's really going on

The same distribution, in disguise

Choose random numbers $U_1, U_2, \ldots$ uniformly from $[0,1]$, and let M be the number needed for their sum to exceed 1. The probability that the first k numbers add up to less than 1 is the volume of the region $\{u_1 + \cdots + u_k < 1, u_i \geq 0\}$, which is $\frac{1}{k!}$ (a triangle of area $\frac{1}{2}$, a tetrahedron of volume $\frac{1}{6}$, and so on). So

$$\mathrm{P}(M > k) = \frac{1}{k!}, \qquad \mathrm{P}(M = k) = \frac{1}{(k-1)!} - \frac{1}{k!} = \frac{k-1}{k!},$$

exactly the distribution of N. In both problems $\mathrm{P}(N > k) = \frac{1}{k!}$: for the cups, that is the probability of $k-1$ correct guesses in a row. Using $\mathrm{E}(N) = \sum_{k \geq 0} \mathrm{P}(N > k)$ (a formula from Section 9.9) gives $\mathrm{E}(N) = \sum \frac{1}{k!} = e$ in one line.

Why e keeps appearing

Whenever a probability involves $\frac{1}{k!}$, whether from k objects in a particular order, k successive events with probabilities $\frac{1}{2}, \frac{1}{3}, \ldots$, or the volume of a simplex, sums over k produce the exponential series. Derangements (Section 9.8) gave $\sum \frac{(-1)^k}{k!} = e^{-1}$; here we have $\sum \frac{1}{k!} = e$. And limits of the form $\left(1 - \frac{1}{n}\right)^n \to e^{-1}$, which come from many independent unlikely events, explain e in the secretary problem and the Poisson distribution.

Monte Carlo e

The "sum exceeds 1" problem gives a way to estimate e by experiment: generate random numbers, count how many are needed to exceed 1, repeat many times and average. It is a cousin of Buffon's needle (Section 12.20), which estimates π. Such *Monte Carlo* methods, named after the casino, use random sampling to estimate quantities that are hard to compute directly; they are now used throughout science, from particle physics to finance, mostly for integrals in many dimensions.

The thinking skill

Recognise a series

When an expectation is a sum involving factorials, simplify the terms until a standard series (exponential, geometric, binomial) appears.

At university, generating functions systematise this: the probability generating function $\mathrm{E}(t^N)$ packages the whole distribution, and its derivatives at $t = 1$ give the mean and variance.

Try this next

1 **The generating function.** Show that $\mathrm{E}(t^N) = 1 + (t-1)e^t$, and use it to find $\mathrm{E}(N)$ and $\mathrm{E}(N(N-1))$.

2 **The tail formula.** Prove that $\mathrm{E}(N) = \sum_{k\geq 0} \mathrm{P}(N > k)$ for a random variable taking non-negative integer values, and use it to find $\mathrm{E}(N)$ in one line.

3 **Starting with one cup.** If the game starts with one cup (which Jim cannot get wrong), how does the distribution change? What is the expected price?

4 **The simplex.** Show that the volume of $\{u_1 + u_2 + u_3 < 1, u_i \geq 0\}$ is $\frac{1}{6}$, and use induction to show the k-dimensional version is $\frac{1}{k!}$.

Further reading

- **Eli Maor, *e: The Story of a Number*** (Princeton University Press). The history of e, including its appearances in probability.
- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press), on generating functions.
- **Paul Nahin, *Digital Dice*** (Princeton University Press). Probability puzzles solved by Monte Carlo simulation.

12.12 When to Stop Looking

The question

On the basis of an interview, the N candidates for admission to a college are ranked in order according to their mathematical potential. The candidates are interviewed in random order (that is, each possible order is equally likely).

(i) Find the probability that the best amongst the first n candidates interviewed is the best overall.

(ii) Find the probability that the best amongst the first n candidates interviewed is the best or second best overall.

Verify your answers for the case $N = 4$, $n = 2$ by listing the possibilities.

The hook

You are interviewing candidates one at a time, and must decide on the spot: accept this one, and stop, or reject them for ever and see the next. You want the very best. If you accept too early you probably miss someone better; if you wait too long the best has probably gone. What should you do?

The answer, one of the most celebrated results in probability, is: reject the first 37% of candidates, then accept the first one who is better than all of those. This succeeds about 37% of the time, and 37% is $\frac{1}{e}$. This question computes the basic probabilities, and its companion from STEP II 2006 (in the exercises) analyses the simplest version of the strategy, rejecting just the first candidate.

Before you start

Ask yourself In a random order, what is the probability that the overall best candidate is among the first n interviewed?

The best candidate is equally likely to be in any of the N positions, so the probability is $\frac{n}{N}$.

Working it through

STAGE 1 The best of the first n is the best

The best among the first n is the best overall exactly when the overall best is among the first n: probability $\frac{n}{N}$. ■

STAGE 2 Best or second best

Ask yourself The best of the first n is the second best overall exactly when the second best is among the first n but the best is not. What is that probability?

The second best is among the first n with probability $\frac{n}{N}$; given that, the best is in one of the other $N-1$ positions, of which $N-n$ are after the first n. So

$$\mathrm{P}(\text{best or second best}) = \frac{n}{N} + \frac{n}{N} \cdot \frac{N-n}{N-1}. \qquad \blacksquare$$

For $N=4$, $n=2$: $\frac{1}{2} + \frac{1}{2} \cdot \frac{2}{3} = \frac{5}{6}$. Listing the positions of the top two candidates among the four places (all 12 ordered placements equally likely), the best of the first two is one of the top two in all but the 2 placements where both top candidates come in positions 3 and 4: $\frac{10}{12} = \frac{5}{6}$.

STAGE 3 The optimal strategy

Ask yourself Strategy: reject the first r candidates, then accept the first candidate better than all of them. If the best candidate is in position $k > r$, when does the strategy pick them?

The strategy picks the best candidate, in position k, if and only if no candidate in positions $r+1, \ldots, k-1$ beats all of the first r, that is, if the best of the first $k-1$ candidates is among the first r. By Stage 1 (with N replaced by $k-1$), this has probability $\frac{r}{k-1}$. Each position k has probability $\frac{1}{N}$, so

$$\mathrm{P}(\text{success}) = \frac{1}{N} \sum_{k=r+1}^{N} \frac{r}{k-1} = \frac{r}{N}\left(\frac{1}{r} + \frac{1}{r+1} + \cdots + \frac{1}{N-1}\right).$$

Ask yourself For large N, write $x = \frac{r}{N}$. The sum is approximately $\ln \frac{N}{r}$. What value of x maximises $x \ln \frac{1}{x}$?

The success probability is approximately $x \ln \frac{1}{x}$. Differentiating, $\ln \frac{1}{x} - 1 = 0$ when $x = \frac{1}{e}$, and the maximum value is $\frac{1}{e} \approx 0.368$ (Figure 12.12.1).

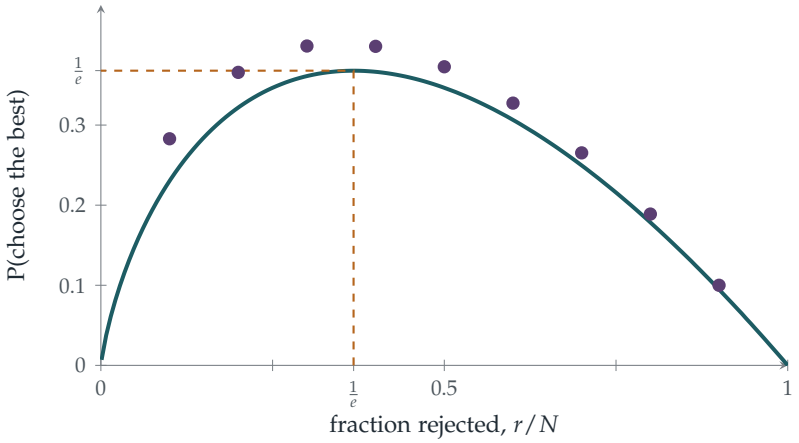


Figure 12.12.1 The probability of choosing the best candidate if the first fraction r/N are rejected: the limiting curve $x \ln \frac{1}{x}$ (teal), and the exact values for $N = 10$ (plum). The best strategy rejects about $\frac{1}{e} \approx 37\%$ of the candidates and succeeds with probability about $\frac{1}{e}$, however large N is.

Looking back

- **Symmetry of positions.** In a random order, each candidate is equally likely to be anywhere, which gave $\frac{n}{N}$ immediately.
- **Condition on where the best is.** The strategy's success was computed by conditioning on the position of the best candidate.
- **Sums become integrals.** For large N the harmonic sum became a logarithm, and an optimisation gave $\frac{1}{e}$.

What's really going on

Optimal stopping

The secretary problem is the founding example of *optimal stopping*: deciding when to take an action, based on information arriving over time, to maximise an expected reward. It can be proved that among *all* strategies (not just "reject r, then take the next best-so-far"), this kind of threshold rule is optimal, by an argument working backwards from the last candidate. The same mathematics governs when to sell a share, when to exercise a financial option, and how long to search for a parking space. A remarkable feature is that the answer does not depend on N in the limit: you succeed with probability $\frac{1}{e}$ whether there are a hundred candidates or a million.

Variations

If you are satisfied with either of the top two candidates, the best strategy succeeds about 57% of the time. If you want to maximise the expected *rank* of the chosen candidate rather than the chance of the very best, the optimal strategy is more complicated, and the expected rank tends to about 3.87 (Chow, Moriguti, Robbins and Samuels, 1964). If you can recall rejected candidates with some probability, or if you know the distribution of candidates' scores (the "full-information" problem), the answers change again. The ice-cream question in the exercises is the simplest case: reject just one, then take the next improvement.

Who solved it?

The problem circulated among mathematicians in the 1950s, and Martin Gardner popularised it in 1960 as the "game of googol" (write numbers on slips of paper; your opponent turns them over one at a time and must stop at the largest). Thomas Ferguson's article "Who solved the secretary problem?" (1989) traces its tangled history; one tradition says that the astronomer Johannes Kepler, choosing a second wife in 1611, interviewed eleven candidates and ended up proposing to the fifth.

The thinking skill

Parametrise a strategy and optimise

When the best strategy is unknown, consider a natural family of strategies with a parameter, compute the success probability as a function of it, and optimise.

At university this approach is used in decision theory, game theory and statistics. Showing that the best strategy in the family is the best overall is a harder and more interesting step, often done by *backward induction*.

Try this next

1 **Ice-creams.** (From STEP II, 2006.) Ice-creams of n different sizes are offered in random order, and you must choose one. You reject the first and take the first one bigger than it (or the last, if the first was the biggest). Show that the probability of getting the biggest is $\frac{1}{n}\left(1+\frac{1}{2}+\cdots+\frac{1}{n-1}\right)$, which is $\frac{11}{24}$ for $n=4$. For $n=4$, find the probabilities of getting the second, third and fourth biggest.

2 **Small N.** For $N=4$, find the best value of r and the probability of success. Check by listing.

3 **Best or second best.** Using part (ii), argue that "the best of the first n is one of the top two" is much more likely than "the best of the first n is the best". What does

this suggest about strategies for getting one of the top two?

4 **An exact maximum.** For $N = 10$, check from the figure that $r = 3$ is optimal. For $N = 100$, show that $r = 37$ is optimal (to four decimal places the success probabilities for $r = 36, 37, 38$ are 0.3710, 0.3710, 0.3708).

Further reading

- **Thomas Ferguson, "Who solved the secretary problem?"**, *Statistical Science* 4 (1989). History and variations; free online.
- **Brian Christian & Tom Griffiths, *Algorithms to Live By*** (William Collins). Optimal stopping and other algorithms applied to everyday decisions.
- **Martin Gardner, *New Mathematical Diversions*** (Mathematical Association of America). The game of googol.

12.13 The Truel

The question

Bread roll throwing duels at the Drones' Club are governed by a strict etiquette. The two duellists throw alternately until one is hit, when the other is declared the winner. If Percy has probability $p > 0$ of hitting his target and Rodney has probability $r > 0$ of hitting his, show that, if Percy throws first, the probability that he beats Rodney is

$$\frac{p}{p + r - pr}.$$

Algernon, Bertie and Cuthbert decide to have a three sided duel in which they throw in order A, B, C, A, B, C, ... except that anyone who is hit must leave the game. Cuthbert always hits his target, Bertie hits his target with probability $\frac{3}{5}$ and Algernon hits his target with probability $\frac{2}{5}$. Bertie and Cuthbert will always aim at each other if they are both still in the duel. Otherwise they aim at Algernon. With his first shot Algernon may aim at either Bertie or Cuthbert or deliberately miss both. Faced with only one opponent Algernon will aim at him. What are Algernon's chances of winning if he:

(i) hits Cuthbert with his first shot?

(ii) hits Bertie with his first shot?

(iii) misses with his first shot?

Advise Algernon as to his best plan and show that, if he uses this plan, his probability of winning is 226/475.

The hook

Three people fight a duel with bread rolls. Cuthbert never misses, Bertie hits three times in five, and Algernon, the worst shot, only two times in five. Algernon throws first. Whom should he aim at? The surprising answer: nobody. He should throw his roll deliberately wide.

The reason is strategic. While all three are in the game, Bertie and Cuthbert, the two dangerous throwers, aim at each other. If Algernon removes one of them, the survivor turns on him. By missing, he lets them fight it out, and then gets the first throw against whoever survives. In fact, with the best plan the worst shot becomes the *most* likely to win. This "truel" is a favourite example in game theory of how weakness can be an advantage.

Before you start

Ask yourself In a two-person duel, Percy throws first. Either he hits at once, or both miss and the situation repeats exactly. Write an equation for Percy's probability of winning.

Let P be Percy's probability of winning. He hits at once with probability p; otherwise (probability $1-p$) Rodney throws, and misses with probability $1-r$, returning to the start. So $P = p + (1-p)(1-r)P$.

Working it through

STAGE 1 The duel

Solving, $P\big(1-(1-p)(1-r)\big) = p$, and $1-(1-p)(1-r) = p+r-pr$, so

$$P = \frac{p}{p+r-pr}. \qquad \blacksquare$$

For Algernon ($p = \frac{2}{5}$) against Bertie ($r = \frac{3}{5}$), with Algernon throwing first, this is $\frac{2/5}{2/5+3/5-6/25} = \frac{10}{19}$; with Bertie throwing first, Algernon wins with probability $\left(1-\frac{3}{5}\right)\cdot\frac{10}{19} = \frac{4}{19}$.

STAGE 2 Algernon's three options

Ask yourself In each case, who is left, and who throws next?

(i) *Algernon hits Cuthbert.* Bertie throws next, at Algernon. Algernon wins the duel with Bertie throwing first: probability $\frac{4}{19}$.

(ii) *Algernon hits Bertie.* Cuthbert throws next, at Algernon, and never misses. Algernon's chance is 0.

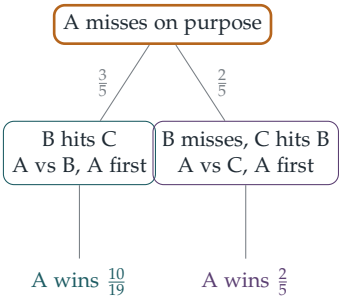
(iii) *Algernon misses.* Bertie throws at Cuthbert. With probability $\frac{3}{5}$ he hits, leaving Algernon and Bertie with Algernon to throw: Algernon wins with probability $\frac{10}{19}$. With probability $\frac{2}{5}$ Bertie misses, Cuthbert hits Bertie, and Algernon faces Cuthbert with one throw: he wins only if he hits, probability $\frac{2}{5}$. Altogether,

$$\tfrac{3}{5}\cdot\tfrac{10}{19} + \tfrac{2}{5}\cdot\tfrac{2}{5} = \tfrac{6}{19} + \tfrac{4}{25} = \tfrac{150+76}{475} = \tfrac{226}{475} \approx 0.476.$$

STAGE 3 Advice

Ask yourself If Algernon aims at Cuthbert, he might hit (probability $\frac{2}{5}$) or miss. What is his overall chance? Compare with deliberately missing.

Aiming at Cuthbert gives $\frac{2}{5} \cdot \frac{4}{19} + \frac{3}{5} \cdot \frac{226}{475} = \frac{200+678}{2375} = \frac{878}{2375} \approx 0.370$. Aiming at Bertie is worse still. Deliberately missing gives $\frac{226}{475} \approx 0.476$, the best. So Algernon should throw wide, and then his probability of winning is $\frac{226}{475}$. ■



with this plan:
A wins $\frac{226}{475} \approx 0.48$
B wins $\frac{135}{475} \approx 0.28$
C wins $\frac{114}{475} \approx 0.24$

Figure 12.13.1 Algernon's best plan. By missing deliberately he lets his two stronger opponents fight first, and is guaranteed the first throw at whoever survives. The worst shot ends up the most likely winner, and the perfect shot the least likely.

Looking back

- **Repeated situations give equations.** The duel "starts again" after two misses, giving a linear equation for the winning probability.
- **Think about the opponents' responses.** Each of Algernon's options changes what Bertie and Cuthbert do next.
- **Compare all options, including doing nothing.** The best move was the one that looked worst.

What's really going on

Strategy and game theory

The truel shows the central idea of *game theory*: the best action depends on how the other players will respond. Hitting Cuthbert, the most dangerous opponent, is the obvious move, but it changes the game in a way that hurts Algernon. In this question Bertie's and Cuthbert's strategies are fixed (they aim at each other); in a full game-theoretic analysis each player would choose their own strategy, knowing that the others are doing the same. Such analyses, starting with the work of John von Neumann and Oskar Morgenstern in 1944 and John Nash in 1950, now underpin economics, evolutionary biology and political science.

Strength as a weakness

Who wins the truel? With Algernon's best plan, the probabilities are $\frac{226}{475}$ for Algernon, $\frac{135}{475}$ for Bertie and $\frac{114}{475}$ for Cuthbert. The best shot is the least likely to survive, because everyone with a choice aims at him. The same pattern appears in elections with several candidates, in competition between firms, and in nature, where the strongest competitor can be the first target. Martin Shubik, who introduced truels into game theory in the 1950s, used them to ask whether the fittest necessarily survive.

The thinking skill

Think one move ahead

Before choosing an action, work out how it changes the situation for everyone else, and what they will do in response.

At university, this becomes backward induction in game theory: analyse the end of the game first, then work back to decide the best opening move.

Try this next

1. **Bertie and Cuthbert.** Verify that, with Algernon missing deliberately, Bertie wins with probability $\frac{135}{475}$ and Cuthbert with $\frac{114}{475}$.
2. **A better shot.** If Algernon hit with probability a instead of $\frac{2}{5}$, for which values of a is deliberately missing still his best first move?
3. **Should Bertie aim at Cuthbert?** Given that Algernon misses deliberately, is it in Bertie's interest to aim at Cuthbert rather than at Algernon? (Assume Cuthbert's behaviour is unchanged.)
4. **A duel with a fixed number of throws.** In a duel where each player has only two throws, find Percy's probability of winning, and compare with the unlimited duel.

Further reading

- **D. Marc Kilgour & Steven Brams, "The truel"**, *Mathematics Magazine* 70 (1997). An accessible survey of three-way duels.
- **Avinash Dixit & Barry Nalebuff, *Thinking Strategically*** (Norton). Game theory for a general reader, including the truel.
- **Ken Binmore, *Game Theory: A Very Short Introduction*** (Oxford University Press).

12.14 Mixed Strategies

The question

A and B play a guessing game. Each simultaneously names one of the numbers $1, 2, 3$. If the numbers differ by 2, whoever guessed the *smaller* pays the opponent £2. If the numbers differ by 1, whoever guessed the *larger* pays the opponent £1. Otherwise no money changes hands. Many rounds of the game are played.

(i) If A says he will always guess the same number N, explain (for each value of N) how B can maximise his winnings.

(ii) In an attempt to improve his play, A announces that he will guess each number at random with probability $\frac{1}{3}$, guesses on different rounds being independent. To counter this, B secretly decides to guess j with probability b_j ($j = 1, 2, 3$, $b_1 + b_2 + b_3 = 1$), guesses on different rounds being independent. Derive an expression for B's expected winnings on any round. How should the probabilities b_j be chosen so as to maximize this expression?

(iii) A now announces that he will guess j with probability a_j ($j = 1, 2, 3$, $a_1 + a_2 + a_3 = 1$). If B guesses j with probability b_j ($j = 1, 2, 3$, $b_1 + b_2 + b_3 = 1$), obtain an expression for his expected winnings in the form

$$Xa_1 + Ya_2 + Za_3.$$

Show that he can choose b_1, b_2 and b_3 such that X, Y and Z are all non-negative. Deduce that, whatever values for a_j are chosen by A, B can ensure that in the long run he loses no money.

The hook

In rock–paper–scissors, any fixed choice can be exploited: play rock every time and your opponent will learn to play paper. The only safe strategy is to randomise. This question is a more intricate version, where the payoffs are unequal and the right randomisation is not obvious. It turns out that by guessing 1, 2, 3 with probabilities $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{4}$, a player can guarantee not to lose in the long run, whatever the opponent does.

This is a first glimpse of John von Neumann's *minimax theorem* of 1928, the founding result of game theory: in every two-player game where one player's gain is the other's loss, there is a randomised strategy for each player that guarantees the best possible result against an opponent who plays perfectly.

Before you start

Ask yourself Write down a table of B's winnings for each pair of guesses (B's guess down the side, A's across the top).

Working through the rules (for instance, if B says 1 and A says 3, they differ by 2 and B, the smaller, pays £2):

B \ A	1	2	3
1	0	+1	−2
2	−1	0	+1
3	+2	−1	0

The table is *antisymmetric* (swapping the players changes the sign), so the game is fair in the sense that neither player has a built-in advantage.

Working it through

STAGE 1 Against a fixed guess

If A always says 1, B should say 3 and win £2 every round. If A always says 2, B says 1 and wins £1. If A always says 3, B says 2 and wins £1. Any fixed guess can be exploited.

STAGE 2 Against random guessing

Ask yourself If A guesses uniformly at random, what is B's expected gain when B says j? (Average along row j of the table.)

The row averages are $\frac{1}{3}(0+1-2) = -\frac{1}{3}$ for $j=1$, 0 for $j=2$ and $\frac{1}{3}(2-1+0) = \frac{1}{3}$ for $j=3$. So B's expected winnings per round are

$$-\tfrac{1}{3}b_1 + 0\cdot b_2 + \tfrac{1}{3}b_3,$$

which is maximised by $b_3 = 1$: always say 3, winning $\frac{1}{3}$ of a pound per round on average. Uniform randomisation does not protect A.

STAGE 3 The general case

Ask yourself Now A plays j with probability a_j. Collect B's expected winnings by A's guess: what is the coefficient of a_1 (the average of column 1 with weights b_j)?

Reading down the columns of the table,

$$\text{B's expected winnings} = (-b_2 + 2b_3)a_1 + (b_1 - b_3)a_2 + (-2b_1 + b_2)a_3,$$

so $X = -b_2 + 2b_3$, $Y = b_1 - b_3$ and $Z = -2b_1 + b_2$.

Ask yourself Can you make all three zero? (Three equations, but they are not independent; why not?)

They are not independent: $X + 2Y + Z = 0$ identically (check it). So $Y = 0$, giving $b_1 = b_3$, and $X = 0$, giving $b_2 = 2b_3$, force $Z = 0$ as well. With $b_1 + b_2 + b_3 = 1$:

$$b_1 = \tfrac{1}{4}, \qquad b_2 = \tfrac{1}{2}, \qquad b_3 = \tfrac{1}{4},$$

giving $X = Y = Z = 0$. Then B's expected winnings are 0 whatever the a_j, and B loses nothing in the long run. ■

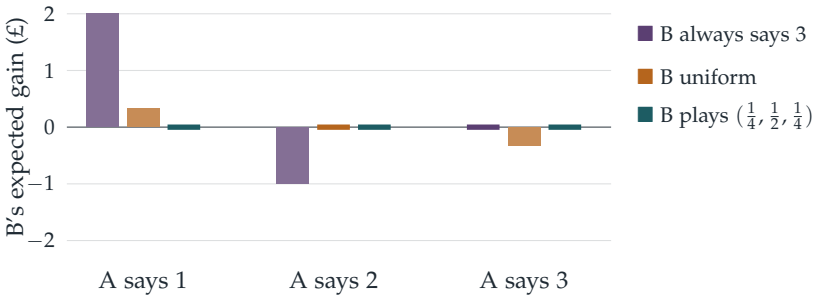


Figure 12.14.1 B's expected gain against each of A's choices, for three strategies of B. Always saying 3 wins against A's 1 but loses against A's 2; uniform guessing loses against A's 3. The strategy $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ gives exactly 0 (drawn as a thin bar) whatever A does, so it can never be exploited.

Looking back

- **Write the payoff table.** All the reasoning was about rows and columns of a 3×3 matrix.
- **Pure strategies are exploitable.** Against any fixed guess, and even against uniform guessing, the opponent has a profitable reply.
- **Equalise.** The safe strategy makes the opponent indifferent: every one of A's choices gives the same expected result.

What's really going on

The minimax theorem

In a two-player *zero-sum* game, given by a payoff matrix M, B choosing probabilities $\mathbf{b}$ and A choosing $\mathbf{a}$, B's expected winnings are $\mathbf{b}^{\mathsf{T}} M \mathbf{a}$. Von Neumann's minimax theorem says

$$\max_{\mathbf{b}} \min_{\mathbf{a}} \mathbf{b}^{\mathsf{T}} M \mathbf{a} = \min_{\mathbf{a}} \max_{\mathbf{b}} \mathbf{b}^{\mathsf{T}} M \mathbf{a}.$$

In words: the amount B can guarantee by announcing a randomised strategy in advance equals the amount A can hold B to by announcing his. This common value is the *value* of the game. Here, by antisymmetry, the value is 0, and the strategy $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ is optimal for both players. The theorem is proved using linear algebra and convexity (it is equivalent to the duality theorem of linear programming), and its generalisation to many players, the *Nash equilibrium*, won John Nash the Nobel Prize in economics.

Why randomise?

Randomness here is not about luck but about information. If B's choice can be predicted, A can exploit it; by randomising with the right probabilities, B makes his choice unpredictable *and* ensures that no prediction would help. Real examples include penalty kicks in football, where studies of professional players have found kickers and goalkeepers randomising between left and right in proportions close to the game-theoretic optimum, and tennis serves.

Von Neumann's 1928 paper

John von Neumann proved the minimax theorem in 1928, in a paper "On the theory of games of strategy", motivated by poker, where bluffing is a form of randomisation. In 1944 he and the economist Oskar Morgenstern published *Theory of Games and Economic Behavior*, which founded game theory as a subject. Von Neumann later recalled that he had regarded the minimax theorem as the key result, without which there would be no theory of games worth publishing.

The thinking skill

Randomise to remove an opponent's advantage

When any fixed choice can be exploited, choose at random, with probabilities that make the opponent indifferent between their options.

At university, game theory, linear programming and the theory of algorithms all use this idea. Randomised algorithms, for example, protect themselves against worst-case inputs by making random choices that no adversary can predict.

Try this next

1. **A's best reply.** If B uses $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$, show that every strategy of A gives expected winnings 0. If instead B uses $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, find A's best reply.

2. **Rock, paper, scissors.** Write the payoff matrix for rock–paper–scissors and find the optimal strategy. Why is it uniform here but not in the question?

3 **Uniqueness.** Show that $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ is the *only* strategy for B that guarantees not losing. (*Hint:* it must make X, Y, Z all non-negative; use $X + 2Y + Z = 0$.)

4 **A modified game.** If the penalty for differing by 2 is £3 instead of £2, find the optimal strategy.

Further reading

- **Ken Binmore, *Playing for Real*** (Oxford University Press). A thorough but friendly introduction to game theory.
- **Avinash Dixit & Barry Nalebuff, *The Art of Strategy*** (Norton). Game theory with examples from sport and business.
- **William Poundstone, *Prisoner's Dilemma*** (Anchor). Von Neumann, game theory and the Cold War.

12.15 Extinction or Explosion

The question

The life of a certain species of elementary particles can be described as follows. Each particle has a life time of T seconds, after which it disintegrates into X particles of the same species, where X is a random variable with binomial distribution $\mathrm{B}(2, p)$. A population of these particles starts with the creation of a single such particle at $t = 0$. Let X_n be the number of particles in existence in the time interval $nT < t < (n+1)T$, where $n = 1, 2, \ldots$.

Show that $\mathrm{P}(X_1 = 2 \text{ and } X_2 = 2) = 6p^4q^2$, where $q = 1 - p$. Find the possible values of p if it is known that $\mathrm{P}(X_1 = 2 \mid X_2 = 2) = 9/25$.

Explain briefly why $\mathrm{E}(X_n) = 2p\,\mathrm{E}(X_{n-1})$ and hence determine $\mathrm{E}(X_n)$ in terms of p. Show that for one of the values of p found above $\lim_{n\to\infty} \mathrm{E}(X_n) = 0$ and that for the other $\lim_{n\to\infty} \mathrm{E}(X_n) = +\infty$.

The hook

In 1873 Francis Galton posed a question in the *Educational Times*: why do aristocratic family names die out so often? Each man has some random number of sons, each of whom passes on the name. Even if each man has on average more than one son, will the name survive for ever? The question founded the theory of *branching processes*, which now models nuclear chain reactions, the spread of epidemics, the growth of cell populations and the survival of mutations.

This question sets up the simplest model: each particle splits into 0, 1 or 2 particles. The expected population grows or shrinks geometrically. But expectation is not the whole story. Even when the expected population explodes, there is a positive chance that the line dies out completely, and a beautiful fixed-point argument finds it.

Before you start

Ask yourself If there are two particles at one stage, what is the distribution of the number at the next stage?

Each of the two particles independently produces a $\mathrm{B}(2, p)$ number of offspring, so the total is $\mathrm{B}(4, p)$: four independent "chances" each succeeding with probability p.

Working it through

STAGE 1 Two, then two

$\mathrm{P}(X_1 = 2) = p^2$, and then $\mathrm{P}(X_2 = 2 \mid X_1 = 2) = \binom{4}{2}p^2q^2 = 6p^2q^2$. So

$$\mathrm{P}(X_1 = 2, X_2 = 2) = 6p^4q^2. \qquad \blacksquare$$

STAGE 2 Reasoning backwards

Ask yourself In how many ways can $X_2 = 2$ happen? (What can X_1 be?)

$X_2 = 2$ needs $X_1 = 1$ (probability $2pq$) followed by 2 offspring (p^2), or $X_1 = 2$ followed by 2 from $\mathrm{B}(4, p)$. ($X_1 = 0$ leaves nothing.) So $\mathrm{P}(X_2 = 2) = 2p^3q + 6p^4q^2$, and

$$\mathrm{P}(X_1 = 2 \mid X_2 = 2) = \frac{6p^4q^2}{2p^3q + 6p^4q^2} = \frac{3pq}{1 + 3pq}.$$

Setting this equal to $\frac{9}{25}$: $75pq = 9 + 27pq$, so $pq = \frac{3}{16}$, that is $p(1-p) = \frac{3}{16}$, giving $p = \frac{1}{4}$ or $p = \frac{3}{4}$. $\blacksquare$

STAGE 3 The expected population

Ask yourself If there are k particles at stage $n-1$, what is the expected number at stage n? Then average over k.

Each particle has on average $\mathrm{E}(X) = 2p$ offspring, so k particles have $2pk$ on average: $\mathrm{E}(X_n \mid X_{n-1} = k) = 2pk$. Averaging over k, $\mathrm{E}(X_n) = 2p\,\mathrm{E}(X_{n-1})$. With $\mathrm{E}(X_0) = 1$,

$$\mathrm{E}(X_n) = (2p)^n.$$

For $p = \frac{1}{4}$, $(2p)^n = 2^{-n} \to 0$; for $p = \frac{3}{4}$, $(2p)^n = \left(\frac{3}{2}\right)^n \to \infty$. $\blacksquare$

Looking back

- **Sums of independent binomials.** k particles each giving $\mathrm{B}(2, p)$ give $\mathrm{B}(2k, p)$ in total.
- **Bayes' theorem in time.** $\mathrm{P}(X_1 = 2 \mid X_2 = 2)$ reasoned backwards from a later observation to an earlier state.
- **Conditional expectation.** Conditioning on the previous generation turned the expectation into a geometric sequence.

What's really going on

The probability of extinction

For $p = \frac{3}{4}$, the expected population grows without limit. Does that mean the population survives? Not necessarily: if the first particle has no offspring (probability $q^2 = \frac{1}{16}$), the line ends at once. Let e be the probability of eventual extinction. The first particle has k offspring, and each of their lines dies out independently with probability e, so

$$e = q^2 + 2pq\,e + p^2e^2 = (q + pe)^2 = \mathrm{G}(e),$$

where $\mathrm{G}(s) = (q + ps)^2$ is the *probability generating function* of the offspring number. The extinction probability is a fixed point of G: in fact the *smallest* non-negative one. For $p = \frac{3}{4}$, $(1 + 3e)^2 = 16e$ gives $9e^2 - 10e + 1 = 0$, so $e = \frac{1}{9}$ or 1, and the extinction probability is $\frac{1}{9}$. For $p = \frac{1}{4}$, the only fixed point in $[0, 1]$ is $e = 1$: extinction is certain (Figure 12.15.1).

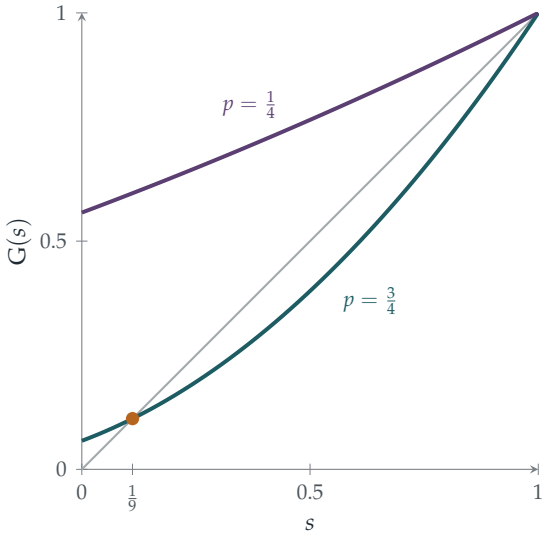


Figure 12.15.1 The offspring generating function $\mathrm{G}(s) = (q + ps)^2$ and the line $y = s$. The probability of extinction is the smallest point where they meet. For $p = \frac{1}{4}$ (plum, mean $\frac{1}{2}$) that is $s = 1$: extinction is certain. For $p = \frac{3}{4}$ (teal, mean $\frac{3}{2}$) it is $s = \frac{1}{9}$ (amber): the population survives for ever with probability $\frac{8}{9}$.

The critical threshold

In general, a branching process dies out with certainty if the mean number of offspring is less than 1, and also if it equals 1, except in the trivial case where every individual has exactly one offspring (when the family line simply continues for ever, one at a time); it survives with positive probability if the mean is greater than 1. The mean offspring number is the *reproduction number* R of epidemiology: an epidemic with $R > 1$ may take off, but even then an early outbreak can fizzle out by chance, with probability given by exactly the fixed-point equation above. In nuclear physics, a chain reaction is *critical*

when each fission causes on average exactly one more, and the art of running a reactor is to keep it there.

Galton, Watson and Bienaymé

Galton's question was answered, partly incorrectly, by the Reverend Henry William Watson in 1874: he found the fixed-point equation but concluded that every family name dies out eventually, missing the second root. The correct analysis had in fact been given in 1845 by the French statistician Irénée-Jules Bienaymé, whose work was forgotten until the 1970s. The model is now called the *Galton–Watson process* (sometimes Bienaymé–Galton–Watson), and is a central example in every course on random processes.

The thinking skill

Condition on the first generation

A process that renews itself (each offspring starts an independent copy of the whole process) satisfies an equation obtained by conditioning on the first step.

The equation $e = \mathrm{G}(e)$ is a fixed-point problem, the same kind met in Section 2.2 and Section 8.7. At university, generating functions and fixed points are the main tools for branching processes, random trees and random networks.

Try this next

1. **Extinction in general.** For offspring distribution $\mathrm{B}(2, p)$, show that the extinction probability is 1 if $p \le \frac{1}{2}$ and $\left(\frac{q}{p}\right)^2$ if $p > \frac{1}{2}$.

2. **The variance.** Show that $\mathrm{Var}(X_n)$ satisfies $\mathrm{Var}(X_n) = (2p)^2\mathrm{Var}(X_{n-1}) + 2pq\,\mathrm{E}(X_{n-1})$, and find it for $p = \frac{1}{2}$.

3. **Extinction by generation n.** Let $e_n = \mathrm{P}(X_n = 0)$. Show that $e_n = \mathrm{G}(e_{n-1})$, and compute e_1, e_2, e_3 for $p = \frac{3}{4}$. Do they approach $\frac{1}{9}$?

4. **Surnames.** If each man has 0, 1, 2 or 3 sons with probabilities $\frac{1}{4}$ each, what is the probability that his surname eventually dies out?

Further reading

- **Geoffrey Grimmett & David Stirzaker, *Probability and Random Processes*** (Oxford University Press), the section on branching processes.
- **Adam Kucharski, *The Rules of Contagion*** (Profile). Reproduction numbers and out-

breaks, for a general reader.

- **David Kendall, "Branching processes since 1873"**, *Journal of the London Mathematical Society* 41 (1966). The history, including Bienaymé.

12.16 Markov Chains Forwards and Backwards

The question

It has been observed that Professor Ecks proves three types of theorems: 1, those that are correct and new; 2, those that are correct, but already known; 3, those that are false. It has also been observed that, if a certain of her theorems is of type i, then her next theorem is of type j with probability p_{ij}, where p_{ij} is the entry in the ith row and jth column of the following array:

$$\begin{pmatrix} 0.3 & 0.3 & 0.4 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{pmatrix}.$$

Let a_i, $i = 1, 2, 3$, be the probability that a given theorem is of type i, and let b_j be the consequent probability that the next theorem is of type j.

(i) Explain why $b_j = a_1 p_{1j} + a_2 p_{2j} + a_3 p_{3j}$.

(ii) Find values of a_1, a_2 and a_3 such that $b_i = a_i$ for $i = 1, 2, 3$.

(iii) For these values of the a_i find the probabilities q_{ij} that, if a particular theorem is of type j, then the *preceding* theorem was of type i.

The hook

Professor Ecks's theorems follow a pattern: what she proves next depends only on what she proved last. A process whose future depends on the present but not on the past is a *Markov chain*, and Markov chains are everywhere: in weather models, in the board game Snakes and Ladders, in models of DNA, in Google's original PageRank algorithm, and in the methods statisticians use to simulate complicated probability distributions.

This question finds the long-run mix of theorem types, the *stationary distribution*: in the long run, one in six of her theorems is new and correct, and half are false. Then it runs the chain backwards in time, and asks: given today's theorem, what was yesterday's?

Before you start

Ask yourself What does the condition "each row adds up to 1" mean? Do the columns add up to 1?

Row i lists the probabilities of the next type, given that the current type is i; one of the three must occur, so each row adds to 1. The columns need not: the second column adds to exactly 1, but the first adds to 0.6 and the third to 1.4.

Working it through

STAGE 1 One step

By the law of total probability, conditioning on the type i of the current theorem,

$$b_j = \sum_i \mathrm{P}(\text{current } i)\,\mathrm{P}(\text{next } j \mid \text{current } i) = a_1 p_{1j} + a_2 p_{2j} + a_3 p_{3j}. \qquad \blacksquare$$

In matrix terms, the row vector $\mathbf{b} = \mathbf{a}P$.

STAGE 2 The stationary distribution

Ask yourself We want $\mathbf{a} = \mathbf{a}P$ with $a_1 + a_2 + a_3 = 1$. Write out the equations. Which are the simplest?

The equations are

$$\begin{aligned} a_1 &= 0.3a_1 + 0.2a_2 + 0.1a_3, \\ a_2 &= 0.3a_1 + 0.4a_2 + 0.3a_3, \\ a_3 &= 0.4a_1 + 0.4a_2 + 0.6a_3. \end{aligned}$$

(They are not independent: adding them gives $a_1 + a_2 + a_3$ on both sides.) The second gives $0.6a_2 = 0.3(a_1 + a_3) = 0.3(1 - a_2)$, so $a_2 = \frac{1}{3}$. The first gives $7a_1 = 2a_2 + a_3 = \frac{2}{3} + a_3$, and $a_1 + a_3 = \frac{2}{3}$, so $8a_1 = \frac{4}{3}$, $a_1 = \frac{1}{6}$, $a_3 = \frac{1}{2}$:

$$(a_1, a_2, a_3) = \left(\tfrac{1}{6}, \tfrac{1}{3}, \tfrac{1}{2}\right). \qquad \blacksquare$$

STAGE 3 Running the chain backwards

Ask yourself $q_{ij} = \mathrm{P}(\text{previous } i \mid \text{current } j)$. Use Bayes' theorem, with the stationary distribution for both the previous and the current theorem.

By Bayes' theorem,

$$q_{ij} = \frac{\mathrm{P}(\text{previous } i)\,\mathrm{P}(\text{current } j \mid \text{previous } i)}{\mathrm{P}(\text{current } j)} = \frac{a_i p_{ij}}{a_j}.$$

For example $q_{11} = \frac{\frac{1}{6}\times 0.3}{\frac{1}{6}} = 0.3$ and $q_{21} = \frac{\frac{1}{3}\times 0.2}{\frac{1}{6}} = 0.4$. Altogether (with i labelling rows and j columns):

$$(q_{ij}) = \begin{pmatrix} 0.3 & 0.15 & \frac{2}{15} \\ 0.4 & 0.4 & \frac{4}{15} \\ 0.3 & 0.45 & 0.6 \end{pmatrix}. \qquad \blacksquare$$

Each *column* now adds to 1, since column j lists the possible types of the previous theorem.

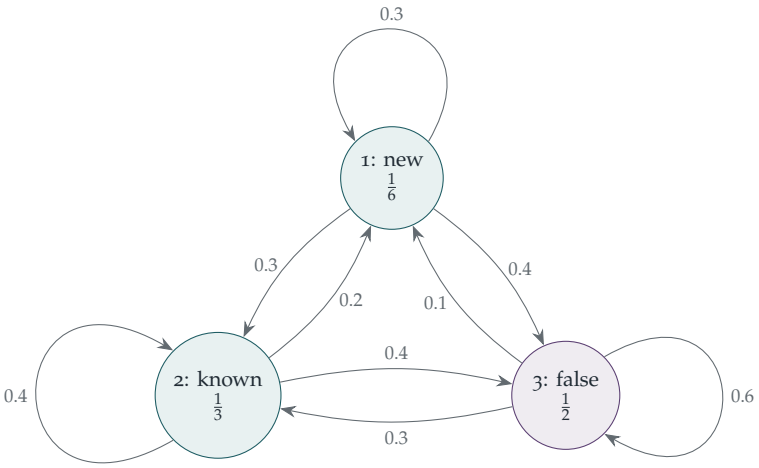


Figure 12.16.1 Professor Ecks's theorems as a Markov chain. Arrows show the probability that the next theorem is of each type. The fractions inside the circles are the stationary distribution: in the long run, one theorem in six is new and correct, one in three is already known, and half are false.

Looking back

- **The law of total probability, as a matrix.** One step of the chain is multiplication of a row vector by the matrix P.
- **Stationarity is an eigenvector.** $\mathbf{a} = \mathbf{a}P$ says that $\mathbf{a}$ is a left eigenvector of P with eigenvalue 1.
- **Reverse with Bayes.** The backwards probabilities are another stochastic matrix, built from the forwards one and the stationary distribution.

What's really going on

Convergence to equilibrium

Start Professor Ecks with any theorem; after n more, the distribution of the type is $\mathbf{a}_0 P^n$. For a chain like this one, in which every type can follow every other, P^n converges to a matrix with every row equal to the stationary distribution $(\frac{1}{6}, \frac{1}{3}, \frac{1}{2})$, whatever the starting point. The speed of convergence is governed by the other eigenvalues of P (here 0.2 and 0.1), which shrink geometrically: after just a few theorems, her past is forgotten. This *ergodic theorem* for Markov chains is a central result of a second-year probability course, and it is what makes Markov chain Monte Carlo methods work: to sample from a complicated distribution, design a Markov chain with that stationary distribution, and run it for a long time.

Reversibility

The backwards chain (q_{ij}) is different from the forwards chain (p_{ij}): you can tell the direction of time from a long record of her theorems. Chains whose backwards and forwards versions agree, $a_i p_{ij} = a_j p_{ji}$ for all i, j, are called *reversible*; they include random walks on networks and the models of physical systems in thermal equilibrium. Reversibility is a powerful property, because it makes the stationary distribution easy to find.

Markov and Pushkin

Andrey Markov introduced his chains in 1906, partly to show that the law of large numbers does not require independence, in a dispute with the mathematician Pavel Nekrasov, who had claimed it did. In 1913 Markov applied his idea to the first 20 000 letters of Pushkin's verse novel *Eugene Onegin*, classifying each as a vowel or consonant and showing that the type of each letter depends on the type of the previous one in a way well described by a two-state chain. It was the first statistical analysis of a text, and the ancestor of every predictive-text system.

The thinking skill

Probabilities as vectors, transitions as matrices

When a system moves between states with fixed probabilities, write the distribution as a vector and the transitions as a matrix. Then long-run behaviour becomes a question about eigenvectors.

At university, this brings the linear algebra of Chapter 7 into probability. The Perron–Frobenius theorem guarantees the stationary distribution, and the other eigenvalues measure how quickly it is reached.

Try this next

1 **A rabbit warren.** (From STEP III, 2003.) Chambers A, B, C, D are at the corners of a square, joined by burrows along the sides, and each has a tunnel to the surface. A rabbit in a chamber runs to one of the two neighbouring chambers, or leaves by the tunnel, each with probability $\frac{1}{3}$. Let p_X be the probability of leaving by the tunnel at A, starting from chamber X. Show that $p_A = \frac{1}{3} + \frac{1}{3}p_B + \frac{1}{3}p_D$, and find p_A. Find the probability that a rabbit starting at A never visits C, given that it leaves by the tunnel at A.

2 **Two steps.** If Professor Ecks's last theorem was new and correct, find the probabilities of the three types for the theorem after next.

3 **Eigenvalues.** Show that the eigenvalues of P are 1, 0.2 and 0.1.

4 **Reversible?** Check whether $a_i p_{ij} = a_j p_{ji}$ for Professor Ecks's chain. What does the answer tell you?

Further reading

- **James Norris, *Markov Chains*** (Cambridge University Press). The standard undergraduate text.
- **Olle Häggström, *Finite Markov Chains and Algorithmic Applications*** (Cambridge University Press). Short, and good on Markov chain Monte Carlo.
- **Brian Hayes, "First links in the Markov chain"**, *American Scientist* 101 (2013). Markov's Pushkin analysis, retold.

12.17 Banach's Matchboxes

The question

A taxi driver keeps a packet of toffees and a packet of mints in her taxi. From time to time she takes either a toffee (with probability p) or mint (with probability $q = 1 - p$). At the beginning of the week she has n toffees and m mints in the packets. On the Nth occasion that she reaches for a sweet, she discovers (for the first time) that she has run out of that kind of sweet. What is the probability that she was reaching for a toffee?

The hook

The mathematician Stefan Banach, so the story goes, was a heavy smoker who kept a box of matches in each pocket. Each time he wanted a match he chose a pocket at random. One day he found a box empty. How many matches were left in the other box? This is *Banach's matchbox problem*, devised by Hugo Steinhaus in Banach's honour, and this question is the taxi driver's version with sweets.

The subtlety is in the word "discovers". The driver does not notice that a packet is empty when she takes its last sweet, only when she next reaches for it. So the event is: she reaches for a toffee, and all n toffees have already gone. Counting the ways this can happen on the Nth reach is a neat exercise in the binomial distribution, and the answer to Banach's problem has a surprising feature: the most likely number of matches left in the other box is 0 (tied with 1).

Before you start

Ask yourself For the driver to discover on the Nth reach that the toffees have run out, what must the Nth reach be? What must the previous $N - 1$ reaches contain?

The Nth reach must be for a toffee, and the previous $N - 1$ reaches must have taken exactly n toffees (all of them) and $N - 1 - n$ mints. Also, she must not have discovered the mints empty earlier.

Working it through

STAGE 1 Discovering the toffees are gone

Ask yourself In what order can the first $N-1$ reaches occur? What condition on $N-1-n$ ensures she never found the mint packet empty?

The n toffee-reaches can be any n of the first $N-1$ reaches. She never reached for a mint after the mints were gone provided she took at most m mints: $N-1-n \le m$. So, for $n+1 \le N \le n+m+1$,

$$\mathrm{P}(T_N) = \binom{N-1}{n} p^{n+1} q^{N-1-n},$$

where T_N is the event "on the Nth reach she discovers the toffees are gone" (the factor p^{n+1} includes the Nth reach itself).

STAGE 2 Discovering the mints are gone

By the same argument with the roles swapped, for $m+1 \le N \le n+m+1$,

$$\mathrm{P}(M_N) = \binom{N-1}{m} q^{m+1} p^{N-1-m}.$$

STAGE 3 The conditional probability

Ask yourself We are told that a discovery happens on the Nth reach. What is the probability that it was a toffee?

The events T_N and M_N are exclusive, so

$$\mathrm{P}(\text{toffee} \mid \text{discovery at } N) = \frac{\binom{N-1}{n} p^{n+1} q^{N-1-n}}{\binom{N-1}{n} p^{n+1} q^{N-1-n} + \binom{N-1}{m} q^{m+1} p^{N-1-m}},$$

where a binomial coefficient is taken to be 0 outside the ranges above. (If $N \le m$, only the toffees can have run out, and the probability is 1; similarly it is 0 if $N \le n$ but $N > m$.) ■

Pause and try Check the formula when $n = m$ and $p = q = \frac{1}{2}$. By symmetry the answer should be $\frac{1}{2}$ for every possible N. Is it?

Looking back

- **Read the event precisely.** "Discovers" meant reaching for an empty packet, not taking the last sweet.

- **Fix the last step, count the rest.** The Nth reach was fixed; the first $N-1$ could be in any order, giving a binomial coefficient.
- **Check the boundaries.** The constraints $N-1-n \le m$ and $N-1-m \le n$ excluded impossible histories.

What's really going on

Banach's matchboxes

In Banach's version, each box starts with n matches and each pocket is equally likely. When he first finds a box empty, let K be the number of matches in the other box. For $K=k$, he must have taken n from the empty box and $n-k$ from the other, in any order, before reaching for the empty box once more; and either box could be the empty one:

$$\mathrm{P}(K=k) = 2\binom{2n-k}{n}\left(\frac{1}{2}\right)^{2n-k+1} = \binom{2n-k}{n}\left(\frac{1}{2}\right)^{2n-k}.$$

These probabilities decrease with k, except that $\mathrm{P}(K=0)=\mathrm{P}(K=1)$ (Figure 12.17.1). The expected number left is about $2\sqrt{n/\pi}-1$: with 50 matches in each box, about 7. The $\sqrt{n}$ is the typical size of the imbalance between the two boxes after $2n$ random choices, the same square-root law as in the birthday problem (Section 9.9) and the random walk.

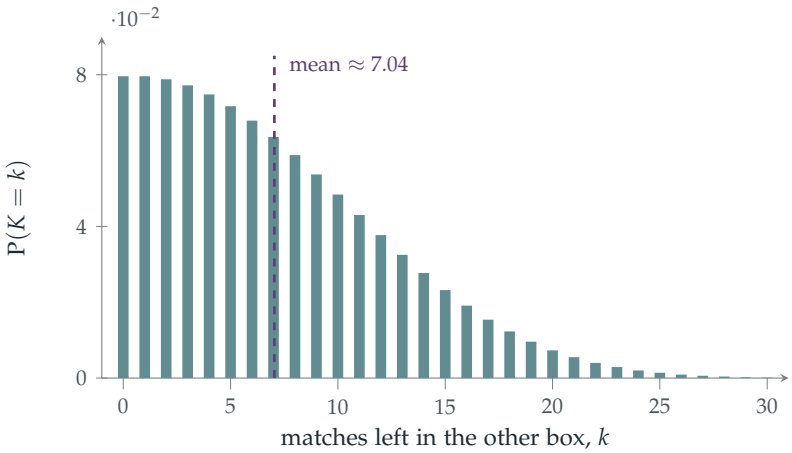


Figure 12.17.1 Banach's matchbox problem with 50 matches in each box: the distribution of the number left in the other box when one is first found empty. The most likely values are 0 and 1, equally, but the mean is about 7.

The negative binomial distribution

The number of trials needed to obtain the rth success in independent trials, each succeeding with probability p, has the *negative binomial distribution*: $\mathrm{P}(N=k) = \binom{k-1}{r-1}p^r q^{k-r}$. The taxi driver's problem is a race between two negative binomial processes

(the toffees and the mints), each needing one more "success" than it has sweets. The negative binomial appears in insurance (numbers of claims), ecology (clustered counts of animals) and genetics, often as a more flexible alternative to the Poisson distribution.

Banach and the Scottish Café

Stefan Banach, one of the founders of functional analysis, did much of his mathematics in the Scottish Café in Lwów (now Lviv), where he and his colleagues wrote problems in a notebook kept by the waiter, the *Scottish Book*, with prizes ranging from a beer to a live goose. Hugo Steinhaus, Banach's mentor, devised the matchbox problem for a speech honouring Banach's smoking. The problem was first analysed in print by William Feller in his celebrated probability textbook of 1950.

The thinking skill

Fix the final event, count the paths to it

To find the probability that a process ends in a particular way at a particular time, fix what happens at the last step and count the histories that lead there.

At university this is the basis of the negative binomial distribution, of first-passage times for random walks, and of the reflection principle in Section 9.10.

Try this next

1. **Small case.** With $n = m = 1$ and $p = \frac{1}{2}$, list the possible sequences and find the distribution of N.
2. **Banach's formula.** Verify that $\sum_{k=0}^{n} \binom{2n-k}{n} 2^{-(2n-k)} = 1$ for $n = 1$ and $n = 2$.
3. **Equal tops.** Show that $\mathrm{P}(K = 0) = \mathrm{P}(K = 1)$ for every n in Banach's problem.
4. **Which runs out first?** For the taxi driver, find the probability that the toffees are the first packet to be discovered empty (at any time), when $n = m = 2$ and $p = \frac{2}{3}$.

Further reading

- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter VI, which introduced Banach's problem to the world.
- **R. Daniel Mauldin (ed.), *The Scottish Book*** (Birkhäuser). The problems from the Scottish Café, with commentary.
- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University

Press), on the negative binomial distribution.

12.18 Memorylessness

The question

The continuous random variable X has probability density function

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0, \\ 0 & \text{otherwise,} \end{cases}$$

where λ is a positive constant. The random variable Y is the greatest integer less than or equal to X, and $Z = X - Y$.

(i) Show that, for any non-negative integer n, $P(Y = n) = (1 - e^{-\lambda})e^{-n\lambda}$.

(ii) Show that $P(Z < z) = \dfrac{1 - e^{-\lambda z}}{1 - e^{-\lambda}}$ for $0 \leq z \leq 1$.

(iii) Evaluate $E(Z)$.

(iv) Obtain an expression for $P(Y = n \text{ and } z_1 < Z < z_2)$, where $0 \leq z_1 < z_2 \leq 1$ and n is a non-negative integer. Determine whether Y and Z are independent.

The hook

A radioactive atom has been sitting in a laboratory for a million years without decaying. Is it "due" to decay soon? No: its chance of decaying in the next second is exactly the same as that of a freshly made atom. Radioactive atoms do not age. The waiting time until decay has the *exponential distribution*, and its defining property is that it has no memory.

This question shows the memorylessness in a striking form. Split an exponential waiting time into its whole number of units and its fractional part. These two pieces turn out to be *independent*: knowing how many whole hours you waited tells you nothing about the minutes. The question also explains, in the exercises, why your wait for a bus can seem longer than the gap between buses.

Before you start

Ask yourself What is $\mathrm{P}(X > x)$ for $x \geq 0$? What is $\mathrm{P}(X > s+t \mid X > s)$?

$\mathrm{P}(X > x) = \int_x^\infty \lambda e^{-\lambda u}\,\mathrm{d}u = e^{-\lambda x}$. Then $\mathrm{P}(X > s+t \mid X > s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathrm{P}(X > t)$: having already waited s, the remaining wait has the same distribution as a fresh one. This is the *memoryless property*.

Working it through

STAGE 1 The whole part

$Y = n$ exactly when $n \leq X < n+1$:

$$\mathrm{P}(Y = n) = e^{-\lambda n} - e^{-\lambda(n+1)} = (1 - e^{-\lambda})e^{-n\lambda}. \qquad \blacksquare$$

So Y has a geometric distribution: each whole unit is "survived" with probability $e^{-\lambda}$.

STAGE 2 The fractional part

Ask yourself $Z < z$ means X lies in one of the intervals $[n, n+z)$. Add up their probabilities.

$$\mathrm{P}(Z < z) = \sum_{n \geq 0} \left(e^{-\lambda n} - e^{-\lambda(n+z)}\right) = (1 - e^{-\lambda z}) \sum_{n \geq 0} e^{-\lambda n} = \frac{1 - e^{-\lambda z}}{1 - e^{-\lambda}}. \qquad \blacksquare$$

STAGE 3 The mean of Z

Ask yourself For a random variable on $[0,1]$, $\mathrm{E}(Z) = \int_0^1 \mathrm{P}(Z > z)\,\mathrm{d}z$. Use it.

With $c = 1 - e^{-\lambda}$, $\mathrm{P}(Z > z) = 1 - \frac{1 - e^{-\lambda z}}{c}$, so

$$\mathrm{E}(Z) = 1 - \frac{1}{c}\left(1 - \frac{1 - e^{-\lambda}}{\lambda}\right) = 1 - \frac{1}{c} + \frac{1}{\lambda} = \frac{1}{\lambda} - \frac{e^{-\lambda}}{1 - e^{-\lambda}} = \frac{1}{\lambda} - \frac{1}{e^{\lambda} - 1}. \qquad \blacksquare$$

(For small λ, long waits, this tends to $\frac{1}{2}$: the fractional part is nearly uniform. For large λ it is close to $\frac{1}{\lambda}$, because X is almost always less than 1.)

STAGE 4 Independence

$$\mathrm{P}(Y = n, z_1 < Z < z_2) = \mathrm{P}(n + z_1 < X < n + z_2) = e^{-\lambda n}\left(e^{-\lambda z_1} - e^{-\lambda z_2}\right).$$

Ask yourself Compare with $\mathrm{P}(Y = n)\,\mathrm{P}(z_1 < Z < z_2)$, using parts (i) and (ii).

From (i) and (ii), $\mathrm{P}(Y = n)\mathrm{P}(z_1 < Z < z_2) = (1 - e^{-\lambda})e^{-n\lambda} \cdot \frac{e^{-\lambda z_1} - e^{-\lambda z_2}}{1 - e^{-\lambda}}$, which is the same. So Y and Z are independent. ■

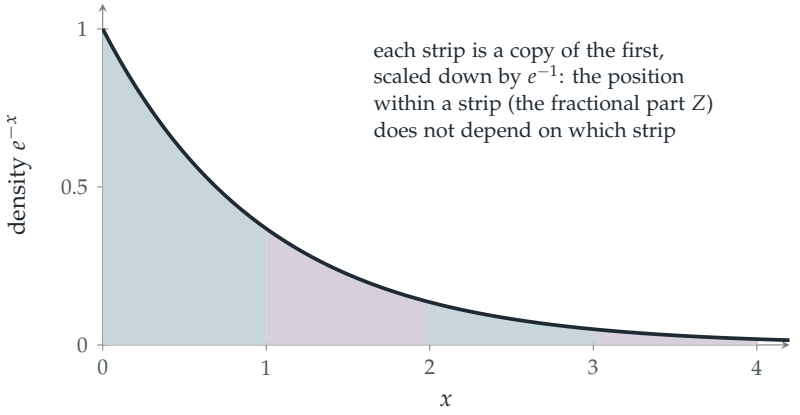


Figure 12.18.1 The exponential density with $\lambda = 1$, divided into unit strips. Because $e^{-(n+z)} = e^{-n}e^{-z}$, every strip has the same shape. Which strip X falls in (the whole part Y) and where it falls within the strip (the fractional part Z) are independent.

Looking back

- **Sum over pieces.** Both the distribution of Z and the joint probabilities came from adding probabilities over the intervals $[n, n+1)$.
- **The tail formula.** $\mathrm{E}(Z) = \int \mathrm{P}(Z > z)\,\mathrm{d}z$ avoided finding the density.
- **Factorisation means independence.** The joint probability split into a function of n times a function of z_1, z_2.

What's really going on

Why the exponential is special

The independence of Y and Z comes straight from $e^{-\lambda(n+z)} = e^{-\lambda n}e^{-\lambda z}$, and that is the memoryless property in disguise. In fact the exponential is the *only* continuous distribution without memory: if $\mathrm{P}(X > s + t) = \mathrm{P}(X > s)\mathrm{P}(X > t)$ for all $s, t \geq 0$, then $g(t) = \ln \mathrm{P}(X > t)$ satisfies $g(s + t) = g(s) + g(t)$, and (for a monotonic g) this forces $g(t) = -\lambda t$. That is why the exponential models waiting times for events that happen "at random", with no build-up: radioactive decays, the arrival of customers, the failure of electronic components that do not wear out.

Minima and the waiting-time paradox

If X_1 and X_2 are independent exponentials with rates λ_1 and λ_2, then $\mathrm{P}(\min > x) = e^{-\lambda_1 x} e^{-\lambda_2 x}$, so the minimum is exponential with rate $\lambda_1 + \lambda_2$. Waiting for the first of two independent random events is like waiting for one event at the combined rate (STEP III 2002 Q13, in the exercises). Memorylessness also explains the *waiting-time paradox*: if buses arrive at random with an average gap of 10 minutes, then a passenger arriving at a random time waits on average 10 minutes, not 5, because the remaining wait is memoryless; and the gap they arrive in is on average 20 minutes long, since long gaps are more likely to be landed in.

Rutherford and the Poisson process

In 1910 Ernest Rutherford and Hans Geiger counted the alpha particles emitted by a radioactive source in intervals of 7.5 seconds, and found that the counts followed the Poisson distribution, as the mathematician Harry Bateman showed they should if the emissions occurred independently at a constant rate. The times between emissions are then exponential. The *Poisson process*, a stream of events with exponential gaps, is now the basic model of randomness in time, used for telephone calls, insurance claims and the firing of neurons; the next section, Section 12.19, extends it to space.

The thinking skill

Test independence by factorising

Two random variables are independent if their joint probabilities factorise into a function of one times a function of the other.

At university, factorisation of a joint density is the standard test for independence, and it is how you will recognise independent components in multivariate problems, such as the radius and angle of a normally distributed point in Section 12.22.

Try this next

1 **Two bus routes.** (From STEP III, 2002.) Show that the minimum of independent exponentials with parameters λ_1 and λ_2 is exponential with parameter $\lambda_1 + \lambda_2$. Buses on route A and route B have exponential gaps with means 15 and 30 minutes, independently. Having just seen one of each leave, show that the median wait for the next bus is about 7 minutes.

2 **Memoryless discrete.** Show that the geometric distribution is memoryless: $\mathrm{P}(Y \geq m + n \mid Y \geq m) = \mathrm{P}(Y \geq n)$.

3 **The waiting-time paradox.** Buses arrive as a Poisson process with rate λ. Show that the gap containing a fixed time t (for large t) has expected length $\frac{2}{\lambda}$, twice the average gap.

4 **The density of Z.** Find the probability density of Z, and sketch it for $\lambda = 1$ and $\lambda = 5$.

Further reading

- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press), on the exponential distribution and the Poisson process.
- **Sheldon Ross, *Introduction to Probability Models*** (Academic Press). Exponential distributions, Poisson processes and queues.
- **David Spiegelhalter, *The Art of Uncertainty*** (Pelican). Randomness and waiting in everyday life.

12.19 Random Points in the Plane

The question

In this question, you may assume that $\int_0^\infty e^{-x^2/2}\,\mathrm{d}x = \sqrt{\frac{1}{2}\pi}$.

The number of supermarkets situated in any given region can be modelled by a Poisson random variable, where the mean is k times the area of the given region. Find the probability that there are no supermarkets within a circle of radius y.

The random variable Y denotes the distance between a randomly chosen point in the region and the nearest supermarket. Write down $\mathrm{P}(Y < y)$ and hence show that the probability density function of Y is $2\pi yke^{-\pi ky^2}$ for $y \geq 0$.

Find $\mathrm{E}(Y)$ and show that $\mathrm{Var}(Y) = \dfrac{4-\pi}{4\pi k}$.

The hook

If supermarkets, trees, stars or mobile-phone masts are scattered "at random" across a region, how far do you have to go to the nearest one? The answer depends only on the density k (the average number per unit area), and it has a characteristic shape, the *Rayleigh distribution*. On average the nearest is at distance $\frac{1}{2\sqrt{k}}$: with one supermarket per square kilometre, 500 metres.

The model is the *spatial Poisson process*, the two-dimensional version of random arrivals in time (Section 12.18). The calculation is short and pretty, but its real interest is as a benchmark: ecologists compare real nearest-neighbour distances with this formula to decide whether trees or nests are clustered, randomly scattered or deliberately spaced out.

Before you start

Ask yourself A Poisson random variable with mean μ takes the value 0 with what probability?

$\mathrm{P}(0) = e^{-\mu}$. A circle of radius y has area πy^2, so the number of supermarkets inside it has mean $k\pi y^2$.

Working it through

STAGE 1 No supermarket nearby

The probability of no supermarket within distance y is $e^{-\pi k y^2}$.

STAGE 2 The distribution of the nearest distance

Ask yourself $Y \geq y$ means there is no supermarket within distance y. So what is $\mathrm{P}(Y < y)$?

$\mathrm{P}(Y < y) = 1 - e^{-\pi k y^2}$, and differentiating gives the density

$$\mathrm{f}(y) = 2\pi k y e^{-\pi k y^2}, \qquad y \geq 0. \qquad \blacksquare$$

STAGE 3 The mean

Ask yourself Either integrate $y\,\mathrm{f}(y)$ by parts, or use $\mathrm{E}(Y) = \int_0^\infty \mathrm{P}(Y > y)\,\mathrm{d}y$. Then substitute to use the given Gaussian integral.

$$\mathrm{E}(Y) = \int_0^\infty e^{-\pi k y^2}\,\mathrm{d}y.$$

Substituting $x = y\sqrt{2\pi k}$, so that $\pi k y^2 = \frac{1}{2}x^2$, gives $\mathrm{E}(Y) = \frac{1}{\sqrt{2\pi k}}\sqrt{\frac{\pi}{2}} = \frac{1}{2\sqrt{k}}$. $\blacksquare$

STAGE 4 The variance

$$\mathrm{E}(Y^2) = \int_0^\infty y^2 \cdot 2\pi k y e^{-\pi k y^2}\,\mathrm{d}y.$$

Substituting $u = \pi k y^2$ (so $\mathrm{d}u = 2\pi k y\,\mathrm{d}y$), this is $\int_0^\infty \frac{u}{\pi k} e^{-u}\,\mathrm{d}u = \frac{1}{\pi k}$. Hence

$$\mathrm{Var}(Y) = \frac{1}{\pi k} - \frac{1}{4k} = \frac{4-\pi}{4\pi k}. \qquad \blacksquare$$

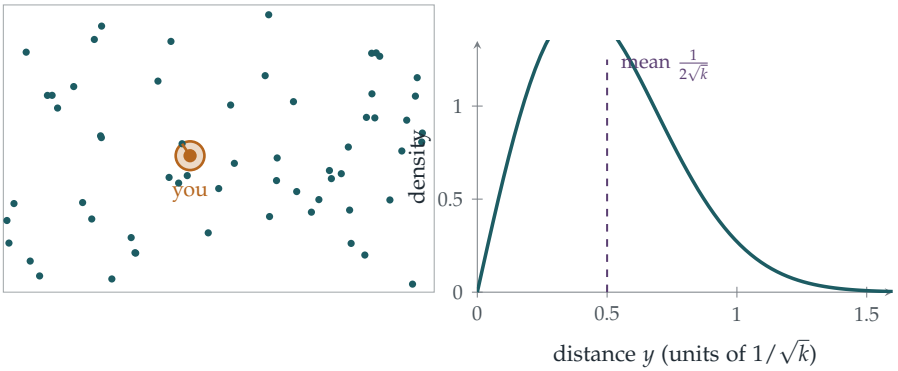


Figure 12.19.1 Left: sixty points scattered at random over a region of area 24 (so $k = 2.5$), and the distance from a chosen point to the nearest of them (amber). Right: the Rayleigh density $2\pi k y e^{-\pi k y^2}$ of that distance, with distance measured in units of $1/\sqrt{k}$. Its mean is $\frac{1}{2\sqrt{k}}$.

Looking back

- **Turn a distance question into a counting question.** "The nearest is further than y" is "there are none within y".
- **Cumulative first, density second.** $\mathrm{P}(Y < y)$ came directly; differentiating gave the density.
- **Substitutions to known integrals.** Each moment reduced to the Gaussian integral or the exponential integral.

What's really going on

The Poisson process in the plane

A spatial Poisson process with intensity k is a random scatter of points such that the numbers in disjoint regions are independent, and the number in a region of area A is Poisson with mean kA. It is what you get from dropping a huge number of points independently and uniformly over a huge area, with the right average density: the "law of rare events" that also explains derangements (Section 9.8). It is the natural model of complete spatial randomness, and real patterns are judged against it. The *Clark–Evans test*, devised by two ecologists in 1954, compares the observed mean nearest-neighbour distance with $\frac{1}{2\sqrt{k}}$: much smaller suggests clustering (seeds falling near their parent tree), much larger suggests regular spacing (territorial birds, or competition between plants for water).

In three dimensions

The same calculation in three dimensions gives $\mathrm{P}(Y > y) = e^{-\frac{4}{3}\pi k y^3}$ for the distance to the nearest star, molecule or galaxy. Subrahmanyan Chandrasekhar used it in 1943 to study the gravitational forces on a star from its random neighbours, and physicists use it for the mean free path of molecules in a gas.

Rayleigh

The density $\frac{y}{\sigma^2} e^{-y^2/(2\sigma^2)}$ is named after Lord Rayleigh, who met it in 1880 in the theory of sound: it describes the amplitude of the sum of many waves with random phases. It also gives the distance from the origin of a point whose two coordinates are independent normal random variables, which is another way to see why it appears here (exercise 3). Today it models the fading of mobile-phone signals, which reach the receiver along many paths with random phases.

The thinking skill

Find the distribution function first

For the distribution of a minimum, a maximum or a nearest distance, $\mathrm{P}(Y > y)$ *is often easy to write down directly. Differentiate to get the density.*

This approach recurs in Section 12.25, for the largest of many random lifetimes. At university it is the standard first step for any "order statistic".

Try this next

1 **The median.** Find the median distance to the nearest supermarket, and compare it with the mean.

2 **The second nearest.** Find the distribution of the distance to the *second* nearest supermarket. (*Hint:* $\mathrm{P}(Y_2 > y)$ is the probability of at most one supermarket within y.)

3 **Normal coordinates.** If U and V are independent $\mathrm{N}(0, \sigma^2)$, show that $\sqrt{U^2 + V^2}$ has density $\frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)}$. Which value of σ gives the supermarket distribution?

4 **Clark–Evans.** In a survey of a 1 km^2 wood containing 400 trees, the mean distance from a tree to its nearest neighbour is 35 m. Is this consistent with complete spatial randomness, or does it suggest clustering or spacing?

Further reading

- **Peter Diggle, *Statistical Analysis of Spatial Point Patterns*** (Arnold). The standard reference, with the Clark–Evans test and its successors.
- **John Kingman, *Poisson Processes*** (Oxford University Press). A beautiful short book on Poisson processes in any number of dimensions.
- **Geoffrey Grimmett & David Stirzaker, *Probability and Random Processes*** (Oxford University Press), the chapter on the Poisson process.

12.20 Buffon's Needle

The question

By making the substitution $y = \cos^{-1} t$, or otherwise, show that

$$\int_0^1 \cos^{-1} t \, \mathrm{d}t = 1.$$

A pin of length $2a$ is thrown onto a floor ruled with parallel lines equally spaced at a distance $2b$ apart. The distance X of its centre from the nearest line is a uniformly distributed random variable taking values between 0 and b and the acute angle Y the pin makes with a direction perpendicular to the line is a uniformly distributed random variable taking values between 0 and $\pi/2$. X and Y are independent. If $X = x$ what is the probability that the pin crosses the line?

If $a < b$ show that the probability that the pin crosses a line for a general throw is $\dfrac{2a}{\pi b}$.

The hook

Drop a needle onto a floor of parallel floorboards. The chance that it lies across a crack involves π, even though there is no circle anywhere in sight. Count the crossings in many throws and you can estimate π by experiment. This is *Buffon's needle*, posed by Georges-Louis Leclerc, Comte de Buffon, in 1733 and solved by him in 1777: the first problem in *geometric probability*, and the ancestor of every Monte Carlo method.

The π comes from the angle: the needle's direction is uniform on a half-turn, and averaging the length of its "shadow" across the lines over all directions gives a factor $\frac{2}{\pi}$. The question does the calculation with an integral of $\cos^{-1}$.

Before you start

Ask yourself The pin's centre is at distance x from the nearest line, and the pin makes angle y with the perpendicular to the lines. How far does the pin reach towards the line?

Each half of the pin has length a, and its component perpendicular to the lines is $a\cos y$. So the pin crosses the nearest line exactly when $a\cos y \geq x$. (Since $a < b$, it cannot reach any other line.)

Working it through

STAGE 1 The integral

With $t = \cos y$, as t goes from 0 to 1, $y = \cos^{-1} t$ goes from $\frac{\pi}{2}$ to 0, and $\mathrm{d}t = -\sin y \,\mathrm{d}y$. So

$$\int_0^1 \cos^{-1} t \,\mathrm{d}t = \int_0^{\pi/2} y \sin y \,\mathrm{d}y = \left[-y\cos y\right]_0^{\pi/2} + \int_0^{\pi/2} \cos y \,\mathrm{d}y = 0 + 1 = 1. \qquad \blacksquare$$

(Or: the integral is the area between the graph of $\cos^{-1}$ and the axes, which is the same as the area under $y = \cos x$ from 0 to $\frac{\pi}{2}$, reflected in $y = x$.)

STAGE 2 Given the position

Ask yourself For fixed x, which angles y give a crossing? What is their probability?

If $x \geq a$, the pin cannot reach the line. If $x < a$, a crossing needs $\cos y \geq \frac{x}{a}$, that is $y \leq \cos^{-1}\frac{x}{a}$. Since Y is uniform on $[0, \frac{\pi}{2}]$,

$$\mathrm{P}(\text{cross} \mid X = x) = \frac{2}{\pi}\cos^{-1}\frac{x}{a} \qquad (0 \leq x < a). \qquad \blacksquare$$

STAGE 3 Averaging over the position

X is uniform on $[0, b]$, with density $\frac{1}{b}$, so

$$\mathrm{P}(\text{cross}) = \frac{1}{b}\int_0^a \frac{2}{\pi}\cos^{-1}\frac{x}{a}\,\mathrm{d}x = \frac{2a}{\pi b}\int_0^1 \cos^{-1} t \,\mathrm{d}t = \frac{2a}{\pi b}, \qquad \blacksquare$$

substituting $x = at$.

Pause and try With a needle exactly as long as the half-spacing ($a = \frac{b}{2}$, so the needle's length $2a$ equals b), the crossing probability is $\frac{1}{\pi}$. How many throws would you need to estimate π to two decimal places?

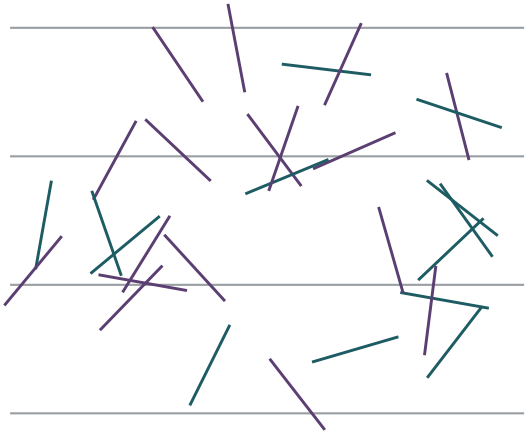


Figure 12.20.1 Thirty needles of length $2a = 1.4$ thrown onto lines $2b = 2$ apart. The ones crossing a line are plum. Here 17 cross, compared with the 13.4 expected: small samples give rough estimates, and Buffon's needle is a famously slow way to compute π.

Looking back

- **Two independent coordinates.** The pin's position and angle were independent and uniform, so the probability was a double integral that could be done one variable at a time.
- **Condition, then average.** First fix x and average over the angle; then average over x.
- **A substitution reveals the constant.** Scaling $x = at$ reduced the answer to a single universal integral.

What's really going on

Buffon's noodle

Here is a way to see the answer with almost no calculation. Bend the needle into any shape, a "noodle", and let C be the number of times it crosses a line. By linearity of expectation, the expected number of crossings of a curve is the sum over its small pieces, so it is proportional to the length: $\mathrm{E}(C) = \kappa L$ for some constant κ. To find κ, take the noodle to be a circle of diameter equal to the line spacing $2b$: it always crosses exactly twice, and its length is $2\pi b$. So $\kappa \cdot 2\pi b = 2$, giving $\kappa = \frac{1}{\pi b}$. For a straight needle of length $2a < 2b$, which crosses at most once, $\mathrm{P}(\text{cross}) = \mathrm{E}(C) = \frac{2a}{\pi b}$. This argument is due to Joseph-Émile Barbier (1860), and it is a gem of probabilistic reasoning: the π comes from the circle that was chosen to calibrate the constant.

Estimating π, and how not to

If n needles are thrown and c cross, then $\frac{c}{n} \approx \frac{2a}{\pi b}$ and $\pi \approx \frac{2an}{bc}$. The error shrinks only like $\frac{1}{\sqrt{n}}$, so to get π to three decimal places needs about a million throws. In 1901 the Italian mathematician Mario Lazzarini claimed to have obtained $\pi \approx \frac{355}{113} = 3.1415929\ldots$ (correct to six decimal places) from 3408 throws. Statisticians have pointed out that the result is far too good to be true: with 3408 throws and a carefully chosen needle, $\frac{355}{113}$ arises from exactly 1808 crossings, and stopping the experiment at the right moment would produce it. It is a classic cautionary tale about selective reporting.

Buffon

Buffon was a naturalist, director of the royal botanical garden in Paris and author of a 36-volume natural history. He presented the needle problem (and a version with square tiles, the "franc-carreau" game) to the Académie des Sciences in 1733, and published the solution in 1777. His aim was to show that geometry, not just arithmetic, could be used in probability. Integral geometry, the theory of random lines and their intersections with curves and surfaces, grew from it, and is now used in stereology, the science of estimating the size and shape of three-dimensional structures (such as cells in a tissue) from two-dimensional slices.

The thinking skill

Probability as an integral over a region

When an outcome depends on several independent continuous random variables, the probability of an event is the integral of the joint density over the region where the event occurs.

At university this becomes the foundation of continuous probability: joint densities, marginal distributions and changes of variable, all of which reappear in Section 12.21 and Section 12.22.

Try this next

1. **A long needle.** If $a > b$, the needle may cross more than one line. Show that the expected number of crossings is still $\frac{2a}{\pi b}$.

2. **The noodle.** Bend a needle of length $2a$ into a right angle (two arms of length a). What is the expected number of crossings?

3. **Square tiles.** A coin of radius r is tossed onto a floor of square tiles of side $s > 2r$. Find the probability that it lies entirely within one tile.

4 Lazzarini's needle. With $a = \frac{5}{6}b$ (so that $\frac{2a}{\pi b} = \frac{5}{3\pi}$), show that 3408 throws with 1808 crossings give the estimate $\pi \approx \frac{355}{113}$.

Further reading

- **Daniel Klain & Gian-Carlo Rota, *Introduction to Geometric Probability*** (Cambridge University Press), which starts from Buffon's needle.
- **Lee Badger, "Lazzarini's lucky approximation of π"**, *Mathematics Magazine* 67 (1994). The statistics of a suspicious experiment.
- **Paul Nahin, *Digital Dice*** (Princeton University Press). Monte Carlo methods for probability problems.

12.21 Breaking a Stick

The question

(i) A rod of unit length is cut into pieces of length X and $1-X$; the latter is then cut in half. The random variable X is uniformly distributed over $[0,1]$. For some values of X a triangle can be formed from the three pieces of the rod. Show that the conditional probability that, if a triangle can be formed, it will be obtuse-angled is $3-2\sqrt{2}$.

(ii) The bivariate distribution of the random variables X and Y is uniform over the triangle with vertices $(1,0)$, $(1,1)$ and $(0,1)$. A pair of values x, y is chosen at random from this distribution and a (perhaps degenerate) triangle ABC is constructed with $BC = x$ and $CA = y$ and $AB = 2-x-y$. Show that the construction is always possible and that $\angle ABC$ is obtuse if and only if

$$y > \frac{x^2-2x+2}{2-x}.$$

Deduce that the probability that $\angle ABC$ is obtuse is $3-4\ln 2$.

The hook

Break a stick at two random points. Can the three pieces form a triangle? The famous answer is: with probability $\frac{1}{4}$. And if they do, is the triangle acute or obtuse? Obtuse, usually: only about a third of the triangles formed are acute. Questions like these, about random shapes, have entertained probabilists since the nineteenth century, and they are a perfect training ground for working with continuous probability: a random point in a region, a condition defining a sub-region, and an integral for its area.

This question takes two different ways of breaking the stick, and in each case asks how likely an obtuse angle is. The answers, $3-2\sqrt{2}$ and $3-4\ln 2$, are unexpected numbers that come from simple geometry.

Before you start

Ask yourself Three lengths form a triangle if and only if each is less than the sum of the other two. If the lengths add up to L, what does this say about the longest one? When is the angle opposite side c obtuse?

$c < a+b$ with $a+b+c = L$ means $c < \frac{L}{2}$; so three pieces form a triangle exactly when every piece is shorter than half the total. By the cosine rule, the angle opposite c is obtuse exactly when $c^2 > a^2 + b^2$.

Working it through

STAGE 1 One cut, then halve

The pieces are X, $\frac{1-X}{2}$, $\frac{1-X}{2}$. The two equal pieces are always less than $\frac{1}{2}$, so a triangle forms if and only if $X < \frac{1}{2}$. The triangle is isosceles; the equal angles are acute, so it is obtuse only if the angle opposite X is obtuse:

$$X^2 > 2\Big(\frac{1-X}{2}\Big)^2 = \frac{(1-X)^2}{2}, \qquad \text{that is} \qquad \sqrt{2}X > 1-X, \quad X > \frac{1}{1+\sqrt{2}} = \sqrt{2}-1.$$

Given $X < \frac{1}{2}$, X is uniform on $[0, \frac{1}{2}]$, so

$$\mathrm{P}(\text{obtuse} \mid \text{triangle}) = \frac{\frac{1}{2} - (\sqrt{2}-1)}{\frac{1}{2}} = 3 - 2\sqrt{2} \approx 0.172. \qquad \blacksquare$$

STAGE 2 The second construction

> **Ask yourself** Check the three triangle inequalities for sides x, y and $2-x-y$, for (x,y) in the triangle with vertices $(1,0)$, $(1,1)$, $(0,1)$.

The region is $\{x \le 1, y \le 1, x+y \ge 1\}$. The inequalities are: $x+y \ge 2-x-y$, i.e. $x+y \ge 1$; $x+(2-x-y) \ge y$, i.e. $y \le 1$; and $y+(2-x-y) \ge x$, i.e. $x \le 1$. All hold in the region (with equality on its edges, where the triangle is degenerate). $\blacksquare$

> **Ask yourself** The angle at B is opposite the side $CA = y$. Write the condition $y^2 > x^2 + (2-x-y)^2$ as an inequality for y.

Expanding, $y^2 > x^2 + 4 + x^2 + y^2 - 4x - 4y + 2xy$, so $4y - 2xy > 2x^2 - 4x + 4$, that is $2y(2-x) > 2(x^2 - 2x + 2)$. Since $2 - x > 0$,

$$\angle ABC \text{ obtuse} \iff y > \frac{x^2 - 2x + 2}{2 - x}. \qquad \blacksquare$$

STAGE 3 The area

> **Ask yourself** Divide $x^2 - 2x + 2$ by $2 - x$. Then find the area of the part of the region above the curve, and divide by the region's area, $\frac{1}{2}$.

Since $x^2 - 2x + 2 = -x(2-x) + 2$, the curve is $y = -x + \frac{2}{2-x}$. It runs from $(0,1)$ to $(1,1)$, and lies above the line $y = 1 - x$ and below $y = 1$ for $0 < x < 1$ (Figure 12.21.1).

The obtuse region lies between the curve and $y = 1$, with area

$$\int_0^1 \Big(1 + x - \frac{2}{2-x}\Big)\,dx = 1 + \frac{1}{2} + \big[2\ln(2-x)\big]_0^1 = \frac{3}{2} - 2\ln 2.$$

Dividing by $\frac{1}{2}$: $\mathrm{P}(\angle ABC \text{ obtuse}) = 3 - 4\ln 2 \approx 0.227$. ■

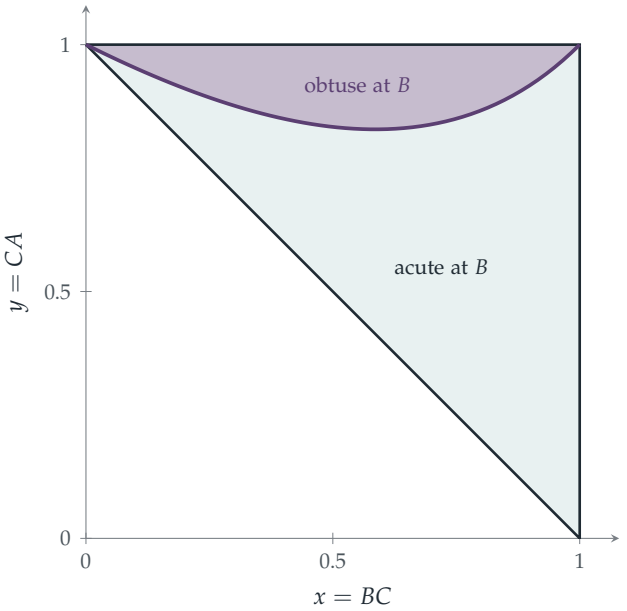


Figure 12.21.1 The triangle of possible (x, y) in part (ii). The angle at B is obtuse above the curve $y = \frac{x^2-2x+2}{2-x}$ (plum region), which occupies a fraction $3 - 4\ln 2 \approx 0.23$ of the triangle.

Looking back

- **Translate geometry into inequalities.** "Forms a triangle" and "is obtuse" both became inequalities between the lengths.
- **Conditional probability for continuous variables.** Given $X < \frac{1}{2}$, X was uniform on the smaller interval.
- **Probability as a ratio of areas.** For a uniform point in a region, probability is area divided by total area.

What's really going on

The classic broken stick

Break a stick at two independent uniform points U and V. The pair (U, V) is uniform on the unit square, and the three pieces form a triangle exactly when each is less than

$\frac{1}{2}$: that cuts out four small triangles of total area $\frac{3}{4}$, leaving probability $\frac{1}{4}$. Given that a triangle forms, the probability that it is acute is $3\ln 2 - 2 \approx 0.079$ out of $\frac{1}{4}$, that is about 32%. The pieces are an example of a *uniform random partition*, and their joint distribution is the *Dirichlet distribution*, which appears in Bayesian statistics and in models of how a resource is shared out at random.

How you break the stick matters

Notice that the two methods in the question give different answers, $3 - 2\sqrt{2}$ and $3 - 4\ln 2$, to similar-sounding questions, and both differ from the classic method. As with Bertrand's chords (Section 12.3), "random" means nothing until the procedure is specified.

Random triangles

Problems about random triangles were popular Victorian puzzles. Lewis Carroll's *Pillow Problems* (1893) asks for the probability that a triangle chosen "at random" is obtuse, and his answer, like those in this section, depends entirely on the model of randomness he chose. Problems about random triangles have continued to produce surprises: for instance, the probability that three uniformly random points in a square form an obtuse triangle is about 0.725, computed exactly only in the twentieth century.

The thinking skill

Geometry of the sample space

Represent the random outcome as a point in a region, the event as a sub-region, and compute probability as a ratio of areas (or an integral of a density).

At university, this picture is the foundation of multivariate probability, and of the change-of-variables formula, which tells you how densities transform when the coordinates are changed.

Try this next

1. **The classic.** Show that if a stick is broken at two independent uniform points, the probability that the pieces form a triangle is $\frac{1}{4}$.

2. **Break the longer piece.** Break a stick at a uniform point, then break the *longer* piece at a uniform point. Show that the probability that the three pieces form a triangle is $2\ln 2 - 1 \approx 0.386$.

3. **Acute or obtuse.** In part (ii), find the probability that the triangle ABC has an obtuse angle at C (opposite $AB = 2 - x - y$). Hence find the probability that it has an

obtuse angle somewhere.

4 Expected longest piece. For the classic broken stick, show that the expected length of the longest piece is $\frac{11}{18}$.

Further reading

- **Martin Gardner, *The Colossal Book of Mathematics*** (Norton), which includes the broken-stick problem.
- **Frederick Mosteller, *Fifty Challenging Problems in Probability*** (Dover).
- **Lewis Carroll, *Pillow Problems*** (Dover, with *A Tangled Tale*). Victorian puzzles, including random triangles.

12.22 A Distribution With No Mean

The question

The random variables X and Y are independently normally distributed with means 0 and variances 1. Show that the joint probability density function for (X, Y) is

$$\mathrm{f}(x,y) = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)} \qquad -\infty < x < \infty,\ -\infty < y < \infty.$$

If (x, y) are the coordinates, referred to rectangular axes, of a point in the plane, explain what is meant by saying that this density is radially symmetrical.

The random variables U and V have a joint probability density function which is radially symmetrical (in the above sense). By considering the straight line with equation $U = kV$, or otherwise, show that

$$\mathrm{P}\left(\frac{U}{V} < k\right) = 2\mathrm{P}(U < kV, V > 0).$$

Hence, or otherwise, show that the probability density function of U/V is

$$\mathrm{g}(k) = \frac{1}{\pi(1+k^2)} \qquad -\infty < k < \infty.$$

The hook

Take the average of many measurements, and the random errors cancel out: the average settles down to the true value. This *law of large numbers* is the foundation of all statistics. But it has a hidden assumption: the measurements must have a finite mean. This question constructs a perfectly natural random variable, the ratio of two independent standard normal variables, which has no mean at all. Its averages never settle down; in fact the average of a million of them has exactly the same distribution as a single one.

This is the *Cauchy distribution*. Its density looks like a harmless bell, but its tails are so heavy that occasional enormous values dominate any average. The question finds it with a lovely geometric argument: a ratio of coordinates is the slope of a line, and radial symmetry makes the angle of that line uniform.

Before you start

Ask yourself If (U, V) is a random point whose density depends only on its distance from the origin, what can you say about the angle Θ that the line from the origin to the point makes with an axis?

Rotating the plane does not change the density, so every direction is equally likely: the angle Θ is uniform on $[0, 2\pi)$. And $\frac{U}{V}$ depends only on the direction, not the distance.

Working it through

STAGE 1 The joint density

Since X and Y are independent, their joint density is the product of their densities:

$$\mathrm{f}(x,y) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-y^2/2} = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)}.$$

It depends on x and y only through $x^2 + y^2$, the square of the distance from the origin: the density is constant on circles centred at the origin. That is what "radially symmetrical" means.

STAGE 2 Using the symmetry

Ask yourself $\frac{U}{V} < k$ happens in two ways, according to the sign of V. How are they related by the rotation through $180°$, $(u, v) \mapsto (-u, -v)$?

$\frac{U}{V} < k$ means: either $V > 0$ and $U < kV$, or $V < 0$ and $U > kV$. The rotation $(u, v) \mapsto (-u, -v)$ maps the first region onto the second, and preserves the density. So the two have the same probability, and

$$\mathrm{P}\Big(\frac{U}{V} < k\Big) = 2\mathrm{P}(U < kV, V > 0). \qquad \blacksquare$$

STAGE 3 An angle

Ask yourself Draw the region $\{V > 0, U < kV\}$ in the plane with u horizontal and v vertical. It is a wedge with its vertex at the origin. What is its angle?

The region $V > 0$ is the upper half-plane. The line $u = kv$ through the origin makes angle $\arctan k$ with the positive v-axis (towards the positive u side when $k > 0$). The part of the upper half-plane to the left of this line is a wedge of angle $\frac{\pi}{2} + \arctan k$. By radial symmetry, the probability of a wedge is its angle divided by 2π, so

$$\mathrm{P}\Big(\frac{U}{V} < k\Big) = 2 \cdot \frac{\frac{\pi}{2} + \arctan k}{2\pi} = \frac{1}{2} + \frac{1}{\pi} \arctan k.$$

Differentiating with respect to k gives the density

$$g(k) = \frac{1}{\pi(1+k^2)}. \qquad \blacksquare$$

Looking back

- **Independent normals are radially symmetric.** The product of two normal densities depends only on distance.
- **Ratio means direction.** $\frac{U}{V}$ depends only on the angle of the point, which is uniform.
- **Probabilities of wedges.** With radial symmetry, the probability of a wedge is proportional to its angle.

What's really going on

No mean

The mean of the Cauchy distribution would be $\int_{-\infty}^{\infty} \frac{k}{\pi(1+k^2)}\,dk$, but $\int_0^K \frac{k}{1+k^2}\,dk = \frac{1}{2}\ln(1+K^2) \to \infty$: the integral does not converge, and the mean does not exist. (Symmetry might suggest the answer is 0, but the positive and negative parts are both infinite, and $\infty - \infty$ has no value.) The variance is infinite too.

Worse, the average of n independent Cauchy variables is again Cauchy, with exactly the same distribution. (One way to see this uses the characteristic function $E(e^{itX}) = e^{-|t|}$.) So averaging does not help at all: the average of a million measurements is no more reliable than one. Figure 12.22.1 shows what happens: the running average of normal samples settles down, while that of Cauchy samples lurches about for ever, jolted by occasional huge values.

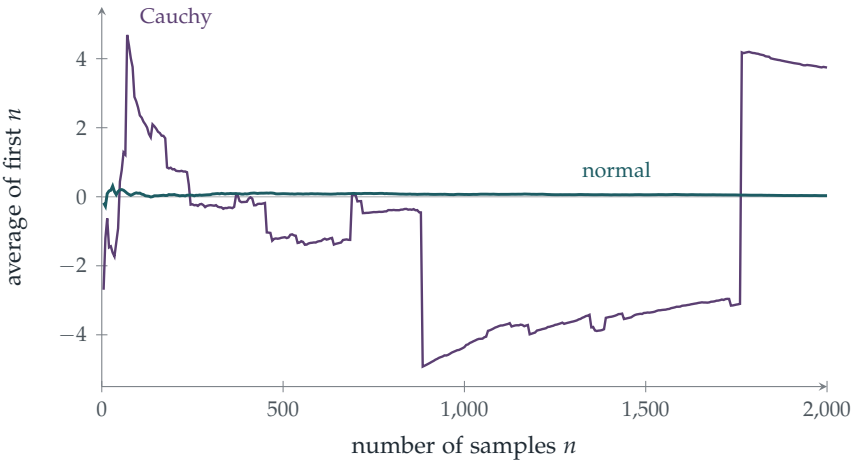


Figure 12.22.1 Running averages of 2000 independent samples. For standard normal samples (teal) the average settles down to 0, as the law of large numbers predicts. For Cauchy samples (plum) it never settles: every so often a single enormous value throws it somewhere new.

Where Cauchy variables come from

Shine a torch, pointing in a uniformly random direction, at a wall: the position of the spot on the wall is Cauchy distributed. In physics the same density, called the *Lorentzian*, describes the shape of spectral lines and resonance peaks (compare Section 11.1). In finance, heavy-tailed distributions like the Cauchy are used to model the occasional crashes that normal models wildly underestimate. The lesson for statistics is serious: methods based on averages and variances fail for heavy-tailed data, and robust methods, such as using the median (which for the Cauchy distribution is perfectly well behaved), are needed.

Poisson, Cauchy and a dispute

The distribution was studied by Siméon Denis Poisson in 1824, who noticed that it violates the law of large numbers. Augustin-Louis Cauchy met it in 1853 during a public dispute with Irénée-Jules Bienaymé over the method of least squares: Cauchy used it to show that least squares can be a poor method when errors are heavy-tailed. The name stuck to Cauchy, an example of what Stephen Stigler calls "Stigler's law of eponymy": no scientific discovery is named after its original discoverer.

The thinking skill

Check that expectations exist

Before using a mean or variance, make sure the integral defining it converges. Symmetry does not make a divergent integral zero.

At university, integrability conditions are part of every theorem in probability: the law of large numbers needs a finite mean, the central limit theorem (Section 12.24) a finite variance. The Cauchy distribution is the standard counterexample when they fail.

Try this next

1 **The torch.** A torch at distance 1 from a long straight wall points in a direction making angle Θ with the perpendicular, where Θ is uniform on $(-\frac{\pi}{2}, \frac{\pi}{2})$. Show that the spot's position $\tan\Theta$ has the Cauchy distribution.

2 **The reciprocal.** Show that if C is Cauchy, so is $\frac{1}{C}$. (*Hint:* $\frac{U}{V}$ and $\frac{V}{U}$.)

3 **The median works.** Show that the median of the Cauchy distribution is 0 and its quartiles are ± 1.

4 **Radius and angle.** For X, Y independent standard normals, show by changing to polar coordinates that $R^2 = X^2 + Y^2$ is exponential with mean 2 and is independent of the angle. (This is the basis of the Box–Muller method for generating normal random numbers.)

Further reading

- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press), on transformations of random variables.
- **Nassim Nicholas Taleb, *The Black Swan*** (Penguin). Heavy tails and why they matter, for a general reader.
- **Stephen Stigler, *The History of Statistics*** (Harvard University Press). Includes the Cauchy–Bienaymé dispute.

12.23 Stirling from Probability

The question

(i) The random variable X has a binomial distribution with parameters n and p, where $n = 16$ and $p = \frac{1}{2}$. Show, using an approximation in terms of the standard normal density function $\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2}$, that

$$\mathrm{P}(X = 8) \approx \frac{1}{2\sqrt{2\pi}}.$$

(ii) By considering a binomial distribution with parameters $2n$ and $\frac{1}{2}$, show that

$$(2n)! \approx \frac{2^{2n}(n!)^2}{\sqrt{n\pi}}.$$

(iii) By considering a Poisson distribution with parameter n, show that

$$n! \approx \sqrt{2\pi n}\, e^{-n} n^n.$$

The hook

How big is 100!? Multiplying out takes a while: it has 158 digits. *Stirling's formula*, $n! \approx \sqrt{2\pi n}\,(n/e)^n$, gives it to within 0.1% in a single line. It is one of the most useful approximations in mathematics, used constantly in probability, statistical physics and the analysis of algorithms.

There are many proofs, using integrals (Section 4.4) or the Euler–Maclaurin formula. This question gives the most surprising one: a proof by probability. The Poisson distribution with mean n is approximately normal, and comparing the two at the peak forces the value of $n!$. The $\sqrt{2\pi}$ in Stirling's formula is literally the $\sqrt{2\pi}$ in the normal density.

Before you start

Ask yourself If a discrete random variable X with mean μ and variance σ^2 is approximately normal, roughly what is $\mathrm{P}(X = \mu)$? (Think of the probability of an interval of width 1 centred at μ.)

$\mathrm{P}(X = \mu) \approx \mathrm{P}(\mu - \frac{1}{2} < X < \mu + \frac{1}{2})$ for the normal approximation, which is about the width 1 times the density at the peak: $\frac{1}{\sigma\sqrt{2\pi}}$.

Working it through

STAGE 1 Sixteen coins

$X \sim \mathrm{B}(16, \frac{1}{2})$ has mean 8 and variance $16 \cdot \frac{1}{2} \cdot \frac{1}{2} = 4$, so $\sigma = 2$. By the normal approximation,

$$\mathrm{P}(X = 8) \approx \frac{1}{2\sqrt{2\pi}} \approx 0.1995.$$

The exact value is $\binom{16}{8}/2^{16} = \frac{12870}{65536} \approx 0.1964$, within 2%.

STAGE 2 The central binomial coefficient

Ask yourself For $\mathrm{B}(2n, \frac{1}{2})$, what are the mean and standard deviation? Write $\mathrm{P}(X = n)$ exactly and approximately.

The mean is n and the variance $\frac{n}{2}$, so $\sigma = \sqrt{n/2}$. Exactly, $\mathrm{P}(X = n) = \binom{2n}{n}\frac{1}{2^{2n}}$; approximately, $\frac{1}{\sqrt{n/2}\sqrt{2\pi}} = \frac{1}{\sqrt{\pi n}}$. Equating,

$$\frac{(2n)!}{(n!)^2\,2^{2n}} \approx \frac{1}{\sqrt{\pi n}}, \qquad (2n)! \approx \frac{2^{2n}(n!)^2}{\sqrt{n\pi}}. \qquad \blacksquare$$

STAGE 3 Stirling's formula

Ask yourself A Poisson distribution with parameter n has mean and variance both equal to n. Write $\mathrm{P}(X = n)$ exactly and approximately.

Exactly, $\mathrm{P}(X = n) = e^{-n}\frac{n^n}{n!}$. For large n the Poisson distribution is approximately normal (it is the sum of n independent Poisson(1) variables), with $\sigma = \sqrt{n}$, so $\mathrm{P}(X = n) \approx \frac{1}{\sqrt{2\pi n}}$. Equating,

$$n! \approx \sqrt{2\pi n}\,e^{-n}n^n. \qquad \blacksquare$$

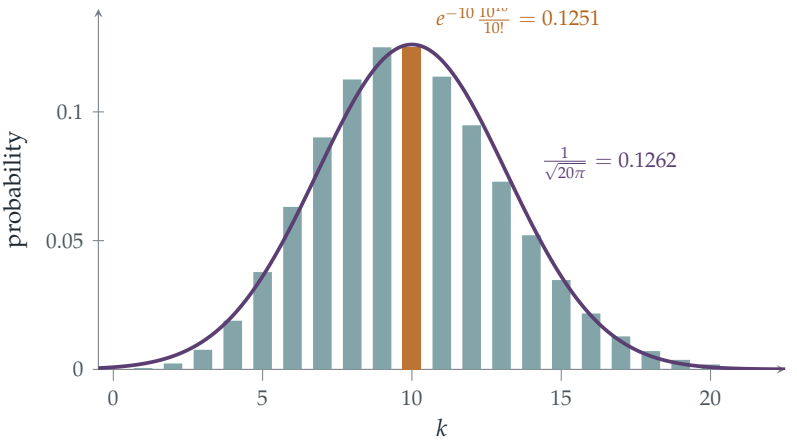


Figure 12.23.1 The Poisson distribution with mean 10 (bars) and the normal density with the same mean and variance (plum). At the peak, the Poisson probability $e^{-10}10^{10}/10!$ (amber) is close to the normal density $\frac{1}{\sqrt{2\pi\cdot 10}}$. Rearranging that near-equality is Stirling's formula.

Looking back

- **Probability at a point ≈ density times width.** For a lattice distribution, the normal density at a point approximates the probability of that point.
- **An exact formula meets an approximation.** The binomial and Poisson probabilities are exact; equating them to their normal approximations gave estimates for factorials.
- **Choose the distribution to fit the formula.** $\binom{2n}{n}$ suggested $\mathrm{B}(2n, \frac{1}{2})$; $\frac{n^n}{n!}$ suggested $\mathrm{Poisson}(n)$.

What's really going on

How good is the approximation?

This argument is a heuristic: it uses the normal approximation *at a single point*, which the usual central limit theorem does not justify. A rigorous version, the *local central limit theorem*, does exactly that, and makes this a genuine proof. Stirling's formula is in fact remarkably accurate: the ratio $\frac{n!}{\sqrt{2\pi n}(n/e)^n}$ is about $1 + \frac{1}{12n}$, so even for $n = 1$ it gives 0.92 instead of 1, and for $n = 10$ it is 0.8% too small. The full series, $1 + \frac{1}{12n} + \frac{1}{288n^2} - \cdots$, is an example of an *asymptotic series*: it gives better and better approximations for large n, but for fixed n it eventually diverges.

Why factorials and normals are linked

The $\sqrt{2\pi}$ in Stirling's formula and in the normal density are the same constant, because $\int_{-\infty}^{\infty} e^{-x^2/2}\,\mathrm{d}x = \sqrt{2\pi}$. Abraham de Moivre found the normal approximation to the

binomial in 1733 in exactly this way, needing an approximation to $\binom{n}{k}$ for large n; he found the form $n! \approx Cn^{n+1/2}e^{-n}$ but not the constant. James Stirling showed that $C = \sqrt{2\pi}$. The normal distribution was born from the asymptotics of factorials.

De Moivre and Stirling

Abraham de Moivre, a French Protestant who fled to London and earned a living tutoring and advising gamblers, published *The Doctrine of Chances* in 1718 and found the normal approximation to the binomial in a pamphlet of 1733, the first appearance of the normal curve. James Stirling, a Scottish mathematician, published the value of the constant in his *Methodus Differentialis* (1730), and de Moivre generously credited him. The approximation of factorials is still an active subject: computing $n!$ for huge n is needed in combinatorics and cryptography.

The thinking skill

Use probability to prove something that is not about probability

An exact probability, compared with an approximation, can give information about the numbers in the formula.

This "probabilistic method" runs both ways. Here probability gave an estimate for factorials; Erdős used it to prove the existence of graphs (Section 9.2). At university you will see probability used to prove theorems in analysis, number theory and combinatorics.

Try this next

1. **Check the formula.** Use Stirling's formula to estimate 10! and 20!, and compare with the true values 3 628 800 and 2 432 902 008 176 640 000.
2. **From (iii) to (ii).** Show that Stirling's formula in part (iii) implies the approximation in part (ii).
3. **Digits of** 100!**.** Use $\log_{10} n! \approx n\log_{10}\frac{n}{e} + \frac{1}{2}\log_{10}(2\pi n)$ to show that 100! has 158 digits.
4. **A tie in coin tossing.** Use part (ii) to show that the probability of exactly 500 heads in 1000 tosses of a fair coin is about 0.025.

Further reading

► **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley),

Chapter II, which proves Stirling's formula.

- **Ronald Graham, Donald Knuth & Oren Patashnik, *Concrete Mathematics*** (Addison-Wesley), Chapter 9, on asymptotics.
- **Stephen Stigler, *The History of Statistics*** (Harvard University Press), on de Moivre and the birth of the normal curve.

12.24 The Central Limit Theorem

The question

Let X be a random variable with a Laplace distribution, so that its probability density function is given by

$$f(x) = \tfrac{1}{2}e^{-|x|}, \qquad -\infty < x < \infty. \qquad (*)$$

Sketch $f(x)$. Show that its moment generating function $M_X(\theta)$ is given by $M_X(\theta) = (1-\theta^2)^{-1}$ and hence find the variance of X.

A frog is jumping up and down, attempting to land on the same spot each time. In fact, in each of n successive jumps he always lands on a fixed straight line but when he lands from the ith jump ($i = 1, 2, \ldots, n$) his displacement from the point from which he jumped is X_i cm, where X_i has the distribution $(*)$. His displacement from his starting point after n jumps is Y cm (so that $Y = \sum_{i=1}^{n} X_i$). Each jump is independent of the others.

Obtain the moment generating function for $Y/\sqrt{2n}$ and, by considering its logarithm, show that this moment generating function tends to $\exp(\frac{1}{2}\theta^2)$ as $n \to \infty$.

Given that $\exp(\frac{1}{2}\theta^2)$ is the moment generating function of the standard Normal random variable, estimate the least number of jumps such that there is a 5% chance that the frog lands 25 cm or more from his starting point.

The hook

Add up many small independent random effects, whatever their individual distribution, and the total follows the bell-shaped normal curve. This is the *central limit theorem*, perhaps the most important theorem in probability. It explains why heights, measurement errors and exam marks are roughly normal, and why statisticians can say how reliable an average is without knowing anything about the underlying distribution, provided it has a finite variance (compare Section 12.22).

This question proves the theorem in a special case. A frog makes random jumps with a two-sided exponential distribution, whose density has a sharp peak, nothing like a bell. After many jumps, his total displacement is normal. The tool is the *moment generating function*, which turns sums of independent variables into products, and products into sums by taking logarithms.

A note on the syllabus

Moment generating functions were removed from the STEP specification in 2019. The moment generating function of a random variable X is $M_X(\theta) = E(e^{\theta X})$. Its derivatives at 0 give the moments: $M'(0) = E(X)$, $M''(0) = E(X^2)$. For independent X and Y, $M_{X+Y} = M_X M_Y$, and $M_{cX}(\theta) = M_X(c\theta)$. If moment generating functions converge, so (in a suitable sense) do the distributions.

Before you start

Ask yourself The density $\frac{1}{2}e^{-|x|}$ is symmetric. What does it look like? What is its mean?

It is a tent-shaped curve with a sharp peak of height $\frac{1}{2}$ at 0, decaying exponentially on both sides (Figure 12.24.1). By symmetry the mean is 0.

Working it through

STAGE 1 The moment generating function

For $|\theta| < 1$, splitting the integral at 0,

$$M_X(\theta) = \frac{1}{2}\int_{-\infty}^{0} e^{(1+\theta)x}\,dx + \frac{1}{2}\int_{0}^{\infty} e^{-(1-\theta)x}\,dx = \frac{1}{2}\left(\frac{1}{1+\theta} + \frac{1}{1-\theta}\right) = \frac{1}{1-\theta^2}. \qquad \blacksquare$$

Expanding, $M_X(\theta) = 1 + \theta^2 + \theta^4 + \cdots$, so $E(X) = 0$ and $E(X^2) = 2$ (the coefficient of $\frac{\theta^2}{2!}$). The variance is 2.

STAGE 2 The frog's displacement

Ask yourself What is the moment generating function of $Y = X_1 + \cdots + X_n$? Of $Y/\sqrt{2n}$?

By independence $M_Y(\theta) = (1-\theta^2)^{-n}$, and scaling,

$$M_{Y/\sqrt{2n}}(\theta) = M_Y\left(\frac{\theta}{\sqrt{2n}}\right) = \left(1 - \frac{\theta^2}{2n}\right)^{-n}.$$

(The scaling by $\sqrt{2n}$, the standard deviation of Y, makes the variance 1.)

STAGE 3 The limit

Ask yourself Take the logarithm and use $\ln(1-u) \approx -u$ for small u.

$\ln \mathrm{M}_{Y/\sqrt{2n}}(\theta) = -n\ln\bigl(1-\tfrac{\theta^2}{2n}\bigr) = -n\bigl(-\tfrac{\theta^2}{2n} - \tfrac{\theta^4}{8n^2} - \cdots\bigr) \to \tfrac{\theta^2}{2}$, so

$$\mathrm{M}_{Y/\sqrt{2n}}(\theta) \to e^{\theta^2/2}, \qquad \blacksquare$$

the moment generating function of the standard normal distribution.

STAGE 4 The frog's chances

Ask yourself For large n, Y is approximately $\mathrm{N}(0, 2n)$. When is $\mathrm{P}(|Y| \geq 25) = 0.05$?

For a standard normal Z, $\mathrm{P}(|Z| \geq 1.96) = 0.05$. So we need $\frac{25}{\sqrt{2n}} = 1.96$, giving $2n = \bigl(\frac{25}{1.96}\bigr)^2 \approx 162.7$ and $n \approx 81.3$. Since the chance of landing far away grows with n, the least number of jumps giving at least a 5% chance is about 82. $\blacksquare$

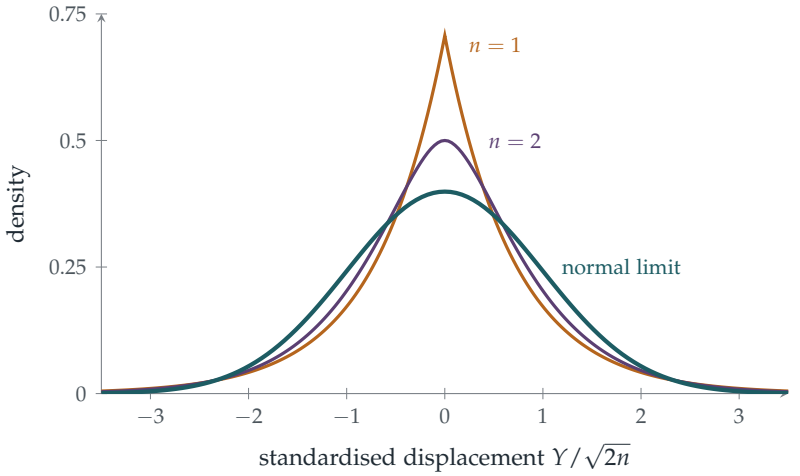


Figure 12.24.1 The density of the frog's standardised displacement after $n = 1$ jump (amber: a sharp Laplace peak) and $n = 2$ jumps (plum), compared with the standard normal density (teal). Already after two jumps the peak has rounded off; the central limit theorem says the curves converge to the normal as $n \to \infty$.

Looking back

- **Transforms turn sums into products.** The moment generating function of a sum of independent variables is the product of theirs.
- **Logarithms turn products into sums.** $\ln\bigl(1-\tfrac{\theta^2}{2n}\bigr)^{-n}$ became a sum whose limit was easy.
- **Standardise.** Dividing by the standard deviation $\sqrt{2n}$ kept the variance at 1, so the limit was neither 0 nor infinite.

What's really going on

The general theorem

The same argument works for any distribution with mean 0 and finite variance σ^2 (and a moment generating function, though this can be avoided). Then $\mathrm{M}(\theta) = 1 + \frac{1}{2}\sigma^2\theta^2 + o(\theta^2)$, and

$$\left(\mathrm{M}\left(\frac{\theta}{\sigma\sqrt{n}}\right)\right)^n = \left(1 + \frac{\theta^2}{2n} + o\left(\frac{1}{n}\right)\right)^n \to e^{\theta^2/2}.$$

Only the variance survives the limit: the shape of the original distribution is forgotten. That is why the normal distribution is universal. At university the theorem is proved with the *characteristic function* $\mathrm{E}(e^{i\theta X})$, which always exists, and the convergence of distributions is made precise by Lévy's continuity theorem.

How fast?

The central limit theorem says what happens in the limit, not how quickly. The *Berry–Esseen theorem* bounds the error by a constant times $\frac{\mathrm{E}|X|^3}{\sigma^3\sqrt{n}}$. For a symmetric distribution like the frog's, the approximation is good after a few dozen jumps. For skewed distributions, or in the extreme tails, it can be poor even for large n: the normal approximation underestimates the chance of very rare events, a fact with consequences for insurance and finance.

From de Moivre to Lindeberg

De Moivre found the normal approximation to the binomial in 1733 (Section 12.23), and Laplace extended it to sums of more general variables in 1810, coining the idea that errors of measurement are sums of many small effects. Rigorous proofs came much later: Aleksandr Lyapunov in 1901, and Jarl Lindeberg in 1922, who found essentially the weakest conditions under which the theorem holds. The name "central limit theorem" was given by George Pólya in 1920, meaning that the theorem is central to probability theory.

The thinking skill

Transform, simplify, transform back

A hard operation (adding independent random variables, which convolves their densities) becomes an easy one (multiplication) after a transform. Do the work there, and interpret the result.

The same strategy underlies generating functions (Section 9.4), Fourier and Laplace transforms in analysis and engineering, and logarithms, which turn

multiplication into addition.

Try this next

1. **Exact for two jumps.** Show that $Y = X_1 + X_2$ has density $\frac{1}{4}(1 + |y|)e^{-|y|}$, and check its variance is 4.

2. **Binomial to normal.** For $X \sim \mathrm{B}(1, \frac{1}{2})$ shifted to have mean 0 (taking values $\pm\frac{1}{2}$), find $\mathrm{M}_X(\theta) = \cosh\frac{\theta}{2}$, and show that the moment generating function of a standardised sum of n such variables tends to $e^{\theta^2/2}$.

3. **The frog, exactly.** Use a computer (or the exact density for $n = 2$ as a check) to see how accurate the normal approximation is for $n = 82$.

4. **Why not Cauchy?** The Cauchy distribution (Section 12.22) has no moment generating function. Which step of the argument fails, and what happens to the average of n Cauchy variables?

Further reading

- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press), the chapter on generating functions and limit theorems.
- **Hans Fischer, *A History of the Central Limit Theorem*** (Springer). From Laplace to Lindeberg.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican), on why the normal distribution is everywhere.

12.25 The Largest of Many

The question

(i) Let $\lambda > 0$. The independent random variables $X_1, X_2, \ldots, X_n$ all have probability density function

$$f(t) = \begin{cases} \lambda e^{-\lambda t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

and cumulative distribution function $F(x)$. The value of random variable Y is the largest of the values $X_1, X_2, \ldots, X_n$. Show that the cumulative distribution function of Y is given, for $y \geq 0$, by

$$G(y) = \left(1 - e^{-\lambda y}\right)^n.$$

(ii) The values $L(\alpha)$ and $U(\alpha)$, where $0 < \alpha \leq \frac{1}{2}$, are such that

$$P\big(Y < L(\alpha)\big) = \alpha \quad \text{and} \quad P\big(Y > U(\alpha)\big) = \alpha.$$

Show that

$$L(\alpha) = -\frac{1}{\lambda} \ln\left(1 - \alpha^{\frac{1}{n}}\right)$$

and write down a similar expression for $U(\alpha)$.

(iii) Use the approximation $e^t \approx 1 + t$, for $|t|$ small, to show that, for sufficiently large n,

$$\lambda L(\alpha) \approx \ln(n) - \ln\left(\ln \frac{1}{\alpha}\right).$$

(iv) Hence show that the median of Y tends to infinity as n increases, but that the width of the interval $U(\alpha) - L(\alpha)$ tends to a value which is independent of n.

(v) You are given that, for $|t|$ small, $\ln(1 + t) \approx t$ and that $e^3 \approx 20$. Show that, for sufficiently large n, there is an interval of width approximately $4\lambda^{-1}$ in which Y lies with probability 0.9.

The hook

How high will the highest flood of the next century be? How long will the longest-lived of a thousand light bulbs last? How fast will the fastest marathon runner in a field of n be? Questions about *extremes*, the largest or smallest of many random values,

matter enormously in engineering, insurance and climate science, and they behave very differently from averages.

This question finds the distribution of the largest of n exponential lifetimes. Its typical size grows like $\ln n$, slowly but without limit. Its spread, however, does not grow at all: the maximum of a thousand lifetimes is just as predictable, give or take, as the maximum of ten. In the limit, the shape of its distribution is the *Gumbel distribution*, the extreme-value counterpart of the normal curve.

Before you start

Ask yourself The largest of $X_1, \dots, X_n$ is at most y exactly when what happens?

$Y \le y$ exactly when every $X_i \le y$. Independence then turns this into a product.

Working it through

STAGE 1 The distribution of the maximum

Each X_i has $\mathrm{F}(y) = 1 - e^{-\lambda y}$ for $y \ge 0$, and by independence

$$\mathrm{G}(y) = \mathrm{P}(X_1 \le y, \dots, X_n \le y) = \left(1 - e^{-\lambda y}\right)^n. \qquad \blacksquare$$

STAGE 2 The quantiles

Ask yourself Solve $\mathrm{G}(L) = \alpha$ for L. Then solve $\mathrm{G}(U) = 1 - \alpha$.

$\left(1 - e^{-\lambda L}\right)^n = \alpha$ gives $e^{-\lambda L} = 1 - \alpha^{1/n}$, so

$$L(\alpha) = -\frac{1}{\lambda}\ln\left(1 - \alpha^{1/n}\right), \qquad U(\alpha) = -\frac{1}{\lambda}\ln\left(1 - (1-\alpha)^{1/n}\right). \qquad \blacksquare$$

STAGE 3 Large n

Ask yourself Write $\alpha^{1/n} = e^{(\ln \alpha)/n}$. For large n the exponent is small. What is $1 - \alpha^{1/n}$ approximately?

$\alpha^{1/n} = e^{(\ln \alpha)/n} \approx 1 + \frac{\ln \alpha}{n}$, so $1 - \alpha^{1/n} \approx \frac{-\ln \alpha}{n} = \frac{\ln(1/\alpha)}{n}$. Hence

$$\lambda L(\alpha) \approx -\ln \frac{\ln(1/\alpha)}{n} = \ln n - \ln\left(\ln \frac{1}{\alpha}\right). \qquad \blacksquare$$

Similarly, $\lambda U(\alpha) \approx \ln n - \ln\left(\ln \frac{1}{1-\alpha}\right)$.

STAGE 4 Growth and spread

The median is $L(\frac{1}{2})$, with $\lambda L(\frac{1}{2}) \approx \ln n - \ln\ln 2 \to \infty$. But

$$\lambda\big(U(\alpha) - L(\alpha)\big) \approx \ln\Big(\ln\frac{1}{\alpha}\Big) - \ln\Big(\ln\frac{1}{1-\alpha}\Big),$$

in which the $\ln n$ terms have cancelled: the width does not depend on n. ■

STAGE 5 A 90% interval

? **Ask yourself** Take $\alpha = 0.05$. What are $\ln\frac{1}{\alpha} = \ln 20$ and $\ln\frac{1}{0.95}$, approximately?

With $\alpha = 0.05$, $\ln 20 \approx 3$, so $\ln\ln\frac{1}{\alpha} \approx \ln 3$. And $\ln\frac{1}{0.95} = -\ln(1-0.05) \approx 0.05 = \frac{1}{20}$, so $\ln\ln\frac{1}{1-\alpha} \approx \ln\frac{1}{20} \approx -3$. The width is therefore about $\frac{1}{\lambda}(\ln 3 + 3) \approx \frac{4.1}{\lambda}$, and Y lies in $[L(0.05), U(0.05)]$, an interval of width about $4\lambda^{-1}$, with probability 0.9. ■

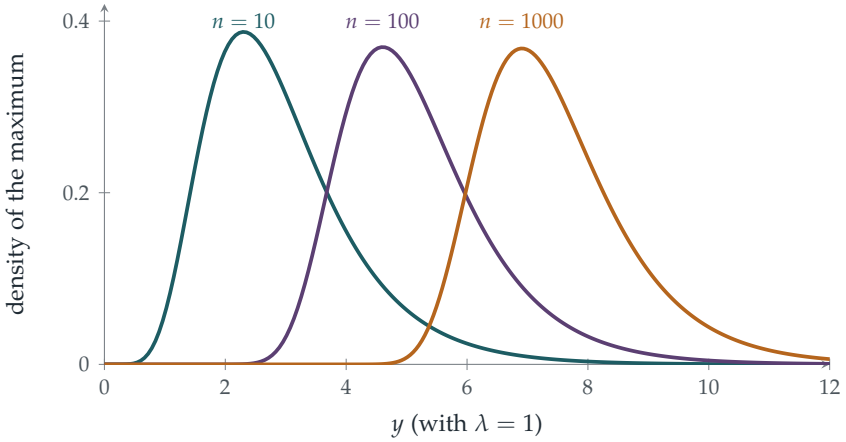


Figure 12.25.1 The density of the largest of n exponential lifetimes (mean 1) for $n = 10$, 100 and 1000. Each tenfold increase in n shifts the curve right by $\ln 10 \approx 2.3$, but its shape and width stay essentially the same: the Gumbel distribution, shifted by $\ln n$.

Looking back

- **The maximum is at most y when all are.** Independence made the distribution of the maximum a power.
- **Quantiles by inversion.** Solving $G(y) = \alpha$ gave exact formulas for the interval ends.
- **Approximate, then compare.** The approximations for L and U both contained $\ln n$, which cancelled in the width.

What's really going on

The Gumbel distribution

Since $\mathrm{P}(\lambda Y - \ln n \le x) = \left(1 - \frac{e^{-x}}{n}\right)^n \to \exp(-e^{-x})$, the shifted maximum $\lambda Y - \ln n$ has, in the limit, the *Gumbel distribution* with distribution function $\exp(-e^{-x})$. Its mean is Euler's constant $\gamma \approx 0.577$. You met it already: the time to collect all n coupons (Section 9.11) is the maximum of n (nearly independent) waiting times, which is why it is about $n \ln n + \gamma n$, with Gumbel fluctuations of order n.

Extreme value theory

A remarkable theorem of Fisher, Tippett and Gnedenko (1928–1943) says that, suitably shifted and scaled, the maximum of many independent identically distributed variables can have only three possible limiting shapes: Gumbel (for light tails, like the exponential or normal), Fréchet (for heavy tails, like the Cauchy) and Weibull (for variables with a hard upper limit). It is the extreme-value analogue of the central limit theorem (Section 12.24). Engineers use it to design flood defences and offshore platforms for the "hundred-year wave", insurers to price catastrophe cover, and climate scientists to estimate how heatwaves are changing.

Gumbel and the floods

Emil Julius Gumbel, a German statistician and an outspoken opponent of the Nazis, who was forced out of his university post in 1932 and eventually fled to the United States, applied extreme value theory to the flooding of rivers in the 1930s and 1940s. His book *Statistics of Extremes* (1958) made the subject a practical tool. The Dutch Delta Works, built after the North Sea flood of 1953 to protect the Netherlands, were designed using extreme value statistics to withstand a storm surge expected about once in 10 000 years.

The thinking skill

Distributions of maxima and minima

For the maximum of independent variables, $\mathrm{P}(\max \le y) = \prod \mathrm{P}(X_i \le y)$*; for the minimum,* $\mathrm{P}(\min > y) = \prod \mathrm{P}(X_i > y)$.

This one idea handles the nearest supermarket (Section 12.19), the first bus (Section 12.18) and the longest-lived bulb. At university it is the starting point of order statistics and extreme value theory.

Try this next

1. **The expected maximum.** Show that $\mathrm{E}(Y) = \frac{1}{\lambda}\left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right)$. (*Hint:* by memorylessness, after the first of the n lifetimes ends, the remaining $n-1$ start afresh.)

2. **The minimum.** Show that the smallest of $X_1, \ldots, X_n$ is exponential with parameter $n\lambda$.

3. **The median exactly.** Find the exact median of Y and compare it with the approximation $\frac{1}{\lambda}(\ln n - \ln \ln 2)$ for $n = 10$.

4. **Uniform maxima.** If $X_1, \ldots, X_n$ are uniform on $[0,1]$, find the distribution of their maximum M, and show that $n(1-M)$ is approximately exponential for large n. Which of the three extreme-value shapes is this?

Further reading

- **Stuart Coles, *An Introduction to Statistical Modeling of Extreme Values*** (Springer). The standard introduction, with real data on floods and sea levels.
- **Emil Julius Gumbel, *Statistics of Extremes*** (Dover). The classic.
- **David Spiegelhalter, *The Art of Uncertainty*** (Pelican), on rare events and how to think about them.

What We Have Learned

We began by saying that uncertainty can be reasoned about exactly, but intuition cannot. We asked three questions: what exactly is the question, what is the clever way in, and what happens in the long run? Here is what the chapter's twenty-five questions showed.

1. What exactly is the question?

The first four sections were about getting the question right. "Given that a numbered disc is taken" and "given that the disc numbered 3 is taken" gave different answers (Section 12.1). A 90% accurate test for a rare disease left a positive patient only 9% likely to be ill, because base rates matter (Section 12.2). "A random chord" had no expected length until we said how it was chosen (Section 12.3), and the broken stick (Section 12.21) made the same point. And "independent" turned out to have several meanings: events can be independent in pairs but not all together (Section 12.4), and uncorrelated variables need not be independent. In every case, the cure was to write down exactly which outcomes are possible and how they arise.

2. What is the clever way in?

Many questions had a long calculation and a short insight.

- **Symmetry.** Two seats that the process cannot tell apart (Section 12.5); "more heads" and "more tails" as complementary mirror images (Section 12.6); every pair of players equally likely to contest any match (Section 12.7); HT and TH equally likely for any coin (Section 12.9); a radially symmetric point whose direction is uniform (Section 12.22).
- **Indicators and linearity.** Counting pairs (Section 12.4), knots (Section 12.10) and matches (Section 12.7) as sums of indicators gave means and variances without the full distribution.
- **States and first-step analysis.** Penney's game (Section 12.8), the rabbit warren and Professor Ecks (Section 12.16), and the extinction of family names (Section 12.15) were all solved by conditioning on the next step.
- **Strategy.** The worst shot in the truel should miss on purpose (Section 12.13); a randomised strategy cannot be exploited (Section 12.14); rejecting the first $\frac{1}{e}$ of the candidates is optimal (Section 12.12).

3. What happens in the long run?

The last parts of the chapter followed randomness over time and across many trials. A branching process either dies out or explodes, and even an exploding one can die by chance (Section 12.15). A Markov chain settles into a stationary distribution and forgets its past (Section 12.16). Exponential waiting times have no memory at all (Section 12.18). Averages usually settle down, but not for the Cauchy distribution, which has no mean (Section 12.22). Sums of many independent effects become normal, by the central limit theorem (Section 12.24), and that normality hides Stirling's formula for $n!$ (Section 12.23). Maxima behave differently again: they grow like $\ln n$ with a spread that does not grow at all (Section 12.25). And the constants e, π and γ kept appearing, in cups and stamps, needles and loops.

4. A toolkit for probability

Method	Where you used it
Write down the sample space	Conditioning (12.1); Bertrand (12.3)
Bayes' theorem and odds	Base rates (12.2); Markov reversal (12.16)
Symmetry	12.5, 12.6, 12.7, 12.22
Indicators and linearity	12.4, 12.7, 12.10
First-step analysis	Penney (12.8); warren (12.16); branching (12.15)
Complement and bounds	Fair decisions (12.9)
Series for expectations	Stamps and e (12.11)
Parametrise and optimise	Secretary (12.12)
Game-theoretic reasoning	Truel (12.13); mixed strategies (12.14)
Counting paths to the end	Matchboxes (12.17)
Distribution function first	12.18, 12.19, 12.25
Probability as area or integral	Buffon (12.20); broken stick (12.21)
Normal approximation	Stirling (12.23); CLT (12.24)

Where next

- Chapter 13 turns from probability to statistics: estimating unknown quantities from data, starting with Section 13.1.
- The counting problems of Section 9.8, Section 9.9 and Section 9.11 are the combinatorial side of this chapter.
- Markov chains use the linear algebra of Chapter 7, and the fixed points of branching processes the dynamics of Section 8.7.

If you want to go further before university

- **Joseph Blitzstein & Jessica Hwang, *Introduction to Probability***: a superb first course, free online, with video lectures.
- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction***: concise and

rigorous, at first-year level.

- **Frederick Mosteller, *Fifty Challenging Problems in Probability***: classic puzzles with full solutions.
- **David Spiegelhalter, *The Art of Statistics*** and ***The Art of Uncertainty***: probability and risk in the real world.

Laplace called probability common sense reduced to calculation. This chapter has shown that the calculation is often needed precisely because common sense fails, and that the best calculations do more than produce numbers: they explain. A symmetry, an indicator, a well-chosen state or a limit theorem turns a puzzle into an insight, and those insights are what a university course in probability is built on.

Answers to Try This Next

These are brief answers and hints, not full solutions: they give the result and the key step, so that you can check your own work. Try each exercise properly before looking.

12.1 Precise Conditioning

1. In both (iv) and (v), the other $k-1$ discs are a random selection from the remaining $b+n-1$, of which $n-1$ are numbered, so both equal $\frac{(n-1)\binom{b}{k-2}}{\binom{b+n-1}{k-1}}$. Answer (vi) is $\frac{\binom{n}{2}\binom{b}{k-2}}{\binom{b+n}{k}-\binom{b}{k}}$. With $N=b+n$, the two agree only when $(2N-nk)\binom{N-1}{k-1}=2k\binom{b}{k}$. That happens only in odd cases, such as $(b,n,k)=(2,4,3)$, $(3,3,4)$ or $(6,11,3)$, or trivially when $n=2$ and every disc is drawn. It does not hold for the question's values.

2. Of the $14^2=196$ equally likely (sex, day) pairs for the two children, $196-13^2=27$ include at least one Tuesday boy. Of those, $7^2-6^2=13$ have two boys. So the probability is $\frac{13}{27}$.

3. You pick door 1. If the car is behind door 1 (probability $\frac{1}{3}$), switching loses; if it is behind door 2 or 3, the host must open the other goat door, and switching wins: $\frac{2}{3}$. If the host opens a door at random and it happens to show a goat, the outcomes "car behind your door" and "car behind the other closed door" are equally likely given what you saw: $\frac{1}{2}$.

4. $\mathrm{P}(\geq 2 \mid \geq 1)=\frac{1-\mathrm{P}(0)-\mathrm{P}(1)}{1-\mathrm{P}(0)}\approx 0.370$, but $\mathrm{P}(\geq 2 \mid \text{ace of spades})=1-\binom{48}{12}/\binom{51}{12}\approx 0.561$. Knowing *which* ace makes a second ace more likely.

12.2 The Base-Rate Fallacy

1. Both say "body" with probability pq if it is there and $(1-p)(1-q)$ if not. With prior odds $1:1$, the posterior is $\frac{pq}{pq+(1-p)(1-q)}=\frac{pq}{1-p-q+2pq}$. For the second part, "$A$ is there" is true in two of the three equally likely cases, so $\mathrm{P}(\text{both} \mid \text{both say } A)=\frac{pq}{2pq+(1-p)(1-q)}$.

2. We need $0.9\pi>\frac{1}{11}(1-\pi)$, that is, $\pi>\frac{10}{109}\approx 9.2\%$.

3. Each positive AL result multiplies the odds by $\frac{0.9}{1/11}=9.9$. From prior odds $\frac{1}{99}$: after one positive, odds 0.1 (probability $\frac{1}{11}$); after two, odds 0.99 (probability about 0.497); after three, odds 9.8 (probability about 0.907).

4. Of the 100 people with the condition, 99 test positive; of the 999 900 without it, about 9999 do. So about 10 100 people are told they may have it, and only 99 of

them (about 1%) actually do.

12.3 What Does "Random" Mean?

1. The chord is longer than the side $\sqrt{3}a$ of the inscribed equilateral triangle when: its far end lies in the third of the circle opposite the near end ($\frac{1}{3}$); its distance from O is less than $\frac{a}{2}$ ($\frac{1}{2}$); its midpoint lies inside the circle of radius $\frac{a}{2}$ ($\frac{1}{4}$).
2. The chord with midpoint C is perpendicular to OC, so it is the chord through C at angle $t = \frac{\pi}{2}$, and $L(\frac{\pi}{2}) = \frac{4a}{3}\big(0 + \frac{1}{2}\sec^2\frac{\pi}{4}\big) = \frac{4a}{3}$.
3. The length is $2\sqrt{a^2 - d^2}$, with mean $\frac{2}{a}\int_0^a \sqrt{a^2 - d^2}\,\mathrm{d}d = \frac{2}{a}\cdot\frac{\pi a^2}{4} = \frac{\pi a}{2}$.
4. Random endpoints: $\frac{2}{\pi\sqrt{4a^2-\ell^2}}$. Random distance: $\frac{\ell}{2a\sqrt{4a^2-\ell^2}}$. Random midpoint: $\frac{\ell}{2a^2}$.

12.4 Independent in Pairs, Not All Together

1. Expanding, $\mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y) = (a - pq)(x_1 - x_2)(y_1 - y_2)$, so $a = pq$; the other three cells then follow, giving independence. Example: put probability $\frac{1}{3}$ on $(0,0)$ and $\frac{1}{6}$ on each of $(\pm1, \pm1)$. Both marginals are uniform and $\mathrm{E}(AB) = 0$, but $\mathrm{P}(A = 0, B = 1) = 0 \neq \frac{1}{9}$.
2. There are 21 pairs, each matching with probability $\frac{1}{6}$, and the indicators are pairwise independent. So the mean is 3.5, the variance is $21 \cdot \frac{1}{6} \cdot \frac{5}{6} \approx 2.92$, and the standard deviation is about 1.71. A score of 7 is about two standard deviations above the mean.
3. The mean is $\binom{k}{2}\frac{1}{365}$. The indicators are pairwise independent, exactly as for the dice, so the variance is $\binom{k}{2}\frac{1}{365}\cdot\frac{364}{365}$. (They are not mutually independent: if 1, 2 and 2, 3 match, so do 1, 3.)
4. Take eight equally likely outcomes, $A = B = \{1,2,3,4\}$ and $C = \{1,5,6,7\}$. Then $\mathrm{P}(A \cap B \cap C) = \frac{1}{8}$, but $\mathrm{P}(A \cap B) = \frac{1}{2} \neq \frac{1}{4}$.

12.5 The Last Passenger's Seat

1. $\frac{1}{3}$, for every $n \geq 3$. The last seat is now one of S_1, S_2 and S_n, and every random choice treats these three symmetrically.
2. If T_1 takes S_1, nobody is displaced. If T_1 takes S_3, T_1 and T_3 are. If T_1 takes S_2, then T_2 chooses: S_1 gives 2 displaced, S_3 gives 3. The expectation is $\frac{1}{3}\big(0 + 2 + \frac{5}{2}\big) = \frac{3}{2}$, which agrees with $\frac{2}{3} + \frac{1}{3} + \frac{1}{2}$ from the formula.
3. With $P_r = \frac{1}{2}$ for $r \geq 2$, $\frac{1}{n}\big(1 + \frac{n-2}{2}\big) = \frac{1}{2}$, so the recurrence is consistent with $P_n = \frac{1}{2}$.
4. The last free seat is always S_1 or S_n (any other S_k is taken by the time T_k boards, if not before). So if it is not S_n, it is S_1.

12.6 One More Coin

1. Let H' be Bob's first $n+1$ tosses and X his last. Bob wins if $H' > H_A$, or if $H' = H_A$ and X is a head. So the probability is $\mathrm{P}(H' > H_A) + \frac{1}{2}\mathrm{P}(H' = H_A)$, and $\mathrm{P}(H' > H_A) = \frac{1}{2}$ by the one-extra-coin result.
2. No. With $n = 0$, Bob wins with probability p. The slick proof swaps heads and tails, which is only a symmetry for a fair coin.
3. $H_A = H_B$ exactly when $H_A + (\text{Bob's tails}) = n$, which is a $\mathrm{B}(2n, \frac{1}{2})$ variable equal to n. Values: $\frac{1}{2}$, $\frac{3}{8}$, $\frac{5}{16}$.
4. No: a six is like a coin with $p = \frac{1}{6}$. For $n = 0$ the probability is $\frac{1}{6}$, and in general it is below $\frac{1}{2}$.

12.7 Who Meets Whom in a Knockout

1. Round k has 2^{n-k} matches, and by symmetry each is equally likely to involve any of the $\binom{2^n}{2}$ pairs. Adding over the $2^n - 1$ matches in the whole tournament gives $\frac{2^n-1}{\binom{2^n}{2}} = \frac{2}{2^n}$.
2. No: random placement in a fixed bracket keeps the symmetry, and the same counting applies.
3. X's round-k opponent comes from a group of 2^{k-1} players, each equally likely to win it. So Y is that opponent with probability $\frac{2^{k-1}}{2^n-1} \cdot \frac{1}{2^{k-1}} = \frac{1}{2^n-1}$, and the total over n rounds is $\frac{n}{2^n-1}$.
4. $\binom{2n}{2} = n(2n-1)$ matches; every pair meets, so the probability is 1.

12.8 Penney's Game

1. HHT wins only if the first two tosses are HH; otherwise a T comes before HH is completed, and THH must appear first. So HHT wins with probability $\frac{1}{4}$. With all four playing: after HH, HHT must win; after TT, TTH must win; from HT and TH, the chances for HTT and THH are $\frac{2}{3}$ and $\frac{1}{3}$ (or vice versa), giving each player $\frac{1}{4} \cdot 1$ or $\frac{1}{4}\left(\frac{2}{3} + \frac{1}{3}\right)$, that is, $\frac{1}{4}$. TTH beats THH with probability $\frac{2}{3}$ (part (i)).
2. For HH: $e_0 = 1 + \frac{1}{2}e_{\mathrm{H}} + \frac{1}{2}e_0$ and $e_{\mathrm{H}} = 1 + \frac{1}{2}e_0$, giving $e_0 = 6$. For HT: once you have an H, you wait for a T (two tosses on average), and the first H takes two tosses: 4.
3. HHH and TTT are beaten by THH and HTT with probability $\frac{7}{8}$; HHT and TTH by THH and HTT with $\frac{3}{4}$; the other four (HTH, HTT, THH, THT) by HHT, HHT, TTH, TTH with $\frac{2}{3}$. HHH and TTT are beaten most easily.
4. With $A = \mathrm{HTT}$ and $B = \mathrm{HHT}$, the correlations are $AA = 4$, $AB = 0$, $BB = 4$, $BA = 2$. The odds that B wins are $(AA - AB) : (BB - BA) = 4 : 2$, that is, probability $\frac{2}{3}$.

12.9 A Fair Decision from a Biased Coin

1. The number of rounds is geometric: mean $\frac{1}{p^2+q^2}$ for part (i) and $\frac{1}{2pq}$ for von Neumann's procedure (HT or TH). Since $p^2+q^2-2pq=(p-q)^2$, von Neumann's is slower for every $p \neq \frac{1}{2}$. It is the fair one, though.
2. Use von Neumann's trick to make fair bits. Take two at a time: 00, 01 and 10 choose the three options, and 11 means start again.
3. Compare the tosses, one by one, with the binary expansion $\frac{1}{3} = 0.010101\ldots$ (reading H as 1 and T as 0). At the first toss that differs, declare "heads" if the toss is *below* the digit (T against a 1) and "tails" otherwise. This gives heads with probability exactly $\frac{1}{3}$, using 2 tosses on average.
4. After an undecided round there are two of one face and one of the other, and the process is a Markov chain on these two states. Solving it gives $\mathrm{P}(\text{hotel}) = \frac{p^3}{p^3+q^3}$, which is $\frac{1}{2}$ when $p=\frac{1}{2}$: fair then, but not otherwise.

12.10 Tying Loose Ends

1. The first tie makes a ring with probability $\frac{1}{3}$ (one of the three other ends is from the same string), and then the last tie always makes a ring. Rings: 2 with probability $\frac{1}{3}$, 1 with probability $\frac{2}{3}$; the mean is $\frac{4}{3}$.
2. With k strings left, a tie avoids making a ring with probability $1-\frac{1}{2k-1}$. The product $\prod_{k=2}^{n} \frac{2k-2}{2k-1}$ equals $\frac{4^n}{2n\binom{2n}{n}}$, which is about $\frac{\sqrt{\pi n}}{2n} = \sqrt{\frac{\pi}{4n}}$.
3. Insert the elements $1,2,\ldots,n$ in turn: element k either starts a new cycle (probability $\frac{1}{k}$) or is placed after one of the $k-1$ earlier elements. Adding the indicators gives H_n.
4. The ring indicators for the successive ties are independent with probabilities $\frac{1}{2k-1}$, so the variance is $\sum \frac{1}{2k-1}\left(1-\frac{1}{2k-1}\right)$. And $\sum_{k\geq 1} \frac{1}{(2k-1)^2} = \frac{\pi^2}{6} - \frac{1}{4}\cdot\frac{\pi^2}{6} = \frac{\pi^2}{8}$.

12.11 Stamps, Cups and e

1. $\sum_{k\geq 2} \frac{k-1}{k!}t^k = t\mathrm{e}^t - (\mathrm{e}^t-1) = 1+(t-1)\mathrm{e}^t$. Differentiating, $\mathrm{E}(N) = \mathrm{G}'(1) = \mathrm{e}$ and $\mathrm{E}(N(N-1)) = \mathrm{G}''(1) = 2\mathrm{e}$, so $\mathrm{Var}(N) = 3\mathrm{e} - \mathrm{e}^2$.
2. $\mathrm{P}(N=j)$ appears in $\mathrm{P}(N>k)$ for $k=0,\ldots,j-1$, that is, j times. Here $\mathrm{P}(N>k) = \frac{1}{k!}$, so $\mathrm{E}(N) = \sum \frac{1}{k!} = \mathrm{e}$.
3. It does not change at all: Jim is certain to "guess" correctly with one cup. The expected price is still e.
4. $\int_0^1 \frac{(1-u)^2}{2}\,\mathrm{d}u = \frac{1}{6}$. In general the slice at $u_k = u$ is a $(k-1)$-dimensional simplex scaled by $1-u$, so $V_k = V_{k-1}\int_0^1 (1-u)^{k-1}\,\mathrm{d}u = \frac{V_{k-1}}{k}$.

12.12 When to Stop Looking

1. If the biggest is in position $k \geq 2$, you get it exactly when the first ice-cream is the biggest of the first $k-1$, with probability $\frac{1}{k-1}$. Averaging over k gives $\frac{1}{n}\sum_{k=2}^{n}\frac{1}{k-1}$, which is $\frac{11}{24}$ for $n=4$. By listing the 24 orders, the second, third and fourth biggest come with probabilities $\frac{7}{24}$, $\frac{1}{6}$ and $\frac{1}{12}$.
2. Rejecting $r = 0, 1, 2, 3$ gives $\frac{1}{4}$, $\frac{11}{24}$, $\frac{5}{12}$, $\frac{1}{4}$. So reject one candidate, and succeed with probability $\frac{11}{24}$.
3. The probability that the best of the first n is in the top two is $\frac{n}{N} + \frac{n}{N}\cdot\frac{N-n}{N-1}$, nearly double $\frac{n}{N}$ when n is small. If one of the top two will do, you can afford to look at fewer candidates before committing. (The best strategy for this goal succeeds with probability about 0.57.)
4. With $\mathrm{P}(r) = \frac{r}{N}\sum_{k=r}^{N-1}\frac{1}{k}$: for $N=10$, $r=2,3,4$ give 0.366, 0.399, 0.398; for $N=100$, $r=36,37,38$ give 0.3710, 0.3710, 0.3708, with $r=37$ just ahead.

12.13 The Truel

1. With Algernon missing, Bertie throws at Cuthbert. If he hits ($\frac{3}{5}$), it is Algernon against Bertie with Algernon throwing first, and Bertie wins with probability $1 - \frac{2/5}{2/5+3/5-6/25} = \frac{9}{19}$. If he misses, Cuthbert hits Bertie, and Cuthbert then wins unless Algernon hits first ($\frac{3}{5}$). So Bertie wins with probability $\frac{3}{5}\cdot\frac{9}{19} = \frac{135}{475}$ and Cuthbert with $\frac{2}{5}\cdot\frac{3}{5} = \frac{114}{475}$.
2. For every $a < 1$. Missing gives $\frac{3}{5}\cdot\frac{a}{a+\frac{3}{5}-\frac{3}{5}a} + \frac{2}{5}a$; hitting Cuthbert gives $1 - \frac{3/5}{3/5+a-\frac{3}{5}a}$ (Bertie then throws first at him), which is always smaller, and hitting Bertie gives 0.
3. No. If Bertie throws at Algernon and hits, Cuthbert then hits Bertie; if he misses, Cuthbert hits Bertie anyway. Either way Bertie gets nothing, against $\frac{27}{95}$ by throwing at Cuthbert.
4. $p + (1-p)(1-r)p$, which is less than $\frac{p}{p+r-pr}$, the sum of the full geometric series.

12.14 Mixed Strategies

1. With $b = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$, $X = Y = Z = 0$, so B wins 0 against everything. Against $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, A's expected winnings are $-\frac{1}{3}$, 0 and $\frac{1}{3}$ for guesses 1, 2 and 3, so A should always guess 3.
2. The payoff matrix has 0 on the diagonal and ± 1 elsewhere, unchanged when the three strategies are permuted cyclically, so the optimal strategy is $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. In the question the penalties for differences of 1 and 2 differ, so the strategies are not interchangeable.
3. If $X, Y, Z \geq 0$ and $X + 2Y + Z = 0$, then all three are 0. This forces $b_1 = b_3$ and $b_2 = 2b_3$.

4. Now $X = -b_2 + 3b_3$, $Y = b_1 - b_3$ and $Z = -3b_1 + b_2$, with $X + 3Y + Z = 0$. Setting all three to zero gives $(\frac{1}{5}, \frac{3}{5}, \frac{1}{5})$.

12.15 Extinction or Explosion

1. The extinction probability is the smallest root in $[0, 1]$ of $s = (q + ps)^2$. The roots are 1 and $\left(\frac{q}{p}\right)^2$, and the second is smaller than 1 exactly when $p > \frac{1}{2}$.
2. Given X_{n-1}, X_n has mean $2pX_{n-1}$ and variance $2pqX_{n-1}$, so $\mathrm{Var}(X_n) = \mathrm{E}(2pqX_{n-1}) + (2p)^2\mathrm{Var}(X_{n-1})$. For $p = \frac{1}{2}$: $\mathrm{Var}(X_n) = \mathrm{Var}(X_{n-1}) + \frac{1}{2}$, so $\mathrm{Var}(X_n) = \frac{n}{2}$.
3. Extinction by generation n means each first-generation particle's line dies within $n - 1$ generations. With $p = \frac{3}{4}$: $e_1 = 0.0625$, $e_2 \approx 0.0881$, $e_3 \approx 0.0999$, increasing towards $\frac{1}{9} \approx 0.111$.
4. $s = \frac{1}{4}(1 + s + s^2 + s^3)$ factorises as $(s - 1)(s^2 + 2s - 1) = 0$, so the extinction probability is $\sqrt{2} - 1 \approx 0.414$.

12.16 Markov Chains Forwards and Backwards

1. By symmetry $p_B = p_D$; also $p_A = \frac{1}{3} + \frac{2}{3}p_B$, $p_B = \frac{1}{3}p_A + \frac{1}{3}p_C$ and $p_C = \frac{2}{3}p_B$. So $p_A = \frac{7}{15}$. Leaving at A without visiting C, starting from A: $x_A = \frac{1}{3} + \frac{2}{3}x_B$ and $x_B = \frac{1}{3}x_A$, so $x_A = \frac{3}{7}$. The conditional probability is $\frac{3/7}{7/15} = \frac{45}{49}$.
2. Multiply the first row by P: $(0.19, 0.33, 0.48)$.
3. The trace is 1.3 and the determinant is 0.02, and 1 is an eigenvalue (rows sum to 1). The other two have sum 0.3 and product 0.02: they are 0.2 and 0.1.
4. $a_1 p_{12} = \frac{1}{20}$ but $a_2 p_{21} = \frac{1}{15}$, so the chain is not reversible: run backwards, it has different transition probabilities (the q_{ij} of part (iii) differ from the p_{ij}).

12.17 Banach's Matchboxes

1. If the second reach is for the same kind as the first, she discovers it empty then ($N = 2$); otherwise she takes the other sweet and discovers an empty packet at the third reach. So $N = 2$ or 3, each with probability $\frac{1}{2}$.
2. $n = 1$: $\frac{1}{2} + \frac{1}{2} = 1$. $n = 2$: $\frac{6}{16} + \frac{6}{16} + \frac{4}{16} = 1$.
3. $\binom{2n-1}{n}2^{-(2n-1)} = \frac{1}{2}\binom{2n}{n} \cdot 2 \cdot 4^{-n} = \binom{2n}{n}4^{-n}$.
4. A first-step recursion on the numbers of toffees and mints left gives $\frac{64}{81} \approx 0.79$.

12.18 Memorylessness

1. $\mathrm{P}(\min > t) = \mathrm{e}^{-\lambda_1 t}\mathrm{e}^{-\lambda_2 t}$. The wait for the next bus is exponential with rate $\frac{1}{15} + \frac{1}{30} = \frac{1}{10}$ per minute, so its median is $10 \ln 2 \approx 6.9$ minutes.

2. $\mathrm{P}(Y \geq k) = (1-p)^k$, so the ratio is $(1-p)^n$.
3. The gap containing t consists of the time back to the last bus and the time forward to the next, both exponential with mean $\frac{1}{\lambda}$ (for large t), so its expected length is $\frac{2}{\lambda}$. Long gaps are more likely to contain a given time.
4. $\frac{\lambda \mathrm{e}^{-\lambda z}}{1-\mathrm{e}^{-\lambda}}$ on $[0,1]$. For $\lambda = 1$ it falls gently from about 1.58 to 0.58; for $\lambda = 5$ it is concentrated near 0, falling from about 5.03 to 0.03.

12.19 Random Points in the Plane

1. $1 - \mathrm{e}^{-\pi k y^2} = \frac{1}{2}$ gives a median of $\sqrt{\frac{\ln 2}{\pi k}} \approx \frac{0.470}{\sqrt{k}}$, a little less than the mean $\frac{1}{2\sqrt{k}}$.
2. $\mathrm{P}(Y_2 > y) = \mathrm{e}^{-\pi k y^2}(1 + \pi k y^2)$, so the density is $2\pi^2 k^2 y^3 \mathrm{e}^{-\pi k y^2}$.
3. In polar coordinates the density of R is $\frac{1}{2\pi\sigma^2}\mathrm{e}^{-r^2/(2\sigma^2)} \cdot 2\pi r$. It matches the supermarket distribution when $\sigma^2 = \frac{1}{2\pi k}$.
4. With $k = 4 \times 10^{-4}$ per m^2, complete randomness predicts a mean of $\frac{1}{2\sqrt{k}} = 25\,\mathrm{m}$. The observed 35 m is larger, which suggests the trees are spaced out (competition), not clustered.

12.20 Buffon's Needle

1. Cut the needle into short pieces. The expected number of crossings is additive, and each short piece contributes in proportion to its length, so the total is $\frac{2a}{\pi b}$ whatever a is.
2. By the same additivity (Buffon's noodle), still $\frac{2a}{\pi b}$.
3. The centre must be at least r from every edge: $\frac{(s-2r)^2}{s^2}$.
4. $\frac{1808}{3408} \approx \frac{5}{3\pi}$ gives $\pi \approx \frac{5\times 3408}{3\times 1808} = \frac{17\,040}{5424} = \frac{355}{113}$: suspiciously perfect.

12.21 Breaking a Stick

1. A triangle needs every piece shorter than $\frac{1}{2}$. In the unit square of break points, this cuts out a region of area $\frac{1}{4}$.
2. Suppose the first break is at $x < \frac{1}{2}$ (the other case is symmetric) and the longer piece $1-x$ is broken uniformly. Both new pieces must be shorter than $\frac{1}{2}$, which has probability $\frac{x}{1-x}$. So the answer is $2\int_0^{1/2} \frac{x}{1-x}\,\mathrm{d}x = 2\ln 2 - 1$.
3. The angle at C is obtuse when $(2-x-y)^2 > x^2 + y^2$, that is, $y < \frac{2-2x}{2-x}$. The area between this curve and $x + y = 1$ is $\frac{3}{2} - 2\ln 2$, so the probability is $3 - 4\ln 2$, the same as at B. This is no accident: the three sides are each at most 1 with sum 2, uniformly, so they are interchangeable. So $\mathrm{P}(\text{obtuse}) = 9 - 12\ln 2 \approx 0.68$.

4. By inclusion–exclusion, the longest piece M has $\mathrm{P}(M > m) = 3(1-m)^2 - 3(1-2m)_+^2 + (1-3m)_+^2$, where $x_+ = \max(x,0)$. Integrating over $[0,1]$ gives $\mathrm{E}(M) = 1 - \frac12 + \frac19 = \frac{11}{18}$.

12.22 A Distribution With No Mean

1. $\mathrm{P}(\tan\Theta \le k) = \frac1\pi\big(\tan^{-1}k + \frac\pi2\big)$, and differentiating gives $\frac{1}{\pi(1+k^2)}$.
2. Swapping U and V preserves a radially symmetric density, so $\frac VU$ has the same distribution as $\frac UV$.
3. The density is symmetric, so the median is 0; and $\mathrm{P}(C \le 1) = \frac12 + \frac1\pi\tan^{-1}1 = \frac34$.
4. $\frac{1}{2\pi}\mathrm{e}^{-r^2/2}\,r\,\mathrm{d}r\,\mathrm{d}\theta$, and with $s = r^2$ this is $\frac{1}{2\pi}\,\mathrm{d}\theta \cdot \frac12\mathrm{e}^{-s/2}\,\mathrm{d}s$: the angle is uniform, R^2 is exponential with mean 2, and they are independent.

12.23 Stirling from Probability

1. $\sqrt{20\pi}(10/\mathrm{e})^{10} \approx 3\,598\,696$ (about 0.8% low) and $\sqrt{40\pi}(20/\mathrm{e})^{20} \approx 2.4228\times10^{18}$ (about 0.4% low).
2. $\frac{(2n)!}{(n!)^2} \approx \frac{\sqrt{4\pi n}(2n/\mathrm{e})^{2n}}{2\pi n(n/\mathrm{e})^{2n}} = \frac{4^n}{\sqrt{\pi n}}$.
3. $100\log_{10}\frac{100}{\mathrm{e}} + \frac12\log_{10}(200\pi) \approx 157.97$, so $100!$ has 158 digits.
4. $\binom{1000}{500}2^{-1000} \approx \frac{1}{\sqrt{500\pi}} \approx 0.0252$.

12.24 The Central Limit Theorem

1. Convolving, $\frac14\int \mathrm{e}^{-|x|-|y-x|}\,\mathrm{d}x = \frac14(1+|y|)\mathrm{e}^{-|y|}$. The variance is $2+2=4$.
2. $\mathrm{E}(\mathrm{e}^{\theta X}) = \frac12(\mathrm{e}^{\theta/2} + \mathrm{e}^{-\theta/2}) = \cosh\frac\theta2$. The standardised sum $\frac{2}{\sqrt n}\sum X_i$ has generating function $\cosh^n\frac{\theta}{\sqrt n}$, whose logarithm is $n\ln\big(1 + \frac{\theta^2}{2n} + \cdots\big) \to \frac{\theta^2}{2}$.
3. Writing Y as a difference of two independent $\Gamma(82,1)$ variables and integrating numerically gives $\mathrm{P}(|Y| \ge 25) \approx 0.051$: the normal approximation is very good.
4. There is no moment generating function, and not even a mean. In fact the average of n independent Cauchy variables is itself Cauchy: averaging does not concentrate it at all.

12.25 The Largest of Many

1. The first of n lifetimes ends after a mean time $\frac{1}{n\lambda}$; then $n-1$ remain, starting afresh, and so on. So $\mathrm{E}(Y) = \frac1\lambda\big(\frac1n + \frac{1}{n-1} + \cdots + 1\big)$.
2. $\mathrm{P}(\min > t) = (\mathrm{e}^{-\lambda t})^n = \mathrm{e}^{-n\lambda t}$.

3. $(1 - \mathrm{e}^{-\lambda y})^n = \frac{1}{2}$ gives $\lambda y = -\ln(1 - 2^{-1/n})$. For $n = 10$ this is 2.704, against the approximation $\ln 10 - \ln\ln 2 \approx 2.669$.

4. $\mathrm{P}(M \le m) = m^n$, so $\mathrm{P}\big(n(1 - M) > s\big) = \big(1 - \frac{s}{n}\big)^n \to \mathrm{e}^{-s}$. This is the Weibull type: the variables are bounded above, and the maximum piles up against the bound.

CHAPTER

13

Statistics and Randomness in the World

Probability goes from a model to data. Statistics goes from data back to the world.

IN THIS CHAPTER

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13.2	Counting Voles	STEP II 2000
13.3	The German Tank Problem	STEP III 1998
13.4	Mean or Median?	STEP II 2000
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Learning from Data

All models are wrong, but some are useful.

— George Box, statistician

In probability, we know how the dice are loaded and ask what will happen. In statistics, we see what happened and ask how the dice must be loaded. Probability reasons from causes to effects; statistics reasons back from effects to causes. Every estimate of a population, every opinion poll, every clinical trial and every scientific measurement is a statistical inference: a conclusion about the world drawn from limited, noisy data.

Statistics is a young subject compared with most of mathematics. Its modern form was created in the early twentieth century by Karl Pearson, Ronald Fisher, Jerzy Neyman and others, often driven by practical problems: brewing beer (the t-test was invented at the Guinness brewery), growing crops, testing medicines. Today, with the explosion of data, statistics and its close relative machine learning are among the most important applications of mathematics.

Why statistics matters

Decisions about medicines, public health, economic policy, climate and justice all rest on statistical evidence. Misreading it has real consequences: a patient frightened by a false positive (Section 12.2), a policy based on a chance fluctuation (Section 13.1), a computer system whose small errors pile up (Section 13.6). Understanding what data can and cannot tell you, and how uncertain any conclusion is, is part of being an informed citizen, and a professional necessity for scientists, doctors, engineers and economists.

What you will meet at university

A first course in statistics covers estimation (finding good estimates of unknown quantities and saying how precise they are), hypothesis testing (deciding whether data are consistent with a hypothesis), confidence intervals, regression (relating one variable to others), and the design of experiments and surveys. The central ideas are *sampling distributions* (an estimate is a random variable), *bias* and *variance*, and *likelihood*. Later courses add Bayesian inference, multivariate analysis, time series and the methods of machine learning, which are statistics at enormous scale.

The questions that run through this chapter

Three big questions

1. **Is it real or is it noise?** How much variation should we expect by chance, and when is a change worth taking seriously?
2. **How should we estimate?** Given data, what is the best estimate of an unknown quantity, and how do we compare different methods?
3. **How does randomness enter the data?** What do the ways data are collected, protected or processed do to our conclusions?

The chapter starts with reading newspaper statistics critically (Section 13.1), then turns to estimation: counting voles by capture–recapture, estimating a total from serial numbers, and comparing the mean with the median. It ends with three questions about how randomness enters data: randomised surveys that protect privacy, rounding errors that accumulate or cancel, and the construction of variables with prescribed correlations.

Ideas in this chapter

§	Topic	Idea
13.1	Reading statistics	Rates, natural variation, trends
13.2	Capture–recapture	Maximum likelihood
13.3	Serial numbers	Bias, variance and sufficiency
13.4	Estimating a centre	Efficiency and robustness
13.5	Randomised response	Mixtures and total variance
13.6	Rounding errors	Systematic versus random error
13.7	Correlation	The Cholesky construction

Section 13.7 uses covariance and correlation, which were removed from the STEP specification in 2019; it explains what it needs.

How to read this chapter

Remember that every estimate is a random variable: if the data had come out differently, so would the estimate. For each method in this chapter, ask what its distribution would be over many repetitions: is it right on average, and how much does it vary? That single question is the heart of statistical thinking.

13.1 Reading the Headlines

The question

The average number of pedestrians killed annually in road accidents in Poldavia during the period 1974–1989 was 1080 and the average number killed annually in commercial flight accidents during the same period was 180. Discuss the following newspaper headlines which appeared in 1991. (The percentage figures in square brackets give a rough indication of the weight of marks attached to each discussion.)

10% Six Times Safer To Fly Than To Walk. 1974–1989 Figures Prove It.

10% Our Skies Are Safer. Only 125 People Killed In Air Accidents In 1990.

30% Road Carnage Increasing. 7 People Killed On Tuesday.

50% Alarming Rise In Pedestrian Casualties. 1350 Pedestrians Killed In Road Accidents During 1990.

The hook

Statistics is the science of learning from data, and its first lesson is that numbers do not speak for themselves. "Six times safer", "alarming rise", "carnage increasing": every day, headlines turn numbers into claims, and the claims are often wrong, not because the numbers are false but because they are compared with the wrong thing, or because ordinary random variation is mistaken for a trend.

This question has no formulas in it at all, which makes it unusual for STEP. It asks you to think like a statistician: what would we need to know to justify each headline, and how much variation should we expect by chance alone? The answers use the Poisson distribution (Section 12.18), and the reasoning is the foundation of the rest of this chapter.

Before you start

? **Ask yourself** If deaths occur independently and at random at a steady rate, how much should the yearly total vary? If the average is 1080 a year, what is a typical deviation from year to year?

Counts of independent rare events are approximately Poisson, and a Poisson count

with mean μ has variance μ and standard deviation $\sqrt{\mu}$. With $\mu = 1080$, the standard deviation is about 33: yearly totals between about 1015 and 1145 would be unremarkable.

Working it through

STAGE 1 **"Six times safer to fly than to walk"**

Ask yourself 1080 pedestrian deaths and 180 air deaths per year. Six times what, exactly?

The headline compares total deaths, but safety is about *risk*: deaths per unit of exposure. Almost everyone walks every day, while relatively few people fly, and then only occasionally. A fair comparison would use deaths per kilometre travelled, per hour of travel, or per journey, and these can give very different answers (air travel is extremely safe per kilometre, less impressively so per journey, since most air deaths happen at take-off and landing). The figures given cannot "prove" anything about relative safety. Also, "pedestrians killed in road accidents" is a narrow category; road deaths overall are much larger.

STAGE 2 **"Our skies are safer: only 125 killed in 1990"**

Ask yourself Are air deaths like independent rare events? How variable would you expect the yearly total to be?

Air deaths come in clusters: a single crash may kill hundreds of people. So the yearly total depends mainly on the number of crashes, which is small, and is extremely variable from year to year. One year with 125 deaths, against an average of 180, is entirely consistent with no change at all: a single extra crash would have changed the figure completely. One year's data cannot show a trend.

STAGE 3 **"Road carnage increasing: 7 killed on Tuesday"**

Ask yourself On average, how many pedestrians are killed per day? If daily deaths are Poisson, how often would you see 7 or more in a day?

The average is $\frac{1080}{365} \approx 2.96$ pedestrian deaths per day. If daily counts are Poisson with this mean, $P(\geq 7) \approx 0.031$: about 11 days a year would have 7 or more pedestrian deaths purely by chance. (And "road carnage" presumably includes drivers and passengers too, making 7 road deaths in a day even less remarkable.) A single day says nothing about a trend. The headline picks an extreme value out of a highly variable series, a classic error.

STAGE 4 **"Alarming rise: 1350 pedestrians killed in 1990"**

Ask yourself Compare 1350 with the Poisson standard deviation. Is this just chance? If not, what else could explain it, and does it justify "alarming rise"?

This is the substantial part. Under a steady Poisson model, 1350 is 270 above the mean of 1080, about 8 standard deviations: it is wildly unlikely to be chance. So something has probably changed. But the headline still overreaches, in several ways:

- *A trend hidden by an average.* The figure 1080 is an average over 16 years. If deaths rose steadily from about 900 in 1974 to about 1260 in 1989, the average would still be 1080, and 1350 in 1990 would simply continue an existing trend, not a sudden "alarming" rise (Figure 13.1.1). We need the year-by-year figures.
- *Exposure.* Population growth, more walking, more traffic or an ageing population would all increase deaths without the roads becoming more dangerous per journey. A rate per head, or per kilometre walked, is the right measure.
- *Definitions.* A change in how deaths are recorded or classified (for example, how long after an accident a death is counted) can produce a jump.
- *Over-dispersion.* Real yearly counts vary more than the Poisson model suggests, because conditions (weather, economic activity) vary. The "8 standard deviations" should be treated with some caution.

The rise is probably real, but whether it is alarming, and what causes it, needs more data.

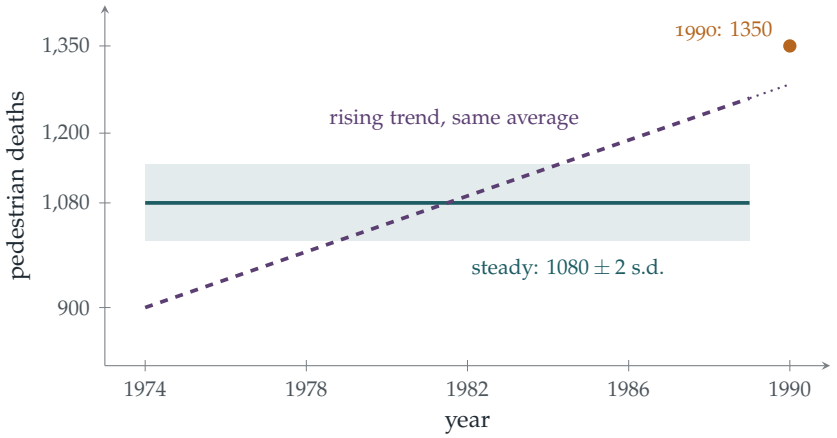


Figure 13.1.1 Two very different histories with the same 16-year average of 1080. If deaths were steady (teal, with the band of typical Poisson variation), 1350 in 1990 would be a dramatic jump. If they had been rising steadily (plum), 1350 would be only a little above the trend. The average alone cannot tell these apart.

Looking back

- **Compare like with like.** Risks need a denominator: deaths per person, per kilometre or per journey.
- **Know the natural variation.** Poisson variability, and the much larger variability of clustered events, tell us when a change is noteworthy.
- **An average hides a trend.** A single summary number cannot describe how a quantity has changed over time.

What's really going on

Signal and noise

Every statistical question is, at bottom, a question of separating *signal* (a real effect or trend) from *noise* (random variation). The Poisson model gives a yardstick for the noise in counts of independent events: a standard deviation of $\sqrt{\mu}$. Rare events have large *relative* variation: a count of 9 varies by about ± 3, a third of its size, while a count of 10 000 varies by only about 1%. That is why small numbers (deaths on one day, crashes in one year) are so often over-interpreted, and why cancer "clusters" in small towns usually turn out to be chance.

Regression to the mean

There is a further trap. Road-safety cameras are often installed at sites that have just had an unusually bad year. The following year, accidents fall, and the cameras get the credit. But an unusually bad year is likely to be followed by a more ordinary one anyway: this is *regression to the mean*, first described by Francis Galton in the 1880s. Separating the effect of a camera from regression to the mean requires comparison with similar sites that did not get cameras, the idea behind controlled experiments.

Florence Nightingale's statistics

Florence Nightingale, famous as a nurse in the Crimean War, was also a pioneering statistician. Her "rose diagrams" of 1858 showed that far more soldiers were dying of preventable disease than of wounds, and persuaded the government to reform army sanitation. She insisted on rates (deaths per thousand soldiers per year) rather than raw totals, and on comparing like with like, the very points this question tests. She was the first woman elected a Fellow of the Royal Statistical Society.

The thinking skill

Ask “compared with what?”

Before accepting a statistical claim, ask what the comparison is, what the denominator is, and how much variation would be expected by chance.

At university, statistics courses turn these questions into precise methods: hypothesis tests ask whether an observation is consistent with chance; confidence intervals say how precisely a quantity has been measured; and the design of experiments ensures that comparisons are fair.

Try this next

1. **Poisson tails.** Check that $\mathrm{P}(X \geq 7) \approx 0.031$ for X Poisson with mean 2.96. How many days a year would you expect 9 or more pedestrian deaths?
2. **Clustered deaths.** Suppose air crashes occur as a Poisson process at 1.2 per year, and each kills 150 people. Find the mean and standard deviation of yearly air deaths. Is a year with 125 deaths or fewer unusual? Could a year have exactly 125 deaths?
3. **A significant rise?** If the 1974–1989 figures had been steady at 1080 ± 33, find (using a normal approximation) the probability of 1350 or more in a single year.
4. **Write a better headline.** For each of the four headlines, write a headline that the data actually support.

Further reading

- **David Spiegelhalter, *The Art of Statistics*** (Pelican). How to learn from data, with many examples from the news.
- **Tim Harford, *How to Make the World Add Up*** (Bridge Street). Ten rules for thinking about numbers in the media.
- **Darrell Huff, *How to Lie with Statistics*** (Penguin). A short classic from 1954, still relevant.

13.2 Counting Voles

The question

A group of biologists attempts to estimate the magnitude, N, of an island population of voles (*Microtus agrestis*). Accordingly, the biologists capture a random sample of 200 voles, mark them and release them. A second random sample of 200 voles is then taken of which 11 are found to be marked. Show that the probability, p_N, of this occurrence is given by

$$p_N = k\frac{\left((N-200)!\right)^2}{N!\,(N-389)!},$$

where k is independent of N.

The biologists then estimate N by calculating the value of N for which p_N is a maximum. Find this estimate.

All unmarked voles in the second sample are marked and then the entire sample is released. Subsequently a third random sample of 200 voles is taken. Write down the probability that this sample contains exactly j marked voles, leaving your answer in terms of binomial coefficients.

Deduce that

$$\sum_{j=0}^{200}\binom{389}{j}\binom{3247}{200-j} = \binom{3636}{200}.$$

The hook

How many voles live on an island? Counting them is impossible: they hide, they move, they look alike. But you can catch some, mark them, let them go, and catch some more. If 200 are marked and, in a second catch of 200, 11 turn out to be marked, then roughly $\frac{11}{200}$ of the population is marked, so the population is about $200 \times \frac{200}{11} \approx 3636$.

This *capture–recapture* method is used to estimate populations of fish, birds and whales, and also of drug users, homeless people and casualties of war, any population that cannot be fully counted. The question puts the intuitive estimate on a firm footing using the principle of *maximum likelihood*, the most important idea in statistical estimation: choose the value of the unknown that makes the observed data most probable.

Before you start

Ask yourself The population has N voles, 200 of them marked. A random sample of 200 is taken. What is the probability that exactly 11 are marked?

Choose 11 of the 200 marked voles and 189 of the $N-200$ unmarked ones, out of $\binom{N}{200}$ equally likely samples:

$$p_N = \frac{\binom{200}{11}\binom{N-200}{189}}{\binom{N}{200}}.$$

This is the *hypergeometric distribution*.

Working it through

STAGE 1 The likelihood

Writing out the binomial coefficients that involve N,

$$\binom{N-200}{189} = \frac{(N-200)!}{189!\,(N-389)!}, \qquad \binom{N}{200} = \frac{N!}{200!\,(N-200)!},$$

so

$$p_N = \binom{200}{11}\frac{200!}{189!}\cdot\frac{\big((N-200)!\big)^2}{N!\,(N-389)!} = k\frac{\big((N-200)!\big)^2}{N!\,(N-389)!}. \qquad \blacksquare$$

Seen as a function of the unknown N, with the data fixed, p_N is called the *likelihood*.

STAGE 2 Maximising over a whole number

Ask yourself N is a whole number, so calculus is awkward. Instead compare p_{N+1} with p_N. When is their ratio bigger than 1?

Most of the factorials cancel:

$$\frac{p_{N+1}}{p_N} = \frac{(N-199)^2}{(N+1)(N-388)}.$$

This exceeds 1 when $(N-199)^2 > (N+1)(N-388)$, that is $N^2 - 398N + 39601 > N^2 - 387N - 388$, or $11N < 39989$, $N < 3635.4$. So p_N increases up to $N = 3636$ and decreases afterwards: the maximum likelihood estimate is

$$\hat{N} = 3636. \qquad \blacksquare$$

This is the integer part of $\frac{200\times 200}{11} = 3636.4$, the intuitive estimate.

STAGE 3 A third sample

After the second sample, the 189 unmarked voles in it are marked too, so 389 voles are now marked. Taking $N = 3636$, there are $3636 - 389 = 3247$ unmarked. A third sample of 200 contains exactly j marked voles with probability

$$\frac{\binom{389}{j}\binom{3247}{200-j}}{\binom{3636}{200}}.$$

These probabilities, for $j = 0, 1, \ldots, 200$, must add up to 1, which gives the identity. ■

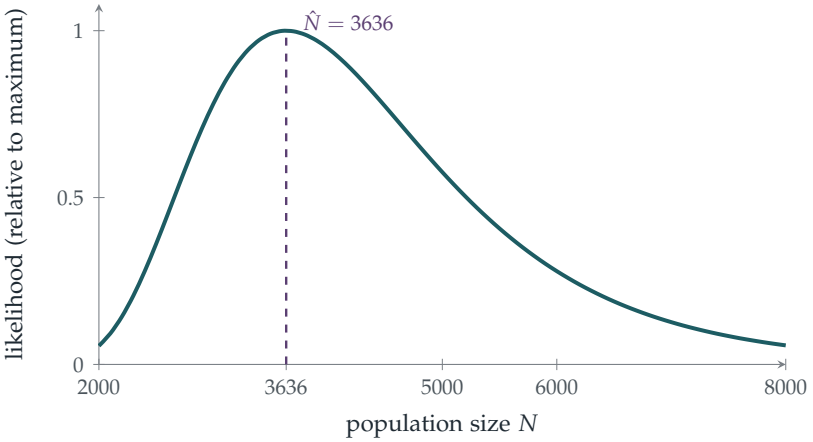


Figure 13.2.1 The likelihood p_N of finding 11 marked voles in the second sample, as a function of the population size N, scaled so that its maximum is 1. It peaks at $N = 3636$, but is broad and skewed: populations from about 2500 to 6500 are all reasonably consistent with the data.

Looking back

- **Probability of the data, as a function of the unknown.** The likelihood reverses the usual viewpoint: the data are fixed and the parameter varies.
- **Ratios for discrete maximisation.** Comparing consecutive terms located the maximum of a function of an integer.
- **Probabilities add to 1.** Summing a distribution over all outcomes proved a combinatorial identity (Vandermonde's identity).

What's really going on

Maximum likelihood

The principle of *maximum likelihood*, developed by Ronald Fisher in the 1910s and 1920s, says: estimate an unknown parameter by the value that makes the observed data most probable. It is by far the most widely used method of estimation, from fitting curves to data to training the neural networks of machine learning, whose

"loss functions" are usually negative log-likelihoods. Fisher showed that, for large samples, maximum likelihood estimates are approximately normal and as precise as any reasonable estimator can be, provided the model is well behaved in ways that hold for most everyday models. There are important exceptions. In the next section (Section 13.3) the unknown is the largest possible value, an endpoint of the distribution, and there the maximum likelihood estimate behaves quite differently: its error is not approximately normal, and it shrinks like $\frac{1}{n}$ rather than $\frac{1}{\sqrt{n}}$.

How precise is the estimate?

Figure 13.2.1 shows that the likelihood is broad: 11 recaptures is not much information. A rough calculation: the number of marked voles in the second sample is approximately Poisson with mean $\frac{200 \times 200}{N}$, so its standard deviation is about $\sqrt{11} \approx 3.3$, a relative error of about 30%, which carries over to the estimate of N. To estimate N to within 10%, the biologists would need about 100 recaptures. Statisticians also correct for the small bias of the simple estimate, using the *Chapman estimator* $\frac{(M+1)(C+1)}{R+1} - 1$, where M and C are the sizes of the two catches and R the number recaptured.

Laplace counts France

Laplace used the same idea in 1802 to estimate the population of France without a census. He knew the number of births registered in the whole country in a year, and, for a sample of communes, both the population and the births. Assuming the ratio of population to births was the same everywhere, he estimated the population of France at about 28 million, and even gave a measure of the uncertainty. The method was rediscovered for animal populations by C. G. J. Petersen, who marked plaice in Danish fjords in the 1890s, and by Frederick Lincoln, who used bird rings in the 1930s; it is still often called the *Lincoln–Petersen* estimator.

The thinking skill

Choose the parameter that makes the data most likely

To estimate an unknown, write the probability of what you observed as a function of the unknown, and maximise it.

At university, maximum likelihood is the backbone of statistical inference, with a theory of how precise the estimates are (the Fisher information) and how to compare competing models (likelihood ratio tests).

Try this next

1. **General formula.** If M animals are marked and a second sample of C contains R marked, show that the maximum likelihood estimate of N is the integer part of $\frac{MC}{R}$.

2. **Vandermonde.** Prove $\sum_j \binom{a}{j}\binom{b}{c-j} = \binom{a+b}{c}$ by counting committees of c people chosen from a women and b men.

3. **Coin tosses.** In n tosses of a coin with unknown probability p of heads, h heads are observed. Show that the maximum likelihood estimate of p is $\frac{h}{n}$.

4. **No recaptures.** What happens to the maximum likelihood estimate if the second sample contains no marked voles? What does that tell you about the method?

Further reading

- **David Spiegelhalter, *The Art of Statistics*** (Pelican), on estimation and uncertainty.
- **George Seber, *The Estimation of Animal Abundance*** (Blackburn). The standard reference on capture–recapture.
- **John Rice, *Mathematical Statistics and Data Analysis*** (Duxbury). A first course in mathematical statistics, including maximum likelihood.

13.3 The German Tank Problem

The question

A hostile naval power possesses a large, unknown number N of submarines. Interception of radio signals yields a small number n of their identification numbers X_i $(i = 1, 2, \ldots, n)$, which are taken to be independent and uniformly distributed over the continuous range from 0 to N. Show that Z_1 and Z_2, defined by

$$Z_1 = \frac{n+1}{n}\max\{X_1, X_2, \ldots, X_n\} \qquad \text{and} \qquad Z_2 = \frac{2}{n}\sum_{i=1}^{n} X_i,$$

both have means equal to N.

Calculate the variance of Z_1 and of Z_2. Which estimator do you prefer, and why?

The hook

During the Second World War, Allied intelligence wanted to know how many tanks Germany was producing. Spies' reports suggested about 1400 a month. Statisticians took a different approach: captured tanks carried serial numbers on their gearboxes and other parts, and serial numbers are, in effect, a random sample from $1, 2, \ldots, N$. From the numbers alone, they estimated production at about 250 a month. In the figures usually quoted, German records examined after the war gave 245. The statisticians had been far more accurate than the spies.

This question compares two ways of estimating N from serial numbers. Both are right on average. But one uses the data much more efficiently than the other, and working out which leads to the central idea of estimation theory: among estimators that are correct on average, prefer the one that varies least.

Before you start

Ask yourself If M is the largest of n independent uniform values on $[0, N]$, what is $\mathrm{P}(M \le m)$?

$M \le m$ exactly when every $X_i \le m$, so $\mathrm{P}(M \le m) = \left(\frac{m}{N}\right)^n$ for $0 \le m \le N$ (compare Section 12.25). Differentiating, M has density $\frac{nm^{n-1}}{N^n}$.

Working it through

STAGE 1 Both are unbiased

For M:

$$\mathrm{E}(M) = \int_0^N m \cdot \frac{n m^{n-1}}{N^n}\,\mathrm{d}m = \frac{n}{n+1}N, \qquad \text{so} \qquad \mathrm{E}(Z_1) = \frac{n+1}{n} \cdot \frac{n}{n+1}N = N.$$

For the mean: each X_i has mean $\frac{N}{2}$, so $\mathrm{E}(Z_2) = \frac{2}{n} \cdot n \cdot \frac{N}{2} = N$. ■

Both estimators are *unbiased*: on average, they give the right answer. The factor $\frac{n+1}{n}$ corrects for the fact that the largest serial number seen is always a little below N.

STAGE 2 Their variances

Ask yourself Find $\mathrm{E}(M^2)$ and hence $\mathrm{Var}(M)$. For Z_2, what is the variance of a single uniform variable on $[0, N]$?

$\mathrm{E}(M^2) = \int_0^N m^2 \cdot \frac{n m^{n-1}}{N^n}\,\mathrm{d}m = \frac{n}{n+2}N^2$, so

$$\mathrm{Var}(M) = \frac{n}{n+2}N^2 - \frac{n^2}{(n+1)^2}N^2 = \frac{nN^2}{(n+1)^2(n+2)},$$
$$\mathrm{Var}(Z_1) = \frac{(n+1)^2}{n^2}\mathrm{Var}(M) = \frac{N^2}{n(n+2)}.$$

A uniform variable on $[0, N]$ has variance $\frac{N^2}{12}$, and the X_i are independent, so

$$\mathrm{Var}(Z_2) = \frac{4}{n^2} \cdot n \cdot \frac{N^2}{12} = \frac{N^2}{3n}.$$ ■

STAGE 3 Which is better?

Ask yourself Compare $\frac{1}{n(n+2)}$ with $\frac{1}{3n}$. How do they behave as n grows?

The ratio is $\frac{\mathrm{Var}(Z_1)}{\mathrm{Var}(Z_2)} = \frac{3}{n+2}$, which is less than 1 for $n \geq 2$ (and equal for $n = 1$, when the two estimators coincide). So Z_1 is better, and much better for large n: its standard deviation shrinks like $\frac{1}{n}$, while that of Z_2 shrinks only like $\frac{1}{\sqrt{n}}$ (Figure 13.3.1). With $n = 10$ serial numbers, Z_1 has a standard deviation about 9% of N, and Z_2 about 18%.

There is another reason to prefer Z_1: Z_2 can be absurd. With the serial numbers 10, 20 and 300, $Z_2 = \frac{2}{3} \cdot 330 = 220$, less than a serial number we have actually seen. $Z_1 = \frac{4}{3} \cdot 300 = 400$ can never be smaller than the largest observation. ■

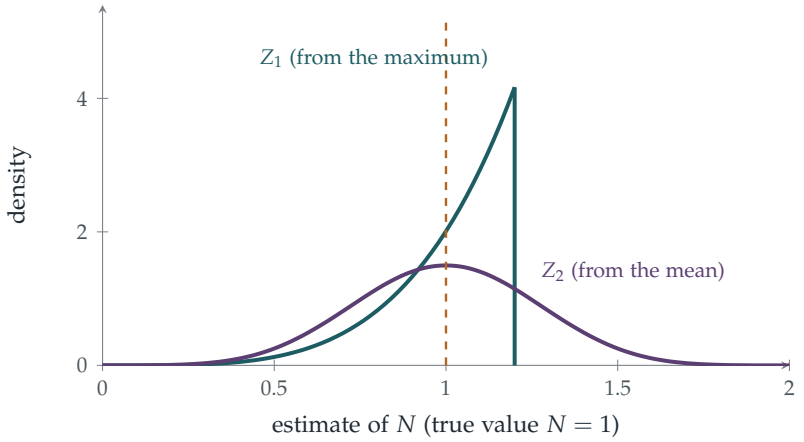


Figure 13.3.1 The sampling distributions of the two estimators with $n = 5$ serial numbers and true value $N = 1$. Both average exactly 1, but Z_1 (teal) is concentrated much more tightly near the truth than Z_2 (plum), and it never falls below the largest serial number observed.

Looking back

- **Unbiasedness.** Each estimator was scaled so that its expected value is exactly N.
- **Compare variances.** Among unbiased estimators, the one with smaller variance is more reliable.
- **Use the most informative statistic.** The maximum contains all the information about N that the sample holds; the mean wastes much of it.

What's really going on

Sufficiency

Why is the maximum so much better? Given the largest value M, the other values are just uniform on $[0, M]$, whatever N is: they carry no further information about N. The maximum is a *sufficient statistic*. A theorem of C. R. Rao and David Blackwell (1945–47) says that any unbiased estimator can be improved, or at least not made worse, by replacing it with its expected value given a sufficient statistic, and here that turns Z_2 into Z_1. The *Lehmann–Scheffé theorem* then shows that Z_1 is the unbiased estimator with the smallest possible variance. Notice too that the variance of Z_1 falls like $\frac{1}{n^2}$, faster than the $\frac{1}{n}$ of most estimators: that happens because N is an endpoint of the distribution, and the data pile up against it.

The discrete version

Real serial numbers are whole numbers $1, 2, \dots, N$, sampled without replacement. The best unbiased estimate is then $\hat{N} = m + \frac{m}{k} - 1$, where m is the largest of k observed serial numbers: the maximum plus the average gap between observations. Using estimates of this kind, based mainly on the serial numbers of gearboxes and wheels, the Allied statisticians estimated German production at about 246 tanks a month, against the 245 later found in German records and the 1400 suggested by intelligence reports.

Serial numbers in history

The method has been used many times since, for example by analysts estimating how many units of a new consumer product had been made from the serial numbers of the first few sold. Manufacturers learned the lesson: many now encode serial numbers so that they cannot be read as a simple sequence. Richard Ruggles and Henry Brodie described the wartime work in the *Journal of the American Statistical Association* in 1947.

The thinking skill

Judge an estimator by its sampling distribution

An estimator is a random variable. Ask whether it is right on average (bias) and how much it varies (variance), and prefer the one with smaller error.

At university this leads to the theory of efficient estimation: the Cramér–Rao bound says how small the variance of an unbiased estimator can be, and sufficiency explains how to reach it.

Try this next

1. **Mean squared error.** The biased estimator M has $\mathrm{E}(M - N)^2 = \mathrm{Var}(M) + \big(\mathrm{E}(M) - N\big)^2$. Compute it and compare with $\mathrm{Var}(Z_1)$. Find the constant c that minimises $\mathrm{E}(cM - N)^2$.

2. **The minimum too.** Show that $(n+1)\min\{X_i\}$ is also an unbiased estimator of N, and find its variance. Is it any good?

3. **The discrete formula.** For serial numbers $1, \dots, N$, sample k without replacement, with maximum m. Show that $\mathrm{P}(\max \le m) = \binom{m}{k} / \binom{N}{k}$, and verify that $m + \frac{m}{k} - 1$ is unbiased when $k = 1$.

4. **Tanks.** If 5 captured tanks have serial numbers 19, 40, 42, 60, 98, estimate the number of tanks made.

Further reading

- **Richard Ruggles & Henry Brodie, "An empirical approach to economic intelligence in World War II"**, *Journal of the American Statistical Association* 42 (1947). The original account.
- **John Rice, *Mathematical Statistics and Data Analysis*** (Duxbury), on bias, variance and sufficiency.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican).

13.4 Mean or Median?

The question

The random variables $X_1, X_2, \ldots, X_{2n+1}$ are independently and uniformly distributed on the interval $0 \leq x \leq 1$. The random variable Y is defined to be the median of $X_1, X_2, \ldots, X_{2n+1}$. Given that the probability density function of Y is $g(y)$, where

$$g(y) = \begin{cases} ky^n(1-y)^n & \text{if } 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

use the result

$$\int_0^1 y^r(1-y)^s \, dy = \frac{r!\, s!}{(r+s+1)!}$$

to show that $k = (2n+1)!/(n!)^2$, and evaluate $\mathrm{E}(Y)$ and $\mathrm{Var}(Y)$. Hence show that, for any given positive number d, the inequality

$$\mathrm{P}\left(|Y - \tfrac{1}{2}| < \frac{d}{\sqrt{n}}\right) < \mathrm{P}\left(|\bar{X} - \tfrac{1}{2}| < \frac{d}{\sqrt{n}}\right)$$

holds provided n is large enough, where $\bar{X}$ is the mean of $X_1, X_2, \ldots, X_{2n+1}$.

[You may assume that Y and $\bar{X}$ are normally distributed for large n.]

The hook

To estimate the centre of a set of measurements, you could take the mean or the median. Which is better? The answer depends on the data. For measurements with occasional wild errors, the median is far more reliable: one absurd value can drag the mean anywhere, but barely moves the median. For well-behaved data, the mean usually wins. This question measures how much it wins by, for uniformly distributed data: the median's variance is three times as large, so you would need three times as many observations to make the median as precise as the mean.

The comparison introduces *efficiency*, one of the main ideas in choosing statistical methods, and the beta integral, which also appeared in Section 4.4.

Before you start

Ask yourself Why is the density of the median proportional to $y^n(1-y)^n$? (For Y to be near y, how many of the $2n+1$ values must be below y, and how many above?)

For the median to be at y, one value must be at y, n values below (each with probability y) and n above (each with probability $1-y$). That gives the factor $y^n(1-y)^n$; the constant counts the ways of choosing which values are which.

Working it through

STAGE 1 The constant

The density must integrate to 1:

$$k\int_0^1 y^n(1-y)^n\,dy = k\cdot\frac{n!\,n!}{(2n+1)!} = 1, \qquad k = \frac{(2n+1)!}{(n!)^2}. \quad \blacksquare$$

STAGE 2 Mean and variance

The density is symmetric about $\frac{1}{2}$ (replace y by $1-y$), so $\mathrm{E}(Y) = \frac{1}{2}$.

Ask yourself Compute $\mathrm{E}(Y^2) = k\int_0^1 y^{n+2}(1-y)^n\,dy$ using the beta integral, and simplify the factorials.

$$\mathrm{E}(Y^2) = \frac{(2n+1)!}{(n!)^2}\cdot\frac{(n+2)!\,n!}{(2n+3)!} = \frac{(n+2)(n+1)}{(2n+3)(2n+2)} = \frac{n+2}{2(2n+3)}.$$

So

$$\mathrm{Var}(Y) = \frac{n+2}{2(2n+3)} - \frac{1}{4} = \frac{2(n+2)-(2n+3)}{4(2n+3)} = \frac{1}{4(2n+3)}. \quad \blacksquare$$

STAGE 3 Comparing with the mean

Ask yourself What are the mean and variance of $\bar{X}$, the average of $2n+1$ independent uniform variables?

Each X_i has mean $\frac{1}{2}$ and variance $\frac{1}{12}$, so $\mathrm{E}(\bar{X}) = \frac{1}{2}$ and $\mathrm{Var}(\bar{X}) = \frac{1}{12(2n+1)}$. For large n,

$$\mathrm{Var}(Y) \approx \frac{1}{8n}, \qquad \mathrm{Var}(\bar{X}) \approx \frac{1}{24n}.$$

Ask yourself If Y and $\bar{X}$ are normal with mean $\frac{1}{2}$ and standard deviations σ_Y and $\sigma_{\bar{X}}$, write both probabilities in terms of the standard normal distribution function Φ.

$\mathrm{P}\big(|Y - \frac{1}{2}| < \frac{d}{\sqrt{n}}\big) = 2\Phi\big(\frac{d}{\sqrt{n}\,\sigma_Y}\big) - 1$, and $\sqrt{n}\,\sigma_Y \to \frac{1}{\sqrt{8}}$; similarly for $\bar{X}$ with $\sqrt{n}\,\sigma_{\bar{X}} \to \frac{1}{\sqrt{24}}$. So the probabilities tend to $2\Phi(d\sqrt{8}) - 1$ and $2\Phi(d\sqrt{24}) - 1$. Since Φ is increasing and $\sqrt{24} > \sqrt{8}$, the second is larger, and so the inequality holds for all sufficiently large n. ■The mean is more likely to be close to the true centre.

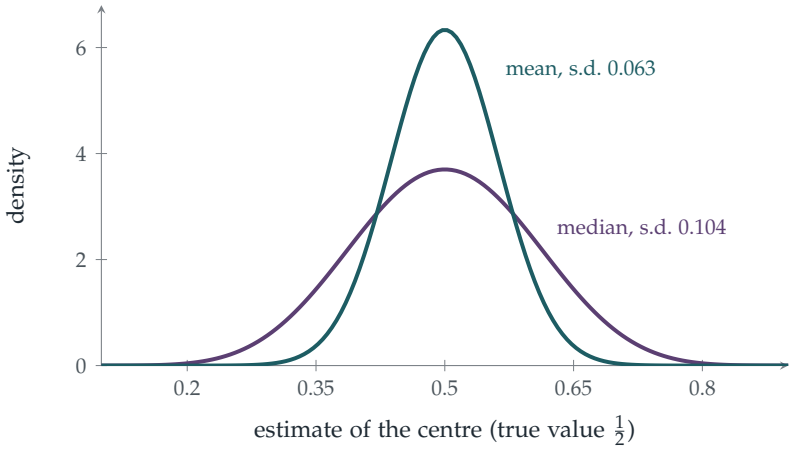


Figure 13.4.1 The sampling distributions of the median (plum) and the mean (teal) of 21 independent uniform observations. Both are centred on the true value $\frac{1}{2}$, but the mean is more tightly concentrated: its variance is about a third of the median's.

Looking back

- **The beta integral.** $\int_0^1 y^r(1-y)^s\,\mathrm{d}y$ gave the normalising constant and the moments in one formula.
- **Symmetry gave the mean.** No integration was needed for $\mathrm{E}(Y)$.
- **Compare spreads on the right scale.** Multiplying standard deviations by $\sqrt{n}$ made the comparison independent of n.

What's really going on

Efficiency

The *relative efficiency* of two unbiased estimators is the ratio of their variances; it says how many times as many observations one needs to match the other. For uniform data

the median has efficiency $\frac{1}{3}$ relative to the mean. For normal data it is $\frac{2}{\pi} \approx 0.64$: the median needs about 57% more observations. But for heavy-tailed data the tables turn. For the Cauchy distribution (Section 12.22) the mean is useless, since the average of n observations is no better than one, while the median works well. For the Laplace distribution (Section 12.24) the median is twice as efficient as the mean.

Robust statistics

The median is *robust*: you can change almost half the data to arbitrary values without moving it far. The mean has no such protection: one wrong value can spoil it. In the 1960s and 1970s John Tukey, Peter Huber and others developed *robust statistics*, estimators that are nearly as efficient as the mean for clean data but resistant to outliers, such as the *trimmed mean* (discard the largest and smallest 10% and average the rest), which is used in judging sports like diving and gymnastics.

Laplace and the median

Laplace showed in 1774 that if errors follow the double exponential (Laplace) distribution, the best estimate of the true value is the median; Gauss showed in 1809 that if they follow the normal distribution, the best estimate is the mean, which led him to the method of least squares. Which is right depends on how errors behave, a question that statisticians still debate for each new kind of data.

The thinking skill

Compare methods by their variability

Two methods may both be right on average; compare how much their answers vary, and how quickly that variation shrinks with more data.

At university, asymptotic efficiency, the comparison of variances for large samples, is a standard tool for choosing between statistical methods, together with robustness for when the model might be wrong.

Try this next

1 **The median's density.** Derive the density $\frac{(2n+1)!}{(n!)^2} y^n (1-y)^n$ by counting: one value in $[y, y+\delta y]$, n below and n above.

2 **Three observations.** For $n = 1$ (three uniform observations), find $\mathrm{Var}(Y)$ and $\mathrm{Var}(\bar{X})$ exactly. Which is smaller?

3 **The midrange.** Another estimator of the centre is the midrange $\frac{1}{2}(\max + \min)$. Show that for uniform data its variance is $\frac{1}{2(m+1)(m+2)}$ with $m = 2n+1$ observations. How does it compare with the mean? (Compare Section 13.3.)

4 **An outlier.** Eleven measurements are 9.8, 9.9, 10.0, 10.0, 10.1, 10.1, 10.2, 10.2, 10.3, 10.4 and 104 (a typing error). Compare the mean and median.

Further reading

- **John Rice, *Mathematical Statistics and Data Analysis*** (Duxbury), on order statistics and efficiency.
- **Rand Wilcox, *Fundamentals of Modern Statistical Methods*** (Springer). Why robust methods matter, for non-specialists.
- **Stephen Stigler, *The Seven Pillars of Statistical Wisdom*** (Harvard University Press). Aggregation, the mean and the median, in historical perspective.

13.5 Asking Awkward Questions

The question

(i) X_1 and X_2 are both random variables which take values $x_1, x_2, \ldots, x_n$, with probabilities $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$ respectively.

The value of random variable Y is defined to be that of X_1 with probability p and that of X_2 with probability $q = 1 - p$.

If X_1 has mean μ_1 and variance σ_1^2, and X_2 has mean μ_2 and variance σ_2^2, find the mean of Y and show that the variance of Y is $p\sigma_1^2 + q\sigma_2^2 + pq(\mu_1 - \mu_2)^2$.

(ii) To find the value of random variable B, a fair coin is tossed and a fair six-sided die is rolled. If the coin shows heads, then $B = 1$ if the die shows a six and $B = 0$ otherwise; if the coin shows tails, then $B = 1$ if the die does not show a six and $B = 0$ if it does.

The value of Z_1 is the sum of n independent values of B, where n is large. Show that Z_1 is a Binomial random variable with probability of success $\frac{1}{2}$. Using a Normal approximation, show that the probability that Z_1 is within 10% of its mean tends to 1 as $n \to \infty$.

(iii) To find the value of random variable Z_2, a fair coin is tossed and n fair six-sided dice are rolled, where n is large. If the coin shows heads, then the value of Z_2 is the number of dice showing a six; if the coin shows tails, then the value of Z_2 is the number of dice not showing a six.

Use part (i) to write down the mean and variance of Z_2. Explain why a Normal distribution with this mean and variance will not be a good approximation to the distribution of Z_2. Show that the probability that Z_2 is within 10% of its mean tends to 0 as $n \to \infty$.

The hook

How do you find out what fraction of students have cheated in an exam, or of employees have stolen from work? Ask them directly and many will lie. In 1965 the statistician Stanley Warner invented an ingenious solution, *randomised response*. Each person secretly tosses a coin or rolls a die, and the outcome decides whether they answer the sensitive question truthfully or answer its opposite. The interviewer never knows which, so no individual answer is incriminating; yet, because the randomisation probabilities are known, the overall proportion can still be estimated.

This question explores the mathematics behind it. Randomising each person *indepen-*

dently (part (ii)) gives a well-behaved total that pins down the proportion. Randomising everyone with a *single* coin toss (part (iii)) produces a total with the same mean but a completely different, two-humped distribution, which tells us almost nothing. Where you put the randomness matters.

Before you start

Ask yourself Y is X_1 with probability p and X_2 with probability q. What is $\mathrm{E}(Y)$? What is $\mathrm{E}(Y^2)$?

Conditioning on which variable is chosen, $\mathrm{E}(Y) = p\mu_1 + q\mu_2$, and $\mathrm{E}(Y^2) = p\,\mathrm{E}(X_1^2) + q\,\mathrm{E}(X_2^2) = p(\sigma_1^2 + \mu_1^2) + q(\sigma_2^2 + \mu_2^2)$.

Working it through

STAGE 1 The variance of a mixture

$$\mathrm{Var}(Y) = p(\sigma_1^2 + \mu_1^2) + q(\sigma_2^2 + \mu_2^2) - (p\mu_1 + q\mu_2)^2.$$

The μ terms are $p\mu_1^2 + q\mu_2^2 - p^2\mu_1^2 - 2pq\mu_1\mu_2 - q^2\mu_2^2 = pq\mu_1^2 + pq\mu_2^2 - 2pq\mu_1\mu_2$, using $p - p^2 = pq$ and $q - q^2 = pq$. So

$$\mathrm{Var}(Y) = p\sigma_1^2 + q\sigma_2^2 + pq(\mu_1 - \mu_2)^2. \qquad \blacksquare$$

The variance of a mixture is the average variance within the components, plus the variance *between* their means.

STAGE 2 Independent randomisation

Ask yourself What is $\mathrm{P}(B = 1)$? Are the n values of B independent?

$\mathrm{P}(B = 1) = \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{2} \cdot \frac{5}{6} = \frac{1}{2}$. The n values are independent, each with success probability $\frac{1}{2}$, so $Z_1 \sim \mathrm{B}(n, \frac{1}{2})$, with mean $\frac{n}{2}$ and standard deviation $\frac{\sqrt{n}}{2}$.

Within 10% of the mean means $|Z_1 - \frac{n}{2}| < \frac{n}{20}$. In standard units, this is $|Z| < \frac{n/20}{\sqrt{n}/2} = \frac{\sqrt{n}}{10}$, which tends to infinity, so the probability tends to 1. $\blacksquare$

STAGE 3 One coin for everybody

Ask yourself Now one coin toss decides for all n dice. Given heads, what is the distribution of Z_2? Given tails? Use part (i).

Given heads, $Z_2 \sim \mathrm{B}(n, \frac{1}{6})$: mean $\frac{n}{6}$, variance $\frac{5n}{36}$. Given tails, $Z_2 \sim \mathrm{B}(n, \frac{5}{6})$: mean $\frac{5n}{6}$,

variance $\frac{5n}{36}$. By part (i) with $p = q = \frac{1}{2}$,

$$\mathrm{E}(Z_2) = \frac{n}{2}, \qquad \mathrm{Var}(Z_2) = \frac{5n}{36} + \frac{1}{4}\left(\frac{4n}{6}\right)^2 = \frac{5n}{36} + \frac{n^2}{9}.$$

Ask yourself Where is most of the probability? Is there much near the mean $\frac{n}{2}$?

Z_2 has two humps, near $\frac{n}{6}$ and $\frac{5n}{6}$, each of width about $\sqrt{n}$ (Figure 13.5.1), and almost nothing in between. A normal distribution with the same mean and variance would be a single hump centred at $\frac{n}{2}$, exactly where Z_2 almost never is. For Z_2 to be within 10% of $\frac{n}{2}$, a $\mathrm{B}(n, \frac{1}{6})$ or $\mathrm{B}(n, \frac{5}{6})$ variable must lie in $[0.45n, 0.55n]$. For $\mathrm{B}(n, \frac{1}{6})$ that is at least $0.28n$ above its mean, which is about $\frac{0.28n}{0.37\sqrt{n}} \approx 0.75\sqrt{n}$ standard deviations, so the probability tends to 0; similarly for $\mathrm{B}(n, \frac{5}{6})$. ■

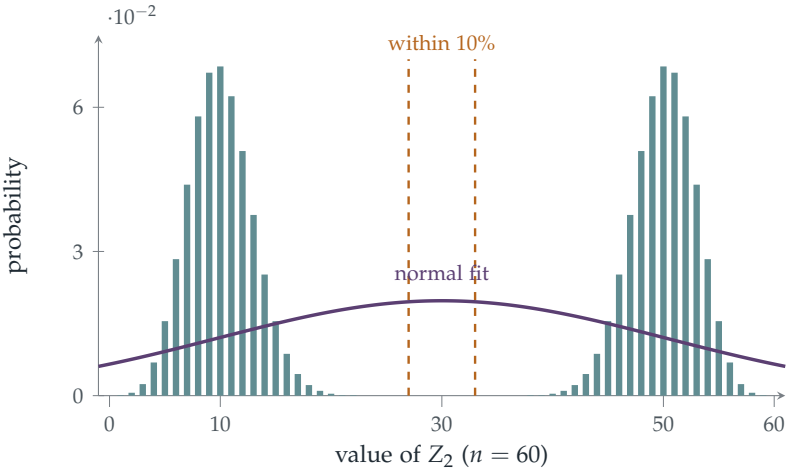


Figure 13.5.1 The distribution of Z_2 for $n = 60$ dice (bars): two humps at 10 and 50, from the two outcomes of the single coin toss. A normal distribution with the same mean (30) and variance (plum) is a poor fit, and almost none of the probability lies within 10% of the mean (between the amber lines).

Looking back

- **Condition on the hidden choice.** The mixture's mean and variance came from conditioning on which component was chosen.
- **Variance within plus variance between.** The extra term $pq(\mu_1 - \mu_2)^2$ measured the spread between the two components.
- **Mean and variance are not the whole story.** Z_2 had the same mean as Z_1, but a completely different shape.

What's really going on

Randomised response

In Warner's scheme, each respondent answers "Do you belong to group A?" with probability p, and "Do you *not* belong to group A?" with probability $1-p$, choosing privately. If a fraction π belong to A, the probability of a "yes" is

$$\lambda = p\pi + (1-p)(1-\pi),$$

so from the observed fraction of yeses, $\hat{\lambda}$, we can estimate $\hat{\pi} = \frac{\hat{\lambda}-(1-p)}{2p-1}$ (provided $p \neq \frac{1}{2}$). Part (ii) of the question is exactly this, with each person's own die deciding. The price of privacy is precision: dividing by $2p-1$ inflates the variance, so a randomised survey needs more respondents than a direct one. But each person randomises independently, so the law of large numbers works. Part (iii) shows what goes wrong if the randomisation is shared: the uncertainty about the single coin never averages out.

Differential privacy

Randomised response is the ancestor of *differential privacy*, a mathematical definition of privacy introduced by Cynthia Dwork and colleagues in 2006: a procedure is private if adding or removing any one person's data changes the probability of any output by at most a small factor. Technology companies use it to collect usage statistics, and the United States Census Bureau used it for the 2020 census, adding carefully calibrated random noise to published figures so that no individual's answers could be reconstructed.

The law of total variance

The identity in part (i) is a special case of the *law of total variance*: $\operatorname{Var}(Y) = \operatorname{E}\big(\operatorname{Var}(Y \mid C)\big) + \operatorname{Var}\big(\operatorname{E}(Y \mid C)\big)$, where C is the hidden choice. It underlies the *analysis of variance* (ANOVA), introduced by Ronald Fisher in the 1920s for agricultural experiments at Rothamsted, which splits the variation in crop yields into the part explained by fertilisers or varieties and the part left unexplained.

The thinking skill

Look at the whole distribution, not just its summaries

A mean and variance describe a single-humped distribution well, but can be badly misleading for a mixture. Ask where the probability actually lies.

At university, mixture models are used to describe populations made of several groups, and the law of total variance is a standard tool in statistics, from survey design to hierarchical models.

Try this next

1 **The twins.** (From STEP II, 2005.) Twins share a computer; the one online is Anna with probability p, who answers truthfully with probability a, or Bella, who does so with probability b. I ask them to toss a coin and say whether it came down heads, and get "yes". Show that the probability that it was heads is $\frac{1}{2}$ if and only if $2(ap + bq) = 1$. If I then ask again and get "no", show that the probability of heads is $\frac{1}{2}$; if I get "yes" again, show that it is $\frac{1}{2}$ provided $2(ap + bq) = 1$.

2 **Estimating π.** In a Warner survey with $p = \frac{3}{4}$, 400 people answer and 160 say yes. Estimate the proportion π in group A, and find the approximate standard error of your estimate.

3 **Total variance.** Prove the law of total variance for a discrete hidden variable C with several values.

4 **Shared coins.** In part (iii), what is the probability that $Z_2 > \frac{n}{2}$? Explain why, for large n, it is close to $\frac{1}{2}$ even though Z_2 is almost never close to $\frac{n}{2}$.

Further reading

- **Stanley Warner, "Randomized response: a survey technique for eliminating evasive answer bias"**, *Journal of the American Statistical Association* 60 (1965).
- **Cynthia Dwork & Aaron Roth, *The Algorithmic Foundations of Differential Privacy*** (free online). The modern theory; the first chapters are accessible.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican), on surveys and their pitfalls.

13.6 Errors That Add Up

The question

Two computers, LEP and VOZ are programmed to add numbers after first approximating each number by an integer. LEP approximates the numbers by rounding: that is, it replaces each number by the nearest integer. VOZ approximates by truncation: that is, it replaces each number by the largest integer less than or equal to the number. The fractional parts of the numbers to be added are uniformly and independently distributed. (The fractional part of a number a is $a - \lfloor a \rfloor$, where $\lfloor a \rfloor$ is the largest integer less than or equal to a.) Both computers approximate and add 1500 numbers. For each computer, find the probability that the magnitude of error in the answer will exceed 15.

How many additions can LEP perform before the probability that the magnitude of error is less than 10 drops below 0.9?

The hook

In 1982 the Vancouver Stock Exchange launched a new index, starting at 1000. Every time a share price changed, the index was recalculated and *truncated* to three decimal places. Each truncation lost on average 0.0005, which seems negligible. But there were about 3000 recalculations a day, and after twenty-two months the index stood at around 520, even though the market had risen. When it was recomputed correctly, overnight it jumped to about 1100.

This question explains what happened. Truncation errors all point the same way, so they add up steadily, in proportion to the number of operations. Rounding errors point both ways and largely cancel: they grow only like the square root of the number of operations, the random-walk behaviour that the central limit theorem (Section 12.24) predicts.

Before you start

Ask yourself If a number's fractional part U is uniform on $[0, 1)$, what is the error when it is truncated? When it is rounded? What are the mean and variance of each?

Truncation replaces a by $\lfloor a \rfloor$, an error (true minus approximate) of U, uniform on $[0, 1)$: mean $\frac{1}{2}$, variance $\frac{1}{12}$. Rounding gives an error uniform on $(-\frac{1}{2}, \frac{1}{2}]$: mean 0, variance $\frac{1}{12}$.

Working it through

STAGE 1 The total errors

The total error after 1500 numbers is the sum of 1500 independent errors, approximately normal by the central limit theorem:

- LEP (rounding): mean 0, variance $\frac{1500}{12} = 125$, standard deviation $\sqrt{125} \approx 11.18$.
- VOZ (truncation): mean $\frac{1500}{2} = 750$, the same standard deviation 11.18.

STAGE 2 The probability of a large error

Ask yourself For LEP, what is $\mathrm{P}(|E| > 15)$ with $E \sim \mathrm{N}(0, 125)$? For VOZ, where is the error almost certainly?

For LEP, $\frac{15}{11.18} \approx 1.342$, so

$$\mathrm{P}(|E| > 15) \approx 2\big(1 - \Phi(1.342)\big) \approx 2 \times 0.090 = 0.18.$$

For VOZ, the error is about 750 ± 11; the probability that it is less than 15 in magnitude is essentially zero, so $\mathrm{P}(|E| > 15) \approx 1$. ■

STAGE 3 How many additions?

Ask yourself After n additions, LEP's error has standard deviation $\sqrt{n/12}$. When is $\mathrm{P}(|E| < 10) \geq 0.9$?

We need $\frac{10}{\sqrt{n/12}} \geq 1.645$ (since $\mathrm{P}(|Z| < 1.645) = 0.9$), that is $\frac{n}{12} \leq \left(\frac{10}{1.645}\right)^2 \approx 36.95$, so $n \leq 443$. LEP can perform about 443 additions. ■

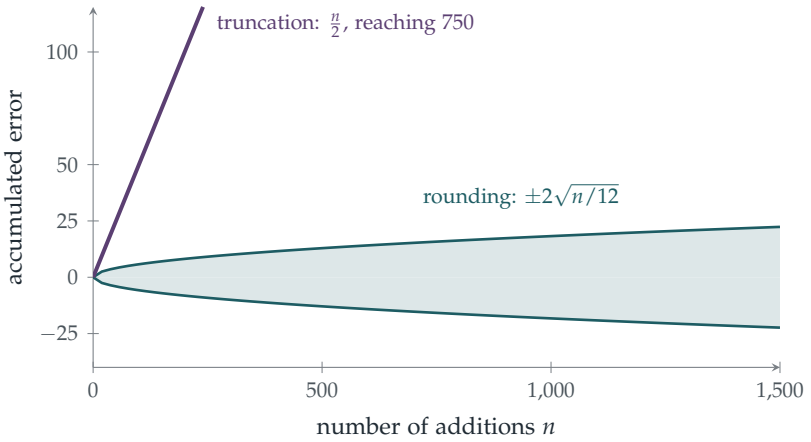


Figure 13.6.1 How errors accumulate. Truncation errors (plum) all have the same sign and grow in proportion to n: after 1500 additions the total is about 750. Rounding errors (teal band, two standard deviations either side of zero) cancel on average and grow only like $\sqrt{n}$: after 1500 additions they are usually within about ± 22.

Looking back

- **Model each error.** Uniform fractional parts gave each error a simple distribution.
- **Add means and variances.** Means add (giving the bias $\frac{n}{2}$) and, for independent errors, so do variances.
- **Normal approximation.** The central limit theorem turned a sum of 1500 uniform errors into a normal distribution.

What's really going on

Bias and variance

Truncation is *biased*: its errors have non-zero mean, and bias accumulates linearly. Rounding is *unbiased*, and its errors behave like a random walk, growing like $\sqrt{n}$. After a million operations, the difference is between an error of order 500 000 and one of order 300. This is the same distinction as between a systematic error in an experiment (a miscalibrated instrument) and random measurement error: averaging many measurements reduces random error but does nothing to systematic error.

Floating point

Every computer stores real numbers to limited precision, and every arithmetic operation rounds its result. Modern computers follow the IEEE 754 standard, which rounds to the nearest representable number, with ties broken towards an even last digit precisely so that rounding is unbiased on average. Even so, careless summation of many numbers

can lose accuracy; William Kahan's *compensated summation* algorithm (1965) keeps track of the lost low-order digits and adds them back. In the Gulf War in 1991, a Patriot missile battery failed to intercept an incoming missile because its clock counted time in tenths of a second, a number that cannot be stored exactly in binary; after 100 hours of operation the truncation error had grown to about a third of a second, enough to misjudge the target's position by over half a kilometre.

The Vancouver index

The Vancouver Stock Exchange index was created in January 1982 with a value of 1000.000. By November 1983 it had fallen to about 524, despite a generally rising market. Consultants found that the index was being recalculated thousands of times a day and truncated (not rounded) to three decimal places each time. Recomputed correctly over a weekend, the index reopened at about 1098. The episode is a standard warning in courses on numerical computation.

The thinking skill

Separate systematic from random error

Errors with a consistent sign accumulate linearly; independent errors with mean zero accumulate like a square root. Identify which kind you have.

At university, numerical analysis studies how errors propagate through calculations, and statistics distinguishes bias from variance; the "bias–variance trade-off" is central to modern machine learning.

Try this next

1 **VOZ corrected.** If VOZ adds $\frac{n}{2}$ to its answer at the end, what is the distribution of its error? Is it now as good as LEP?

2 **Rounding to one decimal place.** If numbers are rounded to one decimal place, with uniformly distributed discarded digits, find the standard deviation of the total error in adding 10 000 numbers.

3 **The Vancouver index.** If the index was recalculated 3000 times a day and each truncation lost on average 0.0005, how much would it lose in a year of 250 trading days?

4 **The exact distribution.** For the sum of two rounding errors, find the exact density (a triangle). For three, sketch it. How quickly does it start to look normal?

Further reading

- **Nicholas Higham, *Accuracy and Stability of Numerical Algorithms*** (SIAM). The authoritative book on rounding errors; the opening chapter is accessible.
- **David Goldberg, "What every computer scientist should know about floating-point arithmetic"**, *ACM Computing Surveys* 23 (1991). A classic, free online.
- **Tim Harford, *How to Make the World Add Up*** (Bridge Street), on systematic errors in data.

13.7 Manufacturing Correlation

The question

In this question, $\mathrm{Corr}(U, V)$ denotes the product moment correlation coefficient between the random variables U and V, defined by

$$\mathrm{Corr}(U, V) \equiv \frac{\mathrm{Cov}(U, V)}{\sqrt{\mathrm{Var}(U)\mathrm{Var}(V)}}.$$

The independent random variables Z_1, Z_2 and Z_3 each have expectation 0 and variance 1. What is the value of $\mathrm{Corr}(Z_1, Z_2)$?

Let $Y_1 = Z_1$ and let

$$Y_2 = \rho_{12} Z_1 + (1 - \rho_{12}^2)^{\frac{1}{2}} Z_2,$$

where ρ_{12} is a given constant with $-1 < \rho_{12} < 1$. Find $\mathrm{E}(Y_2)$, $\mathrm{Var}(Y_2)$ and $\mathrm{Corr}(Y_1, Y_2)$.

Now let $Y_3 = aZ_1 + bZ_2 + cZ_3$, where a, b and c are real constants and $c \geq 0$. Given that $\mathrm{E}(Y_3) = 0$, $\mathrm{Var}(Y_3) = 1$, $\mathrm{Corr}(Y_1, Y_3) = \rho_{13}$ and $\mathrm{Corr}(Y_2, Y_3) = \rho_{23}$, express a, b and c in terms of ρ_{23}, ρ_{13} and ρ_{12}.

Given constants μ_i and σ_i, for $i = 1, 2$ and 3, give expressions in terms of the Y_i for random variables X_i such that $\mathrm{E}(X_i) = \mu_i$, $\mathrm{Var}(X_i) = \sigma_i^2$ and $\mathrm{Corr}(X_i, X_j) = \rho_{ij}$.

The hook

A bank wants to know how likely it is to lose money on a portfolio of shares. Share prices move together: when one falls, others tend to fall too. To simulate thousands of possible futures, the bank must generate random numbers that are correlated in exactly the right way. How do you manufacture random variables with prescribed correlations, starting from independent ones?

This question gives the recipe, step by step, for three variables. Each new variable is a combination of the previous independent ingredients, plus a fresh independent ingredient to make up the variance. The construction is a famous piece of linear algebra in disguise, the *Cholesky decomposition*, and it also reveals a hidden constraint: not every set of correlations is possible.

A note on the syllabus

Covariance and correlation were removed from the STEP specification in 2019. The covariance is $\mathrm{Cov}(U,V) = \mathrm{E}(UV) - \mathrm{E}(U)\mathrm{E}(V)$; it is zero for independent variables, and it is linear in each argument: $\mathrm{Cov}(aU+bV, W) = a\,\mathrm{Cov}(U,W) + b\,\mathrm{Cov}(V,W)$. Also $\mathrm{Var}(U) = \mathrm{Cov}(U,U)$. The correlation always lies between -1 and 1.

Before you start

Ask yourself Z_1 and Z_2 are independent. What is their covariance, and so their correlation?

Independent variables satisfy $\mathrm{E}(Z_1Z_2) = \mathrm{E}(Z_1)\,\mathrm{E}(Z_2)$. So the covariance is 0, and $\mathrm{Corr}(Z_1, Z_2) = 0$.

Working it through

STAGE 1 Two variables

$\mathrm{E}(Y_2) = 0$. Since Z_1 and Z_2 are independent with variance 1,

$$\mathrm{Var}(Y_2) = \rho_{12}^2 + (1-\rho_{12}^2) = 1, \qquad \mathrm{Cov}(Y_1, Y_2) = \mathrm{Cov}\left(Z_1, \rho_{12}Z_1 + \sqrt{1-\rho_{12}^2}\,Z_2\right) = \rho_{12}.$$

Both variances are 1, so $\mathrm{Corr}(Y_1, Y_2) = \rho_{12}$. ■

Ask yourself Think of Z_1 and Z_2 as perpendicular unit vectors. Where is Y_2? What is the cosine of the angle between Y_1 and Y_2?

$Y_2 = \rho_{12}Z_1 + \sqrt{1-\rho_{12}^2}\,Z_2$ is a unit vector making angle θ with $Z_1 = Y_1$, where $\cos\theta = \rho_{12}$. Correlation is the cosine of the angle between the variables, viewed as vectors (Figure 13.7.1).

STAGE 2 The third variable

Ask yourself Write down the three conditions $\mathrm{Var}(Y_3) = 1$, $\mathrm{Cov}(Y_1, Y_3) = \rho_{13}$, $\mathrm{Cov}(Y_2, Y_3) = \rho_{23}$ in terms of a, b, c.

Using linearity and independence:

$$a^2 + b^2 + c^2 = 1, \qquad a = \rho_{13}, \qquad \rho_{12}a + \sqrt{1-\rho_{12}^2}\,b = \rho_{23}.$$

So

$$a = \rho_{13}, \qquad b = \frac{\rho_{23} - \rho_{12}\rho_{13}}{\sqrt{1-\rho_{12}^2}}, \qquad c = \sqrt{1 - \rho_{13}^2 - \frac{(\rho_{23} - \rho_{12}\rho_{13})^2}{1-\rho_{12}^2}}.$$ ■

(Taking the non-negative square root, as $c \geq 0$.)

STAGE 3 Any means and variances

Scaling and shifting do not change correlations, so

$$X_i = \mu_i + \sigma_i Y_i \qquad (i = 1, 2, 3)$$

have means μ_i, variances σ_i^2 and correlations ρ_{ij}. ■

Pause and try The formula for c needs the quantity under the square root to be non-negative. Try $\rho_{12} = \rho_{13} = \rho_{23} = -0.6$. What goes wrong, and why does it make sense that three variables cannot all be strongly negatively correlated with each other?

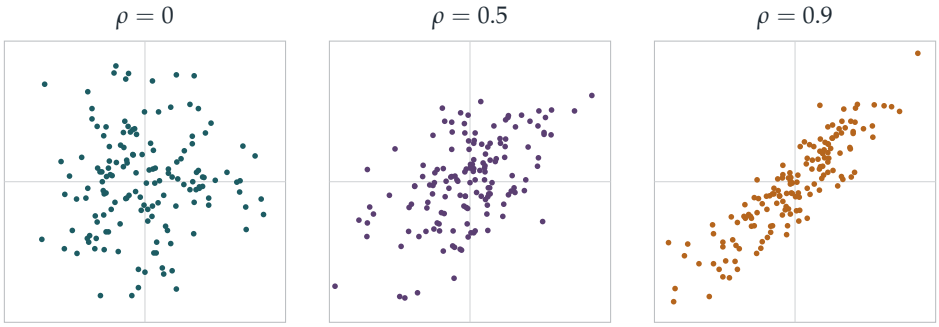


Figure 13.7.1 Samples of (Y_1, Y_2) made from independent standard normal Z_1, Z_2 by the recipe $Y_1 = Z_1$, $Y_2 = \rho Z_1 + \sqrt{1-\rho^2}\, Z_2$, for $\rho = 0$, 0.5 and 0.9. The same independent ingredients produce any correlation we choose.

Looking back

- **Build up one variable at a time.** Each new variable copied the right amount of the previous ingredients and added a fresh, independent one.
- **Covariance is linear.** All the conditions became simple equations in the coefficients.
- **Correlation is a cosine.** Thinking of standardised variables as unit vectors makes correlations into angles.

What's really going on

The Cholesky decomposition

In matrix form, the construction says $\mathbf{Y} = L\mathbf{Z}$ with a *lower triangular* matrix

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \rho_{12} & \sqrt{1-\rho_{12}^2} & 0 \\ a & b & c \end{pmatrix},$$

and the correlation matrix of $\mathbf{Y}$ is $R = LL^{\mathsf{T}}$. Writing a symmetric matrix as LL^{T} is the *Cholesky decomposition*, found by the French military engineer André-Louis Cholesky around 1910 for solving the equations of geodetic surveys (published after his death in the First World War). It works whenever R is *positive definite*, and the computation is exactly what we did by hand: each row is solved in turn. It is the standard way to simulate correlated variables in finance, engineering and climate modelling.

Which correlations are possible?

The number under the square root for c must be non-negative. Multiplying out, the condition is

$$1 - \rho_{12}^2 - \rho_{13}^2 - \rho_{23}^2 + 2\rho_{12}\rho_{13}\rho_{23} \geq 0,$$

which is the condition that the determinant of the correlation matrix is non-negative. Geometrically: three unit vectors with pairwise angles θ_{ij} (where $\cos\theta_{ij} = \rho_{ij}$) must actually fit together in space. If $\rho_{12} = \rho_{13} = \rho_{23} = -0.6$, the angles are all about 127°, and three vectors at 127° to each other would need angles summing to more than 360°: impossible. In general a matrix can be a correlation matrix only if it is positive semi-definite, all its eigenvalues non-negative (Section 7.2). When correlations are estimated separately from incomplete data, this can fail, and statisticians must find the nearest valid correlation matrix, a problem solved by Nicholas Higham in 2002.

Galton, Pearson and correlation

Francis Galton introduced correlation in 1888, while studying how the heights of parents and children are related, and Karl Pearson gave the product moment formula used in this question in 1896. Galton also noticed regression to the mean (Section 13.1). Pearson founded the world's first university statistics department, at University College London, in 1911. The warning that "correlation does not imply causation" is almost as old as correlation itself.

The thinking skill

Construct step by step, using fresh independent ingredients

To build objects with prescribed relationships, add them one at a time, each a combination of what came before plus something new and independent.

This is the Gram–Schmidt process of linear algebra, run in reverse, and the Cholesky decomposition. At university it appears in least squares, in the theory of multivariate normal distributions and in the simulation of random processes.

Try this next

1. **Impossible correlations.** Show that three random variables cannot all have pairwise correlation ρ unless $\rho \geq -\frac{1}{2}$. What is the corresponding statement for n variables?

2. **Check the determinant.** Show that $1 - \rho_{13}^2 - b^2 \geq 0$ is equivalent to $\det R \geq 0$, where R is the 3×3 correlation matrix.

3. **Portfolio risk.** Two shares have returns with standard deviations $\sigma_1 = 0.2$ and $\sigma_2 = 0.3$ and correlation ρ. Find the standard deviation of a portfolio half in each, for $\rho = -1$, 0 and 1. What does diversification achieve?

4. **Recover the ingredients.** Given Y_1, Y_2 as in the question, express Z_2 in terms of Y_1 and Y_2. (This is the Gram–Schmidt step.)

Further reading

- **Gilbert Strang,** ***Introduction to Linear Algebra*** (Wellesley-Cambridge), on positive definite matrices and the Cholesky factorisation.
- **Paul Glasserman,** ***Monte Carlo Methods in Financial Engineering*** (Springer). Simulating correlated variables in practice; advanced but the early chapters are readable.
- **Theodore Porter,** ***Karl Pearson: The Scientific Life in a Statistical Age*** (Princeton University Press).

What We Have Learned

We began by saying that probability goes from a model to data, and statistics goes from data back to the world. We asked three questions: is it real or is it noise, how should we estimate, and how does randomness enter the data? Here is what the chapter's seven questions showed.

1. Is it real or is it noise?

In Section 13.1, four headlines turned numbers into claims. Comparing totals without denominators, reading a trend into one year or one day, and hiding a trend inside an average were all errors, and the Poisson model gave a yardstick for the natural variation of counts. A rise of 8 standard deviations was almost certainly real; seven deaths on a Tuesday almost certainly was not. Regression to the mean and the need for fair comparisons completed the picture.

2. How should we estimate?

Three questions developed the theory of estimation.

- **Maximum likelihood** (Section 13.2): choose the value of the unknown that makes the observed data most probable. For capture–recapture it gave the natural estimate $\frac{200\times200}{11} \approx 3636$, with a broad likelihood showing how uncertain it was.
- **Bias and variance** (Section 13.3): two unbiased estimators of the number of submarines, one based on the maximum and one on the mean. The maximum, a sufficient statistic, was far more precise, with variance shrinking like $\frac{1}{n^2}$ rather than $\frac{1}{n}$. The same reasoning outwitted wartime intelligence about German tanks.
- **Efficiency and robustness** (Section 13.4): for uniform data the mean is three times as efficient as the median, but for heavy-tailed data the median wins, and it is robust to wild errors.

3. How does randomness enter the data?

Randomness can be added deliberately: randomised response (Section 13.5) protects each respondent's privacy while still estimating the population, provided each person randomises independently; a single shared coin would produce a two-humped mixture that tells us almost nothing. Randomness also enters through processing: rounding errors cancel and grow like $\sqrt{n}$, while truncation errors accumulate linearly, as the Vancouver Stock Exchange discovered (Section 13.6). And randomness can be engineered: independent ingredients, combined by the Cholesky construction, produce variables with any valid set of correlations (Section 13.7), but not every set of correlations is valid.

4. A toolkit for statistics

Method	Where you used it
Poisson variation	Headlines (13.1)
Likelihood and its maximum	Voles (13.2)
Unbiasedness and variance	Serial numbers (13.3); median (13.4)
Sufficient statistics	Serial numbers (13.3)
Normal approximation	13.4, 13.5, 13.6
Law of total variance	Randomised response (13.5)
Bias versus random error	Rounding (13.6)
Linearity of covariance	Correlation (13.7)

Where next

- Bayes' theorem (Section 12.2) is the basis of Bayesian statistics, the main alternative to the methods of this chapter.
- The central limit theorem (Section 12.24) justifies the normal approximations used throughout; the Cauchy distribution (Section 12.22) shows when they fail.
- Correlation matrices are positive semi-definite matrices, studied with the eigenvectors of Section 7.2.

If you want to go further before university

- **David Spiegelhalter, *The Art of Statistics***: the best introduction to statistical thinking for a general reader.
- **Tim Harford, *How to Make the World Add Up***: how to read numbers in the news.
- **John Rice, *Mathematical Statistics and Data Analysis***: a first university course in mathematical statistics.
- **Stephen Stigler, *The Seven Pillars of Statistical Wisdom***: the big ideas of statistics and where they came from.

George Box said that all models are wrong, but some are useful. This chapter has tried to show how to make models useful: by knowing how much variation to expect, by judging methods by how they behave over many repetitions, and by thinking about how the data came to be. These are the habits that turn numbers into knowledge.

A last word

This is the end of the book. Over thirteen chapters, we have used STEP questions as doorways into the mathematics of a first university course: number theory and analysis, calculus and complex numbers, algebra and linear algebra, differential equations and dynamics, combinatorics, mechanics, probability and statistics. Each question was an invitation to ask what lies behind it, and almost every time the answer was a whole

subject. If you have worked through even a few of these sections with a pencil in hand, asking the questions in the margins and trying the problems at the end, you have already begun to think like a mathematician. The rest of the journey is yours.

Answers to Try This Next

These are brief answers and hints, not full solutions: they give the result and the key step, so that you can check your own work. Try each exercise properly before looking.

13.1 Reading the Headlines

1. $1 - \sum_{k=0}^{6} \mathrm{e}^{-2.96} \frac{2.96^k}{k!} \approx 0.0315$, so seven or more deaths happen on about 11 days a year. $\mathrm{P}(X \geq 9) \approx 0.0035$: about one day a year.
2. Deaths are $150K$ with K Poisson with mean 1.2: mean 180, variance $150^2 \times 1.2 = 27\,000$, standard deviation about 164. A year with 125 deaths or fewer is a year with no crashes, which has probability $e^{-1.2} \approx 0.30$: entirely ordinary. But exactly 125 deaths is impossible in this model, since deaths come only in multiples of 150. The mean and standard deviation do not show this: clustering makes the distribution lumpy, with all its probability on 0, 150, 300, and so on.
3. $z = \frac{1349.5-1080}{33} \approx 8.2$: a probability of order 10^{-16}. A steady rate cannot explain 1350; something really changed (or the way deaths were counted did).
4. For example: (i) "Totals alone cannot say whether flying or walking is safer"; (ii) "Air deaths in 1990 within the normal range"; (iii) "Seven pedestrian deaths on Tuesday: a bad day, but one that happens about eleven times a year"; (iv) "Pedestrian deaths up sharply in 1990: cause must be investigated".

13.2 Counting Voles

1. $\frac{p_N}{p_{N-1}} = \frac{(N-M)(N-C)}{N(N-M-C+R)}$, which is at least 1 exactly when $NR \leq MC$. So p_N increases up to $N = \frac{MC}{R}$ and then decreases, and the maximum is at its integer part.
2. Count committees of c from a women and b men by the number j of women: $\binom{a}{j}\binom{b}{c-j}$ for each j.
3. $\frac{\mathrm{d}}{\mathrm{d}p}\big(h \ln p + (n-h)\ln(1-p)\big) = \frac{h}{p} - \frac{n-h}{1-p} = 0$ gives $p = \frac{h}{n}$.
4. With $R = 0$, p_N increases for ever, and the estimate is infinite. The method only works if the second sample contains enough marked animals; with few recaptures the estimate is very unstable.

13.3 The German Tank Problem

1. With M the maximum, $\mathrm{E}(M) = \frac{nN}{n+1}$ and $\mathrm{Var}(M) = \frac{nN^2}{(n+1)^2(n+2)}$, so $\mathrm{E}(M-N)^2 = \frac{2N^2}{(n+1)(n+2)}$. This is $\frac{2n}{n+1}$ times $\mathrm{Var}(Z_1) = \frac{N^2}{n(n+2)}$: worse. The best multiple is

$c = \frac{\mathrm{NE}(M)}{\mathrm{E}(M^2)} = \frac{n+2}{n+1}$, with mean squared error $\frac{N^2}{(n+1)^2}$, slightly better than Z_1.

2. $\mathrm{E}(\min) = \frac{N}{n+1}$, and min has the same variance as M, so $\mathrm{Var}\big((n+1)\min\big) = \frac{nN^2}{n+2}$. This does not shrink as n grows: a poor estimator.
3. The maximum is at most m exactly when all k numbers come from $\{1, \ldots, m\}$. For $k = 1$ the estimator is $2X - 1$, with mean $2 \cdot \frac{N+1}{2} - 1 = N$.
4. $98 + \frac{98}{5} - 1 = 116.6$: about 117 tanks.

13.4 Mean or Median?

1. Choose which value is the median ($2n+1$ ways) and which n of the rest are below it ($\binom{2n}{n}$ ways). The probability is $(2n+1)\binom{2n}{n} y^n (1-y)^n \, \delta y = \frac{(2n+1)!}{(n!)^2} y^n (1-y)^n \, \delta y$.
2. $\mathrm{Var}(Y) = \frac{1}{4(2n+3)} = \frac{1}{20}$ and $\mathrm{Var}(\bar{X}) = \frac{1}{12(2n+1)} = \frac{1}{36}$. The mean is better.
3. $\mathrm{Var}(\max) = \mathrm{Var}(\min) = \frac{m}{(m+1)^2(m+2)}$ and $\mathrm{Cov}(\max, \min) = \frac{1}{(m+1)^2(m+2)}$, giving $\frac{1}{2(m+1)(m+2)}$. This shrinks like $\frac{1}{m^2}$, much faster than the mean's $\frac{1}{12m}$: for uniform data the extremes carry the information, as with the tanks.
4. The mean is about 18.6, wrecked by one typing error; the median is 10.1, unaffected.

13.5 Asking Awkward Questions

1. Let $t = ap + bq$, the probability of a true answer. "Yes" has probability $\frac{1}{2}t + \frac{1}{2}(1-t) = \frac{1}{2}$, so $\mathrm{P}(\text{heads} \mid \text{yes}) = t$, which is $\frac{1}{2}$ exactly when $2t = 1$. If the same twin answers twice, "yes" then "no" has probability $pa(1-a) + qb(1-b)$ whether the coin showed heads or tails, so the posterior is $\frac{1}{2}$. Two "yes" answers have probabilities $pa^2 + qb^2$ (heads) and $p(1-a)^2 + q(1-b)^2$ (tails), which are equal exactly when $2(ap + bq) = 1$.
2. $\mathrm{P}(\text{yes}) = \frac{3}{4}\pi + \frac{1}{4}(1-\pi)$, so $\hat{\pi} = \frac{0.4 - 0.25}{0.5} = 0.3$. The standard error is $\frac{\sqrt{0.4 \times 0.6 / 400}}{0.5} \approx 0.049$.
3. $\mathrm{E}(Y^2) = \sum_c \mathrm{P}(C = c)\big(\mathrm{Var}(Y \mid c) + \mathrm{E}(Y \mid c)^2\big)$; subtracting $\mathrm{E}(Y)^2 = \big(\sum_c \mathrm{P}(C = c)\mathrm{E}(Y \mid c)\big)^2$ gives $\mathrm{E}(\mathrm{Var}(Y \mid C)) + \mathrm{Var}(\mathrm{E}(Y \mid C))$.
4. Z_2 is close to $\frac{n}{6}$ (heads) or $\frac{5n}{6}$ (tails), each with probability $\frac{1}{2}$. So $\mathrm{P}(Z_2 > \frac{n}{2}) \to \frac{1}{2}$, although Z_2 is almost never near $\frac{n}{2}$.

13.6 Errors That Add Up

1. Each truncation error is uniform on $(-1, 0]$; adding $\frac{1}{2}$ to each makes it uniform on $(-\frac{1}{2}, \frac{1}{2}]$, exactly like rounding. So the corrected VOZ is as good as LEP.
2. Each error is uniform on $(-0.05, 0.05)$, with variance $\frac{0.1^2}{12}$, so the total has standard deviation $\sqrt{10\,000 \times 0.01/12} \approx 2.9$.

3. $3000 \times 250 \times 0.0005 = 375$ points.
4. For two errors, $1 - |x|$ on $[-1, 1]$. For three: $\frac{3}{4} - x^2$ for $|x| \le \frac{1}{2}$ and $\frac{1}{2}\left(\frac{3}{2} - |x|\right)^2$ for $\frac{1}{2} \le |x| \le \frac{3}{2}$, already close to a normal curve.

13.7 Manufacturing Correlation

1. For standardised variables, $\operatorname{Var}(X_1 + X_2 + X_3) = 3 + 6\rho \ge 0$, so $\rho \ge -\frac{1}{2}$. For n variables, $n + n(n-1)\rho \ge 0$ gives $\rho \ge -\frac{1}{n-1}$.
2. With $b = \frac{\rho_{23} - \rho_{12}\rho_{13}}{\sqrt{1-\rho_{12}^2}}$, $(1-\rho_{12}^2)(1-\rho_{13}^2-b^2) = 1 - \rho_{12}^2 - \rho_{13}^2 - \rho_{23}^2 + 2\rho_{12}\rho_{13}\rho_{23} = \det R$. Since $1 - \rho_{12}^2 > 0$, the two conditions are equivalent.
3. $\operatorname{Var} = \frac{1}{4}(0.04 + 0.09 + 0.12\rho)$: standard deviations 0.05, 0.18 and 0.25 for $\rho = -1$, 0, 1. Unless the shares are perfectly correlated, the mix is less risky than the average of the two risks (0.25).
4. $Z_2 = \dfrac{Y_2 - \rho_{12} Y_1}{\sqrt{1-\rho_{12}^2}}$.

Beyond STEP

STEP is the hardest mathematics examination most school students ever sit. It is also one of the best: its questions are short, unfamiliar and deep, and the best of them are small pieces of university mathematics in disguise.

This book takes the most interesting STEP questions, from the specimen papers that preceded the first examination in 1987 to the papers of 2025, and uses each one as a doorway. Every question is solved carefully, with the questions a good student asks along the way, and then the book steps through the door: where the idea comes from, what it becomes at university, and what to read next.

It is written for strong 17- and 18-year-olds preparing for STEP who want more than practice papers: to start thinking like a mathematics undergraduate, and to find out what is waiting for them.

- 13 chapters across pure mathematics, mechanics, probability and statistics
- 136 past STEP questions, each with a guided solution
- The university mathematics behind every question
- Follow-up exercises with answers, and further reading

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